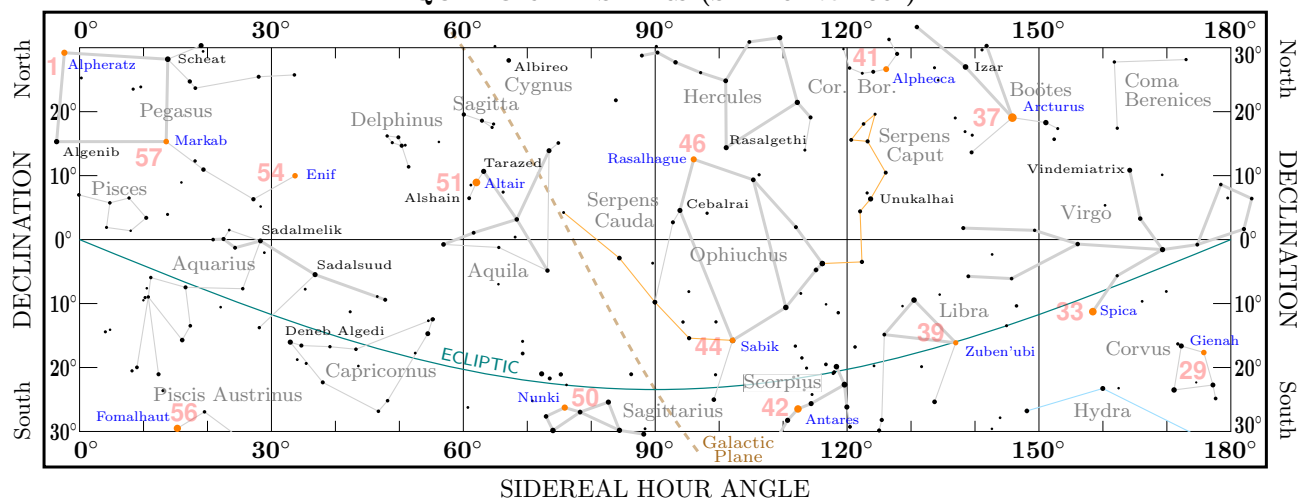


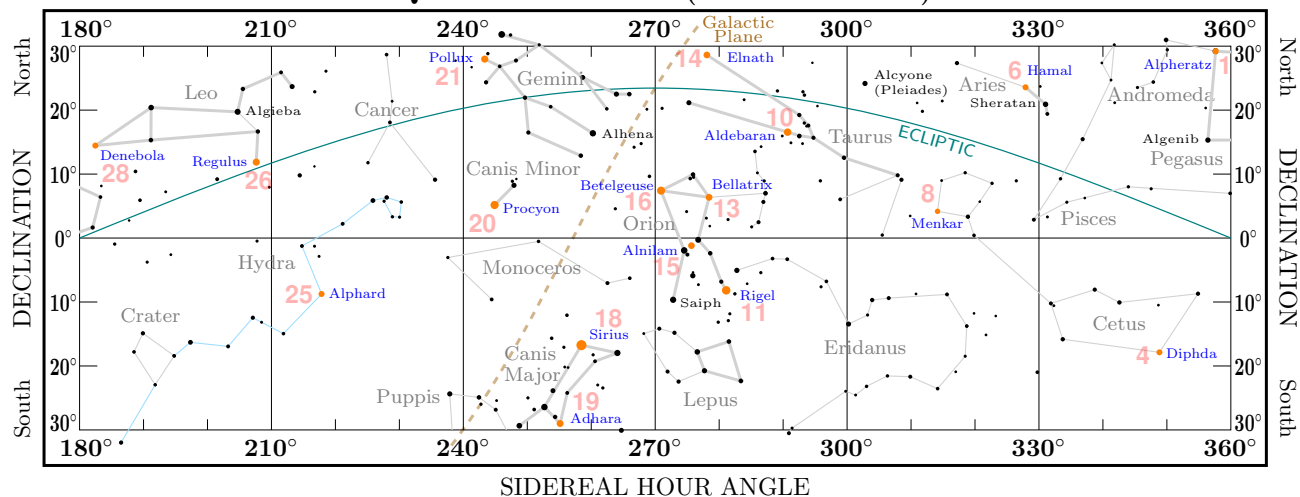
GENERATED USING SKYFIELD

<http://rhodesmill.org/skyfield/>

EQUATORIAL STARS (SHA 0° to 180°)



EQUATORIAL STARS (SHA 180° to 360°)



THE NAUTICAL ALMANAC

2024

Author: Andrew BAUER

Original concept from: Enno RODEGERDTS

August 5, 2024

Disclaimer: These are computer generated tables - use them at your own risk. The accuracy has been randomly checked with JPL HORIZONS System, but cannot be guaranteed. The author claims no liability for any consequences arising from use of these tables. Besides, this publication only contains the 'daily pages' of the Nautical Almanac: an official version of the Nautical Almanac is indispensable.

Information in the data page footers

Information pertaining to the IERS EOP data has been added to the odd data page footers if using MiKTeX or TeX Live (2020 or later). The International Earth Rotation Service (IERS) provides accurate data (updated weekly) on the Earth Orientation Parameters (EOP).

Earth's speed of rotation is not constant, i.e. the day length fluctuates.¹ This is due to *internal torques* caused by relative movements and mass redistribution of Earth's core, mantle, oceans, atmosphere, and cryosphere. This has an immediate impact on the GHA values of all celestial objects.

The IERS monitors and measures several parameters taking the actual speed of Earth's rotation into account. Their measured data begins on 2nd January 1973. Predictive data begins following the last day of (obtained) data and extends about 360 days into the future. (The IERS results are published with a delay of about 18-hours between the date of publication and the last available date with measured EOP.²) These Nautical Almanac daily pages take the (measured or predicted) UT1-UTC values into account providing highly accurate navigational data especially if the predictions are fairly recent.

As long as either measured or predicted data is available the footer will show:

[IERS Earth Orientation data as of dd-mmm-yyyy](#)

This indicates that IERS EOP data is in use - older dates are measured; newer dates are predictions.

If the final date of IERS prediction data is on the current data page, the footer shows:

[IERS Earth Orientation predictions end dd-mmm-yyyy](#)

Pages with dates beyond the final date of IERS prediction data have the following footer:

[No IERS EOP prediction data available](#)

Skyfield then defaults to using the ΔT and leap second files that ship with Skyfield internally.

The footers mentioned are only displayed as long as `'useIERS = True'` is set in `config.py` to enable use of IERS EOP data.

Brief historical overview

The story begins with the XEphem astronomical library, which is declared 'end of life' by its author, Elwood Charles Downey, as no further updates are planned. He generously gave permission for use of XEphem code in Ephem (also known as Pyephem), an astronomical library authored by Brandon Rhodes. Enno Rodergerdts (<https://sv-inua.net/>) created the original Nautical Almanac 'daily pages' in Pyalmanac using Python 2 and LaTeX. After contacting him I obtained permission for its future enhancement and maintenance. Pyalmanac uses Ephem.

Meanwhile Brandon Rhodes was working on a far more sophisticated astronomical library, Skyfield. This was 'state of the art' and clearly surpassed the 'Jean Meeus'-based Pyephem/Ephem. Skyfield uses NASA's NAIF (Navigation and Ancillary Information Facility) SPICE algorithms. The results agree with those from the HORIZONS System (*operated by NASA JPL (Jet Propulsion Laboratory) SSD (Solar System Dynamics) group, not by NAIF*). This in turn implies that celestial positions calculated by Skyfield agree with those generated by the United States Naval Observatory and their *Astronomical Almanac* to within 0.0005 arcseconds (half a milliarcsecond).

Pyephem was then in 'maintenance mode'. Clearly Pyalmanac needed adaptation to use Skyfield, and thus SFalmanac was born. However its performance was poor regarding the calculation of 'events' such as: sunrise, sunset, moonrise, moonset, civil twilight start/end and nautical twilight start/end. An interim (faster) solution was required.

A hybrid application, originally named Skyalmanac, was developed using Ephem to calculate 'events' and Skyfield for the rest. This was indeed much faster at the cost of poorer 'event time' data. It took a while to find a better solution: multiprocessing, which was built into SFalmanac. This now could compare to the execution times in Pyalmanac but with improved data.

New functionality was added to SFalmanac, e.g. lunar phase as a graphic; Lunar Distance tables and charts. The original Skyalmanac is deprecated and has now been replaced with the latest SFalmanac code, so Skyalmanac and SFalmanac are now identical apart from the name. Since April 2019 <http://thenauticalalmanac.com> has been publishing Celestial Navigation related material with software provided here.

¹https://en.wikipedia.org/wiki/Day_length_fluctuations

²<https://hpiers.obspm.fr/eoppc/bul/bulb/explanatory.html>

January 01, 02, 03 UT (Mon., Tue., Wed.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	100°09.1	219°12.0	S18°46.2	193°05.8	S23°57.7	66°27.9	N12°15.8	124°22.4	S11°50.3					Alpheratz	357°35.7	29°13.5	
1	115°11.5	234°11.4	46.8	208°06.2	57.7	81°30.4	15.9	139°24.6	50.3					Ankaa	353°08.0	-42°10.8	
2	130°14.0	249°10.7	47.5	223°06.7	57.8	96°32.9	15.9	154°26.9	50.2					Schedar	349°32.0	56°40.4	
3	145°16.5	264°10.0	· · 48.1	238°07.1	· · 57.9	111°35.3	· · 15.9	169°29.1	· · 50.1					Diphda	348°48.1	-17°51.5	
4	160°18.9	279°09.4	48.8	253°07.6	57.9	126°37.8	15.9	184°31.4	50.0					Achernar	335°20.6	-57°07.2	
5	175°21.4	294°08.7	49.4	268°08.0	58.0	141°40.2	15.9	199°33.6	49.9					Hamal	327°52.0	23°34.6	
6	190°23.9	309°08.0	S18°50.1	283°08.4	S23°58.0	156°42.7	N12°15.9	214°35.9	S11°49.8					Polaris	314°08.6	89°22.2	
7	205°26.3	324°07.3	50.8	298°08.9	58.1	171°45.2	15.9	229°38.2	49.8					Acamar	315°12.2	-40°12.7	
8	220°28.8	339°06.7	51.4	313°09.3	58.2	186°47.6	16.0	244°40.4	49.7					Menkar	314°06.8	4°11.0	
9	235°31.3	354°06.0	· · 52.1	328°09.7	· · 58.2	201°50.1	· · 16.0	259°42.7	· · 49.6					Mirfak	308°29.1	49°56.9	
10	250°33.7	9°05.3	52.7	343°10.2	58.3	216°52.5	16.0	274°44.9	49.5					Aldebaran	290°40.3	16°33.5	
11	265°36.2	24°04.6	53.4	358°10.6	58.3	231°55.0	16.0	289°47.2	49.4					Rigel	281°04.4	-8°10.5	
12	280°38.6	39°04.0	S18°54.0	13°11.0	S23°58.4	246°57.5	N12°16.0	304°49.4	S11°49.3					Capella	280°22.7	46°01.4	
13	295°41.1	54°03.3	54.7	28°11.5	58.5	261°59.9	16.0	319°51.7	49.3					Bellatrix	278°23.5	6°22.3	
14	310°43.6	69°02.6	55.3	43°11.9	58.5	277°02.4	16.1	334°53.9	49.2					Elnath	278°02.6	28°37.7	
15	325°46.0	84°01.9	· · 56.0	58°12.3	· · 58.6	292°04.8	· · 16.1	349°56.2	· · 49.1					Alnilam	275°38.3	-1°11.2	
16	340°48.5	99°01.3	56.6	73°12.8	58.6	307°07.3	16.1	4°58.4	49.0					Betelgeuse	270°52.7	7°24.7	
17	355°51.0	114°00.6	57.3	88°13.2	58.7	322°09.7	16.1	20°00.7	48.9					Canopus	263°52.2	-52°42.5	
18	10°53.4	128°59.9	S18°57.9	103°13.7	S23°58.7	337°12.2	N12°16.1	35°02.9	S11°48.8					Sirius	258°26.6	-16°44.9	
19	25°55.9	143°59.2	58.5	118°14.1	58.8	352°14.6	16.1	50°05.2	48.8					Adhara	255°06.1	-29°00.2	
20	40°58.4	158°58.6	59.2	133°14.5	58.8	7°17.1	16.2	65°07.4	48.7					Procyon	244°51.4	5°09.8	
21	56°00.8	173°57.9	18°59.8	148°15.0	· · 58.9	22°19.5	· · 16.2	80°09.7	· · 48.6					Pollux	243°18.0	27°58.0	
22	71°03.3	188°57.2	19°00.5	163°15.4	58.9	37°22.0	16.2	95°12.0	48.5					Avior	234°14.5	-59°35.0	
23	86°05.7	203°56.5	01.1	178°15.8	59.0	52°24.5	16.2	110°14.2	48.4					Suhail	222°46.6	-43°31.6	
Mer.pass. 17:17			ν -0.7' d 0.7' m -4.02			ν 0.4' d 0.1' m 1.37			ν 2.5' d 0.0' m -2.59			ν 2.3' d -0.1' m 0.95					

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec								
0	101°08.2	218°55.8	S19°01.8	193°16.3	S23°59.1	67°26.9	N12°16.2	125°16.5	S11°48.3					Regulus	207°35.1	11°51.0	
1	116°10.7	233°55.2	02.4	208°16.7	59.1	82°29.4	16.2	140°18.7	48.3					Dubhe	193°41.7	61°37.0	
2	131°13.1	248°54.5	03.0	223°17.1	59.2	97°31.8	16.3	155°21.0	48.2					Denebola	182°25.7	14°26.2	
3	146°15.6	263°53.8	· · 03.7	238°17.6	· · 59.2	112°34.3	· · 16.3	170°23.2	· · 48.1					Gienah	175°44.4	-17°40.4	
4	161°18.1	278°53.1	04.3	253°18.0	59.3	127°36.7	16.3	185°25.5	48.0					Acrux	173°01.0	-63°13.6	
5	176°20.5	293°52.4	04.9	268°18.4	59.3	142°39.2	16.3	200°27.7	47.9					Gacrux	171°52.6	-57°14.5	
6	191°23.0	308°51.7	S19°05.6	283°18.9	S23°59.4	157°41.6	N12°16.3	215°30.0	S11°47.8					Alioth	166°13.7	55°49.5	
7	206°25.5	323°51.1	06.2	298°19.3	59.4	172°44.1	16.3	230°32.2	47.8					Spica	158°23.3	-11°17.1	
8	221°27.9	338°50.4	06.8	313°19.7	59.5	187°46.5	16.4	245°34.5	47.7					Alkaid	152°52.8	49°11.3	
9	236°30.4	353°49.7	· · 07.5	328°20.2	· · 59.5	202°49.0	· · 16.4	260°36.7	· · 47.6					Hadar	148°37.5	-60°29.0	
10	251°32.9	8°49.0	08.1	343°20.6	59.5	217°51.4	16.4	275°39.0	47.5					Menkent	147°58.8	-36°29.1	
11	266°35.3	23°48.3	08.7	358°21.0	59.6	232°53.9	16.4	290°41.2	47.4					Arcturus	145°48.8	19°03.3	
12	281°37.8	38°47.6	S19°09.4	13°21.5	S23°59.6	247°56.3	N12°16.4	305°43.5	S11°47.3					Rigel Kent.	139°41.8	-60°55.8	
13	296°40.2	53°46.9	10.0	28°21.9	59.7	262°58.8	16.5	320°45.7	47.2					Kochab	137°20.4	74°03.1	
14	311°42.7	68°46.2	10.6	43°22.3	59.7	278°01.2	16.5	335°48.0	47.2					Zuben'ubi	136°57.2	-16°08.4	
15	326°45.2	83°45.6	· · 11.3	58°22.8	· · 59.8	293°03.7	· · 16.5	350°50.2	· · 47.1					Alphecca	126°04.7	26°37.9	
16	341°47.6	98°44.9	11.9	73°23.2	59.8	308°06.1	16.5	5°52.5	47.0					Antares	112°17.2	-26°29.0	
17	356°50.1	113°44.2	12.5	88°23.6	59.9	323°08.6	16.5	20°54.7	46.9					Atria	107°12.7	-69°04.1	
18	11°52.6	128°43.5	S19°13.1	103°24.1	S23°59.9	338°11.0	N12°16.6	35°57.0	S11°46.8					Sabik	102°04.1	-15°45.3	
19	26°55.0	143°42.8	13.8	118°24.5	24°00.0	353°13.4	16.6	50°59.2	46.7					Shaula	96°11.9	-37°07.2	
20	41°57.5	158°42.1	14.4	133°24.9	00.0	8°15.9	16.6	66°01.5	46.7					Rasalhague	95°59.7	12°32.5	
21	57°00.0	173°41.4	· · 15.0	148°25.3	· · 00.0	23°18.3	· · 16.6	81°03.7	· · 46.6					Eltanin	90°43.1	51°29.0	
22	72°02.4	188°40.7	15.6	163°25.8	00.1	38°20.8	16.6	96°06.0	46.5					Kaus Aust.	83°34.1	-34°22.4	
23	87°04.9	203°40.0	16.2	178°26.2	00.1	53°23.2	16.7	111°08.2	46.4					Vega	80°34.2	38°48.3	
Mer.pass. 17:13			ν -0.7' d 0.6' m -4.01			ν 0.4' d 0.1' m 1.37			ν 2.5' d 0.0' m -2.58			ν 2.3' d -0.1' m 0.95					

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec								
0	102°07.4	218°39.3	S19°16.9	193°26.6	S24°00.2	68°25.7	N12°16.7	126°10.5	S11°46.3					Jan 01 Mon	SHA	Mer.pass	
1	117°09.8	233°38.6	17.5	208°27.1	00.2	83°28.1	16.7	141°12.7	46.2					Venus	119°03.0	09:24	
2	132°12.3	248°37.9	18.1	223°27.5	00.3	98°30.6	16.7	156°15.0	46.1					Mars	92°56.7	11:07	
3	147°14.7	263°37.2	· · 18.7	238°27.9	· · 00.3	113°33.0	· · 16.7	171°17.2	· · 46.1					Jupiter	326°18.9	19:31	
4	162°17.2	278°36.6	19.3	253°28.4	00.3	128°35.4	16.8	186°19.5	46.0					Saturn	24°13.3	15:40	
5	177°19.7	293°35.9	20.0	268°28.8	00.4	143°37.9	16.8	201°21.7	45.9								
6	192°22.1	308°35.2	S19°20.6	283°29.2	S24°00.4	158°40.3	N12°16.8	216°24.0	S11°45.8								
7	207°24.6	323°34.5	21.2	298°29.7	00.4	173°42.8	16.8	231°26.2	45.7					Jan 02 Tue	SHA	Mer.pass	
8	222°27.1	338°33.8	21.8	313°30.1	00.5	188°45.2	16.8	246°28.5	45.6					Venus	117°47.6	09:25	
9	237°29.5	353°33.1	· · 22.4	328°30.5	· · 00.5												

Sun							Moon							
Mon	GHA	Dec	GHA	ν	Dec	d	HP	Lat.	Twilight		Sunrise	Sunset	Twilight	
									Naut.	Civil			Civil	Naut.
0	179°13.8	S23°03.5	301°02.0	16.1'	N12°37.6	-12.7'	54.1'	N 72°	08:23	10:41	■	■	13:27	15:45
1	194°13.5	03.3	315°37.0	16.1'	12°24.9	-12.8'	54.1'	N 70°	08:05	09:49	■	■	14:19	16:03
2	209°13.2	03.1	330°12.2	16.2'	12°12.1	-12.8'	54.1'	68°	07:50	09:16	■	■	14:52	16:18
3	224°12.9	· · 02.9	344°47.3	16.2'	11°59.3	-12.9'	54.1'	66°	07:37	08:53	10:27	13:41	15:15	16:31
4	239°12.6	02.7	359°22.6	16.3'	11°46.5	-12.9'	54.1'	64°	07:26	08:34	09:49	14:19	15:34	16:41
5	254°12.3	02.5	13°57.8	16.3'	11°33.6	-12.9'	54.1'	62°	07:17	08:18	09:23	14:45	15:49	16:51
6	269°12.0	S23°02.3	28°33.1	16.4'	N11°20.6	-13.0'	54.1'	60°	07:09	08:05	09:02	15:06	16:02	16:59
7	284°11.7	02.1	43°08.5	16.4'	11°07.6	-13.0'	54.1'	N 58°	07:02	07:54	08:45	15:23	16:14	17:06
8	299°11.4	01.9	57°43.9	16.5'	10°54.6	-13.1'	54.1'	56°	06:56	07:44	08:31	15:37	16:24	17:12
9	314°11.2	· · 01.7	72°19.4	16.5'	10°41.6	-13.1'	54.1'	54°	06:50	07:35	08:19	15:49	16:32	17:18
10	329°10.9	01.5	86°54.9	16.5'	10°28.5	-13.1'	54.1'	52°	06:44	07:28	08:08	16:00	16:40	17:24
11	344°10.6	01.3	101°30.4	16.6'	10°15.4	-13.2'	54.1'	50°	06:39	07:20	07:58	16:09	16:47	17:29
12	359°10.3	S23°01.1	116°06.0	16.6'	N10°02.2	-13.2'	54.1'	45°	06:28	07:05	07:38	16:29	17:03	17:40
13	14°10.0	00.9	130°41.6	16.7'	09°49.0	-13.2'	54.1'	N 40°	06:18	06:52	07:22	16:46	17:16	17:50
14	29°09.7	00.7	145°17.2	16.7'	09°35.8	-13.3'	54.1'	35°	06:08	06:40	07:08	17:00	17:28	17:59
15	44°09.4	· · 00.5	159°52.9	16.7'	09°22.5	-13.3'	54.1'	30°	06:00	06:30	06:56	17:12	17:38	18:08
16	59°09.1	00.3	174°28.7	16.8'	09°09.2	-13.3'	54.1'	20°	05:44	06:11	06:35	17:32	17:56	18:24
17	74°08.8	23°00.1	189°04.4	16.8'	08°55.9	-13.4'	54.1'	N 10°	05:28	05:54	06:17	17:50	18:13	18:39
18	89°08.5	S22°59.9	203°40.2	16.8'	N08°42.5	-13.4'	54.1'	0°	05:11	05:38	06:00	18:08	18:30	18:56
19	104°08.2	59.7	218°16.1	16.9'	08°29.1	-13.4'	54.1'	S 10°	04:53	05:20	05:43	18:25	18:48	19:15
20	119°07.9	59.5	232°51.9	16.9'	08°15.7	-13.5'	54.1'	20°	04:31	05:00	05:24	18:43	19:08	19:37
21	134°07.6	· · 59.3	247°27.8	16.9'	08°02.2	-13.5'	54.1'	30°	04:02	04:35	05:03	19:05	19:32	20:05
22	149°07.3	59.1	262°03.7	17.0'	07°48.8	-13.5'	54.1'	35°	03:44	04:20	04:50	19:17	19:47	20:23
23	164°07.0	58.9	276°39.7	17.0'	07°35.3	-13.5'	54.1'	40°	03:21	04:03	04:35	19:32	20:05	20:46
SD = 16.3' d = -0.2'							SD = 14.8'							
Tue	GHA	Dec	GHA	ν	Dec	d	HP	Lat.	Moonrise		Moonset			
									Mon	Tue	Wed	Mon	Tue	Wed
0	179°06.7	S22°58.7	291°15.7	17.0'	N07°21.7	-13.6'	54.1'	N 72°	20:47	22:46	· · ·	12:26	11:51	11:20
1	194°06.4	58.5	305°51.7	17.0'	07°08.2	-13.6'	54.1'	N 70°	21:00	22:49	· · ·	12:11	11:45	11:21
2	209°06.1	58.2	320°27.7	17.1'	06°54.6	-13.6'	54.1'	68°	21:10	22:51	· · ·	11:58	11:40	11:22
3	224°05.9	· · 58.0	335°03.8	17.1'	06°41.0	-13.6'	54.1'	66°	21:18	22:53	· · ·	11:48	11:35	11:23
4	239°05.6	57.8	349°39.9	17.1'	06°27.3	-13.7'	54.1'	64°	21:25	22:55	· · ·	11:40	11:32	11:23
5	254°05.3	57.6	4°16.0	17.1'	06°13.7	-13.7'	54.1'	62°	21:31	22:56	· · ·	11:33	11:28	11:24
6	269°05.0	S22°57.4	18°52.1	17.1'	N06°00.0	-13.7'	54.1'	60°	21:36	22:58	· · ·	11:26	11:26	11:25
7	284°04.7	57.2	33°28.2	17.2'	05°46.3	-13.7'	54.1'	N 58°	21:41	22:59	· · ·	11:21	11:23	11:25
8	299°04.4	57.0	48°04.4	17.2'	05°32.6	-13.7'	54.1'	56°	21:45	23:00	· · ·	11:16	11:21	11:26
9	314°04.1	· · 56.7	62°40.6	17.2'	05°18.8	-13.8'	54.1'	54°	21:48	23:01	· · ·	11:11	11:19	11:26
10	329°03.8	56.5	77°16.8	17.2'	05°05.0	-13.8'	54.1'	52°	21:51	23:01	· · ·	11:07	11:17	11:26
11	344°03.5	56.3	91°53.0	17.2'	04°51.3	-13.8'	54.1'	50°	21:54	23:02	· · ·	11:03	11:15	11:27
12	359°03.2	S22°56.1	106°29.2	17.2'	N04°37.4	-13.8'	54.2'	45°	22:01	23:04	· · ·	10:55	11:12	11:27
13	14°02.9	55.9	121°05.4	17.2'	04°23.6	-13.8'	54.2'	N 40°	22:06	23:05	· · ·	10:48	11:08	11:28
14	29°02.6	55.6	135°41.7	17.3'	04°09.8	-13.9'	54.2'	35°	22:11	23:06	· · ·	10:42	11:06	11:28
15	44°02.3	· · 55.4	150°17.9	17.3'	03°55.9	-13.9'	54.2'	30°	22:15	23:07	· · ·	10:37	11:03	11:29
16	59°02.1	55.2	164°54.2	17.3'	03°42.0	-13.9'	54.2'	20°	22:21	23:09	23:57	10:28	10:59	11:30
17	74°01.8	55.0	179°30.5	17.3'	03°28.1	-13.9'	54.2'	N 10°	22:27	23:11	23:54	10:20	10:55	11:30
18	89°01.5	S22°54.7	194°06.8	17.3'	N03°14.2	-13.9'	54.2'	0°	22:33	23:12	23:51	10:12	10:52	11:31
19	104°01.2	54.5	208°43.1	17.3'	03°00.3	-13.9'	54.2'	S 10°	22:39	23:14	23:49	10:05	10:48	11:31
20	119°00.9	54.3	223°19.4	17.3'	02°46.4	-13.9'	54.2'	20°	22:45	23:15	23:46	09:56	10:45	11:32
21	134°00.6	· · 54.1	237°55.7	17.3'	02°32.4	-14.0'	54.2'	30°	22:51	23:17	23:43	09:47	10:40	11:33
22	149°00.3	53.8	252°32.0	17.3'	02°18.5	-14.0'	54.2'	35°	22:55	23:18	23:41	09:41	10:38	11:33
23	164°00.0	53.6	267°08.3	17.3'	02°04.5	-14.0'	54.2'	40°	22:59	23:19	23:39	09:35	10:35	11:34
SD = 16.3' d = -0.2'							SD = 14.8'							
Wed	GHA	Dec	GHA	ν	Dec	d	HP	Day	Sun		Mer.	Moon		Age
									Eqn.of Time	Pass	Mer.Pass.			
									00 ^h	12 ^h	Pass	Upper	Lower	20-22
									mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	78-61%
0	178°59.7	S22°53.4	281°44.6	17.3'	N01°50.5	-14.0'	54.2'	01	03:05	03:19	12:03	04:03	16:23	
1	193°59.4	53.1	296°20.9	17.3'	01°36.5	-14.0'	54.3'	02	03:33	03:47	12:04	04:42	17:02	
2	208°59.1	52.9	310°57.2	17.3'	01°22.5	-14.0'	54.3'	03	04:01	04:15	12:04	05:21	17:41	
3	223°58.9	· · 52.7	325°33.5	17.3'	01°08.5	-14.0'	54.3'							
4	238°58.6	52.5	340°09.8	17.3'	00°54.5	-14.0'	54.3'							
5	253°58.3	52.2	354°46.0	17.3'	00°40.4	-14.0'	54.3'							
6	268°58.0	S22°52.0	9°22.3	17.3'	N00°26.4	-14.0'	54.3'							
7	283°57.7	51.7	23°58.6	17.3'	N00°12.3	-14.1'	54.3'							
8	298°57.4	51.5	38°34.8	17.2'	S00°01.7	14.1'	54.3'							
9	313°57.1	· · 51.3	53°11.1	17.2'	00°15.8	14.1'	54.3'							
10	328°56.8	51.0	67°47.3	17.2'	00°29.8	14.1'	54.4'							
11	343°56.5	50.8	82°23.5	17.2'	00°43.9	14.1'	54.4'							
12	358°56.3	S22°50.6	96°59.8	17.2'	S00°58.0	14.1'	54.4'							
13	13°56.0	50.3	111°36.0	17.2'	01°12.1	14.1'	54.4'							
14	28°55.7	50.1	126°12.1	17.2'	01°26.1	14.1'	54.4'							
15	43°55.4	· · 49.8	140°48.3	17.1'	01°40.2	14.1'	54.4'							
16	58°55.1	49.6	155°24.4	17.1'	01°54.3	14.1'	54.4'							
17	73°54.8	49.4	170°00.5	17.1'	02°08.4	14.1'	54.4'							
18	88°54.5	S22°49.1	184°36.6	17.1'	S02°22.5	14.1'	54.5'							
19	103°54.2	48.9	199°12.7	17.0'	02°36.5	14.1'	54.5'							
20	118°53.9	48.6	213°48											

January 04, 05, 06 UT (Thu., Fri., Sat.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	103°06.5	218°22.5	S19°31.5	193°37.0	S24°01.0	69°24.2	N12°17.2	127°04.4	S11°44.3	Alpheratz	357°35.7	29°13.5											
1	118°09.0	233°21.8	32.1	208°37.4	01.1	84°26.7	17.2	142°06.7	44.2	Ankaa	353°08.0	-42°10.8											
2	133°11.4	248°21.1	32.7	223°37.8	01.1	99°29.1	17.2	157°08.9	44.1	Schedar	349°32.0	56°40.4											
3	148°13.9	263°20.4	33.3	238°38.3	01.1	114°31.5	17.3	172°11.2	44.0	Diphda	348°48.2	-17°51.5											
4	163°16.3	278°19.7	33.9	253°38.7	01.2	129°34.0	17.3	187°13.4	43.9	Achernar	335°20.7	-57°07.2											
5	178°18.8	293°19.0	34.5	268°39.1	01.2	144°36.4	17.3	202°15.7	43.8	Hamal	327°52.0	23°34.6											
6	193°21.3	308°18.3	S19°35.1	283°39.5	S24°01.2	159°38.8	N12°17.3	217°17.9	S11°43.7	Polaris	314°09.8	89°22.2											
7	208°23.7	323°17.6	35.7	298°40.0	01.2	174°41.3	17.4	232°20.2	43.7	Acamar	315°12.2	-40°12.7											
8	223°26.2	338°16.9	36.2	313°40.4	01.3	189°43.7	17.4	247°22.4	43.6	Menkar	314°06.8	4°11.0											
9	238°28.7	353°16.2	36.8	328°40.8	01.3	204°46.1	17.4	262°24.7	43.5	Mirfak	308°29.1	49°56.9											
10	253°31.1	8°15.5	37.4	343°41.2	01.3	219°48.6	17.4	277°26.9	43.4	Aldebaran	290°40.3	16°33.5											
11	268°33.6	23°14.8	38.0	358°41.7	01.3	234°51.0	17.5	292°29.1	43.3	Rigel	281°04.4	-8°10.5											
12	283°36.1	38°14.0	S19°38.6	13°42.1	S24°01.4	249°53.4	N12°17.5	307°31.4	S11°43.2	Capella	280°22.7	46°01.4											
13	298°38.5	53°13.3	39.2	28°42.5	01.4	264°55.9	17.5	322°33.6	43.1	Bellatrix	278°23.5	6°22.3											
14	313°41.0	68°12.6	39.8	43°43.0	01.4	279°58.3	17.5	337°35.9	43.1	Elnath	278°02.6	28°37.7											
15	328°43.5	83°11.9	40.4	58°43.4	01.4	295°00.7	17.6	352°38.1	43.0	Alnilam	275°38.3	-1°11.2											
16	343°45.9	98°11.2	40.9	73°43.8	01.5	310°03.2	17.6	7°40.4	42.9	Betelgeuse	270°52.7	7°24.7											
17	358°48.4	113°10.5	41.5	88°44.2	01.5	325°05.6	17.6	22°42.6	42.8	Canopus	263°52.2	-52°42.5											
18	13°50.8	128°09.8	S19°42.1	103°44.7	S24°01.5	340°08.0	N12°17.6	37°44.9	S11°42.7	Sirius	258°26.6	-16°44.9											
19	28°53.3	143°09.1	42.7	118°45.1	01.5	355°10.5	17.7	52°47.1	42.6	Adhara	255°06.1	-29°00.3											
20	43°55.8	158°08.3	43.3	133°45.5	01.6	10°12.9	17.7	67°49.3	42.5	Procyon	244°51.4	5°09.8											
21	58°58.2	173°07.6	43.9	148°45.9	01.6	25°15.3	17.7	82°51.6	42.4	Pollux	243°18.0	27°58.0											
22	74°00.7	188°06.9	44.4	163°46.4	01.6	40°17.7	17.7	97°53.8	42.4	Avior	234°14.5	-59°35.0											
23	89°03.2	203°06.2	45.0	178°46.8	01.6	55°20.2	17.8	112°56.1	42.3	Suhail	222°46.6	-43°31.6											
Mer.pass. 17:05				ν -0.7' d 0.6' m -4.00				ν 0.4' d 0.0' m 1.37				ν 2.4' d 0.0' m -2.57				ν 2.2' d -0.1' m 0.96							

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec						
0	104°05.6	218°05.5	S19°45.6	193°47.2	S24°01.6	70°22.6	N12°17.8	127°58.3	S11°42.2	Denebola	182°25.7	14°26.2							
1	119°08.1	233°04.8	46.2	208°47.7	01.7	85°25.0	17.8	143°00.6	42.1	Gienah	175°44.4	-17°40.4							
2	134°10.6	248°04.0	46.7	223°48.1	01.7	100°27.5	17.8	158°02.8	42.0	Acrux	173°01.0	-63°13.6							
3	149°13.0	263°03.3	47.3	238°48.5	01.7	115°29.9	17.9	173°05.1	41.9	Gacrux	171°52.6	-57°14.6							
4	164°15.5	278°02.6	47.9	253°48.9	01.7	130°32.3	17.9	188°07.3	41.8	Alioth	166°13.7	55°49.5							
5	179°17.9	293°01.9	48.5	268°49.4	01.7	145°34.7	17.9	203°09.5	41.7	Spica	158°23.2	-11°17.1							
6	194°20.4	308°01.2	S19°49.0	283°49.8	S24°01.8	160°37.2	N12°17.9	218°11.8	S11°41.7	Alkaid	152°52.8	49°11.3							
7	209°22.9	323°00.4	49.6	298°50.2	01.8	175°39.6	18.0	233°14.0	41.6	Hadar	148°37.4	-60°29.0							
8	224°25.3	337°59.7	50.2	313°50.6	01.8	190°42.0	18.0	248°16.3	41.5	Menkent	147°58.8	-36°29.1							
9	239°27.8	352°59.0	50.8	328°51.1	01.8	205°44.4	18.0	263°18.5	41.4	Arcturus	145°48.8	19°03.3							
10	254°30.3	7°58.3	51.3	343°51.5	01.8	220°46.9	18.1	278°20.8	41.3	Rigil Kent.	139°41.8	-60°55.8							
11	269°32.7	22°57.6	51.9	358°51.9	01.8	235°49.3	18.1	293°23.0	41.2	Kochab	137°20.4	74°03.1							
12	284°35.2	37°56.8	S19°52.5	13°52.3	S24°01.9	250°51.7	N12°18.1	308°25.3	S11°41.1	Zuben'ubi	136°57.1	-16°08.4							
13	299°37.7	52°56.1	53.0	28°52.8	01.9	265°54.1	18.1	323°27.5	41.0	Alphecca	126°04.7	26°37.9							
14	314°40.1	67°55.4	53.6	43°53.2	01.9	280°56.6	18.2	338°29.7	41.0	Antares	112°17.2	-26°29.0							
15	329°42.6	82°54.7	54.2	58°53.6	01.9	295°59.0	18.2	353°32.0	40.9	Atria	107°12.7	-69°04.1							
16	344°45.1	97°53.9	54.7	73°54.0	01.9	311°01.4	18.2	8°34.2	40.8	Sabik	102°04.1	-15°45.3							
17	359°47.5	112°53.2	55.3	88°54.5	01.9	326°03.8	18.3	23°36.5	40.7	Shaula	96°11.9	-37°07.2							
18	14°50.0	127°52.5	S19°55.8	103°54.9	S24°01.9	341°06.2	N12°18.3	38°38.7	S11°40.6	Rasalhague	95°59.7	12°32.5							
19	29°52.4	142°51.8	56.4	118°55.3	01.9	356°08.7	18.3	53°41.0	40.5	Eltanin	90°43.1	51°29.0							
20	44°54.9	157°51.0	57.0	133°55.7	02.0	11°11.1	18.3	68°43.2	40.4	Kaus Aust.	83°34.0	-34°22.4							
21	59°57.4	172°50.3	57.5	148°56.2	02.0	26°13.5	18.4	83°45.4	40.3	Vega	80°34.2	38°48.3							
22	74°59.8	187°49.6	58.1	163°56.6	02.0	41°15.9	18.4	98°47.7	40.3	Nunki	75°49.2	-26°16.1							
23	90°02.3	202°48.9	58.6	178°57.0	02.0	56°18.3	18.4	113°49.9	40.2	Altair	62°01.1	8°55.8							
Mer.pass. 17:01				ν -0.7' d 0.6' m -4.00				ν 0.4' d 0.0' m 1.36				ν 2.4' d 0.0' m -2.56				ν 2.2' d -0.1' m 0.96			

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec						
0	105°04.8	217°48.1	S19°59.2	193°57.4	S24°02.0	71°20.8	N12°18.4	128°52.2	S11°40.1	Jan 04 Thu	SHA	Mer.pass							
1	120°07.2	232°47.4	19°59.7	208°57.9	02.0	86°23.2	18.5	143°54.4	40.0	Venus	115°16.1	09:27							
2	135°09.7	247°46.7	20°00.3	223°58.3	02.0	101°25.6	18.5	158°56.6	39.9	Mars	90°30.5	11:05							
3	150°12.2	262°45.9	00.8	238°58.7	02.0	116°28.0	18.5	173°58.9	39.8	Jupiter	326°17.7	19:19							
4	165°14.6	277°45.2	01.4	253°59.1	02.0	131°30.4	18.6	189°01.1	39.7	Saturn	23°57.9	15:29							
5	180°17.1	292°44.5	02.0	268°59.6	02.0	146°32.9	18.6	204°03.4	39.6										
6	195°19.6	307°43.7	S20°02.5	284°00.0	S24°02.1	161°35.3	N12°18.6	219°05.6	S11°39.6										
7	210°22.0	322°43.0	03.1	299°00.4	02.1	176°37.7	18.7	234°07.9	39.5										
8	225°24.5	337°42.3	03.6	314°00.8	02.1	191°40.1	18.7	249°10.1	39.4										
9	240°26.9	352°41.5	04.1	329°01.3	02.1	206°42.5	18.7	264°12.3	39.3										
10	255°29.4	7°40.8	04.7	344°01.7	02.1	221°44.9	18.7	279°14.6	39.2										
11	270°31.9	22°40.1	05.2	359°02.1	02.1	236°47.4	18.8	294°16.8	39.1										
12	285°34.3	37°39.3	S20°05.8	14°02.5	S24°02.1	251°49.8	N12°18.8	309°19.1	S11°39.0										
13	300°36.8	52°38.6	06.3	29°02.9	02.1	266°52.2	18.8	324°21.3	38.9										
14	315°39.3	67°37.9	06.9	44°03.4	02.1	281°54.6	18.9	339°23.5	38.8										
15	330°41.7	82°37.1	07.4	59°03.8	02.1	296°57.0	18.9	354°25.8	38.8										
16	345°44.2	97°36.4	08.0	74°04.2	02.1	311°59.4	18.9	9°28.0	38.7										
17	0°46.7	112°35.7	08.5	89°04.6	02.1	327°01.8	19.0	24°30.3	38.6										
18	15°49.1	127°34.9	S20°09.0	104°05.1	S24°02.1	342°04.3	N12°19.0	39°32.5	S11°38.5										
19	30°51.6	142°34.2	09.6	119°05.5	02.1	357°06.7	19.0	54°34.7	38.4										
20	45°54.0	157°33.5	10.1	134°05.9	02.1	12°09.1	19.1	69°37.0	38.3										
21	60°56.5	172°32.7	10.6	149°06.3	02.1	27°11.5	19.1	84°39.2	38.2										
22	75°59.0	187°32.0	11.2	164°06.7	02.1	42°13.9	19.1	99°41.5	38.1										
23	91°01.4	202°31.2	11.7	179°07.2	02.1	57°16.3	19.1	114°43.7	38.0										
Mer.pass. 16:57				ν -0.7' d 0.6' m -4.00				ν 0.4' d 0.0' m 1.36				ν 2.4' d 0.0' m -2.55				ν 2.2' d -0.1' m 0.96			

Jan 04 Thu	SHA	Mer.pass
Venus	115°16.1	09:27
Mars	90°30.5	11:05
Jupiter	326°17.7	19:19
Saturn	23°57.9	15:29

Jan 05 Fri	SHA	Mer.pass
Venus	113°59.8	09:28
Mars	89°41.6	11:05
Jupiter	326°17.0	19:15
Saturn	23°52.7	15:26

Jan 06 Sat	SHA	Mer.pass
Venus	112°43.4	09:29
Mars	88°52.7	11:04
Jupiter	326°16.0	19:12
Saturn	23°47.4	15:22

Horizontal parallax		
Venus:	0.1	
Mars:	0.1	

Sun				Moon				
Thu	GHA	Dec		GHA	ν	Dec	d	HP
0	178°52.8	S22°47.6		272°12.7	16.9'	S03°46.9	14.1'	54.6'
1	193°52.5	47.4		286°48.6	16.9'	04°01.0	14.1'	54.6'
2	208°52.2	47.1		301°24.5	16.8'	04°15.0	14.0'	54.6'
3	223°51.9	· · 46.9		316°00.3	16.8'	04°29.1	14.0'	54.6'
4	238°51.7	46.6		330°36.1	16.8'	04°43.1	14.0'	54.6'
5	253°51.4	46.4		345°11.9	16.7'	04°57.2	14.0'	54.6'
6	268°51.1	S22°46.1		359°47.7	16.7'	S05°11.2	14.0'	54.7'
7	283°50.8	45.9		14°23.4	16.7'	05°25.2	14.0'	54.7'
8	298°50.5	45.6		28°59.0	16.6'	05°39.2	14.0'	54.7'
9	313°50.2	· · 45.4		43°34.6	16.6'	05°53.2	14.0'	54.7'
10	328°49.9	45.1		58°10.2	16.5'	06°07.2	14.0'	54.7'
11	343°49.7	44.8		72°45.8	16.5'	06°21.2	14.0'	54.8'
12	358°49.4	S22°44.6		87°21.2	16.4'	S06°35.1	13.9'	54.8'
13	13°49.1	44.3		101°56.7	16.4'	06°49.1	13.9'	54.8'
14	28°48.8	44.1		116°32.1	16.4'	07°03.0	13.9'	54.8'
15	43°48.5	· · 43.8		131°07.4	16.3'	07°16.9	13.9'	54.8'
16	58°48.2	43.5		145°42.7	16.3'	07°30.8	13.9'	54.9'
17	73°48.0	43.3		160°18.0	16.2'	07°44.7	13.9'	54.9'
18	88°47.7	S22°43.0		174°53.2	16.1'	S07°58.6	13.8'	54.9'
19	103°47.4	42.8		189°28.3	16.1'	08°12.4	13.8'	54.9'
20	118°47.1	42.5		204°03.4	16.0'	08°26.2	13.8'	54.9'
21	133°46.8	· · 42.2		218°38.5	16.0'	08°40.0	13.8'	55.0'
22	148°46.5	42.0		233°13.5	15.9'	08°53.8	13.8'	55.0'
23	163°46.3	41.7		247°48.4	15.9'	09°07.6	13.7'	55.0'
SD = 16.3'			d = -0.2'	SD = 14.9'				
Fri	GHA	Dec		GHA	ν	Dec	d	HP
0	178°46.0	S22°41.4		262°23.2	15.8'	S09°21.3	13.7'	55.0'
1	193°45.7	41.2		276°58.0	15.7'	09°35.1	13.7'	55.1'
2	208°45.4	40.9		291°32.8	15.7'	09°48.8	13.7'	55.1'
3	223°45.1	· · 40.6		306°07.4	15.6'	10°02.4	13.6'	55.1'
4	238°44.8	40.3		320°42.1	15.5'	10°16.1	13.6'	55.1'
5	253°44.6	40.1		335°16.6	15.5'	10°29.7	13.6'	55.2'
6	268°44.3	S22°39.8		349°51.1	15.4'	S10°43.3	13.6'	55.2'
7	283°44.0	39.5		4°25.5	15.3'	10°56.8	13.5'	55.2'
8	298°43.7	39.3		18°59.8	15.3'	11°10.4	13.5'	55.2'
9	313°43.4	· · 39.0		33°34.1	15.2'	11°23.9	13.5'	55.3'
10	328°43.2	38.7		48°08.3	15.1'	11°37.3	13.4'	55.3'
11	343°42.9	38.4		62°42.4	15.0'	11°50.8	13.4'	55.3'
12	358°42.6	S22°38.2		77°16.4	15.0'	S12°04.2	13.4'	55.3'
13	13°42.3	37.9		91°50.4	14.9'	12°17.5	13.3'	55.4'
14	28°42.0	37.6		106°24.3	14.8'	12°30.9	13.3'	55.4'
15	43°41.7	· · 37.3		120°58.1	14.7'	12°44.2	13.3'	55.4'
16	58°41.5	37.0		135°31.9	14.7'	12°57.4	13.2'	55.5'
17	73°41.2	36.8		150°05.5	14.6'	13°10.6	13.2'	55.5'
18	88°40.9	S22°36.5		164°39.1	14.5'	S13°23.8	13.1'	55.5'
19	103°40.6	36.2		179°12.6	14.4'	13°36.9	13.1'	55.5'
20	118°40.4	35.9		193°46.0	14.3'	13°50.0	13.1'	55.6'
21	133°40.1	· · 35.6		208°19.3	14.2'	14°03.1	13.0'	55.6'
22	148°39.8	35.3		222°52.5	14.1'	14°16.1	13.0'	55.6'
23	163°39.5	35.1		237°25.6	14.0'	14°29.1	12.9'	55.7'
SD = 16.3'			d = -0.3'	SD = 15.0'				
Sat	GHA	Dec		GHA	ν	Dec	d	HP
0	178°39.2	S22°34.8		251°58.7	14.0'	S14°42.0	12.9'	55.7'
1	193°39.0	34.5		266°31.6	13.9'	14°54.8	12.8'	55.7'
2	208°38.7	34.2		281°04.5	13.8'	15°07.7	12.8'	55.7'
3	223°38.4	· · 33.9		295°37.3	13.7'	15°20.4	12.7'	55.8'
4	238°38.1	33.6		310°10.0	13.6'	15°33.1	12.7'	55.8'
5	253°37.8	33.3		324°42.5	13.5'	15°45.8	12.6'	55.8'
6	268°37.6	S22°33.0		339°15.0	13.4'	S15°58.4	12.6'	55.9'
7	283°37.3	32.7		353°47.4	13.3'	16°11.0	12.5'	55.9'
8	298°37.0	32.5		8°19.7	13.2'	16°23.5	12.4'	55.9'
9	313°36.7	· · 32.2		22°51.9	13.1'	16°35.9	12.4'	56.0'
10	328°36.5	31.9		37°24.0	13.0'	16°48.3	12.3'	56.0'
11	343°36.2	31.6		51°55.9	12.9'	17°00.6	12.3'	56.0'
12	358°35.9	S22°31.3		66°27.8	12.8'	S17°12.9	12.2'	56.1'
13	13°35.6	31.0		80°59.6	12.7'	17°25.1	12.1'	56.1'
14	28°35.4	30.7		95°31.2	12.6'	17°37.2	12.1'	56.1'
15	43°35.1	· · 30.4		110°02.8	12.4'	17°49.3	12.0'	56.2'
16	58°34.8	30.1		124°34.2	12.3'	18°01.3	11.9'	56.2'
17	73°34.5	29.8		139°05.6	12.2'	18°13.2	11.9'	56.2'
18	88°34.3	S22°29.5		153°36.8	12.1'	S18°25.1	11.8'	56.3'
19	103°34.0	29.2		168°07.9	12.0'	18°36.9	11.7'	56.3'
20	118°33.7	28.9		182°38.9	11.9'	18°48.6	11.7'	56.3'
21	133°33.4	· · 28.6		197°09.8	11.8'	19°00.3	11.6'	56.4'
22	148°33.2	28.3		211°40.6	11.7'	19°11.9	11.5'	56.4'
23	163°32.9	28.0		226°11.3	11.5'	19°23.4	11.4'	56.4'
SD = 16.3'			d = -0.3'	SD = 15.2'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:20	10:31	■	■	13:39	15:51
N 70°	08:02	09:44	■	■	14:27	16:09
68°	07:47	09:13	11:33	12:38	14:58	16:23
66°	07:35	08:50	10:21	13:50	15:21	16:35
64°	07:25	08:32	09:45	14:25	15:39	16:46
62°	07:16	08:17	09:20	14:51	15:54	16:55
60°	07:08	08:04	09:00	15:11	16:07	17:02
N 58°	07:01	07:53	08:44	15:27	16:17	17:09
56°	06:55	07:43	08:30	15:41	16:27	17:16
54°	06:49	07:35	08:18	15:53	16:36	17:21
52°	06:44	07:27	08:07	16:03	16:43	17:27
50°	06:39	07:20	07:58	16:13	16:51	17:32
45°	06:28	07:05	07:38	16:32	17:06	17:43
N 40°	06:18	06:52	07:22	16:48	17:19	17:53
35°	06:09	06:41	07:08	17:02	17:30	18:01
30°	06:01	06:30	06:57	17:14	17:40	18:10
20°	05:45	06:12	06:36	17:34	17:58	18:25
N 10°	05:29	05:56	06:18	17:52	18:15	18:41
0°	05:13	05:39	06:01	18:09	18:31	18:57
S 10°	04:55	05:22	05:44	18:26	18:49	19:16
20°	04:33	05:02	05:26	18:44	19:08	19:37
30°	04:05	04:38	05:05	19:05	19:32	20:05
35°	03:47	04:23	04:52	19:18	19:47	20:23
40°	03:24	04:05	04:38	19:32	20:05	20:45
45°	02:55	03:44	04:21	19:49	20:26	21:14
S 50°	02:13	03:15	03:59	20:11	20:54	21:56
52°	01:48	03:01	03:49	20:21	21:08	22:21
54°	01:11	02:44	03:37	20:33	21:25	22:57
56°	-:-	02:24	03:24	20:46	21:45	-:-
58°	-:-	01:57	03:08	21:02	22:12	-:-
S 60°	-:-	01:17	02:49	21:20	22:50	-:-

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	00:43	02:48	05:29	10:48	10:08	08:59
N 70°	00:37	02:31	04:45	10:57	10:28	09:44
68°	00:32	02:18	04:16	11:04	10:43	10:15
66°	00:28	02:07	03:55	11:10	10:56	10:37
64°	00:25	01:58	03:38	11:15	11:06	10:56
62°	00:22	01:50	03:24	11:20	11:15	11:11
60°	00:19	01:43	03:13	11:24	11:23	11:23
N 58°	00:17	01:38	03:03	11:27	11:30	11:34
56°	00:15	01:33	02:54	11:30	11:36	11:44
54°	00:13	01:28	02:46	11:33	11:42	11:52
52°	00:12	01:24	02:39	11:36	11:47	12:00
50°	00:10	01:20	02:33	11:38	11:51	12:07
45°	00:07	01:12	02:20	11:43	12:01	12:22
N 40°	00:05	01:05	02:09	11:48	12:09	12:34
35°	00:02	01:00	02:00	11:51	12:16	12:44
30°	00:00	00:55	01:52	11:55	12:22	12:54
20°	· · ·	00:46	01:38	12:01	12:33	13:10
N 10°	· · ·	00:39	01:26	12:06	12:43	13:24
0°	· · ·	00:32	01:15	12:10	12:52	13:37
S 10°	· · ·	00:25	01:04	12:15	13:01	13:50
20°	· · ·	00:18	00:52	12:20	13:11	14:04
30°	· · ·	00:09	00:39	12:26	13:22	14:20
35°	· · ·	00:05	00:31	12:30	13:28	14:30
40°	23:59	· · ·	00:22	12:34	13:36	14:41
45°	23:53	· · ·	00:12	12:38	13:44	14:54
S 50°	23:46	· · ·	00:00	12:44	13:55	15:10
52°	23:42	23:54	· · ·	12:46	14:00	15:17
54°	23:39	23:48	· · ·	12:49	14:05	15:25
56°	23:35	23:41	23:51	12:52	14:11	15:35
58°	23:30	23:34	23:39	12:55	14:18	15:45
S 60°	23:25	23:25	23:26	12:59	14:25	15:58

Day	Sun		Mer. Pass hh:mm	Moon		Age 23-25 52-32%
	Eqn.of Time	Mer.		Mer.Pass.	Age	
	00 ^h mm:ss	12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
04	04:29	04:43	12:05	06:01	18:21	
05	04:56	05:10	12:05	06:42	19:03	
06	05:23	05:36	12:06	07:26	19:49	

January 07, 08, 09 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars			
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	106°03.9	217°30.5	S20°12.3	194°07.6	S24°02.1	72°18.7	N12°19.2	129°45.9	S11°38.0	Alpheratz	357°35.7	29°13.5	
1	121°06.4	232°29.8	12.8	209°08.0	02.1	87°21.1	19.2	144°48.2	37.9	Ankaa	353°08.0	-42°10.8	
2	136°08.8	247°29.0	13.3	224°08.4	02.1	102°23.5	19.2	159°50.4	37.8	Schedar	349°32.1	56°40.4	
3	151°11.3	262°28.3	· · 13.8	239°08.9	· · 02.1	117°26.0	· · 19.3	174°52.7	· · 37.7	Diphda	348°48.2	-17°51.5	
4	166°13.8	277°27.5	14.4	254°09.3	02.1	132°28.4	19.3	189°54.9	37.6	Achernar	335°20.7	-57°07.2	
5	181°16.2	292°26.8	14.9	269°09.7	02.1	147°30.8	19.3	204°57.1	37.5	Hamal	327°52.0	23°34.6	
6	196°18.7	307°26.1	S20°15.4	284°10.1	S24°02.1	162°33.2	N12°19.4	219°59.4	S11°37.4	Polaris	314°10.8	89°22.2	
7	211°21.2	322°25.3	16.0	299°10.5	02.1	177°35.6	19.4	235°01.6	37.3	Acamar	315°12.2	-40°12.7	
8	226°23.6	337°24.6	16.5	314°11.0	02.1	192°38.0	19.4	250°03.8	37.2	Menkar	314°06.9	4°11.0	
9	241°26.1	352°23.8	· · 17.0	329°11.4	· · 02.1	207°40.4	· · 19.5	265°06.1	· · 37.2	Mirfak	308°29.1	49°56.9	
10	256°28.5	7°23.1	17.5	344°11.8	02.1	222°42.8	19.5	280°08.3	37.1	Aldebaran	290°40.3	16°33.5	
11	271°31.0	22°22.3	18.1	359°12.2	02.1	237°45.2	19.5	295°10.6	37.0	Rigel	281°04.4	-8°10.5	
12	286°33.5	37°21.6	S20°18.6	14°12.7	S24°02.1	252°47.6	N12°19.6	310°12.8	S11°36.9	Capella	280°22.7	46°01.4	
13	301°35.9	52°20.9	19.1	29°13.1	02.1	267°50.0	19.6	325°15.0	36.8	Bellatrix	278°23.5	6°22.3	
14	316°38.4	67°20.1	19.6	44°13.5	02.0	282°52.4	19.6	340°17.3	36.7	Elnath	278°02.6	28°37.7	
15	331°40.9	82°19.4	· · 20.1	59°13.9	· · 02.0	297°54.8	· · 19.7	355°19.5	· · 36.6	Alnilam	275°38.3	-1°11.2	
16	346°43.3	97°18.6	20.7	74°14.3	02.0	312°57.3	19.7	10°21.8	36.5	Betelgeuse	270°52.7	7°24.7	
17	1°45.8	112°17.9	21.2	89°14.8	02.0	327°59.7	19.7	25°24.0	36.4	Canopus	263°52.2	-52°42.5	
18	16°48.3	127°17.1	S20°21.7	104°15.2	S24°02.0	343°02.1	N12°19.8	40°26.2	S11°36.4	Sirius	258°26.6	-16°45.0	
19	31°50.7	142°16.4	22.2	119°15.6	02.0	358°04.5	19.8	55°28.5	36.3	Adhara	255°06.1	-29°00.3	
20	46°53.2	157°15.6	22.7	134°16.0	02.0	13°06.9	19.8	70°30.7	36.2	Procyon	244°51.4	5°09.8	
21	61°55.7	172°14.9	· · 23.2	149°16.4	· · 02.0	28°09.3	· · 19.9	85°32.9	· · 36.1	Pollux	243°18.0	27°58.0	
22	76°58.1	187°14.1	23.8	164°16.9	02.0	43°11.7	19.9	100°35.2	36.0	Avior	234°14.5	-59°35.0	
23	92°00.6	202°13.4	24.3	179°17.3	02.0	58°14.1	19.9	115°37.4	35.9	Suhail	222°46.5	-43°31.6	
Mer.pass. 16:53		ν -0.7' d 0.5' m -3.99		ν 0.4' d -0.0' m 1.36		ν 2.4' d 0.0' m -2.54		ν 2.2' d -0.1' m 0.96		Miaplacidus		221°37.6	-69°48.7
Alphard												217°48.3	-8°45.7
Regulus												207°35.1	11°50.9
Dubhe												193°41.7	61°37.0
Denebola												182°25.7	14°26.2
Gienah												175°44.4	-17°40.4
Acrux												173°00.9	-63°13.6
Gacrux												171°52.5	-57°14.6
Alioth												166°13.7	55°49.5
Spica												158°23.2	-11°17.1
Alkaid												152°52.8	49°11.3
Hadar												148°37.4	-60°29.0
Menkent												147°58.7	-36°29.1
Arcturus												145°48.8	19°03.3
Rigel Kent.												139°41.7	-60°55.8
Kochab												137°20.3	74°03.1
Zuben'ubi												136°57.1	-16°08.4
Alphecca												126°04.7	26°37.9
Antares												112°17.2	-26°29.0
Atria												107°12.6	-69°04.0
Sabik												102°04.1	-15°45.3
Shaula												96°11.9	-37°07.2
Rasalhague												95°59.6	12°32.5
Eltanin												90°43.1	51°29.0
Kaus Aust.												83°34.0	-34°22.4
Vega												80°34.2	38°48.2
Nunki												75°49.2	-26°16.1
Altair												62°01.1	8°55.8
Peacock												53°07.6	-56°39.6
Deneb												49°26.8	45°22.0
Enif												33°39.9	9°59.1
Al Na'ir												27°34.3	-46°50.9
Fomalhaut												15°15.6	-29°29.9
Scheat												13°46.2	28°12.8
Markab												13°30.9	15°20.1
Jan 07 Sun												SHA	Mer.pass
Venus												111°26.6	09:30
Mars												88°03.7	11:03
Jupiter												326°14.8	19:08
Saturn												23°42.0	15:19
Jan 08 Mon												SHA	Mer.pass
Venus												110°09.6	09:32
Mars												87°14.7	11:03
Jupiter												326°13.4	19:04
Saturn												23°36.6	15:15
Jan 09 Tue												SHA	Mer.pass
Venus												108°52.3	09:33
Mars												86°25.6	11:02
Jupiter												326°11.9	19:00
Saturn												23°31.1	15:12
Horizontal parallax													
Venus:												0.1	
Mars:												0.1	
Mer.pass. 16:45		ν -0.8' d 0.5' m -3.98		ν 0.4' d -0.0' m 1.36		ν 2.4' d 0.0' m -2.53		ν 2.2' d -0.1' m 0.97					

h	Sun				Moon										
Sun	GHA	Dec	GHA	ν	Dec	d	HP	Lat.	Twilight		Sunrise	Sunset	Twilight		
0	178°32.6	S22°27.7	240°41.8	11.4'	S19°34.8	11.3'	56.5'	N 72°	Naut.	Civil	■	■	Civil	Naut.	
1	193°32.3	27.4	255°12.3	11.3'	19°46.1	11.3'	56.5'	N 70°	08:16	10:21	■	■	13:53	15:58	
2	208°32.1	27.1	269°42.6	11.2'	19°57.4	11.2'	56.5'	68°	07:58	09:37	■	■	14:36	16:15	
3	223°31.8	26.8	284°12.8	11.1'	20°08.6	11.1'	56.6'	66°	07:44	09:08	11:12	13:01	15:05	16:29	
4	238°31.5	26.4	298°42.9	11.0'	20°19.7	11.0'	56.6'	64°	07:33	08:46	10:14	14:00	15:27	16:41	
5	253°31.2	26.1	313°12.8	10.8'	20°30.7	10.9'	56.6'	62°	07:23	08:29	09:41	14:33	15:44	16:50	
6	268°31.0	S22°25.8	327°42.6	10.7'	S20°41.6	10.8'	56.7'	60°	07:14	08:15	09:16	14:57	15:59	16:59	
7	283°30.7	25.5	342°12.4	10.6'	20°52.4	10.7'	56.7'	58°	07:07	08:02	08:57	15:16	16:11	17:06	
8	298°30.4	25.2	356°42.0	10.5'	21°03.2	10.7'	56.7'	56°	07:00	07:52	08:42	15:32	16:22	17:13	
9	313°30.2	24.9	11°11.4	10.3'	21°13.8	10.6'	56.8'	54°	06:54	07:42	08:28	15:45	16:31	17:19	
10	328°29.9	24.6	25°40.8	10.2'	21°24.4	10.5'	56.8'	52°	06:48	07:34	08:16	15:57	16:39	17:25	
11	343°29.6	24.3	40°10.0	10.1'	21°34.8	10.4'	56.8'	50°	06:43	07:26	08:06	16:07	16:47	17:30	
12	358°29.3	S22°24.0	54°39.1	10.0'	S21°45.2	10.3'	56.9'	48°	06:38	07:19	07:57	16:16	16:54	17:35	
13	13°29.1	23.6	69°08.1	9.8'	21°55.5	10.2'	56.9'	46°	06:28	07:04	07:38	16:36	17:09	17:46	
14	28°28.8	23.3	83°36.9	9.7'	22°05.7	10.1'	56.9'	N 40°	06:18	06:52	07:22	16:51	17:21	17:55	
15	43°28.5	23.0	98°05.6	9.6'	22°15.7	10.0'	57.0'	38°	06:09	06:41	07:09	17:05	17:32	18:04	
16	58°28.3	22.7	112°34.2	9.5'	22°25.7	9.9'	57.0'	36°	06:01	06:31	06:57	17:16	17:42	18:12	
17	73°28.0	22.4	127°02.7	9.3'	22°35.5	9.8'	57.1'	34°	05:46	06:13	06:37	17:36	18:00	18:27	
18	88°27.7	S22°22.1	141°31.1	9.2'	S22°45.3	9.6'	57.1'	32°	05:31	05:57	06:19	17:54	18:16	18:42	
19	103°27.4	21.7	155°59.3	9.1'	22°54.9	9.5'	57.1'	30°	05:14	05:40	06:03	18:10	18:33	18:59	
20	118°27.2	21.4	170°27.4	9.0'	23°04.5	9.4'	57.2'	S 10°	04:56	05:23	05:46	18:27	18:50	19:16	
21	133°26.9	21.1	184°55.3	8.8'	23°13.9	9.3'	57.2'	28°	04:35	05:04	05:28	18:45	19:09	19:38	
22	148°26.6	20.8	199°23.2	8.7'	23°23.2	9.2'	57.2'	26°	04:07	04:40	05:07	19:05	19:33	20:05	
23	163°26.4	20.5	213°50.9	8.6'	23°32.4	9.1'	57.3'	24°	03:50	04:26	04:55	19:18	19:47	20:23	
SD = 16.3' d = -0.3'								SD = 15.4'							
Mon	GHA	Dec	GHA	ν	Dec	d	HP								
0	178°26.1	S22°20.1	228°18.4	8.4'	S23°41.5	9.0'	57.3'	S 50°	03:28	04:08	04:41	19:32	20:04	20:45	
1	193°25.8	19.8	242°45.9	8.3'	23°50.4	8.8'	57.3'	48°	02:59	03:47	04:24	19:49	20:25	21:13	
2	208°25.6	19.5	257°13.2	8.2'	23°59.3	8.7'	57.4'	46°	02:18	03:20	04:03	20:10	20:53	21:54	
3	223°25.3	19.2	271°40.4	8.1'	24°08.0	8.6'	57.4'	44°	01:55	03:06	03:53	20:20	21:06	22:17	
4	238°25.0	18.8	286°07.5	7.9'	24°16.6	8.5'	57.5'	42°	01:21	02:49	03:41	20:31	21:22	22:50	
5	253°24.8	18.5	300°34.4	7.8'	24°25.0	8.3'	57.5'	40°	--	02:30	03:28	20:44	21:42	--	
6	268°24.5	S22°18.2	315°01.2	7.7'	S24°33.4	8.2'	57.5'	38°	--	02:04	03:13	20:59	22:07	--	
7	283°24.2	17.8	329°27.9	7.6'	24°41.6	8.1'	57.6'	S 60°	--	01:28	02:55	21:17	22:42	--	
8	298°23.9	17.5	343°54.5	7.4'	24°49.6	7.9'	57.6'								
9	313°23.7	17.2	358°20.9	7.3'	24°57.6	7.8'	57.6'	Lat.	Moonrise			Moonset			
10	328°23.4	16.9	12°47.2	7.2'	25°05.4	7.7'	57.7'	N 72°	Sun	Mon	Tue	Sun	Mon	Tue	
11	343°23.1	16.5	27°13.4	7.1'	25°13.1	7.5'	57.7'	N 70°	■	■	■	■	■	■	
12	358°22.9	S22°16.2	41°39.4	6.9'	S25°20.6	7.4'	57.7'	68°	06:58	■	■	09:13	■	■	
13	13°22.6	15.9	56°05.4	6.8'	25°28.0	7.3'	57.8'	66°	06:04	■	■	10:08	■	■	
14	28°22.3	15.5	70°31.2	6.7'	25°35.2	7.1'	57.8'	64°	05:32	07:54	■	10:41	10:09	■	
15	43°22.1	15.2	84°56.8	6.6'	25°42.3	7.0'	57.9'	62°	05:08	07:04	09:19	11:05	10:59	10:46	
16	58°21.8	14.9	99°22.4	6.4'	25°49.3	6.8'	57.9'	60°	04:49	06:33	08:18	11:25	11:31	11:47	
17	73°21.6	14.5	113°47.8	6.3'	25°56.1	6.7'	57.9'	N 58°	04:34	06:10	07:45	11:41	11:54	12:21	
18	88°21.3	S22°14.2	128°13.2	6.2'	S26°02.8	6.5'	58.0'	56°	04:21	05:51	07:20	11:55	12:14	12:46	
19	103°21.0	13.8	142°38.4	6.1'	26°09.3	6.4'	58.0'	54°	04:09	05:36	07:00	12:07	12:30	13:06	
20	118°20.8	13.5	157°03.4	6.0'	26°15.7	6.2'	58.0'	52°	03:59	05:22	06:43	12:18	12:44	13:23	
21	133°20.5	13.2	171°28.4	5.8'	26°21.9	6.1'	58.1'	50°	03:50	05:10	06:29	12:27	12:56	13:38	
22	148°20.2	12.8	185°53.3	5.7'	26°27.9	5.9'	58.1'	48°	03:32	04:46	06:00	12:47	13:21	14:07	
23	163°20.0	12.5	200°18.0	5.6'	26°33.8	5.7'	58.1'	N 40°	03:16	04:26	05:37	13:04	13:42	14:31	
SD = 16.3' d = -0.3'								SD = 15.6'							
Tue	GHA	Dec	GHA	ν	Dec	d	HP								
0	178°19.7	S22°12.1	214°42.6	5.5'	S26°39.6	5.6'	58.2'	38°	03:03	04:10	05:19	13:18	13:59	14:50	
1	193°19.4	11.8	229°07.1	5.4'	26°45.2	5.4'	58.2'	36°	02:52	03:56	05:03	13:30	14:13	15:06	
2	208°19.2	11.5	243°31.5	5.3'	26°50.6	5.3'	58.3'	34°	02:33	03:33	04:36	13:51	14:38	15:34	
3	223°18.9	11.1	257°55.8	5.2'	26°55.8	5.1'	58.3'	32°	02:17	03:12	04:12	14:09	15:00	15:58	
4	238°18.6	10.8	272°20.0	5.1'	27°00.9	4.9'	58.3'	30°	02:02	02:54	03:51	14:26	15:20	16:20	
5	253°18.4	10.4	286°44.0	5.0'	27°05.8	4.8'	58.4'	S 10°	01:47	02:35	03:29	14:43	15:41	16:42	
6	268°18.1	S22°10.1	301°08.0	4.9'	S27°10.6	4.6'	58.4'	20°	01:30	02:15	03:07	15:01	16:03	17:06	
7	283°17.9	09.7	315°31.9	4.8'	27°15.2	4.4'	58.4'	30°	01:12	01:52	02:40	15:23	16:28	17:34	
8	298°17.6	09.4	329°55.6	4.7'	27°19.6	4.2'	58.5'	38°	01:02	01:39	02:25	15:35	16:43	17:51	
9	313°17.3	09.0	344°19.3	4.6'	27°23.8	4.1'	58.5'	40°	00:49	01:23	02:06	15:50	17:00	18:10	
10	328°17.1	08.7	358°42.9	4.5'	27°27.9	3.9'	58.5'	45°	00:35	01:05	01:45	16:07	17:21	18:33	
11	343°16.8	08.3	13°06.3	4.4'	27°31.8	3.7'	58.6'	S 50°	00:18	00:42	01:18	16:28	17:48	19:03	
12	358°16.6	S22°08.0	27°29.7	4.3'	S27°35.5	3.5'	58.6'	52°	00:10	00:32	01:04	16:38	18:01	19:18	
13	13°16.3	07.6	41°53.0	4.2'	27°39.0	3.3'	58.6'	54°	00:01	00:20	00:49	16:50	18:16	19:35	
14	28°16.0	07.3	56°16.2	4.1'	27°42.3	3.2'	58.7'	56°	..	00:06	00:31	17:03	18:33	19:55	
15	43°15.8	06.9	70°39.3	4.0'	27°45.5	3.0'	58.7'	58°	23:50	..	00:09	17:19	18:54	20:21	
16	58°15.5	06.6	85°02.3	3.9'	27°48.5	2.8'	58.7'	S 60°	23:30	23:42	..	17:37	19:22	20:57	
17	73°15.2	06.2	99°25.3	3.9'	27°51.3	2.6'	58.8'								
18	88°15.0	S22°05.9	113°48.1	3.8'	S27°53.9	2.4'	58.8'	Day	Sun			Moon			
19	103°14.7	05.5	128°10.9	3.7'	27°56.3	2.2'	58.8'	Eqn.of Time <td>Mer.<td><td>Mer.Pass.<td>Age</td><td></td></td></td></td>	Mer. <td><td>Mer.Pass.<td>Age</td><td></td></td></td>	<td>Mer.Pass.<td>Age</td><td></td></td>	Mer.Pass. <td>Age</td> <td></td>	Age			
20	118°14.5	05.2	142°33.6	3.6'	27°58.6	2.1'	58.9'	00 ^h	Pass		Upper	Lower	26-28		
21	133°14.2	04.8	156°56.3	3.6'	28°00.6	1.9'	58.9'								

January 10, 11, 12 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	109°01.3	216°36.1	S20°48.2	194°37.8	S24°00.9	75°11.4	N12°21.8	132°26.9	S11°31.5	Alpheratz	357°35.7 29°13.5
1	124°03.8	231°35.3	48.6	209°38.2	00.8	90°13.8	21.8	147°29.1	31.4	Ankaa	353°08.0 -42°10.8
2	139°06.3	246°34.6	49.1	224°38.6	00.8	105°16.2	21.8	162°31.4	31.3	Schedar	349°32.1 56°40.4
3	154°08.7	261°33.8	· · 49.6	239°39.0	· · 00.8	120°18.6	· · 21.9	177°33.6	· · 31.2	Diphda	348°48.2 -17°51.5
4	169°11.2	276°33.0	50.0	254°39.5	00.7	135°21.0	21.9	192°35.8	31.1	Achernar	335°20.7 -57°07.2
5	184°13.6	291°32.3	50.5	269°39.9	00.7	150°23.3	22.0	207°38.1	31.0	Hamal	327°52.0 23°34.6
6	199°16.1	306°31.5	S20°50.9	284°40.3	S24°00.7	165°25.7	N12°22.0	222°40.3	S11°30.9	Polaris	314°11.7 89°22.2
7	214°18.6	321°30.7	51.4	299°40.7	00.6	180°28.1	22.0	237°42.5	30.8	Acamar	315°12.2 -40°12.7
8	229°21.0	336°29.9	51.8	314°41.1	00.6	195°30.5	22.1	252°44.8	30.7	Menkar	314°06.9 4°11.0
9	244°23.5	351°29.2	· · 52.3	329°41.5	· · 00.6	210°32.9	· · 22.1	267°47.0	· · 30.6	Mirfak	308°29.1 49°57.0
10	259°26.0	6°28.4	52.8	344°42.0	00.5	225°35.3	22.2	282°49.2	30.5	Aldebaran	290°40.3 16°33.5
11	274°28.4	21°27.6	53.2	359°42.4	00.5	240°37.6	22.2	297°51.5	30.4	Rigel	281°04.4 -8°10.5
12	289°30.9	36°26.8	S20°53.7	14°42.8	S24°00.4	255°40.0	N12°22.3	312°53.7	S11°30.4	Capella	280°22.7 46°01.4
13	304°33.4	51°26.1	54.1	29°43.2	00.4	270°42.4	22.3	327°55.9	30.3	Bellatrix	278°23.5 6°22.3
14	319°35.8	66°25.3	54.6	44°43.6	00.4	285°44.8	22.3	342°58.2	30.2	Elnath	278°02.6 28°37.7
15	334°38.3	81°24.5	· · 55.0	59°44.0	· · 00.3	300°47.2	· · 22.4	358°00.4	· · 30.1	Alnilam	275°38.3 -1°11.2
16	349°40.8	96°23.7	55.5	74°44.4	00.3	315°49.6	22.4	13°02.6	30.0	Betelgeuse	270°52.7 7°24.7
17	4°43.2	111°22.9	55.9	89°44.9	00.2	330°51.9	22.5	28°04.8	29.9	Canopus	263°52.2 -52°42.5
18	19°45.7	126°22.2	S20°56.4	104°45.3	S24°00.2	345°54.3	N12°22.5	43°07.1	S11°29.8	Sirius	258°26.6 -16°45.0
19	34°48.1	141°21.4	56.8	119°45.7	00.2	0°56.7	22.5	58°09.3	29.7	Adhara	255°06.1 -29°00.3
20	49°50.6	156°20.6	57.2	134°46.1	00.1	15°59.1	22.6	73°11.5	29.6	Procyon	244°51.4 5°09.8
21	64°53.1	171°19.8	· · 57.7	149°46.5	· · 00.1	31°01.5	· · 22.6	88°13.8	· · 29.5	Pollux	243°17.9 27°58.0
22	79°55.5	186°19.1	58.1	164°46.9	00.0	46°03.8	22.7	103°16.0	29.4	Avior	234°14.4 -59°35.1
23	94°58.0	201°18.3	58.6	179°47.4	00.0	61°06.2	22.7	118°18.2	29.3	Suhail	222°46.5 -43°31.6
Mer.pass. 16:41		ν -0.8' d 0.5' m -3.98		ν 0.4' d -0.0' m 1.36		ν 2.4' d 0.0' m -2.52		ν 2.2' d -0.1' m 0.97		Miaplacidus	221°37.6 -69°48.7
Alphard 217°48.3 -8°45.7											
Regulus 207°35.1 11°50.9											
Dubhe 193°41.6 61°37.0											
Denebola 182°25.7 14°26.2											
Gienah 175°44.4 -17°40.4											
Acrux 173°00.9 -63°13.6											
Gacrux 171°52.5 -57°14.6											
Alioth 166°13.6 55°49.5											
Spica 158°23.2 -11°17.2											
Alkaid 152°52.7 49°11.3											
Hadar 148°37.3 -60°29.0											
Menkent 147°58.7 -36°29.1											
Arcturus 145°48.8 19°03.3											
Rigel Kent. 139°41.7 -60°55.8											
Kochab 137°20.3 74°03.1											
Zuben'ubi 136°57.1 -16°08.4											
Alphecca 126°04.7 26°37.8											
Antares 112°17.1 -26°29.0											
Atria 107°12.6 -69°04.0											
Sabik 102°04.0 -15°45.3											
Shaula 96°11.9 -37°07.2											
Rasalhague 95°59.6 12°32.4											
Eltanin 90°43.1 51°29.0											
Kaus Aust. 83°34.0 -34°22.4											
Vega 80°34.2 38°48.2											
Nunki 75°49.2 -26°16.1											
Altair 62°01.1 8°55.8											
Peacock 53°07.6 -56°39.6											
Deneb 49°26.8 45°22.0											
Enif 33°39.9 9°59.1											
Al Na'ir 27°34.3 -46°50.9											
Fomalhaut 15°15.6 -29°29.9											
Scheat 13°46.2 28°12.8											
Markab 13°30.9 15°20.1											
Jan 10 Wed SHA Mer.pass											
Venus 107°34.8 09:34											
Mars 85°36.5 11:01											
Jupiter 326°10.1 18:56											
Saturn 23°25.6 15:08											
Jan 11 Thu SHA Mer.pass											
Venus 106°17.0 09:35											
Mars 84°47.3 11:01											
Jupiter 326°08.1 18:52											
Saturn 23°20.0 15:04											
Jan 12 Fri SHA Mer.pass											
Venus 104°59.0 09:37											
Mars 83°58.1 11:00											
Jupiter 326°05.9 18:49											
Saturn 23°14.3 15:01											
Horizontal parallax											
Venus: 0.1											
Mars: 0.1											

h	Sun				Moon			
Wed	GHA	Dec	GHA	ν	Dec	d	HP	
0	178°13.4	S22°03.7	200°03.7	3.4'	S28°05.6	1.3'	59.0'	
1	193°13.2	03.4	214°26.1	3.3'	28°06.9	1.1'	59.0'	
2	208°12.9	03.0	228°48.4	3.2'	28°08.0	0.9'	59.1'	
3	223°12.7	· · 02.6	243°10.7	3.2'	28°08.9	0.7'	59.1'	
4	238°12.4	02.3	257°32.9	3.1'	28°09.6	0.5'	59.1'	
5	253°12.1	01.9	271°55.0	3.1'	28°10.1	0.3'	59.2'	
6	268°11.9	S22°01.5	286°17.1	3.0'	S28°10.4	0.1'	59.2'	
7	283°11.6	01.2	300°39.1	3.0'	28°10.6	-0.1'	59.2'	
8	298°11.4	00.8	315°01.1	3.0'	28°10.5	-0.3'	59.3'	
9	313°11.1	· · 00.5	329°23.1	2.9'	28°10.2	-0.5'	59.3'	
10	328°10.9	22°00.1	343°45.0	2.9'	28°09.7	-0.7'	59.3'	
11	343°10.6	21°59.7	358°06.9	2.9'	28°09.0	-0.9'	59.3'	
12	358°10.4	S21°59.3	12°28.8	2.8'	S28°08.1	-1.1'	59.4'	
13	13°10.1	59.0	26°50.6	2.8'	28°07.0	-1.3'	59.4'	
14	28°09.8	58.6	41°12.4	2.8'	28°05.8	-1.5'	59.4'	
15	43°09.6	· · 58.2	55°34.2	2.8'	28°04.3	-1.7'	59.5'	
16	58°09.3	57.9	69°56.0	2.7'	28°02.6	-1.9'	59.5'	
17	73°09.1	57.5	84°17.7	2.7'	28°00.7	-2.1'	59.5'	
18	88°08.8	S21°57.1	98°39.4	2.7'	S27°58.6	-2.3'	59.5'	
19	103°08.6	56.7	113°01.2	2.7'	27°56.3	-2.5'	59.6'	
20	118°08.3	56.4	127°22.9	2.7'	27°53.8	-2.7'	59.6'	
21	133°08.1	· · 56.0	141°44.6	2.7'	27°51.1	-2.9'	59.6'	
22	148°07.8	55.6	156°06.3	2.7'	27°48.2	-3.1'	59.6'	
23	163°07.6	55.2	170°28.0	2.7'	27°45.1	-3.3'	59.7'	
SD = 16.3'		$d = -0.4'$		SD = 16.1'				
Thu	GHA	Dec	GHA	ν	Dec	d	HP	
0	178°07.3	S21°54.9	184°49.7	2.7'	S27°41.8	-3.5'	59.7'	
1	193°07.1	54.5	199°11.5	2.7'	27°38.2	-3.7'	59.7'	
2	208°06.8	54.1	213°33.2	2.7'	27°34.5	-3.9'	59.7'	
3	223°06.5	· · 53.7	227°54.9	2.8'	27°30.6	-4.1'	59.8'	
4	238°06.3	53.3	242°16.7	2.8'	27°26.5	-4.3'	59.8'	
5	253°06.0	53.0	256°38.5	2.8'	27°22.2	-4.5'	59.8'	
6	268°05.8	S21°52.6	271°00.3	2.8'	S27°17.6	-4.7'	59.8'	
7	283°05.5	52.2	285°22.1	2.9'	27°12.9	-4.9'	59.9'	
8	298°05.3	51.8	299°44.0	2.9'	27°08.0	-5.1'	59.9'	
9	313°05.0	· · 51.4	314°05.9	2.9'	27°02.9	-5.3'	59.9'	
10	328°04.8	51.1	328°27.8	3.0'	26°57.6	-5.5'	59.9'	
11	343°04.5	50.7	342°49.7	3.0'	26°52.1	-5.7'	59.9'	
12	358°04.3	S21°50.3	357°11.7	3.0'	S26°46.4	-5.9'	60.0'	
13	13°04.0	49.9	11°33.8	3.1'	26°40.4	-6.1'	60.0'	
14	28°03.8	49.5	25°55.9	3.1'	26°34.3	-6.3'	60.0'	
15	43°03.5	· · 49.1	40°18.0	3.2'	26°28.1	-6.5'	60.0'	
16	58°03.3	48.7	54°40.2	3.2'	26°21.6	-6.7'	60.0'	
17	73°03.0	48.3	69°02.4	3.3'	26°14.9	-6.9'	60.1'	
18	88°02.8	S21°47.9	83°24.7	3.3'	S26°08.0	-7.1'	60.1'	
19	103°02.5	47.6	97°47.0	3.4'	26°01.0	-7.2'	60.1'	
20	118°02.3	47.2	112°09.4	3.5'	25°53.7	-7.4'	60.1'	
21	133°02.1	· · 46.8	126°31.9	3.5'	25°46.3	-7.6'	60.1'	
22	148°01.8	46.4	140°54.4	3.6'	25°38.6	-7.8'	60.2'	
23	163°01.6	46.0	155°17.0	3.7'	25°30.8	-8.0'	60.2'	
SD = 16.3'		$d = -0.4'$		SD = 16.3'				
Fri	GHA	Dec	GHA	ν	Dec	d	HP	
0	178°01.3	S21°45.6	169°39.6	3.7'	S25°22.8	-8.2'	60.2'	
1	193°01.1	45.2	184°02.4	3.8'	25°14.7	-8.4'	60.2'	
2	208°00.8	44.8	198°25.2	3.9'	25°06.3	-8.5'	60.2'	
3	223°00.6	· · 44.4	212°48.0	3.9'	24°57.8	-8.7'	60.2'	
4	238°00.3	44.0	227°11.0	4.0'	24°49.1	-8.9'	60.2'	
5	253°00.1	43.6	241°34.0	4.1'	24°40.2	-9.1'	60.3'	
6	267°59.8	S21°43.2	255°57.1	4.2'	S24°31.1	-9.2'	60.3'	
7	282°59.6	42.8	270°20.3	4.3'	24°21.8	-9.4'	60.3'	
8	297°59.4	42.4	284°43.5	4.3'	24°12.4	-9.6'	60.3'	
9	312°59.1	· · 42.0	299°06.9	4.4'	24°02.8	-9.8'	60.3'	
10	327°58.9	41.6	313°30.3	4.5'	23°53.1	-9.9'	60.3'	
11	342°58.6	41.2	327°53.8	4.6'	23°43.2	-10.1'	60.3'	
12	357°58.4	S21°40.8	342°17.4	4.7'	S23°33.1	-10.3'	60.3'	
13	12°58.1	40.4	356°41.1	4.8'	23°22.8	-10.4'	60.3'	
14	27°57.9	40.0	11°04.9	4.9'	23°12.4	-10.6'	60.4'	
15	42°57.6	· · 39.6	25°28.7	5.0'	23°01.8	-10.7'	60.4'	
16	57°57.4	39.2	39°52.7	5.1'	22°51.1	-10.9'	60.4'	
17	72°57.2	38.8	54°16.7	5.1'	22°40.2	-11.0'	60.4'	
18	87°56.9	S21°38.3	68°40.9	5.2'	S22°29.2	-11.2'	60.4'	
19	102°56.7	37.9	83°05.1	5.3'	22°18.0	-11.4'	60.4'	
20	117°56.4	37.5	97°29.5	5.4'	22°06.6	-11.5'	60.4'	
21	132°56.2	· · 37.1	111°53.9	5.5'	21°55.1	-11.6'	60.4'	
22	147°56.0	36.7	126°18.4	5.6'	21°43.5	-11.8'	60.4'	
23	162°55.7	36.3	140°43.1	5.7'	21°31.7	-11.9'	60.4'	
SD = 16.3'		$d = -0.4'$		SD = 16.4'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:10	10:09	■	■	14:07	16:06
N 70°	07:54	09:30	■	■	14:46	16:22
68°	07:41	09:03	10:55	13:21	15:13	16:35
66°	07:30	08:42	10:06	14:10	15:34	16:46
64°	07:20	08:25	09:35	14:41	15:50	16:56
62°	07:12	08:12	09:12	15:04	16:04	17:04
60°	07:05	08:00	08:54	15:22	16:16	17:11
N 58°	06:58	07:49	08:39	15:37	16:26	17:18
56°	06:53	07:40	08:26	15:50	16:35	17:23
54°	06:47	07:32	08:15	16:01	16:44	17:29
52°	06:42	07:25	08:05	16:11	16:51	17:34
50°	06:37	07:18	07:56	16:20	16:58	17:38
45°	06:27	07:04	07:37	16:39	17:12	17:49
N 40°	06:18	06:51	07:21	16:54	17:24	17:58
35°	06:09	06:41	07:08	17:07	17:35	18:06
30°	06:01	06:31	06:57	17:19	17:45	18:14
20°	05:46	06:14	06:37	17:38	18:02	18:29
N 10°	05:32	05:58	06:20	17:55	18:18	18:44
0°	05:16	05:42	06:04	18:11	18:34	19:00
S 10°	04:58	05:25	05:48	18:28	18:50	19:17
20°	04:37	05:06	05:30	18:45	19:09	19:38
30°	04:10	04:43	05:10	19:05	19:32	20:05
35°	03:53	04:29	04:58	19:17	19:46	20:22
40°	03:31	04:12	04:44	19:31	20:03	20:43
45°	03:04	03:51	04:27	19:48	20:24	21:11
S 50°	02:24	03:24	04:07	20:08	20:50	21:50
52°	02:02	03:11	03:57	20:18	21:04	22:12
54°	01:31	02:55	03:46	20:29	21:19	22:42
56°	00:25	02:36	03:33	20:41	21:38	23:39
58°	—:—	02:12	03:18	20:56	22:01	—:—
S 60°	—:—	01:39	03:01	21:13	22:34	—:—

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	■	■	■	■
66°	■	■	12:29	■	■	14:14
64°	■	■	11:29	■	■	15:14
62°	■	11:03	10:54	■	13:28	15:47
60°	09:40	10:16	10:29	12:37	14:14	16:12
N 58°	09:01	09:46	10:08	13:16	14:44	16:31
56°	08:34	09:23	09:52	13:43	15:07	16:47
54°	08:12	09:04	09:37	14:04	15:26	17:01
52°	07:54	08:48	09:25	14:22	15:41	17:13
50°	07:39	08:34	09:14	14:37	15:55	17:24
45°	07:08	08:06	08:50	15:08	16:23	17:46
N 40°	06:45	07:43	08:32	15:32	16:44	18:03
35°	06:25	07:25	08:16	15:51	17:02	18:18
30°	06:08	07:09	08:02	16:08	17:18	18:31
20°	05:40	06:42	07:39	16:36	17:44	18:52
N 10°	05:15	06:18	07:18	17:01	18:06	19:11
0°	04:53	05:57	06:59	17:23	18:27	19:28
S 10°	04:30	05:35	06:40	17:46	18:47	19:45
20°	04:06	05:11	06:20	18:10	19:09	20:03
30°	03:38	04:44	05:56	18:37	19:35	20:24
35°	03:21	04:28	05:42	18:54	19:50	20:36
40°	03:02	04:09	05:25	19:13	20:07	20:50
45°	02:38	03:46	05:06	19:36	20:27	21:06
S 50°	02:08	03:17	04:41	20:06	20:53	21:26
52°	01:53	03:03	04:29	20:20	21:05	21:36
54°	01:36	02:46	04:16	20:37	21:19	21:46
56°	01:15	02:26	04:00	20:57	21:36	21:58
58°	00:49	02:01	03:41	21:22	21:55	22:12
S 60°	00:13	01:28	03:18	21:56	22:19</	

January 13, 14, 15 UT (Sat., Sun., Mon.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	111°58.8	215°39.6	S21°19.0	195°07.7	S23°57.3	78°02.3	N12°25.0	135°07.4	S11°24.8	Alpheratz	357°35.7	29°13.5					
1	127°01.2	230°38.8	19.4	210°08.1	57.3	93°04.7	25.0	150°09.6	24.7	Ankaa	353°08.0	-42°10.8					
2	142°03.7	245°38.0	19.7	225°08.5	57.2	108°07.1	25.0	165°11.8	24.6	Schedar	349°32.1	56°40.4					
3	157°06.1	260°37.2	· · 20.1	240°08.9	· · 57.1	123°09.4	· · 25.1	180°14.1	· · 24.5	Diphda	348°48.2	-17°51.5					
4	172°08.6	275°36.4	20.5	255°09.3	57.1	138°11.8	25.1	195°16.3	24.4	Achernar	335°20.7	-57°07.2					
5	187°11.1	290°35.6	20.9	270°09.7	57.0	153°14.1	25.2	210°18.5	24.3	Hamal	327°52.0	23°34.6					
6	202°13.5	305°34.8	S21°21.3	285°10.1	S23°56.9	168°16.5	N12°25.2	225°20.7	S11°24.2	Polaris	314°13.1	89°22.2					
7	217°16.0	320°34.0	21.7	300°10.6	56.9	183°18.9	25.3	240°23.0	24.1	Acamar	315°12.2	-40°12.7					
8	232°18.5	335°33.2	22.1	315°11.0	56.8	198°21.2	25.3	255°25.2	24.0	Menkar	314°06.9	4°11.0					
9	247°20.9	350°32.4	· · 22.4	330°11.4	· · 56.7	213°23.6	· · 25.4	270°27.4	· · 23.9	Mirfak	308°29.2	49°57.0					
10	262°23.4	5°31.6	22.8	345°11.8	56.6	228°25.9	25.4	285°29.6	23.8	Aldebaran	290°40.3	16°33.5					
11	277°25.9	20°30.8	23.2	0°12.2	56.6	243°28.3	25.5	300°31.9	23.7	Rigel	281°04.4	-8°10.5					
12	292°28.3	35°30.0	S21°23.6	15°12.6	S23°56.5	258°30.7	N12°25.5	315°34.1	S11°23.6	Capella	280°22.7	46°01.4					
13	307°30.8	50°29.2	24.0	30°13.0	56.4	273°33.0	25.6	330°36.3	23.5	Bellatrix	278°23.5	6°22.3					
14	322°33.3	65°28.4	24.3	45°13.4	56.4	288°35.4	25.6	345°38.5	23.4	Elnath	278°02.6	28°37.7					
15	337°35.7	80°27.6	· · 24.7	60°13.9	· · 56.3	303°37.7	· · 25.7	0°40.8	· · 23.4	Alnilam	275°38.2	-1°11.2					
16	352°38.2	95°26.8	25.1	75°14.3	56.2	318°40.1	25.7	15°43.0	23.3	Betelgeuse	270°52.6	7°24.7					
17	7°40.6	110°26.0	25.5	90°14.7	56.1	333°42.4	25.8	30°45.2	23.2	Canopus	263°52.2	-52°42.5					
18	22°43.1	125°25.2	S21°25.8	105°15.1	S23°56.1	348°44.8	N12°25.8	45°47.4	S11°23.1	Sirius	258°26.6	-16°45.0					
19	37°45.6	140°24.4	26.2	120°15.5	56.0	3°47.1	25.9	60°49.7	23.0	Adhara	255°06.1	-29°00.3					
20	52°48.0	155°23.6	26.6	135°15.9	55.9	18°49.5	25.9	75°51.9	22.9	Procyon	244°51.3	5°09.8					
21	67°50.5	170°22.7	· · 26.9	150°16.3	· · 55.9	33°51.9	· · 26.0	90°54.1	· · 22.8	Pollux	243°17.9	27°58.0					
22	82°53.0	185°21.9	27.3	165°16.7	55.8	48°54.2	26.0	105°56.3	22.7	Avior	234°14.4	-59°35.1					
23	97°55.4	200°21.1	27.7	180°17.2	55.7	63°56.6	26.1	120°58.6	22.6	Suhail	222°46.5	-43°31.7					
Mer.pass. 16:29		ν -0.8' d 0.4' m -3.97			ν 0.4' d -0.1' m 1.36			ν 2.4' d 0.0' m -2.49			ν 2.2' d -0.1' m 0.97						

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	112°57.9	215°20.3	S21°28.0	195°17.6	S23°55.6	78°58.9	N12°26.1	136°00.8	S11°22.5				
1	128°00.4	230°19.5	28.4	210°18.0	55.5	94°01.3	26.2	151°03.0	22.4				
2	143°02.8	245°18.7	28.8	225°18.4	55.5	109°03.6	26.2	166°05.2	22.3				
3	158°05.3	260°17.9	· · 29.1	240°18.8	· · 55.4	124°06.0	· · 26.3	181°07.5	· · 22.2				
4	173°07.7	275°17.1	29.5	255°19.2	55.3	139°08.3	26.4	196°09.7	22.1				
5	188°10.2	290°16.3	29.9	270°19.6	55.2	154°10.7	26.4	211°11.9	22.0				
6	203°12.7	305°15.5	S21°30.2	285°20.0	S23°55.2	169°13.0	N12°26.5	226°14.1	S11°21.9				
7	218°15.1	320°14.7	30.6	300°20.5	55.1	184°15.4	26.5	241°16.3	21.8				
8	233°17.6	335°13.9	30.9	315°20.9	55.0	199°17.7	26.6	256°18.6	21.7				
9	248°20.1	350°13.1	· · 31.3	330°21.3	· · 54.9	214°20.1	· · 26.6	271°20.8	· · 21.6				
10	263°22.5	5°12.2	31.7	345°21.7	54.8	229°22.4	26.7	286°23.0	21.5				
11	278°25.0	20°11.4	32.0	0°22.1	54.8	244°24.8	26.7	301°25.2	21.5				
12	293°27.5	35°10.6	S21°32.4	15°22.5	S23°54.7	259°27.1	N12°26.8	316°27.5	S11°21.4				
13	308°29.9	50°09.8	32.7	30°22.9	54.6	274°29.5	26.8	331°29.7	21.3				
14	323°32.4	65°09.0	33.1	45°23.3	54.5	289°31.8	26.9	346°31.9	21.2				
15	338°34.9	80°08.2	· · 33.4	60°23.7	· · 54.4	304°34.2	· · 26.9	1°34.1	· · 21.1				
16	353°37.3	95°07.4	33.8	75°24.2	54.3	319°36.5	27.0	16°36.4	21.0				
17	8°39.8	110°06.6	34.1	90°24.6	54.3	334°38.9	27.0	31°38.6	20.9				
18	23°42.2	125°05.8	S21°34.5	105°25.0	S23°54.2	349°41.2	N12°27.1	46°40.8	S11°20.8				
19	38°44.7	140°04.9	34.8	120°25.4	54.1	4°43.6	27.1	61°43.0	20.7				
20	53°47.2	155°04.1	35.2	135°25.8	54.0	19°45.9	27.2	76°45.2	20.6				
21	68°49.6	170°03.3	· · 35.5	150°26.2	· · 53.9	34°48.3	· · 27.2	91°47.5	· · 20.5				
22	83°52.1	185°02.5	35.8	165°26.6	53.8	49°50.6	27.3	106°49.7	20.4				
23	98°54.6	200°01.7	36.2	180°27.0	53.8	64°53.0	27.4	121°51.9	20.3				
Mer.pass. 16:25		ν -0.8' d 0.4' m -3.97			ν 0.4' d -0.1' m 1.35			ν 2.4' d 0.1' m -2.49			ν 2.2' d -0.1' m 0.97		

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	113°57.0	215°00.9	S21°36.5	195°27.5	S23°53.7	79°55.3	N12°27.4	136°54.1	S11°20.2				
1	128°59.5	230°00.1	36.9	210°27.9	53.6	94°57.7	27.5	151°56.3	20.1				
2	144°02.0	244°59.2	37.2	225°28.3	53.5	110°00.0	27.5	166°58.6	20.0				
3	159°04.4	259°58.4	· · 37.5	240°28.7	· · 53.4	125°02.3	· · 27.6	182°00.8	· · 19.9				
4	174°06.9	274°57.6	37.9	255°29.1	53.3	140°04.7	27.6	197°03.0	19.8				
5	189°09.4	289°56.8	38.2	270°29.5	53.2	155°07.0	27.7	212°05.2	19.7				
6	204°11.8	304°56.0	S21°38.5	285°29.9	S23°53.1	170°09.4	N12°27.7	227°07.5	S11°19.6				
7	219°14.3	319°55.2	38.9	300°30.3	53.0	185°11.7	27.8	242°09.7	19.5				
8	234°16.7	334°54.3	39.2	315°30.7	53.0	200°14.1	27.8	257°11.9	19.4				
9	249°19.2	349°53.5	· · 39.5	330°31.2	· · 52.9	215°16.4	· · 27.9	272°14.1	· · 19.3				
10	264°21.7	4°52.7	39.9	345°31.6	52.8	230°18.7	27.9	287°16.3	19.2				
11	279°24.1	19°51.9	40.2	0°32.0	52.7	245°21.1	28.0	302°18.6	19.2				
12	294°26.6	34°51.1	S21°40.5	15°32.4	S23°52.6	260°23.4	N12°28.1	317°20.8	S11°19.1				
13	309°29.1	49°50.3	40.9	30°32.8	52.5	275°25.8	28.1	332°23.0	19.0				
14	324°31.5	64°49.4	41.2	45°33.2	52.4	290°28.1	28.2	347°25.2	18.9				
15	339°34.0	79°48.6	· · 41.5	60°33.6	· · 52.3	305°30.5	· · 28.2	2°27.4	· · 18.8				
16	354°36.5	94°47.8	41.8	75°34.0	52.2	320°32.8	28.3	17°29.7	18.7				
17	9°38.9	109°47.0	42.2	90°34.4	52.1	335°35.1	28.3	32°31.9	18.6				
18	24°41.4	124°46.2	S21°42.5	105°34.9	S23°52.0	350°37.5	N12°28.4	47°34.1	S11°18.5				
19	39°43.8	139°45.3	42.8	120°35.3	51.9	5°39.8	28.4	62°36.3	18.4				
20	54°46.3	154°44.5	43.1	135°35.7	51.8	20°42.1	28.5	77°38.5	18.3				
21	69°48.8	169°43.7	· · 43.4	150°36.1	· · 51.7	35°44.5	· · 28.6	92°40.8	· · 18.2				
22	84°51.2	184°42.9	43.8	165°36.5	51.6	50°46.8	28.6	107°43.0	18.1				
23	99°53.7	199°42.1	44.1	180°36.9	51.5	65°49.2	28.7	122°45.2	18.0				
Mer.pass. 16:22		ν -0.8' d 0.3' m -3.96			ν 0.4' d -0.1' m 1.35			ν 2.3' d 0.1' m -2.48			ν 2.2' d -0.1' m 0.98		

Jan 13 Sat	SHA	Mer.pass
Venus	103°40.8	09:38
Mars	83°08.9	10:59
Jupiter	326°03.6	18:45
Saturn	23°08.6	14:57
Jan 14 Sun	SHA	Mer.pass
Venus	102°22.4	09:39
Mars	82°19.7	10:59
Jupiter	326°01.0	18:41
Saturn	23°02.9	14:54
Jan 15 Mon	SHA	Mer.pass
Venus	101°03.8	09:40
Mars	81°30.4	10:58
Jupiter	325°58.3	18:37
Saturn	22°57.1	14:50
Horizontal parallax		
Venus:		0.1
Mars:		0.1

Sun				Moon				
Sat	GHA	Dec		GHA	ν	Dec	d	HP
0	177°55.5	S21°35.9		155°07.8	5.8'	S21°19.7	-12.1'	60.4'
1	192°55.2	35.5		169°32.6	5.9'	21°07.7	-12.2'	60.4'
2	207°55.0	35.1		183°57.5	6.0'	20°55.4	-12.4'	60.4'
3	222°54.8	· · 34.6		198°22.6	6.1'	20°43.1	-12.5'	60.4'
4	237°54.5	34.2		212°47.7	6.2'	20°30.6	-12.6'	60.4'
5	252°54.3	33.8		227°12.9	6.3'	20°18.0	-12.8'	60.4'
6	267°54.0	S21°33.4		241°38.2	6.4'	S20°05.2	-12.9'	60.4'
7	282°53.8	33.0		256°03.7	6.5'	19°52.3	-13.0'	60.4'
8	297°53.6	32.6		270°29.2	6.6'	19°39.3	-13.1'	60.4'
9	312°53.3	· · 32.1		284°54.8	6.7'	19°26.2	-13.3'	60.4'
10	327°53.1	31.7		299°20.6	6.8'	19°12.9	-13.4'	60.4'
11	342°52.8	31.3		313°46.4	6.9'	18°59.5	-13.5'	60.4'
12	357°52.6	S21°30.9		328°12.3	7.0'	S18°46.0	-13.6'	60.4'
13	12°52.4	30.4		342°38.4	7.1'	18°32.4	-13.7'	60.4'
14	27°52.1	30.0		357°04.5	7.2'	18°18.6	-13.8'	60.4'
15	42°51.9	· · 29.6		11°30.7	7.3'	18°04.8	-14.0'	60.4'
16	57°51.7	29.2		25°57.1	7.4'	17°50.8	-14.1'	60.4'
17	72°51.4	28.8		40°23.5	7.5'	17°36.8	-14.2'	60.4'
18	87°51.2	S21°28.3		54°50.0	7.6'	S17°22.6	-14.3'	60.4'
19	102°51.0	27.9		69°16.7	7.7'	17°08.3	-14.4'	60.4'
20	117°50.7	27.5		83°43.4	7.8'	16°53.9	-14.5'	60.4'
21	132°50.5	· · 27.0		98°10.2	7.9'	16°39.4	-14.6'	60.4'
22	147°50.3	26.6		112°37.2	8.0'	16°24.9	-14.7'	60.4'
23	162°50.0	26.2		127°04.2	8.1'	16°10.2	-14.8'	60.4'
SD = 16.3'			$d = -0.4'$	SD = 16.5'				
Sun	GHA	Dec		GHA	ν	Dec	d	HP
0	177°49.8	S21°25.8		141°31.3	8.2'	S15°55.4	-14.9'	60.4'
1	192°49.6	25.3		155°58.5	8.3'	15°40.5	-15.0'	60.4'
2	207°49.3	24.9		170°25.9	8.4'	15°25.6	-15.0'	60.4'
3	222°49.1	· · 24.5		184°53.3	8.5'	15°10.5	-15.1'	60.4'
4	237°48.9	24.0		199°20.8	8.6'	14°55.4	-15.2'	60.4'
5	252°48.6	23.6		213°48.4	8.7'	14°40.2	-15.3'	60.4'
6	267°48.4	S21°23.2		228°16.1	8.8'	S14°24.9	-15.4'	60.4'
7	282°48.2	22.7		242°43.8	8.9'	14°09.5	-15.5'	60.4'
8	297°47.9	22.3		257°11.7	9.0'	13°54.1	-15.5'	60.4'
9	312°47.7	· · 21.9		271°39.7	9.1'	13°38.6	-15.6'	60.3'
10	327°47.5	21.4		286°07.7	9.1'	13°22.9	-15.7'	60.3'
11	342°47.2	21.0		300°35.9	9.2'	13°07.3	-15.7'	60.3'
12	357°47.0	S21°20.5		315°04.1	9.3'	S12°51.5	-15.8'	60.3'
13	12°46.8	20.1		329°32.4	9.4'	12°35.7	-15.9'	60.3'
14	27°46.6	19.7		344°00.8	9.5'	12°19.8	-15.9'	60.3'
15	42°46.3	· · 19.2		358°29.3	9.6'	12°03.9	-16.0'	60.3'
16	57°46.1	18.8		12°57.8	9.6'	11°47.9	-16.1'	60.3'
17	72°45.9	18.3		27°26.5	9.7'	11°31.9	-16.1'	60.3'
18	87°45.6	S21°17.9		41°55.2	9.8'	S11°15.7	-16.2'	60.2'
19	102°45.4	17.4		56°24.0	9.9'	10°59.6	-16.2'	60.2'
20	117°45.2	17.0		70°52.9	10.0'	10°43.3	-16.3'	60.2'
21	132°45.0	· · 16.6		85°21.9	10.0'	10°27.1	-16.3'	60.2'
22	147°44.7	16.1		99°50.9	10.1'	10°10.7	-16.4'	60.2'
23	162°44.5	15.7		114°20.0	10.2'	09°54.4	-16.4'	60.2'
SD = 16.3'			$d = -0.4'$	SD = 16.5'				
Mon	GHA	Dec		GHA	ν	Dec	d	HP
0	177°44.3	S21°15.2		128°49.2	10.3'	S09°37.9	-16.5'	60.2'
1	192°44.0	14.8		143°18.5	10.3'	09°21.5	-16.5'	60.1'
2	207°43.8	14.3		157°47.8	10.4'	09°05.0	-16.5'	60.1'
3	222°43.6	· · 13.9		172°17.2	10.5'	08°48.4	-16.6'	60.1'
4	237°43.4	13.4		186°46.7	10.5'	08°31.8	-16.6'	60.1'
5	252°43.1	13.0		201°16.2	10.6'	08°15.2	-16.7'	60.1'
6	267°42.9	S21°12.5		215°45.8	10.7'	S07°58.5	-16.7'	60.1'
7	282°42.7	12.1		230°15.5	10.7'	07°41.8	-16.7'	60.1'
8	297°42.5	11.6		244°45.3	10.8'	07°25.1	-16.8'	60.0'
9	312°42.2	· · 11.2		259°15.1	10.9'	07°08.4	-16.8'	60.0'
10	327°42.0	10.7		273°44.9	10.9'	06°51.6	-16.8'	60.0'
11	342°41.8	10.3		288°14.8	11.0'	06°34.8	-16.8'	60.0'
12	357°41.6	S21°09.8		302°44.8	11.0'	S06°18.0	-16.9'	60.0'
13	12°41.4	09.3		317°14.8	11.1'	06°01.1	-16.9'	59.9'
14	27°41.1	08.9		331°44.9	11.1'	05°44.2	-16.9'	59.9'
15	42°40.9	· · 08.4		346°15.1	11.2'	05°27.3	-16.9'	59.9'
16	57°40.7	08.0		0°45.3	11.3'	05°10.4	-16.9'	59.9'
17	72°40.5	07.5		15°15.5	11.3'	04°53.5	-16.9'	59.9'
18	87°40.2	S21°07.1		29°45.8	11.4'	S04°36.6	-16.9'	59.8'
19	102°40.0	06.6		44°16.2	11.4'	04°19.6	-17.0'	59.8'
20	117°39.8	06.1		58°46.6	11.4'	04°02.7	-17.0'	59.8'
21	132°39.6	· · 05.7		73°17.0	11.5'	03°45.7	-17.0'	59.8'
22	147°39.4	05.2		87°47.5	11.5'	03°28.7	-17.0'	59.8'
23	162°39.1	04.7		102°18.1	11.6'	03°11.7	-17.0'	59.7'
SD = 16.3'			$d = -0.4'$	SD = 16.4'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:04	09:58	■	■	14:21	16:15
N 70°	07:49	09:22	■	■	14:56	16:30
68°	07:36	08:57	10:40	13:38	15:22	16:42
66°	07:26	08:37	09:57	14:21	15:41	16:52
64°	07:17	08:21	09:29	14:49	15:57	17:01
62°	07:09	08:08	09:07	15:11	16:10	17:09
60°	07:02	07:57	08:50	15:28	16:22	17:16
N 58°	06:56	07:47	08:35	15:43	16:31	17:22
56°	06:51	07:38	08:23	15:55	16:40	17:28
54°	06:45	07:30	08:12	16:06	16:48	17:33
52°	06:41	07:23	08:02	16:16	16:55	17:38
50°	06:36	07:17	07:54	16:24	17:01	17:42
45°	06:26	07:03	07:36	16:43	17:15	17:52
N 40°	06:17	06:51	07:21	16:57	17:27	18:01
35°	06:09	06:40	07:08	17:10	17:38	18:09
30°	06:02	06:31	06:57	17:21	17:47	18:16
20°	05:47	06:14	06:38	17:40	18:04	18:31
N 10°	05:32	05:58	06:21	17:57	18:19	18:45
0°	05:17	05:43	06:05	18:13	18:35	19:01
S 10°	05:00	05:27	05:49	18:28	18:51	19:18
20°	04:39	05:08	05:32	18:45	19:10	19:38
30°	04:13	04:45	05:12	19:05	19:32	20:04
35°	03:56	04:32	05:01	19:17	19:46	20:21
40°	03:35	04:15	04:47	19:30	20:02	20:42
45°	03:09	03:55	04:31	19:46	20:22	21:08
S 50°	02:30	03:29	04:11	20:06	20:48	21:46
52°	02:09	03:16	04:02	20:15	21:01	22:07
54°	01:41	03:01	03:51	20:26	21:15	22:34
56°	00:53	02:43	03:39	20:38	21:33	23:18
58°	—:—	02:20	03:25	20:52	21:55	—:—
S 60°	—:—	01:50	03:08	21:09	22:25	—:—

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	■	12:01	11:12	■	18:43	21:20
N 70°	12:45	11:38	11:03	16:03	19:03	21:26
68°	12:00	11:20	10:55	16:47	19:19	21:30
66°	11:29	11:06	10:49	17:16	19:32	21:34
64°	11:07	10:54	10:43	17:37	19:42	21:37
62°	10:48	10:43	10:39	17:54	19:51	21:40
60°	10:33	10:34	10:34	18:08	19:58	21:42
N 58°	10:20	10:26	10:31	18:20	20:05	21:44
56°	10:09	10:20	10:27	18:31	20:11	21:46
54°	09:59	10:13	10:24	18:40	20:16	21:48
52°	09:50	10:08	10:22	18:48	20:20	21:49
50°	09:42	10:02	10:19	18:55	20:25	21:51
45°	09:25	09:51	10:14	19:11	20:34	21:54
N 40°	09:10	09:42	10:09	19:23	20:41	21:56
35°	08:58	09:34	10:05	19:34	20:47	21:58
30°	08:48	09:27	10:02	19:43	20:53	22:00

January 16, 17, 18 UT (Tue., Wed., Thu.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	114°56.2	214°41.2	S21°44.4	195°37.3	S23°51.5	80°51.5	N12°28.7	137°47.4	S11°17.9	Alpheratz	357°35.8	29°13.5					
1	129°58.6	229°40.4	44.7	210°37.7	51.4	95°53.8	28.8	152°49.6	17.8	Ankaa	353°08.1	-42°10.8					
2	145°01.1	244°39.6	45.0	225°38.1	51.3	110°56.2	28.8	167°51.9	17.7	Schedar	349°32.1	56°40.4					
3	160°03.6	259°38.8	· · 45.3	240°38.6	· · 51.2	125°58.5	· · 28.9	182°54.1	· · 17.6	Diphda	348°48.2	-17°51.5					
4	175°06.0	274°37.9	45.6	255°39.0	51.1	141°00.8	29.0	197°56.3	17.5	Achernar	335°20.8	-57°07.2					
5	190°08.5	289°37.1	46.0	270°39.4	51.0	156°03.2	29.0	212°58.5	17.4	Hamal	327°52.1	23°34.6					
6	205°11.0	304°36.3	S21°46.3	285°39.8	S23°50.9	171°05.5	N12°29.1	228°00.7	S11°17.3	Polaris	314°14.5	89°22.2					
7	220°13.4	319°35.5	46.6	300°40.2	50.8	186°07.9	29.1	243°02.9	17.2	Acamar	315°12.2	-40°12.7					
8	235°15.9	334°34.6	46.9	315°40.6	50.7	201°10.2	29.2	258°05.2	17.1	Menkar	314°06.9	4°11.0					
9	250°18.3	349°33.8	· · 47.2	330°41.0	· · 50.6	216°12.5	· · 29.2	273°07.4	· · 17.0	Mirfak	308°29.2	49°57.0					
10	265°20.8	4°33.0	47.5	345°41.4	50.4	231°14.9	29.3	288°09.6	16.9	Aldebaran	290°40.3	16°33.5					
11	280°23.3	19°32.2	47.8	0°41.8	50.3	246°17.2	29.4	303°11.8	16.8	Rigel	281°04.4	-8°10.5					
12	295°25.7	34°31.3	S21°48.1	15°42.3	S23°50.2	261°19.5	N12°29.4	318°14.0	S11°16.7	Capella	280°22.7	46°01.4					
13	310°28.2	49°30.5	48.4	30°42.7	50.1	276°21.9	29.5	333°16.3	16.6	Bellatrix	278°23.5	6°22.3					
14	325°30.7	64°29.7	48.7	45°43.1	50.0	291°24.2	29.5	348°18.5	16.5	Elnath	278°02.6	28°37.7					
15	340°33.1	79°28.9	· · 49.0	60°43.5	· · 49.9	306°26.5	· · 29.6	3°20.7	· · 16.4	Alnilam	275°38.3	-1°11.2					
16	355°35.6	94°28.0	49.3	75°43.9	49.8	321°28.9	29.6	18°22.9	16.4	Betelgeuse	270°52.7	7°24.7					
17	10°38.1	109°27.2	49.6	90°44.3	49.7	336°31.2	29.7	33°25.1	16.3	Canopus	263°52.2	-52°42.6					
18	25°40.5	124°26.4	S21°49.9	105°44.7	S23°49.6	351°33.5	N12°29.8	48°27.4	S11°16.2	Sirius	258°26.6	-16°45.0					
19	40°43.0	139°25.6	50.2	120°45.1	49.5	6°35.9	29.8	63°29.6	16.1	Adhara	255°06.1	-29°00.3					
20	55°45.5	154°24.7	50.5	135°45.5	49.4	21°38.2	29.9	78°31.8	16.0	Procyon	244°51.3	5°09.8					
21	70°47.9	169°23.9	· · 50.8	150°46.0	· · 49.3	36°40.5	· · 29.9	93°34.0	· · 15.9	Pollux	243°17.9	27°58.0					
22	85°50.4	184°23.1	51.1	165°46.4	49.2	51°42.8	30.0	108°36.2	15.8	Avior	234°14.4	-59°35.1					
23	100°52.8	199°22.2	51.3	180°46.8	49.1	66°45.2	30.0	123°38.4	15.7	Suhail	222°46.5	-43°31.7					
Mer.pass. 16:18			ν -0.8' d 0.3' m -3.96			ν 0.4' d -0.1' m 1.35			ν 2.3' d 0.1' m -2.47			ν 2.2' d -0.1' m 0.98					

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	115°55.3	214°21.4	S21°51.6	195°47.2	S23°49.0	81°47.5	N12°30.1	138°40.7	S11°15.6					
1	130°57.8	229°20.6	51.9	210°47.6	48.9	96°49.8	30.2	153°42.9	15.5					
2	146°00.2	244°19.8	52.2	225°48.0	48.8	111°52.2	30.2	168°45.1	15.4					
3	161°02.7	259°18.9	· · 52.5	240°48.4	· · 48.6	126°54.5	· · 30.3	183°47.3	· · 15.3					
4	176°05.2	274°18.1	52.8	255°48.8	48.5	141°56.8	30.3	198°49.5	15.2					
5	191°07.6	289°17.3	53.1	270°49.2	48.4	156°59.1	30.4	213°51.7	15.1					
6	206°10.1	304°16.4	S21°53.4	285°49.7	S23°48.3	172°01.5	N12°30.5	228°54.0	S11°15.0					
7	221°12.6	319°15.6	53.6	300°50.1	48.2	187°03.8	30.5	243°56.2	14.9					
8	236°15.0	334°14.8	53.9	315°50.5	48.1	202°06.1	30.6	258°58.4	14.8					
9	251°17.5	349°13.9	· · 54.2	330°50.9	· · 48.0	217°08.5	· · 30.6	274°00.6	· · 14.7					
10	266°19.9	4°13.1	54.5	345°51.3	47.9	232°10.8	30.7	289°02.8	14.6					
11	281°22.4	19°12.3	54.8	0°51.7	47.8	247°13.1	30.8	304°05.0	14.5					
12	296°24.9	34°11.4	S21°55.0	15°52.1	S23°47.6	262°15.4	N12°30.8	319°07.3	S11°14.4					
13	311°27.3	49°10.6	55.3	30°52.5	47.5	277°17.8	30.9	334°09.5	14.3					
14	326°29.8	64°09.8	55.6	45°52.9	47.4	292°20.1	30.9	349°11.7	14.2					
15	341°32.3	79°08.9	· · 55.9	60°53.3	· · 47.3	307°22.4	· · 31.0	4°13.9	· · 14.1					
16	356°34.7	94°08.1	56.1	75°53.8	47.2	322°24.7	31.1	19°16.1	14.0					
17	11°37.2	109°07.3	56.4	90°54.2	47.1	337°27.1	31.1	34°18.3	13.9					
18	26°39.7	124°06.4	S21°56.7	105°54.6	S23°46.9	352°29.4	N12°31.2	49°20.6	S11°13.8					
19	41°42.1	139°05.6	56.9	120°55.0	46.8	7°31.7	31.2	64°22.8	13.7					
20	56°44.6	154°04.8	57.2	135°55.4	46.7	22°34.0	31.3	79°25.0	13.6					
21	71°47.1	169°03.9	· · 57.5	150°55.8	· · 46.6	37°36.4	· · 31.4	94°27.2	· · 13.5					
22	86°49.5	184°03.1	57.7	165°56.2	46.5	52°38.7	31.4	109°29.4	13.4					
23	101°52.0	199°02.3	58.0	180°56.6	46.4	67°41.0	31.5	124°31.6	13.3					
Mer.pass. 16:14			ν -0.8' d 0.3' m -3.96			ν 0.4' d -0.1' m 1.35			ν 2.3' d 0.1' m -2.46			ν 2.2' d -0.1' m 0.98		

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	116°54.4	214°01.4	S21°58.3	195°57.0	S23°46.2	82°43.3	N12°31.6	139°33.9	S11°13.2					
1	131°56.9	229°00.6	58.5	210°57.5	46.1	97°45.6	31.6	154°36.1	13.1					
2	146°59.4	243°59.8	58.8	225°57.9	46.0	112°48.0	31.7	169°38.3	13.0					
3	162°01.8	258°58.9	· · 59.0	240°58.3	· · 45.9	127°50.3	· · 31.7	184°40.5	· · 12.9					
4	177°04.3	273°58.1	59.3	255°58.7	45.8	142°52.6	31.8	199°42.7	12.8					
5	192°06.8	288°57.3	59.6	270°59.1	45.6	157°54.9	31.9	214°44.9	12.7					
6	207°09.2	303°56.4	S21°59.8	285°59.5	S23°45.5	172°57.2	N12°31.9	229°47.1	S11°12.6					
7	222°11.7	318°55.6	22°00.1	300°59.9	45.4	187°59.6	32.0	244°49.4	12.5					
8	237°14.2	333°54.7	00.3	316°00.3	45.3	203°01.9	32.1	259°51.6	12.4					
9	252°16.6	348°53.9	· · 00.6	331°00.7	· · 45.1	218°04.2	· · 32.1	274°53.8	· · 12.3					
10	267°19.1	3°53.1	00.8	346°01.1	45.0	233°06.5	32.2	289°56.0	12.2					
11	282°21.5	18°52.2	01.1	1°01.6	44.9	248°08.8	32.2	304°58.2	12.1					
12	297°24.0	33°51.4	S22°01.3	16°02.0	S23°44.8	263°11.2	N12°32.3	320°00.4	S11°12.0					
13	312°26.5	48°50.6	01.6	31°02.4	44.7	278°13.5	32.4	335°02.7	12.0					
14	327°28.9	63°49.7	01.8	46°02.8	44.5	293°15.8	32.4	350°04.9	11.9					
15	342°31.4	78°48.9	· · 02.1	61°03.2	· · 44.4	308°18.1	· · 32.5	5°07.1	· · 11.8					
16	357°33.9	93°48.0	02.3	76°03.6	44.3	323°20.4	32.6	20°09.3	11.7					
17	12°36.3	108°47.2	02.6	91°04.0	44.1	338°22.7	32.6	35°11.5	11.6					
18	27°38.8	123°46.4	S22°02.8	106°04.4	S23°44.0	353°25.1	N12°32.7	50°13.7	S11°11.5					
19	42°41.3	138°45.5	03.1	121°04.8	43.9	8°27.4	32.7	65°15.9	11.4					
20	57°43.7	153°44.7	03.3	136°05.2	43.8	23°29.7	32.8	80°18.2	11.3					
21	72°46.2	168°43.8	· · 03.5	151°05.7	· · 43.6	38°32.0	· · 32.9	95°20.4	· · 11.2					
22	87°48.7	183°43.0	03.8	166°06.1	43.5	53°34.3	32.9	110°22.6	11.1					
23	102°51.1	198°42.2	04.0	181°06.5	43.4	68°36.6	33.0	125°24.8	11.0					
Mer.pass. 16:10			ν -0.8' d 0.3' m -3.95			ν 0.4' d -0.1' m 1.35			ν 2.3' d 0.1' m -2.45			ν 2.2' d -0.1' m 0.98		

Jan 16 Tue	SHA	Mer.pass
Venus	99°45.1	09:42
Mars	80°41.2	10:57
Jupiter	325°55.3	18:34
Saturn	22°51.2	14:47

Jan 17 Wed	SHA	Mer.pass
Venus	98°26.1	09:43
Mars	79°51.9	10:57
Jupiter	325°52.2	18:30
Saturn	22°45.4	14:43

Jan 18 Thu	SHA	Mer.pass
Venus	97°07.0	09:44
Mars	79°02.6	10:56
Jupiter	325°48.9	18:26
Saturn	22°39.4	14:40

Horizontal parallax		
Venus:	0.1	
Mars:	0.1	

Sun							Moon							
Tue	GHA	Dec	GHA	ν	Dec	d	HP	Lat.	Twilight		Sunrise	Sunset	Twilight	
									Naut.	Civil			Civil	Naut.
0	177°38.9	S21°04.3	116°48.6	11.6'	S02°54.8	-17.0'	59.7'	N 72°	07:57	09:45	■	■	14:35	16:24
1	192°38.7	03.8	131°19.2	11.7'	02°37.8	-17.0'	59.7'	N 70°	07:43	09:13	■	■	15:07	16:38
2	207°38.5	03.3	145°49.9	11.7'	02°20.8	-17.0'	59.7'	68°	07:31	08:50	10:26	13:55	15:31	16:49
3	222°38.3	· · 02.9	160°20.6	11.7'	02°03.8	-17.0'	59.7'	66°	07:21	08:31	09:48	14:32	15:49	16:59
4	237°38.0	02.4	174°51.3	11.8'	01°46.8	-17.0'	59.6'	64°	07:13	08:16	09:22	14:58	16:04	17:07
5	252°37.8	01.9	189°22.1	11.8'	01°29.8	-17.0'	59.6'	62°	07:06	08:04	09:02	15:19	16:17	17:15
6	267°37.6	S21°01.5	203°52.9	11.8'	S01°12.9	-17.0'	59.6'	60°	06:59	07:53	08:45	15:35	16:27	17:21
7	282°37.4	01.0	218°23.7	11.9'	00°55.9	-17.0'	59.6'	N 58°	06:53	07:44	08:31	15:49	16:37	17:27
8	297°37.2	00.5	232°54.6	11.9'	00°38.9	-16.9'	59.6'	56°	06:48	07:35	08:19	16:01	16:45	17:32
9	312°37.0	21°00.1	247°25.5	11.9'	00°22.0	-16.9'	59.5'	54°	06:43	07:28	08:09	16:11	16:53	17:37
10	327°36.7	20°59.6	261°56.4	11.9'	S00°05.1	-16.9'	59.5'	52°	06:39	07:21	08:00	16:21	16:59	17:42
11	342°36.5	59.1	276°27.3	12.0'	N00°11.9	16.9'	59.5'	50°	06:35	07:15	07:51	16:29	17:06	17:46
12	357°36.3	S20°58.7	290°58.3	12.0'	N00°28.8	16.9'	59.5'	45°	06:25	07:01	07:34	16:46	17:19	17:55
13	12°36.1	58.2	305°29.3	12.0'	00°45.7	16.9'	59.4'	N 40°	06:16	06:50	07:19	17:01	17:30	18:04
14	27°35.9	57.7	320°00.3	12.0'	01°02.5	16.9'	59.4'	35°	06:09	06:40	07:07	17:13	17:40	18:11
15	42°35.7	· · 57.2	334°31.3	12.1'	01°19.4	16.8'	59.4'	30°	06:01	06:31	06:56	17:24	17:49	18:19
16	57°35.4	56.8	349°02.4	12.1'	01°36.2	16.8'	59.4'	20°	05:47	06:14	06:38	17:42	18:06	18:33
17	72°35.2	56.3	3°33.5	12.1'	01°53.0	16.8'	59.3'	N 10°	05:33	05:59	06:22	17:58	18:21	18:47
18	87°35.0	S20°55.8	18°04.6	12.1'	N02°09.8	16.8'	59.3'	0°	05:18	05:44	06:06	18:14	18:36	19:01
19	102°34.8	55.3	32°35.7	12.1'	02°26.6	16.7'	59.3'	S 10°	05:02	05:28	05:51	18:29	18:52	19:18
20	117°34.6	54.8	47°06.8	12.1'	02°43.3	16.7'	59.3'	20°	04:42	05:10	05:34	18:46	19:09	19:38
21	132°34.4	· · 54.4	61°37.9	12.1'	03°00.0	16.7'	59.2'	30°	04:16	04:48	05:15	19:05	19:31	20:03
22	147°34.2	53.9	76°09.1	12.2'	03°16.7	16.6'	59.2'	35°	04:00	04:35	05:03	19:16	19:45	20:20
23	162°34.0	53.4	90°40.2	12.2'	03°33.3	16.6'	59.2'	40°	03:39	04:19	04:50	19:29	20:00	20:40
SD = 16.3' d = -0.5'							SD = 16.3'							
Wed	GHA	Dec	GHA	ν	Dec	d	HP	Lat.	Moonrise		Moonset			
									Tue	Wed	Thu	Tue	Wed	Thu
0	177°33.7	S20°52.9	105°11.4	12.2'	N03°49.9	16.6'	59.2'	N 72°	10:34	09:58	09:12	23:43	· · ·	02:11
1	192°33.5	52.4	119°42.6	12.2'	04°06.5	16.5'	59.1'	N 70°	10:34	10:06	09:34	23:38	· · ·	01:53
2	207°33.3	52.0	134°13.7	12.2'	04°23.1	16.5'	59.1'	68°	10:34	10:14	09:50	23:34	· · ·	01:38
3	222°33.1	· · 51.5	148°44.9	12.2'	04°39.6	16.5'	59.1'	66°	10:34	10:20	10:04	23:30	· · ·	01:26
4	237°32.9	51.0	163°16.1	12.2'	04°56.1	16.4'	59.1'	64°	10:34	10:25	10:15	23:27	· · ·	01:17
5	252°32.7	50.5	177°47.3	12.2'	05°12.5	16.4'	59.0'	62°	10:34	10:30	10:25	23:25	· · ·	01:08
6	267°32.5	S20°50.0	192°18.5	12.2'	N05°28.9	16.4'	59.0'	60°	10:34	10:34	10:34	23:22	· · ·	01:01
7	282°32.3	49.5	206°49.7	12.2'	05°45.2	16.3'	59.0'	N 58°	10:34	10:37	10:41	23:20	· · ·	00:55
8	297°32.1	49.1	221°20.9	12.2'	06°01.5	16.3'	59.0'	56°	10:34	10:40	10:48	23:18	· · ·	00:50
9	312°31.8	· · 48.6	235°52.0	12.2'	06°17.8	16.2'	58.9'	54°	10:34	10:43	10:54	23:17	· · ·	00:45
10	327°31.6	48.1	250°23.2	12.2'	06°34.0	16.2'	58.9'	52°	10:34	10:46	10:59	23:15	· · ·	00:40
11	342°31.4	47.6	264°54.4	12.2'	06°50.2	16.1'	58.9'	50°	10:34	10:48	11:04	23:14	· · ·	00:36
12	357°31.2	S20°47.1	279°25.6	12.2'	N07°06.3	16.1'	58.9'	45°	10:34	10:54	11:14	23:11	· · ·	00:28
13	12°31.0	46.6	293°56.7	12.1'	07°22.4	16.0'	58.8'	N 40°	10:34	10:58	11:23	23:09	· · ·	00:21
14	27°30.8	46.1	308°27.9	12.1'	07°38.4	16.0'	58.8'	35°	10:34	11:02	11:31	23:07	· · ·	00:14
15	42°30.6	· · 45.6	322°59.0	12.1'	07°54.4	15.9'	58.8'	30°	10:34	11:05	11:38	23:05	· · ·	00:09
16	57°30.4	45.1	337°30.1	12.1'	08°10.3	15.9'	58.8'	20°	10:34	11:11	11:49	23:02	· · ·	00:00
17	72°30.2	44.6	352°01.3	12.1'	08°26.2	15.8'	58.7'	N 10°	10:34	11:17	12:00	22:59	23:52	· · ·
18	87°30.0	S20°44.2	6°32.4	12.1'	N08°42.0	15.7'	58.7'	0°	10:34	11:22	12:10	22:57	23:44	· · ·
19	102°29.8	43.7	21°03.4	12.1'	08°57.7	15.7'	58.7'	S 10°	10:34	11:27	12:20	22:54	23:37	· · ·
20	117°29.6	43.2	35°34.5	12.0'	09°13.4	15.6'	58.7'	20°	10:34	11:32	12:30	22:51	23:29	· · ·
21	132°29.4	· · 42.7	50°05.6	12.0'	09°29.0	15.6'	58.6'	30°	10:34	11:39	12:43	22:48	23:20	23:53
22	147°29.2	42.2	64°36.6	12.0'	09°44.6	15.5'	58.6'	35°	10:34	11:42	12:50	22:46	23:15	23:45
23	162°28.9	41.7	79°07.6	12.0'	10°00.1	15.4'	58.6'	40°	10:34	11:46	12:58	22:44	23:09	23:35
SD = 16.3' d = -0.5'							SD = 16.1'							
Thu	GHA	Dec	GHA	ν	Dec	d	HP	Day	Sun		Mer.	Moon		Age
									Eqn.of Time	Pass	Upper	Lower	5-7	
									00 ^h	12 ^h	hh:mm	hh:mm	26-48%	
0	177°28.7	S20°41.2	93°38.6	12.0'	N10°15.6	15.4'	58.5'	16	09:24	09:35	12:10	16:45	04:21	
1	192°28.5	40.7	108°09.5	11.9'	10°30.9	15.3'	58.5'	17	09:45	09:55	12:10	17:33	05:09	
2	207°28.3	40.2	122°40.5	11.9'	10°46.3	15.2'	58.5'	18	10:05	10:15	12:10	18:21	05:57	
3	222°28.1	· · 39.7	137°11.4	11.9'	11°01.5	15.2'	58.5'							
4	237°27.9	39.2	151°42.3	11.9'	11°16.7	15.1'	58.4'							
5	252°27.7	38.7	166°13.2	11.8'	11°31.8	15.0'	58.4'							
6	267°27.5	S20°38.2	180°44.0	11.8'	N11°46.8	15.0'	58.4'							
7	282°27.3	37.7	195°14.8	11.8'	12°01.8	14.9'	58.4'							
8	297°27.1	37.2	209°45.6	11.7'	12°16.7	14.8'	58.3'							
9	312°26.9	· · 36.7	224°16.3	11.7'	12°31.5	14.7'	58.3'							
10	327°26.7	36.2	238°47.1	11.7'	12°46.3	14.7'	58.3'							
11	342°26.5	35.7	253°17.7	11.6'	13°00.9	14.6'	58.3'							
12	357°26.3	S20°35.2	267°48.4	11.6'	N13°15.5	14.5'	58.2'							
13	12°26.1	34.7	282°19.0	11.6'	13°30.0	14.4'	58.2'							
14	27°25.9	34.1	296°49.6	11.5'	13°44.5	14.4'	58.2'							
15	42°25.7	· · 33.6	311°20.1	11.5'	13°58.8	14.3'	58.1'							
16	57°25.5	33.1	325°50.6	11.5'	14°13.1	14.2'	58.1'							
17	72°25.3	32.6	340°21.1	11.4'	14°27.3	14.1'	58.1'							
18	87°25.1	S20°32.1	354°51.5	11.4'	N14°41.4	14.0'	58.1'							
19	102°24.9	31.6	9°21.9	11.3'	14°55.4	13.9'	58.0'							
20	117°24.7	31.1	23°52.3	11.3'	15°09.3	13.8'	58.0'							
21	132°24.5	· · 30.6	38°22.6	11.3'	15°23.2	13.7'	58.0'							
22	147°24.3	30.1	52°52.8	11.2'	15°36.9	13.7'	58.0'							
23	162°24.1	29.6	67°23.0	11.2'	15°50.6	13.6'	57.9'							
SD = 16.3' d = -0.5'							SD = 16.0'							

January 19, 20, 21 UT (Fri., Sat., Sun.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	117°53.6	213°41.3	S22°04.2	196°06.9	S23°43.2	83°38.9	N12°33.1	140°27.0	S11°10.9	Alpheratz	357°35.8	29°13.5											
1	132°56.0	228°40.5	04.5	211°07.3	43.1	98°41.3	33.1	155°29.2	10.8	Ankaa	353°08.1	-42°10.8											
2	147°58.5	243°39.6	04.7	226°07.7	43.0	113°43.6	33.2	170°31.4	10.7	Schedar	349°32.2	56°40.4											
3	163°01.0	258°38.8	•• 05.0	241°08.1	•• 42.9	128°45.9	•• 33.3	185°33.6	•• 10.6	Diphda	348°48.2	-17°51.5											
4	178°03.4	273°37.9	05.2	256°08.5	42.7	143°48.2	33.3	200°35.9	10.5	Achernar	335°20.8	-57°07.2											
5	193°05.9	288°37.1	05.4	271°08.9	42.6	158°50.5	33.4	215°38.1	10.4	Hamal	327°52.1	23°34.6											
6	208°08.4	303°36.3	S22°05.6	286°09.4	S23°42.5	173°52.8	N12°33.4	230°40.3	S11°10.3	Polaris	314°15.8	89°22.2											
7	223°10.8	318°35.4	05.9	301°09.8	42.3	188°55.1	33.5	245°42.5	10.2	Acamar	315°12.2	-40°12.7											
8	238°13.3	333°34.6	06.1	316°10.2	42.2	203°57.4	33.6	260°44.7	10.1	Menkar	314°06.9	4°11.0											
9	253°15.8	348°33.7	•• 06.3	331°10.6	•• 42.1	218°59.8	•• 33.6	275°46.9	•• 10.0	Mirfak	308°29.2	49°57.0											
10	268°18.2	3°32.9	06.6	346°11.0	41.9	234°02.1	33.7	290°49.1	09.9	Aldebaran	290°40.3	16°33.5											
11	283°20.7	18°32.0	06.8	1°11.4	41.8	249°04.4	33.8	305°51.4	09.8	Rigel	281°04.4	-8°10.5											
12	298°23.2	33°31.2	S22°07.0	16°11.8	S23°41.7	264°06.7	N12°33.8	320°53.6	S11°09.7	Capella	280°22.7	46°01.4											
13	313°25.6	48°30.3	07.2	31°12.2	41.5	279°09.0	33.9	335°55.8	09.6	Bellatrix	278°23.5	6°22.3											
14	328°28.1	63°29.5	07.4	46°12.6	41.4	294°11.3	34.0	350°58.0	09.5	Elnath	278°02.6	28°37.7											
15	343°30.5	78°28.7	•• 07.7	61°13.0	•• 41.2	309°13.6	•• 34.0	6°00.2	•• 09.4	Alnilam	275°38.3	-1°11.2											
16	358°33.0	93°27.8	07.9	76°13.5	41.1	324°15.9	34.1	21°02.4	09.3	Betelgeuse	270°52.7	7°24.7											
17	13°35.5	108°27.0	08.1	91°13.9	41.0	339°18.2	34.2	36°04.6	09.2	Canopus	263°52.2	-52°42.6											
18	28°37.9	123°26.1	S22°08.3	106°14.3	S23°40.8	354°20.5	N12°34.2	51°06.8	S11°09.1	Sirius	258°26.6	-16°45.0											
19	43°40.4	138°25.3	08.5	121°14.7	40.7	9°22.9	34.3	66°09.1	09.0	Adhara	255°06.1	-29°00.3											
20	58°42.9	153°24.4	08.8	136°15.1	40.6	24°25.2	34.4	81°11.3	08.9	Procyon	244°51.3	5°09.8											
21	73°45.3	168°23.6	•• 09.0	151°15.5	•• 40.4	39°27.5	•• 34.4	96°13.5	•• 08.8	Pollux	243°17.9	27°58.0											
22	88°47.8	183°22.7	09.2	166°15.9	40.3	54°29.8	34.5	111°15.7	08.7	Avior	234°14.4	-59°35.1											
23	103°50.3	198°21.9	09.4	181°16.3	40.1	69°32.1	34.6	126°17.9	08.6	Suhail	222°46.5	-43°31.7											
Mer.pass. 16:06				ν -0.8' d 0.2' m -3.95				ν 0.4' d -0.1' m 1.35				ν 2.3' d 0.1' m -2.45				ν 2.2' d -0.1' m 0.98							

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	118°52.7	213°21.0	S22°09.6	196°16.7	S23°40.0	84°34.4	N12°34.6	141°20.1	S11°08.5										
1	133°55.2	228°20.2	09.8	211°17.1	39.9	99°36.7	34.7	156°22.3	08.4										
2	148°57.7	243°19.3	10.0	226°17.6	39.7	114°39.0	34.8	171°24.5	08.3										
3	164°00.1	258°18.5	•• 10.2	241°18.0	•• 39.6	129°41.3	•• 34.8	186°26.8	•• 08.2										
4	179°02.6	273°17.7	10.4	256°18.4	39.4	144°43.6	34.9	201°29.0	08.1										
5	194°05.0	288°16.8	10.6	271°18.8	39.3	159°45.9	35.0	216°31.2	08.0										
6	209°07.5	303°16.0	S22°10.8	286°19.2	S23°39.1	174°48.2	N12°35.0	231°33.4	S11°07.9										
7	224°10.0	318°15.1	11.0	301°19.6	39.0	189°50.5	35.1	246°35.6	07.8										
8	239°12.4	333°14.3	11.2	316°20.0	38.9	204°52.8	35.2	261°37.8	07.7										
9	254°14.9	348°13.4	•• 11.4	331°20.4	•• 38.7	219°55.1	•• 35.2	276°40.0	•• 07.6										
10	269°17.4	3°12.6	11.6	346°20.8	38.6	234°57.4	35.3	291°42.2	07.5										
11	284°19.8	18°11.7	11.8	1°21.3	38.4	249°59.7	35.4	306°44.4	07.4										
12	299°22.3	33°10.9	S22°12.0	16°21.7	S23°38.3	265°02.0	N12°35.4	321°46.7	S11°07.3										
13	314°24.8	48°10.0	12.2	31°22.1	38.1	280°04.3	35.5	336°48.9	07.2										
14	329°27.2	63°09.2	12.4	46°22.5	38.0	295°06.6	35.6	351°51.1	07.1										
15	344°29.7	78°08.3	•• 12.6	61°22.9	•• 37.8	310°08.9	•• 35.6	6°53.3	•• 07.0										
16	359°32.1	93°07.5	12.8	76°23.3	37.7	325°11.2	35.7	21°55.5	06.9										
17	14°34.6	108°06.6	13.0	91°23.7	37.5	340°13.5	35.8	36°57.7	06.8										
18	29°37.1	123°05.8	S22°13.2	106°24.1	S23°37.4	355°15.8	N12°35.8	51°59.9	S11°06.7										
19	44°39.5	138°04.9	13.4	121°24.5	37.2	10°18.1	35.9	67°02.1	06.6										
20	59°42.0	153°04.1	13.6	136°24.9	37.1	25°20.4	36.0	82°04.3	06.5										
21	74°44.5	168°03.2	•• 13.8	151°25.4	•• 36.9	40°22.7	•• 36.1	97°06.5	•• 06.4										
22	89°46.9	183°02.4	13.9	166°25.8	36.8	55°25.0	36.1	112°08.8	06.3										
23	104°49.4	198°01.5	14.1	181°26.2	36.6	70°27.3	36.2	127°11.0	06.2										
Mer.pass. 16:02				ν -0.8' d 0.2' m -3.95				ν 0.4' d -0.1' m 1.35				ν 2.3' d 0.1' m -2.44				ν 2.2' d -0.1' m 0.98			

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	119°51.9	213°00.7	S22°14.3	196°26.6	S23°36.5	85°29.6	N12°36.3	142°13.2	S11°06.1										
1	134°54.3	227°59.8	14.5	211°27.0	36.3	100°31.9	36.3	157°15.4	06.0										
2	149°56.8	242°58.9	14.7	226°27.4	36.2	115°34.2	36.4	172°17.6	05.9										
3	164°59.3	257°58.1	•• 14.9	241°27.8	•• 36.0	130°36.5	•• 36.5	187°19.8	•• 05.8										
4	180°01.7	272°57.2	15.0	256°28.2	35.9	145°38.8	36.5	202°22.0	05.7										
5	195°04.2	287°56.4	15.2	271°28.6	35.7	160°41.1	36.6	217°24.2	05.6										
6	210°06.6	302°55.5	S22°15.4	286°29.1	S23°35.6	175°43.4	N12°36.7	232°26.4	S11°05.5										
7	225°09.1	317°54.7	15.6	301°29.5	35.4	190°45.7	36.7	247°28.6	05.4										
8	240°11.6	332°53.8	15.7	316°29.9	35.3	205°48.0	36.8	262°30.9	05.3										
9	255°14.0	347°53.0	•• 15.9	331°30.3	•• 35.1	220°50.3	•• 36.9	277°33.1	•• 05.2										
10	270°16.5	2°52.1	16.1	346°30.7	34.9	235°52.6	37.0	292°35.3	05.1										
11	285°19.0	17°51.3	16.2	1°31.1	34.8	250°54.9	37.0	307°37.5	05.0										
12	300°21.4	32°50.4	S22°16.4	16°31.5	S23°34.6	265°57.2	N12°37.1	322°39.7	S11°04.9										
13	315°23.9	47°49.6	16.6	31°31.9	34.5	280°59.5	37.2	337°41.9	04.8										
14	330°26.4	62°48.7	16.8	46°32.3	34.3	296°01.8	37.2	352°44.1	04.7										
15	345°28.8	77°47.8	•• 16.9	61°32.7	•• 34.2	311°04.1	•• 37.3	7°46.3	•• 04.6										
16	0°31.3	92°47.0	17.1	76°33.2	34.0	326°06.4	37.4	22°48.5	04.5										
17	15°33.8	107°46.1	17.3	91°33.6	33.8	341°08.7	37.4	37°50.7	04.4										
18	30°36.2	122°45.3	S22°17.4	106°34.0	S23°33.7	356°11.0	N12°37.5	52°53.0	S11°04.3										
19	45°38.7	137°44.4	17.6	121°34.4	33.5	11°13.3	37.6	67°55.2	04.2										
20	60°41.1	152°43.6	17.7	136°34.8	33.4	26°15.5	37.7	82°57.4	04.1										
21	75°43.6	167°42.7	•• 17.9	151°35.2	•• 33.2	41°17.8	•• 37.7	97°59.6	•• 04.0										
22	90°46.1	182°41.9	18.1	166°35.6	33.0	56°20.1	37.8	113°01.8	03.9										
23	105°48.5	197°41.0	18.2	181°36.0	32.9	71°22.4	37.9	128°04.0	03.8										
Mer.pass. 15:58				ν -0.9' d 0.2' m -3.94				ν 0.4' d -0.2' m 1.34				ν 2.3' d 0.1' m -2.43				ν 2.2' d -0.1' m 0.98			

Jan 19 Fri	SHA	Mer.pass
Venus	95°47.7	09:46
Mars	78°13.3	10:55
Jupiter	325°45.4	18:23
Saturn	22°33.4	14:36

Jan 20 Sat	SHA	Mer.pass
Venus	94°28.3	09:47
Mars	77°24.0	10:55
Jupiter	325°41.7	18:19
Saturn	22°27.4	14:33

Jan 21 Sun	SHA	Mer.pass
Venus	93°08.8	09:49
Mars	76°34.7	10:54
Jupiter	325°37.8	18:15
Saturn	22°21.3	14:29

Horizontal parallax		
Venus:		0.1
Mars:		0.1

h	Sun				Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP	
0	177°23.9	S20°29.0	81°53.2	11.1'	N16°04.1	13.5'	57.9'	
1	192°23.7	28.5	96°23.3	11.1'	16°17.6	13.4'	57.9'	
2	207°23.5	28.0	110°53.4	11.0'	16°31.0	13.3'	57.9'	
3	222°23.3	· · 27.5	125°23.5	11.0'	16°44.3	13.2'	57.8'	
4	237°23.1	27.0	139°53.4	10.9'	16°57.5	13.1'	57.8'	
5	252°22.9	26.5	154°23.4	10.9'	17°10.6	13.0'	57.8'	
6	267°22.8	S20°25.9	168°53.3	10.8'	N17°23.5	12.9'	57.8'	
7	282°22.6	25.4	183°23.1	10.8'	17°36.4	12.8'	57.7'	
8	297°22.4	24.9	197°52.9	10.7'	17°49.2	12.7'	57.7'	
9	312°22.2	· · 24.4	212°22.7	10.7'	18°01.9	12.6'	57.7'	
10	327°22.0	23.9	226°52.4	10.6'	18°14.5	12.5'	57.7'	
11	342°21.8	23.4	241°22.0	10.6'	18°27.0	12.4'	57.6'	
12	357°21.6	S20°22.8	255°51.6	10.5'	N18°39.4	12.3'	57.6'	
13	12°21.4	22.3	270°21.2	10.5'	18°51.7	12.2'	57.6'	
14	27°21.2	21.8	284°50.6	10.4'	19°03.8	12.1'	57.5'	
15	42°21.0	· · 21.3	299°20.1	10.4'	19°15.9	12.0'	57.5'	
16	57°20.8	20.7	313°49.5	10.3'	19°27.8	11.8'	57.5'	
17	72°20.6	20.2	328°18.8	10.3'	19°39.7	11.7'	57.5'	
18	87°20.4	S20°19.7	342°48.1	10.2'	N19°51.4	11.6'	57.4'	
19	102°20.2	19.2	357°17.3	10.2'	20°03.0	11.5'	57.4'	
20	117°20.1	18.6	11°46.5	10.1'	20°14.5	11.4'	57.4'	
21	132°19.9	· · 18.1	26°15.6	10.1'	20°25.9	11.3'	57.4'	
22	147°19.7	17.6	40°44.6	10.0'	20°37.2	11.2'	57.3'	
23	162°19.5	17.0	55°13.7	10.0'	20°48.3	11.0'	57.3'	
SD = 16.2'		d = -0.5'	SD = 15.8'					
Sat	GHA	Dec	GHA	ν	Dec	d	HP	
0	177°19.3	S20°16.5	69°42.6	9.9'	N20°59.4	10.9'	57.3'	
1	192°19.1	16.0	84°11.5	9.8'	21°10.3	10.8'	57.3'	
2	207°18.9	15.5	98°40.3	9.8'	21°21.1	10.7'	57.2'	
3	222°18.7	· · 14.9	113°09.1	9.7'	21°31.8	10.6'	57.2'	
4	237°18.5	14.4	127°37.9	9.7'	21°42.3	10.4'	57.2'	
5	252°18.4	13.9	142°06.5	9.6'	21°52.8	10.3'	57.2'	
6	267°18.2	S20°13.3	156°35.2	9.6'	N22°03.1	10.2'	57.1'	
7	282°18.0	12.8	171°03.7	9.5'	22°13.3	10.1'	57.1'	
8	297°17.8	12.3	185°32.2	9.5'	22°23.3	9.9'	57.1'	
9	312°17.6	· · 11.7	200°00.7	9.4'	22°33.3	9.8'	57.1'	
10	327°17.4	11.2	214°29.1	9.3'	22°43.1	9.7'	57.0'	
11	342°17.2	10.7	228°57.5	9.3'	22°52.8	9.6'	57.0'	
12	357°17.1	S20°10.1	243°25.8	9.2'	N23°02.3	9.4'	57.0'	
13	12°16.9	09.6	257°54.0	9.2'	23°11.7	9.3'	57.0'	
14	27°16.7	09.0	272°22.2	9.1'	23°21.0	9.2'	56.9'	
15	42°16.5	· · 08.5	286°50.3	9.1'	23°30.2	9.0'	56.9'	
16	57°16.3	08.0	301°18.4	9.0'	23°39.2	8.9'	56.9'	
17	72°16.1	07.4	315°46.4	9.0'	23°48.1	8.8'	56.9'	
18	87°15.9	S20°06.9	330°14.4	8.9'	N23°56.8	8.6'	56.9'	
19	102°15.8	06.3	344°42.3	8.9'	24°05.5	8.5'	56.8'	
20	117°15.6	05.8	359°10.2	8.8'	24°14.0	8.3'	56.8'	
21	132°15.4	· · 05.2	13°38.0	8.8'	24°22.3	8.2'	56.8'	
22	147°15.2	04.7	28°05.8	8.7'	24°30.5	8.1'	56.8'	
23	162°15.0	04.2	42°33.5	8.7'	24°38.6	7.9'	56.7'	
SD = 16.2'		d = -0.5'	SD = 15.6'					
Sun	GHA	Dec	GHA	ν	Dec	d	HP	
0	177°14.9	S20°03.6	57°01.2	8.6'	N24°46.5	7.8'	56.7'	
1	192°14.7	03.1	71°28.9	8.6'	24°54.3	7.7'	56.7'	
2	207°14.5	02.5	85°56.4	8.5'	25°02.0	7.5'	56.7'	
3	222°14.3	· · 02.0	100°24.0	8.5'	25°09.5	7.4'	56.6'	
4	237°14.1	01.4	114°51.5	8.4'	25°16.8	7.2'	56.6'	
5	252°14.0	00.9	129°18.9	8.4'	25°24.1	7.1'	56.6'	
6	267°13.8	S20°00.3	143°46.3	8.4'	N25°31.1	6.9'	56.6'	
7	282°13.6	19°59.8	158°13.7	8.3'	25°38.1	6.8'	56.6'	
8	297°13.4	59.2	172°41.0	8.3'	25°44.9	6.6'	56.5'	
9	312°13.2	· · 58.7	187°08.3	8.2'	25°51.5	6.5'	56.5'	
10	327°13.1	58.1	201°35.5	8.2'	25°58.0	6.4'	56.5'	
11	342°12.9	57.6	216°02.7	8.2'	26°04.4	6.2'	56.5'	
12	357°12.7	S19°57.0	230°29.9	8.1'	N26°10.6	6.1'	56.4'	
13	12°12.5	56.5	244°57.0	8.1'	26°16.6	5.9'	56.4'	
14	27°12.4	55.9	259°24.1	8.1'	26°22.5	5.8'	56.4'	
15	42°12.2	· · 55.4	273°51.1	8.0'	26°28.3	5.6'	56.4'	
16	57°12.0	54.8	288°18.1	8.0'	26°33.9	5.5'	56.4'	
17	72°11.8	54.2	302°45.1	8.0'	26°39.3	5.3'	56.3'	
18	87°11.6	S19°53.7	317°12.1	7.9'	N26°44.7	5.2'	56.3'	
19	102°11.5	53.1	331°39.0	7.9'	26°49.8	5.0'	56.3'	
20	117°11.3	52.6	346°05.9	7.9'	26°54.8	4.8'	56.3'	
21	132°11.1	· · 52.0	0°32.8	7.8'	26°59.7	4.7'	56.2'	
22	147°11.0	51.5	14°59.6	7.8'	27°04.4	4.5'	56.2'	
23	162°10.8	50.9	29°26.4	7.8'	27°08.9	4.4'	56.2'	
SD = 16.2'		d = -0.5'	SD = 15.5'					

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:49	09:33	■	■	14:50	16:34
N 70°	07:36	09:04	11:10	13:13	15:19	16:47
68°	07:25	08:42	10:12	14:10	15:40	16:57
66°	07:16	08:25	09:39	14:44	15:58	17:06
64°	07:09	08:11	09:15	15:08	16:12	17:14
62°	07:02	07:59	08:56	15:27	16:23	17:21
60°	06:56	07:49	08:40	15:42	16:34	17:27
N 58°	06:50	07:40	08:27	15:55	16:42	17:32
56°	06:45	07:32	08:16	16:07	16:50	17:37
54°	06:41	07:25	08:06	16:17	16:57	17:42
52°	06:36	07:18	07:57	16:25	17:04	17:46
50°	06:32	07:12	07:49	16:33	17:10	17:50
45°	06:23	06:59	07:32	16:50	17:23	17:59
N 40°	06:15	06:48	07:18	17:04	17:34	18:07
35°	06:08	06:39	07:06	17:16	17:43	18:14
30°	06:01	06:30	06:56	17:26	17:52	18:21
20°	05:47	06:14	06:38	17:44	18:08	18:34
N 10°	05:34	06:00	06:22	18:00	18:22	18:48
0°	05:20	05:45	06:07	18:14	18:36	19:02
S 10°	05:03	05:30	05:52	18:29	18:52	19:18
20°	04:44	05:12	05:36	18:45	19:09	19:38
30°	04:19	04:51	05:17	19:04	19:30	20:02
35°	04:03	04:38	05:06	19:15	19:43	20:18
40°	03:44	04:23	04:54	19:27	19:59	20:37
45°	03:19	04:04	04:39	19:42	20:17	21:02
S 50°	02:44	03:40	04:20	20:01	20:41	21:36
52°	02:25	03:28	04:12	20:09	20:53	21:54
54°	02:02	03:14	04:02	20:19	21:06	22:17
56°	01:28	02:58	03:51	20:30	21:22	22:50
58°	—:—	02:38	03:38	20:43	21:42	—:—
S 60°	—:—	02:13	03:23	20:57	22:06	—:—

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	07:37	□	□	05:31	□	□
N 70°	08:41	□	□	04:28	□	□
68°	09:18	□	□	03:54	□	□
66°	09:44	09:08	□	03:29	05:52	□
64°	10:04	09:48	09:06	03:10	05:12	07:46
62°	10:21	10:16	10:11	02:54	04:45	06:41
60°	10:35	10:38	10:46	02:41	04:24	06:07
N 58°	10:47	10:56	11:11	02:30	04:07	05:42
56°	10:57	11:11	11:32	02:21	03:52	05:22
54°	11:06	11:24	11:49	02:12	03:40	05:05
52°	11:15	11:35	12:03	02:05	03:29	04:51
50°	11:22	11:45	12:16	01:58	03:20	04:39
45°	11:38	12:07	12:42	01:44	02:59	04:13
N 40°	11:51	12:24	13:03	01:32	02:43	03:53
35°	12:03	12:39	13:21	01:22	02:29	03:36
30°	12:13	12:52	13:36	01:13	02:18	03:22
20°	12:30	13:14	14:01	00:58	01:57	02:57
N 10°	12:45	13:33	14:24	00:45	01:40	02:36
0°	12:59	13:51	14:45	00:33	01:23	02:16
S 10°	13:13	14:09	15:05	00:21	01:07	01:57
20°	13:29	14:28	15:28	00:08	00:50	01:36
30°	13:47	14:51	15:54	· · ·	00:30	01:12
35°	13:57	15:04	16:10	· · ·	00:19	00:57
40°	14:09	15:20	16:28	· · ·	00:05	00:41
45°	14:23	15:38	16:50			

January 22, 23, 24 UT (Mon., Tue., Wed.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	120°51.0	212°40.1	S22°18.4	196°36.4	S23°32.7	86°24.7	N12°37.9	143°06.2	S11°03.7					Alpheratz	357°35.8	29°13.4	
1	135°53.5	227°39.3	18.5	211°36.9	32.6	101°27.0	38.0	158°08.4	03.6					Ankaa	353°08.1	-42°10.8	
2	150°55.9	242°38.4	18.7	226°37.3	32.4	116°29.3	38.1	173°10.6	03.5					Schedar	349°32.2	56°40.4	
3	165°58.4	257°37.6	•• 18.8	241°37.7	•• 32.2	131°31.6	•• 38.2	188°12.8	•• 03.4					Diphda	348°48.2	-17°51.5	
4	181°00.9	272°36.7	19.0	256°38.1	32.1	146°33.9	38.2	203°15.0	03.3					Achernar	335°20.8	-57°07.2	
5	196°03.3	287°35.9	19.1	271°38.5	31.9	161°36.2	38.3	218°17.2	03.2					Hamal	327°52.1	23°34.6	
6	211°05.8	302°35.0	S22°19.3	286°38.9	S23°31.7	176°38.5	N12°38.4	233°19.5	S11°03.1					Polaris	314°16.9	89°22.2	
7	226°08.3	317°34.1	19.4	301°39.3	31.6	191°40.7	38.5	248°21.7	03.0					Acamar	315°12.3	-40°12.7	
8	241°10.7	332°33.3	19.6	316°39.7	31.4	206°43.0	38.5	263°23.9	02.9					Menkar	314°06.9	4°11.0	
9	256°13.2	347°32.4	•• 19.7	331°40.1	•• 31.2	221°45.3	•• 38.6	278°26.1	•• 02.8					Mirfak	308°29.2	49°57.0	
10	271°15.6	2°31.6	19.9	346°40.6	31.1	236°47.6	38.7	293°28.3	02.7					Aldebaran	290°40.3	16°33.5	
11	286°18.1	17°30.7	20.0	1°41.0	30.9	251°49.9	38.7	308°30.5	02.6					Rigel	281°04.4	-8°10.5	
12	301°20.6	32°29.9	S22°20.2	16°41.4	S23°30.7	266°52.2	N12°38.8	323°32.7	S11°02.5					Capella	280°22.7	46°01.4	
13	316°23.0	47°29.0	20.3	31°41.8	30.6	281°54.5	38.9	338°34.9	02.4					Bellatrix	278°23.5	6°22.3	
14	331°25.5	62°28.1	20.4	46°42.2	30.4	296°56.8	39.0	353°37.1	02.3					Elnath	278°02.6	28°37.7	
15	346°28.0	77°27.3	•• 20.6	61°42.6	•• 30.2	311°59.0	•• 39.0	8°39.3	•• 02.2					Alnilam	275°38.3	-1°11.2	
16	1°30.4	92°26.4	20.7	76°43.0	30.1	327°01.3	39.1	23°41.5	02.1					Betelgeuse	270°52.6	7°24.7	
17	16°32.9	107°25.6	20.8	91°43.4	29.9	342°03.6	39.2	38°43.7	02.0					Canopus	263°52.2	-52°42.6	
18	31°35.4	122°24.7	S22°21.0	106°43.8	S23°29.7	357°05.9	N12°39.3	53°45.9	S11°01.9					Sirius	258°26.6	-16°45.0	
19	46°37.8	137°23.8	21.1	121°44.3	29.6	12°08.2	39.3	68°48.1	01.8					Adhara	255°06.1	-29°00.3	
20	61°40.3	152°23.0	21.3	136°44.7	29.4	27°10.5	39.4	83°50.4	01.6					Procyon	244°51.3	5°09.8	
21	76°42.8	167°22.1	•• 21.4	151°45.1	•• 29.2	42°12.8	•• 39.5	98°52.6	•• 01.5					Pollux	243°17.9	27°58.0	
22	91°45.2	182°21.3	21.5	166°45.5	29.0	57°15.0	39.5	113°54.8	01.4					Avior	234°14.4	-59°35.1	
23	106°47.7	197°20.4	21.6	181°45.9	28.9	72°17.3	39.6	128°57.0	01.3					Suhail	222°46.5	-43°31.7	
Mer.pass. 15:54			ν -0.9' d 0.2' m -3.94	ν 0.4' d -0.2' m 1.34	ν 2.3' d 0.1' m -2.42	ν 2.2' d -0.1' m 0.98											

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	121°50.1	212°19.5	S22°21.8	196°46.3	S23°28.7	87°19.6	N12°39.7	143°59.2	S11°01.2				
1	136°52.6	227°18.7	21.9	211°46.7	28.5	102°21.9	39.8	159°01.4	01.1				
2	151°55.1	242°17.8	22.0	226°47.1	28.4	117°24.2	39.8	174°03.6	01.0				
3	166°57.5	257°17.0	•• 22.2	241°47.5	•• 28.2	132°26.5	•• 39.9	189°05.8	•• 00.9				
4	182°00.0	272°16.1	22.3	256°48.0	28.0	147°28.7	40.0	204°08.0	00.8				
5	197°02.5	287°15.2	22.4	271°48.4	27.8	162°31.0	40.1	219°10.2	00.7				
6	212°04.9	302°14.4	S22°22.5	286°48.8	S23°27.7	177°33.3	N12°40.1	234°12.4	S11°00.6				
7	227°07.4	317°13.5	22.6	301°49.2	27.5	192°35.6	40.2	249°14.6	00.5				
8	242°09.9	332°12.6	22.8	316°49.6	27.3	207°37.9	40.3	264°16.8	00.4				
9	257°12.3	347°11.8	•• 22.9	331°50.0	•• 27.1	222°40.1	•• 40.4	279°19.0	•• 00.3				
10	272°14.8	2°10.9	23.0	346°50.4	26.9	237°42.4	40.4	294°21.2	00.2				
11	287°17.3	17°10.1	23.1	1°50.8	26.8	252°44.7	40.5	309°23.5	00.1				
12	302°19.7	32°09.2	S22°23.2	16°51.2	S23°26.6	267°47.0	N12°40.6	324°25.7	S11°00.0				
13	317°22.2	47°08.3	23.3	31°51.7	26.4	282°49.3	40.7	339°27.9	10°59.9				
14	332°24.6	62°07.5	23.5	46°52.1	26.2	297°51.5	40.7	354°30.1	59.8				
15	347°27.1	77°06.6	•• 23.6	61°52.5	•• 26.1	312°53.8	•• 40.8	9°32.3	•• 59.7				
16	2°29.6	92°05.7	23.7	76°52.9	25.9	327°56.1	40.9	24°34.5	59.6				
17	17°32.0	107°04.9	23.8	91°53.3	25.7	342°58.4	41.0	39°36.7	59.5				
18	32°34.5	122°04.0	S22°23.9	106°53.7	S23°25.5	358°00.7	N12°41.0	54°38.9	S10°59.4				
19	47°37.0	137°03.2	24.0	121°54.1	25.3	13°02.9	41.1	69°41.1	59.3				
20	62°39.4	152°02.3	24.1	136°54.5	25.2	28°05.2	41.2	84°43.3	59.2				
21	77°41.9	167°01.4	•• 24.2	151°54.9	•• 25.0	43°07.5	•• 41.3	99°45.5	•• 59.1				
22	92°44.4	182°00.6	24.3	166°55.4	24.8	58°09.8	41.3	114°47.7	59.0				
23	107°46.8	196°59.7	24.4	181°55.8	24.6	73°12.0	41.4	129°49.9	58.9				
Mer.pass. 15:50			ν -0.9' d 0.1' m -3.94	ν 0.4' d -0.2' m 1.34	ν 2.3' d 0.1' m -2.42	ν 2.2' d -0.1' m 0.98							

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	122°49.3	211°58.8	S22°24.5	196°56.2	S23°24.4	88°14.3	N12°41.5	144°52.1	S10°58.8				
1	137°51.7	226°58.0	24.6	211°56.6	24.2	103°16.6	41.6	159°54.3	58.7				
2	152°54.2	241°57.1	24.7	226°57.0	24.1	118°18.9	41.7	174°56.5	58.6				
3	167°56.7	256°56.2	•• 24.8	241°57.4	•• 23.9	133°21.1	•• 41.7	189°58.7	•• 58.5				
4	182°59.1	271°55.4	24.9	256°57.8	23.7	148°23.4	41.8	205°00.9	58.4				
5	198°01.6	286°54.5	25.0	271°58.2	23.5	163°25.7	41.9	220°03.2	58.3				
6	213°04.1	301°53.6	S22°25.1	286°58.7	S23°23.3	178°28.0	N12°42.0	235°05.4	S10°58.2				
7	228°06.5	316°52.8	25.2	301°59.1	23.1	193°30.2	42.0	250°07.6	58.1				
8	243°09.0	331°51.9	25.3	316°59.5	22.9	208°32.5	42.1	265°09.8	58.0				
9	258°11.5	346°51.1	•• 25.4	331°59.9	•• 22.8	223°34.8	•• 42.2	280°12.0	•• 57.9				
10	273°13.9	1°50.2	25.5	347°00.3	22.6	238°37.1	42.3	295°14.2	57.8				
11	288°16.4	16°49.3	25.6	2°00.7	22.4	253°39.3	42.3	310°16.4	57.7				
12	303°18.9	31°48.5	S22°25.7	17°01.1	S23°22.2	268°41.6	N12°42.4	325°18.6	S10°57.6				
13	318°21.3	46°47.6	25.7	32°01.5	22.0	283°43.9	42.5	340°20.8	57.5				
14	333°23.8	61°46.7	25.8	47°01.9	21.8	298°46.1	42.6	355°23.0	57.4				
15	348°26.2	76°45.9	•• 25.9	62°02.4	•• 21.6	313°48.4	•• 42.7	10°25.2	•• 57.3				
16	3°28.7	91°45.0	26.0	77°02.8	21.4	328°50.7	42.7	25°27.4	57.2				
17	18°31.2	106°44.1	26.1	92°03.2	21.2	343°53.0	42.8	40°29.6	57.1				
18	33°33.6	121°43.3	S22°26.2	107°03.6	S23°21.0	358°55.2	N12°42.9	55°31.8	S10°57.0				
19	48°36.1	136°42.4	26.2	122°04.0	20.9	13°57.5	43.0	70°34.0	56.9				
20	63°38.6	151°41.5	26.3	137°04.4	20.7	28°59.8	43.0	85°36.2	56.7				
21	78°41.0	166°40.7	•• 26.4	152°04.8	•• 20.5	44°02.0	•• 43.1	100°38.4	•• 56.6				
22	93°43.5	181°39.8	26.5	167°05.2	20.3	59°04.3	43.2	115°40.6	56.5				
23	108°46.0	196°38.9	26.5	182°05.7	20.1	74°06.6	43.3	130°42.8	56.4				
Mer.pass. 15:46			ν -0.9' d 0.1' m -3.94	ν 0.4' d -0.2' m 1.34	ν 2.3' d 0.1' m -2.41	ν 2.2' d -0.1' m 0.98							

Jan 22 Mon	SHA	Mer.pass
Venus	91°49.1	09:50
Mars	75°45.4	10:53
Jupiter	325°33.7	18:12
Saturn	22°15.2	14:25

Jan 23 Tue	SHA	Mer.pass
Venus	90°29.4	09:51
Mars	74°56.2	10:53
Jupiter	325°29.5	18:08
Saturn	22°09.0	14:22

Jan 24 Wed	SHA	Mer.pass
Venus	89°09.5	09:53
Mars	74°06.9	10:52
Jupiter	325°25.0	18:04
Saturn	22°02.8	14:18

Horizontal parallax		
Venus:	0.1	
Mars:	0.1	

Sun				Moon				
Mon	GHA	Dec		GHA	ν	Dec	d	HP
0	177°10.6	S19°50.3		43°53.2	7.8'	N27°13.3	4.2'	56.2'
1	192°10.4	49.8		58°20.0	7.8'	27°17.5	4.1'	56.2'
2	207°10.3	49.2		72°46.8	7.7'	27°21.6	3.9'	56.1'
3	222°10.1	· · 48.7		87°13.5	7.7'	27°25.5	3.8'	56.1'
4	237°09.9	48.1		101°40.2	7.7'	27°29.3	3.6'	56.1'
5	252°09.7	47.5		116°06.9	7.7'	27°32.9	3.5'	56.1'
6	267°09.6	S19°47.0		130°33.6	7.7'	N27°36.4	3.3'	56.1'
7	282°09.4	46.4		145°00.2	7.7'	27°39.7	3.2'	56.0'
8	297°09.2	45.8		159°26.9	7.6'	27°42.8	3.0'	56.0'
9	312°09.1	· · 45.3		173°53.6	7.6'	27°45.8	2.8'	56.0'
10	327°08.9	44.7		188°20.2	7.6'	27°48.7	2.7'	56.0'
11	342°08.7	44.1		202°46.8	7.6'	27°51.4	2.5'	56.0'
12	357°08.6	S19°43.6		217°13.4	7.6'	N27°53.9	2.4'	55.9'
13	12°08.4	43.0		231°40.1	7.6'	27°56.3	2.2'	55.9'
14	27°08.2	42.4		246°06.7	7.6'	27°58.5	2.1'	55.9'
15	42°08.0	· · 41.9		260°33.3	7.6'	28°00.5	1.9'	55.9'
16	57°07.9	41.3		274°59.9	7.6'	28°02.4	1.7'	55.9'
17	72°07.7	40.7		289°26.5	7.6'	28°04.2	1.6'	55.8'
18	87°07.5	S19°40.1		303°53.2	7.6'	N28°05.8	1.4'	55.8'
19	102°07.4	39.6		318°19.8	7.6'	28°07.2	1.3'	55.8'
20	117°07.2	39.0		332°46.4	7.6'	28°08.5	1.1'	55.8'
21	132°07.0	· · 38.4		347°13.1	7.7'	28°09.6	1.0'	55.8'
22	147°06.9	37.8		1°39.7	7.7'	28°10.5	0.8'	55.7'
23	162°06.7	37.3		16°06.4	7.7'	28°11.3	0.6'	55.7'
SD = 16.2'			d = -0.6'	SD = 15.3'				
Tue	GHA	Dec		GHA	ν	Dec	d	HP
0	177°06.5	S19°36.7		30°33.1	7.7'	N28°12.0	0.5'	55.7'
1	192°06.4	36.1		44°59.8	7.7'	28°12.5	0.3'	55.7'
2	207°06.2	35.5		59°26.5	7.7'	28°12.8	0.2'	55.7'
3	222°06.1	· · 35.0		73°53.2	7.8'	28°13.0	0.0'	55.6'
4	237°05.9	34.4		88°20.0	7.8'	28°13.0	-0.1'	55.6'
5	252°05.7	33.8		102°46.8	7.8'	28°12.9	-0.3'	55.6'
6	267°05.6	S19°33.2		117°13.6	7.8'	N28°12.6	-0.4'	55.6'
7	282°05.4	32.6		131°40.4	7.9'	28°12.2	-0.6'	55.6'
8	297°05.2	32.1		146°07.3	7.9'	28°11.6	-0.7'	55.5'
9	312°05.1	· · 31.5		160°34.1	7.9'	28°10.9	-0.9'	55.5'
10	327°04.9	30.9		175°01.1	7.9'	28°10.0	-1.0'	55.5'
11	342°04.8	30.3		189°28.0	8.0'	28°08.9	-1.2'	55.5'
12	357°04.6	S19°29.7		203°55.0	8.0'	N28°07.7	-1.4'	55.5'
13	12°04.4	29.1		218°22.0	8.1'	28°06.4	-1.5'	55.5'
14	27°04.3	28.6		232°49.1	8.1'	28°04.8	-1.7'	55.4'
15	42°04.1	· · 28.0		247°16.1	8.1'	28°03.2	-1.8'	55.4'
16	57°04.0	27.4		261°43.3	8.2'	28°01.4	-2.0'	55.4'
17	72°03.8	26.8		276°10.4	8.2'	27°59.4	-2.1'	55.4'
18	87°03.6	S19°26.2		290°37.7	8.3'	N27°57.3	-2.3'	55.4'
19	102°03.5	25.6		305°04.9	8.3'	27°55.1	-2.4'	55.3'
20	117°03.3	25.0		319°32.2	8.3'	27°52.7	-2.5'	55.3'
21	132°03.2	· · 24.5		333°59.6	8.4'	27°50.1	-2.7'	55.3'
22	147°03.0	23.9		348°27.0	8.4'	27°47.4	-2.8'	55.3'
23	162°02.8	23.3		2°54.4	8.5'	27°44.6	-3.0'	55.3'
SD = 16.2'			d = -0.6'	SD = 15.2'				
Wed	GHA	Dec		GHA	ν	Dec	d	HP
0	177°02.7	S19°22.7		17°21.9	8.6'	N27°41.6	-3.1'	55.3'
1	192°02.5	22.1		31°49.5	8.6'	27°38.5	-3.3'	55.2'
2	207°02.4	21.5		46°17.1	8.7'	27°35.2	-3.4'	55.2'
3	222°02.2	· · 20.9		60°44.7	8.7'	27°31.8	-3.6'	55.2'
4	237°02.1	20.3		75°12.4	8.8'	27°28.2	-3.7'	55.2'
5	252°01.9	19.7		89°40.2	8.8'	27°24.5	-3.8'	55.2'
6	267°01.8	S19°19.1		104°08.0	8.9'	N27°20.7	-4.0'	55.2'
7	282°01.6	18.5		118°35.9	9.0'	27°16.7	-4.1'	55.1'
8	297°01.4	17.9		133°03.9	9.0'	27°12.5	-4.3'	55.1'
9	312°01.3	· · 17.3		147°31.9	9.1'	27°08.3	-4.4'	55.1'
10	327°01.1	16.7		162°00.0	9.1'	27°03.9	-4.5'	55.1'
11	342°01.0	16.1		176°28.1	9.2'	26°59.3	-4.7'	55.1'
12	357°00.8	S19°15.5		190°56.3	9.3'	N26°54.7	-4.8'	55.1'
13	12°00.7	14.9		205°24.6	9.3'	26°49.8	-4.9'	55.0'
14	27°00.5	14.3		219°53.0	9.4'	26°44.9	-5.1'	55.0'
15	42°00.4	· · 13.7		234°21.4	9.5'	26°39.8	-5.2'	55.0'
16	57°00.2	13.1		248°49.9	9.6'	26°34.6	-5.3'	55.0'
17	72°00.1	12.5		263°18.4	9.6'	26°29.3	-5.5'	55.0'
18	86°59.9	S19°11.9		277°47.0	9.7'	N26°23.8	-5.6'	55.0'
19	101°59.8	11.3		292°15.7	9.8'	26°18.2	-5.7'	54.9'
20	116°59.6	10.7		306°44.5	9.8'	26°12.5	-5.9'	54.9'
21	131°59.5	· · 10.1		321°13.4	9.9'	26°06.6	-6.0'	54.9'
22	146°59.3	09.5		335°42.3	10.0'	26°00.6	-6.1'	54.9'
23	161°59.2	08.9		350°11.3	10.1'	25°54.5	-6.2'	54.9'
SD = 16.2'			d = -0.6'	SD = 15.1'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:41	09:20	■	■	15:04	16:44
N 70°	07:29	08:54	10:44	13:40	15:31	16:56
68°	07:19	08:34	09:59	14:26	15:50	17:05
66°	07:11	08:18	09:29	14:55	16:06	17:14
64°	07:04	08:05	09:07	15:17	16:19	17:21
62°	06:57	07:54	08:49	15:35	16:30	17:27
60°	06:52	07:44	08:34	15:50	16:40	17:32
N 58°	06:47	07:36	08:22	16:02	16:48	17:38
56°	06:42	07:28	08:11	16:13	16:56	17:42
54°	06:38	07:21	08:02	16:22	17:03	17:46
52°	06:34	07:15	07:53	16:31	17:09	17:50
50°	06:30	07:10	07:46	16:38	17:14	17:54
45°	06:22	06:57	07:29	16:54	17:26	18:02
N 40°	06:14	06:47	07:16	17:08	17:37	18:10
35°	06:07	06:38	07:05	17:19	17:46	18:17
30°	06:00	06:29	06:55	17:29	17:54	18:23
20°	05:47	06:14	06:38	17:46	18:09	18:36
N 10°	05:34	06:00	06:22	18:01	18:23	18:49
0°	05:21	05:46	06:08	18:15	18:37	19:03
S 10°	05:05	05:31	05:54	18:30	18:52	19:18
20°	04:46	05:14	05:38	18:45	19:09	19:37
30°	04:22	04:54	05:20	19:03	19:29	20:01
35°	04:07	04:41	05:10	19:13	19:42	20:16
40°	03:48	04:26	04:57	19:25	19:56	20:35
45°	03:24	04:08	04:43	19:40	20:14	20:58
S 50°	02:51	03:45	04:25	19:57	20:37	21:31
52°	02:34	03:34	04:17	20:06	20:48	21:48
54°	02:12	03:21	04:07	20:15	21:01	22:09
56°	01:42	03:05	03:57	20:25	21:16	22:37
58°	00:49	02:47	03:45	20:37	21:34	23:25
S 60°	—:—	02:24	03:30	20:51	21:57	—:—

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	□	□	□	□	□	□
N 70°	□	□	□	□	□	□
68°	□	□	□	□	□	□
66°	□	□	□	□	□	□
64°	□	□	□	□	□	□
62°	10:03	□	12:31	08:44	□	10:04
60°	11:06	11:55	13:16	07:41	08:47	09:19
N 58°	11:41	12:32	13:46	07:07	08:10	08:49
56°	12:06	12:58	14:09	06:42	07:44	08:26
54°	12:26	13:19	14:27	06:22	07:23	08:07
52°	12:43	13:37	14:43	06:05	07:06	07:51
50°	12:57	13:51	14:56	05:51	06:51	07:37
45°	13:27	14:21	15:24	05:21	06:21	07:09
N 40°	13:50	14:45	15:45	04:59	05:57	06:47
35°	14:09	15:04	16:03	04:40	05:38	06:29
30°	14:26	15:21	16:19	04:24	05:21	06:13
20°	14:53	15:49	16:45	03:56	04:53	05:46
N 10°	15:17	16:13	17:07	03:33	04:29	05:23
0°	15:40	16:35	17:28	03:11	04:06	05:01
S 10°	16:02	16:57	17:49	02:49	03:44	04:39
20°	16:26	17:21	18:11	02:26	03:20	04:16
30°	16:54	17:49	18:37	01:59	02:51	03:48
35°	17:11	18:06	18:52	01:43	02:35	03:32
40°	17:30	18:25	19:10	01:24	02:15	03:13
45°	17:54	18:48	19:31	01:02	01:51	02:50
S 50°	18:24	19:18	19:57	00:33	01:21	02:21
52°	18:40	19:33	20:10	00:19	01:06	02:06
54°	18:57	19:50	20:25	00:03	00:48	01:49
56°	19:18	20:10	20:42	· · ·	00:27	01:29
58°	19:46	20:36	21:03	· · ·	00:00	01:04
S 60°	20:24	21:11	21:29	23:21	· · ·	00:28

Day	Sun		Mer. Pass hh:mm	Moon		Age 11-13 86-97%
	Eqn.of Time	Mer.		Mer.Pass.	Age	
	00 ^h mm:ss	12 ^h mm:ss		Pass hh:mm	Upper hh:mm	
22	11:18	11:26	12:11	21:53	09:25	
23	11:34	11:42	12:12	22:48	10:21	
24	11:49	11:57	12:12	23:41	11:15	

January 25, 26, 27 UT (Thu., Fri., Sat.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	123°48.4	211°38.1	S22°26.6	197°06.1	S23°19.9	89°08.8	N12°43.4	145°45.0	S10°56.3	Alpheratz	357°35.8	29°13.4											
1	138°50.9	226°37.2	26.7	212°06.5	19.7	104°11.1	43.4	160°47.2	56.2	Ankaa	353°08.1	-42°10.8											
2	153°53.4	241°36.3	26.8	227°06.9	19.5	119°13.4	43.5	175°49.4	56.1	Schedar	349°32.2	56°40.4											
3	168°55.8	256°35.5	· · 26.8	242°07.3	· · 19.3	134°15.7	· · 43.6	190°51.6	· · 56.0	Diphda	348°48.2	-17°51.5											
4	183°58.3	271°34.6	26.9	257°07.7	19.1	149°17.9	43.7	205°53.8	55.9	Achernar	335°20.8	-57°07.2											
5	199°00.7	286°33.7	27.0	272°08.1	18.9	164°20.2	43.8	220°56.0	55.8	Hamal	327°52.1	23°34.6											
6	214°03.2	301°32.9	S22°27.0	287°08.5	S23°18.7	179°22.5	N12°43.8	235°58.3	S10°55.7	Polaris	314°18.3	89°22.3											
7	229°05.7	316°32.0	27.1	302°09.0	18.5	194°24.7	43.9	251°00.5	55.6	Acamar	315°12.3	-40°12.7											
8	244°08.1	331°31.1	27.2	317°09.4	18.3	209°27.0	44.0	266°02.7	55.5	Menkar	314°06.9	4°11.0											
9	259°10.6	346°30.3	· · 27.2	332°09.8	· · 18.1	224°29.3	· · 44.1	281°04.9	· · 55.4	Mirfak	308°29.2	49°57.0											
10	274°13.1	1°29.4	27.3	347°10.2	17.9	239°31.5	44.2	296°07.1	55.3	Aldebaran	290°40.3	16°33.5											
11	289°15.5	16°28.5	27.4	2°10.6	17.7	254°33.8	44.2	311°09.3	55.2	Rigel	281°04.4	-8°10.5											
12	304°18.0	31°27.6	S22°27.4	17°11.0	S23°17.5	269°36.0	N12°44.3	326°11.5	S10°55.1	Capella	280°22.7	46°01.4											
13	319°20.5	46°26.8	27.5	32°11.4	17.3	284°38.3	44.4	341°13.7	55.0	Bellatrix	278°23.5	6°22.3											
14	334°22.9	61°25.9	27.5	47°11.8	17.1	299°40.6	44.5	356°15.9	54.9	Elnath	278°02.6	28°37.7											
15	349°25.4	76°25.0	· · 27.6	62°12.3	· · 16.9	314°42.8	· · 44.6	11°18.1	· · 54.8	Alnilam	275°38.3	-1°11.3											
16	4°27.9	91°24.2	27.6	77°12.7	16.7	329°45.1	44.6	26°20.3	54.7	Betelgeuse	270°52.6	7°24.7											
17	19°30.3	106°23.3	27.7	92°13.1	16.5	344°47.4	44.7	41°22.5	54.6	Canopus	263°52.3	-52°42.6											
18	34°32.8	121°22.4	S22°27.8	107°13.5	S23°16.3	359°49.6	N12°44.8	56°24.7	S10°54.5	Sirius	258°26.6	-16°45.0											
19	49°35.2	136°21.6	27.8	122°13.9	16.1	14°51.9	44.9	71°26.9	54.4	Adhara	255°06.1	-29°00.4											
20	64°37.7	151°20.7	27.9	137°14.3	15.9	29°54.2	45.0	86°29.1	54.3	Procyon	244°51.3	5°09.8											
21	79°40.2	166°19.8	· · 27.9	152°14.7	· · 15.7	44°56.4	· · 45.0	101°31.3	· · 54.2	Pollux	243°17.9	27°58.0											
22	94°42.6	181°19.0	28.0	167°15.2	15.5	59°58.7	45.1	116°33.5	54.1	Avior	234°14.4	-59°35.1											
23	109°45.1	196°18.1	28.0	182°15.6	15.3	75°00.9	45.2	131°35.7	54.0	Suhail	222°46.5	-43°31.7											
Mer.pass. 15:42				ν -0.9' d 0.1' m -3.93				ν 0.4' d -0.2' m 1.34				ν 2.3' d 0.1' m -2.40				ν 2.2' d -0.1' m 0.99							

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	124°47.6	211°17.2	S22°28.1	197°16.0	S23°15.1	90°03.2	N12°45.3	146°37.9	S10°53.9	Denebola	182°25.5	14°26.1							
1	139°50.0	226°16.4	28.1	212°16.4	14.9	105°05.5	45.4	161°40.1	53.8	Gienah	175°44.2	-17°40.5							
2	154°52.5	241°15.5	28.1	227°16.8	14.7	120°07.7	45.4	176°42.3	53.7	Acrux	173°00.7	-63°13.7							
3	169°55.0	256°14.6	· · 28.2	242°17.2	· · 14.5	135°10.0	· · 45.5	191°44.5	· · 53.6	Gacrux	171°52.3	-57°14.6							
4	184°57.4	271°13.7	28.2	257°17.6	14.3	150°12.2	45.6	206°46.7	53.4	Alioth	166°13.4	55°49.5							
5	199°59.9	286°12.9	28.3	272°18.0	14.1	165°14.5	45.7	221°48.9	53.3	Spica	158°23.1	-11°17.2							
6	215°02.4	301°12.0	S22°28.3	287°18.5	S23°13.9	180°16.8	N12°45.8	236°51.1	S10°53.2	Alkaid	152°52.6	49°11.3							
7	230°04.8	316°11.1	28.3	302°18.9	13.7	195°19.0	45.8	251°53.3	53.1	Hadar	148°37.1	-60°29.1							
8	245°07.3	331°10.3	28.4	317°19.3	13.4	210°21.3	45.9	266°55.5	53.0	Menkent	147°58.5	-36°29.1							
9	260°09.7	346°09.4	· · 28.4	332°19.7	· · 13.2	225°23.5	· · 46.0	281°57.7	· · 52.9	Arcturus	145°48.6	19°03.2							
10	275°12.2	1°08.5	28.5	347°20.1	13.0	240°25.8	46.1	296°59.9	52.8	Rigel Kent.	139°41.5	-60°55.8							
11	290°14.7	16°07.7	28.5	2°20.5	12.8	255°28.1	46.2	312°02.1	52.7	Kochab	137°19.9	74°03.0							
12	305°17.1	31°06.8	S22°28.5	17°20.9	S23°12.6	270°30.3	N12°46.3	327°04.3	S10°52.6	Zuben'ubi	136°56.9	-16°08.5							
13	320°19.6	46°05.9	28.5	32°21.4	12.4	285°32.6	46.3	342°06.5	52.5	Alphecca	126°04.5	26°37.8							
14	335°22.1	61°05.0	28.6	47°21.8	12.2	300°34.8	46.4	357°08.7	52.4	Antares	112°17.0	-26°29.1							
15	350°24.5	76°04.2	· · 28.6	62°22.2	· · 12.0	315°37.1	· · 46.5	12°10.9	· · 52.3	Atria	107°12.3	-69°04.0							
16	5°27.0	91°03.3	28.6	77°22.6	11.8	330°39.3	46.6	27°13.1	52.2	Sabik	102°03.9	-15°45.3							
17	20°29.5	106°02.4	28.7	92°23.0	11.6	345°41.6	46.7	42°15.3	52.1	Shaula	96°11.8	-37°07.2							
18	35°31.9	121°01.6	S22°28.7	107°23.4	S23°11.3	0°43.9	N12°46.8	57°17.5	S10°52.0	Rasalhague	95°59.5	12°32.4							
19	50°34.4	136°00.7	28.7	122°23.8	11.1	15°46.1	46.8	72°19.7	51.9	Eltanin	90°43.0	51°28.9							
20	65°36.8	150°59.8	28.7	137°24.3	10.9	30°48.4	46.9	87°21.9	51.8	Kaus Aust.	83°33.9	-34°22.4							
21	80°39.3	165°58.9	· · 28.8	152°24.7	· · 10.7	45°50.6	· · 47.0	102°24.1	· · 51.7	Vega	80°34.1	38°48.2							
22	95°41.8	180°58.1	28.8	167°25.1	10.5	60°52.9	47.1	117°26.3	51.6	Nunki	75°49.1	-26°16.0							
23	110°44.2	195°57.2	28.8	182°25.5	10.3	75°55.1	47.2	132°28.5	51.5	Altair	62°01.1	8°55.8							
Mer.pass. 15:38				ν -0.9' d 0.0' m -3.93				ν 0.4' d -0.2' m 1.34				ν 2.3' d 0.1' m -2.39				ν 2.2' d -0.1' m 0.99			

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	125°46.7	210°56.3	S22°28.8	197°25.9	S23°10.1	90°57.4	N12°47.3	147°30.7	S10°51.4	Fomalhaut	15°15.7	-29°29.9					
1	140°49.2	225°55.5	28.8	212°26.3	09.8	105°59.6	47.3	162°32.9	51.3	Scheat	13°46.2	28°12.8					
2	155°51.6	240°54.6	28.9	227°26.7	09.6	121°01.9	47.4	177°35.1	51.2	Markab	13°30.9	15°20.0					
3	170°54.1	255°53.7	· · 28.9	242°27.1	· · 09.4	136°04.1	· · 47.5	192°37.3	· · 51.1	Jan 25 Thu			SHA	Mer.pass			
4	185°56.6	270°52.8	28.9	257°27.6	09.2	151°06.4	47.6	207°39.5	51.0	Venus	87°49.6	09:54					
5	200°59.0	285°52.0	28.9	272°28.0	09.0	166°08.7	47.7	222°41.7	50.9	Mars	73°17.6	10:51					
6	216°01.5	300°51.1	S22°28.9	287°28.4	S23°08.8	181°10.9	N12°47.8	237°43.9	S10°50.8	Jupiter	325°20.4	18:01					
7	231°04.0	315°50.2	28.9	302°28.8	08.5	196°13.2	47.8	252°46.1	50.6	Saturn	21°56.6	14:15					
8	246°06.4	330°49.4	28.9	317°29.2	08.3	211°15.4	47.9	267°48.3	50.5	Jan 26 Fri			SHA	Mer.pass			
9	261°08.9	345°48.5	· · 28.9	332°29.6	· · 08.1	226°17.7	· · 48.0	282°50.5	· · 50.4	Venus	86°29.7	09:55					
10	276°11.3	0°47.6	28.9	347°30.0	07.9	241°19.9	48.1	297°52.7	50.3	Mars	72°28.4	10:51					
11	291°13.8	15°46.7	28.9	2°30.5	07.7	256°22.2	48.2	312°54.9	50.2	Jupiter	325°15.6	17:57					
12	306°16.3	30°45.9	S22°29.0	17°30.9	S23°07.4	271°24.4	N12°48.3	327°57.1	S10°50.1	Saturn	21°50.3	14:11					
13	321°18.7	45°45.0	29.0	32°31.3	07.2	286°26.7	48.3	342°59.3	50.0	Jan 27 Sat			SHA	Mer.pass			
14	336°21.2	60°44.1	29.0	47°31.7	07.0	301°28.9	48.4	358°01.5	49.9	Venus	85°09.6	09:57					
15	351°23.7	75°43.3	· · 29.0	62°32.1	· · 06.8	316°31.2	· · 48.5	13°03.7	· · 49.8	Mars	71°39.2	10:50					
16	6°26.1	90°42.4	29.0	77°32.5	06.6	331°33.4	48.6										

Sun				Moon				
Thu	GHA	Dec		GHA	ν	Dec	d	HP
0	176°59.0	S19°08.3		4°40.3	10.1'	N25°48.3	-6.4'	54.9'
1	191°58.9	07.7		19°09.5	10.2'	25°41.9	-6.5'	54.9'
2	206°58.7	07.1		33°38.7	10.3'	25°35.4	-6.6'	54.8'
3	221°58.6	· · 06.5		48°08.0	10.4'	25°28.8	-6.7'	54.8'
4	236°58.4	05.9		62°37.4	10.5'	25°22.1	-6.8'	54.8'
5	251°58.3	05.3		77°06.9	10.5'	25°15.2	-7.0'	54.8'
6	266°58.1	S19°04.7		91°36.4	10.6'	N25°08.3	-7.1'	54.8'
7	281°58.0	04.1		106°06.0	10.7'	25°01.2	-7.2'	54.8'
8	296°57.8	03.5		120°35.7	10.8'	24°54.0	-7.3'	54.8'
9	311°57.7	· · 02.8		135°05.5	10.9'	24°46.7	-7.4'	54.7'
10	326°57.6	02.2		149°35.4	10.9'	24°39.2	-7.5'	54.7'
11	341°57.4	01.6		164°05.3	11.0'	24°31.7	-7.7'	54.7'
12	356°57.3	S19°01.0		178°35.3	11.1'	N24°24.0	-7.8'	54.7'
13	11°57.1	19°00.4		193°05.5	11.2'	24°16.3	-7.9'	54.7'
14	26°57.0	18°59.8		207°35.6	11.3'	24°08.4	-8.0'	54.7'
15	41°56.8	· · 59.2		222°05.9	11.4'	24°00.4	-8.1'	54.7'
16	56°56.7	58.5		236°36.3	11.4'	23°52.3	-8.2'	54.6'
17	71°56.5	57.9		251°06.7	11.5'	23°44.1	-8.3'	54.6'
18	86°56.4	S18°57.3		265°37.2	11.6'	N23°35.8	-8.4'	54.6'
19	101°56.3	56.7		280°07.8	11.7'	23°27.4	-8.5'	54.6'
20	116°56.1	56.1		294°38.5	11.8'	23°18.8	-8.6'	54.6'
21	131°56.0	· · 55.5		309°09.3	11.9'	23°10.2	-8.7'	54.6'
22	146°55.8	54.8		323°40.2	11.9'	23°01.5	-8.8'	54.6'
23	161°55.7	54.2		338°11.1	12.0'	22°52.7	-8.9'	54.5'
SD = 16.2' $d = -0.6'$				SD = 15.0'				
Fri	GHA	Dec		GHA	ν	Dec	d	HP
0	176°55.6	S18°53.6		352°42.1	12.1'	N22°43.7	-9.0'	54.5'
1	191°55.4	53.0		7°13.3	12.2'	22°34.7	-9.1'	54.5'
2	206°55.3	52.4		21°44.4	12.3'	22°25.6	-9.2'	54.5'
3	221°55.1	· · 51.7		36°15.7	12.4'	22°16.4	-9.3'	54.5'
4	236°55.0	51.1		50°47.1	12.4'	22°07.1	-9.4'	54.5'
5	251°54.9	50.5		65°18.5	12.5'	21°57.7	-9.5'	54.5'
6	266°54.7	S18°49.9		79°50.1	12.6'	N21°48.2	-9.6'	54.5'
7	281°54.6	49.2		94°21.7	12.7'	21°38.6	-9.7'	54.4'
8	296°54.4	48.6		108°53.4	12.8'	21°28.9	-9.8'	54.4'
9	311°54.3	· · 48.0		123°25.1	12.9'	21°19.1	-9.9'	54.4'
10	326°54.2	47.4		137°57.0	12.9'	21°09.3	-9.9'	54.4'
11	341°54.0	46.7		152°28.9	13.0'	20°59.3	-10.0'	54.4'
12	356°53.9	S18°46.1		167°00.9	13.1'	N20°49.3	-10.1'	54.4'
13	11°53.8	45.5		181°33.0	13.2'	20°39.2	-10.2'	54.4'
14	26°53.6	44.9		196°05.2	13.3'	20°29.0	-10.3'	54.4'
15	41°53.5	· · 44.2		210°37.5	13.3'	20°18.7	-10.4'	54.4'
16	56°53.4	43.6		225°09.8	13.4'	20°08.3	-10.4'	54.3'
17	71°53.2	43.0		239°42.3	13.5'	19°57.9	-10.5'	54.3'
18	86°53.1	S18°42.3		254°14.8	13.6'	N19°47.4	-10.6'	54.3'
19	101°52.9	41.7		268°47.3	13.7'	19°36.7	-10.7'	54.3'
20	116°52.8	41.1		283°20.0	13.7'	19°26.1	-10.8'	54.3'
21	131°52.7	· · 40.4		297°52.7	13.8'	19°15.3	-10.8'	54.3'
22	146°52.6	39.8		312°25.6	13.9'	19°04.5	-10.9'	54.3'
23	161°52.4	39.2		326°58.5	14.0'	18°53.6	-11.0'	54.3'
SD = 16.2' $d = -0.6'$				SD = 14.9'				
Sat	GHA	Dec		GHA	ν	Dec	d	HP
0	176°52.3	S18°38.5		341°31.4	14.0'	N18°42.6	-11.1'	54.3'
1	191°52.2	37.9		356°04.5	14.1'	18°31.5	-11.1'	54.3'
2	206°52.0	37.3		10°37.6	14.2'	18°20.4	-11.2'	54.2'
3	221°51.9	· · 36.6		25°10.8	14.3'	18°09.2	-11.3'	54.2'
4	236°51.8	36.0		39°44.1	14.3'	17°57.9	-11.3'	54.2'
5	251°51.6	35.4		54°17.4	14.4'	17°46.6	-11.4'	54.2'
6	266°51.5	S18°34.7		68°50.8	14.5'	N17°35.2	-11.5'	54.2'
7	281°51.4	34.1		83°24.3	14.6'	17°23.7	-11.5'	54.2'
8	296°51.2	33.4		97°57.9	14.6'	17°12.2	-11.6'	54.2'
9	311°51.1	· · 32.8		112°31.5	14.7'	17°00.5	-11.7'	54.2'
10	326°51.0	32.2		127°05.2	14.8'	16°48.9	-11.7'	54.2'
11	341°50.9	31.5		141°39.0	14.8'	16°37.2	-11.8'	54.2'
12	356°50.7	S18°30.9		156°12.8	14.9'	N16°25.4	-11.9'	54.2'
13	11°50.6	30.2		170°46.7	15.0'	16°13.5	-11.9'	54.2'
14	26°50.5	29.6		185°20.7	15.0'	16°01.6	-12.0'	54.1'
15	41°50.3	· · 29.0		199°54.7	15.1'	15°49.6	-12.0'	54.1'
16	56°50.2	28.3		214°28.9	15.2'	15°37.6	-12.1'	54.1'
17	71°50.1	27.7		229°03.0	15.2'	15°25.5	-12.1'	54.1'
18	86°50.0	S18°27.0		243°37.3	15.3'	N15°13.4	-12.2'	54.1'
19	101°49.8	26.4		258°11.6	15.4'	15°01.2	-12.3'	54.1'
20	116°49.7	25.7		272°45.9	15.4'	14°48.9	-12.3'	54.1'
21	131°49.6	· · 25.1		287°20.4	15.5'	14°36.6	-12.4'	54.1'
22	146°49.5	24.4		301°54.9	15.5'	14°24.3	-12.4'	54.1'
23	161°49.3	23.8		316°29.4	15.6'	14°11.9	-12.5'	54.1'
SD = 16.2' $d = -0.6'$				SD = 14.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:31	09:07	11:53	12:33	15:19	16:55
N 70°	07:21	08:44	10:24	14:02	15:42	17:05
68°	07:12	08:25	09:45	14:40	16:00	17:14
66°	07:05	08:11	09:19	15:07	16:15	17:21
64°	06:58	07:58	08:59	15:27	16:27	17:28
62°	06:52	07:48	08:42	15:44	16:38	17:33
60°	06:47	07:39	08:28	15:57	16:47	17:38
N 58°	06:42	07:31	08:17	16:09	16:54	17:43
56°	06:38	07:24	08:06	16:19	17:01	17:47
54°	06:34	07:18	07:57	16:28	17:08	17:51
52°	06:31	07:12	07:49	16:36	17:14	17:55
50°	06:27	07:07	07:42	16:43	17:19	17:58
45°	06:19	06:55	07:27	16:58	17:30	18:06
N 40°	06:12	06:45	07:14	17:11	17:40	18:13
35°	06:06	06:36	07:03	17:22	17:49	18:20
30°	05:59	06:28	06:54	17:31	17:57	18:26
20°	05:47	06:14	06:37	17:48	18:11	18:38
N 10°	05:35	06:00	06:23	18:02	18:24	18:50
0°	05:22	05:47	06:09	18:16	18:38	19:03
S 10°	05:07	05:33	05:55	18:30	18:52	19:18
20°	04:48	05:16	05:40	18:45	19:08	19:36
30°	04:25	04:57	05:23	19:02	19:28	19:59
35°	04:10	04:45	05:13	19:12	19:40	20:14
40°	03:52	04:30	05:01	19:23	19:54	20:32
45°	03:30	04:13	04:47	19:37	20:11	20:54
S 50°	02:58	03:51	04:30	19:54	20:33	21:25
52°	02:42	03:40	04:22	20:01	20:43	21:41
54°	02:22	03:28	04:13	20:10	20:56	22:00
56°	01:56	03:13	04:03	20:20	21:10	22:26
58°	01:16	02:56	03:52	20:31	21:26	23:03
S 60°	—:—	02:35	03:38	20:45	21:47	—:—

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°			15:45			11:53
N 70°			16:25			11:11
68°		14:31	16:52		11:32	10:43
66°		15:16	17:13		10:47	10:21
64°	13:44	15:45	17:29	10:39	10:17	10:04
62°	14:23	16:06	17:42	09:59	09:54	09:49
60°	14:50	16:24	17:53	09:31	09:35	09:37
N 58°	15:12	16:39	18:03	09:09	09:20	09:26
56°	15:29	16:51	18:12	08:51	09:07	09:17
54°	15:44	17:02	18:19	08:36	08:55	09:09
52°	15:56	17:12	18:26	08:23	08:45	09:01
50°	16:07	17:20	18:32	08:11	08:36	08:55
45°	16:31	17:38	18:45	07:47	08:17	08:40
N 40°	16:49	17:53	18:55	07:28	08:01	08:28
35°	17:05	18:05	19:05	07:12	07:48	08:18
30°	17:18	18:16	19:12	06:58	07:36	08:09
20°	17:41	18:35	19:26	06:33	07:16	07:53
N 10°	18:00	18:51	19:38	06:13	06:58	07:40
0°	18:19	19:05	19:49	05:53	06:42	07:27
S 10°	18:37	19:20	20:00	05:33	06:25	07:14
20°	18:56	19:36	20:12	05:12	06:07	07:00
30°	19:18	19:54	20:25	04:47	05:46	06:44
35°	19:32	20:04	20:33	04:33	05:34	06:34
40°	19:47	20:16	20:41	04:16	05:20	06:23
45°	20:04	20:30	20:52	03:55	05:03	06:11
S 50°	20:26	20:47	21:04	03:30	04:42	05:55
52°	20:37	20:55	21:09	03:17	04:32	05:48
54°	20:48	21:04	21:16	03:03	04:21	05:39
56°	21:02	21:14	21:22	02:46	04:08	05:30
58°	21:17	21:25	21:30	02:25	03:53	05:20
S 60°	21:36	21:38	21:39	02:00	03:35	05:07

Day	Sun		Mer. Pass hh:mm	Moon		Age 14-16 99-98%
	Eqn.of Time	Mer.		Mer.Pass.		
	00 ^h mm:ss	12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
25	12:04	12:11	12:12	—:—	12:06	

January 28, 29, 30 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	126°45.8	210°35.4	S22°28.9	197°35.9	S23°04.8	91°51.4	N12°49.3	148°23.5	S10°48.9	Alpheratz	357°35.8 29°13.4
1	141°48.3	225°34.5	28.9	212°36.3	04.5	106°53.6	49.4	163°25.7	48.8	Ankaa	353°08.1 -42°10.8
2	156°50.8	240°33.7	28.9	227°36.7	04.3	121°55.9	49.4	178°27.9	48.7	Schedar	349°32.2 56°40.3
3	171°53.2	255°32.8	•• 28.9	242°37.1	•• 04.1	136°58.1	•• 49.5	193°30.1	•• 48.6	Diphda	348°48.2 -17°51.5
4	186°55.7	270°31.9	28.9	257°37.5	03.9	152°00.4	49.6	208°32.3	48.5	Achernar	335°20.9 -57°07.2
5	201°58.2	285°31.0	28.9	272°37.9	03.6	167°02.6	49.7	223°34.5	48.3	Hamal	327°52.1 23°34.6
6	217°00.6	300°30.2	S22°28.8	287°38.3	S23°03.4	182°04.9	N12°49.8	238°36.7	S10°48.2	Polaris	314°19.9 89°22.3
7	232°03.1	315°29.3	28.8	302°38.8	03.2	197°07.1	49.9	253°38.9	48.1	Acamar	315°12.3 -40°12.8
8	247°05.6	330°28.4	28.8	317°39.2	02.9	212°09.4	50.0	268°41.1	48.0	Menkar	314°06.9 4°11.0
9	262°08.0	345°27.6	•• 28.8	332°39.6	•• 02.7	227°11.6	•• 50.1	283°43.3	•• 47.9	Mirfak	308°29.2 49°57.0
10	277°10.5	0°26.7	28.8	347°40.0	02.5	242°13.8	50.1	298°45.5	47.8	Aldebaran	290°40.3 16°33.5
11	292°12.9	15°25.8	28.7	2°40.4	02.3	257°16.1	50.2	313°47.7	47.7	Rigel	281°04.4 -8°10.5
12	307°15.4	30°24.9	S22°28.7	17°40.8	S23°02.0	272°18.3	N12°50.3	328°49.9	S10°47.6	Capella	280°22.7 46°01.4
13	322°17.9	45°24.1	28.7	32°41.3	01.8	287°20.6	50.4	343°52.1	47.5	Bellatrix	278°23.5 6°22.3
14	337°20.3	60°23.2	28.7	47°41.7	01.6	302°22.8	50.5	358°54.3	47.4	Elnath	278°02.6 28°37.7
15	352°22.8	75°22.3	•• 28.6	62°42.1	•• 01.3	317°25.1	•• 50.6	13°56.5	•• 47.3	Alnilam	275°38.3 -1°11.3
16	7°25.3	90°21.4	28.6	77°42.5	01.1	332°27.3	50.7	28°58.7	47.2	Betelgeuse	270°52.7 7°24.7
17	22°27.7	105°20.6	28.6	92°42.9	00.9	347°29.5	50.7	44°00.9	47.1	Canopus	263°52.3 -52°42.6
18	37°30.2	120°19.7	S22°28.5	107°43.3	S23°00.6	2°31.8	N12°50.8	59°03.1	S10°47.0	Sirius	258°26.6 -16°45.0
19	52°32.7	135°18.8	28.5	122°43.8	00.4	17°34.0	50.9	74°05.3	46.9	Adhara	255°06.1 -29°00.4
20	67°35.1	150°18.0	28.5	137°44.2	23°00.2	32°36.3	51.0	89°07.5	46.8	Procyon	244°51.3 5°09.7
21	82°37.6	165°17.1	•• 28.5	152°44.6	22°59.9	47°38.5	•• 51.1	104°09.7	•• 46.7	Pollux	243°17.9 27°58.1
22	97°40.1	180°16.2	28.4	167°45.0	59.7	62°40.8	51.2	119°11.9	46.6	Avior	234°14.4 -59°35.2
23	112°42.5	195°15.3	28.4	182°45.4	59.5	77°43.0	51.3	134°14.1	46.5	Suhail	222°46.4 -43°31.7
Mer.pass. 15:30		ν -0.9' d -0.0' m -3.93		ν 0.4' d -0.2' m 1.33		ν 2.2' d 0.1' m -2.38		ν 2.2' d -0.1' m 0.99			

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	127°45.0	210°14.5	S22°28.3	197°45.8	S22°59.2	92°45.2	N12°51.4	149°16.3	S10°46.4		
1	142°47.4	225°13.6	28.3	212°46.3	59.0	107°47.5	51.4	164°18.5	46.3		
2	157°49.9	240°12.7	28.3	227°46.7	58.7	122°49.7	51.5	179°20.7	46.1		
3	172°52.4	255°11.8	•• 28.2	242°47.1	•• 58.5	137°51.9	•• 51.6	194°22.9	•• 46.0		
4	187°54.8	270°11.0	28.2	257°47.5	58.3	152°54.2	51.7	209°25.1	45.9		
5	202°57.3	285°10.1	28.1	272°47.9	58.0	167°56.4	51.8	224°27.3	45.8		
6	217°59.8	300°09.2	S22°28.1	287°48.3	S22°57.8	182°58.7	N12°51.9	239°29.5	S10°45.7		
7	233°02.2	315°08.4	28.1	302°48.7	57.6	198°00.9	52.0	254°31.7	45.6		
8	248°04.7	330°07.5	28.0	317°49.2	57.3	213°03.1	52.1	269°33.9	45.5		
9	263°07.2	345°06.6	•• 28.0	332°49.6	•• 57.1	228°05.4	•• 52.2	284°36.1	•• 45.4		
10	278°09.6	0°05.7	27.9	347°50.0	56.8	243°07.6	52.2	299°38.3	45.3		
11	293°12.1	15°04.9	27.9	2°50.4	56.6	258°09.8	52.3	314°40.5	45.2		
12	308°14.6	30°04.0	S22°27.8	17°50.8	S22°56.3	273°12.1	N12°52.4	329°42.6	S10°45.1		
13	323°17.0	45°03.1	27.8	32°51.3	56.1	288°14.3	52.5	344°44.8	45.0		
14	338°19.5	60°02.2	27.7	47°51.7	55.9	303°16.6	52.6	359°47.0	44.9		
15	353°21.9	75°01.4	•• 27.6	62°52.1	•• 55.6	318°18.8	•• 52.7	14°49.2	•• 44.8		
16	8°24.4	90°00.5	27.6	77°52.5	55.3	333°21.0	52.8	29°51.4	44.7		
17	23°26.9	104°59.6	27.5	92°52.9	55.1	348°23.3	52.9	44°53.6	44.6		
18	38°29.3	119°58.7	S22°27.5	107°53.3	S22°54.9	3°25.5	N12°53.0	59°55.8	S10°44.5		
19	53°31.8	134°57.9	27.4	122°53.8	54.6	18°27.7	53.0	74°58.0	44.4		
20	68°34.3	149°57.0	27.4	137°54.2	54.4	33°30.0	53.1	90°00.2	44.3		
21	83°36.7	164°56.1	•• 27.3	152°54.6	•• 54.2	48°32.2	•• 53.2	105°02.4	•• 44.1		
22	98°39.2	179°55.3	27.2	167°55.0	53.9	63°34.4	53.3	120°04.6	44.0		
23	113°41.7	194°54.4	27.2	182°55.4	53.7	78°36.7	53.4	135°06.8	43.9		
Mer.pass. 15:26		ν -0.9' d -0.0' m -3.92		ν 0.4' d -0.2' m 1.33		ν 2.2' d 0.1' m -2.37		ν 2.2' d -0.1' m 0.99			

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	128°44.1	209°53.5	S22°27.1	197°55.8	S22°53.4	93°38.9	N12°53.5	150°09.0	S10°43.8		
1	143°46.6	224°52.6	27.0	212°56.3	53.2	108°41.1	53.6	165°11.2	43.7		
2	158°49.0	239°51.8	27.0	227°56.7	52.9	123°43.4	53.7	180°13.4	43.6		
3	173°51.5	254°50.9	•• 26.9	242°57.1	•• 52.7	138°45.6	•• 53.8	195°15.6	•• 43.5		
4	188°54.0	269°50.0	26.8	257°57.5	52.4	153°47.8	53.8	210°17.8	43.4		
5	203°56.4	284°49.1	26.8	272°57.9	52.2	168°50.1	53.9	225°20.0	43.3		
6	218°58.9	299°48.3	S22°26.7	287°58.3	S22°51.9	183°52.3	N12°54.0	240°22.2	S10°43.2		
7	234°01.4	314°47.4	26.6	302°58.8	51.7	198°54.5	54.1	255°24.4	43.1		
8	249°03.8	329°46.5	26.5	317°59.2	51.4	213°56.7	54.2	270°26.6	43.0		
9	264°06.3	344°45.7	•• 26.5	332°59.6	•• 51.2	228°59.0	•• 54.3	285°28.8	•• 42.9		
10	279°08.8	359°44.8	26.4	348°00.0	50.9	244°01.2	54.4	300°31.0	42.8		
11	294°11.2	14°43.9	26.3	3°00.4	50.7	259°03.4	54.5	315°33.2	42.7		
12	309°13.7	29°43.0	S22°26.2	18°00.9	S22°50.4	274°05.7	N12°54.6	330°35.4	S10°42.6		
13	324°16.2	44°42.2	26.1	33°01.3	50.2	289°07.9	54.7	345°37.6	42.5		
14	339°18.6	59°41.3	26.1	48°01.7	49.9	304°10.1	54.8	0°39.8	42.3		
15	354°21.1	74°40.4	•• 26.0	63°02.1	•• 49.7	319°12.3	•• 54.8	15°41.9	•• 42.2		
16	9°23.5	89°39.5	25.9	78°02.5	49.4	334°14.6	54.9	30°44.1	42.1		
17	24°26.0	104°38.7	25.8	93°02.9	49.2	349°16.8	55.0	45°46.3	42.0		
18	39°28.5	119°37.8	S22°25.7	108°03.4	S22°48.9	4°19.0	N12°55.1	60°48.5	S10°41.9		
19	54°30.9	134°36.9	25.6	123°03.8	48.6	19°21.3	55.2	75°50.7	41.8		
20	69°33.4	149°36.1	25.6	138°04.2	48.4	34°23.5	55.3	90°52.9	41.7		
21	84°35.9	164°35.2	•• 25.5	153°04.6	•• 48.1	49°25.7	•• 55.4	105°55.1	•• 41.6		
22	99°38.3	179°34.3	25.4	168°05.0	47.9	64°27.9	55.5	120°57.3	41.5		
23	114°40.8	194°33.4	25.3	183°05.5	47.6	79°30.2	55.6	135°59.5	41.4		
Mer.pass. 15:23		ν -0.9' d -0.1' m -3.92		ν 0.4' d -0.2' m 1.33		ν 2.2' d 0.1' m -2.36		ν 2.2' d -0.1' m 0.99			

Jan 28 Sun	SHA	Mer.pass
Venus	83°49.6	09:58
Mars	70°50.0	10:49
Jupiter	325°05.6	17:50
Saturn	21°37.7	14:04

Jan 29 Mon	SHA	Mer.pass
Venus	82°29.5	10:00
Mars	70°00.9	10:49
Jupiter	325°00.2	17:46
Saturn	21°31.3	14:01

Jan 30 Tue	SHA	Mer.pass
Venus	81°09.4	10:01
Mars	69°11.7	10:48
Jupiter	324°54.8	17:43
Saturn	21°24.9	13:57

Horizontal parallax	
Venus:	0.1
Mars:	0.1

2024 January 28 to Jan. 30 UT

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:22	08:54	11:01	13:26	15:33	17:06
N 70°	07:12	08:33	10:05	14:22	15:54	17:15
68°	07:04	08:16	09:32	14:55	16:11	17:23
66°	06:58	08:03	09:09	15:18	16:24	17:29
64°	06:52	07:52	08:50	15:37	16:35	17:35
62°	06:47	07:42	08:35	15:52	16:45	17:40
60°	06:42	07:34	08:22	16:05	16:53	17:45
N 58°	06:38	07:26	08:11	16:16	17:01	17:49
56°	06:34	07:20	08:01	16:25	17:07	17:53
54°	06:31	07:14	07:53	16:34	17:13	17:56
52°	06:27	07:08	07:45	16:41	17:18	17:59
50°	06:24	07:03	07:38	16:48	17:23	18:03
45°	06:17	06:52	07:24	17:03	17:34	18:10
N 40°	06:10	06:43	07:12	17:15	17:44	18:16
35°	06:04	06:35	07:01	17:25	17:52	18:22
30°	05:58	06:27	06:52	17:34	17:59	18:28
20°	05:47	06:13	06:36	17:50	18:13	18:39
N 10°	05:35	06:01	06:23	18:04	18:26	18:51
0°	05:22	05:48	06:09	18:17	18:38	19:04
S 10°	05:08	05:34	05:56	18:30	18:52	19:18
20°	04:51	05:18	05:42	18:44	19:07	19:35
30°	04:28	04:59	05:25	19:00	19:26	19:57
35°	04:14	04:48	05:16	19:10	19:38	20:11
40°	03:57	04:34	05:05	19:21	19:51	20:28
45°	03:35	04:18	04:52	19:34	20:07	20:50
S 50°	03:06	03:57	04:35	19:50	20:28	21:19
52°	02:50	03:46	04:28	19:57	20:38	21:34
54°	02:32	03:35	04:19	20:05	20:50	21:52
56°	02:08	03:21	04:10	20:15	21:03	22:14
58°	01:35	03:05	03:59	20:25	21:18	22:46
S 60°	—:—	02:46	03:47	20:38	21:38	23:50

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	18:15	20:17	22:13	10:51	10:12	09:41
N 70°	18:32	20:23	22:10	10:32	10:04	09:39
68°	18:45	20:28	22:08	10:16	09:56	09:38
66°	18:55	20:32	22:06	10:04	09:50	09:37
64°	19:04	20:35	22:04	09:54	09:45	09:37
62°	19:12	20:38	22:03	09:45	09:40	09:36
60°	19:19	20:41	22:02	09:37	09:36	09:35
N 58°	19:24	20:43	22:01	09:30	09:33	09:35
56°	19:29	20:45	22:00	09:24	09:30	09:34
54°	19:34	20:47	21:59	09:19	09:27	09:34
52°	19:38	20:48	21:58	09:14	09:24	09:34
50°	19:42	20:50	21:57	09:09	09:22	09:33
45°	19:50	20:53	21:56	09:00	09:17	09:32
N 40°	19:56	20:56	21:55	08:52	09:12	09:32
35°	20:02	20:58	21:53	08:45	09:09	09:31
30°	20:07	21:00	21:52	08:38	09:05	09:31
20°	20:15	21:03	21:51	08:28	08:59	09:30
N 10°	20:23	21:07	21:49	08:18	08:54	09:29
0°	20:30	21:09	21:48	08:09	08:49	09:28
S 10°	20:37	21:12	21:47	08:00	08:44	09:27
20°	20:44	21:15	21:46	07:50	08:39	09:26
30°	20:53	21:19	21:44	07:39	08:33	09:25
35°	20:58	21:21	21:43	07:33	08:29	09:25
40°	21:03	21:23	21:42	07:25	08:25	09:24
45°	21:09	21:26	21:41	07:16	08:20	09:23
S 50°	21:17	21:29	21:40	07:06	08:15	09:22
52°	21:20	21:30	21:39	07:01	08:12	09:22
54°	21:24	21:32	21:39	06:55	08:09	09:21
56°	21:28	21:33	21:38	06:49	08:06	09:21
58°	21:33	21:35	21:37	06:43	08:02	09:20
S 60°	21:38	21:37	21:36	06:35	07:58	09:20

Day	Sun			Moon		Age 17-19 95-84%
	Eqn. of Time	Mer.	Mer. Pass.	Upper	Lower	
	00 ^h mm:ss	12 ^h mm:ss	Pass hh:mm	hh:mm	hh:mm	
28	12:43	12:49	12:13	01:59	14:20	
29	12:55	13:00	12:13	02:40	14:59	
30	13:05	13:10	12:13	03:19	15:38	

January 31, 01, 02 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	129°43.3	209°32.6	S22°25.2	198°05.9	S22°47.4	94°32.4	N12°55.7	151°01.7	S10°41.3	Alpheratz	357°35.8	29°13.4
1	144°45.7	224°31.7	25.1	213°06.3	47.1	109°34.6	55.8	166°03.9	41.2	Ankaa	353°08.1	-42°10.8
2	159°48.2	239°30.8	25.0	228°06.7	46.8	124°36.8	55.9	181°06.1	41.1	Schedar	349°32.2	56°40.3
3	174°50.6	254°29.9	•• 24.9	243°07.1	•• 46.6	139°39.1	•• 55.9	196°08.3	•• 41.0	Diphda	348°48.2	-17°51.5
4	189°53.1	269°29.1	24.8	258°07.6	46.3	154°41.3	56.0	211°10.5	40.9	Achernar	335°20.9	-57°07.2
5	204°55.6	284°28.2	24.7	273°08.0	46.1	169°43.5	56.1	226°12.7	40.8	Hamal	327°52.1	23°34.6
6	219°58.0	299°27.3	S22°24.6	288°08.4	S22°45.8	184°45.7	N12°56.2	241°14.9	S10°40.6	Polaris	314°21.5	89°22.3
7	235°00.5	314°26.5	24.5	303°08.8	45.6	199°48.0	56.3	256°17.1	40.5	Acamar	315°12.3	-40°12.8
8	250°03.0	329°25.6	24.4	318°09.2	45.3	214°50.2	56.4	271°19.3	40.4	Menkar	314°06.9	4°11.0
9	265°05.4	344°24.7	•• 24.3	333°09.7	•• 45.0	229°52.4	•• 56.5	286°21.5	•• 40.3	Mirfak	308°29.3	49°57.0
10	280°07.9	359°23.8	24.2	348°10.1	44.8	244°54.6	56.6	301°23.7	40.2	Aldebaran	290°40.3	16°33.5
11	295°10.4	14°23.0	24.1	3°10.5	44.5	259°56.9	56.7	316°25.8	40.1	Rigel	281°04.4	-8°10.5
12	310°12.8	29°22.1	S22°24.0	18°10.9	S22°44.2	274°59.1	N12°56.8	331°28.0	S10°40.0	Capella	280°22.7	46°01.4
13	325°15.3	44°21.2	23.9	33°11.3	44.0	290°01.3	56.9	346°30.2	39.9	Bellatrix	278°23.5	6°22.2
14	340°17.8	59°20.3	23.8	48°11.8	43.7	305°03.5	57.0	1°32.4	39.8	Elnath	278°02.6	28°37.7
15	355°20.2	74°19.5	•• 23.7	63°12.2	•• 43.5	320°05.7	•• 57.1	16°34.6	•• 39.7	Alnilam	275°38.3	-1°11.3
16	10°22.7	89°18.6	23.5	78°12.6	43.2	335°08.0	57.2	31°36.8	39.6	Betelgeuse	270°52.7	7°24.7
17	25°25.1	104°17.7	23.4	93°13.0	42.9	350°10.2	57.3	46°39.0	39.5	Canopus	263°52.3	-52°42.6
18	40°27.6	119°16.9	S22°23.3	108°13.4	S22°42.7	5°12.4	N12°57.3	61°41.2	S10°39.4	Sirius	258°26.6	-16°45.0
19	55°30.1	134°16.0	23.2	123°13.9	42.4	20°14.6	57.4	76°43.4	39.3	Adhara	255°06.1	-29°00.4
20	70°32.5	149°15.1	23.1	138°14.3	42.1	35°16.8	57.5	91°45.6	39.2	Procyon	244°51.3	5°09.7
21	85°35.0	164°14.2	•• 23.0	153°14.7	•• 41.9	50°19.1	•• 57.6	106°47.8	•• 39.1	Pollux	243°17.9	27°58.1
22	100°37.5	179°13.4	22.8	168°15.1	41.6	65°21.3	57.7	121°50.0	38.9	Avior	234°14.4	-59°35.2
23	115°39.9	194°12.5	22.7	183°15.5	41.3	80°23.5	57.8	136°52.2	38.8	Suhail	222°46.4	-43°31.8
Mer.pass. 15:19		ν -0.9' d -0.1' m -3.92		ν 0.4' d -0.3' m 1.33		ν 2.2' d 0.1' m -2.36		ν 2.2' d -0.1' m 0.99				

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	130°42.4	209°11.6	S22°22.6	198°16.0	S22°41.1	95°25.7	N12°57.9	151°54.4	S10°38.7	Denebola	182°25.5	14°26.1
1	145°44.9	224°10.8	22.5	213°16.4	40.8	110°27.9	58.0	166°56.6	38.6	Gienah	175°44.2	-17°40.5
2	160°47.3	239°09.9	22.4	228°16.8	40.5	125°30.2	58.1	181°58.8	38.5	Acrux	173°00.6	-63°13.7
3	175°49.8	254°09.0	•• 22.2	243°17.2	•• 40.3	140°32.4	•• 58.2	197°00.9	•• 38.4	Gacrux	171°52.2	-57°14.7
4	190°52.3	269°08.1	22.1	258°17.6	40.0	155°34.6	58.3	212°03.1	38.3	Alioth	166°13.4	55°49.5
5	205°54.7	284°07.3	22.0	273°18.1	39.7	170°36.8	58.4	227°05.3	38.2	Spica	158°23.0	-11°17.2
6	220°57.2	299°06.4	S22°21.9	288°18.5	S22°39.4	185°39.0	N12°58.5	242°07.5	S10°38.1	Alkaid	152°52.5	49°11.3
7	235°59.6	314°05.5	21.7	303°18.9	39.2	200°41.2	58.6	257°09.7	38.0	Hadar	148°37.0	-60°29.1
8	251°02.1	329°04.7	21.6	318°19.3	38.9	215°43.5	58.7	272°11.9	37.9	Menkent	147°58.5	-36°29.2
9	266°04.6	344°03.8	•• 21.5	333°19.7	•• 38.6	230°45.7	•• 58.8	287°14.1	•• 37.8	Arcturus	145°48.6	19°03.2
10	281°07.0	359°02.9	21.3	348°20.2	38.4	245°47.9	58.9	302°16.3	37.7	Rigel Kent.	139°41.4	-60°55.8
11	296°09.5	14°02.0	21.2	3°20.6	38.1	260°50.1	59.0	317°18.5	37.6	Kochab	137°19.8	74°03.0
12	311°12.0	29°01.2	S22°21.1	18°21.0	S22°37.8	275°52.3	N12°59.0	332°20.7	S10°37.5	Zuben'ubi	136°56.9	-16°08.5
13	326°14.4	44°00.3	20.9	33°21.4	37.5	290°54.5	59.1	347°22.9	37.3	Alphecca	126°04.5	26°37.8
14	341°16.9	58°59.4	20.8	48°21.8	37.3	305°56.8	59.2	2°25.1	37.2	Antares	112°17.0	-26°29.1
15	356°19.4	73°58.6	•• 20.7	63°22.3	•• 37.0	320°59.0	•• 59.3	17°27.3	•• 37.1	Atria	107°12.2	-69°04.0
16	11°21.8	88°57.7	20.5	78°22.7	36.7	336°01.2	59.4	32°29.5	37.0	Sabik	102°03.9	-15°45.3
17	26°24.3	103°56.8	20.4	93°23.1	36.5	351°03.4	59.5	47°31.7	36.9	Shaula	96°11.7	-37°07.2
18	41°26.7	118°56.0	S22°20.2	108°23.5	S22°36.2	6°05.6	N12°59.6	62°33.8	S10°36.8	Rasalhague	95°59.5	12°32.4
19	56°29.2	133°55.1	20.1	123°24.0	35.9	21°07.8	59.7	77°36.0	36.7	Eltanin	90°42.9	51°28.8
20	71°31.7	148°54.2	19.9	138°24.4	35.6	36°10.0	59.8	92°38.2	36.6	Kaus Aust.	83°33.9	-34°22.4
21	86°34.1	163°53.3	•• 19.8	153°24.8	•• 35.3	51°12.2	12°59.9	107°40.4	•• 36.5	Vega	80°34.1	38°48.1
22	101°36.6	178°52.5	19.6	168°25.2	35.1	66°14.5	13°00.0	122°42.6	36.4	Nunki	75°49.0	-26°16.0
23	116°39.1	193°51.6	19.5	183°25.6	34.8	81°16.7	00.1	137°44.8	36.3	Altair	62°01.0	8°55.8
Mer.pass. 15:15		ν -0.9' d -0.1' m -3.92		ν 0.4' d -0.3' m 1.33		ν 2.2' d 0.1' m -2.35		ν 2.2' d -0.1' m 0.99				

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	131°41.5	208°50.7	S22°19.4	198°26.1	S22°34.5	96°18.9	N13°00.2	152°47.0	S10°36.2	Fomalhaut	15°15.7	-29°29.9
1	146°44.0	223°49.9	19.2	213°26.5	34.2	111°21.1	00.3	167°49.2	36.1	Scheat	13°46.3	28°12.8
2	161°46.5	238°49.0	19.1	228°26.9	34.0	126°23.3	00.4	182°51.4	36.0	Markab	13°30.9	15°20.0
3	176°48.9	253°48.1	•• 18.9	243°27.3	•• 33.7	141°25.5	•• 00.5	197°53.6	•• 35.8			
4	191°51.4	268°47.3	18.7	258°27.8	33.4	156°27.7	00.6	212°55.8	35.7			
5	206°53.9	283°46.4	18.6	273°28.2	33.1	171°29.9	00.7	227°58.0	35.6			
6	221°56.3	298°45.5	S22°18.4	288°28.6	S22°32.8	186°32.1	N13°00.8	243°00.2	S10°35.5			
7	236°58.8	313°44.6	18.3	303°29.0	32.6	201°34.4	00.9	258°02.3	35.4			
8	252°01.2	328°43.8	18.1	318°29.4	32.3	216°36.6	01.0	273°04.5	35.3			
9	267°03.7	343°42.9	•• 18.0	333°29.9	•• 32.0	231°38.8	•• 01.1	288°06.7	•• 35.2			
10	282°06.2	358°42.0	17.8	348°30.3	31.7	246°41.0	01.2	303°08.9	35.1			
11	297°08.6	13°41.2	17.6	3°30.7	31.4	261°43.2	01.3	318°11.1	35.0			
12	312°11.1	28°40.3	S22°17.5	18°31.1	S22°31.1	276°45.4	N13°01.4	333°13.3	S10°34.9			
13	327°13.6	43°39.4	17.3	33°31.6	30.9	291°47.6	01.5	348°15.5	34.8			
14	342°16.0	58°38.6	17.1	48°32.0	30.6	306°49.8	01.6	3°17.7	34.7			
15	357°18.5	73°37.7	•• 17.0	63°32.4	•• 30.3	321°52.0	•• 01.6	18°19.9	•• 34.6			
16	12°21.0	88°36.8	16.8	78°32.8	30.0	336°54.2	01.7	33°22.1	34.4			
17	27°23.4	103°36.0	16.6	93°33.3	29.7	351°56.4	01.8	48°24.3	34.3			
18	42°25.9	118°35.1	S22°16.5	108°33.7	S22°29.4	6°58.6	N13°01.9	63°26.5	S10°34.2			
19	57°28.4	133°34.2	16.3	123°34.1	29.2	22°00.9	02.0	78°28.6	34.1			
20	72°30.8	148°33.4	16.1	138°34.5	28.9	37°03.1	02.1	93°30.8	34.0			
21	87°33.3	163°32.5	•• 16.0	153°34.9	•• 28.6	52°05.3	•• 02.2	108°33.0	•• 33.9			
22	102°35.7	178°31.6	15.8	168°35.4	28.3	67°07.5	02.3	123°35.2	33.8			
23	117°38.2	193°30.8	15.6	183°35.8	28.0	82°09.7	02.4	138°37.4	33.7			
Mer.pass. 15:11		ν -0.9' d -0.2' m -3.91		ν 0.4' d -0.3' m 1.32		ν 2.2' d 0.1' m -2.34		ν 2.2' d -0.1' m 0.99				

Jan 31 Wed		SHA	Mer.pass
Venus		79°49.3	10:02
Mars		68°22.6	10:47
Jupiter		324°49.1	17:39
Saturn		21°18.4	13:54

Feb 01 Thu		SHA	Mer.pass
Venus		78°29.2	10:04
Mars		67°33.6	10:47
Jupiter		324°43.3	17:36
Saturn		21°12.0	13:50

Feb 02 Fri		SHA	Mer.pass
Venus		77°09.2	10:05
Mars		66°44.5	10:46
Jupiter		324°37.3	17:32
Saturn		21°05.5	13:47

Horizontal parallax		
Venus:		0.1
Mars:		0.1

DUT1 = UT1-UTC = +0.0049 sec ΔT = TT-UT1 = +69.1791 sec

2024 January 31 to Feb. 02 UT

h	Sun			Moon			
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	176°41.2	S17°35.0	302°10.7	17.3'	S02°17.5	14.0'	54.2'
1	191°41.1	34.3	316°47.1	17.3'	02°31.6	14.0'	54.2'
2	206°41.0	33.6	331°23.4	17.3'	02°45.6	14.0'	54.2'
3	221°40.9	· · 32.9	345°59.6	17.3'	02°59.6	14.0'	54.2'
4	236°40.8	32.2	0°35.9	17.2'	03°13.6	14.0'	54.2'
5	251°40.7	31.5	15°12.1	17.2'	03°27.6	14.0'	54.2'
6	266°40.6	S17°30.8	29°48.4	17.2'	S03°41.6	14.0'	54.2'
7	281°40.5	30.1	44°24.6	17.2'	03°55.6	14.0'	54.2'
8	296°40.4	29.5	59°00.7	17.1'	04°09.5	14.0'	54.2'
9	311°40.3	· · 28.8	73°36.9	17.1'	04°23.5	14.0'	54.3'
10	326°40.2	28.1	88°13.0	17.1'	04°37.4	13.9'	54.3'
11	341°40.1	27.4	102°49.1	17.1'	04°51.4	13.9'	54.3'
12	356°40.1	S17°26.7	117°25.2	17.0'	S05°05.3	13.9'	54.3'
13	11°40.0	26.0	132°01.2	17.0'	05°19.2	13.9'	54.3'
14	26°39.9	25.3	146°37.2	17.0'	05°33.1	13.9'	54.3'
15	41°39.8	· · 24.6	161°13.2	16.9'	05°47.0	13.9'	54.3'
16	56°39.7	23.9	175°49.1	16.9'	06°00.9	13.9'	54.3'
17	71°39.6	23.2	190°25.0	16.9'	06°14.8	13.8'	54.4'
18	86°39.5	S17°22.5	205°00.9	16.8'	S06°28.6	13.8'	54.4'
19	101°39.4	21.8	219°36.8	16.8'	06°42.4	13.8'	54.4'
20	116°39.3	21.1	234°12.6	16.8'	06°56.2	13.8'	54.4'
21	131°39.2	· · 20.4	248°48.3	16.7'	07°10.0	13.8'	54.4'
22	146°39.1	19.7	263°24.0	16.7'	07°23.8	13.8'	54.4'
23	161°39.0	19.0	277°59.7	16.6'	07°37.6	13.7'	54.4'
SD = 16.2' d = -0.7'			SD = 14.8'				

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	176°38.9	S17°18.3	292°35.4	16.6'	S07°51.3	13.7'	54.5'
1	191°38.9	17.6	307°11.0	16.6'	08°05.0	13.7'	54.5'
2	206°38.8	16.9	321°46.5	16.5'	08°18.7	13.7'	54.5'
3	221°38.7	· · 16.2	336°22.0	16.5'	08°32.4	13.6'	54.5'
4	236°38.6	15.5	350°57.5	16.4'	08°46.0	13.6'	54.5'
5	251°38.5	14.8	5°32.9	16.4'	08°59.6	13.6'	54.5'
6	266°38.4	S17°14.1	20°08.2	16.3'	S09°13.2	13.6'	54.6'
7	281°38.3	13.4	34°43.5	16.3'	09°26.8	13.5'	54.6'
8	296°38.2	12.7	49°18.8	16.2'	09°40.3	13.5'	54.6'
9	311°38.1	· · 12.0	63°54.0	16.1'	09°53.9	13.5'	54.6'
10	326°38.1	11.2	78°29.2	16.1'	10°07.3	13.5'	54.6'
11	341°38.0	10.5	93°04.3	16.0'	10°20.8	13.4'	54.7'
12	356°37.9	S17°09.8	107°39.3	16.0'	S10°34.2	13.4'	54.7'
13	11°37.8	09.1	122°14.3	15.9'	10°47.6	13.4'	54.7'
14	26°37.7	08.4	136°49.2	15.9'	11°01.0	13.3'	54.7'
15	41°37.6	· · 07.7	151°24.1	15.8'	11°14.3	13.3'	54.7'
16	56°37.5	07.0	165°58.9	15.7'	11°27.6	13.3'	54.8'
17	71°37.5	06.3	180°33.6	15.7'	11°40.9	13.2'	54.8'
18	86°37.4	S17°05.6	195°08.3	15.6'	S11°54.1	13.2'	54.8'
19	101°37.3	04.9	209°42.9	15.5'	12°07.3	13.2'	54.8'
20	116°37.2	04.1	224°17.4	15.5'	12°20.5	13.1'	54.8'
21	131°37.1	· · 03.4	238°51.9	15.4'	12°33.6	13.1'	54.9'
22	146°37.0	02.7	253°26.3	15.3'	12°46.7	13.0'	54.9'
23	161°37.0	02.0	268°00.6	15.3'	12°59.7	13.0'	54.9'
SD = 16.2' d = -0.7'			SD = 14.9'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	176°36.9	S17°01.3	282°34.9	15.2'	S13°12.7	13.0'	54.9'
1	191°36.8	17°00.6	297°09.1	15.1'	13°25.7	12.9'	54.9'
2	206°36.7	16°59.9	311°43.2	15.0'	13°38.6	12.9'	55.0'
3	221°36.6	· · 59.1	326°17.2	15.0'	13°51.5	12.8'	55.0'
4	236°36.5	58.4	340°51.2	14.9'	14°04.3	12.8'	55.0'
5	251°36.5	57.7	355°25.1	14.8'	14°17.1	12.7'	55.0'
6	266°36.4	S16°57.0	9°58.9	14.7'	S14°29.9	12.7'	55.1'
7	281°36.3	56.3	24°32.6	14.7'	14°42.6	12.6'	55.1'
8	296°36.2	55.6	39°06.3	14.6'	14°55.2	12.6'	55.1'
9	311°36.1	· · 54.8	53°39.8	14.5'	15°07.8	12.5'	55.1'
10	326°36.1	54.1	68°13.3	14.4'	15°20.4	12.5'	55.2'
11	341°36.0	53.4	82°46.7	14.3'	15°32.9	12.4'	55.2'
12	356°35.9	S16°52.7	97°20.0	14.2'	S15°45.3	12.4'	55.2'
13	11°35.8	52.0	111°53.3	14.1'	15°57.7	12.3'	55.2'
14	26°35.8	51.2	126°26.4	14.1'	16°10.0	12.3'	55.3'
15	41°35.7	· · 50.5	140°59.5	14.0'	16°22.3	12.2'	55.3'
16	56°35.6	49.8	155°32.4	13.9'	16°34.5	12.2'	55.3'
17	71°35.5	49.1	170°05.3	13.8'	16°46.7	12.1'	55.4'
18	86°35.5	S16°48.3	184°38.1	13.7'	S16°58.8	12.0'	55.4'
19	101°35.4	47.6	199°10.8	13.6'	17°10.9	12.0'	55.4'
20	116°35.3	46.9	213°43.4	13.5'	17°22.9	11.9'	55.4'
21	131°35.2	· · 46.2	228°15.9	13.4'	17°34.8	11.9'	55.5'
22	146°35.2	45.4	242°48.3	13.3'	17°46.6	11.8'	55.5'
23	161°35.1	44.7	257°20.6	13.2'	17°58.4	11.7'	55.5'
SD = 16.2' d = -0.7'			SD = 15.0'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:11	08:41	10:31	13:57	15:48	17:17
N 70°	07:03	08:22	09:48	14:40	16:06	17:25
68°	06:56	08:07	09:19	15:09	16:21	17:32
66°	06:51	07:55	08:58	15:30	16:34	17:38
64°	06:45	07:44	08:41	15:47	16:44	17:43
62°	06:41	07:35	08:27	16:01	16:53	17:47
60°	06:37	07:28	08:15	16:13	17:00	17:51
N 58°	06:33	07:21	08:05	16:23	17:07	17:55
56°	06:30	07:15	07:56	16:32	17:13	17:58
54°	06:26	07:09	07:48	16:40	17:19	18:01
52°	06:23	07:04	07:41	16:47	17:24	18:04
50°	06:21	06:59	07:34	16:53	17:28	18:07
45°	06:14	06:49	07:21	17:07	17:38	18:14
N 40°	06:08	06:40	07:09	17:18	17:47	18:19
35°	06:02	06:33	06:59	17:28	17:55	18:25
30°	05:57	06:26	06:51	17:37	18:02	18:30
20°	05:46	06:13	06:36	17:52	18:15	18:41
N 10°	05:35	06:01	06:22	18:05	18:27	18:52
0°	05:23	05:48	06:10	18:17	18:39	19:04
S 10°	05:09	05:35	05:57	18:29	18:52	19:17
20°	04:53	05:20	05:44	18:43	19:06	19:34
30°	04:31	05:02	05:28	18:58	19:24	19:55
35°	04:18	04:51	05:19	19:08	19:35	20:08
40°	04:01	04:38	05:08	19:18	19:48	20:25
45°	03:41	04:23	04:56	19:30	20:03	20:45
S 50°	03:13	04:03	04:41	19:45	20:23	21:12
52°	02:59	03:53	04:34	19:52	20:33	21:26
54°	02:42	03:42	04:26	20:00	20:43	21:43
56°	02:20	03:29	04:17	20:09	20:56	22:04
58°	01:52	03:14	04:07	20:19	21:10	22:31
S 60°	01:05	02:57	03:55	20:30	21:28	23:13

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	· · ·	00:12	02:32	09:09	08:34	07:42
N 70°	· · ·	00:00	02:02	09:15	08:49	08:13
68°	23:50	· · ·	01:41	09:20	09:01	08:36
66°	23:42	· · ·	01:24	09:25	09:11	08:55
64°	23:35	· · ·	01:10	09:28	09:19	09:09
62°	23:29	· · ·	00:59	09:31	09:27	09:22
60°	23:24	· · ·	00:49	09:34	09:33	09:33
N 58°	23:19	· · ·	00:41	09:37	09:39	09:42
56°	23:15	· · ·	00:34	09:39	09:44	09:50
54°	23:12	· · ·	00:27	09:41	09:48	09:58
52°	23:09	· · ·	00:21	09:43	09:53	10:04
50°	23:06	· · ·	00:16	09:44	09:56	10:10
45°	22:59	· · ·	00:05	09:48	10:04	10:23
N 40°	22:54	23:55	· · ·	09:51	10:11	10:34
35°	22:50	23:48	· · ·	09:54	10:17	10:43
30°	22:46	23:41	· · ·	09:56	10:22	10:51
20°	22:39	23:29	· · ·	10:00	10:31	11:05
N 10°	22:33	23:18	· · ·	10:04	10:39	11:18
0°	22:28	23:09	23:53	10:07	10:47	11:29
S 10°	22:22	22:59	23:39	10:10	10:55	11:41
20°	22:16	22:49	23:25	10:14	11:03	11:54
30°	22:10	22:37	23:08	10:18	11:12	12:08
35°	22:06	22:31	22:59	10:20	11:17	12:16
40°	22:02	22:23	22:48	10:23	11:23	12:26
45°	21:57	22:15	22:35	10:26	11:31	12:37
S 50°	21:51	22:04	22:20	10:30	11:39	12:51
52°	21:49	21:59	22:13	10:32	11:43	12:57
54°	21:46	21:54	22:05	10:34	11:48	13:05
56°	21:43	21:48	21:56	10:36	11:53	13:13
58°	21:39	21:42	21:46	10:38	11:58	13:22
S 60°	21:35	21:35	21:35	10:41	12:04	13:32

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12h		Mer. Pass.		
	00h			Upper	Lower	20-22
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	77-60%
31	13:15	13:20	12:13	03:58	16:17	
01	13:24	13:28	12:13	04:37	16:58	
02	13:33	13:36	12:14	05:19	17:41	

February 03, 04, 05 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	132°40.7	208°29.9	S22°15.4	198°36.2	S22°27.7	97°11.9	N13°02.5	153°39.6	S10°33.6	Alpheratz	357°35.8	29°13.4
1	147°43.1	223°29.0	15.3	213°36.6	27.4	112°14.1	02.6	168°41.8	33.5	Ankaa	353°08.1	-42°10.8
2	162°45.6	238°28.2	15.1	228°37.1	27.1	127°16.3	02.7	183°44.0	33.4	Schedar	349°32.3	56°40.3
3	177°48.1	253°27.3	•• 14.9	243°37.5	•• 26.8	142°18.5	•• 02.8	198°46.2	•• 33.3	Diphda	348°48.2	-17°51.5
4	192°50.5	268°26.4	14.7	258°37.9	26.6	157°20.7	02.9	213°48.4	33.2	Achernar	335°20.9	-57°07.2
5	207°53.0	283°25.6	14.5	273°38.3	26.3	172°22.9	03.0	228°50.6	33.0	Hamal	327°52.1	23°34.6
6	222°55.5	298°24.7	S22°14.3	288°38.8	S22°26.0	187°25.1	N13°03.1	243°52.8	S10°32.9	Polaris	314°22.8	89°22.3
7	237°57.9	313°23.8	14.2	303°39.2	25.7	202°27.3	03.2	258°54.9	32.8	Acamar	315°12.3	-40°12.8
8	253°00.4	328°22.9	14.0	318°39.6	25.4	217°29.5	03.3	273°57.1	32.7	Menkar	314°06.9	4°11.0
9	268°02.8	343°22.1	•• 13.8	333°40.0	•• 25.1	232°31.7	•• 03.4	288°59.3	•• 32.6	Mirfak	308°29.3	49°57.0
10	283°05.3	358°21.2	13.6	348°40.5	24.8	247°33.9	03.5	304°01.5	32.5	Aldebaran	290°40.3	16°33.5
11	298°07.8	13°20.4	13.4	3°40.9	24.5	262°36.1	03.6	319°03.7	32.4	Rigel	281°04.4	-8°10.5
12	313°10.2	28°19.5	S22°13.2	18°41.3	S22°24.2	277°38.3	N13°03.7	334°05.9	S10°32.3	Capella	280°22.7	46°01.4
13	328°12.7	43°18.6	13.0	33°41.7	23.9	292°40.5	03.8	349°08.1	32.2	Bellatrix	278°23.5	6°22.2
14	343°15.2	58°17.8	12.8	48°42.2	23.6	307°42.7	03.9	4°10.3	32.1	Elnath	278°02.6	28°37.7
15	358°17.6	73°16.9	•• 12.6	63°42.6	•• 23.3	322°44.9	•• 04.0	19°12.5	•• 32.0	Alnilam	275°38.3	-1°11.3
16	13°20.1	88°16.0	12.4	78°43.0	23.0	337°47.1	04.1	34°14.7	31.9	Betelgeuse	270°52.7	7°24.7
17	28°22.6	103°15.2	12.2	93°43.4	22.8	352°49.3	04.2	49°16.8	31.8	Canopus	263°52.3	-52°42.6
18	43°25.0	118°14.3	S22°12.0	108°43.9	S22°22.5	7°51.5	N13°04.3	64°19.0	S10°31.6	Sirius	258°26.6	-16°45.0
19	58°27.5	133°13.4	11.9	123°44.3	22.2	22°53.7	04.4	79°21.2	31.5	Adhara	255°06.1	-29°00.4
20	73°30.0	148°12.6	11.7	138°44.7	21.9	37°55.9	04.5	94°23.4	31.4	Procyon	244°51.3	5°09.7
21	88°32.4	163°11.7	•• 11.4	153°45.1	•• 21.6	52°58.1	•• 04.6	109°25.6	•• 31.3	Pollux	243°17.9	27°58.1
22	103°34.9	178°10.8	11.2	168°45.6	21.3	68°00.3	04.7	124°27.8	31.2	Avior	234°14.4	-59°35.2
23	118°37.3	193°10.0	11.0	183°46.0	21.0	83°02.5	04.8	139°30.0	31.1	Suhail	222°46.4	-43°31.8
Mer.pass. 15:07		ν -0.9' d -0.2' m -3.91		ν 0.4' d -0.3' m 1.32		ν 2.2' d 0.1' m -2.34		ν 2.2' d -0.1' m 0.99				

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	133°39.8	208°09.1	S22°10.8	198°46.4	S22°20.7	98°04.7	N13°04.9	154°32.2	S10°31.0
1	148°42.3	223°08.2	10.6	213°46.8	20.4	113°06.9	05.0	169°34.4	30.9
2	163°44.7	238°07.4	10.4	228°47.8	20.1	128°09.1	05.1	184°36.6	30.8
3	178°47.2	253°06.5	•• 10.2	243°47.7	•• 19.8	143°11.3	•• 05.2	199°38.8	•• 30.7
4	193°49.7	268°05.6	10.0	258°48.1	19.5	158°13.5	05.3	214°40.9	30.6
5	208°52.1	283°04.8	09.8	273°48.5	19.2	173°15.7	05.4	229°43.1	30.5
6	223°54.6	298°03.9	S22°09.6	288°49.0	S22°18.9	188°17.9	N13°05.5	244°45.3	S10°30.3
7	238°57.1	313°03.0	09.4	303°49.4	18.6	203°20.1	05.6	259°47.5	30.2
8	253°59.5	328°02.2	09.2	318°49.8	18.3	218°22.3	05.7	274°49.7	30.1
9	269°02.0	343°01.3	•• 08.9	333°50.2	•• 18.0	233°24.5	•• 05.8	289°51.9	•• 30.0
10	284°04.5	358°00.5	08.7	348°50.7	17.7	248°26.7	05.9	304°54.1	29.9
11	299°06.9	12°59.6	08.5	3°51.1	17.4	263°28.9	06.0	319°56.3	29.8
12	314°09.4	27°58.7	S22°08.3	18°51.5	S22°17.1	278°31.1	N13°06.1	334°58.5	S10°29.7
13	329°11.8	42°57.9	08.1	33°52.0	16.8	293°33.3	06.2	350°00.6	29.6
14	344°14.3	57°57.0	07.9	48°52.4	16.5	308°35.5	06.3	5°02.8	29.5
15	359°16.8	72°56.1	•• 07.6	63°52.8	•• 16.1	323°37.7	•• 06.4	20°05.0	•• 29.4
16	14°19.2	87°55.3	07.4	78°53.2	15.8	338°39.8	06.5	35°07.2	29.3
17	29°21.7	102°54.4	07.2	93°53.7	15.5	353°42.0	06.6	50°09.4	29.2
18	44°24.2	117°53.6	S22°07.0	108°54.1	S22°15.2	8°44.2	N13°06.7	65°11.6	S10°29.0
19	59°26.6	132°52.7	06.7	123°54.5	14.9	23°46.4	06.8	80°13.8	28.9
20	74°29.1	147°51.8	06.5	138°54.9	14.6	38°48.6	06.9	95°16.0	28.8
21	89°31.6	162°51.0	•• 06.3	153°55.4	•• 14.3	53°50.8	•• 07.0	110°18.2	•• 28.7
22	104°34.0	177°50.1	06.0	168°55.8	14.0	68°53.0	07.1	125°20.4	28.6
23	119°36.5	192°49.2	05.8	183°56.2	13.7	83°55.2	07.2	140°22.5	28.5
Mer.pass. 15:03		ν -0.9' d -0.2' m -3.91		ν 0.4' d -0.3' m 1.32		ν 2.2' d 0.1' m -2.33		ν 2.2' d -0.1' m 0.99	

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	134°38.9	207°48.4	S22°05.6	198°56.7	S22°13.4	98°57.4	N13°07.4	155°24.7	S10°28.4
1	149°41.4	222°47.5	05.3	213°57.1	13.1	113°59.6	07.5	170°26.9	28.3
2	164°43.9	237°46.7	05.1	228°57.5	12.8	129°01.8	07.6	185°29.1	28.2
3	179°46.3	252°45.8	•• 04.9	243°57.9	•• 12.5	144°04.0	•• 07.7	200°31.3	•• 28.1
4	194°48.8	267°44.9	04.6	258°58.4	12.1	159°06.1	07.8	215°33.5	28.0
5	209°51.3	282°44.1	04.4	273°58.8	11.8	174°08.3	07.9	230°35.7	27.9
6	224°53.7	297°43.2	S22°04.2	288°59.2	S22°11.5	189°10.5	N13°08.0	245°37.9	S10°27.7
7	239°56.2	312°42.4	03.9	303°59.7	11.2	204°12.7	08.1	260°40.1	27.6
8	254°58.7	327°41.5	03.7	319°00.1	10.9	219°14.9	08.2	275°42.2	27.5
9	270°01.1	342°40.6	•• 03.4	334°00.5	•• 10.6	234°17.1	•• 08.3	290°44.4	•• 27.4
10	285°03.6	357°39.8	03.2	349°00.9	10.3	249°19.3	08.4	305°46.6	27.3
11	300°06.1	12°38.9	03.0	4°01.4	10.0	264°21.5	08.5	320°48.8	27.2
12	315°08.5	27°38.1	S22°02.7	19°01.8	S22°09.6	279°23.7	N13°08.6	335°51.0	S10°27.1
13	330°11.0	42°37.2	02.5	34°02.2	09.3	294°25.9	08.7	350°53.2	27.0
14	345°13.4	57°36.3	02.2	49°02.7	09.0	309°28.0	08.8	5°55.4	26.9
15	0°15.9	72°35.5	•• 02.0	64°03.1	•• 08.7	324°30.2	•• 08.9	20°57.6	•• 26.8
16	15°18.4	87°34.6	01.7	79°03.5	08.4	339°32.4	09.0	35°59.7	26.7
17	30°20.8	102°33.8	01.5	94°03.9	08.1	354°34.6	09.1	51°01.9	26.5
18	45°23.3	117°32.9	S22°01.2	109°04.4	S22°07.8	9°36.8	N13°09.2	66°04.1	S10°26.4
19	60°25.8	132°32.0	01.0	124°04.8	07.4	24°39.0	09.3	81°06.3	26.3
20	75°28.2	147°31.2	00.7	139°05.2	07.1	39°41.2	09.4	96°08.5	26.2
21	90°30.7	162°30.3	•• 00.4	154°05.7	•• 06.8	54°43.3	•• 09.5	111°10.7	•• 26.1
22	105°33.2	177°29.5	22°00.2	169°06.1	06.5	69°45.5	09.6	126°12.9	26.0
23	120°35.6	192°28.6	21°59.9	184°06.5	06.2	84°47.7	09.7	141°15.1	25.9
Mer.pass. 14:59		ν -0.9' d -0.2' m -3.91		ν 0.4' d -0.3' m 1.32		ν 2.2' d 0.1' m -2.32		ν 2.2' d -0.1' m 0.99	

Feb 03 Sat	SHA	Mer.pass
Venus	75°49.2	10:07
Mars	65°55.5	10:45
Jupiter	324°31.2	17:29
Saturn	20°58.9	13:43

Feb 04 Sun	SHA	Mer.pass
Venus	74°29.3	10:08
Mars	65°06.6	10:45
Jupiter	324°24.9	17:25
Saturn	20°52.4	13:40

Feb 05 Mon	SHA	Mer.pass
Venus	73°09.4	10:09
Mars	64°17.7	10:44
Jupiter	324°18.4	17:22
Saturn	20°45.8	13:36

Horizontal parallax	
Venus:	0.1
Mars:	0.1

Sun				Moon				
Sat	GHA	Dec		GHA	ν	Dec	d	HP
0	176°35.0	S16°44.0		271°52.8	13.1'	S18°10.2	11.7'	55.5'
1	191°34.9		43.3	286°24.9	13.0'	18°21.8	11.6'	55.6'
2	206°34.9		42.5	300°56.9	12.9'	18°33.4	11.5'	55.6'
3	221°34.8	· ·	41.8	315°28.8	12.8'	18°45.0	11.5'	55.6'
4	236°34.7		41.1	330°00.6	12.7'	18°56.4	11.4'	55.7'
5	251°34.6		40.4	344°32.3	12.6'	19°07.8	11.3'	55.7'
6	266°34.6	S16°39.6		359°03.9	12.5'	S19°19.1	11.2'	55.7'
7	281°34.5		38.9	13°35.4	12.4'	19°30.4	11.2'	55.8'
8	296°34.4		38.2	28°06.7	12.3'	19°41.5	11.1'	55.8'
9	311°34.4	· ·	37.4	42°38.0	12.2'	19°52.6	11.0'	55.8'
10	326°34.3		36.7	57°09.2	12.1'	20°03.6	10.9'	55.9'
11	341°34.2		36.0	71°40.2	11.9'	20°14.5	10.8'	55.9'
12	356°34.1	S16°35.2		86°11.2	11.8'	S20°25.4	10.8'	55.9'
13	11°34.1		34.5	100°42.0	11.7'	20°36.2	10.7'	55.9'
14	26°34.0		33.8	115°12.7	11.6'	20°46.8	10.6'	56.0'
15	41°33.9	· ·	33.0	129°43.4	11.5'	20°57.4	10.5'	56.0'
16	56°33.9		32.3	144°13.9	11.4'	21°07.9	10.4'	56.0'
17	71°33.8		31.6	158°44.2	11.3'	21°18.4	10.3'	56.1'
18	86°33.7	S16°30.8		173°14.5	11.2'	S21°28.7	10.2'	56.1'
19	101°33.7		30.1	187°44.7	11.0'	21°38.9	10.2'	56.1'
20	116°33.6		29.4	202°14.7	10.9'	21°49.1	10.1'	56.2'
21	131°33.5	· ·	28.6	216°44.6	10.8'	21°59.2	10.0'	56.2'
22	146°33.5		27.9	231°14.4	10.7'	22°09.1	9.9'	56.2'
23	161°33.4		27.1	245°44.1	10.6'	22°19.0	9.8'	56.3'
SD = 16.2' d = -0.7'				SD = 15.1'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP	
0	176°33.3	S16°26.4	260°13.7	10.4'	S22°28.7	9.7'	56.3'	
1	191°33.3		274°43.1	10.3'	22°38.4	9.6'	56.4'	
2	206°33.2		289°12.5	10.2'	22°48.0	9.5'	56.4'	
3	221°33.1	· ·	24.2	303°41.7	10.1'	22°57.4	9.4'	56.4'
4	236°33.1		23.4	318°10.8	10.0'	23°06.8	9.3'	56.5'
5	251°33.0		22.7	332°39.7	9.8'	23°16.1	9.1'	56.5'
6	266°32.9	S16°22.0		347°08.6	9.7'	S23°25.2	9.0'	56.5'
7	281°32.9		21.2	1°37.3	9.6'	23°34.3	8.9'	56.6'
8	296°32.8		20.5	16°05.9	9.5'	23°43.2	8.8'	56.6'
9	311°32.8	· ·	19.7	30°34.4	9.4'	23°52.0	8.7'	56.6'
10	326°32.7		19.0	45°02.7	9.2'	24°00.7	8.6'	56.7'
11	341°32.6		18.2	59°31.0	9.1'	24°09.3	8.5'	56.7'
12	356°32.6	S16°17.5		73°59.1	9.0'	S24°17.8	8.4'	56.7'
13	11°32.5		16.8	88°27.1	8.9'	24°26.1	8.2'	56.8'
14	26°32.5		16.0	102°55.0	8.7'	24°34.4	8.1'	56.8'
15	41°32.4	· ·	15.3	117°22.7	8.6'	24°42.5	8.0'	56.9'
16	56°32.3		14.5	131°50.3	8.5'	24°50.5	7.9'	56.9'
17	71°32.3		13.8	146°17.8	8.4'	24°58.4	7.7'	56.9'
18	86°32.2	S16°13.0		160°45.2	8.3'	S25°06.1	7.6'	57.0'
19	101°32.2		12.3	175°12.5	8.1'	25°13.7	7.5'	57.0'
20	116°32.1		11.5	189°39.6	8.0'	25°21.2	7.4'	57.1'
21	131°32.0	· ·	10.8	204°06.6	7.9'	25°28.6	7.2'	57.1'
22	146°32.0		10.0	218°33.5	7.8'	25°35.8	7.1'	57.1'
23	161°31.9		09.3	233°00.3	7.6'	25°42.9	7.0'	57.2'
SD = 16.2' d = -0.7'				SD = 15.4'				

Mon	GHA	Dec	GHA	ν	Dec	d	HP	
0	176°31.9	S16°08.5	247°26.9	7.5'	S25°49.8	6.8'	57.2'	
1	191°31.8		261°53.4	7.4'	25°56.6	6.7'	57.2'	
2	206°31.8		276°19.8	7.3'	26°03.3	6.5'	57.3'	
3	221°31.7	· ·	06.3	290°46.1	7.2'	26°09.9	6.4'	57.3'
4	236°31.6		05.5	305°12.3	7.0'	26°16.2	6.3'	57.4'
5	251°31.6		04.8	319°38.3	6.9'	26°22.5	6.1'	57.4'
6	266°31.5	S16°04.0		334°04.3	6.8'	S26°28.6	6.0'	57.4'
7	281°31.5		03.3	348°30.1	6.7'	26°34.6	5.8'	57.5'
8	296°31.4		02.5	2°55.8	6.6'	26°40.4	5.7'	57.5'
9	311°31.4	· ·	01.7	17°21.4	6.5'	26°46.0	5.5'	57.6'
10	326°31.3		01.0	31°46.8	6.4'	26°51.5	5.4'	57.6'
11	341°31.3		16°00.2	46°12.2	6.2'	26°56.9	5.2'	57.6'
12	356°31.2	S15°59.5		60°37.4	6.1'	S27°02.1	5.0'	57.7'
13	11°31.1		58.7	75°02.6	6.0'	27°07.1	4.9'	57.7'
14	26°31.1		58.0	89°27.6	5.9'	27°12.0	4.7'	57.8'
15	41°31.0	· ·	57.2	103°52.5	5.8'	27°16.8	4.6'	57.8'
16	56°31.0		56.4	118°17.3	5.7'	27°21.3	4.4'	57.8'
17	71°30.9		55.7	132°42.0	5.6'	27°25.7	4.2'	57.9'
18	86°30.9	S15°54.9		147°06.6	5.5'	S27°30.0	4.1'	57.9'
19	101°30.8		54.2	161°31.1	5.4'	27°34.0	3.9'	58.0'
20	116°30.8		53.4	175°55.5	5.3'	27°38.0	3.7'	58.0'
21	131°30.7	· ·	52.6	190°19.8	5.2'	27°41.7	3.6'	58.0'
22	146°30.7		51.9	204°44.0	5.1'	27°45.3	3.4'	58.1'
23	161°30.6		51.1	219°08.1	5.0'	27°48.7	3.2'	58.1'
SD = 16.2' d = -0.8'				SD = 15.6'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:00	08:27	10:08	14:21	16:02	17:29
N 70°	06:54	08:10	09:32	14:57	16:19	17:36
68°	06:48	07:57	09:07	15:22	16:32	17:41
66°	06:43	07:46	08:47	15:41	16:43	17:46
64°	06:39	07:36	08:32	15:57	16:52	17:50
62°	06:35	07:28	08:19	16:10	17:00	17:54
60°	06:31	07:21	08:08	16:21	17:07	17:58
N 58°	06:28	07:15	07:58	16:30	17:13	18:01
56°	06:25	07:09	07:50	16:38	17:19	18:04
54°	06:22	07:04	07:43	16:46	17:24	18:07
52°	06:19	07:00	07:36	16:52	17:29	18:09
50°	06:17	06:55	07:30	16:58	17:33	18:12
45°	06:11	06:46	07:17	17:11	17:42	18:17
N 40°	06:05	06:38	07:06	17:22	17:50	18:23
35°	06:00	06:30	06:57	17:31	17:58	18:28
30°	05:55	06:24	06:49	17:39	18:04	18:33
20°	05:45	06:12	06:35	17:53	18:16	18:42
N 10°	05:35	06:00	06:22	18:06	18:27	18:53
0°	05:24	05:49	06:10	18:17	18:39	19:04
S 10°	05:11	05:36	05:58	18:29	18:51	19:17
20°	04:55	05:22	05:46	18:42	19:05	19:32
30°	04:34	05:05	05:31	18:57	19:22	19:53
35°	04:22	04:55	05:22	19:05	19:32	20:05
40°	04:06	04:42	05:12	19:15	19:45	20:21
45°	03:46	04:27	05:00	19:27	19:59	20:40
S 50°	03:20	04:08	04:46	19:41	20:18	21:06
52°	03:07	03:59	04:39	19:47	20:27	21:19
54°	02:51	03:49	04:32	19:55	20:37	21:34
56°	02:32	03:37	04:23	20:03	20:49	21:53
58°	02:07	03:24	04:14	20:12	21:02	22:17
S 60°	01:31	03:07	04:03	20:23	21:18	22:51

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	■	■	■	■	■	■
N 70°	04:55	■	■	06:54	■	■
68°	03:54	■	■	07:56	■	■
66°	03:20	06:04	■	08:31	07:29	■
64°	02:56	04:59	■	08:57	08:35	■
62°	02:36	04:24	06:28	09:17	09:10	09:00
60°	02:21	03:59	05:43	09:33	09:36	09:45
N 58°	02:08	03:40	05:14	09:47	09:56	10:15
56°	01:56	03:23	04:52	09:59	10:13	10:37
54°	01:46	03:09	04:33	10:10	10:28	10:56
52°	01:38	02:57	04:18	10:19	10:40	11:12
50°	01:30	02:47	04:05	10:28	10:52	11:26
45°	01:13	02:25	03:37	10:46	11:15	11:54
N 40°	01:00	02:07	03:16	11:01	11:34	12:16
35°	00:48	01:52	02:58	11:13	11:49	12:34
30°	00:38	01:39	02:43	11:24	12:03	12:50
20°	00:21	01:17	02:17	11:43	12:26	13:17
N 10°	00:06	00:58	01:55	12:00	12:47	13:40
0°	· · ·	00:41	01:34	12:15	13:06	14:02
S 10°	· · ·	00:24	01:14	12:31	13:25	14:23
20°	· · ·	00:05	00:52	12:48	13:45	14:47
30°	23:44	· · ·	00:27	13:07	14:09	15:14
35°	23:32	· · ·	00:12	13:18	14:23	15:30
40°	23:18	23:55	· · ·	13:31	14:40	15:48
45°	23:01	23:35	· · ·	13:47	14:59	16:11
S 50°	22:40	23:09	23:51	14:06	15:24	16:40
52°	22:31	22:57	23:36	14:15	15:36	16:54
54°	22:20	22:43	23:19	14:26	15:49	17:11
56°	22:08	22:27	22:59	14:37	16:05	17:30
58°	21:54	22:07	22:35	14:51	16:24	17:55
S 60°	21:37	21:43	22:01	15:06	16:47	18:28

Day	Sun			Moon		
	Eqn.of Time		Mer. Pass	Mer.Pass.		Age
	00 ^h			Upper	Lower	
	mm:ss	12 ^h mm:ss		hh:mm	hh:mm	
03	13:40	13:43	12:14	06:04	18:28	
04	13:47	13:50	12:14	06:53	19:20	
05	13:53	13:55	12:14	07:48	20:17	

February 06, 07, 08 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	135°38.1	207°27.8	S21°59.7	199°06.9	S22°05.9	99°49.9	N13°09.8	156°17.3	S10°25.8	Alpheratz	357°35.8 29°13.4
1	150°40.6	222°26.9	59.4	214°07.4	05.5	114°52.1	09.9	171°19.4	25.7	Ankaa	353°08.1 -42°10.8
2	165°43.0	237°26.0	59.1	229°07.8	05.2	129°54.3	10.0	186°21.6	25.6	Schedar	349°32.3 56°40.3
3	180°45.5	252°25.2	•• 58.9	244°08.2	•• 04.9	144°56.5	•• 10.1	201°23.8	•• 25.5	Diphda	348°48.2 -17°51.5
4	195°47.9	267°24.3	58.6	259°08.7	04.6	159°58.6	10.3	216°26.0	25.3	Achernar	335°20.9 -57°07.2
5	210°50.4	282°23.5	58.3	274°09.1	04.3	175°00.8	10.4	231°28.2	25.2	Hamal	327°52.1 23°34.6
6	225°52.9	297°22.6	S21°58.1	289°09.5	S22°03.9	190°03.0	N13°10.5	246°30.4	S10°25.1	Polaris	314°24.0 89°22.3
7	240°55.3	312°21.8	57.8	304°10.0	03.6	205°05.2	10.6	261°32.6	25.0	Acamar	315°12.3 -40°12.8
8	255°57.8	327°20.9	57.5	319°10.4	03.3	220°07.4	10.7	276°34.8	24.9	Menkar	314°06.9 4°11.0
9	271°00.3	342°20.1	•• 57.3	334°10.8	•• 03.0	235°09.6	•• 10.8	291°36.9	•• 24.8	Mirfak	308°29.3 49°57.0
10	286°02.7	357°19.2	57.0	349°11.3	02.6	250°11.7	10.9	306°39.1	24.7	Aldebaran	290°40.3 16°33.5
11	301°05.2	12°18.3	56.7	4°11.7	02.3	265°13.9	11.0	321°41.3	24.6	Rigel	281°04.4 -8°10.5
12	316°07.7	27°17.5	S21°56.5	19°12.1	S22°02.0	280°16.1	N13°11.1	336°43.5	S10°24.5	Capella	280°22.7 46°01.5
13	331°10.1	42°16.6	56.2	34°12.5	01.7	295°18.3	11.2	351°45.7	24.4	Bellatrix	278°23.5 6°22.2
14	346°12.6	57°15.8	55.9	49°13.0	01.3	310°20.5	11.3	6°47.9	24.3	Elnath	278°02.6 28°37.7
15	1°15.1	72°14.9	•• 55.6	64°13.4	•• 01.0	325°22.6	•• 11.4	21°50.1	•• 24.1	Alnilam	275°38.3 -1°11.3
16	16°17.5	87°14.1	55.4	79°13.8	00.7	340°24.8	11.5	36°52.3	24.0	Betelgeuse	270°52.7 7°24.6
17	31°20.0	102°13.2	55.1	94°14.3	00.4	355°27.0	11.6	51°54.4	23.9	Canopus	263°52.3 -52°42.7
18	46°22.4	117°12.4	S21°54.8	109°14.7	S22°00.0	10°29.2	N13°11.7	66°56.6	S10°23.8	Sirius	258°26.6 -16°45.1
19	61°24.9	132°11.5	54.5	124°15.1	21°59.7	25°31.4	11.8	81°58.8	23.7	Adhara	255°06.1 -29°00.4
20	76°27.4	147°10.6	54.2	139°15.6	59.4	40°33.5	11.9	97°01.0	23.6	Procyon	244°51.3 5°09.7
21	91°29.8	162°09.8	•• 53.9	154°16.0	•• 59.1	55°35.7	•• 12.0	112°03.2	•• 23.5	Pollux	243°17.9 27°58.1
22	106°32.3	177°08.9	53.7	169°16.4	58.7	70°37.9	12.1	127°05.4	23.4	Avior	234°14.4 -59°35.2
23	121°34.8	192°08.1	53.4	184°16.9	58.4	85°40.1	12.3	142°07.6	23.3	Suhail	222°46.4 -43°31.8
Mer.pass. 14:55		ν -0.9' d -0.3' m -3.91		ν 0.4' d -0.3' m 1.32		ν 2.2' d 0.1' m -2.32		ν 2.2' d -0.1' m 0.99			

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	136°37.2	207°07.2	S21°53.1	199°17.3	S21°58.1	100°42.3	N13°12.4	157°09.8	S10°23.2
1	151°39.7	222°06.4	52.8	214°17.7	57.8	115°44.4	12.5	172°11.9	23.1
2	166°42.2	237°05.5	52.5	229°18.2	57.4	130°46.6	12.6	187°14.1	22.9
3	181°44.6	252°04.7	•• 52.2	244°18.6	•• 57.1	145°48.8	•• 12.7	202°16.3	•• 22.8
4	196°47.1	267°03.8	51.9	259°19.0	56.8	160°51.0	12.8	217°18.5	22.7
5	211°49.6	282°03.0	51.6	274°19.5	56.4	175°53.1	12.9	232°20.7	22.6
6	226°52.0	297°02.1	S21°51.3	289°19.9	S21°56.1	190°55.3	N13°13.0	247°22.9	S10°22.5
7	241°54.5	312°01.3	51.0	304°20.3	55.8	205°57.5	13.1	262°25.1	22.4
8	256°56.9	327°00.4	50.7	319°20.8	55.4	220°59.7	13.2	277°27.2	22.3
9	271°59.4	341°59.6	•• 50.4	334°21.2	•• 55.1	236°01.8	•• 13.3	292°29.4	•• 22.2
10	287°01.9	356°58.7	50.2	349°21.6	54.8	251°04.0	13.4	307°31.6	22.1
11	302°04.3	11°57.9	49.9	4°22.1	54.4	266°06.2	13.5	322°33.8	22.0
12	317°06.8	26°57.0	S21°49.6	19°22.5	S21°54.1	281°08.4	N13°13.6	337°36.0	S10°21.8
13	332°09.3	41°56.2	49.2	34°22.9	53.8	296°10.5	13.8	352°38.2	21.7
14	347°11.7	56°55.3	48.9	49°23.4	53.4	311°12.7	13.9	7°40.4	21.6
15	2°14.2	71°54.5	•• 48.6	64°23.8	•• 53.1	326°14.9	•• 14.0	22°42.6	•• 21.5
16	17°16.7	86°53.6	48.3	79°24.2	52.8	341°17.1	14.1	37°44.7	21.4
17	32°19.1	101°52.8	48.0	94°24.7	52.4	356°19.2	14.2	52°46.9	21.3
18	47°21.6	116°51.9	S21°47.7	109°25.1	S21°52.1	11°21.4	N13°14.3	67°49.1	S10°21.2
19	62°24.1	131°51.1	47.4	124°25.5	51.8	26°23.6	14.4	82°51.3	21.1
20	77°26.5	146°50.2	47.1	139°26.0	51.4	41°25.8	14.5	97°53.5	21.0
21	92°29.0	161°49.4	•• 46.8	154°26.4	•• 51.1	56°27.9	•• 14.6	112°55.7	•• 20.9
22	107°31.4	176°48.5	46.5	169°26.8	50.7	71°30.1	14.7	127°57.9	20.7
23	122°33.9	191°47.7	46.2	184°27.3	50.4	86°32.3	14.8	143°00.0	20.6
Mer.pass. 14:51		ν -0.9' d -0.3' m -3.90		ν 0.4' d -0.3' m 1.32		ν 2.2' d 0.1' m -2.31		ν 2.2' d -0.1' m 0.99	

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	137°36.4	206°46.8	S21°45.9	199°27.7	S21°50.1	101°34.5	N13°14.9	158°02.2	S10°20.5
1	152°38.8	221°46.0	45.5	214°28.1	49.7	116°36.6	15.0	173°04.4	20.4
2	167°41.3	236°45.1	45.2	229°28.6	49.4	131°38.8	15.2	188°06.6	20.3
3	182°43.8	251°44.3	•• 44.9	244°29.0	•• 49.0	146°41.0	•• 15.3	203°08.8	•• 20.2
4	197°46.2	266°43.4	44.6	259°29.4	48.7	161°43.1	15.4	218°11.0	20.1
5	212°48.7	281°42.6	44.3	274°29.9	48.4	176°45.3	15.5	233°13.2	20.0
6	227°51.2	296°41.7	S21°43.9	289°30.3	S21°48.0	191°47.5	N13°15.6	248°15.3	S10°19.9
7	242°53.6	311°40.9	43.6	304°30.7	47.7	206°49.6	15.7	263°17.5	19.8
8	257°56.1	326°40.1	43.3	319°31.2	47.3	221°51.8	15.8	278°19.7	19.7
9	272°58.6	341°39.2	•• 43.0	334°31.6	•• 47.0	236°54.0	•• 15.9	293°21.9	•• 19.5
10	288°01.0	356°38.4	42.6	349°32.0	46.7	251°56.2	16.0	308°24.1	19.4
11	303°03.5	11°37.5	42.3	4°32.5	46.3	266°58.3	16.1	323°26.3	19.3
12	318°05.9	26°36.7	S21°42.0	19°32.9	S21°46.0	282°00.5	N13°16.2	338°28.5	S10°19.2
13	333°08.4	41°35.8	41.7	34°33.4	45.6	297°02.7	16.4	353°30.6	19.1
14	348°10.9	56°35.0	41.3	49°33.8	45.3	312°04.8	16.5	8°32.8	19.0
15	3°13.3	71°34.1	•• 41.0	64°34.2	•• 44.9	327°07.0	•• 16.6	23°35.0	•• 18.9
16	18°15.8	86°33.3	40.7	79°34.7	44.6	342°09.2	16.7	38°37.2	18.8
17	33°18.3	101°32.4	40.3	94°35.1	44.2	357°11.3	16.8	53°39.4	18.7
18	48°20.7	116°31.6	S21°40.0	109°35.5	S21°43.9	12°13.5	N13°16.9	68°41.6	S10°18.5
19	63°23.2	131°30.8	39.7	124°36.0	43.6	27°15.7	17.0	83°43.8	18.4
20	78°25.7	146°29.9	39.3	139°36.4	43.2	42°17.8	17.1	98°45.9	18.3
21	93°28.1	161°29.1	•• 39.0	154°36.8	•• 42.9	57°20.0	•• 17.2	113°48.1	•• 18.2
22	108°30.6	176°28.2	38.6	169°37.3	42.5	72°22.2	17.3	128°50.3	18.1
23	123°33.0	191°27.4	38.3	184°37.7	42.2	87°24.3	17.4	143°52.5	18.0
Mer.pass. 14:47		ν -0.8' d -0.3' m -3.90		ν 0.4' d -0.3' m 1.31		ν 2.2' d 0.1' m -2.30		ν 2.2' d -0.1' m 0.99	

Feb 06 Tue	SHA	Mer.pass
Venus	71°49.7	10:11
Mars	63°28.9	10:43
Jupiter	324°11.8	17:18
Saturn	20°39.2	13:33

Feb 07 Wed	SHA	Mer.pass
Venus	70°30.0	10:12
Mars	62°40.1	10:43
Jupiter	324°05.0	17:15
Saturn	20°32.5	13:29

Feb 08 Thu	SHA	Mer.pass
Venus	69°10.5	10:13
Mars	61°51.3	10:42
Jupiter	323°58.1	17:11
Saturn	20°25.9	13:26

Horizontal parallax	
Venus:	0.1
Mars:	0.1

Sun							Moon								
Tue	GHA	Dec	GHA	ν	Dec	d	HP								
0	176°30.6	S15°50.4	233°32.1	4.9'	S27°51.9	3.1'	58.2'								
1	191°30.5	49.6	247°56.0	4.8'	27°54.9	2.9'	58.2'								
2	206°30.5	48.8	262°19.8	4.7'	27°57.8	2.7'	58.2'								
3	221°30.4	· · 48.1	276°43.5	4.6'	28°00.5	2.5'	58.3'								
4	236°30.4	47.3	291°07.2	4.6'	28°03.0	2.3'	58.3'								
5	251°30.3	46.5	305°30.7	4.5'	28°05.4	2.2'	58.4'								
6	266°30.3	S15°45.8	319°54.2	4.4'	S28°07.5	2.0'	58.4'								
7	281°30.3	45.0	334°17.6	4.3'	28°09.5	1.8'	58.4'								
8	296°30.2	44.2	348°40.9	4.2'	28°11.3	1.6'	58.5'								
9	311°30.2	· · 43.5	3°04.1	4.2'	28°12.9	1.4'	58.5'								
10	326°30.1	42.7	17°27.2	4.1'	28°14.4	1.2'	58.6'								
11	341°30.1	41.9	31°50.3	4.0'	28°15.6	1.1'	58.6'								
12	356°30.0	S15°41.2	46°13.3	3.9'	S28°16.7	0.9'	58.6'								
13	11°30.0	40.4	60°36.3	3.9'	28°17.5	0.7'	58.7'								
14	26°29.9	39.6	74°59.1	3.8'	28°18.2	0.5'	58.7'								
15	41°29.9	· · 38.9	89°21.9	3.7'	28°18.7	0.3'	58.8'								
16	56°29.9	38.1	103°44.7	3.7'	28°19.0	0.1'	58.8'								
17	71°29.8	37.3	118°07.4	3.6'	28°19.1	-0.1'	58.8'								
18	86°29.8	S15°36.6	132°30.0	3.6'	S28°19.0	-0.3'	58.9'								
19	101°29.7	35.8	146°52.6	3.5'	28°18.7	-0.5'	58.9'								
20	116°29.7	35.0	161°15.1	3.5'	28°18.3	-0.7'	59.0'								
21	131°29.6	· · 34.2	175°37.6	3.4'	28°17.6	-0.9'	59.0'								
22	146°29.6	33.5	190°00.0	3.4'	28°16.7	-1.1'	59.0'								
23	161°29.6	32.7	204°22.4	3.3'	28°15.7	-1.3'	59.1'								
SD = 16.2'			$d = -0.8'$		SD = 15.9'										
Wed	GHA	Dec	GHA	ν	Dec	d	HP								
0	176°29.5	S15°31.9	218°44.8	3.3'	S28°14.4	-1.5'	59.1'								
1	191°29.5	31.2	233°07.1	3.3'	28°12.9	-1.7'	59.2'								
2	206°29.4	30.4	247°29.3	3.2'	28°11.3	-1.9'	59.2'								
3	221°29.4	· · 29.6	261°51.6	3.2'	28°09.4	-2.1'	59.2'								
4	236°29.4	28.8	276°13.8	3.2'	28°07.4	-2.3'	59.3'								
5	251°29.3	28.1	290°36.0	3.2'	28°05.1	-2.5'	59.3'								
6	266°29.3	S15°27.3	304°58.1	3.1'	S28°02.7	-2.7'	59.3'								
7	281°29.2	26.5	319°20.3	3.1'	28°00.0	-2.9'	59.4'								
8	296°29.2	25.7	333°42.4	3.1'	27°57.2	-3.1'	59.4'								
9	311°29.2	· · 24.9	348°04.5	3.1'	27°54.1	-3.3'	59.5'								
10	326°29.1	24.2	2°26.6	3.1'	27°50.9	-3.5'	59.5'								
11	341°29.1	23.4	16°48.6	3.1'	27°47.4	-3.7'	59.5'								
12	356°29.1	S15°22.6	31°10.7	3.1'	S27°43.8	-3.9'	59.6'								
13	11°29.0	21.8	45°32.8	3.1'	27°39.9	-4.1'	59.6'								
14	26°29.0	21.1	59°54.8	3.1'	27°35.9	-4.3'	59.6'								
15	41°28.9	· · 20.3	74°16.9	3.1'	27°31.6	-4.5'	59.7'								
16	56°28.9	19.5	88°38.9	3.1'	27°27.1	-4.7'	59.7'								
17	71°28.9	18.7	103°01.0	3.1'	27°22.5	-4.9'	59.7'								
18	86°28.8	S15°17.9	117°23.1	3.1'	S27°17.6	-5.1'	59.8'								
19	101°28.8	17.1	131°45.2	3.1'	27°12.6	-5.3'	59.8'								
20	116°28.8	16.4	146°07.3	3.1'	27°07.3	-5.5'	59.8'								
21	131°28.7	· · 15.6	160°29.4	3.1'	27°01.9	-5.7'	59.9'								
22	146°28.7	14.8	174°51.5	3.2'	26°56.2	-5.9'	59.9'								
23	161°28.7	14.0	189°13.7	3.2'	26°50.4	-6.0'	59.9'								
SD = 16.2'			$d = -0.8'$		SD = 16.1'										
Thu	GHA	Dec	GHA	ν	Dec	d	HP								
0	176°28.6	S15°13.2	203°35.9	3.2'	S26°44.3	-6.2'	60.0'								
1	191°28.6	12.4	217°58.1	3.2'	26°38.1	-6.4'	60.0'								
2	206°28.6	11.7	232°20.3	3.3'	26°31.6	-6.6'	60.0'								
3	221°28.5	· · 10.9	246°42.6	3.3'	26°25.0	-6.8'	60.1'								
4	236°28.5	10.1	261°04.9	3.3'	26°18.2	-7.0'	60.1'								
5	251°28.5	09.3	275°27.2	3.4'	26°11.1	-7.2'	60.1'								
6	266°28.4	S15°08.5	289°49.6	3.4'	S26°03.9	-7.4'	60.2'								
7	281°28.4	07.7	304°12.0	3.5'	25°56.5	-7.6'	60.2'								
8	296°28.4	06.9	318°34.4	3.5'	25°48.9	-7.8'	60.2'								
9	311°28.4	· · 06.2	332°56.9	3.5'	25°41.1	-8.0'	60.3'								
10	326°28.3	05.4	347°19.5	3.6'	25°33.1	-8.2'	60.3'								
11	341°28.3	04.6	1°42.1	3.6'	25°24.9	-8.4'	60.3'								
12	356°28.3	S15°03.8	16°04.7	3.7'	S25°16.6	-8.5'	60.3'								
13	11°28.2	03.0	30°27.4	3.8'	25°08.0	-8.7'	60.4'								
14	26°28.2	02.2	44°50.2	3.8'	24°59.3	-8.9'	60.4'								
15	41°28.2	· · 01.4	59°13.0	3.9'	24°50.4	-9.1'	60.4'								
16	56°28.2	15°00.6	73°35.9	3.9'	24°41.3	-9.3'	60.4'								
17	71°28.1	14°59.8	87°58.8	4.0'	24°32.0	-9.5'	60.5'								
18	86°28.1	S14°59.0	102°21.8	4.1'	S24°22.5	-9.6'	60.5'								
19	101°28.1	58.2	116°44.9	4.1'	24°12.9	-9.8'	60.5'								
20	116°28.1	57.4	131°08.0	4.2'	24°03.1	-10.0'	60.6'								
21	131°28.0	· · 56.7	145°31.2	4.3'	23°53.1	-10.2'	60.6'								
22	146°28.0	55.9	159°54.4	4.3'	23°42.9	-10.3'	60.6'								
23	161°28.0	55.1	174°17.8	4.4'	23°32.6	-10.5'	60.6'								
SD = 16.2'			$d = -0.8'$		SD = 16.4'										

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	06:49	08:14	09:46	14:43	16:16	17:41
N 70°	06:44	07:59	09:16	15:13	16:31	17:46
68°	06:39	07:47	08:54	15:35	16:43	17:51
66°	06:35	07:37	08:37	15:53	16:52	17:55
64°	06:31	07:28	08:22	16:07	17:01	17:58
62°	06:28	07:21	08:11	16:18	17:08	18:01
60°	06:25	07:15	08:00	16:29	17:14	18:04
N 58°	06:22	07:09	07:52	16:37	17:20	18:07
56°	06:20	07:04	07:44	16:45	17:25	18:09
54°	06:17	06:59	07:37	16:52	17:30	18:12
52°	06:15	06:55	07:31	16:58	17:34	18:14
50°	06:13	06:51	07:25	17:04	17:38	18:16
45°	06:08	06:42	07:13	17:16	17:46	18:21
N 40°	06:03	06:35	07:03	17:26	17:54	18:26
35°	05:58	06:28	06:54	17:34	18:00	18:31
30°	05:54	06:22	06:47	17:42	18:07	18:35
20°	05:44	06:11	06:33	17:55	18:18	18:44
N 10°	05:35	06:00	06:22	18:07	18:28	18:53
0°	05:24	05:49	06:11	18:18	18:39	19:04
S 10°	05:12	05:37	05:59	18:29	18:50	19:16
20°	04:57	05:24	05:47	18:41	19:04	19:31
30°	04:38	05:08	05:33	18:54		

February 09, 10, 11 UT (Fri., Sat., Sun.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	138°35.5	206°26.5	S21°38.0	199°38.2	S21°41.8	102°26.5	N13°17.6	158°54.7	S10°17.9	Alpheratz	357°35.8	29°13.4											
1	153°38.0	221°25.7	37.6	214°38.6	41.5	117°28.7	17.7	173°56.9	17.8	Ankaa	353°08.1	-42°10.8											
2	168°40.4	236°24.9	37.3	229°39.0	41.1	132°30.8	17.8	188°59.0	17.7	Schedar	349°32.3	56°40.3											
3	183°42.9	251°24.0	•• 36.9	244°39.5	•• 40.8	147°33.0	•• 17.9	204°01.2	•• 17.6	Diphda	348°48.2	-17°51.5											
4	198°45.4	266°23.2	36.6	259°39.9	40.4	162°35.2	18.0	219°03.4	17.4	Achernar	335°20.9	-57°07.2											
5	213°47.8	281°22.3	36.2	274°40.3	40.1	177°37.3	18.1	234°05.6	17.3	Hamal	327°52.1	23°34.6											
6	228°50.3	296°21.5	S21°35.9	289°40.8	S21°39.7	192°39.5	N13°18.2	249°07.8	S10°17.2	Polaris	314°25.5	89°22.3											
7	243°52.8	311°20.7	35.5	304°41.2	39.4	207°41.6	18.3	264°10.0	17.1	Acamar	315°12.3	-40°12.8											
8	258°55.2	326°19.8	35.2	319°41.7	39.0	222°43.8	18.4	279°12.2	17.0	Menkar	314°06.9	4°11.0											
9	273°57.7	341°19.0	•• 34.8	334°42.1	•• 38.7	237°46.0	•• 18.6	294°14.3	•• 16.9	Mirfak	308°29.3	49°57.0											
10	289°00.2	356°18.1	34.5	349°42.5	38.3	252°48.1	18.7	309°16.5	16.8	Aldebaran	290°40.4	16°33.5											
11	304°02.6	11°17.3	34.1	4°43.0	38.0	267°50.3	18.8	324°18.7	16.7	Rigel	281°04.4	-8°10.5											
12	319°05.1	26°16.5	S21°33.8	19°43.4	S21°37.6	282°52.5	N13°18.9	339°20.9	S10°16.6	Capella	280°22.7	46°01.5											
13	334°07.5	41°15.6	33.4	34°43.8	37.3	297°54.6	19.0	354°23.1	16.5	Bellatrix	278°23.5	6°22.2											
14	349°10.0	56°14.8	33.1	49°44.3	36.9	312°56.8	19.1	9°25.3	16.3	Elnath	278°02.6	28°37.7											
15	4°12.5	71°13.9	•• 32.7	64°44.7	•• 36.5	327°58.9	•• 19.2	24°27.5	•• 16.2	Alnilam	275°38.3	-1°11.3											
16	19°14.9	86°13.1	32.4	79°45.2	36.2	343°01.1	19.3	39°29.6	16.1	Betelgeuse	270°52.7	7°24.6											
17	34°17.4	101°12.3	32.0	94°45.6	35.8	358°03.3	19.4	54°31.8	16.0	Canopus	263°52.3	-52°42.7											
18	49°19.9	116°11.4	S21°31.6	109°46.0	S21°35.5	13°05.4	N13°19.6	69°34.0	S10°15.9	Sirius	258°26.6	-16°45.1											
19	64°22.3	131°10.6	31.3	124°46.5	35.1	28°07.6	19.7	84°36.2	15.8	Adhara	255°06.1	-29°00.4											
20	79°24.8	146°09.8	30.9	139°46.9	34.8	43°09.7	19.8	99°38.4	15.7	Procyon	244°51.3	5°09.7											
21	94°27.3	161°08.9	•• 30.5	154°47.4	•• 34.4	58°11.9	•• 19.9	114°40.6	•• 15.6	Pollux	243°17.9	27°58.1											
22	109°29.7	176°08.1	30.2	169°47.8	34.0	73°14.1	20.0	129°42.7	15.5	Avior	234°14.4	-59°35.2											
23	124°32.2	191°07.2	29.8	184°48.2	33.7	88°16.2	20.1	144°44.9	15.4	Suhail	222°46.4	-43°31.8											
Mer.pass. 14:43				ν -0.8' d -0.3' m -3.90				ν 0.4' d -0.3' m 1.31				ν 2.2' d 0.1' m -2.30				ν 2.2' d -0.1' m 0.99							

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	139°34.7	206°06.4	S21°29.4	199°48.7	S21°33.3	103°18.4	N13°20.2	159°47.1	S10°15.2										
1	154°37.1	221°05.6	29.1	214°49.1	33.0	118°20.5	20.3	174°49.3	15.1										
2	169°39.6	236°04.7	28.7	229°49.6	32.6	133°22.7	20.4	189°51.5	15.0										
3	184°42.0	251°03.9	•• 28.3	244°50.0	•• 32.3	148°24.9	•• 20.6	204°53.7	•• 14.9										
4	199°44.5	266°03.1	28.0	259°50.4	31.9	163°27.0	20.7	219°55.8	14.8										
5	214°47.0	281°02.2	27.6	274°50.9	31.5	178°29.2	20.8	234°58.0	14.7										
6	229°49.4	296°01.4	S21°27.2	289°51.3	S21°31.2	193°31.3	N13°20.9	250°00.2	S10°14.6										
7	244°51.9	311°00.6	26.8	304°51.8	30.8	208°33.5	21.0	265°02.4	14.5										
8	259°54.4	325°59.7	26.5	319°52.2	30.5	223°35.6	21.1	280°04.6	14.4										
9	274°56.8	340°58.9	•• 26.1	334°52.6	•• 30.1	238°37.8	•• 21.2	295°06.8	•• 14.2										
10	289°59.3	355°58.1	25.7	349°53.1	29.7	253°40.0	21.3	310°09.0	14.1										
11	305°01.8	10°57.2	25.3	4°53.5	29.4	268°42.1	21.5	325°11.1	14.0										
12	320°04.2	25°56.4	S21°24.9	19°54.0	S21°29.0	283°44.3	N13°21.6	340°13.3	S10°13.9										
13	335°06.7	40°55.6	24.5	34°54.4	28.6	298°46.4	21.7	355°15.5	13.8										
14	350°09.2	55°54.7	24.2	49°54.8	28.3	313°48.6	21.8	10°17.7	13.7										
15	5°11.6	70°53.9	•• 23.8	64°55.3	•• 27.9	328°50.7	•• 21.9	25°19.9	•• 13.6										
16	20°14.1	85°53.1	23.4	79°55.7	27.5	343°52.9	22.0	40°22.1	13.5										
17	35°16.5	100°52.2	23.0	94°56.2	27.2	358°55.0	22.1	55°24.2	13.4										
18	50°19.0	115°51.4	S21°22.6	109°56.6	S21°26.8	13°57.2	N13°22.2	70°26.4	S10°13.2										
19	65°21.5	130°50.6	22.2	124°57.1	26.4	28°59.4	22.4	85°28.6	13.1										
20	80°23.9	145°49.7	21.8	139°57.5	26.1	44°01.5	22.5	100°30.8	13.0										
21	95°26.4	160°48.9	•• 21.4	154°57.9	•• 25.7	59°03.7	•• 22.6	115°33.0	•• 12.9										
22	110°28.9	175°48.1	21.1	169°58.4	25.3	74°05.8	22.7	130°35.2	12.8										
23	125°31.3	190°47.3	20.7	184°58.8	25.0	89°08.0	22.8	145°37.3	12.7										
Mer.pass. 14:39				ν -0.8' d -0.4' m -3.90				ν 0.4' d -0.4' m 1.31				ν 2.2' d 0.1' m -2.29				ν 2.2' d -0.1' m 0.99			

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	140°33.8	205°46.4	S21°20.3	199°59.3	S21°24.6	104°10.1	N13°22.9	160°39.5	S10°12.6										
1	155°36.3	220°45.6	19.9	214°59.7	24.2	119°12.3	23.0	175°41.7	12.5										
2	170°38.7	235°44.8	19.5	230°00.2	23.9	134°14.4	23.2	190°43.9	12.4										
3	185°41.2	250°43.9	•• 19.1	245°00.6	•• 23.5	149°16.6	•• 23.3	205°46.1	•• 12.3										
4	200°43.6	265°43.1	18.7	260°01.0	23.1	164°18.7	23.4	220°48.3	12.1										
5	215°46.1	280°42.3	18.3	275°01.5	22.8	179°20.9	23.5	235°50.4	12.0										
6	230°48.6	295°41.5	S21°17.9	290°01.9	S21°22.4	194°23.0	N13°23.6	250°52.6	S10°11.9										
7	245°51.0	310°40.6	17.5	305°02.4	22.0	209°25.2	23.7	265°54.8	11.8										
8	260°53.5	325°39.8	17.1	320°02.8	21.7	224°27.3	23.8	280°57.0	11.7										
9	275°56.0	340°39.0	•• 16.7	335°03.3	•• 21.3	239°29.5	•• 24.0	295°59.2	•• 11.6										
10	290°58.4	355°38.1	16.3	350°03.7	20.9	254°31.6	24.1	311°01.4	11.5										
11	306°00.9	10°37.3	15.8	5°04.1	20.5	269°33.8	24.2	326°03.5	11.4										
12	321°03.4	25°36.5	S21°15.4	20°04.6	S21°20.2	284°35.9	N13°24.3	341°05.7	S10°11.3										
13	336°05.8	40°35.7	15.0	35°05.0	19.8	299°38.1	24.4	356°07.9	11.1										
14	351°08.3	55°34.8	14.6	50°05.5	19.4	314°40.2	24.5	11°10.1	11.0										
15	6°10.8	70°34.0	•• 14.2	65°05.9	•• 19.0	329°42.4	•• 24.6	26°12.3	•• 10.9										
16	21°13.2	85°33.2	13.8	80°06.4	18.7	344°44.5	24.8	41°14.5	10.8										
17	36°15.7	100°32.4	13.4	95°06.8	18.3	359°46.7	24.9	56°16.6	10.7										
18	51°18.1	115°31.5	S21°13.0	110°07.3	S21°17.9	14°48.8	N13°25.0	71°18.8	S10°10.6										
19	66°20.6	130°30.7	12.6	125°07.7	17.5	29°51.0	25.1	86°21.0	10.5										
20	81°23.1	145°29.9	12.1	140°08.1	17.2	44°53.1	25.2	101°23.2	10.4										
21	96°25.5	160°29.1	•• 11.7	155°08.6	•• 16.8	59°55.3	•• 25.3	116°25.4	•• 10.3										
22	111°28.0	175°28.3	11.3	170°09.0	16.4	74°57.4	25.4	131°27.5	10.1										
23	126°30.5	190°27.4	10.9	185°09.5	16.0	89°59.6	25.6	146°29.7	10.0										
Mer.pass. 14:35				ν -0.8' d -0.4' m -3.90				ν 0.4' d -0.4' m 1.31				ν 2.2' d 0.1' m -2.28				ν 2.2' d -0.1' m 0.99			

Feb 09 Fri	SHA	Mer.pass
Venus	67°51.0	10:15
Mars	61°02.6	10:41
Jupiter	323°51.0	17:08
Saturn	20°19.2	13:22

Feb 10 Sat	SHA	Mer.pass
Venus	66°31.8	10:16
Mars	60°14.0	10:40
Jupiter	323°43.7	17:04
Saturn	20°12.5	13:19

Feb 11 Sun	SHA	Mer.pass
Venus	65°12.6	10:17
Mars	59°25.5	10:40
Jupiter	323°36.3	17:01
Saturn	20°05.7	13:15

Horizontal parallax		
Venus:	0.1	
Mars:	0.1	

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	176°28.0	S14°54.3	188°41.2	4.5'	S23°22.0	-10.7'	60.6'
1	191°27.9	53.5	203°04.7	4.6'	23°11.4	-10.9'	60.7'
2	206°27.9	52.7	217°28.2	4.6'	23°00.5	-11.0'	60.7'
3	221°27.9	· · 51.9	231°51.8	4.7'	22°49.5	-11.2'	60.7'
4	236°27.9	51.1	246°15.5	4.8'	22°38.3	-11.3'	60.7'
5	251°27.8	50.3	260°39.3	4.9'	22°27.0	-11.5'	60.8'
6	266°27.8	S14°49.5	275°03.2	4.9'	S22°15.4	-11.7'	60.8'
7	281°27.8	48.7	289°27.1	5.0'	22°03.8	-11.8'	60.8'
8	296°27.8	47.9	303°51.1	5.1'	21°51.9	-12.0'	60.8'
9	311°27.7	· · 47.1	318°15.2	5.2'	21°40.0	-12.1'	60.8'
10	326°27.7	46.3	332°39.4	5.3'	21°27.8	-12.3'	60.9'
11	341°27.7	45.5	347°03.7	5.4'	21°15.5	-12.4'	60.9'
12	356°27.7	S14°44.7	1°28.1	5.4'	S21°03.1	-12.6'	60.9'
13	11°27.7	43.9	15°52.5	5.5'	20°50.5	-12.7'	60.9'
14	26°27.6	43.1	30°17.0	5.6'	20°37.7	-12.9'	60.9'
15	41°27.6	· · 42.3	44°41.6	5.7'	20°24.9	-13.0'	60.9'
16	56°27.6	41.5	59°06.3	5.8'	20°11.8	-13.2'	61.0'
17	71°27.6	40.7	73°31.1	5.9'	19°58.7	-13.3'	61.0'
18	86°27.6	S14°39.9	87°56.0	6.0'	S19°45.4	-13.4'	61.0'
19	101°27.6	39.1	102°20.9	6.0'	19°31.9	-13.6'	61.0'
20	116°27.5	38.3	116°46.0	6.1'	19°18.3	-13.7'	61.0'
21	131°27.5	· · 37.5	131°11.1	6.2'	19°04.6	-13.8'	61.0'
22	146°27.5	36.7	145°36.3	6.3'	18°50.8	-14.0'	61.0'
23	161°27.5	35.9	160°01.6	6.4'	18°36.8	-14.1'	61.0'
SD = 16.2'			$d = -0.8'$		SD = 16.5'		

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	176°27.5	S14°35.1	174°27.0	6.5'	S18°22.7	-14.2'	61.1'
1	191°27.4	34.3	188°52.5	6.6'	18°08.5	-14.3'	61.1'
2	206°27.4	33.4	203°18.1	6.7'	17°54.1	-14.5'	61.1'
3	221°27.4	· · 32.6	217°43.8	6.8'	17°39.7	-14.6'	61.1'
4	236°27.4	31.8	232°09.5	6.8'	17°25.1	-14.7'	61.1'
5	251°27.4	31.0	246°35.4	6.9'	17°10.4	-14.8'	61.1'
6	266°27.4	S14°30.2	261°01.3	7.0'	S16°55.6	-14.9'	61.1'
7	281°27.4	29.4	275°27.3	7.1'	16°40.6	-15.0'	61.1'
8	296°27.3	28.6	289°53.4	7.2'	16°25.6	-15.1'	61.1'
9	311°27.3	· · 27.8	304°19.6	7.3'	16°10.5	-15.2'	61.1'
10	326°27.3	27.0	318°45.9	7.4'	15°55.2	-15.4'	61.1'
11	341°27.3	26.2	333°12.3	7.5'	15°39.8	-15.5'	61.1'
12	356°27.3	S14°25.4	347°38.7	7.5'	S15°24.4	-15.6'	61.1'
13	11°27.3	24.6	2°05.2	7.6'	15°08.8	-15.6'	61.1'
14	26°27.3	23.7	16°31.9	7.7'	14°53.2	-15.7'	61.1'
15	41°27.3	· · 22.9	30°58.6	7.8'	14°37.5	-15.8'	61.2'
16	56°27.2	22.1	45°25.4	7.9'	14°21.6	-15.9'	61.2'
17	71°27.2	21.3	59°52.2	8.0'	14°05.7	-16.0'	61.2'
18	86°27.2	S14°20.5	74°19.2	8.0'	S13°49.7	-16.1'	61.2'
19	101°27.2	19.7	88°46.2	8.1'	13°33.6	-16.2'	61.2'
20	116°27.2	18.9	103°13.4	8.2'	13°17.4	-16.3'	61.2'
21	131°27.2	· · 18.0	117°40.6	8.3'	13°01.2	-16.3'	61.2'
22	146°27.2	17.2	132°07.8	8.4'	12°44.8	-16.4'	61.2'
23	161°27.2	16.4	146°35.2	8.4'	12°28.4	-16.5'	61.2'
SD = 16.2'			$d = -0.8'$		SD = 16.7'		

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	176°27.2	S14°15.6	161°02.6	8.5'	S12°11.9	-16.6'	61.1'
1	191°27.2	14.8	175°30.2	8.6'	11°55.4	-16.6'	61.1'
2	206°27.2	14.0	189°57.7	8.7'	11°38.7	-16.7'	61.1'
3	221°27.2	· · 13.2	204°25.4	8.7'	11°22.0	-16.8'	61.1'
4	236°27.1	12.3	218°53.2	8.8'	11°05.3	-16.8'	61.1'
5	251°27.1	11.5	233°21.0	8.9'	10°48.5	-16.9'	61.1'
6	266°27.1	S14°10.7	247°48.9	9.0'	S10°31.6	-16.9'	61.1'
7	281°27.1	09.9	262°16.8	9.0'	10°14.6	-17.0'	61.1'
8	296°27.1	09.1	276°44.9	9.1'	09°57.6	-17.1'	61.1'
9	311°27.1	· · 08.2	291°13.0	9.2'	09°40.6	-17.1'	61.1'
10	326°27.1	07.4	305°41.1	9.2'	09°23.5	-17.2'	61.1'
11	341°27.1	06.6	320°09.4	9.3'	09°06.3	-17.2'	61.1'
12	356°27.1	S14°05.8	334°37.7	9.4'	S08°49.1	-17.2'	61.1'
13	11°27.1	05.0	349°06.0	9.4'	08°31.9	-17.3'	61.1'
14	26°27.1	04.1	3°34.5	9.5'	08°14.6	-17.3'	61.1'
15	41°27.1	· · 03.3	18°03.0	9.6'	07°57.2	-17.4'	61.0'
16	56°27.1	02.5	32°31.5	9.6'	07°39.9	-17.4'	61.0'
17	71°27.1	01.7	47°00.2	9.7'	07°22.5	-17.4'	61.0'
18	86°27.1	S14°00.9	61°28.8	9.7'	S07°05.0	-17.5'	61.0'
19	101°27.1	14°00.0	75°57.6	9.8'	06°47.5	-17.5'	61.0'
20	116°27.1	13°59.2	90°26.4	9.9'	06°30.0	-17.5'	61.0'
21	131°27.1	· · 58.4	104°55.2	9.9'	06°12.5	-17.6'	61.0'
22	146°27.1	57.6	119°24.1	10.0'	05°54.9	-17.6'	61.0'
23	161°27.1	56.7	133°53.1	10.0'	05°37.4	-17.6'	60.9'
SD = 16.2'			$d = -0.8'$		SD = 16.7'		

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	06:37	08:00	09:27	15:03	16:30	17:53
N 70°	06:33	07:47	09:01	15:29	16:43	17:57
68°	06:29	07:36	08:41	15:48	16:53	18:00
66°	06:26	07:28	08:26	16:04	17:02	18:03
64°	06:23	07:20	08:13	16:17	17:09	18:06
62°	06:21	07:14	08:02	16:27	17:16	18:09
60°	06:18	07:08	07:53	16:36	17:22	18:11
N 58°	06:16	07:03	07:45	16:44	17:27	18:13
56°	06:14	06:58	07:38	16:52	17:31	18:15
54°	06:12	06:54	07:31	16:58	17:35	18:17
52°	06:10	06:50	07:26	17:04	17:39	18:19
50°	06:08	06:46	07:20	17:09	17:43	18:21
45°	06:04	06:38	07:09	17:20	17:50	18:25
N 40°	06:00	06:32	07:00	17:29	17:57	18:29
35°	05:56	06:25	06:52	17:37	18:03	18:33
30°	05:52	06:20	06:45	17:44	18:09	18:37
20°	05:43	06:09	06:32	17:56	18:19	18:45
N 10°	05:34	06:00	06:21	18:07	18:29	18:54
0°	05:25	05:49	06:11	18:18	18:39	19:04
S 10°	05:13	05:38	06:00	18:28	18:50	19:15
20°	04:59	05:26	05:49	18:39	19:02	19:29
30°	04:40	05:11	05:36	18:52	19:17	19:47
35°	04:29	05:01	05:28	19:00	19:27	19:59
40°	04:15	04:50	05:19	19:08	19:37	20:13
45°	03:57	04:37	05:09	19:18	19:50	20:30
S 50°	03:34	04:20	04:57	19:31	20:07	20:53
52°	03:23	04:12	04:51	19:36	20:15	21:04
54°	03:09	04:03	04:44	19:43	20:23	21:17
56°	02:53	03:53	04:37	19:50	20:33	21:33
58°	02:34	03:41	04:29	19:58	20:45	21:52
S 60°	02:08	03:27	04:20	20:07	20:59	22:16

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	■	10:51	09:43	■	15:21	18:24
N 70°	■	10:14	09:28	■	15:56	18:35
68°	10:57	09:48	09:17	13:11	16:21	18:44
66°	10:01	09:27	09:07	14:05	16:39	18:52
64°	09:28	09:11	08:59	14:38	16:54	18:58
62°	09:03	08:57	08:52	15:01	17:07	19:03
60°	08:43	08:45	08:45	15:20	17:17	19:08
N 58°	08:26	08:35	08:40	15:36	17:26	19:12
56°	08:12	08:26	08:35	15:49	17:34	19:15
54°	08:00	08:18	08:30	16:00	17:41	19:18
52°	07:49	08:11	08:26	16:11	17:47	19:21
50°	07:40	08:04	08:23	16:19	17:53	19:24
45°	07:19	07:50	08:15	16:38	18:05	19:29
N 40°	07:03	07:38	08:08	16:54	18:15	19:34
35°	06:49	07:28	08:02	17:06	18:23	19:38
30°	06:36	07:19	07:57	17:17	18:31	19:41
20°	06:15	07:04	07:48	17:36	18:43	19:47
N 10°	05:57	06:51	07:40	17:53	18:54	19:53
0°	05:40	06:38	07:33	18:08	19:04	19:57
S 10°	05:22	06:25	07:25	18:23	19:14	20:02
20°	05:04	06:12	07:17	18:39	19:25	20:07
30°	04:42	05:56	07:08	18:57	19:37	20:13
35°	04:29	05:47	07:03	19:07	19:44	20:16
40°	04:15	05:36	06:57	19:19	19:52	20:20
45°	03:57	05:24	06:50	19:33	20:01	20:24
S 50°	03:36	05:09	06:41	19:50	20:12	20:29
52°	03:25	05:02	06:37	19:58	20:16	20:31
54°	03:13	04:54	06:33	20:07	20:22	20:33
56°	03:00	04:45	06:28	20:17	20:28	20:36
58°	02:44	04:35	06:22	20:28	20:35	20:39
S 60°	02:25	04:23	06:16	20:41	20:42	20:42

Day	Sun		Mer. Pass hh:mm	Moon		Age 0-2 2-2%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
09	14:08	14:09	12:14	11:54	—:—	
10	14:10	14:11	12:14	12:51	00:23	
11	14:11	14:12	12:14	13:45	01:19	

February 12, 13, 14 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	141°32.9	205°26.6	S21°10.5	200°09.9	S21°15.7	105°01.7	N13°25.7	161°31.9	S10°09.9	Alpheratz	357°35.8	29°13.4
1	156°35.4	220°25.8	10.0	215°10.4	15.3	120°03.9	25.8	176°34.1	09.8	Ankaa	353°08.2	-42°10.8
2	171°37.9	235°25.0	09.6	230°10.8	14.9	135°06.0	25.9	191°36.3	09.7	Schedar	349°32.3	56°40.3
3	186°40.3	250°24.1	•• 09.2	245°11.3	•• 14.5	150°08.1	•• 26.0	206°38.5	•• 09.6	Diphda	348°48.3	-17°51.5
4	201°42.8	265°23.3	08.8	260°11.7	14.1	165°10.3	26.1	221°40.6	09.5	Achernar	335°21.0	-57°07.2
5	216°45.2	280°22.5	08.3	275°12.2	13.8	180°12.4	26.3	236°42.8	09.4	Hamal	327°52.2	23°34.6
6	231°47.7	295°21.7	S21°07.9	290°12.6	S21°13.4	195°14.6	N13°26.4	251°45.0	S10°09.3	Polaris	314°27.2	89°22.3
7	246°50.2	310°20.9	07.5	305°13.0	13.0	210°16.7	26.5	266°47.2	09.1	Acamar	315°12.4	-40°12.8
8	261°52.6	325°20.0	07.0	320°13.5	12.6	225°18.9	26.6	281°49.4	09.0	Menkar	314°07.0	4°11.0
9	276°55.1	340°19.2	•• 06.6	335°13.9	•• 12.2	240°21.0	•• 26.7	296°51.6	•• 08.9	Mirfak	308°29.3	49°57.0
10	291°57.6	355°18.4	06.2	350°14.4	11.9	255°23.2	26.8	311°53.7	08.8	Aldebaran	290°40.4	16°33.5
11	307°00.0	10°17.6	05.8	5°14.8	11.5	270°25.3	27.0	326°55.9	08.7	Rigel	281°04.4	-8°10.5
12	322°02.5	25°16.8	S21°05.3	20°15.3	S21°11.1	285°27.4	N13°27.1	341°58.1	S10°08.6	Capella	280°22.8	46°01.5
13	337°05.0	40°15.9	04.9	35°15.7	10.7	300°29.6	27.2	357°00.3	08.5	Bellatrix	278°23.5	6°22.2
14	352°07.4	55°15.1	04.4	50°16.2	10.3	315°31.7	27.3	12°02.5	08.4	Elnath	278°02.6	28°37.7
15	7°09.9	70°14.3	•• 04.0	65°16.6	•• 09.9	330°33.9	•• 27.4	27°04.7	•• 08.3	Alnilam	275°38.3	-1°11.3
16	22°12.4	85°13.5	03.6	80°17.1	09.6	345°36.0	27.5	42°06.8	08.1	Betelgeuse	270°52.7	7°24.6
17	37°14.8	100°12.7	03.1	95°17.5	09.2	0°38.2	27.6	57°09.0	08.0	Canopus	263°52.3	-52°42.7
18	52°17.3	115°11.9	S21°02.7	110°18.0	S21°08.8	15°40.3	N13°27.8	72°11.2	S10°07.9	Sirius	258°26.6	-16°45.1
19	67°19.7	130°11.0	02.2	125°18.4	08.4	30°42.4	27.9	87°13.4	07.8	Adhara	255°06.2	-29°00.4
20	82°22.2	145°10.2	01.8	140°18.9	08.0	45°44.6	28.0	102°15.6	07.7	Procyon	244°51.3	5°09.7
21	97°24.7	160°09.4	•• 01.4	155°19.3	•• 07.6	60°46.7	•• 28.1	117°17.7	•• 07.6	Pollux	243°17.9	27°58.1
22	112°27.1	175°08.6	00.9	170°19.8	07.3	75°48.9	28.2	132°19.9	07.5	Avior	234°14.4	-59°35.3
23	127°29.6	190°07.8	00.5	185°20.2	06.9	90°51.0	28.3	147°22.1	07.4	Suhail	222°46.4	-43°31.8
Mer.pass. 14:31		ν -0.8' d -0.4' m -3.90		ν 0.4' d -0.4' m 1.31		ν 2.1' d 0.1' m -2.28		ν 2.2' d -0.1' m 0.99				

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	142°32.1	205°07.0	S21°00.0	200°20.7	S21°06.5	105°53.1	N13°28.5	162°24.3	S10°07.2
1	157°34.5	220°06.1	20°59.6	215°21.1	06.1	120°55.3	28.6	177°26.5	07.1
2	172°37.0	235°05.3	59.1	230°21.6	05.7	135°57.4	28.7	192°28.7	07.0
3	187°39.5	250°04.5	•• 58.7	245°22.0	•• 05.3	150°59.6	•• 28.8	207°30.8	•• 06.9
4	202°41.9	265°03.7	58.2	260°22.4	04.9	166°01.7	28.9	222°33.0	06.8
5	217°44.4	280°02.9	57.8	275°22.9	04.5	181°03.8	29.1	237°35.2	06.7
6	232°46.9	295°02.1	S20°57.3	290°23.3	S21°04.1	196°06.0	N13°29.2	252°37.4	S10°06.6
7	247°49.3	310°01.3	56.9	305°23.8	03.8	211°08.1	29.3	267°39.6	06.5
8	262°51.8	325°00.5	56.4	320°24.2	03.4	226°10.3	29.4	282°41.7	06.4
9	277°54.2	339°59.6	•• 55.9	335°24.7	•• 03.0	241°12.4	•• 29.5	297°43.9	•• 06.2
10	292°56.7	354°58.8	55.5	350°25.1	02.6	256°14.5	29.6	312°46.1	06.1
11	307°59.2	9°58.0	55.0	5°25.6	02.2	271°16.7	29.8	327°48.3	06.0
12	323°01.6	24°57.2	S20°54.6	20°26.0	S21°01.8	286°18.8	N13°29.9	342°50.5	S10°05.9
13	338°04.1	39°56.4	54.1	35°26.5	01.4	301°20.9	30.0	357°52.7	05.8
14	353°06.6	54°55.6	53.6	50°26.9	01.0	316°23.1	30.1	12°54.8	05.7
15	8°09.0	69°54.8	•• 53.2	65°27.4	•• 00.6	331°25.2	•• 30.2	27°57.0	•• 05.6
16	23°11.5	84°54.0	52.7	80°27.8	21°00.2	346°27.4	30.3	42°59.2	05.5
17	38°14.0	99°53.2	52.2	95°28.3	20°59.8	1°29.5	30.5	58°01.4	05.4
18	53°16.4	114°52.4	S20°51.8	110°28.7	S20°59.4	16°31.6	N13°30.6	73°03.6	S10°05.2
19	68°18.9	129°51.5	51.3	125°29.2	59.0	31°33.8	30.7	88°05.7	05.1
20	83°21.3	144°50.7	50.8	140°29.7	58.6	46°35.9	30.8	103°07.9	05.0
21	98°23.8	159°49.9	•• 50.4	155°30.1	•• 58.3	61°38.0	•• 30.9	118°10.1	•• 04.9
22	113°26.3	174°49.1	49.9	170°30.6	57.9	76°40.2	31.1	133°12.3	04.8
23	128°28.7	189°48.3	49.4	185°31.0	57.5	91°42.3	31.2	148°14.5	04.7
Mer.pass. 14:27		ν -0.8' d -0.4' m -3.89		ν 0.4' d -0.4' m 1.30		ν 2.1' d 0.1' m -2.27		ν 2.2' d -0.1' m 0.99	

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	143°31.2	204°47.5	S20°49.0	200°31.5	S20°57.1	106°44.4	N13°31.3	163°16.6	S10°04.6
1	158°33.7	219°46.7	48.5	215°31.9	56.7	121°46.6	31.4	178°18.8	04.5
2	173°36.1	234°45.9	48.0	230°32.4	56.3	136°48.7	31.5	193°21.0	04.3
3	188°38.6	249°45.1	•• 47.5	245°32.8	•• 55.9	151°50.8	•• 31.7	208°23.2	•• 04.2
4	203°41.1	264°44.3	47.1	260°33.3	55.5	166°53.0	31.8	223°25.4	04.1
5	218°43.5	279°43.5	46.6	275°33.7	55.1	181°55.1	31.9	238°27.6	04.0
6	233°46.0	294°42.7	S20°46.1	290°34.2	S20°54.7	196°57.2	N13°32.0	253°29.7	S10°03.9
7	248°48.5	309°41.9	45.6	305°34.6	54.3	211°59.4	32.1	268°31.9	03.8
8	263°50.9	324°41.1	45.1	320°35.1	53.9	227°01.5	32.2	283°34.1	03.7
9	278°53.4	339°40.3	•• 44.7	335°35.5	•• 53.5	242°03.6	•• 32.4	298°36.3	•• 03.6
10	293°55.8	354°39.5	44.2	350°36.0	53.1	257°05.8	32.5	313°38.5	03.5
11	308°58.3	9°38.7	43.7	5°36.4	52.7	272°07.9	32.6	328°40.6	03.3
12	324°00.8	24°37.8	S20°43.2	20°36.9	S20°52.3	287°10.0	N13°32.7	343°42.8	S10°03.2
13	339°03.2	39°37.0	42.7	35°37.3	51.9	302°12.2	32.8	358°45.0	03.1
14	354°05.7	54°36.2	42.2	50°37.8	51.5	317°14.3	33.0	13°47.2	03.0
15	9°08.2	69°35.4	•• 41.7	65°38.2	•• 51.1	332°16.4	•• 33.1	28°49.4	•• 02.9
16	24°10.6	84°34.6	41.2	80°38.7	50.7	347°18.6	33.2	43°51.5	02.8
17	39°13.1	99°33.8	40.8	95°39.2	50.3	2°20.7	33.3	58°53.7	02.7
18	54°15.6	114°33.0	S20°40.3	110°39.6	S20°49.9	17°22.8	N13°33.4	73°55.9	S10°02.6
19	69°18.0	129°32.2	39.8	125°40.1	49.5	32°24.9	33.6	88°58.1	02.4
20	84°20.5	144°31.4	39.3	140°40.5	49.0	47°27.1	33.7	104°00.3	02.3
21	99°22.9	159°30.6	•• 38.8	155°41.0	•• 48.6	62°29.2	•• 33.8	119°02.5	•• 02.2
22	114°25.4	174°29.8	38.3	170°41.4	48.2	77°31.3	33.9	134°04.6	02.1
23	129°27.9	189°29.0	37.8	185°41.9	47.8	92°33.5	34.0	149°06.8	02.0
Mer.pass. 14:24		ν -0.8' d -0.5' m -3.89		ν 0.5' d -0.4' m 1.30		ν 2.1' d 0.1' m -2.27		ν 2.2' d -0.1' m 0.98	

Feb 12 Mon	SHA	Mer.pass
Venus	63°53.7	10:19
Mars	58°37.0	10:39
Jupiter	323°28.8	16:57
Saturn	19°59.0	13:12
Feb 13 Tue	SHA	Mer.pass
Venus	62°34.9	10:20
Mars	57°48.6	10:38
Jupiter	323°21.1	16:54
Saturn	19°52.2	13:08
Feb 14 Wed	SHA	Mer.pass
Venus	61°16.3	10:21
Mars	57°00.3	10:38
Jupiter	323°13.2	16:51
Saturn	19°45.4	13:05
Horizontal parallax		
Venus:		0.1
Mars:		0.1

Sun				Moon				
Mon	GHA	Dec		GHA	ν	Dec	d	HP
0	176°27.1	S13°55.9		148°22.1	10.1'	S05°19.7	-17.6'	60.9'
1	191°27.1	55.1		162°51.2	10.1'	05°02.1	-17.6'	60.9'
2	206°27.1	54.3		177°20.3	10.2'	04°44.5	-17.7'	60.9'
3	221°27.1	· · 53.4		191°49.5	10.2'	04°26.8	-17.7'	60.9'
4	236°27.1	52.6		206°18.7	10.3'	04°09.1	-17.7'	60.9'
5	251°27.1	51.8		220°47.9	10.3'	03°51.4	-17.7'	60.8'
6	266°27.1	S13°51.0		235°17.2	10.3'	S03°33.7	-17.7'	60.8'
7	281°27.1	50.1		249°46.6	10.4'	03°16.0	-17.7'	60.8'
8	296°27.1	49.3		264°16.0	10.4'	02°58.3	-17.7'	60.8'
9	311°27.1	· · 48.5		278°45.4	10.5'	02°40.6	-17.7'	60.8'
10	326°27.1	47.6		293°14.9	10.5'	02°22.9	-17.7'	60.7'
11	341°27.1	46.8		307°44.4	10.5'	02°05.2	-17.7'	60.7'
12	356°27.1	S13°46.0		322°13.9	10.6'	S01°47.4	-17.7'	60.7'
13	11°27.1	45.1		336°43.5	10.6'	01°29.7	-17.7'	60.7'
14	26°27.1	44.3		351°13.1	10.6'	01°12.0	-17.7'	60.7'
15	41°27.1	· · 43.5		5°42.8	10.7'	00°54.3	-17.7'	60.6'
16	56°27.1	42.7		20°12.4	10.7'	00°36.6	-17.7'	60.6'
17	71°27.1	41.8		34°42.2	10.7'	00°18.9	-17.7'	60.6'
18	86°27.1	S13°41.0		49°11.9	10.8'	S00°01.3	-17.7'	60.6'
19	101°27.1	40.2		63°41.7	10.8'	N00°16.4	17.6'	60.5'
20	116°27.1	39.3		78°11.5	10.8'	00°34.0	17.6'	60.5'
21	131°27.2	· · 38.5		92°41.3	10.8'	00°51.7	17.6'	60.5'
22	146°27.2	37.7		107°11.1	10.9'	01°09.3	17.6'	60.5'
23	161°27.2	36.8		121°41.0	10.9'	01°26.9	17.6'	60.4'
SD = 16.2'			d = -0.8'		SD = 16.6'			
Tue	GHA	Dec		GHA	ν	Dec	d	HP
0	176°27.2	S13°36.0		136°10.9	10.9'	N01°44.4	17.5'	60.4'
1	191°27.2	35.2		150°40.8	10.9'	02°01.9	17.5'	60.4'
2	206°27.2	34.3		165°10.7	10.9'	02°19.4	17.5'	60.4'
3	221°27.2	· · 33.5		179°40.7	11.0'	02°36.9	17.4'	60.3'
4	236°27.2	32.6		194°10.6	11.0'	02°54.4	17.4'	60.3'
5	251°27.2	31.8		208°40.6	11.0'	03°11.8	17.4'	60.3'
6	266°27.2	S13°31.0		223°10.6	11.0'	N03°29.2	17.3'	60.3'
7	281°27.2	30.1		237°40.6	11.0'	03°46.5	17.3'	60.2'
8	296°27.3	29.3		252°10.6	11.0'	04°03.8	17.3'	60.2'
9	311°27.3	· · 28.5		266°40.6	11.0'	04°21.1	17.2'	60.2'
10	326°27.3	27.6		281°10.7	11.0'	04°38.3	17.2'	60.1'
11	341°27.3	26.8		295°40.7	11.0'	04°55.5	17.1'	60.1'
12	356°27.3	S13°25.9		310°10.7	11.1'	N05°12.6	17.1'	60.1'
13	11°27.3	25.1		324°40.8	11.1'	05°29.7	17.0'	60.1'
14	26°27.3	24.3		339°10.8	11.1'	05°46.8	17.0'	60.0'
15	41°27.3	· · 23.4		353°40.9	11.1'	06°03.8	16.9'	60.0'
16	56°27.3	22.6		8°11.0	11.1'	06°20.7	16.9'	60.0'
17	71°27.4	21.7		22°41.0	11.1'	06°37.6	16.8'	59.9'
18	86°27.4	S13°20.9		37°11.1	11.1'	N06°54.5	16.8'	59.9'
19	101°27.4	20.1		51°41.1	11.1'	07°11.2	16.7'	59.9'
20	116°27.4	19.2		66°11.2	11.0'	07°28.0	16.7'	59.8'
21	131°27.4	· · 18.4		80°41.2	11.0'	07°44.7	16.6'	59.8'
22	146°27.4	17.5		95°11.3	11.0'	08°01.3	16.6'	59.8'
23	161°27.4	16.7		109°41.3	11.0'	08°17.8	16.5'	59.7'
SD = 16.2'			d = -0.8'		SD = 16.5'			
Wed	GHA	Dec		GHA	ν	Dec	d	HP
0	176°27.5	S13°15.8		124°11.3	11.0'	N08°34.3	16.4'	59.7'
1	191°27.5	15.0		138°41.3	11.0'	08°50.7	16.4'	59.7'
2	206°27.5	14.2		153°11.3	11.0'	09°07.1	16.3'	59.7'
3	221°27.5	· · 13.3		167°41.3	11.0'	09°23.4	16.2'	59.6'
4	236°27.5	12.5		182°11.3	11.0'	09°39.6	16.2'	59.6'
5	251°27.5	11.6		196°41.3	11.0'	09°55.8	16.1'	59.6'
6	266°27.6	S13°10.8		211°11.2	10.9'	N10°11.9	16.0'	59.5'
7	281°27.6	09.9		225°41.2	10.9'	10°27.9	15.9'	59.5'
8	296°27.6	09.1		240°11.1	10.9'	10°43.8	15.9'	59.5'
9	311°27.6	· · 08.2		254°41.0	10.9'	10°59.7	15.8'	59.4'
10	326°27.6	07.4		269°10.9	10.9'	11°15.5	15.7'	59.4'
11	341°27.7	06.5		283°40.7	10.8'	11°31.2	15.6'	59.3'
12	356°27.7	S13°05.7		298°10.6	10.8'	N11°46.8	15.6'	59.3'
13	11°27.7	04.8		312°40.4	10.8'	12°02.4	15.5'	59.3'
14	26°27.7	04.0		327°10.2	10.8'	12°17.8	15.4'	59.2'
15	41°27.7	· · 03.1		341°40.0	10.7'	12°33.2	15.3'	59.2'
16	56°27.8	02.3		356°09.7	10.7'	12°48.5	15.2'	59.2'
17	71°27.8	01.4		10°39.4	10.7'	13°03.7	15.1'	59.1'
18	86°27.8	S13°00.6		25°09.1	10.7'	N13°18.9	15.0'	59.1'
19	101°27.8	12°59.7		39°38.8	10.6'	13°33.9	14.9'	59.1'
20	116°27.9	58.9		54°08.4	10.6'	13°48.9	14.9'	59.0'
21	131°27.9	· · 58.0		68°38.0	10.6'	14°03.7	14.8'	59.0'
22	146°27.9	57.2		83°07.6	10.5'	14°18.5	14.7'	59.0'
23	161°27.9	56.3		97°37.2	10.5'	14°33.1	14.6'	58.9'
SD = 16.2'			d = -0.8'		SD = 16.3'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	06:25	07:46	09:08	15:22	16:44	18:05
N 70°	06:22	07:35	08:46	15:44	16:55	18:08
68°	06:20	07:26	08:29	16:01	17:04	18:10
66°	06:17	07:18	08:15	16:15	17:12	18:12
64°	06:15	07:11	08:03	16:26	17:18	18:14
62°	06:13	07:06	07:53	16:36	17:24	18:16
60°	06:12	07:01	07:45	16:44	17:29	18:18
N 58°	06:10	06:56	07:38	16:52	17:33	18:20
56°	06:08	06:52	07:31	16:58	17:37	18:21
54°	06:07	06:48	07:25	17:04	17:41	18:23
52°	06:05	06:45	07:20	17:09	17:44	18:24
50°	06:04	06:42	07:15	17:14	17:48	18:26
45°	06:00	06:34	07:05	17:24	17:55	18:29
N 40°	05:56	06:28	06:56	17:33	18:01	18:33
35°	05:53	06:23	06:49	17:40	18:06	18:36
30°	05:49	06:18	06:42	17:47	18:11	18:39
20°	05:42	06:08	06:31	17:58	18:21	18:47
N 10°	05:34	05:59	06:20	18:08	18:30	18:55
0°	05:25	05:50	06:11	18:18	18:39	19:04
S 10°	05:14	05:39	06:01	18:27	18:49	19:14
20°	05:01	05:28	05:50	18:38	19:00	19:27
30°	04:43	05:13	05:38	18:50	19:15	19:44
35°	04:32	05:04	05:31	18:57	19:23	19:55
40°	04:19	04:54	05:23	19:05	19:33	20:08
45°	04:03	04:42	05:14	19:14	19:46	20:25
S 50°	03:41	04:26	05:02	19:25	20:01	20:46
52°	03:30	04:19	04:57	19:31	20:08	20:56
54°	03:18	04:10	04:51	19:36	20:16	21:08
56°	03:03	04:01	04:44	19:43	20:26	21:23
58°	02:46	03:50	04:37	19:50	20:36	21:40
S 60°	02:23	03:37	04:28	19:59	20:49	22:01

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	09:00	08:22	07:40	20:57	23:28	· · ·
N 70°	08:57	08:28	07:56	20:56	23:15	· · ·
68°	08:53	08:32	08:09	20:56	23:04	· · ·
66°	08:51	08:36	08:20	20:55	22:56	· · ·
64°	08:49	08:39	08:29	20:54	22:48	· · ·
62°	08:47	08:42	08:37	20:54	22:42	· · ·
60°	08:45	08:44	08:44	20:53	22:37	· · ·
N 58°	08:43	08:47	08:50	20:53	22:32	· · ·
56°	08:42	08:49	08:55	20:53	22:28	· · ·
54°	08:41	08:50	09:00	20:52	22:24	23:55
52°	08:40	08:52	09:05	20:52	22:21	23:49
50°	08:39	08:53	09:09	20:52	22:18	23:43
45°	08:36	08:57	09:18	20:51	22:11	23:30
N 40°	08:34	08:59	09:25	20:51	22:06	23:20
35°	08:33	09:02	09:31	20:50	22:01	23:11
30°	08:31	09:04	09:37	20:50	21:57	23:03
20°	08:29	09:08	09:47	20:49	21:50	22:50
N 10°	08:27	09:11	09:56	20:49	21:44	22:39
0°	08:24	09:14	10:04	20:48	21:38	22:28
S 10°	08:22	09:18	10:12	20:48	21:32	22:17
20°	08:20	09:21	10:21	20:47	21:26	22:06
30°	08:18	09:25	10:32	20:46	21:19	21:53
35°	08:16	09:28	10:38	20:46	21:15	21:45
40°	08:15	09:30	10:45	20:45	21:10	21:37
45°	08:13	09:33	10:53	20:45	21:05	21:27
S 50°	08:10	09:37	11:03	20:44	20:59	21:15
52°	08:09	09:39	11:07	20:44	20:56	21:10
54°	08:08	09:41	11:12	20:43	20:53	21:04
56°	08:07	09:43	11:18	20:43	20:50	20:57
58°	08:06	09:45	11:24	20:42	20:46	20:50
S 60°	08:04	09:48	11:31	20:42	20:41	20:41

Day	Sun		Mer. Pass hh:mm	Moon		Age 3-5 6-22%	
	Eqn.of Time	00 ^h mm:ss		Mer.Pass.	Upper hh:mm		Lower hh:mm
	12 ^h mm:ss						
12	14:12	14:12	12:14	14:36	02:11		
13	14:11	14:11	12:14	15:26	03:01		
14	14:10	14:09	12:14	16:16	03:51		

February 15, 16, 17 UT (Thu., Fri., Sat.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	144°30.3	204°28.2	S20°37.3	200°42.3	S20°47.4	107°35.6	N13°34.2	164°09.0	S10°01.9	Alpheratz	357°35.8 29°13.4
1	159°32.8	219°27.4	36.8	215°42.8	47.0	122°37.7	34.3	179°11.2	01.8	Ankaa	353°08.2 -42°10.7
2	174°35.3	234°26.6	36.3	230°43.2	46.6	137°39.8	34.4	194°13.4	01.7	Schedar	349°32.3 56°40.3
3	189°37.7	249°25.9	· · 35.8	245°43.7	· · 46.2	152°42.0	· · 34.5	209°15.5	· · 01.6	Diphda	348°48.3 -17°51.5
4	204°40.2	264°25.1	35.3	260°44.2	45.8	167°44.1	34.6	224°17.7	01.4	Achernar	335°21.0 -57°07.2
5	219°42.7	279°24.3	34.8	275°44.6	45.4	182°46.2	34.8	239°19.9	01.3	Hamal	327°52.2 23°34.6
6	234°45.1	294°23.5	S20°34.3	290°45.1	S20°45.0	197°48.4	N13°34.9	254°22.1	S10°01.2	Polaris	314°28.6 89°22.3
7	249°47.6	309°22.7	33.8	305°45.5	44.6	212°50.5	35.0	269°24.3	01.1	Acamar	315°12.4 -40°12.8
8	264°50.1	324°21.9	33.3	320°46.0	44.2	227°52.6	35.1	284°26.4	01.0	Menkar	314°07.0 4°11.0
9	279°52.5	339°21.1	· · 32.7	335°46.4	· · 43.8	242°54.7	· · 35.3	299°28.6	· · 00.9	Mirfak	308°29.3 49°57.0
10	294°55.0	354°20.3	32.2	350°46.9	43.3	257°56.9	35.4	314°30.8	00.8	Aldebaran	290°40.4 16°33.5
11	309°57.4	9°19.5	31.7	5°47.3	42.9	272°59.0	35.5	329°33.0	00.7	Rigel	281°04.4 -8°10.5
12	324°59.9	24°18.7	S20°31.2	20°47.8	S20°42.5	288°01.1	N13°35.6	344°35.2	S10°00.5	Capella	280°22.8 46°01.5
13	340°02.4	39°17.9	30.7	35°48.3	42.1	303°03.2	35.7	359°37.3	00.4	Bellatrix	278°23.5 6°22.2
14	355°04.8	54°17.1	30.2	50°48.7	41.7	318°05.4	35.9	14°39.5	00.3	Elnath	278°02.6 28°37.7
15	10°07.3	69°16.3	· · 29.7	65°49.2	· · 41.3	333°07.5	· · 36.0	29°41.7	· · 00.2	Alnilam	275°38.3 -1°11.3
16	25°09.8	84°15.5	29.2	80°49.6	40.9	348°09.6	36.1	44°43.9	00.1	Betelgeuse	270°52.7 7°24.6
17	40°12.2	99°14.7	28.6	95°50.1	40.5	3°11.7	36.2	59°46.1	10°00.0	Canopus	263°52.4 -52°42.7
18	55°14.7	114°13.9	S20°28.1	110°50.5	S20°40.1	18°13.9	N13°36.3	74°48.2	S09°59.9	Sirius	258°26.6 -16°45.1
19	70°17.2	129°13.1	27.6	125°51.0	39.6	33°16.0	36.5	89°50.4	59.8	Adhara	255°06.2 -29°00.4
20	85°19.6	144°12.3	27.1	140°51.5	39.2	48°18.1	36.6	104°52.6	59.6	Procyon	244°51.3 5°09.7
21	100°22.1	159°11.6	· · 26.6	155°51.9	· · 38.8	63°20.2	· · 36.7	119°54.8	· · 59.5	Pollux	243°17.9 27°58.1
22	115°24.6	174°10.8	26.0	170°52.4	38.4	78°22.3	36.8	134°57.0	59.4	Avior	234°14.5 -59°35.3
23	130°27.0	189°10.0	25.5	185°52.8	38.0	93°24.5	37.0	149°59.1	59.3	Suhail	222°46.4 -43°31.8
Mer.pass. 14:20		ν -0.8' d -0.5' m -3.89		ν 0.5' d -0.4' m 1.30		ν 2.1' d 0.1' m -2.26		ν 2.2' d -0.1' m 0.98		Miaplacidus	221°37.6 -69°49.0
										Alphard	217°48.2 -8°45.9
										Regulus	207°34.9 11°50.9
										Dubhe	193°41.2 61°37.1
										Denebola	182°25.4 14°26.1
										Gienah	175°44.1 -17°40.6
										Acrux	173°00.5 -63°13.8
										Gacrux	171°52.1 -57°14.7
										Alioth	166°13.2 55°49.5
										Spica	158°22.9 -11°17.3
										Alkaid	152°52.3 49°11.3
										Hadar	148°36.8 -60°29.1
										Menkent	147°58.4 -36°29.2
										Arcturus	145°48.5 19°03.2
										Rigel Kent.	139°41.2 -60°55.9
										Kochab	137°19.5 74°03.0
										Zuben'ubi	136°56.8 -16°08.5
										Alphecca	126°04.4 26°37.7
										Antares	112°16.8 -26°29.1
										Atria	107°11.9 -69°04.0
										Sabik	102°03.8 -15°45.3
										Shaula	96°11.6 -37°07.2
										Rasalhague	95°59.4 12°32.3
										Eltanin	90°42.8 51°28.8
										Kaus Aust.	83°33.8 -34°22.4
										Vega	80°34.0 38°48.1
										Nunki	75°48.9 -26°16.0
										Altair	62°01.0 8°55.7
										Peacock	53°07.4 -56°39.4
										Deneb	49°26.8 45°21.8
										Enif	33°39.9 9°59.0
										Al Na'ir	27°34.3 -46°50.8
										Fomalhaut	15°15.7 -29°29.9
										Scheat	13°46.3 28°12.7
										Markab	13°30.9 15°20.0
Feb 15 Thu		SHA		Mer.pass							
Venus		59°57.9		10:23							
Mars		56°12.0		10:37							
Jupiter		323°05.2		16:47							
Saturn		19°38.7		13:02							
Feb 16 Fri		SHA		Mer.pass							
Venus		58°39.7		10:24							
Mars		55°23.8		10:36							
Jupiter		322°57.1		16:44							
Saturn		19°31.8		12:58							
Feb 17 Sat		SHA		Mer.pass							
Venus		57°21.7		10:25							
Mars		54°35.7		10:35							
Jupiter		322°48.8		16:40							
Saturn		19°25.0		12:55							

Sun				Moon			
h	GHA	Dec		GHA	ν	Dec	d HP
Thu							
0	176°27.9	S12°55.5		112°06.7	10.5'	N14°47.7	14.5' 58.9'
1	191°28.0	54.6		126°36.2	10.5'	15°02.2	14.4' 58.9'
2	206°28.0	53.8		141°05.6	10.4'	15°16.6	14.3' 58.8'
3	221°28.0	· · 52.9		155°35.1	10.4'	15°30.9	14.2' 58.8'
4	236°28.0	52.1		170°04.4	10.3'	15°45.1	14.1' 58.8'
5	251°28.1	51.2		184°33.8	10.3'	15°59.1	14.0' 58.7'
6	266°28.1	S12°50.4		199°03.1	10.3'	N16°13.1	13.9' 58.7'
7	281°28.1	49.5		213°32.4	10.2'	16°27.0	13.8' 58.6'
8	296°28.1	48.6		228°01.6	10.2'	16°40.8	13.7' 58.6'
9	311°28.2	· · 47.8		242°30.8	10.2'	16°54.4	13.6' 58.6'
10	326°28.2	46.9		257°00.0	10.1'	17°08.0	13.5' 58.5'
11	341°28.2	46.1		271°29.1	10.1'	17°21.5	13.3' 58.5'
12	356°28.3	S12°45.2		285°58.2	10.0'	N17°34.8	13.2' 58.5'
13	11°28.3	44.4		300°27.2	10.0'	17°48.0	13.1' 58.4'
14	26°28.3	43.5		314°56.3	10.0'	18°01.2	13.0' 58.4'
15	41°28.3	· · 42.6		329°25.2	9.9'	18°14.2	12.9' 58.4'
16	56°28.4	41.8		343°54.1	9.9'	18°27.1	12.8' 58.3'
17	71°28.4	40.9		358°23.0	9.8'	18°39.9	12.7' 58.3'
18	86°28.4	S12°40.1		12°51.9	9.8'	N18°52.5	12.6' 58.3'
19	101°28.5	39.2		27°20.7	9.8'	19°05.1	12.4' 58.2'
20	116°28.5	38.4		41°49.4	9.7'	19°17.5	12.3' 58.2'
21	131°28.5	· · 37.5		56°18.1	9.7'	19°29.8	12.2' 58.1'
22	146°28.5	36.6		70°46.8	9.6'	19°42.0	12.1' 58.1'
23	161°28.6	35.8		85°15.4	9.6'	19°54.1	12.0' 58.1'
SD = 16.2' $d = -0.9'$				SD = 16.1'			

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	176°28.6	S12°34.9	99°44.0	9.5'	N20°06.0	11.8'	58.0'
1	191°28.6	34.1	114°12.6	9.5'	20°17.9	11.7'	58.0'
2	206°28.7	33.2	128°41.1	9.5'	20°29.6	11.6'	58.0'
3	221°28.7	· · 32.3	143°09.5	9.4'	20°41.2	11.5'	57.9'
4	236°28.7	31.5	157°37.9	9.4'	20°52.6	11.3'	57.9'
5	251°28.8	30.6	172°06.3	9.3'	21°04.0	11.2'	57.9'
6	266°28.8	S12°29.7	186°34.6	9.3'	N21°15.2	11.1'	57.8'
7	281°28.8	28.9	201°02.9	9.2'	21°26.2	10.9'	57.8'
8	296°28.9	28.0	215°31.1	9.2'	21°37.2	10.8'	57.8'
9	311°28.9	· · 27.1	229°59.3	9.1'	21°48.0	10.7'	57.7'
10	326°28.9	26.3	244°27.4	9.1'	21°58.7	10.6'	57.7'
11	341°29.0	25.4	258°55.5	9.1'	22°09.2	10.4'	57.7'
12	356°29.0	S12°24.6	273°23.6	9.0'	N22°19.7	10.3'	57.6'
13	11°29.1	23.7	287°51.6	9.0'	22°29.9	10.2'	57.6'
14	26°29.1	22.8	302°19.6	8.9'	22°40.1	10.0'	57.6'
15	41°29.1	· · 22.0	316°47.5	8.9'	22°50.1	9.9'	57.5'
16	56°29.2	21.1	331°15.4	8.8'	23°00.0	9.7'	57.5'
17	71°29.2	20.2	345°43.2	8.8'	23°09.7	9.6'	57.5'
18	86°29.2	S12°19.4	0°11.0	8.8'	N23°19.4	9.5'	57.4'
19	101°29.3	18.5	14°38.7	8.7'	23°28.8	9.3'	57.4'
20	116°29.3	17.6	29°06.5	8.7'	23°38.1	9.2'	57.4'
21	131°29.3	· · 16.7	43°34.1	8.6'	23°47.3	9.0'	57.3'
22	146°29.4	15.9	58°01.8	8.6'	23°56.4	8.9'	57.3'
23	161°29.4	15.0	72°29.3	8.5'	24°05.3	8.8'	57.3'
SD = 16.2' $d = -0.9'$				SD = 15.8'			

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	176°29.5	S12°14.1	86°56.9	8.5'	N24°14.1	8.6'	57.2'
1	191°29.5	13.3	101°24.4	8.5'	24°22.7	8.5'	57.2'
2	206°29.5	12.4	115°51.9	8.4'	24°31.2	8.3'	57.2'
3	221°29.6	· · 11.5	130°19.3	8.4'	24°39.5	8.2'	57.1'
4	236°29.6	10.7	144°46.7	8.4'	24°47.7	8.0'	57.1'
5	251°29.7	09.8	159°14.0	8.3'	24°55.7	7.9'	57.1'
6	266°29.7	S12°08.9	173°41.4	8.3'	N25°03.6	7.8'	57.0'
7	281°29.7	08.0	188°08.6	8.2'	25°11.4	7.6'	57.0'
8	296°29.8	07.2	202°35.9	8.2'	25°19.0	7.5'	57.0'
9	311°29.8	· · 06.3	217°03.1	8.2'	25°26.4	7.3'	56.9'
10	326°29.9	05.4	231°30.3	8.1'	25°33.7	7.2'	56.9'
11	341°29.9	04.6	245°57.4	8.1'	25°40.9	7.0'	56.9'
12	356°30.0	S12°03.7	260°24.5	8.1'	N25°47.9	6.9'	56.8'
13	11°30.0	02.8	274°51.6	8.1'	25°54.8	6.7'	56.8'
14	26°30.0	01.9	289°18.7	8.0'	26°01.5	6.6'	56.8'
15	41°30.1	· · 01.1	303°45.7	8.0'	26°08.0	6.4'	56.7'
16	56°30.1	12°00.2	318°12.7	8.0'	26°14.4	6.3'	56.7'
17	71°30.2	11°59.3	332°39.7	7.9'	26°20.7	6.1'	56.7'
18	86°30.2	S11°58.4	347°06.6	7.9'	N26°26.8	5.9'	56.6'
19	101°30.3	57.6	1°33.5	7.9'	26°32.7	5.8'	56.6'
20	116°30.3	56.7	16°00.4	7.9'	26°38.5	5.6'	56.6'
21	131°30.4	· · 55.8	30°27.3	7.9'	26°44.2	5.5'	56.6'
22	146°30.4	54.9	44°54.2	7.8'	26°49.6	5.3'	56.5'
23	161°30.5	54.1	59°21.0	7.8'	26°55.0	5.2'	56.5'
SD = 16.2' $d = -0.9'$				SD = 15.6'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	06:12	07:32	08:51	15:39	16:58	18:18
N 70°	06:11	07:23	08:31	15:58	17:07	18:19
68°	06:10	07:15	08:16	16:13	17:15	18:20
66°	06:08	07:08	08:04	16:26	17:21	18:22
64°	06:07	07:02	07:53	16:36	17:27	18:23
62°	06:06	06:58	07:44	16:45	17:32	18:24
60°	06:04	06:53	07:37	16:52	17:36	18:25
N 58°	06:03	06:49	07:30	16:59	17:40	18:26
56°	06:02	06:46	07:24	17:05	17:44	18:27
54°	06:01	06:42	07:19	17:10	17:47	18:28
52°	06:00	06:39	07:14	17:15	17:50	18:29
50°	05:59	06:36	07:10	17:19	17:52	18:30
45°	05:56	06:30	07:00	17:28	17:59	18:33
N 40°	05:53	06:25	06:52	17:36	18:04	18:36
35°	05:50	06:20	06:46	17:43	18:09	18:39
30°	05:47	06:15	06:40	17:49	18:13	18:42
20°	05:41	06:06	06:29	17:59	18:22	18:48
N 10°	05:33	05:58	06:20	18:09	18:30	18:55
0°	05:25	05:50	06:11	18:17	18:39	19:03
S 10°	05:15	05:40	06:02	18:26	18:48	19:13
20°	05:02	05:29	05:52	18:36	18:59	19:25
30°	04:46	05:16	05:41	18:47	19:12	19:41
35°	04:36	05:08	05:34	18:53	19:20	19:51
40°	04:23	04:58	05:27	19:01	19:29	20:04
45°	04:08	04:46	05:18	19:09	19:41	20:19
S 50°	03:48	04:32	05:07	19:20	19:55	20:39
52°	03:38	04:25	05:02	19:25	20:02	20:49
54°	03:26	04:17	04:57	19:30	20:09	21:00
56°	03:13	04:09	04:51	19:36	20:18	21:13
58°	02:57	03:59	04:44	19:43	20:28	21:28
S 60°	02:37	03:47	04:36	19:50	20:39	21:47

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	06:31	□	□	02:24	□	□
N 70°	07:11	□	□	01:46	□	□
68°	07:39	06:39	□	01:20	04:09	□
66°	08:01	07:30	□	01:00	03:18	□
64°	08:18	08:03	07:32	00:44	02:47	05:09
62°	08:32	08:27	08:21	00:31	02:24	04:21
60°	08:44	08:46	08:52	00:20	02:05	03:51
N 58°	08:55	09:02	09:15	00:11	01:50	03:28
56°	09:04	09:16	09:34	00:03	01:37	03:09
54°	09:12	09:28	09:50	· · ·	01:26	02:54
52°	09:20	09:39	10:04	· · ·	01:16	02:40
50°	09:26	09:48	10:16	· · ·	01:07	02:29
45°	09:41	10:08	10:42	· · ·	00:49	02:04
N 40°	09:53	10:24	11:02	· · ·	00:33	01:45
35°	10:03	10:38	11:19	· · ·	00:21	01:29
30°	10:12	10:50	11:33	· · ·	00:10	01:15
20°	10:28	11:11	11:58	23:51	· · ·	00:51
N 10°	10:41	11:29	12:20	23:34	· · ·	00:31
0°	10:54	11:47	12:40	23:19	· · ·	00:12
S 10°	11:08	12:04	13:01	23:04	23:53	· · ·
20°	11:22	12:22	13:23	22:48	23:33	· · ·
30°	11:38	12:44	13:48	22:29	23:10	23:56
35°	11:48	12:57	14:04	22:19	22:56	23:40
40°	11:59	13:11	14:21	22:06	22:41	23:22
45°	12:11	13:29	14:43	21:52	22:22	23:00
S 50°	12:27	13:51	15:10	21:34	21:59	22:32
52°	12:35	14:01	15:23	21:26	21:48	22:18
54°	12:43	14:13	15:38	21:17	21:36	22:03
56°	12:53	14:27	15:56	21:07	21:21	21:44
58°	13:03	14:43	16:18	20:55	21:05	21:22
S 60°	13:16	15:02	16:47	20:42	20:45	20:53

Day	Sun		Mer. Pass	Moon		Age 6-8 33-54%
	Eqn. of Time 00 ^h mm:ss	Time 12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
15	14:08	14:07	12:14	17:07	04:41	
16	14:06	14:04	12:14	17:59	05:33	
17	14:02	14:00	12:14	18:54	06:26	

February 18, 19, 20 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	147°27.8	203°31.7	S19°58.6	201°15.4	S20°17.2	110°08.2	N13°43.0	166°45.9	S09°53.8	Alpheratz	357°35.8 29°13.4
1	162°30.2	218°31.0	58.1	216°15.9	16.7	125°10.3	43.1	181°48.1	53.7	Ankaa	353°08.2 -42°10.7
2	177°32.7	233°30.2	57.5	231°16.4	16.3	140°12.4	43.2	196°50.3	53.6	Schedar	349°32.3 56°40.3
3	192°35.2	248°29.4	•• 56.9	246°16.8	•• 15.9	155°14.5	•• 43.4	211°52.5	•• 53.5	Diphda	348°48.3 -17°51.5
4	207°37.6	263°28.6	56.3	261°17.3	15.4	170°16.6	43.5	226°54.7	53.4	Achernar	335°21.0 -57°07.1
5	222°40.1	278°27.9	55.7	276°17.8	15.0	185°18.7	43.6	241°56.8	53.2	Hamal	327°52.2 23°34.6
6	237°42.5	293°27.1	S19°55.2	291°18.2	S20°14.6	200°20.9	N13°43.8	256°59.0	S09°53.1	Polaris	314°29.8 89°22.3
7	252°45.0	308°26.3	54.6	306°18.7	14.1	215°23.0	43.9	272°01.2	53.0	Acamar	315°12.4 -40°12.7
8	267°47.5	323°25.6	54.0	321°19.2	13.7	230°25.1	44.0	287°03.4	52.9	Menkar	314°07.0 4°11.0
9	282°49.9	338°24.8	•• 53.4	336°19.6	•• 13.3	245°27.2	•• 44.1	302°05.6	•• 52.8	Mirfak	308°29.4 49°57.0
10	297°52.4	353°24.0	52.8	351°20.1	12.8	260°29.3	44.3	317°07.7	52.7	Aldebaran	290°40.4 16°33.5
11	312°54.9	8°23.3	52.3	6°20.6	12.4	275°31.4	44.4	332°09.9	52.6	Rigel	281°04.4 -8°10.6
12	327°57.3	23°22.5	S19°51.7	21°21.0	S20°11.9	290°33.5	N13°44.5	347°12.1	S09°52.4	Capella	280°22.8 46°01.5
13	342°59.8	38°21.7	51.1	36°21.5	11.5	305°35.6	44.6	2°14.3	52.3	Bellatrix	278°23.5 6°22.2
14	358°02.3	53°21.0	50.5	51°22.0	11.1	320°37.7	44.8	17°16.5	52.2	Elnath	278°02.6 28°37.7
15	13°04.7	68°20.2	•• 49.9	66°22.4	•• 10.6	335°39.8	•• 44.9	32°18.6	•• 52.1	Alnilam	275°38.3 -1°11.3
16	28°07.2	83°19.4	49.3	81°22.9	10.2	350°41.9	45.0	47°20.8	52.0	Betelgeuse	270°52.7 7°24.6
17	43°09.7	98°18.7	48.7	96°23.4	09.8	5°44.0	45.1	62°23.0	51.9	Canopus	263°52.4 -52°42.7
18	58°12.1	113°17.9	S19°48.1	111°23.8	S20°09.3	20°46.1	N13°45.3	77°25.2	S09°51.8	Sirius	258°26.7 -16°45.1
19	73°14.6	128°17.1	47.5	126°24.3	08.9	35°48.3	45.4	92°27.3	51.7	Adhara	255°06.2 -29°00.4
20	88°17.0	143°16.4	47.0	141°24.8	08.4	50°50.4	45.5	107°29.5	51.5	Procyon	244°51.3 5°09.7
21	103°19.5	158°15.6	•• 46.4	156°25.2	•• 08.0	65°52.5	•• 45.6	122°31.7	•• 51.4	Pollux	243°17.9 27°58.1
22	118°22.0	173°14.8	45.8	171°25.7	07.5	80°54.6	45.8	137°33.9	51.3	Avior	234°14.5 -59°35.3
23	133°24.4	188°14.1	45.2	186°26.2	07.1	95°56.7	45.9	152°36.1	51.2	Suhail	222°46.4 -43°31.9
Mer.pass. 14:08		ν -0.8' d -0.6' m -3.89		ν 0.5' d -0.4' m 1.29		ν 2.1' d 0.1' m -2.24		ν 2.2' d -0.1' m 0.98			

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	148°26.9	203°13.3	S19°44.6	201°26.6	S20°06.7	110°58.8	N13°46.0	167°38.2	S09°51.1		
1	163°29.4	218°12.6	44.0	216°27.1	06.2	126°00.9	46.1	182°40.4	51.0		
2	178°31.8	233°11.8	43.4	231°27.6	05.8	141°03.0	46.3	197°42.6	50.9		
3	193°34.3	248°11.0	•• 42.8	246°28.0	•• 05.3	156°05.1	•• 46.4	212°44.8	•• 50.8		
4	208°36.8	263°10.3	42.2	261°28.5	04.9	171°07.2	46.5	227°47.0	50.6		
5	223°39.2	278°09.5	41.6	276°29.0	04.4	186°09.3	46.6	242°49.1	50.5		
6	238°41.7	293°08.8	S19°41.0	291°29.4	S20°04.0	201°11.4	N13°46.8	257°51.3	S09°50.4		
7	253°44.1	308°08.0	40.4	306°29.9	03.6	216°13.5	46.9	272°53.5	50.3		
8	268°46.6	323°07.2	39.8	321°30.4	03.1	231°15.6	47.0	287°55.7	50.2		
9	283°49.1	338°06.5	•• 39.1	336°30.9	•• 02.7	246°17.7	•• 47.2	302°57.8	•• 50.1		
10	298°51.5	353°05.7	38.5	351°31.3	02.2	261°19.8	47.3	318°00.0	50.0		
11	313°54.0	8°05.0	37.9	6°31.8	01.8	276°21.9	47.4	333°02.2	49.9		
12	328°56.5	23°04.2	S19°37.3	21°32.3	S20°01.3	291°24.0	N13°47.5	348°04.4	S09°49.7		
13	343°58.9	38°03.5	36.7	36°32.7	00.9	306°26.1	47.7	3°06.6	49.6		
14	359°01.4	53°02.7	36.1	51°33.2	00.4	321°28.2	47.8	18°08.7	49.5		
15	14°03.9	68°01.9	•• 35.5	66°33.7	20°00.0	336°30.3	•• 47.9	33°10.9	•• 49.4		
16	29°06.3	83°01.2	34.9	81°34.1	19°59.5	351°32.4	48.0	48°13.1	49.3		
17	44°08.8	98°00.4	34.3	96°34.6	59.1	6°34.5	48.2	63°15.3	49.2		
18	59°11.3	112°59.7	S19°33.6	111°35.1	S19°58.6	21°36.6	N13°48.3	78°17.5	S09°49.1		
19	74°13.7	127°58.9	33.0	126°35.6	58.2	36°38.7	48.4	93°19.6	49.0		
20	89°16.2	142°58.2	32.4	141°36.0	57.7	51°40.8	48.6	108°21.8	48.8		
21	104°18.6	157°57.4	•• 31.8	156°36.5	•• 57.3	66°42.9	•• 48.7	123°24.0	•• 48.7		
22	119°21.1	172°56.7	31.2	171°37.0	56.8	81°45.0	48.8	138°26.2	48.6		
23	134°23.6	187°55.9	30.5	186°37.4	56.4	96°47.1	48.9	153°28.3	48.5		
Mer.pass. 14:04		ν -0.8' d -0.6' m -3.89		ν 0.5' d -0.4' m 1.29		ν 2.1' d 0.1' m -2.24		ν 2.2' d -0.1' m 0.98			

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	149°26.0	202°55.2	S19°29.9	201°37.9	S19°55.9	111°49.2	N13°49.1	168°30.5	S09°48.4		
1	164°28.5	217°54.4	29.3	216°38.4	55.5	126°51.3	49.2	183°32.7	48.3		
2	179°31.0	232°53.7	28.7	231°38.9	55.0	141°53.4	49.3	198°34.9	48.2		
3	194°33.4	247°52.9	•• 28.1	246°39.3	•• 54.6	156°55.5	•• 49.5	213°37.1	•• 48.0		
4	209°35.9	262°52.2	27.4	261°39.8	54.1	171°57.6	49.6	228°39.2	47.9		
5	224°38.4	277°51.4	26.8	276°40.3	53.7	186°59.7	49.7	243°41.4	47.8		
6	239°40.8	292°50.7	S19°26.2	291°40.7	S19°53.2	202°01.8	N13°49.8	258°43.6	S09°47.7		
7	254°43.3	307°49.9	25.5	306°41.2	52.8	217°03.9	50.0	273°45.8	47.6		
8	269°45.8	322°49.2	24.9	321°41.7	52.3	232°06.0	50.1	288°48.0	47.5		
9	284°48.2	337°48.4	•• 24.3	336°42.2	•• 51.8	247°08.1	•• 50.2	303°50.1	•• 47.4		
10	299°50.7	352°47.7	23.7	351°42.6	51.4	262°10.2	50.8	318°52.3	47.3		
11	314°53.1	7°46.9	23.0	6°43.1	50.9	277°12.3	50.5	333°54.5	47.1		
12	329°55.6	22°46.2	S19°22.4	21°43.6	S19°50.5	292°14.4	N13°50.6	348°56.7	S09°47.0		
13	344°58.1	37°45.4	21.8	36°44.1	50.0	307°16.5	50.7	3°58.8	46.9		
14	0°00.5	52°44.7	21.1	51°44.5	49.6	322°18.6	50.9	19°01.0	46.8		
15	15°03.0	67°43.9	•• 20.5	66°45.0	•• 49.1	337°20.7	•• 51.0	34°03.2	•• 46.7		
16	30°05.5	82°43.2	19.9	81°45.5	48.7	352°22.8	51.1	49°05.4	46.6		
17	45°07.9	97°42.4	19.2	96°46.0	48.2	7°24.9	51.3	64°07.6	46.5		
18	60°10.4	112°41.7	S19°18.6	111°46.4	S19°47.7	22°27.0	N13°51.4	79°09.7	S09°46.4		
19	75°12.9	127°41.0	17.9	126°46.9	47.3	37°29.1	51.5	94°11.9	46.2		
20	90°15.3	142°40.2	17.3	141°47.4	46.8	52°31.2	51.6	109°14.1	46.1		
21	105°17.8	157°39.5	•• 16.7	156°47.9	•• 46.4	67°33.3	•• 51.8	124°16.3	•• 46.0		
22	120°20.3	172°38.7	16.0	171°48.3	45.9	82°35.4	51.9	139°18.4	45.9		
23	135°22.7	187°38.0	15.4	186°48.8	45.4	97°37.5	52.0	154°20.6	45.8		
Mer.pass. 14:00		ν -0.8' d -0.6' m -3.89		ν 0.5' d -0.5' m 1.29		ν 2.1' d 0.1' m -2.23		ν 2.2' d -0.1' m 0.98			

Feb 18 Sun			SHA	Mer.pass
Venus			56°04.0	10:26
Mars			53°47.7	10:35
Jupiter			322°40.4	16:37
Saturn			19°18.2	12:51

Feb 19 Mon			SHA	Mer.pass
Venus			54°46.4	10:28
Mars			52°59.7	10:34
Jupiter			322°31.9	16:34
Saturn			19°11.3	12:48

Feb 20 Tue			SHA	Mer.pass
Venus			53°29.1	10:29
Mars			52°11.9	10:33
Jupiter			322°23.2	16:30
Saturn			19°04.5	12:44

Horizontal parallax			
Venus:			0.1
Mars:			0.1

Sun				Moon			
h	GHA	Dec		GHA	ν	Dec	HP
Sun							
0	176°30.5	S11°53.2		73°47.8	7.8'	N27°00.2	5.0'
1	191°30.5	52.3		88°14.6	7.8'	27°05.2	4.9'
2	206°30.6	51.4		102°41.4	7.8'	27°10.1	4.7'
3	221°30.6	· · 50.5		117°08.1	7.7'	27°14.8	4.6'
4	236°30.7	49.7		131°34.9	7.7'	27°19.3	4.4'
5	251°30.7	48.8		146°01.6	7.7'	27°23.7	4.2'
6	266°30.8	S11°47.9		160°28.3	7.7'	N27°28.0	4.1'
7	281°30.8	47.0		174°55.0	7.7'	27°32.1	3.9'
8	296°30.9	46.1		189°21.7	7.7'	27°36.0	3.8'
9	311°30.9	· · 45.3		203°48.4	7.7'	27°39.8	3.6'
10	326°31.0	44.4		218°15.1	7.7'	27°43.4	3.5'
11	341°31.0	43.5		232°41.8	7.7'	27°46.8	3.3'
12	356°31.1	S11°42.6		247°08.5	7.7'	N27°50.2	3.1'
13	11°31.1	41.7		261°35.1	7.7'	27°53.3	3.0'
14	26°31.2	40.9		276°01.8	7.7'	27°56.3	2.8'
15	41°31.2	· · 40.0		290°28.5	7.7'	27°59.1	2.7'
16	56°31.3	39.1		304°55.1	7.7'	28°01.8	2.5'
17	71°31.3	38.2		319°21.8	7.7'	28°04.3	2.4'
18	86°31.4	S11°37.3		333°48.5	7.7'	N28°06.7	2.2'
19	101°31.4	36.5		348°15.2	7.7'	28°08.9	2.0'
20	116°31.5	35.6		2°41.9	7.7'	28°10.9	1.9'
21	131°31.5	· · 34.7		17°08.6	7.7'	28°12.8	1.7'
22	146°31.6	33.8		31°35.3	7.7'	28°14.6	1.6'
23	161°31.7	32.9		46°02.0	7.7'	28°16.2	1.4'
SD = 16.2'			d = -0.9'	SD = 15.4'			

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	176°31.7	S11°32.0	60°28.8	7.8'	N28°17.6	1.3'	55.8'
1	191°31.8	31.1	74°55.5	7.8'	28°18.8	1.1'	55.8'
2	206°31.8	30.3	89°22.3	7.8'	28°20.0	1.0'	55.8'
3	221°31.9	· · 29.4	103°49.1	7.8'	28°20.9	0.8'	55.7'
4	236°31.9	28.5	118°15.9	7.8'	28°21.7	0.6'	55.7'
5	251°32.0	27.6	132°42.7	7.8'	28°22.4	0.5'	55.7'
6	266°32.0	S11°26.7	147°09.5	7.9'	N28°22.8	0.3'	55.7'
7	281°32.1	25.8	161°36.4	7.9'	28°23.2	0.2'	55.6'
8	296°32.2	24.9	176°03.3	7.9'	28°23.4	0.0'	55.6'
9	311°32.2	· · 24.1	190°30.2	8.0'	28°23.4	-0.1'	55.6'
10	326°32.3	23.2	204°57.2	8.0'	28°23.3	-0.3'	55.6'
11	341°32.3	22.3	219°24.2	8.0'	28°23.0	-0.4'	55.5'
12	356°32.4	S11°21.4	233°51.2	8.0'	N28°22.5	-0.6'	55.5'
13	11°32.4	20.5	248°18.2	8.1'	28°22.0	-0.7'	55.5'
14	26°32.5	19.6	262°45.3	8.1'	28°21.2	-0.9'	55.5'
15	41°32.6	· · 18.7	277°12.4	8.1'	28°20.3	-1.0'	55.4'
16	56°32.6	17.8	291°39.6	8.2'	28°19.3	-1.2'	55.4'
17	71°32.7	16.9	306°06.8	8.2'	28°18.1	-1.3'	55.4'
18	86°32.7	S11°16.0	320°34.0	8.3'	N28°16.8	-1.5'	55.4'
19	101°32.8	15.2	335°01.2	8.3'	28°15.3	-1.6'	55.4'
20	116°32.9	14.3	349°28.6	8.4'	28°13.6	-1.8'	55.3'
21	131°32.9	· · 13.4	3°55.9	8.4'	28°11.8	-1.9'	55.3'
22	146°33.0	12.5	18°23.3	8.4'	28°09.9	-2.1'	55.3'
23	161°33.0	11.6	32°50.7	8.5'	28°07.8	-2.2'	55.3'
SD = 16.2'			d = -0.9'	SD = 15.2'			

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	176°33.1	S11°10.7	47°18.2	8.5'	N28°05.6	-2.4'	55.3'
1	191°33.2	09.8	61°45.8	8.6'	28°03.2	-2.5'	55.2'
2	206°33.2	08.9	76°13.4	8.6'	28°00.7	-2.7'	55.2'
3	221°33.3	· · 08.0	90°41.0	8.7'	27°58.0	-2.8'	55.2'
4	236°33.3	07.1	105°08.7	8.7'	27°55.2	-3.0'	55.2'
5	251°33.4	06.2	119°36.4	8.8'	27°52.2	-3.1'	55.1'
6	266°33.5	S11°05.3	134°04.2	8.9'	N27°49.1	-3.2'	55.1'
7	281°33.5	04.4	148°32.1	8.9'	27°45.9	-3.4'	55.1'
8	296°33.6	03.6	163°00.0	9.0'	27°42.5	-3.5'	55.1'
9	311°33.7	· · 02.7	177°28.0	9.0'	27°39.0	-3.7'	55.1'
10	326°33.7	01.8	191°56.0	9.1'	27°35.3	-3.8'	55.1'
11	341°33.8	00.9	206°24.1	9.1'	27°31.5	-3.9'	55.0'
12	356°33.9	S11°00.0	220°52.2	9.2'	N27°27.6	-4.1'	55.0'
13	11°33.9	10°59.1	235°20.5	9.3'	27°23.5	-4.2'	55.0'
14	26°34.0	58.2	249°48.7	9.3'	27°19.3	-4.4'	55.0'
15	41°34.0	· · 57.3	264°17.1	9.4'	27°14.9	-4.5'	55.0'
16	56°34.1	56.4	278°45.5	9.5'	27°10.4	-4.6'	54.9'
17	71°34.2	55.5	293°13.9	9.5'	27°05.8	-4.8'	54.9'
18	86°34.2	S10°54.6	307°42.5	9.6'	N27°01.1	-4.9'	54.9'
19	101°34.3	53.7	322°11.1	9.7'	26°56.2	-5.0'	54.9'
20	116°34.4	52.8	336°39.7	9.7'	26°51.1	-5.2'	54.9'
21	131°34.4	· · 51.9	351°08.5	9.8'	26°46.0	-5.3'	54.9'
22	146°34.5	51.0	5°37.3	9.9'	26°40.7	-5.4'	54.8'
23	161°34.6	50.1	20°06.2	10.0'	26°35.3	-5.5'	54.8'
SD = 16.2'			d = -0.9'	SD = 15.1'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	05:59	07:18	08:34	15:56	17:11	18:31
N 70°	05:59	07:10	08:17	16:12	17:19	18:30
68°	05:59	07:04	08:03	16:26	17:26	18:31
66°	05:59	06:58	07:52	16:37	17:31	18:31
64°	05:58	06:53	07:43	16:46	17:36	18:31
62°	05:58	06:49	07:35	16:53	17:40	18:31
60°	05:57	06:45	07:29	17:00	17:43	18:32
N 58°	05:56	06:42	07:23	17:06	17:47	18:32
56°	05:56	06:39	07:17	17:11	17:50	18:33
54°	05:55	06:36	07:12	17:16	17:52	18:34
52°	05:54	06:34	07:08	17:20	17:55	18:34
50°	05:54	06:31	07:04	17:24	17:57	18:35
45°	05:51	06:26	06:56	17:33	18:03	18:37
N 40°	05:49	06:21	06:48	17:40	18:07	18:39
35°	05:47	06:17	06:42	17:46	18:12	18:41
30°	05:44	06:12	06:37	17:51	18:16	18:44
20°	05:39	06:05	06:27	18:01	18:23	18:49
N 10°	05:32	05:57	06:19	18:09	18:31	18:55
0°	05:25	05:49	06:10	18:17	18:38	19:03
S 10°	05:16	05:41	06:02	18:25	18:47	19:12
20°	05:04	05:31	05:53	18:34	18:57	19:23
30°	04:49	05:18	05:43	18:44	19:09	19:38
35°	04:39	05:11	05:37	18:50	19:16	19:48
40°	04:28	05:02	05:30	18:57	19:25	19:59
45°	04:13	04:51	05:22	19:05	19:36	20:13
S 50°	03:54	04:38	05:13	19:14	19:49	20:32
52°	03:45	04:31	05:08	19:19	19:55	20:41
54°	03:35	04:24	05:03	19:23	20:02	20:51
56°	03:22	04:16	04:58	19:29	20:10	21:03
58°	03:08	04:07	04:51	19:35	20:19	21:17
S 60°	02:50	03:56	04:44	19:42	20:29	21:34

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	□	□	□	□	□	□
N 70°	□	□	□	□	□	□
68°	□	□	□	□	□	□
66°	□	□	□	□	□	□
64°	□	□	□	□	□	□
62°	08:10	□	10:00	06:27	□	08:26
60°	09:06	09:45	10:58	05:31	06:48	07:27
N 58°	09:39	10:23	11:31	04:58	06:09	06:54
56°	10:04	10:51	11:56	04:34	05:42	06:29
54°	10:24	11:12	12:16	04:14	05:21	06:09
52°	10:40	11:30	12:32	03:58	05:03	05:52
50°	10:55	11:45	12:46	03:44	04:48	05:38
45°	11:24	12:15	13:15	03:15	04:17	05:09
N 40°	11:47	12:39	13:38	02:53	03:54	04:46
35°	12:05	12:59	13:56	02:34	03:34	04:27
30°	12:22	13:15	14:12	02:18	03:17	04:10
20°	12:49	13:44	14:39	01:51	02:49	03:43
N 10°	13:13	14:08	15:03	01:28	02:25	03:19
0°	13:35	14:31	15:24	01:07	02:02	02:57
S 10°	13:58	14:53	15:46	00:45	01:39	02:34
20°	14:22	15:18	16:09	00:22	01:15	02:10
30°	14:50	15:46	16:36	· · ·	00:47	01:42
35°	15:06	16:03	16:51	· · ·	00:30	01:26
40°	15:26	16:22	17:10	· · ·	00:10	01:06
45°	15:49	16:46	17:32	23:47	· · ·	00:43
S 50°	16:20	17:17	18:00	23:16	· · ·	00:12
52°	16:35	17:32	18:13	23:01	23:57	· · ·
54°	16:52	17:50	18:29	22:43	23:40	· · ·
56°	17:14	18:11	18:48	22:22	23:18	· · ·
58°	17:41	18:38	19:10	21:55	22:51	· · ·
S 60°	18:19	19:17	19:40	21:16	22:12	23:40

Day	Sun		Mer. Pass	Moon		Age 9-11 64-82%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
18	13:58	13:56	12:14	19:49	07:21	
19	13:53	13:50	12:14	20:44	08:16	
20	13:48	13:45	12:14	21:37	09:11	

February 21, 22, 23 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	150°25.2	202°37.2	S19°14.7	201°49.3	S19°45.0	112°39.6	N13°52.2	169°22.8	S09°45.7	Alpheratz	357°35.8 29°13.4	
1	165°27.6	217°36.5	14.1	216°49.8	44.5	127°41.7	52.3	184°25.0	45.6	Ankaa	353°08.2 -42°10.7	
2	180°30.1	232°35.8	13.4	231°50.2	44.1	142°43.8	52.4	199°27.2	45.4	Schedar	349°32.3 56°40.3	
3	195°32.6	247°35.0	•• 12.8	246°50.7	•• 43.6	157°45.9	•• 52.5	214°29.3	•• 45.3	Diphda	348°48.3 -17°51.4	
4	210°35.0	262°34.3	12.1	261°51.2	43.1	172°47.9	52.7	229°31.5	45.2	Achernar	335°21.0 -57°07.1	
5	225°37.5	277°33.5	11.5	276°51.7	42.7	187°50.0	52.8	244°33.7	45.1	Hamal	327°52.2 23°34.6	
6	240°40.0	292°32.8	S19°10.8	291°52.1	S19°42.2	202°52.1	N13°52.9	259°35.9	S09°45.0	Polaris	314°31.2 89°22.3	
7	255°42.4	307°32.1	10.2	306°52.6	41.8	217°54.2	53.1	274°38.0	44.9	Acamar	315°12.4 -40°12.7	
8	270°44.9	322°31.3	09.5	321°53.1	41.3	232°56.3	53.2	289°40.2	44.8	Menkar	314°07.0 4°11.0	
9	285°47.4	337°30.6	•• 08.9	336°53.6	•• 40.8	247°58.4	•• 53.3	304°42.4	•• 44.7	Mirfak	308°29.4 49°57.0	
10	300°49.8	352°29.8	08.2	351°54.0	40.4	263°00.5	53.4	319°44.6	44.5	Aldebaran	290°40.4 16°33.5	
11	315°52.3	7°29.1	07.6	6°54.5	39.9	278°02.6	53.6	334°46.8	44.4	Rigel	281°04.5 -8°10.6	
12	330°54.7	22°28.4	S19°06.9	21°55.0	S19°39.4	293°04.7	N13°53.7	349°48.9	S09°44.3	Capella	280°22.8 46°01.5	
13	345°57.2	37°27.6	06.2	36°55.5	39.0	308°06.8	53.8	4°51.1	44.2	Bellatrix	278°23.5 6°22.2	
14	0°59.7	52°26.9	05.6	51°55.9	38.5	323°08.9	54.0	19°53.3	44.1	Elnath	278°02.6 28°37.7	
15	16°02.1	67°26.2	•• 04.9	66°56.4	•• 38.0	338°11.0	•• 54.1	34°55.5	•• 44.0	Alnilam	275°38.3 -1°11.3	
16	31°04.6	82°25.4	04.3	81°56.9	37.6	353°13.0	54.2	49°57.6	43.9	Betelgeuse	270°52.7 7°24.6	
17	46°07.1	97°24.7	03.6	96°57.4	37.1	8°15.1	54.4	64°59.8	43.8	Canopus	263°52.4 -52°42.7	
18	61°09.5	112°24.0	S19°02.9	111°57.9	S19°36.6	23°17.2	N13°54.5	80°02.0	S09°43.6	Sirius	258°26.7 -16°45.1	
19	76°12.0	127°23.2	02.3	126°58.3	36.2	38°19.3	54.6	95°04.2	43.5	Adhara	255°06.2 -29°00.4	
20	91°14.5	142°22.5	01.6	141°58.8	35.7	53°21.4	54.8	110°06.4	43.4	Procyon	244°51.3 5°09.7	
21	106°16.9	157°21.8	•• 01.0	156°59.3	•• 35.2	68°23.5	•• 54.9	125°08.5	•• 43.3	Pollux	243°17.9 27°58.1	
22	121°19.4	172°21.0	19°00.3	171°59.8	34.8	83°25.6	55.0	140°10.7	43.2	Avior	234°14.5 -59°35.3	
23	136°21.9	187°20.3	18°59.6	187°00.2	34.3	98°27.7	55.1	155°12.9	43.1	Suhail	222°46.4 -43°31.9	
Mer.pass. 13:56		ν -0.7' d -0.6' m -3.88		ν 0.5' d -0.5' m 1.29		ν 2.1' d 0.1' m -2.23		ν 2.2' d -0.1' m 0.97		Miaplacidus	221°37.6 -69°49.0	
											Alphard	217°48.1 -8°45.9
											Regulus	207°34.9 11°50.9
											Dubhe	193°41.2 61°37.1
											Denebola	182°25.4 14°26.1
											Gienah	175°44.1 -17°40.6
											Acrux	173°00.4 -63°13.8
											Gacrux	171°52.0 -57°14.8
											Alioth	166°13.1 55°49.5
											Spica	158°22.9 -11°17.3
											Alkaid	152°52.3 49°11.3
											Hadar	148°36.8 -60°29.2
											Menkent	147°58.3 -36°29.2
											Arcturus	145°48.4 19°03.2
											Rigel Kent.	139°41.1 -60°55.9
											Kochab	137°19.4 74°03.0
											Zuben'ubi	136°56.7 -16°08.6
											Alphecca	126°04.3 26°37.7
											Antares	112°16.8 -26°29.1
											Atria	107°11.8 -69°04.0
											Sabik	102°03.7 -15°45.3
											Shaula	96°11.5 -37°07.2
											Rasalhague	95°59.3 12°32.3
											Eltanin	90°42.8 51°28.8
											Kaus Aust.	83°33.7 -34°22.4
											Vega	80°34.0 38°48.0
											Nunki	75°48.9 -26°16.0
											Altair	62°00.9 8°55.7
											Peacock	53°07.4 -56°39.4
											Deneb	49°26.7 45°21.8
											Enif	33°39.8 9°59.0
											Al Na'ir	27°34.3 -46°50.8
											Fomalhaut	15°15.7 -29°29.8
											Scheat	13°46.3 28°12.7
											Markab	13°30.9 15°20.0
Feb 21 Wed SHA Mer.pass												
											Venus	52°12.1 10:30
											Mars	51°24.1 10:32
											Jupiter	322°14.4 16:27
											Saturn	18°57.6 12:41
Feb 22 Thu SHA Mer.pass												
											Venus	50°55.2 10:31
											Mars	50°36.4 10:32
											Jupiter	322°05.4 16:24
											Saturn	18°50.8 12:37
Feb 23 Fri SHA Mer.pass												
											Venus	49°38.7 10:32
											Mars	49°48.8 10:31
											Jupiter	321°56.4 16:20
											Saturn	18°43.9 12:34
Horizontal parallax												
											Venus:	0.1
											Mars:	0.1
Mer.pass. 13:48		ν -0.7' d -0.7' m -3.88		ν 0.5' d -0.5' m 1.28		ν 2.1' d 0.1' m -2.22		ν 2.2' d -0.1' m 0.97				

Sun				Moon				
Wed	GHA	Dec		GHA	ν	Dec	d	HP
0	176°34.7	S10°49.2		34°35.1	10.0'	N26°29.8	-5.7'	54.8'
1	191°34.7	48.3		49°04.2	10.1'	26°24.1	-5.8'	54.8'
2	206°34.8	47.4		63°33.2	10.2'	26°18.3	-5.9'	54.8'
3	221°34.9	· · 46.5		78°02.4	10.2'	26°12.4	-6.0'	54.8'
4	236°34.9	45.6		92°31.7	10.3'	26°06.3	-6.2'	54.7'
5	251°35.0	44.7		107°01.0	10.4'	26°00.2	-6.3'	54.7'
6	266°35.1	S10°43.8		121°30.4	10.5'	N25°53.9	-6.4'	54.7'
7	281°35.1	42.9		135°59.8	10.5'	25°47.5	-6.5'	54.7'
8	296°35.2	42.0		150°29.4	10.6'	25°40.9	-6.7'	54.7'
9	311°35.3	· · 41.1		164°59.0	10.7'	25°34.3	-6.8'	54.7'
10	326°35.3	40.2		179°28.7	10.8'	25°27.5	-6.9'	54.6'
11	341°35.4	39.3		193°58.5	10.9'	25°20.6	-7.0'	54.6'
12	356°35.5	S10°38.4		208°28.3	10.9'	N25°13.6	-7.1'	54.6'
13	11°35.6	37.5		222°58.3	11.0'	25°06.5	-7.2'	54.6'
14	26°35.6	36.6		237°28.3	11.1'	24°59.2	-7.4'	54.6'
15	41°35.7	· · 35.7		251°58.4	11.2'	24°51.9	-7.5'	54.6'
16	56°35.8	34.8		266°28.5	11.2'	24°44.4	-7.6'	54.6'
17	71°35.9	33.9		280°58.8	11.3'	24°36.8	-7.7'	54.5'
18	86°35.9	S10°33.0		295°29.1	11.4'	N24°29.1	-7.8'	54.5'
19	101°36.0	32.1		309°59.5	11.5'	24°21.3	-7.9'	54.5'
20	116°36.1	31.2		324°30.0	11.6'	24°13.4	-8.0'	54.5'
21	131°36.1	· · 30.2		339°00.5	11.6'	24°05.4	-8.1'	54.5'
22	146°36.2	29.3		353°31.2	11.7'	23°57.3	-8.2'	54.5'
23	161°36.3	28.4		8°01.9	11.8'	23°49.0	-8.3'	54.5'
SD = 16.2'			$d = -0.9'$	SD = 14.9'				
Thu	GHA	Dec		GHA	ν	Dec	d	HP
0	176°36.4	S10°27.5		22°32.7	11.9'	N23°40.7	-8.4'	54.5'
1	191°36.4	26.6		37°03.6	12.0'	23°32.3	-8.5'	54.4'
2	206°36.5	25.7		51°34.6	12.0'	23°23.7	-8.6'	54.4'
3	221°36.6	· · 24.8		66°05.6	12.1'	23°15.1	-8.7'	54.4'
4	236°36.7	23.9		80°36.7	12.2'	23°06.3	-8.8'	54.4'
5	251°36.7	23.0		95°07.9	12.3'	22°57.5	-8.9'	54.4'
6	266°36.8	S10°22.1		109°39.2	12.4'	N22°48.6	-9.0'	54.4'
7	281°36.9	21.2		124°10.6	12.4'	22°39.5	-9.1'	54.4'
8	296°37.0	20.3		138°42.0	12.5'	22°30.4	-9.2'	54.4'
9	311°37.1	· · 19.4		153°13.6	12.6'	22°21.1	-9.3'	54.3'
10	326°37.1	18.5		167°45.2	12.7'	22°11.8	-9.4'	54.3'
11	341°37.2	17.5		182°16.8	12.8'	22°02.4	-9.5'	54.3'
12	356°37.3	S10°16.6		196°48.6	12.8'	N21°52.9	-9.6'	54.3'
13	11°37.4	15.7		211°20.4	12.9'	21°43.3	-9.7'	54.3'
14	26°37.4	14.8		225°52.4	13.0'	21°33.6	-9.8'	54.3'
15	41°37.5	· · 13.9		240°24.4	13.1'	21°23.8	-9.9'	54.3'
16	56°37.6	13.0		254°56.4	13.2'	21°13.9	-10.0'	54.3'
17	71°37.7	12.1		269°28.6	13.2'	21°03.9	-10.0'	54.3'
18	86°37.8	S10°11.2		284°00.8	13.3'	N20°53.9	-10.1'	54.2'
19	101°37.8	10.3		298°33.1	13.4'	20°43.8	-10.2'	54.2'
20	116°37.9	09.3		313°05.5	13.5'	20°33.5	-10.3'	54.2'
21	131°38.0	· · 08.4		327°38.0	13.5'	20°23.2	-10.4'	54.2'
22	146°38.1	07.5		342°10.5	13.6'	20°12.9	-10.5'	54.2'
23	161°38.2	06.6		356°43.1	13.7'	20°02.4	-10.5'	54.2'
SD = 16.2'			$d = -0.9'$	SD = 14.8'				
Fri	GHA	Dec		GHA	ν	Dec	d	HP
0	176°38.3	S10°05.7		11°15.8	13.8'	N19°51.8	-10.6'	54.2'
1	191°38.3	04.8		25°48.6	13.8'	19°41.2	-10.7'	54.2'
2	206°38.4	03.9		40°21.4	13.9'	19°30.5	-10.8'	54.2'
3	221°38.5	· · 03.0		54°54.4	14.0'	19°19.7	-10.9'	54.2'
4	236°38.6	02.0		69°27.3	14.1'	19°08.9	-10.9'	54.2'
5	251°38.7	01.1		84°00.4	14.1'	18°57.9	-11.0'	54.1'
6	266°38.7	S10°00.2		98°33.5	14.2'	N18°46.9	-11.1'	54.1'
7	281°38.8	09°59.3		113°06.7	14.3'	18°35.8	-11.2'	54.1'
8	296°38.9	58.4		127°40.0	14.3'	18°24.7	-11.2'	54.1'
9	311°39.0	· · 57.5		142°13.4	14.4'	18°13.5	-11.3'	54.1'
10	326°39.1	56.6		156°46.8	14.5'	18°02.2	-11.4'	54.1'
11	341°39.2	55.6		171°20.3	14.6'	17°50.8	-11.4'	54.1'
12	356°39.3	S09°54.7		185°53.8	14.6'	N17°39.4	-11.5'	54.1'
13	11°39.3	53.8		200°27.4	14.7'	17°27.9	-11.6'	54.1'
14	26°39.4	52.9		215°01.1	14.8'	17°16.3	-11.6'	54.1'
15	41°39.5	· · 52.0		229°34.9	14.8'	17°04.7	-11.7'	54.1'
16	56°39.6	51.1		244°08.7	14.9'	16°53.0	-11.8'	54.1'
17	71°39.7	50.1		258°42.6	15.0'	16°41.2	-11.8'	54.1'
18	86°39.8	S09°49.2		273°16.6	15.0'	N16°29.4	-11.9'	54.0'
19	101°39.9	48.3		287°50.6	15.1'	16°17.5	-11.9'	54.0'
20	116°39.9	47.4		302°24.7	15.1'	16°05.5	-12.0'	54.0'
21	131°40.0	· · 46.5		316°58.8	15.2'	15°53.5	-12.1'	54.0'
22	146°40.1	45.6		331°33.0	15.3'	15°41.5	-12.1'	54.0'
23	161°40.2	44.6		346°07.3	15.3'	15°29.3	-12.2'	54.0'
SD = 16.2'			$d = -0.9'$	SD = 14.8'				
Lat.		Twilight		Sunrise	Sunset	Twilight		
		Naut.	Civil			Civil	Naut.	
N 72°	05:46	07:04	08:17	16:12	17:25	18:44		
N 70°	05:47	06:58	08:03	16:26	17:31	18:42		
68°	05:48	06:52	07:51	16:38	17:36	18:41		
66°	05:49	06:48	07:41	16:47	17:41	18:40		
64°	05:49	06:44	07:33	16:55	17:44	18:40		
62°	05:49	06:41	07:26	17:02	17:48	18:39		
60°	05:49	06:37	07:20	17:08	17:51	18:39		
N 58°	05:49	06:35	07:15	17:13	17:53	18:39		
56°	05:49	06:32	07:10	17:18	17:56	18:39		
54°	05:49	06:30	07:06	17:22	17:58	18:39		
52°	05:49	06:28	07:02	17:26	18:00	18:39		
50°	05:48	06:26	06:58	17:29	18:02	18:40		
45°	05:47	06:21	06:51	17:37	18:07	18:41		
N 40°	05:45	06:17	06:44	17:43	18:11	18:42		
35°	05:44	06:13	06:39	17:49	18:14	18:44		
30°	05:42	06:10	06:34	17:54	18:18	18:46		
20°	05:37	06:03	06:25	18:02	18:24	18:50		
N 10°	05:32	05:56	06:18	18:10	18:31	18:56		
0°	05:25	05:49	06:10	18:17	18:38	19:02		
S 10°	05:16	05:41	06:03	18:24	18:46	19:10		
20°	05:06	05:32	05:55	18:32	18:55	19:21		
30°	04:52	05:21	05:45	18:41	19:06	19:35		
35°	04:43	05:14	05:40	18:47	19:13	19:44		
40°	04:32	05:06	05:34	18:53	19:21	19:54		
45°	04:18	04:56	05:26	19:00	19:30	20:08		
S 50°	04:01	04:43	05:18	19:08	19:42	20:25		
52°	03:52	04:37	05:14	19:12	19:48	20:33		
54°	03:43	04:31	05:09	19:17	19:55	20:43		
56°	03:31	04:24	05:04	19:21	20:02	20:53		
58°	03:18	04:15	04:59	19:27	20:10	21:06		
S 60°	03:02	04:05	04:52	19:33	20:19	21:22		
Lat.		Moonrise			Moonset			
		Wed	Thu	Fri	Wed	Thu	Fri	
N 72°			12:36				10:57	
N 70°			13:46				09:45	
68°		11:12	14:22		10:44		09:08	
66°		12:41	14:47		09:14		08:41	
64°	11:04	13:18	15:07	09:10	08:36		08:20	
62°	11:58	13:45	15:23	08:15	08:09		08:03	
60°	12:31	14:05	15:36	07:42	07:47		07:49	
N 58°	12:55	14:22	15:47	07:18	07:30		07:37	
56°	13:14	14:36	15:57	06:58	07:15		07:27	
54°	13:30	14:48	16:06	06:41	07:03		07:17	
52°	13:44	14:59	16:13	06:27	06:52		07:09	
50°	13:56	15:08	16:20	06:15	06:42		07:01	
45°	14:21	15:28	16:35	05:49	06:21		06:45	
N 40°	14:40	15:44	16:47	05:29	06:04		06:32	
35°	14:57	15:57	16:57	05:12	05:49		06:21	
30°	15:11	16:09	17:06	04:57				

February 24, 25, 26 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	153°22.6	201°44.9	S18°25.8	202°23.9	S19°10.9	115°09.8	N14°01.6	171°59.6	S09°37.5	Alpheratz	357°35.8 29°13.4
1	168°25.1	216°44.2	25.1	217°24.4	10.4	130°11.9	01.7	187°01.8	37.4	Ankaa	353°08.2 -42°10.7
2	183°27.5	231°43.5	24.4	232°24.9	09.9	145°13.9	01.9	202°03.9	37.3	Schedar	349°32.4 56°40.3
3	198°30.0	246°42.8	•• 23.6	247°25.3	•• 09.4	160°16.0	•• 02.0	217°06.1	•• 37.2	Diphda	348°48.3 -17°51.4
4	213°32.5	261°42.1	22.9	262°25.8	09.0	175°18.1	02.1	232°08.3	37.1	Achernar	335°21.1 -57°07.1
5	228°34.9	276°41.4	22.2	277°26.3	08.5	190°20.2	02.3	247°10.5	37.0	Hamal	327°52.2 23°34.6
6	243°37.4	291°40.7	S18°21.5	292°26.8	S19°08.0	205°22.2	N14°02.4	262°12.7	S09°36.8	Polaris	314°32.8 89°22.3
7	258°39.8	306°40.0	20.8	307°27.3	07.5	220°24.3	02.5	277°14.8	36.7	Acamar	315°12.4 -40°12.7
8	273°42.3	321°39.3	20.0	322°27.8	07.0	235°26.4	02.7	292°17.0	36.6	Menkar	314°07.0 4°11.0
9	288°44.8	336°38.6	•• 19.3	337°28.3	•• 06.5	250°28.5	•• 02.8	307°19.2	•• 36.5	Mirfak	308°29.4 49°57.0
10	303°47.2	351°37.9	18.6	352°28.8	06.0	265°30.5	02.9	322°21.4	36.4	Aldebaran	290°40.4 16°33.5
11	318°49.7	6°37.2	17.9	7°29.2	05.5	280°32.6	03.1	337°23.5	36.3	Rigel	281°04.5 -8°10.6
12	333°52.2	21°36.4	S18°17.2	22°29.7	S19°05.0	295°34.7	N14°03.2	352°25.7	S09°36.2	Capella	280°22.8 46°01.5
13	348°54.6	36°35.7	16.4	37°30.2	04.5	310°36.8	03.3	7°27.9	36.0	Bellatrix	278°23.5 6°22.2
14	3°57.1	51°35.0	15.7	52°30.7	04.1	325°38.8	03.5	22°30.1	35.9	Elnath	278°02.7 28°37.7
15	18°59.6	66°34.3	•• 15.0	67°31.2	•• 03.6	340°40.9	•• 03.6	37°32.2	•• 35.8	Alnilam	275°38.3 -1°11.3
16	34°02.0	81°33.6	14.3	82°31.7	03.1	355°43.0	03.7	52°34.4	35.7	Betelgeuse	270°52.7 7°24.6
17	49°04.5	96°32.9	13.5	97°32.2	02.6	10°45.1	03.9	67°36.6	35.6	Canopus	263°52.4 -52°42.7
18	64°06.9	111°32.2	S18°12.8	112°32.7	S19°02.1	25°47.1	N14°04.0	82°38.8	S09°35.5	Sirius	258°26.7 -16°45.1
19	79°09.4	126°31.5	12.1	127°33.1	01.6	40°49.2	04.1	97°41.0	35.4	Adhara	255°06.2 -29°00.5
20	94°11.9	141°30.8	11.3	142°33.6	01.1	55°51.3	04.3	112°43.1	35.3	Procyon	244°51.3 5°09.7
21	109°14.3	156°30.1	•• 10.6	157°34.1	•• 00.6	70°53.4	•• 04.4	127°45.3	•• 35.1	Pollux	243°17.9 27°58.1
22	124°16.8	171°29.4	09.9	172°34.6	19°00.1	85°55.4	04.5	142°47.5	35.0	Avior	234°14.5 -59°35.3
23	139°19.3	186°28.7	09.1	187°35.1	18°59.6	100°57.5	04.7	157°49.7	34.9	Suhail	222°46.4 -43°31.9
Mer.pass. 13:44		ν -0.7' d -0.7' m -3.88		ν 0.5' d -0.5' m 1.28		ν 2.1' d 0.1' m -2.21		ν 2.2' d -0.1' m 0.96		Miaplacidus 221°37.6 -69°49.0	
										Alphard 217°48.1 -8°45.9	
										Regulus 207°34.9 11°50.9	
										Dubhe 193°41.2 61°37.2	
										Denebola 182°25.4 14°26.1	
										Gienah 175°44.0 -17°40.6	
										Acrux 173°00.4 -63°13.8	
										Gacrux 171°52.0 -57°14.8	
										Alioth 166°13.1 55°49.5	
										Spica 158°22.8 -11°17.3	
										Alkaid 152°52.2 49°11.3	
										Hadar 148°36.7 -60°29.2	
										Menkent 147°58.3 -36°29.3	
										Arcturus 145°48.4 19°03.2	
										Rigel Kent. 139°41.0 -60°55.9	
										Kochab 137°19.3 74°03.0	
										Zuben'ubi 136°56.7 -16°08.6	
										Alphecca 126°04.3 26°37.7	
										Antares 112°16.7 -26°29.1	
										Atria 107°11.7 -69°04.0	
										Sabik 102°03.7 -15°45.3	
										Shaula 96°11.5 -37°07.2	
										Rasalhague 95°59.3 12°32.3	
										Eltanin 90°42.7 51°28.8	
										Kaus Aust. 83°33.7 -34°22.4	
										Vega 80°33.9 38°48.0	
										Nunki 75°48.9 -26°16.0	
										Altair 62°00.9 8°55.7	
										Peacock 53°07.3 -56°39.4	
										Deneb 49°26.7 45°21.7	
										Enif 33°39.8 9°59.0	
										Al Na'ir 27°34.2 -46°50.8	
										Fomalhaut 15°15.6 -29°29.8	
										Scheat 13°46.3 28°12.7	
										Markab 13°30.9 15°20.0	
										Feb 24 Sat SHA Mer.pass	
										Venus 48°22.3 10:34	
										Mars 49°01.3 10:30	
										Jupiter 321°47.2 16:17	
										Saturn 18°37.0 12:30	
										Feb 25 Sun SHA Mer.pass	
										Venus 47°06.3 10:35	
										Mars 48°13.9 10:29	
										Jupiter 321°37.9 16:14	
										Saturn 18°30.1 12:27	
										Feb 26 Mon SHA Mer.pass	
										Venus 45°50.5 10:36	
										Mars 47°26.5 10:28	
										Jupiter 321°28.4 16:10	
										Saturn 18°23.2 12:23	
										Horizontal parallax	
										Venus: 0.1	
										Mars: 0.1	
Mer.pass. 13:36		ν -0.7' d -0.8' m -3.88		ν 0.5' d -0.5' m 1.27		ν 2.1' d 0.1' m -2.20		ν 2.2' d -0.1' m 0.96			

Sun								Moon							
Sat	GHA	Dec		GHA	ν	Dec	d	HP							
0	176°40.3	S09°43.7		0°41.6	15.4'	N15°17.1	-12.2'	54.0'							
1	191°40.4	42.8		15°16.0	15.5'	15°04.9	-12.3'	54.0'							
2	206°40.5	41.9		29°50.5	15.5'	14°52.6	-12.4'	54.0'							
3	221°40.6	· · 41.0		44°25.0	15.6'	14°40.3	-12.4'	54.0'							
4	236°40.6	40.0		58°59.5	15.6'	14°27.8	-12.5'	54.0'							
5	251°40.7	39.1		73°34.2	15.7'	14°15.4	-12.5'	54.0'							
6	266°40.8	S09°38.2		88°08.9	15.7'	N14°02.9	-12.6'	54.0'							
7	281°40.9	37.3		102°43.6	15.8'	13°50.3	-12.6'	54.0'							
8	296°41.0	36.4		117°18.4	15.8'	13°37.7	-12.7'	54.0'							
9	311°41.1	· · 35.4		131°53.2	15.9'	13°25.1	-12.7'	54.0'							
10	326°41.2	34.5		146°28.1	15.9'	13°12.3	-12.8'	54.0'							
11	341°41.3	33.6		161°03.1	16.0'	12°59.6	-12.8'	54.0'							
12	356°41.4	S09°32.7		175°38.1	16.1'	N12°46.8	-12.8'	54.0'							
13	11°41.5	31.8		190°13.1	16.1'	12°33.9	-12.9'	54.0'							
14	26°41.6	30.8		204°48.2	16.2'	12°21.1	-12.9'	53.9'							
15	41°41.6	· · 29.9		219°23.4	16.2'	12°08.1	-13.0'	53.9'							
16	56°41.7	29.0		233°58.6	16.2'	11°55.1	-13.0'	53.9'							
17	71°41.8	28.1		248°33.8	16.3'	11°42.1	-13.1'	53.9'							
18	86°41.9	S09°27.1		263°09.1	16.3'	N11°29.1	-13.1'	53.9'							
19	101°42.0	26.2		277°44.5	16.4'	11°15.9	-13.1'	53.9'							
20	116°42.1	25.3		292°19.8	16.4'	11°02.8	-13.2'	53.9'							
21	131°42.2	· · 24.4		306°55.3	16.5'	10°49.6	-13.2'	53.9'							
22	146°42.3	23.4		321°30.7	16.5'	10°36.4	-13.3'	53.9'							
23	161°42.4	22.5		336°06.2	16.6'	10°23.2	-13.3'	53.9'							
SD = 16.2' $d = -0.9'$								SD = 14.7'							
Sun	GHA	Dec		GHA	ν	Dec	d	HP							
0	176°42.5	S09°21.6		350°41.8	16.6'	N10°09.9	-13.3'	53.9'							
1	191°42.6	20.7		5°17.4	16.6'	09°56.5	-13.4'	53.9'							
2	206°42.7	19.7		19°53.0	16.7'	09°43.2	-13.4'	53.9'							
3	221°42.8	· · 18.8		34°28.7	16.7'	09°29.8	-13.4'	53.9'							
4	236°42.9	17.9		49°04.4	16.7'	09°16.3	-13.5'	53.9'							
5	251°43.0	17.0		63°40.1	16.8'	09°02.9	-13.5'	53.9'							
6	266°43.1	S09°16.0		78°15.9	16.8'	N08°49.4	-13.5'	53.9'							
7	281°43.1	15.1		92°51.7	16.8'	08°35.9	-13.6'	53.9'							
8	296°43.2	14.2		107°27.6	16.9'	08°22.3	-13.6'	53.9'							
9	311°43.3	· · 13.3		122°03.5	16.9'	08°08.7	-13.6'	53.9'							
10	326°43.4	12.3		136°39.4	16.9'	07°55.1	-13.6'	53.9'							
11	341°43.5	11.4		151°15.3	17.0'	07°41.5	-13.7'	53.9'							
12	356°43.6	S09°10.5		165°51.3	17.0'	N07°27.8	-13.7'	53.9'							
13	11°43.7	09.5		180°27.3	17.0'	07°14.1	-13.7'	53.9'							
14	26°43.8	08.6		195°03.3	17.1'	07°00.4	-13.7'	53.9'							
15	41°43.9	· · 07.7		209°39.4	17.1'	06°46.7	-13.8'	53.9'							
16	56°44.0	06.8		224°15.5	17.1'	06°32.9	-13.8'	53.9'							
17	71°44.1	05.8		238°51.6	17.1'	06°19.1	-13.8'	53.9'							
18	86°44.2	S09°04.9		253°27.7	17.2'	N06°05.3	-13.8'	53.9'							
19	101°44.3	04.0		268°03.9	17.2'	05°51.5	-13.9'	53.9'							
20	116°44.4	03.0		282°40.0	17.2'	05°37.6	-13.9'	53.9'							
21	131°44.5	· · 02.1		297°16.2	17.2'	05°23.8	-13.9'	53.9'							
22	146°44.6	01.2		311°52.5	17.2'	05°09.9	-13.9'	53.9'							
23	161°44.7	00.3		326°28.7	17.3'	04°56.0	-13.9'	53.9'							
SD = 16.2' $d = -0.9'$								SD = 14.7'							
Mon	GHA	Dec		GHA	ν	Dec	d	HP							
0	176°44.8	S08°59.3		341°04.9	17.3'	N04°42.0	-13.9'	53.9'							
1	191°44.9	58.4		355°41.2	17.3'	04°28.1	-14.0'	53.9'							
2	206°45.0	57.5		10°17.5	17.3'	04°14.1	-14.0'	53.9'							
3	221°45.1	· · 56.5		24°53.8	17.3'	04°00.2	-14.0'	53.9'							
4	236°45.2	55.6		39°30.1	17.3'	03°46.2	-14.0'	53.9'							
5	251°45.3	54.7		54°06.5	17.3'	03°32.2	-14.0'	53.9'							
6	266°45.4	S08°53.7		68°42.8	17.4'	N03°18.2	-14.0'	53.9'							
7	281°45.5	52.8		83°19.2	17.4'	03°04.1	-14.0'	53.9'							
8	296°45.6	51.9		97°55.6	17.4'	02°50.1	-14.0'	53.9'							
9	311°45.7	· · 50.9		112°31.9	17.4'	02°36.1	-14.1'	53.9'							
10	326°45.8	50.0		127°08.3	17.4'	02°22.0	-14.1'	53.9'							
11	341°45.9	49.1		141°44.7	17.4'	02°07.9	-14.1'	53.9'							
12	356°46.0	S08°48.1		156°21.1	17.4'	N01°53.9	-14.1'	53.9'							
13	11°46.1	47.2		170°57.5	17.4'	01°39.8	-14.1'	53.9'							
14	26°46.2	46.3		185°34.0	17.4'	01°25.7	-14.1'	53.9'							
15	41°46.4	· · 45.3		200°10.4	17.4'	01°11.6	-14.1'	53.9'							
16	56°46.5	44.4		214°46.8	17.4'	00°57.5	-14.1'	54.0'							
17	71°46.6	43.5		229°23.2	17.4'	00°43.4	-14.1'	54.0'							
18	86°46.7	S08°42.5		243°59.6	17.4'	N00°29.3	-14.1'	54.0'							
19	101°46.8	41.6		258°36.1	17.4'	00°15.2	-14.1'	54.0'							
20	116°46.9	40.7		273°12.5	17.4'	N00°01.0	-14.1'	54.0'							
21	131°47.0	· · 39.7		287°48.9	17.4'	S00°13.1	-14.1'	54.0'							
22	146°47.1	38.8		302°25.3	17.4'	00°27.2	-14.1'	54.0'							
23	161°47.2	37.9		317°01.7	17.4'	00°41.3	-14.1'	54.0'							
SD = 16.2' $d = -0.9'$								SD = 14.7'							
Twilight								Sunrise							
Lat.	Naut.	Civil	Sunrise	Sunset	Civil	Naut.	Lat.	Naut.	Civil	Sunrise	Sunset	Civil	Naut.		
N 72°	05:32	06:50	08:01	16:27	17:39	18:57	N 70°	05:35	06:45	07:48	16:39	17:43	18:54		
68°	05:37	06:41	07:38	16:49	17:47	18:51	66°	05:38	06:37	07:30	16:58	17:50	18:50		
64°	05:40	06:34	07:23	17:04	17:53	18:48	62°	05:41	06:32	07:17	17:10	17:56	18:47		
60°	05:41	06:29	07:12	17:16	17:58	18:46	N 58°	05:42	06:27	07:07	17:20	18:00	18:46		
56°	05:42	06:25	07:03	17:24	18:02	18:45	54°	05:42	06:23	06:59	17:28	18:04	18:45		
54°	05:42	06:23	06:59	17:28	18:04	18:45	52°	05:43	06:22	06:56	17:31	18:06	18:45		
50°	05:43	06:20	06:53	17:34	18:07	18:45	48°	05:43	06:19	06:46	17:34	18:07	18:45		
46°	05:44	06:16	06:40	17:41	18:11	18:45	N 40°	05:41	06:13	06:40	17:47	18:14	18:45		
38°	05:40	06:10	06:35	17:51	18:17	18:46	36°	05:39	06:07	06:31	17:56	18:20	18:48		
34°	05:35	06:01	06:23	18:03	18:26	18:51	30°	05:30	05:55	06:16	18:10	18:31	18:56		
N 10°	05:30	05:55	06:16	18:10	18:31	18:56	0°	05:24	05:49	06:10	18:16	18:37	19:02		
S 10°	05:17	05:42	06:03	18:23	18:44	19:09	20°	05:07	05:33	05:56	18:30	18:52	19:19		
30°	04:54	05:23	05:48	18:38	19:03	19:31	35°	04:46	05:17	05:43	18:43	19:09	19:40		
40°	04:36	05:09	05:37	18:48	19:16	19:49	45°	04:23	05:00	05:31	18:55	19:25	20:02		
S 50°	04:07	04:49	05:23	19:02	19:36	20:18	52°	03:59	04:43	05:19	19:06	19:41	20:25		
54°	03:50	04													

February 27, 28, 29 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	156°20.0	200°54.9	S17°32.1	202°59.3	S18°35.0	117°38.9	N14°11.3	174°36.3	S09°29.4	Alpheratz	357°35.9	29°13.4
1	171°22.5	215°54.3	31.3	217°59.8	34.5	132°40.9	11.5	189°38.5	29.2	Ankaa	353°08.2	-42°10.7
2	186°24.9	230°53.6	30.5	233°00.3	34.0	147°43.0	11.6	204°40.7	29.1	Schedar	349°32.4	56°40.2
3	201°27.4	245°52.9	•• 29.7	248°00.8	•• 33.5	162°45.0	•• 11.7	219°42.9	•• 29.0	Diphda	348°48.3	-17°51.4
4	216°29.9	260°52.2	29.0	263°01.3	32.9	177°47.1	11.9	234°45.0	28.9	Achernar	335°21.1	-57°07.1
5	231°32.3	275°51.6	28.2	278°01.8	32.4	192°49.2	12.0	249°47.2	28.8	Hamal	327°52.2	23°34.6
6	246°34.8	290°50.9	S17°27.4	293°02.3	S18°31.9	207°51.2	N14°12.1	264°49.4	S09°28.7	Polaris	314°34.3	89°22.2
7	261°37.3	305°50.2	26.6	308°02.8	31.4	222°53.3	12.3	279°51.6	28.6	Acamar	315°12.5	-40°12.7
8	276°39.7	320°49.5	25.8	323°03.3	30.9	237°55.4	12.4	294°53.8	28.5	Menkar	314°07.0	4°11.0
9	291°42.2	335°48.9	•• 25.0	338°03.8	•• 30.4	252°57.4	•• 12.5	309°55.9	•• 28.3	Mirfak	308°29.4	49°57.0
10	306°44.6	350°48.2	24.3	353°04.3	29.9	267°59.5	12.7	324°58.1	28.2	Aldebaran	290°40.4	16°33.5
11	321°47.1	5°47.5	23.5	8°04.8	29.3	283°01.5	12.8	340°00.3	28.1	Rigel	281°04.5	-8°10.6
12	336°49.6	20°46.8	S17°22.7	23°05.3	S18°28.8	298°03.6	N14°13.0	355°02.5	S09°28.0	Capella	280°22.8	46°01.5
13	351°52.0	35°46.2	21.9	38°05.8	28.3	313°05.7	13.1	10°04.6	27.9	Bellatrix	278°23.6	6°22.2
14	6°54.5	50°45.5	21.1	53°06.3	27.8	328°07.7	13.2	25°06.8	27.8	Elnath	278°02.7	28°37.7
15	21°57.0	65°44.8	•• 20.3	68°06.8	•• 27.3	343°09.8	•• 13.4	40°09.0	•• 27.7	Alnilam	275°38.4	-1°11.3
16	36°59.4	80°44.2	19.5	83°07.3	26.8	358°11.8	13.5	55°11.2	27.5	Betelgeuse	270°52.7	7°24.6
17	52°01.9	95°43.5	18.7	98°07.8	26.2	13°13.9	13.6	70°13.3	27.4	Canopus	263°52.5	-52°42.7
18	67°04.4	110°42.8	S17°17.9	113°08.3	S18°25.7	28°16.0	N14°13.8	85°15.5	S09°27.3	Sirius	258°26.7	-16°45.1
19	82°06.8	125°42.1	17.1	128°08.8	25.2	43°18.0	13.9	100°17.7	27.2	Adhara	255°06.2	-29°00.5
20	97°09.3	140°41.5	16.3	143°09.3	24.7	58°20.1	14.1	115°19.9	27.1	Procyon	244°51.3	5°09.7
21	112°11.8	155°40.8	•• 15.6	158°09.8	•• 24.2	73°22.1	•• 14.2	130°22.1	•• 27.0	Pollux	243°17.9	27°58.1
22	127°14.2	170°40.1	14.8	173°10.3	23.7	88°24.2	14.3	145°24.2	26.9	Avior	234°14.5	-59°35.3
23	142°16.7	185°39.5	14.0	188°10.8	23.1	103°26.2	14.5	160°26.4	26.7	Suhail	222°46.4	-43°31.9
Mer.pass. 13:32		ν -0.7' d -0.8' m -3.88		ν 0.5' d -0.5' m 1.27		ν 2.1' d 0.1' m -2.20		ν 2.2' d -0.1' m 0.95				

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	157°19.1	200°38.8	S17°13.2	203°11.3	S18°22.6	118°28.3	N14°14.6	175°28.6	S09°26.6		
1	172°21.6	215°38.1	12.4	218°11.8	22.1	133°30.4	14.7	190°30.8	26.5		
2	187°24.1	230°37.5	11.6	233°12.3	21.6	148°32.4	14.9	205°32.9	26.4		
3	202°26.5	245°36.8	•• 10.8	248°12.8	•• 21.1	163°34.5	•• 15.0	220°35.1	•• 26.3		
4	217°29.0	260°36.1	10.0	263°13.3	20.5	178°36.5	15.2	235°37.3	26.2		
5	232°31.5	275°35.5	09.2	278°13.8	20.0	193°38.6	15.3	250°39.5	26.1		
6	247°33.9	290°34.8	S17°08.4	293°14.3	S18°19.5	208°40.6	N14°15.4	265°41.6	S09°26.0		
7	262°36.4	305°34.1	07.6	308°14.8	19.0	223°42.7	15.6	280°43.8	25.8		
8	277°38.9	320°33.5	06.7	323°15.3	18.5	238°44.8	15.7	295°46.0	25.7		
9	292°41.3	335°32.8	•• 05.9	338°15.8	•• 17.9	253°46.8	•• 15.9	310°48.2	•• 25.6		
10	307°43.8	350°32.2	05.1	353°16.3	17.4	268°48.9	16.0	325°50.4	25.5		
11	322°46.2	5°31.5	04.3	8°16.8	16.9	283°50.9	16.1	340°52.5	25.4		
12	337°48.7	20°30.8	S17°03.5	23°17.3	S18°16.4	298°53.0	N14°16.3	355°54.7	S09°25.3		
13	352°51.2	35°30.2	02.7	38°17.8	15.8	313°55.0	16.4	10°56.9	25.2		
14	7°53.6	50°29.5	01.9	53°18.3	15.3	328°57.1	16.5	25°59.1	25.0		
15	22°56.1	65°28.8	•• 01.1	68°18.8	•• 14.8	343°59.1	•• 16.7	41°01.2	•• 24.9		
16	37°58.6	80°28.2	17°00.3	83°19.3	14.3	359°01.2	16.8	56°03.4	24.8		
17	53°01.0	95°27.5	16°59.5	98°19.8	13.7	14°03.3	17.0	71°05.6	24.7		
18	68°03.5	110°26.9	S16°58.7	113°20.3	S18°13.2	29°05.3	N14°17.1	86°07.8	S09°24.6		
19	83°06.0	125°26.2	57.8	128°20.8	12.7	44°07.4	17.2	101°09.9	24.5		
20	98°08.4	140°25.5	57.0	143°21.3	12.2	59°09.4	17.4	116°12.1	24.4		
21	113°10.9	155°24.9	•• 56.2	158°21.8	•• 11.6	74°11.5	•• 17.5	131°14.3	•• 24.3		
22	128°13.4	170°24.2	55.4	173°22.3	11.1	89°13.5	17.7	146°16.5	24.1		
23	143°15.8	185°23.6	54.6	188°22.8	10.6	104°15.6	17.8	161°18.7	24.0		
Mer.pass. 13:29		ν -0.7' d -0.8' m -3.88		ν 0.5' d -0.5' m 1.27		ν 2.1' d 0.1' m -2.19		ν 2.2' d -0.1' m 0.95			

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	158°18.3	200°22.9	S16°53.8	203°23.3	S18°10.1	119°17.6	N14°17.9	176°20.8	S09°23.9		
1	173°20.7	215°22.3	52.9	218°23.8	09.5	134°19.7	18.1	191°23.0	23.8		
2	188°23.2	230°21.6	52.1	233°24.3	09.0	149°21.7	18.2	206°25.2	23.7		
3	203°25.7	245°21.0	•• 51.3	248°24.8	•• 08.5	164°23.8	•• 18.3	221°27.4	•• 23.6		
4	218°28.1	260°20.3	50.5	263°25.3	08.0	179°25.8	18.5	236°29.5	23.5		
5	233°30.6	275°19.6	49.7	278°25.9	07.4	194°27.9	18.6	251°31.7	23.3		
6	248°33.1	290°19.0	S16°48.8	293°26.4	S18°06.9	209°29.9	N14°18.8	266°33.9	S09°23.2		
7	263°35.5	305°18.3	48.0	308°26.9	06.4	224°32.0	18.9	281°36.1	23.1		
8	278°38.0	320°17.7	47.2	323°27.4	05.8	239°34.1	19.0	296°38.2	23.0		
9	293°40.5	335°17.0	•• 46.4	338°27.9	•• 05.3	254°36.1	•• 19.2	311°40.4	•• 22.9		
10	308°42.9	350°16.4	45.5	353°28.4	04.8	269°38.2	19.3	326°42.6	22.8		
11	323°45.4	5°15.7	44.7	8°28.9	04.2	284°40.2	19.5	341°44.8	22.7		
12	338°47.8	20°15.1	S16°43.9	23°29.4	S18°03.7	299°42.3	N14°19.6	356°47.0	S09°22.5		
13	353°50.3	35°14.4	43.0	38°29.9	03.2	314°44.3	19.7	11°49.1	22.4		
14	8°52.8	50°13.8	42.2	53°30.4	02.7	329°46.4	19.9	26°51.3	22.3		
15	23°55.2	65°13.1	•• 41.4	68°30.9	•• 02.1	344°48.4	•• 20.0	41°53.5	•• 22.2		
16	38°57.7	80°12.5	40.6	83°31.4	01.6	359°50.5	20.2	56°55.7	22.1		
17	54°00.2	95°11.8	39.7	98°31.9	01.1	14°52.5	20.3	71°57.8	22.0		
18	69°02.6	110°11.2	S16°38.9	113°32.4	S18°00.5	29°54.6	N14°20.4	87°00.0	S09°21.9		
19	84°05.1	125°10.5	38.1	128°32.9	18°00.0	44°56.6	20.6	102°02.2	21.8		
20	99°07.6	140°09.9	37.2	143°33.5	17°59.5	59°58.7	20.7	117°04.4	21.6		
21	114°10.0	155°09.2	•• 36.4	158°34.0	•• 58.9	75°00.7	•• 20.9	132°06.5	•• 21.5		
22	129°12.5	170°08.6	35.5	173°34.5	58.4	90°02.8	21.0	147°08.7	21.4		
23	144°15.0	185°07.9	34.7	188°35.0	57.9	105°04.8	21.1	162°10.9	21.3		
Mer.pass. 13:25		ν -0.7' d -0.8' m -3.88		ν 0.5' d -0.5' m 1.27		ν 2.1' d 0.1' m -2.19		ν 2.2' d -0.1' m 0.95			

Feb 27 Tue		SHA	Mer.pass
Venus		44°34.9	10:37
Mars		46°39.3	10:28
Jupiter		321°18.8	16:07
Saturn		18°16.3	12:20

Feb 28 Wed		SHA	Mer.pass
Venus		43°19.7	10:38
Mars		45°52.1	10:27
Jupiter		321°09.2	16:04
Saturn		18°09.4	12:16

Feb 29 Thu		SHA	Mer.pass
Venus		42°04.6	10:39
Mars		45°05.0	10:26
Jupiter		320°59.4	16:01
Saturn		18°02.5	12:13

Horizontal parallax		
Venus: 0.1		
Mars: 0.1		

h	Sun			Moon			
Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	176°47.3	S08°36.9	331°38.1	17.4'	S00°55.4	14.1'	54.0'
1	191°47.4	36.0	346°14.5	17.4'	01°09.5	14.1'	54.0'
2	206°47.5	35.0	0°50.9	17.4'	01°23.7	14.1'	54.0'
3	221°47.6	· · 34.1	15°27.3	17.4'	01°37.8	14.1'	54.0'
4	236°47.7	33.2	30°03.6	17.4'	01°51.9	14.1'	54.0'
5	251°47.8	32.2	44°40.0	17.3'	02°06.0	14.1'	54.0'
6	266°47.9	S08°31.3	59°16.4	17.3'	S02°20.1	14.1'	54.0'
7	281°48.0	30.4	73°52.7	17.3'	02°34.2	14.1'	54.0'
8	296°48.2	29.4	88°29.0	17.3'	02°48.3	14.1'	54.0'
9	311°48.3	· · 28.5	103°05.3	17.3'	03°02.4	14.1'	54.0'
10	326°48.4	27.5	117°41.6	17.3'	03°16.4	14.1'	54.1'
11	341°48.5	26.6	132°17.9	17.3'	03°30.5	14.1'	54.1'
12	356°48.6	S08°25.7	146°54.1	17.2'	S03°44.5	14.0'	54.1'
13	11°48.7	24.7	161°30.4	17.2'	03°58.6	14.0'	54.1'
14	26°48.8	23.8	176°06.6	17.2'	04°12.6	14.0'	54.1'
15	41°48.9	· · 22.9	190°42.8	17.2'	04°26.7	14.0'	54.1'
16	56°49.0	21.9	205°18.9	17.1'	04°40.7	14.0'	54.1'
17	71°49.1	21.0	219°55.1	17.1'	04°54.7	14.0'	54.1'
18	86°49.2	S08°20.0	234°31.2	17.1'	S05°08.6	14.0'	54.1'
19	101°49.4	19.1	249°07.3	17.1'	05°22.6	14.0'	54.1'
20	116°49.5	18.2	263°43.4	17.0'	05°36.6	13.9'	54.1'
21	131°49.6	· · 17.2	278°19.4	17.0'	05°50.5	13.9'	54.1'
22	146°49.7	16.3	292°55.4	17.0'	06°04.4	13.9'	54.2'
23	161°49.8	15.3	307°31.4	17.0'	06°18.3	13.9'	54.2'
SD = 16.1'			SD = 14.7'				

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	176°49.9	S08°14.4	322°07.4	16.9'	S06°32.2	13.9'	54.2'
1	191°50.0	13.4	336°43.3	16.9'	06°46.1	13.8'	54.2'
2	206°50.1	12.5	351°19.2	16.9'	06°59.9	13.8'	54.2'
3	221°50.2	· · 11.6	5°55.1	16.8'	07°13.8	13.8'	54.2'
4	236°50.4	10.6	20°30.9	16.8'	07°27.6	13.8'	54.2'
5	251°50.5	09.7	35°06.7	16.7'	07°41.4	13.8'	54.2'
6	266°50.6	S08°08.7	49°42.4	16.7'	S07°55.1	13.7'	54.2'
7	281°50.7	07.8	64°18.1	16.7'	08°08.9	13.7'	54.2'
8	296°50.8	06.8	78°53.8	16.6'	08°22.6	13.7'	54.3'
9	311°50.9	· · 05.9	93°29.4	16.6'	08°36.3	13.7'	54.3'
10	326°51.0	05.0	108°05.0	16.5'	08°49.9	13.6'	54.3'
11	341°51.2	04.0	122°40.5	16.5'	09°03.6	13.6'	54.3'
12	356°51.3	S08°03.1	137°16.0	16.5'	S09°17.2	13.6'	54.3'
13	11°51.4	02.1	151°51.5	16.4'	09°30.7	13.5'	54.3'
14	26°51.5	01.2	166°26.9	16.4'	09°44.3	13.5'	54.3'
15	41°51.6	08°00.2	181°02.3	16.3'	09°57.8	13.5'	54.3'
16	56°51.7	07°59.3	195°37.6	16.3'	10°11.3	13.5'	54.4'
17	71°51.8	58.4	210°12.9	16.2'	10°24.7	13.4'	54.4'
18	86°52.0	S07°57.4	224°48.1	16.2'	S10°38.2	13.4'	54.4'
19	101°52.1	56.5	239°23.2	16.1'	10°51.6	13.4'	54.4'
20	116°52.2	55.5	253°58.4	16.1'	11°04.9	13.3'	54.4'
21	131°52.3	· · 54.6	268°33.4	16.0'	11°18.2	13.3'	54.4'
22	146°52.4	53.6	283°08.4	16.0'	11°31.5	13.2'	54.4'
23	161°52.5	52.7	297°43.4	15.9'	11°44.8	13.2'	54.5'
SD = 16.1'			SD = 14.8'				

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	176°52.7	S07°51.7	312°18.3	15.8'	S11°58.0	13.2'	54.5'
1	191°52.8	50.8	326°53.1	15.8'	12°11.1	13.1'	54.5'
2	206°52.9	49.8	341°27.9	15.7'	12°24.3	13.1'	54.5'
3	221°53.0	· · 48.9	356°02.6	15.7'	12°37.4	13.0'	54.5'
4	236°53.1	47.9	10°37.3	15.6'	12°50.4	13.0'	54.5'
5	251°53.2	47.0	25°11.9	15.5'	13°03.4	13.0'	54.6'
6	266°53.4	S07°46.1	39°46.4	15.5'	S13°16.4	12.9'	54.6'
7	281°53.5	45.1	54°20.9	15.4'	13°29.3	12.9'	54.6'
8	296°53.6	44.2	68°55.3	15.3'	13°42.2	12.8'	54.6'
9	311°53.7	· · 43.2	83°29.6	15.3'	13°55.0	12.8'	54.6'
10	326°53.8	42.3	98°03.9	15.2'	14°07.8	12.7'	54.6'
11	341°54.0	41.3	112°38.1	15.1'	14°20.5	12.7'	54.7'
12	356°54.1	S07°40.4	127°12.2	15.1'	S14°33.2	12.6'	54.7'
13	11°54.2	39.4	141°46.3	15.0'	14°45.8	12.6'	54.7'
14	26°54.3	38.5	156°20.3	14.9'	14°58.4	12.5'	54.7'
15	41°54.4	· · 37.5	170°54.2	14.8'	15°11.0	12.5'	54.7'
16	56°54.6	36.6	185°28.0	14.8'	15°23.4	12.4'	54.7'
17	71°54.7	35.6	200°01.8	14.7'	15°35.9	12.4'	54.8'
18	86°54.8	S07°34.7	214°35.5	14.6'	S15°48.2	12.3'	54.8'
19	101°54.9	33.7	229°09.1	14.5'	16°00.6	12.3'	54.8'
20	116°55.0	32.8	243°42.7	14.5'	16°12.8	12.2'	54.8'
21	131°55.2	· · 31.8	258°16.2	14.4'	16°25.0	12.1'	54.8'
22	146°55.3	30.9	272°49.5	14.3'	16°37.2	12.1'	54.9'
23	161°55.4	29.9	287°22.8	14.2'	16°49.3	12.0'	54.9'
SD = 16.1'			SD = 14.9'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	05:17	06:35	07:45	16:42	17:52	19:10
N 70°	05:22	06:32	07:34	16:53	17:55	19:06
68°	05:25	06:29	07:26	17:01	17:58	19:02
66°	05:28	06:27	07:19	17:08	18:00	18:59
64°	05:30	06:25	07:13	17:14	18:02	18:57
62°	05:32	06:23	07:08	17:19	18:04	18:55
60°	05:33	06:21	07:03	17:23	18:05	18:54
N 58°	05:34	06:19	06:59	17:27	18:07	18:52
56°	05:35	06:18	06:55	17:31	18:08	18:51
54°	05:36	06:17	06:52	17:34	18:10	18:51
52°	05:36	06:15	06:49	17:37	18:11	18:50
50°	05:37	06:14	06:47	17:39	18:12	18:49
45°	05:37	06:11	06:41	17:45	18:15	18:49
N 40°	05:37	06:09	06:36	17:50	18:17	18:49
35°	05:37	06:06	06:31	17:54	18:20	18:49
30°	05:36	06:04	06:28	17:58	18:22	18:50
20°	05:33	05:59	06:21	18:04	18:27	18:52
N 10°	05:29	05:54	06:15	18:10	18:31	18:56
0°	05:24	05:48	06:09	18:16	18:37	19:01
S 10°	05:17	05:42	06:03	18:22	18:43	19:08
20°	05:09	05:35	05:57	18:28	18:50	19:16
30°	04:57	05:25	05:50	18:35	18:59	19:28
35°	04:49	05:20	05:45	18:39	19:05	19:35
40°	04:40	05:13	05:41	18:44	19:12	19:45
45°	04:28	05:05	05:35	18:49	19:20	19:56
S 50°	04:13	04:54	05:28	18:56	19:30	20:11
52°	04:06	04:49	05:25	18:59	19:34	20:18
54°	03:58	04:44	05:21	19:03	19:40	20:26
56°	03:48	04:38	05:17	19:06	19:46	20:35
58°	03:37	04:31	05:13	19:11	19:52	20:45
S 60°	03:25	04:23	05:08	19:15	20:00	20:58

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	21:45	23:55	· · ·	07:31	06:57	06:12
N 70°	21:36	23:32	· · ·	07:35	07:09	06:37
68°	21:28	23:15	· · ·	07:38	07:18	06:56
66°	21:22	23:02	· · ·	07:40	07:27	07:11
64°	21:17	22:50	· · ·	07:42	07:33	07:24
62°	21:13	22:41	· · ·	07:44	07:39	07:34
60°	21:09	22:33	· · ·	07:46	07:45	07:44
N 58°	21:05	22:26	23:50	07:47	07:49	07:52
56°	21:02	22:20	23:40	07:49	07:53	07:59
54°	21:00	22:14	23:31	07:50	07:57	08:05
52°	20:57	22:09	23:23	07:51	08:00	08:11
50°	20:55	22:05	23:16	07:52	08:03	08:16
45°	20:50	21:55	23:02	07:54	08:10	08:28
N 40°	20:46	21:47	22:49	07:56	08:16	08:37
35°	20:43	21:40	22:39	07:58	08:21	08:45
30°	20:40	21:34	22:30	07:59	08:25	08:53
20°	20:35	21:24	22:15	08:02	08:32	09:05
N 10°	20:30	21:15	22:01	08:04	08:39	09:16
0°	20:26	21:06	21:49	08:06	08:45	09:26
S 10°	20:22	20:58	21:36	08:08	08:52	09:37
20°	20:18	20:49	21:23	08:10	08:58	09:48
30°	20:13	20:39	21:08	08:13	09:06	10:01
35°	20:10	20:34	21:00	08:14	09:10	10:08
40°	20:07	20:27	20:50	08:16	09:15	10:17
45°	20:03	20:20	20:39	08:18	09:21	10:27
S 50°	19:59	20:11	20:25	08:20	09:28	10:39
52°	19:57	20:07	20:19	08:21	09:32	10:44
54°	19:55	20:02	20:12	08:22	09:35	10:51
56°	19:52	19:57	20:04	08:23	09:39	10:58
58°	19:50	19:52	19:55	08:25	09:44	11:06
S 60°	19:47	19:46	19:45	08:26	09:49	11:15

Day	Sun		Mer. Pass	Moon		Age 18-20 95-83%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
27	12:51	12:46	12:13	01:57	14:16	
28	12:40	12:35	12:13	02:36	14:56	
29	12:29	12:24	12:12	03:16	15:37	

March 01, 02, 03 UT (Fri., Sat., Sun.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	159°17.4	200°07.3	S16°33.9	203°35.5	S17°57.3	120°06.8	N14°21.3	177°13.1	S09°21.2	Alpheratz	357°35.9 29°13.3
1	174°19.9	215°06.7	33.0	218°36.0	56.8	135°08.9	21.4	192°15.3	21.1	Ankaa	353°08.2 -42°10.7
2	189°22.3	230°06.0	32.2	233°36.5	56.2	150°10.9	21.6	207°17.4	21.0	Schedar	349°32.4 56°40.2
3	204°24.8	245°05.4	· · 31.4	248°37.0	· · 55.7	165°13.0	· · 21.7	222°19.6	· · 20.8	Diphda	348°48.3 -17°51.4
4	219°27.3	260°04.7	30.5	263°37.5	55.2	180°15.0	21.8	237°21.8	20.7	Achernar	335°21.1 -57°07.1
5	234°29.7	275°04.1	29.7	278°38.0	54.6	195°17.1	22.0	252°24.0	20.6	Hamal	327°52.2 23°34.6
6	249°32.2	290°03.4	S16°28.8	293°38.5	S17°54.1	210°19.1	N14°22.1	267°26.1	S09°20.5	Polaris	314°35.6 89°22.2
7	264°34.7	305°02.8	28.0	308°39.1	53.6	225°21.2	22.3	282°28.3	20.4	Acamar	315°12.5 -40°12.7
8	279°37.1	320°02.2	27.1	323°39.6	53.0	240°23.2	22.4	297°30.5	20.3	Menkar	314°07.0 4°11.0
9	294°39.6	335°01.5	· · 26.3	338°40.1	· · 52.5	255°25.3	· · 22.5	312°32.7	· · 20.2	Mirfak	308°29.4 49°57.0
10	309°42.1	350°00.9	25.4	353°40.6	52.0	270°27.3	22.7	327°34.8	20.1	Aldebaran	290°40.4 16°33.5
11	324°44.5	5°00.2	24.6	8°41.1	51.4	285°29.4	22.8	342°37.0	19.9	Rigel	281°04.5 -8°10.6
12	339°47.0	19°59.6	S16°23.8	23°41.6	S17°50.9	300°31.4	N14°23.0	357°39.2	S09°19.8	Capella	280°22.9 46°01.5
13	354°49.5	34°59.0	22.9	38°42.1	50.3	315°33.5	23.1	12°41.4	19.7	Bellatrix	278°23.6 6°22.2
14	9°51.9	49°58.3	22.1	53°42.6	49.8	330°35.5	23.2	27°43.6	19.6	Elnath	278°02.7 28°37.7
15	24°54.4	64°57.7	· · 21.2	68°43.1	· · 49.3	345°37.5	· · 23.4	42°45.7	· · 19.5	Alnilam	275°38.4 -1°11.3
16	39°56.8	79°57.0	20.4	83°43.6	48.7	0°39.6	23.5	57°47.9	19.4	Betelgeuse	270°52.8 7°24.6
17	54°59.3	94°56.4	19.5	98°44.2	48.2	15°41.6	23.7	72°50.1	19.3	Canopus	263°52.5 -52°42.7
18	70°01.8	109°55.8	S16°18.7	113°44.7	S17°47.6	30°43.7	N14°23.8	87°52.3	S09°19.1	Sirius	258°26.7 -16°45.1
19	85°04.2	124°55.1	17.8	128°45.2	47.1	45°45.7	23.9	102°54.4	19.0	Adhara	255°06.2 -29°00.5
20	100°06.7	139°54.5	16.9	143°45.7	46.6	60°47.8	24.1	117°56.6	18.9	Procyon	244°51.4 5°09.7
21	115°09.2	154°53.9	· · 16.1	158°46.2	· · 46.0	75°49.8	· · 24.2	132°58.8	· · 18.8	Pollux	243°17.9 27°58.1
22	130°11.6	169°53.2	15.2	173°46.7	45.5	90°51.9	24.4	148°01.0	18.7	Avior	234°14.5 -59°35.3
23	145°14.1	184°52.6	14.4	188°47.2	44.9	105°53.9	24.5	163°03.1	18.6	Suhail	222°46.5 -43°31.9
Mer.pass. 13:21		ν -0.6' d -0.8' m -3.88		ν 0.5' d -0.5' m 1.27		ν 2.0' d 0.1' m -2.18		ν 2.2' d -0.1' m 0.96		Miaplacidus	221°37.6 -69°49.0
Alphard	217°48.2 -8°45.9										
Regulus	207°34.9 11°50.9										
Dubhe	193°41.2 61°37.2										
Denebola	182°25.4 14°26.1										
Gienah	175°44.0 -17°40.6										
Acrux	173°00.3 -63°13.9										
Gacrux	171°52.0 -57°14.8										
Alioth	166°13.1 55°49.5										
Spica	158°22.8 -11°17.3										
Alkaid	152°52.2 49°11.3										
Hadar	148°36.7 -60°29.2										
Menkent	147°58.2 -36°29.3										
Arcturus	145°48.4 19°03.2										
Rigel Kent.	139°41.0 -60°55.9										
Kochab	137°19.2 74°03.0										
Zuben'ubi	136°56.7 -16°08.6										
Alphecca	126°04.2 26°37.7										
Antares	112°16.7 -26°29.1										
Atria	107°11.6 -69°04.0										
Sabik	102°03.6 -15°45.3										
Shaula	96°11.4 -37°07.2										
Rasalhague	95°59.3 12°32.3										
Eltanin	90°42.7 51°28.7										
Kaus Aust.	83°33.6 -34°22.3										
Vega	80°33.9 38°48.0										
Nunki	75°48.8 -26°16.0										
Altair	62°00.9 8°55.7										
Peacock	53°07.3 -56°39.4										
Deneb	49°26.7 45°21.7										
Enif	33°39.8 9°59.0										
Al Na'ir	27°34.2 -46°50.7										
Fomalhaut	15°15.6 -29°29.8										
Scheat	13°46.3 28°12.7										
Markab	13°30.9 15°20.0										
Mar 01 Fri		SHA		Mer.pass							
Venus		40°49.9		10:40							
Mars		44°18.1		10:25							
Jupiter		320°49.4		15:57							
Saturn		17°55.7		12:09							
Mar 02 Sat		SHA		Mer.pass							
Venus		39°35.4		10:41							
Mars		43°31.2		10:24							
Jupiter		320°39.4		15:54							
Saturn		17°48.8		12:06							
Mar 03 Sun		SHA		Mer.pass							
Venus		38°21.2		10:42							
Mars		42°44.4		10:24							
Jupiter		320°29.2		15:51							
Saturn		17°41.9		12:02							
Horizontal parallax											
Venus:										0.1	
Mars:										0.1	

Sun				Moon				
Fri	GHA	Dec		GHA	ν	Dec	d	HP
0	176°55.5	S07°29.0		301°56.1	14.1'	S17°01.3	12.0'	54.9'
1	191°55.7	28.0		316°29.2	14.1'	17°13.2	11.9'	54.9'
2	206°55.8	27.1		331°02.3	14.0'	17°25.1	11.8'	54.9'
3	221°55.9	· · 26.1		345°35.3	13.9'	17°37.0	11.8'	55.0'
4	236°56.0	25.2		0°08.1	13.8'	17°48.7	11.7'	55.0'
5	251°56.1	24.2		14°41.0	13.7'	18°00.5	11.6'	55.0'
6	266°56.3	S07°23.3		29°13.7	13.6'	S18°12.1	11.6'	55.0'
7	281°56.4	22.3		43°46.3	13.5'	18°23.7	11.5'	55.0'
8	296°56.5	21.3		58°18.9	13.5'	18°35.2	11.4'	55.1'
9	311°56.6	· · 20.4		72°51.3	13.4'	18°46.6	11.4'	55.1'
10	326°56.8	19.4		87°23.7	13.3'	18°58.0	11.3'	55.1'
11	341°56.9	18.5		101°55.9	13.2'	19°09.2	11.2'	55.1'
12	356°57.0	S07°17.5		116°28.1	13.1'	S19°20.5	11.1'	55.2'
13	11°57.1	16.6		131°00.2	13.0'	19°31.6	11.1'	55.2'
14	26°57.3	15.6		145°32.2	12.9'	19°42.7	11.0'	55.2'
15	41°57.4	· · 14.7		160°04.1	12.8'	19°53.6	10.9'	55.2'
16	56°57.5	13.7		174°35.9	12.7'	20°04.6	10.8'	55.3'
17	71°57.6	12.8		189°07.6	12.6'	20°15.4	10.7'	55.3'
18	86°57.8	S07°11.8		203°39.2	12.5'	S20°26.1	10.7'	55.3'
19	101°57.9	10.9		218°10.7	12.4'	20°36.8	10.6'	55.3'
20	116°58.0	09.9		232°42.2	12.3'	20°47.4	10.5'	55.4'
21	131°58.1	· · 09.0		247°13.5	12.2'	20°57.9	10.4'	55.4'
22	146°58.3	08.0		261°44.7	12.1'	21°08.3	10.3'	55.4'
23	161°58.4	07.0		276°15.8	12.0'	21°18.6	10.2'	55.4'
SD = 16.1' d = -1.0'				SD = 15.0'				
Sat	GHA	Dec		GHA	ν	Dec	d	HP
0	176°58.5	S07°06.1		290°46.8	11.9'	S21°28.9	10.1'	55.5'
1	191°58.7	05.1		305°17.7	11.8'	21°39.0	10.1'	55.5'
2	206°58.8	04.2		319°48.6	11.7'	21°49.1	10.0'	55.5'
3	221°58.9	· · 03.2		334°19.3	11.6'	21°59.0	9.9'	55.5'
4	236°59.0	02.3		348°49.9	11.5'	22°08.9	9.8'	55.6'
5	251°59.2	01.3		3°20.4	11.4'	22°18.7	9.7'	55.6'
6	266°59.3	S07°00.3		17°50.8	11.3'	S22°28.4	9.6'	55.6'
7	281°59.4	06°59.4		32°21.1	11.2'	22°38.0	9.5'	55.6'
8	296°59.6	58.4		46°51.3	11.1'	22°47.4	9.4'	55.7'
9	311°59.7	· · 57.5		61°21.4	11.0'	22°56.8	9.3'	55.7'
10	326°59.8	56.5		75°51.3	10.9'	23°06.1	9.2'	55.7'
11	341°59.9	55.6		90°21.2	10.8'	23°15.3	9.1'	55.8'
12	357°00.1	S06°54.6		104°51.0	10.7'	S23°24.4	9.0'	55.8'
13	12°00.2	53.6		119°20.6	10.5'	23°33.4	8.9'	55.8'
14	27°00.3	52.7		133°50.2	10.4'	23°42.2	8.8'	55.8'
15	42°00.5	· · 51.7		148°19.6	10.3'	23°51.0	8.7'	55.9'
16	57°00.6	50.8		162°49.0	10.2'	23°59.7	8.5'	55.9'
17	72°00.7	49.8		177°18.2	10.1'	24°08.2	8.4'	55.9'
18	87°00.9	S06°48.9		191°47.3	10.0'	S24°16.6	8.3'	56.0'
19	102°01.0	47.9		206°16.3	9.9'	24°25.0	8.2'	56.0'
20	117°01.1	46.9		220°45.2	9.8'	24°33.2	8.1'	56.0'
21	132°01.2	· · 46.0		235°14.0	9.7'	24°41.3	8.0'	56.1'
22	147°01.4	45.0		249°42.7	9.6'	24°49.2	7.9'	56.1'
23	162°01.5	44.1		264°11.3	9.5'	24°57.1	7.7'	56.1'
SD = 16.1' d = -1.0'				SD = 15.1'				
Sun	GHA	Dec		GHA	ν	Dec	d	HP
0	177°01.6	S06°43.1		278°39.7	9.4'	S25°04.8	7.6'	56.2'
1	192°01.8	42.1		293°08.1	9.2'	25°12.5	7.5'	56.2'
2	207°01.9	41.2		307°36.3	9.1'	25°20.0	7.4'	56.2'
3	222°02.0	· · 40.2		322°04.5	9.0'	25°27.3	7.2'	56.3'
4	237°02.2	39.3		336°32.5	8.9'	25°34.6	7.1'	56.3'
5	252°02.3	38.3		351°00.4	8.8'	25°41.7	7.0'	56.3'
6	267°02.4	S06°37.3		5°28.2	8.7'	S25°48.7	6.9'	56.3'
7	282°02.6	36.4		19°55.9	8.6'	25°55.5	6.7'	56.4'
8	297°02.7	35.4		34°23.5	8.5'	26°02.3	6.6'	56.4'
9	312°02.8	· · 34.5		48°51.0	8.4'	26°08.9	6.5'	56.4'
10	327°03.0	33.5		63°18.4	8.3'	26°15.3	6.3'	56.5'
11	342°03.1	32.5		77°45.7	8.2'	26°21.7	6.2'	56.5'
12	357°03.2	S06°31.6		92°12.9	8.1'	S26°27.9	6.1'	56.5'
13	12°03.4	30.6		106°39.9	8.0'	26°33.9	5.9'	56.6'
14	27°03.5	29.7		121°06.9	7.9'	26°39.8	5.8'	56.6'
15	42°03.6	· · 28.7		135°33.8	7.8'	26°45.6	5.6'	56.7'
16	57°03.8	27.7		150°00.5	7.7'	26°51.3	5.5'	56.7'
17	72°03.9	26.8		164°27.2	7.6'	26°56.8	5.3'	56.7'
18	87°04.1	S06°25.8		178°53.7	7.5'	S27°02.1	5.2'	56.8'
19	102°04.2	24.8		193°20.2	7.4'	27°07.3	5.1'	56.8'
20	117°04.3	23.9		207°46.6	7.3'	27°12.4	4.9'	56.8'
21	132°04.5	· · 22.9		222°12.8	7.2'	27°17.3	4.8'	56.9'
22	147°04.6	22.0		236°39.0	7.1'	27°22.0	4.6'	56.9'
23	162°04.7	21.0		251°05.0	7.0'	27°26.6	4.5'	56.9'
SD = 16.1' d = -1.0'				SD = 15.3'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	05:03	06:21	07:29	16:57	18:06	19:24
N 70°	05:08	06:19	07:20	17:05	18:07	19:18
68°	05:13	06:17	07:13	17:12	18:09	19:13
66°	05:17	06:16	07:07	17:18	18:10	19:09
64°	05:20	06:15	07:02	17:23	18:11	19:06
62°	05:22	06:14	06:58	17:27	18:12	19:03
60°	05:25	06:13	06:54	17:31	18:13	19:01
N 58°	05:26	06:12	06:51	17:34	18:14	18:59
56°	05:28	06:11	06:48	17:37	18:14	18:58
54°	05:29	06:10	06:45	17:40	18:15	18:56
52°	05:30	06:09	06:43	17:42	18:16	18:55
50°	05:31	06:08	06:40	17:44	18:17	18:54
45°	05:32	06:06	06:35	17:49	18:19	18:53
N 40°	05:33	06:04	06:31	17:53	18:20	18:52
35°	05:33	06:02	06:28	17:57	18:22	18:52
30°	05:33	06:00	06:24	18:00	18:24	18:52
20°	05:31	05:57	06:19	18:06	18:28	18:53
N 10°	05:28	05:52	06:14	18:11	18:32	18:56
0°	05:24	05:48	06:09	18:15	18:36	19:00
S 10°	05:18	05:42	06:04	18:20	18:41	19:06
20°	05:10	05:36	05:58	18:25	18:48	19:14
30°	04:59	05:28	05:52	18:32	18:56	19:24
35°	04:52	05:22	05:48	18:35	19:01	19:31
40°	04:44	05:16	05:44	18:39	19:07	19:40
45°	04:33	05:09	05:39	18:44	19:14	19:50
S 50°	04:19	05:00	05:33	18:50	19:23	20:03
52°	04:13	04:55	05:30	18:53	19:27	20:10
54°	04:05	04:50	05:27	18:56	19:32	20:17
56°	03:57	04:45	05:24	18:59	19:37	20:26
58°	03:47	04:39	05:20	19:02	19:44	20:35
S 60°	03:35	04:32	05:16	19:06	19:50	20:46

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	03:22	■	■	04:16	■	■
N 70°	01:56	■	■	05:43	■	■
68°	01:18	■	■	06:23	■	■
66°	00:51	03:06	■	06:50	06:12	■
64°	00:31	02:25	05:08	07:12	06:54	05:57
62°	00:15	01:57	03:52	07:29	07:22	07:13
60°	00:01	01:36	03:17	07:43	07:44	07:49
N 58°	· · ·	01:19	02:51	07:56	08:02	08:15
56°	· · ·	01:04	02:31	08:06	08:18	08:36
54°	· · ·	00:52	02:14	08:16	08:31	08:53
52°	· · ·	00:41	02:00	08:24	08:42	09:08
50°	· · ·	00:31	01:47	08:32	08:52	09:21
45°	· · ·	00:11	01:22	08:48	09:14	09:47
N 40°	23:54	· · ·	01:01	09:02	09:31	10:08
35°	23:41	· · ·	00:44	09:13	09:46	10:26
30°	23:29	· · ·	00:30	09:23	09:59	10:41
20°	23:08	· · ·	00:05	09:41	10:21	11:07
N 10°	22:51	23:44	· · ·	09:56	10:40	11:29
0°	22:35	23:25	· · ·	10:10	10:58	11:50
S 10°	22:18	23:05	23:57	10:25	11:16	12:11
20°	22:01	22:44	23:34	10:40	11:35	12:34
30°	21:42	22				

March 04, 05, 06 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars			
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	162°14.8	199°22.0	S15°31.5	204°12.5	S17°18.0	122°33.8	N14°31.5	179°49.8	S09°13.0	Alpheratz	357°35.9 29°13.3		
1	177°17.3	214°21.4	30.6	219°13.0	17.4	137°35.8	31.6	194°52.0	12.9	Ankaa	353°08.2 -42°10.7		
2	192°19.8	229°20.8	29.7	234°13.6	16.8	152°37.9	31.8	209°54.2	12.8	Schedar	349°32.4 56°40.2		
3	207°22.2	244°20.2	•• 28.8	249°14.1	•• 16.3	167°39.9	•• 31.9	224°56.4	•• 12.7	Diphda	348°48.3 -17°51.4		
4	222°24.7	259°19.6	27.9	264°14.6	15.7	182°41.9	32.0	239°58.5	12.6	Achernar	335°21.1 -57°07.1		
5	237°27.2	274°19.0	27.0	279°15.1	15.2	197°44.0	32.2	255°00.7	12.4	Hamal	327°52.2 23°34.6		
6	252°29.6	289°18.4	S15°26.1	294°15.6	S17°14.6	212°46.0	N14°32.3	270°02.9	S09°12.3	Polaris	314°36.7 89°22.2		
7	267°32.1	304°17.8	25.2	309°16.2	14.0	227°48.0	32.5	285°05.1	12.2	Acamar	315°12.5 -40°12.7		
8	282°34.5	319°17.2	24.3	324°16.7	13.5	242°50.1	32.6	300°07.2	12.1	Menkar	314°07.0 4°11.0		
9	297°37.0	334°16.6	•• 23.4	339°17.2	•• 12.9	257°52.1	•• 32.8	315°09.4	•• 12.0	Mirfak	308°29.5 49°57.0		
10	312°39.5	349°15.9	22.5	354°17.7	12.4	272°54.1	32.9	330°11.6	11.9	Aldebaran	290°40.5 16°33.5		
11	327°41.9	4°15.3	21.6	9°18.3	11.8	287°56.2	33.0	345°13.8	11.8	Rigel	281°04.5 -8°10.6		
12	342°44.4	19°14.7	S15°20.7	24°18.8	S17°11.2	302°58.2	N14°33.2	0°16.0	S09°11.7	Capella	280°22.9 46°01.5		
13	357°46.9	34°14.1	19.8	39°19.3	10.7	318°00.2	33.3	15°18.1	11.5	Bellatrix	278°23.6 6°22.2		
14	12°49.3	49°13.5	18.9	54°19.8	10.1	333°02.3	33.5	30°20.3	11.4	Elnath	278°02.7 28°37.7		
15	27°51.8	64°12.9	•• 17.9	69°20.3	•• 09.6	348°04.3	•• 33.6	45°22.5	•• 11.3	Alnilam	275°38.4 -1°11.3		
16	42°54.3	79°12.3	17.0	84°20.9	09.0	3°06.3	33.8	60°24.7	11.2	Betelgeuse	270°52.8 7°24.6		
17	57°56.7	94°11.7	16.1	99°21.4	08.4	18°08.3	33.9	75°26.8	11.1	Canopus	263°52.5 -52°42.7		
18	72°59.2	109°11.1	S15°15.2	114°21.9	S17°07.9	33°10.4	N14°34.0	90°29.0	S09°11.0	Sirius	258°26.7 -16°45.1		
19	88°01.7	124°10.5	14.3	129°22.4	07.3	48°12.4	34.2	105°31.2	10.9	Adhara	255°06.2 -29°00.5		
20	103°04.1	139°09.9	13.4	144°23.0	06.7	63°14.4	34.3	120°33.4	10.7	Procyon	244°51.4 5°09.7		
21	118°06.6	154°09.3	•• 12.5	159°23.5	•• 06.2	78°16.5	•• 34.5	135°35.5	•• 10.6	Pollux	243°17.9 27°58.1		
22	133°09.0	169°08.7	11.6	174°24.0	05.6	93°18.5	34.6	150°37.7	10.5	Avior	234°14.6 -59°35.4		
23	148°11.5	184°08.1	10.7	189°24.5	05.0	108°20.5	34.8	165°39.9	10.4	Suhail	222°46.5 -43°31.9		
Mer.pass. 13:09		ν -0.6' d -0.9' m -3.88		ν 0.5' d -0.6' m 1.26		ν 2.0' d 0.1' m -2.17		ν 2.2' d -0.1' m 0.97		Miaplacidus	221°37.7 -69°49.1		
											Alphard	217°48.2 -8°45.9	
											Regulus	207°34.9 11°50.9	
											Dubhe	193°41.2 61°37.2	
											Denebola	182°25.3 14°26.1	
											Gienah	175°44.0 -17°40.6	
											Acrux	173°00.3 -63°13.9	
											Gacrux	171°51.9 -57°14.8	
											Alioth	166°13.1 55°49.5	
											Spica	158°22.8 -11°17.3	
											Alkaid	152°52.2 49°11.3	
											Hadar	148°36.6 -60°29.2	
											Menkent	147°58.2 -36°29.3	
											Arcturus	145°48.3 19°03.2	
											Rigel Kent.	139°40.9 -60°55.9	
											Kochab	137°19.1 74°03.0	
											Zuben'ubi	136°56.6 -16°08.6	
											Alphecca	126°04.2 26°37.7	
											Antares	112°16.7 -26°29.1	
											Atria	107°11.6 -69°04.0	
											Sabik	102°03.6 -15°45.3	
											Shaula	96°11.4 -37°07.2	
											Rasalhague	95°59.3 12°32.3	
											Eltanin	90°42.6 51°28.7	
											Kaus Aust.	83°33.6 -34°22.3	
											Vega	80°33.9 38°48.0	
											Nunki	75°48.8 -26°16.0	
											Altair	62°00.9 8°55.7	
											Peacock	53°07.3 -56°39.4	
											Deneb	49°26.7 45°21.7	
											Enif	33°39.8 9°59.0	
											Al Na'ir	27°34.2 -46°50.7	
											Fomalhaut	15°15.6 -29°29.8	
											Scheat	13°46.3 28°12.7	
											Markab	13°30.9 15°20.0	
											Mar 04 Mon	SHA	Mer.pass
											Venus	37°07.2 10:43	
											Mars	41°57.7 10:23	
											Jupiter	320°19.0 15:48	
											Saturn	17°35.0 11:59	
											Mar 05 Tue	SHA	Mer.pass
											Venus	35°53.5 10:44	
											Mars	41°11.1 10:22	
											Jupiter	320°08.6 15:44	
											Saturn	17°28.1 11:55	
											Mar 06 Wed	SHA	Mer.pass
											Venus	34°40.1 10:45	
											Mars	40°24.6 10:21	
											Jupiter	319°58.1 15:41	
											Saturn	17°21.2 11:52	
											Horizontal parallax		
											Venus:	0.1	
											Mars:	0.1	
Mer.pass. 13:01		ν -0.6' d -0.9' m -3.88		ν 0.5' d -0.6' m 1.25		ν 2.0' d 0.1' m -2.16		ν 2.2' d -0.1' m 0.99					

Sun			Moon				
Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	177°04.9	S06°20.0	265°31.0	6.9'	S27°31.1	4.3'	57.0'
1	192°05.0	19.1	279°56.9	6.8'	27°35.4	4.1'	57.0'
2	207°05.1	18.1	294°22.6	6.7'	27°39.5	4.0'	57.0'
3	222°05.3	· · 17.1	308°48.3	6.6'	27°43.5	3.8'	57.1'
4	237°05.4	16.2	323°13.9	6.5'	27°47.4	3.7'	57.1'
5	252°05.6	15.2	337°39.4	6.4'	27°51.0	3.5'	57.2'
6	267°05.7	S06°14.2	352°04.8	6.3'	S27°54.5	3.4'	57.2'
7	282°05.8	13.3	6°30.1	6.2'	27°57.9	3.2'	57.2'
8	297°06.0	12.3	20°55.4	6.1'	28°01.1	3.0'	57.3'
9	312°06.1	· · 11.3	35°20.5	6.1'	28°04.1	2.9'	57.3'
10	327°06.2	10.4	49°45.6	6.0'	28°07.0	2.7'	57.3'
11	342°06.4	09.4	64°10.5	5.9'	28°09.7	2.5'	57.4'
12	357°06.5	S06°08.5	78°35.4	5.8'	S28°12.2	2.4'	57.4'
13	12°06.7	07.5	93°00.3	5.7'	28°14.6	2.2'	57.5'
14	27°06.8	06.5	107°25.0	5.7'	28°16.8	2.0'	57.5'
15	42°06.9	· · 05.6	121°49.7	5.6'	28°18.8	1.9'	57.5'
16	57°07.1	04.6	136°14.3	5.5'	28°20.6	1.7'	57.6'
17	72°07.2	03.6	150°38.8	5.4'	28°22.3	1.5'	57.6'
18	87°07.4	S06°02.7	165°03.2	5.4'	S28°23.8	1.3'	57.6'
19	102°07.5	01.7	179°27.6	5.3'	28°25.1	1.2'	57.7'
20	117°07.6	06°00.7	193°51.9	5.2'	28°26.3	1.0'	57.7'
21	132°07.8	05°59.8	208°16.2	5.2'	28°27.3	0.8'	57.8'
22	147°07.9	58.8	222°40.4	5.1'	28°28.1	0.6'	57.8'
23	162°08.1	57.8	237°04.5	5.1'	28°28.7	0.4'	57.8'
SD = 16.1' $d = -1.0'$			SD = 15.5'				
Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	177°08.2	S05°56.9	251°28.5	5.0'	S28°29.1	0.3'	57.9'
1	192°08.3	55.9	265°52.6	4.9'	28°29.4	0.1'	57.9'
2	207°08.5	54.9	280°16.5	4.9'	28°29.5	-0.1'	58.0'
3	222°08.6	· · 54.0	294°40.4	4.8'	28°29.4	-0.3'	58.0'
4	237°08.8	53.0	309°04.2	4.8'	28°29.1	-0.5'	58.0'
5	252°08.9	52.0	323°28.0	4.7'	28°28.7	-0.6'	58.1'
6	267°09.0	S05°51.1	337°51.8	4.7'	S28°28.0	-0.8'	58.1'
7	282°09.2	50.1	352°15.5	4.7'	28°27.2	-1.0'	58.1'
8	297°09.3	49.1	6°39.1	4.6'	28°26.2	-1.2'	58.2'
9	312°09.5	· · 48.1	21°02.7	4.6'	28°25.0	-1.4'	58.2'
10	327°09.6	47.2	35°26.3	4.5'	28°23.6	-1.6'	58.3'
11	342°09.8	46.2	49°49.9	4.5'	28°22.0	-1.8'	58.3'
12	357°09.9	S05°45.2	64°13.4	4.5'	S28°20.3	-1.9'	58.3'
13	12°10.0	44.3	78°36.8	4.4'	28°18.3	-2.1'	58.4'
14	27°10.2	43.3	93°00.3	4.4'	28°16.2	-2.3'	58.4'
15	42°10.3	· · 42.3	107°23.7	4.4'	28°13.9	-2.5'	58.5'
16	57°10.5	41.4	121°47.1	4.4'	28°11.4	-2.7'	58.5'
17	72°10.6	40.4	136°10.5	4.3'	28°08.7	-2.9'	58.5'
18	87°10.8	S05°39.4	150°33.8	4.3'	S28°05.8	-3.1'	58.6'
19	102°10.9	38.5	164°57.1	4.3'	28°02.7	-3.3'	58.6'
20	117°11.1	37.5	179°20.4	4.3'	27°59.4	-3.5'	58.7'
21	132°11.2	· · 36.5	193°43.7	4.3'	27°56.0	-3.6'	58.7'
22	147°11.3	35.5	208°07.0	4.3'	27°52.3	-3.8'	58.7'
23	162°11.5	34.6	222°30.3	4.3'	27°48.5	-4.0'	58.8'
SD = 16.1' $d = -1.0'$			SD = 15.8'				
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	177°11.6	S05°33.6	236°53.6	4.3'	S27°44.5	-4.2'	58.8'
1	192°11.8	32.6	251°16.8	4.3'	27°40.3	-4.4'	58.9'
2	207°11.9	31.7	265°40.1	4.3'	27°35.8	-4.6'	58.9'
3	222°12.1	· · 30.7	280°03.3	4.3'	27°31.2	-4.8'	58.9'
4	237°12.2	29.7	294°26.6	4.3'	27°26.5	-5.0'	59.0'
5	252°12.4	28.7	308°49.8	4.3'	27°21.5	-5.2'	59.0'
6	267°12.5	S05°27.8	323°13.1	4.3'	S27°16.3	-5.4'	59.1'
7	282°12.7	26.8	337°36.4	4.3'	27°11.0	-5.5'	59.1'
8	297°12.8	25.8	351°59.7	4.3'	27°05.4	-5.7'	59.1'
9	312°12.9	· · 24.9	6°23.0	4.3'	26°59.7	-5.9'	59.2'
10	327°13.1	23.9	20°46.3	4.3'	26°53.8	-6.1'	59.2'
11	342°13.2	22.9	35°09.6	4.3'	26°47.7	-6.3'	59.2'
12	357°13.4	S05°21.9	49°32.9	4.4'	S26°41.4	-6.5'	59.3'
13	12°13.5	21.0	63°56.3	4.4'	26°34.9	-6.7'	59.3'
14	27°13.7	20.0	78°19.7	4.4'	26°28.2	-6.9'	59.4'
15	42°13.8	· · 19.0	92°43.1	4.4'	26°21.4	-7.0'	59.4'
16	57°14.0	18.1	107°06.5	4.5'	26°14.3	-7.2'	59.4'
17	72°14.1	17.1	121°30.0	4.5'	26°07.1	-7.4'	59.5'
18	87°14.3	S05°16.1	135°53.5	4.5'	S25°59.7	-7.6'	59.5'
19	102°14.4	15.1	150°17.0	4.6'	25°52.1	-7.8'	59.5'
20	117°14.6	14.2	164°40.6	4.6'	25°44.3	-8.0'	59.6'
21	132°14.7	· · 13.2	179°04.1	4.6'	25°36.3	-8.1'	59.6'
22	147°14.9	12.2	193°27.8	4.7'	25°28.2	-8.3'	59.7'
23	162°15.0	11.2	207°51.4	4.7'	25°19.9	-8.5'	59.7'
SD = 16.1' $d = -1.0'$			SD = 16.0'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	04:47	06:06	07:13	17:11	18:19	19:39
N 70°	04:55	06:06	07:06	17:18	18:19	19:31
68°	05:01	06:05	07:01	17:24	18:19	19:24
66°	05:06	06:05	06:56	17:28	18:20	19:19
64°	05:10	06:05	06:52	17:32	18:20	19:15
62°	05:13	06:04	06:49	17:36	18:20	19:11
60°	05:16	06:04	06:45	17:38	18:20	19:08
N 58°	05:18	06:04	06:43	17:41	18:20	19:06
56°	05:20	06:03	06:40	17:44	18:21	19:04
54°	05:22	06:03	06:38	17:46	18:21	19:02
52°	05:23	06:02	06:36	17:48	18:21	19:00
50°	05:25	06:02	06:34	17:49	18:22	18:59
45°	05:27	06:01	06:30	17:53	18:23	18:57
N 40°	05:28	06:00	06:27	17:57	18:24	18:55
35°	05:29	05:58	06:24	17:59	18:25	18:54
30°	05:29	05:57	06:21	18:02	18:26	18:54
20°	05:29	05:54	06:16	18:07	18:29	18:54
N 10°	05:27	05:51	06:12	18:11	18:32	18:56
0°	05:23	05:47	06:08	18:15	18:35	19:00
S 10°	05:18	05:43	06:04	18:19	18:40	19:04
20°	05:11	05:37	05:59	18:23	18:45	19:11
30°	05:01	05:30	05:54	18:28	18:52	19:21
35°	04:55	05:25	05:51	18:31	18:57	19:27
40°	04:47	05:20	05:47	18:35	19:02	19:35
45°	04:37	05:13	05:43	18:39	19:09	19:44
S 50°	04:25	05:05	05:38	18:44	19:17	19:56
52°	04:19	05:01	05:36	18:46	19:20	20:02
54°	04:12	04:57	05:33	18:48	19:25	20:09
56°	04:04	04:52	05:30	18:51	19:29	20:16
58°	03:55	04:46	05:27	18:54	19:35	20:25
S 60°	03:45	04:40	05:24	18:57	19:41	20:35

March 07, 08, 09 UT (Thu., Fri., Sat.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	165°12.3	198°39.2	S14°25.1	204°50.4	S16°37.0	124°59.8	N14°41.9	182°26.6	S09°04.9					Alpheratz	357°35.9	29°13.3	
1	180°14.7	213°38.6	24.2	219°50.9	36.4	140°01.8	42.0	197°28.8	04.7					Ankaa	353°08.2	-42°10.7	
2	195°17.2	228°38.0	23.2	234°51.5	35.8	155°03.8	42.2	212°31.0	04.6					Schedar	349°32.4	56°40.2	
3	210°19.6	243°37.4	•• 22.3	249°52.0	•• 35.2	170°05.8	•• 42.3	227°33.1	•• 04.5					Diphda	348°48.3	-17°51.4	
4	225°22.1	258°36.8	21.3	264°52.5	34.7	185°07.8	42.4	242°35.3	04.4					Achernar	335°21.1	-57°07.1	
5	240°24.6	273°36.3	20.4	279°53.1	34.1	200°09.8	42.6	257°37.5	04.3					Hamal	327°52.2	23°34.6	
6	255°27.0	288°35.7	S14°19.4	294°53.6	S16°33.5	215°11.9	N14°42.7	272°39.7	S09°04.2					Polaris	314°38.0	89°22.2	
7	270°29.5	303°35.1	18.5	309°54.1	32.9	230°13.9	42.9	287°41.9	04.1					Acamar	315°12.5	-40°12.7	
8	285°32.0	318°34.5	17.5	324°54.7	32.3	245°15.9	43.0	302°44.0	04.0					Menkar	314°07.0	4°11.0	
9	300°34.4	333°34.0	•• 16.6	339°55.2	•• 31.8	260°17.9	•• 43.2	317°46.2	•• 03.8					Mirfak	308°29.5	49°56.9	
10	315°36.9	348°33.4	15.6	354°55.7	31.2	275°19.9	43.3	332°48.4	03.7					Aldebaran	290°40.5	16°33.5	
11	330°39.4	3°32.8	14.7	9°56.3	30.6	290°22.0	43.5	347°50.6	03.6					Rigel	281°04.5	-8°10.6	
12	345°41.8	18°32.2	S14°13.7	24°56.8	S16°30.0	305°24.0	N14°43.6	2°52.7	S09°03.5					Capella	280°22.9	46°01.5	
13	0°44.3	33°31.7	12.8	39°57.3	29.4	320°26.0	43.8	17°54.9	03.4					Bellatrix	278°23.6	6°22.2	
14	15°46.8	48°31.1	11.8	54°57.9	28.8	335°28.0	43.9	32°57.1	03.3					Elnath	278°02.7	28°37.7	
15	30°49.2	63°30.5	•• 10.8	69°58.4	•• 28.3	350°30.0	•• 44.1	47°59.3	•• 03.2					Alnilam	275°38.4	-1°11.3	
16	45°51.7	78°29.9	09.9	84°58.9	27.7	5°32.1	44.2	63°01.5	03.0					Betelgeuse	270°52.8	7°24.6	
17	60°54.1	93°29.4	08.9	99°59.5	27.1	20°34.1	44.3	78°03.6	02.9					Canopus	263°52.5	-52°42.7	
18	75°56.6	108°28.8	S14°08.0	115°00.0	S16°26.5	35°36.1	N14°44.5	93°05.8	S09°02.8					Sirius	258°26.7	-16°45.1	
19	90°59.1	123°28.2	07.0	130°00.5	25.9	50°38.1	44.6	108°08.0	02.7					Adhara	255°06.2	-29°00.5	
20	106°01.5	138°27.7	06.1	145°01.1	25.3	65°40.1	44.8	123°10.2	02.6					Procyon	244°51.4	5°09.7	
21	121°04.0	153°27.1	•• 05.1	160°01.6	•• 24.7	80°42.1	•• 44.9	138°12.4	•• 02.5					Pollux	243°17.9	27°58.1	
22	136°06.5	168°26.5	04.1	175°02.1	24.2	95°44.2	45.1	153°14.5	02.4					Avior	234°14.6	-59°35.4	
23	151°08.9	183°25.9	03.2	190°02.7	23.6	110°46.2	45.2	168°16.7	02.3					Suhail	222°46.5	-43°31.9	
Mer.pass. 12:57			ν -0.6' d -0.9' m -3.88	ν 0.5' d -0.6' m 1.25	ν 2.0' d 0.1' m -2.15	ν 2.2' d -0.1' m 0.99											

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	166°11.4	198°25.4	S14°02.2	205°03.2	S16°23.0	125°48.2	N14°45.4	183°18.9	S09°02.1
1	181°13.9	213°24.8	01.2	220°03.8	22.4	140°50.2	45.5	198°21.1	02.0
2	196°16.3	228°24.2	14°00.3	235°04.3	21.8	155°52.2	45.7	213°23.2	01.9
3	211°18.8	243°23.7	13°59.3	250°04.8	•• 21.2	170°54.2	•• 45.8	228°25.4	•• 01.8
4	226°21.3	258°23.1	58.3	265°05.4	20.6	185°56.2	46.0	243°27.6	01.7
5	241°23.7	273°22.5	57.4	280°05.9	20.1	200°58.3	46.1	258°29.8	01.6
6	256°26.2	288°22.0	S13°56.4	295°06.4	S16°19.5	216°00.3	N14°46.3	273°32.0	S09°01.5
7	271°28.6	303°21.4	55.4	310°07.0	18.9	231°02.3	46.4	288°34.1	01.4
8	286°31.1	318°20.8	54.5	325°07.5	18.3	246°04.3	46.5	303°36.3	01.2
9	301°33.6	333°20.3	•• 53.5	340°08.0	•• 17.7	261°06.3	•• 46.7	318°38.5	•• 01.1
10	316°36.0	348°19.7	52.5	355°08.6	17.1	276°08.3	46.8	333°40.7	01.0
11	331°38.5	3°19.1	51.6	10°09.1	16.5	291°10.3	47.0	348°42.9	00.9
12	346°41.0	18°18.6	S13°50.6	25°09.7	S16°15.9	306°12.4	N14°47.1	3°45.0	S09°00.8
13	1°43.4	33°18.0	49.6	40°10.2	15.3	321°14.4	47.3	18°47.2	00.7
14	16°45.9	48°17.5	48.7	55°10.7	14.7	336°16.4	47.4	33°49.4	00.6
15	31°48.4	63°16.9	•• 47.7	70°11.3	•• 14.2	351°18.4	•• 47.6	48°51.6	•• 00.5
16	46°50.8	78°16.3	46.7	85°11.8	13.6	6°20.4	47.7	63°53.7	00.3
17	61°53.3	93°15.8	45.7	100°12.4	13.0	21°22.4	47.9	78°55.9	00.2
18	76°55.8	108°15.2	S13°44.8	115°12.9	S16°12.4	36°24.4	N14°48.0	93°58.1	S09°00.1
19	91°58.2	123°14.6	43.8	130°13.4	11.8	51°26.5	48.2	109°00.3	09°00.0
20	107°00.7	138°14.1	42.8	145°14.0	11.2	66°28.5	48.3	124°02.5	08°59.9
21	122°03.1	153°13.5	•• 41.8	160°14.5	•• 10.6	81°30.5	•• 48.5	139°04.6	•• 59.8
22	137°05.6	168°13.0	40.9	175°15.1	10.0	96°32.5	48.6	154°06.8	59.7
23	152°08.1	183°12.4	39.9	190°15.6	09.4	111°34.5	48.8	169°09.0	59.5
Mer.pass. 12:53			ν -0.6' d -1.0' m -3.88	ν 0.5' d -0.6' m 1.25	ν 2.0' d 0.1' m -2.15	ν 2.2' d -0.1' m 0.99			

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	167°10.5	198°11.9	S13°38.9	205°16.1	S16°08.8	126°36.5	N14°48.9	184°11.2	S08°59.4
1	182°13.0	213°11.3	37.9	220°16.7	08.2	141°38.5	49.0	199°13.4	59.3
2	197°15.5	228°10.7	36.9	235°17.2	07.6	156°40.5	49.2	214°15.5	59.2
3	212°17.9	243°10.2	•• 36.0	250°17.8	•• 07.0	171°42.5	•• 49.3	229°17.7	•• 59.1
4	227°20.4	258°09.6	35.0	265°18.3	06.4	186°44.6	49.5	244°19.9	59.0
5	242°22.9	273°09.1	34.0	280°18.8	05.9	201°46.6	49.6	259°22.1	58.9
6	257°25.3	288°08.5	S13°33.0	295°19.4	S16°05.3	216°48.6	N14°49.8	274°24.2	S08°58.8
7	272°27.8	303°08.0	32.0	310°19.9	04.7	231°50.6	49.9	289°26.4	58.6
8	287°30.2	318°07.4	31.0	325°20.5	04.1	246°52.6	50.1	304°28.6	58.5
9	302°32.7	333°06.8	•• 30.1	340°21.0	•• 03.5	261°54.6	•• 50.2	319°30.8	•• 58.4
10	317°35.2	348°06.3	29.1	355°21.5	02.9	276°56.6	50.4	334°33.0	58.3
11	332°37.6	3°05.7	28.1	10°22.1	02.3	291°58.6	50.5	349°35.1	58.2
12	347°40.1	18°05.2	S13°27.1	25°22.6	S16°01.7	307°00.6	N14°50.7	4°37.3	S08°58.1
13	2°42.6	33°04.6	26.1	40°23.2	01.1	322°02.6	50.8	19°39.5	58.0
14	17°45.0	48°04.1	25.1	55°23.7	16°00.5	337°04.7	51.0	34°41.7	57.9
15	32°47.5	63°03.5	•• 24.1	70°24.3	15°59.9	352°06.7	•• 51.1	49°43.9	•• 57.7
16	47°50.0	78°03.0	23.1	85°24.8	59.3	7°08.7	51.3	64°46.0	57.6
17	62°52.4	93°02.4	22.2	100°25.3	58.7	22°10.7	51.4	79°48.2	57.5
18	77°54.9	108°01.9	S13°21.2	115°25.9	S15°58.1	37°12.7	N14°51.6	94°50.4	S08°57.4
19	92°57.4	123°01.3	20.2	130°26.4	57.5	52°14.7	51.7	109°52.6	57.3
20	107°59.8	138°00.8	19.2	145°27.0	56.9	67°16.7	51.9	124°54.8	57.2
21	123°02.3	153°00.2	•• 18.2	160°27.5	•• 56.3	82°18.7	•• 52.0	139°56.9	•• 57.1
22	138°04.7	167°59.7	17.2	175°28.1	55.7	97°20.7	52.2	154°59.1	57.0
23	153°07.2	182°59.1	16.2	190°28.6	55.1	112°22.7	52.3	170°01.3	56.8
Mer.pass. 12:49			ν -0.6' d -1.0' m -3.88	ν 0.5' d -0.6' m 1.25	ν 2.0' d 0.1' m -2.14	ν 2.2' d -0.1' m 1.00			

Mar 07 Thu	SHA	Mer.pass
Venus	33°26.9	10:46
Mars	39°38.1	10:20
Jupiter	319°47.5	15:38
Saturn	17°14.4	11:49

Mar 08 Fri	SHA	Mer.pass
Venus	32°14.0	10:47
Mars	38°51.8	10:19
Jupiter	319°36.8	15:35
Saturn	17°07.5	11:45

Mar 09 Sat	SHA	Mer.pass
Venus	31°01.3	10:48
Mars	38°05.6	10:19
Jupiter	319°26.0	15:31
Saturn	17°00.6	11:42

Horizontal parallax		
Venus:	0.1	
Mars:	0.1	

Sun				Moon			
h	GHA	Dec		GHA	ν	Dec	d HP
Thu							
0	177°15.2	S05°10.3		222°15.1	4.7'	S25°11.3	-8.7' 59.7'
1	192°15.3	09.3		236°38.9	4.8'	25°02.7	-8.9' 59.8'
2	207°15.5	08.3		251°02.7	4.8'	24°53.8	-9.0' 59.8'
3	222°15.6	· · 07.4		265°26.5	4.9'	24°44.8	-9.2' 59.8'
4	237°15.8	06.4		279°50.4	4.9'	24°35.5	-9.4' 59.9'
5	252°15.9	05.4		294°14.3	5.0'	24°26.1	-9.6' 59.9'
6	267°16.1	S05°04.4		308°38.3	5.0'	S24°16.6	-9.7' 59.9'
7	282°16.2	03.5		323°02.3	5.1'	24°06.8	-9.9' 60.0'
8	297°16.4	02.5		337°26.4	5.1'	23°56.9	-10.1' 60.0'
9	312°16.5	· · 01.5		351°50.5	5.2'	23°46.8	-10.3' 60.0'
10	327°16.7	05°00.5		6°14.7	5.2'	23°36.6	-10.4' 60.1'
11	342°16.8	04°59.6		20°39.0	5.3'	23°26.2	-10.6' 60.1'
12	357°17.0	S04°58.6		35°03.3	5.4'	S23°15.6	-10.8' 60.1'
13	12°17.1	57.6		49°27.6	5.4'	23°04.8	-10.9' 60.2'
14	27°17.3	56.6		63°52.0	5.5'	22°53.9	-11.1' 60.2'
15	42°17.4	· · 55.7		78°16.5	5.5'	22°42.8	-11.2' 60.2'
16	57°17.6	54.7		92°41.0	5.6'	22°31.6	-11.4' 60.3'
17	72°17.7	53.7		107°05.6	5.7'	22°20.2	-11.6' 60.3'
18	87°17.9	S04°52.7		121°30.3	5.7'	S22°08.6	-11.7' 60.3'
19	102°18.0	51.8		135°55.0	5.8'	21°56.9	-11.9' 60.4'
20	117°18.2	50.8		150°19.8	5.8'	21°45.0	-12.0' 60.4'
21	132°18.3	· · 49.8		164°44.6	5.9'	21°32.9	-12.2' 60.4'
22	147°18.5	48.8		179°09.5	6.0'	21°20.8	-12.3' 60.5'
23	162°18.6	47.9		193°34.5	6.0'	21°08.4	-12.5' 60.5'
	SD = 16.1'	$d = -1.0'$		SD = 16.3'			

Fri	GHA	Dec	GHA	ν	Dec	d HP
0	177°18.8	S04°46.9	207°59.5	6.1'	S20°55.9	-12.6' 60.5'
1	192°18.9	45.9	222°24.7	6.2'	20°43.3	-12.8' 60.5'
2	207°19.1	44.9	236°49.8	6.2'	20°30.5	-12.9' 60.6'
3	222°19.3	· · 43.9	251°15.1	6.3'	20°17.6	-13.1' 60.6'
4	237°19.4	43.0	265°40.4	6.4'	20°04.5	-13.2' 60.6'
5	252°19.6	42.0	280°05.8	6.4'	19°51.3	-13.4' 60.7'
6	267°19.7	S04°41.0	294°31.2	6.5'	S19°37.9	-13.5' 60.7'
7	282°19.9	40.0	308°56.7	6.6'	19°24.4	-13.6' 60.7'
8	297°20.0	39.1	323°22.3	6.7'	19°10.8	-13.8' 60.7'
9	312°20.2	· · 38.1	337°48.0	6.7'	18°57.0	-13.9' 60.8'
10	327°20.3	37.1	352°13.7	6.8'	18°43.1	-14.0' 60.8'
11	342°20.5	36.1	6°39.5	6.9'	18°29.1	-14.2' 60.8'
12	357°20.6	S04°35.2	21°05.3	6.9'	S18°14.9	-14.3' 60.8'
13	12°20.8	34.2	35°31.3	7.0'	18°00.6	-14.4' 60.9'
14	27°21.0	33.2	49°57.3	7.1'	17°46.2	-14.5' 60.9'
15	42°21.1	· · 32.2	64°23.3	7.1'	17°31.6	-14.7' 60.9'
16	57°21.3	31.2	78°49.5	7.2'	17°17.0	-14.8' 60.9'
17	72°21.4	30.3	93°15.7	7.3'	17°02.2	-14.9' 60.9'
18	87°21.6	S04°29.3	107°41.9	7.3'	S16°47.3	-15.0' 61.0'
19	102°21.7	28.3	122°08.3	7.4'	16°32.3	-15.1' 61.0'
20	117°21.9	27.3	136°34.7	7.5'	16°17.1	-15.2' 61.0'
21	132°22.0	· · 26.3	151°01.2	7.5'	16°01.9	-15.4' 61.0'
22	147°22.2	25.4	165°27.7	7.6'	15°46.5	-15.5' 61.0'
23	162°22.4	24.4	179°54.3	7.7'	15°31.1	-15.6' 61.1'
	SD = 16.1'	$d = -1.0'$	SD = 16.5'			

Sat	GHA	Dec	GHA	ν	Dec	d HP
0	177°22.5	S04°23.4	194°21.0	7.7'	S15°15.5	-15.7' 61.1'
1	192°22.7	22.4	208°47.8	7.8'	14°59.8	-15.8' 61.1'
2	207°22.8	21.5	223°14.6	7.9'	14°44.0	-15.9' 61.1'
3	222°23.0	· · 20.5	237°41.5	7.9'	14°28.2	-16.0' 61.1'
4	237°23.1	19.5	252°08.4	8.0'	14°12.2	-16.1' 61.1'
5	252°23.3	18.5	266°35.4	8.1'	13°56.1	-16.2' 61.2'
6	267°23.5	S04°17.5	281°02.5	8.1'	S13°40.0	-16.3' 61.2'
7	282°23.6	16.6	295°29.6	8.2'	13°23.7	-16.3' 61.2'
8	297°23.8	15.6	309°56.8	8.3'	13°07.3	-16.4' 61.2'
9	312°23.9	· · 14.6	324°24.1	8.3'	12°50.9	-16.5' 61.2'
10	327°24.1	13.6	338°51.4	8.4'	12°34.4	-16.6' 61.2'
11	342°24.2	12.6	353°18.7	8.4'	12°17.8	-16.7' 61.2'
12	357°24.4	S04°11.7	7°46.2	8.5'	S12°01.1	-16.8' 61.3'
13	12°24.6	10.7	22°13.7	8.6'	11°44.3	-16.8' 61.3'
14	27°24.7	09.7	36°41.2	8.6'	11°27.5	-16.9' 61.3'
15	42°24.9	· · 08.7	51°08.8	8.7'	11°10.6	-17.0' 61.3'
16	57°25.0	07.7	65°36.5	8.7'	10°53.6	-17.1' 61.3'
17	72°25.2	06.8	80°04.2	8.8'	10°36.5	-17.1' 61.3'
18	87°25.4	S04°05.8	94°32.0	8.8'	S10°19.4	-17.2' 61.3'
19	102°25.5	04.8	108°59.8	8.9'	10°02.2	-17.3' 61.3'
20	117°25.7	03.8	123°27.7	8.9'	09°44.9	-17.3' 61.3'
21	132°25.8	· · 02.8	137°55.7	9.0'	09°27.6	-17.4' 61.3'
22	147°26.0	01.9	152°23.6	9.0'	09°10.2	-17.4' 61.3'
23	162°26.2	00.9	166°51.7	9.1'	08°52.7	-17.5' 61.3'
	SD = 16.1'	$d = -1.0'$	SD = 16.7'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	04:31	05:51	06:58	17:26	18:33	19:54
N 70°	04:40	05:52	06:53	17:31	18:31	19:44
68°	04:48	05:53	06:48	17:35	18:30	19:36
66°	04:54	05:54	06:45	17:38	18:29	19:29
64°	04:59	05:54	06:42	17:41	18:29	19:24
62°	05:03	05:55	06:39	17:44	18:28	19:20
60°	05:07	05:55	06:37	17:46	18:28	19:16
N 58°	05:10	05:55	06:34	17:48	18:27	19:13
56°	05:13	05:56	06:33	17:50	18:27	19:10
54°	05:15	05:56	06:31	17:51	18:27	19:08
52°	05:17	05:56	06:29	17:53	18:27	19:06
50°	05:18	05:56	06:28	17:54	18:27	19:04
45°	05:22	05:55	06:25	17:57	18:27	19:01
N 40°	05:24	05:55	06:22	18:00	18:27	18:58
35°	05:25	05:54	06:20	18:02	18:27	18:57
30°	05:26	05:54	06:18	18:04	18:28	18:56
20°	05:26	05:52	06:14	18:08	18:30	18:55
N 10°	05:25	05:50	06:11	18:11	18:32	18:56
0°	05:23	05:47	06:07	18:14	18:35	18:59
S 10°	05:18	05:43	06:04	18:17	18:38	19:03
20°	05:12	05:38	06:00	18:21	18:43	19:09
30°	05:04	05:32	05:56	18:25	18:49	19:17
35°	04:58	05:28	05:53	18:27	18:53	19:23
40°	04:51	05:23	05:51	18:30	18:57	19:29
45°	04:42	05:17	05:47	18:33	19:03	19:38
S 50°	04:31	05:10	05:43	18:37	19:10	19:49
52°	04:25	05:07	05:41	18:39	19:13	19:55
54°	04:19	05:03	05:39	18:41	19:17	20:01
56°	04:12	04:59	05:37	18:43	19:21	20:07
58°	04:04	04:54	05:34	18:46	19:26	20:15
S 60°	03:54	04:48	05:31	18:48	19:31	20:24

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	■	■	08:26	■	■	15:03
N 70°	■	09:16	08:02	■	12:13	15:24
68°	■	08:27	07:44	■	13:00	15:40
66°	09:12	07:55	07:29	10:12	13:31	15:53
64°	07:57	07:31	07:17	11:26	13:53	16:03
62°	07:20	07:12	07:06	12:02	14:11	16:12
60°	06:53	06:56	06:57	12:27	14:25	16:20
N 58°	06:32	06:43	06:49	12:47	14:38	16:26
56°	06:15	06:31	06:42	13:04	14:49	16:32
54°	06:00	06:21	06:36	13:18	14:58	16:37
52°	05:48	06:12	06:30	13:30	15:06	16:42
50°	05:36	06:04	06:25	13:41	15:14	16:46
45°	05:13	05:46	06:13	14:04	15:29	16:55
N 40°	04:53	05:32	06:04	14:22	15:42	17:03
35°	04:37	05:20	05:56	14:37	15:53	17:09
30°	04:24	05:09	05:49	14:50	16:03	17:15
20°	04:00	04:50	05:36	15:12	16:19	17:24
N 10°	03:39	04:34	05:25	15:31	16:33	17:33
0°	03:20	04:19	05:15	15:48	16:46	17:40
S 10°	03:00	04:03	05:05	16:06	16:59	17:48
20°	02:39	03:47	04:53	16:24	17:12	17:56
30°	02:15	03:28	04:41	16:45	17:28	18:05
35°	02:01	03:17	04:33	16:58	17:37	18:11
40°	01:44	03:04	04:25	17:12	17:47	18:16
45°	01:24	02:49	04:15	17:28	17:58	18:23
S 50°	00:59	02:30	04:03	17:49	18:12	18:31
52°	00:47	02:21	03:57	17:58	18:19	18:35
54°	00:33	02:11	03:51	18:09	18:26	18:39
56°	00:17	01:59	03:44	18:21	18:34	18:43
58°	· · ·	01:46	03:36	18:35	18:43	18:48
S 60°	· · ·	01:31	03:27	18:51	18:54	18:54

Day	Sun		Mer. Pass	Moon		Age
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
07	10:59	10:52	12:11	09:34	22:04	
08	10:45	10:37	12:11	10:32	23:00	
09	10:30	10:22	12:10	11:28	23:54	

March 10, 11, 12 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	168°09.7	197°58.6	S13°15.2	205°29.1	S15°54.5	127°24.7	N14°52.4	185°03.5	S08°56.7	Alpheratz	357°35.9	29°13.3
1	183°12.1	212°58.0	14.2	220°29.7	53.9	142°26.7	52.6	200°05.6	56.6	Ankaa	353°08.2	-42°10.7
2	198°14.6	227°57.5	13.2	235°30.2	53.3	157°28.8	52.7	215°07.8	56.5	Schedar	349°32.4	56°40.2
3	213°17.1	242°56.9	•• 12.2	250°30.8	•• 52.7	172°30.8	•• 52.9	230°10.0	•• 56.4	Diphda	348°48.3	-17°51.4
4	228°19.5	257°56.4	11.2	265°31.3	52.1	187°32.8	53.0	245°12.2	56.3	Achernar	335°21.1	-57°07.1
5	243°22.0	272°55.9	10.2	280°31.9	51.5	202°34.8	53.2	260°14.4	56.2	Hamal	327°52.2	23°34.6
6	258°24.5	287°55.3	S13°09.2	295°32.4	S15°50.9	217°36.8	N14°53.3	275°16.5	S08°56.1	Polaris	314°39.4	89°22.2
7	273°26.9	302°54.8	08.2	310°33.0	50.3	232°38.8	53.5	290°18.7	55.9	Acamar	315°12.5	-40°12.7
8	288°29.4	317°54.2	07.2	325°33.5	49.7	247°40.8	53.6	305°20.9	55.8	Menkar	314°07.0	4°11.0
9	303°31.8	332°53.7	•• 06.2	340°34.0	•• 49.1	262°42.8	•• 53.8	320°23.1	•• 55.7	Mirfak	308°29.5	49°56.9
10	318°34.3	347°53.1	05.2	355°34.6	48.5	277°44.8	53.9	335°25.3	55.6	Aldebaran	290°40.5	16°33.5
11	333°36.8	2°52.6	04.2	10°35.1	47.9	292°46.8	54.1	350°27.4	55.5	Rigel	281°04.5	-8°10.6
12	348°39.2	17°52.0	S13°03.2	25°35.7	S15°47.3	307°48.8	N14°54.2	5°29.6	S08°55.4	Capella	280°22.9	46°01.5
13	3°41.7	32°51.5	02.2	40°36.2	46.7	322°50.8	54.4	20°31.8	55.3	Bellatrix	278°23.6	6°22.2
14	18°44.2	47°51.0	01.2	55°36.8	46.1	337°52.8	54.5	35°34.0	55.2	Elnath	278°02.7	28°37.7
15	33°46.6	62°50.4	13°00.2	70°37.3	•• 45.5	352°54.8	•• 54.7	50°36.2	•• 55.0	Alnilam	275°38.4	-1°11.3
16	48°49.1	77°49.9	12°59.2	85°37.9	44.9	7°56.8	54.8	65°38.3	54.9	Betelgeuse	270°52.8	7°24.6
17	63°51.6	92°49.3	58.2	100°38.4	44.2	22°58.8	55.0	80°40.5	54.8	Canopus	263°52.6	-52°42.7
18	78°54.0	107°48.8	S12°57.2	115°39.0	S15°43.6	38°00.8	N14°55.1	95°42.7	S08°54.7	Sirius	258°26.7	-16°45.1
19	93°56.5	122°48.3	56.2	130°39.5	43.0	53°02.8	55.3	110°44.9	54.6	Adhara	255°06.3	-29°00.5
20	108°59.0	137°47.7	55.2	145°40.1	42.4	68°04.9	55.4	125°47.1	54.5	Procyon	244°51.4	5°09.7
21	124°01.4	152°47.2	•• 54.2	160°40.6	•• 41.8	83°06.9	•• 55.6	140°49.2	•• 54.4	Pollux	243°17.9	27°58.1
22	139°03.9	167°46.6	53.2	175°41.2	41.2	98°08.9	55.7	155°51.4	54.3	Avior	234°14.6	-59°35.4
23	154°06.3	182°46.1	52.2	190°41.7	40.6	113°10.9	55.9	170°53.6	54.1	Suhail	222°46.5	-43°32.0
Mer.pass. 12:45		ν -0.5' d -1.0' m -3.88		ν 0.5' d -0.6' m 1.24		ν 2.0' d 0.1' m -2.14		ν 2.2' d -0.1' m 1.00				

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	169°08.8	197°45.6	S12°51.2	205°42.2	S15°40.0	128°12.9	N14°56.0	185°55.8	S08°54.0	Alphard	217°48.2	-8°45.9
1	184°11.3	212°45.0	50.2	220°42.8	39.4	143°14.9	56.2	200°58.0	53.9	Regulus	207°34.9	11°50.9
2	199°13.7	227°44.5	49.1	235°43.3	38.8	158°16.9	56.3	216°00.1	53.8	Dubhe	193°41.1	61°37.2
3	214°16.2	242°44.0	•• 48.1	250°43.9	•• 38.2	173°18.9	•• 56.5	231°02.3	•• 53.7	Denebola	182°25.3	14°26.1
4	229°18.7	257°43.4	47.1	265°44.4	37.6	188°20.9	56.6	246°04.5	53.6	Gienah	175°44.0	-17°40.7
5	244°21.1	272°42.9	46.1	280°45.0	37.0	203°22.9	56.8	261°06.7	53.5	Acrux	173°00.3	-63°13.9
6	259°23.6	287°42.3	S12°45.1	295°45.5	S15°36.4	218°24.9	N14°56.9	276°08.9	S08°53.4	Gacrux	171°51.9	-57°14.9
7	274°26.1	302°41.8	44.1	310°46.1	35.7	233°26.9	57.1	291°11.0	53.2	Alioth	166°13.0	55°49.5
8	289°28.5	317°41.3	43.1	325°46.6	35.1	248°28.9	57.2	306°13.2	53.1	Spica	158°22.8	-11°17.3
9	304°31.0	332°40.7	•• 42.1	340°47.2	•• 34.5	263°30.9	•• 57.4	321°15.4	•• 53.0	Alkaid	152°52.1	49°11.3
10	319°33.4	347°40.2	41.0	355°47.7	33.9	278°32.9	57.5	336°17.6	52.9	Hadar	148°36.6	-60°29.2
11	334°35.9	2°39.7	40.0	10°48.3	33.3	293°34.9	57.6	351°19.8	52.8	Menkent	147°58.2	-36°29.3
12	349°38.4	17°39.1	S12°39.0	25°48.8	S15°32.7	308°36.9	N14°57.8	6°21.9	S08°52.7	Arcturus	145°48.3	19°03.2
13	4°40.8	32°38.6	38.0	40°49.4	32.1	323°38.9	57.9	21°24.1	52.6	Rigil Kent.	139°40.9	-60°56.0
14	19°43.3	47°38.1	37.0	55°49.9	31.5	338°40.9	58.1	36°26.3	52.5	Kochab	137°19.0	74°03.0
15	34°45.8	62°37.6	•• 36.0	70°50.5	•• 30.9	353°42.9	•• 58.2	51°28.5	•• 52.4	Zuben'ubi	136°56.6	-16°08.6
16	49°48.2	77°37.0	34.9	85°51.0	30.3	8°44.9	58.4	66°30.7	52.2	Alphecca	126°04.2	26°37.7
17	64°50.7	92°36.5	33.9	100°51.6	29.6	23°46.9	58.5	81°32.8	52.1	Antares	112°16.6	-26°29.1
18	79°53.2	107°36.0	S12°32.9	115°52.1	S15°29.0	38°48.9	N14°58.7	96°35.0	S08°52.0	Atria	107°11.4	-69°04.0
19	94°55.6	122°35.4	31.9	130°52.7	28.4	53°50.9	58.8	111°37.2	51.9	Sabik	102°03.6	-15°45.4
20	109°58.1	137°34.9	30.9	145°53.2	27.8	68°52.9	59.0	126°39.4	51.8	Shaula	96°11.3	-37°07.2
21	125°00.6	152°34.4	•• 29.8	160°53.8	•• 27.2	83°54.9	•• 59.1	141°41.6	•• 51.7	Rasalhague	95°59.2	12°32.3
22	140°03.0	167°33.8	28.8	175°54.3	26.6	98°56.9	59.3	156°43.7	51.6	Eltanin	90°42.6	51°28.7
23	155°05.5	182°33.3	27.8	190°54.9	26.0	113°58.9	59.4	171°45.9	51.5	Kaus Aust.	83°33.5	-34°22.3
Mer.pass. 12:41		ν -0.5' d -1.0' m -3.88		ν 0.5' d -0.6' m 1.24		ν 2.0' d 0.1' m -2.14		ν 2.2' d -0.1' m 1.01		Vega	80°33.8	38°48.0

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	170°07.9	197°32.8	S12°26.8	205°55.5	S15°25.4	129°00.9	N14°59.6	186°48.1	S08°51.3	Nunki	75°48.8	-26°16.0
1	185°10.4	212°32.3	25.7	220°56.0	24.7	144°02.9	59.7	201°50.3	51.2	Altair	62°00.8	8°55.7
2	200°12.9	227°31.7	24.7	235°56.6	24.1	159°04.9	14°59.9	216°52.5	51.1	Peacock	53°07.2	-56°39.4
3	215°15.3	242°31.2	•• 23.7	250°57.1	•• 23.5	174°06.9	15°00.0	231°54.6	•• 51.0	Deneb	49°26.6	45°21.7
4	230°17.8	257°30.7	22.7	265°57.7	22.9	189°08.9	00.2	246°56.8	50.9	Enif	33°39.8	9°59.0
5	245°20.3	272°30.2	21.6	280°58.2	22.3	204°10.9	00.3	261°59.0	50.8	Al Na'ir	27°34.2	-46°50.7
6	260°22.7	287°29.6	S12°20.6	295°58.8	S15°21.7	219°12.9	N15°00.5	277°01.2	S08°50.7	Fomalhaut	15°15.6	-29°29.8
7	275°25.2	302°29.1	19.6	310°59.3	21.1	234°14.9	00.6	292°03.4	50.6	Scheat	13°46.2	28°12.7
8	290°27.7	317°28.6	18.6	325°59.9	20.4	249°16.9	00.8	307°05.5	50.4	Markab	13°30.9	15°20.0
9	305°30.1	332°28.1	•• 17.5	341°00.4	•• 19.8	264°18.9	•• 00.9	322°07.7	•• 50.3			
10	320°32.6	347°27.5	16.5	356°01.0	19.2	279°20.9	01.1	337°09.9	50.2			
11	335°35.1	2°27.0	15.5	11°01.5	18.6	294°22.9	01.2	352°12.1	50.1			
12	350°37.5	17°26.5	S12°14.4	26°02.1	S15°18.0	309°24.9	N15°01.4	7°14.3	S08°50.0			
13	5°40.0	32°26.0	13.4	41°02.6	17.4	324°26.9	01.5	22°16.4	49.9			
14	20°42.4	47°25.5	12.4	56°03.2	16.7	339°28.9	01.7	37°18.6	49.8			
15	35°44.9	62°24.9	•• 11.3	71°03.8	•• 16.1	354°30.9	•• 01.8	52°20.8	•• 49.7			
16	50°47.4	77°24.4	10.3	86°04.3	15.5	9°32.8	02.0	67°23.0	49.5			
17	65°49.8	92°23.9	09.3	101°04.9	14.9	24°34.8	02.1	82°25.2	49.4			
18	80°52.3	107°23.4	S12°08.2	116°05.4	S15°14.3	39°36.8	N15°02.3	97°27.3	S08°49.3			
19	95°54.8	122°22.9	07.2	131°06.0	13.7	54°38.8	02.4	112°29.5	49.2			
20	110°57.2	137°22.3	06.2	146°06.5	13.0	69°40.8	02.6	127°31.7	49.1			
21	125°59.7	152°21.8	•• 05.1	161°07.1	•• 12.4	84°42.8	•• 02.7	142°33.9	•• 49.0			
22	141°02.2	167°21.3	04.1	176°07.6	11.8	99°44.8	02.9	157°36.1	48.9			
23	156°04.6	182°20.8	03.1	191°08.2	11.2	114°46.8	03.0	172°38.2	48.8			
Mer.pass. 12:37		ν -0.5' d -1.0' m -3.88		ν 0.6' d -0.6' m 1.24		ν 2.0' d 0.1' m -2.13		ν 2.2' d -0.1' m 1.01				

h								Moon							
Sun															
Sun	GHA	Dec		GHA	ν	Dec	d	HP							
0	177°26.3	S03°59.9		181°19.8	9.1'	S08°35.2	-17.6'	61.3'							
1	192°26.5	58.9		195°47.9	9.2'	08°17.7	-17.6'	61.4'							
2	207°26.6	57.9		210°16.1	9.2'	08°00.1	-17.7'	61.4'							
3	222°26.8	· · 56.9		224°44.3	9.3'	07°42.4	-17.7'	61.4'							
4	237°27.0	56.0		239°12.6	9.3'	07°24.7	-17.7'	61.4'							
5	252°27.1	55.0		253°40.9	9.4'	07°07.0	-17.8'	61.4'							
6	267°27.3	S03°54.0		268°09.2	9.4'	S06°49.2	-17.8'	61.4'							
7	282°27.4	53.0		282°37.6	9.4'	06°31.4	-17.9'	61.4'							
8	297°27.6	52.0		297°06.1	9.5'	06°13.5	-17.9'	61.4'							
9	312°27.8	· · 51.1		311°34.5	9.5'	05°55.6	-17.9'	61.4'							
10	327°27.9	50.1		326°03.0	9.6'	05°37.7	-18.0'	61.4'							
11	342°28.1	49.1		340°31.6	9.6'	05°19.7	-18.0'	61.4'							
12	357°28.2	S03°48.1		355°00.2	9.6'	S05°01.7	-18.0'	61.4'							
13	12°28.4	47.1		9°28.8	9.7'	04°43.7	-18.1'	61.4'							
14	27°28.6	46.1		23°57.5	9.7'	04°25.6	-18.1'	61.3'							
15	42°28.7	· · 45.2		38°26.1	9.7'	04°07.5	-18.1'	61.3'							
16	57°28.9	44.2		52°54.9	9.7'	03°49.5	-18.1'	61.3'							
17	72°29.1	43.2		67°23.6	9.8'	03°31.3	-18.1'	61.3'							
18	87°29.2	S03°42.2		81°52.4	9.8'	S03°13.2	-18.1'	61.3'							
19	102°29.4	41.2		96°21.2	9.8'	02°55.1	-18.2'	61.3'							
20	117°29.5	40.3		110°50.0	9.9'	02°36.9	-18.2'	61.3'							
21	132°29.7	· · 39.3		125°18.9	9.9'	02°18.7	-18.2'	61.3'							
22	147°29.9	38.3		139°47.8	9.9'	02°00.6	-18.2'	61.3'							
23	162°30.0	37.3		154°16.7	9.9'	01°42.4	-18.2'	61.3'							
SD = 16.1'				$d = -1.0'$				SD = 16.7'							
Mon	GHA	Dec		GHA	ν	Dec	d	HP							
0	177°30.2	S03°36.3		168°45.6	9.9'	S01°24.2	-18.2'	61.3'							
1	192°30.4	35.3		183°14.5	10.0'	01°06.0	-18.2'	61.3'							
2	207°30.5	34.4		197°43.5	10.0'	00°47.8	-18.2'	61.3'							
3	222°30.7	· · 33.4		212°12.5	10.0'	00°29.6	-18.2'	61.2'							
4	237°30.8	32.4		226°41.5	10.0'	S00°11.4	-18.2'	61.2'							
5	252°31.0	31.4		241°10.5	10.0'	N00°06.7	18.2'	61.2'							
6	267°31.2	S03°30.4		255°39.5	10.0'	N00°24.9	18.2'	61.2'							
7	282°31.3	29.4		270°08.6	10.1'	00°43.1	18.1'	61.2'							
8	297°31.5	28.5		284°37.6	10.1'	01°01.2	18.1'	61.2'							
9	312°31.7	· · 27.5		299°06.7	10.1'	01°19.3	18.1'	61.2'							
10	327°31.8	26.5		313°35.8	10.1'	01°37.4	18.1'	61.1'							
11	342°32.0	25.5		328°04.9	10.1'	01°55.5	18.1'	61.1'							
12	357°32.2	S03°24.5		342°34.0	10.1'	N02°13.6	18.0'	61.1'							
13	12°32.3	23.5		357°03.1	10.1'	02°31.6	18.0'	61.1'							
14	27°32.5	22.6		11°32.2	10.1'	02°49.7	18.0'	61.1'							
15	42°32.7	· · 21.6		26°01.3	10.1'	03°07.7	18.0'	61.1'							
16	57°32.8	20.6		40°30.4	10.1'	03°25.6	17.9'	61.0'							
17	72°33.0	19.6		54°59.5	10.1'	03°43.6	17.9'	61.0'							
18	87°33.2	S03°18.6		69°28.6	10.1'	N04°01.5	17.9'	61.0'							
19	102°33.3	17.6		83°57.7	10.1'	04°19.3	17.8'	61.0'							
20	117°33.5	16.6		98°26.8	10.1'	04°37.2	17.8'	61.0'							
21	132°33.7	· · 15.7		112°55.9	10.1'	04°55.0	17.8'	60.9'							
22	147°33.8	14.7		127°25.0	10.1'	05°12.7	17.7'	60.9'							
23	162°34.0	13.7		141°54.1	10.1'	05°30.4	17.7'	60.9'							
SD = 16.1'				$d = -1.0'$				SD = 16.7'							
Tue	GHA	Dec		GHA	ν	Dec	d	HP							
0	177°34.2	S03°12.7		156°23.2	10.1'	N05°48.1	17.6'	60.9'							
1	192°34.3	11.7		170°52.3	10.1'	06°05.7	17.6'	60.8'							
2	207°34.5	10.7		185°21.4	10.1'	06°23.3	17.5'	60.8'							
3	222°34.7	· · 09.8		199°50.4	10.0'	06°40.8	17.5'	60.8'							
4	237°34.8	08.8		214°19.5	10.0'	06°58.3	17.4'	60.8'							
5	252°35.0	07.8		228°48.5	10.0'	07°15.7	17.4'	60.8'							
6	267°35.2	S03°06.8		243°17.5	10.0'	N07°33.0	17.3'	60.7'							
7	282°35.3	05.8		257°46.5	10.0'	07°50.3	17.2'	60.7'							
8	297°35.5	04.8		272°15.5	10.0'	08°07.6	17.2'	60.7'							
9	312°35.7	· · 03.8		286°44.5	10.0'	08°24.7	17.1'	60.6'							
10	327°35.8	02.9		301°13.5	9.9'	08°41.9	17.0'	60.6'							
11	342°36.0	01.9		315°42.4	9.9'	08°58.9	17.0'	60.6'							
12	357°36.2	S03°00.9		330°11.3	9.9'	N09°15.9	16.9'	60.6'							
13	12°36.3	02°59.9		344°40.2	9.9'	09°32.8	16.8'	60.5'							
14	27°36.5	58.9		359°09.1	9.9'	09°49.6	16.8'	60.5'							
15	42°36.7	· · 57.9		13°38.0	9.8'	10°06.4	16.7'	60.5'							
16	57°36.8	56.9		28°06.8	9.8'	10°23.1	16.6'	60.4'							
17	72°37.0	56.0		42°35.6	9.8'	10°39.7	16.5'	60.4'							
18	87°37.2	S02°55.0		57°04.4	9.8'	N10°56.3	16.5'	60.4'							
19	102°37.3	54.0		71°33.1	9.7'	11°12.7	16.4'	60.4'							
20	117°37.5	53.0		86°01.9	9.7'	11°29.1	16.3'	60.3'							
21	132°37.7	· · 52.0		100°30.6	9.7'	11°45.4	16.2'	60.3'							
22	147°37.9	51.0		114°59.3	9.6'	12°01.6	16.1'	60.3'							
23	162°38.0	50.0		129°27.9	9.6'	12°17.8	16.0'	60.2'							
SD = 16.1'				$d = -1.0'$				SD = 16.6'							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	04:14	05:35	06:42	17:40	18:47	20:09
N 70°	04:26	05:38	06:39	17:43	18:44	19:57
68°	04:35	05:41	06:36	17:46	18:41	19:48
66°	04:42	05:42	06:33	17:48	18:39	19:40
64°	04:48	05:44	06:31	17:50	18:37	19:34
62°	04:53	05:45	06:29	17:52	18:36	19:28
60°	04:58	05:46	06:28	17:53	18:35	19:24
N 58°	05:01	05:47	06:26	17:55	18:34	19:20
56°	05:05	05:48	06:25	17:56	18:33	19:17
54°	05:07	05:48	06:24	17:57	18:32	19:14
52°	05:10	05:49	06:22	17:58	18:32	19:11
50°	05:12	05:49	06:21	17:59	18:31	19:09
45°	05:16	05:50	06:19	18:01	18:31	19:05
N 40°	05:19	05:50	06:17	18:03	18:30	19:01
35°	05:21	05:50	06:16	18:05	18:30	18:59
30°	05:23	05:50	06:14	18:06	18:30	18:58
20°	05:24	05:49	06:11	18:09	18:31	18:56
N 10°	05:24	05:48	06:09	18:11	18:32	18:56
0°	05:22	05:46	06:07	18:13	18:34	18:58
S 10°	05:19	05:43	06:04	18:16	18:37	19:01
20°	05:13	05:39	06:01	18:18	18:40	19:06
30°	05:06	05:34	05:58	18:21	18:45	19:13
35°	05:01	05:31	05:56	18:23	18:48	19:18
40°	04:54	05:27	05:54	18:25	18:52	19:24
45°	04:46	05:22	05:51	18:28	18:57	19:32
S 50°	04:36	05:15	05:48	18:31	19:03	19:42
52°	04:31	05:12	05:46	18:32	19:06	19:47
54°	04:26	05:09	05:45	18:34	19:09	19:52
56°	04:19	05:05	05:43	18:35	19:13	19:59
58°	04:12	05:01	05:41	18:37	19:17	20:06
S 60°	04:04	04:56	05:39	18:39	19:22	20:14

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	07:33	06:53	06:13	17:49	20:24	23:09
N 70°	07:24	06:54	06:23	17:54	20:17	22:45
68°	07:17	06:54	06:32	17:58	20:11	22:28

March 13, 14, 15 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	171°07.1	197°20.3	S12°02.0	206°08.8	S15°10.6	129°48.8	N15°03.2	187°40.4	S08°48.7	Alpheratz	357°35.9	29°13.3
1	186°09.5	212°19.7	12°01.0	221°09.3	09.9	144°50.8	03.3	202°42.6	48.5	Ankaa	353°08.2	-42°10.6
2	201°12.0	227°19.2	11°59.9	236°09.9	09.3	159°52.8	03.5	217°44.8	48.4	Schedar	349°32.4	56°40.2
3	216°14.5	242°18.7	•• 58.9	251°10.4	•• 08.7	174°54.8	•• 03.6	232°47.0	•• 48.3	Diphda	348°48.3	-17°51.4
4	231°16.9	257°18.2	57.9	266°11.0	08.1	189°56.8	03.8	247°49.2	48.2	Achernar	335°21.2	-57°07.0
5	246°19.4	272°17.7	56.8	281°11.5	07.5	204°58.8	03.9	262°51.3	48.1	Hamal	327°52.3	23°34.6
6	261°21.9	287°17.2	S11°55.8	296°12.1	S15°06.8	220°00.8	N15°04.1	277°53.5	S08°48.0	Polaris	314°40.6	89°22.2
7	276°24.3	302°16.7	54.7	311°12.7	06.2	235°02.8	04.2	292°55.7	47.9	Acamar	315°12.5	-40°12.7
8	291°26.8	317°16.1	53.7	326°13.2	05.6	250°04.8	04.4	307°57.9	47.8	Menkar	314°07.1	4°11.0
9	306°29.3	332°15.6	•• 52.7	341°13.8	•• 05.0	265°06.8	•• 04.5	323°00.1	•• 47.6	Mirfak	308°29.5	49°56.9
10	321°31.7	347°15.1	51.6	356°14.3	04.3	280°08.8	04.7	338°02.2	47.5	Aldebaran	290°40.5	16°33.4
11	336°34.2	2°14.6	50.6	11°14.9	03.7	295°10.7	04.8	353°04.4	47.4	Rigel	281°04.6	-8°10.6
12	351°36.7	17°14.1	S11°49.5	26°15.4	S15°03.1	310°12.7	N15°05.0	8°06.6	S08°47.3	Capella	280°22.9	46°01.5
13	6°39.1	32°13.6	48.5	41°16.0	02.5	325°14.7	05.1	23°08.8	47.2	Bellatrix	278°23.6	6°22.2
14	21°41.6	47°13.1	47.4	56°16.6	01.9	340°16.7	05.3	38°11.0	47.1	Elnath	278°02.7	28°37.7
15	36°44.0	62°12.6	•• 46.4	71°17.1	•• 01.2	355°18.7	•• 05.4	53°13.1	•• 47.0	Alnilam	275°38.4	-1°11.3
16	51°46.5	77°12.0	45.3	86°17.7	00.6	10°20.7	05.6	68°15.3	46.9	Betelgeuse	270°52.8	7°24.6
17	66°49.0	92°11.5	44.3	101°18.2	15°00.0	25°22.7	05.7	83°17.5	46.8	Canopus	263°52.6	-52°42.8
18	81°51.4	107°11.0	S11°43.3	116°18.8	S14°59.4	40°24.7	N15°05.9	98°19.7	S08°46.6	Sirius	258°26.8	-16°45.1
19	96°53.9	122°10.5	42.2	131°19.4	58.7	55°26.7	06.0	113°21.9	46.5	Adhara	255°06.3	-29°00.5
20	111°56.4	137°10.0	41.2	146°19.9	58.1	70°28.7	06.2	128°24.0	46.4	Procyon	244°51.4	5°09.7
21	126°58.8	152°09.5	•• 40.1	161°20.5	•• 57.5	85°30.7	•• 06.3	143°26.2	•• 46.3	Pollux	243°18.0	27°58.1
22	142°01.3	167°09.0	39.1	176°21.0	56.9	100°32.7	06.5	158°28.4	46.2	Avior	234°14.6	-59°35.4
23	157°03.8	182°08.5	38.0	191°21.6	56.2	115°34.7	06.6	173°30.6	46.1	Suhail	222°46.5	-43°32.0
Mer.pass. 12:33		ν -0.5' d -1.0' m -3.88		ν 0.6' d -0.6' m 1.24		ν 2.0' d 0.2' m -2.13		ν 2.2' d -0.1' m 1.01		Miaplacidus	221°37.7	-69°49.1
										Alphard	217°48.2	-8°45.9
										Regulus	207°34.9	11°50.9
										Dubhe	193°41.1	61°37.2
										Denebola	182°25.3	14°26.1
										Gienah	175°44.0	-17°40.7
										Acrux	173°00.3	-63°14.0
										Gacrux	171°51.9	-57°14.9
										Alioth	166°13.0	55°49.6
										Spica	158°22.8	-11°17.3
										Alkaid	152°52.1	49°11.3
										Hadar	148°36.5	-60°29.3
										Menkent	147°58.2	-36°29.3
										Arcturus	145°48.3	19°03.2
										Rigil Kent.	139°40.8	-60°56.0
										Kochab	137°19.0	74°03.0
										Zuben'ubi	136°56.6	-16°08.6
										Alphecca	126°04.1	26°37.7
										Antares	112°16.6	-26°29.1
										Atria	107°11.4	-69°04.0
										Sabik	102°03.5	-15°45.4
										Shaula	96°11.3	-37°07.2
										Rasalhague	95°59.2	12°32.3
										Eltanin	90°42.6	51°28.7
										Kaus Aust.	83°33.5	-34°22.3
										Vega	80°33.8	38°48.0
										Nunki	75°48.7	-26°16.0
										Altair	62°00.8	8°55.7
										Peacock	53°07.2	-56°39.3
										Deneb	49°26.6	45°21.7
										Enif	33°39.8	9°58.9
										Al Na'ir	27°34.2	-46°50.7
										Fomalhaut	15°15.6	-29°29.8
										Scheat	13°46.2	28°12.7
										Markab	13°30.9	15°19.9
										Mar 13 Wed		
										Venus	26°13.2	10:51
										Mars	35°01.7	10:15
										Jupiter	318°41.7	15:19
										Saturn	16°33.3	11:28
										Mar 14 Thu		
										Venus	25°01.8	10:52
										Mars	34°15.9	10:14
										Jupiter	318°30.4	15:16
										Saturn	16°26.6	11:24
										Mar 15 Fri		
										Venus	23°50.6	10:53
										Mars	33°30.3	10:13
										Jupiter	318°19.0	15:12
										Saturn	16°19.8	11:21
										Horizontal parallax		
										Venus:	0.1	
										Mars:	0.1	
Mer.pass. 12:26		ν -0.5' d -1.1' m -3.88		ν 0.6' d -0.6' m 1.23		ν 2.0' d 0.2' m -2.12		ν 2.2' d -0.1' m 1.02				

h Sun				Moon			
Wed	GHA	Dec		GHA	ν	Dec	d HP
0	177°38.2	S02°49.1		143°56.5	9.6'	N12°33.8	16.0' 60.2'
1	192°38.4	48.1		158°25.1	9.6'	12°49.8	15.9' 60.2'
2	207°38.5	47.1		172°53.6	9.5'	13°05.6	15.8' 60.1'
3	222°38.7	· · 46.1		187°22.2	9.5'	13°21.4	15.7' 60.1'
4	237°38.9	45.1		201°50.7	9.5'	13°37.1	15.6' 60.1'
5	252°39.0	44.1		216°19.1	9.4'	13°52.6	15.5' 60.0'
6	267°39.2	S02°43.1		230°47.5	9.4'	N14°08.1	15.4' 60.0'
7	282°39.4	42.2		245°15.9	9.3'	14°23.5	15.3' 60.0'
8	297°39.6	41.2		259°44.2	9.3'	14°38.8	15.2' 59.9'
9	312°39.7	· · 40.2		274°12.6	9.3'	14°54.0	15.1' 59.9'
10	327°39.9	39.2		288°40.8	9.2'	15°09.1	15.0' 59.9'
11	342°40.1	38.2		303°09.1	9.2'	15°24.0	14.9' 59.8'
12	357°40.2	S02°37.2		317°37.3	9.2'	N15°38.9	14.8' 59.8'
13	12°40.4	36.2		332°05.4	9.1'	15°53.7	14.7' 59.8'
14	27°40.6	35.3		346°33.5	9.1'	16°08.3	14.5' 59.7'
15	42°40.8	· · 34.3		1°01.6	9.0'	16°22.9	14.4' 59.7'
16	57°40.9	33.3		15°29.6	9.0'	16°37.3	14.3' 59.6'
17	72°41.1	32.3		29°57.6	9.0'	16°51.6	14.2' 59.6'
18	87°41.3	S02°31.3		44°25.6	8.9'	N17°05.8	14.1' 59.6'
19	102°41.4	30.3		58°53.5	8.9'	17°19.9	14.0' 59.5'
20	117°41.6	29.3		73°21.4	8.8'	17°33.9	13.8' 59.5'
21	132°41.8	· · 28.3		87°49.2	8.8'	17°47.7	13.7' 59.5'
22	147°42.0	27.4		102°17.0	8.7'	18°01.4	13.6' 59.4'
23	162°42.1	26.4		116°44.7	8.7'	18°15.1	13.5' 59.4'
SD = 16.1'			d = -1.0'	SD = 16.4'			

Thu	GHA	Dec		GHA	ν	Dec	d HP
0	177°42.3	S02°25.4		131°12.4	8.7'	N18°28.5	13.4' 59.3'
1	192°42.5	24.4		145°40.1	8.6'	18°41.9	13.2' 59.3'
2	207°42.6	23.4		160°07.7	8.6'	18°55.1	13.1' 59.3'
3	222°42.8	· · 22.4		174°35.3	8.5'	19°08.2	13.0' 59.2'
4	237°43.0	21.4		189°02.8	8.5'	19°21.2	12.9' 59.2'
5	252°43.2	20.4		203°30.3	8.4'	19°34.1	12.7' 59.1'
6	267°43.3	S02°19.5		217°57.7	8.4'	N19°46.8	12.6' 59.1'
7	282°43.5	18.5		232°25.1	8.4'	19°59.4	12.5' 59.1'
8	297°43.7	17.5		246°52.5	8.3'	20°11.9	12.3' 59.0'
9	312°43.9	· · 16.5		261°19.8	8.3'	20°24.2	12.2' 59.0'
10	327°44.0	15.5		275°47.0	8.2'	20°36.4	12.1' 59.0'
11	342°44.2	14.5		290°14.2	8.2'	20°48.4	11.9' 58.9'
12	357°44.4	S02°13.5		304°41.4	8.1'	N21°00.3	11.8' 58.9'
13	12°44.5	12.5		319°08.5	8.1'	21°12.1	11.6' 58.8'
14	27°44.7	11.6		333°35.6	8.0'	21°23.8	11.5' 58.8'
15	42°44.9	· · 10.6		348°02.7	8.0'	21°35.3	11.4' 58.8'
16	57°45.1	09.6		2°29.7	8.0'	21°46.6	11.2' 58.7'
17	72°45.2	08.6		16°56.6	7.9'	21°57.8	11.1' 58.7'
18	87°45.4	S02°07.6		31°23.6	7.9'	N22°08.9	10.9' 58.6'
19	102°45.6	06.6		45°50.4	7.8'	22°19.8	10.8' 58.6'
20	117°45.8	05.6		60°17.3	7.8'	22°30.6	10.6' 58.6'
21	132°45.9	· · 04.6		74°44.1	7.8'	22°41.3	10.5' 58.5'
22	147°46.1	03.7		89°10.8	7.7'	22°51.8	10.3' 58.5'
23	162°46.3	02.7		103°37.5	7.7'	23°02.1	10.2' 58.4'
SD = 16.1'			d = -1.0'	SD = 16.2'			

Fri	GHA	Dec		GHA	ν	Dec	d HP
0	177°46.5	S02°01.7		118°04.2	7.6'	N23°12.3	10.0' 58.4'
1	192°46.6	02°00.7		132°30.9	7.6'	23°22.3	9.9' 58.4'
2	207°46.8	01°59.7		146°57.4	7.6'	23°32.2	9.7' 58.3'
3	222°47.0	· · 58.7		161°24.0	7.5'	23°42.0	9.6' 58.3'
4	237°47.2	57.7		175°50.5	7.5'	23°51.6	9.4' 58.2'
5	252°47.3	56.7		190°17.0	7.5'	24°01.0	9.3' 58.2'
6	267°47.5	S01°55.8		204°43.5	7.4'	N24°10.3	9.1' 58.2'
7	282°47.7	54.8		219°09.9	7.4'	24°19.4	9.0' 58.1'
8	297°47.9	53.8		233°36.3	7.3'	24°28.4	8.8' 58.1'
9	312°48.1	· · 52.8		248°02.6	7.3'	24°37.2	8.7' 58.0'
10	327°48.2	51.8		262°28.9	7.3'	24°45.9	8.5' 58.0'
11	342°48.4	50.8		276°55.2	7.3'	24°54.4	8.4' 58.0'
12	357°48.6	S01°49.8		291°21.5	7.2'	N25°02.8	8.2' 57.9'
13	12°48.8	48.8		305°47.7	7.2'	25°10.9	8.0' 57.9'
14	27°48.9	47.8		320°13.9	7.2'	25°19.0	7.9' 57.8'
15	42°49.1	· · 46.9		334°40.0	7.1'	25°26.9	7.7' 57.8'
16	57°49.3	45.9		349°06.2	7.1'	25°34.6	7.6' 57.8'
17	72°49.5	44.9		3°32.3	7.1'	25°42.1	7.4' 57.7'
18	87°49.6	S01°43.9		17°58.4	7.1'	N25°49.5	7.2' 57.7'
19	102°49.8	42.9		32°24.4	7.0'	25°56.7	7.1' 57.6'
20	117°50.0	41.9		46°50.5	7.0'	26°03.8	6.9' 57.6'
21	132°50.2	· · 40.9		61°16.5	7.0'	26°10.7	6.7' 57.6'
22	147°50.3	39.9		75°42.5	7.0'	26°17.5	6.6' 57.5'
23	162°50.5	39.0		90°08.5	7.0'	26°24.1	6.4' 57.5'
SD = 16.1'			d = -1.0'	SD = 15.9'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	03:57	05:20	06:27	17:54	19:01	20:25
N 70°	04:10	05:24	06:25	17:55	18:56	20:11
68°	04:21	05:28	06:23	17:57	18:52	20:00
66°	04:30	05:31	06:22	17:58	18:49	19:51
64°	04:37	05:33	06:21	17:59	18:46	19:43
62°	04:43	05:35	06:19	18:00	18:44	19:37
60°	04:48	05:37	06:19	18:01	18:42	19:32
N 58°	04:53	05:39	06:18	18:02	18:41	19:27
56°	04:57	05:40	06:17	18:02	18:39	19:23
54°	05:00	05:41	06:16	18:03	18:38	19:20
52°	05:03	05:42	06:16	18:03	18:37	19:17
50°	05:05	05:43	06:15	18:04	18:36	19:14
45°	05:10	05:44	06:14	18:05	18:34	19:09
N 40°	05:14	05:46	06:13	18:06	18:33	19:05
35°	05:17	05:46	06:12	18:07	18:32	19:02
30°	05:19	05:47	06:11	18:08	18:32	19:00
20°	05:21	05:47	06:09	18:09	18:31	18:57
N 10°	05:22	05:46	06:07	18:11	18:32	18:56
0°	05:21	05:45	06:06	18:12	18:33	18:57
S 10°	05:19	05:43	06:04	18:14	18:35	18:59
20°	05:14	05:40	06:02	18:16	18:38	19:03
30°	05:08	05:36	06:00	18:18	18:42	19:10
35°	05:03	05:33	05:59	18:19	18:44	19:14
40°	04:58	05:30	05:57	18:20	18:48	19:19
45°	04:51	05:26	05:55	18:22	18:52	19:26
S 50°	04:42	05:20	05:53	18:24	18:57	19:35
52°	04:37	05:18	05:52	18:25	18:59	19:39
54°	04:32	05:15	05:51	18:26	19:02	19:44
56°	04:27	05:12	05:49	18:28	19:05	19:50
58°	04:20	05:08	05:48	18:29	19:08	19:56
S 60°	04:12	05:04	05:46	18:30	19:12	20:04

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	05:19	□	□	□	□	□
N 70°	05:45	04:28	□	· ·	01:55	□
68°	06:05	05:22	□	· ·	01:02	□
66°	06:21	05:55	04:46	· ·	00:30	03:34
64°	06:34	06:20	05:57	· ·	00:07	02:24
62°	06:45	06:40	06:33	23:48	· ·	01:49
60°	06:55	06:56	06:59	23:33	· ·	01:24
N 58°	07:03	07:09	07:20	23:20	· ·	01:04
56°	07:11	07:21	07:37	23:09	· ·	00:47
54°	07:18	07:32	07:52	23:00	· ·	00:33
52°	07:24	07:41	08:04	22:51	· ·	00:21
50°	07:29	07:49	08:16	22:44	· ·	00:10
45°	07:41	08:07	08:39	22:27	23:48	· ·
N 40°	07:51	08:22	08:58	22:14	23:30	· ·
35°	08:00	08:34	09:14	22:03	23:15	· ·
30°	08:07	08:45	09:28	21:53	23:02	· ·
20°	08:21	09:04	09:52	21:36	22:40	23:42
N 10°	08:32	09:21	10:12	21:22	22:21	23:20
0°	08:43	09:37	10:32	21:08	22:03	22:59
S 10°	08:54	09:52	10:51	20:55	21:45	22:38
20°	09:06	10:09	11:12	20:41	21:26	22:16
30°	09:20	10:29	11:37	20:24	21:05	21:50
35°	09:28	10:41	11:51	20:15	20:52	21:35
40°	09:38	10:54	12:08	20:04	20:38	21:17
45°	09:48	11:10	12:28	19:52	20:20	20:56
S 50°	10:02	11:30	12:54	19:37	19:59	20:30
52°	10:08	11:39	13:06	19:29	19:49	20:17
54°	10:15	11:49	13:21	19:22	19:38	20:02
56°	10:23	12:01	13:37	19:13	19:25	19:45
58°	10:31	12:15	13:57	19:03	19:11	19:24
S 60°	10:42	12:32	14:23	18:52	18:53	18:58

Day	Sun		Mer. Pass hh:mm	Moon		Age 3-5 10-28%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
13	09:27	09:19	12:09	14:56	02:29	
14	09:11	09:02	12:09	15:50	03:22	
15	08:54	08:46	12:09	16:45	04:17	

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	176°02.8	196°21.1	S09°53.7	207°16.7	S13°54.4	133°47.0	N15°21.4	192°02.3	S08°35.3
1	191°05.2	211°20.6	52.6	222°17.3	53.7	148°49.0	21.5	207°04.5	35.2
2	206°07.7	226°20.1	51.5	237°17.9	53.1	163°51.0	21.7	222°06.7	35.1
3	221°10.2	241°19.7	· · 50.4	252°18.5	· · 52.4	178°53.0	· · 21.8	237°08.9	· · 35.0
4	236°12.6	256°19.2	49.3	267°19.0	51.8	193°54.9	22.0	252°11.1	34.9
5	251°15.1	271°18.7	48.2	282°19.6	51.1	208°56.9	22.1	267°18.2	34.8
6	266°17.6	286°18.2	S09°47.1	297°20.2	S13°50.5	223°58.9	N15°22.3	282°15.4	S08°34.7
7	281°20.0	301°17.8	46.0	312°20.8	49.8	239°00.9	22.4	297°17.6	34.5
8	296°22.5	316°17.3	44.9	327°21.3	49.2	254°02.8	22.6	312°19.8	34.4
9	311°25.0	331°16.8	· · 43.8	342°21.9	· · 48.5	269°04.8	· · 22.8	327°22.0	· · 34.3
10	326°27.4	346°16.4	42.7	357°22.5	47.9	284°06.8	22.9	342°24.2	34.2
11	341°29.9	1°15.9	41.6	12°23.1	47.2	299°08.8	23.1	357°26.4	34.1
12	356°32.3	16°15.4	S09°40.4	27°23.7	S13°46.5	314°10.7	N15°23.2	12°28.5	S08°34.0
13	11°34.8	31°15.0	39.3	42°24.2	45.9	329°12.7	23.4	27°30.7	33.9
14	26°37.3	46°14.5	38.2	57°24.8	45.2	344°14.7	23.5	42°32.9	33.8
15	41°39.7	61°14.0	· · 37.1	72°25.4	· · 44.6	359°16.6	· · 23.7	57°35.1	· · 33.7
16	56°42.2	76°13.6	36.0	87°26.0	43.9	14°18.6	23.8	72°37.3	33.5
17	71°44.7	91°13.1	34.9	102°26.6	43.3	29°20.6	24.0	87°39.5	33.4
18	86°47.1	106°12.7	S09°33.8	117°27.1	S13°42.6	44°22.6	N15°24.1	102°41.7	S08°33.3
19	101°49.6	121°12.2	32.7	132°27.7	42.0	59°24.5	24.3	117°43.8	33.2
20	116°52.1	136°11.7	31.6	147°28.3	41.3	74°26.5	24.4	132°46.0	33.1
21	131°54.5	151°11.3	· · 30.5	162°28.9	· · 40.7	89°28.5	· · 24.6	147°48.2	· · 33.0
22	146°57.0	166°10.8	29.4	177°29.5	40.0	104°30.5	24.7	162°50.4	32.9
23	161°59.5	181°10.3	28.3	192°30.0	39.4	119°32.4	24.9	177°52.6	32.8
Mer.pass. 12:14		ν -0.5' d -1.1' m -3.88		ν 0.6' d -0.6' m 1.23		ν 2.0' d 0.2' m -2.11		ν 2.2' d -0.1' m 1.03	

Mar 16 Sat	SHA	Mer.pass
Venus	22° 39.6	10:53
Mars	32° 44.8	10:12
Jupiter	318° 07.5	15:09
Saturn	16° 13.0	11:17

Mar 17 Sun	SHA	Mer.pass
Venus	21° 28.8	10:54
Mars	31° 59.3	10:11
Jupiter	317° 55.9	15:06
Saturn	16° 06.3	11:14

Mar 18 Mon	SHA	Mer.pass
Venus	20° 18.3	10:55
Mars	31° 14.0	10:10
Jupiter	317° 44.2	15:03
Saturn	15° 59.6	11:10

Horizontal parallax	
Venus:	0.1
Mars:	0.1

h Sun Moon						
Sat	GHA	Dec	GHA	ν	Dec	d HP
0	177°50.7	S01°38.0	104°34.4	6.9'	N26°30.5	6.3' 57.4'
1	192°50.9	37.0	119°00.4	6.9'	26°36.7	6.1' 57.4'
2	207°51.1	36.0	133°26.3	6.9'	26°42.8	5.9' 57.4'
3	222°51.2	· · 35.0	147°52.2	6.9'	26°48.8	5.8' 57.3'
4	237°51.4	34.0	162°18.1	6.9'	26°54.5	5.6' 57.3'
5	252°51.6	33.0	176°44.0	6.9'	27°00.1	5.4' 57.2'
6	267°51.8	S01°32.0	191°09.9	6.9'	N27°05.5	5.3' 57.2'
7	282°51.9	31.0	205°35.8	6.9'	27°10.8	5.1' 57.2'
8	297°52.1	30.1	220°01.7	6.9'	27°15.9	4.9' 57.1'
9	312°52.3	· · 29.1	234°27.5	6.9'	27°20.9	4.8' 57.1'
10	327°52.5	28.1	248°53.4	6.9'	27°25.6	4.6' 57.1'
11	342°52.7	27.1	263°19.2	6.9'	27°30.2	4.4' 57.0'
12	357°52.8	S01°26.1	277°45.1	6.9'	N27°34.7	4.3' 57.0'
13	12°53.0	25.1	292°10.9	6.9'	27°39.0	4.1' 57.0'
14	27°53.2	24.1	306°36.8	6.9'	27°43.1	3.9' 56.9'
15	42°53.4	· · 23.1	321°02.6	6.9'	27°47.0	3.8' 56.9'
16	57°53.6	22.1	335°28.5	6.9'	27°50.8	3.6' 56.8'
17	72°53.7	21.2	349°54.4	6.9'	27°54.4	3.4' 56.8'
18	87°53.9	S01°20.2	4°20.2	6.9'	N27°57.8	3.3' 56.8'
19	102°54.1	19.2	18°46.1	6.9'	28°01.1	3.1' 56.7'
20	117°54.3	18.2	33°12.0	6.9'	28°04.2	3.0' 56.7'
21	132°54.5	· · 17.2	47°37.9	6.9'	28°07.2	2.8' 56.7'
22	147°54.6	16.2	62°03.8	6.9'	28°10.0	2.6' 56.6'
23	162°54.8	15.2	76°29.8	6.9'	28°12.6	2.5' 56.6'
SD = 16.1' d = -1.0' SD = 15.7'						
Sun	GHA	Dec	GHA	ν	Dec	d HP
0	177°55.0	S01°14.2	90°55.7	7.0'	N28°15.1	2.3' 56.6'
1	192°55.2	13.3	105°21.7	7.0'	28°17.3	2.1' 56.5'
2	207°55.4	12.3	119°47.7	7.0'	28°19.5	2.0' 56.5'
3	222°55.5	· · 11.3	134°13.7	7.0'	28°21.4	1.8' 56.5'
4	237°55.7	10.3	148°39.7	7.1'	28°23.2	1.6' 56.4'
5	252°55.9	09.3	163°05.8	7.1'	28°24.9	1.5' 56.4'
6	267°56.1	S01°08.3	177°31.8	7.1'	N28°26.3	1.3' 56.4'
7	282°56.3	07.3	191°58.0	7.1'	28°27.7	1.1' 56.3'
8	297°56.4	06.3	206°24.1	7.2'	28°28.8	1.0' 56.3'
9	312°56.6	· · 05.3	220°50.3	7.2'	28°29.8	0.8' 56.3'
10	327°56.8	04.4	235°16.5	7.2'	28°30.6	0.7' 56.2'
11	342°57.0	03.4	249°42.7	7.3'	28°31.3	0.5' 56.2'
12	357°57.2	S01°02.4	264°09.0	7.3'	N28°31.8	0.3' 56.2'
13	12°57.3	01.4	278°35.3	7.3'	28°32.1	0.2' 56.1'
14	27°57.5	01°00.4	293°01.6	7.4'	28°32.3	0.0' 56.1'
15	42°57.7	00°59.4	307°28.0	7.4'	28°32.3	-0.1' 56.1'
16	57°57.9	58.4	321°54.4	7.5'	28°32.2	-0.3' 56.0'
17	72°58.1	57.4	336°20.9	7.5'	28°31.9	-0.5' 56.0'
18	87°58.2	S00°56.4	350°47.4	7.6'	N28°31.4	-0.6' 56.0'
19	102°58.4	55.5	5°14.0	7.6'	28°30.8	-0.8' 55.9'
20	117°58.6	54.5	19°40.6	7.6'	28°30.1	-0.9' 55.9'
21	132°58.8	· · 53.5	34°07.2	7.7'	28°29.1	-1.1' 55.9'
22	147°59.0	52.5	48°33.9	7.7'	28°28.1	-1.2' 55.8'
23	162°59.1	51.5	63°00.7	7.8'	28°26.8	-1.4' 55.8'
SD = 16.1' d = -1.0' SD = 15.4'						
Mon	GHA	Dec	GHA	ν	Dec	d HP
0	177°59.3	S00°50.5	77°27.5	7.9'	N28°25.5	-1.5' 55.8'
1	192°59.5	49.5	91°54.3	7.9'	28°23.9	-1.7' 55.7'
2	207°59.7	48.5	106°21.2	8.0'	28°22.2	-1.8' 55.7'
3	222°59.9	· · 47.5	120°48.2	8.0'	28°20.4	-2.0' 55.7'
4	238°00.1	46.6	135°15.2	8.1'	28°18.4	-2.1' 55.7'
5	253°00.2	45.6	149°42.3	8.1'	28°16.2	-2.3' 55.6'
6	268°00.4	S00°44.6	164°09.4	8.2'	N28°14.0	-2.4' 55.6'
7	283°00.6	43.6	178°36.6	8.3'	28°11.5	-2.6' 55.6'
8	298°00.8	42.6	193°03.9	8.3'	28°08.9	-2.7' 55.5'
9	313°01.0	· · 41.6	207°31.2	8.4'	28°06.2	-2.9' 55.5'
10	328°01.2	40.6	221°58.6	8.5'	28°03.3	-3.0' 55.5'
11	343°01.3	39.6	236°26.1	8.5'	28°00.3	-3.2' 55.5'
12	358°01.5	S00°38.6	250°53.6	8.6'	N27°57.1	-3.3' 55.4'
13	13°01.7	37.7	265°21.2	8.7'	27°53.8	-3.5' 55.4'
14	28°01.9	36.7	279°48.8	8.7'	27°50.3	-3.6' 55.4'
15	43°02.1	· · 35.7	294°16.6	8.8'	27°46.7	-3.7' 55.4'
16	58°02.2	34.7	308°44.4	8.9'	27°43.0	-3.9' 55.3'
17	73°02.4	33.7	323°12.2	8.9'	27°39.1	-4.0' 55.3'
18	88°02.6	S00°32.7	337°40.2	9.0'	N27°35.1	-4.2' 55.3'
19	103°02.8	31.7	352°08.2	9.1'	27°30.9	-4.3' 55.3'
20	118°03.0	30.7	6°36.3	9.2'	27°26.6	-4.4' 55.2'
21	133°03.2	· · 29.7	21°04.4	9.2'	27°22.2	-4.6' 55.2'
22	148°03.3	28.8	35°32.7	9.3'	27°17.6	-4.7' 55.2'
23	163°03.5	27.8	50°01.0	9.4'	27°12.9	-4.8' 55.2'
SD = 16.1' d = -1.0' SD = 15.2'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	03:38	05:04	06:11	18:07	19:15	20:43
N 70°	03:54	05:10	06:11	18:08	19:09	20:26
68°	04:07	05:15	06:11	18:08	19:03	20:12
66°	04:17	05:19	06:10	18:08	18:59	20:02
64°	04:26	05:23	06:10	18:08	18:55	19:53
62°	04:33	05:26	06:10	18:08	18:52	19:46
60°	04:39	05:28	06:09	18:08	18:50	19:39
N 58°	04:44	05:30	06:09	18:08	18:48	19:34
56°	04:48	05:32	06:09	18:08	18:46	19:30
54°	04:52	05:34	06:09	18:09	18:44	19:26
52°	04:56	05:35	06:09	18:09	18:42	19:22
50°	04:59	05:36	06:09	18:09	18:41	19:19
45°	05:05	05:39	06:08	18:09	18:38	19:13
N 40°	05:09	05:41	06:08	18:09	18:36	19:08
35°	05:13	05:42	06:07	18:10	18:35	19:04
30°	05:15	05:43	06:07	18:10	18:34	19:01
20°	05:19	05:44	06:06	18:10	18:32	18:58
N 10°	05:20	05:45	06:06	18:11	18:32	18:56
0°	05:20	05:44	06:05	18:11	18:32	18:56
S 10°	05:19	05:43	06:04	18:12	18:33	18:58
20°	05:15	05:41	06:03	18:13	18:35	19:01
30°	05:10	05:38	06:02	18:14	18:38	19:06
35°	05:06	05:36	06:01	18:15	18:40	19:10
40°	05:01	05:33	06:00	18:16	18:43	19:14
45°	04:55	05:30	05:59	18:17	18:46	19:20
S 50°	04:47	05:25	05:58	18:18	18:50	19:28
52°	04:43	05:23	05:57	18:18	18:52	19:32
54°	04:39	05:21	05:56	18:19	18:54	19:36
56°	04:33	05:18	05:56	18:20	18:57	19:41
58°	04:28	05:15	05:55	18:20	19:00	19:47
S 60°	04:21	05:12	05:54	18:21	19:03	19:53

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	☐	☐	☐	☐	☐	☐
64°	☐	☐	☐	☐	☐	☐
62°	06:23	☐	☐	03:57	☐	☐
60°	07:09	07:37	08:41	03:11	04:42	05:34
N 58°	07:39	08:16	09:18	02:41	04:03	04:57
56°	08:02	08:43	09:44	02:19	03:36	04:30
54°	08:21	09:05	10:05	02:00	03:14	04:09
52°	08:37	09:23	10:22	01:45	02:57	03:52
50°	08:51	09:38	10:37	01:31	02:41	03:37
45°	09:19	10:08	11:07	01:03	02:11	03:07
N 40°	09:41	10:32	11:30	00:42	01:47	02:43
35°	10:00	10:52	11:49	00:24	01:27	02:24
30°	10:16	11:09	12:06	00:08	01:11	02:07
20°	10:43	11:37	12:34	· ·	00:42	01:39
N 10°	11:06	12:02	12:58	· ·	00:18	01:14
0°	11:28	12:25	13:20	23:56	· ·	00:52
S 10°	11:50	12:48	13:42	23:33	· ·	00:29
20°	12:14	13:12	14:06	23:09	· ·	00:04
30°	12:42	13:41	14:33	22:41	23:36	· ·
35°	12:58	13:58	14:50	22:24	23:19	· ·
40°	13:17	14:18	15:09	22:04	22:59	23:59
45°	13:40	14:42	15:32	21:41	22:35	23:37
S 50°	14:10	15:13	16:01	21:10	22:04	23:08
52°	14:25	15:29	16:15	20:56	21:48	22:54
54°	14:42	15:47	16:32	20:38	21:30	22:38
56°	15:03	16:09	16:52	20:17	21:08	22:18
58°	15:29	16:38	17:17	19:51	20:40	21:53
S 60°	16:06	17:20	17:51	19:14	19:58	21:20

Day	Sun		Mer. Pass	Moon		Age 6-8 38-59%
	Eqn. of Time 00 ^h mm:ss	Time 12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
16	08:37	08:29	12:08	17:42	05:14	
17	08:20	08:11	12:08	18:38	06:10	
18	08:03	07:54	12:08	19:33	07:06	

March 19, 20, 21 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	177°01.9	196°09.9	S09°27.2	207°30.6	S13°38.7	134°34.4	N15°25.1	192°54.8	S08°32.7	Alpheratz	357°35.8	29°13.3
1	192°04.4	211°09.4	26.0	222°31.2	38.0	149°36.4	25.2	207°57.0	32.6	Ankaa	353°08.2	-42°10.6
2	207°06.8	226°08.9	24.9	237°31.8	37.4	164°38.3	25.4	222°59.1	32.4	Schedar	349°32.4	56°40.2
3	222°09.3	241°08.5	•• 23.8	252°32.4	•• 36.7	179°40.3	•• 25.5	238°01.3	•• 32.3	Diphda	348°48.3	-17°51.4
4	237°11.8	256°08.0	22.7	267°32.9	36.1	194°42.3	25.7	253°03.5	32.2	Achernar	335°21.2	-57°07.0
5	252°14.2	271°07.6	21.6	282°33.5	35.4	209°44.3	25.8	268°05.7	32.1	Hamal	327°52.3	23°34.5
6	267°16.7	286°07.1	S09°20.5	297°34.1	S13°34.8	224°46.2	N15°26.0	283°07.9	S08°32.0	Polaris	314°42.4	89°22.2
7	282°19.2	301°06.6	19.4	312°34.7	34.1	239°48.2	26.1	298°10.1	31.9	Acamar	315°12.6	-40°12.7
8	297°21.6	316°06.2	18.3	327°35.3	33.5	254°50.2	26.3	313°12.3	31.8	Menkar	314°07.1	4°11.0
9	312°24.1	331°05.7	•• 17.1	342°35.8	•• 32.8	269°52.1	•• 26.4	328°14.4	•• 31.7	Mirfak	308°29.5	49°56.9
10	327°26.6	346°05.3	16.0	357°36.4	32.1	284°54.1	26.6	343°16.6	31.6	Aldebaran	290°40.5	16°33.4
11	342°29.0	1°04.8	14.9	12°37.0	31.5	299°56.1	26.7	358°18.8	31.5	Rigel	281°04.6	-8°10.6
12	357°31.5	16°04.3	S09°13.8	27°37.6	S13°30.8	314°58.1	N15°26.9	13°21.0	S08°31.3	Capella	280°23.0	46°01.5
13	12°34.0	31°03.9	12.7	42°38.2	30.2	330°00.0	27.0	28°23.2	31.2	Bellatrix	278°23.6	6°22.2
14	27°36.4	46°03.4	11.6	57°38.8	29.5	345°02.0	27.2	43°25.4	31.1	Elnath	278°02.8	28°37.7
15	42°38.9	61°03.0	•• 10.5	72°39.3	•• 28.9	0°04.0	•• 27.4	58°27.6	•• 31.0	Alnilam	275°38.4	-1°11.3
16	57°41.3	76°02.5	09.3	87°39.9	28.2	15°05.9	27.5	73°29.7	30.9	Betelgeuse	270°52.8	7°24.6
17	72°43.8	91°02.1	08.2	102°40.5	27.5	30°07.9	27.7	88°31.9	30.8	Canopus	263°52.6	-52°42.8
18	87°46.3	106°01.6	S09°07.1	117°41.1	S13°26.9	45°09.9	N15°27.8	103°34.1	S08°30.7	Sirius	258°26.8	-16°45.1
19	102°48.7	121°01.2	06.0	132°41.7	26.2	60°11.8	28.0	118°36.3	30.6	Adhara	255°06.3	-29°00.5
20	117°51.2	136°00.7	04.9	147°42.3	25.6	75°13.8	28.1	133°38.5	30.5	Procyon	244°51.4	5°09.7
21	132°53.7	151°00.2	•• 03.7	162°42.8	•• 24.9	90°15.8	•• 28.3	148°40.7	•• 30.4	Pollux	243°18.0	27°58.1
22	147°56.1	165°59.8	02.6	177°43.4	24.2	105°17.7	28.4	163°42.9	30.3	Avior	234°14.7	-59°35.4
23	162°58.6	180°59.3	01.5	192°44.0	23.6	120°19.7	28.6	178°45.0	30.1	Suhail	222°46.5	-43°32.0
Mer.pass. 12:10		ν -0.5' d -1.1' m -3.88		ν 0.6' d -0.7' m 1.22		ν 2.0' d 0.2' m -2.11		ν 2.2' d -0.1' m 1.03				

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	178°01.1	195°58.9	S09°00.4	207°44.6	S13°22.9	135°21.7	N15°28.7	193°47.2	S08°30.0		
1	193°03.5	210°58.4	08°59.3	222°45.2	22.3	150°23.7	28.9	208°49.4	29.9		
2	208°06.0	225°58.0	58.1	237°45.8	21.6	165°25.6	29.0	223°51.6	29.8		
3	223°08.5	240°57.5	•• 57.0	252°46.3	•• 20.9	180°27.6	•• 29.2	238°53.8	•• 29.7		
4	238°10.9	255°57.1	55.9	267°46.9	20.3	195°29.6	29.4	253°56.0	29.6		
5	253°13.4	270°56.6	54.8	282°47.5	19.6	210°31.5	29.5	268°58.2	29.5		
6	268°15.8	285°56.2	S08°53.7	297°48.1	S13°19.0	225°33.5	N15°29.7	284°00.3	S08°29.4		
7	283°18.3	300°55.7	52.5	312°48.7	18.3	240°35.5	29.8	299°02.5	29.3		
8	298°20.8	315°55.3	51.4	327°49.3	17.6	255°37.4	30.0	314°04.7	29.2		
9	313°23.2	330°54.8	•• 50.3	342°49.9	•• 17.0	270°39.4	•• 30.1	329°06.9	•• 29.0		
10	328°25.7	345°54.4	49.2	357°50.4	16.3	285°41.4	30.3	344°09.1	28.9		
11	343°28.2	0°53.9	48.0	12°51.0	15.6	300°43.3	30.4	359°11.3	28.8		
12	358°30.6	15°53.5	S08°46.9	27°51.6	S13°15.0	315°45.3	N15°30.6	14°13.5	S08°28.7		
13	13°33.1	30°53.0	45.8	42°52.2	14.3	330°47.3	30.7	29°15.6	28.6		
14	28°35.6	45°52.6	44.7	57°52.8	13.6	345°49.2	30.9	44°17.8	28.5		
15	43°38.0	60°52.1	•• 43.5	72°53.4	•• 13.0	0°51.2	•• 31.0	59°20.0	•• 28.4		
16	58°40.5	75°51.7	42.4	87°54.0	12.3	15°53.2	31.2	74°22.2	28.3		
17	73°42.9	90°51.2	41.3	102°54.5	11.7	30°55.1	31.4	89°24.4	28.2		
18	88°45.4	105°50.8	S08°40.1	117°55.1	S13°11.0	45°57.1	N15°31.5	104°26.6	S08°28.0		
19	103°47.9	120°50.3	39.0	132°55.7	10.3	60°59.1	31.7	119°28.8	28.1		
20	118°50.3	135°49.9	37.9	147°56.3	09.7	76°01.0	31.8	134°31.0	27.8		
21	133°52.8	150°49.4	•• 36.8	162°56.9	•• 09.0	91°03.0	•• 32.0	149°33.1	•• 27.6		
22	148°55.3	165°49.0	35.6	177°57.5	08.3	106°05.0	32.1	164°35.3	27.7		
23	163°57.7	180°48.5	34.5	192°58.1	07.7	121°06.9	32.3	179°37.5	27.5		
Mer.pass. 12:06		ν -0.5' d -1.1' m -3.88		ν 0.6' d -0.7' m 1.22		ν 2.0' d 0.2' m -2.10		ν 2.2' d -0.1' m 1.03			

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	179°00.2	195°48.1	S08°33.4	207°58.7	S13°07.0	136°08.9	N15°32.4	194°39.7	S08°27.4		
1	194°02.7	210°47.6	32.2	222°59.2	06.3	151°10.9	32.6	209°41.9	27.3		
2	209°05.1	225°47.2	31.1	237°59.8	05.7	166°12.8	32.7	224°44.1	27.2		
3	224°07.6	240°46.7	•• 30.0	253°00.4	•• 05.0	181°14.8	•• 32.9	239°46.3	•• 27.1		
4	239°10.1	255°46.3	28.8	268°01.0	04.3	196°16.7	33.0	254°48.5	27.0		
5	254°12.5	270°45.8	27.7	283°01.6	03.7	211°18.7	33.2	269°50.6	26.9		
6	269°15.0	285°45.4	S08°26.6	298°02.2	S13°03.0	226°20.7	N15°33.4	284°52.8	S08°26.8		
7	284°17.4	300°45.0	25.5	313°02.8	02.3	241°22.6	33.5	299°55.0	26.6		
8	299°19.9	315°44.5	24.3	328°03.4	01.7	256°24.6	33.7	314°57.2	26.5		
9	314°22.4	330°44.1	•• 23.2	343°04.0	•• 01.0	271°26.6	•• 33.8	329°59.4	•• 26.4		
10	329°24.8	345°43.6	22.0	358°04.5	13°00.3	286°28.5	34.0	345°01.6	26.3		
11	344°27.3	0°43.2	20.9	13°05.1	12°59.7	301°30.5	34.1	0°03.8	26.2		
12	359°29.8	15°42.7	S08°19.8	28°05.7	S12°59.0	316°32.5	N15°34.3	15°05.9	S08°26.1		
13	14°32.2	30°42.3	18.6	43°06.3	58.3	331°34.4	34.4	30°08.1	26.0		
14	29°34.7	45°41.9	17.5	58°06.9	57.7	346°36.4	34.6	45°10.3	25.9		
15	44°37.2	60°41.4	•• 16.4	73°07.5	•• 57.0	1°38.3	•• 34.7	60°12.5	•• 25.8		
16	59°39.6	75°41.0	15.2	88°08.1	56.3	16°40.3	34.9	75°14.7	25.7		
17	74°42.1	90°40.5	14.1	103°08.7	55.7	31°42.3	35.1	90°16.9	25.6		
18	89°44.6	105°40.1	S08°13.0	118°09.3	S12°55.0	46°44.2	N15°35.2	105°19.1	S08°25.4		
19	104°47.0	120°39.7	11.8	133°09.9	54.3	61°46.2	35.4	120°21.3	25.3		
20	119°49.5	135°39.2	10.7	148°10.5	53.6	76°48.2	35.5	135°23.5	25.2		
21	134°51.9	150°38.8	•• 09.5	163°11.0	•• 53.0	91°50.1	•• 35.7	150°25.6	•• 25.1		
22	149°54.4	165°38.3	08.4	178°11.6	52.3	106°52.1	35.8	165°27.8	25.0		
23	164°56.9	180°37.9	07.3	193°12.2	51.6	121°54.0	36.0	180°30.0	24.9		
Mer.pass. 12:02		ν -0.4' d -1.1' m -3.88		ν 0.6' d -0.7' m 1.22		ν 2.0' d 0.2' m -2.10		ν 2.2' d -0.1' m 1.03			

Mar 19 Tue		SHA	Mer.pass
Venus		19°08.0	10:56
Mars		30°28.7	10:10
Jupiter		317°32.5	15:00
Saturn		15°52.8	11:07

Mar 20 Wed		SHA	Mer.pass
Venus		17°57.8	10:56
Mars		29°43.5	10:09
Jupiter		317°20.6	14:57
Saturn		15°46.2	11:03

Mar 21 Thu		SHA	Mer.pass
Venus		16°47.9	10:57
Mars		28°58.5	10:08
Jupiter		317°08.7	14:53
Saturn		15°39.5	11:00

Horizontal parallax		
Venus:		0.1
Mars:		0.1

h Sun Moon						
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	178°03.7	S00°26.8	64°29.4	9.5'	N27°08.1	-5.0' 55.1'
1	193°03.9	25.8	78°57.8	9.5'	27°03.1	-5.1' 55.1'
2	208°04.1	24.8	93°26.4	9.6'	26°58.0	-5.2' 55.1'
3	223°04.3	· · 23.8	107°55.0	9.7'	26°52.7	-5.4' 55.1'
4	238°04.5	22.8	122°23.7	9.8'	26°47.4	-5.5' 55.0'
5	253°04.6	21.8	136°52.5	9.9'	26°41.9	-5.6' 55.0'
6	268°04.8	S00°20.9	151°21.3	9.9'	N26°36.3	-5.7' 55.0'
7	283°05.0	19.9	165°50.3	10.0'	26°30.5	-5.9' 55.0'
8	298°05.2	18.9	180°19.3	10.1'	26°24.6	-6.0' 55.0'
9	313°05.4	· · 17.9	194°48.4	10.2'	26°18.6	-6.1' 54.9'
10	328°05.6	16.9	209°17.6	10.3'	26°12.5	-6.2' 54.9'
11	343°05.7	15.9	223°46.8	10.4'	26°06.3	-6.4' 54.9'
12	358°05.9	S00°14.9	238°16.2	10.4'	N25°59.9	-6.5' 54.9'
13	13°06.1	13.9	252°45.6	10.5'	25°53.4	-6.6' 54.8'
14	28°06.3	12.9	267°15.2	10.6'	25°46.8	-6.7' 54.8'
15	43°06.5	· · 12.0	281°44.8	10.7'	25°40.1	-6.8' 54.8'
16	58°06.7	11.0	296°14.4	10.8'	25°33.2	-7.0' 54.8'
17	73°06.8	10.0	310°44.2	10.9'	25°26.3	-7.1' 54.8'
18	88°07.0	S00°09.0	325°14.1	10.9'	N25°19.2	-7.2' 54.7'
19	103°07.2	08.0	339°44.0	11.0'	25°12.0	-7.3' 54.7'
20	118°07.4	07.0	354°14.0	11.1'	25°04.7	-7.4' 54.7'
21	133°07.6	· · 06.0	8°44.1	11.2'	24°57.2	-7.5' 54.7'
22	148°07.8	05.0	23°14.3	11.3'	24°49.7	-7.6' 54.7'
23	163°08.0	04.1	37°44.6	11.4'	24°42.1	-7.8' 54.7'
SD = 16.1' $d = -1.0'$			SD = 15.0'			
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	178°08.1	S00°03.1	52°15.0	11.5'	N24°34.3	-7.9' 54.6'
1	193°08.3	02.1	66°45.4	11.5'	24°26.5	-8.0' 54.6'
2	208°08.5	01.1	81°16.0	11.6'	24°18.5	-8.1' 54.6'
3	223°08.7	S00°00.1	95°46.6	11.7'	24°10.4	-8.2' 54.6'
4	238°08.9	N00°00.9	110°17.3	11.8'	24°02.2	-8.3' 54.6'
5	253°09.1	01.9	124°48.1	11.9'	23°53.9	-8.4' 54.5'
6	268°09.3	N00°02.9	139°19.0	12.0'	N23°45.6	-8.5' 54.5'
7	283°09.4	03.9	153°49.9	12.0'	23°37.1	-8.6' 54.5'
8	298°09.6	04.8	168°21.0	12.1'	23°28.5	-8.7' 54.5'
9	313°09.8	· · 05.8	182°52.1	12.2'	23°19.8	-8.8' 54.5'
10	328°10.0	06.8	197°23.3	12.3'	23°11.0	-8.9' 54.5'
11	343°10.2	07.8	211°54.6	12.4'	23°02.1	-9.0' 54.5'
12	358°10.4	N00°08.8	226°26.0	12.5'	N22°53.1	-9.1' 54.4'
13	13°10.6	09.8	240°57.4	12.5'	22°44.0	-9.2' 54.4'
14	28°10.7	10.8	255°29.0	12.6'	22°34.8	-9.3' 54.4'
15	43°10.9	· · 11.8	270°00.6	12.7'	22°25.6	-9.4' 54.4'
16	58°11.1	12.7	284°32.3	12.8'	22°16.2	-9.5' 54.4'
17	73°11.3	13.7	299°04.1	12.9'	22°06.7	-9.6' 54.4'
18	88°11.5	N00°14.7	313°36.0	13.0'	N21°57.2	-9.6' 54.3'
19	103°11.7	15.7	328°08.0	13.0'	21°47.5	-9.7' 54.3'
20	118°11.9	16.7	342°40.0	13.1'	21°37.8	-9.8' 54.3'
21	133°12.0	· · 17.7	357°12.1	13.2'	21°28.0	-9.9' 54.3'
22	148°12.2	18.7	11°44.3	13.3'	21°18.1	-10.0' 54.3'
23	163°12.4	19.7	26°16.6	13.4'	21°08.1	-10.1' 54.3'
SD = 16.1' $d = -1.0'$			SD = 14.9'			
Thu	GHA	Dec	GHA	ν	Dec	d HP
0	178°12.6	N00°20.6	40°49.0	13.4'	N20°58.0	-10.2' 54.3'
1	193°12.8	21.6	55°21.4	13.5'	20°47.9	-10.2' 54.3'
2	208°13.0	22.6	69°53.9	13.6'	20°37.6	-10.3' 54.2'
3	223°13.2	· · 23.6	84°26.5	13.7'	20°27.3	-10.4' 54.2'
4	238°13.3	24.6	98°59.2	13.7'	20°16.9	-10.5' 54.2'
5	253°13.5	25.6	113°31.9	13.8'	20°06.4	-10.6' 54.2'
6	268°13.7	N00°26.6	128°04.8	13.9'	N19°55.8	-10.6' 54.2'
7	283°13.9	27.6	142°37.7	14.0'	19°45.2	-10.7' 54.2'
8	298°14.1	28.5	157°10.6	14.1'	19°34.5	-10.8' 54.2'
9	313°14.3	· · 29.5	171°43.7	14.1'	19°23.7	-10.9' 54.2'
10	328°14.5	30.5	186°16.8	14.2'	19°12.8	-11.0' 54.2'
11	343°14.7	31.5	200°50.0	14.3'	19°01.8	-11.0' 54.1'
12	358°14.8	N00°32.5	215°23.3	14.3'	N18°50.8	-11.1' 54.1'
13	13°15.0	33.5	229°56.6	14.4'	18°39.7	-11.2' 54.1'
14	28°15.2	34.5	244°30.0	14.5'	18°28.5	-11.2' 54.1'
15	43°15.4	· · 35.4	259°03.5	14.6'	18°17.3	-11.3' 54.1'
16	58°15.6	36.4	273°37.1	14.6'	18°06.0	-11.4' 54.1'
17	73°15.8	37.4	288°10.7	14.7'	17°54.6	-11.4' 54.1'
18	88°16.0	N00°38.4	302°44.4	14.8'	N17°43.2	-11.5' 54.1'
19	103°16.2	39.4	317°18.2	14.8'	17°31.7	-11.6' 54.1'
20	118°16.3	40.4	331°52.0	14.9'	17°20.1	-11.6' 54.1'
21	133°16.5	· · 41.4	346°25.9	15.0'	17°08.4	-11.7' 54.1'
22	148°16.7	42.4	0°59.8	15.0'	16°56.7	-11.8' 54.1'
23	163°16.9	43.3	15°33.9	15.1'	16°45.0	-11.8' 54.0'
SD = 16.0' $d = 1.0'$			SD = 14.8'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	03:18	04:48	05:56	18:21	19:30	21:01
N 70°	03:38	04:56	05:57	18:20	19:21	20:41
68°	03:52	05:02	05:58	18:19	19:15	20:25
66°	04:04	05:08	05:59	18:18	19:09	20:13
64°	04:14	05:12	05:59	18:17	19:05	20:03
62°	04:22	05:16	06:00	18:16	19:01	19:55
60°	04:29	05:19	06:00	18:16	18:57	19:48
N 58°	04:35	05:22	06:01	18:15	18:54	19:42
56°	04:40	05:24	06:01	18:15	18:52	19:36
54°	04:44	05:26	06:01	18:14	18:50	19:32
52°	04:48	05:28	06:02	18:14	18:48	19:28
50°	04:52	05:30	06:02	18:14	18:46	19:24
45°	04:59	05:33	06:02	18:13	18:42	19:17
N 40°	05:04	05:36	06:03	18:12	18:39	19:11
35°	05:08	05:38	06:03	18:12	18:37	19:07
30°	05:12	05:40	06:03	18:12	18:36	19:03
20°	05:16	05:42	06:04	18:11	18:33	18:59
N 10°	05:19	05:43	06:04	18:11	18:32	18:56
0°	05:19	05:43	06:04	18:11	18:31	18:55
S 10°	05:19	05:43	06:04	18:10	18:31	18:56
20°	05:16	05:42	06:04	18:10	18:32	18:58
30°	05:12	05:40	06:04	18:10	18:34	19:02
35°	05:09	05:38	06:03	18:11	18:36	19:05
40°	05:05	05:36	06:03	18:11	18:38	19:09
45°	04:59	05:34	06:03	18:11	18:40	19:14
S 50°	04:52	05:30	06:02	18:11	18:43	19:21
52°	04:49	05:29	06:02	18:11	18:45	19:25
54°	04:45	05:27	06:02	18:12	18:47	19:28
56°	04:40	05:25	06:02	18:12	18:49	19:33
58°	04:35	05:22	06:01	18:12	18:51	19:38
S 60°	04:29	05:19	06:01	18:12	18:54	19:43

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	□	□	□	□	□	□
N 70°	□	□	10:54	□	□	08:33
68°	□	□	11:49	□	□	07:37
66°	□	09:57	12:21	□	07:52	07:03
64°	□	10:51	12:45	□	06:57	06:38
62°	09:32	11:23	13:04	06:33	06:25	06:18
60°	10:11	11:47	13:19	05:53	06:00	06:02
N 58°	10:38	12:06	13:32	05:26	05:41	05:49
56°	11:00	12:21	13:43	05:04	05:24	05:37
54°	11:17	12:35	13:53	04:46	05:10	05:26
52°	11:32	12:47	14:01	04:31	04:58	05:17
50°	11:45	12:57	14:09	04:18	04:47	05:09
45°	12:11	13:19	14:26	03:51	04:25	04:51
N 40°	12:32	13:36	14:39	03:29	04:06	04:36
35°	12:50	13:50	14:50	03:11	03:51	04:24
30°	13:05	14:03	15:00	02:56	03:37	04:13
20°	13:30	14:25	15:17	02:29	03:14	03:54
N 10°	13:52	14:43	15:32	02:07	02:54	03:38
0°	14:12	15:00	15:45	01:45	02:36	03:23
S 10°	14:32	15:18	15:59	01:24	02:17	03:07
20°	14:54	15:36	16:14	01:01	01:56	02:50
30°	15:18	15:57	16:30	00:34	01:33	02:31
35°	15:33	16:09	16:40	00:18	01:19	02:20
40°	15:50	16:23	16:51	· · ·	01:03	02:06
45°	16:10	16:40	17:03	· · ·	00:43	01:51
S 50°	16:35	17:00	17:19	· · ·	00:19	01:32
52°	16:48	17:10	17:26	· · ·	00:07	01:23
54°	17:02	17:21	17:34	23:54	· · ·	01:12
56°	17:18	17:33	17:43	23:38	· · ·	01:01
58°	17:37	17:47	17:53	23:19	· · ·	00:47
S 60°	18:01	18:04	18:04	22:56	· · ·	00:31

Day	Sun		Mer. Pass hh:mm	Moon		Age 9-11 68-84%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
19	07:45	07:36	12:08	20:24	07:59	
20	07:27	07:19	12:07	21:12	08:48	
21	07:10	07:01	12:07	21:56	09:34	

March 22, 23, 24 UT (Fri., Sat., Sun.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	179°59.3	195°37.5	S08°06.1	208°12.8	S12°51.0	136°56.0	N15°36.1	195°32.2	S08°24.8	Alpheratz	357°35.8	29°13.3											
1	195°01.8	210°37.0	05.0	223°13.4	50.3	151°58.0	36.3	210°34.4	24.7	Ankaa	353°08.2	-42°10.6											
2	210°04.3	225°36.6	03.8	228°14.0	49.6	166°59.9	36.4	225°36.6	24.6	Schedar	349°32.4	56°40.1											
3	225°06.7	240°36.1	•• 02.7	253°14.6	•• 48.9	182°01.9	•• 36.6	240°38.8	•• 24.5	Diphda	348°48.3	-17°51.4											
4	240°09.2	255°35.7	01.6	268°15.2	48.3	197°03.9	36.8	255°41.0	24.4	Achernar	335°21.2	-57°07.0											
5	255°11.7	270°35.3	08°00.4	283°15.8	47.6	212°05.8	36.9	270°43.1	24.2	Hamal	327°52.3	23°34.5											
6	270°14.1	285°34.8	S07°59.3	298°16.4	S12°46.9	227°07.8	N15°37.1	285°45.3	S08°24.1	Polaris	314°43.6	89°22.2											
7	285°16.6	300°34.4	58.1	313°17.0	46.3	242°09.7	37.2	300°47.5	24.0	Acamar	315°12.6	-40°12.7											
8	300°19.0	315°34.0	57.0	328°17.6	45.6	257°11.7	37.4	315°49.7	23.9	Menkar	314°07.1	4°11.0											
9	315°21.5	330°33.5	•• 55.9	343°18.2	•• 44.9	272°13.7	•• 37.5	330°51.9	•• 23.8	Mirfak	308°29.6	49°56.9											
10	330°24.0	345°33.1	54.7	358°18.7	44.2	287°15.6	37.7	345°54.1	23.7	Aldebaran	290°40.5	16°33.4											
11	345°26.4	0°32.6	53.6	13°19.3	43.6	302°17.6	37.8	0°56.3	23.6	Rigel	281°04.6	-8°10.6											
12	0°28.9	15°32.2	S07°52.4	28°19.9	S12°42.9	317°19.5	N15°38.0	15°58.5	S08°23.5	Capella	280°23.0	46°01.5											
13	15°31.4	30°31.8	51.3	43°20.5	42.2	332°21.5	38.1	31°00.7	23.4	Bellatrix	278°23.7	6°22.2											
14	30°33.8	45°31.3	50.1	58°21.1	41.5	347°23.5	38.3	46°02.8	23.3	Elnath	278°02.8	28°37.7											
15	45°36.3	60°30.9	•• 49.0	73°21.7	•• 40.9	2°25.4	•• 38.5	61°05.0	•• 23.2	Alnilam	275°38.5	-1°11.3											
16	60°38.8	75°30.5	47.8	88°22.3	40.2	17°27.4	38.6	76°07.2	23.1	Betelgeuse	270°52.8	7°24.6											
17	75°41.2	90°30.0	46.7	103°22.9	39.5	32°29.3	38.8	91°09.4	22.9	Canopus	263°52.7	-52°42.8											
18	90°43.7	105°29.6	S07°45.6	118°23.5	S12°38.9	47°31.3	N15°38.9	106°11.6	S08°22.8	Sirius	258°26.8	-16°45.1											
19	105°46.2	120°29.2	44.4	133°24.1	38.2	62°33.3	39.1	121°13.8	22.7	Adhara	255°06.3	-29°00.5											
20	120°48.6	135°28.7	43.3	148°24.7	37.5	77°35.2	39.2	136°16.0	22.6	Procyon	244°51.4	5°09.7											
21	135°51.1	150°28.3	•• 42.1	163°25.3	•• 36.8	92°37.2	•• 39.4	151°18.2	•• 22.5	Pollux	243°18.0	27°58.1											
22	150°53.5	165°27.9	41.0	178°25.9	36.2	107°39.1	39.5	166°20.3	22.4	Avior	234°14.7	-59°35.4											
23	165°56.0	180°27.4	39.8	193°26.5	35.5	122°41.1	39.7	181°22.5	22.3	Suhail	222°46.5	-43°32.0											
Mer.pass. 11:58				ν -0.4' d -1.1' m -3.88				ν 0.6' d -0.7' m 1.22				ν 2.0' d 0.2' m -2.10				ν 2.2' d -0.1' m 1.04							

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	180°58.5	195°27.0	S07°38.7	208°27.1	S12°34.8	137°43.0	N15°39.8	196°24.7	S08°22.2										
1	196°00.9	210°26.6	37.5	223°27.7	34.1	152°45.0	40.0	211°26.9	22.1										
2	211°03.4	225°26.1	36.4	238°28.3	33.4	167°47.0	40.2	226°29.1	22.0										
3	226°05.9	240°25.7	•• 35.2	253°28.9	•• 32.8	182°48.9	•• 40.3	241°31.3	•• 21.9										
4	241°08.3	255°25.3	34.1	268°29.5	32.1	197°50.9	40.5	256°33.5	21.8										
5	256°10.8	270°24.9	32.9	283°30.1	31.4	212°52.8	40.6	271°35.7	21.6										
6	271°13.3	285°24.4	S07°31.8	298°30.7	S12°30.7	227°54.8	N15°40.8	286°37.9	S08°21.5										
7	286°15.7	300°24.0	30.6	313°31.2	30.1	242°56.8	40.9	301°40.1	21.4										
8	301°18.2	315°23.6	29.5	328°31.8	29.4	257°58.7	41.1	316°42.2	21.3										
9	316°20.6	330°23.1	•• 28.3	343°32.4	•• 28.7	273°00.7	•• 41.2	331°44.4	•• 21.2										
10	331°23.1	345°22.7	27.2	358°33.0	28.0	288°02.6	41.4	346°46.6	21.1										
11	346°25.6	0°22.3	26.0	13°33.6	27.4	303°04.6	41.6	1°48.8	21.0										
12	1°28.0	15°21.8	S07°24.9	28°34.2	S12°26.7	318°06.5	N15°41.7	16°51.0	S08°20.9										
13	16°30.5	30°21.4	23.7	43°34.8	26.0	333°08.5	41.9	31°53.2	20.8										
14	31°33.0	45°21.0	22.5	58°35.4	25.3	348°10.4	42.0	46°55.4	20.7										
15	46°35.4	60°20.6	•• 21.4	73°36.0	•• 24.6	3°12.4	•• 42.2	61°57.6	•• 20.6										
16	61°37.9	75°20.1	20.2	88°36.6	24.0	18°14.4	42.3	76°59.8	20.5										
17	76°40.4	90°19.7	19.1	103°37.2	23.3	33°16.3	42.5	92°01.9	20.3										
18	91°42.8	105°19.3	S07°17.9	118°37.8	S12°22.6	48°18.3	N15°42.6	107°04.1	S08°20.2										
19	106°45.3	120°18.9	16.8	133°38.4	21.9	63°20.2	42.8	122°06.3	20.1										
20	121°47.8	135°18.4	15.6	148°39.0	21.2	78°22.2	42.9	137°08.5	20.0										
21	136°50.2	150°18.0	•• 14.5	163°39.6	•• 20.6	93°24.1	•• 43.1	152°10.7	•• 19.9										
22	151°52.7	165°17.6	13.3	178°40.2	19.9	108°26.1	43.3	167°12.9	19.8										
23	166°55.1	180°17.2	12.2	193°40.8	19.2	123°28.0	43.4	182°15.1	19.7										
Mer.pass. 11:54				ν -0.4' d -1.1' m -3.88				ν 0.6' d -0.7' m 1.21				ν 2.0' d 0.2' m -2.09				ν 2.2' d -0.1' m 1.04			

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	181°57.6	195°16.7	S07°11.0	208°41.4	S12°18.5	138°30.0	N15°43.6	197°17.3	S08°19.6										
1	197°00.1	210°16.3	09.8	223°42.0	17.8	153°32.0	43.7	212°19.5	19.5										
2	212°02.5	225°15.9	08.7	238°42.6	17.2	168°33.9	43.9	227°21.7	19.4										
3	227°05.0	240°15.5	•• 07.5	253°43.2	•• 16.5	183°35.9	•• 44.0	242°23.8	•• 19.3										
4	242°07.5	255°15.0	06.4	268°43.8	15.8	198°37.8	44.2	257°26.0	19.2										
5	257°09.9	270°14.6	05.2	283°44.4	15.1	213°39.8	44.3	272°28.2	19.1										
6	272°12.4	285°14.2	S07°04.0	298°45.0	S12°14.4	228°41.7	N15°44.5	287°30.4	S08°18.9										
7	287°14.9	300°13.8	02.9	313°45.6	13.7	243°43.7	44.7	302°32.6	18.8										
8	302°17.3	315°13.3	01.7	328°46.2	13.1	258°45.6	44.8	317°34.8	18.7										
9	317°19.8	330°12.9	07°00.6	343°46.8	•• 12.4	273°47.6	•• 45.0	332°37.0	•• 18.6										
10	332°22.2	345°12.5	06°59.4	358°47.4	11.7	288°49.5	45.1	347°39.2	18.5										
11	347°24.7	0°12.1	58.2	13°48.0	11.0	303°51.5	45.3	2°41.4	18.4										
12	2°27.2	15°11.6	S06°57.1	28°48.6	S12°10.3	318°53.5	N15°45.4	17°43.6	S08°18.3										
13	17°29.6	30°11.2	55.9	43°49.2	09.6	333°55.4	45.6	32°45.7	18.2										
14	32°32.1	45°10.8	54.8	58°49.8	09.0	348°57.4	45.7	47°47.9	18.1										
15	47°34.6	60°10.4	•• 53.6	73°50.4	•• 08.3	3°59.3	•• 45.9	62°50.1	•• 18.0										
16	62°37.0	75°10.0	52.4	88°51.0	07.6	19°01.3	46.0	77°52.3	17.9										
17	77°39.5	90°09.5	51.3	103°51.6	06.9	34°03.2	46.2	92°54.5	17.8										
18	92°42.0	105°09.1	S06°50.1	118°52.2	S12°06.2	49°05.2	N15°46.4	107°56.7	S08°17.7										
19	107°44.4	120°08.7	48.9	133°52.8	05.5	64°07.1	46.5	122°58.9	17.5										
20	122°46.9	135°08.3	47.8	148°53.4	04.9	79°09.1	46.7	138°01.1	17.4										
21	137°49.4	150°07.9	•• 46.6	163°54.0	•• 04.2	94°11.0	•• 46.8	153°03.3	•• 17.3										
22	152°51.8	165°07.4	45.5	178°54.6	03.5	109°13.0	47.0	168°05.5	17.2										
23	167°54.3	180°07.0	44.3	193°55.2	02.8	124°14.9	47.1	183°07.7	17.1										
Mer.pass. 11:50				ν -0.4' d -1.2' m -3.88				ν 0.6' d -0.7' m 1.21				ν 2.0' d 0.2' m -2.09				ν 2.2' d -0.1' m 1.04			

Mar 22 Fri	SHA	Mer.pass
Venus	15°38.1	10:58
Mars	28°13.5	10:07
Jupiter	316°56.7	14:50
Saturn	15°32.9	10:56

Mar 23 Sat	SHA	Mer.pass
Venus	14°28.5	10:59
Mars	27°28.6	10:06
Jupiter	316°44.6	14:47
Saturn	15°26.3	10:53

Mar 24 Sun	SHA	Mer.pass
Venus	13°19.1	10:59
Mars	26°43.8	10:05
Jupiter	316°32.4	14:44
Saturn	15°19.7	10:49

Horizontal parallax		
Venus:	0.1	
Mars:	0.1	

Sun			Moon				
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	178°17.1	N00°44.3	30°07.9	15.2'	N16°33.1	-11.9'	54.0'
1	193°17.3	45.3	44°42.1	15.2'	16°21.2	-12.0'	54.0'
2	208°17.5	46.3	59°16.3	15.3'	16°09.3	-12.0'	54.0'
3	223°17.7	· · 47.3	73°50.6	15.3'	15°57.2	-12.1'	54.0'
4	238°17.8	48.3	88°24.9	15.4'	15°45.2	-12.1'	54.0'
5	253°18.0	49.3	102°59.3	15.5'	15°33.0	-12.2'	54.0'
6	268°18.2	N00°50.2	117°33.8	15.5'	N15°20.8	-12.2'	54.0'
7	283°18.4	51.2	132°08.3	15.6'	15°08.6	-12.3'	54.0'
8	298°18.6	52.2	146°42.9	15.6'	14°56.3	-12.4'	54.0'
9	313°18.8	· · 53.2	161°17.5	15.7'	14°43.9	-12.4'	54.0'
10	328°19.0	54.2	175°52.2	15.7'	14°31.5	-12.5'	54.0'
11	343°19.2	55.2	190°26.9	15.8'	14°19.1	-12.5'	54.0'
12	358°19.3	N00°56.2	205°01.7	15.8'	N14°06.5	-12.6'	54.0'
13	13°19.5	57.1	219°36.6	15.9'	13°54.0	-12.6'	54.0'
14	28°19.7	58.1	234°11.5	16.0'	13°41.3	-12.7'	54.0'
15	43°19.9	00°59.1	248°46.4	16.0'	13°28.7	-12.7'	54.0'
16	58°20.1	01°00.1	263°21.4	16.1'	13°16.0	-12.8'	53.9'
17	73°20.3	01.1	277°56.5	16.1'	13°03.2	-12.8'	53.9'
18	88°20.5	N01°02.1	292°31.6	16.2'	N12°50.4	-12.9'	53.9'
19	103°20.7	03.1	307°06.7	16.2'	12°37.5	-12.9'	53.9'
20	118°20.8	04.0	321°41.9	16.2'	12°24.6	-13.0'	53.9'
21	133°21.0	· · 05.0	336°17.2	16.3'	12°11.7	-13.0'	53.9'
22	148°21.2	06.0	350°52.5	16.3'	11°58.7	-13.0'	53.9'
23	163°21.4	07.0	5°27.8	16.4'	11°45.6	-13.1'	53.9'
SD = 16.0'			$d = 1.0'$		SD = 14.7'		
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	178°21.6	N01°08.0	20°03.2	16.4'	N11°32.5	-13.1'	53.9'
1	193°21.8	09.0	34°38.6	16.5'	11°19.4	-13.2'	53.9'
2	208°22.0	10.0	49°14.1	16.5'	11°06.3	-13.2'	53.9'
3	223°22.2	· · 10.9	63°49.6	16.5'	10°53.1	-13.2'	53.9'
4	238°22.4	11.9	78°25.1	16.6'	10°39.8	-13.3'	53.9'
5	253°22.5	12.9	93°00.7	16.6'	10°26.5	-13.3'	53.9'
6	268°22.7	N01°13.9	107°36.4	16.7'	N10°13.2	-13.4'	53.9'
7	283°22.9	14.9	122°12.0	16.7'	09°59.9	-13.4'	53.9'
8	298°23.1	15.9	136°47.7	16.7'	09°46.5	-13.4'	53.9'
9	313°23.3	· · 16.8	151°23.5	16.8'	09°33.1	-13.5'	53.9'
10	328°23.5	17.8	165°59.2	16.8'	09°19.6	-13.5'	53.9'
11	343°23.7	18.8	180°35.0	16.8'	09°06.1	-13.5'	53.9'
12	358°23.9	N01°19.8	195°10.9	16.9'	N08°52.6	-13.6'	53.9'
13	13°24.1	20.8	209°46.7	16.9'	08°39.0	-13.6'	53.9'
14	28°24.2	21.8	224°22.6	16.9'	08°25.4	-13.6'	53.9'
15	43°24.4	· · 22.8	238°58.6	17.0'	08°11.8	-13.6'	53.9'
16	58°24.6	23.7	253°34.5	17.0'	07°58.2	-13.7'	53.9'
17	73°24.8	24.7	268°10.5	17.0'	07°44.5	-13.7'	53.9'
18	88°25.0	N01°25.7	282°46.5	17.0'	N07°30.8	-13.7'	53.9'
19	103°25.2	26.7	297°22.6	17.1'	07°17.1	-13.8'	53.9'
20	118°25.4	27.7	311°58.6	17.1'	07°03.3	-13.8'	53.9'
21	133°25.6	· · 28.7	326°34.7	17.1'	06°49.5	-13.8'	53.9'
22	148°25.8	29.6	341°10.8	17.1'	06°35.7	-13.8'	53.9'
23	163°25.9	30.6	355°47.0	17.2'	06°21.9	-13.9'	53.9'
SD = 16.0'			$d = 1.0'$		SD = 14.7'		
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	178°26.1	N01°31.6	10°23.1	17.2'	N06°08.0	-13.9'	53.9'
1	193°26.3	32.6	24°59.3	17.2'	05°54.2	-13.9'	53.9'
2	208°26.5	33.6	39°35.5	17.2'	05°40.2	-13.9'	53.9'
3	223°26.7	· · 34.6	54°11.7	17.2'	05°26.3	-13.9'	53.9'
4	238°26.9	35.5	68°47.9	17.2'	05°12.4	-14.0'	53.9'
5	253°27.1	36.5	83°24.2	17.3'	04°58.4	-14.0'	53.9'
6	268°27.3	N01°37.5	98°00.5	17.3'	N04°44.4	-14.0'	53.9'
7	283°27.5	38.5	112°36.7	17.3'	04°30.4	-14.0'	53.9'
8	298°27.6	39.5	127°13.0	17.3'	04°16.4	-14.0'	53.9'
9	313°27.8	· · 40.5	141°49.3	17.3'	04°02.4	-14.0'	53.9'
10	328°28.0	41.4	156°25.6	17.3'	03°48.3	-14.1'	53.9'
11	343°28.2	42.4	171°02.0	17.3'	03°34.3	-14.1'	53.9'
12	358°28.4	N01°43.4	185°38.3	17.3'	N03°20.2	-14.1'	53.9'
13	13°28.6	44.4	200°14.7	17.4'	03°06.1	-14.1'	53.9'
14	28°28.8	45.4	214°51.0	17.4'	02°52.0	-14.1'	53.9'
15	43°29.0	· · 46.4	229°27.4	17.4'	02°37.9	-14.1'	53.9'
16	58°29.2	47.3	244°03.7	17.4'	02°23.7	-14.1'	53.9'
17	73°29.3	48.3	258°40.1	17.4'	02°09.6	-14.2'	54.0'
18	88°29.5	N01°49.3	273°16.5	17.4'	N01°55.5	-14.2'	54.0'
19	103°29.7	50.3	287°52.9	17.4'	01°41.3	-14.2'	54.0'
20	118°29.9	51.3	302°29.2	17.4'	01°27.1	-14.2'	54.0'
21	133°30.1	· · 52.2	317°05.6	17.4'	01°13.0	-14.2'	54.0'
22	148°30.3	53.2	331°42.0	17.4'	00°58.8	-14.2'	54.0'
23	163°30.5	54.2	346°18.4	17.4'	00°44.6	-14.2'	54.0'
SD = 16.0'			$d = 1.0'$		SD = 14.7'		

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	02:57	04:31	05:41	18:35	19:45	21:22
N 70°	03:20	04:41	05:43	18:32	19:34	20:57
68°	03:37	04:49	05:45	18:29	19:26	20:39
66°	03:51	04:56	05:47	18:27	19:19	20:25
64°	04:02	05:01	05:49	18:26	19:14	20:13
62°	04:11	05:06	05:50	18:24	19:09	20:04
60°	04:19	05:10	05:51	18:23	19:05	19:56
N 58°	04:26	05:13	05:52	18:22	19:01	19:49
56°	04:31	05:16	05:53	18:21	18:58	19:43
54°	04:36	05:19	05:54	18:20	18:55	19:38
52°	04:41	05:21	05:55	18:19	18:53	19:33
50°	04:45	05:23	05:55	18:18	18:51	19:29
45°	04:53	05:28	05:57	18:17	18:46	19:21
N 40°	04:59	05:31	05:58	18:15	18:43	19:14
35°	05:04	05:34	05:59	18:14	18:40	19:09
30°	05:08	05:36	06:00	18:13	18:37	19:05
20°	05:14	05:39	06:01	18:12	18:34	19:00
N 10°	05:17	05:41	06:02	18:11	18:32	18:56
0°	05:18	05:42	06:03	18:10	18:30	18:54
S 10°	05:19	05:43	06:04	18:09	18:30	18:54
20°	05:17	05:43	06:05	18:08	18:30	18:55
30°	05:14	05:42	06:05	18:07	18:31	18:58
35°	05:11	05:41	06:06	18:06	18:32	19:01
40°	05:08	05:39	06:06	18:06	18:33	19:04
45°	05:03	05:37	06:07	18:05	18:35	19:09
S 50°	04:57	05:35	06:07	18:05	18:37	19:15
52°	04:54	05:34	06:07	18:04	18:38	19:17
54°	04:51	05:32	06:08	18:04	18:39	19:21
56°	04:47	05:31	06:08	18:04	18:41	19:24
58°	04:42	05:29	06:08	18:03	18:42	19:29
S 60°	04:37	05:27	06:08	18:03	18:44	19:34

March 25, 26, 27 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	182°56.7	195°06.6	S06°43.1	208°55.8	S12°02.1	139°16.9	N15°47.3	198°09.8	S08°17.0	Alpheratz	357°35.8	29°13.3
1	197°59.2	210°06.2	42.0	223°56.4	01.4	154°18.8	47.4	213°12.0	16.9	Ankaa	353°08.2	-42°10.6
2	213°01.7	225°05.8	40.8	238°57.1	00.7	169°20.8	47.6	228°14.2	16.8	Schedar	349°32.4	56°40.1
3	228°04.1	240°05.4	· · 39.6	253°57.7	12°00.1	184°22.7	· · 47.8	243°16.4	· · 16.7	Diphda	348°48.3	-17°51.4
4	243°06.6	255°04.9	38.5	268°58.3	11°59.4	199°24.7	47.9	258°18.6	16.6	Achernar	335°21.2	-57°07.0
5	258°09.1	270°04.5	37.3	283°58.9	58.7	214°26.6	48.1	273°20.8	16.5	Hamal	327°52.3	23°34.5
6	273°11.5	285°04.1	S06°36.1	298°59.5	S11°58.0	229°28.6	N15°48.2	288°23.0	S08°16.4	Polaris	314°44.6	89°22.2
7	288°14.0	300°03.7	35.0	314°00.1	57.3	244°30.5	48.4	303°25.2	16.3	Acamar	315°12.6	-40°12.7
8	303°16.5	315°03.3	33.8	329°00.7	56.6	259°32.5	48.5	318°27.4	16.2	Menkar	314°07.1	4°11.0
9	318°18.9	330°02.8	· · 32.6	344°01.3	· · 55.9	274°34.4	· · 48.7	333°29.6	· · 16.0	Mirfak	308°29.6	49°56.9
10	333°21.4	345°02.4	31.5	359°01.9	55.3	289°36.4	48.8	348°31.8	15.9	Aldebaran	290°40.5	16°33.4
11	348°23.8	0°02.0	30.3	14°02.5	54.6	304°38.3	49.0	3°34.0	15.8	Rigel	281°04.6	-8°10.6
12	3°26.3	15°01.6	S06°29.1	29°03.1	S11°53.9	319°40.3	N15°49.2	18°36.1	S08°15.7	Capella	280°23.0	46°01.5
13	18°28.8	30°01.2	28.0	44°03.7	53.2	334°42.2	49.3	33°38.3	15.6	Bellatrix	278°23.7	6°22.2
14	33°31.2	45°00.8	26.8	59°04.3	52.5	349°44.2	49.5	48°40.5	15.5	Elnath	278°02.8	28°37.7
15	48°33.7	60°00.4	· · 25.6	74°04.9	· · 51.8	4°46.1	· · 49.6	63°42.7	· · 15.4	Alnilam	275°38.5	-1°11.3
16	63°36.2	74°59.9	24.4	89°05.5	51.1	19°48.1	49.8	78°44.9	15.3	Betelgeuse	270°52.8	7°24.6
17	78°38.6	89°59.5	23.3	104°06.1	50.4	34°50.0	49.9	93°47.1	15.2	Canopus	263°52.7	-52°42.8
18	93°41.1	104°59.1	S06°22.1	119°06.7	S11°49.7	49°52.0	N15°50.1	108°49.3	S08°15.1	Sirius	258°26.8	-16°45.1
19	108°43.6	119°58.7	20.9	134°07.3	49.1	64°53.9	50.2	123°51.5	15.0	Adhara	255°06.3	-29°00.5
20	123°46.0	134°58.3	19.8	149°07.9	48.4	79°55.9	50.4	138°53.7	14.9	Procyon	244°51.4	5°09.7
21	138°48.5	149°57.9	· · 18.6	164°08.5	· · 47.7	94°57.8	· · 50.6	153°55.9	· · 14.8	Pollux	243°18.0	27°58.1
22	153°51.0	164°57.5	17.4	179°09.1	47.0	109°59.8	50.7	168°58.1	14.7	Avior	234°14.7	-59°35.4
23	168°53.4	179°57.0	16.2	194°09.8	46.3	125°01.7	50.9	184°00.3	14.5	Suhail	222°46.5	-43°32.0
Mer.pass. 11:46		ν -0.4' d -1.2' m -3.88		ν 0.6' d -0.7' m 1.21		ν 2.0' d 0.2' m -2.09		ν 2.2' d -0.1' m 1.04				

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	183°55.9	194°56.6	S06°15.1	209°10.4	S11°45.6	140°03.7	N15°51.0	199°02.4	S08°14.4	Denebola	182°25.3	14°26.1
1	198°58.3	209°56.2	13.9	224°11.0	44.9	155°05.6	51.2	214°04.6	14.3	Gienah	175°44.0	-17°40.7
2	214°00.8	224°55.8	12.7	239°11.6	44.2	170°07.6	51.3	229°06.8	14.2	Acrux	173°00.2	-63°14.0
3	229°03.3	239°55.4	· · 11.6	254°12.2	· · 43.5	185°09.5	· · 51.5	244°09.0	· · 14.1	Gacrux	171°51.9	-57°15.0
4	244°05.7	254°55.0	10.4	269°12.8	42.8	200°11.5	51.6	259°11.2	14.0	Alioth	166°13.0	55°49.6
5	259°08.2	269°54.6	09.2	284°13.4	42.2	215°13.4	51.8	274°13.4	13.9	Spica	158°22.7	-11°17.4
6	274°10.7	284°54.2	S06°08.0	299°14.0	S11°41.5	230°15.4	N15°52.0	289°15.6	S08°13.8	Alkaid	152°52.1	49°11.4
7	289°13.1	299°53.7	06.9	314°14.6	40.8	245°17.3	52.1	304°17.8	13.7	Hadar	148°36.4	-60°29.3
8	304°15.6	314°53.3	05.7	329°15.2	40.1	260°19.3	52.3	319°20.0	13.6	Menkent	147°58.1	-36°29.4
9	319°18.1	329°52.9	· · 04.5	344°15.8	· · 39.4	275°21.2	· · 52.4	334°22.2	· · 13.5	Arcturus	145°48.2	19°03.2
10	334°20.5	344°52.5	03.3	359°16.4	38.7	290°23.2	52.6	349°24.4	13.4	Rigil Kent.	139°40.7	-60°56.0
11	349°23.0	359°52.1	02.2	14°17.0	38.0	305°25.1	52.7	4°26.6	13.3	Kochab	137°18.8	74°03.1
12	4°25.5	14°51.7	S06°01.0	29°17.7	S11°37.3	320°27.1	N15°52.9	19°28.8	S08°13.2	Zuben'ubi	136°56.5	-16°08.6
13	19°27.9	29°51.3	05°59.8	44°18.3	36.6	335°29.0	53.0	34°31.0	13.0	Alphecca	126°04.1	26°37.7
14	34°30.4	44°50.9	58.6	59°18.9	35.9	350°30.9	53.2	49°33.1	12.9	Antares	112°16.5	-26°29.2
15	49°32.8	59°50.5	· · 57.5	74°19.5	· · 35.2	5°32.9	· · 53.4	64°35.3	· · 12.8	Atria	107°11.2	-69°04.0
16	64°35.3	74°50.1	56.3	89°20.1	34.5	20°34.8	53.5	79°37.5	12.7	Sabik	102°03.5	-15°45.4
17	79°37.8	89°49.7	55.1	104°20.7	33.8	35°36.8	53.7	94°39.7	12.6	Shaula	96°11.2	-37°07.2
18	94°40.2	104°49.2	S05°53.9	119°21.3	S11°33.1	50°38.7	N15°53.8	109°41.9	S08°12.5	Rasalhague	95°59.1	12°32.3
19	109°42.7	119°48.8	52.7	134°21.9	32.5	65°40.7	54.4	124°44.1	12.4	Eltanin	90°42.4	51°28.7
20	124°45.2	134°48.4	51.6	149°22.5	31.8	80°42.6	54.1	139°46.3	12.3	Kaus Aust.	83°33.4	-34°22.3
21	139°47.6	149°48.0	· · 50.4	164°23.1	· · 31.1	95°44.6	· · 54.3	154°48.5	· · 12.2	Vega	80°33.7	38°48.0
22	154°50.1	164°47.6	49.2	179°23.7	30.4	110°46.5	54.4	169°50.7	12.1	Nunki	75°48.6	-26°16.0
23	169°52.6	179°47.2	48.0	194°24.4	29.7	125°48.5	54.6	184°52.9	12.0	Altair	62°00.7	8°55.7
Mer.pass. 11:42		ν -0.4' d -1.2' m -3.88		ν 0.6' d -0.7' m 1.21		ν 1.9' d 0.2' m -2.08		ν 2.2' d -0.1' m 1.04				

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	184°55.0	194°46.8	S05°46.8	209°25.0	S11°29.0	140°50.4	N15°54.8	199°55.1	S08°11.9	Denebola	182°25.3	14°26.1
1	199°57.5	209°46.4	45.7	224°25.6	28.3	155°52.4	54.9	214°57.3	11.8	Gienah	175°44.0	-17°40.7
2	214°59.9	224°46.0	44.5	239°26.2	27.6	170°54.3	55.1	229°59.5	11.7	Acrux	173°00.2	-63°14.0
3	230°02.4	239°45.6	· · 43.3	254°26.8	· · 26.9	185°56.2	· · 55.2	245°01.7	· · 11.6	Gacrux	171°51.9	-57°15.0
4	245°04.9	254°45.2	42.1	269°27.4	26.2	200°58.2	55.4	260°03.8	11.4	Alioth	166°13.0	55°49.6
5	260°07.3	269°44.8	40.9	284°28.0	25.5	216°00.1	55.5	275°06.0	11.3	Spica	158°22.7	-11°17.4
6	275°09.8	284°44.4	S05°39.8	299°28.6	S11°24.8	231°02.1	N15°55.7	290°08.2	S08°11.2	Alkaid	152°52.1	49°11.4
7	290°12.3	299°44.0	38.6	314°29.2	24.1	246°04.0	55.8	305°10.4	11.1	Hadar	148°36.4	-60°29.3
8	305°14.7	314°43.6	37.4	329°29.8	23.4	261°06.0	56.0	320°12.6	11.0	Menkent	147°58.1	-36°29.4
9	320°17.2	329°43.1	· · 36.2	344°30.5	· · 22.7	276°07.9	· · 56.2	335°14.8	· · 10.9	Arcturus	145°48.2	19°03.2
10	335°19.7	344°42.7	35.0	359°31.1	22.0	291°09.9	56.3	350°17.0	10.8	Rigil Kent.	139°40.7	-60°56.0
11	350°22.1	359°42.3	33.9	14°31.7	21.3	306°11.8	56.5	5°19.2	10.7	Kochab	137°18.8	74°03.1
12	5°24.6	14°41.9	S05°32.7	29°32.3	S11°20.6	321°13.7	N15°56.6	20°21.4	S08°10.6	Zuben'ubi	136°56.5	-16°08.6
13	20°27.1	29°41.5	31.5	44°32.9	19.9	336°15.7	56.8	35°23.6	10.5	Alphecca	126°04.1	26°37.7
14	35°29.5	44°41.1	30.3	59°33.5	19.2	351°17.6	56.9	50°25.8	10.4	Antares	112°16.5	-26°29.2
15	50°32.0	59°40.7	· · 29.1	74°34.1	· · 18.5	6°19.6	· · 57.1	65°28.0	· · 10.3	Atria	107°11.2	-69°04.0
16	65°34.4	74°40.3	27.9	89°34.7	17.8	21°21.5	57.2	80°30.2	10.2	Sabik	102°03.5	-15°45.4
17	80°36.9	89°39.9	26.8	104°35.4	17.1	36°23.5	57.4	95°32.4	10.1	Shaula	96°11.2	-37°07.2
18	95°39.4	104°39.5	S05°25.6	119°36.0	S11°16.4	51°25.4	N15°57.6	110°34.6	S08°10.0	Rasalhague	95°59.1	12°32.3
19	110°41.8	119°39.1	24.4	134°36.6	15.7	66°27.3	57.7	125°36.8	09.9	Eltanin	90°42.4	51°28.7
20	125°44.3	134°38.7	23.2	149°37.2	15.0	81°29.3	57.9	140°39.0	09.8	Kaus Aust.	83°33.4	-34°22.3
21	140°46.8	149°38.3	· · 22.0	164°37.8	· · 14.4	96°31.2	· · 58.0	155°41.1	· · 09.6	Vega	80°33.7	38°48.0
22	155°49.2	164°37.9	20.8	179°38.4	13.7	111°33.2	58.2	170°43.3	09.5	Nunki	75°48.6	-26°16.0
23	170°51.7	179°37.5	19.6	194°39.0	13.0	126°35.1	58.3	185°45.5	09.4	Altair	62°00.7	8°55.7
Mer.pass. 11:38		ν -0.4' d -1.2' m -3.88		ν 0.6' d -0.7' m 1.20		ν 1.9' d 0.2' m -2.08		ν 2.2' d -0.1' m 1.05				

Mar 25 Mon		SHA	Mer.pass
Venus		12°09.9	11:00
Mars		25°59.1	10:04
Jupiter		316°20.1	14:41
Saturn		15°13.1	10:46

Mar 26 Tue		SHA	Mer.pass
Venus		11°00.7	11:01
Mars		25°14.5	10:03
Jupiter		316°07.8	14:38
Saturn		15°06.6	10:42

Mar 27 Wed		SHA	Mer.pass
Venus		9°51.8	11:01
Mars		24°29.9	10:02
Jupiter		315°55.4	14:35
Saturn		15°00.1	10:39

Horizontal parallax		
Venus:		0.1
Mars:		0.1

Sun			Moon			
h						
Mon	GHA	Dec	GHA	ν	Dec	HP
0	178°30.7	N01°55.2	0°54.8	17.4'	N00°30.4	-14.2' 54.0'
1	193°30.9	56.2	15°31.1	17.4'	00°16.2	-14.2' 54.0'
2	208°31.1	57.2	30°07.5	17.4'	N00°02.0	-14.2' 54.0'
3	223°31.2	· · 58.1	44°43.9	17.4'	S00°12.2	14.2' 54.0'
4	238°31.4	01°59.1	59°20.2	17.4'	00°26.4	14.2' 54.0'
5	253°31.6	02°00.1	73°56.6	17.3'	00°40.6	14.2' 54.0'
6	268°31.8	N02°01.1	88°32.9	17.3'	S00°54.8	14.2' 54.0'
7	283°32.0	02.1	103°09.3	17.3'	01°09.1	14.2' 54.0'
8	298°32.2	03.0	117°45.6	17.3'	01°23.3	14.2' 54.0'
9	313°32.4	· · 04.0	132°21.9	17.3'	01°37.5	14.2' 54.0'
10	328°32.6	05.0	146°58.2	17.3'	01°51.7	14.2' 54.0'
11	343°32.8	06.0	161°34.5	17.3'	02°05.9	14.2' 54.0'
12	358°32.9	N02°07.0	176°10.8	17.3'	S02°20.1	14.2' 54.1'
13	13°33.1	08.0	190°47.0	17.2'	02°34.3	14.2' 54.1'
14	28°33.3	08.9	205°23.3	17.2'	02°48.5	14.2' 54.1'
15	43°33.5	· · 09.9	219°59.5	17.2'	03°02.6	14.2' 54.1'
16	58°33.7	10.9	234°35.7	17.2'	03°16.8	14.2' 54.1'
17	73°33.9	11.9	249°11.9	17.2'	03°31.0	14.2' 54.1'
18	88°34.1	N02°12.9	263°48.1	17.1'	S03°45.1	14.1' 54.1'
19	103°34.3	13.8	278°24.2	17.1'	03°59.3	14.1' 54.1'
20	118°34.5	14.8	293°00.3	17.1'	04°13.4	14.1' 54.1'
21	133°34.7	· · 15.8	307°36.4	17.1'	04°27.5	14.1' 54.1'
22	148°34.8	16.8	322°12.5	17.1'	04°41.7	14.1' 54.1'
23	163°35.0	17.8	336°48.6	17.0'	04°55.8	14.1' 54.1'
SD = 16.0' d = 1.0'			SD = 14.7'			

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	178°35.2	N02°18.7	351°24.6	17.0'	S05°09.9	14.1'	54.1'
1	193°35.4	19.7	6°00.6	17.0'	05°23.9	14.1'	54.2'
2	208°35.6	20.7	20°36.6	16.9'	05°38.0	14.0'	54.2'
3	223°35.8	· · 21.7	35°12.5	16.9'	05°52.0	14.0'	54.2'
4	238°36.0	22.7	49°48.4	16.9'	06°06.1	14.0'	54.2'
5	253°36.2	23.6	64°24.3	16.9'	06°20.1	14.0'	54.2'
6	268°36.4	N02°24.6	79°00.2	16.8'	S06°34.1	14.0'	54.2'
7	283°36.5	25.6	93°36.0	16.8'	06°48.0	14.0'	54.2'
8	298°36.7	26.6	108°11.8	16.7'	07°02.0	13.9'	54.2'
9	313°36.9	· · 27.6	122°47.5	16.7'	07°15.9	13.9'	54.2'
10	328°37.1	28.5	137°23.2	16.7'	07°29.8	13.9'	54.2'
11	343°37.3	29.5	151°58.9	16.6'	07°43.7	13.9'	54.2'
12	358°37.5	N02°30.5	166°34.5	16.6'	S07°57.6	13.8'	54.3'
13	13°37.7	31.5	181°10.1	16.6'	08°11.4	13.8'	54.3'
14	28°37.9	32.5	195°45.7	16.5'	08°25.3	13.8'	54.3'
15	43°38.1	· · 33.4	210°21.2	16.5'	08°39.1	13.8'	54.3'
16	58°38.3	34.4	224°56.7	16.4'	08°52.8	13.7'	54.3'
17	73°38.4	35.4	239°32.1	16.4'	09°06.6	13.7'	54.3'
18	88°38.6	N02°36.4	254°07.5	16.3'	S09°20.3	13.7'	54.3'
19	103°38.8	37.3	268°42.8	16.3'	09°33.9	13.6'	54.3'
20	118°39.0	38.3	283°18.1	16.2'	09°47.6	13.6'	54.3'
21	133°39.2	· · 39.3	297°53.4	16.2'	10°01.2	13.6'	54.4'
22	148°39.4	40.3	312°28.6	16.2'	10°14.8	13.6'	54.4'
23	163°39.6	41.3	327°03.7	16.1'	10°28.3	13.5'	54.4'
	SD = 16.0'	d = 1.0'			SD = 14.8'		

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	178°39.8	N02°42.2	341°38.8	16.1'	S10°41.9	13.5'	54.4'
1	193°40.0	43.2	356°13.9	16.0'	10°55.3	13.4'	54.4'
2	208°40.1	44.2	10°48.9	15.9'	11°08.8	13.4'	54.4'
3	223°40.3	· · 45.2	25°23.8	15.9'	11°22.2	13.4'	54.4'
4	238°40.5	46.1	39°58.7	15.8'	11°35.6	13.3'	54.4'
5	253°40.7	47.1	54°33.6	15.8'	11°48.9	13.3'	54.4'
6	268°40.9	N02°48.1	69°08.4	15.7'	S12°02.2	13.3'	54.5'
7	283°41.1	49.1	83°43.1	15.7'	12°15.5	13.2'	54.5'
8	298°41.3	50.1	98°17.8	15.6'	12°28.7	13.2'	54.5'
9	313°41.5	· · 51.0	112°52.4	15.6'	12°41.9	13.1'	54.5'
10	328°41.7	52.0	127°26.9	15.5'	12°55.0	13.1'	54.5'
11	343°41.9	53.0	142°01.4	15.4'	13°08.1	13.0'	54.5'
12	358°42.0	N02°54.0	156°35.8	15.4'	S13°21.1	13.0'	54.5'
13	13°42.2	54.9	171°10.2	15.3'	13°34.1	12.9'	54.5'
14	28°42.4	55.9	185°44.5	15.2'	13°47.1	12.9'	54.6'
15	43°42.6	· · 56.9	200°18.7	15.2'	14°00.0	12.9'	54.6'
16	58°42.8	57.9	214°52.9	15.1'	14°12.8	12.8'	54.6'
17	73°43.0	58.8	229°27.0	15.0'	14°25.6	12.8'	54.6'
18	88°43.2	N02°59.8	244°01.1	15.0'	S14°38.4	12.7'	54.6'
19	103°43.4	03°00.8	258°35.0	14.9'	14°51.1	12.6'	54.6'
20	118°43.6	01.8	273°08.9	14.8'	15°03.7	12.6'	54.6'
21	133°43.7	· · 02.8	287°42.8	14.8'	15°16.3	12.5'	54.7'
22	148°43.9	03.7	302°16.5	14.7'	15°28.9	12.5'	54.7'
23	163°44.1	04.7	316°50.2	14.6'	15°41.3	12.4'	54.7'
	SD = 16.0'	d = 1.0'			SD = 14.8'		

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	02:33	04:14	05:25	18:49	20:00	21:44
N 70°	03:01	04:26	05:29	18:44	19:48	21:15
68°	03:21	04:36	05:33	18:40	19:38	20:54
66°	03:37	04:43	05:36	18:37	19:30	20:37
64°	03:49	04:50	05:38	18:35	19:23	20:24
62°	04:00	04:55	05:40	18:32	19:17	20:13
60°	04:09	05:00	05:42	18:30	19:13	20:04
N 58°	04:16	05:04	05:44	18:28	19:08	19:57
56°	04:23	05:08	05:45	18:27	19:05	19:50
54°	04:28	05:11	05:47	18:26	19:01	19:44
52°	04:33	05:14	05:48	18:24	18:58	19:39
50°	04:38	05:16	05:49	18:23	18:56	19:34
45°	04:47	05:22	05:51	18:21	18:50	19:25
N 40°	04:54	05:26	05:53	18:19	18:46	19:18
35°	05:00	05:29	05:55	18:17	18:42	19:12
30°	05:04	05:32	05:56	18:15	18:39	19:07
20°	05:11	05:36	05:59	18:13	18:35	19:01
N 10°	05:15	05:39	06:00	18:11	18:32	18:56
0°	05:18	05:42	06:02	18:09	18:29	18:53
S 10°	05:19	05:43	06:04	18:07	18:28	18:52
20°	05:18	05:44	06:06	18:05	18:27	18:53
30°	05:16	05:43	06:07	18:03	18:27	18:55
35°	05:14	05:43	06:08	18:02	18:27	18:57
40°	05:11	05:42	06:09	18:01	18:28	18:59
45°	05:07	05:41	06:11	18:00	18:29	19:03
S 50°	05:02	05:40	06:12	17:58	18:30	19:08
52°	04:59	05:39	06:13	17:57	18:31	19:10
54°	04:57	05:38	06:13	17:57	18:32	19:13
56°	04:53	05:37	06:14	17:56	18:33	19:16
58°	04:50	05:36	06:15	17:55	18:34	19:20
S 60°	04:45	05:34	06:16	17:54	18:35	19:24

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	19:19	21:24	· · ·	05:54	05:21	04:41
N 70°	19:12	21:06	23:18	05:55	05:30	05:00
68°	19:08	20:53	22:49	05:56	05:37	05:16
66°	19:04	20:42	22:28	05:57	05:43	05:28
64°	19:00	20:33	22:11	05:58	05:49	05:39
62°	18:57	20:25	21:57	05:58	05:53	05:48
60°	18:55	20:18	21:46	05:59	05:57	05:56
N 58°	18:52	20:12	21:36	05:59	06:01	06:03
56°	18:50	20:07	21:27	06:00	06:04	06:09
54°	18:48	20:03	21:19	06:00	06:07	06:15
52°	18:47	19:58	21:12	06:01	06:10	06:20
50°	18:45	19:55	21:06	06:01	06:12	06:24
45°	18:42	19:46	20:53	06:02	06:17	06:34
N 40°	18:39	19:40	20:42	06:02	06:22	06:42
35°	18:37	19:34	20:33	06:03	06:25	06:49
30°	18:35	19:29	20:25	06:03	06:29	06:56
20°	18:32	19:20	20:11	06:04	06:35	07:07
N 10°	18:29	19:13	19:59	06:05	06:40	07:16
0°	18:26	19:06	19:47	06:06	06:45	07:25
S 10°	18:23	18:59	19:36	06:06	06:50	07:35
20°	18:20	18:51	19:25	06:07	06:55	07:44
30°	18:17	18:43	19:11	06:08	07:01	07:56
35°	18:15	18:38	19:03	06:08	07:04	08:02
40°	18:13	18:33	18:55	06:09	07:08	08:10
45°	18:10	18:26	18:45	06:09	07:13	08:18
S 50°	18:07	18:19	18:32	06:10	07:19	08:29
52°	18:06	18:16	18:27	06:10	07:21	08:34
54°	18:05	18:12	18:20	06:11	07:24	08:39
56°	18:03	18:08	18:14	06:11	07:27	08:45
58°	18:01	18:03	18:06	06:12	07:31	08:52
S 60°	17:59	17:58	17:57	06:12	07:34	09:00

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
25	05:57	05:48	12:06	—:—	12:16	
26	05:39	05:30	12:06	00:35	12:55	
27	05:21	05:12	12:05	01:16	13:36	

March 28, 29, 30 UT (Thu., Fri., Sat.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	185°54.2	194°37.1	S05°18.5	209°39.6	S11°12.3	141°37.1	N15°58.5	200°47.7	S08°09.3	Alpheratz	357°35.8 29°13.3
1	200°56.6	209°36.7	17.3	224°40.3	11.6	156°39.0	58.7	215°49.9	09.2	Ankaa	353°08.2 -42°10.6
2	215°59.1	224°36.3	16.1	239°40.9	10.9	171°40.9	58.8	230°52.1	09.1	Schedar	349°32.4 56°40.1
3	231°01.5	239°35.9	•• 14.9	254°41.5	•• 10.2	186°42.9	•• 59.0	245°54.3	•• 09.0	Diphda	348°48.3 -17°51.4
4	246°04.0	254°35.5	13.7	269°42.1	09.5	201°44.8	59.1	260°56.5	08.9	Achernar	335°21.2 -57°07.0
5	261°06.5	269°35.1	12.5	284°42.7	08.8	216°46.8	59.3	275°58.7	08.8	Hamal	327°52.3 23°34.5
6	276°08.9	284°34.7	S05°11.3	299°43.3	S11°08.1	231°48.7	N15°59.4	291°00.9	S08°08.7	Polaris	314°45.4 89°22.2
7	291°11.4	299°34.3	10.2	314°43.9	07.4	246°50.7	59.6	306°03.1	08.6	Acamar	315°12.6 -40°12.7
8	306°13.9	314°33.9	09.0	329°44.6	06.7	261°52.6	59.7	321°05.3	08.5	Menkar	314°07.1 4°11.0
9	321°16.3	329°33.5	•• 07.8	344°45.2	•• 06.0	276°54.5	15°59.9	336°07.5	•• 08.4	Mirfak	308°29.6 49°56.9
10	336°18.8	344°33.1	06.6	359°45.8	05.3	291°56.5	16°00.1	351°09.7	08.3	Aldebaran	290°40.6 16°33.4
11	351°21.3	359°32.7	05.4	14°46.4	04.6	306°58.4	00.2	6°11.9	08.2	Rigel	281°04.6 -8°10.6
12	6°23.7	14°32.3	S05°04.2	29°47.0	S11°03.8	322°00.4	N16°00.4	21°14.1	S08°08.1	Capella	280°23.0 46°01.5
13	21°26.2	29°31.9	03.0	44°47.6	03.1	337°02.3	00.5	36°16.3	08.0	Bellatrix	278°23.7 6°22.2
14	36°28.7	44°31.5	01.8	59°48.3	02.4	352°04.2	00.7	51°18.5	07.8	Elnath	278°02.8 28°37.7
15	51°31.1	59°31.1	05°00.6	74°48.9	•• 01.7	7°06.2	•• 00.8	66°20.7	•• 07.7	Alnilam	275°38.5 -1°11.3
16	66°33.6	74°30.7	04°59.5	89°49.5	01.0	22°08.1	01.0	81°22.9	07.6	Betelgeuse	270°52.9 7°24.6
17	81°36.0	89°30.3	58.3	104°50.1	11°00.3	37°10.1	01.1	96°25.0	07.5	Canopus	263°52.7 -52°42.8
18	96°38.5	104°29.9	S04°57.1	119°50.7	S10°59.6	52°12.0	N16°01.3	111°27.2	S08°07.4	Sirius	258°26.8 -16°45.1
19	111°41.0	119°29.5	55.9	134°51.3	58.9	67°13.9	01.5	126°29.4	07.3	Adhara	255°06.4 -29°00.5
20	126°43.4	134°29.1	54.7	149°51.9	58.2	82°15.9	01.6	141°31.6	07.2	Procyon	244°51.5 5°09.7
21	141°45.9	149°28.7	•• 53.5	164°52.6	•• 57.5	97°17.8	•• 01.8	156°33.8	•• 07.1	Pollux	243°18.0 27°58.1
22	156°48.4	164°28.3	52.3	179°53.2	56.8	112°19.8	01.9	171°36.0	07.0	Avior	234°14.8 -59°35.4
23	171°50.8	179°27.9	51.1	194°53.8	56.1	127°21.7	02.1	186°38.2	06.9	Suhail	222°46.6 -43°32.0
Mer.pass. 11:34		ν -0.4' d -1.2' m -3.88		ν 0.6' d -0.7' m 1.20		ν 1.9' d 0.2' m -2.08		ν 2.2' d -0.1' m 1.05		Miaplacidus	221°37.9 -69°49.2
										Alphard	217°48.2 -8°45.9
										Regulus	207°34.9 11°50.9
										Dubhe	193°41.2 61°37.3
										Denebola	182°25.3 14°26.1
										Gienah	175°44.0 -17°40.7
										Acrux	173°00.2 -63°14.0
										Gacrux	171°51.9 -57°15.0
										Alioth	166°13.0 55°49.6
										Spica	158°22.7 -11°17.4
										Alkaid	152°52.0 49°11.4
										Hadar	148°36.4 -60°29.3
										Menkent	147°58.1 -36°29.4
										Arcturus	145°48.2 19°03.2
										Rigel Kent.	139°40.7 -60°56.0
										Kochab	137°18.8 74°03.1
										Zuben'ubi	136°56.5 -16°08.6
										Alphecca	126°04.0 26°37.7
										Antares	112°16.5 -26°29.2
										Atria	107°11.1 -69°04.0
										Sabik	102°03.4 -15°45.4
										Shaula	96°11.2 -37°07.2
										Rasalhague	95°59.1 12°32.3
										Eltanin	90°42.4 51°28.7
										Kaus Aust.	83°33.4 -34°22.3
										Vega	80°33.7 38°48.0
										Nunki	75°48.6 -26°16.0
										Altair	62°00.7 8°55.7
										Peacock	53°07.0 -56°39.3
										Deneb	49°26.5 45°21.7
										Enif	33°39.7 9°58.9
										Al Na'ir	27°34.1 -46°50.6
										Fomalhaut	15°15.6 -29°29.7
										Scheat	13°46.2 28°12.6
										Markab	13°30.9 15°19.9
										Mar 28 Thu SHA Mer.pass	
										Venus	8°42.9 11:02
										Mars	23°45.5 10:01
										Jupiter	315°42.9 14:32
										Saturn	14°53.6 10:35
										Mar 29 Fri SHA Mer.pass	
										Venus	7°34.2 11:02
										Mars	23°01.1 10:00
										Jupiter	315°30.3 14:29
										Saturn	14°47.1 10:32
										Mar 30 Sat SHA Mer.pass	
										Venus	6°25.6 11:03
										Mars	22°16.8 09:59
										Jupiter	315°17.7 14:25
										Saturn	14°40.7 10:28
										Horizontal parallax	
										Venus:	0.1
										Mars:	0.1
Mer.pass. 11:27		ν -0.4' d -1.2' m -3.88		ν 0.6' d -0.7' m 1.20		ν 1.9' d 0.2' m -2.07		ν 2.2' d -0.1' m 1.05			

Sun				Moon			
h	GHA	Dec		GHA	ν	Dec	d HP
Thu							
0	178°44.3	N03°05.7		331°23.8	14.5'	S15°53.8	12.4' 54.7'
1	193°44.5	06.7		345°57.4	14.5'	16°06.1	12.3' 54.7'
2	208°44.7	07.6		0°30.9	14.4'	16°18.5	12.2' 54.7'
3	223°44.9	· · 08.6		15°04.3	14.3'	16°30.7	12.2' 54.8'
4	238°45.1	09.6		29°37.6	14.2'	16°42.9	12.1' 54.8'
5	253°45.3	10.6		44°10.8	14.2'	16°55.0	12.1' 54.8'
6	268°45.4	N03°11.5		58°44.0	14.1'	S17°07.1	12.0' 54.8'
7	283°45.6	12.5		73°17.1	14.0'	17°19.1	11.9' 54.8'
8	298°45.8	13.5		87°50.1	13.9'	17°31.0	11.9' 54.8'
9	313°46.0	· · 14.5		102°23.0	13.9'	17°42.9	11.8' 54.9'
10	328°46.2	15.4		116°55.9	13.8'	17°54.7	11.7' 54.9'
11	343°46.4	16.4		131°28.6	13.7'	18°06.4	11.7' 54.9'
12	358°46.6	N03°17.4		146°01.3	13.6'	S18°18.1	11.6' 54.9'
13	13°46.8	18.4		160°33.9	13.5'	18°29.7	11.5' 54.9'
14	28°47.0	19.3		175°06.5	13.4'	18°41.2	11.4' 54.9'
15	43°47.1	· · 20.3		189°38.9	13.4'	18°52.6	11.4' 55.0'
16	58°47.3	21.3		204°11.3	13.3'	19°04.0	11.3' 55.0'
17	73°47.5	22.3		218°43.5	13.2'	19°15.3	11.2' 55.0'
18	88°47.7	N03°23.2		233°15.7	13.1'	S19°26.5	11.1' 55.0'
19	103°47.9	24.2		247°47.8	13.0'	19°37.7	11.1' 55.0'
20	118°48.1	25.2		262°19.8	12.9'	19°48.7	11.0' 55.0'
21	133°48.3	· · 26.1		276°51.7	12.8'	19°59.7	10.9' 55.1'
22	148°48.5	27.1		291°23.6	12.7'	20°10.6	10.8' 55.1'
23	163°48.7	28.1		305°55.3	12.7'	20°21.4	10.7' 55.1'
SD = 16.0'				SD = 14.9'			

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	178°48.8	N03°29.1	320°26.9	12.6'	S20°32.2	10.7'	55.1'
1	193°49.0	30.0	334°58.5	12.5'	20°42.8	10.6'	55.1'
2	208°49.2	31.0	349°30.0	12.4'	20°53.4	10.5'	55.2'
3	223°49.4	· · 32.0	4°01.4	12.3'	21°03.9	10.4'	55.2'
4	238°49.6	33.0	18°32.6	12.2'	21°14.2	10.3'	55.2'
5	253°49.8	33.9	33°03.8	12.1'	21°24.5	10.2'	55.2'
6	268°50.0	N03°34.9	47°34.9	12.0'	S21°34.8	10.1'	55.2'
7	283°50.2	35.9	62°05.9	11.9'	21°44.9	10.0'	55.3'
8	298°50.4	36.8	76°36.9	11.8'	21°54.9	9.9'	55.3'
9	313°50.5	· · 37.8	91°07.7	11.7'	22°04.8	9.8'	55.3'
10	328°50.7	38.8	105°38.4	11.6'	22°14.7	9.7'	55.3'
11	343°50.9	39.8	120°09.0	11.5'	22°24.4	9.6'	55.3'
12	358°51.1	N03°40.7	134°39.6	11.4'	S22°34.1	9.5'	55.4'
13	13°51.3	41.7	149°10.0	11.3'	22°43.6	9.4'	55.4'
14	28°51.5	42.7	163°40.4	11.2'	22°53.1	9.3'	55.4'
15	43°51.7	· · 43.6	178°10.6	11.2'	23°02.4	9.2'	55.4'
16	58°51.9	44.6	192°40.8	11.1'	23°11.6	9.1'	55.4'
17	73°52.0	45.6	207°10.8	11.0'	23°20.8	9.0'	55.5'
18	88°52.2	N03°46.6	221°40.8	10.9'	S23°29.8	8.9'	55.5'
19	103°52.4	47.5	236°10.6	10.8'	23°38.7	8.8'	55.5'
20	118°52.6	48.5	250°40.4	10.7'	23°47.5	8.7'	55.5'
21	133°52.8	· · 49.5	265°10.1	10.6'	23°56.2	8.6'	55.5'
22	148°53.0	50.4	279°39.6	10.5'	24°04.8	8.5'	55.6'
23	163°53.2	51.4	294°09.1	10.4'	24°13.3	8.4'	55.6'
SD = 16.0'				SD = 15.0'			

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	178°53.4	N03°52.4	308°38.5	10.3'	S24°21.7	8.3'	55.6'
1	193°53.5	53.4	323°07.8	10.2'	24°30.0	8.1'	55.6'
2	208°53.7	54.3	337°37.0	10.1'	24°38.1	8.0'	55.7'
3	223°53.9	· · 55.3	352°06.1	10.0'	24°46.1	7.9'	55.7'
4	238°54.1	56.3	6°35.0	9.9'	24°54.1	7.8'	55.7'
5	253°54.3	57.2	21°03.9	9.8'	25°01.8	7.7'	55.7'
6	268°54.5	N03°58.2	35°32.7	9.7'	S25°09.5	7.6'	55.8'
7	283°54.7	03°59.2	50°01.4	9.6'	25°17.1	7.4'	55.8'
8	298°54.9	04°00.1	64°30.0	9.5'	25°24.5	7.3'	55.8'
9	313°55.0	· · 01.1	78°58.6	9.4'	25°31.8	7.2'	55.8'
10	328°55.2	02.1	93°27.0	9.3'	25°39.0	7.1'	55.9'
11	343°55.4	03.1	107°55.3	9.2'	25°46.0	6.9'	55.9'
12	358°55.6	N04°04.0	122°23.5	9.1'	S25°53.0	6.8'	55.9'
13	13°55.8	05.0	136°51.6	9.0'	25°59.8	6.7'	55.9'
14	28°56.0	06.0	151°19.7	8.9'	26°06.4	6.5'	56.0'
15	43°56.2	· · 06.9	165°47.6	8.9'	26°13.0	6.4'	56.0'
16	58°56.4	07.9	180°15.5	8.8'	26°19.4	6.3'	56.0'
17	73°56.5	08.9	194°43.2	8.7'	26°25.6	6.1'	56.0'
18	88°56.7	N04°09.8	209°10.9	8.6'	S26°31.8	6.0'	56.1'
19	103°56.9	10.8	223°38.5	8.5'	26°37.8	5.9'	56.1'
20	118°57.1	11.8	238°06.0	8.4'	26°43.6	5.7'	56.1'
21	133°57.3	· · 12.7	252°33.4	8.3'	26°49.3	5.6'	56.1'
22	148°57.5	13.7	267°00.7	8.2'	26°54.9	5.4'	56.2'
23	163°57.7	14.7	281°27.9	8.1'	27°00.4	5.3'	56.2'
SD = 16.0'				SD = 15.2'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	02:06	03:57	05:09	19:03	20:16	22:12
N 70°	02:40	04:11	05:15	18:56	20:02	21:34
68°	03:04	04:22	05:20	18:51	19:50	21:09
66°	03:22	04:31	05:24	18:47	19:41	20:50
64°	03:37	04:39	05:28	18:43	19:33	20:35
62°	03:48	04:45	05:31	18:40	19:26	20:23
60°	03:58	04:51	05:33	18:38	19:20	20:13
N 58°	04:07	04:56	05:35	18:35	19:15	20:04
56°	04:14	05:00	05:37	18:33	19:11	19:57
54°	04:20	05:03	05:39	18:31	19:07	19:50
52°	04:26	05:07	05:41	18:29	19:04	19:45
50°	04:31	05:10	05:42	18:28	19:01	19:40
45°	04:41	05:16	05:46	18:24	18:54	19:29
N 40°	04:49	05:21	05:48	18:22	18:49	19:21
35°	04:55	05:25	05:51	18:19	18:45	19:15
30°	05:00	05:29	05:53	18:17	18:41	19:09
20°	05:08	05:34	05:56	18:14	18:36	19:01
N 10°	05:13	05:38	05:59	18:11	18:32	18:56
0°	05:17	05:41	06:01	18:08	18:29	18:53
S 10°	05:18	05:43	06:04	18:05	18:26	18:51
20°	05:19	05:44	06:06	18:03	18:25	18:50
30°	05:17	05:45	06:09	18:00	18:23	18:51
35°	05:16	05:45	06:11	17:58	18:23	18:53
40°	05:14	05:45	06:12	17:56	18:23	18:54
45°	05:11	05:45	06:14	17:54	18:23	18:57
S 50°	05:07	05:44	06:17	17:52	18:24	19:01
52°	05:05	05:44	06:18	17:51	18:24	19:03
54°	05:02	05:44	06:19	17:49	18:25	19:06
56°	05:00	05:43	06:20	17:48	18:25	19:08
58°	04:56	05:42	06:21	17:47	18:26	19:11
S 60°	04:53	05:41	06:23	17:45	18:26	19:15

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	00:02	■	■	03:31	■	■
N 70°	■	■	■	04:16	■	■
68°	· · ·	01:24	■	04:47	03:46	■
66°	· · ·	00:32	■	05:09	04:39	■
64°	· · ·	00:00	02:15	05:28	05:12	04:40
62°	23:37	· · ·	01:27	05:43	05:36	05:28
60°	23:19	· · ·	00:57	05:55	05:56	05:58
N 58°	23:03	· · ·	00:35	06:06	06:12	06:22
56°	22:50	· · ·	00:16	06:16	06:25	06:41
54°	22:39	· · ·	00:01	06:24	06:37	06:57
52°	22:29	23:47	· · ·	06:32	06:48	07:10
50°	22:20	23:36	· · ·	06:39	06:57	07:22
45°	22:01	23:12	· · ·	06:53	07:17	07:47
N 40°	21:46	22:52	23:59	07:06	07:33	08:08
35°	21:34	22:36	23:40	07:16	07:47	08:24
30°	21:22	22:23	23:24	07:25	07:59	08:39
20°	21:04	21:59	22:57	07:41	08:20	09:04
N 10°	20:47	21:39	22:34	07:55	08:38	09:25
0°	20:32	21:20	22:13	08:09	08:55	09:45
S 10°	20:17	21:02	21:52	08:22	09:12	10:05
20°	20:01	20:42	21:29	08:36	09:30	10:27
30°	19:43	20:19	21:03	08:52	09:51	10:52
35°	19:32	20:06	20:47	09:02	10:04	11:07
40°	19:20	19:51	20:29	09:13	10:18	11:24
45°	19:06	19:33	20:08	09:26	10:35	11:45
S 50°	18:49	19:11	19:41	09:41	10:56	12:11
52°	18:41	19:00	19:27	09:49	11:06	12:24
54°	18:32	18:48	19:12	09:57	11:18	12:39
56°	18:22	18:34	18:55	10:06	11:31	12:56
58°	18:10	18:18	18:33	10:17	11:46	13:17
S 60°	17:57	17:59	18:06	10:29	12:05	13:44

Day	Sun		Mer. Pass	Moon		Age 18-20 93-81%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper Mer. Pass hh:mm	Lower Mer. Pass hh:mm	
28	05:03	04:54	12:05	01:58	14:20	
29	04:45	04:36	12:05	02:43	15:08	
30	04:27	04:18	12:04	03:33	15:59	

March 31, 01, 02 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	188°51.6	194°08.7	S03°52.5	210°24.2	S10°21.5	143°56.6	N16°09.7	203°25.9	S08°01.8	Alpheratz	357°35.8 29°13.3
1	203°54.0	209°08.3	51.2	225°24.8	20.7	158°58.5	09.9	218°28.1	01.6	Ankaa	353°08.2 -42°10.6
2	218°56.5	224°07.9	50.0	240°25.4	20.0	174°00.4	10.1	233°30.3	01.5	Schedar	349°32.4 56°40.1
3	233°59.0	239°07.5	· · 48.8	255°26.1	· · 19.3	189°02.4	· · 10.2	248°32.5	· · 01.4	Diphda	348°48.3 -17°51.4
4	249°01.4	254°07.1	47.6	270°26.7	18.6	204°04.3	10.4	263°34.7	01.3	Achernar	335°21.2 -57°07.0
5	264°03.9	269°06.7	46.4	285°27.3	17.9	219°06.2	10.5	278°36.9	01.2	Hamal	327°52.3 23°34.5
6	279°06.4	284°06.4	S03°45.2	300°27.9	S10°17.2	234°08.2	N16°10.7	293°39.1	S08°01.1	Polaris	314°46.0 89°22.1
7	294°08.8	299°06.0	44.0	315°28.6	16.5	249°10.1	10.8	308°41.3	01.0	Acamar	315°12.6 -40°12.6
8	309°11.3	314°05.6	42.8	330°29.2	15.8	264°12.0	11.0	323°43.5	00.9	Menkar	314°07.1 4°11.0
9	324°13.8	329°05.2	· · 41.6	345°29.8	· · 15.0	279°14.0	· · 11.1	338°45.7	· · 00.8	Mirfak	308°29.6 49°56.9
10	339°16.2	344°04.8	40.4	0°30.4	14.3	294°15.9	11.3	353°47.9	00.7	Aldebaran	290°40.6 16°33.4
11	354°18.7	359°04.4	39.2	15°31.1	13.6	309°17.8	11.5	8°50.1	00.6	Rigel	281°04.6 -8°10.6
12	9°21.1	14°04.0	S03°38.0	30°31.7	S10°12.9	324°19.8	N16°11.6	23°52.3	S08°00.5	Capella	280°23.0 46°01.5
13	24°23.6	29°03.6	36.8	45°32.3	12.2	339°21.7	11.8	38°54.5	00.4	Bellatrix	278°23.7 6°22.2
14	39°26.1	44°03.3	35.6	60°32.9	11.5	354°23.6	11.9	53°56.7	00.3	Elnath	278°02.8 28°37.7
15	54°28.5	59°02.9	· · 34.4	75°33.6	· · 10.8	9°25.6	· · 12.1	68°58.9	· · 00.2	Alnilam	275°38.5 -1°11.3
16	69°31.0	74°02.5	33.2	90°34.2	10.0	24°27.5	12.2	84°01.1	00.1	Betelgeuse	270°52.9 7°24.6
17	84°33.5	89°02.1	32.0	105°34.8	09.3	39°29.4	12.4	99°03.3	08°00.0	Canopus	263°52.7 -52°42.8
18	99°35.9	104°01.7	S03°30.8	120°35.4	S10°08.6	54°31.4	N16°12.6	114°05.5	S07°59.9	Sirius	258°26.8 -16°45.1
19	114°38.4	119°01.3	29.6	135°36.1	07.9	69°33.3	12.7	129°07.7	59.8	Adhara	255°06.4 -29°00.5
20	129°40.9	134°00.9	28.4	150°36.7	07.2	84°35.2	12.9	144°09.9	59.7	Procyon	244°51.5 5°09.7
21	144°43.3	149°00.6	· · 27.2	165°37.3	· · 06.5	99°37.1	· · 13.0	159°12.0	· · 59.6	Pollux	243°18.0 27°58.1
22	159°45.8	164°00.2	25.9	180°37.9	05.8	114°39.1	13.2	174°14.2	59.5	Avior	234°14.8 -59°35.5
23	174°48.2	178°59.8	24.7	195°38.6	05.0	129°41.0	13.3	189°16.4	59.4	Suhail	222°46.6 -43°32.0
Mer.pass. 11:23		ν -0.4' d -1.2' m -3.88		ν 0.6' d -0.7' m 1.19		ν 1.9' d 0.2' m -2.07		ν 2.2' d -0.1' m 1.05			

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	189°50.7	193°59.4	S03°23.5	210°39.2	S10°04.3	144°42.9	N16°13.5	204°18.6	S07°59.3		
1	204°53.2	208°59.0	22.3	225°39.8	03.6	159°44.9	13.7	219°20.8	59.2		
2	219°55.6	223°58.6	21.1	240°40.5	02.9	174°46.8	13.8	234°23.0	59.0		
3	234°58.1	238°58.3	· · 19.9	255°41.1	· · 02.2	189°48.7	· · 14.0	249°25.2	· · 58.9		
4	250°00.6	253°57.9	18.7	270°41.7	01.5	204°50.7	14.1	264°27.4	58.8		
5	265°03.0	268°57.5	17.5	285°42.3	00.7	219°52.6	14.3	279°29.6	58.7		
6	280°05.5	283°57.1	S03°16.3	300°43.0	S10°00.0	234°54.5	N16°14.4	294°31.8	S07°58.6		
7	295°08.0	298°56.7	15.1	315°43.6	09°59.3	249°56.4	14.6	309°34.0	58.5		
8	310°10.4	313°56.3	13.9	330°44.2	58.6	264°58.4	14.7	324°36.2	58.4		
9	325°12.9	328°55.9	· · 12.7	345°44.9	· · 57.0	280°00.3	· · 14.9	339°38.4	· · 58.3		
10	340°15.4	343°55.6	11.5	0°45.5	57.2	295°02.2	15.1	354°40.6	58.2		
11	355°17.8	358°55.2	10.2	15°46.1	56.4	310°04.2	15.2	9°42.8	58.1		
12	10°20.3	13°54.8	S03°09.0	30°46.7	S09°55.7	325°06.1	N16°15.4	24°45.0	S07°58.0		
13	25°22.7	28°54.4	07.8	45°47.4	55.0	340°08.0	15.5	39°47.2	57.9		
14	40°25.2	43°54.0	06.6	60°48.0	54.3	355°09.9	15.7	54°49.4	57.8		
15	55°27.7	58°53.6	· · 05.4	75°48.6	· · 53.6	10°11.9	· · 15.8	69°51.6	· · 57.7		
16	70°30.1	73°53.3	04.2	90°49.3	52.9	25°13.8	16.0	84°53.9	57.6		
17	85°32.6	88°52.9	03.0	105°49.9	52.1	40°15.7	16.2	99°56.1	57.5		
18	100°35.1	103°52.5	S03°01.8	120°50.5	S09°51.4	55°17.7	N16°16.3	114°58.3	S07°57.4		
19	115°37.5	118°52.1	03°00.6	135°51.0	50.7	70°19.6	16.5	130°00.5	57.3		
20	130°40.0	133°51.7	02°59.4	150°51.8	50.0	85°21.5	16.6	145°02.7	57.2		
21	145°42.5	148°51.4	· · 58.1	165°52.4	· · 49.3	100°23.4	· · 16.8	160°04.9	· · 57.1		
22	160°44.9	163°51.0	56.9	180°53.0	48.5	115°25.4	16.9	175°07.1	57.0		
23	175°47.4	178°50.6	55.7	195°53.7	47.8	130°27.3	17.1	190°09.3	56.9		
Mer.pass. 11:19		ν -0.4' d -1.2' m -3.88		ν 0.6' d -0.7' m 1.19		ν 1.9' d 0.2' m -2.07		ν 2.2' d -0.1' m 1.06			

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	190°49.9	193°50.2	S02°54.5	210°54.3	S09°47.1	145°29.2	N16°17.3	205°11.5	S07°56.8		
1	205°52.3	208°49.8	53.3	225°54.9	46.4	160°31.2	17.4	220°13.7	56.7		
2	220°54.8	223°49.4	52.1	240°55.5	45.7	175°33.1	17.6	235°15.9	56.6		
3	235°57.2	238°49.1	· · 50.9	255°56.2	· · 44.9	190°35.0	· · 17.7	250°18.1	· · 56.5		
4	250°59.7	253°48.7	49.7	270°56.8	44.2	205°36.9	17.9	265°20.3	56.4		
5	266°02.2	268°48.3	48.5	285°57.4	43.5	220°38.9	18.0	280°22.5	56.3		
6	281°04.6	283°47.9	S02°47.2	300°58.1	S09°42.8	235°40.8	N16°18.2	295°24.7	S07°56.2		
7	296°07.1	298°47.5	46.0	315°58.7	42.1	250°42.7	18.3	310°26.9	56.1		
8	311°09.6	313°47.2	44.8	330°59.3	41.3	265°44.6	18.5	325°29.1	55.9		
9	326°12.0	328°46.8	· · 43.6	346°00.0	· · 40.6	280°46.6	· · 18.7	340°31.3	· · 55.8		
10	341°14.5	343°46.4	42.4	1°00.6	39.9	295°48.5	18.8	355°33.5	55.7		
11	356°17.0	358°46.0	41.2	16°01.2	39.2	310°50.4	19.0	10°35.7	55.6		
12	11°19.4	13°45.6	S02°40.0	31°01.9	S09°38.5	325°52.4	N16°19.1	25°37.9	S07°55.5		
13	26°21.9	28°45.3	38.8	46°02.5	37.7	340°54.3	19.3	40°40.1	55.4		
14	41°24.4	43°44.9	37.5	61°03.1	37.0	355°56.2	19.4	55°42.3	55.3		
15	56°26.8	58°44.5	· · 36.3	76°03.8	· · 36.3	10°58.1	· · 19.6	70°44.5	· · 55.2		
16	71°29.3	73°44.1	35.1	91°04.4	35.6	26°00.1	19.8	85°46.7	55.1		
17	86°31.7	88°43.7	33.9	106°05.0	34.9	41°02.0	19.9	100°48.9	55.0		
18	101°34.2	103°43.4	S02°32.7	121°05.7	S09°34.1	56°03.9	N16°20.1	115°51.1	S07°54.9		
19	116°36.7	118°43.0	31.5	136°06.3	33.4	71°05.8	20.2	130°53.3	54.8		
20	131°39.1	133°42.6	30.3	151°06.9	32.7	86°07.8	20.4	145°55.5	54.7		
21	146°41.6	148°42.2	· · 29.0	166°07.6	· · 32.0	101°09.7	· · 20.5	160°57.7	· · 54.6		
22	161°44.1	163°41.9	27.8	181°08.2	31.2	116°11.6	20.7	175°59.9	54.5		
23	176°46.5	178°41.5	26.6	196°08.8	30.5	131°13.5	20.9	191°02.1	54.4		
Mer.pass. 11:15		ν -0.4' d -1.2' m -3.88		ν 0.6' d -0.7' m 1.19		ν 1.9' d 0.2' m -2.06		ν 2.2' d -0.1' m 1.06			

Mar 31 Sun	SHA	Mer.pass
Venus	5°17.1	11:04
Mars	21°32.6	09:58
Jupiter	315°05.0	14:22
Saturn	14°34.3	10:25

Apr 01 Mon	SHA	Mer.pass
Venus	4°08.7	11:04
Mars	20°48.5	09:57
Jupiter	314°52.2	14:19
Saturn	14°27.9	10:21

Apr 02 Tue	SHA	Mer.pass
Venus	3°00.4	11:05
Mars	20°04.4	09:56
Jupiter	314°39.4	14:16
Saturn	14°21.6	10:18

Horizontal parallax	
Venus:	0.1
Mars:	0.1

Sun							Moon							
Sun	GHA	Dec	GHA	ν	Dec	d	HP	Lat.	Twilight		Sunrise	Sunset	Twilight	
									Naut.	Civil			Civil	Naut.
0	178°57.9	N04°15.6	295°55.0	8.0'	S27°05.7	5.2'	56.2'	N 72°	01:32	03:38	04:54	19:17	20:33	22:47
1	193°58.0	16.6	310°22.1	8.0'	27°10.8	5.0'	56.2'	N 70°	02:17	03:55	05:01	19:09	20:16	21:57
2	208°58.2	17.6	324°49.0	7.9'	27°15.9	4.9'	56.3'	68°	02:46	04:08	05:07	19:02	20:02	21:26
3	223°58.4	•• 18.5	339°15.9	7.8'	27°20.7	4.7'	56.3'	66°	03:07	04:19	05:13	18:57	19:51	21:04
4	238°58.6	19.5	353°42.7	7.7'	27°25.4	4.6'	56.3'	64°	03:23	04:27	05:17	18:52	19:42	20:47
5	253°58.8	20.5	8°09.4	7.6'	27°30.0	4.4'	56.3'	62°	03:37	04:35	05:21	18:48	19:35	20:33
6	268°59.0	N04°21.4	22°36.1	7.5'	S27°34.4	4.3'	56.4'	60°	03:48	04:41	05:24	18:45	19:28	20:22
7	283°59.2	22.4	37°02.6	7.5'	27°38.7	4.1'	56.4'	N 58°	03:57	04:47	05:27	18:42	19:22	20:12
8	298°59.3	23.4	51°29.1	7.4'	27°42.9	4.0'	56.4'	56°	04:05	04:52	05:30	18:39	19:17	20:04
9	313°59.5	•• 24.3	65°55.5	7.3'	27°46.8	3.8'	56.5'	54°	04:12	04:56	05:32	18:37	19:13	19:57
10	328°59.7	25.3	80°21.8	7.2'	27°50.6	3.7'	56.5'	52°	04:18	05:00	05:34	18:35	19:09	19:51
11	343°59.9	26.3	94°48.0	7.2'	27°54.3	3.5'	56.5'	50°	04:24	05:03	05:36	18:33	19:05	19:45
12	359°00.1	N04°27.2	109°14.2	7.1'	S27°57.8	3.4'	56.5'	45°	04:35	05:10	05:40	18:28	18:58	19:33
13	14°00.3	28.2	123°40.3	7.0'	28°01.2	3.2'	56.6'	N 40°	04:44	05:16	05:43	18:25	18:52	19:24
14	29°00.5	29.2	138°06.3	6.9'	28°04.4	3.0'	56.6'	35°	04:51	05:21	05:46	18:22	18:47	19:17
15	44°00.6	•• 30.1	152°32.3	6.9'	28°07.4	2.9'	56.6'	30°	04:57	05:25	05:49	18:19	18:43	19:11
16	59°00.8	31.1	166°58.1	6.8'	28°10.3	2.7'	56.7'	20°	05:05	05:31	05:53	18:14	18:37	19:02
17	74°01.0	32.1	181°23.9	6.7'	28°13.0	2.6'	56.7'	N 10°	05:11	05:36	05:57	18:11	18:32	18:56
18	89°01.2	N04°33.0	195°49.7	6.7'	S28°15.6	2.4'	56.7'	0°	05:16	05:40	06:00	18:07	18:28	18:52
19	104°01.4	34.0	210°15.4	6.6'	28°18.0	2.2'	56.7'	S 10°	05:18	05:43	06:04	18:04	18:25	18:49
20	119°01.6	35.0	224°41.0	6.6'	28°20.3	2.1'	56.8'	20°	05:20	05:45	06:07	18:00	18:22	18:48
21	134°01.8	•• 35.9	239°06.5	6.5'	28°22.3	1.9'	56.8'	30°	05:19	05:47	06:11	17:56	18:20	18:48
22	149°01.9	36.9	253°32.0	6.4'	28°24.2	1.7'	56.8'	35°	05:18	05:48	06:13	17:54	18:19	18:48
23	164°02.1	37.8	267°57.5	6.4'	28°26.0	1.6'	56.9'	40°	05:17	05:48	06:15	17:51	18:18	18:50
SD = 16.0' d = 1.0'							SD = 15.3'							
Mon	GHA	Dec	GHA	ν	Dec	d	HP	Lat.	Moonrise			Moonset		
									Sun	Mon	Tue	Sun	Mon	Tue
0	179°02.3	N04°38.8	282°22.9	6.3'	S28°27.6	1.4'	56.9'	N 72°	■	■	■	■	■	■
1	194°02.5	39.8	296°48.2	6.3'	28°29.0	1.3'	56.9'	N 70°	■	■	■	■	■	■
2	209°02.7	40.7	311°13.4	6.2'	28°30.2	1.1'	57.0'	68°	■	■	■	■	■	■
3	224°02.9	•• 41.7	325°38.7	6.2'	28°31.3	0.9'	57.0'	66°	■	■	■	■	■	■
4	239°03.1	42.7	340°03.8	6.1'	28°32.2	0.7'	57.0'	64°	■	■	■	■	■	■
5	254°03.2	43.6	354°28.9	6.1'	28°33.0	0.6'	57.0'	62°	03:38	■	■	05:08	■	■
6	269°03.4	N04°44.6	8°54.0	6.0'	S28°33.6	0.4'	57.1'	60°	02:38	04:05	04:49	06:08	06:40	07:59
7	284°03.6	45.6	23°19.0	6.0'	28°34.0	0.2'	57.1'	N 58°	02:05	03:22	04:12	06:42	07:23	08:36
8	299°03.8	46.5	37°44.0	5.9'	28°34.2	0.1'	57.1'	56°	01:40	02:54	03:46	07:07	07:52	09:02
9	314°04.0	•• 47.5	52°09.0	5.9'	28°34.3	-0.1'	57.2'	54°	01:21	02:31	03:25	07:27	08:14	09:22
10	329°04.2	48.4	66°33.9	5.9'	28°34.2	-0.3'	57.2'	52°	01:04	02:13	03:08	07:43	08:32	09:40
11	344°04.4	49.4	80°58.7	5.8'	28°33.9	-0.5'	57.2'	50°	00:50	01:57	02:53	07:58	08:48	09:54
12	359°04.5	N04°50.4	95°23.5	5.8'	S28°33.4	-0.6'	57.3'	45°	00:21	01:26	02:23	08:27	09:19	10:24
13	14°04.7	51.3	109°48.3	5.8'	28°32.8	-0.8'	57.3'	N 40°	•• ••	01:02	01:59	08:50	09:44	10:47
14	29°04.9	52.3	124°13.1	5.7'	28°32.0	-1.0'	57.3'	35°	•• ••	00:42	01:39	09:09	10:03	11:06
15	44°05.1	•• 53.3	138°37.8	5.7'	28°31.0	-1.2'	57.4'	30°	•• ••	00:25	01:23	09:26	10:21	11:23
16	59°05.3	54.2	153°02.5	5.7'	28°29.9	-1.3'	57.4'	20°	23:56	•• ••	00:55	09:53	10:49	11:50
17	74°05.5	55.2	167°27.2	5.6'	28°28.5	-1.5'	57.4'	N 10°	23:32	•• ••	00:30	10:17	11:14	12:14
18	89°05.6	N04°56.1	181°51.8	5.6'	S28°27.0	-1.7'	57.5'	0°	23:09	•• ••	00:08	10:39	11:37	12:36
19	104°05.8	57.1	196°16.5	5.6'	28°25.3	-1.9'	57.5'	S 10°	22:46	23:45	•• ••	11:02	12:00	12:58
20	119°06.0	58.1	210°41.1	5.6'	28°23.5	-2.0'	57.5'	20°	22:22	23:21	•• ••	11:26	12:24	13:21
21	134°06.2	04°59.0	225°05.7	5.6'	28°21.4	-2.2'	57.6'	30°	21:54	22:52	23:58	11:54	12:53	13:48
22	149°06.4	05°00.0	239°30.2	5.6'	28°19.2	-2.4'	57.6'	35°	21:37	22:36	23:42	12:10	13:10	14:04
23	164°06.6	00.9	253°54.8	5.5'	28°16.8	-2.6'	57.6'	40°	21:17	22:16	23:25	12:29	13:30	14:22
SD = 16.0' d = 1.0'							SD = 15.5'							
Tue	GHA	Dec	GHA	ν	Dec	d	HP	Day	Sun		Mer.	Moon		Age
									Eqn. of Time	12 ^h	Pass	Upper	Lower	21-23
									mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	72-52%
0	179°06.8	N05°01.9	268°19.3	5.5'	S28°14.3	-2.7'	57.7'	31	04:09	04:00	12:04	04:26	16:54	
1	194°06.9	02.9	282°43.8	5.5'	28°11.5	-2.9'	57.7'	01	03:51	03:42	12:04	05:23	17:52	
2	209°07.1	03.8	297°08.3	5.5'	28°08.6	-3.1'	57.7'	02	03:33	03:24	12:03	06:22	18:51	
3	224°07.3	•• 04.8	311°32.9	5.5'	28°05.5	-3.3'	57.8'							
4	239°07.5	05.7	325°57.4	5.5'	28°02.2	-3.5'	57.8'							
5	254°07.7	06.7	340°21.8	5.5'	27°58.8	-3.6'	57.8'							
6	269°07.9	N05°07.7	354°46.3	5.5'	S27°55.2	-3.8'	57.9'							
7	284°08.0	08.6	9°10.8	5.5'	27°51.3	-4.0'	57.9'							
8	299°08.2	09.6	23°35.3	5.5'	27°47.4	-4.2'	57.9'							
9	314°08.4	•• 10.5	37°59.8	5.5'	27°43.2	-4.3'	58.0'							
10	329°08.6	11.5	52°24.3	5.5'	27°38.8	-4.5'	58.0'							
11	344°08.8	12.5	66°48.9	5.5'	27°34.3	-4.7'	58.0'							
12	359°09.0	N05°13.4	81°13.4	5.5'	S27°29.6	-4.9'	58.1'							
13	14°09.1	14.4	95°37.9	5.5'	27°24.8	-5.1'	58.1'							
14	29°09.3	15.3	110°02.4	5.6'	27°19.7	-5.2'	58.1'							
15	44°09.5	•• 16.3	124°27.0	5.6'	27°14.5	-5.4'	58.2'							
16	59°09.7	17.3	138°51.6	5.6'	27°09.1	-5.6'	58.2'							
17	74°09.9	18.2	153°16.2	5.6'	27°03.5	-5.8'	58.2'							
18	89°10.1	N05°19.2	167°40.8	5.6'	S26°57.7	-5.9'	58.3'							
19	104°10.2	20.1	182°05.4	5.7'	26°51.8	-6.1'	58.3'							
20	119°10.4	21.1	196°30.1	5.7'	26°45.7	-6.3'	58.3'							
21	134°10.6	•• 22.0	210°54.7	5.7'	26°39.4	-6.5'	58.4'							
22	149°10.8	23.0	225°19.4	5.7'	26°33.0	-6.6'	58.4'							
23	164°11.0	24.0	239°44.2	5.8'	26°26.3	-6.8'	58.4'							
SD = 16.0' d = 1.0'							SD = 15.7'							

April 03, 04, 05 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	191°49.0	193°41.1	S02°25.4	211°09.5	S09°29.8	146°15.5	N16°21.0	206°04.3	S07°54.3	Alpheratz	357°35.8 29°13.3
1	206°51.5	208°40.7	24.2	226°10.1	29.1	161°17.4	21.2	221°06.5	54.2	Ankaa	353°08.2 -42°10.5
2	221°53.9	223°40.3	23.0	241°10.7	28.4	176°19.3	21.3	236°08.7	54.1	Schedar	349°32.4 56°40.5
3	236°56.4	238°40.0	· · 21.8	256°11.4	· · 27.6	191°21.2	· · 21.5	251°10.9	· · 54.0	Diphda	348°48.3 -17°51.4
4	251°58.9	253°39.6	20.5	271°12.0	26.9	206°23.2	21.6	266°13.1	53.9	Achernar	335°21.2 -57°06.9
5	267°01.3	268°39.2	19.3	286°12.6	26.2	221°25.1	21.8	281°15.3	53.8	Hamal	327°52.3 23°34.5
6	282°03.8	283°38.8	S02°18.1	301°13.3	S09°25.5	236°27.0	N16°21.9	296°17.5	S07°53.7	Polaris	314°46.7 89°22.1
7	297°06.2	298°38.4	16.9	316°13.9	24.7	251°28.9	22.1	311°19.7	53.6	Acamar	315°12.6 -40°12.6
8	312°08.7	313°38.1	15.7	331°14.5	24.0	266°30.9	22.3	326°21.9	53.5	Menkar	314°07.1 4°11.0
9	327°11.2	328°37.7	· · 14.5	346°15.2	· · 23.3	281°32.8	· · 22.4	341°24.1	· · 53.4	Mirfak	308°29.6 49°56.9
10	342°13.6	343°37.3	13.3	1°15.8	22.6	296°34.7	22.6	356°26.3	53.3	Aldebaran	290°40.6 16°33.4
11	357°16.1	358°36.9	12.0	16°16.4	21.8	311°36.6	22.7	11°28.5	53.2	Rigel	281°04.6 -8°10.6
12	12°18.6	13°36.6	S02°10.8	31°17.1	S09°21.1	326°38.5	N16°22.9	26°30.7	S07°53.1	Capella	280°23.1 46°01.5
13	27°21.0	28°36.2	09.6	46°17.7	20.4	341°40.5	23.0	41°32.9	53.0	Bellatrix	278°23.7 6°22.2
14	42°23.5	43°35.8	08.4	61°18.3	19.7	356°42.4	23.2	56°35.1	52.9	Elnath	278°02.8 28°37.7
15	57°26.0	58°35.4	· · 07.2	76°19.0	· · 18.9	11°44.3	· · 23.4	71°37.3	· · 52.8	Alnilam	275°38.5 -1°11.3
16	72°28.4	73°35.1	06.0	91°19.6	18.2	26°46.2	23.5	86°39.5	52.7	Betelgeuse	270°52.9 7°24.6
17	87°30.9	88°34.7	04.7	106°20.2	17.5	41°48.2	23.7	101°41.7	52.6	Canopus	263°52.8 -52°42.8
18	102°33.3	103°34.3	S02°03.5	121°20.9	S09°16.8	56°50.1	N16°23.8	116°44.0	S07°52.5	Sirius	258°26.8 -16°45.1
19	117°35.8	118°33.9	02.3	136°21.5	16.0	71°52.0	24.0	131°46.2	52.4	Adhara	255°06.4 -29°00.5
20	132°38.3	133°33.5	02°01.1	151°22.1	15.3	86°53.9	24.1	146°48.4	52.3	Procyon	244°51.5 5°09.7
21	147°40.7	148°33.2	01°59.9	166°22.8	· · 14.6	101°55.9	· · 24.3	161°50.6	· · 52.2	Pollux	243°18.0 27°58.1
22	162°43.2	163°32.8	58.7	181°23.4	13.9	116°57.8	24.5	176°52.8	52.1	Avior	234°14.8 -59°35.5
23	177°45.7	178°32.4	57.4	196°24.1	13.1	131°59.7	24.6	191°55.0	52.0	Suhail	222°46.6 -43°32.0
Mer.pass. 11:11		ν -0.4' d -1.2' m -3.88		ν 0.6' d -0.7' m 1.19		ν 1.9' d 0.2' m -2.06		ν 2.2' d -0.1' m 1.06		Miaplacidus	221°38.0 -69°49.2
										Alphard	217°48.2 -8°45.9
										Regulus	207°34.9 11°50.9
										Dubhe	193°41.2 61°37.3
										Denebola	182°25.3 14°26.1
										Gienah	175°43.9 -17°40.7
										Acrux	17

h		Sun			Moon		
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	179° 11.1	N05° 24.9	254° 08.9	5.8'	S26° 19.5	-7.0'	58.5'
1	194° 11.3	25.9	268° 33.7	5.8'	26° 12.6	-7.1'	58.5'
2	209° 11.5	26.8	282° 58.5	5.8'	26° 05.4	-7.3'	58.5'
3	224° 11.7	· · 27.8	297° 23.4	5.9'	25° 58.1	-7.5'	58.6'
4	239° 11.9	28.7	311° 48.2	5.9'	25° 50.6	-7.7'	58.6'
5	254° 12.1	29.7	326° 13.2	6.0'	25° 42.9	-7.8'	58.6'
6	269° 12.2	N05° 30.7	340° 38.1	6.0'	S25° 35.1	-8.0'	58.7'
7	284° 12.4	31.6	355° 03.1	6.0'	25° 27.1	-8.2'	58.7'
8	299° 12.6	32.6	9° 28.1	6.1'	25° 19.0	-8.3'	58.7'
9	314° 12.8	· · 33.5	23° 53.2	6.1'	25° 10.6	-8.5'	58.8'
10	329° 13.0	34.5	38° 18.3	6.2'	25° 02.1	-8.7'	58.8'
11	344° 13.2	35.4	52° 43.5	6.2'	24° 53.5	-8.8'	58.8'
12	359° 13.3	N05° 36.4	67° 08.7	6.2'	S24° 44.6	-9.0'	58.9'
13	14° 13.5	37.3	81° 33.9	6.3'	24° 35.6	-9.2'	58.9'
14	29° 13.7	38.3	95° 59.2	6.3'	24° 26.5	-9.3'	58.9'
15	44° 13.9	· · 39.2	110° 24.6	6.4'	24° 17.2	-9.5'	59.0'
16	59° 14.1	40.2	124° 50.0	6.4'	24° 07.7	-9.6'	59.0'
17	74° 14.2	41.2	139° 15.4	6.5'	23° 58.1	-9.8'	59.0'
18	89° 14.4	N05° 42.1	153° 40.9	6.5'	S23° 48.3	-10.0'	59.1'
19	104° 14.6	43.1	168° 06.4	6.6'	23° 38.3	-10.1'	59.1'
20	119° 14.8	44.0	182° 32.0	6.6'	23° 28.2	-10.3'	59.1'
21	134° 15.0	· · 45.0	196° 57.6	6.7'	23° 17.9	-10.4'	59.2'
22	149° 15.1	45.9	211° 23.3	6.7'	23° 07.5	-10.6'	59.2'
23	164° 15.3	46.9	225° 49.1	6.8'	22° 56.9	-10.7'	59.2'
SD = 16.0'		$d = 1.0'$	SD = 15.9'				

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	179° 15.5	N05° 47.8	240° 14.9	6.9'	S22° 46.2	-10.9'	59.3'
1	194° 15.7	48.8	254° 40.7	6.9'	22° 35.3	-11.0'	59.3'
2	209° 15.9	49.7	269° 06.6	7.0'	22° 24.3	-11.2'	59.3'
3	224° 16.0	· · 50.7	283° 32.6	7.0'	22° 13.1	-11.3'	59.4'
4	239° 16.2	51.6	297° 58.6	7.1'	22° 01.8	-11.5'	59.4'
5	254° 16.4	52.6	312° 24.7	7.1'	21° 50.3	-11.6'	59.4'
6	269° 16.6	N05° 53.5	326° 50.8	7.2'	S21° 38.6	-11.8'	59.5'
7	284° 16.8	54.5	341° 17.0	7.2'	21° 26.9	-11.9'	59.5'
8	299° 16.9	55.4	355° 43.2	7.3'	21° 15.0	-12.1'	59.5'
9	314° 17.1	· · 56.4	10° 09.5	7.4'	21° 02.9	-12.2'	59.6'
10	329° 17.3	57.3	24° 35.9	7.4'	20° 50.7	-12.3'	59.6'
11	344° 17.5	58.3	39° 02.3	7.5'	20° 38.4	-12.5'	59.6'
12	359° 17.7	N05° 59.3	53° 28.8	7.5'	S20° 25.9	-12.6'	59.6'
13	14° 17.8	06° 00.2	67° 55.3	7.6'	20° 13.3	-12.7'	59.7'
14	29° 18.0	01.2	82° 21.9	7.7'	20° 00.5	-12.9'	59.7'
15	44° 18.2	· · 02.1	96° 48.6	7.7'	19° 47.7	-13.0'	59.7'
16	59° 18.4	03.1	111° 15.3	7.8'	19° 34.6	-13.1'	59.8'
17	74° 18.6	04.0	125° 42.0	7.8'	19° 21.5	-13.3'	59.8'
18	89° 18.7	N06° 05.0	140° 08.9	7.9'	S19° 08.2	-13.4'	59.8'
19	104° 18.9	05.9	154° 35.8	7.9'	18° 54.8	-13.5'	59.9'
20	119° 19.1	06.9	169° 02.7	8.0'	18° 41.3	-13.7'	59.9'
21	134° 19.3	· · 07.8	183° 29.7	8.1'	18° 27.6	-13.8'	59.9'
22	149° 19.5	08.7	197° 56.8	8.1'	18° 13.9	-13.9'	59.9'
23	164° 19.6	09.7	212° 23.9	8.2'	18° 00.0	-14.0'	60.0'
SD = 16.0'		$d = 1.0'$	SD = 16.2'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	179° 19.8	N06° 10.6	226° 51.0	8.2'	S17° 45.9	-14.1'	60.0'
1	194° 20.0	11.6	241° 18.3	8.3'	17° 31.8	-14.3'	60.0'
2	209° 20.2	12.5	255° 45.6	8.3'	17° 17.5	-14.4'	60.1'
3	224° 20.3	· · 13.5	270° 12.9	8.4'	17° 03.2	-14.5'	60.1'
4	239° 20.5	14.4	284° 40.3	8.5'	16° 48.7	-14.6'	60.1'
5	254° 20.7	15.4	299° 07.8	8.5'	16° 34.1	-14.7'	60.1'
6	269° 20.9	N06° 16.3	313° 35.3	8.6'	S16° 19.4	-14.8'	60.2'
7	284° 21.1	17.3	328° 02.8	8.6'	16° 04.5	-14.9'	60.2'
8	299° 21.2	18.2	342° 30.5	8.7'	15° 49.6	-15.0'	60.2'
9	314° 21.4	· · 19.2	356° 58.1	8.7'	15° 34.6	-15.1'	60.2'
10	329° 21.6	20.1	11° 25.8	8.8'	15° 19.4	-15.2'	60.3'
11	344° 21.8	21.1	25° 53.6	8.8'	15° 04.2	-15.3'	60.3'
12	359° 21.9	N06° 22.0	40° 21.5	8.9'	S14° 48.8	-15.4'	60.3'
13	14° 22.1	23.0	54° 49.3	8.9'	14° 33.4	-15.5'	60.3'
14	29° 22.3	23.9	69° 17.3	9.0'	14° 17.8	-15.6'	60.4'
15	44° 22.5	· · 24.9	83° 45.2	9.0'	14° 02.2	-15.7'	60.4'
16	59° 22.7	25.8	98° 13.3	9.1'	13° 46.5	-15.8'	60.4'
17	74° 22.8	26.7	112° 41.4	9.1'	13° 30.6	-15.9'	60.4'
18	89° 23.0	N06° 27.7	127° 09.5	9.2'	S13° 14.7	-16.0'	60.5'
19	104° 23.2	28.6	141° 37.7	9.2'	12° 58.7	-16.1'	60.5'
20	119° 23.4	29.6	156° 05.9	9.3'	12° 42.6	-16.2'	60.5'
21	134° 23.5	· · 30.5	170° 34.1	9.3'	12° 26.4	-16.3'	60.5'
22	149° 23.7	31.5	185° 02.5	9.4'	12° 10.2	-16.4'	60.5'
23	164° 23.9	32.4	199° 30.8	9.4'	11° 53.8	-16.4'	60.6'
SD = 16.0'		$d = 0.9'$	SD = 16.4'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	00:37	03:19	04:38	19:31	20:51	—:—
N 70°	01:51	03:39	04:47	19:21	20:31	22:23
68°	02:26	03:54	04:55	19:13	20:15	21:45
66°	02:51	04:06	05:01	19:07	20:02	21:19
64°	03:09	04:16	05:06	19:01	19:52	20:59
62°	03:24	04:24	05:11	18:56	19:43	20:44
60°	03:37	04:32	05:15	18:52	19:36	20:31
N 58°	03:47	04:38	05:19	18:49	19:29	20:21
56°	03:56	04:43	05:22	18:45	19:24	20:11
54°	04:04	04:48	05:25	18:42	19:19	20:04
52°	04:11	04:53	05:27	18:40	19:14	19:57
50°	04:17	04:56	05:29	18:37	19:10	19:50
45°	04:29	05:05	05:34	18:32	19:02	19:38
N 40°	04:39	05:11	05:39	18:28	18:55	19:28
35°	04:47	05:17	05:42	18:24	18:50	19:20
30°	04:53	05:21	05:45	18:21	18:45	19:13
20°	05:03	05:29	05:51	18:15	18:37	19:03
N 10°	05:10	05:34	05:55	18:10	18:32	18:56
0°	05:15	05:39	06:00	18:06	18:27	18:51
S 10°	05:18	05:43	06:04	18:02	18:23	18:47
20°	05:20	05:46	06:08	17:57	18:19	18:45
30°	05:21	05:49	06:13	17:53	18:16	18:44
35°	05:21	05:50	06:15	17:50	18:15	18:44
40°	05:20	05:51	06:18	17:47	18:14	18:45
45°	05:19	05:52	06:22	17:43	18:12	18:46
S 50°	05:16	05:54	06:26	17:39	18:11	18:48
52°	05:15	05:54	06:28	17:37	18:11	18:50
54°	05:14	05:54	06:30	17:35	18:10	18:51
56°	05:12	05:55	06:32	17:32	18:10	18:53
58°	05:10	05:55	06:35	17:30	18:09	18:54
S 60°	05:07	05:56	06:37	17:27	18:09	18:57

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	■	■	07:21	■	■	11:33
N 70°	■	■	06:41	■	■	12:10
68°	■	07:25	06:14	■	09:29	12:36
66°	■	06:27	05:52	■	10:26	12:56
64°	06:38	05:53	05:35	08:14	10:59	13:11
62°	05:38	05:28	05:21	09:14	11:23	13:24
60°	05:04	05:08	05:09	09:48	11:42	13:35
N 58°	04:38	04:51	04:59	10:12	11:58	13:44
56°	04:18	04:37	04:49	10:32	12:11	13:52
54°	04:01	04:25	04:41	10:48	12:23	13:59
52°	03:47	04:14	04:34	11:02	12:33	14:06
50°	03:34	04:05	04:27	11:14	12:42	14:12
45°	03:08	03:44	04:13	11:40	13:01	14:24
N 40°	02:47	03:28	04:01	11:59	13:16	14:34
35°	02:30	03:14	03:51	12:16	13:29	14:43
30°	02:15	03:01	03:42	12:30	13:40	14:51
20°	01:49	02:40	03:26	12:54	13:59	15:04
N 10°	01:27	02:22	03:13	13:15	14:16	15:15
0°	01:07	02:04	03:00	13:34	14:31	15:25
S 10°	00:46	01:47	02:47	13:54	14:46	15:36
20°	00:24	01:28	02:33	14:14	15:02	15:46
30°	· · ·	01:07	02:17	14:37	15:21	15:59
35°	· · ·	00:54	02:08	14:51	15:31	16:06
40°	· · ·	00:39	01:57	15:07	15:43	16:14
45°	· · ·	00:22	01:45	15:25	15:57	16:23
S 50°	· · ·	00:00	01:29	15:48	16:14	16:34
52°	23:50	· · ·	01:22	15:59	16:22	16:40
54°	23:38	· · ·	01:14	16:12	16:31	16:45
56°	23:24	· · ·	01:05	16:26	16:41	16:51
58°	23:08	· · ·	00:54	16:42	16:52	16:58
S 60°	22:49	· · ·	00:42	17:02	17:05	17:06

Day	Sun		Mer. Pass hh:mm	Moon		Age 24-26 41-20%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
03	03:15	03:07	12:03	07:21	19:49	
04	02:58	02:49	12:03	08:18	20:45	
05	02:41	02:32	12:03	09:13	21:39	

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	196°44.7	192°56.3	N00°01.1	212°26.4	S08°02.1	150°05.6	N16°39.8	210°29.0	S07°42.2
1	211°47.1	207°55.9	02.3	227°27.0	01.3	165°07.5	39.9	225°31.3	42.1
2	226°49.6	202°55.5	03.5	242°27.7	08°00.6	180°09.5	40.1	240°33.5	42.0
3	241°52.1	237°55.2	· · 04.8	257°28.3	07°59.8	195°11.4	· · 40.3	255°35.7	· · 41.9
4	256°54.5	252°54.8	06.0	272°29.0	59.1	210°13.3	40.4	270°37.9	41.8
5	271°57.0	267°54.4	07.2	287°29.6	58.4	225°15.2	40.6	285°40.1	41.7
6	286°59.5	282°54.1	N00°08.4	302°30.3	S07°57.6	240°17.1	N16°40.7	300°42.3	S07°41.6
7	302°01.9	297°53.7	09.7	317°30.9	56.9	255°19.0	40.9	315°44.5	41.5
8	317°04.4	312°53.3	10.9	332°31.6	56.2	270°20.9	41.0	330°46.7	41.4
9	332°06.9	327°52.9	· · 12.1	347°32.2	· · 55.4	285°22.8	· · 41.2	345°48.9	· · 41.3
10	347°09.3	342°52.6	13.3	2°32.9	54.7	300°24.7	41.3	0°51.2	41.2
11	2°11.8	357°52.2	14.6	17°33.5	53.9	153°26.6	41.5	15°53.4	41.1
12	17°14.3	12°51.8	N00°15.8	32°34.2	S07°53.2	330°28.6	N16°41.7	30°55.6	S07°41.0
13	32°16.7	27°51.5	17.0	47°34.8	52.5	345°30.5	41.8	45°57.8	40.9
14	47°19.2	42°51.1	18.2	62°35.5	51.7	0°32.4	42.0	61°00.0	40.8
15	62°21.6	57°50.7	· · 19.5	77°36.1	· · 51.0	15°34.3	· · 42.1	76°02.2	· · 40.7
16	77°24.1	72°50.4	20.7	92°36.8	50.2	30°36.2	42.3	91°04.4	40.6
17	92°26.6	87°50.0	21.9	107°37.4	49.5	45°38.1	42.4	106°06.6	40.5
18	107°29.0	102°49.6	N00°23.1	122°38.1	S07°48.8	60°40.0	N16°42.6	121°08.8	S07°40.4
19	122°31.5	117°49.3	24.4	137°38.7	48.0	75°41.9	42.8	136°11.1	40.3
20	137°34.0	132°48.9	25.6	152°39.4	47.3	90°43.8	42.9	151°13.3	40.2
21	152°36.4	147°48.5	· · 26.8	167°40.0	· · 46.5	105°45.7	· · 43.1	166°15.5	· · 40.1
22	167°38.9	162°48.1	28.0	182°40.7	45.8	120°47.7	43.2	181°17.7	40.0
23	182°41.4	177°47.8	29.3	197°41.3	45.0	135°49.6	43.4	196°19.9	39.9
Mer.pass. 10:51		ν -0.4' d 1.2' m -3.89		ν 0.6' d -0.7' m 1.17		ν 1.9' d 0.2' m -2.05		ν 2.2' d -0.1' m 1.06	

Apr 06 Sat	SHA	Mer.pass
Venus	358° 27.7	11:07
Mars	17° 09.0	09:52
Jupiter	313° 47.3	14:04
Saturn	13° 56.6	10:04

Apr 07 Sun	SHA	Mer.pass
Venus	357° 19.6	11:08
Mars	16° 25.3	09:51
Jupiter	313° 34.2	14:01
Saturn	13° 50.5	10:00

Apr 08 Mon	SHA	Mer.pass
Venus	356° 11.6	11:09
Mars	15° 41.7	09:50
Jupiter	313° 20.9	13:58
Saturn	13° 44.4	09:57

Horizontal parallax		
Venus:		0.1
Mars:		0.1

h Sun Moon						
Sat	GHA	Dec	GHA	ν	Dec	d HP
0	179°24.1	N06°33.4	213°59.2	9.4'	S11°37.4	-16.5' 60.6'
1	194°24.2	34.3	228°27.6	9.5'	11°20.9	-16.6' 60.6'
2	209°24.4	35.2	242°56.1	9.5'	11°04.3	-16.7' 60.6'
3	224°24.6	· · 36.2	257°24.6	9.6'	10°47.6	-16.7' 60.6'
4	239°24.8	37.1	271°53.2	9.6'	10°30.9	-16.8' 60.7'
5	254°24.9	38.1	286°21.8	9.6'	10°14.1	-16.9' 60.7'
6	269°25.1	N06°39.0	300°50.4	9.7'	S09°57.2	-16.9' 60.7'
7	284°25.3	40.0	315°19.1	9.7'	09°40.3	-17.0' 60.7'
8	299°25.5	40.9	329°47.8	9.7'	09°23.2	-17.1' 60.7'
9	314°25.6	· · 41.8	344°16.5	9.8'	09°06.2	-17.1' 60.8'
10	329°25.8	42.8	358°45.3	9.8'	08°49.0	-17.2' 60.8'
11	344°26.0	43.7	13°14.1	9.8'	08°31.8	-17.3' 60.8'
12	359°26.2	N06°44.7	27°43.0	9.9'	S08°14.6	-17.3' 60.8'
13	14°26.3	45.6	42°11.8	9.9'	07°57.3	-17.4' 60.8'
14	29°26.5	46.6	56°40.7	9.9'	07°39.9	-17.4' 60.8'
15	44°26.7	· · 47.5	71°09.7	10.0'	07°22.5	-17.5' 60.8'
16	59°26.9	48.4	85°38.6	10.0'	07°05.0	-17.5' 60.9'
17	74°27.0	49.4	100°07.6	10.0'	06°47.5	-17.6' 60.9'
18	89°27.2	N06°50.3	114°36.6	10.0'	S06°30.0	-17.6' 60.9'
19	104°27.4	51.3	129°05.6	10.0'	06°12.3	-17.6' 60.9'
20	119°27.6	52.2	143°34.7	10.1'	05°54.7	-17.7' 60.9'
21	134°27.7	· · 53.1	158°03.7	10.1'	05°37.0	-17.7' 60.9'
22	149°27.9	54.1	172°32.8	10.1'	05°19.3	-17.8' 60.9'
23	164°28.1	55.0	187°01.9	10.1'	05°01.5	-17.8' 60.9'
SD = 16.0' $d = 0.9'$			SD = 16.5'			
Sun	GHA	Dec	GHA	ν	Dec	d HP
0	179°28.3	N06°56.0	201°31.0	10.1'	S04°43.7	-17.8' 60.9'
1	194°28.4	56.9	216°00.2	10.2'	04°25.9	-17.9' 61.0'
2	209°28.6	57.8	230°29.3	10.2'	04°08.0	-17.9' 61.0'
3	224°28.8	· · 58.8	244°58.5	10.2'	03°50.1	-17.9' 61.0'
4	239°29.0	06°59.7	259°27.7	10.2'	03°32.2	-17.9' 61.0'
5	254°29.1	07°00.7	273°56.9	10.2'	03°14.2	-18.0' 61.0'
6	269°29.3	N07°01.6	288°26.1	10.2'	S02°56.3	-18.0' 61.0'
7	284°29.5	02.5	302°55.3	10.2'	02°38.3	-18.0' 61.0'
8	299°29.7	03.5	317°24.5	10.2'	02°20.3	-18.0' 61.0'
9	314°29.8	· · 04.4	331°53.8	10.2'	02°02.3	-18.0' 61.0'
10	329°30.0	05.3	346°23.0	10.2'	01°44.2	-18.1' 61.0'
11	344°30.2	06.3	0°52.2	10.2'	01°26.2	-18.1' 61.0'
12	359°30.3	N07°07.2	15°21.5	10.2'	S01°08.1	-18.1' 61.0'
13	14°30.5	08.2	29°50.7	10.2'	00°50.0	-18.1' 61.0'
14	29°30.7	09.1	44°20.0	10.2'	00°31.9	-18.1' 61.0'
15	44°30.9	· · 10.0	58°49.2	10.2'	S00°13.8	-18.1' 61.0'
16	59°31.0	11.0	73°18.5	10.2'	N00°04.2	18.1' 61.0'
17	74°31.2	11.9	87°47.7	10.2'	00°22.3	18.1' 61.0'
18	89°31.4	N07°12.8	102°16.9	10.2'	N00°40.4	18.1' 61.0'
19	104°31.6	13.8	116°46.2	10.2'	00°58.5	18.1' 61.0'
20	119°31.7	14.7	131°15.4	10.2'	01°16.6	18.1' 61.0'
21	134°31.9	· · 15.6	145°44.6	10.2'	01°34.7	18.1' 61.0'
22	149°32.1	16.6	160°13.8	10.2'	01°52.7	18.1' 61.0'
23	164°32.2	17.5	174°43.0	10.2'	02°10.8	18.0' 61.0'
SD = 16.0' $d = 0.9'$			SD = 16.6'			
Mon	GHA	Dec	GHA	ν	Dec	d HP
0	179°32.4	N07°18.4	189°12.2	10.2'	N02°28.8	18.0' 61.0'
1	194°32.6	19.4	203°41.4	10.2'	02°46.8	18.0' 61.0'
2	209°32.8	20.3	218°10.6	10.1'	03°04.8	18.0' 61.0'
3	224°32.9	· · 21.2	232°39.7	10.1'	03°22.8	18.0' 61.0'
4	239°33.1	22.2	247°08.8	10.1'	03°40.8	17.9' 61.0'
5	254°33.3	23.1	261°38.0	10.1'	03°58.7	17.9' 61.0'
6	269°33.4	N07°24.0	276°07.1	10.1'	N04°16.6	17.9' 61.0'
7	284°33.6	25.0	290°36.1	10.1'	04°34.5	17.9' 61.0'
8	299°33.8	25.9	305°05.2	10.0'	04°52.4	17.8' 61.0'
9	314°33.9	· · 26.8	319°34.2	10.0'	05°10.2	17.8' 61.0'
10	329°34.1	27.8	334°03.2	10.0'	05°28.0	17.8' 61.0'
11	344°34.3	28.7	348°32.2	10.0'	05°45.8	17.7' 60.9'
12	359°34.5	N07°29.6	3°01.2	9.9'	N06°03.5	17.7' 60.9'
13	14°34.6	30.6	17°30.1	9.9'	06°21.2	17.6' 60.9'
14	29°34.8	31.5	31°59.1	9.9'	06°38.8	17.6' 60.9'
15	44°35.0	· · 32.4	46°27.9	9.9'	06°56.4	17.5' 60.9'
16	59°35.1	33.4	60°56.8	9.8'	07°13.9	17.5' 60.9'
17	74°35.3	34.3	75°25.6	9.8'	07°31.4	17.4' 60.9'
18	89°35.5	N07°35.2	89°54.4	9.8'	N07°48.9	17.4' 60.9'
19	104°35.6	36.2	104°23.2	9.7'	08°06.3	17.3' 60.9'
20	119°35.8	37.1	118°51.9	9.7'	08°23.6	17.3' 60.8'
21	134°36.0	· · 38.0	133°20.6	9.7'	08°40.9	17.2' 60.8'
22	149°36.2	38.9	147°49.2	9.6'	08°58.2	17.2' 60.8'
23	164°36.3	39.9	162°17.9	9.6'	09°15.3	17.1' 60.8'
SD = 16.0' $d = 0.9'$			SD = 16.6'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	02:59	04:21	19:46	21:10	—:—
N 70°	01:17	03:22	04:33	19:34	20:46	23:00
68°	02:04	03:39	04:42	19:25	20:28	22:06
66°	02:34	03:53	04:49	19:17	20:14	21:35
64°	02:55	04:04	04:56	19:10	20:02	21:13
62°	03:12	04:14	05:01	19:04	19:52	20:55
60°	03:26	04:22	05:06	19:00	19:44	20:41
N 58°	03:37	04:29	05:10	18:55	19:37	20:29
56°	03:47	04:35	05:14	18:51	19:30	20:19
54°	03:55	04:41	05:17	18:48	19:25	20:10
52°	04:03	04:45	05:20	18:45	19:20	20:03
50°	04:09	04:50	05:23	18:42	19:15	19:56
45°	04:23	04:59	05:29	18:36	19:06	19:42
N 40°	04:34	05:06	05:34	18:31	18:58	19:31
35°	04:42	05:12	05:38	18:26	18:52	19:23
30°	04:49	05:18	05:42	18:23	18:47	19:15
20°	05:00	05:26	05:48	18:16	18:38	19:04
N 10°	05:08	05:32	05:54	18:10	18:32	18:56
0°	05:14	05:38	05:59	18:05	18:26	18:50
S 10°	05:18	05:43	06:04	18:00	18:21	18:46
20°	05:21	05:47	06:09	17:55	18:17	18:43
30°	05:23	05:50	06:15	17:49	18:13	18:41
40°	05:23	05:52	06:18	17:46	18:11	18:40
45°	05:23	05:54	06:21	17:42	18:09	18:40
50°	05:22	05:56	06:26	17:38	18:07	18:41
S 50°	05:21	05:58	06:31	17:33	18:05	18:42
52°	05:20	05:59	06:33	17:30	18:04	18:43
54°	05:19	06:00	06:35	17:28	18:03	18:44
56°	05:18	06:01	06:38	17:25	18:02	18:45
58°	05:16	06:02	06:41	17:22	18:01	18:46
S 60°	05:15	06:03	06:45	17:18	18:00	18:48

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	06:10	05:26	04:47	14:37	17:12	19:49
N 70°	05:54	05:22	04:52	14:49	17:12	19:36
68°	05:42	05:18	04:56	14:59	17:12	19:25
66°	05:32	05:15	04:59	15:07	17:11	19:17
64°	05:23	05:12	05:02	15:13	17:11	19:09
62°	05:15	05:10	05:05	15:19	17:11	19:03
60°	05:09	05:08	05:07	15:24	17:11	18:58
N 58°	05:03	05:06	05:09	15:28	17:11	18:53
56°	04:58	05:05	05:11	15:32	17:10	18:49
54°	04:53	05:03	05:13	15:35	17:10	18:46
52°	04:49	05:02	05:14	15:38	17:10	18:42
50°	04:45	05:01	05:16	15:41	17:10	18:39
45°	04:37	04:58	05:19	15:47	17:10	18:33
N 40°	04:30	04:56	05:21	15:52	17:10	18:27
35°	04:24	04:54	05:24	15:56	17:09	18:23
30°	04:18	04:52	05:26	16:00	17:09	18:19
20°	04:09	04:49	05:29	16:07	17:09	18:12
N 10°	04:01	04:47	05:32	16:12	17:09	18:05
0°	03:53	04:44	05:35	16:17	17:08	18:00
S 10°	03:45	04:42	05:38	16:22	17:08	17:54
20°	03:37	04:39	05:42	16:28	17:08	17:48
30°	03:27	04:36	05:46	16:34	17:07	17:41
35°	03:22	04:35	05:48	16:37	17:07	17:37
40°	03:15	04:33	05:50	16:41	17:07	17:33
45°	03:08	04:31	05:53	16:46	17:06	17:27
S 50°	02:59	04:28	05:57	16:51	17:06	17:21
52°	02:55	04:27	05:59	16:53	17:06	17:18
54°	02:50	04:25	06:00	16:56	17:06	17:15
56°	02:45	04:24	06:02	16:59	17:05	17:12
58°	02:39	04:22	06:05	17:02	17:05	17:08
S 60°	02:33	04:20	06:07	17:06	17:05	17:04

Day	Sun		Mer. Pass hh:mm	Moon		Age 27-0 11-1%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
06	02:24	02:15	12:02	10:05	22:31	
07	02:07	01:59	12:02	10:56	23:22	
08	01:50	01:42	12:02	11:47	—:—	

April 09, 10, 11 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	197°43.8	192°47.4	N00°30.5	212°42.0	S07°44.3	150°51.5	N16°43.5	211°22.1	S07°39.8	Alpheratz	357°35.8 29°13.3
1	212°46.3	207°47.0	31.7	227°42.6	43.6	165°53.4	43.7	226°24.3	39.7	Ankaa	353°08.2 -42°10.5
2	227°48.7	222°46.7	32.9	242°43.3	42.8	180°55.3	43.8	241°26.5	39.6	Schedar	349°32.4 56°40.1
3	242°51.2	237°46.3	· · 34.2	257°43.9	· · 42.1	195°57.2	· · 44.0	256°28.8	· · 39.5	Diphda	348°48.3 -17°51.3
4	257°53.7	252°45.9	35.4	272°44.6	41.3	210°59.1	44.2	271°31.0	39.4	Achernar	335°21.2 -57°06.9
5	272°56.1	267°45.6	36.6	287°45.2	40.6	226°01.0	44.3	286°33.2	39.3	Hamal	327°52.3 23°34.5
6	287°58.6	282°45.2	N00°37.9	302°45.9	S07°39.9	241°02.9	N16°44.5	301°35.4	S07°39.2	Polaris	314°48.1 89°22.1
7	303°01.1	297°44.8	39.1	317°46.5	39.1	256°04.8	44.6	316°37.6	39.1	Acamar	315°12.6 -40°12.6
8	318°03.5	312°44.4	40.3	332°47.2	38.4	271°06.7	44.8	331°39.8	39.0	Menkar	314°07.1 4°11.0
9	333°06.0	327°44.1	· · 41.5	347°47.8	· · 37.6	286°08.7	· · 44.9	346°42.0	· · 38.9	Mirfak	308°29.6 49°56.9
10	348°08.5	342°43.7	42.8	2°48.5	36.9	301°10.6	45.1	1°44.2	38.8	Aldebaran	290°40.6 16°33.4
11	3°10.9	357°43.3	44.0	17°49.1	36.1	316°12.5	45.2	16°46.5	38.8	Rigel	281°04.7 -8°10.5
12	18°13.4	12°43.0	N00°45.2	32°49.8	S07°35.4	331°14.4	N16°45.4	31°48.7	S07°38.7	Capella	280°23.1 46°01.4
13	33°15.9	27°42.6	46.4	47°50.4	34.7	346°16.3	45.6	46°50.9	38.6	Bellatrix	278°23.7 6°22.2
14	48°18.3	42°42.2	47.7	62°51.1	33.9	1°18.2	45.7	61°53.1	38.5	Elnath	278°02.9 28°37.7
15	63°20.8	57°41.9	· · 48.9	77°51.8	· · 33.2	16°20.1	· · 45.9	76°55.3	· · 38.4	Alnilam	275°38.5 -1°11.3
16	78°23.2	72°41.5	50.1	92°52.4	32.4	31°22.0	46.0	91°57.5	38.3	Betelgeuse	270°52.9 7°24.6
17	93°25.7	87°41.1	51.3	107°53.1	31.7	46°23.9	46.2	106°59.7	38.2	Canopus	263°52.8 -52°42.7
18	108°28.2	102°40.8	N00°52.6	122°53.7	S07°30.9	61°25.8	N16°46.3	122°01.9	S07°38.1	Sirius	258°26.9 -16°45.1
19	123°30.6	117°40.4	53.8	137°54.4	30.2	76°27.7	46.5	137°04.2	38.0	Adhara	255°06.4 -29°00.5
20	138°33.1	132°40.0	55.0	152°55.0	29.5	91°29.6	46.7	152°06.4	37.9	Procyon	244°51.5 5°09.7
21	153°35.6	147°39.6	· · 56.2	167°55.7	· · 28.7	106°31.5	· · 46.8	167°08.6	· · 37.8	Pollux	243°18.1 27°58.1
22	168°38.0	162°39.3	57.5	182°56.3	28.0	121°33.5	47.0	182°10.8	37.7	Avior	234°14.9 -59°35.5
23	183°40.5	177°38.9	58.7	197°57.0	27.2	136°35.4	47.1	197°13.0	37.6	Suhail	222°46.6 -43°32.0
Mer.pass. 10:47		ν -0.4' d 1.2' m -3.89		ν 0.7' d -0.7' m 1.17		ν 1.9' d 0.2' m -2.05		ν 2.2' d -0.1' m 1.07		Miaplacidus	221°38.1 -69°49.2
										Alphard	217°48.2 -8°45.9
										Regulus	207°34.9 11°50.9
										Dubhe	193°41.2 61°37.3
										Denebola	182°25.3 14°26.1
										Gienah	175°43.9 -17°40.7
										Acrux	173°00.2 -63°14.1
										Gacrux	171°51.8 -57°15.0
										Alioth	166°12.9 55°49.7
										Spica	158°22.7 -11°17.4
										Alkaid	152°52.0 49°11.4
										Hadar	148°36.3 -60°29.4
										Menkent	147°58.0 -36°29.4
										Arcturus	145°48.2 19°03.2
										Rigil Kent.	139°40.6 -60°56.1
										Kochab	137°18.6 74°03.2
										Zuben'ubi	136°56.4 -16°08.6
										Alphecca	126°04.0 26°37.8
										Antares	112°16.4 -26°29.2
										Atria	107°10.9 -69°04.1
										Sabik	102°03.3 -15°45.4
										Shaula	96°11.1 -37°07.2
										Rasalhague	95°59.0 12°32.3
										Eltanin	90°42.3 51°28.8
										Kaus Aust.	83°33.3 -34°22.3
										Vega	80°33.6 38°48.0
										Nunki	75°48.5 -26°16.0
										Altair	62°00.6 8°55.7
										Peacock	53°06.9 -56°39.3
										Deneb	49°26.4 45°21.6
										Enif	33°39.6 9°59.0
										Al Na'ir	27°34.0 -46°50.6
										Fomalhaut	15°15.5 -29°29.7
										Scheat	13°46.1 28°12.6
										Markab	13°30.8 15°19.9
										Apr 09 Tue SHA Mer.pass	
										Venus	355°03.6 11:09
										Mars	14°58.1 09:49
										Jupiter	313°07.7 13:55
										Saturn	13°38.3 09:53
										Apr 10 Wed SHA Mer.pass	
										Venus	353°55.6 11:10
										Mars	14°14.7 09:48
										Jupiter	312°54.3 13:52
										Saturn	13°32.3 09:50
										Apr 11 Thu SHA Mer.pass	
										Venus	352°47.6 11:10
										Mars	13°31.3 09:47
										Jupiter	312°40.9 13:49
										Saturn	13°26.3 09:46
										Horizontal parallax Venus: 0.1 Mars: 0.1	
Mer.pass. 10:39		ν -0.4' d 1.2' m -3.89		ν 0.7' d -0.7' m 1.17		ν 1.9' d 0.2' m -2.04		ν 2.2' d -0.1' m 1.07			

Sun				Moon				
Tue	GHA	Dec		GHA	ν	Dec	d	HP
0	179°36.5	N07°40.8		176°46.4	9.5'	N09°32.4	17.0'	60.8'
1	194°36.7	41.7		191°15.0	9.5'	09°49.5	17.0'	60.8'
2	209°36.8	42.7		205°43.5	9.5'	10°06.4	16.9'	60.7'
3	224°37.0	· · 43.6		220°11.9	9.4'	10°23.4	16.8'	60.7'
4	239°37.2	44.5		234°40.4	9.4'	10°40.2	16.8'	60.7'
5	254°37.3	45.4		249°08.7	9.3'	10°57.0	16.7'	60.7'
6	269°37.5	N07°46.4		263°37.1	9.3'	N11°13.7	16.6'	60.7'
7	284°37.7	47.3		278°05.4	9.2'	11°30.3	16.5'	60.7'
8	299°37.8	48.2		292°33.6	9.2'	11°46.8	16.5'	60.6'
9	314°38.0	· · 49.2		307°01.8	9.2'	12°03.3	16.4'	60.6'
10	329°38.2	50.1		321°30.0	9.1'	12°19.7	16.3'	60.6'
11	344°38.3	51.0		335°58.1	9.1'	12°36.0	16.2'	60.6'
12	359°38.5	N07°51.9		350°26.2	9.0'	N12°52.2	16.1'	60.6'
13	14°38.7	52.9		4°54.2	9.0'	13°08.3	16.0'	60.5'
14	29°38.8	53.8		19°22.1	8.9'	13°24.3	15.9'	60.5'
15	44°39.0	· · 54.7		33°50.1	8.9'	13°40.3	15.9'	60.5'
16	59°39.2	55.6		48°17.9	8.8'	13°56.1	15.8'	60.5'
17	74°39.3	56.6		62°45.7	8.8'	14°11.9	15.7'	60.4'
18	89°39.5	N07°57.5		77°13.5	8.7'	N14°27.6	15.6'	60.4'
19	104°39.7	58.4		91°41.2	8.7'	14°43.1	15.5'	60.4'
20	119°39.8	07°59.3		106°08.9	8.6'	14°58.6	15.4'	60.4'
21	134°40.0	08°00.3		120°36.5	8.6'	15°14.0	15.3'	60.3'
22	149°40.2	01.2		135°04.0	8.5'	15°29.2	15.2'	60.3'
23	164°40.3	02.1		149°31.5	8.4'	15°44.4	15.0'	60.3'
SD = 16.0'			$d = 0.9'$	SD = 16.6'				
Wed	GHA	Dec		GHA	ν	Dec	d	HP
0	179°40.5	N08°03.0		163°59.0	8.4'	N15°59.4	14.9'	60.3'
1	194°40.7	04.0		178°26.4	8.3'	16°14.4	14.8'	60.2'
2	209°40.8	04.9		192°53.7	8.3'	16°29.2	14.7'	60.2'
3	224°41.0	· · 05.8		207°21.0	8.2'	16°43.9	14.6'	60.2'
4	239°41.2	06.7		221°48.2	8.2'	16°58.5	14.5'	60.1'
5	254°41.3	07.6		236°15.4	8.1'	17°13.0	14.4'	60.1'
6	269°41.5	N08°08.6		250°42.5	8.1'	N17°27.4	14.3'	60.1'
7	284°41.7	09.5		265°09.6	8.0'	17°41.6	14.1'	60.1'
8	299°41.8	10.4		279°36.6	7.9'	17°55.8	14.0'	60.0'
9	314°42.0	· · 11.3		294°03.5	7.9'	18°09.8	13.9'	60.0'
10	329°42.2	12.3		308°30.4	7.8'	18°23.7	13.8'	60.0'
11	344°42.3	13.2		322°57.2	7.8'	18°37.4	13.6'	59.9'
12	359°42.5	N08°14.1		337°24.0	7.7'	N18°51.0	13.5'	59.9'
13	14°42.7	15.0		351°50.7	7.7'	19°04.5	13.4'	59.9'
14	29°42.8	15.9		6°17.4	7.6'	19°17.9	13.2'	59.8'
15	44°43.0	· · 16.9		20°44.0	7.5'	19°31.2	13.1'	59.8'
16	59°43.1	17.8		35°10.5	7.5'	19°44.3	13.0'	59.8'
17	74°43.3	18.7		49°37.0	7.4'	19°57.2	12.8'	59.7'
18	89°43.5	N08°19.6		64°03.4	7.4'	N20°10.1	12.7'	59.7'
19	104°43.6	20.5		78°29.8	7.3'	20°22.8	12.6'	59.7'
20	119°43.8	21.4		92°56.1	7.3'	20°35.3	12.4'	59.6'
21	134°44.0	· · 22.4		107°22.4	7.2'	20°47.8	12.3'	59.6'
22	149°44.1	23.3		121°48.6	7.2'	21°00.1	12.1'	59.6'
23	164°44.3	24.2		136°14.8	7.1'	21°12.2	12.0'	59.5'
SD = 16.0'			$d = 0.9'$	SD = 16.4'				
Thu	GHA	Dec		GHA	ν	Dec	d	HP
0	179°44.5	N08°25.1		150°40.9	7.0'	N21°24.2	11.8'	59.5'
1	194°44.6	26.0		165°06.9	7.0'	21°36.0	11.7'	59.5'
2	209°44.8	27.0		179°32.9	6.9'	21°47.7	11.6'	59.4'
3	224°44.9	· · 27.9		193°58.8	6.9'	21°59.3	11.4'	59.4'
4	239°45.1	28.8		208°24.7	6.8'	22°10.7	11.3'	59.4'
5	254°45.3	29.7		222°50.6	6.8'	22°21.9	11.1'	59.3'
6	269°45.4	N08°30.6		237°16.3	6.7'	N22°33.0	10.9'	59.3'
7	284°45.6	31.5		251°42.1	6.7'	22°44.0	10.8'	59.3'
8	299°45.8	32.5		266°07.8	6.6'	22°54.8	10.6'	59.2'
9	314°45.9	· · 33.4		280°33.4	6.6'	23°05.4	10.5'	59.2'
10	329°46.1	34.3		294°59.0	6.5'	23°15.9	10.3'	59.1'
11	344°46.2	35.2		309°24.5	6.5'	23°26.2	10.2'	59.1'
12	359°46.4	N08°36.1		323°50.0	6.4'	N23°36.4	10.0'	59.1'
13	14°46.6	37.0		338°15.5	6.4'	23°46.4	9.8'	59.0'
14	29°46.7	37.9		352°40.9	6.4'	23°56.2	9.7'	59.0'
15	44°46.9	· · 38.9		7°06.2	6.3'	24°05.9	9.5'	59.0'
16	59°47.0	39.8		21°31.6	6.3'	24°15.4	9.4'	58.9'
17	74°47.2	40.7		35°56.8	6.2'	24°24.8	9.2'	58.9'
18	89°47.4	N08°41.6		50°22.1	6.2'	N24°34.0	9.0'	58.8'
19	104°47.5	42.5		64°47.3	6.2'	24°43.0	8.9'	58.8'
20	119°47.7	43.4		79°12.4	6.1'	24°51.9	8.7'	58.8'
21	134°47.9	· · 44.3		93°37.5	6.1'	25°00.6	8.5'	58.7'
22	149°48.0	45.2		108°02.6	6.1'	25°09.1	8.4'	58.7'
23	164°48.2	46.2		122°27.7	6.0'	25°17.5	8.2'	58.7'
SD = 16.0'			$d = 0.9'$	SD = 16.2'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	02:37	04:05	20:01	21:32	—:—
N 70°	—:—	03:04	04:18	19:47	21:03	—:—
68°	01:38	03:24	04:29	19:36	20:42	22:32
66°	02:15	03:40	04:38	19:27	20:26	21:53
64°	02:40	03:52	04:45	19:19	20:12	21:27
62°	02:59	04:03	04:51	19:13	20:01	21:07
60°	03:14	04:12	04:57	19:07	19:52	20:51
N 58°	03:27	04:20	05:02	19:02	19:44	20:38
56°	03:38	04:27	05:06	18:57	19:37	20:27
54°	03:47	04:33	05:10	18:54	19:31	20:17
52°	03:55	04:38	05:14	18:50	19:25	20:09
50°	04:02	04:43	05:17	18:47	19:20	20:02
45°	04:17	04:53	05:24	18:40	19:10	19:47
N 40°	04:28	05:02	05:29	18:34	19:02	19:35
35°	04:38	05:08	05:34	18:29	18:55	19:25
30°	04:45	05:14	05:38	18:24	18:49	19:18
20°	04:57	05:23	05:46	18:17	18:39	19:05
N 10°	05:06	05:31	05:52	18:10	18:32	18:56
0°	05:13	05:37	05:58	18:04	18:25	18:50
S 10°	05:18	05:42	06:04	17:59	18:20	18:44
20°	05:22	05:47	06:10	17:52	18:15	18:40
30°	05:24	05:52	06:16	17:46	18:10	18:37
35°	05:25	05:55	06:20	17:42	18:07	18:37
40°	05:26	05:57	06:24	17:37	18:05	18:36
45°	05:26	06:00	06:29	17:32	18:02	18:36
S 50°	05:25	06:03	06:35	17:26	17:59	18:36
52°	05:25	06:04	06:38	17:24	17:57	18:36
54°	05:24	06:05	06:41	17:20	17:56	18:37
56°	05:24	06:07	06:44	17:17	17:55	18:38
58°	05:23	06:08	06:48	17:14	17:53	18:38
S 60°	05:22	06:10	06:52	17:09	17:51	18:39

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	04:02	02:47	☐	22:58	☐	☐
N 70°	04:18	03:30	☐	22:17	☐	☐
68°	04:32	03:59	02:38	21:49	· · ·	01:07
66°	04:42	04:22	03:46	21:29	· · ·	00:01
64°	04:52	04:39	04:22	21:12	23:26	· · ·
62°	05:00	04:54	04:48	20:59	23:00	· · ·
60°	05:07	05:07	05:09	20:48	22:41	· · ·
N 58°	05:13	05:18	05:26	20:38	22:25	· · ·
56°	05:18	05:27	05:40	20:29	22:11	23:50
54°	05:23	05:36	05:53	20:22	21:59	23:33
52°	05:27	05:43	06:04	20:15	21:49	23:19
50°	05:32	05:50	06:14	20:09	21:39	23:06
45°	05:40	06:05	06:35	19:56	21:20	22:41
N 40°	05:48	06:17	06:52	19:46	21:04	22:21
35°	05:54	06:28	07:06	19:37	20:51	22:04
30°	06:00	06:37	07:19	19:29	20:39	21:50
20°	06:10	06:53	07:40	19:15	20:20	21:25
N 10°	06:19	07:07	07:59	19:03	20:03	21:04
0°	06:27	07:21	08:17	18:52	19:47	20:44
S 10°	06:36	07:34	08:35	18:41	19:31	20:25
20°	06:45	07:49	08:54	18:30	19:15	20:04
30°	06:55	08:06	09:16	18:16	18:56	19:40
35°	07:01	08:16	09:30	18:09	18:44	19:26
40°	07:08	08:27	09:45	18:00	18:32	19:09
45°	07:17	08:40	10:03	17:50	18:17	18:50
S 50°	07:26	08:57	10:26	17:38	17:59	18:26
52°	07:31	09:05	10:37	17:33	17:50	18:14
54°	07:36	09:13	10:50	17:26	17:41	18:01
56°	07:42	09:23	11:04	17:20	17:30	17:46
58°	07:48	09:34	11:21	17:12	17:18	17:28
S 60°	07:56	09:47	11:42	17:03	17:04	17:07

Day	Sun		Mer. Pass hh:mm	Moon		Age 1-3 0-7%
	Eqn.of Time 00 ^h	12 ^h		Mer.Pass. Upper hh:mm	Lower hh:mm	
	mm:ss	mm:ss				
09	01:34	01:26	12:01	12:40	00:13	
10	01:18	01:10	12:01	13:34	01:06	
11	01:02	00:54	12:01	14:30	02:02	

April 12, 13, 14 UT (Fri., Sat., Sun.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	200°41.2	192°20.8	N01°58.7	213°29.2	S06°50.7	153°08.7	N16°54.8	214°01.6	S07°32.8	Alpheratz	357°35.8	29°13.3					
1	215°43.7	207°20.4	01°59.9	228°29.8	49.9	168°10.6	54.9	229°03.8	32.7	Ankaa	353°08.2	-42°10.5					
2	230°46.2	222°20.0	02°01.2	243°30.5	49.2	183°12.5	55.1	244°06.0	32.6	Schedar	349°32.4	56°40.1					
3	245°48.6	237°19.7	02.4	258°31.2	48.4	198°14.4	55.2	259°08.2	32.5	Diphda	348°48.2	-17°51.3					
4	260°51.1	252°19.3	03.6	273°31.8	47.7	213°16.3	55.4	274°10.4	32.4	Achernar	335°21.2	-57°06.9					
5	275°53.6	267°18.9	04.8	288°32.5	46.9	228°18.2	55.5	289°12.7	32.3	Hamal	327°52.3	23°34.5					
6	290°56.0	282°18.6	N02°06.1	303°33.1	S06°46.2	243°20.1	N16°55.7	304°14.9	S07°32.3	Polaris	314°48.3	89°22.1					
7	305°58.5	297°18.2	07.3	318°33.8	45.4	258°22.0	55.8	319°17.1	32.2	Acamar	315°12.6	-40°12.6					
8	321°00.9	312°17.8	08.5	333°34.5	44.7	273°23.9	56.0	334°19.3	32.1	Menkar	314°07.1	4°11.0					
9	336°03.4	327°17.5	09.7	348°35.1	43.9	288°25.8	56.2	349°21.5	32.0	Mirfak	308°29.6	49°56.9					
10	351°05.9	342°17.1	11.0	358.8	43.2	303°27.7	56.3	4°23.7	31.9	Aldebaran	290°40.6	16°33.4					
11	6°08.3	357°16.7	12.2	18°36.4	42.4	318°29.6	56.5	19°26.0	31.8	Rigel	281°04.7	-8°10.5					
12	21°10.8	12°16.3	N02°13.4	33°37.1	S06°41.7	333°31.5	N16°56.6	34°28.2	S07°31.7	Capella	280°23.1	46°01.4					
13	36°13.3	27°16.0	14.6	48°37.8	40.9	348°33.4	56.8	49°30.4	31.6	Bellatrix	278°23.7	6°22.2					
14	51°15.7	42°15.6	15.8	63°38.4	40.2	3°35.3	56.9	64°32.6	31.5	Elnath	278°02.9	28°37.7					
15	66°18.2	57°15.2	17.1	78°39.1	39.4	18°37.2	57.1	79°34.8	31.4	Alnilam	275°38.5	-1°11.3					
16	81°20.7	72°14.9	18.3	93°39.8	38.7	33°39.1	57.2	94°37.1	31.3	Betelgeuse	270°52.9	7°24.6					
17	96°23.1	87°14.5	19.5	108°40.4	37.9	48°41.0	57.4	109°39.3	31.2	Canopus	263°52.9	-52°42.7					
18	111°25.6	102°14.1	N02°20.7	123°41.1	S06°37.2	63°42.9	N16°57.6	124°41.5	S07°31.1	Sirius	258°26.9	-16°45.1					
19	126°28.1	117°13.7	22.0	138°41.7	36.4	78°44.8	57.7	139°43.7	31.0	Adhara	255°06.4	-29°00.5					
20	141°30.5	132°13.4	23.2	153°42.4	35.7	93°46.7	57.9	154°45.9	30.9	Procyon	244°51.5	5°09.7					
21	156°33.0	147°13.0	24.4	168°43.1	34.9	108°48.6	58.0	169°48.1	30.8	Pollux	243°18.1	27°58.1					
22	171°35.4	162°12.6	25.6	183°43.7	34.2	123°50.5	58.2	184°50.4	30.7	Avior	234°14.9	-59°35.5					
23	186°37.9	177°12.3	26.9	198°44.4	33.4	138°52.4	58.3	199°52.6	30.6	Suhail	222°46.6	-43°32.1					
Mer.pass. 10:36			ν -0.4' d 1.2' m -3.89			ν 0.7' d -0.7' m 1.16			ν 1.9' d 0.2' m -2.04			ν 2.2' d -0.1' m 1.07					

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	201°40.4	192°11.9	N02°28.1	213°45.0	S06°32.7	153°54.3	N16°58.5	214°54.8	S07°30.5					
1	216°42.8	207°11.5	29.3	228°45.7	31.9	168°56.2	58.6	229°57.0	30.4					
2	231°45.3	222°11.1	30.5	243°46.4	31.2	183°58.1	58.8	244°59.2	30.3					
3	246°47.8	237°10.8	31.7	258°47.0	30.4	199°00.0	59.0	260°01.5	30.2					
4	261°50.2	252°10.4	33.0	273°47.7	29.6	214°01.9	59.1	275°03.7	30.2					
5	276°52.7	267°10.0	34.2	288°48.4	28.9	229°03.8	59.3	290°05.9	30.1					
6	291°55.2	282°09.7	N02°35.4	303°49.0	S06°28.1	244°05.7	N16°59.4	305°08.1	S07°30.0					
7	306°57.6	297°09.3	36.6	318°49.7	27.4	259°07.6	59.6	320°10.3	29.9					
8	322°00.1	312°08.9	37.9	333°50.4	26.6	274°09.5	59.7	335°12.6	29.8					
9	337°02.6	327°08.5	39.1	348°51.0	25.9	289°11.4	60.0	350°14.8	29.7					
10	352°05.0	342°08.2	40.3	358.7	25.1	304°13.3	60.0	5°17.0	29.6					
11	7°07.5	357°07.8	41.5	18°52.3	24.4	319°15.2	60.2	20°19.2	29.5					
12	22°09.9	12°07.4	N02°42.7	33°53.0	S06°23.6	334°17.1	N17°00.3	35°21.4	S07°29.4					
13	37°12.4	27°07.1	44.0	48°53.7	22.9	349°19.0	60.5	50°23.7	29.3					
14	52°14.9	42°06.7	45.2	63°54.3	22.1	4°20.9	60.7	65°25.9	29.2					
15	67°17.3	57°06.3	46.4	78°55.0	21.4	19°22.8	60.8	80°28.1	29.1					
16	82°19.8	72°05.9	47.6	93°55.7	20.6	34°24.7	61.0	95°30.3	29.0					
17	97°22.3	87°05.6	48.9	108°56.3	19.9	49°26.6	61.1	110°32.5	28.9					
18	112°24.7	102°05.2	N02°50.1	123°57.0	S06°19.1	64°28.5	N17°01.3	125°34.8	S07°28.8					
19	127°27.2	117°04.8	51.3	138°57.7	18.4	79°30.4	61.4	140°37.0	28.7					
20	142°29.7	132°04.4	52.5	153°58.3	17.6	94°32.3	61.6	155°39.2	28.6					
21	157°32.1	147°04.1	53.7	168°59.0	16.9	109°34.2	61.7	170°41.4	28.5					
22	172°34.6	162°03.7	55.0	183°59.7	16.1	124°36.1	61.9	185°43.6	28.4					
23	187°37.1	177°03.3	56.2	199°00.3	15.4	139°38.0	62.1	200°45.9	28.4					
Mer.pass. 10:32			ν -0.4' d 1.2' m -3.89			ν 0.7' d -0.8' m 1.16			ν 1.9' d 0.2' m -2.04			ν 2.2' d -0.1' m 1.07		

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	202°39.5	192°03.0	N02°57.4	214°01.0	S06°14.6	154°39.9	N17°02.2	215°48.1	S07°28.3					
1	217°42.0	207°02.6	58.6	229°01.6	13.9	169°41.8	62.4	230°50.3	28.2					
2	232°44.4	222°02.2	02°59.8	244°02.3	13.1	184°43.7	62.5	245°52.5	28.1					
3	247°46.9	237°01.8	03°01.1	259°03.0	12.3	199°45.6	62.7	260°54.7	28.0					
4	262°49.4	252°01.5	02.3	274°03.6	11.6	214°47.5	62.8	275°57.0	27.9					
5	277°51.8	267°01.1	03.5	289°04.3	10.8	229°49.4	63.0	290°59.2	27.8					
6	292°54.3	282°00.7	N03°04.7	304°05.0	S06°10.1	244°51.3	N17°03.1	306°01.4	S07°27.7					
7	307°56.8	297°00.3	05.9	319°05.6	09.3	259°53.2	63.3	321°03.6	27.6					
8	322°59.2	312°00.0	07.2	334°06.3	08.6	274°55.1	63.5	336°05.9	27.5					
9	338°01.7	326°59.6	08.4	349°07.0	07.8	289°57.0	63.6	351°08.1	27.4					
10	353°04.2	341°59.2	09.6	4°07.6	07.1	304°58.9	63.8	6°10.3	27.3					
11	8°06.6	356°58.8	10.8	19°08.3	06.3	320°00.8	63.9	21°12.5	27.2					
12	23°09.1	11°58.5	N03°12.0	34°09.0	S06°05.6	335°02.7	N17°04.1	36°14.7	S07°27.1					
13	38°11.6	26°58.1	13.3	49°09.6	04.8	350°04.6	64.2	51°17.0	27.0					
14	53°14.0	41°57.7	14.5	64°10.3	04.1	5°06.5	64.4	66°19.2	26.9					
15	68°16.5	56°57.4	15.7	79°11.0	03.3	20°08.4	64.5	81°21.4	26.8					
16	83°18.9	71°57.0	16.9	94°11.6	02.6	35°10.3	64.7	96°23.6	26.7					
17	98°21.4	86°56.6	18.1	109°12.3	01.8	50°12.1	64.8	111°25.9	26.7					
18	113°23.9	101°56.2	N03°19.4	124°13.0	S06°01.0	65°14.0	N17°05.0	126°28.1	S07°26.6					
19	128°26.3	116°55.9	20.6	139°13.6	06°00.3	80°15.9	65.2	141°30.3	26.5					
20	143°28.8	131°55.5	21.8	154°14.3	05°59.5	95°17.8	65.3	156°32.5	26.4					
21	158°31.3	146°55.1	23.0	169°15.0	58.8	110°19.7	65.5	171°34.7	26.3					
22	173°33.7	161°54.7	24.2	184°15.6	58.0	125°21.6	65.6	186°37.0	26.2					
23	188°36.2	176°54.4	25.5	199°16.3	57.3	140°23.5	65.8	201°39.2	26.1					
Mer.pass. 10:28			ν -0.4' d 1.2' m -3.89			ν 0.7' d -0.8' m 1.16			ν 1.9' d 0.2' m -2.04			ν 2.2' d -0.1' m 1.07		

Apr 12 Fri	SHA	Mer.pass
Venus	351°39.6	11:11
Mars	12°47.9	09:46
Jupiter	312°27.5	13:46
Saturn	13°20.3	09:42

Apr 13 Sat	SHA	Mer.pass
Venus	350°31.5	11:11
Mars	12°04.7	09:45
Jupiter	312°13.9	13:43
Saturn	13°14.4	09:39

Apr 14 Sun	SHA	Mer.pass
Venus	349°23.4	11:12
Mars	11°21.5	09:44
Jupiter	312°00.4	13:40
Saturn	13°08.6	09:35

Horizontal parallax		
Venus:	0.1	
Mars:	0.1	

h Sun				Moon			
Fri	GHA	Dec		GHA	ν	Dec	HP
0	179°48.3	N08°47.1		136°52.7	6.0'	N25°25.7	8.0' 58.6'
1	194°48.5	48.0		151°17.7	6.0'	25°33.7	7.9' 58.6'
2	209°48.7	48.9		165°42.7	5.9'	25°41.5	7.7' 58.5'
3	224°48.8	· · 49.8		180°07.6	5.9'	25°49.2	7.5' 58.5'
4	239°49.0	50.7		194°32.5	5.9'	25°56.7	7.3' 58.5'
5	254°49.1	51.6		208°57.4	5.9'	26°04.1	7.2' 58.4'
6	269°49.3	N08°52.5		223°22.2	5.8'	N26°11.2	7.0' 58.4'
7	284°49.4	53.4		237°47.1	5.8'	26°18.2	6.8' 58.3'
8	299°49.6	54.4		252°11.9	5.8'	26°25.0	6.6' 58.3'
9	314°49.8	· · 55.3		266°36.7	5.8'	26°31.7	6.5' 58.3'
10	329°49.9	56.2		281°01.4	5.8'	26°38.1	6.3' 58.2'
11	344°50.1	57.1		295°26.2	5.7'	26°44.4	6.1' 58.2'
12	359°50.2	N08°58.0		309°50.9	5.7'	N26°50.5	5.9' 58.1'
13	14°50.4	58.9		324°15.7	5.7'	26°56.5	5.8' 58.1'
14	29°50.6	08°59.8		338°40.4	5.7'	27°02.3	5.6' 58.1'
15	44°50.7	09°00.7		353°05.1	5.7'	27°07.8	5.4' 58.0'
16	59°50.9	01.6		7°29.8	5.7'	27°13.3	5.2' 58.0'
17	74°51.0	02.5		21°54.5	5.7'	27°18.5	5.1' 57.9'
18	89°51.2	N09°03.4		36°19.2	5.7'	N27°23.6	4.9' 57.9'
19	104°51.3	04.3		50°43.9	5.7'	27°28.4	4.7' 57.9'
20	119°51.5	05.3		65°08.5	5.7'	27°33.1	4.5' 57.8'
21	134°51.7	· · 06.2		79°33.2	5.7'	27°37.7	4.4' 57.8'
22	149°51.8	07.1		93°57.9	5.7'	27°42.0	4.2' 57.7'
23	164°52.0	08.0		108°22.6	5.7'	27°46.2	4.0' 57.7'
SD = 15.9' d = 0.9'				SD = 16.0'			
Sat	GHA	Dec		GHA	ν	Dec	HP
0	179°52.1	N09°08.9		122°47.3	5.7'	N27°50.2	3.8' 57.7'
1	194°52.3	09.8		137°12.0	5.7'	27°54.0	3.6' 57.6'
2	209°52.4	10.7		151°36.8	5.7'	27°57.7	3.5' 57.6'
3	224°52.6	· · 11.6		166°01.5	5.8'	28°01.1	3.3' 57.5'
4	239°52.8	12.5		180°26.2	5.8'	28°04.4	3.1' 57.5'
5	254°52.9	13.4		194°51.0	5.8'	28°07.6	2.9' 57.5'
6	269°53.1	N09°14.3		209°15.8	5.8'	N28°10.5	2.8' 57.4'
7	284°53.2	15.2		223°40.6	5.8'	28°13.3	2.6' 57.4'
8	299°53.4	16.1		238°05.4	5.9'	28°15.9	2.4' 57.4'
9	314°53.5	· · 17.0		252°30.3	5.9'	28°18.3	2.2' 57.3'
10	329°53.7	17.9		266°55.2	5.9'	28°20.5	2.1' 57.3'
11	344°53.9	18.8		281°20.1	5.9'	28°22.6	1.9' 57.2'
12	359°54.0	N09°19.7		295°45.0	6.0'	N28°24.5	1.7' 57.2'
13	14°54.2	20.6		310°10.0	6.0'	28°26.2	1.5' 57.2'
14	29°54.3	21.5		324°35.0	6.0'	28°27.7	1.4' 57.1'
15	44°54.5	· · 22.4		339°00.0	6.1'	28°29.1	1.2' 57.1'
16	59°54.6	23.3		353°25.1	6.1'	28°30.3	1.0' 57.0'
17	74°54.8	24.2		7°50.2	6.2'	28°31.3	0.9' 57.0'
18	89°54.9	N09°25.1		22°15.3	6.2'	N28°32.2	0.7' 57.0'
19	104°55.1	26.0		36°40.5	6.2'	28°32.8	0.5' 56.9'
20	119°55.2	26.9		51°05.8	6.3'	28°33.4	0.3' 56.9'
21	134°55.4	· · 27.8		65°31.1	6.3'	28°33.7	0.2' 56.9'
22	149°55.6	28.7		79°56.4	6.4'	28°33.9	0.0' 56.8'
23	164°55.7	29.6		94°21.8	6.4'	28°33.9	-0.2' 56.8'
SD = 15.9' d = 0.9'				SD = 15.7'			
Sun	GHA	Dec		GHA	ν	Dec	HP
0	179°55.9	N09°30.5		108°47.2	6.5'	N28°33.7	-0.3' 56.7'
1	194°56.0	31.4		123°12.7	6.5'	28°33.4	-0.5' 56.7'
2	209°56.2	32.3		137°38.2	6.6'	28°32.9	-0.7' 56.7'
3	224°56.3	· · 33.2		152°03.8	6.7'	28°32.2	-0.8' 56.6'
4	239°56.5	34.1		166°29.5	6.7'	28°31.4	-1.0' 56.6'
5	254°56.6	35.0		180°55.2	6.8'	28°30.4	-1.2' 56.6'
6	269°56.8	N09°35.9		195°21.0	6.8'	N28°29.3	-1.3' 56.5'
7	284°56.9	36.8		209°46.8	6.9'	28°27.9	-1.5' 56.5'
8	299°57.1	37.7		224°12.7	7.0'	28°26.5	-1.6' 56.5'
9	314°57.2	· · 38.6		238°38.7	7.0'	28°24.8	-1.8' 56.4'
10	329°57.4	39.5		253°04.7	7.1'	28°23.0	-2.0' 56.4'
11	344°57.5	40.4		267°30.8	7.2'	28°21.1	-2.1' 56.4'
12	359°57.7	N09°41.3		281°57.0	7.2'	N28°19.0	-2.3' 56.3'
13	14°57.8	42.2		296°23.2	7.3'	28°16.7	-2.4' 56.3'
14	29°58.0	43.1		310°49.5	7.4'	28°14.3	-2.6' 56.3'
15	44°58.1	· · 44.0		325°15.9	7.5'	28°11.7	-2.7' 56.2'
16	59°58.3	44.9		339°42.4	7.5'	28°09.0	-2.9' 56.2'
17	74°58.5	45.8		354°08.9	7.6'	28°06.1	-3.0' 56.1'
18	89°58.6	N09°46.7		8°35.5	7.7'	N28°03.0	-3.2' 56.1'
19	104°58.8	47.6		23°02.2	7.8'	27°59.8	-3.3' 56.1'
20	119°58.9	48.4		37°29.0	7.9'	27°56.5	-3.5' 56.1'
21	134°59.1	· · 49.3		51°55.8	7.9'	27°53.0	-3.6' 56.0'
22	149°59.2	50.2		66°22.8	8.0'	27°49.4	-3.8' 56.0'
23	164°59.4	51.1		80°49.8	8.1'	27°45.6	-3.9' 56.0'
SD = 15.9' d = 0.9'				SD = 15.5'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	02:12	03:48	20:17	21:56	—:—
N 70°	—:—	02:45	04:04	20:00	21:21	—:—
68°	01:04	03:08	04:16	19:48	20:57	23:11
66°	01:54	03:26	04:26	19:37	20:38	22:13
64°	02:23	03:40	04:35	19:28	20:23	21:42
62°	02:45	03:52	04:42	19:21	20:11	21:19
60°	03:02	04:02	04:48	19:14	20:00	21:01
N 58°	03:16	04:11	04:54	19:09	19:51	20:47
56°	03:28	04:19	04:59	19:04	19:44	20:35
54°	03:38	04:25	05:03	18:59	19:37	20:24
52°	03:47	04:31	05:07	18:55	19:31	20:15
50°	03:55	04:37	05:10	18:51	19:25	20:07
45°	04:11	04:48	05:18	18:43	19:14	19:51
N 40°	04:23	04:57	05:25	18:37	19:05	19:38
35°	04:33	05:04	05:30	18:31	18:57	19:28
30°	04:42	05:11	05:35	18:26	18:51	19:20
20°	04:55	05:21	05:43	18:18	18:40	19:06
N 10°	05:04	05:29	05:50	18:10	18:32	18:57
0°	05:12	05:36	05:57	18:04	18:25	18:49
S 10°	05:18	05:42	06:04	17:57	18:18	18:43
20°	05:22	05:48	06:10	17:50	18:12	18:38
30°	05:26	05:54	06:18	17:42	18:06	18:34
35°	05:28	05:57	06:22	17:38	18:03	18:33
40°	05:29	06:00	06:27	17:33	18:00	18:32
45°	05:29	06:03	06:33	17:27	17:57	18:31
S 50°	05:30	06:07	06:40	17:20	17:53	18:30
52°	05:30	06:09	06:43	17:17	17:51	18:30
54°	05:30	06:11	06:46	17:13	17:49	18:30
56°	05:29	06:12	06:50	17:10	17:47	18:30
58°	05:29	06:14	06:54	17:05	17:45	18:31
S 60°	05:28	06:17	06:59	17:01	17:43	18:31

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	☐	☐	☐	☐	☐	☐
64°	03:35	☐	☐	02:13	☐	☐
62°	04:40	☐	☐	01:09	☐	☐
60°	05:15	05:33	06:23	00:34	02:19	03:31
N 58°	05:41	06:09	07:02	00:09	01:43	02:51
56°	06:01	06:35	07:30	· · ·	01:17	02:24
54°	06:18	06:56	07:51	· · ·	00:57	02:02
52°	06:33	07:13	08:09	· · ·	00:40	01:44
50°	06:45	07:28	08:24	· · ·	00:25	01:29
45°	07:12	07:58	08:55	23:55	· · ·	00:58
N 40°	07:33	08:22	09:19	23:32	· · ·	00:34
35°	07:50	08:41	09:39	23:13	· · ·	00:14
30°	08:05	08:58	09:55	22:56	23:57	· · ·
20°	08:31	09:26	10:24	22:29	23:29	· · ·
N 10°	08:54	09:51	10:48	22:05	23:04	· · ·
0°	09:15	10:13	11:11	21:43	22:41	23:38
S 10°	09:36	10:36	11:34	21:21	22:18	23:15
20°	09:59	11:01	11:58	20:57	21:54	22:52
30°	10:25	11:29	12:26	20:30	21:25	22:24
35°	10:41	11:46	12:43	20:13	21:08	22:07
40°	10:59	12:06	13:03	19:55	20:48	21:48
45°	11:21	12:30	13:26	19:32	20:24	21:25
S 50°	11:50	13:01	13:57	19:03	19:52	20:55
52°	12:04	13:17	14:12	18:49	19:37	20:40
54°	12:20	13:35	14:30	18:32	19:19	20:22
56°	12:39	13:57	14:51	18:13	18:56	20:01
58°	13:03	14:26	15:18	17:48	18:28	19:35
S 60°	13:35	15:08	15:56	17:16	17:46	18:57

Day	Sun		Mer. Pass hh:mm	Moon		Age 4-6 14-32%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
12	00:47	00:39	12:01	15:29	02:59	
13	00:31	00:24	12:00	16:27	03:58	
14	00:17	00:09	12:00	17:24	04:56	

April 15, 16, 17 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	203°38.7	191°54.0	N03°26.7	214°17.0	S05°56.5	155°25.4	N17°05.9	216°41.4	S07°26.0	Alpheratz	357°35.8 29°13.3
1	218°41.1	206°53.6		229°17.6	55.8	170°27.3	06.1	231°43.6	25.9	Ankaa	353°08.1 -42°10.5
2	233°43.6	221°53.2	29.1	244°18.3	55.0	185°29.2	06.2	246°45.9	25.8	Schedar	349°32.3 56°40.1
3	248°46.1	236°52.9	· · 30.3	259°19.0	· · 54.3	200°31.1	· · 06.4	261°48.1	· · 25.7	Diphda	348°48.2 -17°51.3
4	263°48.5	251°52.5	31.5	274°19.7	53.5	215°33.0	06.5	276°50.3	25.6	Achernar	335°21.2 -57°06.9
5	278°51.0	266°52.1	32.8	289°20.3	52.7	230°34.9	06.7	291°52.5	25.5	Hamal	327°52.3 23°34.5
6	293°53.4	281°51.7	N03°34.0	304°21.0	S05°52.0	245°36.8	N17°06.9	306°54.7	S07°25.4	Polaris	314°48.6 89°22.1
7	308°55.9	296°51.4	35.2	319°21.7	51.2	260°38.7	07.0	321°57.0	25.3	Acamar	315°12.6 -40°12.6
8	323°58.4	311°51.0	36.4	334°22.3	50.5	275°40.6	07.2	336°59.2	25.3	Menkar	314°07.1 4°11.0
9	339°00.8	326°50.6	· · 37.6	349°23.0	· · 49.7	290°42.5	· · 07.3	352°01.4	· · 25.2	Mirfak	308°29.6 49°56.9
10	354°03.3	341°50.2	38.8	4°23.7	49.0	305°44.4	07.5	7°03.6	25.1	Aldebaran	290°40.6 16°33.4
11	9°05.8	356°49.9	40.1	19°24.3	48.2	320°46.3	07.6	22°05.9	25.0	Rigel	281°04.7 -8°10.5
12	24°08.2	11°49.5	N03°41.3	34°25.0	S05°47.5	335°48.2	N17°07.8	37°08.1	S07°24.9	Capella	280°23.1 46°01.4
13	39°10.7	26°49.1	42.5	49°25.7	46.7	350°50.1	07.9	52°10.3	24.8	Bellatrix	278°23.7 6°22.2
14	54°13.2	41°48.7	43.7	64°26.3	45.9	5°52.0	08.1	67°12.5	24.7	Elnath	278°02.9 28°37.7
15	69°15.6	56°48.3	· · 44.9	79°27.0	· · 45.2	20°53.9	· · 08.2	82°14.8	· · 24.6	Alnilam	275°38.5 -1°11.3
16	84°18.1	71°48.0	46.1	94°27.7	44.4	35°55.7	08.4	97°17.0	24.5	Betelgeuse	270°52.9 7°24.6
17	99°20.5	86°47.6	47.4	109°28.3	43.7	50°57.6	08.6	112°19.2	24.4	Canopus	263°52.9 -52°42.7
18	114°23.0	101°47.2	N03°48.6	124°29.0	S05°42.9	65°59.5	N17°08.7	127°21.4	S07°24.3	Sirius	258°26.9 -16°45.1
19	129°25.5	116°46.8	49.8	139°29.7	42.2	81°01.4	08.9	142°23.7	24.2	Adhara	255°06.4 -29°00.5
20	144°27.9	131°46.5	51.0	154°30.4	41.4	96°03.3	09.0	157°25.9	24.1	Procyon	244°51.5 5°09.7
21	159°30.4	146°46.1	· · 52.2	169°31.0	· · 40.6	111°05.2	· · 09.2	172°28.1	· · 24.0	Pollux	243°18.1 27°58.1
22	174°32.9	161°45.7	53.4	184°31.7	39.9	126°07.1	09.3	187°30.3	24.0	Avior	234°14.9 -59°35.5
23	189°35.3	176°45.3	54.7	199°32.4	39.1	141°09.0	09.5	202°32.6	23.9	Suhail	222°46.7 -43°32.1
Mer.pass. 10:24		ν -0.4' d 1.2' m -3.90		ν 0.7' d -0.8' m 1.16		ν 1.9' d 0.2' m -2.03		ν 2.2' d -0.1' m 1.07		Miaplacidus	221°38.1 -69°49.2
										Alphard	217°48.2 -8°45.9
										Regulus	207°34.9 11°50.9
										Dubhe	193°41.2 61°37.4
										Denebola	182°25.3 14°26.1
										Gienah	175°43.9 -17°40.7
										Acrux	173°00.2 -63°14.1
										Gacrux	171°51.8 -57°15.1
										Alioth	166°12.9 55°49.7
										Spica	158°22.6 -11°17.4
										Alkaid	152°52.0 49°11.4
										Hadar	148°36.3 -60°29.4
										Menkent	147°58.0 -36°29.4
										Arcturus	145°48.1 19°03.2
										Rigel Kent.	139°40.6 -60°56.1
										Kochab	137°18.6 74°03.2
										Zuben'ubi	136°56.4 -16°08.7
										Alphecca	126°03.9 26°37.8
										Antares	112°16.3 -26°29.2
										Atria	107°10.8 -69°04.1
										Sabik	102°03.3 -15°45.4
										Shaula	96°11.0 -37°07.2
										Rasalhague	95°58.9 12°32.3
										Eltanin	90°42.2 51°28.8
										Kaus Aust.	83°33.2 -34°22.3
										Vega	80°33.5 38°48.0
										Nunki	75°48.4 -26°16.0
										Altair	62°00.6 8°55.7
										Peacock	53°06.8 -56°39.2
										Deneb	49°26.3 45°21.6
										Enif	33°39.6 9°59.0
										Al Na'ir	27°33.9 -46°50.5
										Fomalhaut	15°15.5 -29°29.7
										Scheat	13°46.1 28°12.6
										Markab	13°30.8 15°19.9
										Apr 15 Mon	SHA Mer.pass
										Venus	348°15.3 11:13
										Mars	10°38.3 09:42
										Jupiter	311°46.8 13:37
										Saturn	13°02.8 09:32
										Apr 16 Tue	SHA Mer.pass
										Venus	347°07.2 11:13
										Mars	9°55.2 09:41
										Jupiter	311°33.1 13:34
										Saturn	12°57.0 09:28
										Apr 17 Wed	SHA Mer.pass
										Venus	345°58.9 11:14
										Mars	9°12.2 09:40
										Jupiter	311°19.4 13:31
										Saturn	12°51.3 09:25
										Horizontal parallax	
										Venus:	0.1
										Mars:	0.1
Mer.pass. 10:16		ν -0.4' d 1.2' m -3.90		ν 0.7' d -0.8' m 1.15		ν 1.9' d 0.2' m -2.03		ν 2.2' d -0.1' m 1.07			

Sun				Moon				
Mon	GHA	Dec		GHA	ν	Dec	d	HP
0	179°59.5	N09°52.0		95°16.9	8.2'	N27°41.6	-4.1'	55.9'
1	194°59.7	52.9		109°44.1	8.3'	27°37.6	-4.2'	55.9'
2	209°59.8	53.8		124°11.3	8.4'	27°33.4	-4.4'	55.9'
3	225°00.0	• • 54.7		138°38.7	8.4'	27°29.0	-4.5'	55.8'
4	240°00.1	55.6		153°06.1	8.5'	27°24.5	-4.6'	55.8'
5	255°00.3	56.5		167°33.7	8.6'	27°19.8	-4.8'	55.8'
6	270°00.4	N09°57.4		182°01.3	8.7'	N27°15.1	-4.9'	55.7'
7	285°00.6	58.3		196°29.0	8.8'	27°10.1	-5.1'	55.7'
8	300°00.7	09°59.1		210°56.8	8.9'	27°05.1	-5.2'	55.7'
9	315°00.8	10°00.0		225°24.7	9.0'	26°59.9	-5.3'	55.6'
10	330°01.0	00.9		239°52.7	9.1'	26°54.6	-5.5'	55.6'
11	345°01.1	01.8		254°20.7	9.2'	26°49.1	-5.6'	55.6'
12	0°01.3	N10°02.7		268°48.9	9.3'	N26°43.5	-5.7'	55.6'
13	15°01.4	03.6		283°17.2	9.4'	26°37.8	-5.9'	55.5'
14	30°01.6	04.5		297°45.5	9.4'	26°31.9	-6.0'	55.5'
15	45°01.7	• • 05.4		312°14.0	9.5'	26°25.9	-6.1'	55.5'
16	60°01.9	06.3		326°42.5	9.6'	26°19.8	-6.2'	55.4'
17	75°02.0	07.1		341°11.1	9.7'	26°13.6	-6.4'	55.4'
18	90°02.2	N10°08.0		355°39.9	9.8'	N26°07.2	-6.5'	55.4'
19	105°02.3	08.9		10°08.7	9.9'	26°00.8	-6.6'	55.4'
20	120°02.5	09.8		24°37.6	10.0'	25°54.1	-6.7'	55.3'
21	135°02.6	• • 10.7		39°06.6	10.1'	25°47.4	-6.9'	55.3'
22	150°02.8	11.6		53°35.8	10.2'	25°40.6	-7.0'	55.3'
23	165°02.9	12.5		68°05.0	10.3'	25°33.6	-7.1'	55.2'
SD = 15.9' $d = 0.9'$				SD = 15.3'				
Tue	GHA	Dec		GHA	ν	Dec	d	HP
0	180°03.1	N10°13.3		82°34.3	10.4'	N25°26.5	-7.2'	55.2'
1	195°03.2	14.2		97°03.7	10.5'	25°19.3	-7.3'	55.2'
2	210°03.4	15.1		111°33.2	10.6'	25°12.0	-7.4'	55.2'
3	225°03.5	• • 16.0		126°02.8	10.7'	25°04.5	-7.5'	55.1'
4	240°03.6	16.9		140°32.5	10.8'	24°57.0	-7.7'	55.1'
5	255°03.8	17.8		155°02.3	10.9'	24°49.3	-7.8'	55.1'
6	270°03.9	N10°18.6		169°32.1	11.0'	N24°41.6	-7.9'	55.1'
7	285°04.1	19.5		184°02.1	11.1'	24°33.7	-8.0'	55.0'
8	300°04.2	20.4		198°32.2	11.2'	24°25.7	-8.1'	55.0'
9	315°04.4	• • 21.3		213°02.4	11.3'	24°17.6	-8.2'	55.0'
10	330°04.5	22.2		227°32.7	11.4'	24°09.4	-8.3'	55.0'
11	345°04.7	23.1		242°03.0	11.5'	24°01.1	-8.4'	55.0'
12	0°04.8	N10°23.9		256°33.5	11.6'	N23°52.6	-8.5'	54.9'
13	15°05.0	24.8		271°04.0	11.7'	23°44.1	-8.6'	54.9'
14	30°05.1	25.7		285°34.7	11.7'	23°35.5	-8.7'	54.9'
15	45°05.2	• • 26.6		300°05.4	11.8'	23°26.8	-8.8'	54.9'
16	60°05.4	27.5		314°36.3	11.9'	23°18.0	-8.9'	54.8'
17	75°05.5	28.3		329°07.2	12.0'	23°09.0	-9.0'	54.8'
18	90°05.7	N10°29.2		343°38.3	12.1'	N23°00.0	-9.1'	54.8'
19	105°05.8	30.1		358°09.4	12.2'	22°50.9	-9.2'	54.8'
20	120°06.0	31.0		12°40.6	12.3'	22°41.7	-9.3'	54.8'
21	135°06.1	• • 31.9		27°11.9	12.4'	22°32.4	-9.4'	54.7'
22	150°06.2	32.7		41°43.3	12.5'	22°23.0	-9.5'	54.7'
23	165°06.4	33.6		56°14.8	12.6'	22°13.5	-9.6'	54.7'
SD = 15.9' $d = 0.9'$				SD = 15.1'				
Wed	GHA	Dec		GHA	ν	Dec	d	HP
0	180°06.5	N10°34.5		70°46.4	12.7'	N22°03.9	-9.7'	54.7'
1	195°06.7	35.4		85°18.1	12.8'	21°54.3	-9.8'	54.7'
2	210°06.8	36.2		99°49.8	12.9'	21°44.5	-9.8'	54.6'
3	225°07.0	• • 37.1		114°21.7	12.9'	21°34.7	-9.9'	54.6'
4	240°07.1	38.0		128°53.6	13.0'	21°24.7	-10.0'	54.6'
5	255°07.2	38.9		143°25.7	13.1'	21°14.7	-10.1'	54.6'
6	270°07.4	N10°39.8		157°57.8	13.2'	N21°04.6	-10.2'	54.6'
7	285°07.5	40.6		172°30.0	13.3'	20°54.4	-10.3'	54.5'
8	300°07.7	41.5		187°02.3	13.4'	20°44.2	-10.3'	54.5'
9	315°07.8	• • 42.4		201°34.7	13.5'	20°33.8	-10.4'	54.5'
10	330°08.0	43.3		216°07.1	13.5'	20°23.4	-10.5'	54.5'
11	345°08.1	44.1		230°39.7	13.6'	20°12.9	-10.6'	54.5'
12	0°08.2	N10°45.0		245°12.3	13.7'	N20°02.3	-10.7'	54.5'
13	15°08.4	45.9		259°45.0	13.8'	19°51.6	-10.7'	54.4'
14	30°08.5	46.8		274°17.8	13.9'	19°40.9	-10.8'	54.4'
15	45°08.7	• • 47.6		288°50.7	14.0'	19°30.0	-10.9'	54.4'
16	60°08.8	48.5		303°23.6	14.0'	19°19.2	-11.0'	54.4'
17	75°08.9	49.4		317°56.7	14.1'	19°08.2	-11.0'	54.4'
18	90°09.1	N10°50.2		332°29.8	14.2'	N18°57.2	-11.1'	54.4'
19	105°09.2	51.1		347°03.0	14.3'	18°46.0	-11.2'	54.4'
20	120°09.4	52.0		1°36.3	14.4'	18°34.9	-11.2'	54.3'
21	135°09.5	• • 52.9		16°09.6	14.4'	18°23.6	-11.3'	54.3'
22	150°09.6	53.7		30°43.1	14.5'	18°12.3	-11.4'	54.3'
23	165°09.8	54.6		45°16.6	14.6'	18°00.9	-11.5'	54.3'
SD = 15.9' $d = 0.9'$				SD = 14.9'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	01:43	03:31	20:33	22:26	—:—
N 70°	—:—	02:24	03:49	20:14	21:41	—:—
68°	—:—	02:52	04:03	19:59	21:12	—:—
66°	01:29	03:12	04:14	19:47	20:51	22:39
64°	02:06	03:28	04:24	19:37	20:34	21:59
62°	02:31	03:42	04:32	19:29	20:20	21:32
60°	02:50	03:53	04:39	19:22	20:09	21:12
N 58°	03:06	04:02	04:45	19:15	19:59	20:56
56°	03:19	04:10	04:51	19:10	19:50	20:43
54°	03:30	04:18	04:56	19:05	19:43	20:32
52°	03:39	04:24	05:00	19:00	19:36	20:22
50°	03:48	04:30	05:04	18:56	19:31	20:13
45°	04:05	04:42	05:13	18:47	19:18	19:56
N 40°	04:18	04:52	05:20	18:40	19:08	19:42
35°	04:29	05:00	05:26	18:34	19:00	19:31
30°	04:38	05:07	05:32	18:28	18:53	19:22
20°	04:52	05:18	05:41	18:19	18:41	19:08
N 10°	05:03	05:28	05:49	18:11	18:32	18:57
0°	05:11	05:35	05:56	18:03	18:24	18:48
S 10°	05:18	05:42	06:04	17:56	18:17	18:42
20°	05:23	05:49	06:11	17:48	18:10	18:36
30°	05:28	05:56	06:20	17:39	18:03	18:31
35°	05:30	05:59	06:25	17:34	18:00	18:29
40°	05:31	06:03	06:30	17:28	17:56	18:27
45°	05:33	06:07	06:37	17:22	17:52	18:26
S 50°	05:34	06:12	06:45	17:14	17:47	18:24
52°	05:34	06:14	06:48	17:11	17:45	18:24
54°	05:35	06:16	06:52	17:07	17:43	18:24
56°	05:35	06:18	06:56	17:02	17:40	18:23
58°	05:35	06:21	07:01	16:57	17:38	18:23
S 60°	05:35	06:23	07:06	16:52	17:35	18:23

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°						
N 70°						
68°			09:04			06:14
66°			09:50			05:27
64°		08:15	10:20		05:22	04:56
62°	06:58	08:57	10:42	04:51	04:40	04:33
60°	07:48	09:25	11:00	04:01	04:11	04:14
N 58°	08:19	09:47	11:15	03:30	03:49	03:59
56°	08:43	10:04	11:27	03:06	03:31	03:46
54°	09:01	10:19	11:38	02:47	03:15	03:34
52°	09:17	10:32	11:48	02:31	03:02	03:24
50°	09:31	10:43	11:57	02:17	02:50	03:15
45°	09:59	11:07	12:15	01:48	02:26	02:55
N 40°	10:21	11:26	12:30	01:25	02:06	02:39
35°	10:39	11:41	12:42	01:07	01:50	02:26
30°	10:55	11:55	12:53	00:50	01:36	02:14
20°	11:22	12:18	13:12	00:23	01:11	01:53
N 10°	11:44	12:38	13:28	00:00	00:50	01:36
0°	12:05	12:56	13:43	• •	00:30	01:19
S 10°	12:27	13:14	13:58	• •	00:10	01:02
20°	12:49	13:34	14:14	23:49	• •	00:44
30°	13:15	13:57	14:32	23:24	• •	00:23
35°	13:31	14:10	14:42	23:09	• •	00:11
40°	13:48	14:25	14:54	22:52	23:57	• •
45°	14:10	14:43	15:08	22:31	23:40	• •
S 50°	14:37	15:05	15:25	22:05	23:19	• •
52°	14:50	15:16	15:33	21:52	23:09	• •
54°	15:05	15:27	15:42	21:38	22:57	• •
56°	15:23	15:41	15:52	21:20	22:44	• •
58°	15:44	15:57	16:04	21:00	22:29	23:56
S 60°	16:11	16:16	16:17	20:33	22:11	23:43

Day	Sun		Mer. Pass hh:mm	Moon		Age 7-9 42-62%
	Eqn.of Time	Mer.		Mer.Pass.	Age	
	00 ^h	12 ^h		Upper	Lower	
	mm:ss	mm:ss		hh:mm	hh:mm	
15	00:02	00:05	12:00	18:18	05:52	
16	00:12	00:19	12:00	19:08	06:43	
17	00:26	00:33	11:59	19:53	07:31	

April 18, 19, 20 UT (Thu., Fri., Sat.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Thu	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec			
0	206°36.1	191°26.7	N04°54.1	215°05.3	S05°02.0	157°41.7	N17°17.0	219°21.7	S07°19.4				Alpheratz	357°35.8	29°13.3		
1	221°38.5	206°26.3	55.3	230°06.0	01.2	172°43.6	17.2	234°23.9	19.3				Ankaa	353°08.1	-42°10.5		
2	236°41.0	221°26.0	56.5	245°06.7	05°00.5	187°45.5	17.3	249°26.1	19.2				Schedar	349°32.3	56°40.0		
3	251°43.5	236°25.6	· · 57.7	260°07.3	04°59.7	202°47.4	· · 17.5	264°28.4	· · 19.1				Diphda	348°48.2	-17°51.3		
4	266°45.9	251°25.2	04°58.9	275°08.0	58.9	217°49.3	17.7	279°30.6	19.0				Achernar	335°21.2	-57°06.8		
5	281°48.4	266°24.8	05°00.1	290°08.7	58.2	232°51.2	17.8	294°32.8	18.9				Hamal	327°52.3	23°34.5		
6	296°50.9	281°24.4	N05°01.3	305°09.4	S04°57.4	247°53.1	N17°18.0	309°35.0	S07°18.8				Polaris	314°49.1	89°22.1		
7	311°53.3	296°24.0	02.5	320°10.1	56.7	262°54.9	18.1	324°37.3	18.7				Acamar	315°12.6	-40°12.6		
8	326°55.8	311°23.6	03.7	335°10.7	55.9	277°56.8	18.3	339°39.5	18.6				Menkar	314°07.1	4°11.0		
9	341°58.3	326°23.3	· · 04.9	350°11.4	· · 55.1	292°58.7	· · 18.4	354°41.7	· · 18.5				Mirfak	308°29.6	49°56.8		
10	357°00.7	341°22.9	06.1	5°12.1	54.4	308°00.6	18.6	9°44.0	18.5				Aldebaran	290°40.6	16°33.4		
11	12°03.2	356°22.5	07.3	20°12.8	53.6	323°02.5	18.7	24°46.2	18.4				Rigel	281°04.7	-8°10.5		
12	27°05.6	11°22.1	N05°08.5	35°13.4	S04°52.9	338°04.4	N17°18.9	39°48.4	S07°18.3				Capella	280°23.1	46°01.4		
13	42°08.1	26°21.7	09.7	50°14.1	52.1	353°06.3	19.0	54°50.7	18.2				Bellatrix	278°23.8	6°22.2		
14	57°10.6	41°21.3	10.9	65°14.8	51.3	8°08.2	19.2	69°52.9	18.1				Elnath	278°02.9	28°37.7		
15	72°13.0	56°21.0	· · 12.2	80°15.5	· · 50.6	23°10.1	· · 19.3	84°55.1	· · 18.0				Alnilam	275°38.6	-1°11.3		
16	87°15.5	71°20.6	13.4	95°16.1	49.8	38°11.9	19.5	99°57.3	17.9				Betelgeuse	270°52.9	7°24.6		
17	102°18.0	86°20.2	14.6	110°16.8	49.1	53°13.8	19.6	114°59.6	17.8				Canopus	263°52.9	-52°42.7		
18	117°20.4	101°19.8	N05°15.8	125°17.5	S04°48.3	68°15.7	N17°19.8	130°01.8	S07°17.7				Sirius	258°26.9	-16°45.1		
19	132°22.9	116°19.4	17.0	140°18.2	47.5	83°17.6	20.0	145°04.0	17.6				Adhara	255°06.5	-29°00.5		
20	147°25.4	131°19.0	18.2	155°18.8	46.8	98°19.5	20.1	160°06.3	17.5				Procyon	244°51.5	5°09.7		
21	162°27.8	146°18.6	· · 19.4	170°19.5	· · 46.0	113°21.4	· · 20.3	175°08.5	· · 17.5				Pollux	243°18.1	27°58.1		
22	177°30.3	161°18.3	20.6	185°20.2	45.3	128°23.3	20.4	190°10.7	17.4				Avior	234°15.0	-59°35.5		
23	192°32.7	176°17.9	21.8	200°20.9	44.5	143°25.2	20.6	205°13.0	17.3				Suhail	222°46.7	-43°32.1		
Mer.pass. 10:12			ν -0.4' d 1.2' m -3.90			ν 0.7' d -0.8' m 1.15			ν 1.9' d 0.2' m -2.03			ν 2.2' d -0.1' m 1.07					

Fri	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec						
0	207°35.2	191°17.5	N05°23.0	215°21.5	S04°43.7	158°27.1	N17°20.7	220°15.2	S07°17.2					
1	222°37.7	206°17.1	24.2	230°22.2	43.0	173°28.9	20.9	235°17.4	17.1					
2	237°40.1	221°16.7	25.4	245°22.9	42.2	188°30.8	21.0	250°19.7	17.0					
3	252°42.6	236°16.3	· · 26.6	260°23.6	· · 41.5	203°32.7	· · 21.2	265°21.9	· · 16.9					
4	267°45.1	251°15.9	27.8	275°24.3	40.7	218°34.6	21.3	280°24.1	16.8					
5	282°47.5	266°15.5	29.0	290°24.9	39.9	233°36.5	21.5	295°26.3	16.7					
6	297°50.0	281°15.2	N05°30.2	305°25.6	S04°39.2	248°38.4	N17°21.6	310°28.6	S07°16.6					
7	312°52.5	296°14.8	31.4	320°26.3	38.4	263°40.3	21.8	325°30.8	16.6					
8	327°54.9	311°14.4	32.6	335°27.0	37.6	278°42.2	21.9	340°33.0	16.5					
9	342°57.4	326°14.0	· · 33.8	350°27.6	· · 36.9	293°44.0	· · 22.1	355°35.3	· · 16.4					
10	357°59.9	341°13.6	35.0	5°28.3	36.1	308°45.9	22.3	10°37.5	16.3					
11	13°02.3	356°13.2	36.2	20°29.0	35.4	323°47.8	22.4	25°39.7	16.2					
12	28°04.8	11°12.8	N05°37.4	35°29.7	S04°34.6	338°49.7	N17°22.6	40°42.0	S07°16.1					
13	43°07.2	26°12.4	38.6	50°30.4	33.8	353°51.6	22.7	55°44.2	16.0					
14	58°09.7	41°12.1	39.8	65°31.0	33.1	8°53.5	22.9	70°46.4	15.9					
15	73°12.2	56°11.7	· · 41.0	80°31.7	· · 32.3	23°55.4	· · 23.0	85°48.7	· · 15.8					
16	88°14.6	71°11.3	42.2	95°32.4	31.6	38°57.3	23.2	100°50.9	15.8					
17	103°17.1	86°10.9	43.4	110°33.1	30.8	53°59.1	23.3	115°53.1	15.7					
18	118°19.6	101°10.5	N05°44.6	125°33.8	S04°30.0	69°01.0	N17°23.5	130°55.4	S07°15.6					
19	133°22.0	116°10.1	45.8	140°34.4	29.3	84°02.9	23.6	145°57.6	15.5					
20	148°24.5	131°09.7	47.0	155°35.1	28.5	99°04.8	23.8	160°59.8	15.4					
21	163°27.0	146°09.3	· · 48.2	170°35.8	· · 27.7	114°06.7	· · 23.9	176°02.1	· · 15.3					
22	178°29.4	161°08.9	49.4	185°36.5	27.0	129°08.6	24.1	191°04.3	15.2					
23	193°31.9	176°08.5	50.6	200°37.1	26.2	144°10.5	24.2	206°06.5	15.1					
Mer.pass. 10:08			ν -0.4' d 1.2' m -3.90			ν 0.7' d -0.8' m 1.15			ν 1.9' d 0.2' m -2.03			ν 2.2' d -0.1' m 1.07		

Sat	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec						
0	208°34.3	191°08.2	N05°51.8	215°37.8	S04°25.5	159°12.4	N17°24.4	221°08.8	S07°15.0					
1	223°36.8	206°07.8	53.0	230°38.5	24.7	174°14.2	24.5	236°11.0	15.0					
2	238°39.3	221°07.4	54.2	245°39.2	23.9	189°16.1	24.7	251°13.2	14.9					
3	253°41.7	236°07.0	· · 55.4	260°39.9	· · 23.2	204°18.0	· · 24.9	266°15.5	· · 14.8					
4	268°44.2	251°06.6	56.6	275°40.5	22.4	219°19.9	25.0	281°17.7	14.7					
5	283°46.7	266°06.2	57.8	290°41.2	21.6	234°21.8	25.2	296°19.9	14.6					
6	298°49.1	281°05.8	N05°59.0	305°41.9	S04°20.9	249°23.7	N17°25.3	311°22.2	S07°14.5					
7	313°51.6	296°05.4	06°00.2	320°42.6	20.1	264°25.6	25.5	326°24.4	14.4					
8	328°54.1	311°05.0	01.4	335°43.3	19.4	279°27.4	25.6	341°26.6	14.3					
9	343°56.5	326°04.6	· · 02.6	350°43.9	· · 18.6	294°29.3	· · 25.8	356°28.9	· · 14.2					
10	358°59.0	341°04.2	03.8	5°44.6	17.8	309°31.2	25.9	11°31.1	14.2					
11	14°01.5	356°03.8	05.0	20°45.3	17.1	324°33.1	26.1	26°33.3	14.1					
12	29°03.9	11°03.5	N06°06.2	35°46.0	S04°16.3	339°35.0	N17°26.2	41°35.6	S07°14.0					
13	44°06.4	26°03.1	07.4	50°46.7	15.5	354°36.9	26.4	56°37.8	13.9					
14	59°08.8	41°02.7	08.6	65°47.3	14.8	9°38.8	26.5	71°40.0	13.8					
15	74°11.3	56°02.3	· · 09.8	80°48.0	· · 14.0	24°40.6	· · 26.7	86°42.3	· · 13.7					
16	89°13.8	71°01.9	11.0	95°48.7	13.3	39°42.5	26.8	101°44.5	13.6					
17	104°16.2	86°01.5	12.2	110°49.4	12.5	54°44.4	27.0	116°46.7	13.5					
18	119°18.7	101°01.1	N06°13.4	125°50.1	S04°11.7	69°46.3	N17°27.1	131°49.0	S07°13.4					
19	134°21.2	116°00.7	14.6	140°50.7	11.0	84°48.2	27.3	146°51.2	13.4					
20	149°23.6	131°00.3	15.8	155°51.4	10.2	99°50.1	27.5	161°53.4	13.3					
21	164°26.1	145°59.9	· · 17.0	170°52.1	· · 09.4	114°52.0	· · 27.6	176°55.7	· · 13.2					
22	179°28.6	160°59.5	18.2	185°52.8	08.7	129°53.8	27.8	191°57.9	13.1					
23	194°31.0	175°59.1	19.3	200°53.5	07.9	144°55.7	27.9	207°00.1	13.0					
Mer.pass. 10:04			ν -0.4' d 1.2' m -3.90			ν 0.7' d -0.8' m 1.14			ν 1.9' d 0.2' m -2.02			ν 2.2' d -0.1' m 1.07		

Apr 18 Thu	SHA	Mer.pass
Venus	344°50.6	11:15
Mars	8°29.2	09:39
Jupiter	311°05.6	13:28
Saturn	12°45.6	09:21

Apr 19 Fri	SHA	Mer.pass
Venus	343°42.3	11:15
Mars	7°46.3	09:38
Jupiter	310°51.8	13:25
Saturn	12°40.0	09:18

Apr 20 Sat	SHA	Mer.pass
Venus	342°33.8	11:16
Mars	7°03.5	09:37
Jupiter	310°38.0	13:21
Saturn	12°34.4	09:14

Horizontal parallax	
Venus:	0.1
Mars:	0.1

2024 April 18 to Apr. 20 UT

IERS Earth Orientation data as of 01-Aug-2024

April 21, 22, 23 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	209°33.5	190°58.7	N06°20.5	215°54.2	S04°07.2	159°57.6	N17°28.1	222°02.4	S07°12.9	Alpheratz	357°35.7	29°13.3
1	224°36.0	205°58.3	21.7	230°54.8	06.4	174°59.5	28.2	237°04.6	12.8	Ankaa	353°08.1	-42°10.4
2	239°38.4	220°57.9	22.9	245°55.5	05.6	190°01.4	28.4	252°06.9	12.7	Schedar	349°32.3	56°40.0
3	254°40.9	235°57.5	•• 24.1	260°56.2	•• 04.9	205°03.3	•• 28.5	267°09.1	•• 12.6	Diphda	348°48.2	-17°51.3
4	269°43.3	250°57.1	25.3	275°56.9	04.1	220°05.1	28.7	282°11.3	12.6	Achernar	335°21.2	-57°06.8
5	284°45.8	265°56.7	26.5	290°57.6	03.3	235°07.0	28.8	297°13.6	12.5	Hamal	327°52.3	23°34.5
6	299°48.3	280°56.3	N06°27.7	305°58.2	S04°02.6	250°08.9	N17°29.0	312°15.8	S07°12.4	Polaris	314°49.5	89°22.0
7	314°50.7	295°56.0	28.9	320°58.9	01.8	265°10.8	29.1	327°18.0	12.3	Acamar	315°12.6	-40°12.5
8	329°53.2	310°55.6	30.1	335°59.6	01.0	280°12.7	29.3	342°20.3	12.2	Menkar	314°07.1	4°11.0
9	344°55.7	325°55.2	•• 31.3	351°00.3	04°00.3	295°14.6	•• 29.4	357°22.5	•• 12.1	Mirfak	308°29.6	49°56.8
10	359°58.1	340°54.8	32.5	6°01.0	03°59.5	310°16.5	29.6	12°24.7	12.0	Aldebaran	290°40.6	16°33.4
11	15°00.6	355°54.4	33.7	21°01.7	58.8	325°18.3	29.7	27°27.0	11.9	Rigel	281°04.7	-8°10.5
12	30°03.1	10°54.0	N06°34.8	36°02.3	S03°58.0	340°20.2	N17°29.9	42°29.2	S07°11.9	Capella	280°23.1	46°01.4
13	45°05.5	25°53.6	36.0	51°03.0	57.2	355°22.1	30.0	57°31.4	11.8	Bellatrix	278°23.8	6°22.2
14	60°08.0	40°53.2	37.2	66°03.7	56.5	10°24.0	30.2	72°33.7	11.7	Elnath	278°02.9	28°37.7
15	75°10.4	55°52.8	•• 38.4	81°04.4	•• 55.7	25°25.9	•• 30.3	87°35.9	•• 11.6	Alnilam	275°38.6	-1°11.3
16	90°12.9	70°52.4	39.6	96°05.1	54.9	40°27.8	30.5	102°38.2	11.5	Betelgeuse	270°52.9	7°24.6
17	105°15.4	85°52.0	40.8	111°05.8	54.2	55°29.6	30.7	117°40.4	11.4	Canopus	263°52.9	-52°42.7
18	120°17.8	100°51.6	N06°42.0	126°06.4	S03°53.4	70°31.5	N17°30.8	132°42.6	S07°11.3	Sirius	258°26.9	-16°45.1
19	135°20.3	115°51.2	43.2	141°07.1	52.6	85°33.4	31.0	147°44.9	11.2	Adhara	255°06.5	-29°00.5
20	150°22.8	130°50.8	44.4	156°07.8	51.9	100°35.3	31.1	162°47.1	11.2	Procyon	244°51.5	5°09.7
21	165°25.2	145°50.4	•• 45.6	171°08.5	•• 51.1	115°37.2	•• 31.3	177°49.3	•• 11.1	Pollux	243°18.1	27°58.1
22	180°27.7	160°50.0	46.7	186°09.2	50.3	130°39.1	31.4	192°51.6	11.0	Avior	234°15.0	-59°35.5
23	195°30.2	175°49.6	47.9	201°09.8	49.6	145°40.9	31.6	207°53.8	10.9	Suhail	222°46.7	-43°32.1
Mer.pass. 10:00		ν -0.4' d 1.2' m -3.90		ν 0.7' d -0.8' m 1.14		ν 1.9' d 0.2' m -2.02		ν 2.2' d -0.1' m 1.07		Miaplacidus	221°38.2	-69°49.3
										Alphard	217°48.3	-8°45.9
										Regulus	207°34.9	11°50.9
										Dubhe	193°41.3	61°37.4
										Denebola	182°25.3	14°26.1
										Gienah	175°43.9	-17°40.7
										Acrux	173°00.2	-63°14.2
										Gacrux	171°51.9	-57°15.1
										Alioth	166°12.9	55°49.7
										Spica	158°22.6	-11°17.4
										Alkaid	152°52.0	49°11.5
										Hadar	148°36.3	-60°29.5
										Menkent	147°58.0	-36°29.5
										Arcturus	145°48.1	19°03.2
										Rigel Kent.	139°40.5	-60°56.2
										Kochab	137°18.6	74°03.2
										Zuben'ubi	136°56.4	-16°08.7
										Alphecca	126°03.9	26°37.8
										Antares	112°16.3	-26°29.2
										Atria	107°10.7	-69°04.1
										Sabik	102°03.3	-15°45.4
										Shaula	96°11.0	-37°07.2
										Rasalhague	95°58.9	12°32.3
										Eltanin	90°42.2	51°28.8
										Kaus Aust.	83°33.2	-34°22.3
										Vega	80°33.5	38°48.0
										Nunki	75°48.4	-26°16.0
										Altair	62°00.5	8°55.7
										Peacock	53°06.7	-56°39.2
										Deneb	49°26.3	45°21.6
										Enif	33°39.5	9°59.0
										Al Na'ir	27°33.9	-46°50.5
										Fomalhaut	15°15.4	-29°29.6
										Scheat	13°46.1	28°12.6
										Markab	13°30.7	15°19.9
Apr 21 Sun		SHA		Mer.pass								
		Venus		341°25.2		11:16						
		Mars		6°20.7		09:36						
		Jupiter		310°24.1		13:18						
		Saturn		12°28.9		09:10						
Apr 22 Mon		SHA		Mer.pass								
		Venus		340°16.6		11:17						
		Mars		5°37.9		09:35						
		Jupiter		310°10.2		13:15						
		Saturn		12°23.4		09:07						
Apr 23 Tue		SHA		Mer.pass								
		Venus		339°07.8		11:18						
		Mars		4°55.2		09:34						
		Jupiter		309°56.2		13:12						
		Saturn		12°18.0		09:03						
		</										

Sun				Moon				
Sun	GHA	Dec		GHA	ν	Dec	d	HP
0	180°19.5	N11°57.3		30°21.3	17.4'	N02°03.0	-14.2'	54.0'
1	195°19.6	58.1		44°57.7	17.4'	01°48.8	-14.2'	54.0'
2	210°19.7	59.0		59°34.0	17.4'	01°34.7	-14.2'	54.0'
3	225°19.8	11°59.8		74°10.4	17.4'	01°20.5	-14.2'	54.1'
4	240°20.0	12°00.7		88°46.8	17.4'	01°06.3	-14.2'	54.1'
5	255°20.1	01.5		103°23.2	17.4'	00°52.1	-14.2'	54.1'
6	270°20.2	N12°02.4		117°59.5	17.4'	N00°37.9	-14.2'	54.1'
7	285°20.3	03.2		132°35.9	17.4'	00°23.7	-14.2'	54.1'
8	300°20.5	04.0		147°12.2	17.3'	N00°09.5	-14.2'	54.1'
9	315°20.6	· · 04.9		161°48.6	17.3'	S00°04.7	14.2'	54.1'
10	330°20.7	05.7		176°24.9	17.3'	00°18.9	14.2'	54.1'
11	345°20.8	06.6		191°01.2	17.3'	00°33.1	14.2'	54.1'
12	0°20.9	N12°07.4		205°37.6	17.3'	S00°47.3	14.2'	54.1'
13	15°21.1	08.3		220°13.9	17.3'	01°01.6	14.2'	54.1'
14	30°21.2	09.1		234°50.2	17.3'	01°15.8	14.2'	54.1'
15	45°21.3	· · 09.9		249°26.4	17.3'	01°30.0	14.2'	54.1'
16	60°21.4	10.8		264°02.7	17.3'	01°44.2	14.2'	54.1'
17	75°21.6	11.6		278°39.0	17.2'	01°58.5	14.2'	54.1'
18	90°21.7	N12°12.5		293°15.2	17.2'	S02°12.7	14.2'	54.2'
19	105°21.8	13.3		307°51.4	17.2'	02°26.9	14.2'	54.2'
20	120°21.9	14.1		322°27.6	17.2'	02°41.1	14.2'	54.2'
21	135°22.1	· · 15.0		337°03.8	17.2'	02°55.3	14.2'	54.2'
22	150°22.2	15.8		351°39.9	17.1'	03°09.5	14.2'	54.2'
23	165°22.3	16.7		6°16.1	17.1'	03°23.7	14.2'	54.2'
SD = 15.9' $d = 0.8'$				SD = 14.7'				
Mon	GHA	Dec		GHA	ν	Dec	d	HP
0	180°22.4	N12°17.5		20°52.2	17.1'	S03°37.9	14.2'	54.2'
1	195°22.5	18.3		35°28.3	17.1'	03°52.1	14.2'	54.2'
2	210°22.7	19.2		50°04.3	17.0'	04°06.3	14.2'	54.2'
3	225°22.8	· · 20.0		64°40.4	17.0'	04°20.4	14.2'	54.2'
4	240°22.9	20.8		79°16.4	17.0'	04°34.6	14.1'	54.2'
5	255°23.0	21.7		93°52.4	17.0'	04°48.8	14.1'	54.2'
6	270°23.1	N12°22.5		108°28.3	16.9'	S05°02.9	14.1'	54.3'
7	285°23.3	23.4		123°04.2	16.9'	05°17.0	14.1'	54.3'
8	300°23.4	24.2		137°40.1	16.9'	05°31.1	14.1'	54.3'
9	315°23.5	· · 25.0		152°16.0	16.8'	05°45.2	14.1'	54.3'
10	330°23.6	25.9		166°51.8	16.8'	05°59.3	14.1'	54.3'
11	345°23.7	26.7		181°27.6	16.8'	06°13.4	14.0'	54.3'
12	0°23.9	N12°27.5		196°03.4	16.7'	S06°27.4	14.0'	54.3'
13	15°24.0	28.4		210°39.1	16.7'	06°41.4	14.0'	54.3'
14	30°24.1	29.2		225°14.8	16.6'	06°55.5	14.0'	54.3'
15	45°24.2	· · 30.0		239°50.4	16.6'	07°09.5	14.0'	54.4'
16	60°24.3	30.9		254°26.0	16.6'	07°23.4	14.0'	54.4'
17	75°24.5	31.7		269°01.6	16.5'	07°37.4	13.9'	54.4'
18	90°24.6	N12°32.5		283°37.1	16.5'	S07°51.3	13.9'	54.4'
19	105°24.7	33.4		298°12.6	16.4'	08°05.2	13.9'	54.4'
20	120°24.8	34.2		312°48.0	16.4'	08°19.1	13.9'	54.4'
21	135°24.9	· · 35.0		327°23.4	16.3'	08°33.0	13.8'	54.4'
22	150°25.0	35.9		341°58.7	16.3'	08°46.8	13.8'	54.4'
23	165°25.2	36.7		356°34.0	16.3'	09°00.6	13.8'	54.4'
SD = 15.9' $d = 0.8'$				SD = 14.8'				
Tue	GHA	Dec		GHA	ν	Dec	d	HP
0	180°25.3	N12°37.5		11°09.3	16.2'	S09°14.4	13.8'	54.5'
1	195°25.4	38.3		25°44.5	16.2'	09°28.2	13.7'	54.5'
2	210°25.5	39.2		40°19.6	16.1'	09°41.9	13.7'	54.5'
3	225°25.6	· · 40.0		54°54.7	16.0'	09°55.6	13.7'	54.5'
4	240°25.7	40.8		69°29.8	16.0'	10°09.3	13.6'	54.5'
5	255°25.9	41.7		84°04.8	15.9'	10°22.9	13.6'	54.5'
6	270°26.0	N12°42.5		98°39.7	15.9'	S10°36.5	13.6'	54.5'
7	285°26.1	43.3		113°14.6	15.8'	10°50.1	13.5'	54.5'
8	300°26.2	44.1		127°49.4	15.8'	11°03.6	13.5'	54.6'
9	315°26.3	· · 45.0		142°24.2	15.7'	11°17.1	13.5'	54.6'
10	330°26.4	45.8		156°58.9	15.7'	11°30.6	13.4'	54.6'
11	345°26.6	46.6		171°33.6	15.6'	11°44.0	13.4'	54.6'
12	0°26.7	N12°47.4		186°08.2	15.5'	S11°57.4	13.4'	54.6'
13	15°26.8	48.3		200°42.7	15.5'	12°10.8	13.3'	54.6'
14	30°26.9	49.1		215°17.2	15.4'	12°24.1	13.3'	54.6'
15	45°27.0	· · 49.9		229°51.6	15.4'	12°37.4	13.2'	54.7'
16	60°27.1	50.7		244°26.0	15.3'	12°50.6	13.2'	54.7'
17	75°27.2	51.6		259°00.3	15.2'	13°03.8	13.1'	54.7'
18	90°27.3	N12°52.4		273°34.5	15.2'	S13°16.9	13.1'	54.7'
19	105°27.5	53.2		288°08.7	15.1'	13°30.0	13.1'	54.7'
20	120°27.6	54.0		302°42.8	15.0'	13°43.1	13.0'	54.7'
21	135°27.7	· · 54.9		317°16.8	15.0'	13°56.1	13.0'	54.7'
22	150°27.8	55.7		331°50.7	14.9'	14°09.0	12.9'	54.8'
23	165°27.9	56.5		346°24.6	14.8'	14°21.9	12.9'	54.8'
SD = 15.9' $d = 0.8'$				SD = 14.9'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	—:—	02:54	21:08	—:—	—:—
N 70°	—:—	01:35	03:18	20:42	22:30	—:—
68°	—:—	02:16	03:36	20:23	21:46	—:—
66°	—:—	02:43	03:51	20:08	21:18	—:—
64°	01:23	03:03	04:03	19:56	20:57	22:42
62°	01:59	03:20	04:13	19:46	20:40	22:02
60°	02:24	03:33	04:22	19:37	20:26	21:36
N 58°	02:44	03:44	04:29	19:29	20:14	21:16
56°	02:59	03:54	04:36	19:22	20:04	21:00
54°	03:12	04:03	04:42	19:16	19:55	20:46
52°	03:24	04:10	04:48	19:10	19:48	20:35
50°	03:33	04:17	04:52	19:05	19:41	20:25
45°	03:53	04:32	05:03	18:55	19:26	20:05
N 40°	04:09	04:43	05:12	18:46	19:15	19:49
35°	04:21	04:52	05:19	18:38	19:05	19:37
30°	04:31	05:01	05:25	18:32	18:57	19:26
20°	04:47	05:14	05:37	18:21	18:43	19:10
N 10°	04:59	05:25	05:46	18:11	18:32	18:58
0°	05:09	05:34	05:55	18:02	18:23	18:48
S 10°	05:18	05:42	06:04	17:53	18:14	18:39
20°	05:25	05:51	06:13	17:43	18:06	18:32
30°	05:31	05:59	06:24	17:33	17:57	18:25
35°	05:34	06:04	06:30	17:27	17:53	18:22
40°	05:37	06:09	06:36	17:20	17:48	18:19
45°	05:40	06:14	06:44	17:12	17:42	18:16
S 50°	05:42	06:20	06:54	17:02	17:36	18:14
52°	05:44	06:23	06:58	16:58	17:33	18:12
54°	05:45	06:26	07:03	16:53	17:30	18:11
56°	05:46	06:29	07:08	16:48	17:27	18:10
58°	05:47	06:33	07:14	16:42	17:23	18:09
S 60°	05:48	06:37	07:21	16:35	17:19	18:08

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	16:50	18:52	21:13	04:17	03:45	03:08
N 70°	16:47	18:39	20:43	04:16	03:51	03:24
68°	16:45	18:28	20:21	04:15	03:56	03:36
66°	16:43	18:20	20:04	04:14	04:01	03:46
64°	16:41	18:13	19:50	04:13	04:04	03:55
62°	16:40	18:07	19:38	04:12	04:07	04:02
60°	16:39	18:02	19:28	04:12	04:10	04:09
N 58°	16:38	17:57	19:20	04:11	04:13	04:15
56°	16:37	17:53	19:12	04:11	04:15	04:20
54°	16:36	17:49	19:06	04:10	04:17	04:24
52°	16:35	17:46	19:00	04:10	04:19	04:29
50°	16:34	17:43	18:54	04:10	04:21	04:32
45°	16:33	17:37	18:43	04:09	04:24	04:41
N 40°	16:31	17:31	18:33	04:08	04:27	04:48
35°	16:30	17:27	18:25	04:08	04:30	04:54
30°	16:29	17:23	18:18	04:07	04:32	04:59
20°	16:28	17:16	18:06	04:06	04:37	05:08
N 10°	16:26	17:10	17:56	04:05	04:40	05:16
0°	16:25	17:04	17:46	04:05	04:44	05:24
S 10°	16:24	16:59	17:36	04:04	04:47	05:32
20°	16:22	16:53	17:26	04:03	04:51	05:40
30°	16:21	16:46	17:14	04:02	04:55	05:49
35°	16:20	16:43	17:07	04:01	04:57	05:55
40°	16:19	16:38	17:00	04:01	05:00	06:01
45°	16:18	16:33	16:51	04:00	05:03	06:08
S 50°	16:16	16:28	16:40	03:59	05:07	06:17
52°	16:16	16:25	16:35	03:59	05:09	06:21
54°	16:15	16:22	16:30	03:58	05:11	06:26
56°	16:14	16:19	16:24	03:58	05:13	06:31
58°	16:13	16:15	16:17	03:57	05:16	06:36
S 60°	16:12	16:11	16:10	03:56	05:18	06:43

Day	Sun		Mer. Pass hh:mm	Moon		Age 92-99%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
21	01:18	01:24	11:59	22:34	10:15	
22	01:30	01:35	11:58	23:14	10:54	
23	01:41	01:47	11:58	23:56	11:35	

April 24, 25, 26 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars			
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	212° 30.9	190° 29.7	N07° 45.8	216° 43.4	S03° 12.1	162° 13.1	N17° 39.0	224° 43.6	S07° 06.7	Alpheratz	357° 35.7 29° 13.3		
1	227° 33.4	205° 29.3	47.0	231° 44.1	11.3	177° 15.0	39.1	239° 45.8	06.6	Ankaa	353° 08.1 -42° 10.4		
2	242° 35.8	220° 28.9	48.2	246° 44.8	10.6	192° 16.9	39.3	254° 48.0	06.5	Schedar	349° 32.3 56° 40.0		
3	257° 38.3	235° 28.5	· · 49.3	261° 45.5	· · 09.8	207° 18.8	· · 39.4	269° 50.3	· · 06.4	Diphda	348° 48.2 -17° 51.3		
4	272° 40.8	250° 28.1	50.5	276° 46.2	09.0	222° 20.7	39.6	284° 52.5	06.3	Achernar	335° 21.2 -57° 06.8		
5	287° 43.2	265° 27.7	51.7	291° 46.9	08.3	237° 22.5	39.7	299° 54.8	06.2	Hamal	327° 52.3 23° 34.5		
6	302° 45.7	280° 27.3	N07° 52.9	306° 47.5	S03° 07.5	252° 24.4	N17° 39.9	314° 57.0	S07° 06.2	Polaris	314° 49.6 89° 22.0		
7	317° 48.1	295° 26.8	54.0	321° 48.2	06.7	267° 26.3	40.0	329° 59.2	06.1	Acamar	315° 12.6 -40° 12.5		
8	332° 50.6	310° 26.4	55.2	336° 48.9	06.0	282° 28.2	40.2	345° 01.5	06.0	Menkar	314° 07.1 4° 11.0		
9	347° 53.1	325° 26.0	· · 56.4	351° 49.6	· · 05.2	297° 30.1	· · 40.3	0° 03.7	· · 05.9	Mirfak	308° 29.6 49° 56.8		
10	2° 55.5	340° 25.6	57.6	6° 50.3	04.4	312° 31.9	40.5	15° 06.0	05.8	Aldebaran	290° 40.6 16° 33.4		
11	17° 58.0	355° 25.2	58.7	21° 51.0	03.7	327° 33.8	40.7	30° 08.2	05.7	Rigel	281° 04.7 -8° 10.5		
12	33° 00.5	10° 24.8	N07° 59.9	36° 51.7	S03° 02.9	342° 35.7	N17° 40.8	45° 10.5	S07° 05.7	Capella	280° 23.1 46° 01.4		
13	48° 02.9	25° 24.4	08° 01.1	51° 52.4	02.1	357° 37.6	41.0	60° 12.7	05.6	Bellatrix	278° 23.8 6° 22.2		
14	63° 05.4	40° 24.0	02.2	66° 53.0	01.4	12° 39.4	41.1	75° 14.9	05.5	Elnath	278° 02.9 28° 37.7		
15	78° 07.9	55° 23.5	· · 03.4	81° 53.7	03° 00.6	27° 41.3	· · 41.3	90° 17.2	· · 05.4	Alnilam	275° 38.6 -1° 11.3		
16	93° 10.3	70° 23.1	04.6	96° 54.4	02° 59.8	42° 43.2	41.4	105° 19.4	05.3	Betelgeuse	270° 53.0 7° 24.6		
17	108° 12.8	85° 22.7	05.8	111° 55.1	59.1	57° 45.1	41.6	120° 21.7	05.2	Canopus	263° 52.9 -52° 42.7		
18	123° 15.3	100° 22.3	N08° 06.9	126° 55.8	S02° 58.3	72° 47.0	N17° 41.7	135° 23.9	S07° 05.1	Sirius	258° 26.9 -16° 45.1		
19	138° 17.7	115° 21.9	08.1	141° 56.5	57.5	87° 48.8	41.9	150° 26.2	05.1	Adhara	255° 06.5 -29° 00.5		
20	153° 20.2	130° 21.5	09.3	156° 57.2	56.8	102° 50.7	42.0	165° 28.4	05.0	Procyon	244° 51.6 5° 09.7		
21	168° 22.6	145° 21.0	· · 10.4	171° 57.9	· · 56.0	117° 52.6	· · 42.2	180° 30.7	· · 04.9	Pollux	243° 18.1 27° 58.1		
22	183° 25.1	160° 20.6	11.6	186° 58.6	55.2	132° 54.5	42.3	195° 32.9	04.8	Avior	234° 15.0 -59° 35.5		
23	198° 27.6	175° 20.2	12.8	201° 59.2	54.5	147° 56.4	42.5	210° 35.1	04.7	Suhail	222° 46.7 -43° 32.1		
Mer.pass. 09:48		ν -0.4' d 1.2' m -3.90		ν 0.7' d -0.8' m 1.13		ν 1.9' d 0.2' m -2.02		ν 2.2' d -0.1' m 1.07		Miaplacidus	221° 38.3 -69° 49.3		
											Alphard	217° 48.3 -8° 45.9	
											Regulus	207° 35.0 11° 50.9	
											Dubhe	193° 41.3 61° 37.4	
											Denebola	182° 25.3 14° 26.2	
											Gienah	175° 44.0 -17° 40.7	
											Acrux	173° 00.2 -63° 14.2	
											Gacrux	171° 51.9 -57° 15.1	
											Alioth	166° 13.0 55° 49.7	
											Spica	158° 22.6 -11° 17.4	
											Alkaid	152° 52.0 49° 11.5	
											Hadar	148° 36.3 -60° 29.5	
											Menkent	147° 58.0 -36° 29.5	
											Arcturus	145° 48.1 19° 03.3	
											Rigel Kent.	139° 40.5 -60° 56.2	
											Kochab	137° 18.6 74° 03.2	
											Zuben'ubi	136° 56.4 -16° 08.7	
											Alphecca	126° 03.9 26° 37.8	
											Antares	112° 16.3 -26° 29.2	
											Atria	107° 10.7 -69° 04.1	
											Sabik	102° 03.2 -15° 45.4	
											Shaula	96° 10.9 -37° 07.2	
											Rasalhague	95° 58.9 12° 32.3	
											Eltanin	90° 42.1 51° 28.8	
											Kaus Aust.	83° 33.1 -34° 22.3	
											Vega	80° 33.4 38° 48.0	
											Nunki	75° 48.4 -26° 16.0	
											Altair	62° 00.5 8° 55.7	
											Peacock	53° 06.7 -56° 39.2	
											Deneb	49° 26.3 45° 21.6	
											Enif	33° 39.5 9° 59.0	
											Al Na'ir	27° 33.9 -46° 50.5	
											Fomalhaut	15° 15.4 -29° 29.6	
											Scheat	13° 46.0 28° 12.6	
											Markab	13° 30.7 15° 19.9	
											Apr 24 Wed	SHA	Mer.pass
											Venus	337° 58.8 11:18	
											Mars	4° 12.5 09:33	
											Jupiter	309° 42.2 13:09	
											Saturn	12° 12.7 09:00	
											Apr 25 Thu	SHA	Mer.pass
											Venus	336° 49.8 11:19	
											Mars	3° 29.9 09:32	
											Jupiter	309° 28.2 13:06	
											Saturn	12° 07.3 08:56	
											Apr 26 Fri	SHA	Mer.pass
											Venus	335° 40.5 11:20	
											Mars	2° 47.3 09:30	
											Jupiter	309° 14.1 13:03	
											Saturn	12° 02.1 08:53	
											Horizontal parallax		
											Venus:	0.1	
											Mars:	0.1	
Mer.pass. 09:40		ν -0.4' d 1.2' m -3.91		ν 0.7' d -0.8' m 1.13		ν 1.9' d 0.2' m -2.01		ν 2.2' d -0.1' m 1.07					

h	Sun				Moon			
Wed	GHA	Dec	GHA	ν	Dec	d	HP	
0	180°28.0	N12°57.3	0°58.4	14.7'	S14°34.8	12.8'	54.8'	
1	195°28.1	58.1	15°32.2	14.7'	14°47.6	12.8'	54.8'	
2	210°28.2	59.0	30°05.9	14.6'	15°00.3	12.7'	54.8'	
3	225°28.4	12°59.8	44°39.5	14.5'	15°13.0	12.6'	54.8'	
4	240°28.5	13°00.6	59°13.0	14.4'	15°25.7	12.6'	54.8'	
5	255°28.6	01.4	73°46.4	14.4'	15°38.3	12.5'	54.9'	
6	270°28.7	N13°02.2	88°19.8	14.3'	S15°50.8	12.5'	54.9'	
7	285°28.8	03.1	102°53.1	14.2'	16°03.3	12.4'	54.9'	
8	300°28.9	03.9	117°26.3	14.1'	16°15.7	12.4'	54.9'	
9	315°29.0	· · 04.7	131°59.5	14.1'	16°28.0	12.3'	54.9'	
10	330°29.1	05.5	146°32.5	14.0'	16°40.3	12.2'	54.9'	
11	345°29.2	06.3	161°05.5	13.9'	16°52.6	12.2'	55.0'	
12	0°29.3	N13°07.1	175°38.4	13.8'	S17°04.7	12.1'	55.0'	
13	15°29.5	08.0	190°11.2	13.7'	17°16.8	12.0'	55.0'	
14	30°29.6	08.8	204°44.0	13.7'	17°28.9	12.0'	55.0'	
15	45°29.7	· · 09.6	219°16.6	13.6'	17°40.8	11.9'	55.0'	
16	60°29.8	10.4	233°49.2	13.5'	17°52.7	11.8'	55.0'	
17	75°29.9	11.2	248°21.7	13.4'	18°04.6	11.8'	55.1'	
18	90°30.0	N13°12.0	262°54.1	13.3'	S18°16.3	11.7'	55.1'	
19	105°30.1	12.8	277°26.4	13.2'	18°28.0	11.6'	55.1'	
20	120°30.2	13.7	291°58.7	13.1'	18°39.6	11.5'	55.1'	
21	135°30.3	· · 14.5	306°30.8	13.1'	18°51.2	11.5'	55.1'	
22	150°30.4	15.3	321°02.9	13.0'	19°02.6	11.4'	55.1'	
23	165°30.5	16.1	335°34.9	12.9'	19°14.0	11.3'	55.2'	
SD = 15.9'			d = 0.8'			SD = 14.9'		
Thu	GHA	Dec	GHA	ν	Dec	d	HP	
0	180°30.6	N13°16.9	350°06.7	12.8'	S19°25.3	11.2'	55.2'	
1	195°30.8	17.7	4°38.5	12.7'	19°36.5	11.2'	55.2'	
2	210°30.9	18.5	19°10.2	12.6'	19°47.7	11.1'	55.2'	
3	225°31.0	· · 19.3	33°41.9	12.5'	19°58.8	11.0'	55.2'	
4	240°31.1	20.2	48°13.4	12.4'	20°09.8	10.9'	55.2'	
5	255°31.2	21.0	62°44.8	12.3'	20°20.7	10.8'	55.3'	
6	270°31.3	N13°21.8	77°16.2	12.3'	S20°31.5	10.7'	55.3'	
7	285°31.4	22.6	91°47.4	12.2'	20°42.2	10.6'	55.3'	
8	300°31.5	23.4	106°18.6	12.1'	20°52.8	10.6'	55.3'	
9	315°31.6	· · 24.2	120°49.7	12.0'	21°03.4	10.5'	55.3'	
10	330°31.7	25.0	135°20.6	11.9'	21°13.9	10.4'	55.3'	
11	345°31.8	25.8	149°51.5	11.8'	21°24.2	10.3'	55.4'	
12	0°31.9	N13°26.6	164°22.3	11.7'	S21°34.5	10.2'	55.4'	
13	15°32.0	27.4	178°53.0	11.6'	21°44.7	10.1'	55.4'	
14	30°32.1	28.2	193°23.6	11.5'	21°54.8	10.0'	55.4'	
15	45°32.2	· · 29.0	207°54.1	11.4'	22°04.8	9.9'	55.4'	
16	60°32.3	29.9	222°24.5	11.3'	22°14.7	9.8'	55.5'	
17	75°32.4	30.7	236°54.8	11.2'	22°24.5	9.7'	55.5'	
18	90°32.5	N13°31.5	251°25.0	11.1'	S22°34.2	9.6'	55.5'	
19	105°32.6	32.3	265°55.2	11.0'	22°43.8	9.5'	55.5'	
20	120°32.7	33.1	280°25.2	10.9'	22°53.3	9.4'	55.5'	
21	135°32.8	· · 33.9	294°55.1	10.8'	23°02.7	9.3'	55.5'	
22	150°32.9	34.7	309°25.0	10.7'	23°11.9	9.2'	55.6'	
23	165°33.0	35.5	323°54.7	10.6'	23°21.1	9.1'	55.6'	
SD = 15.9'			d = 0.8'			SD = 15.0'		
Fri	GHA	Dec	GHA	ν	Dec	d	HP	
0	180°33.1	N13°36.3	338°24.4	10.6'	S23°30.2	9.0'	55.6'	
1	195°33.2	37.1	352°53.9	10.5'	23°39.1	8.8'	55.6'	
2	210°33.3	37.9	7°23.4	10.4'	23°48.0	8.7'	55.6'	
3	225°33.5	· · 38.7	21°52.7	10.3'	23°56.7	8.6'	55.7'	
4	240°33.6	39.5	36°22.0	10.2'	24°05.3	8.5'	55.7'	
5	255°33.7	40.3	50°51.2	10.1'	24°13.8	8.4'	55.7'	
6	270°33.8	N13°41.1	65°20.2	10.0'	S24°22.2	8.3'	55.7'	
7	285°33.9	41.9	79°49.2	9.9'	24°30.5	8.2'	55.7'	
8	300°34.0	42.7	94°18.1	9.8'	24°38.7	8.0'	55.8'	
9	315°34.1	· · 43.5	108°46.9	9.7'	24°46.7	7.9'	55.8'	
10	330°34.2	44.3	123°15.6	9.6'	24°54.6	7.8'	55.8'	
11	345°34.3	45.1	137°44.2	9.5'	25°02.4	7.7'	55.8'	
12	0°34.4	N13°45.9	152°12.7	9.4'	S25°10.1	7.5'	55.8'	
13	15°34.4	46.7	166°41.1	9.3'	25°17.6	7.4'	55.9'	
14	30°34.5	47.5	181°09.5	9.2'	25°25.0	7.3'	55.9'	
15	45°34.6	· · 48.3	195°37.7	9.1'	25°32.3	7.2'	55.9'	
16	60°34.7	49.1	210°05.8	9.1'	25°39.5	7.0'	55.9'	
17	75°34.8	49.9	224°33.9	9.0'	25°46.5	6.9'	55.9'	
18	90°34.9	N13°50.7	239°01.9	8.9'	S25°53.4	6.8'	56.0'	
19	105°35.0	51.5	253°29.7	8.8'	26°00.2	6.6'	56.0'	
20	120°35.1	52.3	267°57.5	8.7'	26°06.8	6.5'	56.0'	
21	135°35.2	· · 53.1	282°25.2	8.6'	26°13.3	6.4'	56.0'	
22	150°35.3	53.8	296°52.8	8.5'	26°19.7	6.2'	56.0'	
23	165°35.4	54.6	311°20.4	8.4'	26°25.9	6.1'	56.1'	
SD = 15.9'			d = 0.8'			SD = 15.2'		

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	—:—	02:34	21:27	—:—	—:—
N 70°	—:—	01:00	03:02	20:58	23:10	—:—
68°	—:—	01:55	03:23	20:36	22:07	—:—
66°	—:—	02:27	03:39	20:19	21:33	—:—
64°	00:53	02:50	03:53	20:05	21:09	23:16
62°	01:41	03:08	04:04	19:54	20:50	22:20
60°	02:11	03:23	04:13	19:44	20:35	21:49
N 58°	02:32	03:36	04:22	19:36	20:22	21:26
56°	02:49	03:46	04:29	19:28	20:11	21:09
54°	03:04	03:56	04:36	19:21	20:02	20:54
52°	03:16	04:04	04:41	19:16	19:53	20:42
50°	03:26	04:11	04:47	19:10	19:46	20:31
45°	03:47	04:26	04:58	18:59	19:30	20:09
N 40°	04:04	04:39	05:07	18:49	19:18	19:53
35°	04:17	04:49	05:15	18:41	19:08	19:40
30°	04:28	04:57	05:22	18:34	18:59	19:29
20°	04:45	05:12	05:34	18:22	18:44	19:11
N 10°	04:58	05:23	05:45	18:11	18:33	18:58
0°	05:09	05:33	05:54	18:01	18:23	18:47
S 10°	05:18	05:43	06:04	17:52	18:13	18:38
20°	05:26	05:52	06:14	17:41	18:04	18:30
30°	05:33	06:01	06:25	17:30	17:54	18:23
35°	05:36	06:06	06:32	17:23	17:49	18:19
40°	05:40	06:11	06:39	17:16	17:44	18:16
45°	05:43	06:18	06:48	17:07	17:38	18:12
S 50°	05:47	06:25	06:58	16:57	17:30	18:08
52°	05:48	06:28	07:03	16:52	17:27	18:07
54°	05:50	06:31	07:08	16:47	17:24	18:05
56°	05:51	06:35	07:14	16:41	17:20	18:04
58°	05:53	06:39	07:20	16:34	17:16	18:02
S 60°	05:54	06:43	07:28	16:27	17:11	18:00

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	■	■	■	02:14	■	■
N 70°	23:44	■	■	02:47	01:19	■
68°	22:36	■	■	03:10	02:28	■
66°	22:01	· · ·	00:52	03:29	03:04	01:54
64°	21:36	23:39	■	03:44	03:31	03:07
62°	21:16	23:03	· · ·	03:57	03:51	03:44
60°	21:00	22:38	· · ·	04:08	04:08	04:10
N 58°	20:47	22:18	23:50	04:17	04:22	04:31
56°	20:35	22:01	23:27	04:26	04:35	04:48
54°	20:25	21:47	23:09	04:33	04:45	05:03
52°	20:16	21:35	22:53	04:40	04:55	05:15
50°	20:08	21:24	22:40	04:46	05:03	05:26
45°	19:51	21:02	22:13	04:59	05:22	05:50
N 40°	19:38	20:44	21:51	05:10	05:37	06:09
35°	19:26	20:29	21:33	05:20	05:49	06:25
30°	19:16	20:16	21:18	05:28	06:00	06:39
20°	18:59	19:54	20:52	05:42	06:20	07:02
N 10°	18:44	19:35	20:30	05:55	06:36	07:23
0°	18:30	19:17	20:09	06:06	06:52	07:42
S 10°	18:16	19:00	19:48	06:18	07:08	08:01
20°	18:01	18:41	19:27	06:31	07:25	08:22
30°	17:45	18:20	19:01	06:46	07:45	08:46
35°	17:35	18:07	18:46	06:54	07:56	09:00
40°	17:24	17:53	18:29	07:04	08:09	09:16
45°	17:11	17:36	18:09	07:15	08:25	09:36
S 50°	16:56	17:16	17:43	07:29	08:44	10:01
52°	16:48	17:06	17:31	07:36	08:54	10:13
54°	16:40	16:55	17:17	07:43	09:04	10:26
56°	16:31	16:42	17:00	07:51	09:16	10:42
58°	16:21	16:28	16:41	08:01	09:30	11:01
S 60°	16:10	16:11	16:16	08:11	09:46	11:25

Day	Sun			Moon		
	Eqn.of Time		Mer.	Mer.Pass.		Age
	00 ^h	12 ^h	Pass	Upper	Lower	16-18
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	100-96%
24	01:52	01:57	11:58	—:—	12:18	
25	02:03	02:08	11:58	00:41	13:05	
26	02:13	02:17	11:58	01:29	13:55	

April 27, 28, 29 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	215°28.3	189°59.5	N09°09.6	217°33.1	S02°16.9	164°28.3	N17°49.8	227°25.2	S07°00.6	Alpheratz	357°35.7	29°13.3
1	230°30.8	204°59.0	10.7	232°33.8	16.1	179°30.2	50.0	242°27.5	00.6	Ankaa	353°08.1	-42°10.4
2	245°33.2	219°58.6	11.9	247°34.4	15.3	194°32.1	50.1	257°29.7	00.5	Schedar	349°32.3	56°40.0
3	260°35.7	234°58.2	•• 13.0	262°35.1	•• 14.6	209°33.9	•• 50.3	272°32.0	•• 00.4	Diphda	348°48.2	-17°51.3
4	275°38.2	249°57.8	14.2	277°35.8	13.8	224°35.8	50.4	287°34.2	00.3	Achernar	335°21.2	-57°06.8
5	290°40.6	264°57.3	15.3	292°36.5	13.0	239°37.7	50.6	302°36.5	00.2	Hamal	327°52.2	23°34.5
6	305°43.1	279°56.9	N09°16.5	307°37.2	S02°12.3	254°39.6	N17°50.7	317°38.7	S07°00.2	Polaris	314°49.4	89°22.0
7	320°45.6	294°56.5	17.6	322°37.9	11.5	269°41.4	50.9	332°41.0	00.1	Acamar	315°12.6	-40°12.5
8	335°48.0	309°56.0	18.8	337°38.6	10.7	284°43.3	51.0	347°43.2	07°00.0	Menkar	314°07.1	4°11.0
9	350°50.5	324°55.6	•• 19.9	352°39.3	•• 10.0	299°45.2	•• 51.2	2°45.5	06°59.9	Mirfak	308°29.6	49°56.8
10	5°53.0	339°55.2	21.1	7°40.0	09.2	314°47.1	51.3	17°47.7	59.8	Aldebaran	290°40.6	16°33.4
11	20°55.4	354°54.7	22.2	22°40.7	08.4	329°48.9	51.5	32°50.0	59.7	Rigel	281°04.7	-8°10.5
12	35°57.9	9°54.3	N09°23.4	37°41.4	S02°07.7	344°50.8	N17°51.6	47°52.2	S06°59.7	Capella	280°23.2	46°01.4
13	51°00.4	24°53.9	24.5	52°42.1	06.9	359°52.7	51.8	62°54.5	59.6	Bellatrix	278°23.8	6°22.2
14	66°02.8	39°53.4	25.7	67°42.8	06.1	14°54.6	51.9	77°56.7	59.5	Elnath	278°02.9	28°37.7
15	81°05.3	54°53.0	•• 26.8	82°43.4	•• 05.3	29°56.4	•• 52.1	92°59.0	•• 59.4	Alnilam	275°38.6	-1°11.3
16	96°07.7	69°52.6	28.0	97°44.1	04.6	44°58.3	52.2	108°01.2	59.3	Betelgeuse	270°53.0	7°24.6
17	111°10.2	84°52.1	29.1	112°44.8	03.8	60°00.2	52.4	123°03.5	59.3	Canopus	263°53.0	-52°42.7
18	126°12.7	99°51.7	N09°30.3	127°45.5	S02°03.0	75°02.1	N17°52.5	138°05.7	S06°59.2	Sirius	258°26.9	-16°45.1
19	141°15.1	114°51.2	31.4	142°46.2	02.3	90°03.9	52.7	153°08.0	59.1	Adhara	255°06.5	-29°00.5
20	156°17.6	129°50.8	32.5	157°46.9	01.5	105°05.8	52.8	168°10.2	59.0	Procyon	244°51.6	5°09.7
21	171°20.1	144°50.4	•• 33.7	172°47.6	•• 00.7	120°07.7	•• 52.9	183°12.5	•• 58.9	Pollux	243°18.1	27°58.1
22	186°22.5	159°49.9	34.8	187°48.3	02°00.0	135°09.6	53.1	198°14.7	58.8	Avior	234°15.1	-59°35.5
23	201°25.0	174°49.5	36.0	202°49.0	01°59.2	150°11.4	53.2	213°17.0	58.8	Suhail	222°46.7	-43°32.1
Mer.pass. 09:37		ν -0.4' d 1.2' m -3.91		ν 0.7' d -0.8' m 1.13		ν 1.9' d 0.1' m -2.01		ν 2.2' d -0.1' m 1.07				

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	216°27.5	189°49.1	N09°37.1	217°49.7	S01°58.4	165°13.3	N17°53.4	228°19.2	S06°58.7
1	231°29.9	204°48.6	38.3	232°50.4	57.7	180°15.2	53.5	243°21.5	58.6
2	246°32.4	219°48.2	39.4	247°51.1	56.9	195°17.1	53.7	258°23.7	58.5
3	261°34.8	234°47.8	•• 40.5	262°51.8	•• 56.1	210°18.9	•• 53.8	273°26.0	•• 58.4
4	276°37.3	249°47.3	41.7	277°52.5	55.4	225°20.8	54.0	288°28.2	58.4
5	291°39.8	264°46.9	42.8	292°53.2	54.6	240°22.7	54.1	303°30.5	58.3
6	306°42.2	279°46.4	N09°44.0	307°53.8	S01°53.8	255°24.6	N17°54.3	318°32.7	S06°58.2
7	321°44.7	294°46.0	45.1	322°54.5	53.1	270°26.4	54.4	333°35.0	58.1
8	336°47.2	309°45.6	46.3	337°55.2	52.3	285°28.3	54.6	348°37.2	58.0
9	351°49.6	324°45.1	•• 47.4	352°55.9	•• 51.5	300°30.2	•• 54.7	3°39.5	•• 58.0
10	6°52.1	339°44.7	48.5	7°56.6	50.8	315°32.0	54.9	18°41.7	57.9
11	21°54.6	354°44.2	49.7	22°57.3	50.0	330°33.9	55.0	33°44.0	57.8
12	36°57.0	9°43.8	N09°50.8	37°58.0	S01°49.2	345°35.8	N17°55.2	48°46.2	S06°57.7
13	51°59.5	24°43.4	51.9	52°58.7	48.5	0°37.7	55.3	63°48.5	57.6
14	67°02.0	39°42.9	53.1	67°59.4	47.7	15°39.5	55.5	78°50.7	57.6
15	82°04.4	54°42.5	•• 54.2	83°00.1	•• 46.9	30°41.4	•• 55.6	93°53.0	•• 57.5
16	97°06.9	69°42.0	55.4	98°00.8	46.2	45°43.3	55.8	108°55.2	57.4
17	112°09.3	84°41.6	56.5	113°01.5	45.4	60°45.2	55.9	123°57.5	57.3
18	127°11.8	99°41.1	N09°57.6	128°02.2	S01°44.6	75°47.0	N17°56.1	138°59.8	S06°57.2
19	142°14.3	114°40.7	58.8	143°02.9	43.8	90°48.9	56.2	154°02.0	57.2
20	157°16.7	129°40.3	09°59.9	158°03.6	43.1	105°50.8	56.4	169°04.3	57.1
21	172°19.2	144°39.8	10°01.0	173°04.3	•• 42.3	120°52.7	•• 56.5	184°06.5	•• 57.0
22	187°21.7	159°39.4	02.2	188°05.0	41.5	135°54.5	56.7	199°08.8	56.9
23	202°24.1	174°38.9	03.3	203°05.6	40.8	150°56.4	56.8	214°11.0	56.8
Mer.pass. 09:33		ν -0.4' d 1.1' m -3.91		ν 0.7' d -0.8' m 1.13		ν 1.9' d 0.1' m -2.01		ν 2.3' d -0.1' m 1.07	

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	217°26.6	189°38.5	N10°04.4	218°06.3	S01°40.0	165°58.3	N17°57.0	229°13.3	S06°56.8
1	232°29.1	204°38.0	05.6	233°07.0	39.2	181°00.1	57.1	244°15.5	56.7
2	247°31.5	219°37.6	06.7	248°07.7	38.5	196°02.0	57.3	259°17.8	56.6
3	262°34.0	234°37.1	•• 07.8	263°08.4	•• 37.7	211°03.9	•• 57.4	274°20.0	•• 56.5
4	277°36.5	249°36.7	09.0	278°09.1	36.9	226°05.8	57.6	289°22.3	56.4
5	292°38.9	264°36.3	10.1	293°09.8	36.2	241°07.6	57.7	304°24.5	56.4
6	307°41.4	279°35.8	N10°11.2	308°10.5	S01°35.4	256°09.5	N17°57.9	319°26.8	S06°56.3
7	322°43.8	294°35.4	12.4	323°11.2	34.6	271°11.4	58.0	334°29.0	56.2
8	337°46.3	309°34.9	13.5	338°11.9	33.9	286°13.2	58.2	349°31.3	56.1
9	352°48.8	324°34.5	•• 14.6	353°12.6	•• 33.1	301°15.1	•• 58.3	4°33.6	•• 56.0
10	7°51.2	339°34.0	15.8	8°13.3	32.3	316°17.0	58.4	19°35.8	56.0
11	22°53.7	354°33.6	16.9	23°14.0	31.6	331°18.9	58.6	34°38.1	55.9
12	37°56.2	9°33.1	N10°18.0	38°14.7	S01°30.8	346°20.7	N17°58.7	49°40.3	S06°55.8
13	52°58.6	24°32.7	19.1	53°15.4	30.0	1°22.6	58.9	64°42.6	55.7
14	68°01.1	39°32.2	20.3	68°16.1	29.3	16°24.5	59.0	79°44.8	55.6
15	83°03.6	54°31.8	•• 21.4	83°16.8	•• 28.5	31°26.4	•• 59.2	94°47.1	•• 55.6
16	98°06.0	69°31.3	22.5	98°17.5	27.7	46°28.2	59.3	109°49.3	55.5
17	113°08.5	84°30.9	23.6	113°18.2	26.9	61°30.1	59.5	124°51.6	55.4
18	128°11.0	99°30.4	N10°24.8	128°18.9	S01°26.2	76°32.0	N17°59.6	139°53.9	S06°55.3
19	143°13.4	114°30.0	25.9	143°19.5	25.4	91°33.8	59.8	154°56.1	55.2
20	158°15.9	129°29.5	27.0	158°20.2	24.6	106°35.7	17°59.9	169°58.4	55.2
21	173°18.3	144°29.1	•• 28.2	173°20.9	•• 23.9	121°37.6	18°00.1	185°00.6	•• 55.1
22	188°20.8	159°28.6	29.3	188°21.6	23.1	136°39.5	00.2	200°02.9	55.0
23	203°23.3	174°28.2	30.4	203°22.3	22.3	151°41.3	00.4	215°05.1	54.9
Mer.pass. 09:29		ν -0.4' d 1.1' m -3.91		ν 0.7' d -0.8' m 1.12		ν 1.9' d 0.1' m -2.01		ν 2.3' d -0.1' m 1.07	

Apr 27 Sat		SHA	Mer.pass
Venus	334°31.2	11:20	
Mars	2°04.7	09:29	
Jupiter	309°00.0	13:00	
Saturn	11°56.9	08:49	

Apr 28 Sun		SHA	Mer.pass
Venus	333°21.6	11:21	
Mars	1°22.2	09:28	
Jupiter	308°45.9	12:57	
Saturn	11°51.8	08:45	

Apr 29 Mon		SHA	Mer.pass
Venus	332°11.9	11:22	
Mars	0°39.7	09:27	
Jupiter	308°31.7	12:55	
Saturn	11°46.7	08:42	

Horizontal parallax		
Venus:	0.1	
Mars:	0.1	

h Sun				Moon			
Sat	GHA	Dec		GHA	ν	Dec	HP
0	180°35.5	N13°55.4		325°47.8	8.4'	S26°32.0	6.0'
1	195°35.6	56.2		340°15.2	8.3'	26°38.0	5.8'
2	210°35.7	57.0		354°42.4	8.2'	26°43.8	5.7'
3	225°35.8	· · 57.8		9°09.6	8.1'	26°49.5	5.5'
4	240°35.9	58.6		23°36.7	8.0'	26°55.0	5.4'
5	255°36.0	13°59.4		38°03.8	7.9'	27°00.4	5.2'
6	270°36.1	N14°00.2		52°30.7	7.9'	S27°05.6	5.1'
7	285°36.2	01.0		66°57.6	7.8'	27°10.7	4.9'
8	300°36.3	01.8		81°24.4	7.7'	27°15.7	4.8'
9	315°36.4	· · 02.6		95°51.1	7.6'	27°20.5	4.7'
10	330°36.5	03.3		110°17.7	7.6'	27°25.1	4.5'
11	345°36.6	04.1		124°44.3	7.5'	27°29.6	4.4'
12	0°36.7	N14°04.9		139°10.8	7.4'	S27°34.0	4.2'
13	15°36.8	05.7		153°37.2	7.4'	27°38.2	4.0'
14	30°36.9	06.5		168°03.5	7.3'	27°42.2	3.9'
15	45°36.9	· · 07.3		182°29.8	7.2'	27°46.1	3.7'
16	60°37.0	08.1		196°56.0	7.1'	27°49.9	3.6'
17	75°37.1	08.9		211°22.2	7.1'	27°53.4	3.4'
18	90°37.2	N14°09.7		225°48.3	7.0'	S27°56.9	3.3'
19	105°37.3	10.4		240°14.3	7.0'	28°00.1	3.1'
20	120°37.4	11.2		254°40.2	6.9'	28°03.2	3.0'
21	135°37.5	· · 12.0		269°06.1	6.8'	28°06.2	2.8'
22	150°37.6	12.8		283°32.0	6.8'	28°09.0	2.6'
23	165°37.7	13.6		297°57.8	6.7'	28°11.6	2.5'
SD = 15.9'			d = 0.8'	SD = 15.3'			

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	180°37.8	N14°14.4	312°23.5	6.7'	S28°14.1	2.3'	56.6'
1	195°37.9	15.1	326°49.2	6.6'	28°16.4	2.1'	56.6'
2	210°38.0	15.9	341°14.8	6.6'	28°18.5	2.0'	56.6'
3	225°38.0	· · 16.7	355°40.4	6.5'	28°20.5	1.8'	56.7'
4	240°38.1	17.5	10°05.9	6.5'	28°22.3	1.6'	56.7'
5	255°38.2	18.3	24°31.4	6.4'	28°24.0	1.5'	56.7'
6	270°38.3	N14°19.1	38°56.8	6.4'	S28°25.5	1.3'	56.7'
7	285°38.4	19.8	53°22.2	6.3'	28°26.8	1.2'	56.8'
8	300°38.5	20.6	67°47.5	6.3'	28°27.9	1.0'	56.8'
9	315°38.6	· · 21.4	82°12.9	6.3'	28°28.9	0.8'	56.8'
10	330°38.7	22.2	96°38.1	6.2'	28°29.7	0.6'	56.8'
11	345°38.8	23.0	111°03.4	6.2'	28°30.4	0.5'	56.8'
12	0°38.8	N14°23.7	125°28.6	6.2'	S28°30.9	0.3'	56.9'
13	15°38.9	24.5	139°53.7	6.1'	28°31.2	0.1'	56.9'
14	30°39.0	25.3	154°18.9	6.1'	28°31.3	-0.0'	56.9'
15	45°39.1	· · 26.1	168°44.0	6.1'	28°31.3	-0.2'	56.9'
16	60°39.2	26.8	183°09.1	6.1'	28°31.1	-0.4'	57.0'
17	75°39.3	27.6	197°34.1	6.0'	28°30.7	-0.5'	57.0'
18	90°39.4	N14°28.4	211°59.2	6.0'	S28°30.1	-0.7'	57.0'
19	105°39.5	29.2	226°24.2	6.0'	28°29.4	-0.9'	57.0'
20	120°39.5	30.0	240°49.2	6.0'	28°28.5	-1.1'	57.1'
21	135°39.6	· · 30.7	255°14.2	6.0'	28°27.5	-1.2'	57.1'
22	150°39.7	31.5	269°39.2	6.0'	28°26.3	-1.4'	57.1'
23	165°39.8	32.3	284°04.1	5.9'	28°24.8	-1.6'	57.1'
	SD = 15.9'	d = 0.8'			SD = 15.4'		

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	180°39.9	N14°33.1	298°29.1	5.9'	S28°23.3	-1.7'	57.2'
1	195°40.0	33.8	312°54.0	5.9'	28°21.5	-1.9'	57.2'
2	210°40.1	34.6	327°18.9	5.9'	28°19.6	-2.1'	57.2'
3	225°40.1	· · 35.4	341°43.8	5.9'	28°17.5	-2.3'	57.2'
4	240°40.2	36.1	356°08.8	5.9'	28°15.2	-2.4'	57.2'
5	255°40.3	36.9	10°33.7	5.9'	28°12.8	-2.6'	57.3'
6	270°40.4	N14°37.7	24°58.6	5.9'	S28°10.2	-2.8'	57.3'
7	285°40.5	38.5	39°23.5	5.9'	28°07.4	-3.0'	57.3'
8	300°40.6	39.2	53°48.5	5.9'	28°04.4	-3.1'	57.3'
9	315°40.6	· · 40.0	68°13.4	5.9'	28°01.3	-3.3'	57.4'
10	330°40.7	40.8	82°38.3	6.0'	27°58.0	-3.5'	57.4'
11	345°40.8	41.5	97°03.3	6.0'	27°54.5	-3.6'	57.4'
12	0°40.9	N14°42.3	111°28.2	6.0'	S27°50.9	-3.8'	57.4'
13	15°41.0	43.1	125°53.2	6.0'	27°47.1	-4.0'	57.5'
14	30°41.1	43.8	140°18.2	6.0'	27°43.1	-4.2'	57.5'
15	45°41.1	· · 44.6	154°43.2	6.0'	27°38.9	-4.3'	57.5'
16	60°41.2	45.4	169°08.3	6.0'	27°34.6	-4.5'	57.5'
17	75°41.3	46.1	183°33.3	6.1'	27°30.1	-4.7'	57.6'
18	90°41.4	N14°46.9	197°58.4	6.1'	S27°25.4	-4.8'	57.6'
19	105°41.5	47.7	212°23.5	6.1'	27°20.5	-5.0'	57.6'
20	120°41.5	48.4	226°48.6	6.1'	27°15.5	-5.2'	57.6'
21	135°41.6	· · 49.2	241°13.7	6.2'	27°10.3	-5.4'	57.7'
22	150°41.7	50.0	255°38.9	6.2'	27°05.0	-5.5'	57.7'
23	165°41.8	50.7	270°04.1	6.2'	26°59.5	-5.7'	57.7'
	SD = 15.9'	d = 0.8'			SD = 15.6'		

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	—:—	02:13	21:48	—:—	—:—
N 70°	—:—	—:—	02:46	21:13	—:—	—:—
68°	—:—	01:32	03:09	20:49	22:31	—:—
66°	—:—	02:10	03:27	20:30	21:49	—:—
64°	—:—	02:37	03:42	20:15	21:21	—:—
62°	01:21	02:57	03:55	20:02	21:00	22:41
60°	01:56	03:13	04:05	19:52	20:44	22:03
N 58°	02:20	03:27	04:14	19:42	20:30	21:38
56°	02:39	03:38	04:22	19:34	20:18	21:18
54°	02:55	03:48	04:29	19:27	20:08	21:02
52°	03:08	03:57	04:35	19:21	19:59	20:49
50°	03:19	04:05	04:41	19:15	19:51	20:37
45°	03:42	04:21	04:53	19:02	19:34	20:14
N 40°	03:59	04:34	05:03	18:52	19:21	19:57
35°	04:13	04:45	05:12	18:43	19:10	19:43
30°	04:24	04:54	05:20	18:36	19:01	19:31
20°	04:43	05:10	05:32	18:23	18:46	19:13
N 10°	04:57	05:22	05:44	18:11	18:33	18:58
0°	05:08	05:33	05:54	18:01	18:22	18:47
S 10°	05:18	05:43	06:04	17:50	18:12	18:37
20°	05:26	05:52	06:15	17:39	18:02	18:28
30°	05:34	06:03	06:27	17:27	17:52	18:20
35°	05:38	06:08	06:34	17:20	17:46	18:16
40°	05:42	06:14	06:42	17:12	17:40	18:12
45°	05:46	06:21	06:52	17:03	17:33	18:08
S 50°	05:51	06:29	07:03	16:51	17:25	18:03
52°	05:52	06:32	07:08	16:46	17:22	18:02
54°	05:54	06:36	07:14	16:40	17:18	18:00
56°	05:56	06:40	07:20	16:34	17:14	17:58
58°	05:58	06:45	07:27	16:27	17:09	17:56
S 60°	06:00	06:50	07:35	16:19	17:04	17:53

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	■	■	■	■
66°	■	■	■	■	■	■
64°	■	■	■	■	■	■
62°	01:05	■	■	03:31	■	■
60°	00:20	01:53	02:51	04:17	04:40	05:44
N 58°	· ·	01:13	02:12	04:47	05:21	06:23
56°	· ·	00:45	01:44	05:10	05:49	06:51
54°	· ·	00:23	01:23	05:29	06:11	07:12
52°	· ·	00:06	01:05	05:45	06:29	07:30
50°	23:50	· ·	00:49	05:59	06:44	07:45
45°	23:20	· ·	00:19	06:27	07:15	08:16
N 40°	22:56	23:55	· ·	06:49	07:39	08:39
35°	22:36	23:35	· ·	07:07	07:59	08:59
30°	22:19	23:18	· ·	07:23	08:16	09:15
20°	21:51	22:49	23:45	07:50	08:44	09:44
N 10°	21:27	22:25	23:22	08:13	09:09	10:08
0°	21:04	22:02	23:00	08:35	09:32	10:30
S 10°	20:42	21:39	22:39	08:57	09:55	10:52
20°	20:18	21:15	22:16	09:20	10:19	11:16
30°	19:50	20:46	21:49	09:47	10:48	11:44
35°	19:33	20:29	21:33	10:03	11:05	12:00
40°	19:14	20:10	21:15	10:22	11:24	12:19
45°	18:51	19:46	20:52	10:45	11:48	12:42
S 50°	18:22	19:15	20:24	11:14	12:19	13:11
52°	18:07	19:00	20:09	11:28	12:34	13:26
54°	17:50	18:42	19:53	11:45	12:52	13:42
56°	17:30	18:20	19:34	12:05	13:14	14:02
58°	17:05	17:53	19:10	12:30	13:42	14:26
S 60°	16:31	17:13	18:37	13:04	14:21	14:59

Day	Sun		Mer. Pass hh:mm	Moon		Age 19-21 91-76%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
27	02:22	02:27	11:58	02:22	14:50	
28	02:31	02:35	11:57	03:18	15:47	
29	02:40	02:44	11:57	04:16	16:45	

April 30, 01, 02 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	218°25.7	189°27.7	N10°31.5	218°23.0	S01°21.6	166°43.2	N18°00.5	230°07.4	S06°54.8	Alpheratz	357°35.7	29°13.3
1	233°28.2	204°27.3	32.6	233°23.7	20.8	181°45.1	00.7	245°09.6	54.8	Ankaa	353°08.1	-42°10.4
2	248°30.7	219°26.8	33.8	248°24.4	20.0	196°46.9	00.8	260°11.9	54.7	Schedar	349°32.3	56°40.0
3	263°33.1	234°26.3	· · 34.9	263°25.1	· · 19.3	211°48.8	· · 01.0	275°14.2	· · 54.6	Diphda	348°48.2	-17°51.3
4	278°35.6	249°25.9	36.0	278°25.8	18.5	226°50.7	01.1	290°16.4	54.5	Achernar	335°21.2	-57°06.8
5	293°38.1	264°25.4	37.1	293°26.5	17.7	241°52.5	01.3	305°18.7	54.4	Hamal	327°52.2	23°34.5
6	308°40.5	279°25.0	N10°38.3	308°27.2	S01°17.0	256°54.4	N18°01.4	320°20.9	S06°54.4	Polaris	314°49.3	89°22.0
7	323°43.0	294°24.5	39.4	323°27.9	16.2	271°56.3	01.6	335°23.2	54.3	Acamar	315°12.6	-40°12.5
8	338°45.5	309°24.1	40.5	338°28.6	15.4	286°58.2	01.7	350°25.4	54.2	Menkar	314°07.1	4°11.0
9	353°47.9	324°23.6	· · 41.6	353°29.3	· · 14.7	302°00.0	· · 01.9	5°27.7	· · 54.1	Mirfak	308°29.6	49°56.8
10	8°50.4	339°23.2	42.7	8°30.0	13.9	317°01.9	02.0	20°30.0	54.1	Aldebaran	290°40.6	16°33.4
11	23°52.8	354°22.7	43.9	23°30.7	13.1	332°03.8	02.1	35°32.2	54.0	Rigel	281°04.7	-8°10.5
12	38°55.3	9°22.3	N10°45.0	38°31.4	S01°12.4	347°05.6	N18°02.3	50°34.5	S06°53.9	Capella	280°23.2	46°01.4
13	53°57.8	24°21.8	46.1	53°32.1	11.6	2°07.5	02.4	65°36.7	53.8	Bellatrix	278°23.8	6°22.2
14	69°00.2	39°21.3	47.2	68°32.8	10.8	17°09.4	02.6	80°39.0	53.7	Elnath	278°02.9	28°37.7
15	84°02.7	54°20.9	· · 48.3	83°33.5	· · 10.1	32°11.3	· · 02.7	95°41.2	· · 53.7	Alnilam	275°38.6	-1°11.3
16	99°05.2	69°20.4	49.5	98°34.2	09.3	47°13.1	02.9	110°43.5	53.6	Betelgeuse	270°53.0	7°24.7
17	114°07.6	84°20.0	50.6	113°34.9	08.5	62°15.0	03.0	125°45.8	53.5	Canopus	263°53.0	-52°42.7
18	129°10.1	99°19.5	N10°51.7	128°35.6	S01°07.7	77°16.9	N18°03.2	140°48.0	S06°53.4	Sirius	258°27.0	-16°45.1
19	144°12.6	114°19.0	52.8	143°36.3	07.0	92°18.7	03.3	155°50.3	53.3	Adhara	255°06.5	-29°00.5
20	159°15.0	129°18.6	53.9	158°37.0	06.2	107°20.6	03.5	170°52.5	53.3	Procyon	244°51.6	5°09.7
21	174°17.5	144°18.1	· · 55.0	173°37.7	· · 05.4	122°22.5	· · 03.6	185°54.8	· · 53.2	Pollux	243°18.1	27°58.1
22	189°19.9	159°17.7	56.1	188°38.4	04.7	137°24.3	03.8	200°57.0	53.1	Avior	234°15.1	-59°35.5
23	204°22.4	174°17.2	57.3	203°39.0	03.9	152°26.2	03.9	215°59.3	53.0	Suhail	222°46.7	-43°32.1
Mer.pass. 09:25		ν -0.5' d 1.1' m -3.91		ν 0.7' d -0.8' m 1.12		ν 1.9' d 0.1' m -2.01		ν 2.3' d -0.1' m 1.07		Miaplacidus	221°38.4	-69°49.3
										Alphard	217°48.3	-8°45.9
										Regulus	207°35.0	11°50.9
										Dubhe	193°41.3	61°37.4
										Denebola	182°25.3	14°26.2
										Gienah	175°43.9	-17°40.7
										Acrux	173°00.2	-63°14.2
										Gacrux	171°51.9	-57°15.1
										Alioth	166°13.0	55°49.8
										Spica	158°22.6	-11°17.4
										Alkaid	152°52.0	49°11.5
										Hadar	148°36.2	-60°29.5
										Menkent	147°57.9	-36°29.5
										Arcturus	145°48.1	19°03.3
										Rigil Kent.	139°40.5	-60°56.2
										Kochab	137°18.5	74°03.3
										Zuben'ubi	136°56.3	-16°08.7
										Alphecca	126°03.9	26°37.8
										Antares	112°16.2	-26°29.2
										Atria	107°10.6	-69°04.1
										Sabik	102°03.2	-15°45.4
										Shaula	96°10.9	-37°07.2
										Rasalhague	95°58.9	12°32.4
										Eltanin	90°42.1	51°28.8
										Kaus Aust.	83°33.1	-34°22.3
										Vega	80°33.4	38°48.1
										Nunki	75°48.3	-26°16.0
										Altair	62°00.5	8°55.8
										Peacock	53°06.6	-56°39.2
										Deneb	49°26.2	45°21.7
										Enif	33°39.5	9°59.0
										Al Na'ir	27°33.8	-46°50.5
										Fomalhaut	15°15.4	-29°29.6
										Scheat	13°46.0	28°12.6
										Markab	13°30.7	15°20.0
Apr 30 Tue SHA Mer.pass												
										Venus	331°02.0	11:22
										Mars	359°57.3	09:26
										Jupiter	308°17.5	12:52
										Saturn	11°41.6	08:38
May 01 Wed SHA Mer.pass												
										Venus	329°51.9	11:23
										Mars	359°14.9	09:25
										Jupiter	308°03.2	12:49
										Saturn	11°36.7	08:35
May 02 Thu SHA Mer.pass												
										Venus	328°41.6	11:24
										Mars	358°32.5	09:24
										Jupiter	307°48.9	12:46
										Saturn	11°31.8	08:31
Horizontal parallax Venus: 0.1 Mars: 0.1												
Mer.pass. 09:17		ν -0.5' d 1.1' m -3.91		ν 0.7' d -0.8' m 1.12		ν 1.9' d 0.1' m -2.01		ν 2.3' d -0.1' m 1.07				

h Sun Moon						
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	180°41.9	N14°51.5	284°29.4	6.3'	S26°53.8	-5.9' 57.7'
1	195°41.9	52.3	298°54.6	6.3'	26°47.9	-6.0' 57.8'
2	210°42.0	53.0	313°19.9	6.3'	26°41.9	-6.2' 57.8'
3	225°42.1	· · 53.8	327°45.3	6.4'	26°35.7	-6.4' 57.8'
4	240°42.2	54.6	342°10.7	6.4'	26°29.4	-6.5' 57.8'
5	255°42.3	55.3	356°36.1	6.5'	26°22.8	-6.7' 57.9'
6	270°42.3	N14°56.1	11°01.5	6.5'	S26°16.1	-6.8' 57.9'
7	285°42.4	56.8	25°27.0	6.5'	26°09.3	-7.0' 57.9'
8	300°42.5	57.6	39°52.6	6.6'	26°02.3	-7.2' 57.9'
9	315°42.6	· · 58.4	54°18.2	6.6'	25°55.1	-7.3' 58.0'
10	330°42.7	59.1	68°43.8	6.7'	25°47.8	-7.5' 58.0'
11	345°42.7	14°59.9	83°09.5	6.7'	25°40.3	-7.7' 58.0'
12	0°42.8	N15°00.6	97°35.2	6.8'	S25°32.6	-7.8' 58.0'
13	15°42.9	01.4	112°01.0	6.8'	25°24.8	-8.0' 58.1'
14	30°43.0	02.2	126°26.8	6.9'	25°16.9	-8.1' 58.1'
15	45°43.0	· · 02.9	140°52.7	6.9'	25°08.7	-8.3' 58.1'
16	60°43.1	03.7	155°18.6	7.0'	25°00.4	-8.4' 58.1'
17	75°43.2	04.4	169°44.6	7.0'	24°52.0	-8.6' 58.2'
18	90°43.3	N15°05.2	184°10.6	7.1'	S24°43.4	-8.7' 58.2'
19	105°43.3	05.9	198°36.7	7.1'	24°34.7	-8.9' 58.2'
20	120°43.4	06.7	213°02.8	7.2'	24°25.8	-9.1' 58.2'
21	135°43.5	· · 07.5	227°29.0	7.3'	24°16.7	-9.2' 58.3'
22	150°43.6	08.2	241°55.3	7.3'	24°07.5	-9.4' 58.3'
23	165°43.6	09.0	256°21.6	7.4'	23°58.1	-9.5' 58.3'
SD = 15.9' d = 0.8'			SD = 15.7'			
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	180°43.7	N15°09.7	270°48.0	7.4'	S23°48.6	-9.7' 58.3'
1	195°43.8	10.5	285°14.4	7.5'	23°39.0	-9.8' 58.4'
2	210°43.9	11.2	299°40.9	7.5'	23°29.2	-9.9' 58.4'
3	225°43.9	· · 12.0	314°07.4	7.6'	23°19.2	-10.1' 58.4'
4	240°44.0	12.7	328°34.0	7.7'	23°09.2	-10.2' 58.4'
5	255°44.1	13.5	343°00.7	7.7'	22°58.9	-10.4' 58.5'
6	270°44.2	N15°14.2	357°27.4	7.8'	S22°48.5	-10.5' 58.5'
7	285°44.2	15.0	11°54.2	7.8'	22°38.0	-10.7' 58.5'
8	300°44.3	15.7	26°21.1	7.9'	22°27.4	-10.8' 58.5'
9	315°44.4	· · 16.5	40°48.0	8.0'	22°16.6	-10.9' 58.6'
10	330°44.4	17.2	55°14.9	8.0'	22°05.6	-11.1' 58.6'
11	345°44.5	18.0	69°42.0	8.1'	21°54.6	-11.2' 58.6'
12	0°44.6	N15°18.7	84°09.1	8.2'	S21°43.4	-11.3' 58.6'
13	15°44.7	19.5	98°36.2	8.2'	21°32.0	-11.5' 58.7'
14	30°44.7	20.2	113°03.5	8.3'	21°20.6	-11.6' 58.7'
15	45°44.8	· · 21.0	127°30.7	8.3'	21°09.0	-11.7' 58.7'
16	60°44.9	21.7	141°58.1	8.4'	20°57.2	-11.9' 58.7'
17	75°44.9	22.5	156°25.5	8.5'	20°45.4	-12.0' 58.8'
18	90°45.0	N15°23.2	170°53.0	8.5'	S20°33.4	-12.1' 58.8'
19	105°45.1	24.0	185°20.5	8.6'	20°21.2	-12.2' 58.8'
20	120°45.1	24.7	199°48.1	8.7'	20°09.0	-12.4' 58.8'
21	135°45.2	· · 25.4	214°15.8	8.7'	19°56.6	-12.5' 58.9'
22	150°45.3	26.2	228°43.5	8.8'	19°44.1	-12.6' 58.9'
23	165°45.3	26.9	243°11.3	8.8'	19°31.5	-12.7' 58.9'
SD = 15.9' d = 0.8'			SD = 15.9'			
Thu	GHA	Dec	GHA	ν	Dec	d HP
0	180°45.4	N15°27.7	257°39.1	8.9'	S19°18.8	-12.9' 58.9'
1	195°45.5	28.4	272°07.0	9.0'	19°05.9	-13.0' 58.9'
2	210°45.6	29.2	286°35.0	9.0'	18°53.0	-13.1' 59.0'
3	225°45.6	· · 29.9	301°03.0	9.1'	18°39.9	-13.2' 59.0'
4	240°45.7	30.6	315°31.1	9.2'	18°26.7	-13.3' 59.0'
5	255°45.8	31.4	329°59.3	9.2'	18°13.3	-13.4' 59.0'
6	270°45.8	N15°32.1	344°27.5	9.3'	S17°59.9	-13.5' 59.1'
7	285°45.9	32.9	358°55.8	9.3'	17°46.4	-13.7' 59.1'
8	300°46.0	33.6	13°24.1	9.4'	17°32.7	-13.8' 59.1'
9	315°46.0	· · 34.3	27°52.5	9.4'	17°19.0	-13.9' 59.1'
10	330°46.1	35.1	42°20.9	9.5'	17°05.1	-14.0' 59.2'
11	345°46.1	35.8	56°49.4	9.6'	16°51.1	-14.1' 59.2'
12	0°46.2	N15°36.6	71°18.0	9.6'	S16°37.0	-14.2' 59.2'
13	15°46.3	37.3	85°46.6	9.7'	16°22.9	-14.3' 59.2'
14	30°46.3	38.0	100°15.3	9.7'	16°08.6	-14.4' 59.3'
15	45°46.4	· · 38.8	114°44.0	9.8'	15°54.2	-14.5' 59.3'
16	60°46.5	39.5	129°12.8	9.8'	15°39.7	-14.6' 59.3'
17	75°46.5	40.2	143°41.6	9.9'	15°25.1	-14.7' 59.3'
18	90°46.6	N15°41.0	158°10.5	9.9'	S15°10.5	-14.8' 59.3'
19	105°46.7	41.7	172°39.4	10.0'	14°55.7	-14.9' 59.4'
20	120°46.7	42.4	187°08.4	10.0'	14°40.9	-14.9' 59.4'
21	135°46.8	· · 43.2	201°37.5	10.1'	14°25.9	-15.0' 59.4'
22	150°46.9	43.9	216°06.5	10.1'	14°10.9	-15.1' 59.4'
23	165°46.9	44.7	230°35.7	10.2'	13°55.7	-15.2' 59.4'
SD = 15.9' d = 0.7'			SD = 16.1'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	—:—	01:49	22:13	—:—	—:—
N 70°	—:—	—:—	02:28	21:30	—:—	—:—
68°	—:—	01:02	02:55	21:02	23:04	—:—
66°	—:—	01:53	03:16	20:41	22:07	—:—
64°	—:—	02:23	03:32	20:24	21:35	—:—
62°	00:54	02:46	03:45	20:11	21:11	23:10
60°	01:40	03:03	03:57	19:59	20:53	22:19
N 58°	02:08	03:18	04:07	19:49	20:38	21:49
56°	02:29	03:31	04:15	19:40	20:25	21:28
54°	02:46	03:41	04:23	19:33	20:14	21:10
52°	03:00	03:51	04:30	19:26	20:05	20:56
50°	03:12	03:59	04:36	19:19	19:56	20:43
45°	03:36	04:16	04:49	19:06	19:39	20:19
N 40°	03:55	04:30	05:00	18:55	19:25	20:00
35°	04:09	04:42	05:09	18:46	19:13	19:46
30°	04:21	04:52	05:17	18:38	19:03	19:33
20°	04:40	05:08	05:31	18:24	18:47	19:14
N 10°	04:55	05:21	05:43	18:12	18:33	18:59
0°	05:07	05:32	05:54	18:00	18:22	18:47
S 10°	05:18	05:43	06:05	17:49	18:11	18:36
20°	05:27	05:53	06:16	17:38	18:00	18:27
30°	05:36	06:04	06:29	17:24	17:49	18:18
35°	05:40	06:10	06:37	17:17	17:43	18:13
40°	05:45	06:17	06:45	17:08	17:36	18:09
45°	05:50	06:24	06:55	16:58	17:29	18:04
S 50°	05:55	06:33	07:07	16:46	17:20	17:59
52°	05:57	06:37	07:13	16:41	17:16	17:57
54°	05:59	06:41	07:19	16:34	17:12	17:54
56°	06:01	06:46	07:26	16:28	17:08	17:52
58°	06:04	06:51	07:33	16:20	17:03	17:49
S 60°	06:06	06:56	07:42	16:11	16:57	17:47

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	■	■	06:49	■	■	07:46
N 70°	■	■	05:22	■	■	09:11
68°	■	■	04:42	■	■	09:49
66°	■	05:02	04:14	■	07:35	10:15
64°	■	04:13	03:53	■	08:22	10:35
62°	03:53	03:42	03:35	06:44	08:53	10:52
60°	03:12	03:19	03:21	07:25	09:15	11:05
N 58°	02:44	03:00	03:08	07:52	09:34	11:16
56°	02:22	02:44	02:57	08:14	09:49	11:26
54°	02:03	02:30	02:48	08:32	10:02	11:35
52°	01:48	02:18	02:39	08:47	10:13	11:43
50°	01:34	02:07	02:31	09:00	10:23	11:50
45°	01:07	01:45	02:15	09:27	10:45	12:05
N 40°	00:45	01:27	02:01	09:48	11:02	12:17
35°	00:27	01:12	01:50	10:05	11:16	12:27
30°	00:11	00:58	01:39	10:20	11:28	12:36
20°	· · ·	00:36	01:22	10:46	11:49	12:51
N 10°	· · ·	00:16	01:06	11:08	12:07	13:05
0°	23:57	· · ·	00:52	11:28	12:24	13:17
S 10°	23:39	· · ·	00:37	11:48	12:40	13:29
20°	23:19	· · ·	00:22	12:09	12:58	13:42
30°	22:56	· · ·	00:04	12:34	13:18	13:57
35°	22:42	23:53	· · ·	12:49	13:30	14:05
40°	22:26	23:41	· · ·	13:05	13:43	14:15
45°	22:07	23:27	· · ·	13:25	13:59	14:26
S 50°	21:44	23:09	· · ·	13:50	14:18	14:39
52°	21:32	23:00	· · ·	14:02	14:27	14:45
54°	21:19	22:51	· · ·	14:15	14:37	14:52
56°	21:04	22:40	· · ·	14:31	14:48	15:00
58°	20:46	22:28	· · ·	14:49	15:01	15:08
S 60°	20:24	22:14	· · ·	15:12	15:17	15:18

Day	Sun		Mer. Pass hh:mm	Moon		Age 22-24 66-44%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
30	02:47	02:51	11:57	05:14	17:43	
01	02:55	02:58	11:57	06:11	18:38	
02	03:02	03:05	11:57	07:04	19:30	

May 03, 04, 05 UT (Fri., Sat., Sun.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	221°23.2	188°54.2	N11°51.3	219°13.3	S00°26.3	168°57.8	N18°11.1	232°50.1	S06°49.3					Alpheratz	357°35.7	29°13.3	
1	236°25.6	203°53.7		52.4	234°14.0		25.5	183°59.7		11.2	247°52.4		49.2	Ankaa	353°08.0	-42°10.4	
2	251°28.1	218°53.2		53.5	249°14.7		24.8	199°01.5		11.4	262°54.6		49.1	Schedar	349°32.2	56°40.0	
3	266°30.5	233°52.8	· ·	54.6	264°15.4	· ·	24.0	214°03.4	· ·	11.5	277°56.9	· ·	49.0	Diphda	348°48.2	-17°51.2	
4	281°33.0	248°52.3		55.7	279°16.1		23.2	229°05.3		11.7	292°59.1		49.0	Achernar	335°21.2	-57°06.8	
5	296°35.5	263°51.8		56.7	294°16.8		22.5	244°07.1		11.8	308°01.4		48.9	Hamal	327°52.2	23°34.5	
6	311°37.9	278°51.3	N11°57.8	309°17.5	S00°21.7	259°09.0	N18°12.0	323°03.7	S06°48.8					Polaris	314°49.4	89°22.0	
7	326°40.4	293°50.8	11°58.9	324°18.2		20.9		338°05.9		48.7				Acamar	315°12.6	-40°12.5	
8	341°42.9	308°50.4	12°00.0	339°18.9		20.2		353°08.2		48.7				Menkar	314°07.1	4°11.0	
9	356°45.3	323°49.9	· ·	01.1	354°19.6	· ·	19.4	8°10.5	· ·	48.6				Mirfak	308°29.6	49°56.8	
10	11°47.8	338°49.4		02.2	9°20.3		18.6	23°12.7		48.5				Aldebaran	290°40.6	16°33.4	
11	26°50.3	353°48.9		03.3	24°21.0		17.9	38°15.0		48.4				Rigel	281°04.7	-8°10.5	
12	41°52.7	8°48.4	N12°04.4	39°21.7	S00°17.1	349°20.2	N18°12.9	53°17.3	S06°48.4					Capella	280°23.2	46°01.4	
13	56°55.2	23°48.0		05.4	54°22.4		16.3	68°19.5		48.3				Bellatrix	278°23.8	6°22.3	
14	71°57.7	38°47.5		06.5	69°23.1		15.6	83°21.8		48.2				Elnath	278°02.9	28°37.7	
15	87°00.1	53°47.0	· ·	07.6	84°23.8	· ·	14.8	98°24.1	· ·	48.1				Alnilam	275°38.6	-1°11.3	
16	102°02.6	68°46.5		08.7	99°24.5		14.0	113°26.3		48.1				Betelgeuse	270°53.0	7°24.7	
17	117°05.0	83°46.0		09.8	114°25.2		13.3	128°28.6		48.0				Canopus	263°53.0	-52°42.7	
18	132°07.5	98°45.5	N12°10.9	129°25.9	S00°12.5	79°31.4	N18°13.7	143°30.9	S06°47.9					Sirius	258°27.0	-16°45.1	
19	147°10.0	113°45.0		11.9	144°26.6		11.7	158°33.1		47.8				Adhara	255°06.5	-29°00.5	
20	162°12.4	128°44.6		13.0	159°27.3		11.0	173°35.4		47.8				Procyon	244°51.6	5°09.7	
21	177°14.9	143°44.1	· ·	14.1	174°28.0	· ·	10.2	188°37.6	· ·	47.7				Pollux	243°18.2	27°58.1	
22	192°17.4	158°43.6		15.2	189°28.7		09.4	203°39.9		47.6				Avior	234°15.1	-59°35.5	
23	207°19.8	173°43.1		16.3	204°29.4		08.7	218°42.2		47.5				Suhail	222°46.8	-43°32.1	
Mer.pass. 09:13			ν -0.5' d 1.1' m -3.91			ν 0.7' d -0.8' m 1.11			ν 1.9' d 0.1' m -2.01			ν 2.3' d -0.1' m 1.07					

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec								
0	222°22.3	188°42.6	N12°17.4	219°30.1	S00°07.9	169°42.6	N18°14.6	233°44.4	S06°47.5								
1	237°24.8	203°42.1		18.4	234°30.8		07.1	248°46.7	47.4								
2	252°27.2	218°41.6		19.5	249°31.5		06.4	263°49.0	47.3								
3	267°29.7	233°41.1	· ·	20.6	264°32.2	· ·	05.6	278°51.2	· · 47.3								
4	282°32.1	248°40.7		21.7	279°32.9		04.8	293°53.5	47.2								
5	297°34.6	263°40.2		22.7	294°33.6		04.1	308°55.8	47.1								
6	312°37.1	278°39.7	N12°23.8	309°34.3	S00°03.3	259°53.8	N18°15.5	323°58.0	S06°47.0								
7	327°39.5	293°39.2		24.9	324°35.0		02.5	339°00.3	47.0								
8	342°42.0	308°38.7		26.0	339°35.7		01.8	354°02.6	46.9								
9	357°44.5	323°38.2	· ·	27.0	354°36.4	· ·	01.0	9°04.8	· · 46.8								
10	12°46.9	338°37.7		28.1	9°37.1	S00°00.2		24°07.1	46.7								
11	27°49.4	353°37.2		29.2	24°37.8	N00°00.5		39°09.4	46.7								
12	42°51.9	8°36.7	N12°30.3	39°38.5	N00°01.3	350°05.0	N18°16.3	54°11.6	S06°46.6								
13	57°54.3	23°36.2		31.3	54°39.2		02.1	69°13.9	46.5								
14	72°56.8	38°35.7		32.4	69°39.9		02.8	84°16.2	46.4								
15	87°59.3	53°35.3	· ·	33.5	84°40.6	· ·	03.6	99°18.5	· · 46.4								
16	103°01.7	68°34.8		34.6	99°41.3		04.4	114°20.7	46.3								
17	118°04.2	83°34.3		35.6	114°42.0		05.1	129°23.0	46.2								
18	133°06.6	98°33.8	N12°36.7	129°42.7	N00°05.9	80°16.2	N18°17.2	144°25.3	S06°46.1								
19	148°09.1	113°33.3		37.8	144°43.4		06.7	159°27.5	46.1								
20	163°11.6	128°32.8		38.8	159°44.1		07.4	174°29.8	46.0								
21	178°14.0	143°32.3	· ·	39.9	174°44.8	· ·	08.2	189°32.1	· · 45.9								
22	193°16.5	158°31.8		41.0	189°45.5		09.0	204°34.3	45.8								
23	208°19.0	173°31.3		42.0	204°46.2		09.7	219°36.6	45.8								
Mer.pass. 09:09			ν -0.5' d 1.1' m -3.91			ν 0.7' d -0.8' m 1.11			ν 1.9' d 0.1' m -2.01			ν 2.3' d -0.1' m 1.07					

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec								
0	223°21.4	188°30.8	N12°43.1	219°46.9	N00°10.5	170°27.4	N18°18.1	234°38.9	S06°45.7								
1	238°23.9	203°30.3		44.2	234°47.6		11.3	249°41.1	45.6								
2	253°26.4	218°29.8		45.2	249°48.3		12.0	264°43.4	45.6								
3	268°28.8	233°29.3	· ·	46.3	264°49.0	· ·	12.8	279°45.7	· · 45.5								
4	283°31.3	248°28.8		47.4	279°49.7		13.6	294°47.9	45.4								
5	298°33.8	263°28.3		48.4	294°50.4		14.3	309°50.2	45.3								
6	313°36.2	278°27.8	N12°49.5	309°51.1	N00°15.1	260°38.6	N18°18.9	324°52.5	S06°45.3								
7	328°38.7	293°27.3		50.6	324°51.8		15.9	339°54.8	45.2								
8	343°41.1	308°26.8		51.6	339°52.5		16.6	354°57.0	45.1								
9	358°43.6	323°26.3	· ·	52.7	354°53.2	· ·	17.4	9°59.3	· · 45.0								
10	13°46.1	338°25.8		53.8	9°53.9		18.2	25°01.6	45.0								
11	28°48.5	353°25.3		54.8	24°54.6		18.9	40°03.8	44.9								
12	43°51.0	8°24.8	N12°55.9	39°55.3	N00°19.7	350°49.8	N18°19.8	55°06.1	S06°44.8								
13	58°53.5	23°24.3		56.9	54°56.0		20.5	70°08.4	44.8								
14	73°55.9	38°23.8		58.0	69°56.7		21.2	85°10.6	44.7								
15	88°58.4	53°23.3	12°59.1	84°57.4	· ·	22.0		100°12.9	· · 44.6								
16	104°00.9	68°22.8	13°00.1	99°58.1		22.8		115°15.2	44.5								
17	119°03.3	83°22.3		01.2	114°58.8		23.5	130°17.5	44.5								
18	134°05.8	98°21.8	N13°02.2	129°59.5	N00°24.3	81°00.9	N18°20.7	145°19.7	S06°44.4								
19	149°08.2	113°21.3		03.3	145°00.2		25.1	160°22.0	44.3								
20	164°10.7	128°20.8		04.4	160°00.9		25.8	175°24.3	44.2								
21	179°13.2	143°20.3	· ·	05.4	175°01.6	· ·	26.6	190°26.5	· · 44.2								
22	194°15.6	158°19.8		06.5	190°02.3		27.3	205°28.8	44.1								
23	209°18.1	173°19.3		07.5	205°03.0		28.1	220°31.1	44.0								
Mer.pass. 09:05			ν -0.5' d 1.1' m -3.91			ν 0.7' d 0.8' m 1.11			ν 1.9' d 0.1' m -2.00			ν 2.3' d -0.1' m 1.07					

May 03 Fri	SHA	Mer.pass
Venus	327°31.0	11:25
Mars	357°50.1	09:23
Jupiter	307°34.6	12:43
Saturn	11°26.9	08:27

May 04 Sat	SHA	Mer.pass
Venus	326°20.3	11:26
Mars	357°07.8	09:22
Jupiter	307°20.3	12:40
Saturn	11°22.2	08:24

May 05 Sun	SHA	Mer.pass
Venus	325°09.4	11:26
Mars	356°25.5	09:20
Jupiter	307°05.9	12:37
Saturn	11°17.4	08:20

Horizontal parallax		
Venus:		0.1
Mars:		0.1

h								Moon							
Sun								Moon							
Fri	GHA	Dec	GHA	ν	Dec	d	HP								
0	180°47.0	N15°45.4	245°04.9	10.2'	S13°40.5	-15.3'	59.5'								
1	195°47.0	46.1	259°34.1	10.3'	13°25.2	-15.4'	59.5'								
2	210°47.1	46.8	274°03.4	10.3'	13°09.8	-15.5'	59.5'								
3	225°47.2	· · 47.6	288°32.7	10.4'	12°54.4	-15.5'	59.5'								
4	240°47.2	48.3	303°02.0	10.4'	12°38.8	-15.6'	59.6'								
5	255°47.3	49.0	317°31.5	10.4'	12°23.2	-15.7'	59.6'								
6	270°47.3	N15°49.8	332°00.9	10.5'	S12°07.5	-15.8'	59.6'								
7	285°47.4	50.5	346°30.4	10.5'	11°51.7	-15.9'	59.6'								
8	300°47.5	51.2	0°59.9	10.6'	11°35.9	-15.9'	59.6'								
9	315°47.5	· · 52.0	15°29.5	10.6'	11°19.9	-16.0'	59.7'								
10	330°47.6	52.7	29°59.1	10.6'	11°03.9	-16.1'	59.7'								
11	345°47.6	53.4	44°28.8	10.7'	10°47.9	-16.1'	59.7'								
12	0°47.7	N15°54.1	58°58.4	10.7'	S10°31.7	-16.2'	59.7'								
13	15°47.8	54.9	73°28.1	10.7'	10°15.5	-16.3'	59.7'								
14	30°47.8	55.6	87°57.9	10.8'	09°59.2	-16.3'	59.7'								
15	45°47.9	· · 56.3	102°27.7	10.8'	09°42.9	-16.4'	59.8'								
16	60°47.9	57.0	116°57.5	10.8'	09°26.5	-16.5'	59.8'								
17	75°48.0	57.8	131°27.3	10.9'	09°10.0	-16.5'	59.8'								
18	90°48.1	N15°58.5	145°57.2	10.9'	S08°53.5	-16.6'	59.8'								
19	105°48.1	59.2	160°27.1	10.9'	08°37.0	-16.6'	59.8'								
20	120°48.2	15°59.9	174°57.0	11.0'	08°20.3	-16.7'	59.9'								
21	135°48.2	16°00.7	189°27.0	11.0'	08°03.6	-16.7'	59.9'								
22	150°48.3	01.4	203°57.0	11.0'	07°46.9	-16.8'	59.9'								
23	165°48.3	02.1	218°27.0	11.0'	07°30.1	-16.8'	59.9'								
SD = 15.9' $d = 0.7'$								SD = 16.2'							
Sat	GHA	Dec	GHA	ν	Dec	d	HP								
0	180°48.4	N16°02.8	232°57.0	11.0'	S07°13.3	-16.9'	59.9'								
1	195°48.5	03.6	247°27.0	11.1'	06°56.4	-16.9'	59.9'								
2	210°48.5	04.3	261°57.1	11.1'	06°39.4	-17.0'	59.9'								
3	225°48.6	· · 05.0	276°27.2	11.1'	06°22.5	-17.0'	60.0'								
4	240°48.6	05.7	290°57.3	11.1'	06°05.4	-17.1'	60.0'								
5	255°48.7	06.4	305°27.4	11.1'	05°48.4	-17.1'	60.0'								
6	270°48.7	N16°07.2	319°57.5	11.1'	S05°31.3	-17.1'	60.0'								
7	285°48.8	07.9	334°27.7	11.2'	05°14.1	-17.2'	60.0'								
8	300°48.8	08.6	348°57.8	11.2'	04°56.9	-17.2'	60.0'								
9	315°48.9	· · 09.3	3°28.0	11.2'	04°39.7	-17.2'	60.0'								
10	330°48.9	10.0	17°58.2	11.2'	04°22.5	-17.3'	60.1'								
11	345°49.0	10.7	32°28.4	11.2'	04°05.2	-17.3'	60.1'								
12	0°49.1	N16°11.5	46°58.6	11.2'	S03°47.9	-17.3'	60.1'								
13	15°49.1	12.2	61°28.8	11.2'	03°30.6	-17.4'	60.1'								
14	30°49.2	12.9	75°59.0	11.2'	03°13.2	-17.4'	60.1'								
15	45°49.2	· · 13.6	90°29.2	11.2'	02°55.8	-17.4'	60.1'								
16	60°49.3	14.3	104°59.4	11.2'	02°38.4	-17.4'	60.1'								
17	75°49.3	15.0	119°29.6	11.2'	02°21.0	-17.5'	60.1'								
18	90°49.4	N16°15.7	133°59.8	11.2'	S02°03.5	-17.5'	60.2'								
19	105°49.4	16.5	148°30.0	11.2'	01°46.0	-17.5'	60.2'								
20	120°49.5	17.2	163°00.2	11.2'	01°28.6	-17.5'	60.2'								
21	135°49.5	· · 17.9	177°30.4	11.2'	01°11.0	-17.5'	60.2'								
22	150°49.6	18.6	192°00.6	11.2'	00°53.5	-17.5'	60.2'								
23	165°49.6	19.3	206°30.8	11.2'	00°36.0	-17.5'	60.2'								
SD = 15.9' $d = 0.7'$								SD = 16.3'							
Sun	GHA	Dec	GHA	ν	Dec	d	HP								
0	180°49.7	N16°20.0	221°01.0	11.2'	S00°18.5	-17.5'	60.2'								
1	195°49.7	20.7	235°31.1	11.2'	S00°00.9	-17.5'	60.2'								
2	210°49.8	21.4	250°01.3	11.1'	N00°16.6	17.6'	60.2'								
3	225°49.8	· · 22.1	264°31.4	11.1'	00°34.2	17.6'	60.2'								
4	240°49.9	22.8	279°01.6	11.1'	00°51.7	17.6'	60.2'								
5	255°49.9	23.6	293°31.7	11.1'	01°09.3	17.6'	60.2'								
6	270°50.0	N16°24.3	308°01.8	11.1'	N01°26.8	17.6'	60.3'								
7	285°50.0	25.0	322°31.9	11.1'	01°44.4	17.5'	60.3'								
8	300°50.1	25.7	337°01.9	11.0'	02°01.9	17.5'	60.3'								
9	315°50.1	· · 26.4	351°32.0	11.0'	02°19.5	17.5'	60.3'								
10	330°50.2	27.1	6°02.0	11.0'	02°37.0	17.5'	60.3'								
11	345°50.2	27.8	20°32.0	11.0'	02°54.5	17.5'	60.3'								
12	0°50.3	N16°28.5	35°02.0	11.0'	N03°12.0	17.5'	60.3'								
13	15°50.3	29.2	49°32.0	10.9'	03°29.5	17.5'	60.3'								
14	30°50.4	29.9	64°01.9	10.9'	03°47.0	17.5'	60.3'								
15	45°50.4	· · 30.6	78°31.8	10.9'	04°04.5	17.4'	60.3'								
16	60°50.4	31.3	93°01.6	10.8'	04°21.9	17.4'	60.3'								
17	75°50.5	32.0	107°31.5	10.8'	04°39.4	17.4'	60.3'								
18	90°50.5	N16°32.7	122°01.3	10.8'	N04°56.8	17.4'	60.3'								
19	105°50.6	33.4	136°31.1	10.7'	05°14.1	17.3'	60.3'								
20	120°50.6	34.1	151°00.8	10.7'	05°31.5	17.3'	60.3'								
21	135°50.7	· · 34.8	165°30.5	10.7'	05°48.8	17.3'	60.3'								
22	150°50.7	35.5	180°00.2	10.6'	06°06.1	17.3'	60.3'								
23	165°50.8	36.2	194°29.8	10.6'	06°23.3	17.2'	60.3'								
SD = 15.9' $d = 0.7'$								SD = 16.4'							
Twilight								Moonrise							
Lat.	Naut.	Civil	Sunrise	Sunset	Civil	Naut.		Lat.	Fri	Sat	Sun	Fri	Sat	Sun	
N 72°	—:—	—:—	01:20	22:44	—:—	—:—		N 72°	04:43	03:55	03:17	11:40	14:16	16:44	
N 70°	—:—	—:—	02:10	21:49	—:—	—:—		N 70°	04:22	03:47	03:17	11:59	14:20	16:38	
68°	—:—	—:—	02:41	21:16	—:—	—:—		68°	04:05	03:40	03:18	12:14	14:24	16:32	
66°	—:—	01:32	03:04	20:52	22:27	—:—		66°	03:51	03:34	03:18	12:26	14:28	16:28	
64°	—:—	02:09	03:22	20:34	21:49	—:—		64°	03:39	03:28	03:19	12:36	14:30	16:24	
62°	—:—	02:34	03:36	20:19	21:22	—:—		62°	03:29	03:24	03:19	12:44	14:33	16:21	
60°	01:22	02:54	03:49	20:06	21:02	22:37		60°	03:21	03:20	03:19	12:51	14:35	16:18	
N 58°	01:55	03:10	03:59	19:56	20:46	22:02		N 58°	03:13	03:17	03:20	12:57	14:37	16:16	
56°	02:19	03:23	04:09	19:46	20:32	21:37		56°	03:06	03:14	03:20	13:03	14:38	16:14	
54°	02:37	03:34	04:17	19:38	20:21	21:18		54°	03:00	03:11	03:20	13:08	14:40	16:12	
52°	02:52	03:44	04:24	19:31	20:10	21:03		52°	02:55	03:08	03:20	13:12	14:41	16:10	
50°	03:05	03:53	04:31	19:24	20:01	20:50		50°	02:50	03:06	03:20	13:16	14:42	16:08	
45°	03:31	04:12	04:45	19:10	19:43	20:24		45°	02:39	03:01	03:21	13:25	14:45	16:05	
N 40°	03:50	04:26	04:56	18:58	19:28	20:04		N 40°	02:30	02:56	03:21	13:32	14:47	16:02	
35°	04:06	04:38	05:06	18:48	19:16	19:49		35°	02:23	02:53	03:22	13:38	14:49	15:59	
30°	04:18	04:49	05:14	18:40	19:05	19:36		30°	02:16	02:49	03:22	13:43	14:50	15:57	
20°	04:38	05:06	05:29	18:25	18:48	19:15		20°	02:04	02:44	03:22	13:52	14:53	15:53	

May 06, 07, 08 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	224°20.6	188°18.8	N13°08.6	220°03.7	N00°28.9	171°12.1	N18°21.5	235°33.4	S06°44.0	Alpheratz	357°35.7 29°13.3
1	239°23.0	203°18.2	09.6	235°04.5	29.6	186°14.0	21.7	250°35.6	43.9	Ankaa	353°08.0 -42°10.4
2	254°25.5	218°17.7	10.7	250°05.2	30.4	201°15.9	21.8	265°37.9	43.8	Schedar	349°32.2 56°40.0
3	269°28.0	233°17.2	•• 11.7	265°05.9	•• 31.2	216°17.7	•• 22.0	280°40.2	•• 43.7	Diphda	348°48.2 -17°51.2
4	284°30.4	248°16.7	12.8	280°06.6	31.9	231°19.6	22.1	295°42.4	43.7	Achernar	335°21.1 -57°06.7
5	299°32.9	263°16.2	13.8	295°07.3	32.7	246°21.5	22.3	310°44.7	43.6	Hamal	327°52.2 23°34.5
6	314°35.4	278°15.7	N13°14.9	310°08.0	N00°33.5	261°23.3	N18°22.4	325°47.0	S06°43.5	Polaris	314°49.3 89°22.0
7	329°37.8	293°15.2	15.9	325°08.7	34.2	276°25.2	22.5	340°49.3	43.5	Acamar	315°12.6 -40°12.5
8	344°40.3	308°14.7	17.0	340°09.4	35.0	291°27.0	22.7	355°51.5	43.4	Menkar	314°07.1 4°11.0
9	359°42.7	323°14.2	•• 18.0	355°10.1	•• 35.8	306°28.9	•• 22.8	10°53.8	•• 43.3	Mirfak	308°29.6 49°56.8
10	14°45.2	338°13.7	19.1	10°10.8	36.5	321°30.8	23.0	25°56.1	43.2	Aldebaran	290°40.6 16°33.4
11	29°47.7	353°13.2	20.1	25°11.5	37.3	336°32.6	23.1	40°58.3	43.2	Rigel	281°04.7 -8°10.5
12	44°50.1	8°12.6	N13°21.2	40°12.2	N00°38.1	351°34.5	N18°23.3	56°00.6	S06°43.1	Capella	280°23.2 46°01.4
13	59°52.6	23°12.1	22.2	55°12.9	38.8	6°36.4	23.4	71°02.9	43.0	Bellatrix	278°23.8 6°22.3
14	74°55.1	38°11.6	23.3	70°13.6	39.6	21°38.2	23.6	86°05.2	43.0	Elnath	278°02.9 28°37.7
15	89°57.5	53°11.1	•• 24.3	85°14.3	•• 40.4	36°40.1	•• 23.7	101°07.4	•• 42.9	Alnilam	275°38.6 -1°11.2
16	105°00.0	68°10.6	25.4	100°15.0	41.1	51°42.0	23.8	116°09.7	42.8	Betelgeuse	270°53.0 7°24.7
17	120°02.5	83°10.1	26.4	115°15.7	41.9	66°43.8	24.0	131°12.0	42.7	Canopus	263°53.0 -52°42.7
18	135°04.9	98°09.6	N13°27.5	130°16.4	N00°42.6	81°45.7	N18°24.1	146°14.3	S06°42.7	Sirius	258°27.0 -16°45.1
19	150°07.4	113°09.1	28.5	145°17.1	43.4	96°47.5	24.3	161°16.5	42.6	Adhara	255°06.5 -29°00.5
20	165°09.8	128°08.5	29.6	160°17.8	44.2	111°49.4	24.4	176°18.8	42.5	Procyon	244°51.6 5°09.7
21	180°12.3	143°08.0	•• 30.6	175°18.5	•• 44.9	126°51.3	•• 24.6	191°21.1	•• 42.5	Pollux	243°18.2 27°58.1
22	195°14.8	158°07.5	31.6	190°19.2	45.7	141°53.1	24.7	206°23.4	42.4	Avior	234°15.2 -59°35.5
23	210°17.2	173°07.0	32.7	205°19.9	46.5	156°55.0	24.8	221°25.6	42.3	Suhail	222°46.8 -43°32.1
Mer.pass. 09:01		ν -0.5' d 1.1' m -3.91		ν 0.7' d 0.8' m 1.11		ν 1.9' d 0.1' m -2.00		ν 2.3' d -0.1' m 1.07		Miaplacidus	221°38.5 -69°49.3
Alphard 217°48.3 -8°45.9 Regulus 207°35.0 11°50.9 Dubhe 193°41.4 61°37.4 Denebola 182°25.3 14°26.2 Gienah 175°44.0 -17°40.8 Acrux 173°00.3 -63°14.2 Gacrux 171°51.9 -57°15.2 Alioth 166°13.0 55°49.8 Spica 158°22.6 -11°17.4 Alkaid 152°52.0 49°11.5 Hadar 148°36.2 -60°29.5 Menkent 147°57.9 -36°29.5 Arcturus 145°48.1 19°03.3 Rigel Kent. 139°40.5 -60°56.2 Kochab 137°18.5 74°03.3 Zuben'ubi 136°56.3 -16°08.7 Alphecca 126°03.9 26°37.8 Antares 112°16.2 -26°29.2 Atria 107°10.5 -69°04.2 Sabik 102°03.2 -15°45.4 Shaula 96°10.8 -37°07.3 Rasalhague 95°58.8 12°32.4 Eltanin 90°42.1 51°28.8 Kaus Aust. 83°33.0 -34°22.3 Vega 80°33.4 38°48.1 Nunki 75°48.3 -26°16.0 Altair 62°00.4 8°55.8 Peacock 53°06.5 -56°39.2 Deneb 49°26.1 45°21.7 Enif 33°39.4 9°59.0 Al Na'ir 27°33.7 -46°50.5 Fomalhaut 15°15.3 -29°29.6 Scheat 13°46.0 28°12.6 Markab 13°30.6 15°20.0											
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	225°19.7	188°06.5	N13°33.7	220°20.6	N00°47.2	171°56.9	N18°25.0	236°27.9	S06°42.2		
1	240°22.2	203°06.0	34.8	235°21.3	48.0	186°58.7	25.1	251°30.2	42.2		
2	255°24.6	218°05.4	35.8	250°22.0	48.8	202°00.6	25.3	266°32.5	42.1		
3	270°27.1	233°04.9	•• 36.8	265°22.7	•• 49.5	217°02.4	•• 25.4	281°34.7	•• 42.0		
4	285°29.6	248°04.4	37.9	280°23.4	50.3	232°04.3	25.6	296°37.0	42.0		
5	300°32.0	263°03.9	38.9	295°24.1	51.1	247°06.2	25.7	311°39.3	41.9		
6	315°34.5	278°03.4	N13°40.0	310°24.8	N00°51.8	262°08.0	N18°25.8	326°41.6	S06°41.8		
7	330°37.0	293°02.9	41.0	325°25.5	52.6	277°09.9	26.0	341°43.8	41.7		
8	345°39.4	308°02.3	42.0	340°26.3	53.3	292°11.8	26.1	356°46.1	41.7		
9	0°41.9	323°01.8	•• 43.1	355°27.0	•• 54.1	307°13.6	•• 26.3	11°48.4	•• 41.6		
10	15°44.3	338°01.3	44.1	10°27.7	54.9	322°15.5	26.4	26°50.7	41.5		
11	30°46.8	353°00.8	45.1	25°28.4	55.6	337°17.4	26.6	41°52.9	41.5		
12	45°49.3	8°00.2	N13°46.2	40°29.1	N00°56.4	352°19.2	N18°26.7	56°55.2	S06°41.4		
13	60°51.7	22°59.7	47.2	55°29.8	57.2	7°21.1	26.8	71°57.5	41.3		
14	75°54.2	37°59.2	48.2	70°30.5	57.9	22°22.9	27.0	86°59.8	41.3		
15	90°56.7	52°58.7	•• 49.3	85°31.2	•• 58.7	37°24.8	•• 27.1	102°02.0	•• 41.2		
16	105°59.1	67°58.2	50.3	100°31.9	00°59.5	52°26.7	27.3	117°04.3	41.1		
17	121°01.6	82°57.6	51.3	115°32.6	01°00.2	67°28.5	27.4	132°06.6	41.0		
18	136°04.1	97°57.1	N13°52.4	130°33.3	N01°01.0	82°30.4	N18°27.6	147°08.9	S06°41.0		
19	151°06.5	112°56.6	53.4	145°34.0	01.7	97°32.3	27.7	162°11.1	40.9		
20	166°09.0	127°56.1	54.4	160°34.7	02.5	112°34.1	27.8	177°13.4	40.8		
21	181°11.5	142°55.5	•• 55.5	175°35.4	•• 03.3	127°36.0	•• 28.0	192°15.7	•• 40.8		
22	196°13.9	157°55.0	56.5	190°36.1	04.0	142°37.8	28.1	207°18.0	40.7		
23	211°16.4	172°54.5	57.5	205°36.8	04.8	157°39.7	28.3	222°20.2	40.6		
Mer.pass. 08:57		ν -0.5' d 1.0' m -3.91		ν 0.7' d 0.8' m 1.10		ν 1.9' d 0.1' m -2.00		ν 2.3' d -0.1' m 1.07			
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	226°18.8	187°54.0	N13°58.5	220°37.5	N01°05.6	172°41.6	N18°28.4	237°22.5	S06°40.6		
1	241°21.3	202°53.4	13°59.6	235°38.2	06.3	187°43.4	28.6	252°24.8	40.5		
2	256°23.8	217°52.9	14°00.6	250°38.9	07.1	202°45.3	28.7	267°27.1	40.4		
3	271°26.2	232°52.4	•• 01.6	265°39.6	•• 07.9	217°47.2	•• 28.8	282°29.4	•• 40.3		
4	286°28.7	247°51.8	02.6	280°40.3	08.6	232°49.0	29.0	297°31.6	40.3		
5	301°31.2	262°51.3	03.7	295°41.0	09.4	247°50.9	29.1	312°33.9	40.2		
6	316°33.6	277°50.8	N14°04.7	310°41.7	N01°10.1	262°52.7	N18°29.3	327°36.2	S06°40.1		
7	331°36.1	292°50.3	05.7	325°42.4	10.9	277°54.6	29.4	342°38.5	40.1		
8	346°38.6	307°49.7	06.7	340°43.2	11.7	292°56.5	29.5	357°40.7	40.0		
9	1°41.0	322°49.2	•• 07.8	355°43.9	•• 12.4	307°58.3	•• 29.7	12°43.0	•• 39.9		
10	16°43.5	337°48.7	08.8	10°44.6	13.2	323°00.2	29.8	27°45.3	39.9		
11	31°46.0	352°48.1	09.8	25°45.3	14.0	338°02.1	30.0	42°47.6	39.8		
12	46°48.4	7°47.6	N14°10.8	40°46.0	N01°14.7	353°03.9	N18°30.1	57°49.9	S06°39.7		
13	61°50.9	22°47.1	11.8	55°46.7	15.5	8°05.8	30.3	72°52.1	39.7		
14	76°53.3	37°46.5	12.9	70°47.4	16.2	23°07.6	30.4	87°54.4	39.6		
15	91°55.8	52°46.0	•• 13.9	85°48.1	•• 17.0	38°09.5	•• 30.5	102°56.7	•• 39.5		
16	106°58.3	67°45.5	14.9	100°48.8	17.8	53°11.4	30.7	117°59.0	39.4		
17	122°00.7	82°44.9	15.9	115°49.5	18.5	68°13.2	30.8	133°01.2	39.4		
18	137°03.2	97°44.4	N14°16.9	130°50.2	N01°19.3	83°15.1	N18°31.0	148°03.5	S06°39.3		
19	152°05.7	112°43.9	18.0	145°50.9	20.1	98°16.9	31.1	163°05.8	39.2		
20	167°08.1	127°43.3	19.0	160°51.6	20.8	113°18.8	31.3	178°08.1	39.2		
21	182°10.6	142°42.8	•• 20.0	175°52.3	•• 21.6	128°20.7	•• 31.4	193°10.4	•• 39.1		
22	197°13.1	157°42.3	21.0	190°53.0	22.3	143°22.5	31.5	208°12.6	39.0		
23	212°15.										

Sun				Moon				
Mon	GHA	Dec		GHA	ν	Dec	d	HP
0	180°50.8	N16°36.9		208°59.4	10.6'	N06°40.6	17.2'	60.3'
1	195°50.9	37.6		223°29.0	10.5'	06°57.8	17.1'	60.3'
2	210°50.9	38.3		237°58.5	10.5'	07°14.9	17.1'	60.3'
3	225°50.9	· · 39.0		252°28.0	10.4'	07°32.0	17.1'	60.3'
4	240°51.0	39.7		266°57.4	10.4'	07°49.1	17.0'	60.3'
5	255°51.0	40.4		281°26.8	10.3'	08°06.1	17.0'	60.3'
6	270°51.1	N16°41.1		295°56.1	10.3'	N08°23.1	16.9'	60.3'
7	285°51.1	41.8		310°25.4	10.2'	08°40.0	16.9'	60.3'
8	300°51.2	42.5		324°54.7	10.2'	08°56.9	16.8'	60.3'
9	315°51.2	· · 43.2		339°23.9	10.1'	09°13.7	16.8'	60.3'
10	330°51.2	43.9		353°53.0	10.1'	09°30.5	16.7'	60.3'
11	345°51.3	44.6		8°22.1	10.0'	09°47.2	16.7'	60.3'
12	0°51.3	N16°45.3		22°51.1	10.0'	N10°03.9	16.6'	60.3'
13	15°51.4	46.0		37°20.1	9.9'	10°20.5	16.5'	60.3'
14	30°51.4	46.7		51°49.1	9.9'	10°37.1	16.5'	60.3'
15	45°51.4	· · 47.4		66°18.0	9.8'	10°53.5	16.4'	60.2'
16	60°51.5	48.0		80°46.8	9.8'	11°10.0	16.4'	60.2'
17	75°51.5	48.7		95°15.6	9.7'	11°26.3	16.3'	60.2'
18	90°51.6	N16°49.4		109°44.3	9.7'	N11°42.6	16.2'	60.2'
19	105°51.6	50.1		124°12.9	9.6'	11°58.8	16.1'	60.2'
20	120°51.6	50.8		138°41.5	9.5'	12°15.0	16.1'	60.2'
21	135°51.7	· · 51.5		153°10.1	9.5'	12°31.0	16.0'	60.2'
22	150°51.7	52.2		167°38.5	9.4'	12°47.0	15.9'	60.2'
23	165°51.8	52.9		182°06.9	9.4'	13°03.0	15.8'	60.2'
SD = 15.8' d = 0.7'				SD = 16.4'				
Tue	GHA	Dec		GHA	ν	Dec	d	HP
0	180°51.8	N16°53.6		196°35.3	9.3'	N13°18.8	15.8'	60.2'
1	195°51.8	54.2		211°03.6	9.2'	13°34.6	15.7'	60.2'
2	210°51.9	54.9		225°31.8	9.2'	13°50.2	15.6'	60.1'
3	225°51.9	· · 55.6		240°00.0	9.1'	14°05.8	15.5'	60.1'
4	240°51.9	56.3		254°28.1	9.0'	14°21.3	15.4'	60.1'
5	255°52.0	57.0		268°56.1	9.0'	14°36.7	15.3'	60.1'
6	270°52.0	N16°57.7		283°24.0	8.9'	N14°52.1	15.2'	60.1'
7	285°52.1	58.4		297°51.9	8.8'	15°07.3	15.1'	60.1'
8	300°52.1	59.0		312°19.8	8.8'	15°22.4	15.0'	60.1'
9	315°52.1	16°59.7		326°47.5	8.7'	15°37.5	14.9'	60.0'
10	330°52.2	17°00.4		341°15.2	8.6'	15°52.4	14.8'	60.0'
11	345°52.2	01.1		355°42.8	8.5'	16°07.3	14.7'	60.0'
12	0°52.2	N17°01.8		10°10.4	8.5'	N16°22.0	14.6'	60.0'
13	15°52.3	02.4		24°37.9	8.4'	16°36.6	14.5'	60.0'
14	30°52.3	03.1		39°05.3	8.3'	16°51.2	14.4'	60.0'
15	45°52.3	· · 03.8		53°32.6	8.3'	17°05.6	14.3'	60.0'
16	60°52.4	04.5		67°59.9	8.2'	17°19.9	14.2'	59.9'
17	75°52.4	05.2		82°27.1	8.1'	17°34.1	14.1'	59.9'
18	90°52.4	N17°05.8		96°54.2	8.1'	N17°48.2	14.0'	59.9'
19	105°52.5	06.5		111°21.2	8.0'	18°02.2	13.9'	59.9'
20	120°52.5	07.2		125°48.2	7.9'	18°16.0	13.7'	59.9'
21	135°52.5	· · 07.9		140°15.1	7.8'	18°29.8	13.6'	59.8'
22	150°52.6	08.6		154°42.0	7.8'	18°43.4	13.5'	59.8'
23	165°52.6	09.2		169°08.7	7.7'	18°56.9	13.4'	59.8'
SD = 15.8' d = 0.7'				SD = 16.4'				
Wed	GHA	Dec		GHA	ν	Dec	d	HP
0	180°52.6	N17°09.9		183°35.4	7.6'	N19°10.3	13.3'	59.8'
1	195°52.7	10.6		198°02.0	7.5'	19°23.5	13.1'	59.8'
2	210°52.7	11.3		212°28.6	7.5'	19°36.6	13.0'	59.7'
3	225°52.7	· · 11.9		226°55.0	7.4'	19°49.6	12.9'	59.7'
4	240°52.8	12.6		241°21.4	7.3'	20°02.5	12.7'	59.7'
5	255°52.8	13.3		255°47.8	7.3'	20°15.2	12.6'	59.7'
6	270°52.8	N17°14.0		270°14.0	7.2'	N20°27.9	12.5'	59.7'
7	285°52.9	14.6		284°40.2	7.1'	20°40.3	12.3'	59.6'
8	300°52.9	15.3		299°06.3	7.0'	20°52.6	12.2'	59.6'
9	315°52.9	· · 16.0		313°32.4	7.0'	21°04.8	12.1'	59.6'
10	330°53.0	16.6		327°58.4	6.9'	21°16.9	11.9'	59.6'
11	345°53.0	17.3		342°24.3	6.8'	21°28.8	11.8'	59.5'
12	0°53.0	N17°18.0		356°50.1	6.8'	N21°40.6	11.6'	59.5'
13	15°53.0	18.6		11°15.9	6.7'	21°52.2	11.5'	59.5'
14	30°53.1	19.3		25°41.6	6.6'	22°03.7	11.3'	59.5'
15	45°53.1	· · 20.0		40°07.2	6.6'	22°15.0	11.2'	59.4'
16	60°53.1	20.7		54°32.8	6.5'	22°26.2	11.0'	59.4'
17	75°53.2	21.3		68°58.3	6.4'	22°37.2	10.9'	59.4'
18	90°53.2	N17°22.0		83°23.7	6.4'	N22°48.1	10.7'	59.4'
19	105°53.2	22.7		97°49.1	6.3'	22°58.8	10.6'	59.3'
20	120°53.2	23.3		112°14.4	6.2'	23°09.4	10.4'	59.3'
21	135°53.3	· · 24.0		126°39.6	6.2'	23°19.8	10.3'	59.3'
22	150°53.3	24.6		141°04.8	6.1'	23°30.1	10.1'	59.3'
23	165°53.3	25.3		155°30.0	6.1'	23°40.2	9.9'	59.2'
SD = 15.8' d = 0.7'				SD = 16.3'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	—:—	00:37	23:51	—:—	—:—
N 70°	—:—	—:—	01:50	22:09	—:—	—:—
68°	—:—	—:—	02:27	21:30	—:—	—:—
66°	—:—	01:08	02:52	21:04	22:53	—:—
64°	—:—	01:53	03:12	20:44	22:04	—:—
62°	—:—	02:22	03:28	20:27	21:34	—:—
60°	01:01	02:44	03:41	20:14	21:12	23:00
N 58°	01:42	03:01	03:52	20:02	20:54	22:15
56°	02:08	03:15	04:02	19:52	20:39	21:48
54°	02:28	03:28	04:11	19:43	20:27	21:27
52°	02:45	03:38	04:19	19:36	20:16	21:10
50°	02:59	03:48	04:26	19:28	20:06	20:56
45°	03:26	04:07	04:40	19:13	19:47	20:29
N 40°	03:46	04:23	04:53	19:01	19:31	20:08
35°	04:02	04:35	05:03	18:51	19:18	19:52
30°	04:15	04:46	05:12	18:42	19:07	19:38
20°	04:37	05:04	05:27	18:26	18:49	19:17
N 10°	04:53	05:19	05:41	18:12	18:35	19:00
0°	05:06	05:31	05:53	18:00	18:22	18:47
S 10°	05:18	05:43	06:05	17:48	18:10	18:35
20°	05:29	05:55	06:18	17:35	17:58	18:24
30°	05:39	06:08	06:33	17:20	17:45	18:13
35°	05:45	06:15	06:42	17:11	17:38	18:08
40°	05:50	06:23	06:51	17:01	17:30	18:02
45°	05:56	06:31	07:02	16:50	17:21	17:57
S 50°	06:02	06:41	07:16	16:36	17:11	17:50
52°	06:05	06:46	07:22	16:30	17:07	17:47
54°	06:08	06:51	07:29	16:23	17:02	17:44
56°	06:11	06:56	07:37	16:15	16:56	17:41
58°	06:14	07:02	07:46	16:06	16:50	17:38
S 60°	06:18	07:09	07:56	15:56	16:43	17:34

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	02:37	01:45	☐	19:26	☐	☐
N 70°	02:47	02:10	00:55	19:04	22:13	☐
68°	02:56	02:29	01:47	18:47	21:22	☐
66°	03:03	02:45	02:20	18:33	20:50	☐ · ·
64°	03:09	02:58	02:44	18:22	20:27	22:50
62°	03:14	03:09	03:03	18:12	20:09	22:14
60°	03:19	03:18	03:19	18:04	19:54	21:48
N 58°	03:23	03:27	03:33	17:57	19:42	21:28
56°	03:26	03:34	03:45	17:51	19:31	21:12
54°	03:30	03:41	03:55	17:45	19:21	20:57
52°	03:33	03:47	04:05	17:40	19:13	20:45
50°	03:35	03:52	04:13	17:36	19:05	20:34
45°	03:41	04:04	04:31	17:26	18:49	20:12
N 40°	03:46	04:14	04:45	17:18	18:36	19:54
35°	03:51	04:22	04:58	17:11	18:25	19:39
30°	03:55	04:30	05:09	17:05	18:15	19:26
20°	04:01	04:43	05:28	16:55	17:58	19:04
N 10°	04:08	04:54	05:44	16:46	17:44	18:45
0°	04:13	05:05	06:00	16:38	17:31	18:27
S 10°	04:19	05:16	06:15	16:29	17:17	18:09
20°	04:25	05:28	06:32	16:20	17:03	17:50
30°	04:33	05:42	06:52	16:10	16:47	17:28
35°	04:37	05:50	07:04	16:04	16:38	17:16
40°	04:42	05:59	07:17	15:58	16:27	17:01
45°	04:47	06:09	07:33	15:50	16:15	16:44
S 50°	04:54	06:23	07:52	15:41	16:00	16:23
52°	04:57	06:29	08:02	15:37	15:53	16:13
54°	05:01	06:36	08:12	15:32	15:45	16:02
56°	05:04	06:43	08:24	15:27	15:36	15:49
58°	05:09	06:52	08:38	15:22	15:27	15:35
S 60°	05:14	07:02	08:55	15:16	15:16	15:17

Day	Sun		Mer. Pass hh:mm	Moon		Age 28-0 6-0%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
06	03:23	03:25	11:57	10:25	22:51	
07	03:27	03:29	11:57	11:18	23:45	
08	03:31	03:32	11:56	12:13	—:—	

May 09, 10, 11 UT (Thu., Fri., Sat.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Thu	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec			
0	227°18.0	187°41.2	N14°23.0	220°54.4	N01°23.9	173°26.3	N18°31.8	238°17.2	S06°38.9				Alpheratz	357°35.6	29°13.3		
1	242°20.4	202°40.7	24.0	235°55.1	24.6	188°28.1	32.0	253°19.5	38.8				Ankaa	353°08.0	-42°10.4		
2	257°22.9	217°40.1	25.0	250°55.8	25.4	203°30.0	32.1	268°21.8	38.8				Schedar	349°32.2	56°40.0		
3	272°25.4	232°39.6	· · 26.1	265°56.6	· · 26.1	218°31.8	· · 32.2	283°24.0	· · 38.7				Diphda	348°48.1	-17°51.2		
4	287°27.8	247°39.0	27.1	280°57.3	26.9	233°33.7	32.4	298°26.3	38.6				Achernar	335°21.1	-57°06.7		
5	302°30.3	262°38.5	28.1	295°58.0	27.7	248°35.6	32.5	313°28.6	38.6				Hamal	327°52.2	23°34.5		
6	317°32.8	277°38.0	N14°29.1	310°58.7	N01°28.4	263°37.4	N18°32.7	328°30.9	S06°38.5				Polaris	314°48.8	89°22.0		
7	332°35.2	292°37.4	30.1	325°59.4	29.2	278°39.3	32.8	343°33.2	38.4				Acamar	315°12.6	-40°12.5		
8	347°37.7	307°36.9	31.1	341°00.1	30.0	293°41.1	33.0	358°35.4	38.4				Menkar	314°07.1	4°11.0		
9	2°40.2	322°36.3	· · 32.1	356°00.8	· · 30.7	308°43.0	· · 33.1	13°37.7	· · 38.3				Mirfak	308°29.6	49°56.8		
10	17°42.6	337°35.8	33.1	11°01.5	31.5	323°44.9	33.2	28°40.0	38.2				Aldebaran	290°40.6	16°33.4		
11	32°45.1	352°35.3	34.1	26°02.2	32.2	338°46.7	33.4	43°42.3	38.1				Rigel	281°04.7	-8°10.5		
12	47°47.6	7°34.7	N14°35.1	41°02.9	N01°33.0	353°48.6	N18°33.5	58°44.6	S06°38.1				Capella	280°23.2	46°01.4		
13	62°50.0	22°34.2	36.1	56°03.6	33.8	8°50.5	33.7	73°46.9	38.0				Bellatrix	278°23.8	6°22.3		
14	77°52.5	37°33.6	37.1	71°04.3	34.5	23°52.3	33.8	88°49.1	37.9				Elnath	278°02.9	28°37.7		
15	92°54.9	52°33.1	· · 38.1	86°05.0	· · 35.3	38°54.2	· · 33.9	103°51.4	· · 37.9				Alnilam	275°38.6	-1°11.2		
16	107°57.4	67°32.5	39.2	101°05.7	36.0	53°56.0	34.1	118°53.7	37.8				Betelgeuse	270°53.0	7°24.7		
17	122°59.9	82°32.0	40.2	116°06.4	36.8	68°57.9	34.2	133°56.0	37.7				Canopus	263°53.0	-52°42.7		
18	138°02.3	97°31.5	N14°41.2	131°07.1	N01°37.6	83°59.8	N18°34.4	148°58.3	S06°37.7				Sirius	258°27.0	-16°45.1		
19	153°04.8	112°30.9	42.2	146°07.8	38.3	99°01.6	34.5	164°00.5	37.6				Adhara	255°06.6	-29°00.5		
20	168°07.3	127°30.4	43.2	161°08.5	39.1	114°03.5	34.6	179°02.8	37.5				Procyon	244°51.6	5°09.7		
21	183°09.7	142°29.8	· · 44.2	176°09.3	· · 39.9	129°05.3	· · 34.8	194°05.1	· · 37.5				Pollux	243°18.2	27°58.1		
22	198°12.2	157°29.3	45.2	191°10.0	40.6	144°07.2	34.9	209°07.4	37.4				Avior	234°15.2	-59°35.5		
23	213°14.7	172°28.7	46.2	206°10.7	41.4	159°09.1	35.1	224°09.7	37.3				Suhail	222°46.8	-43°32.1		
Mer.pass. 08:49			ν -0.5' d 1.0' m -3.91			ν 0.7' d 0.8' m 1.10			ν 1.9' d 0.1' m -2.00			ν 2.3' d -0.1' m 1.06					

Fri	GHA	Dec	GHA	Dec		GHA	Dec		GHA	Dec		GHA	Dec				
0	228°17.1	187°28.2	N14°47.2	221°11.4	N01°42.1	174°10.9	N18°35.2	239°12.0	S06°37.3			239°12.0	S06°37.3		Alphard	217°48.3	-8°45.9
1	243°19.6	202°27.6	48.2	236°12.1	42.9	189°12.8	35.4	254°14.2	37.2			254°14.2	37.2		Regulus	207°35.0	11°50.9
2	258°22.1	217°27.1	49.1	251°12.8	43.7	204°14.6	35.5	269°16.5	37.1			269°16.5	37.1		Dubhe	193°41.4	61°37.4
3	273°24.5	232°26.5	· · 50.1	266°13.5	· · 44.4	219°16.5	· · 35.6	284°18.8	· · 37.1			284°18.8	· · 37.1		Denebola	182°25.3	14°26.2
4	288°27.0	247°26.0	51.1	281°14.2	45.2	234°18.4	35.8	299°21.1	37.0			299°21.1	37.0		Gienah	175°44.0	-17°40.8
5	303°29.4	262°25.4	52.1	296°14.9	45.9	249°20.2	35.9	314°23.4	36.9			314°23.4	36.9		Acrux	173°00.3	-63°14.3
6	318°31.9	277°24.9	N14°53.1	311°15.6	N01°46.7	264°22.1	N18°36.1	329°25.7	S06°36.9			329°25.7	S06°36.9		Gacrux	171°51.9	-57°15.2
7	333°34.4	292°24.3	54.1	326°16.3	47.5	279°23.9	36.2	344°27.9	36.8			344°27.9	36.8		Alioth	166°13.0	55°49.8
8	348°36.8	307°23.8	55.1	341°17.0	48.2	294°25.8	36.3	359°30.2	36.7			359°30.2	36.7		Spica	158°22.6	-11°17.4
9	3°39.3	322°23.2	· · 56.1	356°17.7	· · 49.0	309°27.7	· · 36.5	14°32.5	· · 36.7			14°32.5	· · 36.7		Alkaid	152°52.0	49°11.6
10	18°41.8	337°22.7	57.1	11°18.4	49.7	324°29.5	36.6	29°34.8	36.6			29°34.8	36.6		Hadar	148°36.2	-60°29.5
11	33°44.2	352°22.1	58.1	26°19.1	50.5	339°31.4	36.8	44°37.1	36.5			44°37.1	36.5		Menkent	147°57.9	-36°29.5
12	48°46.7	7°21.6	N14°59.1	41°19.8	N01°51.3	354°33.3	N18°36.9	59°39.4	S06°36.5			59°39.4	S06°36.5		Arcturus	145°48.1	19°03.3
13	63°49.2	22°21.0	15°00.1	56°20.6	52.0	9°35.1	37.0	74°41.6	36.4			74°41.6	36.4		Rigil Kent.	139°40.5	-60°56.2
14	78°51.6	37°20.5	01.1	71°21.3	52.8	24°37.0	37.2	89°43.9	36.3			89°43.9	36.3		Kochab	137°18.6	74°03.3
15	93°54.1	52°19.9	· · 02.1	86°22.0	· · 53.5	39°38.8	· · 37.3	104°46.2	· · 36.3			104°46.2	· · 36.3		Zuben'ubi	136°56.3	-16°08.7
16	108°56.6	67°19.4	03.0	101°22.7	54.3	54°40.7	37.5	119°48.5	36.2			119°48.5	36.2		Alphecca	126°03.9	26°37.9
17	123°59.0	82°18.8	04.0	116°23.4	55.1	69°42.6	37.6	134°50.8	36.1			134°50.8	36.1		Antares	112°16.2	-26°29.2
18	139°01.5	97°18.3	N15°05.0	131°24.1	N01°55.8	84°44.4	N18°37.7	149°53.1	S06°36.1			149°53.1	S06°36.1		Atria	107°10.5	-69°04.2
19	154°03.9	112°17.7	06.0	146°24.8	56.6	99°46.3	37.9	164°55.3	36.0			164°55.3	36.0		Sabik	102°03.1	-15°45.4
20	169°06.4	127°17.1	07.0	161°25.5	57.3	114°48.1	38.0	179°57.6	35.9			179°57.6	35.9		Shaula	96°10.8	-37°07.3
21	184°08.9	142°16.6	· · 08.0	176°26.2	· · 58.1	129°50.0	· · 38.2	194°59.9	· · 35.9			194°59.9	· · 35.9		Rasalhague	95°58.8	12°32.4
22	199°11.3	157°16.0	09.0	191°26.9	58.9	144°51.9	38.3	210°02.2	35.8			210°02.2	35.8		Eltanin	90°42.0	51°28.9
23	214°13.8	172°15.5	09.9	206°27.6	59.6	159°53.7	38.4	225°04.5	35.7			225°04.5	35.7		Kaus Aust.	83°33.0	-34°22.3
Mer.pass. 08:45			ν -0.5' d 1.0' m -3.92			ν 0.7' d 0.8' m 1.10			ν 1.9' d 0.1' m -2.00			ν 2.3' d -0.1' m 1.06					

Sat	GHA	Dec	GHA	Dec		GHA	Dec		GHA	Dec		GHA	Dec				
0	229°16.3	187°14.9	N15°10.9	221°28.3	N02°00.4	174°55.6	N18°38.6	240°06.8	S06°35.7			240°06.8	S06°35.7		May 09 Thu	SHA	Mer.pass
1	244°18.7	202°14.4	11.9	236°29.0	01.1	189°57.4	38.7	255°09.1	35.6			255°09.1	35.6		Venus	320°23.2	11:30
2	259°21.2	217°13.8	12.9	251°29.7	01.9	204°59.3	38.9	270°11.3	35.5			270°11.3	35.5		Mars	353°36.4	09:16
3	274°23.7	232°13.2	· · 13.9	266°30.4	· · 02.6	220°01.2	· · 39.0	285°13.6	· · 35.5			285°13.6	· · 35.5		Jupiter	306°08.3	12:25
4	289°26.1	247°12.7	14.8	281°31.2	03.4	235°03.0	39.1	300°15.9	35.4			300°15.9	35.4		Saturn	10°59.2	08:06
5	304°28.6	262°12.1	15.8	296°31.9	04.2	250°04.9	39.3	315°18.2	35.3			315°18.2	35.3				
6	319°31.1	277°11.6	N15°16.8	311°32.6	N02°04.9	265°06.7	N18°39.4	330°20.5	S06°35.3			330°20.5	S06°35.3				

h	Sun				Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP	
0	180°53.3	N17°26.0	169°55.0	6.0'	N23°50.1	9.8'	59.2'	
1	195°53.4	26.6	184°20.0	6.0'	23°59.9	9.6'	59.2'	
2	210°53.4	27.3	198°45.0	5.9'	24°09.5	9.5'	59.1'	
3	225°53.4	· · 28.0	213°09.9	5.8'	24°19.0	9.3'	59.1'	
4	240°53.4	28.6	227°34.7	5.8'	24°28.3	9.1'	59.1'	
5	255°53.5	29.3	241°59.5	5.7'	24°37.4	9.0'	59.1'	
6	270°53.5	N17°29.9	256°24.2	5.7'	N24°46.4	8.8'	59.0'	
7	285°53.5	30.6	270°48.9	5.6'	24°55.2	8.6'	59.0'	
8	300°53.5	31.3	285°13.5	5.6'	25°03.8	8.5'	59.0'	
9	315°53.6	· · 31.9	299°38.1	5.5'	25°12.3	8.3'	58.9'	
10	330°53.6	32.6	314°02.7	5.5'	25°20.6	8.1'	58.9'	
11	345°53.6	33.2	328°27.2	5.5'	25°28.7	7.9'	58.9'	
12	0°53.6	N17°33.9	342°51.6	5.4'	N25°36.6	7.8'	58.8'	
13	15°53.7	34.6	357°16.0	5.4'	25°44.4	7.6'	58.8'	
14	30°53.7	35.2	11°40.4	5.3'	25°52.0	7.4'	58.8'	
15	45°53.7	· · 35.9	26°04.7	5.3'	25°59.4	7.2'	58.7'	
16	60°53.7	36.5	40°29.0	5.3'	26°06.7	7.1'	58.7'	
17	75°53.8	37.2	54°53.3	5.2'	26°13.8	6.9'	58.7'	
18	90°53.8	N17°37.8	69°17.5	5.2'	N26°20.6	6.7'	58.6'	
19	105°53.8	38.5	83°41.7	5.2'	26°27.4	6.5'	58.6'	
20	120°53.8	39.1	98°05.9	5.1'	26°33.9	6.4'	58.6'	
21	135°53.8	· · 39.8	112°30.0	5.1'	26°40.3	6.2'	58.6'	
22	150°53.9	40.4	126°54.1	5.1'	26°46.5	6.0'	58.5'	
23	165°53.9	41.1	141°18.2	5.1'	26°52.5	5.8'	58.5'	
SD = 15.8'		d = 0.7'		SD = 16.1'				
Fri	GHA	Dec	GHA	ν	Dec	d	HP	
0	180°53.9	N17°41.7	155°42.3	5.1'	N26°58.3	5.6'	58.5'	
1	195°53.9	42.4	170°06.4	5.0'	27°03.9	5.5'	58.4'	
2	210°53.9	43.0	184°30.4	5.0'	27°09.4	5.3'	58.4'	
3	225°54.0	· · 43.7	198°54.4	5.0'	27°14.7	5.1'	58.4'	
4	240°54.0	44.3	213°18.4	5.0'	27°19.8	4.9'	58.3'	
5	255°54.0	45.0	227°42.4	5.0'	27°24.7	4.7'	58.3'	
6	270°54.0	N17°45.6	242°06.4	5.0'	N27°29.4	4.6'	58.3'	
7	285°54.0	46.3	256°30.4	5.0'	27°34.0	4.4'	58.2'	
8	300°54.1	46.9	270°54.3	5.0'	27°38.4	4.2'	58.2'	
9	315°54.1	· · 47.6	285°18.3	5.0'	27°42.5	4.0'	58.1'	
10	330°54.1	48.2	299°42.3	5.0'	27°46.5	3.8'	58.1'	
11	345°54.1	48.9	314°06.3	5.0'	27°50.4	3.6'	58.1'	
12	0°54.1	N17°49.5	328°30.2	5.0'	N27°54.0	3.5'	58.0'	
13	15°54.2	50.2	342°54.2	5.0'	27°57.5	3.3'	58.0'	
14	30°54.2	50.8	357°18.2	5.0'	28°00.7	3.1'	58.0'	
15	45°54.2	· · 51.5	11°42.2	5.0'	28°03.8	2.9'	57.9'	
16	60°54.2	52.1	26°06.2	5.0'	28°06.7	2.7'	57.9'	
17	75°54.2	52.7	40°30.3	5.1'	28°09.5	2.5'	57.9'	
18	90°54.2	N17°53.4	54°54.3	5.1'	N28°12.0	2.4'	57.8'	
19	105°54.2	54.0	69°18.4	5.1'	28°14.4	2.2'	57.8'	
20	120°54.3	54.7	83°42.5	5.1'	28°16.6	2.0'	57.8'	
21	135°54.3	· · 55.3	98°06.6	5.1'	28°18.6	1.8'	57.7'	
22	150°54.3	55.9	112°30.7	5.2'	28°20.4	1.6'	57.7'	
23	165°54.3	56.6	126°54.9	5.2'	28°22.0	1.5'	57.7'	
SD = 15.8'		d = 0.7'		SD = 15.9'				
Sat	GHA	Dec	GHA	ν	Dec	d	HP	
0	180°54.3	N17°57.2	141°19.1	5.2'	N28°23.5	1.3'	57.6'	
1	195°54.3	57.9	155°43.4	5.3'	28°24.7	1.1'	57.6'	
2	210°54.4	58.5	170°07.6	5.3'	28°25.8	0.9'	57.6'	
3	225°54.4	· · 59.1	184°31.9	5.3'	28°26.8	0.7'	57.5'	
4	240°54.4	17°59.8	198°56.3	5.4'	28°27.5	0.6'	57.5'	
5	255°54.4	18°00.4	213°20.7	5.4'	28°28.1	0.4'	57.4'	
6	270°54.4	N18°01.0	227°45.1	5.5'	N28°28.4	0.2'	57.4'	
7	285°54.4	01.7	242°09.6	5.5'	28°28.7	0.0'	57.4'	
8	300°54.4	02.3	256°34.1	5.6'	28°28.7	-0.1'	57.3'	
9	315°54.4	· · 03.0	270°58.7	5.6'	28°28.5	-0.3'	57.3'	
10	330°54.5	03.6	285°23.3	5.7'	28°28.2	-0.5'	57.3'	
11	345°54.5	04.2	299°48.0	5.7'	28°27.7	-0.7'	57.2'	
12	0°54.5	N18°04.8	314°12.8	5.8'	N28°27.1	-0.8'	57.2'	
13	15°54.5	05.5	328°37.6	5.9'	28°26.2	-1.0'	57.2'	
14	30°54.5	06.1	343°02.4	5.9'	28°25.2	-1.2'	57.1'	
15	45°54.5	· · 06.7	357°27.3	6.0'	28°24.0	-1.3'	57.1'	
16	60°54.5	07.4	11°52.3	6.1'	28°22.7	-1.5'	57.1'	
17	75°54.5	08.0	26°17.4	6.1'	28°21.2	-1.7'	57.0'	
18	90°54.5	N18°08.6	40°42.5	6.2'	N28°19.5	-1.9'	57.0'	
19	105°54.6	09.3	55°07.7	6.3'	28°17.6	-2.0'	57.0'	
20	120°54.6	09.9	69°32.9	6.3'	28°15.6	-2.2'	56.9'	
21	135°54.6	· · 10.5	83°58.3	6.4'	28°13.4	-2.3'	56.9'	
22	150°54.6	11.1	98°23.7	6.5'	28°11.1	-2.5'	56.9'	
23	165°54.6	11.8	112°49.2	6.6'	28°08.6	-2.7'	56.8'	
SD = 15.8'		d = 0.6'		SD = 15.7'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°	—:—	—:—	01:28	22:33	—:—	—:—
68°	—:—	—:—	02:11	21:46	—:—	—:—
66°	—:—	00:34	02:40	21:16	23:41	—:—
64°	—:—	01:37	03:02	20:53	22:21	—:—
62°	—:—	02:10	03:19	20:36	21:46	—:—
60°	00:30	02:34	03:33	20:21	21:21	23:42
N 58°	01:27	02:53	03:46	20:09	21:02	22:30
56°	01:57	03:08	03:56	19:58	20:46	21:58
54°	02:20	03:21	04:05	19:49	20:33	21:36
52°	02:37	03:32	04:13	19:40	20:22	21:17
50°	02:52	03:42	04:21	19:33	20:12	21:02
45°	03:21	04:03	04:37	19:17	19:51	20:33
N 40°	03:42	04:19	04:49	19:04	19:34	20:12
35°	03:59	04:32	05:00	18:53	19:21	19:55
30°	04:13	04:44	05:10	18:43	19:09	19:40
20°	04:35	05:02	05:26	18:27	18:51	19:18
N 10°	04:52	05:18	05:40	18:13	18:35	19:01
0°	05:06	05:31	05:53	18:00	18:22	18:47
S 10°	05:18	05:44	06:06	17:47	18:09	18:34
20°	05:30	05:56	06:19	17:33	17:56	18:23
30°	05:41	06:10	06:35	17:18	17:43	18:11
35°	05:47	06:17	06:44	17:09	17:35	18:06
40°	05:53	06:25	06:54	16:58	17:27	18:00
45°	05:59	06:34	07:06	16:46	17:18	17:53
S 50°	06:06	06:45	07:20	16:32	17:07	17:46
52°	06:09	06:50	07:27	16:25	17:02	17:43
54°	06:12	06:55	07:35	16:18	16:57	17:40
56°	06:16	07:01	07:43	16:09	16:51	17:36
58°	06:20	07:08	07:52	16:00	16:44	17:32
S 60°	06:24	07:15	08:03	15:49	16:37	17:28

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°						
N 70°						
68°						
66°	01:08			00:01		
64°	02:21					
62°	02:57	02:47		· · ·	00:28	
60°	03:24	03:35	04:09	23:40	· · ·	01:13
N 58°	03:44	04:06	04:48	23:10	· · ·	00:34
56°	04:02	04:29	05:15	22:47	· · ·	00:06
54°	04:16	04:48	05:36	22:28	23:45	· · ·
52°	04:29	05:04	05:54	22:13	23:27	· · ·
50°	04:40	05:18	06:09	21:59	23:12	· · ·
45°	05:04	05:47	06:40	21:31	22:41	23:38
N 40°	05:23	06:09	07:04	21:09	22:17	23:15
35°	05:39	06:28	07:24	20:51	21:58	22:56
30°	05:53	06:44	07:41	20:36	21:41	22:39
20°	06:17	07:11	08:09	20:09	21:13	22:11
N 10°	06:38	07:35	08:34	19:47	20:48	21:47
0°	06:57	07:57	08:57	19:26	20:26	21:25
S 10°	07:17	08:19	09:19	19:05	20:03	21:02
20°	07:38	08:43	09:44	18:42	19:39	20:38
30°	08:03	09:11	10:13	18:16	19:10	20:09
35°	08:17	09:27	10:30	18:01	18:53	19:52
40°	08:34	09:46	10:49	17:43	18:34	19:33
45°	08:55	10:10	11:14	17:22	18:10	19:09
S 50°	09:20	10:40	11:45	16:55	17:40	18:38
52°	09:33	10:55	12:00	16:42	17:24	18:22
54°	09:47	11:12	12:18	16:27	17:	

May 12, 13, 14 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars			
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	230°15.4	187°01.4	N15°34.3	221°45.3	N02°18.6	175°40.2	N18°41.9	241°01.7	S06°34.1	Alpheratz	357°35.6	29°13.3	
1	245°17.9	202°00.8	35.3	236°46.0	19.3	190°42.1	42.1	256°04.0	34.0	Ankaa	353°08.0	-42°10.3	
2	260°20.3	217°00.2	36.2	251°46.7	20.1	205°43.9	42.2	271°06.2	34.0	Schedar	349°32.1	56°40.0	
3	275°22.8	231°59.7	· · 37.2	266°47.4	· · 20.8	220°45.8	· · 42.4	286°08.5	· · 33.9	Diphda	348°48.1	-17°51.2	
4	290°25.3	246°59.1	38.2	281°48.1	21.6	235°47.6	42.5	301°10.8	33.8	Achernar	335°21.1	-57°06.7	
5	305°27.7	261°58.5	39.1	296°48.8	22.4	250°49.5	42.6	316°13.1	33.8	Hamal	327°52.2	23°34.5	
6	320°30.2	276°58.0	N15°40.1	311°49.5	N02°23.1	265°51.4	N18°42.8	331°15.4	S06°33.7	Polaris	314°48.3	89°21.9	
7	335°32.7	291°57.4	41.1	326°50.2	23.9	280°53.2	42.9	346°17.7	33.7	Acamar	315°12.6	-40°12.4	
8	350°35.1	306°56.8	42.0	341°51.0	24.6	295°55.1	43.1	1°20.0	33.6	Menkar	314°07.1	4°11.0	
9	5°37.6	321°56.2	· · 43.0	356°51.7	· · 25.4	310°56.9	· · 43.2	16°22.3	· · 33.5	Mirfak	308°29.6	49°56.8	
10	20°40.0	336°55.7	43.9	11°52.4	26.1	325°58.8	43.3	31°24.6	33.5	Aldebaran	290°40.6	16°33.4	
11	35°42.5	351°55.1	44.9	26°53.1	26.9	341°00.7	43.5	46°26.8	33.4	Rigel	281°04.7	-8°10.5	
12	50°45.0	6°54.5	N15°45.9	41°53.8	N02°27.7	356°02.5	N18°43.6	61°29.1	S06°33.3	Capella	280°23.2	46°01.4	
13	65°47.4	21°53.9	46.8	56°54.5	28.4	11°04.4	43.8	76°31.4	33.3	Bellatrix	278°23.8	6°22.3	
14	80°49.9	36°53.4	47.8	71°55.2	29.2	26°06.2	43.9	91°33.7	33.2	Elnath	278°02.9	28°37.7	
15	95°52.4	51°52.8	· · 48.7	86°55.9	· · 29.9	41°08.1	· · 44.0	106°36.0	· · 33.1	Alnilam	275°38.6	-1°11.2	
16	110°54.8	66°52.2	49.7	101°56.6	30.7	56°10.0	44.2	121°38.3	33.1	Betelgeuse	270°53.0	7°24.7	
17	125°57.3	81°51.6	50.6	116°57.3	31.4	71°11.8	44.3	136°40.6	33.0	Canopus	263°53.1	-52°42.7	
18	140°59.8	96°51.1	N15°51.6	131°58.0	N02°32.2	86°13.7	N18°44.4	151°42.9	S06°32.9	Sirius	258°27.0	-16°45.1	
19	156°02.2	111°50.5	52.6	146°58.7	33.0	101°15.5	44.6	166°45.2	32.9	Adhara	255°06.6	-29°00.4	
20	171°04.7	126°49.9	53.5	161°59.4	33.7	116°17.4	44.7	181°47.5	32.8	Procyon	244°51.6	5°09.7	
21	186°07.2	141°49.3	· · 54.5	177°00.2	· · 34.5	131°19.2	· · 44.9	196°49.7	· · 32.8	Pollux	243°18.2	27°58.1	
22	201°09.6	156°48.8	55.4	192°00.9	35.2	146°21.1	45.0	211°52.0	32.7	Avior	234°15.2	-59°35.5	
23	216°12.1	171°48.2	56.4	207°01.6	36.0	161°23.0	45.1	226°54.3	32.6	Suhail	222°46.8	-43°32.1	
Mer.pass. 08:38		ν -0.6' d 1.0' m -3.92		ν 0.7' d 0.8' m 1.09		ν 1.9' d 0.1' m -2.00		ν 2.3' d -0.1' m 1.06		Miaplacidus		221°38.5	-69°49.3
										Alphard		217°48.3	-8°45.9
										Regulus		207°35.0	11°50.9
										Dubhe		193°41.4	61°37.4
										Denebola		182°25.3	14°26.2
										Gienah		175°44.0	-17°40.8
										Acrux		173°00.3	-63°14.3
										Gacrux		171°51.9	-57°15.2
										Alioth		166°13.0	55°49.8
										Spica		158°22.6	-11°17.4
										Alkaid		152°52.0	49°11.6
										Hadar		148°36.2	-60°29.6
										Menkent		147°57.9	-36°29.5
										Arcturus		145°48.1	19°03.3
										Rigil Kent.		139°40.5	-60°56.3
										Kochab		137°18.6	74°03.3
										Zuben'ubi		136°56.3	-16°08.7
										Alphecca		126°03.9	26°37.9
										Antares		112°16.2	-26°29.2
										Atria		107°10.4	-69°04.2
										Sabik		102°03.1	-15°45.4
										Shaula		96°10.8	-37°07.3
										Rasalhague		95°58.8	12°32.4
										Eltanin		90°42.0	51°28.9
										Kaus Aust.		83°33.0	-34°22.3
										Vega		80°33.3	38°48.1
										Nunki		75°48.2	-26°16.0
										Altair		62°00.4	8°55.8
										Peacock		53°06.4	-56°39.2
										Deneb		49°26.1	45°21.7
										Enif		33°39.4	9°59.0
										Al Na'ir		27°33.7	-46°50.4
										Fomalhaut		15°15.3	-29°29.5
										Scheat		13°45.9	28°12.6
										Markab		13°30.6	15°20.0
										May 12 Sun		SHA	Mer.pass
										Venus		316°46.0	11:32
										Mars		351°29.9	09:13
										Jupiter		305°24.8	12:16
										Saturn		10°46.3	07:55
										May 13 Mon		SHA	Mer.pass
										Venus		315°33.0	11:33
										Mars		350°47.7	09:11
										Jupiter		305°10.3	12:13
										Saturn		10°42.1	07:51
										May 14 Tue		SHA	Mer.pass
										Venus		314°19.8	11:34
										Mars		350°05.6	09:10
										Jupiter		304°55.7	12:10
										Saturn		10°38.0	07:47
										Horizontal parallax			
										Venus:		0.1	
										Mars:		0.1	

Sun				Moon				
Sun	GHA	Dec		GHA	ν	Dec	d	HP
0	180° 54.6	N18° 12.4		127° 14.7	6.6'	N28° 05.9	-2.8'	56.8'
1	195° 54.6	13.0		141° 40.4	6.7'	28° 03.1	-3.0'	56.7'
2	210° 54.6	13.6		156° 06.1	6.8'	28° 00.1	-3.2'	56.7'
3	225° 54.6	· · 14.3		170° 31.9	6.9'	27° 56.9	-3.3'	56.7'
4	240° 54.6	14.9		184° 57.8	7.0'	27° 53.6	-3.5'	56.6'
5	255° 54.6	15.5		199° 23.7	7.1'	27° 50.1	-3.6'	56.6'
6	270° 54.7	N18° 16.1		213° 49.8	7.1'	N27° 46.5	-3.8'	56.6'
7	285° 54.7	16.8		228° 15.9	7.2'	27° 42.7	-3.9'	56.5'
8	300° 54.7	17.4		242° 42.2	7.3'	27° 38.8	-4.1'	56.5'
9	315° 54.7	· · 18.0		257° 08.5	7.4'	27° 34.7	-4.2'	56.5'
10	330° 54.7	18.6		271° 34.9	7.5'	27° 30.5	-4.4'	56.4'
11	345° 54.7	19.3		286° 01.4	7.6'	27° 26.1	-4.5'	56.4'
12	0° 54.7	N18° 19.9		300° 28.0	7.7'	N27° 21.6	-4.7'	56.4'
13	15° 54.7	20.5		314° 54.7	7.8'	27° 16.9	-4.8'	56.3'
14	30° 54.7	21.1		329° 21.5	7.9'	27° 12.1	-5.0'	56.3'
15	45° 54.7	· · 21.7		343° 48.4	8.0'	27° 07.1	-5.1'	56.3'
16	60° 54.7	22.3		358° 15.3	8.1'	27° 02.0	-5.2'	56.2'
17	75° 54.7	23.0		12° 42.4	8.2'	26° 56.8	-5.4'	56.2'
18	90° 54.7	N18° 23.6		27° 09.6	8.3'	N26° 51.4	-5.5'	56.2'
19	105° 54.7	24.2		41° 36.9	8.4'	26° 45.8	-5.7'	56.1'
20	120° 54.7	24.8		56° 04.2	8.5'	26° 40.2	-5.8'	56.1'
21	135° 54.7	· · 25.4		70° 31.7	8.6'	26° 34.4	-5.9'	56.1'
22	150° 54.7	26.0		84° 59.3	8.7'	26° 28.4	-6.1'	56.1'
23	165° 54.7	26.7		99° 27.0	8.8'	26° 22.4	-6.2'	56.0'
SD = 15.8' $d = 0.6'$				SD = 15.5'				
Mon	GHA	Dec		GHA	ν	Dec	d	HP
0	180° 54.7	N18° 27.3		113° 54.8	8.9'	N26° 16.2	-6.3'	56.0'
1	195° 54.7	27.9		128° 22.6	9.0'	26° 09.8	-6.5'	56.0'
2	210° 54.7	28.5		142° 50.6	9.1'	26° 03.4	-6.6'	55.9'
3	225° 54.8	· · 29.1		157° 18.7	9.2'	25° 56.8	-6.7'	55.9'
4	240° 54.8	29.7		171° 46.9	9.3'	25° 50.0	-6.8'	55.9'
5	255° 54.8	30.3		186° 15.2	9.4'	25° 43.2	-7.0'	55.8'
6	270° 54.8	N18° 30.9		200° 43.6	9.5'	N25° 36.2	-7.1'	55.8'
7	285° 54.8	31.5		215° 12.1	9.6'	25° 29.1	-7.2'	55.8'
8	300° 54.8	32.1		229° 40.8	9.7'	25° 21.9	-7.3'	55.8'
9	315° 54.8	· · 32.8		244° 09.5	9.8'	25° 14.6	-7.5'	55.7'
10	330° 54.8	33.4		258° 38.3	9.9'	25° 07.1	-7.6'	55.7'
11	345° 54.8	34.0		273° 07.2	10.0'	24° 59.5	-7.7'	55.7'
12	0° 54.8	N18° 34.6		287° 36.3	10.1'	N24° 51.8	-7.8'	55.6'
13	15° 54.8	35.2		302° 05.4	10.3'	24° 44.0	-7.9'	55.6'
14	30° 54.8	35.8		316° 34.7	10.4'	24° 36.1	-8.0'	55.6'
15	45° 54.8	· · 36.4		331° 04.0	10.5'	24° 28.1	-8.1'	55.5'
16	60° 54.8	37.0		345° 33.5	10.6'	24° 19.9	-8.3'	55.5'
17	75° 54.8	37.6		0° 03.1	10.7'	24° 11.7	-8.4'	55.5'
18	90° 54.8	N18° 38.2		14° 32.7	10.8'	N24° 03.3	-8.5'	55.5'
19	105° 54.8	38.8		29° 02.5	10.9'	23° 54.8	-8.6'	55.4'
20	120° 54.8	39.4		43° 32.4	11.0'	23° 46.3	-8.7'	55.4'
21	135° 54.8	· · 40.0		58° 02.4	11.1'	23° 37.6	-8.8'	55.4'
22	150° 54.7	40.6		72° 32.5	11.2'	23° 28.8	-8.9'	55.4'
23	165° 54.7	41.2		87° 02.7	11.3'	23° 19.9	-9.0'	55.3'
SD = 15.8' $d = 0.6'$				SD = 15.3'				
Tue	GHA	Dec		GHA	ν	Dec	d	HP
0	180° 54.7	N18° 41.8		101° 33.0	11.4'	N23° 10.9	-9.1'	55.3'
1	195° 54.7	42.4		116° 03.4	11.5'	23° 01.8	-9.2'	55.3'
2	210° 54.7	43.0		130° 33.9	11.6'	22° 52.6	-9.3'	55.3'
3	225° 54.7	· · 43.6		145° 04.5	11.7'	22° 43.4	-9.4'	55.2'
4	240° 54.7	44.2		159° 35.3	11.8'	22° 34.0	-9.5'	55.2'
5	255° 54.7	44.8		174° 06.1	11.9'	22° 24.5	-9.6'	55.2'
6	270° 54.7	N18° 45.4		188° 37.0	12.0'	N22° 14.9	-9.7'	55.2'
7	285° 54.7	46.0		203° 08.0	12.1'	22° 05.3	-9.8'	55.1'
8	300° 54.7	46.6		217° 39.1	12.2'	21° 55.5	-9.8'	55.1'
9	315° 54.7	· · 47.2		232° 10.4	12.3'	21° 45.7	-9.9'	55.1'
10	330° 54.7	47.8		246° 41.7	12.4'	21° 35.7	-10.0'	55.1'
11	345° 54.7	48.4		261° 13.1	12.5'	21° 25.7	-10.1'	55.0'
12	0° 54.7	N18° 49.0		275° 44.6	12.6'	N21° 15.6	-10.2'	55.0'
13	15° 54.7	49.6		290° 16.2	12.7'	21° 05.4	-10.3'	55.0'
14	30° 54.7	50.2		304° 47.9	12.8'	20° 55.2	-10.4'	55.0'
15	45° 54.7	· · 50.8		319° 19.7	12.9'	20° 44.8	-10.4'	54.9'
16	60° 54.7	51.3		333° 51.6	13.0'	20° 34.4	-10.5'	54.9'
17	75° 54.7	51.9		348° 23.6	13.1'	20° 23.8	-10.6'	54.9'
18	90° 54.7	N18° 52.5		2° 55.7	13.2'	N20° 13.2	-10.7'	54.9'
19	105° 54.6	53.1		17° 27.9	13.3'	20° 02.6	-10.8'	54.9'
20	120° 54.6	53.7		32° 00.2	13.4'	19° 51.8	-10.8'	54.8'
21	135° 54.6	· · 54.3		46° 32.6	13.5'	19° 41.0	-10.9'	54.8'
22	150° 54.6	54.9		61° 05.0	13.5'	19° 30.1	-11.0'	54.8'
23	165° 54.6	55.5		75° 37.6	13.6'	19° 19.1	-11.1'	54.8'
SD = 15.8' $d = 0.6'$				SD = 15.1'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°	—:—	—:—	01:00	23:04	—:—	—:—
68°	—:—	—:—	01:56	22:02	—:—	—:—
66°	—:—	—:—	02:28	21:28	—:—	—:—
64°	—:—	01:19	02:52	21:03	22:40	—:—
62°	—:—	01:58	03:11	20:44	21:58	—:—
60°	—:—	02:24	03:26	20:28	21:31	—:—
N 58°	01:11	02:44	03:39	20:15	21:10	22:48
56°	01:46	03:01	03:50	20:04	20:53	22:10
54°	02:11	03:15	04:00	19:54	20:39	21:44
52°	02:30	03:27	04:09	19:45	20:27	21:25
50°	02:46	03:37	04:16	19:37	20:16	21:09
45°	03:16	03:59	04:33	19:21	19:55	20:38
N 40°	03:38	04:16	04:46	19:07	19:38	20:15
35°	03:56	04:30	04:58	18:55	19:23	19:57
30°	04:10	04:42	05:08	18:45	19:12	19:43
20°	04:33	05:01	05:25	18:28	18:52	19:20
N 10°	04:51	05:17	05:39	18:14	18:36	19:02
0°	05:06	05:31	05:53	18:00	18:22	18:47
S 10°	05:19	05:44	06:06	17:46	18:08	18:34
20°	05:31	05:57	06:21	17:32	17:55	18:22
30°	05:43	06:12	06:37	17:16	17:41	18:10
35°	05:49	06:19	06:46	17:06	17:33	18:04
40°	05:55	06:28	06:57	16:55	17:24	17:57
45°	06:02	06:38	07:09	16:43	17:15	17:50
S 50°	06:10	06:49	07:25	16:28	17:03	17:42
52°	06:13	06:54	07:32	16:20	16:58	17:39
54°	06:17	07:00	07:40	16:13	16:52	17:35
56°	06:20	07:06	07:48	16:04	16:46	17:32
58°	06:24	07:13	07:58	15:54	16:39	17:27
S 60°	06:29	07:21	08:10	15:42	16:31	17:23

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°						
N 70°						
68°						
66°			07:08			03:55
64°		05:20	07:48		03:57	03:14
62°	04:13	06:24	08:15	03:10	02:53	02:46
60°	05:22	06:58	08:36	02:01	02:18	02:24
N 58°	05:57	07:23	08:53	01:26	01:53	02:07
56°	06:22	07:43	09:08	01:00	01:33	01:52
54°	06:42	07:59	09:20	00:40	01:16	01:39
52°	06:59	08:14	09:31	00:23	01:01	01:27
50°	07:14	08:26	09:40	00:08	00:49	01:17
45°	07:43	08:51	10:01	· · ·	00:22	00:56
N 40°	08:06	09:11	10:17	· · ·	00:02	00:38
35°	08:25	09:28	10:31	23:44	· · ·	00:24
30°	08:41	09:42	10:43	23:29	· · ·	00:11
20°	09:09	10:07	11:03	23:03	23:49	· · ·
N 10°	09:32	10:28	11:20			

May 15, 16, 17 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	233°12.8	186°19.2	N16°42.1	222°36.3	N03°12.9	177°54.0	N18°51.9	243°46.8	S06°29.6	Alpheratz	357°35.6	29°13.3
1	248°15.3	201°18.6	43.0	237°37.0	13.7	192°55.9	52.0	258°49.0	29.5	Ankaa	353°08.0	-42°10.3
2	263°17.8	216°18.0	43.9	252°37.7	14.4	207°57.7	52.2	273°51.3	29.5	Schedar	349°32.1	56°40.0
3	278°20.2	231°17.4	· · 44.8	267°38.4	· · 15.2	222°59.6	· · 52.3	288°53.6	· · 29.4	Diphda	348°48.1	-17°51.2
4	293°22.7	246°16.8	45.8	282°39.1	15.9	238°01.4	52.4	303°55.9	29.3	Achernar	335°21.1	-57°06.7
5	308°25.1	261°16.2	46.7	297°39.8	16.7	253°03.3	52.6	318°58.2	29.3	Hamal	327°52.2	23°34.5
6	323°27.6	276°15.6	N16°47.6	312°40.5	N03°17.4	268°05.2	N18°52.7	334°00.5	S06°29.2	Polaris	314°48.0	89°21.9
7	338°30.1	291°15.0	48.5	327°41.3	18.2	283°07.0	52.8	349°02.8	29.2	Acamar	315°12.6	-40°12.4
8	353°32.5	306°14.4	49.4	342°42.0	18.9	298°08.9	53.0	4°05.1	29.1	Menkar	314°07.1	4°11.0
9	8°35.0	321°13.8	· · 50.3	357°42.7	· · 19.7	313°10.7	· · 53.1	19°07.4	· · 29.0	Mirfak	308°29.6	49°56.8
10	23°37.5	336°13.2	51.2	12°43.4	20.4	328°12.6	53.3	34°09.7	29.0	Aldebaran	290°40.6	16°33.4
11	38°39.9	351°12.6	52.1	27°44.1	21.2	343°14.5	53.4	49°12.0	28.9	Rigel	281°04.7	-8°10.5
12	53°42.4	6°11.9	N16°53.0	42°44.8	N03°21.9	358°16.3	N18°53.5	64°14.3	S06°28.9	Capella	280°23.2	46°01.4
13	68°44.9	21°11.3	53.9	57°45.5	22.7	13°18.2	53.7	79°16.6	28.8	Bellatrix	278°23.8	6°22.3
14	83°47.3	36°10.7	54.8	72°46.2	23.4	28°20.0	53.8	94°18.9	28.7	Elnath	278°02.9	28°37.7
15	98°49.8	51°10.1	· · 55.8	87°46.9	· · 24.2	43°21.9	· · 53.9	109°21.2	· · 28.7	Alnilam	275°38.6	-1°11.2
16	113°52.3	66°09.5	56.7	102°47.6	24.9	58°23.7	54.1	124°23.5	28.6	Betelgeuse	270°53.0	7°24.7
17	128°54.7	81°08.9	57.6	117°48.3	25.7	73°25.6	54.2	139°25.8	28.6	Canopus	263°53.1	-52°42.7
18	143°57.2	96°08.3	N16°58.5	132°49.1	N03°26.4	88°27.5	N18°54.3	154°28.1	S06°28.5	Sirius	258°27.0	-16°45.1
19	158°59.6	111°07.7	16°59.4	147°49.8	27.2	103°29.3	54.5	169°30.4	28.4	Adhara	255°06.6	-29°00.4
20	174°02.1	126°07.1	17°00.3	162°50.5	27.9	118°31.2	54.6	184°32.7	28.4	Procyon	244°51.6	5°09.7
21	189°04.6	141°06.5	· · 01.2	177°51.2	· · 28.7	133°33.0	· · 54.8	199°35.0	· · 28.3	Pollux	243°18.2	27°58.1
22	204°07.0	156°05.8	02.1	192°51.9	29.4	148°34.9	54.9	214°37.3	28.3	Avior	234°15.2	-59°35.5
23	219°09.5	171°05.2	03.0	207°52.6	30.2	163°36.7	55.0	229°39.6	28.2	Suhail	222°46.8	-43°32.1
Mer.pass. 08:26		ν -0.6' d 0.9' m -3.92		ν 0.7' d 0.8' m 1.08		ν 1.9' d 0.1' m -2.00		ν 2.3' d -0.1' m 1.06				

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	234°12.0	186°04.6	N17°03.9	222°53.3	N03°30.9	178°38.6	N18°55.2	244°41.9	S06°28.1
1	249°14.4	201°04.0	04.8	237°54.0	31.7	193°40.5	55.3	259°44.2	28.1
2	264°16.9	216°03.4	05.7	252°54.7	32.4	208°42.3	55.4	274°46.5	28.0
3	279°19.4	231°02.8	· · 06.6	267°55.4	· · 33.2	223°44.2	· · 55.6	289°48.8	· · 28.0
4	294°21.8	246°02.2	07.5	282°56.1	33.9	238°46.0	55.7	304°51.1	27.9
5	309°24.3	261°01.5	08.3	297°56.9	34.7	253°47.9	55.8	319°53.4	27.9
6	324°26.8	276°00.9	N17°09.2	312°57.6	N03°35.4	268°49.7	N18°56.0	334°55.7	S06°27.8
7	339°29.2	291°00.3	10.1	327°58.3	36.2	283°51.6	56.1	349°58.0	27.7
8	354°31.7	305°59.7	11.0	342°59.0	36.9	298°53.5	56.2	5°00.3	27.7
9	9°34.1	320°59.1	· · 11.9	357°59.7	· · 37.7	313°55.3	· · 56.4	20°02.6	· · 27.6
10	24°36.6	335°58.5	12.8	13°00.4	38.4	328°57.2	56.5	35°04.9	27.6
11	39°39.1	350°57.8	13.7	28°01.1	39.2	343°59.0	56.7	50°07.2	27.5
12	54°41.5	5°57.2	N17°14.6	43°01.8	N03°39.9	359°00.9	N18°56.8	65°09.5	S06°27.4
13	69°44.0	20°56.6	15.5	58°02.5	40.7	14°02.7	56.9	80°11.8	27.4
14	84°46.5	35°56.0	16.4	73°03.2	41.4	29°04.6	57.1	95°14.1	27.3
15	99°48.9	50°55.4	· · 17.2	88°04.0	· · 42.2	44°06.5	· · 57.2	110°16.4	· · 27.3
16	114°51.4	65°54.7	18.1	103°04.7	42.9	59°08.3	57.3	125°18.7	27.2
17	129°53.9	80°54.1	19.0	118°05.4	43.7	74°10.2	57.5	140°21.0	27.1
18	144°56.3	95°53.5	N17°19.9	133°06.1	N03°44.4	89°12.0	N18°57.6	155°23.4	S06°27.1
19	159°58.8	110°52.9	20.8	148°06.8	45.2	104°13.9	57.7	170°25.7	27.0
20	175°01.2	125°52.2	21.7	163°07.5	45.9	119°15.7	57.9	185°28.0	27.0
21	190°03.7	140°51.6	· · 22.6	178°08.2	· · 46.7	134°17.6	· · 58.0	200°30.3	· · 26.9
22	205°06.2	155°51.0	23.4	193°08.9	47.4	149°19.5	58.1	215°32.6	26.9
23	220°08.6	170°50.4	24.3	208°09.6	48.2	164°21.3	58.3	230°34.9	26.8
Mer.pass. 08:22		ν -0.6' d 0.9' m -3.92		ν 0.7' d 0.7' m 1.08		ν 1.9' d 0.1' m -2.00		ν 2.3' d -0.1' m 1.05	

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	235°11.1	185°49.7	N17°25.2	223°10.3	N03°48.9	179°23.2	N18°58.4	245°37.2	S06°26.7
1	250°13.6	200°49.1	26.1	238°11.0	49.7	194°25.0	58.5	260°39.5	26.7
2	265°16.0	215°48.5	26.9	253°11.8	50.4	209°26.9	58.7	275°41.8	26.6
3	280°18.5	230°47.9	· · 27.8	268°12.5	· · 51.2	224°28.7	· · 58.8	290°44.1	· · 26.6
4	295°21.0	245°47.2	28.7	283°13.2	51.9	239°30.6	59.0	305°46.4	26.5
5	310°23.4	260°46.6	29.6	298°13.9	52.6	254°32.4	59.1	320°48.7	26.5
6	325°25.9	275°46.0	N17°30.5	313°14.6	N03°53.4	269°34.3	N18°59.2	335°51.0	S06°26.4
7	340°28.4	290°45.4	31.3	328°15.3	54.1	284°36.2	59.4	350°53.3	26.3
8	355°30.8	305°44.7	32.2	343°16.0	54.9	299°38.0	59.5	5°55.6	26.3
9	10°33.3	320°44.1	· · 33.1	358°16.7	· · 55.6	314°39.9	· · 59.6	20°57.9	· · 26.2
10	25°35.7	335°43.5	33.9	13°17.4	56.4	329°41.7	59.8	36°00.2	26.2
11	40°38.2	350°42.8	34.8	28°18.1	57.1	344°43.6	18°59.9	51°02.5	26.1
12	55°40.7	5°42.2	N17°35.7	43°18.9	N03°57.9	359°45.4	N19°00.0	66°04.8	S06°26.0
13	70°43.1	20°41.6	36.6	58°19.6	58.6	14°47.3	00.2	81°07.1	26.0
14	85°45.6	35°40.9	37.4	73°20.3	03°59.4	29°49.2	00.3	96°09.4	25.9
15	100°48.1	50°40.3	· · 38.3	88°21.0	04°00.1	44°51.0	· · 00.4	111°11.7	· · 25.9
16	115°50.5	65°39.7	39.2	103°21.7	00.9	59°52.9	00.6	126°14.0	25.8
17	130°53.0	80°39.0	40.0	118°22.4	01.6	74°54.7	00.7	141°16.3	25.8
18	145°55.5	95°38.4	N17°40.9	133°23.1	N04°02.4	89°56.6	N19°00.8	156°18.7	S06°25.7
19	160°57.9	110°37.8	41.8	148°23.8	03.1	104°58.4	01.0	171°21.0	25.6
20	176°00.4	125°37.1	42.6	163°24.5	03.8	120°00.3	01.1	186°23.3	25.6
21	191°02.8	140°36.5	· · 43.5	178°25.2	· · 04.6	135°02.2	· · 01.2	201°25.6	· · 25.5
22	206°05.3	155°35.9	44.3	193°26.0	05.3	150°04.0	01.4	216°27.9	25.5
23	221°07.8	170°35.2	45.2	208°26.7	06.1	165°05.9	01.5	231°30.2	25.4
Mer.pass. 08:18		ν -0.6' d 0.9' m -3.92		ν 0.7' d 0.7' m 1.08		ν 1.9' d 0.1' m -2.00		ν 2.3' d -0.1' m 1.05	

May 15 Wed		SHA	Mer.pass
Venus	313°06.4		11:35
Mars	349°23.5		09:09
Jupiter	304°41.2		12:07
Saturn	10°33.9		07:44

May 16 Thu		SHA	Mer.pass
Venus	311°52.6		11:36
Mars	348°41.3		09:08
Jupiter	304°26.6		12:04
Saturn	10°30.0		07:40

May 17 Fri		SHA	Mer.pass
Venus	310°38.6		11:37
Mars	347°59.2		09:07
Jupiter	304°12.1		12:01
Saturn	10°26.1		07:36

Horizontal parallax		
Venus:		0.1
Mars:		0.1

Sun				Moon				
Wed	GHA	Dec		GHA	ν	Dec	d	HP
0	180°54.6	N18°56.1		90°10.2	13.7'	N19°08.0	-11.1'	54.8'
1	195°54.6	56.6		104°42.9	13.8'	18°56.9	-11.2'	54.7'
2	210°54.6	57.2		119°15.8	13.9'	18°45.7	-11.3'	54.7'
3	225°54.6	•• 57.8		133°48.7	14.0'	18°34.4	-11.3'	54.7'
4	240°54.6	58.4		148°21.6	14.1'	18°23.1	-11.4'	54.7'
5	255°54.6	59.0		162°54.7	14.2'	18°11.7	-11.5'	54.7'
6	270°54.6	N18°59.6		177°27.9	14.2'	N18°00.2	-11.5'	54.7'
7	285°54.5	19°00.1		192°01.1	14.3'	17°48.7	-11.6'	54.6'
8	300°54.5	00.7		206°34.4	14.4'	17°37.1	-11.7'	54.6'
9	315°54.5	•• 01.3		221°07.8	14.5'	17°25.4	-11.7'	54.6'
10	330°54.5	01.9		235°41.3	14.6'	17°13.7	-11.8'	54.6'
11	345°54.5	02.5		250°14.9	14.6'	17°01.9	-11.8'	54.6'
12	0°54.5	N19°03.0		264°48.5	14.7'	N16°50.1	-11.9'	54.6'
13	15°54.5	03.6		279°22.2	14.8'	16°38.2	-12.0'	54.5'
14	30°54.5	04.2		293°56.0	14.9'	16°26.2	-12.0'	54.5'
15	45°54.5	•• 04.8		308°29.9	14.9'	16°14.2	-12.1'	54.5'
16	60°54.4	05.4		323°03.8	15.0'	16°02.1	-12.1'	54.5'
17	75°54.4	05.9		337°37.8	15.1'	15°50.0	-12.2'	54.5'
18	90°54.4	N19°06.5		352°11.9	15.2'	N15°37.8	-12.2'	54.5'
19	105°54.4	07.1		6°46.1	15.2'	15°25.5	-12.3'	54.4'
20	120°54.4	07.7		21°20.3	15.3'	15°13.2	-12.4'	54.4'
21	135°54.4	•• 08.2		35°54.6	15.4'	15°00.9	-12.4'	54.4'
22	150°54.4	08.8		50°29.0	15.4'	14°48.5	-12.5'	54.4'
23	165°54.3	09.4		65°03.4	15.5'	14°36.0	-12.5'	54.4'
SD = 15.8' d = 0.6'				SD = 14.9'				
Thu	GHA	Dec		GHA	ν	Dec	d	HP
0	180°54.3	N19°10.0		79°37.9	15.6'	N14°23.5	-12.6'	54.4'
1	195°54.3	10.5		94°12.4	15.6'	14°10.9	-12.6'	54.4'
2	210°54.3	11.1		108°47.1	15.7'	13°58.3	-12.7'	54.4'
3	225°54.3	•• 11.7		123°21.7	15.7'	13°45.7	-12.7'	54.3'
4	240°54.3	12.3		137°56.5	15.8'	13°33.0	-12.7'	54.3'
5	255°54.3	12.8		152°31.3	15.9'	13°20.2	-12.8'	54.3'
6	270°54.2	N19°13.4		167°06.2	15.9'	N13°07.4	-12.8'	54.3'
7	285°54.2	14.0		181°41.1	16.0'	12°54.6	-12.9'	54.3'
8	300°54.2	14.5		196°16.1	16.0'	12°41.7	-12.9'	54.3'
9	315°54.2	•• 15.1		210°51.1	16.1'	12°28.8	-13.0'	54.3'
10	330°54.2	15.7		225°26.2	16.1'	12°15.9	-13.0'	54.3'
11	345°54.2	16.2		240°01.3	16.2'	12°02.9	-13.0'	54.3'
12	0°54.1	N19°16.8		254°36.5	16.2'	N11°49.8	-13.1'	54.3'
13	15°54.1	17.4		269°11.8	16.3'	11°36.7	-13.1'	54.2'
14	30°54.1	17.9		283°47.1	16.3'	11°23.6	-13.2'	54.2'
15	45°54.1	•• 18.5		298°22.4	16.4'	11°10.5	-13.2'	54.2'
16	60°54.1	19.1		312°57.8	16.4'	10°57.3	-13.2'	54.2'
17	75°54.1	19.6		327°33.3	16.5'	10°44.0	-13.3'	54.2'
18	90°54.0	N19°20.2		342°08.8	16.5'	N10°30.8	-13.3'	54.2'
19	105°54.0	20.7		356°44.3	16.6'	10°17.5	-13.3'	54.2'
20	120°54.0	21.3		11°19.9	16.6'	10°04.1	-13.4'	54.2'
21	135°54.0	•• 21.9		25°55.5	16.7'	09°50.8	-13.4'	54.2'
22	150°54.0	22.4		40°31.1	16.7'	09°37.4	-13.4'	54.2'
23	165°53.9	23.0		55°06.8	16.7'	09°23.9	-13.5'	54.2'
SD = 15.8' d = 0.6'				SD = 14.8'				
Fri	GHA	Dec		GHA	ν	Dec	d	HP
0	180°53.9	N19°23.6		69°42.6	16.8'	N09°10.5	-13.5'	54.2'
1	195°53.9	24.1		84°18.4	16.8'	08°57.0	-13.5'	54.2'
2	210°53.9	24.7		98°54.2	16.8'	08°43.4	-13.6'	54.2'
3	225°53.9	•• 25.2		113°30.0	16.9'	08°29.9	-13.6'	54.2'
4	240°53.8	25.8		128°05.9	16.9'	08°16.3	-13.6'	54.1'
5	255°53.8	26.3		142°41.8	16.9'	08°02.7	-13.6'	54.1'
6	270°53.8	N19°26.9		157°17.8	17.0'	N07°49.0	-13.7'	54.1'
7	285°53.8	27.5		171°53.8	17.0'	07°35.4	-13.7'	54.1'
8	300°53.8	28.0		186°29.8	17.0'	07°21.7	-13.7'	54.1'
9	315°53.7	•• 28.6		201°05.8	17.1'	07°08.0	-13.7'	54.1'
10	330°53.7	29.1		215°41.9	17.1'	06°54.2	-13.8'	54.1'
11	345°53.7	29.7		230°18.0	17.1'	06°40.5	-13.8'	54.1'
12	0°53.7	N19°30.2		244°54.1	17.1'	N06°26.7	-13.8'	54.1'
13	15°53.6	30.8		259°30.2	17.2'	06°12.9	-13.8'	54.1'
14	30°53.6	31.3		274°06.4	17.2'	05°59.1	-13.9'	54.1'
15	45°53.6	•• 31.9		288°42.6	17.2'	05°45.2	-13.9'	54.1'
16	60°53.6	32.4		303°18.8	17.2'	05°31.3	-13.9'	54.1'
17	75°53.6	33.0		317°55.0	17.2'	05°17.4	-13.9'	54.1'
18	90°53.5	N19°33.5		332°31.3	17.3'	N05°03.5	-13.9'	54.1'
19	105°53.5	34.1		347°07.5	17.3'	04°49.6	-13.9'	54.1'
20	120°53.5	34.6		1°43.8	17.3'	04°35.7	-14.0'	54.1'
21	135°53.5	•• 35.2		16°20.1	17.3'	04°21.7	-14.0'	54.1'
22	150°53.4	35.7		30°56.4	17.3'	04°07.7	-14.0'	54.1'
23	165°53.4	36.3		45°32.7	17.3'	03°53.7	-14.0'	54.1'
SD = 15.8' d = 0.6'				SD = 14.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°	—:—	—:—	00:00	—:—	—:—	—:—
68°	—:—	—:—	01:39	22:19	—:—	—:—
66°	—:—	—:—	02:16	21:40	—:—	—:—
64°	—:—	00:57	02:42	21:13	23:04	—:—
62°	—:—	01:45	03:03	20:52	22:12	—:—
60°	—:—	02:14	03:19	20:35	21:41	—:—
N 58°	00:51	02:36	03:33	20:21	21:18	23:09
56°	01:35	02:54	03:45	20:09	21:00	22:22
54°	02:02	03:09	03:55	19:59	20:45	21:53
52°	02:23	03:22	04:04	19:50	20:32	21:32
50°	02:39	03:33	04:12	19:41	20:21	21:15
45°	03:11	03:55	04:29	19:24	19:58	20:43
N 40°	03:35	04:13	04:44	19:10	19:41	20:19
35°	03:53	04:27	04:55	18:58	19:26	20:00
30°	04:08	04:40	05:06	18:47	19:14	19:45
20°	04:32	05:00	05:23	18:30	18:53	19:21
N 10°	04:50	05:16	05:39	18:14	18:37	19:03
0°	05:06	05:31	05:53	18:00	18:22	18:47
S 10°	05:19	05:45	06:07	17:46	18:08	18:34
20°	05:32	05:58	06:22	17:31	17:54	18:21
30°	05:44	06:13	06:39	17:14	17:39	18:08
35°	05:51	06:21	06:49	17:04	17:31	18:02
40°	05:58	06:30	07:00	16:53	17:22	17:55
45°	06:05	06:41	07:13	16:40	17:12	17:47
S 50°	06:13	06:53	07:29	16:24	16:59	17:39
52°	06:17	06:58	07:36	16:16	16:54	17:35
54°	06:21	07:04	07:44	16:08	16:48	17:32
56°	06:25	07:11	07:54	15:58	16:41	17:27
58°	06:29	07:18	08:04	15:48	16:34	17:23
S 60°	06:34	07:26	08:16	15:36	16:26	17:18

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	06:52	10:13	12:22	05:50	03:58	03:14
N 70°	08:15	10:36	12:31	04:26	03:33	03:02
68°	08:53	10:53	12:39	03:46	03:14	02:52
66°	09:19	11:07	12:45	03:19	02:59	02:44
64°	09:39	11:18	12:51	02:58	02:46	02:37
62°	09:55	11:28	12:55	02:41	02:35	02:31
60°	10:09	11:36	12:59	02:26	02:26	02:25
N 58°	10:20	11:43	13:03	02:14	02:18	02:21
56°	10:30	11:49	13:06	02:03	02:11	02:16
54°	10:39	11:55	13:09	01:54	02:04	02:13
52°	10:47	12:00	13:11	01:45	01:58	02:09
50°	10:54	12:05	13:13	01:38	01:53	02:06
45°	11:09	12:14	13:18	01:21	01:42	01:59
N 40°	11:21	12:22	13:22	01:08	01:32	01:53
35°	11:31	12:29	13:26	00:56	01:24	01:48
30°	11:40	12:36	13:29	00:46	01:17	01:44
20°	11:56	12:46	13:34	00:29	01:04	01:36
N 10°	12:09	12:55	13:39	00:13	00:53	01:29
0°	12:22	13:04	13:44	•• ••	00:42	01:23

May 18, 19, 20 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	236°10.2	185°34.6	N17°46.1	223°27.4	N04°06.8	180°07.7	N19°01.6	246°32.5	S06°25.4	Alpheratz	357°35.6	29°13.3
1	251°12.7	200°34.0	46.9	238°28.1	07.6	195°09.6	01.8	261°34.8	25.3	Ankaa	353°07.9	-42°10.3
2	266°15.2	215°33.3	47.8	253°28.8	08.3	210°11.4	01.9	276°37.1	25.3	Schedar	349°32.1	56°40.0
3	281°17.6	230°32.7	· · 48.6	268°29.5	· · 09.1	225°13.3	· · 02.0	291°39.4	· · 25.2	Diphda	348°48.1	-17°51.2
4	296°20.1	245°32.0	49.5	283°30.2	09.8	240°15.2	02.2	306°41.7	25.1	Achernar	335°21.1	-57°06.7
5	311°22.6	260°31.4	50.4	298°30.9	10.6	255°17.0	02.3	321°44.0	25.1	Hamal	327°52.2	23°34.5
6	326°25.0	275°30.8	N17°51.2	313°31.6	N04°11.3	270°18.9	N19°02.4	336°46.3	S06°25.0	Polaris	314°47.7	89°21.9
7	341°27.5	290°30.1	52.1	328°32.3	12.0	285°20.7	02.6	351°48.6	25.0	Acamar	315°12.6	-40°12.4
8	356°30.0	305°29.5	52.9	343°33.1	12.8	300°22.6	02.7	6°50.9	24.9	Menkar	314°07.1	4°11.0
9	11°32.4	320°28.8	· · 53.8	358°33.8	· · 13.5	315°24.4	· · 02.8	21°53.3	· · 24.9	Mirfak	308°29.6	49°56.8
10	26°34.9	335°28.2	54.6	13°34.5	14.3	330°26.3	03.0	36°55.6	24.8	Aldebaran	290°40.6	16°33.4
11	41°37.3	350°27.6	55.5	28°35.2	15.0	345°28.1	03.1	51°57.9	24.7	Rigel	281°04.7	-8°10.5
12	56°39.8	5°26.9	N17°56.3	43°35.9	N04°15.8	0°30.0	N19°03.2	67°00.2	S06°24.7	Capella	280°23.2	46°01.4
13	71°42.3	20°26.3	57.2	58°36.6	16.5	15°31.9	03.4	82°02.5	24.6	Bellatrix	278°23.8	6°22.3
14	86°44.7	35°25.6	58.0	73°37.3	17.3	30°33.7	03.5	97°04.8	24.6	Elnath	278°02.9	28°37.7
15	101°47.2	50°25.0	· · 58.9	88°38.0	· · 18.0	45°35.6	· · 03.6	112°07.1	· · 24.5	Alnilam	275°38.6	-1°11.2
16	116°49.7	65°24.3	17°59.7	103°38.7	18.7	60°37.4	03.8	127°09.4	24.5	Betelgeuse	270°53.0	7°24.7
17	131°52.1	80°23.7	18°00.6	118°39.4	19.5	75°39.3	03.9	142°11.7	24.4	Canopus	263°53.1	-52°42.6
18	146°54.6	95°23.1	N18°01.4	133°40.2	N04°20.2	90°41.1	N19°04.0	157°14.0	S06°24.4	Sirius	258°27.0	-16°45.1
19	161°57.1	110°22.4	02.3	148°40.9	21.0	105°43.0	04.2	172°16.3	24.3	Adhara	255°06.6	-29°00.4
20	176°59.5	125°21.8	03.1	163°41.6	21.7	120°44.9	04.3	187°18.6	24.2	Procyon	244°51.6	5°09.7
21	192°02.0	140°21.1	· · 04.0	178°42.3	· · 22.5	135°46.7	· · 04.4	202°21.0	· · 24.2	Pollux	243°18.2	27°58.1
22	207°04.5	155°20.5	04.8	193°43.0	23.2	150°48.6	04.6	217°23.3	24.1	Avior	234°15.3	-59°35.5
23	222°06.9	170°19.8	05.6	208°43.7	23.9	165°50.4	04.7	232°25.6	24.1	Suhail	222°46.8	-43°32.1
Mer.pass. 08:14		ν -0.6' d 0.9' m -3.92		ν 0.7' d 0.7' m 1.08		ν 1.9' d 0.1' m -2.00		ν 2.3' d -0.1' m 1.05				

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	237°09.4	185°19.2	N18°06.5	223°44.4	N04°24.7	180°52.3	N19°04.8	247°27.9	S06°24.0
1	252°11.8	200°18.5	07.3	238°45.1	25.4	195°54.1	05.0	262°30.2	24.0
2	267°14.3	215°17.9	08.2	253°45.8	26.2	210°56.0	05.1	277°32.5	23.9
3	282°16.8	230°17.2	· · 09.0	268°46.5	· · 26.9	225°57.9	· · 05.2	292°34.8	· · 23.9
4	297°19.2	245°16.6	09.8	283°47.3	27.7	240°59.7	05.4	307°37.1	23.8
5	312°21.7	260°15.9	10.7	298°48.0	28.4	256°01.6	05.5	322°39.4	23.7
6	327°24.2	275°15.3	N18°11.5	313°48.7	N04°29.1	271°03.4	N19°05.6	337°41.7	S06°23.7
7	342°26.6	290°14.6	12.4	328°49.4	29.9	286°05.3	05.8	352°44.1	23.6
8	357°29.1	305°14.0	13.2	343°50.1	30.6	301°07.1	05.9	7°46.4	23.6
9	12°31.6	320°13.3	· · 14.0	358°50.8	· · 31.4	316°09.0	· · 06.0	22°48.7	· · 23.5
10	27°34.0	335°12.7	14.9	13°51.5	32.1	331°10.8	06.2	37°51.0	23.5
11	42°36.5	350°12.0	15.7	28°52.2	32.9	346°12.7	06.3	52°53.3	23.4
12	57°38.9	5°11.4	N18°16.5	43°52.9	N04°33.6	1°14.6	N19°06.4	67°55.6	S06°23.4
13	72°41.4	20°10.7	17.4	58°53.6	34.3	16°16.4	06.6	82°57.9	23.3
14	87°43.9	35°10.1	18.2	73°54.4	35.1	31°18.3	06.7	98°00.2	23.3
15	102°46.3	50°09.4	· · 19.0	88°55.1	· · 35.8	46°20.1	· · 06.8	113°02.5	· · 23.2
16	117°48.8	65°08.7	19.8	103°55.8	36.6	61°22.0	07.0	128°04.9	23.1
17	132°51.3	80°08.1	20.7	118°56.5	37.3	76°23.8	07.1	143°07.2	23.1
18	147°53.7	95°07.4	N18°21.5	133°57.2	N04°38.0	91°25.7	N19°07.2	158°09.5	S06°23.0
19	162°56.2	110°06.8	22.3	148°57.9	38.8	106°27.6	07.4	173°11.8	23.0
20	177°58.7	125°06.1	23.1	163°58.6	39.5	121°29.4	07.5	188°14.1	22.9
21	193°01.1	140°05.5	· · 24.0	178°59.3	· · 40.3	136°31.3	· · 07.6	203°16.4	· · 22.9
22	208°03.6	155°04.8	24.8	194°00.0	41.0	151°33.1	07.8	218°18.7	22.8
23	223°06.1	170°04.1	25.6	209°00.7	41.7	166°35.0	07.9	233°21.0	22.8
Mer.pass. 08:10		ν -0.6' d 0.8' m -3.92		ν 0.7' d 0.7' m 1.08		ν 1.9' d 0.1' m -2.00		ν 2.3' d -0.1' m 1.05	

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	238°08.5	185°03.5	N18°26.4	224°01.5	N04°42.5	181°36.8	N19°08.0	248°23.4	S06°22.7
1	253°11.0	200°02.8	27.3	239°02.2	43.2	196°38.7	08.2	263°25.7	22.7
2	268°13.4	215°02.2	28.1	254°02.9	44.0	211°40.5	08.3	278°28.0	22.6
3	283°15.9	230°01.5	· · 28.9	269°03.6	· · 44.7	226°42.4	· · 08.4	293°30.3	· · 22.6
4	298°18.4	245°00.8	29.7	284°04.3	45.4	241°44.3	08.6	308°32.6	22.5
5	313°20.8	260°00.2	30.5	299°05.0	46.2	256°46.1	08.7	323°34.9	22.4
6	328°23.3	274°59.5	N18°31.3	314°05.7	N04°46.9	271°48.0	N19°08.8	338°37.2	S06°22.4
7	343°25.8	289°58.8	32.2	329°06.4	47.7	286°49.8	09.0	353°39.5	22.3
8	358°28.2	304°58.2	33.0	344°07.1	48.4	301°51.7	09.1	8°41.9	22.3
9	13°30.7	319°57.5	· · 33.8	359°07.8	· · 49.1	316°53.5	· · 09.2	23°44.2	· · 22.2
10	28°33.2	334°56.8	34.6	14°08.6	49.9	331°55.4	09.4	38°46.5	22.2
11	43°35.6	349°56.2	35.4	29°09.3	50.6	346°57.2	09.5	53°48.8	22.1
12	58°38.1	4°55.5	N18°36.2	44°10.0	N04°51.4	1°59.1	N19°09.6	68°51.1	S06°22.1
13	73°40.5	19°54.9	37.0	59°10.7	52.1	17°01.0	09.8	83°53.4	22.0
14	88°43.0	34°54.2	37.8	74°11.4	52.8	32°02.8	09.9	98°55.7	22.0
15	103°45.5	49°53.5	· · 38.7	89°12.1	· · 53.6	47°04.7	· · 10.0	113°58.1	· · 21.9
16	118°47.9	64°52.9	39.5	104°12.8	54.3	62°06.5	10.1	129°00.4	21.9
17	133°50.4	79°52.2	40.3	119°13.5	55.1	77°08.4	10.3	144°02.7	21.8
18	148°52.9	94°51.5	N18°41.1	134°14.2	N04°55.8	92°10.2	N19°10.4	159°05.0	S06°21.8
19	163°55.3	109°50.8	41.9	149°14.9	56.5	107°12.1	10.5	174°07.3	21.7
20	178°57.8	124°50.2	42.7	164°15.7	57.3	122°13.9	10.7	189°09.6	21.6
21	194°00.3	139°49.5	· · 43.5	179°16.4	· · 58.0	137°15.8	· · 10.8	204°12.0	· · 21.6
22	209°02.7	154°48.8	44.3	194°17.1	58.8	152°17.7	10.9	219°14.3	21.5
23	224°05.2	169°48.2	45.1	209°17.8	59.5	167°19.5	11.1	234°16.6	21.5
Mer.pass. 08:06		ν -0.7' d 0.8' m -3.92		ν 0.7' d 0.7' m 1.07		ν 1.9' d 0.1' m -2.00		ν 2.3' d -0.1' m 1.05	

May 18 Sat	SHA	Mer.pass
Venus	309°24.4	11:38
Venus	347°17.1	09:06
Jupiter	303°57.5	11:58
Saturn	10°22.2	07:33

May 19 Sun	SHA	Mer.pass
Venus	308°09.8	11:39
Mars	346°35.0	09:05
Jupiter	303°42.9	11:55
Saturn	10°18.5	07:29

May 20 Mon	SHA	Mer.pass
Venus	306°55.0	11:40
Mars	345°52.9	09:03
Jupiter	303°28.3	11:52
Saturn	10°14.8	07:25

Horizontal parallax	
Venus:	0.1
Mars:	0.1

2024 May 18 to May. 20 UT

IERS Earth Orientation data as of 01-Aug-2024
 Page 95
<https://pypi.org/project/sfalmanac/>

Aries Venus

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	240°06.8	184°31.2	N19°04.9	224°35.5	N05°17.9	183°05.9	N19°14.3	250°14.5	S06°20.2
1	255°09.3	199°30.6	05.6	239°36.3	18.6	198°07.8	14.5	265°16.8	20.1
2	270°11.7	214°29.9	06.4	254°37.0	19.4	213°09.6	14.6	280°19.2	20.1
3	285°14.2	229°29.2	· · 07.2	269°37.7	· · 20.1	228°11.5	· · 14.7	295°21.5	· · 20.0
4	300°16.7	244°28.5	08.0	284°38.4	20.8	243°13.3	14.9	310°23.8	20.0
5	315°19.1	259°27.8	08.8	299°39.1	21.6	258°15.2	15.0	325°26.1	19.9
6	330°21.6	274°27.1	N19°09.5	314°39.8	N05°22.3	273°17.0	N19°15.1	340°28.4	S06°19.9
7	345°24.0	289°26.4	10.3	329°40.5	23.0	288°18.9	15.3	355°30.8	19.8
8	0°26.5	304°25.8	11.1	344°41.2	23.8	303°20.8	15.4	10°33.1	19.8
9	15°29.0	319°25.1	· · 11.8	359°41.9	· · 24.5	318°22.6	· · 15.5	25°35.4	· · 19.7
10	30°31.4	334°24.4	12.6	14°42.6	25.2	333°24.5	15.6	40°37.7	19.7
11	45°33.9	349°23.7	13.4	29°43.4	26.0	348°26.3	15.8	55°40.0	19.6
12	60°36.4	4°23.0	N19°14.2	44°44.1	N05°26.7	3°28.2	N19°15.9	70°42.4	S06°19.6
13	75°38.8	19°22.3	14.9	59°44.8	27.4	18°30.0	16.0	85°44.7	19.5
14	90°41.3	34°21.6	15.7	74°45.5	28.2	33°31.9	16.2	100°47.0	19.5
15	105°43.8	49°20.9	· · 16.5	89°46.2	· · 28.9	48°33.7	· · 16.3	115°49.3	· · 19.4
16	120°46.2	64°20.2	17.2	104°46.9	29.6	63°35.6	16.4	130°51.6	19.4
17	135°48.7	79°19.6	18.0	119°47.6	30.4	78°37.5	16.6	145°54.0	19.3
18	150°51.1	94°18.9	N19°18.8	134°48.3	N05°31.1	93°39.3	N19°16.7	160°56.3	S06°19.3
19	165°53.6	109°18.2	19.5	149°49.0	31.8	108°41.2	16.8	175°58.6	19.2
20	180°56.1	124°17.5	20.3	164°49.7	32.6	123°43.0	16.9	191°00.9	19.2
21	195°58.5	139°16.8	· · 21.0	179°50.5	· · 33.3	138°44.9	· · 17.1	206°03.2	· · 19.1
22	211°01.0	154°16.1	21.8	194°51.2	34.0	153°46.7	17.2	221°05.6	19.1
23	226°03.5	169°15.4	22.6	209°51.9	34.8	168°48.6	17.3	236°07.9	19.0
Mer.pass. 07:58		ν -0.7' d 0.8' m -3.92		ν 0.7' d 0.7' m 1.07		ν 1.9' d 0.1' m -2.00		ν 2.3' d -0.1' m 1.04	

<https://pypi.org/project/sfalmanac/>

May 21 Tue	SHA	Mer.pass
Venus	305° 39.8	11:41
Mars	345° 10.8	09:02
Jupiter	303° 13.7	11:49
Saturn	10° 11.2	07:22

May 22 Wed	SHA	Mer.pass
Venus	304° 24.4	11:42
Mars	344° 28.7	09:01
Jupiter	302° 59.1	11:46
Saturn	10° 07.7	07:18

May 23 Thu	SHA	Mer.pass
Venus	303° 08.8	11:44
Mars	343° 46.6	09:00
Jupiter	302° 44.5	11:43
Saturn	10° 04.3	07:14

Horizontal parallax		
Venus:		0.1
Mars:		0.1

May 22 Wed	SHA	Mer.pass
Venus	304°24.4	11:42
Mars	344°28.7	09:01
Jupiter	302°59.1	11:46
Saturn	10°07.7	07:18

Horizontal parallax	
Venus:	0.1
Mars:	0.1

h Sun Moon						
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	180°50.9	N20°14.5	31°01.6	15.1'	S13°02.6	13.1' 54.8'
1	195°50.9	15.1	45°35.8	15.1'	13°15.8	13.1' 54.8'
2	210°50.9	15.6	60°09.8	15.0'	13°28.9	13.0' 54.9'
3	225°50.8	· · 16.1	74°43.8	14.9'	13°41.9	13.0' 54.9'
4	240°50.8	16.6	89°17.7	14.8'	13°54.9	13.0' 54.9'
5	255°50.7	17.1	103°51.5	14.8'	14°07.9	12.9' 54.9'
6	270°50.7	N20°17.6	118°25.3	14.7'	S14°20.8	12.9' 54.9'
7	285°50.6	18.1	132°59.0	14.6'	14°33.6	12.8' 55.0'
8	300°50.6	18.6	147°32.6	14.5'	14°46.4	12.8' 55.0'
9	315°50.6	· · 19.0	162°06.1	14.5'	14°59.2	12.7' 55.0'
10	330°50.5	19.5	176°39.6	14.4'	15°11.9	12.7' 55.0'
11	345°50.5	20.0	191°12.9	14.3'	15°24.6	12.6' 55.0'
12	0°50.4	N20°20.5	205°46.2	14.2'	S15°37.1	12.5' 55.0'
13	15°50.4	21.0	220°19.4	14.1'	15°49.7	12.5' 55.1'
14	30°50.3	21.5	234°52.6	14.0'	16°02.2	12.4' 55.1'
15	45°50.3	· · 22.0	249°25.6	14.0'	16°14.6	12.4' 55.1'
16	60°50.2	22.5	263°58.6	13.9'	16°27.0	12.3' 55.1'
17	75°50.2	23.0	278°31.5	13.8'	16°39.3	12.2' 55.1'
18	90°50.2	N20°23.5	293°04.2	13.7'	S16°51.5	12.2' 55.2'
19	105°50.1	24.0	307°36.9	13.6'	17°03.7	12.1' 55.2'
20	120°50.1	24.5	322°09.6	13.5'	17°15.8	12.1' 55.2'
21	135°50.0	· · 25.0	336°42.1	13.4'	17°27.9	12.0' 55.2'
22	150°50.0	25.5	351°14.5	13.4'	17°39.9	11.9' 55.2'
23	165°49.9	26.0	5°46.9	13.3'	17°51.8	11.9' 55.2'
SD = 15.8' $d = 0.5'$ SD = 15.0'						
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	180°49.9	N20°26.4	20°19.2	13.2'	S18°03.7	11.8' 55.3'
1	195°49.8	26.9	34°51.3	13.1'	18°15.4	11.7' 55.3'
2	210°49.8	27.4	49°23.4	13.0'	18°27.2	11.6' 55.3'
3	225°49.7	· · 27.9	63°55.4	12.9'	18°38.8	11.6' 55.3'
4	240°49.7	28.4	78°27.3	12.8'	18°50.4	11.5' 55.3'
5	255°49.6	28.9	92°59.1	12.7'	19°01.9	11.4' 55.4'
6	270°49.6	N20°29.4	107°30.8	12.6'	S19°13.3	11.3' 55.4'
7	285°49.5	29.8	122°02.4	12.5'	19°24.6	11.3' 55.4'
8	300°49.5	30.3	136°33.9	12.4'	19°35.9	11.2' 55.4'
9	315°49.4	· · 30.8	151°05.4	12.3'	19°47.1	11.1' 55.4'
10	330°49.4	31.3	165°36.7	12.2'	19°58.2	11.0' 55.5'
11	345°49.3	31.8	180°07.9	12.1'	20°09.2	10.9' 55.5'
12	0°49.3	N20°32.3	194°39.0	12.0'	S20°20.1	10.8' 55.5'
13	15°49.2	32.7	209°10.1	11.9'	20°31.0	10.8' 55.5'
14	30°49.2	33.2	223°41.0	11.8'	20°41.7	10.7' 55.6'
15	45°49.1	· · 33.7	238°11.8	11.7'	20°52.4	10.6' 55.6'
16	60°49.1	34.2	252°42.6	11.6'	21°03.0	10.5' 55.6'
17	75°49.0	34.7	267°13.2	11.5'	21°13.5	10.4' 55.6'
18	90°49.0	N20°35.1	281°43.7	11.4'	S21°23.9	10.3' 55.6'
19	105°48.9	35.6	296°14.2	11.3'	21°34.2	10.2' 55.7'
20	120°48.9	36.1	310°44.5	11.2'	21°44.4	10.1' 55.7'
21	135°48.8	· · 36.6	325°14.7	11.1'	21°54.5	10.0' 55.7'
22	150°48.8	37.0	339°44.9	11.0'	22°04.5	9.9' 55.7'
23	165°48.7	37.5	354°14.9	10.9'	22°14.4	9.8' 55.7'
SD = 15.8' $d = 0.5'$ SD = 15.1'						
Thu	GHA	Dec	GHA	ν	Dec	d HP
0	180°48.7	N20°38.0	8°44.8	10.8'	S22°24.3	9.7' 55.8'
1	195°48.6	38.5	23°14.7	10.7'	22°34.0	9.6' 55.8'
2	210°48.6	38.9	37°44.4	10.6'	22°43.6	9.5' 55.8'
3	225°48.5	· · 39.4	52°14.0	10.5'	22°53.1	9.4' 55.8'
4	240°48.5	39.9	66°43.5	10.4'	23°02.5	9.3' 55.8'
5	255°48.4	40.3	81°12.9	10.3'	23°11.8	9.2' 55.9'
6	270°48.4	N20°40.8	95°42.2	10.2'	S23°21.0	9.1' 55.9'
7	285°48.3	41.3	110°11.5	10.1'	23°30.1	9.0' 55.9'
8	300°48.2	41.8	124°40.6	10.0'	23°39.1	8.9' 55.9'
9	315°48.2	· · 42.2	139°09.6	9.9'	23°47.9	8.7' 56.0'
10	330°48.1	42.7	153°38.5	9.8'	23°56.7	8.6' 56.0'
11	345°48.1	43.2	168°07.3	9.7'	24°05.3	8.5' 56.0'
12	0°48.0	N20°43.6	182°36.0	9.6'	S24°13.8	8.4' 56.0'
13	15°48.0	44.1	197°04.6	9.5'	24°22.2	8.3' 56.0'
14	30°47.9	44.6	211°33.1	9.4'	24°30.5	8.2' 56.1'
15	45°47.9	· · 45.0	226°01.5	9.3'	24°38.6	8.0' 56.1'
16	60°47.8	45.5	240°29.8	9.2'	24°46.6	7.9' 56.1'
17	75°47.7	45.9	254°58.0	9.1'	24°54.6	7.8' 56.1'
18	90°47.7	N20°46.4	269°26.1	9.0'	S25°02.3	7.7' 56.2'
19	105°47.6	46.9	283°54.1	8.9'	25°10.0	7.5' 56.2'
20	120°47.6	47.3	298°22.0	8.8'	25°17.5	7.4' 56.2'
21	135°47.5	· · 47.8	312°49.8	8.7'	25°24.9	7.3' 56.2'
22	150°47.5	48.3	327°17.6	8.6'	25°32.2	7.1' 56.2'
23	165°47.4	48.7	341°45.2	8.5'	25°39.3	7.0' 56.3'
SD = 15.8' $d = 0.5'$ SD = 15.2'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	--	--	00:58	23:03	--	--
66°	--	--	01:52	22:05	--	--
64°	--	--	02:24	21:32	--	--
62°	--	01:17	02:48	21:08	22:41	--
60°	--	01:55	03:06	20:49	22:01	--
N 58°	--	02:21	03:22	20:33	21:34	--
56°	01:10	02:41	03:35	20:20	21:14	22:48
54°	01:45	02:58	03:46	20:09	20:57	22:11
52°	02:09	03:12	03:56	19:58	20:43	21:46
50°	02:28	03:24	04:05	19:49	20:31	21:27
45°	03:03	03:48	04:24	19:30	20:06	20:51
N 40°	03:28	04:07	04:39	19:15	19:47	20:26
35°	03:48	04:23	04:52	19:02	19:31	20:06
30°	04:04	04:36	05:03	18:51	19:18	19:50
20°	04:29	04:58	05:22	18:32	18:56	19:24
N 10°	04:49	05:15	05:38	18:16	18:38	19:05
0°	05:05	05:31	05:53	18:00	18:22	18:48
S 10°	05:20	05:46	06:08	17:45	18:08	18:33
20°	05:34	06:01	06:24	17:29	17:53	18:20
30°	05:47	06:17	06:42	17:11	17:37	18:06
35°	05:55	06:26	06:53	17:00	17:28	17:59
40°	06:02	06:35	07:05	16:48	17:18	17:51
45°	06:10	06:47	07:19	16:34	17:06	17:43
S 50°	06:20	07:00	07:36	16:17	16:53	17:33
52°	06:24	07:06	07:45	16:08	16:47	17:29
54°	06:28	07:13	07:54	15:59	16:40	17:25
56°	06:33	07:20	08:04	15:49	16:33	17:20
58°	06:38	07:28	08:16	15:37	16:25	17:15
S 60°	06:44	07:37	08:29	15:24	16:16	17:09

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	■	■	■	00:50	■	■
N 70°	20:32	■	■	01:14	00:20	■
68°	19:54	■	■	01:33	01:00	■
66°	19:27	21:45	■	01:48	01:27	00:48
64°	19:07	21:03	23:50	02:01	01:49	01:31
62°	18:51	20:35	22:32	02:11	02:06	02:00
60°	18:38	20:14	21:56	02:20	02:20	02:22
N 58°	18:26	19:57	21:30	02:28	02:32	02:40
56°	18:16	19:42	21:10	02:35	02:43	02:55
54°	18:08	19:29	20:53	02:42	02:53	03:08
52°	18:00	19:18	20:38	02:48	03:01	03:20
50°	17:53	19:09	20:26	02:53	03:09	03:30
45°	17:38	18:48	20:00	03:04	03:25	03:51
N 40°	17:26	18:32	19:40	03:14	03:39	04:09
35°	17:15	18:18	19:23	03:22	03:50	04:24
30°	17:06	18:06	19:08	03:29	04:00	04:37
20°	16:51	17:46	18:44	03:41	04:18	04:59
N 10°	16:38	17:28	18:23	03:52	04:33	05:18
0°	16:25	17:12	18:03	04:02	04:47	05:36
S 10°	16:13	16:56	17:43	04:13	05:02	05:54
20°	16:00	16:39	17:22	04:24	05:17	06:13
30°	15:45	16:19	16:58	04:37	05:35	06:36
35°	15:36	16:07	16:44	04:44	05:45	06:49
40°	15:27	15:54	16:28	04:52	05:57	07:04
45°	15:15	15:39	16:09	05:02	06:11	07:23
S 50°	15:02	15:20	15:45	05:14	06:29	07:46
52°	14:55	15:11	15:34	05:20	06:37	07:57
54°	14:48	15:02	15:21	05:26	06:46	08:09
56°	14:41	14:50	15:06	05:33	06:57	08:23
58°	14:32	14:38	14:48	05:41	07:09	08:40
S 60°	14:22	14:23	14:27	05:50	07:23	09:01

Day	Sun		Mer. Pass hh:mm	Moon		Age 13-15 93-100%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
21	03:24	03:22	11:57	22:36	10:14	
22	03:19	03:17	11:57	23:24	10:59	
23	03:15	03:12	11:57	--	11:49	

May 24, 25, 26 UT (Fri., Sat., Sun.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	242°05.1	183°57.9	N19°41.3	225°09.6	N05°53.0	184°35.0	N19°20.6	252°06.0	S06°17.8			Alpheratz	357°35.5	29°13.3			
1	257°07.5	198°57.2	42.0	240°10.3	53.8	199°36.8	20.7	267°08.3	17.8			Ankaa	353°07.9	-42°10.3			
2	272°10.0	213°56.5	42.7	255°11.0	54.5	214°38.7	20.8	282°10.6	17.7			Schedar	349°32.0	56°40.0			
3	287°12.5	228°55.8	· · 43.5	270°11.7	· · 55.2	229°40.5	· · 20.9	297°13.0	· · 17.7			Diphda	348°48.0	-17°51.2			
4	302°14.9	243°55.1	44.2	285°12.5	56.0	244°42.4	21.1	312°15.3	17.6			Achernar	335°21.0	-57°06.6			
5	317°17.4	258°54.3	44.9	300°13.2	56.7	259°44.2	21.2	327°17.6	17.6			Hamal	327°52.1	23°34.5			
6	332°19.9	273°53.6	N19°45.7	315°13.9	N05°57.4	274°46.1	N19°21.3	342°19.9	S06°17.5			Polaris	314°46.2	89°21.9			
7	347°22.3	288°52.9	46.4	330°14.6	58.2	289°48.0	21.5	357°22.3	17.5			Acamar	315°12.6	-40°12.4			
8	2°24.8	303°52.2	47.1	345°15.3	58.9	304°49.8	21.6	12°24.6	17.4			Menkar	314°07.0	4°11.1			
9	17°27.3	318°51.5	· · 47.9	0°16.0	05°59.6	319°51.7	· · 21.7	27°26.9	· · 17.4			Mirfak	308°29.5	49°56.7			
10	32°29.7	333°50.8	48.6	15°16.7	06°00.3	334°53.5	21.8	42°29.3	17.3			Aldebaran	290°40.6	16°33.4			
11	47°32.2	348°50.1	49.3	30°17.4	01.1	349°55.4	22.0	57°31.6	17.3			Rigel	281°04.7	-8°10.5			
12	62°34.6	3°49.4	N19°50.0	45°18.1	N06°01.8	4°57.2	N19°22.1	72°33.9	S06°17.2			Capella	280°23.2	46°01.4			
13	77°37.1	18°48.7	50.8	60°18.8	02.5	19°59.1	22.2	87°36.2	17.2			Bellatrix	278°23.8	6°22.3			
14	92°39.6	33°48.0	51.5	75°19.6	03.2	35°00.9	22.4	102°38.6	17.1			Elnath	278°02.9	28°37.7			
15	107°42.0	48°47.2	· · 52.2	90°20.3	· · 04.0	50°02.8	· · 22.5	117°40.9	· · 17.1			Alnilam	275°38.6	-1°11.2			
16	122°44.5	63°46.5	52.9	105°21.0	04.7	65°04.7	22.6	132°43.2	17.0			Betelgeuse	270°53.0	7°24.7			
17	137°47.0	78°45.8	53.7	120°21.7	05.4	80°06.5	22.7	147°45.5	17.0			Canopus	263°53.1	-52°42.6			
18	152°49.4	93°45.1	N19°54.4	135°22.4	N06°06.2	95°08.4	N19°22.9	162°47.9	S06°16.9			Sirius	258°27.0	-16°45.0			
19	167°51.9	108°44.4	55.1	150°23.1	06.9	110°10.2	23.0	177°50.2	16.9			Adhara	255°06.6	-29°00.4			
20	182°54.4	123°43.7	55.8	165°23.8	07.6	125°12.1	23.1	192°52.5	16.9			Procyon	244°51.6	5°09.7			
21	197°56.8	138°43.0	· · 56.5	180°24.5	· · 08.3	140°13.9	· · 23.3	207°54.9	· · 16.8			Pollux	243°18.2	27°58.1			
22	212°59.3	153°42.2	57.3	195°25.2	09.1	155°15.8	23.4	222°57.2	16.8			Avior	234°15.3	-59°35.5			
23	228°01.8	168°41.5	58.0	210°25.9	09.8	170°17.6	23.5	237°59.5	16.7			Suhail	222°46.9	-43°32.1			
Mer.pass. 07:50			ν -0.7' d 0.7' m -3.92			ν 0.7' d 0.7' m 1.06			ν 1.9' d 0.1' m -2.00			ν 2.3' d -0.0' m 1.04					
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec						
0	243°04.2	183°40.8	N19°58.7	225°26.6	N06°10.5	185°19.5	N19°23.6	253°01.8	S06°16.7			Alphard	217°48.4	-8°45.9			
1	258°06.7	198°40.1	19°59.4	240°27.4	11.2	200°21.3	23.8	268°04.2	16.6			Regulus	207°35.0	11°50.9			
2	273°09.1	213°39.4	20°00.1	255°28.1	12.0	215°23.2	23.9	283°06.5	16.6			Dubhe	193°41.5	61°37.5			
3	288°11.6	228°38.6	· · 00.8	270°28.8	· · 12.7	230°25.1	· · 24.0	298°08.8	· · 16.5			Denebola	182°25.4	14°26.2			
4	303°14.1	243°37.9	01.5	285°29.5	13.4	245°26.9	24.1	313°11.2	16.5			Gienah	175°44.0	-17°40.8			
5	318°16.5	258°37.2	02.3	300°30.2	14.1	260°28.8	24.3	328°13.5	16.4			Acrux	173°00.4	-63°14.3			
6	333°19.0	273°36.5	N20°03.0	315°30.9	N06°14.9	275°30.6	N19°24.4	343°15.8	S06°16.4			Gacrux	171°52.0	-57°15.2			
7	348°21.5	288°35.8	03.7	330°31.6	15.6	290°32.5	24.5	358°18.1	16.3			Alioth	166°13.1	55°49.8			
8	3°23.9	303°35.0	04.4	345°32.3	16.3	305°34.3	24.7	13°20.5	16.3			Spica	158°22.6	-11°17.4			
9	18°26.4	318°34.3	· · 05.1	0°33.0	· · 17.0	320°36.2	· · 24.8	28°22.8	· · 16.2			Alkaid	152°52.0	49°11.6			
10	33°28.9	333°33.6	05.8	15°33.7	17.8	335°38.0	24.9	43°25.1	16.2			Hadar	148°36.2	-60°29.6			
11	48°31.3	348°32.9	06.5	30°34.4	18.5	350°39.9	25.0	58°27.5	16.2			Menkent	147°57.9	-36°29.5			
12	63°33.8	3°32.2	N20°07.2	45°35.2	N06°19.2	5°41.8	N19°25.2	73°29.8	S06°16.1			Arcturus	145°48.1	19°03.3			
13	78°36.2	18°31.4	07.9	60°35.9	19.9	20°43.6	25.3	88°32.1	16.1			Rigel Kent.	139°40.5	-60°56.3			
14	93°38.7	33°30.7	08.6	75°36.6	20.7	35°45.5	25.4	103°34.5	16.0			Kochab	137°18.6	74°03.4			
15	108°41.2	48°30.0	· · 09.3	90°37.3	· · 21.4	50°47.3	· · 25.5	118°36.8	· · 16.0			Zuben'ubi	136°56.3	-16°08.7			
16	123°43.6	63°29.3	10.0	105°38.0	22.1	65°49.2	25.7	133°39.1	15.9			Alphecca	126°03.8	26°37.9			
17	138°46.1	78°28.5	10.7	120°38.7	22.8	80°51.0	25.8	148°41.5	15.9			Antares	112°16.1	-26°29.2			
18	153°48.6	93°27.8	N20°11.4	135°39.4	N06°23.6	95°52.9	N19°25.9	163°43.8	S06°15.8			Atria	107°10.3	-69°04.2			
19	168°51.0	108°27.1	12.1	150°40.1	24.3	110°54.7	26.1	178°46.1	15.8			Sabik	102°03.1	-15°45.3			
20	183°53.5	123°26.4	12.8	165°40.8	25.0	125°56.6	26.2	193°48.4	15.7			Shaula	96°10.7	-37°07.3			
21	198°56.0	138°25.6	· · 13.5	180°41.5	· · 25.7	140°58.5	· · 26.3	208°50.8	· · 15.7			Rasalhague	95°58.7	12°32.4			
22	213°58.4	153°24.9	14.2	195°42.2	26.5	156°00.3	26.4	223°53.1	15.6			Eltanin	90°41.9	51°28.9			
23	229°00.9	168°24.2	14.9	210°43.0	27.2	171°02.2	26.6	238°55.4	15.6			Kaus Aust.	83°32.9	-34°22.3			
Mer.pass. 07:46			ν -0.7' d 0.7' m -3.92			ν 0.7' d 0.7' m 1.06			ν 1.9' d 0.1' m -2.00			ν 2.3' d -0.0' m 1.04					
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec						
0	244°03.4	183°23.4	N20°15.6	225°43.7	N06°27.9	186°04.0	N19°26.7	253°57.8	S06°15.6			May 24 Fri	SHA	Mer.pass			
1	259°05.8	198°22.7	16.3	240°44.4	28.6	201°05.9	26.8	269°00.1	15.5			Venus	301°52.8	11:45			
2	274°08.3	213°22.0	17.0	255°45.1	29.4	216°07.7	26.9	284°02.4	15.5			Mars	343°04.5	08:59			
3	289°10.7	228°21.2	· · 17.6	270°45.8	· · 30.1	231°09.6	· · 27.1	299°04.8	· · 15.4			Jupiter	302°29.9	11:40			
4	304°13.2	243°20.5	18.3	285°46.5	30.8	246°11.4	27.2	314°07.1	15.4			Saturn	10°00.9	07:10			
5	319°15.7	258°19.8	19.0	300°47.2	31.5	261°13.3	27.3	329°09.4	15.3								
6	334°18.1	273°19.1	N20°19.7	315°47.9	N06°32.2	276°15.2	N19°27.5	344°11.8	S06°15.3								
7	349°20.6	288°18.3	20.4	330°48.6	33.0	291°17.0	27.6	359°14.1	15.2								
8	4°23.1	303°17.6	21.1	345°49.3	33.7	306°18.9	27.7	14°16.4	15.2			May 25 Sat	SHA	Mer.pass			
9	19°25.5	318°16.9	· · 21.8	0°50.0	· · 34.4	321°20.7	· · 27.8	29°18.8	· · 15.1			Venus	300°36.6	11:46			
10	34°28.0	333°16.1	22.4	15°50.8	35.1	336°22.6	28.0	44°21.1	15.1			Mars	342°22.4	08:58			
11	49°30.5	348°15.4	23.1	30°51.5	35.9	351°24.4	28.1	59°23.4	15.1			Jupiter	302°15.3	11:37			
12	64°32.9	3°14.7	N20°23.8	45°52.2	N06°36.6	6°26.3	N19°28.2	74°25.8	S06°15.0			Saturn	9°57.6	07:07			
13	79°35.4	18°13.9	24.5	60°52.9	37.3	21°28.1	28.3	89°28.1	15.0								
14	94°37.9	33°13.2	25.2	75°53.6	38.0	36°30.0	28.5	104°30.4	14.9			May 26 Sun	SHA	Mer.pass			
15																	

h Sun Moon						
Fri	GHA	Dec	GHA	ν	Dec	d HP
0	180°47.3	N20°49.2	356°12.7	8.4'	S25°46.3	6.9' 56.3'
1	195°47.3	49.6	10°40.1	8.3'	25°53.2	6.7' 56.3'
2	210°47.2	50.1	25°07.4	8.2'	25°59.9	6.6' 56.3'
3	225°47.2	· · 50.5	39°34.7	8.1'	26°06.5	6.5' 56.3'
4	240°47.1	51.0	54°01.8	8.1'	26°13.0	6.3' 56.4'
5	255°47.1	51.5	68°28.9	8.0'	26°19.3	6.2' 56.4'
6	270°47.0	N20°51.9	82°55.9	7.9'	S26°25.5	6.0' 56.4'
7	285°46.9	52.4	97°22.7	7.8'	26°31.5	5.9' 56.4'
8	300°46.9	52.8	111°49.5	7.7'	26°37.4	5.7' 56.5'
9	315°46.8	· · 53.3	126°16.2	7.6'	26°43.1	5.6' 56.5'
10	330°46.8	53.7	140°42.8	7.5'	26°48.7	5.5' 56.5'
11	345°46.7	54.2	155°09.4	7.4'	26°54.2	5.3' 56.5'
12	0°46.6	N20°54.6	169°35.8	7.4'	S26°59.5	5.2' 56.5'
13	15°46.6	55.1	184°02.2	7.3'	27°04.6	5.0' 56.6'
14	30°46.5	55.5	198°28.4	7.2'	27°09.6	4.9' 56.6'
15	45°46.4	· · 56.0	212°54.6	7.1'	27°14.5	4.7' 56.6'
16	60°46.4	56.4	227°20.8	7.0'	27°19.2	4.5' 56.6'
17	75°46.3	56.9	241°46.8	7.0'	27°23.7	4.4' 56.6'
18	90°46.3	N20°57.3	256°12.8	6.9'	S27°28.1	4.2' 56.7'
19	105°46.2	57.8	270°38.7	6.8'	27°32.3	4.1' 56.7'
20	120°46.1	58.2	285°04.5	6.8'	27°36.4	3.9' 56.7'
21	135°46.1	· · 58.7	299°30.3	6.7'	27°40.3	3.8' 56.7'
22	150°46.0	59.1	313°55.9	6.6'	27°44.1	3.6' 56.8'
23	165°46.0	59.6	328°21.6	6.6'	27°47.7	3.4' 56.8'
SD = 15.8' $d = 0.5'$ SD = 15.3'						
Sat	GHA	Dec	GHA	ν	Dec	d HP
0	180°45.9	N21°00.0	342°47.1	6.5'	S27°51.1	3.3' 56.8'
1	195°45.8	00.4	357°12.6	6.4'	27°54.4	3.1' 56.8'
2	210°45.8	00.9	11°38.0	6.4'	27°57.5	2.9' 56.8'
3	225°45.7	· · 01.3	26°03.4	6.3'	28°00.5	2.8' 56.9'
4	240°45.6	01.8	40°28.7	6.2'	28°03.3	2.6' 56.9'
5	255°45.6	02.2	54°53.9	6.2'	28°05.9	2.5' 56.9'
6	270°45.5	N21°02.7	69°19.1	6.1'	S28°08.3	2.3' 56.9'
7	285°45.4	03.1	83°44.3	6.1'	28°10.6	2.1' 56.9'
8	300°45.4	03.5	98°09.4	6.0'	28°12.7	2.0' 57.0'
9	315°45.3	· · 04.0	112°34.4	6.0'	28°14.7	1.8' 57.0'
10	330°45.2	04.4	126°59.4	5.9'	28°16.5	1.6' 57.0'
11	345°45.2	04.8	141°24.3	5.9'	28°18.1	1.4' 57.0'
12	0°45.1	N21°05.3	155°49.2	5.9'	S28°19.5	1.3' 57.1'
13	15°45.0	05.7	170°14.1	5.8'	28°20.8	1.1' 57.1'
14	30°45.0	06.2	184°38.9	5.8'	28°21.9	0.9' 57.1'
15	45°44.9	· · 06.6	199°03.7	5.7'	28°22.8	0.8' 57.1'
16	60°44.8	07.0	213°28.4	5.7'	28°23.6	0.6' 57.1'
17	75°44.8	07.5	227°53.2	5.7'	28°24.2	0.4' 57.2'
18	90°44.7	N21°07.9	242°17.8	5.7'	S28°24.6	0.2' 57.2'
19	105°44.6	08.3	256°42.5	5.6'	28°24.8	0.1' 57.2'
20	120°44.6	08.8	271°07.1	5.6'	28°24.8	-0.1' 57.2'
21	135°44.5	· · 09.2	285°31.7	5.6'	28°24.7	-0.3' 57.2'
22	150°44.4	09.6	299°56.3	5.6'	28°24.4	-0.5' 57.3'
23	165°44.4	10.0	314°20.9	5.5'	28°24.0	-0.6' 57.3'
SD = 15.8' $d = 0.4'$ SD = 15.5'						
Sun	GHA	Dec	GHA	ν	Dec	d HP
0	180°44.3	N21°10.5	328°45.4	5.5'	S28°23.3	-0.8' 57.3'
1	195°44.2	10.9	343°10.0	5.5'	28°22.5	-1.0' 57.3'
2	210°44.2	11.3	357°34.5	5.5'	28°21.5	-1.2' 57.3'
3	225°44.1	· · 11.8	11°59.0	5.5'	28°20.4	-1.3' 57.4'
4	240°44.0	12.2	26°23.5	5.5'	28°19.0	-1.5' 57.4'
5	255°44.0	12.6	40°48.0	5.5'	28°17.5	-1.7' 57.4'
6	270°43.9	N21°13.0	55°12.4	5.5'	S28°15.8	-1.9' 57.4'
7	285°43.8	13.5	69°36.9	5.5'	28°13.9	-2.1' 57.4'
8	300°43.8	13.9	84°01.4	5.5'	28°11.9	-2.2' 57.5'
9	315°43.7	· · 14.3	98°25.9	5.5'	28°09.6	-2.4' 57.5'
10	330°43.6	14.7	112°50.4	5.5'	28°07.2	-2.6' 57.5'
11	345°43.5	15.1	127°14.9	5.5'	28°04.6	-2.8' 57.5'
12	0°43.5	N21°15.6	141°39.4	5.5'	S28°01.9	-2.9' 57.5'
13	15°43.4	16.0	156°03.9	5.5'	27°59.0	-3.1' 57.6'
14	30°43.3	16.4	170°28.4	5.5'	27°55.8	-3.3' 57.6'
15	45°43.3	· · 16.8	184°53.0	5.6'	27°52.6	-3.5' 57.6'
16	60°43.2	17.2	199°17.5	5.6'	27°49.1	-3.6' 57.6'
17	75°43.1	17.7	213°42.1	5.6'	27°45.5	-3.8' 57.6'
18	90°43.0	N21°18.1	228°06.7	5.6'	S27°41.7	-4.0' 57.7'
19	105°43.0	18.5	242°31.3	5.6'	27°37.7	-4.2' 57.7'
20	120°42.9	18.9	256°55.9	5.7'	27°33.5	-4.3' 57.7'
21	135°42.8	· · 19.3	271°20.6	5.7'	27°29.2	-4.5' 57.7'
22	150°42.8	19.7	285°45.3	5.7'	27°24.7	-4.7' 57.7'
23	165°42.7	20.2	300°10.0	5.8'	27°20.0	-4.8' 57.8'
SD = 15.8' $d = 0.4'$ SD = 15.6'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°	--	--	00:26	--	--	--
66°	--	--	01:40	22:18	--	--
64°	--	--	02:15	21:41	--	--
62°	--	01:01	02:41	21:15	22:58	--
60°	--	01:46	03:00	20:55	22:11	--
N 58°	--	02:14	03:16	20:39	21:42	--
56°	00:56	02:36	03:30	20:25	21:20	23:03
54°	01:36	02:53	03:42	20:13	21:02	22:20
52°	02:03	03:08	03:52	20:02	20:48	21:53
50°	02:23	03:20	04:02	19:53	20:35	21:33
45°	02:59	03:45	04:21	19:34	20:09	20:55
N 40°	03:26	04:05	04:37	19:18	19:49	20:29
35°	03:46	04:21	04:50	19:04	19:33	20:09
30°	04:02	04:35	05:01	18:53	19:20	19:52
20°	04:28	04:57	05:21	18:33	18:57	19:26
N 10°	04:49	05:15	05:38	18:16	18:39	19:06
0°	05:05	05:31	05:53	18:01	18:23	18:49
S 10°	05:21	05:46	06:09	17:45	18:07	18:33
20°	05:35	06:02	06:25	17:29	17:52	18:19
30°	05:49	06:18	06:44	17:10	17:35	18:05
35°	05:56	06:27	06:55	16:59	17:26	17:57
40°	06:04	06:38	07:07	16:46	17:16	17:49
45°	06:13	06:49	07:22	16:32	17:04	17:41
S 50°	06:23	07:03	07:40	16:14	16:51	17:31
52°	06:27	07:09	07:48	16:05	16:44	17:27
54°	06:32	07:16	07:58	15:56	16:37	17:22
56°	06:37	07:24	08:09	15:45	16:30	17:17
58°	06:42	07:32	08:21	15:33	16:21	17:11
S 60°	06:48	07:42	08:35	15:19	16:11	17:05

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°						
N 70°						
68°						
66°						
64°				00:32		
62°				01:51		
60°	23:35	· ·	00:47	02:27	02:44	03:34
N 58°	22:59	· ·	00:07	02:53	03:21	04:15
56°	22:32	23:39	· ·	03:14	03:47	04:43
54°	22:12	23:17	· ·	03:31	04:08	05:04
52°	21:54	22:59	23:48	03:46	04:26	05:22
50°	21:40	22:44	23:34	03:59	04:41	05:38
45°	21:10	22:13	23:05	04:26	05:11	06:08
N 40°	20:47	21:49	22:42	04:47	05:34	06:32
35°	20:28	21:29	22:24	05:04	05:54	06:52
30°	20:11	21:12	22:08	05:20	06:10	07:09
20°	19:44	20:43	21:40	05:46	06:39	07:37
N 10°	19:20	20:19	21:17	06:08	07:03	08:02
0°	18:58	19:56	20:55	06:29	07:25	08:24
S 10°	18:36	19:33	20:33	06:50	07:48	08:47
20°	18:13	19:09	20:10	07:12	08:12	09:11
30°	17:45	18:40	19:42	07:38	08:41	09:39
35°	17:29	18:23	19:26	07:54	08:57	09:56
40°	17:11	18:04	19:07	08:12	09:17	10:15
45°	16:49	17:40	18:44	08:34	09:40	10:38
S 50°	16:20	17:09	18:15	09:01	10:11	11:08
52°	16:06	16:54	18:00	09:15	10:26	11:23
54°	15:50	16:37	17:43	09:31	10:44	11:40
56°	15:32	16:15	17:23	09:49	11:05	12:00
58°	15:08	15:48	16:58	10:12	11:32	12:26
S 60°	14:38	15:10	16:23	10:42	12:10	13:01

Day	Sun			Moon		Age 16-18 100-93%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss	Mer. Pass hh:mm	Mer.Pass. Upper hh:mm	Lower hh:mm	
24	03:09	03:07	11:57	00:16	12:43	
25	03:04	03:00	11:57	01:12	13:41	
26	02:57	02:54	11:57	02:10	14:40	

May 27, 28, 29 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars							
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	245°02.5	183°05.8	N20°31.9	226°00.7	N06°45.2	186°48.5	N19°29.7	254°53.8	S06°14.5	Alpheratz	357°35.5	29°13.3					
1	260°05.0	198°05.1	32.6	241°01.4	45.9	201°50.4	29.8	269°56.1	14.4	Ankaa	353°07.9	-42°10.3					
2	275°07.4	213°04.3	33.2	256°02.1	46.7	216°52.3	30.0	284°58.5	14.4	Schedar	349°32.0	56°40.0					
3	290°09.9	228°03.6	•• 33.9	271°02.8	•• 47.4	231°54.1	•• 30.1	300°00.8	•• 14.3	Diphda	348°48.0	-17°51.1					
4	305°12.4	243°02.8	34.6	286°03.5	48.1	246°56.0	30.2	315°03.1	14.3	Achernar	335°21.0	-57°06.6					
5	320°14.8	258°02.1	35.2	301°04.2	48.8	261°57.8	30.3	330°05.5	14.3	Hamal	327°52.1	23°34.5					
6	335°17.3	273°01.4	N20°35.9	316°04.9	N06°49.5	276°59.7	N19°30.5	345°07.8	S06°14.2	Polaris	314°45.4	89°21.9					
7	350°19.7	288°00.6	36.6	331°05.6	50.3	292°01.5	30.6	0°10.1	14.2	Acamar	315°12.6	-40°12.4					
8	5°22.2	302°59.9	37.2	346°06.3	51.0	307°03.4	30.7	15°12.5	14.1	Menkar	314°07.0	4°11.1					
9	20°24.7	317°59.1	•• 37.9	1°07.1	•• 51.7	322°05.2	•• 30.8	30°14.8	•• 14.1	Mirfak	308°29.5	49°56.7					
10	35°27.1	332°58.4	38.5	16°07.8	52.4	337°07.1	31.0	45°17.1	14.0	Aldebaran	290°40.6	16°33.5					
11	50°29.6	347°57.6	39.2	31°08.5	53.1	352°09.0	31.1	60°19.5	14.0	Rigel	281°04.7	-8°10.4					
12	65°32.1	2°56.9	N20°39.9	46°09.2	N06°53.9	7°10.8	N19°31.2	75°21.8	S06°14.0	Capella	280°23.2	46°01.3					
13	80°34.5	17°56.1	40.5	61°09.9	54.6	22°12.7	31.4	90°24.2	13.9	Bellatrix	278°23.8	6°22.3					
14	95°37.0	32°55.4	41.2	76°10.6	55.3	37°14.5	31.5	105°26.5	13.9	Elnath	278°02.9	28°37.7					
15	110°39.5	47°54.7	•• 41.8	91°11.3	•• 56.0	52°16.4	•• 31.6	120°28.8	•• 13.8	Alnilam	275°38.6	-1°11.2					
16	125°41.9	62°53.9	42.5	106°12.0	56.7	67°18.2	31.7	135°31.2	13.8	Betelgeuse	270°53.0	7°24.7					
17	140°44.4	77°53.2	43.1	121°12.7	57.4	82°20.1	31.9	150°33.5	13.7	Canopus	263°53.1	-52°42.6					
18	155°46.9	92°52.4	N20°43.8	136°13.4	N06°58.2	97°21.9	N19°32.0	165°35.8	S06°13.7	Sirius	258°27.0	-16°45.0					
19	170°49.3	107°51.7	44.4	151°14.1	58.9	112°23.8	32.1	180°38.2	13.7	Adhara	255°06.6	-29°00.4					
20	185°51.8	122°50.9	45.1	166°14.8	06°59.6	127°25.7	32.2	195°40.5	13.6	Procyon	244°51.6	5°09.7					
21	200°54.2	137°50.2	•• 45.7	181°15.6	07°00.3	142°27.5	•• 32.4	210°42.9	•• 13.6	Pollux	243°18.2	27°58.1					
22	215°56.7	152°49.4	46.4	196°16.3	01.0	157°29.4	32.5	225°45.2	13.5	Avior	234°15.3	-59°35.5					
23	230°59.2	167°48.7	47.0	211°17.0	01.7	172°31.2	32.6	240°47.5	13.5	Suhail	222°46.9	-43°32.1					
Mer.pass. 07:39		ν -0.7' d 0.7' m -3.92		ν 0.7' d 0.7' m 1.06		ν 1.9' d 0.1' m -2.00		ν 2.3' d -0.0' m 1.03		Miaplacidus	221°38.8	-69°49.3					
Alphard	217°48.4	-8°45.9	Regulus	207°35.0	11°50.9	Dubhe	193°41.5	61°37.5	Denebola	182°25.4	14°26.2						
Gienah	175°44.0	-17°40.8	Acrux	173°00.4	-63°14.3	Gacrux	171°52.0	-57°15.2	Alioth	166°13.1	55°49.9						
Spica	158°22.6	-11°17.4	Alkaid	152°52.0	49°11.6	Hadar	148°36.2	-60°29.6	Menkent	147°57.9	-36°29.5						
Arcturus	145°48.1	19°03.3	Rigil Kent.	139°40.5	-60°56.3	Kochab	137°18.7	74°03.4	Zuben'ubi	136°56.3	-16°08.7						
Alphecca	126°03.8	26°37.9	Antares	112°16.1	-26°29.2	Atria	107°10.3	-69°04.2	Sabik	102°03.0	-15°45.3						
Shaula	96°10.7	-37°07.3	Rasalhague	95°58.7	12°32.4	Eltanin	90°41.9	51°28.9	Kaus Aust.	83°32.9	-34°22.3						
Vega	80°33.2	38°48.2	Nunki	75°48.1	-26°16.0	Altair	62°00.3	8°55.8	Peacock	53°06.2	-56°39.2						
Deneb	49°25.9	45°21.7	Enif	33°39.2	9°59.1	Al Na'ir	27°33.5	-46°50.4	Fomalhaut	15°15.1	-29°29.5						
Scheat	13°45.8	28°12.7	Markab	13°30.5	15°20.0												
May 27 Mon		SHA	Mer.pass														
Venus		298°03.3	11:48														
Mars		340°58.2	08:56														
Jupiter		301°46.1	11:31														
Saturn		9°51.3	06:59														
May 28 Tue		SHA	Mer.pass														
Venus		296°46.3	11:49														
Mars		340°16.0	08:54														
Jupiter		301°31.4	11:28														
Saturn		9°48.2	06:56														
May 29 Wed		SHA	Mer.pass														
Venus		295°29.0	11:51														
Mars		339°33.9	08:53														
Jupiter		301°16.8	11:25														
Saturn		9°45.3	06:52														
Horizontal parallax																	
Venus:		0.1															
Mars:		0.1															
Mer.pass. 07:31		ν -0.8' d 0.6' m -3.92		ν 0.7' d 0.7' m 1.05		ν 1.9' d 0.1' m -2.00		ν 2.3' d -0.0' m 1.03									

Sun				Moon				
Mon	GHA	Dec		GHA	ν	Dec	d	HP
0	180°42.6	N21°20.6		314°34.8	5.8'	S27°15.2	-5.0'	57.8'
1	195°42.5	21.0		328°59.6	5.8'	27°10.1	-5.2'	57.8'
2	210°42.5	21.4		343°24.4	5.9'	27°05.0	-5.4'	57.8'
3	225°42.4	· · 21.8		357°49.3	5.9'	26°59.6	-5.5'	57.8'
4	240°42.3	22.2		12°14.2	5.9'	26°54.1	-5.7'	57.8'
5	255°42.2	22.6		26°39.1	6.0'	26°48.4	-5.9'	57.9'
6	270°42.2	N21°23.0		41°04.1	6.0'	S26°42.5	-6.0'	57.9'
7	285°42.1	23.5		55°29.2	6.1'	26°36.5	-6.2'	57.9'
8	300°42.0	23.9		69°54.3	6.1'	26°30.3	-6.4'	57.9'
9	315°41.9	· · 24.3		84°19.4	6.2'	26°23.9	-6.5'	57.9'
10	330°41.9	24.7		98°44.6	6.2'	26°17.4	-6.7'	58.0'
11	345°41.8	25.1		113°09.8	6.3'	26°10.7	-6.9'	58.0'
12	0°41.7	N21°25.5		127°35.1	6.3'	S26°03.8	-7.0'	58.0'
13	15°41.6	25.9		142°00.4	6.4'	25°56.8	-7.2'	58.0'
14	30°41.6	26.3		156°25.8	6.4'	25°49.6	-7.3'	58.0'
15	45°41.5	· · 26.7		170°51.2	6.5'	25°42.3	-7.5'	58.0'
16	60°41.4	27.1		185°16.7	6.6'	25°34.8	-7.7'	58.1'
17	75°41.3	27.5		199°42.3	6.6'	25°27.1	-7.8'	58.1'
18	90°41.3	N21°27.9		214°07.9	6.7'	S25°19.3	-8.0'	58.1'
19	105°41.2	28.3		228°33.6	6.7'	25°11.3	-8.1'	58.1'
20	120°41.1	28.7		242°59.3	6.8'	25°03.2	-8.3'	58.1'
21	135°41.0	· · 29.1		257°25.1	6.9'	24°54.9	-8.4'	58.1'
22	150°40.9	29.5		271°51.0	6.9'	24°46.5	-8.6'	58.2'
23	165°40.9	29.9		286°16.9	7.0'	24°37.9	-8.7'	58.2'
SD = 15.8' d = 0.4'				SD = 15.8'				
Tue	GHA	Dec		GHA	ν	Dec	d	HP
0	180°40.8	N21°30.3		300°42.9	7.1'	S24°29.1	-8.9'	58.2'
1	195°40.7	30.7		315°09.0	7.1'	24°20.2	-9.0'	58.2'
2	210°40.6	31.1		329°35.1	7.2'	24°11.2	-9.2'	58.2'
3	225°40.6	· · 31.5		344°01.3	7.3'	24°02.0	-9.3'	58.3'
4	240°40.5	31.9		358°27.6	7.3'	23°52.7	-9.5'	58.3'
5	255°40.4	32.3		12°53.9	7.4'	23°43.2	-9.6'	58.3'
6	270°40.3	N21°32.7		27°20.3	7.5'	S23°33.6	-9.8'	58.3'
7	285°40.2	33.1		41°46.8	7.5'	23°23.8	-9.9'	58.3'
8	300°40.2	33.5		56°13.4	7.6'	23°13.9	-10.1'	58.3'
9	315°40.1	· · 33.9		70°40.0	7.7'	23°03.8	-10.2'	58.4'
10	330°40.0	34.3		85°06.7	7.8'	22°53.6	-10.3'	58.4'
11	345°39.9	34.7		99°33.5	7.8'	22°43.3	-10.5'	58.4'
12	0°39.8	N21°35.0		114°00.3	7.9'	S22°32.8	-10.6'	58.4'
13	15°39.8	35.4		128°27.2	8.0'	22°22.2	-10.7'	58.4'
14	30°39.7	35.8		142°54.2	8.1'	22°11.5	-10.9'	58.4'
15	45°39.6	· · 36.2		157°21.2	8.1'	22°00.6	-11.0'	58.4'
16	60°39.5	36.6		171°48.4	8.2'	21°49.6	-11.1'	58.5'
17	75°39.4	37.0		186°15.6	8.3'	21°38.5	-11.3'	58.5'
18	90°39.3	N21°37.4		200°42.9	8.4'	S21°27.2	-11.4'	58.5'
19	105°39.3	37.8		215°10.2	8.4'	21°15.8	-11.5'	58.5'
20	120°39.2	38.1		229°37.7	8.5'	21°04.3	-11.6'	58.5'
21	135°39.1	· · 38.5		244°05.2	8.6'	20°52.6	-11.8'	58.5'
22	150°39.0	38.9		258°32.7	8.7'	20°40.9	-11.9'	58.6'
23	165°38.9	39.3		273°00.4	8.7'	20°29.0	-12.0'	58.6'
SD = 15.8' d = 0.4'				SD = 15.9'				
Wed	GHA	Dec		GHA	ν	Dec	d	HP
0	180°38.9	N21°39.7		287°28.1	8.8'	S20°17.0	-12.1'	58.6'
1	195°38.8	40.1		301°55.9	8.9'	20°04.8	-12.2'	58.6'
2	210°38.7	40.4		316°23.8	8.9'	19°52.6	-12.4'	58.6'
3	225°38.6	· · 40.8		330°51.7	9.0'	19°40.2	-12.5'	58.6'
4	240°38.5	41.2		345°19.8	9.1'	19°27.7	-12.6'	58.6'
5	255°38.4	41.6		359°47.9	9.2'	19°15.1	-12.7'	58.7'
6	270°38.3	N21°42.0		14°16.0	9.2'	S19°02.4	-12.8'	58.7'
7	285°38.3	42.3		28°44.3	9.3'	18°49.6	-12.9'	58.7'
8	300°38.2	42.7		43°12.6	9.4'	18°36.7	-13.0'	58.7'
9	315°38.1	· · 43.1		57°41.0	9.5'	18°23.6	-13.1'	58.7'
10	330°38.0	43.5		72°09.4	9.5'	18°10.5	-13.3'	58.7'
11	345°37.9	43.9		86°37.9	9.6'	17°57.2	-13.4'	58.7'
12	0°37.8	N21°44.2		101°06.5	9.7'	S17°43.9	-13.5'	58.8'
13	15°37.8	44.6		115°35.2	9.7'	17°30.4	-13.6'	58.8'
14	30°37.7	45.0		130°03.9	9.8'	17°16.9	-13.7'	58.8'
15	45°37.6	· · 45.3		144°32.7	9.9'	17°03.2	-13.8'	58.8'
16	60°37.5	45.7		159°01.6	9.9'	16°49.5	-13.9'	58.8'
17	75°37.4	46.1		173°30.6	10.0'	16°35.6	-13.9'	58.8'
18	90°37.3	N21°46.5		187°59.6	10.1'	S16°21.7	-14.0'	58.8'
19	105°37.2	46.8		202°28.6	10.1'	16°07.6	-14.1'	58.9'
20	120°37.1	47.2		216°57.8	10.2'	15°53.5	-14.2'	58.9'
21	135°37.1	· · 47.6		231°27.0	10.3'	15°39.3	-14.3'	58.9'
22	150°37.0	47.9		245°56.2	10.3'	15°24.9	-14.4'	58.9'
23	165°36.9	48.3		260°25.5	10.4'	15°10.5	-14.5'	58.9'
SD = 15.8' d = 0.4'				SD = 16.0'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°						
66°	—:—	—:—	01:27	22:31	—:—	—:—
64°	—:—	—:—	02:07	21:50	—:—	—:—
62°	—:—	00:43	02:34	21:22	23:19	—:—
60°	—:—	01:37	02:55	21:01	22:21	—:—
N 58°	—:—	02:07	03:12	20:44	21:49	—:—
56°	00:39	02:30	03:26	20:30	21:26	23:23
54°	01:28	02:48	03:38	20:17	21:07	22:29
52°	01:57	03:04	03:49	20:06	20:52	22:00
50°	02:18	03:17	03:59	19:57	20:39	21:38
45°	02:56	03:43	04:19	19:36	20:12	20:59
N 40°	03:23	04:03	04:35	19:20	19:52	20:32
35°	03:44	04:20	04:49	19:06	19:35	20:11
30°	04:01	04:34	05:00	18:55	19:21	19:54
20°	04:28	04:56	05:20	18:34	18:59	19:27
N 10°	04:48	05:15	05:38	18:17	18:40	19:06
0°	05:06	05:32	05:54	18:01	18:23	18:49
S 10°	05:21	05:47	06:10	17:45	18:08	18:34
20°	05:36	06:03	06:27	17:28	17:52	18:19
30°	05:50	06:20	06:46	17:09	17:35	18:04
35°	05:58	06:29	06:57	16:57	17:25	17:56
40°	06:06	06:40	07:10	16:45	17:15	17:48
45°	06:15	06:52	07:25	16:29	17:02	17:39
S 50°	06:26	07:06	07:43	16:11	16:48	17:29
52°	06:30	07:13	07:52	16:02	16:42	17:24
54°	06:35	07:20	08:02	15:52	16:34	17:19
56°	06:40	07:28	08:13	15:41	16:26	17:14
58°	06:46	07:37	08:26	15:29	16:18	17:08
S 60°	06:52	07:47	08:40	15:14	16:07	17:02

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°						
N 70°			04:10			06:14
68°			03:09			07:13
66°			02:33			07:47
64°		02:33	02:08		05:53	08:12
62°	02:07	01:55	01:48	04:18	06:30	08:31
60°	01:18	01:28	01:31	05:07	06:57	08:46
N 58°	00:47	01:07	01:17	05:38	07:17	08:59
56°	00:23	00:49	01:05	06:01	07:34	09:11
54°	00:04	00:34	00:54	06:20	07:48	09:20
52°	· · ·	00:21	00:45	06:36	08:01	09:29
50°	· · ·	00:10	00:36	06:50	08:12	09:37
45°	23:46	· · ·	00:18	07:18	08:34	09:53
N 40°	23:27	· · ·	00:03	07:40	08:52	10:07
35°	23:11	23:51	· · ·	07:58	09:08	10:18
30°	22:57	23:40	· · ·	08:13	09:21	10:28
20°	22:33	23:21	· · ·	08:40	09:43	10:45
N 10°	22:12	23:04	23:51	09:02	10:02	11:00
0°	21:53	22:48	23:40	09:23	10:20	11:13
S 10°	21:33	22:32	23:29	09:44	10:37	11:27
20°	21:13	22:16	23:17	10:06	10:56	11:41
30°	20:48	21:56	23:03	10:32	11:18	11:57
35°	20:34	21:45	22:55	10:47	11:30	12:07
40°	20:17	21:31	22:46	11:04	11:44	12:17
45°	19:57					

May 30, 31, 01 UT (Thu., Fri., Sat.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Thu	GHA	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	SHA	Dec		
0	247°59.9	182°11.3	N21°17.5	226°51.7	N07°36.7	189°02.1	N19°38.7	257°42.3	S06°11.5					Alpheratz	357°35.5 29°13.3		
1	263°02.4	197°10.6	18.1	241°52.4	37.4	204°04.0	38.8	272°44.6	11.4					Ankaa	353°07.8 -42°10.2		
2	278°04.8	212°09.8	18.7	256°53.1	38.1	219°05.9	38.9	287°47.0	11.4					Schedar	349°32.0 56°40.0		
3	293°07.3	227°09.0	· · 19.3	271°53.8	· · 38.8	234°07.7	· · 39.0	302°49.3	· · 11.4					Diphda	348°48.0 -17°51.1		
4	308°09.8	242°08.2	19.9	286°54.5	39.5	249°09.6	39.2	317°51.7	11.3					Achernar	335°21.0 -57°06.6		
5	323°12.2	257°07.5	20.5	301°55.2	40.2	264°11.4	39.3	332°54.0	11.3					Hamal	327°52.1 23°34.5		
6	338°14.7	272°06.7	N21°21.1	316°55.9	N07°40.9	279°13.3	N19°39.4	347°56.4	S06°11.2					Polaris	314°44.8 89°21.9		
7	353°17.2	287°05.9	21.7	331°56.6	41.6	294°15.1	39.5	2°58.7	11.2					Acamar	315°12.6 -40°12.3		
8	8°19.6	302°05.1	22.2	346°57.3	42.4	309°17.0	39.6	18°01.1	11.2					Menkar	314°07.0 4°11.1		
9	23°22.1	317°04.4	· · 22.8	1°58.0	· · 43.1	324°18.8	· · 39.8	33°03.4	· · 11.1					Mirfak	308°29.5 49°56.7		
10	38°24.6	332°03.6	23.4	16°58.7	43.8	339°20.7	39.9	48°05.8	11.1					Aldebaran	290°40.6 16°33.5		
11	53°27.0	347°02.8	24.0	31°59.4	44.5	354°22.6	40.0	63°08.1	11.0					Rigel	281°04.7 -8°10.4		
12	68°29.5	2°02.0	N21°24.6	47°00.1	N07°45.2	9°24.4	N19°40.1	78°10.5	S06°11.0					Capella	280°23.1 46°01.3		
13	83°32.0	17°01.3	25.2	62°00.8	45.9	24°26.3	40.3	93°12.8	11.0					Bellatrix	278°23.8 6°22.3		
14	98°34.4	32°00.5	25.8	77°01.5	46.6	39°28.1	40.4	108°15.2	10.9					Elnath	278°02.9 28°37.7		
15	113°36.9	46°59.7	· · 26.3	92°02.3	· · 47.3	54°30.0	· · 40.5	123°17.5	· · 10.9					Alnilam	275°38.6 -1°11.2		
16	128°39.3	61°58.9	26.9	107°03.0	48.0	69°31.8	40.6	138°19.8	10.8					Betelgeuse	270°53.0 7°24.7		
17	143°41.8	76°58.1	27.5	122°03.7	48.7	84°33.7	40.7	153°22.2	10.8					Canopus	263°53.1 -52°42.6		
18	158°44.3	91°57.4	N21°28.1	137°04.4	N07°49.4	99°35.6	N19°40.9	168°24.5	S06°10.8					Sirius	258°27.0 -16°45.0		
19	173°46.7	106°56.6	28.7	152°05.1	50.1	114°37.4	41.0	183°26.9	10.7					Adhara	255°06.6 -29°00.4		
20	188°49.2	121°55.8	29.2	167°05.8	50.8	129°39.3	41.1	198°29.2	10.7					Procyon	244°51.6 5°09.8		
21	203°51.7	136°55.0	· · 29.8	182°06.5	· · 51.5	144°41.1	· · 41.2	213°31.6	· · 10.7					Pollux	243°18.2 27°58.1		
22	218°54.1	151°54.2	30.4	197°07.2	52.3	159°43.0	41.4	228°33.9	10.6					Avior	234°15.4 -59°35.4		
23	233°56.6	166°53.5	31.0	212°07.9	53.0	174°44.8	41.5	243°36.3	10.6					Suhail	222°46.9 -43°32.0		
Mer.pass. 07:27			ν -0.8' d 0.6' m -3.92			ν 0.7' d 0.7' m 1.05			ν 1.9' d 0.1' m -2.00			ν 2.3' d -0.0' m 1.02			Miaaplacidus 221°38.8 -69°49.3		
															Alphard 217°48.4 -8°45.9		
															Regulus 207°35.1 11°50.9		
															Dubhe 193°41.6 61°37.5		
															Denebola 182°25.4 16°26.2		
															Gienah 175°44.0 -17°40.8		
															Acrux 173°00.4 -63°14.3		
															Gacrux 171°52.0 -57°15.2		
															Alioth 166°13.1 55°49.9		
															Spica 158°22.6 -11°17.4		
															Alkaid 152°52.0 49°11.6		
															Hadar 148°36.3 -60°29.6		
															Menkent 147°57.9 -36°29.5		
															Arcturus 145°48.1 19°03.3		
															Rigel Kent. 139°40.5 -60°56.3		
															Kochab 137°18.7 74°03.4		
															Zuben'ubi 136°56.3 -16°08.7		
															Alphecca 126°03.8 26°37.9		
															Antares 112°16.1 -26°29.2		
															Atria 107°10.2 -69°04.3		
															Sabik 102°03.0 -15°45.3		
															Shaula 96°10.7 -37°07.3		
															Rasalhague 95°58.7 12°32.4		
															Eltanin 90°41.9 51°29.0		
															Kaus Aust. 83°32.8 -34°22.3		
															Vega 80°33.2 38°48.2		
															Nunki 75°48.1 -26°16.0		
															Altair 62°00.2 8°55.8		
															Peacock 53°06.2 -56°39.2		
															Deneb 49°25.9 45°21.7		
															Enif 33°39.2 9°59.1		
															Al Na'ir 27°33.5 -46°50.4		
															Fomalhaut 15°15.1 -29°29.5		
															Scheat 13°45.8 28°12.7		
															Markab 13°30.5 15°20.0		
May 30 Thu			SHA			Mer.pass											
			Venus 294°11.4			11:52											
			Mars 338°51.7			08:52											
			Jupiter 301°02.2			11:22											
			Saturn 9°42.4			06:48											
May 31 Fri			SHA			Mer.pass											
			Venus 292°53.6			11:53											
			Mars 338°09.6			08:51											
			Jupiter 300°47.6			11:19											
			Saturn 9°39.6			06:44											
Jun 01 Sat			SHA			Mer.pass											
			Venus 291°35.6			11:54											
			Mars 337°27.4			08:50											
			Jupiter 300°33.0			11:17											
			Saturn 9°36.9			06:41											

Sun			Moon				
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	180°36.8	N21°48.7	274°54.9	10.4'	S14°56.1	-14.6'	58.9'
1	195°36.7	49.0	289°24.4	10.5'	14°41.5	-14.7'	58.9'
2	210°36.6	49.4	303°53.9	10.6'	14°26.8	-14.7'	58.9'
3	225°36.5	· · 49.8	318°23.5	10.6'	14°12.1	-14.8'	59.0'
4	240°36.4	50.1	332°53.1	10.7'	13°57.3	-14.9'	59.0'
5	255°36.4	50.5	347°22.8	10.7'	13°42.4	-15.0'	59.0'
6	270°36.3	N21°50.9	1°52.5	10.8'	S13°27.4	-15.1'	59.0'
7	285°36.2	51.2	16°22.3	10.8'	13°12.3	-15.1'	59.0'
8	300°36.1	51.6	30°52.1	10.9'	12°57.2	-15.2'	59.0'
9	315°36.0	· · 51.9	45°22.1	11.0'	12°42.0	-15.3'	59.0'
10	330°35.9	52.3	59°52.0	11.0'	12°26.7	-15.3'	59.0'
11	345°35.8	52.7	74°22.0	11.1'	12°11.4	-15.4'	59.1'
12	0°35.7	N21°53.0	88°52.1	11.1'	S11°56.0	-15.5'	59.1'
13	15°35.6	53.4	103°22.2	11.2'	11°40.5	-15.5'	59.1'
14	30°35.6	53.7	117°52.3	11.2'	11°24.9	-15.6'	59.1'
15	45°35.5	· · 54.1	132°22.5	11.2'	11°09.3	-15.7'	59.1'
16	60°35.4	54.5	146°52.8	11.3'	10°53.6	-15.7'	59.1'
17	75°35.3	54.8	161°23.1	11.3'	10°37.9	-15.8'	59.1'
18	90°35.2	N21°55.2	175°53.4	11.4'	S10°22.1	-15.9'	59.1'
19	105°35.1	55.5	190°23.8	11.4'	10°06.2	-15.9'	59.1'
20	120°35.0	55.9	204°54.2	11.5'	09°50.3	-16.0'	59.2'
21	135°34.9	· · 56.2	219°24.6	11.5'	09°34.4	-16.0'	59.2'
22	150°34.8	56.6	233°55.1	11.5'	09°18.3	-16.1'	59.2'
23	165°34.7	56.9	248°25.7	11.6'	09°02.3	-16.1'	59.2'
SD = 15.8' $d = 0.4'$			SD = 16.1'				
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	180°34.6	N21°57.3	262°56.2	11.6'	S08°46.1	-16.2'	59.2'
1	195°34.5	57.6	277°26.9	11.6'	08°29.9	-16.2'	59.2'
2	210°34.5	58.0	291°57.5	11.7'	08°13.7	-16.3'	59.2'
3	225°34.4	· · 58.3	306°28.2	11.7'	07°57.4	-16.3'	59.2'
4	240°34.3	58.7	320°58.9	11.7'	07°41.1	-16.4'	59.2'
5	255°34.2	59.0	335°29.6	11.8'	07°24.8	-16.4'	59.2'
6	270°34.1	N21°59.4	350°00.4	11.8'	S07°08.3	-16.5'	59.3'
7	285°34.0	21°59.7	4°31.2	11.8'	06°51.9	-16.5'	59.3'
8	300°33.9	22°00.1	19°02.0	11.8'	06°35.4	-16.5'	59.3'
9	315°33.8	· · 00.4	33°32.8	11.9'	06°18.9	-16.6'	59.3'
10	330°33.7	00.8	48°03.7	11.9'	06°02.3	-16.6'	59.3'
11	345°33.6	01.1	62°34.6	11.9'	05°45.7	-16.6'	59.3'
12	0°33.5	N22°01.5	77°05.5	11.9'	S05°29.1	-16.7'	59.3'
13	15°33.4	01.8	91°36.4	11.9'	05°12.4	-16.7'	59.3'
14	30°33.3	02.1	106°07.3	12.0'	04°55.7	-16.7'	59.3'
15	45°33.2	· · 02.5	120°38.3	12.0'	04°39.0	-16.8'	59.3'
16	60°33.1	02.8	135°09.3	12.0'	04°22.2	-16.8'	59.3'
17	75°33.0	03.2	149°40.3	12.0'	04°05.4	-16.8'	59.3'
18	90°32.9	N22°03.5	164°11.3	12.0'	S03°48.6	-16.8'	59.4'
19	105°32.8	03.8	178°42.3	12.0'	03°31.8	-16.9'	59.4'
20	120°32.8	04.2	193°13.3	12.0'	03°14.9	-16.9'	59.4'
21	135°32.7	· · 04.5	207°44.4	12.0'	02°58.1	-16.9'	59.4'
22	150°32.6	04.8	222°15.4	12.1'	02°41.2	-16.9'	59.4'
23	165°32.5	05.2	236°46.5	12.1'	02°24.3	-16.9'	59.4'
SD = 15.8' $d = 0.4'$			SD = 16.1'				
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	180°32.4	N22°05.5	251°17.5	12.1'	S02°07.3	-16.9'	59.4'
1	195°32.3	05.9	265°48.6	12.1'	01°50.4	-17.0'	59.4'
2	210°32.2	06.2	280°19.6	12.1'	01°33.4	-17.0'	59.4'
3	225°32.1	· · 06.5	294°50.7	12.1'	01°16.5	-17.0'	59.4'
4	240°32.0	06.9	309°21.8	12.1'	00°59.5	-17.0'	59.4'
5	255°31.9	07.2	323°52.8	12.1'	00°42.5	-17.0'	59.4'
6	270°31.8	N22°07.5	338°23.9	12.1'	S00°25.5	-17.0'	59.4'
7	285°31.7	07.8	352°54.9	12.0'	S00°08.5	-17.0'	59.4'
8	300°31.6	08.2	7°26.0	12.0'	N00°08.5	17.0'	59.4'
9	315°31.5	· · 08.5	21°57.0	12.0'	00°25.5	17.0'	59.4'
10	330°31.4	08.8	36°28.0	12.0'	00°42.5	17.0'	59.4'
11	345°31.3	09.2	50°59.1	12.0'	00°59.5	17.0'	59.5'
12	0°31.2	N22°09.5	65°30.1	12.0'	N01°16.5	17.0'	59.5'
13	15°31.1	09.8	80°01.1	12.0'	01°33.5	17.0'	59.5'
14	30°31.0	10.1	94°32.0	12.0'	01°50.5	17.0'	59.5'
15	45°30.9	· · 10.5	109°03.0	11.9'	02°07.5	17.0'	59.5'
16	60°30.8	10.8	123°34.0	11.9'	02°24.5	17.0'	59.5'
17	75°30.7	11.1	138°04.9	11.9'	02°41.5	17.0'	59.5'
18	90°30.6	N22°11.4	152°35.8	11.9'	N02°58.4	17.0'	59.5'
19	105°30.5	11.8	167°06.7	11.9'	03°15.4	16.9'	59.5'
20	120°30.4	12.1	181°37.5	11.8'	03°32.3	16.9'	59.5'
21	135°30.3	· · 12.4	196°08.4	11.8'	03°49.2	16.9'	59.5'
22	150°30.2	12.7	210°39.2	11.8'	04°06.2	16.9'	59.5'
23	165°30.1	13.0	225°10.0	11.8'	04°23.0	16.9'	59.5'
SD = 15.8' $d = 0.3'$			SD = 16.2'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°						
66°	—:—	—:—	01:15	22:45	—:—	—:—
64°	—:—	—:—	01:59	21:58	—:—	—:—
62°	—:—	00:12	02:28	21:29	—:—	—:—
60°	—:—	01:28	02:50	21:07	22:30	—:—
N 58°	—:—	02:01	03:08	20:49	21:56	—:—
56°	00:10	02:25	03:23	20:34	21:31	—:—
54°	01:20	02:44	03:35	20:21	21:12	22:38
52°	01:51	03:00	03:46	20:10	20:56	22:06
50°	02:14	03:14	03:56	20:00	20:43	21:43
45°	02:53	03:41	04:17	19:39	20:15	21:03
N 40°	03:21	04:01	04:34	19:22	19:54	20:35
35°	03:43	04:18	04:47	19:08	19:38	20:13
30°	04:00	04:33	05:00	18:56	19:23	19:56
20°	04:27	04:56	05:20	18:36	19:00	19:29
N 10°	04:48	05:15	05:38	18:18	18:41	19:07
0°	05:06	05:32	05:54	18:01	18:24	18:50
S 10°	05:22	05:48	06:10	17:45	18:08	18:34
20°	05:37	06:04	06:28	17:28	17:51	18:19
30°	05:52	06:21	06:47	17:08	17:34	18:04
35°	06:00	06:31	06:59	16:56	17:24	17:56
40°	06:08	06:42	07:12	16:43	17:13	17:47
45°	06:18	06:54	07:28	16:28	17:01	17:38
S 50°	06:28	07:09	07:47	16:09	16:46	17:27
52°	06:33	07:16	07:56	16:00	16:39	17:22
54°	06:38	07:23	08:06	15:50	16:32	17:17
56°	06:43	07:31	08:17	15:38	16:24	17:12
58°	06:49	07:41	08:30	15:25	16:15	17:06
S 60°	06:56	07:51	08:46	15:10	16:04	16:59

Aries Venus

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	251°56.5	180°55.2	N22°10.0	227°59.4	N08°44.0	192°00.4	N19°50.2	261°28.1	S06°08.0
1	266°58.9	195°54.4	10.5	243°00.1	44.7	207°02.2	50.4	276°30.5	07.9
2	282°01.4	210°53.6	11.0	258°00.8	45.4	222°01.0	50.5	291°32.9	07.9
3	297°03.9	225°52.8	• • 11.5	273°01.5	• • 46.1	237°05.9	• • 50.6	306°35.2	• • 07.9
4	312°06.3	240°52.0	12.0	288°02.2	46.8	252°07.8	50.7	321°37.6	07.8
5	327°08.8	255°51.2	12.5	303°02.9	47.5	267°09.6	50.8	336°39.9	07.8
6	342°11.3	270°50.3	N22°13.0	318°03.7	N08°48.2	282°11.5	N19°51.0	351°42.3	S06°07.8
7	357°13.7	285°49.5	13.4	333°04.4	48.9	297°13.4	51.1	6°44.7	07.7
8	12°16.2	300°48.7	13.9	348°05.1	49.6	312°15.2	51.2	21°47.0	07.7
9	27°18.6	315°47.9	• • 14.4	3°05.8	• • 50.3	327°17.1	• • 51.3	36°49.4	• • 07.7
10	42°21.1	330°47.1	14.9	18°06.5	51.0	342°18.9	51.4	51°51.7	07.6
11	57°23.6	345°46.3	15.4	33°07.2	51.7	357°20.8	51.5	66°54.1	07.6
12	72°26.0	0°45.5	N22°15.9	48°07.9	N08°52.4	12°22.6	N19°51.7	81°56.5	S06°07.6
13	87°28.5	15°44.6	16.3	63°08.6	53.0	27°24.5	51.8	96°58.8	07.5
14	102°31.0	30°43.8	16.8	78°09.3	53.7	42°26.4	51.9	112°01.2	07.5
15	117°33.4	45°43.0	• • 17.3	93°10.0	• • 54.4	57°28.2	• • 52.0	127°03.6	• • 07.5
16	132°35.9	60°42.2	17.8	108°10.7	55.1	72°30.1	52.1	142°05.9	07.4
17	147°38.4	75°41.4	18.3	123°11.4	55.8	87°31.9	52.3	157°08.3	07.4
18	162°40.8	90°40.6	N22°18.7	138°12.1	N08°56.5	102°33.8	N19°52.4	172°10.6	S06°07.4
19	177°43.3	105°39.7	19.2	153°12.8	57.2	117°35.6	52.5	187°13.0	07.3
20	192°45.8	120°38.9	19.7	168°13.5	57.9	132°37.5	52.6	202°15.4	07.3
21	207°48.2	135°38.1	• • 20.2	183°14.2	• • 58.6	147°39.4	• • 52.7	217°17.7	• • 07.3
22	222°50.7	150°37.3	20.6	198°14.9	59.3	162°41.2	52.8	232°20.1	07.3
23	237°53.1	165°36.5	21.1	213°15.6	59.9	177°43.1	53.0	247°22.5	07.2
Mer.pass. 07:11		ν -0.8' d 0.5' m -3.92		ν 0.7' d 0.7' m 1.04		ν 1.9' d 0.1' m -2.00		ν 2.4' d -0.0' m 1.01	

<https://pypi.org/project/sfalmanac/>

Jun 02 Sun	SHA	Mer.pass
Venus	290° 17.3	11:56
Mars	336° 45.2	08:49
Jupiter	300° 18.5	11:14
Saturn	9° 34.2	06:37

Jun 03 Mon	SHA	Mer.pass
Venus	288° 58.8	11:57
Mars	336° 03.0	08:48
Jupiter	300° 03.9	11:11
Saturn	9° 31.7	06:33

Jun 04 Tue	SHA	Mer.pass
Venus	287° 40.0	11:58
Mars	335° 20.7	08:46
Jupiter	299° 49.3	11:08
Saturn	9° 29.2	06:29

Horizontal parallax		
Venus:	0.1	
Mars:	0.1	

Horizontal parallax	
Venus:	0.1
Mars:	0.1

2024 June 02 to Jun. 04 UT

IERS Earth Orientation data as of 01-Aug-2024
 Page 105
 <https://pypi.org/project/sfalmanac/>

June 05, 06, 07 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	253°54.7	180°15.8	N22°32.5	228°33.2	N09°17.1	193°29.5	N19°55.9	263°21.6	S06°06.4	Alpheratz	357°35.4	29°13.3
1	268°57.2	195°14.9	33.0	243°33.9	17.8	208°31.4	56.0	278°23.9	06.4	Ankaa	353°07.8	-42°10.2
2	283°59.7	210°14.1	33.4	258°34.6	18.5	223°33.2	56.1	293°26.3	06.4	Schedar	349°31.9	56°40.0
3	299°02.1	225°13.3	•• 33.8	273°35.3	•• 19.2	238°35.1	•• 56.2	308°28.7	•• 06.4	Diphda	348°48.0	-17°51.1
4	314°04.6	240°12.5	34.3	288°36.0	19.9	253°36.9	56.3	323°31.0	06.3	Achernar	335°20.9	-57°06.6
5	329°07.1	255°11.6	34.7	303°36.7	20.6	268°38.8	56.5	338°33.4	06.3	Hamal	327°52.0	23°34.5
6	344°09.5	270°10.8	N22°35.2	318°37.4	N09°21.2	283°40.7	N19°56.6	353°35.8	S06°06.3	Polaris	314°42.9	89°21.8
7	359°12.0	285°10.0	35.6	333°38.1	21.9	298°42.5	56.7	8°38.1	06.2	Acamar	315°12.5	-40°12.3
8	14°14.5	300°09.1	36.0	348°38.8	22.6	313°44.4	56.8	23°40.5	06.2	Menkar	314°07.0	4°11.1
9	29°16.9	315°08.3	•• 36.5	3°39.6	•• 23.3	328°46.2	•• 56.9	38°42.9	•• 06.2	Mirfak	308°29.5	49°56.7
10	44°19.4	330°07.5	36.9	18°40.3	24.0	343°48.1	57.0	53°45.2	06.1	Aldebaran	290°40.6	16°33.5
11	59°21.9	345°06.6	37.3	33°41.0	24.7	358°50.0	57.2	68°47.6	06.1	Rigel	281°04.7	-8°10.4
12	74°24.3	0°05.8	N22°37.8	48°41.7	N09°25.3	13°51.8	N19°57.3	83°50.0	S06°06.1	Capella	280°23.1	46°01.3
13	89°26.8	15°05.0	38.2	63°42.4	26.0	28°53.7	57.4	98°52.4	06.1	Bellatrix	278°23.8	6°22.3
14	104°29.2	30°04.1	38.6	78°43.1	26.7	43°55.5	57.5	113°54.7	06.0	Elnath	278°02.9	28°37.7
15	119°31.7	45°03.3	•• 39.0	93°43.8	•• 27.4	58°57.4	•• 57.6	128°57.1	•• 06.0	Alnilam	275°38.6	-1°11.2
16	134°34.2	60°02.5	39.5	108°44.5	28.1	73°59.2	57.7	143°59.5	06.0	Betelgeuse	270°53.0	7°24.7
17	149°36.6	75°01.6	39.9	123°45.2	28.8	89°01.1	57.9	159°01.8	05.9	Canopus	263°53.2	-52°42.6
18	164°39.1	90°00.8	N22°40.3	138°45.9	N09°29.4	104°03.0	N19°58.0	174°04.2	S06°05.9	Sirius	258°27.0	-16°45.0
19	179°41.6	105°00.0	40.7	153°46.6	30.1	119°04.8	58.1	189°06.6	05.9	Adhara	255°06.6	-29°00.4
20	194°44.0	119°59.1	41.2	168°47.3	30.8	134°06.7	58.2	204°08.9	05.8	Procyon	244°51.6	5°09.8
21	209°46.5	134°58.3	•• 41.6	183°48.0	•• 31.5	149°08.5	•• 58.3	219°11.3	•• 05.8	Pollux	243°18.2	27°58.1
22	224°49.0	149°57.5	42.0	198°48.7	32.2	164°10.4	58.4	234°13.7	05.8	Avior	234°15.4	-59°35.4
23	239°51.4	164°56.6	42.4	213°49.4	32.8	179°12.3	58.5	249°16.0	05.8	Suhail	222°46.9	-43°32.0
Mer.pass. 07:03		ν -0.8' d 0.4' m -3.92		ν 0.7' d 0.7' m 1.04		ν 1.9' d 0.1' m -2.00		ν 2.4' d -0.0' m 1.01				

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	254°53.9	179°55.8	N22°42.8	228°50.1	N09°33.5	194°14.1	N19°58.7	264°18.4	S06°05.7
1	269°56.4	194°54.9	43.2	243°50.8	34.2	209°16.0	58.8	279°20.8	05.7
2	284°58.8	209°54.1	43.7	258°51.5	34.9	224°17.8	58.9	294°23.2	05.7
3	300°01.3	224°53.3	•• 44.1	273°52.2	•• 35.6	239°19.7	•• 59.0	309°25.5	•• 05.6
4	315°03.7	239°52.4	44.5	288°52.9	36.2	254°21.5	59.1	324°27.9	05.6
5	330°06.2	254°51.6	44.9	303°53.6	36.9	269°23.4	59.2	339°30.3	05.6
6	345°08.7	269°50.7	N22°45.3	318°54.3	N09°37.6	284°25.3	N19°59.4	354°32.6	S06°05.6
7	0°11.1	284°49.9	45.7	333°55.0	38.3	299°27.1	59.5	9°35.0	05.5
8	15°13.6	299°49.1	46.1	348°55.7	39.0	314°29.0	59.6	24°37.4	05.5
9	30°16.1	314°48.2	•• 46.5	3°56.4	•• 39.6	329°30.8	•• 59.7	39°39.8	•• 05.5
10	45°18.5	329°47.4	46.9	18°57.1	40.3	344°32.7	59.8	54°42.1	05.4
11	60°21.0	344°46.5	47.3	33°57.8	41.0	359°34.6	19°59.9	69°44.5	05.4
12	75°23.5	359°45.7	N22°47.7	48°58.5	N09°41.7	14°36.4	N20°00.0	84°46.9	S06°05.4
13	90°25.9	14°44.9	48.1	63°59.2	42.4	29°38.3	00.2	99°49.2	05.4
14	105°28.4	29°44.0	48.5	78°59.9	43.0	44°40.1	00.3	114°51.6	05.3
15	120°30.9	44°43.2	•• 48.9	94°00.6	•• 43.7	59°42.0	•• 00.4	129°54.0	•• 05.3
16	135°33.3	59°42.3	49.3	109°01.3	44.4	74°43.9	00.5	144°56.4	05.3
17	150°35.8	74°41.5	49.7	124°02.0	45.1	89°45.7	00.6	159°58.7	05.3
18	165°38.2	89°40.6	N22°50.1	139°02.7	N09°45.7	104°47.6	N20°00.7	175°01.1	S06°05.2
19	180°40.7	104°39.8	50.5	154°03.4	46.4	119°49.4	00.8	190°03.5	05.2
20	195°43.2	119°39.0	50.9	169°04.1	47.1	134°51.3	01.0	205°05.9	05.2
21	210°45.6	134°38.1	•• 51.3	184°04.9	•• 47.8	149°53.2	•• 01.1	220°08.2	•• 05.1
22	225°48.1	149°37.3	51.7	199°05.6	48.5	164°55.0	01.2	235°10.6	05.1
23	240°50.6	164°36.4	52.1	214°06.3	49.1	179°56.9	01.3	250°13.0	05.1
Mer.pass. 06:59		ν -0.8' d 0.4' m -3.92		ν 0.7' d 0.7' m 1.03		ν 1.9' d 0.1' m -2.00		ν 2.4' d -0.0' m 1.00	

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	255°53.0	179°35.6	N22°52.5	229°07.0	N09°49.8	194°58.7	N20°01.4	265°15.3	S06°05.1
1	270°55.5	194°34.7	52.9	244°07.7	50.5	210°00.6	01.5	280°17.7	05.0
2	285°58.0	209°33.9	53.3	259°08.4	51.2	225°02.4	01.6	295°20.1	05.0
3	301°00.4	224°33.0	•• 53.6	274°09.1	•• 51.8	240°04.3	•• 01.8	310°22.5	•• 05.0
4	316°02.9	239°32.2	54.0	289°09.8	52.5	255°06.2	01.9	325°24.8	05.0
5	331°05.4	254°31.3	54.4	304°10.5	53.2	270°08.0	02.0	340°27.2	04.9
6	346°07.8	269°30.5	N22°54.8	319°11.2	N09°53.9	285°09.9	N20°02.1	355°29.6	S06°04.9
7	1°10.3	284°29.6	55.2	334°11.9	54.5	300°11.7	02.2	10°32.0	04.9
8	16°12.7	299°28.8	55.5	349°12.6	55.2	315°13.6	02.3	25°34.3	04.8
9	31°15.2	314°28.0	•• 55.9	4°13.3	•• 55.9	330°15.5	•• 02.4	40°36.7	•• 04.8
10	46°17.7	329°27.1	56.3	19°14.0	56.6	345°17.3	02.6	55°39.1	04.8
11	61°20.1	344°26.3	56.7	34°14.7	57.2	0°19.2	02.7	70°41.5	04.8
12	76°22.6	359°25.4	N22°57.1	49°15.4	N09°57.9	15°21.0	N20°02.8	85°43.8	S06°04.7
13	91°25.1	14°24.6	57.4	64°16.1	58.6	30°22.9	02.9	100°46.2	04.7
14	106°27.5	29°23.7	57.8	79°16.8	59.3	45°24.8	03.0	115°48.6	04.7
15	121°30.0	44°22.9	•• 58.2	94°17.5	09°59.9	60°26.6	•• 03.1	130°51.0	•• 04.7
16	136°32.5	59°22.0	58.5	109°18.2	10°00.6	75°28.5	03.2	145°53.3	04.6
17	151°34.9	74°21.2	58.9	124°18.9	01.3	90°30.3	03.4	160°55.7	04.6
18	166°37.4	89°20.3	N22°59.3	139°19.6	N10°01.9	105°32.2	N20°03.5	175°58.1	S06°04.6
19	181°39.9	104°19.5	22°59.6	154°20.3	02.6	120°34.1	03.6	191°00.5	04.6
20	196°42.3	119°18.6	23°00.0	169°21.0	03.3	135°35.9	03.7	206°02.9	04.5
21	211°44.8	134°17.8	•• 00.4	184°21.7	•• 04.0	150°37.8	•• 03.8	221°05.2	•• 04.5
22	226°47.2	149°16.9	00.7	199°22.4	04.6	165°39.6	03.9	236°07.6	04.5
23	241°49.7	164°16.0	01.1	214°23.1	05.3	180°41.5	04.0	251°10.0	04.5
Mer.pass. 06:55		ν -0.8' d 0.4' m -3.92		ν 0.7' d 0.7' m 1.03		ν 1.9' d 0.1' m -2.00		ν 2.4' d -0.0' m 1.00	

Jun 05 Wed	SHA	Mer.pass
Venus	286°21.0	12:00
Mars	334°38.5	08:45
Jupiter	299°34.8	11:05
Saturn	9°26.8	06:26

Jun 06 Thu	SHA	Mer.pass
Venus	285°01.9	12:01
Mars	333°56.2	08:44
Jupiter	299°20.2	11:02
Saturn	9°24.5	06:22

Jun 07 Fri	SHA	Mer.pass
Venus	283°42.5	12:02
Mars	333°13.9	08:43
Jupiter	299°05.7	10:59
Saturn	9°22.3	06:18

Horizontal parallax	
Venus:	0.1
Mars:	0.1

Sun				Moon				
Wed	GHA	Dec		GHA	ν	Dec	d	HP
0	180°22.3	N22°34.6		202°11.4	7.1'	N22°14.4	10.9'	59.0'
1	195°22.2	34.8		216°37.6	7.0'	22°25.3	10.8'	58.9'
2	210°22.1	35.1		231°03.6	7.0'	22°36.1	10.6'	58.9'
3	225°22.0	· · 35.4		245°29.6	6.9'	22°46.7	10.5'	58.9'
4	240°21.8	35.6		259°55.5	6.8'	22°57.2	10.3'	58.9'
5	255°21.7	35.9		274°21.3	6.8'	23°07.5	10.2'	58.9'
6	270°21.6	N22°36.2		288°47.1	6.7'	N23°17.7	10.1'	58.9'
7	285°21.5	36.4		303°12.8	6.6'	23°27.8	9.9'	58.8'
8	300°21.4	36.7		317°38.4	6.5'	23°37.7	9.7'	58.8'
9	315°21.3	· · 37.0		332°03.9	6.5'	23°47.4	9.6'	58.8'
10	330°21.2	37.2		346°29.4	6.4'	23°57.0	9.4'	58.8'
11	345°21.1	37.5		0°54.8	6.3'	24°06.5	9.3'	58.8'
12	0°20.9	N22°37.8		15°20.1	6.3'	N24°15.8	9.1'	58.7'
13	15°20.8	38.0		29°45.4	6.2'	24°24.9	9.0'	58.7'
14	30°20.7	38.3		44°10.6	6.1'	24°33.9	8.8'	58.7'
15	45°20.6	· · 38.5		58°35.7	6.1'	24°42.7	8.7'	58.7'
16	60°20.5	38.8		73°00.8	6.0'	24°51.3	8.5'	58.7'
17	75°20.4	39.1		87°25.8	6.0'	24°59.8	8.3'	58.6'
18	90°20.3	N22°39.3		101°50.8	5.9'	N25°08.2	8.2'	58.6'
19	105°20.1	39.6		116°15.7	5.8'	25°16.3	8.0'	58.6'
20	120°20.0	39.8		130°40.5	5.8'	25°24.3	7.8'	58.6'
21	135°19.9	· · 40.1		145°05.3	5.7'	25°32.2	7.7'	58.6'
22	150°19.8	40.3		159°30.1	5.7'	25°39.8	7.5'	58.5'
23	165°19.7	40.6		173°54.7	5.6'	25°47.3	7.3'	58.5'
	SD = 15.8'	$d = 0.3'$		SD = 16.1'				
Thu	GHA	Dec		GHA	ν	Dec	d	HP
0	180°19.6	N22°40.8		188°19.4	5.6'	N25°54.6	7.2'	58.5'
1	195°19.4	41.1		202°43.9	5.5'	26°01.8	7.0'	58.5'
2	210°19.3	41.4		217°08.5	5.5'	26°08.8	6.8'	58.4'
3	225°19.2	· · 41.6		231°33.0	5.4'	26°15.6	6.6'	58.4'
4	240°19.1	41.9		245°57.4	5.4'	26°22.3	6.5'	58.4'
5	255°19.0	42.1		260°21.8	5.4'	26°28.7	6.3'	58.4'
6	270°18.9	N22°42.4		274°46.2	5.3'	N26°35.0	6.1'	58.3'
7	285°18.7	42.6		289°10.5	5.3'	26°41.2	5.9'	58.3'
8	300°18.6	42.9		303°34.8	5.3'	26°47.1	5.8'	58.3'
9	315°18.5	· · 43.1		317°59.0	5.2'	26°52.9	5.6'	58.3'
10	330°18.4	43.3		332°23.3	5.2'	26°58.5	5.4'	58.3'
11	345°18.3	43.6		346°47.4	5.2'	27°03.9	5.2'	58.2'
12	0°18.2	N22°43.8		1°11.6	5.1'	N27°09.1	5.1'	58.2'
13	15°18.0	44.1		15°35.7	5.1'	27°14.2	4.9'	58.2'
14	30°17.9	44.3		29°59.9	5.1'	27°19.1	4.7'	58.2'
15	45°17.8	· · 44.6		44°23.9	5.1'	27°23.8	4.5'	58.1'
16	60°17.7	44.8		58°48.0	5.1'	27°28.3	4.3'	58.1'
17	75°17.6	45.1		73°12.1	5.0'	27°32.7	4.2'	58.1'
18	90°17.5	N22°45.3		87°36.1	5.0'	N27°36.8	4.0'	58.1'
19	105°17.3	45.5		102°00.1	5.0'	27°40.8	3.8'	58.0'
20	120°17.2	45.8		116°24.1	5.0'	27°44.6	3.6'	58.0'
21	135°17.1	· · 46.0		130°48.1	5.0'	27°48.2	3.4'	58.0'
22	150°17.0	46.3		145°12.1	5.0'	27°51.7	3.3'	57.9'
23	165°16.9	46.5		159°36.1	5.0'	27°55.0	3.1'	57.9'
	SD = 15.8'	$d = 0.3'$		SD = 16.0'				
Fri	GHA	Dec		GHA	ν	Dec	d	HP
0	180°16.7	N22°46.7		174°00.1	5.0'	N27°58.0	2.9'	57.9'
1	195°16.6	47.0		188°24.1	5.0'	28°00.9	2.7'	57.9'
2	210°16.5	47.2		202°48.1	5.0'	28°03.6	2.5'	57.8'
3	225°16.4	· · 47.4		217°12.1	5.0'	28°06.2	2.4'	57.8'
4	240°16.3	47.7		231°36.1	5.0'	28°08.5	2.2'	57.8'
5	255°16.2	47.9		246°00.1	5.0'	28°10.7	2.0'	57.8'
6	270°16.0	N22°48.1		260°24.2	5.0'	N28°12.7	1.8'	57.7'
7	285°15.9	48.4		274°48.2	5.1'	28°14.5	1.6'	57.7'
8	300°15.8	48.6		289°12.3	5.1'	28°16.1	1.4'	57.7'
9	315°15.7	· · 48.8		303°36.3	5.1'	28°17.6	1.3'	57.6'
10	330°15.6	49.1		318°00.4	5.1'	28°18.9	1.1'	57.6'
11	345°15.4	49.3		332°24.6	5.2'	28°19.9	0.9'	57.6'
12	0°15.3	N22°49.5		346°48.7	5.2'	N28°20.8	0.7'	57.6'
13	15°15.2	49.7		1°12.9	5.2'	28°21.6	0.5'	57.5'
14	30°15.1	50.0		15°37.1	5.2'	28°22.1	0.4'	57.5'
15	45°15.0	· · 50.2		30°01.4	5.3'	28°22.5	0.2'	57.5'
16	60°14.8	50.4		44°25.6	5.3'	28°22.7	0.0'	57.4'
17	75°14.7	50.6		58°50.0	5.4'	28°22.7	-0.2'	57.4'
18	90°14.6	N22°50.9		73°14.3	5.4'	N28°22.5	-0.3'	57.4'
19	105°14.5	51.1		87°38.7	5.4'	28°22.2	-0.5'	57.4'
20	120°14.3	51.3		102°03.2	5.5'	28°21.7	-0.7'	57.3'
21	135°14.2	· · 51.5		116°27.7	5.5'	28°21.0	-0.9'	57.3'
22	150°14.1	51.8		130°52.2	5.6'	28°20.1	-1.0'	57.3'
23	165°14.0	52.0		145°16.8	5.6'	28°19.1	-1.2'	57.2'
	SD = 15.8'	$d = 0.2'$		SD = 15.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	--	--	00:47	23:15	--	--
64°	--	--	01:46	22:14	--	--
62°	--	--	02:18	21:40	--	--
60°	--	01:11	02:42	21:16	22:49	--
N 58°	--	01:51	03:01	20:57	22:08	--
56°	--	02:17	03:17	20:41	21:41	--
54°	01:05	02:38	03:31	20:28	21:20	22:55
52°	01:42	02:55	03:42	20:16	21:03	22:17
50°	02:06	03:09	03:53	20:05	20:49	21:52
45°	02:49	03:37	04:14	19:44	20:21	21:09
N 40°	03:18	03:59	04:31	19:26	19:59	20:40
35°	03:40	04:17	04:46	19:12	19:41	20:18
30°	03:58	04:31	04:59	18:59	19:26	20:00
20°	04:26	04:55	05:20	18:38	19:02	19:31
N 10°	04:48	05:15	05:38	18:19	18:42	19:09
0°	05:07	05:33	05:55	18:02	18:25	18:51
S 10°	05:23	05:49	06:12	17:46	18:08	18:34
20°	05:39	06:06	06:30	17:28	17:51	18:19
30°	05:54	06:24	06:50	17:07	17:33	18:03
35°	06:03	06:34	07:02	16:55	17:23	17:55
40°	06:12	06:46	07:16	16:41	17:12	17:46
45°	06:22	06:59	07:32	16:25	16:59	17:36
S 50°	06:33	07:14	07:52	16:05	16:43	1

June 08, 09, 10 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	256°52.2	179°15.2	N23°01.5	229°23.8	N10°06.0	195°43.4	N20°04.1	266°12.4	S06°04.4	Alpheratz	357°35.4 29°13.3
1	271°54.6	194°14.3	01.8	244°24.5	06.7	210°45.2	04.3	281°14.7	04.4	Ankaa	353°07.8 -42°10.2
2	286°57.1	209°13.5	02.2	259°25.2	07.3	225°47.1	04.4	296°17.1	04.4	Schedar	349°31.8 56°40.0
3	301°59.6	224°12.6	•• 02.5	274°25.9	•• 08.0	240°48.9	•• 04.5	311°19.5	•• 04.4	Diphda	348°47.9 -17°51.1
4	317°02.0	239°11.8	02.9	289°26.6	08.7	255°50.8	04.6	326°21.9	04.3	Achernar	335°20.9 -57°06.5
5	332°04.5	254°10.9	03.3	304°27.3	09.3	270°52.7	04.7	341°24.3	04.3	Hamal	327°52.0 23°34.5
6	347°07.0	269°10.1	N23°03.6	319°28.0	N10°10.0	285°54.5	N20°04.8	356°26.6	S06°04.3	Polaris	314°41.7 89°21.8
7	2°09.4	284°09.2	04.0	334°28.7	10.7	300°56.4	04.9	11°29.0	04.2	Acamar	315°12.5 -40°12.3
8	17°11.9	299°08.4	04.3	349°29.4	11.3	315°58.2	05.0	26°31.4	04.2	Menkar	314°07.0 4°11.1
9	32°14.3	314°07.5	•• 04.7	4°30.1	•• 12.0	331°00.1	•• 05.2	41°33.8	•• 04.2	Mirfak	308°29.4 49°56.7
10	47°16.8	329°06.6	05.0	19°30.8	12.7	346°02.0	05.3	56°36.1	04.2	Aldebaran	290°40.5 16°33.5
11	62°19.3	344°05.8	05.4	34°31.5	13.4	1°03.8	05.4	71°38.5	04.1	Rigel	281°04.7 -8°10.4
12	77°21.7	359°04.9	N23°05.7	49°32.2	N10°14.0	16°05.7	N20°05.5	86°40.9	S06°04.1	Capella	280°23.1 46°01.3
13	92°24.2	14°04.1	06.1	64°32.9	14.7	31°07.5	05.6	101°43.3	04.1	Bellatrix	278°23.8 6°22.3
14	107°26.7	29°03.2	06.4	79°33.6	15.4	46°09.4	05.7	116°45.7	04.1	Elnath	278°02.9 28°37.7
15	122°29.1	44°02.4	•• 06.8	94°34.3	•• 16.0	61°11.3	•• 05.8	131°48.0	•• 04.1	Alnilam	275°38.6 -1°11.2
16	137°31.6	59°01.5	07.1	109°35.0	16.7	76°13.1	05.9	146°50.4	04.0	Betelgeuse	270°53.0 7°24.7
17	152°34.1	74°00.7	07.4	124°35.7	17.4	91°15.0	06.1	161°52.8	04.0	Canopus	263°53.2 -52°42.5
18	167°36.5	88°59.8	N23°07.8	139°36.4	N10°18.0	106°16.8	N20°06.2	176°55.2	S06°04.0	Sirius	258°27.0 -16°45.0
19	182°39.0	103°58.9	08.1	154°37.1	18.7	121°18.7	06.3	191°57.6	04.0	Adhara	255°06.6 -29°00.4
20	197°41.5	118°58.1	08.5	169°37.8	19.4	136°20.6	06.4	206°59.9	03.9	Procyon	244°51.6 5°09.8
21	212°43.9	133°57.2	•• 08.8	184°38.5	•• 20.0	151°22.4	•• 06.5	222°02.3	•• 03.9	Pollux	243°18.2 27°58.1
22	227°46.4	148°56.4	09.1	199°39.2	20.7	166°24.3	06.6	237°04.7	03.9	Avior	234°15.4 -59°35.4
23	242°48.8	163°55.5	09.5	214°39.9	21.4	181°26.1	06.7	252°07.1	03.9	Suhail	222°46.9 -43°32.0
Mer.pass. 06:51		ν -0.9' d 0.4' m -3.92		ν 0.7' d 0.7' m 1.03		ν 1.9' d 0.1' m -2.00		ν 2.4' d -0.0' m 1.00		Miaplacidus 221°38.9 -69°49.3	
Alphard 217°48.4 -8°45.9											
Regulus 207°35.1 11°50.9											
Dubhe 193°41.6 61°37.5											
Denebola 182°25.4 14°26.2											
Gienah 175°44.0 -17°40.8											
Acrux 173°00.5 -63°14.3											
Gacrux 171°52.0 -57°15.3											
Alioth 166°13.1 55°49.9											
Spica 158°22.6 -11°17.4											
Alkaid 152°52.1 49°11.7											
Hadar 148°36.3 -60°29.7											
Menkent 147°57.9 -36°29.6											
Arcturus 145°48.1 19°03.4											
Rigel Kent. 139°40.5 -60°56.4											
Kochab 137°18.8 74°03.4											
Zuben'ubi 136°56.3 -16°08.7											
Alphecca 126°03.8 26°38.0											
Antares 112°16.1 -26°29.2											
Atria 107°10.2 -69°04.3											
Sabik 102°03.0 -15°45.3											
Shaula 96°10.6 -37°07.3											
Rasalhague 95°58.7 12°32.5											
Eltanin 90°41.9 51°29.0											
Kaus Aust. 83°32.8 -34°22.3											
Vega 80°33.2 38°48.2											
Nunki 75°48.0 -26°16.0											
Altair 62°00.2 8°55.9											
Peacock 53°06.1 -56°39.2											
Deneb 49°25.8 45°21.8											
Enif 33°39.2 9°59.1											
Al Na'ir 27°33.4 -46°50.4											
Fomalhaut 15°15.0 -29°29.5											
Scheat 13°45.7 28°12.7											
Markab 13°30.4 15°20.1											
Jun 08 Sat SHA Mer.pass											
Venus 282°23.0 12:04											
Mars 332°31.6 08:42											
Jupiter 298°51.2 10:56											
Saturn 9°20.2 06:14											
Jun 09 Sun SHA Mer.pass											
Venus 281°03.3 12:05											
Mars 331°49.3 08:41											
Jupiter 298°36.7 10:53											
Saturn 9°18.2 06:10											
Jun 10 Mon SHA Mer.pass											
Venus 279°43.5 12:06											
Mars 331°07.0 08:40											
Jupiter 298°22.2 10:50											
Saturn 9°16.2 06:07											
Horizontal parallax											
Venus: 0.1											
Mars: 0.1											
Mer.pass. 06:44		ν -0.9' d 0.3' m -3.92		ν 0.7' d 0.7' m 1.02		ν 1.9' d 0.1' m -2.00		ν 2.4' d -0.0' m 0.99			

2024 June 08 to Jun. 10 UT

IERS Earth Orientation data as of 01-Aug-2024
 Page 109
 <https://pypi.org/project/sfalmanac/>

June 11, 12, 13 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	259°49.6	178°13.1	N23°24.4	230°14.2	N10°53.8	197°57.4	N20°12.2	269°03.9	S06°02.8	Alpheratz	357°35.4	29°13.3
1	274°52.1	193°12.2	24.7	245°14.9	54.5	212°59.2	12.3	284°06.3	02.7	Ankaa	353°07.7	-42°10.2
2	289°54.5	208°11.3	25.0	260°15.6	55.1	228°01.1	12.4	299°08.7	02.7	Schedar	349°31.8	56°40.0
3	304°57.0	223°10.5	•• 25.3	275°16.3	•• 55.8	243°03.0	•• 12.5	314°11.1	•• 02.7	Diphda	348°47.9	-17°51.1
4	319°59.5	238°09.6	25.5	290°17.0	56.4	258°04.8	12.6	329°13.5	02.7	Achernar	335°20.9	-57°06.5
5	335°01.9	253°08.7	25.8	305°17.7	57.1	273°06.7	12.7	344°15.9	02.7	Hamal	327°52.0	23°34.5
6	350°04.4	268°07.8	N23°26.1	320°18.4	N10°57.7	288°08.5	N20°12.8	359°18.3	S06°02.6	Polaris	314°40.8	89°21.8
7	5°06.8	283°07.0	26.3	335°19.1	58.4	303°10.4	12.9	14°20.7	02.6	Acamar	315°12.5	-40°12.3
8	20°09.3	298°06.1	26.6	350°19.8	59.1	318°12.3	13.0	29°23.1	02.6	Menkar	314°06.9	4°11.1
9	35°11.8	313°05.2	•• 26.9	5°20.5	10°59.7	333°14.1	•• 13.2	44°25.4	•• 02.6	Mirfak	308°29.4	49°56.7
10	50°14.2	328°04.4	27.1	20°21.2	11°00.4	348°16.0	13.3	59°27.8	02.6	Aldebaran	290°40.5	16°33.5
11	65°16.7	343°03.5	27.4	35°21.9	01.0	3°17.9	13.4	74°30.2	02.5	Rigel	281°04.7	-8°10.4
12	80°19.2	358°02.6	N23°27.7	50°22.6	N11°01.7	18°19.7	N20°13.5	89°32.6	S06°02.5	Capella	280°23.1	46°01.3
13	95°21.6	13°01.7	27.9	65°23.3	02.3	33°21.6	13.6	104°35.0	02.5	Bellatrix	278°23.7	6°22.3
14	110°24.1	28°00.9	28.2	80°24.0	03.0	48°23.4	13.7	119°37.4	02.5	Elnath	278°02.9	28°37.7
15	125°26.6	43°00.0	•• 28.4	95°24.7	•• 03.6	63°25.3	•• 13.8	134°39.8	•• 02.5	Alnilam	275°38.6	-1°11.2
16	140°29.0	57°59.1	28.7	110°25.4	04.3	78°27.2	13.9	149°42.2	02.4	Betelgeuse	270°53.0	7°24.7
17	155°31.5	72°58.2	29.0	125°26.1	04.9	93°29.0	14.0	164°44.6	02.4	Canopus	263°53.2	-52°42.5
18	170°33.9	87°57.4	N23°29.2	140°26.8	N11°05.6	108°30.9	N20°14.1	179°47.0	S06°02.4	Sirius	258°27.0	-16°45.0
19	185°36.4	102°56.5	29.5	155°27.5	06.3	123°32.8	14.3	194°49.4	02.4	Adhara	255°06.6	-29°00.3
20	200°38.9	117°55.6	29.7	170°28.2	06.9	138°34.6	14.4	209°51.7	02.4	Procyon	244°51.6	5°09.8
21	215°41.3	132°54.7	•• 30.0	185°28.9	•• 07.6	153°36.5	•• 14.5	224°54.1	•• 02.3	Pollux	243°18.2	27°58.1
22	230°43.8	147°53.9	30.2	200°29.6	08.2	168°38.3	14.6	239°56.5	02.3	Avior	234°15.4	-59°35.4
23	245°46.3	162°53.0	30.5	215°30.3	08.9	183°40.2	14.7	254°58.9	02.3	Suhail	222°47.0	-43°32.0
Mer.pass. 06:40		ν -0.9' d 0.3' m -3.92		ν 0.7' d 0.7' m 1.02		ν 1.9' d 0.1' m -2.00		ν 2.4' d -0.0' m 0.99				
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	260°48.7	177°52.1	N23°30.7	230°31.0	N11°09.5	198°42.1	N20°14.8	270°01.3	S06°02.3	Alphard	217°48.4	-8°45.9
1	275°51.2	192°51.2	31.0	245°31.7	10.2	213°43.9	14.9	285°03.7	02.3	Regulus	207°35.1	11°50.9
2	290°53.7	207°50.3	31.2	260°32.4	10.8	228°45.8	15.0	300°06.1	02.2	Dubhe	193°41.7	61°37.5
3	305°56.1	222°49.5	•• 31.5	275°33.1	•• 11.5	243°47.7	•• 15.1	315°08.5	•• 02.2	Denebola	182°25.4	14°26.2
4	320°58.6	237°48.6	31.7	290°33.8	12.1	258°49.5	15.2	330°10.9	02.2	Gienah	175°44.0	-17°40.8
5	336°01.1	252°47.7	32.0	305°34.5	12.8	273°51.4	15.3	345°13.3	02.2	Acrux	173°00.5	-63°14.4
6	351°03.5	267°46.8	N23°32.2	320°35.2	N11°13.4	288°53.3	N20°15.4	0°15.7	S06°02.2	Gacrux	171°52.0	-57°15.3
7	6°06.0	282°46.0	32.4	335°35.9	14.1	303°55.1	15.6	15°18.1	02.1	Alioth	166°13.2	55°49.9
8	21°08.4	297°45.1	32.7	350°36.6	14.7	318°57.0	15.7	30°20.5	02.1	Spica	158°22.7	-11°17.4
9	36°10.9	312°44.2	•• 32.9	5°37.3	•• 15.4	333°58.8	•• 15.8	45°22.9	•• 02.1	Alkaid	152°52.1	49°11.7
10	51°13.4	327°43.3	33.2	20°38.0	16.0	349°00.7	15.9	60°25.2	02.1	Hadar	148°36.3	-60°29.7
11	66°15.8	342°42.4	33.4	35°38.7	16.7	4°02.6	16.0	75°27.6	02.1	Menkent	147°57.9	-36°29.6
12	81°18.3	357°41.6	N23°33.6	50°39.4	N11°17.3	19°04.4	N20°16.1	90°30.0	S06°02.1	Arcturus	145°48.1	19°03.4
13	96°20.8	12°40.7	33.9	65°40.1	18.0	34°06.3	16.2	105°32.4	02.0	Rigel Kent.	139°40.5	-60°56.4
14	111°23.2	27°39.8	34.1	80°40.8	18.6	49°08.2	16.3	120°34.8	02.0	Kochab	137°18.8	74°03.5
15	126°25.7	42°38.9	•• 34.3	95°41.5	•• 19.3	64°10.0	•• 16.4	135°37.2	•• 02.0	Zuben'ubi	136°56.3	-16°08.7
16	141°28.2	57°38.0	34.5	110°42.2	19.9	79°11.9	16.5	150°39.6	02.0	Alphecca	126°03.8	26°38.0
17	156°30.6	72°37.2	34.8	125°42.9	20.6	94°13.8	16.6	165°42.0	02.0	Antares	112°16.1	-26°29.2
18	171°33.1	87°36.3	N23°35.0	140°43.6	N11°21.2	109°15.6	N20°16.7	180°44.4	S06°01.9	Atria	107°10.2	-69°04.3
19	186°35.6	102°35.4	35.2	155°44.3	21.9	124°17.5	16.8	195°46.8	01.9	Sabik	102°03.0	-15°45.3
20	201°38.0	117°34.5	35.5	170°45.0	22.5	139°19.4	17.0	210°49.2	01.9	Shaula	96°10.6	-37°07.3
21	216°40.5	132°33.6	•• 35.7	185°45.7	•• 23.2	154°21.2	•• 17.1	225°51.6	•• 01.9	Rasalhague	95°58.7	12°32.5
22	231°42.9	147°32.8	35.9	200°46.4	23.8	169°23.1	17.2	240°54.0	01.9	Eltanin	90°41.9	51°29.0
23	246°45.4	162°31.9	36.1	215°47.1	24.4	184°24.9	17.3	255°56.4	01.9	Kaus Aust.	83°32.8	-34°22.4
Mer.pass. 06:36		ν -0.9' d 0.2' m -3.92		ν 0.7' d 0.7' m 1.02		ν 1.9' d 0.1' m -2.00		ν 2.4' d -0.0' m 0.98		Vega	80°33.1	38°48.2
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	261°47.9	177°31.0	N23°36.3	230°47.8	N11°25.1	199°26.8	N20°17.4	270°58.8	S06°01.8	Nunki	75°48.0	-26°16.0
1	276°50.3	192°30.1	36.6	245°48.5	25.7	214°28.7	17.5	286°01.2	01.8	Altair	62°00.2	8°55.9
2	291°52.8	207°29.2	36.8	260°49.2	26.4	229°30.5	17.6	301°03.6	01.8	Peacock	53°00.6	-56°39.2
3	306°55.3	222°28.4	•• 37.0	275°49.9	•• 27.0	244°32.4	•• 17.7	316°06.0	•• 01.8	Deneb	49°25.8	45°21.8
4	321°57.7	237°27.5	37.2	290°50.6	27.7	259°34.3	17.8	331°08.4	01.8	Enif	33°39.1	9°59.1
5	337°00.2	252°26.6	37.4	305°51.2	28.3	274°36.1	17.9	346°10.8	01.7	Al Na'ir	27°33.3	-46°50.4
6	352°02.7	267°25.7	N23°37.6	320°51.9	N11°29.0	289°38.0	N20°18.0	1°13.2	S06°01.7	Fomalhaut	15°15.0	-29°29.4
7	7°05.1	282°24.8	37.8	335°52.6	29.6	304°39.9	18.1	16°15.5	01.7	Scheat	13°45.6	28°12.7
8	22°07.6	297°23.9	38.1	350°53.3	30.3	319°41.7	18.2	31°17.9	01.7	Markab	13°30.4	15°20.1
9	37°10.0	312°23.1	•• 38.3	5°54.0	•• 30.9	334°43.6	•• 18.4	46°20.3	•• 01.7			
10	52°12.5	327°22.2	38.5	20°54.7	31.6	349°45.5	18.5	61°22.7	01.7	Jun 11 Tue	SHA	Mer.pass
11	67°15.0	342°21.3	38.7	35°55.4	32.2	4°47.3	18.6	76°25.1	01.6	Venus	278°23.5	12:08
12	82°17.4	357°20.4	N23°38.9	50°56.1	N11°32.8	19°49.2	N20°18.7	91°27.5	S06°01.6	Mars	330°24.6	08:39
13	97°19.9	12°19.5	39.1	65°56.8	33.5	34°51.1	18.8	106°29.9	01.6	Jupiter	298°07.8	10:47
14	112°22.4	27°18.6	39.3	80°57.5	34.1	49°52.9	18.9	121°32.3	01.6	Saturn	9°14.3	06:03
15	127°24.8	42°17.8	•• 39.5	95°58.2	•• 34.8	64°54.8	•• 19.0	136°34.7	•• 01.6			
16	142°27.3	57°16.9	39.7	110°58.9	35.4	79°56.6	19.1	151°37.1	01.6	Jun 12 Wed	SHA	Mer.pass
17	157°29.8	72°16.0	39.9	125°59.6	36.1	94°58.5	19.2	166°39.5	01.5	Venus	277°03.4	12:09
18	172°32.2	87°15.1	N23°40.1	141°00.3	N11°36.7	110°00.4	N20°19.3	181°41.9	S06°01.5	Mars	329°42.3	08:38
19	187°34.7	102°14.2	40.3	156°01.0	37.3	125°02.2	19.4	196°44.3	01.5	Jupiter	297°53.3	10:44
20	202°37.2	117°13.3	40.5	171°01.7	38.0	140°04.1	19.5	211°46.7	01.5	Saturn	9°12.6	05:59
21	217°39.6	132°12.5	•• 40.7	186°02.4	•• 38.6	155°06.0	•• 19.6</					

Sun							Moon							
Tue	GHA	Dec	GHA	ν	Dec	d	HP	Lat.	Twilight		Sunrise	Sunset	Twilight	
									Naut.	Civil			Civil	Naut.
0	180°04.9	N23°06.2	121°06.0	12.6'	N20°28.1	-10.7'	55.2'	N 72°	☐	☐	☐	☐	☐	☐
1	195°04.7	06.4	135°37.6	12.7'	20°17.4	-10.7'	55.2'	N 70°	☐	☐	☐	☐	☐	☐
2	210°04.6	06.6	150°09.2	12.8'	20°06.7	-10.8'	55.2'	68°	☐	☐	☐	☐	☐	☐
3	225°04.5	· · 06.7	164°41.0	12.9'	19°55.8	-10.9'	55.2'	66°	—:—	—:—	00:02	—:—	—:—	—:—
4	240°04.3	06.9	179°12.9	13.0'	19°44.9	-11.0'	55.1'	64°	—:—	—:—	01:36	22:25	—:—	—:—
5	255°04.2	07.1	193°44.8	13.1'	19°34.0	-11.1'	55.1'	62°	—:—	—:—	02:12	21:49	—:—	—:—
6	270°04.1	N23°07.2	208°16.9	13.2'	N19°22.9	-11.1'	55.1'	60°	—:—	00:57	02:37	21:23	23:04	—:—
7	285°04.0	07.4	222°49.0	13.2'	19°11.8	-11.2'	55.1'	N 58°	—:—	01:44	02:57	21:03	22:17	—:—
8	300°03.8	07.6	237°21.3	13.3'	19°00.6	-11.3'	55.1'	56°	—:—	02:12	03:14	20:47	21:48	—:—
9	315°03.7	· · 07.7	251°53.6	13.4'	18°49.3	-11.3'	55.0'	54°	00:53	02:34	03:28	20:33	21:27	23:09
10	330°03.6	07.9	266°26.0	13.5'	18°38.0	-11.4'	55.0'	52°	01:35	02:52	03:40	20:20	21:09	22:26
11	345°03.4	08.0	280°58.6	13.6'	18°26.5	-11.5'	55.0'	50°	02:02	03:06	03:51	20:10	20:54	21:59
12	0°03.3	N23°08.2	295°31.2	13.7'	N18°15.1	-11.6'	55.0'	45°	02:46	03:36	04:13	19:47	20:25	21:14
13	15°03.2	08.4	310°03.9	13.8'	18°03.5	-11.6'	55.0'	N 40°	03:16	03:58	04:31	19:30	20:02	20:44
14	30°03.1	08.5	324°36.7	13.9'	17°51.9	-11.7'	54.9'	35°	03:39	04:16	04:45	19:15	19:44	20:21
15	45°02.9	· · 08.7	339°09.6	14.0'	17°40.2	-11.7'	54.9'	30°	03:58	04:31	04:58	19:02	19:29	20:02
16	60°02.8	08.8	353°42.5	14.1'	17°28.5	-11.8'	54.9'	20°	04:26	04:56	05:20	18:40	19:04	19:34
17	75°02.7	09.0	8°15.6	14.1'	17°16.7	-11.9'	54.9'	N 10°	04:49	05:16	05:39	18:21	18:44	19:11
18	90°02.5	N23°09.2	22°48.7	14.2'	N17°04.8	-11.9'	54.9'	0°	05:08	05:34	05:56	18:04	18:26	18:52
19	105°02.4	09.3	37°22.0	14.3'	16°52.9	-12.0'	54.8'	S 10°	05:24	05:51	06:14	17:46	18:09	18:35
20	120°02.3	09.5	51°55.3	14.4'	16°40.9	-12.1'	54.8'	20°	05:40	06:08	06:32	17:28	17:52	18:19
21	135°02.1	· · 09.6	66°28.7	14.5'	16°28.8	-12.1'	54.8'	30°	05:57	06:27	06:53	17:07	17:33	18:03
22	150°02.0	09.8	81°02.1	14.6'	16°16.7	-12.2'	54.8'	35°	06:05	06:37	07:05	16:55	17:23	17:55
23	165°01.9	09.9	95°35.7	14.6'	16°04.5	-12.2'	54.8'	40°	06:15	06:49	07:19	16:41	17:11	17:45
SD = 15.7' $d = 0.2'$							SD = 15.1'							
Wed	GHA	Dec	GHA	ν	Dec	d	HP	Lat.	Moonrise		Moonset			
									Tue	Wed	Thu	Tue	Wed	Thu
0	180°01.8	N23°10.1	110°09.3	14.7'	N15°52.3	-12.3'	54.7'	N 72°	☐	07:27	09:47	☐	02:32	01:39
1	195°01.6	10.2	124°43.0	14.8'	15°40.0	-12.3'	54.7'	N 70°	05:00	07:58	10:01	03:25	01:59	01:23
2	210°01.5	10.4	139°16.8	14.9'	15°27.7	-12.4'	54.7'	68°	06:08	08:21	10:11	02:16	01:35	01:10
3	225°01.4	· · 10.5	153°50.7	14.9'	15°15.3	-12.4'	54.7'	66°	06:43	08:38	10:20	01:39	01:16	00:59
4	240°01.2	10.7	168°24.6	15.0'	15°02.9	-12.5'	54.7'	64°	07:09	08:53	10:28	01:13	01:00	00:50
5	255°01.1	10.8	182°58.7	15.1'	14°50.4	-12.5'	54.7'	62°	07:28	09:04	10:34	00:52	00:48	00:43
6	270°01.0	N23°11.0	197°32.7	15.2'	N14°37.8	-12.6'	54.6'	60°	07:44	09:14	10:39	00:36	00:36	00:36
7	285°00.8	11.1	212°06.9	15.2'	14°25.3	-12.6'	54.6'	N 58°	07:58	09:23	10:44	00:21	00:27	00:30
8	300°00.7	11.3	226°41.1	15.3'	14°12.6	-12.7'	54.6'	56°	08:09	09:31	10:48	00:09	00:18	00:25
9	315°00.6	· · 11.4	241°15.5	15.4'	13°59.9	-12.7'	54.6'	54°	08:19	09:37	10:52	· · ·	00:11	00:20
10	330°00.5	11.6	255°49.8	15.4'	13°47.2	-12.8'	54.6'	52°	08:28	09:43	10:56	· · ·	00:04	00:16
11	345°00.3	11.7	270°24.3	15.5'	13°34.4	-12.8'	54.6'	50°	08:36	09:49	10:59	23:58	· · ·	00:12
12	0°00.2	N23°11.9	284°58.8	15.6'	N13°21.6	-12.9'	54.6'	45°	08:53	10:00	11:05	23:45	· · ·	00:03
13	15°00.1	12.0	299°33.4	15.6'	13°08.8	-12.9'	54.5'	N 40°	09:07	10:10	11:11	23:34	23:56	· · ·
14	29°59.9	12.1	314°08.0	15.7'	12°55.8	-12.9'	54.5'	35°	09:18	10:18	11:16	23:24	23:50	· · ·
15	44°59.8	· · 12.3	328°42.7	15.8'	12°42.9	-13.0'	54.5'	30°	09:29	10:26	11:20	23:16	23:45	· · ·
16	59°59.7	12.4	343°17.5	15.8'	12°29.9	-13.0'	54.5'	20°	09:46	10:38	11:27	23:01	23:35	· · ·
17	74°59.5	12.6	357°52.3	15.9'	12°16.9	-13.1'	54.5'	N 10°	10:01	10:49	11:34	22:49	23:27	· · ·
18	89°59.4	N23°12.7	12°27.2	15.9'	N12°03.8	-13.1'	54.5'	0°	10:15	10:59	11:40	22:37	23:19	23:59
19	104°59.3	12.8	27°02.1	16.0'	11°50.7	-13.1'	54.5'	S 10°	10:29	11:09	11:45	22:24	23:11	23:55
20	119°59.1	13.0	41°37.1	16.1'	11°37.6	-13.2'	54.4'	20°	10:44	11:19	11:52	22:11	23:02	23:51
21	134°59.0	· · 13.1	56°12.2	16.1'	11°24.4	-13.2'	54.4'	30°	11:01	11:32	11:59	21:56	22:52	23:46
22	149°58.9	13.3	70°47.3	16.2'	11°11.2	-13.3'	54.4'	35°	11:11	11:38	12:03	21:48	22:46	23:43
23	164°58.8	13.4	85°22.5	16.2'	10°57.9	-13.3'	54.4'	40°	11:22	11:46	12:08	21:38	22:40	23:40
SD = 15.7' $d = 0.2'$							SD = 14.9'							
Thu	GHA	Dec	GHA	ν	Dec	d	HP	Day	Sun		Mer.	Moon		Age
									Eqn.of Time	Pass	Mer.Pass.	Upper	Lower	5-7
0	179°58.6	N23°13.5	99°57.7	16.3'	N10°44.6	-13.3'	54.4'		00 ^h	12 ^h	Pass	hh:mm	hh:mm	21-39%
1	194°58.5	13.7	114°33.0	16.3'	10°31.3	-13.4'	54.4'	11	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	
2	209°58.4	13.8	129°08.3	16.4'	10°17.9	-13.4'	54.4'	12	00:19	00:13	12:00	16:26	04:03	
3	224°58.2	· · 13.9	143°43.7	16.4'	10°04.6	-13.4'	54.4'	13	00:07	00:01	12:00	17:09	04:48	
4	239°58.1	14.1	158°19.1	16.5'	09°51.1	-13.4'	54.4'		00:06	00:12	12:00	17:49	05:29	
5	254°58.0	14.2	172°54.6	16.5'	09°37.7	-13.5'	54.4'							
6	269°57.8	N23°14.3	187°30.1	16.6'	N09°24.2	-13.5'	54.3'							
7	284°57.7	14.5	202°05.7	16.6'	09°10.7	-13.5'	54.3'							
8	299°57.6	14.6	216°41.3	16.7'	08°57.2	-13.6'	54.3'							
9	314°57.4	· · 14.7	231°16.9	16.7'	08°43.6	-13.6'	54.3'							
10	329°57.3	14.8	245°52.6	16.7'	08°30.0	-13.6'	54.3'							
11	344°57.2	15.0	260°28.3	16.8'	08°16.4	-13.6'	54.3'							
12	359°57.0	N23°15.1	275°04.1	16.8'	N08°02.8	-13.7'	54.3'							
13	14°56.9	15.2	289°39.9	16.8'	07°49.1	-13.7'	54.3'							
14	29°56.8	15.4	304°15.8	16.9'	07°35.4	-13.7'	54.3'							
15	44°56.6	· · 15.5	318°51.7	16.9'	07°21.7	-13.7'	54.3'							
16	59°56.5	15.6	333°27.6	17.0'	07°08.0	-13.8'	54.3'							
17	74°56.4	15.7	348°03.5	17.0'	06°54.2	-13.8'	54.3'							
18	89°56.2	N23°15.8	2°39.5	17.0'	N06°40.4	-13.8'	54.2'							
19	104°56.1	16.0	17°15.5	17.0'	06°26.6	-13.8'	54.2'							
20	119°56.0	16.1	31°51.6	17.1'	06°12.8	-13.8'	54.2'							
21	134°55.													

June 14, 15, 16 UT (Fri., Sat., Sun.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	262°47.0	177°09.8	N23°41.3	231°04.5	N11°40.6	200°11.6	N20°20.0	271°56.3	S06°01.4					Alpheratz	357°35.3	29°13.3	
1	277°49.5	192°08.9		41.4	246°05.2		41.2	215°13.4		20.1	286°58.7		01.4	Ankaa	353°07.7	-42°10.2	
2	292°51.9	207°08.0		41.6	261°05.9		41.8	230°15.3		20.2	302°01.1		01.4	Schedar	349°31.8	56°40.0	
3	307°54.4	222°07.1	· ·	41.8	276°06.6	· ·	42.5	245°17.2	· ·	20.3	317°03.5	· ·	01.4	Diphda	348°47.9	-17°51.1	
4	322°56.9	237°06.3		42.0	291°07.3		43.1	260°19.0		20.4	332°05.9		01.4	Achernar	335°20.8	-57°06.5	
5	337°59.3	252°05.4		42.2	306°08.0		43.8	275°20.9		20.5	347°08.3		01.4	Hamal	327°52.0	23°34.5	
6	353°01.8	267°04.5	N23°42.4	321°08.7	N11°44.4	290°22.8	N20°20.6	2°10.7	S06°01.3					Polaris	314°39.9	89°21.8	
7	8°04.3	282°03.6		42.6	336°09.4		45.0	305°24.6		20.7	17°13.1		01.3	Acamar	315°12.5	-40°12.3	
8	23°06.7	297°02.7		42.7	351°10.1		45.7	320°26.5		20.8	32°15.5		01.3	Menkar	314°06.9	4°11.1	
9	38°09.2	312°01.8	· ·	42.9	6°10.8	· ·	46.3	335°28.4	· ·	20.9	47°17.9	· ·	01.3	Mirfak	308°29.4	49°56.7	
10	53°11.7	327°00.9		43.1	21°11.5		47.0	350°30.2		21.0	62°20.3		01.3	Aldebaran	290°40.5	16°33.5	
11	68°14.1	342°00.0		43.3	36°12.2		47.6	5°32.1		21.1	77°22.7		01.3	Rigel	281°04.7	-8°10.4	
12	83°16.6	356°59.2	N23°43.5	51°12.9	N11°48.2	20°34.0	N20°21.2	92°25.1	S06°01.2					Capella	280°23.1	46°01.3	
13	98°19.0	11°58.3		43.6	66°13.6		48.9	35°35.8		21.3	107°27.5		01.2	Bellatrix	278°23.7	6°22.3	
14	113°21.5	26°57.4		43.8	81°14.3		49.5	50°37.7		21.4	122°29.9		01.2	Elnath	278°02.9	28°37.7	
15	128°24.0	41°56.5	· ·	44.0	96°14.9	· ·	50.1	65°39.6	· ·	21.5	137°32.3	· ·	01.2	Alnilam	275°38.6	-1°11.2	
16	143°26.4	56°55.6		44.1	111°15.6		50.8	80°41.4		21.7	152°34.7		01.2	Betelgeuse	270°53.0	7°24.7	
17	158°28.9	71°54.7		44.3	126°16.3		51.4	95°43.3		21.8	167°37.1		01.2	Canopus	263°53.2	-52°42.5	
18	173°31.4	86°53.8	N23°44.5	141°17.0	N11°52.1	110°45.2	N20°21.9	182°39.5	S06°01.2					Sirius	258°27.0	-16°45.0	
19	188°33.8	101°52.9		44.7	156°17.7		52.7	125°47.0		22.0	197°41.9		01.1	Adhara	255°06.6	-29°00.3	
20	203°36.3	116°52.1		44.8	171°18.4		53.3	140°48.9		22.1	212°44.3		01.1	Procyon	244°51.6	5°09.8	
21	218°38.8	131°51.2	· ·	45.0	186°19.1	· ·	54.0	155°50.8	· ·	22.2	227°46.8	· ·	01.1	Pollux	243°18.2	27°58.1	
22	233°41.2	146°50.3		45.2	201°19.8		54.6	170°52.6		22.3	242°49.2		01.1	Avior	234°15.5	-59°35.4	
23	248°43.7	161°49.4		45.3	216°20.5		55.2	185°54.5		22.4	257°51.6		01.1	Suhail	222°47.0	-43°32.0	
Mer.pass. 06:28			ν -0.9' d 0.2' m -3.91			ν 0.7' d 0.6' m 1.01			ν 1.9' d 0.1' m -2.01			ν 2.4' d -0.0' m 0.98					

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	263°46.1	176°48.5	N23°45.5	231°21.2	N11°55.9	200°56.4	N20°22.5	272°54.0	S06°01.1					
1	278°48.6	191°47.6	45.6	246°21.9	56.5	215°58.2	22.6	287°56.4	01.1					
2	293°51.1	206°46.7	45.8	261°22.6	57.2	231°00.1	22.7	302°58.8	01.0					
3	308°53.5	221°45.8	· ·	46.0	276°23.3	· ·	57.8	318°01.2	· ·	01.0				
4	323°56.0	236°44.9	46.1	291°24.0	58.4	261°03.8	22.9	333°03.6	01.0					
5	338°58.5	251°44.1	46.3	306°24.7	59.1	276°05.7	23.0	348°06.0	01.0					
6	354°00.9	266°43.2	N23°46.4	321°25.4	N11°59.7	291°07.6	N20°23.1	3°08.4	S06°01.0					
7	9°03.4	281°42.3	46.6	336°26.1	12°00.3	306°09.4	23.2	18°10.8	01.0					
8	24°05.9	296°41.4	46.7	351°26.8	01.0	321°11.3	23.3	33°13.2	01.0					
9	39°08.3	311°40.5	· ·	46.9	6°27.5	· ·	01.6	48°15.6	· ·	00.9				
10	54°10.8	326°39.6	47.0	21°28.2	02.2	351°15.0	23.5	63°18.0	00.9					
11	69°13.3	341°38.7	47.2	36°28.9	02.9	6°16.9	23.6	78°20.4	00.9					
12	84°15.7	356°37.8	N23°47.3	51°29.6	N12°03.5	21°18.8	N20°23.8	93°22.8	S06°00.9					
13	99°18.2	11°36.9	47.5	66°30.3	04.1	36°20.6	23.9	108°25.2	00.9					
14	114°20.6	26°36.0	47.6	81°31.0	04.8	51°22.5	24.0	123°27.6	00.9					
15	129°23.1	41°35.2	· ·	47.8	96°31.7	· ·	05.4	138°30.0	· ·	00.9				
16	144°25.6	56°34.3	47.9	111°32.3	06.0	81°26.2	24.2	153°32.4	00.9					
17	159°28.0	71°33.4	48.0	126°33.0	06.7	96°28.1	24.3	168°34.8	00.8					
18	174°30.5	86°32.5	N23°48.2	141°33.7	N12°07.3	111°30.0	N20°24.4	183°37.3	S06°00.8					
19	189°33.0	101°31.6	48.3	156°34.4	07.9	126°31.8	24.5	198°39.7	00.8					
20	204°35.4	116°30.7	48.5	171°35.1	08.6	141°33.7	24.6	213°42.1	00.8					
21	219°37.9	131°29.8	· ·	48.6	186°35.8	· ·	09.2	228°44.5	· ·	00.8				
22	234°40.4	146°28.9	48.7	201°36.5	09.8	171°37.4	24.8	243°46.9	00.8					
23	249°42.8	161°28.0	48.9	216°37.2	10.5	186°39.3	24.9	258°49.3	00.8					
Mer.pass. 06:24			ν -0.9' d 0.2' m -3.91			ν 0.7' d 0.6' m 1.01			ν 1.9' d 0.1' m -2.01			ν 2.4' d -0.0' m 0.97		

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	264°45.3	176°27.1	N23°49.0	231°37.9	N12°11.1	201°41.2	N20°25.0	273°51.7	S06°00.7					
1	279°47.7	191°26.2	49.1	246°38.6	11.7	216°43.0	25.1	288°54.1	00.7					
2	294°50.2	206°25.4	49.3	261°39.3	12.3	231°44.9	25.2	303°56.5	00.7					
3	309°52.7	221°24.5	· ·	49.4	276°40.0	· ·	13.0	318°58.9	· ·	00.7				
4	324°55.1	236°23.6	49.5	291°40.7	13.6	261°48.6	25.4	334°01.3	00.7					
5	339°57.6	251°22.7	49.6	306°41.4	14.2	276°50.5	25.5	349°03.7	00.7					
6	355°00.1	266°21.8	N23°49.8	321°42.1	N12°14.9	291°52.4	N20°25.6	4°06.1	S06°00.7					
7	10°02.5	281°20.9	49.9	336°42.8	15.5	306°54.2	25.7	19°08.5	00.7					
8	25°05.0	296°20.0	50.0	351°43.5	16.1	321°56.1	25.8	34°11.0	00.6					
9	40°07.5	311°19.1	· ·	50.1	6°44.2	· ·	16.8	49°13.4	· ·	00.6				
10	55°09.9	326°18.2	50.3	21°44.9	17.4	351°59.8	26.0	64°15.8	00.6					
11	70°12.4	341°17.3	50.4	36°45.6	18.0	7°01.7	26.1	79°18.2	00.6					
12	85°14.9	356°16.4	N23°50.5	51°46.2	N12°18.6	22°03.6	N20°26.3	94°20.6	S06°00.6					
13	100°17.3	11°15.5	50.6	66°46.9	19.3	37°05.5	26.4	109°23.0	00.6					
14	115°19.8	26°14.6	50.7	81°47.6	19.9	52°07.3	26.5	124°25.4	00.6					
15	130°22.2	41°13.7	· ·	50.8	96°48.3	· ·	20.5	139°27.8	· ·	00.6				
16	145°24.7	56°12.9	51.0	111°49.0	21.1	82°11.1	26.7	154°30.2	00.6					
17	160°27.2	71°12.0	51.1	126°49.7	21.8	97°12.9	26.8	169°32.6	00.5					
18	175°29.6	86°11.1	N23°51.2	141°50.4	N12°22.4	112°14.8	N20°26.9	184°35.0	S06°00.5					
19	190°32.1	101°10.2	51.3	156°51.1	23.0	127°16.7	27.0	199°37.5	00.5					
20	205°34.6	116°09.3	51.4	171°51.8	23.7	142°18.5	27.1	214°39.9	00.5					
21	220°37.0	131°08.4	· ·	51.5	186°52.5	· ·	24.3	229°42.3	· ·	00.5				
22	235°39.5	146°07.5	51.6	201°53.2	24.9	172°22.3	27.3	244°44.7	00.5					
23	250°42.0	161°06.6	51.7	216°53.9	25.5	187°24.1	27.4	259°47.1	00.5					
Mer.pass. 06:20			ν -0.9' d 0.1' m -3.91			ν 0.7' d 0.6' m 1.01			ν 1.9' d 0.1' m -2.01			ν 2.4' d -0.0' m 0.97		

Stars		
Alpheratz	357°35.3	29°13.3
Ankaa	353°07.7	-42°10.2
Schedar	349°31.8	56°40.0
Diphda	348°47.9	-17°51.1
Achernar	335°20.8	-57°06.5
Hamal	327°52.0	23°34.5
Polaris	314°39.9	89°21.8
Acamar	315°12.5	-40°12.3
Menkar	314°06.9	4°11.1
Mirfak	308°29.4	49°56.7
Aldebaran	290°40.5	16°33.5
Rigel	281°04.7	-8°10.4
Capella	280°23.1	46°01.3
Bellatrix	278°23.7	6°22.3
Elnath	278°02.9	28°37.7
Alnilam	275°38.6	-1°11.2
Betelgeuse	270°53.0	7°24.7
Canopus	263°53.2	-52°42.5
Sirius	258°27.0	-16°45.0
Adhara	255°06.6	-29°00.3
Procyon	244°51.6	5°09.8
Pollux	243°18.2	27°58.1
Avior	234°15.5	-59°35.4
Suhail	222°47.0	-43°32°32

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	179°55.5	N23°16.6	90°16.0	17.2'	N05°17.4	-13.9'	54.2'
1	194°55.3	16.7	104°52.2	17.2'	05°03.5	-13.9'	54.2'
2	209°55.2	16.8	119°28.4	17.2'	04°49.5	-13.9'	54.2'
3	224°55.1	•• 16.9	134°04.6	17.2'	04°35.6	-14.0'	54.2'
4	239°54.9	17.0	148°40.8	17.3'	04°21.7	-14.0'	54.2'
5	254°54.8	17.1	163°17.1	17.3'	04°07.7	-14.0'	54.2'
6	269°54.7	N23°17.3	177°53.4	17.3'	N03°53.7	-14.0'	54.2'
7	284°54.5	17.4	192°29.7	17.3'	03°39.7	-14.0'	54.2'
8	299°54.4	17.5	207°06.0	17.3'	03°25.7	-14.0'	54.2'
9	314°54.3	•• 17.6	221°42.3	17.3'	03°11.7	-14.0'	54.2'
10	329°54.1	17.7	236°18.6	17.3'	02°57.7	-14.0'	54.2'
11	344°54.0	17.8	250°55.0	17.4'	02°43.7	-14.0'	54.2'
12	359°53.9	N23°17.9	265°31.3	17.4'	N02°29.6	-14.0'	54.2'
13	14°53.7	18.0	280°07.7	17.4'	02°15.6	-14.1'	54.2'
14	29°53.6	18.1	294°44.1	17.4'	02°01.5	-14.1'	54.2'
15	44°53.5	•• 18.3	309°20.5	17.4'	01°47.5	-14.1'	54.2'
16	59°53.3	18.4	323°56.8	17.4'	01°33.4	-14.1'	54.2'
17	74°53.2	18.5	338°33.2	17.4'	01°19.3	-14.1'	54.2'
18	89°53.1	N23°18.6	353°09.6	17.4'	N01°05.2	-14.1'	54.2'
19	104°52.9	18.7	7°46.0	17.4'	00°51.1	-14.1'	54.2'
20	119°52.8	18.8	22°22.4	17.4'	00°37.0	-14.1'	54.2'
21	134°52.7	•• 18.9	36°58.8	17.4'	00°22.9	-14.1'	54.2'
22	149°52.5	19.0	51°35.2	17.4'	N00°08.8	-14.1'	54.2'
23	164°52.4	19.1	66°11.6	17.4'	S00°05.2	14.1'	54.2'
SD = 15.7'			SD = 14.8'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	179°52.3	N23°19.2	80°48.0	17.4'	S00°19.4	14.1'	54.2'
1	194°52.1	19.3	95°24.4	17.4'	00°33.5	14.1'	54.2'
2	209°52.0	19.4	110°00.8	17.4'	00°47.6	14.1'	54.2'
3	224°51.9	•• 19.5	124°37.2	17.4'	01°01.7	14.1'	54.2'
4	239°51.7	19.6	139°13.5	17.4'	01°15.8	14.1'	54.2'
5	254°51.6	19.7	153°49.9	17.3'	01°29.9	14.1'	54.2'
6	269°51.4	N23°19.8	168°26.2	17.3'	S01°43.9	14.1'	54.2'
7	284°51.3	19.9	183°02.6	17.3'	01°58.0	14.1'	54.2'
8	299°51.2	20.0	197°38.9	17.3'	02°12.1	14.1'	54.2'
9	314°51.0	•• 20.1	212°15.2	17.3'	02°26.2	14.1'	54.2'
10	329°50.9	20.2	226°51.5	17.3'	02°40.3	14.1'	54.3'
11	344°50.8	20.3	241°27.8	17.3'	02°54.4	14.1'	54.3'
12	359°50.6	N23°20.4	256°04.0	17.2'	S03°08.4	14.1'	54.3'
13	14°50.5	20.4	270°40.3	17.2'	03°22.5	14.1'	54.3'
14	29°50.4	20.5	285°16.5	17.2'	03°36.6	14.0'	54.3'
15	44°50.2	•• 20.6	299°52.7	17.2'	03°50.6	14.0'	54.3'
16	59°50.1	20.7	314°28.8	17.1'	04°04.6	14.0'	54.3'
17	74°50.0	20.8	329°05.0	17.1'	04°18.7	14.0'	54.3'
18	89°49.8	N23°20.9	343°41.1	17.1'	S04°32.7	14.0'	54.3'
19	104°49.7	21.0	358°17.2	17.1'	04°46.7	14.0'	54.3'
20	119°49.6	21.1	12°53.3	17.0'	05°00.7	14.0'	54.3'
21	134°49.4	•• 21.2	27°29.3	17.0'	05°14.6	14.0'	54.3'
22	149°49.3	21.2	42°05.3	17.0'	05°28.6	14.0'	54.3'
23	164°49.2	21.3	56°41.3	16.9'	05°42.6	13.9'	54.4'
SD = 15.7'			SD = 14.8'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	179°49.0	N23°21.4	71°17.2	16.9'	S05°56.5	13.9'	54.4'
1	194°48.9	21.5	85°53.2	16.9'	06°10.4	13.9'	54.4'
2	209°48.8	21.6	100°29.0	16.8'	06°24.3	13.9'	54.4'
3	224°48.6	•• 21.7	115°04.9	16.8'	06°38.2	13.9'	54.4'
4	239°48.5	21.7	129°40.7	16.8'	06°52.1	13.9'	54.4'
5	254°48.4	21.8	144°16.4	16.7'	07°05.9	13.8'	54.4'
6	269°48.2	N23°21.9	158°52.2	16.7'	S07°19.8	13.8'	54.4'
7	284°48.1	22.0	173°27.9	16.6'	07°33.6	13.8'	54.4'
8	299°48.0	22.1	188°03.5	16.6'	07°47.4	13.8'	54.5'
9	314°47.8	•• 22.1	202°39.1	16.6'	08°01.2	13.8'	54.5'
10	329°47.7	22.2	217°14.7	16.5'	08°14.9	13.7'	54.5'
11	344°47.5	22.3	231°50.2	16.5'	08°28.6	13.7'	54.5'
12	359°47.4	N23°22.4	246°25.6	16.4'	S08°42.3	13.7'	54.5'
13	14°47.3	22.4	261°01.0	16.4'	08°56.0	13.7'	54.5'
14	29°47.1	22.5	275°36.4	16.3'	09°09.7	13.6'	54.5'
15	44°47.0	•• 22.6	290°11.7	16.3'	09°23.3	13.6'	54.5'
16	59°46.9	22.7	304°47.0	16.2'	09°36.9	13.6'	54.6'
17	74°46.7	22.7	319°22.2	16.2'	09°50.5	13.5'	54.6'
18	89°46.6	N23°22.8	333°57.4	16.1'	S10°04.0	13.5'	54.6'
19	104°46.5	22.9	348°32.5	16.0'	10°17.5	13.5'	54.6'
20	119°46.3	22.9	3°07.5	16.0'	10°31.0	13.5'	54.6'
21	134°46.2	•• 23.0	17°42.5	15.9'	10°44.5	13.4'	54.6'
22	149°46.1	23.1	32°17.4	15.9'	10°57.9	13.4'	54.7'
23	164°45.9	23.1	46°52.3	15.8'	11°11.3	13.4'	54.7'
SD = 15.7'			SD = 14.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	☐	☐	☐	☐	☐	☐
64°	--	--	01:33	22:29	--	--
62°	--	--	02:10	21:52	--	--
60°	--	00:53	02:36	21:26	23:10	--
N 58°	--	01:41	02:56	21:05	22:20	--
56°	--	02:11	03:13	20:48	21:51	--
54°	00:48	02:33	03:27	20:34	21:29	23:14
52°	01:33	02:51	03:39	20:22	21:11	22:29
50°	02:01	03:06	03:50	20:11	20:56	22:01
45°	02:46	03:35	04:13	19:49	20:26	21:16
N 40°	03:16	03:58	04:31	19:31	20:04	20:45
35°	03:39	04:16	04:46	19:16	19:45	20:22
30°	03:58	04:31	04:59	19:03	19:30	20:04
20°	04:27	04:56	05:21	18:41	19:05	19:35
N 10°	04:49	05:16	05:39	18:22	18:45	19:12
0°	05:08	05:34	05:57	18:04	18:27	18:53
S 10°	05:25	05:52	06:14	17:47	18:10	18:36
20°	05:41	06:09	06:33	17:28	17:52	18:20
30°	05:58	06:28	06:54	17:07	17:34	18:03
35°	06:06	06:38	07:06	16:55	17:23	17:55
40°	06:16	06:50	07:20	16:41	17:11	17:45
45°	06:26	07:03	07:37	16:24	16:58	17:35
S 50°	06:38	07:19	07:58	16:03	16:42	17:23
52°	06:43	07:27	08:08	15:54	16:34	17:18
54°	06:49	07:35	08:19	15:42	16:26	17:12
56°	06:55	07:44	08:31	15:30	16:17	17:06
58°	07:02	07:54	08:46	15:15	16:07	17:00
S 60°	07:09	08:06	09:03	14:58	15:55	16:52

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	11:47	13:44	15:47	01:02	00:31 23:59	23:20
N 70°	11:51	13:38	15:31	00:56	00:32	00:07 23:39
68°	11:54	13:34	15:18	00:50	00:32	00:14 23:53
66°	11:56	13:31	15:08	00:46	00:33	00:20
64°	11:58	13:28	14:59	00:42	00:33	00:25
62°	12:00	13:25	14:52	00:38	00:34	00:29
60°	12:01	13:23	14:46	00:35	00:34	00:33
N 58°	12:03	13:21	14:40	00:32	00:34	00:36
56°	12:04	13:19	14:35	00:30	00:34	00:39
54°	12:05	13:17	14:31	00:28	00:35	00:41
52°	12:06	13:16	14:27	00:26	00:35	00:44
50°	12:07	13:15	14:24	00:24	00:35	00:46
45°	12:09	13:12	14:16	00:20	00:35	00:51
N 40°	12:10	13:10	14:10	00:17	00:36	00:55
35°	12:12	13:08	14:04	00:14	00:36	00:59
30°	12:13	13:06	13:59	00:11	00:36	01:02
20°	12:15	13:03	13:51	00:06	00:37	01:07
N 10°	12:17	13:00	13:44	00:02	00:37	01:12
0°	12:19	12:58	13:37	•• ••	00:37	01:17
S 10°	12:21	12:55	13:31	•• ••	00:38	01:21
20°	12:22	12:53	13:24	•• ••	00:38	01:26
30°	12:25	12:50	13:16	•• ••	00:38	01:31
35°	12:26	12:48	13:11	•• ••	00:39	01:35
40°	12:27	12:46	13:06	•• ••	00:39	01:38
45°	12:29	12:44	13:00	•• ••	00:39	01:43
S 50°	12:31	12:42	12:53	•• ••	00:40	01:48
52°	12:31	12:40	12:50	•• ••	00:40	01:50
54°	12:32	12:39	12:47	•• ••	00:40	01:53
56°	12:33	12:38	12:43	•• ••	00:40	01:55
58°	12:35	12:36	12:38	•• ••	00:40	01:59
S 60°	12:36	12:35	12:34	•• ••	00:40	02:02

Day	Sun		Mer. Pass hh:mm	Moon		Age 8-10 48-67%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
14	00:18	00:25	12:00	18:28	06:09	
15	00:31	00:37	12:01	19:07	06:47	
16	00:44	00:50	12:01	19:47	07:27	

June 17, 18, 19 UT (Mon., Tue., Wed.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	265°44.4	176°05.7	N23°51.8	231°54.6	N12°26.2	202°26.0	N20°27.5	274°49.5	S06°00.5					Alpheratz	357°35.3	29°13.3	
1	280°46.9	191°04.8	51.9	246°55.3	26.8	217°27.9	27.6	289°51.9	00.5					Ankaa	353°07.7	-42°10.2	
2	295°49.4	206°03.9	52.0	261°56.0	27.4	232°29.7	27.7	304°54.3	00.4					Schedar	349°31.7	56°40.0	
3	310°51.8	221°03.0	•• 52.1	276°56.7	•• 28.0	247°31.6	•• 27.8	319°56.7	•• 00.4					Diphda	348°47.9	-17°51.1	
4	325°54.3	236°02.1	52.2	291°57.4	28.7	262°33.5	27.9	334°59.2	00.4					Achernar	335°20.8	-57°06.5	
5	340°56.7	251°01.2	52.3	306°58.1	29.3	277°35.4	28.0	350°01.6	00.4					Hamal	327°52.0	23°34.6	
6	355°59.2	266°00.3	N23°52.4	321°58.7	N12°29.9	292°37.2	N20°28.1	5°04.0	S06°00.4					Polaris	314°38.7	89°21.8	
7	11°01.7	280°59.4	52.5	336°59.4	30.5	307°39.1	28.2	20°06.4	00.4					Acamar	315°12.5	-40°12.2	
8	26°04.1	295°58.6	52.6	352°00.1	31.1	322°41.0	28.3	35°08.8	00.4					Menkar	314°06.9	4°11.1	
9	41°06.6	310°57.7	•• 52.7	7°00.8	•• 31.8	337°42.8	•• 28.4	50°11.2	•• 00.4					Mirfak	308°29.4	49°56.7	
10	56°09.1	325°56.8	52.8	22°01.5	32.4	352°44.7	28.5	65°13.6	00.4					Aldebaran	290°40.5	16°33.5	
11	71°11.5	340°55.9	52.9	37°02.2	33.0	7°46.6	28.6	80°16.0	00.3					Rigel	281°04.7	-8°10.4	
12	86°14.0	355°55.0	N23°53.0	52°02.9	N12°33.6	22°48.4	N20°28.7	95°18.5	S06°00.3					Capella	280°23.1	46°01.3	
13	101°16.5	10°54.1	53.1	67°03.6	34.3	37°50.3	28.8	110°20.9	00.3					Bellatrix	278°23.7	6°22.3	
14	116°18.9	25°53.2	53.1	82°04.3	34.9	52°52.2	28.9	125°23.3	00.3					Elnath	278°02.9	28°37.7	
15	131°21.4	40°52.3	•• 53.2	97°05.0	•• 35.5	67°54.1	•• 29.0	140°25.7	•• 00.3					Alnilam	275°38.6	-1°11.2	
16	146°23.8	55°51.4	53.3	112°05.7	36.1	82°55.9	29.1	155°28.1	00.3					Betelgeuse	270°52.9	7°24.7	
17	161°26.3	70°50.5	53.4	127°06.4	36.7	97°57.8	29.2	170°30.5	00.3					Canopus	263°53.2	-52°42.5	
18	176°28.8	85°49.6	N23°53.5	142°07.1	N12°37.4	112°59.7	N20°29.3	185°32.9	S06°00.3					Sirius	258°27.0	-16°45.0	
19	191°31.2	100°48.7	53.6	157°07.8	38.0	128°01.5	29.4	200°35.3	00.3					Adhara	255°06.6	-29°00.3	
20	206°33.7	115°47.8	53.6	172°08.5	38.6	143°03.4	29.5	215°37.8	00.3					Procyon	244°51.6	5°09.8	
21	221°36.2	130°46.9	•• 53.7	187°09.2	•• 39.2	158°05.3	•• 29.6	230°40.2	•• 00.2					Pollux	243°18.2	27°58.1	
22	236°38.6	145°46.0	53.8	202°09.8	39.9	173°07.1	29.7	245°42.6	00.2					Avior	234°15.5	-59°35.4	
23	251°41.1	160°45.1	53.9	217°10.5	40.5	188°09.0	29.8	260°45.0	00.2					Suhail	222°47.0	-43°32.0	
Mer.pass. 06:16			ν -0.9' d 0.1' m -3.91	ν 0.7' d 0.6' m 1.01	ν 1.9' d 0.1' m -2.01	ν 2.4' d -0.0' m 0.97											

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	266°43.6	175°44.2	N23°53.9	232°11.2	N12°41.1	203°10.9	N20°29.9	275°47.4	S06°00.2		
1	281°46.0	190°43.3	54.0	247°11.9	41.7	218°12.8	30.0	290°49.8	00.2		
2	296°48.5	205°42.4	54.1	262°12.6	42.3	233°14.6	30.1	305°52.3	00.2		
3	311°51.0	220°41.5	•• 54.1	277°13.3	•• 42.9	248°16.5	•• 30.2	320°54.7	•• 00.2		
4	326°53.4	235°40.6	54.2	292°14.0	43.6	263°18.4	30.4	335°57.1	00.2		
5	341°55.9	250°39.7	54.3	307°14.7	44.2	278°20.2	30.5	350°59.5	00.2		
6	356°58.3	265°38.9	N23°54.4	322°15.4	N12°44.8	293°22.1	N20°30.6	6°01.9	S06°00.2		
7	12°00.8	280°38.0	54.4	337°16.1	45.4	308°24.0	30.7	21°04.3	00.2		
8	27°03.3	295°37.1	54.5	352°16.8	46.0	323°25.8	30.8	36°06.7	00.1		
9	42°05.7	310°36.2	•• 54.5	7°17.5	•• 46.7	338°27.7	•• 30.9	51°09.2	•• 00.1		
10	57°08.2	325°35.3	54.6	22°18.2	47.3	353°29.6	31.0	66°11.6	00.1		
11	72°10.7	340°34.4	54.7	37°18.9	47.9	8°31.5	31.1	81°14.0	00.1		
12	87°13.1	355°33.5	N23°54.7	52°19.6	N12°48.5	23°33.3	N20°31.2	96°16.4	S06°00.1		
13	102°15.6	10°32.6	54.8	67°20.2	49.1	38°35.2	31.3	111°18.8	00.1		
14	117°18.1	25°31.7	54.8	82°20.9	49.7	53°37.1	31.4	126°21.2	00.1		
15	132°20.5	40°30.8	•• 54.9	97°21.6	•• 50.4	68°38.9	•• 31.5	141°23.7	•• 00.1		
16	147°23.0	55°29.9	55.0	112°22.3	51.0	83°40.8	31.6	156°26.1	00.1		
17	162°25.5	70°29.0	55.0	127°23.0	51.6	98°42.7	31.7	171°28.5	00.1		
18	177°27.9	85°28.1	N23°55.1	142°23.7	N12°52.2	113°44.6	N20°31.8	186°30.9	S06°00.1		
19	192°30.4	100°27.2	55.1	157°24.4	52.8	128°46.4	31.9	201°33.3	00.0		
20	207°32.8	115°26.3	55.2	172°25.1	53.4	143°48.3	32.0	216°35.7	00.0		
21	222°35.3	130°25.4	•• 55.2	187°25.8	•• 54.1	158°50.2	•• 32.1	231°38.2	•• 00.0		
22	237°37.8	145°24.5	55.3	202°26.5	54.7	173°52.0	32.2	246°40.6	00.0		
23	252°40.2	160°23.6	55.3	217°27.2	55.3	188°53.9	32.3	261°43.0	00.0		
Mer.pass. 06:12			ν -0.9' d 0.1' m -3.91	ν 0.7' d 0.6' m 1.00	ν 1.9' d 0.1' m -2.01	ν 2.4' d -0.0' m 0.96					

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	267°42.7	175°22.7	N23°55.3	232°27.9	N12°55.9	203°55.8	N20°32.4	276°45.4	S06°00.0		
1	282°45.2	190°21.8	55.4	247°28.6	56.5	218°57.7	32.5	291°47.8	00.0		
2	297°47.6	205°20.9	55.4	262°29.3	57.1	233°59.5	32.6	306°50.3	00.0		
3	312°50.1	220°20.0	•• 55.5	277°29.9	•• 57.7	249°01.4	•• 32.7	321°52.7	•• 00.0		
4	327°52.6	235°19.1	55.5	292°30.6	58.3	264°03.3	32.8	336°55.1	00.0		
5	342°55.0	250°18.2	55.5	307°31.3	59.0	279°05.1	32.9	351°57.5	00.0		
6	357°57.5	265°17.3	N23°55.6	322°32.0	N12°59.6	294°07.0	N20°33.0	6°59.9	S06°00.0		
7	13°00.0	280°16.4	55.6	337°32.7	13°00.2	309°08.9	33.1	22°02.4	06°00.0		
8	28°02.4	295°15.5	55.7	352°33.4	00.8	324°10.8	33.2	37°04.8	05°59.9		
9	43°04.9	310°14.6	•• 55.7	7°34.1	•• 01.4	339°12.6	•• 33.3	52°07.2	•• 59.9		
10	58°07.3	325°13.7	55.7	22°34.8	02.0	354°14.5	33.4	67°09.6	59.9		
11	73°09.8	340°12.9	55.8	37°35.5	02.6	9°16.4	33.5	82°12.0	59.9		
12	88°12.3	355°12.0	N23°55.8	52°36.2	N13°03.2	24°18.2	N20°33.6	97°14.5	S05°59.9		
13	103°14.7	10°11.1	55.8	67°36.9	03.9	39°20.1	33.7	112°16.9	59.9		
14	118°17.2	25°10.2	55.8	82°37.6	04.5	54°22.0	33.8	127°19.3	59.9		
15	133°19.7	40°09.3	•• 55.9	97°38.2	•• 05.1	69°23.9	•• 33.9	142°21.7	•• 59.9		
16	148°22.1	55°08.4	55.9	112°38.9	05.7	84°25.7	34.0	157°24.1	59.9		
17	163°24.6	70°07.5	55.9	127°39.6	06.3	99°27.6	34.1	172°26.6	59.9		
18	178°27.1	85°06.6	N23°55.9	142°40.3	N13°06.9	114°29.5	N20°34.2	187°29.0	S05°59.9		
19	193°29.5	100°05.7	56.0	157°41.0	07.5	129°31.4	34.3	202°31.4	59.9		
20	208°32.0	115°04.8	56.0	172°41.7	08.1	144°33.2	34.4	217°33.8	59.9		
21	223°34.4	130°03.9	•• 56.0	187°42.4	•• 08.7	159°35.1	•• 34.5	232°36.2	•• 59.9		
22	238°36.9	145°03.0	56.0	202°43.1	09.3	174°37.0	34.6	247°38.7	59.9		
23	253°39.4	160°02.1	56.0	217°43.8	10.0	189°38.8	34.7	262°41.1	59.8		
Mer.pass. 06:08			ν -0.9' d 0.0' m -3.91	ν 0.7' d 0.6' m 1.00	ν 1.9' d 0.1' m -2.01	ν 2.4' d -0.0' m 0.96					

Jun 17 Mon	SHA	Mer.pass
Venus	270°21.3	12:16
Mars	326°10.2	08:32
Jupiter	296°41.6	10:29
Saturn	9°05.1	05:40

Jun 18 Tue	SHA	Mer.pass
Venus	269°00.7	12:18
Mars	325°27.7	08:31
Jupiter	296°27.3	10:26
Saturn	9°03.9	05:36

Jun 19 Wed	SHA	Mer.pass
Venus	267°40.0	12:19
Mars	324°45.2	08:30
Jupiter	296°13.1	10:23
Saturn	9°02.7	05:32

Horizontal parallax	
Venus:	0.1
Mars:	0.1

Sun				Moon				
Mon	GHA	Dec		GHA	ν	Dec	d	HP
0	179°45.8	N23°23.2		61°27.1	15.8'	S11°24.7	13.3'	54.7'
1	194°45.7	23.3		76°01.9	15.7'	11°38.0	13.3'	54.7'
2	209°45.5	23.3		90°36.6	15.6'	11°51.3	13.3'	54.7'
3	224°45.4	· · 23.4		105°11.2	15.6'	12°04.5	13.2'	54.7'
4	239°45.3	23.5		119°45.7	15.5'	12°17.7	13.2'	54.7'
5	254°45.1	23.5		134°20.2	15.4'	12°30.9	13.1'	54.8'
6	269°45.0	N23°23.6		148°54.7	15.4'	S12°44.0	13.1'	54.8'
7	284°44.8	23.7		163°29.0	15.3'	12°57.1	13.1'	54.8'
8	299°44.7	23.7		178°03.3	15.2'	13°10.2	13.0'	54.8'
9	314°44.6	· · 23.8		192°37.5	15.1'	13°23.2	13.0'	54.8'
10	329°44.4	23.8		207°11.7	15.1'	13°36.2	12.9'	54.9'
11	344°44.3	23.9		221°45.7	15.0'	13°49.1	12.9'	54.9'
12	359°44.2	N23°24.0		236°19.7	14.9'	S14°02.0	12.8'	54.9'
13	14°44.0	24.0		250°53.7	14.8'	14°14.8	12.8'	54.9'
14	29°43.9	24.1		265°27.5	14.8'	14°27.6	12.7'	54.9'
15	44°43.8	· · 24.1		280°01.3	14.7'	14°40.3	12.7'	55.0'
16	59°43.6	24.2		294°35.0	14.6'	14°53.0	12.6'	55.0'
17	74°43.5	24.2		309°08.6	14.5'	15°05.7	12.6'	55.0'
18	89°43.4	N23°24.3		323°42.1	14.4'	S15°18.3	12.5'	55.0'
19	104°43.2	24.3		338°15.5	14.4'	15°30.8	12.5'	55.0'
20	119°43.1	24.4		352°48.9	14.3'	15°43.3	12.4'	55.1'
21	134°43.0	· · 24.4		7°22.2	14.2'	15°55.7	12.4'	55.1'
22	149°42.8	24.5		21°55.4	14.1'	16°08.1	12.3'	55.1'
23	164°42.7	24.5		36°28.5	14.0'	16°20.4	12.3'	55.1'
SD = 15.7' d = 0.1'				SD = 14.9'				
Tue	GHA	Dec		GHA	ν	Dec	d	HP
0	179°42.5	N23°24.6		51°01.5	13.9'	S16°32.7	12.2'	55.1'
1	194°42.4	24.6		65°34.4	13.8'	16°44.9	12.1'	55.2'
2	209°42.3	24.7		80°07.3	13.8'	16°57.0	12.1'	55.2'
3	224°42.1	· · 24.7		94°40.1	13.7'	17°09.1	12.0'	55.2'
4	239°42.0	24.8		109°12.7	13.6'	17°21.1	12.0'	55.2'
5	254°41.9	24.8		123°45.3	13.5'	17°33.0	11.9'	55.2'
6	269°41.7	N23°24.9		138°17.8	13.4'	S17°44.9	11.8'	55.3'
7	284°41.6	24.9		152°50.2	13.3'	17°56.8	11.8'	55.3'
8	299°41.5	25.0		167°22.5	13.2'	18°08.5	11.7'	55.3'
9	314°41.3	· · 25.0		181°54.7	13.1'	18°20.2	11.6'	55.3'
10	329°41.2	25.1		196°26.8	13.0'	18°31.8	11.5'	55.4'
11	344°41.1	25.1		210°58.8	12.9'	18°43.4	11.5'	55.4'
12	359°40.9	N23°25.1		225°30.7	12.8'	S18°54.8	11.4'	55.4'
13	14°40.8	25.2		240°02.5	12.7'	19°06.2	11.3'	55.4'
14	29°40.7	25.2		254°34.2	12.6'	19°17.6	11.3'	55.5'
15	44°40.5	· · 25.3		269°05.8	12.5'	19°28.8	11.2'	55.5'
16	59°40.4	25.3		283°37.4	12.4'	19°40.0	11.1'	55.5'
17	74°40.2	25.3		298°08.8	12.3'	19°51.1	11.0'	55.5'
18	89°40.1	N23°25.4		312°40.1	12.2'	S20°02.1	10.9'	55.5'
19	104°40.0	25.4		327°11.3	12.1'	20°13.0	10.9'	55.6'
20	119°39.8	25.4		341°42.4	12.0'	20°23.9	10.8'	55.6'
21	134°39.7	· · 25.5		356°13.4	11.9'	20°34.7	10.7'	55.6'
22	149°39.6	25.5		10°44.3	11.8'	20°45.4	10.6'	55.6'
23	164°39.4	25.5		25°15.1	11.7'	20°56.0	10.5'	55.7'
SD = 15.7' d = 0.0'				SD = 15.0'				
Wed	GHA	Dec		GHA	ν	Dec	d	HP
0	179°39.3	N23°25.6		39°45.8	11.6'	S21°06.5	10.4'	55.7'
1	194°39.2	25.6		54°16.4	11.5'	21°16.9	10.3'	55.7'
2	209°39.0	25.6		68°46.9	11.4'	21°27.2	10.2'	55.7'
3	224°38.9	· · 25.7		83°17.2	11.3'	21°37.4	10.1'	55.8'
4	239°38.8	25.7		97°47.5	11.2'	21°47.6	10.1'	55.8'
5	254°38.6	25.7		112°17.6	11.0'	21°57.6	10.0'	55.8'
6	269°38.5	N23°25.8		126°47.7	10.9'	S22°07.6	9.9'	55.8'
7	284°38.3	25.8		141°17.6	10.8'	22°17.5	9.8'	55.9'
8	299°38.2	25.8		155°47.5	10.7'	22°27.2	9.7'	55.9'
9	314°38.1	· · 25.8		170°17.2	10.6'	22°36.9	9.6'	55.9'
10	329°37.9	25.9		184°46.8	10.5'	22°46.4	9.5'	55.9'
11	344°37.8	25.9		199°16.3	10.4'	22°55.9	9.3'	56.0'
12	359°37.7	N23°25.9		213°45.7	10.3'	S23°05.2	9.2'	56.0'
13	14°37.5	25.9		228°15.0	10.2'	23°14.5	9.1'	56.0'
14	29°37.4	26.0		242°44.2	10.1'	23°23.6	9.0'	56.1'
15	44°37.3	· · 26.0		257°13.2	10.0'	23°32.6	8.9'	56.1'
16	59°37.1	26.0		271°42.2	9.8'	23°41.5	8.8'	56.1'
17	74°37.0	26.0		286°11.0	9.7'	23°50.3	8.7'	56.1'
18	89°36.9	N23°26.0		300°39.8	9.6'	S23°59.0	8.6'	56.2'
19	104°36.7	26.1		315°08.4	9.5'	24°07.6	8.5'	56.2'
20	119°36.6	26.1		329°36.9	9.4'	24°16.1	8.3'	56.2'
21	134°36.4	· · 26.1		344°05.3	9.3'	24°24.4	8.2'	56.2'
22	149°36.3	26.1		358°33.6	9.2'	24°32.6	8.1'	56.3'
23	164°36.2	26.1		13°01.8	9.1'	24°40.7	8.0'	56.3'
SD = 15.7' d = 0.0'				SD = 15.2'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°						
66°						
64°	—:—	—:—	01:31	22:32	—:—	—:—
62°	—:—	—:—	02:09	21:54	—:—	—:—
60°	—:—	00:50	02:36	21:27	23:13	—:—
N 58°	—:—	01:40	02:56	21:07	22:22	—:—
56°	—:—	02:10	03:13	20:50	21:52	—:—
54°	00:46	02:33	03:27	20:36	21:30	23:18
52°	01:32	02:51	03:39	20:23	21:12	22:31
50°	02:00	03:06	03:50	20:12	20:57	22:03
45°	02:46	03:35	04:13	19:50	20:27	21:17
N 40°	03:16	03:58	04:31	19:32	20:05	20:46
35°	03:39	04:16	04:46	19:17	19:46	20:23
30°	03:58	04:32	04:59	19:04	19:31	20:04
20°	04:27	04:56	05:21	18:42	19:06	19:35
N 10°	04:50	05:17	05:40	18:23	18:46	19:13
0°	05:09	05:35	05:58	18:05	18:27	18:54
S 10°	05:26	05:52	06:15	17:47	18:10	18:37
20°	05:42	06:10	06:34	17:29	17:53	18:20
30°	05:59	06:28	06:55	17:08	17:34	18:04
35°	06:07	06:39	07:07	16:55	17:23	17:55
40°	06:17	06:51	07:21	16:41	17:12	17:46
45°	06:27	07:04	07:38	16:24	16:58	17:35
S 50°	06:39	07:20	07:59	16:04	16:42	17:24
52°	06:44	07:28	08:09	15:54	16:35	17:18
54°	06:50	07:36	08:20	15:42	16:26	17:13
56°	06:56	07:45	08:33	15:30	16:17	17:06
58°	07:03	07:56	08:47	15:15	16:07	17:00
S 60°	07:10	08:07	09:05	14:58	15:55	16:52

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	18:19			22:16		
N 70°	17:40			22:57		
68°	17:13	19:42		23:26	22:32	
66°	16:53	18:56		00:05 23:47	23:19	
64°	16:37	18:26	20:39	00:15	00:05 23:50	23:22
62°	16:24	18:04	19:55	00:24	00:19	00:13
60°	16:13	17:46	19:26	00:32	00:31	00:32
N 58°	16:03	17:32	19:04	00:38	00:42	00:47
56°	15:55	17:19	18:46	00:44	00:51	01:01
54°	15:48	17:08	18:31	00:49	00:59	01:12
52°	15:41	16:58	18:18	00:54	01:06	01:23
50°	15:35	16:50	18:07	00:58	01:13	01:32
45°	15:22	16:31	17:43	01:08	01:27	01:51
N 40°	15:12	16:17	17:24	01:16	01:39	02:07
35°	15:03	16:04	17:08	01:23	01:49	02:21
30°	14:55	15:53	16:55	01:29	01:58	02:33
20°	14:42	15:35	16:32	01:39	02:14	02:53
N 10°	14:30	15:19	16:12	01:48	02:28	03:11
0°	14:19	15:04	15:53	01:57	02:40	03:27
S 10°	14:08	14:49	15:35	02:06	02:53	03:44
20°	13:57	14:34	15:16	02:15	03:07	04:02
30°	13:44	14:16	14:53	02:26	03:23	04:23
35°	13:37	14:06	14:40	02:32	03:32	04:35
40°	13:28	13:54	14:25	02:39	03:43	04:49
45°	13:18	13:40	14:07	02:48	03:55	05:05
S 50°	13:07	13:23	13:46	02:58	04:10	05:26
52°	13:01	13:16	13:35	03:02	04:18	05:36
54°	12:55	13:07	13:23	03:07	04:26	05:47
56°	12:49	12:57	13:10	03:13	04:35	06:00
58°	12:41	12:46	12:55	03:20	04:45	06:15
S 60°	12:33	12:33	12:36	03:27	04:57	06:33

Day	Sun			Moon		
	Eqn.of Time 00 ^h mm:ss	Time 12 ^h mm:ss	Mer. Pass hh:mm	Mer.Pass. Upper hh:mm	Lower hh:mm	Age 11-13 75-90%
17	00:57	01:03	12:01	20:30	08:08	
18	01:10	01:16	12:01	21:16	08:52	
19	01:23	01:29				

June 20, 21, 22 UT (Thu., Fri., Sat.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	268°41.8	175°01.2	N23°56.0	232°44.5	N13°10.6	204°40.7	N20°34.8	277°43.5	S05°59.8	Alpheratz	357°35.3	29°13.3
1	283°44.3	190°00.3	56.1	247°45.2	11.2	219°42.6	34.9	292°45.9	59.8	Ankaa	353°07.6	-42°10.2
2	298°46.8	204°59.4	56.1	262°45.9	11.8	234°44.5	35.0	307°48.4	59.8	Schedar	349°31.7	56°40.0
3	313°49.2	219°58.5	· · 56.1	277°46.5	· · 12.4	249°46.3	· · 35.1	322°50.8	· · 59.8	Diphda	348°47.8	-17°51.1
4	328°51.7	234°57.6	56.1	292°47.2	13.0	264°48.2	35.2	337°53.2	59.8	Achernar	335°20.8	-57°06.5
5	343°54.2	249°56.7	56.1	307°47.9	13.6	279°50.1	35.3	352°55.6	59.8	Hamal	327°51.9	23°34.6
6	358°56.6	264°55.8	N23°56.1	322°48.6	N13°14.2	294°52.0	N20°35.4	7°58.1	S05°59.8	Polaris	314°37.3	89°21.8
7	13°59.1	279°54.9	56.1	337°49.3	14.8	309°53.8	35.5	23°00.5	59.8	Acamar	315°12.4	-40°12.2
8	29°01.6	294°54.0	56.1	352°50.0	15.4	324°55.7	35.6	38°02.9	59.8	Menkar	314°06.9	4°11.1
9	44°04.0	309°53.1	· · 56.1	7°50.7	· · 16.0	339°57.6	· · 35.7	53°05.3	· · 59.8	Mirfak	308°29.3	49°56.7
10	59°06.5	324°52.2	56.1	22°51.4	16.6	354°59.5	35.8	68°07.7	59.8	Aldebaran	290°40.5	16°33.5
11	74°08.9	339°51.3	56.1	37°52.1	17.2	10°01.3	35.9	83°10.2	59.8	Rigel	281°04.7	-8°10.4
12	89°11.4	354°50.4	N23°56.1	52°52.8	N13°17.8	25°03.2	N20°36.0	98°12.6	S05°59.8	Capella	280°23.1	46°01.3
13	104°13.9	9°49.5	56.1	67°53.5	18.4	40°05.1	36.1	113°15.0	59.8	Bellatrix	278°23.7	6°22.3
14	119°16.3	24°48.6	56.1	82°54.2	19.1	55°06.9	36.2	128°17.4	59.8	Elnath	278°02.8	28°37.7
15	134°18.8	39°47.7	· · 56.1	97°54.8	· · 19.7	70°08.8	· · 36.3	143°19.9	· · 59.8	Alnilam	275°38.5	-1°11.2
16	149°21.3	54°46.8	56.1	112°55.5	20.3	85°10.7	36.4	158°22.3	59.7	Betelgeuse	270°52.9	7°24.7
17	164°23.7	69°45.9	56.1	127°56.2	20.9	100°12.6	36.5	173°24.7	59.7	Canopus	263°53.2	-52°42.5
18	179°26.2	84°45.0	N23°56.1	142°56.9	N13°21.5	115°14.4	N20°36.6	188°27.1	S05°59.7	Sirius	258°27.0	-16°45.0
19	194°28.7	99°44.1	56.1	157°57.6	22.1	130°16.3	36.7	203°29.6	59.7	Adhara	255°06.6	-29°00.3
20	209°31.1	114°43.2	56.1	172°58.3	22.7	145°18.2	36.8	218°32.0	59.7	Procyon	244°51.6	5°09.8
21	224°33.6	129°42.3	· · 56.1	187°59.0	· · 23.3	160°20.1	· · 36.9	233°34.4	· · 59.7	Pollux	243°18.2	27°58.1
22	239°36.1	144°41.4	56.1	202°59.7	23.9	175°21.9	36.9	248°36.8	59.7	Avior	234°15.5	-59°35.4
23	254°38.5	159°40.6	56.0	218°00.4	24.5	190°23.8	37.0	263°39.3	59.7	Suhail	222°47.0	-43°32.0
Mer.pass. 06:04		ν -0.9' d 0.0' m -3.91		ν 0.7' d 0.6' m 1.00		ν 1.9' d 0.1' m -2.01		ν 2.4' d -0.0' m 0.95		Miaplacidus	221°39.1	-69°49.2
										Alphard	217°48.4	-8°45.9
										Regulus	207°35.1	11°50.9
										Dubhe	193°41.7	61°37.5
										Denebola	182°25.4	14°26.2
										Gienah	175°44.1	-17°40.7
										Acrux	173°00.6	-63°14.4
										Gacrux	171°52.1	-57°15.3
										Alioth	166°13.2	55°49.9
										Spica	158°22.7	-11°17.4
										Alkaid	152°52.1	49°11.7
										Hadar	148°36.3	-60°29.7
										Menkent	147°58.0	-36°29.6
										Arcturus	145°48.1	19°03.4
										Rigel Kent.	139°40.5	-60°56.4
										Kochab	137°18.9	74°03.5
										Zuben'ubi	136°56.3	-16°08.7
										Alphecca	126°03.8	26°38.0
										Antares	112°16.1	-26°29.2
										Atria	107°10.2	-69°04.4
										Sabik	102°03.0	-15°45.3
										Shaula	96°10.6	-37°07.3
										Rasalhague	95°58.6	12°32.5
										Eltanin	90°41.9	51°29.1
										Kaus Aust.	83°32.7	-34°22.4
										Vega	80°33.1	38°48.3
										Nunki	75°48.0	-26°15.9
										Altair	62°00.1	8°55.9
										Peacock	53°06.0	-56°39.2
										Deneb	49°25.8	45°21.8
										Enif	33°39.1	9°59.1
										Al Na'ir	27°33.3	-46°50.4
										Fomalhaut	15°14.9	-29°29.4
										Scheat	13°45.6	28°12.7
										Markab	13°30.3	15°20.1
Jun 20 Thu		SHA		Mer.pass								
Venus		266°19.3		12:21								
Mars		324°02.6		08:29								
Jupiter		295°58.9		10:20								
Saturn		9°01.7		05:28								
Jun 21 Fri		SHA		Mer.pass								
Venus		264°58.7		12:22								
Mars		323°20.1		08:28								
Jupiter		295°44.7		10:17								
Saturn		9°00.7		05:24								
Jun 22 Sat		SHA		Mer.pass								
Venus		263°38.0		12:24								
Mars		322°37.5		08:26								
Jupiter		295°30.6		10:14								
Saturn		8°59.8		05:20								
Horizontal parallax												
Venus: 0.1												
Mars: 0.1												

Sun				Moon				
Thu	GHA	Dec		GHA	ν	Dec	d	HP
0	179°36.0	N23°26.1		27°29.9	9.0'	S24°48.7	7.9'	56.3'
1	194°35.9	26.2		41°57.8	8.9'	24°56.5	7.7'	56.3'
2	209°35.8	26.2		56°25.7	8.8'	25°04.3	7.6'	56.4'
3	224°35.6	· · 26.2		70°53.5	8.6'	25°11.9	7.5'	56.4'
4	239°35.5	26.2		85°21.1	8.5'	25°19.4	7.3'	56.4'
5	254°35.4	26.2		99°48.6	8.4'	25°26.7	7.2'	56.5'
6	269°35.2	N23°26.2		114°16.1	8.3'	S25°33.9	7.1'	56.5'
7	284°35.1	26.2		128°43.4	8.2'	25°41.0	6.9'	56.5'
8	299°35.0	26.2		143°10.6	8.1'	25°47.9	6.8'	56.5'
9	314°34.8	· · 26.2		157°37.8	8.0'	25°54.7	6.7'	56.6'
10	329°34.7	26.3		172°04.8	7.9'	26°01.4	6.5'	56.6'
11	344°34.6	26.3		186°31.7	7.8'	26°08.0	6.4'	56.6'
12	359°34.4	N23°26.3		200°58.5	7.7'	S26°14.3	6.3'	56.6'
13	14°34.3	26.3		215°25.2	7.6'	26°20.6	6.1'	56.7'
14	29°34.1	26.3		229°51.8	7.5'	26°26.7	6.0'	56.7'
15	44°34.0	· · 26.3		244°18.4	7.4'	26°32.7	5.8'	56.7'
16	59°33.9	26.3		258°44.8	7.3'	26°38.5	5.7'	56.8'
17	74°33.7	26.3		273°11.1	7.2'	26°44.2	5.5'	56.8'
18	89°33.6	N23°26.3		287°37.3	7.1'	S26°49.7	5.4'	56.8'
19	104°33.5	26.3		302°03.5	7.0'	26°55.1	5.2'	56.8'
20	119°33.3	26.3		316°29.5	6.9'	27°00.3	5.1'	56.9'
21	134°33.2	· · 26.3		330°55.4	6.9'	27°05.4	4.9'	56.9'
22	149°33.1	26.3		345°21.3	6.8'	27°10.3	4.8'	56.9'
23	164°32.9	26.3		359°47.1	6.7'	27°15.1	4.6'	56.9'
SD = 15.7'			d = 0.0'	SD = 15.4'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	179°32.8	N23°26.3	14°12.7	6.6'	S27°19.7	4.5'	57.0'
1	194°32.7	26.3	28°38.3	6.5'	27°24.2	4.3'	57.0'
2	209°32.5	26.3	43°03.8	6.4'	27°28.5	4.1'	57.0'
3	224°32.4	· · 26.3	57°29.3	6.3'	27°32.6	4.0'	57.1'
4	239°32.3	26.3	71°54.6	6.3'	27°36.6	3.8'	57.1'
5	254°32.1	26.3	86°19.9	6.2'	27°40.4	3.7'	57.1'
6	269°32.0	N23°26.3	100°45.1	6.1'	S27°44.0	3.5'	57.1'
7	284°31.8	26.3	115°10.2	6.0'	27°47.5	3.3'	57.2'
8	299°31.7	26.2	129°35.2	6.0'	27°50.9	3.2'	57.2'
9	314°31.6	· · 26.2	144°00.2	5.9'	27°54.0	3.0'	57.2'
10	329°31.4	26.2	158°25.1	5.8'	27°57.0	2.8'	57.2'
11	344°31.3	26.2	172°49.9	5.8'	27°59.8	2.7'	57.3'
12	359°31.2	N23°26.2	187°14.6	5.7'	S28°02.5	2.5'	57.3'
13	14°31.0	26.2	201°39.3	5.6'	28°05.0	2.3'	57.3'
14	29°30.9	26.2	216°04.0	5.6'	28°07.3	2.1'	57.3'
15	44°30.8	· · 26.2	230°28.5	5.5'	28°09.4	2.0'	57.4'
16	59°30.6	26.2	244°53.0	5.4'	28°11.4	1.8'	57.4'
17	74°30.5	26.1	259°17.5	5.4'	28°13.2	1.6'	57.4'
18	89°30.4	N23°26.1	273°41.9	5.3'	S28°14.8	1.4'	57.4'
19	104°30.2	26.1	288°06.2	5.3'	28°16.3	1.3'	57.5'
20	119°30.1	26.1	302°30.5	5.2'	28°17.6	1.1'	57.5'
21	134°30.0	· · 26.1	316°54.7	5.2'	28°18.6	0.9'	57.5'
22	149°29.8	26.1	331°18.9	5.2'	28°19.6	0.7'	57.5'
23	164°29.7	26.0	345°43.1	5.1'	28°20.3	0.6'	57.6'
SD = 15.7'			d = -0.0'	SD = 15.5'			

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	179°29.6	N23°26.0	0°07.2	5.1'	S28°20.9	0.4'	57.6'
1	194°29.4	26.0	14°31.3	5.0'	28°21.3	0.2'	57.6'
2	209°29.3	26.0	28°55.3	5.0'	28°21.5	0.0'	57.6'
3	224°29.2	· · 26.0	43°19.3	5.0'	28°21.5	-0.2'	57.7'
4	239°29.0	25.9	57°43.3	4.9'	28°21.3	-0.3'	57.7'
5	254°28.9	25.9	72°07.2	4.9'	28°21.0	-0.5'	57.7'
6	269°28.8	N23°25.9	86°31.1	4.9'	S28°20.5	-0.7'	57.7'
7	284°28.6	25.9	100°55.0	4.9'	28°19.8	-0.9'	57.8'
8	299°28.5	25.8	115°18.9	4.8'	28°18.9	-1.1'	57.8'
9	314°28.3	· · 25.8	129°42.7	4.8'	28°17.8	-1.2'	57.8'
10	329°28.2	25.8	144°06.4	4.8'	28°16.6	-1.4'	57.8'
11	344°28.1	25.8	158°30.4	4.8'	28°15.1	-1.6'	57.9'
12	359°27.9	N23°25.7	172°54.2	4.8'	S28°13.5	-1.8'	57.9'
13	14°27.8	25.7	187°18.0	4.8'	28°11.7	-2.0'	57.9'
14	29°27.7	25.7	201°41.8	4.8'	28°09.7	-2.2'	57.9'
15	44°27.5	· · 25.7	216°05.5	4.8'	28°07.6	-2.3'	58.0'
16	59°27.4	25.6	230°29.3	4.8'	28°05.2	-2.5'	58.0'
17	74°27.3	25.6	244°53.1	4.8'	28°02.7	-2.7'	58.0'
18	89°27.1	N23°25.6	259°16.9	4.8'	S28°00.0	-2.9'	58.0'
19	104°27.0	25.5	273°40.6	4.8'	27°57.1	-3.1'	58.0'
20	119°26.9	25.5	288°04.4	4.8'	27°54.0	-3.3'	58.1'
21	134°26.7	· · 25.5	302°28.2	4.8'	27°50.8	-3.4'	58.1'
22	149°26.6	25.4	316°52.0	4.8'	27°47.3	-3.6'	58.1'
23	164°26.5	25.4	331°15.9	4.8'	27°43.7	-3.8'	58.1'
SD = 15.7'			d = -0.0'	SD = 15.7'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°						
66°						
64°	—:—	—:—	01:31	22:33	—:—	—:—
62°	—:—	—:—	02:09	21:54	—:—	—:—
60°	—:—	00:49	02:36	21:28	23:14	—:—
N 58°	—:—	01:41	02:56	21:07	22:23	—:—
56°	—:—	02:11	03:13	20:51	21:53	—:—
54°	00:45	02:33	03:28	20:36	21:31	23:19
52°	01:32	02:51	03:40	20:24	21:13	22:31
50°	02:00	03:06	03:51	20:13	20:58	22:03
45°	02:46	03:36	04:13	19:50	20:28	21:18
N 40°	03:17	03:59	04:31	19:32	20:05	20:47
35°	03:40	04:17	04:47	19:17	19:47	20:24
30°	03:59	04:32	05:00	19:04	19:32	20:05
20°	04:28	04:57	05:22	18:42	19:07	19:36
N 10°	04:50	05:18	05:41	18:23	18:46	19:13
0°	05:09	05:36	05:58	18:06	18:28	18:54
S 10°	05:27	05:53	06:16	17:48	18:11	18:37
20°	05:43	06:10	06:34	17:30	17:54	18:21
30°	05:59	06:29	06:56	17:08	17:35	18:05
35°	06:08	06:40	07:08	16:56	17:24	17:56
40°	06:17	06:52	07:22	16:42	17:12	17:46
45°	06:28	07:05	07:39	16:25	16:59	17:36
S 50°	06:40	07:21	08:00	16:04	16:43	17:24
52°	06:45	07:29	08:10	15:54	16:35	17:19
54°	06:51	07:37	08:21	15:43	16:27	17:13
56°	06:57	07:46	08:34	15:30	16:18	17:07
58°	07:04	07:56	08:48	15:16	16:08	17:00
S 60°	07:11	08:08	09:06	14:58	15:56	16:53

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°						
N 70°						
68°						
66°						
64°						
62°	22:03			00:06 23:52		
60°	21:09	22:36	23:21	00:36	00:47	01:23
N 58°	20:37	21:55	22:46	00:58	01:20	02:03
56°	20:13	21:27	22:20	01:17	01:44	02:31
54°	19:53	21:06	22:00	01:32	02:04	02:53
52°	19:37	20:48	21:43	01:46	02:20	03:11
50°	19:23	20:32	21:28	01:58	02:34	03:26
45°	18:54	20:01	20:59	02:22	03:03	03:57
N 40°	18:32	19:37	20:35	02:42	03:26	04:21
35°	18:14	19:18	20:16	02:59	03:45	04:41
30°	17:58	19:01	20:00	03:13	04:01	04:58
20°	17:31	18:32	19:32	03:38	04:29	05:27
N 10°	17:09	18:08	19:08	03:59	04:53	05:51
0°	16:47	17:45	18:45	04:19	05:15	06:14
S 10°	16:26	17:23	18:23	04:39	05:37	06:37
20°	16:04	16:58	17:59	05:00	06:01	07:01
30°	15:38	16:30	17:31	05:25	06:28	07:29
35°	15:22	16:14	17:14	05:40	06:45	07:46
40°	15:05	15:54	16:55	05:57	07:04	08:06
45°	14:43	15:31	16:31	06:17	07:27	08:29
S 50°	14:17	15:01	16:01	06:43	07:56	09:00
52°	14:04	14:46	15:46	06:56	08:11	09:15
54°	13:49	14:29	15:29	07:10	08:28	09:32
56°	13:31	14:08	15:08	07:27	08:49	09:53
58°	13:10	13:42	14:42	07:48	09:14	10:20
S 60°	12:44	13:06	14:05	08:14	09:50	10:57

Day	Sun		Mer. Pass hh:mm	Moon		Age 14-16 95-100%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
20	01:36	01:42	12:02	23:01	10:33	
21	01:49	01:55	12:02	23:59	11:30	
22	02:02	02:08	12:02	—:—	12:30	

June 23, 24, 25 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	271°39.3	173°56.6	N23°53.9	233°34.2	N13°53.7	206°55.7	N20°41.8	280°38.3	S05°59.6	Alpheratz	357°35.2	29°13.4
1	286°41.7	188°55.7	53.8	248°34.8	54.3	221°57.6	41.9	295°40.8	59.6	Ankaa	353°07.6	-42°10.2
2	301°44.2	203°54.8	53.7	263°35.5	54.9	236°59.5	42.0	310°43.2	59.6	Schedar	349°31.7	56°40.0
3	316°46.7	218°53.9	· · 53.7	278°36.2	· · 55.5	252°01.4	· · 42.1	325°45.6	· · 59.6	Diphda	348°47.8	-17°51.0
4	331°49.1	233°53.0	53.6	293°36.9	56.1	267°03.2	42.2	340°48.1	59.6	Achernar	335°20.7	-57°06.5
5	346°51.6	248°52.2	53.5	308°37.6	56.7	282°05.1	42.3	355°50.5	59.6	Hamal	327°51.9	23°34.6
6	1°54.1	263°51.3	N23°53.4	323°38.3	N13°57.3	297°07.0	N20°42.4	10°52.9	S05°59.6	Polaris	314°36.0	89°21.8
7	16°56.5	278°50.4	53.3	338°39.0	57.9	312°08.9	42.5	25°55.4	59.6	Acamar	315°12.4	-40°12.2
8	31°59.0	293°49.5	53.3	353°39.7	58.4	327°10.7	42.6	40°57.8	59.6	Menkar	314°06.9	4°11.1
9	47°01.4	308°48.6	· · 53.2	8°40.3	· · 59.0	342°12.6	· · 42.7	56°00.2	· · 59.6	Mirfak	308°29.3	49°56.7
10	62°03.9	323°47.7	53.1	23°41.0	13°59.6	357°14.5	42.8	71°02.7	59.6	Aldebaran	290°40.5	16°33.5
11	77°06.4	338°46.8	53.0	38°41.7	14°00.2	12°16.4	42.9	86°05.1	59.6	Rigel	281°04.6	-8°10.4
12	92°08.8	353°45.9	N23°52.9	53°42.4	N14°00.8	27°18.3	N20°43.0	101°07.6	S05°59.5	Capella	280°23.0	46°01.3
13	107°11.3	8°45.0	52.8	68°43.1	01.4	42°20.1	43.1	116°10.0	59.5	Bellatrix	278°23.7	6°22.3
14	122°13.8	23°44.1	52.7	83°43.8	02.0	57°22.0	43.1	131°12.4	59.5	Elnath	278°02.8	28°37.7
15	137°16.2	38°43.2	· · 52.6	98°44.5	· · 02.6	72°23.9	· · 43.2	146°14.9	· · 59.5	Alnilam	275°38.5	-1°11.1
16	152°18.7	53°42.3	52.5	113°45.2	03.1	87°25.8	43.3	161°17.3	59.5	Betelgeuse	270°52.9	7°24.7
17	167°21.2	68°41.4	52.4	128°45.8	03.7	102°27.7	43.4	176°19.7	59.5	Canopus	263°53.2	-52°42.5
18	182°23.6	83°40.5	N23°52.3	143°46.5	N14°04.3	117°29.5	N20°43.5	191°22.2	S05°59.5	Sirius	258°27.0	-16°45.0
19	197°26.1	98°39.6	52.3	158°47.2	04.9	132°31.4	43.6	206°24.6	59.5	Adhara	255°06.6	-29°00.3
20	212°28.6	113°38.7	52.2	173°47.9	05.5	147°33.3	43.7	221°27.0	59.5	Procyon	244°51.6	5°09.8
21	227°31.0	128°37.8	· · 52.1	188°48.6	· · 06.1	162°35.2	· · 43.8	236°29.5	· · 59.5	Pollux	243°18.2	27°58.1
22	242°33.5	143°37.0	52.0	203°49.3	06.7	177°37.0	43.9	251°31.9	59.5	Avior	234°15.5	-59°35.4
23	257°35.9	158°36.1	51.9	218°50.0	07.2	192°38.9	44.0	266°34.4	59.5	Suhail	222°47.0	-43°32.0
Mer.pass. 05:52		ν -0.9' d -0.1' m -3.91		ν 0.7' d 0.6' m 0.99		ν 1.9' d 0.1' m -2.02		ν 2.4' d -0.0' m 0.94				

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	272°38.4	173°35.2	N23°51.7	233°50.7	N14°07.8	207°40.8	N20°44.1	281°36.8	S05°59.5
1	287°40.9	188°34.3	51.6	248°51.4	08.4	222°42.7	44.2	296°39.2	59.5
2	302°43.3	203°33.4	51.5	263°52.0	09.0	237°44.6	44.3	311°41.7	59.5
3	317°45.8	218°32.5	· · 51.4	278°52.7	· · 09.6	252°46.4	· · 44.4	326°44.1	· · 59.5
4	332°48.3	233°31.6	51.3	293°53.4	10.2	267°48.3	44.5	341°46.5	59.5
5	347°50.7	248°30.7	51.2	308°54.1	10.7	282°50.2	44.6	356°49.0	59.5
6	2°53.2	263°29.8	N23°51.1	323°54.8	N14°11.3	297°52.1	N20°44.7	11°51.4	S05°59.6
7	17°55.7	278°28.9	51.0	338°55.5	11.9	312°54.0	44.8	26°53.9	59.6
8	32°58.1	293°28.0	50.9	353°56.2	12.5	327°55.8	44.9	41°56.3	59.6
9	48°00.6	308°27.1	· · 50.8	8°56.8	· · 13.1	342°57.7	· · 45.0	56°58.7	· · 59.6
10	63°03.0	323°26.2	50.6	23°57.5	13.7	357°59.6	45.0	72°01.2	59.6
11	78°05.5	338°25.3	50.5	38°58.2	14.2	13°01.5	45.1	87°03.6	59.6
12	93°08.0	353°24.5	N23°50.4	53°58.9	N14°14.8	28°03.4	N20°45.2	102°06.1	S05°59.6
13	108°10.4	8°23.6	50.3	68°59.6	15.4	43°05.2	45.3	117°08.5	59.6
14	123°12.9	23°22.7	50.2	84°00.3	16.0	58°07.1	45.4	132°10.9	59.6
15	138°15.4	38°21.8	· · 50.1	99°01.0	· · 16.6	73°09.0	· · 45.5	147°13.4	· · 59.6
16	153°17.8	53°20.9	49.9	114°01.7	17.1	88°10.9	45.6	162°15.8	59.6
17	168°20.3	68°20.0	49.8	129°02.3	17.7	103°12.8	45.7	177°18.3	59.6
18	183°22.8	83°19.1	N23°49.7	144°03.0	N14°18.3	118°14.6	N20°45.8	192°20.7	S05°59.6
19	198°25.2	98°18.2	49.6	159°03.7	18.9	133°16.5	45.9	207°23.1	59.6
20	213°27.7	113°17.3	49.4	174°04.4	19.5	148°18.4	46.0	222°25.6	59.6
21	228°30.2	128°16.4	· · 49.3	189°05.1	· · 20.0	163°20.3	· · 46.1	237°28.0	· · 59.6
22	243°32.6	143°15.5	49.2	204°05.8	20.6	178°22.2	46.2	252°30.5	59.6
23	258°35.1	158°14.6	49.0	219°06.5	21.2	193°24.0	46.3	267°32.9	59.6
Mer.pass. 05:48		ν -0.9' d -0.1' m -3.91		ν 0.7' d 0.6' m 0.99		ν 1.9' d 0.1' m -2.02		ν 2.4' d 0.0' m 0.94	

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	273°37.5	173°13.8	N23°48.9	234°07.1	N14°21.8	208°25.9	N20°46.4	282°35.3	S05°59.6
1	288°40.0	188°12.9	48.8	249°07.8	22.4	223°27.8	46.5	297°37.8	59.6
2	303°42.5	203°12.0	48.6	264°08.5	22.9	238°29.7	46.6	312°40.2	59.6
3	318°44.9	218°11.1	· · 48.5	279°09.2	· · 23.5	253°31.6	· · 46.6	327°42.7	· · 59.6
4	333°47.4	233°10.2	48.4	294°09.9	24.1	268°33.4	46.7	342°45.1	59.6
5	348°49.9	248°09.3	48.2	309°10.6	24.7	283°35.3	46.8	357°47.6	59.6
6	3°52.3	263°08.4	N23°48.1	324°11.3	N14°25.3	298°37.2	N20°46.9	12°50.0	S05°59.6
7	18°54.8	278°07.5	47.9	339°12.0	25.8	313°39.1	47.0	27°52.4	59.6
8	33°57.3	293°06.6	47.8	354°12.6	26.4	328°41.0	47.1	42°54.9	59.6
9	48°59.7	308°05.7	· · 47.7	9°13.3	· · 27.0	343°42.8	· · 47.2	57°57.3	· · 59.6
10	64°02.2	323°04.9	47.5	24°14.0	27.6	358°44.7	47.3	72°59.8	59.6
11	79°04.7	338°04.0	47.4	39°14.7	28.1	13°46.6	47.4	88°02.2	59.6
12	94°07.1	353°03.1	N23°47.2	54°15.4	N14°28.7	28°48.5	N20°47.5	103°04.7	S05°59.6
13	109°09.6	8°02.2	47.1	69°16.1	29.3	43°50.4	47.6	118°07.1	59.6
14	124°12.0	23°01.3	46.9	84°16.8	29.9	58°52.2	47.7	133°09.5	59.6
15	139°14.5	38°00.4	· · 46.8	99°17.4	· · 30.4	73°54.1	· · 47.8	148°12.0	· · 59.6
16	154°17.0	52°59.5	46.6	114°18.1	31.0	88°56.0	47.9	163°14.4	59.6
17	169°19.4	67°58.6	46.5	129°18.8	31.6	103°57.9	48.0	178°16.9	59.6
18	184°21.9	82°57.7	N23°46.3	144°19.5	N14°32.2	118°59.8	N20°48.0	193°19.3	S05°59.6
19	199°24.4	97°56.8	46.1	159°20.2	32.7	134°01.7	48.1	208°21.8	59.6
20	214°26.8	112°56.0	46.0	174°20.9	33.3	149°03.5	48.2	223°24.2	59.6
21	229°29.3	127°55.1	· · 45.8	189°21.6	· · 33.9	164°05.4	· · 48.3	238°26.6	· · 59.6
22	244°31.8	142°54.2	45.7	204°22.2	34.4	179°07.3	48.4	253°29.1	59.6
23	259°34.2	157°53.3	45.5	219°22.9	35.0	194°09.2	48.5	268°31.5	59.6
Mer.pass. 05:45		ν -0.9' d -0.1' m -3.91		ν 0.7' d 0.6' m 0.98		ν 1.9' d 0.1' m -2.02		ν 2.4' d 0.0' m 0.93	

Jun 23 Sun		SHA	Mer.pass
Venus		262°17.4	12:25
Mars		321°54.9	08:25
Jupiter		295°16.5	10:11
Saturn		8°59.1	05:17

Jun 24 Mon		SHA	Mer.pass
Venus		260°56.8	12:26
Mars		321°12.3	08:24
Jupiter		295°02.4	10:08
Saturn		8°58.4	05:13

Jun 25 Tue		SHA	Mer.pass
Venus		259°36.2	12:28
Mars		320°29.6	08:23
Jupiter		294°48.4	10:05
Saturn		8°57.8	05:09

Horizontal parallax		
Venus:		0.1
Mars:		0.1

Sun				Moon				
Sun	GHA	Dec		GHA	ν	Dec	d	HP
0	179°26.3	N23°25.4		345°39.7	4.9'	S27°39.9	-4.0'	58.2'
1	194°26.2	25.3		0°03.6	4.9'	27°35.9	-4.2'	58.2'
2	209°26.1	25.3		14°27.4	4.9'	27°31.7	-4.4'	58.2'
3	224°25.9	· · 25.2		28°51.3	4.9'	27°27.4	-4.5'	58.2'
4	239°25.8	25.2		43°15.3	5.0'	27°22.8	-4.7'	58.2'
5	254°25.7	25.2		57°39.2	5.0'	27°18.1	-4.9'	58.3'
6	269°25.5	N23°25.1		72°03.2	5.0'	S27°13.2	-5.1'	58.3'
7	284°25.4	25.1		86°27.2	5.0'	27°08.2	-5.2'	58.3'
8	299°25.3	25.0		100°51.3	5.1'	27°02.9	-5.4'	58.3'
9	314°25.1	· · 25.0		115°15.3	5.1'	26°57.5	-5.6'	58.3'
10	329°25.0	25.0		129°39.5	5.2'	26°51.9	-5.8'	58.4'
11	344°24.9	24.9		144°03.6	5.2'	26°46.1	-6.0'	58.4'
12	359°24.7	N23°24.9		158°27.8	5.2'	S26°40.2	-6.1'	58.4'
13	14°24.6	24.8		172°52.1	5.3'	26°34.0	-6.3'	58.4'
14	29°24.5	24.8		187°16.4	5.3'	26°27.7	-6.5'	58.4'
15	44°24.3	· · 24.7		201°40.7	5.4'	26°21.3	-6.6'	58.5'
16	59°24.2	24.7		216°05.1	5.4'	26°14.6	-6.8'	58.5'
17	74°24.1	24.6		230°29.6	5.5'	26°07.8	-7.0'	58.5'
18	89°23.9	N23°24.6		244°54.1	5.6'	S26°00.8	-7.2'	58.5'
19	104°23.8	24.5		259°18.6	5.6'	25°53.6	-7.3'	58.5'
20	119°23.7	24.5		273°43.2	5.7'	25°46.3	-7.5'	58.6'
21	134°23.5	· · 24.4		288°07.9	5.7'	25°38.8	-7.7'	58.6'
22	149°23.4	24.4		302°32.6	5.8'	25°31.2	-7.8'	58.6'
23	164°23.3	24.3		316°57.4	5.8'	25°23.3	-8.0'	58.6'
SD = 15.7' d = -0.0'				SD = 15.9'				
Mon	GHA	Dec		GHA	ν	Dec	d	HP
0	179°23.1	N23°24.3		331°22.2	5.9'	S25°15.3	-8.2'	58.6'
1	194°23.0	24.2		345°47.2	6.0'	25°07.2	-8.3'	58.6'
2	209°22.9	24.2		0°12.1	6.0'	24°58.9	-8.5'	58.7'
3	224°22.7	· · 24.1		14°37.2	6.1'	24°50.4	-8.6'	58.7'
4	239°22.6	24.0		29°02.3	6.2'	24°41.8	-8.8'	58.7'
5	254°22.5	24.0		43°27.5	6.2'	24°33.0	-9.0'	58.7'
6	269°22.3	N23°23.9		57°52.7	6.3'	S24°24.0	-9.1'	58.7'
7	284°22.2	23.9		72°18.0	6.4'	24°14.9	-9.3'	58.7'
8	299°22.1	23.8		86°43.4	6.5'	24°05.6	-9.4'	58.8'
9	314°21.9	· · 23.8		101°08.9	6.5'	23°56.2	-9.6'	58.8'
10	329°21.8	23.7		115°34.4	6.6'	23°46.7	-9.7'	58.8'
11	344°21.7	23.6		130°00.0	6.7'	23°36.9	-9.9'	58.8'
12	359°21.5	N23°23.6		144°25.7	6.8'	S23°27.1	-10.0'	58.8'
13	14°21.4	23.5		158°51.5	6.8'	23°17.1	-10.2'	58.8'
14	29°21.3	23.4		173°17.3	6.9'	23°06.9	-10.3'	58.8'
15	44°21.1	· · 23.4		187°43.3	7.0'	22°56.6	-10.5'	58.9'
16	59°21.0	23.3		202°09.3	7.1'	22°46.1	-10.6'	58.9'
17	74°20.9	23.3		216°35.3	7.2'	22°35.5	-10.7'	58.9'
18	89°20.7	N23°23.2		231°01.5	7.2'	S22°24.8	-10.9'	58.9'
19	104°20.6	23.1		245°27.7	7.3'	22°13.9	-11.0'	58.9'
20	119°20.5	23.1		259°54.1	7.4'	22°02.9	-11.1'	58.9'
21	134°20.4	· · 23.0		274°20.5	7.5'	21°51.8	-11.3'	58.9'
22	149°20.2	22.9		288°47.0	7.6'	21°40.5	-11.4'	58.9'
23	164°20.1	22.8		303°13.5	7.6'	21°29.1	-11.5'	59.0'
SD = 15.7' d = -0.1'				SD = 16.0'				
Tue	GHA	Dec		GHA	ν	Dec	d	HP
0	179°20.0	N23°22.8		317°40.2	7.7'	S21°17.5	-11.7'	59.0'
1	194°19.8	22.7		332°06.9	7.8'	21°05.9	-11.8'	59.0'
2	209°19.7	22.6		346°33.7	7.9'	20°54.0	-11.9'	59.0'
3	224°19.6	· · 22.6		1°00.6	8.0'	20°42.1	-12.1'	59.0'
4	239°19.4	22.5		15°27.6	8.1'	20°30.1	-12.2'	59.0'
5	254°19.3	22.4		29°54.7	8.1'	20°17.9	-12.3'	59.0'
6	269°19.2	N23°22.3		44°21.8	8.2'	S20°05.6	-12.4'	59.0'
7	284°19.0	22.3		58°49.0	8.3'	19°53.1	-12.5'	59.0'
8	299°18.9	22.2		73°16.3	8.4'	19°40.6	-12.7'	59.1'
9	314°18.8	· · 22.1		87°43.7	8.5'	19°27.9	-12.8'	59.1'
10	329°18.6	22.0		102°11.2	8.6'	19°15.2	-12.9'	59.1'
11	344°18.5	22.0		116°38.8	8.6'	19°02.3	-13.0'	59.1'
12	359°18.4	N23°21.9		131°06.4	8.7'	S18°49.3	-13.1'	59.1'
13	14°18.2	21.8		145°34.1	8.8'	18°36.2	-13.2'	59.1'
14	29°18.1	21.7		160°01.9	8.9'	18°22.9	-13.3'	59.1'
15	44°18.0	· · 21.6		174°29.8	9.0'	18°09.6	-13.4'	59.1'
16	59°17.9	21.5		188°57.8	9.0'	17°56.2	-13.5'	59.1'
17	74°17.7	21.5		203°25.8	9.1'	17°42.6	-13.6'	59.1'
18	89°17.6	N23°21.4		217°54.0	9.2'	S17°29.0	-13.7'	59.1'
19	104°17.5	21.3		232°22.2	9.3'	17°15.2	-13.8'	59.2'
20	119°17.3	21.2		246°50.4	9.4'	17°01.4	-13.9'	59.2'
21	134°17.2	· · 21.1		261°18.8	9.4'	16°47.4	-14.0'	59.2'
22	149°17.1	21.0		275°47.2	9.5'	16°33.4	-14.1'	59.2'
23	164°16.9	21.0		290°15.8	9.6'	16°19.3	-14.2'	59.2'
SD = 15.7' d = -0.1'				SD = 16.1'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°						
66°						
64°	--	--	01:33	22:32	--	--
62°	--	--	02:11	21:54	--	--
60°	--	00:51	02:37	21:28	23:13	--
N 58°	--	01:42	02:57	21:07	22:23	--
56°	--	02:12	03:14	20:51	21:53	--
54°	00:47	02:34	03:28	20:36	21:31	23:17
52°	01:34	02:52	03:41	20:24	21:13	22:31
50°	02:01	03:07	03:52	20:13	20:58	22:03
45°	02:47	03:37	04:14	19:51	20:28	21:18
N 40°	03:18	03:59	04:32	19:33	20:06	20:47
35°	03:41	04:18	04:47	19:18	19:47	20:24
30°	03:59	04:33	05:00	19:05	19:32	20:06
20°	04:29	04:58	05:22	18:43	19:07	19:37
N 10°	04:51	05:18	05:41	18:24	18:47	19:14
0°	05:10	05:36	05:59	18:06	18:29	18:55
S 10°	05:27	05:53	06:16	17:49	18:12	18:38
20°	05:43	06:11	06:35	17:30	17:54	18:22
30°	06:00	06:30	06:56	17:09	17:35	18:05
35°	06:09	06:40	07:08	16:57	17:25	17:57
40°	06:18	06:52	07:23	16:43	17:13	17:47
45°	06:28	07:06	07:39	16:26	17:00	17:37
S 50°	06:40	07:22	08:00	16:05	16:44	17:25
52°	06:45	07:29	08:10	15:55	16:36	17:20
54°	06:51	07:37	08:21	15:44	16:28	17:14
56°	06:57	07:46	08:34	15:31	16:19	17:08
58°	07:04	07:57	08:49	15:17	16:09	17:01
S 60°	07:11	08:08	09:06	14:59	15:57	16:54

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°						
N 70°						
68°			01:44			04:26
66°			00:55			05:14
64°		00:58	00:24		03:12	05:45
62°	00:27	00:08	00:00 23:54	01:38	04:02	06:07
60°	23:36	23:41	23:42	02:44	04:33	06:25
N 58°	23:12	23:25	23:32	03:19	04:56	06:40
56°	22:53	23:11	23:23	03:44	05:15	06:53
54°	22:36	23:00	23:15	04:04	05:31	07:04
52°	22:22	22:49	23:08	04:21	05:44	07:14
50°	22:10	22:40	23:02	04:35	05:56	07:23
45°	21:44	22:20	22:48	05:04	06:21	07:41
N 40°	21:24	22:04	22:36	05:27	06:40	07:56
35°	21:07	21:50	22:26	05:46	06:56	08:09
30°	20:52	21:38	22:18	06:02	07:10	08:19
20°	20:27	21:17	22:02	06:29	07:34	08:38
N 10°	20:06	21:00	21:49	06:52	07:54	08:54
0°	19:45	20:43	21:37	07:14	08:13	09:09
S 10°	19:25	20:26	21:24	07:36	08:32</	

June 26, 27, 28 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	274°36.7	172°52.4	N23°45.4	234°23.6	N14°35.6	209°11.1	N20°48.6	283°34.0	S05°59.6	Alpheratz	357°35.2	29°13.4
1	289°39.2	187°51.5	45.2	249°24.3	36.2	224°13.0	48.7	298°36.4	59.6	Ankaa	353°07.6	-42°10.1
2	304°41.6	202°50.6	45.0	264°25.0	36.7	239°14.8	48.8	313°38.9	59.6	Schedar	349°31.6	56°40.0
3	319°44.1	217°49.7	•• 44.9	279°25.7	•• 37.3	254°16.7	•• 48.9	328°41.3	•• 59.6	Diphda	348°47.8	-17°51.0
4	334°46.5	232°48.9	44.7	294°26.4	37.9	269°18.6	49.0	343°43.8	59.7	Achernar	335°20.7	-57°06.5
5	349°49.0	247°48.0	44.5	309°27.0	38.4	284°20.5	49.1	358°46.2	59.7	Hamal	327°51.9	23°34.6
6	4°51.5	262°47.1	N23°44.4	324°27.7	N14°39.0	299°22.4	N20°49.2	13°48.7	S05°59.7	Polaris	314°34.8	89°21.8
7	19°53.9	277°46.2	44.2	339°28.4	39.6	314°24.2	49.2	28°51.1	59.7	Acamar	315°12.4	-40°12.2
8	34°56.4	292°45.3	44.0	354°29.1	40.2	329°26.1	49.3	43°53.5	59.7	Menkar	314°06.8	4°11.1
9	49°58.9	307°44.4	•• 43.8	9°29.8	•• 40.7	344°28.0	•• 49.4	58°56.0	•• 59.7	Mirfak	308°29.3	49°56.7
10	65°01.3	322°43.5	43.7	24°30.5	41.3	359°29.9	49.5	73°58.4	59.7	Aldebaran	290°40.5	16°33.5
11	80°03.8	337°42.7	43.5	39°31.1	41.9	14°31.8	49.6	89°00.9	59.7	Rigel	281°04.6	-8°10.3
12	95°06.3	352°41.8	N23°43.3	54°31.8	N14°42.4	29°33.7	N20°49.7	104°03.3	S05°59.7	Capella	280°23.0	46°01.3
13	110°08.7	7°40.9	43.1	69°32.5	43.0	44°35.5	49.8	119°05.8	59.7	Bellatrix	278°23.7	6°22.3
14	125°11.2	22°40.0	43.0	84°33.2	43.6	59°37.4	49.9	134°08.2	59.7	Elnath	278°02.8	28°37.7
15	140°13.6	37°39.1	•• 42.8	99°33.9	•• 44.1	74°39.3	•• 50.0	149°10.7	•• 59.7	Alnilam	275°38.5	-1°11.1
16	155°16.1	52°38.2	42.6	114°34.6	44.7	89°41.2	50.1	164°13.1	59.7	Betelgeuse	270°52.9	7°24.7
17	170°18.6	67°37.3	42.4	129°35.3	45.3	104°43.1	50.2	179°15.6	59.7	Canopus	263°53.2	-52°42.4
18	185°21.0	82°36.5	N23°42.2	144°35.9	N14°45.8	119°45.0	N20°50.3	194°18.0	S05°59.7	Sirius	258°27.0	-16°44.9
19	200°23.5	97°35.6	42.0	159°36.6	46.4	134°46.8	50.3	209°20.5	59.7	Adhara	255°06.6	-29°00.3
20	215°26.0	112°34.7	41.9	174°37.3	47.0	149°48.7	50.4	224°22.9	59.7	Procyon	244°51.6	5°09.8
21	230°28.4	127°33.8	•• 41.7	189°38.0	•• 47.6	164°50.6	•• 50.5	239°25.4	•• 59.7	Pollux	243°18.2	27°58.1
22	245°30.9	142°32.9	41.5	204°38.7	48.1	179°52.5	50.6	254°27.8	59.7	Avior	234°15.5	-59°35.3
23	260°33.4	157°32.0	41.3	219°39.4	48.7	194°54.4	50.7	269°30.3	59.7	Suhail	222°47.0	-43°32.0
Mer.pass. 05:41		ν -0.9' d -0.2' m -3.91		ν 0.7' d 0.6' m 0.98		ν 1.9' d 0.1' m -2.02		ν 2.4' d 0.0' m 0.93				

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	275°35.8	172°31.1	N23°41.1	234°40.0	N14°49.2	209°56.3	N20°50.8	284°32.7	S05°59.7	Alphard	217°48.4	-8°45.9
1	290°38.3	187°30.3	40.9	249°40.7	49.8	224°58.1	50.9	299°35.2	59.7	Regulus	207°35.1	11°50.9
2	305°40.8	202°29.4	40.7	264°41.4	50.4	240°00.0	51.0	314°37.6	59.8	Dubhe	193°41.8	61°37.4
3	320°43.2	217°28.5	•• 40.5	279°42.1	•• 50.9	255°01.9	•• 51.1	329°40.1	•• 59.8	Denebola	182°25.5	14°26.2
4	335°45.7	232°27.6	40.3	294°42.8	51.5	270°03.8	51.2	344°42.5	59.8	Gienah	175°44.1	-17°40.7
5	350°48.1	247°26.7	40.1	309°43.5	52.1	285°05.7	51.3	359°45.0	59.8	Acrux	173°00.6	-63°14.4
6	5°50.6	262°25.8	N23°39.9	324°44.2	N14°52.6	300°07.6	N20°51.4	14°47.4	S05°59.8	Gacrux	171°52.1	-57°15.3
7	20°53.1	277°25.0	39.7	339°44.8	53.2	315°09.5	51.4	29°49.9	59.8	Alioth	166°13.3	55°49.9
8	35°55.5	292°24.1	39.5	354°45.5	53.8	330°11.3	51.5	44°52.3	59.8	Spica	158°22.7	-11°17.4
9	50°58.0	307°23.2	•• 39.3	9°46.2	•• 54.3	345°13.2	•• 51.6	59°54.8	•• 59.8	Alkaid	152°52.2	49°11.7
10	66°00.5	322°22.3	39.1	24°46.9	54.9	0°15.1	51.7	74°57.2	59.8	Hadar	148°36.4	-60°29.7
11	81°02.9	337°21.4	38.9	39°47.6	55.5	15°17.0	51.8	89°59.7	59.8	Menkent	147°58.0	-36°29.6
12	96°05.4	352°20.5	N23°38.7	54°48.3	N14°56.0	30°18.9	N20°51.9	105°02.1	S05°59.8	Arcturus	145°48.1	19°03.4
13	111°07.9	7°19.7	38.5	69°48.9	56.6	45°20.8	52.0	120°04.6	59.8	Rigil Kent.	139°40.6	-60°56.4
14	126°10.3	22°18.8	38.3	84°49.6	57.1	60°22.6	52.1	135°07.0	59.8	Kochab	137°19.0	74°03.5
15	141°12.8	37°17.9	•• 38.1	99°50.3	•• 57.7	75°24.5	•• 52.2	150°09.5	•• 59.8	Zuben'ubi	136°56.3	-16°08.7
16	156°15.3	52°17.0	37.9	114°51.0	58.3	90°26.4	52.3	165°11.9	59.8	Alphecca	126°03.8	26°38.0
17	171°17.7	67°16.1	37.7	129°51.7	58.8	105°28.3	52.4	180°14.4	59.8	Antares	112°16.1	-26°29.2
18	186°20.2	82°15.3	N23°37.4	144°52.4	N14°59.4	120°30.2	N20°52.4	195°16.8	S05°59.8	Atria	107°10.1	-69°04.4
19	201°22.6	97°14.4	37.2	159°53.0	15°00.0	135°32.1	52.5	210°19.3	59.9	Sabik	102°03.0	-15°45.3
20	216°25.1	112°13.5	37.0	174°53.7	00.5	150°34.0	52.6	225°21.7	59.9	Shaula	96°10.6	-37°07.3
21	231°27.6	127°12.6	•• 36.8	189°54.4	•• 01.1	165°35.8	•• 52.7	240°24.2	•• 59.9	Rasalhague	95°58.6	12°32.5
22	246°30.0	142°11.7	36.6	204°55.1	01.6	180°37.7	52.8	255°26.6	59.9	Eltanin	90°41.8	51°29.1
23	261°32.5	157°10.9	36.4	219°55.8	02.2	195°39.6	52.9	270°29.1	59.9	Kaus Aust.	83°32.7	-34°22.4
Mer.pass. 05:37		ν -0.9' d -0.2' m -3.91		ν 0.7' d 0.6' m 0.98		ν 1.9' d 0.1' m -2.02		ν 2.4' d 0.0' m 0.93				

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	276°35.0	172°10.0	N23°36.1	234°56.5	N15°02.8	210°41.5	N20°53.0	285°31.5	S05°59.9	Nunki	75°47.9	-26°16.0
1	291°37.4	187°09.1	35.9	249°57.1	03.3	225°43.4	53.1	300°34.0	59.9	Altair	62°00.1	8°55.9
2	306°39.9	202°08.2	35.7	264°57.8	03.9	240°45.3	53.2	315°36.4	59.9	Peacock	53°05.9	-56°39.2
3	321°42.4	217°07.3	•• 35.5	279°58.5	•• 04.4	255°47.2	•• 53.3	330°38.9	•• 59.9	Deneb	49°25.7	45°21.9
4	336°44.8	232°06.5	35.2	294°59.2	05.0	270°49.0	53.3	345°41.4	59.9	Enif	33°39.0	9°59.2
5	351°47.3	247°05.6	35.0	309°59.9	05.5	285°50.9	53.4	0°43.8	59.9	AI Na'ir	27°33.2	-46°50.3
6	6°49.7	262°04.7	N23°34.8	325°00.6	N15°06.1	300°52.8	N20°53.5	15°46.3	S05°59.9	Fomalhaut	15°14.9	-29°29.4
7	21°52.2	277°03.8	34.6	340°01.2	06.7	315°54.7	53.6	30°48.7	59.9	Scheat	13°45.5	28°12.8
8	36°54.7	292°02.9	34.3	355°01.9	07.2	330°56.6	53.7	45°51.2	59.9	Markab	13°30.2	15°20.1
9	51°57.1	307°02.1	•• 34.1	10°02.6	•• 07.8	345°58.5	•• 53.8	60°53.6	05°59.9			
10	66°59.6	322°01.2	33.9	25°03.3	08.3	1°00.4	53.9	75°56.1	06°00.0			
11	82°02.1	337°00.3	33.6	40°04.0	08.9	16°02.2	54.0	90°58.5	00.0			
12	97°04.5	351°59.4	N23°33.4	55°04.7	N15°09.4	31°04.1	N20°54.1	106°01.0	S06°00.0			
13	112°07.0	6°58.5	33.2	70°05.3	10.0	46°06.0	54.2	121°03.4	00.0			
14	127°09.5	21°57.7	32.9	85°06.0	10.6	61°07.9	54.2	136°05.9	00.0			
15	142°11.9	36°56.8	•• 32.7	100°06.7	•• 11.1	76°09.8	•• 54.3	151°08.4	•• 00.0			
16	157°14.4	51°55.9	32.4	115°07.4	11.7	91°11.7	54.4	166°10.8	00.0			
17	172°16.9	66°55.0	32.2	130°08.1	12.2	106°13.6	54.5	181°13.3	00.0			
18	187°19.3	81°54.2	N23°32.0	145°08.8	N15°12.8	121°15.5	N20°54.6	196°15.7	S06°00.0			
19	202°21.8	96°53.3	31.7	160°09.4	13.3	136°17.3	54.7	211°18.2	00.0			
20	217°24.2	111°52.4	31.5	175°10.1	13.9	151°19.2	54.8	226°20.6	00.0			
21	232°26.7	126°51.5	•• 31.2	190°10.8	•• 14.4	166°21.1	•• 54.9	241°23.1	•• 00.0			
22	247°29.2	141°50.7	31.0	205°11.5	15.0	181°23.0	55.0	256°25.5	00.1			
23	262°31.6	156°49.8	30.7	220°12.2	15.6	196						

h	Sun				Moon			
Wed	GHA	Dec	GHA	ν	Dec	d	HP	
0	179°16.8	N23°20.9	304°44.4	9.7'	S16°05.0	-14.3'	59.2'	
1	194°16.7	20.8	319°13.0	9.7'	15°50.7	-14.4'	59.2'	
2	209°16.5	20.7	333°41.8	9.8'	15°36.3	-14.5'	59.2'	
3	224°16.4	· · 20.6	348°10.6	9.9'	15°21.8	-14.6'	59.2'	
4	239°16.3	20.5	2°39.5	10.0'	15°07.2	-14.7'	59.2'	
5	254°16.2	20.4	17°08.4	10.0'	14°52.6	-14.7'	59.2'	
6	269°16.0	N23°20.3	31°37.5	10.1'	S14°37.8	-14.8'	59.2'	
7	284°15.9	20.2	46°06.6	10.2'	14°23.0	-14.9'	59.2'	
8	299°15.8	20.1	60°35.8	10.2'	14°08.1	-15.0'	59.2'	
9	314°15.6	· · 20.0	75°05.0	10.3'	13°53.1	-15.1'	59.2'	
10	329°15.5	20.0	89°34.3	10.4'	13°38.0	-15.1'	59.2'	
11	344°15.4	19.9	104°03.7	10.5'	13°22.9	-15.2'	59.3'	
12	359°15.2	N23°19.8	118°33.2	10.5'	S13°07.7	-15.3'	59.3'	
13	14°15.1	19.7	133°02.7	10.6'	12°52.4	-15.4'	59.3'	
14	29°15.0	19.6	147°32.3	10.6'	12°37.1	-15.4'	59.3'	
15	44°14.9	· · 19.5	162°01.9	10.7'	12°21.6	-15.5'	59.3'	
16	59°14.7	19.4	176°31.6	10.8'	12°06.1	-15.6'	59.3'	
17	74°14.6	19.3	191°01.4	10.8'	11°50.6	-15.6'	59.3'	
18	89°14.5	N23°19.2	205°31.2	10.9'	S11°35.0	-15.7'	59.3'	
19	104°14.3	19.1	220°01.1	10.9'	11°19.3	-15.7'	59.3'	
20	119°14.2	19.0	234°31.0	11.0'	11°03.6	-15.8'	59.3'	
21	134°14.1	· · 18.9	249°01.0	11.1'	10°47.8	-15.9'	59.3'	
22	149°13.9	18.8	263°31.1	11.1'	10°31.9	-15.9'	59.3'	
23	164°13.8	18.7	278°01.2	11.2'	10°16.0	-16.0'	59.3'	
SD = 15.7' d = -0.1'								
SD = 16.1'								
Thu	GHA	Dec	GHA	ν	Dec	d	HP	
0	179°13.7	N23°18.6	292°31.4	11.2'	S10°00.0	-16.0'	59.3'	
1	194°13.6	18.4	307°01.6	11.3'	09°44.0	-16.1'	59.3'	
2	209°13.4	18.3	321°31.9	11.3'	09°28.0	-16.1'	59.3'	
3	224°13.3	· · 18.2	336°02.2	11.4'	09°11.8	-16.2'	59.3'	
4	239°13.2	18.1	350°32.6	11.4'	08°55.7	-16.2'	59.3'	
5	254°13.0	18.0	5°03.0	11.5'	08°39.5	-16.3'	59.3'	
6	269°12.9	N23°17.9	19°33.5	11.5'	S08°23.2	-16.3'	59.3'	
7	284°12.8	17.8	34°04.0	11.6'	08°06.9	-16.3'	59.3'	
8	299°12.7	17.7	48°34.5	11.6'	07°50.6	-16.4'	59.3'	
9	314°12.5	· · 17.6	63°05.1	11.6'	07°34.2	-16.4'	59.3'	
10	329°12.4	17.5	77°35.8	11.7'	07°17.8	-16.5'	59.3'	
11	344°12.3	17.4	92°06.4	11.7'	07°01.4	-16.5'	59.3'	
12	359°12.1	N23°17.2	106°37.2	11.8'	S06°44.9	-16.5'	59.3'	
13	14°12.0	17.1	121°07.9	11.8'	06°28.4	-16.6'	59.3'	
14	29°11.9	17.0	135°38.7	11.8'	06°11.8	-16.6'	59.3'	
15	44°11.8	· · 16.9	150°09.5	11.9'	05°55.2	-16.6'	59.3'	
16	59°11.6	16.8	164°40.4	11.9'	05°38.6	-16.6'	59.3'	
17	74°11.5	16.7	179°11.3	11.9'	05°22.0	-16.7'	59.3'	
18	89°11.4	N23°16.5	193°42.2	12.0'	S05°05.3	-16.7'	59.3'	
19	104°11.3	16.4	208°13.2	12.0'	04°48.6	-16.7'	59.3'	
20	119°11.1	16.3	222°44.1	12.0'	04°31.9	-16.7'	59.3'	
21	134°11.0	· · 16.2	237°15.2	12.0'	04°15.2	-16.8'	59.3'	
22	149°10.9	16.1	251°46.2	12.1'	03°58.4	-16.8'	59.3'	
23	164°10.7	15.9	266°17.2	12.1'	03°41.7	-16.8'	59.3'	
SD = 15.7' d = -0.1'								
SD = 16.2'								
Fri	GHA	Dec	GHA	ν	Dec	d	HP	
0	179°10.6	N23°15.8	280°48.3	12.1'	S03°24.9	-16.8'	59.3'	
1	194°10.5	15.7	295°19.4	12.1'	03°08.1	-16.8'	59.3'	
2	209°10.4	15.6	309°50.6	12.1'	02°51.2	-16.8'	59.3'	
3	224°10.2	· · 15.5	324°21.7	12.2'	02°34.4	-16.8'	59.3'	
4	239°10.1	15.3	338°52.9	12.2'	02°17.6	-16.9'	59.3'	
5	254°10.0	15.2	353°24.0	12.2'	02°00.7	-16.9'	59.3'	
6	269°09.9	N23°15.1	7°55.2	12.2'	S01°43.9	-16.9'	59.3'	
7	284°09.7	15.0	22°26.4	12.2'	01°27.0	-16.9'	59.3'	
8	299°09.6	14.8	36°57.6	12.2'	01°10.1	-16.9'	59.3'	
9	314°09.5	· · 14.7	51°28.9	12.2'	00°53.3	-16.9'	59.3'	
10	329°09.3	14.6	66°00.1	12.2'	00°36.4	-16.9'	59.3'	
11	344°09.2	14.4	80°31.3	12.2'	00°19.5	-16.9'	59.3'	
12	359°09.1	N23°14.3	95°02.6	12.3'	S00°02.6	-16.9'	59.3'	
13	14°09.0	14.2	109°33.8	12.3'	N00°14.2	16.9'	59.3'	
14	29°08.8	14.0	124°05.1	12.3'	00°31.1	16.9'	59.3'	
15	44°08.7	· · 13.9	138°36.3	12.3'	00°48.0	16.9'	59.2'	
16	59°08.6	13.8	153°07.6	12.3'	01°04.8	16.9'	59.2'	
17	74°08.5	13.6	167°38.8	12.3'	01°21.7	16.8'	59.2'	
18	89°08.3	N23°13.5	182°10.1	12.3'	N01°38.5	16.8'	59.2'	
19	104°08.2	13.4	196°41.4	12.2'	01°55.4	16.8'	59.2'	
20	119°08.1	13.2	211°12.6	12.2'	02°12.2	16.8'	59.2'	
21	134°08.0	· · 13.1	225°43.8	12.2'	02°29.0	16.8'	59.2'	
22	149°07.8	13.0	240°15.1	12.2'	02°45.8	16.8'	59.2'	
23	164°07.7	12.8	254°46.3	12.2'	03°02.5	16.8'	59.2'	
SD = 15.7' d = -0.1'								
SD = 16.2'								

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°						
66°						
64°	—:—	—:—	01:36	22:30	—:—	—:—
62°	—:—	—:—	02:13	21:53	—:—	—:—
60°	—:—	00:56	02:39	21:27	23:09	—:—
N 58°	—:—	01:44	02:59	21:07	22:21	—:—
56°	—:—	02:14	03:16	20:50	21:52	—:—
54°	00:51	02:36	03:30	20:36	21:30	23:14
52°	01:36	02:54	03:42	20:24	21:13	22:30
50°	02:03	03:08	03:53	20:13	20:58	22:03
45°	02:48	03:38	04:15	19:51	20:28	21:18
N 40°	03:19	04:00	04:33	19:33	20:06	20:47
35°	03:42	04:19	04:48	19:18	19:48	20:24
30°	04:00	04:34	05:01	19:05	19:33	20:06
20°	04:29	04:59	05:23	18:43	19:08	19:37
N 10°	04:52	05:19	05:42	18:24	18:47	19:14
0°	05:11	05:37	06:00	18:07	18:29	18:56
S 10°	05:28	05:54	06:17	17:50	18:12	18:39
20°	05:44	06:11	06:35	17:31	17:55	18:23
30°	06:00	06:30	06:56	17:10	17:36	18:06
35°	06:09	06:41	07:09	16:58	17:26	17:58
40°	06:18	06:52	07:23	16:44	17:14	17:48
45°	06:29	07:06	07:40	16:27	17:01	17:38
S 50°	06:40	07:22	08:00	16:06	16:45	17:26
52°	06:45	07:29	08:10	15:57	16:37	17:21
54°	06:51	07:37	08:21	15:46	16:29	17:15
56°	06:57	07:46	08:34	15:33	16:20	17:09
58°	07:04	07:56	08:48	15:18	16:10	17:03
S 60°	07:11	08:08	09:05	15:01	15:59	16:55

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	01:47	00:42	00:01 23:25	06:19	09:10	11:35
N 70°	01:11	00:27 23:56	23:29	06:53	09:22	11:36
68°	00:44	00:15 23:52	23:32	07:17	09:32	11:36
66°	00:24	00:05 23:49	23:35	07:36	09:40	11:36
64°	00:08 23:56	23:46	23:37	07:51	09:46	11:36
62°	23:49	23:44	23:39	08:03	09:52	11:37
60°	23:42	23:42	23:41	08:14	09:57	11:37
N 58°	23:37	23:40	23:43	08:23	10:01	11:37
56°	23:32	23:38	23:44	08:31	10:05	11:37
54°	23:27	23:37	23:45	08:38	10:08	11:37
52°	23:23	23:35	23:47	08:44	10:12	11:37
50°	23:19	23:34	23:48	08:50	10:14	11:37
45°	23:11	23:31	23:50	09:02	10:20	11:38
N 40°	23:04	23:29	23:52	09:12	10:25	11:38
35°	22:58	23:26	23:54	09:20	10:30	11:38
30°	22:52	23:25	23:56	09:27	10:33	11:38
20°	22:43	23:21	23:59	09:40	10:40	11:38
N 10°	22:35	23:19	· · ·	09:51	10:45	11:38
0°	22:27	23:16	· · ·	10:01	10:51	11:39
S 10°	22:20	23:13	· · ·	10:11	10:56	11:39
20°	22:12	23:11	· · ·	10:22	11:01	11:39
30°	22:02	23:07	· · ·	10:34	11:07	11:39
35°	21:57	23:05	· · ·	10:41	11:11	11:39
40°	21:50	23:03	· · ·	10:49	11:15	11:39
45°	21:43	23:01	· · ·	10:58	11:19	11:39
S 50°	21:34	22:58	· · ·	11:09	11:25	11:39
52°	21:30	22:57	· · ·	11:14	11:27	11:39
54°	21:26	22:55	· · ·	11:19	11:30	11:39
56°	21:21	22:54	· · ·	11:25	11:33	11:39
58°	21:15	22:52	· · ·	11:32	11:36	11:39
S 60°	21:09	22:50	· · ·	11:40	11:40	11:39

Day	Sun		Mer.	Moon		Age
	Eqn.of Time		Pass	Mer.Pass.		
	00 ^h	12 ^h	Pass	Upper	Lower	
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	
26	02:53	02:59	12:03	03:49	16:14	
27	03:05	03:11	12:03	04:39	17:03	
28	03:18	03:24	12:03	05:27	17:51	

June 29, 30, 01 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	277°34.1	171°48.9	N23°30.5	235°12.9	N15°16.1	211°26.8	N20°55.1	286°30.5	S06°00.1	Alpheratz	357°35.2 29°13.4	
1	292°36.6	186°48.0	30.2	250°13.5	16.7	226°28.7	55.2	301°32.9	00.1	Ankaa	353°07.6 -42°10.1	
2	307°39.0	201°47.2	30.0	265°14.2	17.2	241°30.6	55.3	316°35.4	00.1	Schedar	349°31.6 56°40.0	
3	322°41.5	216°46.3	•• 29.7	280°14.9	•• 17.8	256°32.4	•• 55.4	331°37.8	•• 00.1	Diphda	348°47.8 -17°51.0	
4	337°44.0	231°45.4	29.5	295°15.6	18.3	271°34.3	55.5	346°40.3	00.1	Achernar	335°20.7 -57°06.5	
5	352°46.4	246°44.5	29.2	310°16.3	18.9	286°36.2	55.6	1°42.7	00.1	Hamal	327°51.9 23°34.6	
6	7°48.9	261°43.7	N23°29.0	325°16.9	N15°19.4	301°38.1	N20°55.7	16°45.2	S06°00.1	Polaris	314°33.6 89°21.8	
7	22°51.4	276°42.8	28.7	340°17.6	20.0	316°40.0	55.8	31°47.7	00.1	Acamar	315°12.4 -40°12.2	
8	37°53.8	291°41.9	28.4	355°18.3	20.5	331°41.9	55.8	46°50.1	00.1	Menkar	314°06.8 4°11.2	
9	52°56.3	306°41.0	•• 28.2	10°19.0	•• 21.1	346°43.8	•• 55.9	61°52.6	•• 00.1	Mirfak	308°29.3 49°56.7	
10	67°58.7	321°40.2	27.9	25°19.7	21.6	1°45.7	56.0	76°55.0	00.2	Aldebaran	290°40.4 16°33.5	
11	83°01.2	336°39.3	27.7	40°20.4	22.2	16°47.5	56.1	91°57.5	00.2	Rigel	281°04.6 -8°10.3	
12	98°03.7	351°38.4	N23°27.4	55°21.0	N15°22.7	31°49.4	N20°56.2	107°00.0	S06°00.2	Capella	280°23.0 46°01.3	
13	113°06.1	6°37.5	27.1	70°21.7	23.3	46°51.3	56.3	122°02.4	00.2	Bellatrix	278°23.7 6°22.3	
14	128°08.6	21°36.7	26.9	85°22.4	23.8	61°53.2	56.4	137°04.9	00.2	Elnath	278°02.8 28°37.7	
15	143°11.1	36°35.8	•• 26.6	100°23.1	•• 24.4	76°55.1	•• 56.5	152°07.3	•• 00.2	Alnilam	275°38.5 -1°11.1	
16	158°13.5	51°34.9	26.3	115°23.8	24.9	91°57.0	56.6	167°09.8	00.2	Betelgeuse	270°52.9 7°24.7	
17	173°16.0	66°34.1	26.0	130°24.4	25.5	106°58.9	56.6	182°12.2	00.2	Canopus	263°53.2 -52°42.4	
18	188°18.5	81°33.2	N23°25.8	145°25.1	N15°26.0	122°00.8	N20°56.7	197°14.7	S06°00.2	Sirius	258°27.0 -16°44.9	
19	203°20.9	96°32.3	25.5	160°25.8	26.6	137°02.7	56.8	212°17.2	00.2	Adhara	255°06.6 -29°00.3	
20	218°23.4	111°31.4	25.2	175°26.5	27.1	152°04.5	56.9	227°19.6	00.3	Procyon	244°51.6 5°09.8	
21	233°25.8	126°30.6	•• 25.0	190°27.2	•• 27.7	167°06.4	•• 57.0	242°22.1	•• 00.3	Pollux	243°18.2 27°58.1	
22	248°28.3	141°29.7	24.7	205°27.9	28.2	182°08.3	57.1	257°24.5	00.3	Avior	234°15.5 -59°35.3	
23	263°30.8	156°28.8	24.4	220°28.5	28.8	197°10.2	57.2	272°27.0	00.3	Suhail	222°47.0 -43°32.0	
Mer.pass. 05:29		ν -0.9' d -0.3' m -3.90		ν 0.7' d 0.6' m 0.97		ν 1.9' d 0.1' m -2.03		ν 2.5' d 0.0' m 0.92		Miaplacidus	221°39.2 -69°49.2	
											Alphard	217°48.4 -8°45.9
											Regulus	207°35.1 11°51.0
											Dubhe	193°41.8 61°37.4
											Denebola	182°25.5 14°26.2
											Gienah	175°44.1 -17°40.7
											Acrux	173°00.6 -63°14.4
											Gacrux	171°52.2 -57°15.3
											Alioth	166°13.3 55°49.9
											Spica	158°22.7 -11°17.4
											Alkaid	152°52.2 49°11.7
											Hadar	148°36.4 -60°29.7
											Menkent	147°58.0 -36°29.6
											Arcturus	145°48.2 19°03.4
											Rigel Kent.	139°40.6 -60°56.4
											Kochab	137°19.0 74°03.5
											Zuben'ubi	136°56.3 -16°08.7
											Alphecca	126°03.8 26°38.0
											Antares	112°16.1 -26°29.2
											Atria	107°10.2 -69°04.4
											Sabik	102°03.0 -15°45.3
											Shaula	96°10.6 -37°07.3
											Rasalhague	95°58.6 12°32.5
											Eltanin	90°41.8 51°29.1
											Kaus Aust.	83°32.7 -34°22.4
											Vega	80°33.1 38°48.3
											Nunki	75°47.9 -26°16.0
											Altair	62°00.1 8°55.9
											Peacock	53°05.9 -56°39.2
											Deneb	49°25.7 45°21.9
											Enif	33°39.0 9°59.2
											Al Na'ir	27°33.2 -46°50.3
											Fomalhaut	15°14.8 -29°29.4
											Scheat	13°45.5 28°12.8
											Markab	13°30.2 15°20.1
											Jun 29 Sat SHA Mer.pass	
											Venus	254°14.8 12:33
											Mars	317°38.7 08:19
											Jupiter	293°52.7 09:53
											Saturn	8°56.4 04:53
											Jun 30 Sun SHA Mer.pass	
											Venus	252°54.7 12:35
											Mars	316°56.0 08:18
											Jupiter	293°38.9 09:50
											Saturn	8°56.2 04:49
											Jul 01 Mon SHA Mer.pass	
											Venus	251°34.8 12:36
											Mars	316°13.2 08:17
											Jupiter	293°25.1 09:47
											Saturn	8°56.2 04:45
											Horizontal parallax	
											Venus:	0.1
											Mars:	0.1
Mer.pass. 05:21		ν -0.9' d -0.3' m -3.90		ν 0.7' d 0.5' m 0.97		ν 1.9' d 0.1' m -2.03		ν 2.5' d 0.0' m 0.91				

Sun				Moon			
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	179°07.6	N23°12.7	269°17.5	12.2'	N03°19.3	16.7'	59.2'
1	194°07.5	12.5	283°48.7	12.2'	03°36.0	16.7'	59.2'
2	209°07.3	12.4	298°19.9	12.2'	03°52.7	16.7'	59.2'
3	224°07.2	•• 12.3	312°51.1	12.2'	04°09.4	16.7'	59.2'
4	239°07.1	12.1	327°22.2	12.1'	04°26.1	16.6'	59.2'
5	254°07.0	12.0	341°53.4	12.1'	04°42.8	16.6'	59.2'
6	269°06.8	N23°11.8	356°24.5	12.1'	N04°59.4	16.6'	59.2'
7	284°06.7	11.7	10°55.6	12.1'	05°16.0	16.6'	59.2'
8	299°06.6	11.6	25°26.7	12.1'	05°32.5	16.5'	59.2'
9	314°06.5	•• 11.4	39°57.7	12.0'	05°49.0	16.5'	59.1'
10	329°06.3	11.3	54°28.7	12.0'	06°05.5	16.5'	59.1'
11	344°06.2	11.1	68°59.7	12.0'	06°22.0	16.4'	59.1'
12	359°06.1	N23°11.0	83°30.7	12.0'	N06°38.4	16.4'	59.1'
13	14°06.0	10.8	98°01.7	11.9'	06°54.8	16.3'	59.1'
14	29°05.8	10.7	112°32.6	11.9'	07°11.1	16.3'	59.1'
15	44°05.7	•• 10.5	127°03.5	11.9'	07°27.5	16.3'	59.1'
16	59°05.6	10.4	141°34.4	11.8'	07°43.7	16.2'	59.1'
17	74°05.5	10.2	156°05.2	11.8'	08°00.0	16.2'	59.1'
18	89°05.3	N23°10.1	170°36.0	11.8'	N08°16.1	16.1'	59.1'
19	104°05.2	09.9	185°06.7	11.7'	08°32.3	16.1'	59.1'
20	119°05.1	09.8	199°37.5	11.7'	08°48.4	16.0'	59.1'
21	134°05.0	•• 09.6	214°08.1	11.6'	09°04.4	16.0'	59.1'
22	149°04.8	09.5	228°38.8	11.6'	09°20.4	15.9'	59.0'
23	164°04.7	09.3	243°09.4	11.6'	09°36.3	15.9'	59.0'
	SD = 15.7'	$d = -0.1'$	SD = 16.1'				
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	179°04.6	N23°09.1	257°39.9	11.5'	N09°52.2	15.8'	59.0'
1	194°04.5	09.0	272°10.5	11.5'	10°08.0	15.8'	59.0'
2	209°04.4	08.8	286°40.9	11.4'	10°23.8	15.7'	59.0'
3	224°04.2	•• 08.7	301°11.4	11.4'	10°39.5	15.7'	59.0'
4	239°04.1	08.5	315°41.7	11.3'	10°55.2	15.6'	59.0'
5	254°04.0	08.4	330°12.1	11.3'	11°10.8	15.5'	59.0'
6	269°03.9	N23°08.2	344°42.4	11.2'	N11°26.3	15.5'	59.0'
7	284°03.7	08.0	359°12.6	11.2'	11°41.8	15.4'	59.0'
8	299°03.6	07.9	13°42.8	11.1'	11°57.2	15.3'	59.0'
9	314°03.5	•• 07.7	28°12.9	11.1'	12°12.6	15.3'	58.9'
10	329°03.4	07.6	42°43.0	11.0'	12°27.8	15.2'	58.9'
11	344°03.3	07.4	57°13.0	11.0'	12°43.0	15.1'	58.9'
12	359°03.1	N23°07.2	71°43.0	10.9'	N12°58.2	15.1'	58.9'
13	14°03.0	07.1	86°12.9	10.9'	13°13.2	15.0'	58.9'
14	29°02.9	06.9	100°42.8	10.8'	13°28.2	14.9'	58.9'
15	44°02.8	•• 06.7	115°12.6	10.7'	13°43.2	14.8'	58.9'
16	59°02.6	06.6	129°42.3	10.7'	13°58.0	14.8'	58.9'
17	74°02.5	06.4	144°12.0	10.6'	14°12.8	14.7'	58.9'
18	89°02.4	N23°06.2	158°41.6	10.6'	N14°27.4	14.6'	58.9'
19	104°02.3	06.1	173°11.2	10.5'	14°42.0	14.5'	58.8'
20	119°02.2	05.9	187°40.7	10.4'	14°56.6	14.4'	58.8'
21	134°02.0	•• 05.7	202°10.1	10.4'	15°11.0	14.4'	58.8'
22	149°01.9	05.5	216°39.5	10.3'	15°25.4	14.3'	58.8'
23	164°01.8	05.4	231°08.8	10.2'	15°39.6	14.2'	58.8'
	SD = 15.7'	$d = -0.2'$	SD = 16.1'				
Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	179°01.7	N23°05.2	245°38.0	10.2'	N15°53.8	14.1'	58.8'
1	194°01.6	05.0	260°07.2	10.1'	16°07.9	14.0'	58.8'
2	209°01.4	04.9	274°36.3	10.0'	16°21.9	13.9'	58.8'
3	224°01.3	•• 04.7	289°05.3	10.0'	16°35.8	13.8'	58.7'
4	239°01.2	04.5	303°34.3	9.9'	16°49.6	13.7'	58.7'
5	254°01.1	04.3	318°03.2	9.8'	17°03.3	13.6'	58.7'
6	269°01.0	N23°04.1	332°32.0	9.8'	N17°16.9	13.5'	58.7'
7	284°00.8	04.0	347°00.8	9.7'	17°30.4	13.4'	58.7'
8	299°00.7	03.8	1°29.4	9.6'	17°43.8	13.3'	58.7'
9	314°00.6	•• 03.6	15°58.1	9.5'	17°57.2	13.2'	58.7'
10	329°00.5	03.4	30°26.6	9.5'	18°10.4	13.1'	58.7'
11	344°00.4	03.3	44°55.1	9.4'	18°23.5	13.0'	58.6'
12	359°00.2	N23°03.1	59°23.5	9.3'	N18°36.5	12.9'	58.6'
13	14°00.1	02.9	73°51.8	9.2'	18°49.4	12.8'	58.6'
14	29°00.0	02.7	88°20.0	9.2'	19°02.2	12.7'	58.6'
15	43°59.9	•• 02.5	102°48.2	9.1'	19°14.8	12.6'	58.6'
16	58°59.8	02.3	117°16.3	9.0'	19°27.4	12.4'	58.6'
17	73°59.6	02.2	131°44.3	8.9'	19°39.8	12.3'	58.6'
18	88°59.5	N23°02.0	146°12.3	8.9'	N19°52.2	12.2'	58.6'
19	103°59.4	01.8	160°40.2	8.8'	20°04.4	12.1'	58.5'
20	118°59.3	01.6	175°08.0	8.7'	20°16.5	12.0'	58.5'
21	133°59.2	•• 01.4	189°35.7	8.6'	20°28.5	11.9'	58.5'
22	148°59.0	01.2	204°03.3	8.6'	20°40.3	11.7'	58.5'
23	163°58.9	01.0	218°30.9	8.5'	20°52.1	11.6'	58.5'
	SD = 15.7'	$d = -0.2'$	SD = 16.0'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°						
66°	—:—	—:—	00:13	23:44	—:—	—:—
64°	—:—	—:—	01:40	22:26	—:—	—:—
62°	—:—	—:—	02:16	21:51	—:—	—:—
60°	—:—	01:02	02:42	21:25	23:04	—:—
N 58°	—:—	01:48	03:02	21:06	22:19	—:—
56°	—:—	02:16	03:18	20:49	21:50	—:—
54°	00:57	02:38	03:32	20:35	21:29	23:09
52°	01:39	02:56	03:44	20:23	21:12	22:27
50°	02:06	03:10	03:55	20:13	20:57	22:01
45°	02:50	03:40	04:17	19:51	20:28	21:17
N 40°	03:20	04:02	04:35	19:33	20:05	20:47
35°	03:43	04:20	04:49	19:18	19:48	20:24
30°	04:02	04:35	05:02	19:05	19:33	20:06
20°	04:30	05:00	05:24	18:43	19:08	19:37
N 10°	04:53	05:20	05:43	18:25	18:48	19:15
0°	05:11	05:38	06:00	18:07	18:30	18:56
S 10°	05:28	05:55	06:17	17:50	18:13	18:39
20°	05:44	06:12	06:36	17:32	17:56	18:23
30°	06:00	06:30	06:57	17:11	17:37	18:07
35°	06:09	06:41	07:09	16:59	17:27	17:59
40°	06:18	06:52	07:23	16:45	17:15	17:49
45°	06:28	07:06	07:39	16:28	17:02	17:39
S 50°	06:40	07:21	08:00	16:08	16:46	17:28
52°	06:45	07:29	08:09	15:58	16:39	17:23
54°	06:51	07:37	08:20	15:47	16:31	17:17
56°	06:57	07:46	08:33	15:35	16:22	17:11
58°	07:03	07:56	08:47	15:21	16:12	17:04
S 60°	07:11	08:07	09:04	15:04	16:01	16:57

July 02, 03, 04 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	280°31.5	170°46.5	N23°09.3	236°01.9	N15°55.2	213°42.9	N21°01.4	289°27.8	S06°00.8	Alpheratz	357°35.2 29°13.4
1	295°34.0	185°45.6	09.0	251°02.6	55.7	228°44.8	01.5	304°30.2	00.9	Ankaa	353°07.5 -42°10.1
2	310°36.4	200°44.8	08.7	266°03.3	56.3	243°46.7	01.6	319°32.7	00.9	Schedar	349°31.6 56°40.0
3	325°38.9	215°43.9	•• 08.3	281°03.9	•• 56.8	258°48.6	•• 01.7	334°35.2	•• 00.9	Diphda	348°47.7 -17°51.0
4	340°41.4	230°43.1	08.0	296°04.6	57.3	273°50.5	01.8	349°37.6	00.9	Achernar	335°20.6 -57°06.4
5	355°43.8	245°42.2	07.6	311°05.3	57.9	288°52.4	01.8	4°40.1	00.9	Hamal	327°51.8 23°34.6
6	10°46.3	260°41.4	N23°07.3	326°06.0	N15°58.4	303°54.3	N21°01.9	19°42.6	S06°00.9	Polaris	314°32.1 89°21.8
7	25°48.8	275°40.5	07.0	341°06.7	58.9	318°56.1	02.0	34°45.1	00.9	Acamar	315°12.3 -40°12.2
8	40°51.2	290°39.6	06.6	356°07.3	15°59.5	333°58.0	02.1	49°47.5	01.0	Menkar	314°06.8 4°11.2
9	55°53.7	305°38.8	•• 06.3	11°08.0	16°00.0	348°59.9	•• 02.2	64°50.0	•• 01.0	Mirfak	308°29.2 49°56.7
10	70°56.2	320°37.9	05.9	26°08.7	00.5	4°01.8	02.3	79°52.5	01.0	Aldebaran	290°40.4 16°33.5
11	85°58.6	335°37.1	05.6	41°09.4	01.1	19°03.7	02.3	94°54.9	01.0	Rigel	281°04.6 -8°10.3
12	101°01.1	350°36.2	N23°05.2	56°10.1	N16°01.6	34°05.6	N21°02.4	109°57.4	S06°01.0	Capella	280°23.0 46°01.3
13	116°03.6	5°35.4	04.9	71°10.7	02.1	49°07.5	02.5	124°59.9	01.0	Bellatrix	278°23.7 6°22.3
14	131°06.0	20°34.5	04.5	86°11.4	02.6	64°09.4	02.6	140°02.3	01.0	Elnath	278°02.8 28°37.7
15	146°08.5	35°33.7	•• 04.2	101°12.1	•• 03.2	79°11.3	•• 02.7	155°04.8	•• 01.1	Alnilam	275°38.5 -1°11.1
16	161°10.9	50°32.8	03.8	116°12.8	03.7	94°13.2	02.8	170°07.3	01.1	Betelgeuse	270°52.9 7°24.7
17	176°13.4	65°32.0	03.5	131°13.5	04.2	109°15.1	02.9	185°09.8	01.1	Canopus	263°53.1 -52°42.4
18	191°15.9	80°31.1	N23°03.1	146°14.1	N16°04.8	124°17.0	N21°02.9	200°12.2	S06°01.1	Sirius	258°27.0 -16°44.9
19	206°18.3	95°30.3	02.7	161°14.8	05.3	139°18.9	03.0	215°14.7	01.1	Adhara	255°06.6 -29°00.3
20	221°20.8	110°29.4	02.4	176°15.5	05.8	154°20.8	03.1	230°17.2	01.1	Procyon	244°51.6 5°09.8
21	236°23.3	125°28.5	•• 02.0	191°16.2	•• 06.3	169°22.7	•• 03.2	245°19.6	•• 01.1	Pollux	243°18.2 27°58.1
22	251°25.7	140°27.7	01.7	206°16.8	06.9	184°24.6	03.3	260°22.1	01.2	Avior	234°15.5 -59°35.3
23	266°28.2	155°26.8	01.3	221°17.5	07.4	199°26.5	03.4	275°24.6	01.2	Suhail	222°47.0 -43°32.0
Mer.pass. 05:17		ν -0.9' d -0.3' m -3.90		ν 0.7' d 0.5' m 0.96		ν 1.9' d 0.1' m -2.03		ν 2.5' d 0.0' m 0.90		Miaplacidus	221°39.2 -69°49.2
Alphard 217°48.4 -8°45.9											
Regulus 207°35.1 11°51.0											
Dubhe 193°41.8 61°37.4											
Denebola 182°25.5 14°26.2											
Gienah 175°44.1 -17°40.7											
Acrux 173°00.7 -63°14.4											
Gacrux 171°52.2 -57°15.3											
Alioth 166°13.3 55°49.9											
Spica 158°22.7 -11°17.4											
Alkaid 152°52.2 49°11.7											
Hadar 148°36.4 -60°29.7											
Menkent 147°58.0 -36°29.6											
Arcturus 145°48.2 19°03.4											
Rigel Kent. 139°40.6 -60°56.4											
Kochab 137°19.1 74°03.5											
Zuben'ubi 136°56.3 -16°08.7											
Alphecca 126°03.8 26°38.0											
Antares 112°16.1 -26°29.2											
Atria 107°10.2 -69°04.4											
Sabik 102°03.0 -15°45.3											
Shaula 96°10.6 -37°07.3											
Rasalhague 95°58.6 12°32.5											
Eltanin 90°41.9 51°29.1											
Kaus Aust. 83°32.7 -34°22.4											
Vega 80°33.1 38°48.4											
Nunki 75°47.9 -26°16.0											
Altair 62°00.1 8°56.0											
Peacock 53°05.8 -56°39.2											
Deneb 49°25.7 45°21.9											
Enif 33°39.0 9°59.2											
Al Na'ir 27°33.1 -46°50.3											
Fomalhaut 15°14.8 -29°29.4											
Scheat 13°45.5 28°12.8											
Markab 13°30.2 15°20.1											
Jul 02 Tue		SHA		Mer.pass							
Venus		250°15.0		12:38							
Mars		315°30.4		08:15							
Jupiter		293°11.4		09:44							
Saturn		8°56.2		04:41							
Jul 03 Wed		SHA		Mer.pass							
Venus		248°55.3		12:39							
Mars		314°47.5		08:14							
Jupiter		292°57.7		09:41							
Saturn		8°56.4		04:37							
Jul 04 Thu		SHA		Mer.pass							
Venus		247°35.9		12:40							
Mars		314°04.7		08:13							
Jupiter		292°44.1		09:38							
Saturn		8°56.6		04:33							
Horizontal parallax											
Venus: 0.1											
Mars: 0.1											
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	282°29.8	170°05.7	N22°51.8	236°34.5	N16°20.5	215°13.9	N21°05.5	291°26.4	S06°01.6		
1	297°32.3	185°04.8	51.4	251°35.2	21.0	230°15.8	05.5	306°28.9	01.6		
2	312°34.7	200°04.0	51.0	266°35.9	21.5	245°17.7	05.6	321°31.4	01.6		
3	327°37.2	215°03.1	•• 50.6	281°36.5	•• 22.0	260°19.6	•• 05.7	336°33.9	•• 01.6		
4	342°39.7	230°02.3	50.3	296°37.2	22.5	275°21.5	05.8	351°36.3	01.6		
5	357°42.1	245°01.5	49.9	311°37.9	23.1	290°23.4	05.9	6°38.8	01.6		
6	12°44.6	260°00.6	N22°49.5	326°38.6	N16°23.6	305°25.3	N21°06.0	21°41.3	S06°01.7		
7	27°47.1	274°59.8	49.1	341°39.2	24.1	320°27.2	06.0	36°43.8	01.7		
8	42°49.5	289°58.9	48.7	356°39.9	24.6	335°29.1	06.1	51°46.3	01.7		
9	57°52.0	304°58.1	•• 48.2	11°40.6	•• 25.1	350°31.0	•• 06.2	66°48.7	•• 01.7		
10	72°54.4	319°57.3	47.8	26°41.3	25.7	5°32.9	06.3	81°51.2	01.7		
11	87°56.9	334°56.4	47.4	41°42.0	26.2	20°34.8	06.4	96°53.7	01.7		
12	102°59.4	349°55.6	N22°47.0	56°42.6	N16°26.7	35°36.7	N21°06.5	111°56.2	S06°01.8		
13	118°01.8	4°54.7	46.6	71°43.3	27.2	50°38.6	06.5	126°58.6	01.8		
14	133°04.3	19°53.9	46.2	86°44.0	27.7	65°40.5	06.6	142°01.1	01.8		
15	148°06.8	34°53.1	•• 45.8	101°44.7	•• 28.2	80°42.4	•• 06.7	157°03.6	•• 01.8		
16	163°09.2	49°52.2	45.4	116°45.4	28.7	95°44.3	06.8	172°06.1	01.8		
17	178°11.7	64°51.4	45.0	131°46.0	29.3	110°46.2	06.9	187°08.6	01.9		
18	193°14.2	79°50.5	N22°44.6	146°46.7	N16°29.8	125°48.1	N21°06.9	202°11.0	S06°01.9		
19	208°16.6	94°49.7	44.2	161°47.4	30.3	140°50.0	07.0	217°13.5	01.9		
20	223°19.1	109°48.9	43.7	176°48.1	30.8	155°51.9	07.1	232°16.0	01.9		
21	238°21.5	124°48.0	•• 43.3	191°48.7	•• 31.3	170°53.8	•• 07.2	247°18.5	•• 01.9		
22	253°24.0	139°47.2	42.9	206°49.4	31.8	185°55.7	07.3	262°21.0	01.9		
23	268°26.5	154°46.4	42.5	221°50.1	32.3	200°57.6	07.4	277°23.4	02.0		
Mer.pass. 05:09		ν -0.8' d -0.4' m -3.90		ν 0.7' d 0.5' m 0.96		ν 1.9' d 0.1' m -2.04		ν 2.5' d 0.0' m 0.89			

Sun				Moon					
Tue	GHA	Dec		GHA	ν	Dec	d	HP	
0	178°58.8	N23°00.8		232°58.4	8.4'	N21°03.7	11.5'	58.5'	
1	193°58.7	00.7		247°25.8	8.3'	21°15.2	11.4'	58.5'	
2	208°58.6	00.5		261°53.2	8.3'	21°26.5	11.2'	58.4'	
3	223°58.5	· · 00.3		276°20.4	8.2'	21°37.8	11.1'	58.4'	
4	238°58.3	23°00.1		290°47.6	8.1'	21°48.9	11.0'	58.4'	
5	253°58.2	22°59.9		305°14.7	8.0'	21°59.8	10.8'	58.4'	
6	268°58.1	N22°59.7		319°41.8	8.0'	N22°10.7	10.7'	58.4'	
7	283°58.0	59.5		334°08.8	7.9'	22°21.4	10.6'	58.4'	
8	298°57.9	59.3		348°35.7	7.8'	22°32.0	10.4'	58.3'	
9	313°57.7	· · 59.1		3°02.5	7.7'	22°42.4	10.3'	58.3'	
10	328°57.6	58.9		17°29.2	7.7'	22°52.7	10.2'	58.3'	
11	343°57.5	58.7		31°55.9	7.6'	23°02.9	10.0'	58.3'	
12	358°57.4	N22°58.5		46°22.5	7.5'	N23°12.9	9.9'	58.3'	
13	13°57.3	58.3		60°49.0	7.5'	23°22.8	9.7'	58.3'	
14	28°57.2	58.1		75°15.5	7.4'	23°32.6	9.6'	58.2'	
15	43°57.1	· · 57.9		89°41.9	7.3'	23°42.2	9.5'	58.2'	
16	58°56.9	57.7		104°08.2	7.3'	23°51.6	9.3'	58.2'	
17	73°56.8	57.5		118°34.5	7.2'	24°00.9	9.2'	58.2'	
18	88°56.7	N22°57.3		133°00.6	7.1'	N24°10.1	9.0'	58.2'	
19	103°56.6	57.1		147°26.8	7.0'	24°19.1	8.9'	58.2'	
20	118°56.5	56.9		161°52.8	7.0'	24°28.0	8.7'	58.1'	
21	133°56.4	· · 56.7		176°18.8	6.9'	24°36.7	8.6'	58.1'	
22	148°56.2	56.5		190°44.7	6.9'	24°45.2	8.4'	58.1'	
23	163°56.1	56.3		205°10.6	6.8'	24°53.6	8.3'	58.1'	
SD = 15.7'				$d = -0.2'$				SD = 15.9'	
Wed	GHA	Dec		GHA	ν	Dec	d	HP	
0	178°56.0	N22°56.1		219°36.3	6.7'	N25°01.9	8.1'	58.1'	
1	193°55.9	55.9		234°02.1	6.7'	25°10.0	7.9'	58.1'	
2	208°55.8	55.7		248°27.7	6.6'	25°18.0	7.8'	58.0'	
3	223°55.7	· · 55.5		262°53.3	6.5'	25°25.7	7.6'	58.0'	
4	238°55.5	55.3		277°18.9	6.5'	25°33.4	7.5'	58.0'	
5	253°55.4	55.0		291°44.4	6.4'	25°40.8	7.3'	58.0'	
6	268°55.3	N22°54.8		306°09.8	6.4'	N25°48.2	7.1'	58.0'	
7	283°55.2	54.6		320°35.2	6.3'	25°55.3	7.0'	58.0'	
8	298°55.1	54.4		335°00.5	6.3'	26°02.3	6.8'	57.9'	
9	313°55.0	· · 54.2		349°25.8	6.2'	26°09.1	6.7'	57.9'	
10	328°54.9	54.0		3°51.0	6.2'	26°15.8	6.5'	57.9'	
11	343°54.7	53.8		18°16.2	6.1'	26°22.3	6.3'	57.9'	
12	358°54.6	N22°53.6		32°41.3	6.1'	N26°28.6	6.2'	57.9'	
13	13°54.5	53.3		47°06.4	6.0'	26°34.8	6.0'	57.8'	
14	28°54.4	53.1		61°31.4	6.0'	26°40.8	5.8'	57.8'	
15	43°54.3	· · 52.9		75°56.4	6.0'	26°46.6	5.7'	57.8'	
16	58°54.2	52.7		90°21.4	5.9'	26°52.3	5.5'	57.8'	
17	73°54.1	52.5		104°46.3	5.9'	26°57.8	5.3'	57.8'	
18	88°54.0	N22°52.3		119°11.2	5.8'	N27°03.1	5.2'	57.7'	
19	103°53.8	52.0		133°36.0	5.8'	27°08.2	5.0'	57.7'	
20	118°53.7	51.8		148°00.8	5.8'	27°13.2	4.8'	57.7'	
21	133°53.6	· · 51.6		162°25.6	5.7'	27°18.0	4.6'	57.7'	
22	148°53.5	51.4		176°50.4	5.7'	27°22.7	4.5'	57.7'	
23	163°53.4	51.2		191°15.1	5.7'	27°27.1	4.3'	57.6'	
SD = 15.7'				$d = -0.2'$				SD = 15.8'	
Thu	GHA	Dec		GHA	ν	Dec	d	HP	
0	178°53.3	N22°50.9		205°39.8	5.7'	N27°31.4	4.1'	57.6'	
1	193°53.2	50.7		220°04.5	5.6'	27°35.6	3.9'	57.6'	
2	208°53.1	50.5		234°29.1	5.6'	27°39.5	3.8'	57.6'	
3	223°52.9	· · 50.3		248°53.7	5.6'	27°43.3	3.6'	57.6'	
4	238°52.8	50.0		263°18.3	5.6'	27°46.9	3.4'	57.5'	
5	253°52.7	49.8		277°42.9	5.6'	27°50.3	3.3'	57.5'	
6	268°52.6	N22°49.6		292°07.5	5.6'	N27°53.6	3.1'	57.5'	
7	283°52.5	49.4		306°32.1	5.6'	27°56.6	2.9'	57.5'	
8	298°52.4	49.1		320°56.7	5.6'	27°59.5	2.7'	57.5'	
9	313°52.3	· · 48.9		335°21.2	5.5'	28°02.3	2.5'	57.4'	
10	328°52.2	48.7		349°45.8	5.5'	28°04.8	2.4'	57.4'	
11	343°52.1	48.4		4°10.3	5.5'	28°07.2	2.2'	57.4'	
12	358°51.9	N22°48.2		18°34.9	5.5'	N28°09.4	2.0'	57.4'	
13	13°51.8	48.0		32°59.4	5.6'	28°11.4	1.8'	57.4'	
14	28°51.7	47.7		47°24.0	5.6'	28°13.2	1.7'	57.3'	
15	43°51.6	· · 47.5		61°48.5	5.6'	28°14.9	1.5'	57.3'	
16	58°51.5	47.3		76°13.1	5.6'	28°16.4	1.3'	57.3'	
17	73°51.4	47.0		90°37.7	5.6'	28°17.7	1.1'	57.3'	
18	88°51.3	N22°46.8		105°02.2	5.6'	N28°18.9	1.0'	57.2'	
19	103°51.2	46.6		119°26.9	5.6'	28°19.8	0.8'	57.2'	
20	118°51.1	46.3		133°51.5	5.6'	28°20.6	0.6'	57.2'	
21	133°51.0	· · 46.1		148°16.1	5.7'	28°21.2	0.4'	57.2'	
22	148°50.9	45.9		162°40.8	5.7'	28°21.7	0.3'	57.2'	
23	163°50.7	45.6		177°05.5	5.7'	28°21.9	0.1'	57.1'	
SD = 15.7'				$d = -0.2'$				SD = 15.7'	

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°						
66°	--	--	00:38	23:25	--	--
64°	--	--	01:46	22:21	--	--
62°	--	--	02:20	21:47	--	--
60°	--	01:10	02:45	21:23	22:57	--
N 58°	--	01:52	03:04	21:04	22:15	--
56°	--	02:20	03:21	20:48	21:48	--
54°	01:04	02:41	03:34	20:34	21:27	23:03
52°	01:43	02:58	03:46	20:22	21:10	22:24
50°	02:09	03:13	03:57	20:12	20:55	21:59
45°	02:53	03:42	04:19	19:50	20:27	21:16
N 40°	03:22	04:04	04:36	19:32	20:05	20:46
35°	03:45	04:21	04:51	19:18	19:47	20:24
30°	04:03	04:36	05:03	19:05	19:32	20:05
20°	04:31	05:01	05:25	18:44	19:08	19:37
N 10°	04:54	05:21	05:44	18:25	18:48	19:15
0°	05:12	05:38	06:01	18:08	18:30	18:57
S 10°	05:29	05:55	06:18	17:51	18:14	18:40
20°	05:45	06:12	06:36	17:33	17:57	18:24
30°	06:01	06:30	06:57	17:12	17:39	18:08
35°	06:09	06:41	07:09	17:00	17:28	18:00
40°	06:18	06:52	07:22	16:46	17:17	17:51
45°	06:28	07:05	07:39	16:30	17:04	17:41
S 50°	06:39	07:21	07:59	16:10	16:48	17:29
52°	06:44	07:28	08:08	16:01	16:41	17:24
54°	06:50	07:36	08:19	15:50	16:33	17:19
56°	06:56	07:45	08:31	15:38	16:24	17:13
58°	07:02	07:54	08:45	15:24	16:15	17:07
S 60°	07:10	08:06	09:02	15:07	16:03	16:59

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°						
N 70°						
68°						
66°						
64°	22:36			19:43		
62°	23:19	23:10		19:01	21:11	
60°	23:48	· ·	00:02	18:32	20:19	21:42
N 58°	· ·	00:10	00:34	18:11	19:47	21:04
56°	00:10	00:28	00:58	17:53	19:23	20:37
54°	00:22	00:44	01:18	17:38	19:04	20:16
52°	00:32	00:57	01:34	17:25	18:48	19:58
50°	00:41	01:09	01:48	17:14	18:34	19:43
45°	01:00	01:34	02:17	16:50	18:05	19:12
N 40°	01:15	01:53	02:40	16:31	17:43	18:48
35°	01:29	02:10	02:59	16:16	17:25	18:29
30°	01:40	02:24	03:15	16:02	17:09	18:12
20°	02:00	02:49	03:43	15:39	16:42	17:44
N 10°	02:18	03:10	04:06	15:19	16:19	17:19
0°	02:34	03:30	04:28	15:01	15:58	16:57
S 10°	02:51	03:50	04:51	14:42	15:37	16:34
20°	03:09	04:12	05:15	14:23	15:14	16:10
30°	03:30	04:37	05:43	14:00	14:47	15:41
35°	03:42	04:52	05:59	13:47	14:32	15:2

July 05, 06, 07 UT (Fri., Sat., Sun.)

Aries				Venus		Mars		Jupiter		Saturn	
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	283°28.9	169°45.5	N22°42.1	236°50.8	N16°32.9	215°59.5	N21°07.4	292°25.9		S06°02.0	
1	298°31.4	184°44.7	41.6	251°51.5	33.4	231°01.4	07.5	307°28.4		02.0	
2	313°33.9	199°43.9	41.2	266°52.1	33.9	246°03.3	07.6	322°30.9		02.0	
3	328°36.3	214°43.0	•• 40.8	281°52.8	•• 34.4	261°05.2	•• 07.7	337°33.4	••	02.0	
4	343°38.8	229°42.2	40.4	296°53.5	34.9	276°07.1	07.8	352°35.8		02.1	
5	358°41.3	244°41.4	39.9	311°54.2	35.4	291°09.0	07.8	7°38.3		02.1	
6	13°43.7	259°40.5	N22°39.5	326°54.8	N16°35.9	306°10.9	N21°07.9	22°40.8		S06°02.1	
7	28°46.2	274°39.7	39.1	341°55.5	36.4	321°12.8	08.0	37°43.3		02.1	
8	43°48.7	289°38.9	38.7	356°56.2	36.9	336°14.7	08.1	52°45.8		02.1	
9	58°51.1	304°38.0	•• 38.2	11°56.9	•• 37.5	351°16.6	•• 08.2	67°48.2	••	02.1	
10	73°53.6	319°37.2	37.8	26°57.6	38.0	6°18.5	08.3	82°50.7		02.2	
11	88°56.0	334°36.4	37.4	41°58.2	38.5	21°20.4	08.3	97°53.2		02.2	
12	103°58.5	349°35.5	N22°36.9	56°58.9	N16°39.0	36°22.3	N21°08.4	112°55.7		S06°02.2	
13	119°01.0	4°34.7	36.5	71°59.6	39.5	51°24.2	08.5	127°58.2		02.2	
14	134°03.4	19°33.9	36.1	87°00.3	40.0	66°26.1	08.6	143°00.6		02.2	
15	149°05.9	34°33.0	•• 35.6	102°00.9	•• 40.5	81°28.0	•• 08.7	158°03.1	••	02.3	
16	164°08.4	49°32.2	35.2	117°01.6	41.0	96°29.9	08.7	173°05.6		02.3	
17	179°10.8	64°31.4	34.7	132°02.3	41.5	111°31.8	08.8	188°08.1		02.3	
18	194°13.3	79°30.5	N22°34.3	147°03.0	N16°42.0	126°33.7	N21°08.9	203°10.6		S06°02.3	
19	209°15.8	94°29.7	33.9	162°03.7	42.5	141°35.6	09.0	218°13.1		02.3	
20	224°18.2	109°28.9	33.4	177°04.3	43.0	156°37.5	09.1	233°15.5		02.4	
21	239°20.7	124°28.1	•• 33.0	192°05.0	•• 43.6	171°39.4	•• 09.1	248°18.0	••	02.4	
22	254°23.2	139°27.2	32.5	207°05.7	44.1	186°41.3	09.2	263°20.5		02.4	
23	269°25.6	154°26.4	32.1	222°06.4	44.6	201°43.2	09.3	278°23.0		02.4	
Mer.pass. 05:05				ν -0.8' d -0.4' m -3.90		ν 0.7' d 0.5' m 0.96		ν 1.9' d 0.1' m -2.04		ν 2.5' d 0.0' m 0.89	

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	284°28.1	169°25.6	N22°31.6	237°07.0	N16°45.1	216°45.1	N21°09.4	293°25.5	S06°02.4
1	299°30.5	184°24.8	31.2	252°07.7	45.6	231°47.0	09.5	308°28.0	02.5
2	314°33.0	199°23.9	30.7	267°08.4	46.1	246°48.9	09.6	323°30.4	02.5
3	329°35.5	214°23.1	•• 30.3	282°09.1	•• 46.6	261°50.8	•• 09.6	338°32.9	•• 02.5
4	344°37.9	229°22.3	29.8	297°09.8	47.1	276°52.7	09.7	353°35.4	02.5
5	359°40.4	244°21.4	29.4	312°10.4	47.6	291°54.6	09.8	8°37.9	02.5
6	14°42.9	259°20.6	N22°28.9	327°11.1	N16°48.1	306°56.5	N21°09.9	23°40.4	S06°02.6
7	29°45.3	274°19.8	28.5	342°11.8	48.6	321°58.4	10.0	38°42.9	02.6
8	44°47.8	289°19.0	28.0	357°12.5	49.1	337°00.3	10.0	53°45.4	02.6
9	59°50.3	304°18.2	•• 27.6	12°13.1	•• 49.6	352°02.2	•• 10.1	68°47.8	•• 02.6
10	74°52.7	319°17.3	27.1	27°13.8	50.1	7°04.1	10.2	83°50.3	02.6
11	89°55.2	334°16.5	26.6	42°14.5	50.6	22°06.0	10.3	98°52.8	02.7
12	104°57.7	349°15.7	N22°26.2	57°15.2	N16°51.1	37°07.9	N21°10.4	113°55.3	S06°02.7
13	120°00.1	4°14.9	25.7	72°15.8	51.6	52°09.8	10.4	128°57.8	02.7
14	135°02.6	19°14.0	25.2	87°16.5	52.1	67°11.7	10.5	144°00.3	02.7
15	150°05.0	34°13.2	•• 24.8	102°17.2	•• 52.6	82°13.6	•• 10.6	159°02.8	•• 02.7
16	165°07.5	49°12.4	24.3	117°17.9	53.1	97°15.5	10.7	174°05.2	02.8
17	180°10.0	64°11.6	23.9	132°18.6	53.6	112°17.4	10.8	189°07.7	02.8
18	195°12.4	79°10.8	N22°23.4	147°19.2	N16°54.1	127°19.3	N21°10.8	204°10.2	S06°02.8
19	210°14.9	94°09.9	22.9	162°19.9	54.6	142°21.2	10.9	219°12.7	02.8
20	225°17.4	109°09.1	22.4	177°20.6	55.1	157°23.2	11.0	234°15.2	02.8
21	240°19.8	124°08.3	•• 22.0	192°21.3	•• 55.6	172°25.1	•• 11.1	249°17.7	•• 02.9
22	255°22.3	139°07.5	21.5	207°21.9	56.1	187°27.0	11.2	264°20.2	02.9
23	270°24.8	154°06.7	21.0	222°22.6	56.6	202°28.9	11.2	279°22.7	02.9
Mer.pass. 05:01		ν -0.8' d -0.4' m -3.90		ν 0.7' d 0.5' m 0.95		ν 1.9' d 0.1' m -2.04		ν 2.5' d 0.0' m 0.88	

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	285°27.2	169°05.8	N22°20.6	237°23.3	N16°57.1	217°30.8	N21°11.3	294°25.1	S06°02.9
1	300°29.7	184°05.0	20.1	252°24.0	57.6	232°32.7	11.4	309°27.6	02.9
2	315°32.2	199°04.2	19.6	267°24.7	58.1	247°34.6	11.5	324°30.1	03.0
3	330°34.6	214°03.4	•• 19.1	282°25.3	•• 58.6	262°36.5	•• 11.6	339°32.6	•• 03.0
4	345°37.1	229°02.6	18.6	297°26.0	59.1	277°38.4	11.6	354°35.1	03.0
5	0°39.5	244°01.8	18.2	312°26.7	16°59.6	292°40.3	11.7	9°37.6	03.0
6	15°42.0	259°00.9	N22°17.7	327°27.4	N17°00.1	307°42.2	N21°11.8	24°40.1	S06°03.1
7	30°44.5	274°00.1	17.2	342°28.0	00.6	322°44.1	11.9	39°42.6	03.1
8	45°46.9	288°59.3	16.7	357°28.7	01.1	337°46.0	11.9	54°45.0	03.1
9	60°49.4	303°58.5	•• 16.2	12°29.4	•• 01.6	352°47.9	•• 12.0	69°47.5	•• 03.1
10	75°51.9	318°57.7	15.7	27°30.1	02.1	7°49.8	12.1	84°50.0	03.1
11	90°54.3	333°56.9	15.3	42°30.7	02.6	22°51.7	12.2	99°52.5	03.2
12	105°56.8	348°56.1	N22°14.8	57°31.4	N17°03.1	37°53.6	N21°12.3	114°55.0	S06°03.2
13	120°59.3	3°55.2	14.3	72°32.1	03.6	52°55.5	12.3	129°57.5	03.2
14	136°01.7	18°54.4	13.8	87°32.8	04.1	67°57.4	12.4	145°00.0	03.2
15	151°04.2	33°53.6	•• 13.3	102°33.5	•• 04.6	82°59.4	•• 12.5	160°02.5	•• 03.3
16	166°06.7	48°52.8	12.8	117°34.1	05.1	98°01.3	12.6	175°05.0	03.3
17	181°09.1	63°52.0	12.3	132°34.8	05.5	113°03.2	12.7	190°07.5	03.3
18	196°11.6	78°51.2	N22°11.8	147°35.5	N17°06.0	128°05.1	N21°12.7	205°09.9	S06°03.3
19	211°14.0	93°50.4	11.3	162°36.2	06.5	143°07.0	12.8	220°12.4	03.3
20	226°16.5	108°49.6	10.8	177°36.8	07.0	158°08.9	12.9	235°14.9	03.4
21	241°19.0	123°48.8	•• 10.3	192°37.5	•• 07.5	173°10.8	•• 13.0	250°17.4	•• 03.4
22	256°21.4	138°47.9	09.8	207°38.2	08.0	188°12.7	13.1	265°19.9	03.4
23	271°23.9	153°47.1	09.3	222°38.9	08.5	203°14.6	13.1	280°22.4	03.4
Mer.pass. 04:57		ν -0.8' d -0.5' m -3.90		ν 0.7' d 0.5' m 0.95		ν 1.9' d 0.1' m -2.04		ν 2.5' d 0.0' m 0.88	

	Stars	
	SHA	Dec
Alpheratz	357° 35.1	29° 13.4
Ankaa	353° 07.5	-42° 10.1
Schedar	349° 31.5	56° 40.0
Diphda	348° 47.7	-17° 51.0
Achernar	335° 20.6	-57° 06.4
Hamal	327° 51.8	23° 34.6
Polaris	314° 30.5	89° 21.8
Acamar	315° 12.3	-40° 12.2
Menkar	314° 06.8	4° 11.2
Mirfak	308° 29.2	49° 56.7
Aldebaran	290° 40.4	16° 33.5
Rigel	281° 04.6	-8° 10.3
Capella	280° 23.0	46° 01.3
Bellatrix	278° 23.6	6° 22.3
Elnath	278° 02.8	28° 37.7
Anilam	275° 38.5	-1° 11.1
Betelgeuse	270° 52.9	7° 24.7
Canopus	263° 53.1	-52° 42.4
Sirius	258° 27.0	-16° 44.9
Adhara	255° 06.6	-29° 00.2
Procyon	244° 51.6	5° 09.8
Pollux	243° 18.2	27° 58.1
Avior	234° 15.5	-59° 35.3
Suhail	222° 47.0	-43° 31.9
Miaplacidus	221° 39.2	-69° 49.2
Alphard	217° 48.4	-8° 45.9
Regulus	207° 35.1	11° 51.0
Dubhe	193° 41.8	61° 37.4
Denebola	182° 25.5	14° 26.2
Gienah	175° 44.1	-17° 40.7
Acrux	173° 00.7	-63° 14.4
Gacrux	171° 52.2	-57° 15.3
Alioth	166° 13.3	55° 49.9
Spica	158° 22.7	-11° 17.4
Alkaid	152° 52.2	49° 11.7
Hadar	148° 36.4	-60° 29.7
Menkent	147° 58.0	-36° 29.6
Arcturus	145° 48.2	19° 03.4
Rigel Kent.	139° 40.6	-60° 56.4
Kochab	137° 19.2	74° 03.5
Zuben 'ubi	136° 56.3	-16° 08.7
Alphecca	126° 03.9	26° 38.1
Antares	112° 16.0	-26° 29.2
Atria	107° 10.1	-69° 04.4
Sabik	102° 02.9	-15° 45.3
Shaula	96° 10.5	-37° 07.3
Rasalhague	95° 58.6	12° 32.6
Eltanin	90° 41.9	51° 29.2
Kaus Aust.	83° 32.7	-34° 22.4
Vega	80° 33.1	38° 48.4
Nunki	75° 47.9	-26° 15.9
Altair	62° 00.0	8° 56.0
Peacock	53° 05.8	-56° 39.2
Deneb	49° 25.7	45° 21.9
Enif	33° 39.0	9° 59.2
Al Na'ir	27° 33.1	-46° 50.3
Fomalhaut	15° 14.8	-29° 29.4
Scheat	13° 45.4	28° 12.8
Markab	13° 30.2	15° 20.2

Jul 05 Fri	SHA	Mer.pass
Venus	246°16.6	12:42
Mars	313°21.8	08:12
Jupiter	292°30.5	09:35
Saturn	8°57.0	04:30

Jul 06 Sat	SHA	Mer.pass
Venus	244°57.5	12:43
Mars	312°39.0	08:11
Jupiter	292°17.0	09:32
Saturn	8°57.4	04:26

Jul 07 Sun	SHA	Mer.pass
Venus	243°38.6	12:44
Mars	311°56.1	08:10
Jupiter	292°03.5	09:29
Saturn	8°57.9	04:22

Horizontal parallax	
Venus:	0.1
Mars:	0.1

2024 July 05 to Jul. 07 UT

IERS Earth Orientation data as of 01-Aug-2024

July 08, 09, 10 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	286°26.4	168°46.3	N22°08.8	237°39.5	N17°09.0	218°16.5	N21°13.2	295°24.9	S06°03.5	Alpheratz	357°35.1 29°13.4	
1	301°28.8	183°45.5	08.3	252°40.2	09.5	233°18.4	13.3	310°27.4	03.5	Ankaa	353°07.5 -42°10.1	
2	316°31.3	198°44.7	07.8	267°40.9	10.0	248°20.3	13.4	325°29.9	03.5	Schedar	349°31.5 56°40.0	
3	331°33.8	213°43.9	•• 07.3	282°41.6	•• 10.5	263°22.2	•• 13.4	340°32.4	•• 03.5	Diphda	348°47.7 -17°51.0	
4	346°36.2	228°43.1	06.8	297°42.2	11.0	278°24.1	13.5	355°34.9	03.5	Achernar	335°20.6 -57°06.4	
5	1°38.7	243°42.3	06.3	312°42.9	11.4	293°26.1	13.6	10°37.3	03.6	Hamal	327°51.8 23°34.6	
6	16°41.1	258°41.5	N22°05.8	327°43.6	N17°11.9	308°28.0	N21°13.7	25°39.8	S06°03.6	Polaris	314°29.1 89°21.7	
7	31°43.6	273°40.7	05.3	342°44.3	12.4	323°29.9	13.8	40°42.3	03.6	Acamar	315°12.3 -40°12.1	
8	46°46.1	288°39.9	04.8	357°45.0	12.9	338°31.8	13.8	55°44.8	03.6	Menkar	314°06.7 4°11.2	
9	61°48.5	303°39.1	•• 04.2	12°45.6	•• 13.4	353°33.7	•• 13.9	70°47.3	•• 03.7	Mirfak	308°29.2 49°56.7	
10	76°51.0	318°38.3	03.7	27°46.3	13.9	8°35.6	14.0	85°49.8	03.7	Aldebaran	290°40.4 16°33.5	
11	91°53.5	333°37.5	03.2	42°47.0	14.4	23°37.5	14.1	100°52.3	03.7	Rigel	281°04.6 -8°10.3	
12	106°55.9	348°36.6	N22°02.7	57°47.7	N17°14.9	38°39.4	N21°14.1	115°54.8	S06°03.7	Capella	280°22.9 46°01.3	
13	121°58.4	3°35.8	02.2	72°48.3	15.4	53°41.3	14.2	130°57.3	03.8	Bellatrix	278°23.6 6°22.4	
14	137°00.9	18°35.0	01.7	87°49.0	15.8	68°43.2	14.3	145°59.8	03.8	Elnath	278°02.7 28°37.7	
15	152°03.3	33°34.2	•• 01.2	102°49.7	•• 16.3	83°45.1	•• 14.4	161°02.3	•• 03.8	Alnilam	275°38.5 -1°11.1	
16	167°05.8	48°33.4	00.6	117°50.4	16.8	98°47.0	14.5	176°04.8	03.8	Betelgeuse	270°52.9 7°24.7	
17	182°08.3	63°32.6	22°00.1	132°51.0	17.3	113°49.0	14.5	191°07.3	03.9	Canopus	263°53.1 -52°42.4	
18	197°10.7	78°31.8	N21°59.6	147°51.7	N17°17.8	128°50.9	N21°14.6	206°09.8	S06°03.9	Sirius	258°27.0 -16°44.9	
19	212°13.2	93°31.0	59.1	162°52.4	18.3	143°52.8	14.7	221°12.3	03.9	Adhara	255°06.6 -29°00.2	
20	227°15.6	108°30.2	58.5	177°53.1	18.8	158°54.7	14.8	236°14.8	03.9	Procyon	244°51.6 5°09.8	
21	242°18.1	123°29.4	•• 58.0	192°53.7	•• 19.2	173°56.6	•• 14.8	251°17.2	•• 04.0	Pollux	243°18.2 27°58.1	
22	257°20.6	138°28.6	57.5	207°54.4	19.7	188°58.5	14.9	266°19.7	04.0	Avior	234°15.6 -59°35.3	
23	272°23.0	153°27.8	57.0	222°55.1	20.2	204°00.4	15.0	281°22.2	04.0	Suhail	222°47.0 -43°31.9	
Mer.pass. 04:53		ν -0.8' d -0.5' m -3.89		ν 0.7' d 0.5' m 0.95		ν 1.9' d 0.1' m -2.04		ν 2.5' d 0.0' m 0.87		Miaplacidus	221°39.2 -69°49.2	
											Alphard	217°48.4 -8°45.9
											Regulus	207°35.1 11°51.0
											Dubhe	193°41.8 61°37.4
											Denebola	182°25.5 14°26.2
											Gienah	175°44.1 -17°40.7
											Acrux	173°00.7 -63°14.4
											Gacrux	171°52.2 -57°15.3
											Alioth	166°13.3 55°49.9
											Spica	158°22.7 -11°17.4
											Alkaid	152°52.2 49°11.7
											Hadar	148°36.4 -60°29.7
											Menkent	147°58.0 -36°29.6
											Arcturus	145°48.2 19°03.4
											Rigel Kent.	139°40.6 -60°56.4
											Kochab	137°19.2 74°03.5
											Zuben'ubi	136°56.3 -16°08.7
											Alphecca	126°03.9 26°38.1
											Antares	112°16.0 -26°29.2
											Atria	107°10.2 -69°04.4
											Sabik	102°02.9 -15°45.3
											Shaula	96°10.5 -37°07.3
											Rasalhague	95°58.6 12°32.6
											Eltanin	90°41.9 51°29.2
											Kaus Aust.	83°32.7 -34°22.4
											Vega	80°33.1 38°48.4
											Nunki	75°47.9 -26°16.0
											Altair	62°00.0 8°56.0
											Peacock	53°05.8 -56°39.3
											Deneb	49°25.7 45°21.9
											Enif	33°38.9 9°59.2
											Al Na'ir	27°33.1 -46°50.3
											Fomalhaut	15°14.8 -29°29.4
											Scheat	13°45.4 28°12.8
											Markab	13°30.1 15°20.2
											Jul 08 Mon	SHA Mer.pass
											Venus	242°20.0 12:46
											Mars	311°13.2 08:09
											Jupiter	291°50.2 09:26
											Saturn	8°58.5 04:18
											Jul 09 Tue	SHA Mer.pass
											Venus	241°01.5 12:47
											Mars	310°30.3 08:08
											Jupiter	291°36.8 09:23
											Saturn	8°59.2 04:14
											Jul 10 Wed	SHA Mer.pass
											Venus	239°43.3 12:48
											Mars	309°47.4 08:07
											Jupiter	291°23.5 09:20
											Saturn	9°00.0 04:10
											Horizontal parallax	
											Venus:	0.1
											Mars:	0.1
Mer.pass. 04:46		ν -0.8' d -0.6' m -3.89		ν 0.7' d 0.5' m 0.94		ν 1.9' d 0.1' m -2.05		ν 2.5' d 0.0' m 0.86				

h	Sun			Moon			
Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	178°43.2	N22°26.4	152°08.6	11.6'	N21°45.2	-10.1'	55.5'
1	193°43.1	26.1	166°39.2	11.7'	21°35.2	-10.2'	55.4'
2	208°43.0	25.8	181°09.9	11.8'	21°25.0	-10.3'	55.4'
3	223°42.9	· · 25.5	195°40.7	11.9'	21°14.7	-10.3'	55.4'
4	238°42.8	25.2	210°11.6	12.0'	21°04.4	-10.4'	55.4'
5	253°42.8	24.9	224°42.6	12.1'	20°54.0	-10.5'	55.4'
6	268°42.7	N22°24.6	239°13.8	12.2'	N20°43.4	-10.6'	55.3'
7	283°42.6	24.3	253°45.0	12.3'	20°32.8	-10.7'	55.3'
8	298°42.5	24.0	268°16.3	12.4'	20°22.1	-10.8'	55.3'
9	313°42.4	· · 23.7	282°47.7	12.5'	20°11.4	-10.9'	55.3'
10	328°42.3	23.4	297°19.2	12.6'	20°00.5	-10.9'	55.2'
11	343°42.2	23.1	311°50.8	12.7'	19°49.6	-11.0'	55.2'
12	358°42.1	N22°22.8	326°22.5	12.8'	N19°38.5	-11.1'	55.2'
13	13°42.0	22.5	340°54.3	12.9'	19°27.4	-11.2'	55.2'
14	28°41.9	22.2	355°26.2	13.0'	19°16.3	-11.3'	55.2'
15	43°41.8	· · 21.9	9°58.2	13.1'	19°05.0	-11.3'	55.1'
16	58°41.7	21.6	24°30.2	13.2'	18°53.7	-11.4'	55.1'
17	73°41.6	21.4	39°02.4	13.3'	18°42.3	-11.5'	55.1'
18	88°41.5	N22°21.0	53°34.7	13.4'	N18°30.8	-11.5'	55.1'
19	103°41.4	20.7	68°07.0	13.4'	18°19.3	-11.6'	55.1'
20	118°41.3	20.4	82°39.5	13.5'	18°07.7	-11.7'	55.0'
21	133°41.2	· · 20.1	97°12.0	13.6'	17°56.0	-11.7'	55.0'
22	148°41.2	19.8	111°44.6	13.7'	17°44.2	-11.8'	55.0'
23	163°41.1	19.5	126°17.3	13.8'	17°32.4	-11.9'	55.0'
SD = 15.7'			SD = 15.1'				

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	178°41.0	N22°19.2	140°50.1	13.9'	N17°20.5	-11.9'	55.0'
1	193°40.9	18.9	155°23.0	14.0'	17°08.6	-12.0'	55.0'
2	208°40.8	18.6	169°56.0	14.1'	16°56.6	-12.1'	54.9'
3	223°40.7	· · 18.3	184°29.0	14.1'	16°44.5	-12.1'	54.9'
4	238°40.6	18.0	199°02.2	14.2'	16°32.4	-12.2'	54.9'
5	253°40.5	17.7	213°35.4	14.3'	16°20.2	-12.2'	54.9'
6	268°40.4	N22°17.4	228°08.7	14.4'	N16°07.9	-12.3'	54.9'
7	283°40.3	17.1	242°42.1	14.5'	15°55.6	-12.4'	54.8'
8	298°40.2	16.8	257°15.5	14.5'	15°43.3	-12.4'	54.8'
9	313°40.1	· · 16.5	271°49.1	14.6'	15°30.9	-12.5'	54.8'
10	328°40.1	16.2	286°22.7	14.7'	15°18.4	-12.5'	54.8'
11	343°40.0	15.8	300°56.4	14.8'	15°05.9	-12.6'	54.8'
12	358°39.9	N22°15.5	315°30.2	14.9'	N14°53.3	-12.6'	54.8'
13	13°39.8	15.2	330°04.0	14.9'	14°40.7	-12.7'	54.7'
14	28°39.7	14.9	344°38.0	15.0'	14°28.0	-12.7'	54.7'
15	43°39.6	· · 14.6	359°12.0	15.1'	14°15.3	-12.8'	54.7'
16	58°39.5	14.3	13°46.0	15.1'	14°02.5	-12.8'	54.7'
17	73°39.4	14.0	28°20.2	15.2'	13°49.7	-12.9'	54.7'
18	88°39.3	N22°13.6	42°54.4	15.3'	N13°36.8	-12.9'	54.7'
19	103°39.2	13.3	57°28.7	15.4'	13°23.9	-13.0'	54.6'
20	118°39.2	13.0	72°03.0	15.4'	13°11.0	-13.0'	54.6'
21	133°39.1	· · 12.7	86°37.5	15.5'	12°58.0	-13.0'	54.6'
22	148°39.0	12.4	101°11.9	15.6'	12°44.9	-13.1'	54.6'
23	163°38.9	12.0	115°46.5	15.6'	12°31.8	-13.1'	54.6'
SD = 15.7'			SD = 15.0'				

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	178°38.8	N22°11.7	130°21.1	15.7'	N12°18.7	-13.2'	54.6'
1	193°38.7	11.4	144°55.8	15.7'	12°05.6	-13.2'	54.6'
2	208°38.6	11.1	159°30.5	15.8'	11°52.4	-13.2'	54.5'
3	223°38.5	· · 10.8	174°05.3	15.9'	11°39.1	-13.3'	54.5'
4	238°38.4	10.4	188°40.2	15.9'	11°25.9	-13.3'	54.5'
5	253°38.4	10.1	203°15.1	16.0'	11°12.5	-13.3'	54.5'
6	268°38.3	N22°09.8	217°50.1	16.0'	N10°59.2	-13.4'	54.5'
7	283°38.2	09.5	232°25.2	16.1'	10°45.8	-13.4'	54.5'
8	298°38.1	09.1	247°00.3	16.1'	10°32.4	-13.4'	54.5'
9	313°38.0	· · 08.8	261°35.4	16.2'	10°19.0	-13.5'	54.5'
10	328°37.9	08.5	276°10.6	16.3'	10°05.5	-13.5'	54.4'
11	343°37.9	08.2	290°45.9	16.3'	09°52.0	-13.5'	54.4'
12	358°37.8	N22°07.8	305°21.2	16.4'	N09°38.5	-13.6'	54.4'
13	13°37.7	07.5	319°56.5	16.4'	09°24.9	-13.6'	54.4'
14	28°37.6	07.2	334°31.9	16.5'	09°11.3	-13.6'	54.4'
15	43°37.5	· · 06.8	349°07.4	16.5'	08°57.7	-13.6'	54.4'
16	58°37.4	06.5	3°42.9	16.5'	08°44.1	-13.7'	54.4'
17	73°37.3	06.2	18°18.4	16.6'	08°30.4	-13.7'	54.4'
18	88°37.3	N22°05.8	32°54.0	16.6'	N08°16.7	-13.7'	54.4'
19	103°37.2	05.5	47°29.7	16.7'	08°03.0	-13.7'	54.3'
20	118°37.1	05.2	62°05.3	16.7'	07°49.3	-13.8'	54.3'
21	133°37.0	· · 04.8	76°41.1	16.8'	07°35.5	-13.8'	54.3'
22	148°36.9	04.5	91°16.8	16.8'	07°21.7	-13.8'	54.3'
23	163°36.8	04.2	105°52.6	16.8'	07°07.9	-13.8'	54.3'
SD = 15.7'			SD = 14.9'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°						
66°	--	--	01:10	22:57	--	--
64°	--	--	02:00	22:08	--	--
62°	--	--	02:31	21:38	--	--
60°	--	01:27	02:54	21:16	22:41	--
N 58°	--	02:04	03:12	20:58	22:05	--
56°	--	02:29	03:27	20:43	21:41	--
54°	01:20	02:49	03:40	20:30	21:21	22:48
52°	01:54	03:05	03:52	20:18	21:05	22:15
50°	02:17	03:19	04:02	20:08	20:51	21:52
45°	02:58	03:46	04:23	19:47	20:24	21:12
N 40°	03:27	04:08	04:40	19:31	20:03	20:43
35°	03:49	04:25	04:54	19:16	19:46	20:22
30°	04:06	04:39	05:06	19:04	19:31	20:04
20°	04:34	05:03	05:27	18:43	19:08	19:37
N 10°	04:55	05:22	05:45	18:26	18:48	19:15
0°	05:13	05:39	06:02	18:09	18:31	18:57
S 10°	05:29	05:56	06:18	17:53	18:15	18:41
20°	05:45	06:12	06:36	17:35	17:59	18:26
30°	06:00	06:30	06:56	17:15	17:41	18:11
35°	06:08	06:40	07:07	17:03	17:31	18:03
40°	06:17	06:51	07:21	16:50	17:20	17:54
45°	06:26	07:03	07:37	16:34	17:08	17:44
S 50°	06:37	07:18	07:56	16:15	16:53	17:34
52°	06:42	07:25	08:05	16:06	16:46	17:29
54°	06:47	07:33	08:15	15:56	16:38	17:24
56°	06:53	07:41	08:27	15:44	16:30	17:18
58°	06:59	07:50	08:40	15:31	16:21	17:12
S 60°	07:06	08:01	08:56	15:15	16:10	17:05

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°		04:26	07:09		01:19	00:06 23:25
N 70°		05:14	07:28		00:29 23:45	23:16
68°	03:05	05:45	07:42	01:01 23:57	23:29	23:08
66°	04:02	06:07	07:54	00:04 23:34	23:15	23:01
64°	04:35	06:25	08:03	23:15	23:04	22:55
62°	04:59	06:39	08:12	23:00	22:55	22:50
60°	05:18	06:51	08:19	22:47	22:47	22:46
N 58°	05:34	07:01	08:25	22:35	22:39	22:42
56°	05:47	07:10	08:30	22:26	22:33	22:39
54°	05:58	07:18	08:35	22:17	22:27	22:35
52°	06:08	07:25	08:39	22:09	22:22	22:33
50°	06:18	07:32	08:43	22:02	22:17	22:30
45°	06:37	07:46	08:52	21:47	22:07	22:24
N 40°	06:52	07:57	08:59	21:34	21:58	22:20
35°	07:05	08:06	09:05	21:24	21:51	22:15
30°	07:16	08:15	09:10	21:14	21:44	22:12
20°	07:35	08:29	09:20	20:58	21:33	22:05
N 10°	07:52	08:41	09:28	20:43	21:23	22:00
0°	08:08	08:53	09:35	20:30	21:13	21:54
S 10°	08:23	09:04	09:42	20:16	21:04	21:49
20°	08:40	09:17	09:50	20:02	20:54	21:43
30°	08:58	09:31	09:59	19:45	20:42	21:36
35°	09:09	09:39	10:04	19:35	20:35	21:33
40°	09:22	09:48	10:10	19:23	20:27	21:28
45°	09:36	09:59	10:17	19:10	20:18	21:23
S 50°	09:54	10:11	10:25	18:54	20:07	21:17
52°	10:03	10:17	10:29	18:46	20:01	21:14
54°	10:12	10:24	10:33	18:37	19:56	21:11
56°	10:23	10:31	10:37	18:27	19:49	21:07
58°	10:34	10:39	10:42	18:16	19:42	21:03
S 60°	10:48	10:49	10:48	18:03	19:34	20:59

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	Time 12 ^h mm:ss	Pass hh:mm	Upper Mer. Pass. hh:mm	Lower Mer. Pass. hh:mm	3-5 5-16%
08	05:07	05:12	12:05	14:19	01:55	
09	05:16	05:21	12:05	15:03	02:42	
10	05:25	05:29	12:05	15:45	03:24	

Aries Venus

<https://pypi.org/project/sfalmanac/>

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	178°36.8	N22°03.8	120°28.4	16.9'	N06°54.1	-13.8'	54.3'
1	193°36.7	03.5	135°04.3	16.9'	06°40.2	-13.9'	54.3'
2	208°36.6	03.2	149°40.2	16.9'	06°26.4	-13.9'	54.3'
3	223°36.5	· · 02.8	164°16.2	17.0'	06°12.5	-13.9'	54.3'
4	238°36.4	02.5	178°52.1	17.0'	05°58.6	-13.9'	54.3'
5	253°36.3	02.1	193°28.1	17.0'	05°44.7	-13.9'	54.3'
6	268°36.3	N22°01.8	208°04.2	17.1'	N05°30.8	-13.9'	54.3'
7	283°36.2	01.5	222°40.2	17.1'	05°16.8	-14.0'	54.2'
8	298°36.1	01.1	237°16.3	17.1'	05°02.9	-14.0'	54.2'
9	313°36.0	· · 00.8	251°52.5	17.1'	04°48.9	-14.0'	54.2'
10	328°35.9	00.4	266°28.6	17.2'	04°34.9	-14.0'	54.2'
11	343°35.9	22°00.1	281°04.8	17.2'	04°20.9	-14.0'	54.2'
12	358°35.8	N21°59.8	295°41.0	17.2'	N04°06.9	-14.0'	54.2'
13	13°35.7	59.4	310°17.2	17.2'	03°52.9	-14.0'	54.2'
14	28°35.6	59.1	324°53.4	17.3'	03°38.8	-14.0'	54.2'
15	43°35.5	· · 58.7	339°29.7	17.3'	03°24.8	-14.0'	54.2'
16	58°35.5	58.4	354°06.0	17.3'	03°10.8	-14.1'	54.2'
17	73°35.4	58.0	8°42.3	17.3'	02°56.7	-14.1'	54.2'
18	88°35.3	N21°57.7	23°18.6	17.3'	N02°42.7	-14.1'	54.2'
19	103°35.2	57.3	37°54.9	17.3'	02°28.6	-14.1'	54.2'
20	118°35.1	57.0	52°31.3	17.4'	02°14.5	-14.1'	54.2'
21	133°35.1	· · 56.6	67°07.6	17.4'	02°00.4	-14.1'	54.2'
22	148°35.0	56.3	81°44.0	17.4'	01°46.3	-14.1'	54.2'
23	163°34.9	55.9	96°20.4	17.4'	01°32.3	-14.1'	54.2'
SD = 15.7'			SD = 14.8'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	178°34.8	N21°55.6	110°56.8	17.4'	N01°18.2	-14.1'	54.2'
1	193°34.7	55.2	125°33.2	17.4'	01°04.1	-14.1'	54.2'
2	208°34.7	54.9	140°09.6	17.4'	00°50.0	-14.1'	54.2'
3	223°34.6	· · 54.5	154°46.0	17.4'	00°35.9	-14.1'	54.2'
4	238°34.5	54.2	169°22.4	17.4'	00°21.8	-14.1'	54.2'
5	253°34.4	53.8	183°58.9	17.4'	N00°07.7	-14.1'	54.2'
6	268°34.4	N21°53.4	198°35.3	17.4'	S00°06.4	14.1'	54.2'
7	283°34.3	53.1	213°11.7	17.4'	00°20.5	14.1'	54.2'
8	298°34.2	52.7	227°48.1	17.4'	00°34.6	14.1'	54.2'
9	313°34.1	· · 52.4	242°24.6	17.4'	00°48.7	14.1'	54.2'
10	328°34.1	52.0	257°01.0	17.4'	01°02.8	14.1'	54.2'
11	343°34.0	51.7	271°37.4	17.4'	01°16.9	14.1'	54.2'
12	358°33.9	N21°51.3	286°13.8	17.4'	S01°30.9	14.1'	54.2'
13	13°33.8	50.9	300°50.3	17.4'	01°45.0	14.1'	54.2'
14	28°33.8	50.6	315°26.7	17.4'	01°59.1	14.1'	54.2'
15	43°33.7	· · 50.2	330°03.1	17.4'	02°13.1	14.1'	54.2'
16	58°33.6	49.9	344°39.5	17.4'	02°27.2	14.0'	54.2'
17	73°33.5	49.5	359°15.8	17.4'	02°41.2	14.0'	54.2'
18	88°33.5	N21°49.1	13°52.2	17.4'	S02°55.3	14.0'	54.2'
19	103°33.4	48.8	28°28.6	17.3'	03°09.3	14.0'	54.2'
20	118°33.3	48.4	43°04.9	17.3'	03°23.3	14.0'	54.2'
21	133°33.2	· · 48.0	57°41.2	17.3'	03°37.3	14.0'	54.2'
22	148°33.2	47.7	72°17.5	17.3'	03°51.3	14.0'	54.2'
23	163°33.1	47.3	86°53.8	17.3'	04°05.3	14.0'	54.2'
SD = 15.7'			SD = 14.8'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	178°33.0	N21°46.9	101°30.1	17.3'	S04°19.3	14.0'	54.2'
1	193°32.9	46.6	116°06.4	17.2'	04°33.3	13.9'	54.2'
2	208°32.9	46.2	130°42.6	17.2'	04°47.2	13.9'	54.2'
3	223°32.8	· · 45.8	145°18.8	17.2'	05°01.1	13.9'	54.2'
4	238°32.7	45.5	159°55.0	17.2'	05°15.1	13.9'	54.2'
5	253°32.7	45.1	174°31.2	17.1'	05°29.0	13.9'	54.2'
6	268°32.6	N21°44.7	189°07.3	17.1'	S05°42.9	13.9'	54.2'
7	283°32.5	44.3	203°43.4	17.1'	05°56.7	13.9'	54.2'
8	298°32.4	44.0	218°19.5	17.1'	06°10.6	13.8'	54.2'
9	313°32.4	· · 43.6	232°55.5	17.0'	06°24.4	13.8'	54.2'
10	328°32.3	43.2	247°31.6	17.0'	06°38.2	13.8'	54.3'
11	343°32.2	42.8	262°07.6	17.0'	06°52.0	13.8'	54.3'
12	358°32.2	N21°42.5	276°43.5	16.9'	S07°05.8	13.8'	54.3'
13	13°32.1	42.1	291°19.4	16.9'	07°19.5	13.7'	54.3'
14	28°32.0	41.7	305°55.3	16.9'	07°33.3	13.7'	54.3'
15	43°31.9	· · 41.3	320°31.2	16.8'	07°47.0	13.7'	54.3'
16	58°31.9	41.0	335°07.0	16.8'	08°00.7	13.7'	54.3'
17	73°31.8	40.6	349°42.8	16.7'	08°14.3	13.6'	54.3'
18	88°31.7	N21°40.2	4°18.5	16.7'	S08°28.0	13.6'	54.3'
19	103°31.7	39.8	18°54.2	16.7'	08°41.6	13.6'	54.3'
20	118°31.6	39.4	33°29.9	16.6'	08°55.2	13.6'	54.3'
21	133°31.5	· · 39.1	48°05.5	16.6'	09°08.7	13.5'	54.4'
22	148°31.5	38.7	62°41.0	16.5'	09°22.3	13.5'	54.4'
23	163°31.4	38.3	77°16.6	16.5'	09°35.8	13.5'	54.4'
SD = 15.7'			SD = 14.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	--	--	01:24	22:44	--	--
64°	--	--	02:08	22:01	--	--
62°	--	00:24	02:37	21:33	23:36	--
60°	--	01:37	02:59	21:11	22:32	--
N 58°	--	02:10	03:16	20:54	22:00	--
56°	00:22	02:34	03:31	20:39	21:36	23:38
54°	01:29	02:53	03:44	20:27	21:17	22:40
52°	02:00	03:09	03:55	20:16	21:02	22:10
50°	02:22	03:22	04:05	20:06	20:48	21:48
45°	03:02	03:49	04:25	19:46	20:22	21:09
N 40°	03:30	04:10	04:42	19:29	20:01	20:41
35°	03:51	04:27	04:56	19:15	19:45	20:20
30°	04:08	04:41	05:08	19:03	19:30	20:03
20°	04:35	05:04	05:28	18:43	19:07	19:36
N 10°	04:56	05:23	05:46	18:26	18:48	19:15
0°	05:14	05:40	06:02	18:09	18:32	18:58
S 10°	05:30	05:56	06:18	17:53	18:16	18:42
20°	05:45	06:12	06:35	17:36	18:00	18:27
30°	06:00	06:29	06:55	17:17	17:43	18:12
35°	06:08	06:39	07:07	17:05	17:33	18:04
40°	06:16	06:50	07:20	16:52	17:22	17:56
45°	06:25	07:02	07:35	16:37	17:10	17:47
S 50°	06:36	07:16	07:54	16:18	16:55	17:36
52°	06:40	07:23	08:03	16:09	16:49	17:32
54°	06:45	07:30	08:13	15:59	16:41	17:27
56°	06:51	07:39	08:24	15:48	16:33	17:21
58°	06:57	07:48	08:37	15:35	16:24	17:15
S 60°	07:03	07:58	08:52	15:20	16:14	17:09

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	09:15	11:11	13:10	22:53	22:21	21:46
N 70°	09:22	11:09	12:58	22:51	22:27	22:00
68°	09:27	11:08	12:49	22:49	22:31	22:12
66°	09:32	11:06	12:42	22:48	22:35	22:21
64°	09:36	11:05	12:35	22:47	22:38	22:29
62°	09:39	11:04	12:30	22:46	22:41	22:36
60°	09:42	11:03	12:25	22:45	22:43	22:42
N 58°	09:45	11:03	12:21	22:44	22:45	22:47
56°	09:47	11:02	12:17	22:43	22:47	22:52
54°	09:49	11:01	12:14	22:42	22:49	22:56
52°	09:51	11:01	12:11	22:42	22:51	23:00
50°	09:52	11:00	12:08	22:41	22:52	23:04
45°	09:56	10:59	12:02	22:40	22:55	23:12
N 40°	09:59	10:58	11:57	22:39	22:58	23:18
35°	10:02	10:57	11:53	22:38	23:00	23:24
30°	10:04	10:57	11:50	22:37	23:03	23:29
20°	10:08	10:56	11:43	22:36	23:06	23:37
N 10°	10:12	10:55	11:38	22:35	23:09	23:45
0°	10:15	10:54	11:33	22:34	23:12	23:52
S 10°	10:18	10:53	11:28	22:32	23:15	23:59
20°	10:22	10:52	11:22	22:31	23:19	· ·
30°	10:26	10:51	11:16	22:30	23:22	· ·
35°	10:28	10:50	11:13	22:29	23:24	· ·
40°	10:30	10:50	11:09	22:28	23:27	· ·
45°	10:33	10:49	11:04	22:27	23:30	· ·
S 50°	10:37	10:48	10:59	22:25	23:33	· ·
52°	10:38	10:47	10:57	22:25	23:34	· ·
54°	10:40	10:47	10:54	22:24	23:36	· ·
56°	10:42	10:47	10:51	22:23	23:38	· ·
58°	10:44	10:46	10:48	22:22	23:40	· ·
S 60°	10:47	10:45	10:44	22:21	23:42	· ·

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss	hh:mm	Mer. Pass. Upper hh:mm	Lower hh:mm	6-8 24-41%
11	05:33	05:37	12:06	16:24	04:05	
12	05:41	05:44	12:06	17:03	04:44	
13	05:48	05:51	12:06	17:42	05:23	

July 14, 15, 16 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	292°21.2	166°54.3	N20°45.2	239°16.8	N18°16.6	222°52.3	N21°24.0	301°25.3	S06°07.4	Alpheratz	357°35.1 29°13.4	
1	307°23.7	181°53.6	44.5	254°17.5	17.0	237°54.2	24.1	316°27.8	07.5	Ankaa	353°07.4 -42°10.1	
2	322°26.1	196°52.9	43.9	269°18.2	17.5	252°56.2	24.1	331°30.3	07.5	Schedar	349°31.4 56°40.1	
3	337°28.6	211°52.1	· · 43.2	284°18.9	· · 17.9	267°58.1	· · 24.2	346°32.8	· · 07.5	Diphda	348°47.6 -17°51.0	
4	352°31.1	226°51.4	42.5	299°19.5	18.4	283°00.0	24.3	1°35.4	07.6	Achernar	335°20.5 -57°06.4	
5	7°33.5	241°50.6	41.9	314°20.2	18.8	298°01.9	24.3	16°37.9	07.6	Hamal	327°51.7 23°34.6	
6	22°36.0	256°49.9	N20°41.2	329°20.9	N18°19.3	313°03.9	N21°24.4	31°40.4	S06°07.6	Polaris	314°26.3 89°21.7	
7	37°38.4	271°49.1	40.6	344°21.6	19.7	328°05.8	24.5	46°42.9	07.7	Acamar	315°12.2 -40°12.1	
8	52°40.9	286°48.4	39.9	359°22.2	20.2	343°07.7	24.6	61°45.4	07.7	Menkar	314°06.7 4°11.2	
9	67°43.4	301°47.7	· · 39.2	14°22.9	· · 20.6	358°09.6	· · 24.6	76°47.9	· · 07.7	Mirfak	308°29.1 49°56.7	
10	82°45.8	316°46.9	38.6	29°23.6	21.0	13°11.6	24.7	91°50.4	07.8	Aldebaran	290°40.3 16°33.5	
11	97°48.3	331°46.2	37.9	44°24.3	21.5	28°13.5	24.8	106°53.0	07.8	Rigel	281°04.5 -8°10.3	
12	112°50.8	346°45.4	N20°37.2	59°24.9	N18°21.9	43°15.4	N21°24.8	121°55.5	S06°07.8	Capella	280°22.9 46°01.3	
13	127°53.2	1°44.7	36.6	74°25.6	22.4	58°17.3	24.9	136°58.0	07.9	Bellatrix	278°23.6 6°22.4	
14	142°55.7	16°44.0	35.9	89°26.3	22.8	73°19.3	25.0	152°00.5	07.9	Elnath	278°02.7 28°37.7	
15	157°58.2	31°43.2	· · 35.2	104°27.0	· · 23.3	88°21.2	· · 25.1	167°03.0	· · 07.9	Alnilam	275°38.4 -1°11.1	
16	173°00.6	46°42.5	34.5	119°27.7	23.7	103°23.1	25.1	182°05.5	08.0	Betelgeuse	270°52.8 7°24.8	
17	188°03.1	61°41.8	33.9	134°28.3	24.1	118°25.0	25.2	197°08.1	08.0	Canopus	263°53.1 -52°42.3	
18	203°05.5	76°41.0	N20°33.2	149°29.0	N18°24.6	133°27.0	N21°25.3	212°10.6	S06°08.0	Sirius	258°26.9 -16°44.9	
19	218°08.0	91°40.3	32.5	164°29.7	25.0	148°28.9	25.3	227°13.1	08.1	Adhara	255°06.6 -29°00.2	
20	233°10.5	106°39.5	31.8	179°30.4	25.5	163°30.8	25.4	242°15.6	08.1	Procyon	244°51.6 5°09.8	
21	248°12.9	121°38.8	· · 31.2	194°31.0	· · 25.9	178°32.7	· · 25.5	257°18.1	· · 08.1	Pollux	243°18.2 27°58.1	
22	263°15.4	136°38.1	30.5	209°31.7	26.4	193°34.7	25.5	272°20.6	08.2	Avior	234°15.6 -59°35.3	
23	278°17.9	151°37.3	29.8	224°32.4	26.8	208°36.6	25.6	287°23.2	08.2	Suhail	222°47.0 -43°31.9	
Mer.pass. 04:30		ν -0.7' d -0.7' m -3.89		ν 0.7' d 0.4' m 0.93		ν 1.9' d 0.1' m -2.06		ν 2.5' d 0.0' m 0.84		Miaplacidus	221°39.3 -69°49.1	
											Alphard	217°48.4 -8°45.8
											Regulus	207°35.1 11°51.0
											Dubhe	193°41.9 61°37.4
											Denebola	182°25.5 14°26.2
											Gienah	175°44.1 -17°40.7
											Acrux	173°00.8 -63°14.4
											Gacrux	171°52.3 -59°15.3
											Alioth	166°13.4 55°49.9
											Spica	158°22.7 -11°17.4
											Alkaid	152°52.3 49°11.7
											Hadar	148°36.5 -60°29.7
											Menkent	147°58.0 -36°29.6
											Arcturus	145°48.2 19°03.4
											Rigel Kent.	139°40.7 -60°56.4
											Kochab	137°19.3 74°03.5
											Zuben'ubi	136°56.3 -16°08.7
											Alphecca	126°03.9 26°38.1
											Antares	112°16.1 -26°29.2
											Atria	107°10.2 -69°04.4
											Sabik	102°03.0 -15°45.3
											Shaula	96°10.5 -37°07.3
											Rasalhague	95°58.6 12°32.6
											Eltanin	90°41.9 51°29.2
											Kaus Aust.	83°32.7 -34°22.4
											Vega	80°33.1 38°48.4
											Nunki	75°47.9 -26°16.0
											Altair	62°00.0 8°56.0
											Peacock	53°05.8 -56°39.3
											Deneb	49°25.6 45°22.0
											Enif	33°38.9 9°59.2
											Al Na'ir	27°33.0 -46°50.4
											Fomalhaut	15°14.7 -29°29.4
											Scheat	13°45.4 28°12.8
											Markab	13°30.1 15°20.2
											Jul 14 Sun	SHA Mer.pass
											Venus	234°33.1 12:53
											Mars	306°55.7 08:03
											Jupiter	290°31.1 09:07
											Saturn	9°04.1 03:54
											Jul 15 Mon	SHA Mer.pass
											Venus	233°16.3 12:54
											Mars	306°12.7 08:01
											Jupiter	290°18.2 09:04
											Saturn	9°05.3 03:50
											Jul 16 Tue	SHA Mer.pass
											Venus	231°59.7 12:55
											Mars	305°29.8 08:00
											Jupiter	290°05.3 09:01
											Saturn	9°06.7 03:46
											Horizontal parallax	
											Venus:	0.1
											Mars:	0.1
Mer.pass. 04:22		ν -0.7' d -0.7' m -3.88		ν 0.7' d 0.4' m 0.92		ν 1.9' d 0.1' m -2.06		ν 2.5' d 0.0' m 0.83				

h	Sun			Moon			
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	178°31.3	N21°37.9	91°52.0	16.4'	S09°49.2	13.4'	54.4'
1	193°31.3	37.5	106°27.5	16.4'	10°02.7	13.4'	54.4'
2	208°31.2	37.1	121°02.8	16.3'	10°16.1	13.4'	54.4'
3	223°31.1	· · 36.8	135°38.2	16.3'	10°29.5	13.3'	54.4'
4	238°31.1	36.4	150°13.4	16.2'	10°42.8	13.3'	54.4'
5	253°31.0	36.0	164°48.7	16.2'	10°56.1	13.3'	54.4'
6	268°30.9	N21°35.6	179°23.8	16.1'	S11°09.4	13.2'	54.5'
7	283°30.9	35.2	193°58.9	16.1'	11°22.7	13.2'	54.5'
8	298°30.8	34.8	208°34.0	16.0'	11°35.9	13.2'	54.5'
9	313°30.7	· · 34.4	223°09.0	15.9'	11°49.1	13.1'	54.5'
10	328°30.7	34.1	237°43.9	15.9'	12°02.2	13.1'	54.5'
11	343°30.6	33.7	252°18.8	15.8'	12°15.3	13.1'	54.5'
12	358°30.5	N21°33.3	266°53.6	15.8'	S12°28.4	13.0'	54.5'
13	13°30.5	32.9	281°28.4	15.7'	12°41.4	13.0'	54.6'
14	28°30.4	32.5	296°03.0	15.6'	12°54.4	12.9'	54.6'
15	43°30.3	· · 32.1	310°37.7	15.6'	13°07.3	12.9'	54.6'
16	58°30.3	31.7	325°12.2	15.5'	13°20.2	12.9'	54.6'
17	73°30.2	31.3	339°46.7	15.4'	13°33.1	12.8'	54.6'
18	88°30.1	N21°30.9	354°21.1	15.4'	S13°45.9	12.8'	54.6'
19	103°30.1	30.5	8°55.5	15.3'	13°58.7	12.7'	54.7'
20	118°30.0	30.1	23°29.8	15.2'	14°11.4	12.7'	54.7'
21	133°30.0	· · 29.7	38°04.0	15.1'	14°24.1	12.6'	54.7'
22	148°29.9	29.3	52°38.1	15.1'	14°36.7	12.6'	54.7'
23	163°29.8	28.9	67°12.2	15.0'	14°49.3	12.5'	54.7'
SD = 15.7'			SD = 14.8'				

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	178°29.8	N21°28.5	81°46.2	14.9'	S15°01.8	12.5'	54.8'
1	193°29.7	28.1	96°20.1	14.8'	15°14.3	12.4'	54.8'
2	208°29.6	27.7	110°53.9	14.8'	15°26.7	12.4'	54.8'
3	223°29.6	· · 27.3	125°27.7	14.7'	15°39.1	12.3'	54.8'
4	238°29.5	26.9	140°01.4	14.6'	15°51.4	12.3'	54.8'
5	253°29.5	26.5	154°35.0	14.5'	16°03.7	12.2'	54.8'
6	268°29.4	N21°26.1	169°08.5	14.4'	S16°15.9	12.2'	54.9'
7	283°29.3	25.7	183°41.9	14.4'	16°28.0	12.1'	54.9'
8	298°29.3	25.3	198°15.3	14.3'	16°40.1	12.0'	54.9'
9	313°29.2	· · 24.9	212°48.5	14.2'	16°52.2	12.0'	54.9'
10	328°29.2	24.5	227°21.7	14.1'	17°04.2	11.9'	55.0'
11	343°29.1	24.1	241°54.8	14.0'	17°16.1	11.9'	55.0'
12	358°29.0	N21°23.7	256°27.8	13.9'	S17°28.0	11.8'	55.0'
13	13°29.0	23.3	271°00.7	13.8'	17°39.7	11.7'	55.0'
14	28°28.9	22.9	285°33.6	13.7'	17°51.5	11.7'	55.0'
15	43°28.9	· · 22.5	300°06.3	13.6'	18°03.2	11.6'	55.1'
16	58°28.8	22.1	314°39.0	13.6'	18°14.8	11.5'	55.1'
17	73°28.7	21.7	329°11.5	13.5'	18°26.3	11.5'	55.1'
18	88°28.7	N21°21.3	343°44.0	13.4'	S18°37.8	11.4'	55.1'
19	103°28.6	20.8	358°16.3	13.3'	18°49.2	11.3'	55.2'
20	118°28.6	20.4	12°48.6	13.2'	19°00.5	11.3'	55.2'
21	133°28.5	· · 20.0	27°20.8	13.1'	19°11.7	11.2'	55.2'
22	148°28.5	19.6	41°52.9	13.0'	19°22.9	11.1'	55.2'
23	163°28.4	19.2	56°24.8	12.9'	19°34.0	11.0'	55.2'
SD = 15.7'			SD = 14.9'				

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	178°28.3	N21°18.8	70°56.7	12.8'	S19°45.1	11.0'	55.3'
1	193°28.3	18.4	85°28.5	12.7'	19°56.0	10.9'	55.3'
2	208°28.2	18.0	100°00.2	12.6'	20°06.9	10.8'	55.3'
3	223°28.2	· · 17.5	114°31.8	12.5'	20°17.7	10.7'	55.3'
4	238°28.1	17.1	129°03.2	12.4'	20°28.4	10.6'	55.4'
5	253°28.1	16.7	143°34.6	12.3'	20°39.1	10.6'	55.4'
6	268°28.0	N21°16.3	158°05.9	12.2'	S20°49.6	10.5'	55.4'
7	283°28.0	15.9	172°37.1	12.1'	21°00.1	10.4'	55.5'
8	298°27.9	15.5	187°08.1	12.0'	21°10.5	10.3'	55.5'
9	313°27.8	· · 15.0	201°39.1	11.9'	21°20.8	10.2'	55.5'
10	328°27.8	14.6	216°09.9	11.7'	21°31.0	10.1'	55.5'
11	343°27.7	14.2	230°40.7	11.6'	21°41.1	10.0'	55.6'
12	358°27.7	N21°13.8	245°11.3	11.5'	S21°51.1	9.9'	55.6'
13	13°27.6	13.4	259°41.8	11.4'	22°01.1	9.8'	55.6'
14	28°27.6	12.9	274°12.3	11.3'	22°10.9	9.8'	55.6'
15	43°27.5	· · 12.5	288°42.6	11.2'	22°20.7	9.7'	55.7'
16	58°27.5	12.1	303°12.8	11.1'	22°30.3	9.6'	55.7'
17	73°27.4	11.7	317°42.9	11.0'	22°39.9	9.5'	55.7'
18	88°27.4	N21°11.2	332°12.8	10.9'	S22°49.3	9.4'	55.8'
19	103°27.3	10.8	346°42.7	10.8'	22°58.7	9.3'	55.8'
20	118°27.3	10.4	1°12.5	10.6'	23°08.0	9.2'	55.8'
21	133°27.2	· · 10.0	15°42.1	10.5'	23°17.1	9.1'	55.8'
22	148°27.2	09.5	30°11.7	10.4'	23°26.2	8.9'	55.9'
23	163°27.1	09.1	44°41.1	10.3'	23°35.1	8.8'	55.9'
SD = 15.7'			SD = 15.1'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	—:—	—:—	01:37	22:31	—:—	—:—
64°	—:—	—:—	02:17	21:53	—:—	—:—
62°	—:—	00:53	02:44	21:27	23:13	—:—
60°	—:—	01:46	03:04	21:06	22:23	—:—
N 58°	—:—	02:17	03:21	20:50	21:53	—:—
56°	00:49	02:40	03:36	20:36	21:31	23:18
54°	01:38	02:58	03:48	20:23	21:13	22:32
52°	02:06	03:13	03:59	20:13	20:58	22:04
50°	02:27	03:26	04:08	20:03	20:45	21:43
45°	03:05	03:52	04:28	19:44	20:19	21:06
N 40°	03:32	04:12	04:44	19:28	19:59	20:39
35°	03:53	04:29	04:58	19:14	19:43	20:18
30°	04:10	04:43	05:09	19:02	19:29	20:02
20°	04:37	05:05	05:29	18:43	19:07	19:35
N 10°	04:57	05:24	05:47	18:26	18:48	19:15
0°	05:14	05:40	06:02	18:10	18:32	18:58
S 10°	05:30	05:56	06:18	17:54	18:16	18:42
20°	05:44	06:11	06:35	17:37	18:01	18:28
30°	05:59	06:28	06:54	17:18	17:44	18:13
35°	06:07	06:38	07:05	17:07	17:35	18:06
40°	06:15	06:48	07:18	16:54	17:24	17:58
45°	06:24	07:00	07:33	16:39	17:12	17:49
S 50°	06:34	07:14	07:51	16:21	16:58	17:39
52°	06:38	07:21	08:00	16:13	16:52	17:34
54°	06:43	07:28	08:10	16:03	16:45	17:30
56°	06:48	07:36	08:21	15:52	16:37	17:24
58°	06:54	07:44	08:33	15:40	16:28	17:19
S 60°	07:00	07:54	08:48	15:25	16:18	17:13

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	15:25	■	■	20:57	■	■
N 70°	14:58	17:37	■	21:26	20:18	■
68°	14:38	16:46	■	21:48	21:10	■
66°	14:22	16:15	18:44	22:05	21:42	20:52
64°	14:09	15:52	17:51	22:19	22:06	21:46
62°	13:59	15:34	17:19	22:31	22:25	22:19
60°	13:50	15:19	16:56	22:41	22:41	22:43
N 58°	13:42	15:07	16:37	22:50	22:55	23:03
56°	13:35	14:56	16:21	22:58	23:06	23:19
54°	13:28	14:46	16:08	23:05	23:16	23:33
52°	13:23	14:38	15:56	23:11	23:26	23:45
50°	13:18	14:30	15:46	23:17	23:34	23:56
45°	13:07	14:14	15:24	23:30	23:51	· ·
N 40°	12:58	14:01	15:07	23:40	· ·	00:06
35°	12:51	13:50	14:52	23:49	· ·	00:18
30°	12:44	13:40	14:40	23:57	· ·	00:29
20°	12:32	13:24	14:19	· ·	00:10	00:47
N 10°	12:22	13:10	14:00	· ·	00:22	01:03
0°	12:13	12:56	13:43	· ·	00:33	01:18
S 10°	12:04	12:43	13:26	· ·	00:45	01:33
20°	11:54	12:29	13:08	00:07	00:57	01:50
30°	11:43	12:13	12:47	00:16	01:11	02:09
35°	11:37	12:04	12:35	00:21	01:19	02:20
40°	11:30	11:53	12:22	00:27	01:28	02:32
45°	11:21	11:41	12:05	00:33	01:39	02:47
S 50°	11:11	11:26	11:46	00:41	01:52	03:06
52°	11:07	11:19	11:36	00:45	01:58	03:15
54°	11:02	11:12	11:26	00:49	02:05	03:25
56°	10:56	11:03	11:14	00:54	02:13	03:36
58°	10:50	10:54	11:00	00:59	02:22	03:49
S 60°	10:43	10:43	10:44	01:05	02:32	04:04

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12 ^h		Mer. Pass.		
	00 ^h	mm:ss	hh:mm	Upper	Lower	9-11
				hh:mm	hh:mm	51-69%
14	05:55	05:58	12:06	18:23	06:02	
15	06:01	06:04	12:06	19:07	06:45	
16	06:07	06:09	12:06	19:55	07:31	

July 17, 18, 19 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	295° 18.6	166° 02.0	N19° 55.3	240° 05.5	N18° 48.0	225° 11.1	N21° 29.0	304° 26.7	S06° 09.9	Alpheratz	357° 35.0	29° 13.4
1	310° 21.1	181° 01.3	54.5	255° 06.1	48.4	240° 13.1	29.1	319° 29.2	09.9	Ankaa	353° 07.4	-42° 10.1
2	325° 23.5	196° 00.6	53.8	270° 06.8	48.8	255° 15.0	29.1	334° 31.8	10.0	Schedar	349° 31.4	56° 40.1
3	340° 26.0	210° 59.9	•• 53.1	285° 07.5	•• 49.3	270° 16.9	•• 29.2	349° 34.3	•• 10.0	Diphda	348° 47.6	-17° 51.0
4	355° 28.5	225° 59.2	52.3	300° 08.2	49.7	285° 18.9	29.3	4° 36.8	10.1	Achernar	335° 20.5	-57° 06.4
5	10° 30.9	240° 58.4	51.6	315° 08.8	50.1	300° 20.8	29.3	19° 39.3	10.1	Hamal	327° 51.7	23° 34.6
6	25° 33.4	255° 57.7	N19° 50.9	330° 09.5	N18° 50.5	315° 22.7	N21° 29.4	34° 41.9	S06° 10.1	Polaris	314° 24.6	89° 21.7
7	40° 35.9	270° 57.0	50.1	345° 10.2	50.9	330° 24.7	29.5	49° 44.4	10.2	Acamar	315° 12.2	-40° 12.1
8	55° 38.3	285° 56.3	49.4	0° 10.9	51.4	345° 26.6	29.5	64° 46.9	10.2	Menkar	314° 06.7	4° 11.2
9	70° 40.8	300° 55.6	•• 48.7	15° 11.5	•• 51.8	0° 28.5	•• 29.6	79° 49.4	•• 10.2	Mirfak	308° 29.1	49° 56.7
10	85° 43.3	315° 54.9	47.9	30° 12.2	52.2	15° 30.5	29.7	94° 52.0	10.3	Aldebaran	290° 40.3	16° 33.5
11	100° 45.7	330° 54.2	47.2	45° 12.9	52.6	30° 32.4	29.7	109° 54.5	10.3	Rigel	281° 04.5	-8° 10.3
12	115° 48.2	345° 53.5	N19° 46.5	60° 13.6	N18° 53.1	45° 34.3	N21° 29.8	124° 57.0	S06° 10.4	Capella	280° 22.9	46° 01.3
13	130° 50.6	0° 52.8	45.7	75° 14.2	53.5	60° 36.3	29.9	139° 59.5	10.4	Bellatrix	278° 23.6	6° 22.4
14	145° 53.1	15° 52.1	45.0	90° 14.9	53.9	75° 38.2	29.9	155° 02.1	10.4	Elnath	278° 02.7	28° 37.7
15	160° 55.6	30° 51.4	•• 44.2	105° 15.6	•• 54.3	90° 40.1	•• 30.0	170° 04.6	•• 10.5	Alnilam	275° 38.4	-1° 11.1
16	175° 58.0	45° 50.7	43.5	120° 16.3	54.7	105° 42.1	30.1	185° 07.1	10.5	Betelgeuse	270° 52.8	7° 24.8
17	191° 00.5	60° 50.0	42.7	135° 16.9	55.2	120° 44.0	30.1	200° 09.6	10.5	Canopus	263° 53.1	-52° 42.3
18	206° 03.0	75° 49.3	N19° 42.0	150° 17.6	N18° 55.6	135° 45.9	N21° 30.2	215° 12.2	S06° 10.6	Sirius	258° 26.9	-16° 44.9
19	221° 05.4	90° 48.6	41.3	165° 18.3	56.0	150° 47.9	30.3	230° 14.7	10.6	Adhara	255° 06.6	-29° 00.2
20	236° 07.9	105° 47.9	40.5	180° 19.0	56.4	165° 49.8	30.3	245° 17.2	10.7	Procyon	244° 51.6	5° 09.8
21	251° 10.4	120° 47.2	•• 39.8	195° 19.6	•• 56.8	180° 51.7	•• 30.4	260° 19.8	•• 10.7	Pollux	243° 18.2	27° 58.1
22	266° 12.8	135° 46.5	39.0	210° 20.3	57.3	195° 53.7	30.5	275° 22.3	10.7	Avior	234° 15.6	-59° 35.2
23	281° 15.3	150° 45.8	38.3	225° 21.0	57.7	210° 55.6	30.5	290° 24.8	10.8	Suhail	222° 47.0	-43° 31.9
Mer.pass. 04:18		ν -0.7' d -0.7' m -3.88		ν 0.7' d 0.4' m 0.92		ν 1.9' d 0.1' m -2.07		ν 2.5' d 0.0' m 0.83		Miaplacidus	221° 39.3	-69° 49.1
Alphard	217° 48.4									Regulus	207° 35.1	11° 51.0
Dubhe	193° 41.9									Denebola	182° 25.5	14° 26.2
Gienah	175° 44.1									Acrux	173° 00.8	-63° 14.3
Gacrux	171° 52.3									Alioth	166° 13.4	55° 49.9
Spica	158° 22.7									Alkaid	152° 52.3	49° 11.7
Hadar	148° 36.5									Menkent	147° 58.0	-36° 29.6
Arcturus	145° 48.2									Rigil Kent.	139° 40.7	-60° 56.4
Kochab	137° 19.4									Zuben'ubi	136° 56.3	-16° 08.7
Alphecca	126° 03.9									Antares	112° 16.1	-26° 29.2
Atria	107° 10.2									Sabik	102° 03.0	-15° 45.3
Shaula	96° 10.5									Rasalhague	95° 58.6	12° 32.6
Eltanin	90° 41.9									Kaus Aust.	83° 32.6	-34° 22.4
Vega	80° 33.1									Nunki	75° 47.9	-26° 16.0
Altair	62° 00.0									Peacock	53° 05.7	-56° 39.3
Deneb	49° 25.6									Enif	33° 38.9	9° 59.2
Al Na'ir	27° 33.0									Fomalhaut	15° 14.7	-29° 29.4
Scheat	13° 45.4									Markab	13° 30.1	15° 20.2
Jul 17 Wed		SHA		Mer.pass								
Venus	230° 43.4	12:56										
Mars	304° 46.8	07:59										
Jupiter	289° 52.5	08:58										
Saturn	9° 08.1	03:42										
Jul 18 Thu		SHA		Mer.pass								
Venus	229° 27.4	12:58										
Mars	304° 03.9	07:58										
Jupiter	289° 39.8	08:55										
Saturn	9° 09.6	03:38										
Jul 19 Fri		SHA		Mer.pass								
Venus	228° 11.6	12:59										
Mars	303° 21.0	07:57										
Jupiter	289° 27.1	08:52										
Saturn	9° 11.2	03:34										
Horizontal parallax												
Venus:										0.1		
Mars:										0.1		

h	Sun			Moon			
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	178°27.1	N21°08.7	59°10.4	10.2'	S23°44.0	8.7'	55.9'
1	193°27.0	08.2	73°39.6	10.1'	23°52.7	8.6'	56.0'
2	208°27.0	07.8	88°08.7	10.0'	24°01.3	8.5'	56.0'
3	223°26.9	· · 07.4	102°37.7	9.9'	24°09.8	8.4'	56.0'
4	238°26.9	07.0	117°06.5	9.8'	24°18.2	8.3'	56.0'
5	253°26.8	06.5	131°35.3	9.6'	24°26.5	8.2'	56.1'
6	268°26.8	N21°06.1	146°03.9	9.5'	S24°34.7	8.1'	56.1'
7	283°26.7	05.7	160°32.5	9.4'	24°42.7	7.9'	56.1'
8	298°26.7	05.2	175°00.9	9.3'	24°50.7	7.8'	56.2'
9	313°26.6	· · 04.8	189°29.2	9.2'	24°58.5	7.7'	56.2'
10	328°26.6	04.4	203°57.4	9.1'	25°06.2	7.6'	56.2'
11	343°26.5	03.9	218°25.4	9.0'	25°13.7	7.4'	56.3'
12	358°26.5	N21°03.5	232°53.4	8.9'	S25°21.2	7.3'	56.3'
13	13°26.4	03.1	247°21.3	8.7'	25°28.5	7.2'	56.3'
14	28°26.4	02.6	261°49.0	8.6'	25°35.7	7.1'	56.4'
15	43°26.3	· · 02.2	276°16.6	8.5'	25°42.7	6.9'	56.4'
16	58°26.3	01.7	290°44.2	8.4'	25°49.7	6.8'	56.4'
17	73°26.2	01.3	305°11.6	8.3'	25°56.5	6.7'	56.5'
18	88°26.2	N21°00.9	319°38.9	8.2'	S26°03.1	6.5'	56.5'
19	103°26.1	00.4	334°06.1	8.1'	26°09.7	6.4'	56.5'
20	118°26.1	21°00.0	348°33.2	8.0'	26°16.1	6.3'	56.5'
21	133°26.0	20°59.5	3°00.1	7.9'	26°22.3	6.1'	56.6'
22	148°26.0	59.1	17°27.0	7.8'	26°28.4	6.0'	56.6'
23	163°25.9	58.7	31°53.8	7.7'	26°34.4	5.8'	56.6'
SD = 15.7'			SD = 15.3'				

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	178°25.9	N20°58.2	46°20.4	7.6'	S26°40.2	5.7'	56.7'
1	193°25.9	57.8	60°47.0	7.5'	26°45.9	5.5'	56.7'
2	208°25.8	57.3	75°13.5	7.4'	26°51.5	5.4'	56.7'
3	223°25.8	· · 56.9	89°39.8	7.2'	26°56.9	5.3'	56.8'
4	238°25.7	56.4	104°06.1	7.1'	27°02.1	5.1'	56.8'
5	253°25.7	56.0	118°32.2	7.0'	27°07.2	5.0'	56.8'
6	268°25.6	N20°55.5	132°58.3	7.0'	S27°12.2	4.8'	56.9'
7	283°25.6	55.1	147°24.2	6.9'	27°17.0	4.7'	56.9'
8	298°25.5	54.6	161°50.1	6.8'	27°21.7	4.5'	56.9'
9	313°25.5	· · 54.2	176°15.8	6.7'	27°26.2	4.3'	57.0'
10	328°25.5	53.7	190°41.5	6.6'	27°30.5	4.2'	57.0'
11	343°25.4	53.3	205°07.1	6.5'	27°34.7	4.0'	57.0'
12	358°25.4	N20°52.8	219°32.5	6.4'	S27°38.7	3.9'	57.1'
13	13°25.3	52.4	233°57.9	6.3'	27°42.6	3.7'	57.1'
14	28°25.3	51.9	248°23.2	6.2'	27°46.3	3.5'	57.1'
15	43°25.2	· · 51.5	262°48.4	6.1'	27°49.8	3.4'	57.2'
16	58°25.2	51.0	277°13.6	6.0'	27°53.2	3.2'	57.2'
17	73°25.2	50.6	291°38.6	6.0'	27°56.4	3.1'	57.2'
18	88°25.1	N20°50.1	306°03.6	5.9'	S27°59.5	2.9'	57.3'
19	103°25.1	49.7	320°28.4	5.8'	28°02.4	2.7'	57.3'
20	118°25.0	49.2	334°53.2	5.7'	28°05.1	2.5'	57.3'
21	133°25.0	· · 48.8	349°18.0	5.6'	28°07.6	2.4'	57.4'
22	148°25.0	48.3	3°42.6	5.6'	28°10.0	2.2'	57.4'
23	163°24.9	47.9	18°07.2	5.5'	28°12.2	2.0'	57.4'
SD = 15.7'			SD = 15.5'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	178°24.9	N20°47.4	32°31.7	5.4'	S28°14.2	1.9'	57.5'
1	193°24.8	46.9	46°56.1	5.4'	28°16.1	1.7'	57.5'
2	208°24.8	46.5	61°20.5	5.3'	28°17.8	1.5'	57.5'
3	223°24.8	· · 46.0	75°44.8	5.2'	28°19.3	1.3'	57.6'
4	238°24.7	45.6	90°09.0	5.2'	28°20.6	1.2'	57.6'
5	253°24.7	45.1	104°33.2	5.1'	28°21.8	1.0'	57.6'
6	268°24.7	N20°44.6	118°57.3	5.1'	S28°22.8	0.8'	57.7'
7	283°24.6	44.2	133°21.4	5.0'	28°23.6	0.6'	57.7'
8	298°24.6	43.7	147°45.4	5.0'	28°24.2	0.4'	57.7'
9	313°24.5	· · 43.2	162°09.3	4.9'	28°24.7	0.3'	57.8'
10	328°24.5	42.8	176°33.2	4.9'	28°24.9	0.1'	57.8'
11	343°24.5	42.3	190°57.1	4.8'	28°25.0	-0.1'	57.8'
12	358°24.4	N20°41.9	205°20.9	4.8'	S28°24.9	-0.3'	57.9'
13	13°24.4	41.4	219°44.7	4.7'	28°24.6	-0.5'	57.9'
14	28°24.4	40.9	234°08.4	4.7'	28°24.2	-0.6'	57.9'
15	43°24.3	· · 40.5	248°32.1	4.7'	28°23.5	-0.8'	58.0'
16	58°24.3	40.0	262°55.7	4.6'	28°22.7	-1.0'	58.0'
17	73°24.3	39.5	277°19.4	4.6'	28°21.7	-1.2'	58.0'
18	88°24.2	N20°39.1	291°42.9	4.6'	S28°20.5	-1.4'	58.1'
19	103°24.2	38.6	306°06.5	4.5'	28°19.1	-1.6'	58.1'
20	118°24.1	38.1	320°30.0	4.5'	28°17.5	-1.8'	58.1'
21	133°24.1	· · 37.6	334°53.5	4.5'	28°15.7	-1.9'	58.2'
22	148°24.1	37.2	349°17.0	4.5'	28°13.8	-2.1'	58.2'
23	163°24.0	36.7	3°40.5	4.4'	28°11.7	-2.3'	58.2'
SD = 15.7'			SD = 15.7'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	—:—	—:—	00:39	23:22	—:—	—:—
66°	—:—	—:—	01:50	22:19	—:—	—:—
64°	—:—	—:—	02:26	21:44	—:—	—:—
62°	—:—	01:12	02:51	21:20	22:55	—:—
60°	—:—	01:56	03:10	21:01	22:14	—:—
N 58°	—:—	02:24	03:27	20:45	21:46	—:—
56°	01:06	02:46	03:40	20:31	21:25	23:02
54°	01:47	03:03	03:52	20:20	21:08	22:24
52°	02:13	03:18	04:02	20:09	20:54	21:58
50°	02:33	03:30	04:11	20:00	20:42	21:38
45°	03:09	03:55	04:31	19:41	20:17	21:02
N 40°	03:35	04:15	04:46	19:26	19:57	20:36
35°	03:56	04:31	05:00	19:13	19:41	20:16
30°	04:12	04:44	05:11	19:01	19:28	20:00
20°	04:38	05:06	05:30	18:42	19:06	19:34
N 10°	04:58	05:25	05:47	18:25	18:48	19:14
0°	05:15	05:41	06:03	18:10	18:32	18:58
S 10°	05:30	05:56	06:18	17:55	18:17	18:43
20°	05:44	06:11	06:34	17:38	18:02	18:29
30°	05:58	06:27	06:53	17:20	17:46	18:15
35°	06:05	06:36	07:04	17:09	17:36	18:07
40°	06:13	06:47	07:16	16:57	17:26	18:00
45°	06:22	06:58	07:31	16:42	17:15	17:51
S 50°	06:31	07:12	07:48	16:25	17:01	17:42
52°	06:36	07:18	07:57	16:16	16:55	17:37
54°	06:40	07:25	08:06	16:07	16:48	17:33
56°	06:45	07:32	08:17	15:56	16:41	17:28
58°	06:51	07:41	08:29	15:44	16:32	17:23
S 60°	06:56	07:50	08:43	15:30	16:23	17:17

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	☐	☐	☐	☐	☐	☐
64°	☐	☐	☐	☐	☐	☐
62°	19:18	☐	☐	22:09	☐	☐
60°	18:37	20:15	21:19	22:50	23:12	· · ·
N 58°	18:09	19:36	20:40	23:18	23:50	· · ·
56°	17:48	19:09	20:12	23:40	· · ·	00:18
54°	17:30	18:48	19:50	23:58	· · ·	00:39
52°	17:15	18:30	19:33	· · ·	00:14	00:57
50°	17:02	18:15	19:17	· · ·	00:27	01:12
45°	16:35	17:44	18:47	00:19	00:55	01:43
N 40°	16:14	17:21	18:23	00:37	01:16	02:07
35°	15:57	17:01	18:03	00:52	01:35	02:26
30°	15:42	16:45	17:46	01:06	01:50	02:43
20°	15:16	16:17	17:17	01:29	02:17	03:12
N 10°	14:55	15:53	16:53	01:49	02:39	03:36
0°	14:34	15:31	16:30	02:07	03:01	03:59
S 10°	14:14	15:08	16:07	02:26	03:22	04:21
20°	13:53	14:45	15:43	02:46	03:45	04:46
30°	13:28	14:17	15:14	03:09	04:12	05:14
35°	13:14	14:01	14:58	03:23	04:28	05:31
40°	12:57	13:42	14:38	03:39	04:46	05:51
45°	12:37	13:19	14:14	03:58	05:09	06:15
S 50°	12:12	12:50	13:44	04:22	05:37	06:45
52°	12:00	12:36	13:28	04:34	05:51	07:00
54°	11:47	12:19	13:11	04:47	06:07	07:18
56°	11:31	12:00	12:49	05:02	06:27	07:40
58°	11:12	11:35	12:22	05:20	06:51	08:07
S 60°	10:49	11:03	11:43	05:43	07:23	08:46

Day	Sun		Mer. Pass hh:mm	Moon		Age 12-14 78-93%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper Mer.Pass. hh:mm	Lower Mer.Pass. hh:mm	
17	06:12	06:14	12:06	20:48	08:21	
18	06:16	06:19	12:06	21:45	09:16	
19	06:20	06:22	12:06	22:45	10:14	

July 20, 21, 22 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	298°16.0	165°12.2	N19°00.3	240°54.1	N19°17.7	227°30.6	N21°33.8	307°28.9	S06°12.7	Alpheratz	357°35.0 29°13.5	
1	313°18.5	180°11.6	18°59.5	255°54.8	18.1	242°32.6	33.8	322°31.4	12.7	Ankaa	353°07.4 -42°10.1	
2	328°21.0	195°10.9	58.7	270°55.4	18.5	257°34.5	33.9	337°34.0	12.8	Schedar	349°31.3 56°40.1	
3	343°23.4	210°10.2	•• 57.9	285°56.1	•• 18.9	272°36.4	•• 33.9	352°36.5	•• 12.8	Diphda	348°47.6 -17°51.0	
4	358°25.9	225°09.6	57.1	300°56.8	19.3	287°38.4	34.0	7°39.0	12.9	Achernar	335°20.4 -57°06.4	
5	13°28.4	240°08.9	56.3	315°57.5	19.8	302°40.3	34.1	22°41.6	12.9	Hamal	327°51.7 23°34.6	
6	28°30.8	255°08.2	N18°55.5	330°58.1	N19°20.2	317°42.3	N21°34.1	37°44.1	S06°12.9	Polaris	314°22.9 89°21.7	
7	43°33.3	270°07.5	54.7	345°58.8	20.6	332°44.2	34.2	52°46.6	13.0	Acamar	315°12.2 -40°12.1	
8	58°35.7	285°06.9	53.9	0°59.5	20.9	347°46.2	34.3	67°49.2	13.0	Menkar	314°06.7 4°11.2	
9	73°38.2	300°06.2	•• 53.1	16°00.2	•• 21.3	2°48.1	•• 34.3	82°51.7	•• 13.1	Mirfak	308°29.0 49°56.7	
10	88°40.7	315°05.5	52.3	31°00.8	21.7	17°50.0	34.4	97°54.2	13.1	Aldebaran	290°40.3 16°33.5	
11	103°43.1	330°04.9	51.5	46°01.5	22.1	32°52.0	34.5	112°56.8	13.2	Rigel	281°04.5 -8°10.3	
12	118°45.6	345°04.2	N18°50.7	61°02.2	N19°22.5	47°53.9	N21°34.5	127°59.3	S06°13.2	Capella	280°22.8 46°01.3	
13	133°48.1	0°03.5	49.9	76°02.9	22.9	62°55.9	34.6	143°01.9	13.2	Bellatrix	278°23.6 6°22.4	
14	148°50.5	15°02.9	49.1	91°03.5	23.3	77°57.8	34.7	158°04.4	13.3	Elnath	278°02.7 28°37.7	
15	163°53.0	30°02.2	•• 48.3	106°04.2	•• 23.7	92°59.8	•• 34.7	173°06.9	•• 13.3	Alnilam	275°38.4 -1°11.1	
16	178°55.5	45°01.6	47.5	121°04.9	24.1	108°01.7	34.8	188°09.5	13.4	Betelgeuse	270°52.8 7°24.8	
17	193°57.9	60°00.9	46.7	136°05.6	24.5	123°03.6	34.8	203°12.0	13.4	Canopus	263°53.1 -52°42.3	
18	209°00.4	75°00.2	N18°45.8	151°06.2	N19°24.9	138°05.6	N21°34.9	218°14.6	S06°13.4	Sirius	258°26.9 -16°44.9	
19	224°02.9	89°59.6	45.0	166°06.9	25.3	153°07.5	35.0	233°17.1	13.5	Adhara	255°06.5 -29°00.2	
20	239°05.3	104°58.9	44.2	181°07.6	25.7	168°09.5	35.0	248°19.6	13.5	Procyon	244°51.6 5°09.8	
21	254°07.8	119°58.2	•• 43.4	196°08.3	•• 26.1	183°11.4	•• 35.1	263°22.2	•• 13.6	Pollux	243°18.1 27°58.1	
22	269°10.2	134°57.6	42.6	211°08.9	26.5	198°13.4	35.2	278°24.7	13.6	Avior	234°15.6 -59°35.2	
23	284°12.7	149°56.9	41.8	226°09.6	26.9	213°15.3	35.2	293°27.2	13.7	Suhail	222°47.0 -43°31.9	
Mer.pass. 04:06		ν -0.7' d -0.8' m -3.88		ν 0.7' d 0.4' m 0.91		ν 1.9' d 0.1' m -2.07		ν 2.5' d 0.0' m 0.81		Miaplacidus	221°39.3 -69°49.1	
											Alphard	217°48.4 -8°45.8
											Regulus	207°35.1 11°51.0
											Dubhe	193°41.9 61°37.4
											Denebola	182°25.5 14°26.2
											Gienah	175°44.1 -17°40.7
											Acrux	173°00.8 -63°14.3
											Gacrux	171°52.3 -57°15.3
											Alioth	166°13.4 55°49.9
											Spica	158°22.7 -11°17.4
											Alkaid	152°52.3 49°11.7
											Hadar	148°36.5 -60°29.7
											Menkent	147°58.0 -36°29.6
											Arcturus	145°48.2 19°03.4
											Rigel Kent.	139°40.7 -60°56.4
											Kochab	137°19.4 74°03.5
											Zuben'ubi	136°56.3 -16°08.7
											Alphecca	126°03.9 26°38.1
											Antares	112°16.1 -26°29.2
											Atria	107°10.2 -69°04.5
											Sabik	102°03.0 -15°45.3
											Shaula	96°10.5 -37°07.4
											Rasalhague	95°58.6 12°32.6
											Eltanin	90°41.9 51°29.2
											Kaus Aust.	83°32.6 -34°22.4
											Vega	80°33.1 38°48.4
											Nunki	75°47.9 -26°16.0
											Altair	62°00.0 8°56.0
											Peacock	53°05.7 -56°39.3
											Deneb	49°25.6 45°22.0
											Enif	33°38.9 9°59.3
											Al Na'ir	27°33.0 -46°50.4
											Fomalhaut	15°14.7 -29°29.4
											Scheat	13°45.3 28°12.9
											Markab	13°30.1 15°20.2
Jul 20 Sat		SHA		Mer.pass								
		Venus		226°56.2		13:00						
		Mars		302°38.0		07:56						
		Jupiter		289°14.6		08:49						
		Saturn		9°12.9		03:29						
Jul 21 Sun		SHA		Mer.pass								
		Venus		225°41.1		13:01						
		Mars		301°55.1		07:55						
		Jupiter		289°02.1		08:46						
		Saturn		9°14.6		03:25						
Jul 22 Mon		SHA		Mer.pass								
		Venus		224°26.2		13:02						
		Mars		301°12.2		07:54						
		Jupiter		288°49.7		08:43						
		Saturn		9°16.4		03:21						
											Horizontal parallax	
											Venus:	0.1
											Mars:	0.1

Sun				Moon												
Sat	GHA	Dec		GHA	ν	Dec	d	HP	Lat.	Twilight		Sunrise	Sunset	Twilight		
									Naut.	Civil				Civil	Naut.	
0	178°24.0	N20°36.2		18°03.9	4.4'	S28°09.3	-2.5'	58.2'	N 72°							
1	193°24.0	35.8		32°27.4	4.4'	28°06.8	-2.7'	58.3'	N 70°							
2	208°23.9	35.3		46°50.8	4.4'	28°04.1	-2.9'	58.3'	68°	--	--	01:10	22:56	--	--	
3	223°23.9	34.8		61°14.2	4.4'	28°01.2	-3.1'	58.3'	66°	--	--	02:03	22:06	--	--	
4	238°23.9	34.3		75°37.6	4.4'	27°58.2	-3.3'	58.4'	64°	--	--	02:35	21:36	--	--	
5	253°23.8	33.9		90°01.0	4.4'	27°54.9	-3.4'	58.4'	62°	--	01:28	02:58	21:13	22:40	--	
6	268°23.8	N20°33.4		104°24.4	4.4'	S27°51.5	-3.6'	58.4'	60°	--	02:06	03:17	20:55	22:04	--	
7	283°23.8	32.9		118°47.8	4.4'	27°47.8	-3.8'	58.5'	N 58°	--	02:32	03:32	20:40	21:39	--	
8	298°23.7	32.4		133°11.2	4.4'	27°44.0	-4.0'	58.5'	56°	01:21	02:52	03:45	20:27	21:19	22:48	
9	313°23.7	32.0		147°34.7	4.4'	27°40.0	-4.2'	58.5'	54°	01:55	03:08	03:56	20:16	21:03	22:15	
10	328°23.7	31.5		161°58.1	4.4'	27°35.8	-4.4'	58.6'	52°	02:20	03:22	04:06	20:06	20:50	21:52	
11	343°23.7	31.0		176°21.5	4.4'	27°31.4	-4.6'	58.6'	50°	02:38	03:34	04:15	19:57	20:38	21:33	
12	358°23.6	N20°30.5		190°45.0	4.5'	S27°26.8	-4.8'	58.6'	45°	03:13	03:59	04:34	19:38	20:14	20:58	
13	13°23.6	30.0		205°08.4	4.5'	27°22.1	-4.9'	58.6'	N 40°	03:39	04:18	04:49	19:24	19:55	20:34	
14	28°23.6	29.6		219°31.9	4.5'	27°17.1	-5.1'	58.7'	35°	03:58	04:33	05:02	19:11	19:39	20:14	
15	43°23.5	29.1		233°55.4	4.5'	27°12.0	-5.3'	58.7'	30°	04:14	04:46	05:13	19:00	19:26	19:58	
16	58°23.5	28.6		248°18.9	4.5'	27°06.7	-5.5'	58.7'	20°	04:39	05:08	05:32	18:41	19:05	19:33	
17	73°23.5	28.1		262°42.5	4.6'	27°01.2	-5.7'	58.8'	N 10°	04:59	05:25	05:48	18:25	18:48	19:14	
18	88°23.4	N20°27.6		277°06.0	4.6'	S26°55.5	-5.9'	58.8'	0°	05:15	05:41	06:03	18:10	18:32	18:58	
19	103°23.4	27.1		291°29.6	4.6'	26°49.7	-6.0'	58.8'	S 10°	05:30	05:56	06:18	17:55	18:17	18:43	
20	118°23.4	26.7		305°53.3	4.7'	26°43.6	-6.2'	58.8'	20°	05:43	06:10	06:34	17:39	18:03	18:30	
21	133°23.4	26.2		320°16.9	4.7'	26°37.4	-6.4'	58.9'	30°	05:57	06:26	06:52	17:21	17:47	18:16	
22	148°23.3	25.7		334°40.6	4.7'	26°31.0	-6.6'	58.9'	35°	06:04	06:35	07:02	17:11	17:38	18:09	
23	163°23.3	25.2		349°04.4	4.8'	26°24.4	-6.8'	58.9'	40°	06:11	06:45	07:14	16:59	17:29	18:02	
SD = 15.7' $d = -0.5'$				SD = 15.9'												
Sun	GHA	Dec		GHA	ν	Dec	d	HP								
0	178°23.3	N20°24.7		3°28.1	4.8'	S26°17.6	-7.0'	58.9'	S 50°	06:29	07:09	07:45	16:28	17:05	17:45	
1	193°23.2	24.2		17°52.0	4.9'	26°10.7	-7.1'	59.0'	52°	06:33	07:15	07:53	16:20	16:59	17:41	
2	208°23.2	23.7		32°15.8	4.9'	26°03.5	-7.3'	59.0'	54°	06:37	07:21	08:02	16:11	16:52	17:36	
3	223°23.2	23.3		46°39.7	5.0'	25°56.2	-7.5'	59.0'	56°	06:42	07:29	08:12	16:01	16:45	17:32	
4	238°23.2	22.8		61°03.7	5.0'	25°48.7	-7.7'	59.0'	58°	06:47	07:37	08:24	15:50	16:37	17:27	
5	253°23.1	22.3		75°27.7	5.1'	25°41.1	-7.8'	59.1'	S 60°	06:52	07:46	08:37	15:36	16:28	17:21	
6	268°23.1	N20°21.8		89°51.7	5.1'	S25°33.2	-8.0'	59.1'								
7	283°23.1	21.3		104°15.8	5.2'	25°25.2	-8.2'	59.1'	Lat.	Moonrise			Moonset			
8	298°23.1	20.8		118°40.0	5.2'	25°17.0	-8.4'	59.1'	Sat	Sun	Mon	Sat	Sun	Mon		
9	313°23.0	20.3		133°04.2	5.3'	25°08.7	-8.5'	59.2'	N 72°							
10	328°23.0	19.8		147°28.5	5.3'	25°00.2	-8.7'	59.2'	N 70°			23:48				
11	343°23.0	19.3		161°52.8	5.4'	24°51.5	-8.9'	59.2'	68°			23:11				
12	358°23.0	N20°18.8		176°17.2	5.5'	S24°42.6	-9.0'	59.2'	66°		23:31	22:44			02:15	
13	13°22.9	18.3		190°41.7	5.5'	24°33.6	-9.2'	59.3'	64°		22:44	22:24			03:01	
14	28°22.9	17.8		205°06.2	5.6'	24°24.4	-9.4'	59.3'	62°	22:24	22:14	22:07		01:16	03:30	
15	43°22.9	17.4		219°30.8	5.6'	24°15.0	-9.5'	59.3'	60°	21:43	21:51	21:53	00:13	01:57	03:52	
16	58°22.9	16.9		233°55.4	5.7'	24°05.5	-9.7'	59.3'	N 58°	21:15	21:32	21:41	00:52	02:25	04:10	
17	73°22.8	16.4		248°20.1	5.8'	23°55.8	-9.8'	59.4'	56°	20:52	21:16	21:30	01:20	02:46	04:25	
18	88°22.8	N20°15.9		262°44.9	5.9'	S23°46.0	-10.0'	59.4'	54°	20:34	21:03	21:21	01:41	03:04	04:38	
19	103°22.8	15.4		277°09.7	5.9'	23°36.0	-10.2'	59.4'	52°	20:19	20:51	21:13	01:59	03:19	04:50	
20	118°22.8	14.9		291°34.7	6.0'	23°25.8	-10.3'	59.4'	50°	20:05	20:40	21:05	02:14	03:32	04:59	
21	133°22.8	14.4		305°59.7	6.1'	23°15.5	-10.5'	59.4'	45°	19:38	20:18	20:49	02:45	03:59	05:20	
22	148°22.7	13.9		320°24.7	6.1'	23°05.0	-10.6'	59.5'	N 40°	19:16	20:00	20:36	03:08	04:20	05:37	
23	163°22.7	13.4		334°49.9	6.2'	22°54.4	-10.8'	59.5'	35°	18:58	19:45	20:24	03:28	04:38	05:51	
SD = 15.7' $d = -0.5'$				SD = 16.1'												
Mon	GHA	Dec		GHA	ν	Dec	d	HP								
0	178°22.7	N20°12.9		349°15.1	6.3'	S22°43.6	-10.9'	59.5'	S 10°	17:10	18:12	19:13	05:22	06:20	07:15	
1	193°22.7	12.4		3°40.4	6.4'	22°32.6	-11.1'	59.5'	20°	16:47	17:53	18:58	05:45	06:41	07:32	
2	208°22.6	11.9		18°05.7	6.4'	22°21.6	-11.2'	59.5'	30°	16:20	17:29	18:40	06:13	07:06	07:52	
3	223°22.6	11.4		32°31.2	6.5'	22°10.3	-11.4'	59.6'	35°	16:04	17:16	18:30	06:30	07:20	08:04	
4	238°22.6	10.9		46°56.7	6.6'	21°59.0	-11.5'	59.6'	40°	15:45	17:00	18:18	06:48	07:37	08:17	
5	253°22.6	10.4		61°22.3	6.7'	21°47.4	-11.7'	59.6'	45°	15:23	16:41	18:04	07:11	07:57	08:32	
6	268°22.6	N20°09.9		75°48.0	6.8'	S21°35.8	-11.8'	59.6'	S 50°	14:54	16:18	17:47	07:41	08:22	08:51	
7	283°22.5	09.3		90°13.7	6.8'	21°24.0	-11.9'	59.6'	52°	14:40	16:06	17:39	07:55	08:34	09:00	
8	298°22.5	08.8		104°39.6	6.9'	21°12.0	-12.1'	59.6'	54°	14:24	15:53	17:29	08:11	08:47	09:10	
9	313°22.5	08.3		119°05.5	7.0'	20°59.9	-12.2'	59.7'	56°	14:04	15:38	17:19	08:31	09:03	09:21	
10	328°22.5	07.8		133°31.5	7.1'	20°47.7	-12.4'	59.7'	58°	13:40	15:21	17:07	08:56	09:21	09:34	
11	343°22.5	07.3		147°57.5	7.2'	20°35.4	-12.5'	59.7'	S 60°	13:08	14:59	16:53	09:28	09:43	09:49	
12	358°22.4	N20°06.8		162°23.7	7.2'	S20°22.9	-12.6'	59.7'								
13	13°22.4	06.3		176°49.9	7.3'	20°10.3	-12.7'	59.7'	Day	Sun		Mer.	Moon		Age	
14	28°22.4	05.8		191°16.3	7.4'	19°57.5	-12.9'	59.7'	Eqn.of Time	12h	Pass	Mer.Pass.	Upper	Lower	15-17	
15	43°22.4	05.3		205°42.7	7.5'	19°44.6	-13.0'	59.7'	00h	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	97-99%	
16	58°22.4	04.8		220°09.1	7.6'	19°31.6	-13.1'	59.8'	20	06:24	06:26	12:06	23:46	11:15		
17	73°22.4	04.3		234°35.7	7.6'	19°18.5	-13.2'	59.8'	21	06:27	06:28	12:06	--	12:15		
18	88°22.3	N20°03.8		249°02.4	7.7'	S19°05.3	-13.4'	59.8'	22	06:29	06:30	12:07	00:45	13:13		
19	103°22.3	03.2		263°29.1	7.8'	18°51.9	-13.5'	59.8'								
20	118°22.3	02.7		277°55.9	7.9'	18°38.4	-13.6'	59.8'								
21	133°22.3	02.2		292°22.8	8.0'	18°24.8	-13.7'	59.8'								
22	148°22.3	01.7		306°49.8	8.1'	18°11.1	-13.8'	59.8'								
23	163°22.3	01.2		321°16.8	8.1'	17°57.3	-13.9'	59.8'								
SD = 15.7' $d = -0.5'$				SD = 16.2'												

July 23, 24, 25 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	301°13.5	164°25.2	N18°00.6	241°42.7	N19°45.9	229°50.8	N21°38.3	310°31.8	S06°15.8	Alpheratz	357°35.0	29°13.5
1	316°15.9	179°24.5	17°59.8	256°43.4	46.2	244°52.8	38.3	325°34.4	15.8	Ankaa	353°07.3	-42°10.1
2	331°18.4	194°23.9	58.9	271°44.1	46.6	259°54.7	38.4	340°36.9	15.9	Schedar	349°31.3	56°40.1
3	346°20.8	209°23.3	•• 58.1	286°44.8	•• 47.0	274°56.7	•• 38.5	355°39.5	•• 15.9	Diphda	348°47.6	-17°51.0
4	1°23.3	224°22.6	57.2	301°45.5	47.4	289°58.6	38.5	10°42.0	16.0	Achernar	335°20.4	-57°06.4
5	16°25.8	239°22.0	56.3	316°46.1	47.7	305°00.6	38.6	25°44.6	16.0	Hamal	327°51.6	23°34.6
6	31°28.2	254°21.4	N17°55.5	331°46.8	N19°48.1	320°02.5	N21°38.6	40°47.1	S06°16.1	Polaris	314°21.5	89°21.7
7	46°30.7	269°20.7	54.6	346°47.5	48.5	335°04.5	38.7	55°49.7	16.1	Acamar	315°12.2	-40°12.1
8	61°33.2	284°20.1	53.7	1°48.2	48.9	350°06.4	38.8	70°52.2	16.2	Menkar	314°06.6	4°11.2
9	76°35.6	299°19.5	•• 52.9	16°48.8	•• 49.3	5°08.4	•• 38.8	85°54.7	•• 16.2	Mirfak	308°29.0	49°56.7
10	91°38.1	314°18.9	52.0	31°49.5	49.6	20°10.3	38.9	100°57.3	16.2	Aldebaran	290°40.3	16°33.5
11	106°40.6	329°18.2	51.1	46°50.2	50.0	35°12.3	38.9	115°59.8	16.3	Rigel	281°04.5	-8°10.3
12	121°43.0	344°17.6	N17°50.3	61°50.9	N19°50.4	50°14.2	N21°39.0	131°02.4	S06°16.3	Capella	280°22.8	46°01.3
13	136°45.5	359°17.0	49.4	76°51.5	50.8	65°16.2	39.1	146°04.9	16.4	Bellatrix	278°23.5	6°22.4
14	151°48.0	14°16.3	48.5	91°52.2	51.1	80°18.1	39.1	161°07.5	16.4	Elnath	278°02.6	28°37.7
15	166°50.4	29°15.7	•• 47.6	106°52.9	•• 51.5	95°20.1	•• 39.2	176°10.0	•• 16.5	Alnilam	275°38.4	-1°11.1
16	181°52.9	44°15.1	46.8	121°53.6	51.9	110°22.1	39.2	191°12.6	16.5	Betelgeuse	270°52.8	7°24.8
17	196°55.3	59°14.5	45.9	136°54.3	52.2	125°24.0	39.3	206°15.1	16.6	Canopus	263°53.1	-52°42.3
18	211°57.8	74°13.8	N17°45.0	151°54.9	N19°52.6	140°26.0	N21°39.4	221°17.7	S06°16.6	Sirius	258°26.9	-16°44.8
19	227°00.3	89°13.2	44.1	166°55.6	53.0	155°27.9	39.4	236°20.2	16.7	Adhara	255°06.5	-29°00.2
20	242°02.7	104°12.6	43.3	181°56.3	53.4	170°29.9	39.5	251°22.8	16.7	Procyon	244°51.6	5°09.8
21	257°05.2	119°12.0	•• 42.4	196°57.0	•• 53.7	185°31.8	•• 39.5	266°25.3	•• 16.8	Pollux	243°18.1	27°58.1
22	272°07.7	134°11.3	41.5	211°57.6	54.1	200°33.8	39.6	281°27.9	16.8	Avior	234°15.6	-59°35.2
23	287°10.1	149°10.7	40.6	226°58.3	54.5	215°35.7	39.7	296°30.4	16.8	Suhail	222°47.0	-43°31.9
Mer.pass. 03:54		ν -0.6' d -0.9' m -3.88		ν 0.7' d 0.4' m 0.90		ν 2.0' d 0.1' m -2.08		ν 2.5' d 0.0' m 0.80				

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	302°12.6	164°10.1	N17°39.7	241°59.0	N19°54.9	230°37.7	N21°39.7	311°33.0	S06°16.9	Denebola	182°25.5	14°26.2
1	317°15.1	179°09.5	38.9	256°59.7	55.2	245°39.6	39.8	326°35.5	16.9	Gienah	175°44.1	-17°40.7
2	332°17.5	194°08.8	38.0	272°00.3	55.6	260°41.6	39.8	341°38.1	17.0	Acrux	173°00.8	-63°14.3
3	347°20.0	209°08.2	•• 37.1	287°01.0	•• 56.0	275°43.6	•• 39.9	356°40.6	•• 17.0	Gacrux	171°52.3	-57°15.3
4	2°22.5	224°07.6	36.2	302°01.7	56.3	290°45.5	40.0	11°43.2	17.1	Alioth	166°13.4	55°49.9
5	17°24.9	239°07.0	35.3	317°02.4	56.7	305°47.5	40.0	26°45.7	17.1	Spica	158°22.7	-11°17.4
6	32°27.4	254°06.4	N17°34.4	332°03.1	N19°57.1	320°49.4	N21°40.1	41°48.3	S06°17.2	Alkaid	152°52.3	49°11.7
7	47°29.8	269°05.7	33.6	347°03.7	57.4	335°51.4	40.1	56°50.8	17.2	Hadar	148°36.5	-60°29.7
8	62°32.3	284°05.1	32.7	2°04.4	57.8	350°53.3	40.2	71°53.4	17.3	Menkent	147°58.1	-36°29.6
9	77°34.8	299°04.5	•• 31.8	17°05.1	•• 58.2	5°55.3	•• 40.3	86°55.9	•• 17.3	Arcturus	145°48.2	19°03.4
10	92°37.2	314°03.9	30.9	32°05.8	58.5	20°57.3	40.3	101°58.5	17.4	Rigel Kent.	139°40.7	-60°56.4
11	107°39.7	329°03.3	30.0	47°06.4	58.9	35°59.2	40.4	117°01.0	17.4	Kochab	137°19.5	74°03.5
12	122°42.2	344°02.7	N17°29.1	62°07.1	N19°59.3	51°01.2	N21°40.4	132°03.6	S06°17.5	Zuben'ubi	136°56.4	-16°08.7
13	137°44.6	359°02.0	28.2	77°07.8	19°59.6	66°03.1	40.5	147°06.1	17.5	Alphecca	126°03.9	26°38.1
14	152°47.1	14°01.4	27.3	92°08.5	20°00.0	81°05.1	40.6	162°08.7	17.5	Antares	112°16.1	-26°29.2
15	167°49.6	29°00.8	•• 26.4	107°09.1	•• 00.4	96°07.0	•• 40.6	177°11.2	•• 17.6	Atria	107°10.2	-69°04.5
16	182°52.0	44°00.2	25.5	122°09.8	00.7	111°09.0	40.7	192°13.8	17.6	Sabik	102°03.0	-15°45.3
17	197°54.5	58°59.6	24.6	137°10.5	01.1	126°11.0	40.7	207°16.3	17.7	Shaula	96°10.5	-37°07.4
18	212°56.9	73°59.0	N17°23.7	152°11.2	N20°01.5	141°12.9	N21°40.8	222°18.9	S06°17.7	Rasalhague	95°58.6	12°32.6
19	227°59.4	88°58.4	22.9	167°11.9	01.8	156°14.9	40.9	237°21.4	17.8	Eltanin	90°41.9	51°29.2
20	243°01.9	103°57.7	22.0	182°12.5	02.2	171°16.8	40.9	252°24.0	17.8	Kaus Aust.	83°32.6	-34°22.4
21	258°04.3	118°57.1	•• 21.1	197°13.2	•• 02.6	186°18.8	•• 41.0	267°26.5	•• 17.9	Vega	80°33.1	38°48.5
22	273°06.8	133°56.5	20.2	212°13.9	02.9	201°20.8	41.0	282°29.1	17.9	Nunki	75°47.9	-26°16.0
23	288°09.3	148°55.9	19.3	227°14.6	03.3	216°22.7	41.1	297°31.6	18.0	Altair	62°00.0	8°56.0
Mer.pass. 03:51		ν -0.6' d -0.9' m -3.88		ν 0.7' d 0.4' m 0.89		ν 2.0' d 0.1' m -2.09		ν 2.5' d 0.0' m 0.79				

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	303°11.7	163°55.3	N17°18.4	242°15.2	N20°03.7	231°24.7	N21°41.2	312°34.2	S06°18.0	Jul 23 Tue	SHA	Mer.pass
1	318°14.2	178°54.7	17.5	257°15.9	04.0	246°26.6	41.2	327°36.8	18.1	Venus	223°11.7	13:03
2	333°16.7	193°54.1	16.6	272°16.6	04.4	261°28.6	41.3	342°39.3	18.1	Mars	300°29.3	07:53
3	348°19.1	208°53.5	•• 15.6	287°17.3	•• 04.8	276°30.5	•• 41.3	357°41.9	•• 18.2	Jupiter	288°37.3	08:39
4	3°21.6	223°52.9	14.7	302°18.0	05.1	291°32.5	41.4	12°44.4	18.2	Saturn	9°18.4	03:17
5	18°24.1	238°52.3	13.8	317°18.6	05.5	306°34.5	41.4	27°47.0	18.3			
6	33°26.5	253°51.6	N17°12.9	332°19.3	N20°05.8	321°36.4	N21°41.5	42°49.5	S06°18.3			
7	48°29.0	268°51.0	12.0	347°20.0	06.2	336°38.4	41.6	57°52.1	18.4			
8	63°31.4	283°50.4	11.1	2°20.7	06.6	351°40.3	41.6	72°54.6	18.4			
9	78°33.9	298°49.8	•• 10.2	17°21.3	•• 06.9	6°42.3	•• 41.7	87°57.2	•• 18.5			
10	93°36.4	313°49.2	09.3	32°22.0	07.3	21°44.3	41.7	102°59.7	18.5	Jul 24 Wed	SHA	Mer.pass
11	108°38.8	328°48.6	08.4	47°22.7	07.6	36°46.2	41.8	118°02.3	18.5	Venus	221°57.5	13:04
12	123°41.3	343°48.0	N17°07.5	62°23.4	N20°08.0	51°48.2	N21°41.9	133°04.8	S06°18.6	Mars	299°46.4	07:52
13	138°43.8	358°47.4	06.6	77°24.1	08.4	66°50.2	41.9	148°07.4	18.6	Jupiter	288°25.1	08:36
14	153°46.2	13°46.8	05.7	92°24.7	08.7	81°52.1	42.0	163°10.0	18.7	Saturn	9°20.4	03:13
15	168°48.7	28°46.2	•• 04.7	107°25.4	•• 09.1	96°54.1	•• 42.0	178°12.5	•• 18.7			
16	183°51.2	43°45.6	03.8	122°26.1	09.4	111°56.0	42.1	193°15.1	18.8			
17	198°53.6	58°45.0	02.9	137°26.8	09.8	126°58.0	42.1	208°17.6	18.8			
18	213°56.1	73°44.4	N17°02.0	152°27.4	N20°10.2	142°00.0	N21°42.2	223°20.2	S06°18.9			
19	228°58.5	88°43.8	01.1	167°28.1	10.5	157°01.9	42.3	238°22.7	18.9			
20	244°01.0	103°43.2	17°00.2	182°28.8	10.9	172°03.9	42.3	253°25.3	19.0			
21	259°03.5	118°42.6	16°59.3	197°29.5	•• 11.2	187°05.8	•• 42.4	268°27.8	•• 19.0			
22	274°05.9	133°42.0	58.3	212°30.2	11.6	202°07.8	42.4	283°30.4	19.1			
23	289°08.4	148°41.4	57.4	227°30.8	11.9	217°09.8	42.5	298°33.0	19.1			
Mer.pass. 03:47		ν -0.6' d -0.9' m -3.87		ν 0.7' d 0.4' m 0.89		ν 2.0' d 0.1' m -2.09		ν 2.6' d 0.0' m 0.79				

			Horizontal parallax		
			Venus:	0.1	
			Mars:	0.1	

DUT1 = UT1-UTC = +0.0205 sec $\Delta T = TT-UT1 = +69.1635$ sec

2024 July 23 to Jul. 25 UT

h Sun Moon						
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	178°22.2	N20°00.7	335°44.0	8.2'	S17°43.3	-14.1' 59.9'
1	193°22.2	20°00.2	350°11.2	8.3'	17°29.3	-14.2' 59.9'
2	208°22.2	19°59.7	4°38.5	8.4'	17°15.1	-14.3' 59.9'
3	223°22.2	· · 59.1	19°05.9	8.5'	17°00.8	-14.4' 59.9'
4	238°22.2	58.6	33°33.3	8.5'	16°46.5	-14.5' 59.9'
5	253°22.2	58.1	48°00.9	8.6'	16°32.0	-14.6' 59.9'
6	268°22.1	N19°57.6	62°28.5	8.7'	S16°17.4	-14.7' 59.9'
7	283°22.1	57.1	76°56.2	8.8'	16°02.7	-14.8' 59.9'
8	298°22.1	56.5	91°24.0	8.9'	15°48.0	-14.9' 59.9'
9	313°22.1	· · 56.0	105°51.8	8.9'	15°33.1	-15.0' 59.9'
10	328°22.1	55.5	120°19.7	9.0'	15°18.1	-15.0' 59.9'
11	343°22.1	55.0	134°47.7	9.1'	15°03.1	-15.1' 60.0'
12	358°22.1	N19°54.5	149°15.8	9.2'	S14°48.0	-15.2' 60.0'
13	13°22.1	53.9	163°44.0	9.2'	14°32.7	-15.3' 60.0'
14	28°22.0	53.4	178°12.2	9.3'	14°17.4	-15.4' 60.0'
15	43°22.0	· · 52.9	192°40.5	9.4'	14°02.0	-15.5' 60.0'
16	58°22.0	52.4	207°08.9	9.4'	13°46.5	-15.6' 60.0'
17	73°22.0	51.8	221°37.3	9.5'	13°31.0	-15.6' 60.0'
18	88°22.0	N19°51.3	236°05.9	9.6'	S13°15.4	-15.7' 60.0'
19	103°22.0	50.8	250°34.4	9.7'	12°59.6	-15.8' 60.0'
20	118°22.0	50.3	265°03.1	9.7'	12°43.9	-15.9' 60.0'
21	133°22.0	· · 49.7	279°31.8	9.8'	12°28.0	-15.9' 60.0'
22	148°22.0	49.2	294°00.6	9.9'	12°12.1	-16.0' 60.0'
23	163°21.9	48.7	308°29.5	9.9'	11°56.1	-16.1' 60.0'
SD = 15.7' $d = -0.5'$			SD = 16.3'			
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	178°21.9	N19°48.2	322°58.4	10.0'	S11°40.0	-16.1' 60.0'
1	193°21.9	47.6	337°27.4	10.1'	11°23.9	-16.2' 60.0'
2	208°21.9	47.1	351°56.5	10.1'	11°07.7	-16.3' 60.0'
3	223°21.9	· · 46.6	6°25.6	10.2'	10°51.4	-16.3' 60.0'
4	238°21.9	46.0	20°54.8	10.2'	10°35.1	-16.4' 60.0'
5	253°21.9	45.5	35°24.0	10.3'	10°18.7	-16.4' 60.0'
6	268°21.9	N19°45.0	49°53.3	10.4'	S10°02.3	-16.5' 60.0'
7	283°21.9	44.4	64°22.7	10.4'	09°45.8	-16.5' 60.0'
8	298°21.9	43.9	78°52.1	10.5'	09°29.3	-16.6' 60.0'
9	313°21.9	· · 43.4	93°21.6	10.5'	09°12.7	-16.6' 60.0'
10	328°21.9	42.8	107°51.1	10.6'	08°56.1	-16.7' 60.0'
11	343°21.8	42.3	122°20.7	10.6'	08°39.4	-16.7' 60.0'
12	358°21.8	N19°41.8	136°50.3	10.7'	S08°22.7	-16.8' 60.0'
13	13°21.8	41.2	151°20.0	10.7'	08°05.9	-16.8' 60.0'
14	28°21.8	40.7	165°49.7	10.8'	07°49.1	-16.8' 60.0'
15	43°21.8	· · 40.2	180°19.5	10.8'	07°32.3	-16.9' 60.0'
16	58°21.8	39.6	194°49.4	10.9'	07°15.4	-16.9' 60.0'
17	73°21.8	39.1	209°19.3	10.9'	06°58.5	-17.0' 60.0'
18	88°21.8	N19°38.6	223°49.2	11.0'	S06°41.5	-17.0' 60.0'
19	103°21.8	38.0	238°19.2	11.0'	06°24.5	-17.0' 60.0'
20	118°21.8	37.5	252°49.2	11.1'	06°07.5	-17.0' 60.0'
21	133°21.8	· · 36.9	267°19.2	11.1'	05°50.5	-17.1' 60.0'
22	148°21.8	36.4	281°49.3	11.1'	05°33.4	-17.1' 60.0'
23	163°21.8	35.9	296°19.5	11.2'	05°16.3	-17.1' 60.0'
SD = 15.7' $d = -0.5'$			SD = 16.4'			
Thu	GHA	Dec	GHA	ν	Dec	d HP
0	178°21.8	N19°35.3	310°49.7	11.2'	S04°59.2	-17.1' 60.0'
1	193°21.8	34.8	325°19.9	11.3'	04°42.0	-17.2' 60.0'
2	208°21.8	34.2	339°50.1	11.3'	04°24.9	-17.2' 59.9'
3	223°21.8	· · 33.7	354°20.4	11.3'	04°07.7	-17.2' 59.9'
4	238°21.8	33.1	8°50.7	11.3'	03°50.5	-17.2' 59.9'
5	253°21.8	32.6	23°21.1	11.4'	03°33.3	-17.2' 59.9'
6	268°21.8	N19°32.0	37°51.5	11.4'	S03°16.0	-17.2' 59.9'
7	283°21.8	31.5	52°21.9	11.4'	02°58.8	-17.2' 59.9'
8	298°21.8	31.0	66°52.3	11.5'	02°41.5	-17.3' 59.9'
9	313°21.8	· · 30.4	81°22.8	11.5'	02°24.3	-17.3' 59.9'
10	328°21.8	29.9	95°53.2	11.5'	02°07.0	-17.3' 59.9'
11	343°21.8	29.3	110°23.8	11.5'	01°49.8	-17.3' 59.9'
12	358°21.8	N19°28.8	124°54.3	11.6'	S01°32.5	-17.3' 59.9'
13	13°21.8	28.2	139°24.8	11.6'	01°15.2	-17.3' 59.9'
14	28°21.8	27.7	153°55.4	11.6'	00°57.9	-17.3' 59.8'
15	43°21.8	· · 27.1	168°26.0	11.6'	00°40.7	-17.3' 59.8'
16	58°21.8	26.6	182°56.6	11.6'	00°23.4	-17.3' 59.8'
17	73°21.8	26.0	197°27.2	11.6'	S00°06.1	-17.3' 59.8'
18	88°21.8	N19°25.5	211°57.9	11.7'	N00°11.1	17.2' 59.8'
19	103°21.8	24.9	226°28.5	11.7'	00°28.4	17.2' 59.8'
20	118°21.8	24.4	240°59.2	11.7'	00°45.6	17.2' 59.8'
21	133°21.8	· · 23.8	255°29.9	11.7'	01°02.8	17.2' 59.8'
22	148°21.8	23.2	270°00.6	11.7'	01°20.1	17.2' 59.8'
23	163°21.8	22.7	284°31.2	11.7'	01°37.3	17.2' 59.7'
SD = 15.7' $d = -0.5'$			SD = 16.4'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°	—:—	—:—	01:32	22:36	—:—	—:—
66°	—:—	—:—	02:16	21:54	—:—	—:—
64°	—:—	00:37	02:44	21:26	23:24	—:—
62°	—:—	01:43	03:06	21:05	22:26	—:—
60°	—:—	02:16	03:23	20:48	21:55	—:—
N 58°	00:34	02:40	03:38	20:34	21:31	23:28
56°	01:34	02:58	03:50	20:22	21:13	22:36
54°	02:04	03:14	04:01	20:11	20:58	22:06
52°	02:27	03:27	04:10	20:02	20:45	21:45
50°	02:44	03:39	04:19	19:53	20:33	21:27
45°	03:18	04:02	04:37	19:35	20:10	20:54
N 40°	03:42	04:20	04:52	19:21	19:52	20:30
35°	04:01	04:35	05:04	19:09	19:37	20:12
30°	04:16	04:48	05:14	18:58	19:25	19:56
20°	04:41	05:09	05:33	18:40	19:04	19:32
N 10°	05:00	05:26	05:48	18:25	18:47	19:13
0°	05:15	05:41	06:03	18:10	18:32	18:58
S 10°	05:29	05:55	06:17	17:56	18:18	18:44
20°	05:43	06:09	06:33	17:40	18:04	18:31
30°	05:56	06:25	06:50	17:23	17:49	18:18
35°	06:02	06:33	07:00	17:13	17:40	18:11
40°	06:10	06:42	07:12	17:02	17:31	18:04
45°	06:17	06:53	07:25	16:48	17:20	17:56
S 50°	06:26	07:06	07:42	16:32	17:08	17:48
52°	06:30	07:11	07:49	16:24	17:02	17:44
54°	06:34	07:18	07:58	16:16	16:56	17:40
56°	06:38	07:24	08:08	16:06	16:49	17:36
58°	06:43	07:32	08:19	15:55	16:42	17:31
S 60°	06:48	07:41	08:31	15:42	16:33	17:26

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	00:54 23:11	22:24	21:46	02:54	06:29	09:03
N 70°	22:51	22:16	21:47	03:58	06:46	09:07
68°	22:35	22:10	21:49	04:34	07:00	09:10
66°	22:22	22:05	21:50	04:58	07:11	09:12
64°	22:11	22:00	21:50	05:18	07:20	09:14
62°	22:01	21:56	21:51	05:33	07:27	09:16
60°	21:53	21:53	21:52	05:46	07:34	09:18
N 58°	21:46	21:50	21:53	05:57	07:40	09:19
56°	21:40	21:47	21:53	06:07	07:45	09:20
54°	21:34	21:44	21:54	06:15	07:50	09:21
52°	21:29	21:42	21:54	06:22	07:54	09:22
50°	21:24	21:40	21:55	06:29	07:57	09:23
45°	21:14	21:36	21:55	06:44	08:05	09:25
N 40°	21:06	21:32	21:56	06:55	08:12	09:27
35°	20:58	21:28	21:57	07:05	08:18	09:28
30°	20:52	21:26	21:58	07:14	08:23	09:29
20°	20:41	21:20	21:59	07:29	08:31	09:32
N 10°	20:31	21:16	22:00	07:42	08:39	09:33
0°	20:21	21:12	22:01	07:54	08:45	09:35
S 10°	20:12	21:08	22:02	08:05	08:52	09:37
20°	20:02	21:03	22:03	08:18	08:59	09:38
30°	19:50	20:58	22:04	08:32	09:07	09:40
35°	19:43	20:55	22:05	08:40	09:12	09:41
40°	19:36	20:51	22:06	08:49	09:17	09:42
45°	19:27	20:47	22:07	09:00	09:23	09:44
S 50°	19:16	20:43	22:08	09:13	09:30	09:45
52°	19:11	20:41	22:08	09:19	09:34	09:46
54°	19:05	20:38	22:09	09:26	09:37	09:47
56°	18:59	20:35	22:10	09:33	09:41	09:48
58°	18:52	20:32	22:10	09:41	09:46	09:49
S 60°	18:44	20:29	22:11	09:50	09:50	09:50

Day	Sun		Mer. Pass hh:mm	Moon		Age 18-20 97-83%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
23	06:31	06:32	12:07	01:41	14:07	
24	06:32	06:33	12:07	02:33	14:59	
25	06:33	06:33	12:07	03:23	15:48	

July 26, 27, 28 UT (Fri., Sat., Sun.)

Aries		Venus		Mars		Jupiter		Saturn	
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	304°10.9	163°40.8	N16°56.5	242°31.5	N20°12.3	232°11.7	N21°42.6	313°35.5	S06°19.2
1	310°13.3	178°40.2	55.6	257°32.2	12.6	247°13.7	42.6	328°38.1	19.2
2	334°15.8	193°39.6	54.7	272°32.9	13.0	262°15.7	42.7	343°40.6	19.3
3	349°18.3	208°39.0	· · 53.7	287°33.5	· · 13.4	277°17.6	· · 42.7	358°43.2	· · 19.3
4	4°20.7	223°38.4	52.8	302°34.2	13.7	292°19.6	42.8	13°45.7	19.4
5	19°23.2	238°37.8	51.9	317°34.9	14.1	307°21.5	42.8	28°48.3	19.4
6	34°25.7	253°37.2	N16°51.0	332°35.6	N20°14.4	322°23.5	N21°42.9	43°50.8	S06°19.5
7	49°28.1	268°36.6	50.0	347°36.3	14.8	337°25.5	43.0	58°53.4	19.5
8	64°30.6	283°36.0	49.1	2°36.9	15.1	352°27.4	43.0	73°56.0	19.6
9	79°33.0	298°35.4	· · 48.2	17°37.6	· · 15.5	7°29.4	· · 43.1	88°58.5	· · 19.6
10	94°35.5	313°34.9	47.3	32°38.3	15.8	22°31.4	43.1	104°01.1	19.7
11	109°38.0	328°34.3	46.3	47°39.0	16.2	37°33.3	43.2	119°03.6	19.7
12	124°40.4	343°33.7	N16°45.4	62°39.6	N20°16.5	52°35.3	N21°43.2	134°06.2	S06°19.8
13	139°42.9	358°33.1	44.5	77°40.3	16.9	67°37.3	43.3	149°08.8	19.8
14	154°45.4	13°32.5	43.5	92°41.0	17.2	82°39.2	43.4	164°11.3	19.9
15	169°47.8	28°31.9	· · 42.6	107°41.7	· · 17.6	97°41.2	· · 43.4	179°13.9	· · 19.9
16	184°50.3	43°31.3	41.7	122°42.4	17.9	112°43.2	43.5	194°16.4	20.0
17	199°52.8	58°30.7	40.7	137°43.0	18.3	127°45.1	43.5	209°19.0	20.0
18	214°55.2	73°30.1	N16°39.8	152°43.7	N20°18.6	142°47.1	N21°43.6	224°21.5	S06°20.1
19	229°57.7	88°29.5	38.9	167°44.4	19.0	157°49.1	43.6	239°24.1	20.1
20	245°00.2	103°29.0	37.9	182°45.1	19.3	172°51.0	43.7	254°26.7	20.2
21	260°02.6	118°28.4	· · 37.0	197°45.8	· · 19.7	187°53.0	· · 43.8	269°29.2	· · 20.2
22	275°05.1	133°27.8	36.1	212°46.4	20.0	202°55.0	43.8	284°31.8	20.3
23	290°07.5	148°27.2	35.1	227°47.1	20.4	217°56.9	43.9	299°34.3	20.3
Mer.pass. 03:43		ν -0.6' d -0.9' m -3.87		ν 0.7' d 0.4' m 0.89		ν 2.0' d 0.1' m -2.09		ν 2.6' d 0.0' m 0.78	

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	305°10.0	163°26.6	N16°34.2	242°47.8	N20°20.7	232°58.9	N21°43.9	314°36.9	S06°20.4
1	320°12.5	178°26.0	33.2	257°48.5	21.1	248°00.9	44.0	329°39.5	20.4
2	335°14.9	193°25.4	32.3	272°49.1	21.4	263°02.8	44.0	344°42.0	20.5
3	350°17.4	208°24.9	· · 31.4	287°49.8	· · 21.8	278°04.8	· · 44.1	359°44.6	· · 20.5
4	5°19.9	223°24.3	30.4	302°50.5	22.1	293°06.8	44.2	14°47.1	20.6
5	20°22.3	238°23.7	29.5	317°51.2	22.5	308°08.7	44.2	29°49.7	20.6
6	35°24.8	253°23.1	N16°28.5	332°51.9	N20°22.8	323°10.7	N21°44.3	44°52.3	S06°20.7
7	50°27.3	268°22.5	27.6	347°52.5	23.1	338°12.7	44.3	59°54.8	20.7
8	65°29.7	283°21.9	26.6	2°53.2	23.5	353°14.6	44.4	74°57.4	20.8
9	80°32.2	298°21.4	· · 25.7	17°53.9	· · 23.8	8°16.6	· · 44.4	89°59.9	· · 20.8
10	95°34.6	313°20.8	24.7	32°54.6	24.2	23°18.6	44.5	105°02.5	20.9
11	110°37.1	328°20.2	23.8	47°55.3	24.5	38°20.5	44.5	120°05.1	20.9
12	125°39.6	343°19.6	N16°22.9	62°55.9	N20°24.9	53°22.5	N21°44.6	135°07.6	S06°21.0
13	140°42.0	358°19.0	21.9	77°56.6	25.2	68°24.5	44.7	150°10.2	21.0
14	155°44.5	13°18.5	21.0	92°57.3	25.6	83°26.4	44.7	165°12.7	21.1
15	170°47.0	28°17.9	· · 20.0	107°58.0	· · 25.9	98°28.4	· · 44.8	180°15.3	· · 21.1
16	185°49.4	43°17.3	19.1	122°58.7	26.2	113°30.4	44.8	195°17.9	21.2
17	200°51.9	58°16.7	18.1	137°59.3	26.6	128°32.3	44.9	210°20.4	21.2
18	215°54.4	73°16.2	N16°17.1	153°00.0	N20°26.9	143°34.3	N21°44.9	225°23.0	S06°21.3
19	230°56.8	88°15.6	16.2	168°00.7	27.3	158°36.3	45.0	240°25.6	21.3
20	245°59.3	103°15.0	15.2	183°01.4	27.6	173°38.2	45.0	255°28.1	21.4
21	261°01.8	118°14.4	· · 14.3	198°02.1	· · 28.0	188°40.2	· · 45.1	270°30.7	· · 21.4
22	276°04.2	133°13.9	13.3	213°02.7	28.3	203°42.2	45.2	285°33.2	21.5
23	291°06.7	148°13.3	12.4	228°03.4	28.6	218°44.2	45.2	300°35.8	21.5
Mer.pass. 03:39		ν -0.6' d -0.9' m -3.87		ν 0.7' d 0.3' m 0.88		ν 2.0' d 0.1' m -2.10		ν 2.6' d 0.1' m 0.78	

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	306°09.1	163°12.7	N16°11.4	243°04.1	N20°29.0	233°46.1	N21°45.3	315°38.4	S06°21.6
1	321°11.6	178°12.1	10.5	258°04.8	29.3	248°48.1	45.3	330°40.9	21.7
2	336°14.1	193°11.6	09.5	273°05.5	29.7	263°50.1	45.4	345°43.5	21.7
3	351°16.5	208°11.0	· · 08.5	288°06.1	· · 30.0	278°52.0	· · 45.4	0°46.1	· · 21.8
4	6°19.0	223°10.4	07.6	303°06.8	30.3	293°54.0	45.5	15°48.6	21.8
5	21°21.5	238°09.8	06.6	318°07.5	30.7	308°56.0	45.5	30°51.2	21.9
6	36°23.9	253°09.3	N16°05.7	333°08.2	N20°31.0	323°58.0	N21°45.6	45°53.7	S06°21.9
7	51°26.4	268°08.7	04.7	348°08.9	31.3	338°59.9	45.7	60°56.3	22.0
8	66°28.9	283°08.1	03.7	3°09.5	31.7	354°01.9	45.7	75°58.9	22.0
9	81°31.3	298°07.6	· · 02.8	18°10.2	· · 32.0	9°03.9	· · 45.8	91°01.4	· · 22.1
10	96°33.8	313°07.0	01.8	33°10.9	32.4	24°05.8	45.8	106°04.0	22.1
11	111°36.3	328°06.4	16°00.8	48°11.6	32.7	39°07.8	45.9	121°06.6	22.2
12	126°38.7	343°05.9	N15°59.9	63°12.3	N20°33.0	54°09.8	N21°45.9	136°09.1	S06°22.2
13	141°41.2	358°05.3	58.9	78°12.9	33.4	69°11.8	46.0	151°11.7	22.3
14	156°43.6	13°04.7	57.9	93°13.6	33.7	84°13.7	46.0	166°14.3	22.3
15	171°46.1	28°04.2	· · 57.0	108°14.3	· · 34.0	99°15.7	· · 46.1	181°16.8	· · 22.4
16	186°48.6	43°03.6	56.0	123°15.0	34.4	114°17.7	46.2	196°19.4	22.4
17	201°51.0	58°03.0	55.0	138°15.7	34.7	129°19.6	46.2	211°22.0	22.5
18	216°53.5	73°02.5	N15°54.0	153°16.3	N20°35.0	144°21.6	N21°46.3	226°24.5	S06°22.5
19	231°56.0	88°01.9	53.1	168°17.0	35.4	159°23.6	46.3	241°27.1	22.6
20	246°58.4	103°01.3	52.1	183°17.7	35.7	174°25.6	46.4	256°29.6	22.6
21	262°00.9	118°00.8	· · 51.1	198°18.4	· · 36.0	189°27.5	· · 46.4	271°32.2	· · 22.7
22	277°03.4	133°00.2	50.2	213°19.1	36.4	204°29.5	46.5	286°34.8	22.7
23	292°05.8	147°59.7	49.2	228°19.7	36.7	219°31.5	46.5	301°37.3	22.8
Mer.pass. 03:35		ν -0.6' d -1.0' m -3.87		ν 0.7' d 0.3' m 0.88		ν 2.0' d 0.1' m -2.10		ν 2.6' d 0.1' m 0.77	

	SHA	Dec
Alpheratz	357°35.0	29°13.5
Ankaa	353°07.3	-42°10.1
Schedar	349°31.3	56°40.1
Diphda	348°47.5	-17°50.9
Achernar	335°20.3	-57°06.4
Hamal	327°51.6	23°34.7
Polaris	314°20.1	89°21.7
Acamar	315°12.1	-40°12.1
Menkar	314°06.6	4°11.2
Mirfak	308°29.0	49°56.7
Aldebaran	290°40.3	16°33.5
Rigel	281°04.5	-8°10.3
Capella	280°22.8	46°01.2
Bellatrix	278°23.5	6°22.4
Elnath	278°02.6	28°37.7
Alnilam	275°38.4	-1°11.1
Betelgeuse	270°52.8	7°24.8
Canopus	263°53.1	-52°42.3
Sirius	258°26.9	-16°44.8
Adhara	255°06.5	-29°00.2
Procyon	244°51.6	5°09.8
Pollux	243°18.1	27°58.1
Avior	234°15.6	-59°35.2
Suhail	222°47.0	-43°31.9
Miaplacidus	221°39.3	-69°49.1
Alphard	217°48.4	-8°45.8
Regulus	207°35.1	11°51.0
Dubhe	193°41.9	61°37.4
Denebola	182°25.5	14°26.2
Gienah	175°44.1	-17°40.7
Acrux	173°00.9	-63°14.3
Gacrux	171°52.3	-57°15.2
Alioth	166°13.5	55°49.9
Spica	158°22.8	-11°17.4
Alkaid	152°52.3	49°11.7
Hadar	148°36.6	-60°29.7
Menkent	147°58.1	-36°29.6
Arcturus	145°48.2	19°03.4
Rigel Kent.	139°40.8	-60°56.4
Kochab	137°19.5	74°03.5
Zuben'ubi	136°56.4	-16°08.7
Alphecca	126°03.9	26°38.1
Antares	112°16.1	-26°29.2
Atria	107°10.3	-69°04.5
Sabik	102°03.0	-15°45.3
Shaula	96°10.5	-37°07.4
Rasalhague	95°58.6	12°32.6
Eltanin	90°41.9	51°29.3
Kaus Aust.	83°32.6	-34°22.4
Vega	80°33.1	38°48.5
Nunki	75°47.9	-26°16.0
Altair	62°00.0	8°56.0
Peacock	53°05.7	-56°39.3
Deneb	49°25.6	45°22.0
Enif	33°38.9	9°59.3
Al Na'ir	27°32.9	-46°50.4
Fomalhaut	15°14.6	-29°29.4
Scheat	13°45.3	28°12.9
Markab	13°30.0	15°20.2

Jul 26 Fri	SHA	Mer.pass
Venus	219°29.9	13:06
Mars	298°20.6	07:50
Jupiter	288°00.9	08:30
Saturn	9°24.6	03:05

Jul 27 Sat	SHA	Mer.pass
Venus	218°16.6	13:07
Mars	297°37.8	07:48
Jupiter	287°48.9	08:27
Saturn	9°26.9	03:01

Jul 28 Sun	SHA	Mer.pass
Venus	217°03.6	13:08
Mars	296°54.9	07:47
Jupiter	287°37.0	08:24
Saturn	9°29.2	02:57

Horizontal parallax	
Venus:	0.1
Mars:	0.1

h		Sun		Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	178°21.8	N19°22.1	299°01.9	11.7'	N01°54.4	17.2'	59.7'
1	193°21.8	21.6	313°32.6	11.7'	02°11.6	17.1'	59.7'
2	208°21.8	21.0	328°03.3	11.7'	02°28.7	17.1'	59.7'
3	223°21.8	· · 20.5	342°34.0	11.7'	02°45.9	17.1'	59.7'
4	238°21.8	19.9	357°04.7	11.7'	03°03.0	17.1'	59.7'
5	253°21.8	19.3	11°35.5	11.7'	03°20.1	17.1'	59.7'
6	268°21.8	N19°18.8	26°06.2	11.7'	N03°37.1	17.0'	59.7'
7	283°21.8	18.2	40°36.9	11.7'	03°54.1	17.0'	59.6'
8	298°21.8	17.7	55°07.5	11.7'	04°11.1	17.0'	59.6'
9	313°21.8	· · 17.1	69°38.2	11.7'	04°28.1	16.9'	59.6'
10	328°21.8	16.5	84°08.9	11.7'	04°45.0	16.9'	59.6'
11	343°21.8	16.0	98°39.6	11.7'	05°01.9	16.9'	59.6'
12	358°21.8	N19°15.4	113°10.3	11.7'	N05°18.8	16.8'	59.6'
13	13°21.8	14.9	127°40.9	11.6'	05°35.6	16.8'	59.6'
14	28°21.8	14.3	142°11.5	11.6'	05°52.4	16.8'	59.5'
15	43°21.8	· · 13.7	156°42.2	11.6'	06°09.2	16.7'	59.5'
16	58°21.8	13.2	171°12.8	11.6'	06°25.9	16.7'	59.5'
17	73°21.8	12.6	185°43.4	11.6'	06°42.6	16.6'	59.5'
18	88°21.8	N19°12.0	200°13.9	11.6'	N06°59.2	16.6'	59.5'
19	103°21.9	11.5	214°44.5	11.5'	07°15.8	16.5'	59.5'
20	118°21.9	10.9	229°15.0	11.5'	07°32.3	16.5'	59.4'
21	133°21.9	· · 10.3	243°45.5	11.5'	07°48.8	16.4'	59.4'
22	148°21.9	09.8	258°16.0	11.5'	08°05.2	16.4'	59.4'
23	163°21.9	09.2	272°46.5	11.4'	08°21.6	16.3'	59.4'
SD = 15.7'		$d = -0.6'$	SD = 16.3'				
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	178°21.9	N19°08.6	287°16.9	11.4'	N08°38.0	16.3'	59.4'
1	193°21.9	08.1	301°47.4	11.4'	08°54.2	16.2'	59.4'
2	208°21.9	07.5	316°17.7	11.4'	09°10.5	16.2'	59.3'
3	223°21.9	· · 06.9	330°48.1	11.3'	09°26.6	16.1'	59.3'
4	238°21.9	06.4	345°18.4	11.3'	09°42.7	16.0'	59.3'
5	253°21.9	05.8	359°48.7	11.3'	09°58.8	16.0'	59.3'
6	268°21.9	N19°05.2	14°19.0	11.2'	N10°14.8	15.9'	59.3'
7	283°22.0	04.6	28°49.2	11.2'	10°30.7	15.9'	59.3'
8	298°22.0	04.1	43°19.4	11.2'	10°46.5	15.8'	59.2'
9	313°22.0	· · 03.5	57°49.6	11.1'	11°02.3	15.7'	59.2'
10	328°22.0	02.9	72°19.7	11.1'	11°18.1	15.7'	59.2'
11	343°22.0	02.3	86°49.8	11.0'	11°33.7	15.6'	59.2'
12	358°22.0	N19°01.8	101°19.8	11.0'	N11°49.3	15.5'	59.2'
13	13°22.0	01.2	115°49.9	11.0'	12°04.8	15.4'	59.1'
14	28°22.0	00.6	130°19.8	10.9'	12°20.3	15.4'	59.1'
15	43°22.0	19°00.0	144°49.7	10.9'	12°35.6	15.3'	59.1'
16	58°22.1	18°59.5	159°19.6	10.8'	12°50.9	15.2'	59.1'
17	73°22.1	58.9	173°49.5	10.8'	13°06.1	15.1'	59.1'
18	88°22.1	N18°58.3	188°19.2	10.7'	N13°21.3	15.1'	59.0'
19	103°22.1	57.7	202°49.0	10.7'	13°36.3	15.0'	59.0'
20	118°22.1	57.1	217°18.7	10.6'	13°51.3	14.9'	59.0'
21	133°22.1	· · 56.6	231°48.3	10.6'	14°06.2	14.8'	59.0'
22	148°22.1	56.0	246°17.9	10.6'	14°21.0	14.7'	59.0'
23	163°22.2	55.4	260°47.5	10.5'	14°35.7	14.6'	58.9'
SD = 15.7'		$d = -0.6'$	SD = 16.2'				
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	178°22.2	N18°54.8	275°17.0	10.4'	N14°50.4	14.5'	58.9'
1	193°22.2	54.2	289°46.4	10.4'	15°04.9	14.5'	58.9'
2	208°22.2	53.7	304°15.8	10.3'	15°19.4	14.4'	58.9'
3	223°22.2	· · 53.1	318°45.2	10.3'	15°33.7	14.3'	58.9'
4	238°22.2	52.5	333°14.5	10.2'	15°48.0	14.2'	58.8'
5	253°22.2	51.9	347°43.7	10.2'	16°02.2	14.1'	58.8'
6	268°22.3	N18°51.3	2°12.9	10.1'	N16°16.3	14.0'	58.8'
7	283°22.3	50.7	16°42.0	10.1'	16°30.3	13.9'	58.8'
8	298°22.3	50.1	31°11.1	10.0'	16°44.2	13.8'	58.8'
9	313°22.3	· · 49.6	45°40.1	9.9'	16°57.9	13.7'	58.7'
10	328°22.3	49.0	60°09.9	9.9'	17°11.6	13.6'	58.7'
11	343°22.3	48.4	74°37.9	9.8'	17°25.2	13.5'	58.7'
12	358°22.4	N18°47.8	89°06.7	9.8'	N17°38.7	13.4'	58.7'
13	13°22.4	47.2	103°35.5	9.7'	17°52.1	13.3'	58.7'
14	28°22.4	46.6	118°04.2	9.6'	18°05.4	13.2'	58.6'
15	43°22.4	· · 46.0	132°32.9	9.6'	18°18.5	13.1'	58.6'
16	58°22.4	45.4	147°01.5	9.5'	18°31.6	12.9'	58.6'
17	73°22.5	44.8	161°30.0	9.5'	18°44.5	12.8'	58.6'
18	88°22.5	N18°44.3	175°58.4	9.4'	N18°57.4	12.7'	58.6'
19	103°22.5	43.7	190°26.8	9.3'	19°10.1	12.6'	58.5'
20	118°22.5	43.1	204°55.2	9.3'	19°22.7	12.5'	58.5'
21	133°22.5	· · 42.5	219°23.4	9.2'	19°35.2	12.4'	58.5'
22	148°22.5	41.9	233°51.7	9.1'	19°47.6	12.3'	58.5'
23	163°22.6	41.3	248°19.8	9.1'	19°59.9	12.1'	58.4'
SD = 15.7'		$d = -0.6'$	SD = 16.1'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	—:—	—:—	00:19	23:29	—:—	—:—
68°	—:—	—:—	01:50	22:18	—:—	—:—
66°	—:—	—:—	02:28	21:42	—:—	—:—
64°	—:—	01:09	02:54	21:17	22:57	—:—
62°	—:—	01:56	03:14	20:57	22:13	—:—
60°	—:—	02:26	03:30	20:41	21:45	—:—
N 58°	01:02	02:47	03:44	20:28	21:24	23:04
56°	01:46	03:05	03:56	20:16	21:06	22:24
54°	02:13	03:20	04:06	20:06	20:52	21:58
52°	02:34	03:32	04:15	19:57	20:39	21:38
50°	02:50	03:43	04:23	19:49	20:29	21:21
45°	03:22	04:06	04:40	19:32	20:06	20:50
N 40°	03:45	04:23	04:54	19:18	19:49	20:27
35°	04:04	04:38	05:06	19:07	19:35	20:09
30°	04:19	04:50	05:16	18:56	19:23	19:54
20°	04:42	05:10	05:34	18:39	19:03	19:31
N 10°	05:00	05:27	05:49	18:24	18:46	19:12
0°	05:16	05:41	06:03	18:10	18:32	18:57
S 10°	05:29	05:55	06:17	17:56	18:18	18:44
20°	05:42	06:08	06:32	17:42	18:05	18:32
30°	05:54	06:23	06:48	17:25	17:50	18:19
35°	06:01	06:31	06:58	17:15	17:42	18:13
40°	06:07	06:40	07:09	17:04	17:33	18:06
45°	06:15	06:50	07:22	16:51	17:23	17:59
S 50°	06:23	07:02	07:38	16:36	17:11	17:51
52°	06:26	07:08	07:45	16:28	17:06	17:47
54°	06:30	07:13	07:53	16:20	17:00	17:44
56°	06:34	07:20	08:03	16:11	16:54	17:40
58°	06:38	07:27	08:13	16:01	16:47	17:35
S 60°	06:43	07:35	08:25	15:49	16:38	17:31

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	21:07	20:15	☐	11:29	14:06	☐
N 70°	21:18	20:41	19:30	11:21	13:43	16:42
68°	21:27	21:01	20:21	11:15	13:25	15:52
66°	21:34	21:17	20:53	11:10	13:11	15:22
64°	21:41	21:30	21:17	11:06	12:59	14:59
62°	21:46	21:41	21:36	11:02	12:49	14:41
60°	21:51	21:51	21:52	10:59	12:41	14:26
N 58°	21:56	21:59	22:05	10:56	12:34	14:13
56°	21:59	22:07	22:17	10:54	12:28	14:03
54°	22:03	22:14	22:27	10:52	12:22	13:53
52°	22:06	22:20	22:36	10:50	12:17	13:45
50°	22:09	22:25	22:45	10:48	12:12	13:37
45°	22:15	22:37	23:02	10:44	12:02	13:21
N 40°	22:21	22:47	23:17	10:41	11:54	13:08
35°	22:25	22:55	23:29	10:38	11:47	12:57
30°	22:29	23:03	23:40	10:35	11:41	12:47
20°	22:37	23:16	23:59	10:31	11:30	12:31
N 10°	22:43	23:28	· · ·	10:27	11:21	12:16
0°	22:49	23:39	· · ·	10:24	11:12	12:03
S 10°	22:55	23:50	· · ·	10:20	11:04	11:50
20°	23:02	· · ·	00:02	10:16	10:55	11:36
30°	23:09	· · ·	00:15	10:12	10:44	11:19
35°	23:14	· · ·	00:23	10:09	10:39	11:10
40°	23:19	· · ·	00:33	10:07	10:32	11:00
45°	23:25	· · ·	00:43	10:03	10:24	10:47
S 50°	23:32	· · ·	00:57	10:00	10:15	10:32
52°	23:35	· · ·	01:03	09:58	10:10	10:25
54°	23:39	· · ·	01:10	09:56	10:06	10:18
56°	23:43	· · ·	01:17	09:54	10:01	10:09
58°	23:48	· · ·	01:26	09:52	09:55	09:59
S 60°	23:53	· · ·	01:36	09:49	09:48	09:48

Day	Sun		Mer. Pass hh:mm	Moon		Age 21-23 74-51%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
26	06:33	06:33	12:07	04:12	16:36	
27	06:32	06:32	12:07	05:01	17:26	
28	06:31	06:31	12:07	05:51	18:17	

July 29, 30, 31 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	307°08.3	162°59.1	N15°48.2	243°20.4	N20°37.0	234°33.5	N21°46.6	316°39.9	S06°22.9	Alpheratz	357°35.0	29°13.5
1	322°10.7	177°58.5	47.2	258°21.1	37.4	249°35.4	46.6	331°42.5	22.9	Ankaa	353°07.3	-42°10.1
2	337°13.2	192°58.0	46.3	273°21.8	37.7	264°37.4	46.7	346°45.0	23.0	Schedar	349°31.2	56°40.1
3	352°15.7	207°57.4	•• 45.3	288°22.5	•• 38.0	279°39.4	•• 46.8	1°47.6	•• 23.0	Diphda	348°47.5	-17°50.9
4	7°18.1	222°56.9	44.3	303°23.1	38.4	294°41.4	46.8	16°50.2	23.1	Achernar	335°20.3	-57°06.4
5	22°20.6	237°56.3	43.3	318°23.8	38.7	309°43.3	46.9	31°52.7	23.1	Hamal	327°51.6	23°34.7
6	37°23.1	252°55.7	N15°42.3	333°24.5	N20°39.0	324°45.3	N21°46.9	46°55.3	S06°23.2	Polaris	314°18.3	89°21.7
7	52°25.5	267°55.2	41.4	348°25.2	39.3	339°47.3	47.0	61°57.9	23.2	Acamar	315°12.1	-40°12.1
8	67°28.0	282°54.6	40.4	3°25.9	39.7	354°49.3	47.0	77°00.4	23.3	Menkar	314°06.6	4°11.2
9	82°30.5	297°54.1	•• 39.4	18°26.5	•• 40.0	9°51.2	•• 47.1	92°03.0	•• 23.3	Mirfak	308°28.9	49°56.7
10	97°32.9	312°53.5	38.4	33°27.2	40.3	24°53.2	47.1	107°05.6	23.4	Aldebaran	290°40.2	16°33.5
11	112°35.4	327°53.0	37.4	48°27.9	40.7	39°55.2	47.2	122°08.1	23.4	Rigel	281°04.4	-8°10.2
12	127°37.9	342°52.4	N15°36.4	63°28.6	N20°41.0	54°57.2	N21°47.2	137°10.7	S06°23.5	Capella	280°22.8	46°01.2
13	142°40.3	357°51.8	35.5	78°29.3	41.3	69°59.1	47.3	152°13.3	23.5	Bellatrix	278°23.5	6°22.4
14	157°42.8	12°51.3	34.5	93°30.0	41.6	85°01.1	47.3	167°15.8	23.6	Elnath	278°02.6	28°37.7
15	172°45.2	27°50.7	•• 33.5	108°30.6	•• 42.0	100°03.1	•• 47.4	182°18.4	•• 23.7	Alnilam	275°38.3	-1°11.1
16	187°47.7	42°50.2	32.5	123°31.3	42.3	115°05.1	47.5	197°21.0	23.7	Betelgeuse	270°52.7	7°24.8
17	202°50.2	57°49.6	31.5	138°32.0	42.6	130°07.0	47.5	212°23.6	23.8	Canopus	263°53.0	-52°42.3
18	217°52.6	72°49.1	N15°30.5	153°32.7	N20°42.9	145°09.0	N21°47.6	227°26.1	S06°23.8	Sirius	258°26.9	-16°44.8
19	232°55.1	87°48.5	29.5	168°33.4	43.3	160°11.0	47.6	242°28.7	23.9	Adhara	255°06.5	-29°00.1
20	247°57.6	102°48.0	28.5	183°34.0	43.6	175°13.0	47.7	257°31.3	23.9	Procyon	244°51.5	5°09.8
21	263°00.0	117°47.4	•• 27.5	198°34.7	•• 43.9	190°15.0	•• 47.7	272°33.8	•• 24.0	Pollux	243°18.1	27°58.1
22	278°02.5	132°46.9	26.6	213°35.4	44.2	205°16.9	47.8	287°36.4	24.0	Avior	234°15.6	-59°35.2
23	293°05.0	147°46.3	25.6	228°36.1	44.6	220°18.9	47.8	302°39.0	24.1	Suhail	222°47.0	-43°31.8
Mer.pass. 03:31		ν -0.6' d -1.0' m -3.87		ν 0.7' d 0.3' m 0.87		ν 2.0' d 0.1' m -2.10		ν 2.6' d 0.1' m 0.77				

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	308°07.4	162°45.8	N15°24.6	243°36.8	N20°44.9	235°20.9	N21°47.9	317°41.5	S06°24.1	Denebola	182°25.5	14°26.2
1	323°09.9	177°45.2	23.6	258°37.5	45.2	250°22.9	47.9	332°44.1	24.2	Gienah	175°44.1	-17°40.7
2	338°12.4	192°44.7	22.6	273°38.1	45.5	265°24.8	48.0	347°46.7	24.2	Acrux	173°00.9	-63°14.3
3	353°14.8	207°44.1	•• 21.6	288°38.8	•• 45.9	280°26.8	•• 48.0	2°29.2	•• 24.3	Gacrux	171°52.4	-57°15.2
4	8°17.3	222°43.6	20.6	303°39.5	46.2	295°28.8	48.1	17°51.8	24.4	Alioth	166°13.5	55°49.9
5	23°19.7	237°43.0	19.6	318°40.2	46.5	310°30.8	48.2	32°54.4	24.4	Spica	158°22.8	-11°17.4
6	38°22.2	252°42.5	N15°18.6	333°40.9	N20°46.8	325°32.8	N21°48.2	47°57.0	S06°24.5	Alkaid	152°52.3	49°11.7
7	53°24.7	267°41.9	17.6	348°41.5	47.2	340°34.7	48.3	62°59.5	24.5	Hadar	148°36.6	-60°29.7
8	68°27.1	282°41.4	16.6	3°42.2	47.5	355°36.7	48.3	78°02.1	24.6	Menkent	147°58.1	-36°29.6
9	83°29.6	297°40.9	•• 15.6	18°42.9	•• 47.8	10°38.7	•• 48.4	93°04.7	•• 24.6	Arcturus	145°48.2	19°03.4
10	98°32.1	312°40.3	14.6	33°43.6	48.1	25°40.7	48.4	108°07.2	24.7	Rigel Kent.	139°40.8	-60°56.4
11	113°34.5	327°39.8	13.6	48°44.3	48.4	40°42.7	48.5	123°09.8	24.7	Kochab	137°19.6	74°03.5
12	128°37.0	342°39.2	N15°12.6	63°45.0	N20°48.8	55°44.6	N21°48.5	138°12.4	S06°24.8	Zuben'ubi	136°56.4	-16°08.7
13	143°39.5	357°38.7	11.6	78°45.6	49.1	70°46.6	48.6	153°14.9	24.8	Alphecca	126°03.9	26°38.1
14	158°41.9	12°38.1	10.6	93°46.3	49.4	85°48.6	48.6	168°17.5	24.9	Antares	112°16.1	-26°29.2
15	173°44.4	27°37.6	•• 09.6	108°47.0	•• 49.7	100°50.6	•• 48.7	183°20.1	•• 25.0	Atria	107°10.3	-69°04.5
16	188°46.9	42°37.1	08.6	123°47.7	50.0	115°52.6	48.7	198°22.7	25.0	Sabik	102°03.0	-15°45.3
17	203°49.3	57°36.5	07.6	138°48.4	50.4	130°54.5	48.8	213°25.2	25.1	Shaula	96°10.6	-37°07.4
18	218°51.8	72°36.0	N15°06.6	153°49.1	N20°50.7	145°56.5	N21°48.8	228°27.8	S06°25.1	Rasalhague	95°58.6	12°32.6
19	233°54.2	87°35.4	05.6	168°49.7	51.0	160°58.5	48.9	243°30.4	25.2	Eltanin	90°41.9	51°29.3
20	248°56.7	102°34.9	04.6	183°50.4	51.3	176°00.5	48.9	258°32.9	25.2	Kaus Aust.	83°32.6	-34°22.4
21	263°59.2	117°34.4	•• 03.6	198°51.1	•• 51.6	191°02.5	•• 49.0	273°35.5	•• 25.3	Vega	80°33.1	38°48.5
22	279°01.6	132°33.8	02.5	213°51.8	51.9	206°04.0	49.0	288°38.1	25.3	Nunki	75°47.9	-26°16.0
23	294°04.1	147°33.3	01.5	228°52.5	52.3	221°06.4	49.1	303°40.7	25.4	Altair	62°00.0	8°56.0
Mer.pass. 03:27		ν -0.5' d -1.0' m -3.87		ν 0.7' d 0.3' m 0.87		ν 2.0' d 0.1' m -2.11		ν 2.6' d 0.1' m 0.76				

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	309°06.6	162°32.7	N15°00.5	243°53.2	N20°52.6	236°08.4	N21°49.2	318°43.2	S06°25.4	0		
1	324°09.0	177°32.2	14°59.5	258°53.8	52.9	251°10.4	49.2	333°45.8	25.5	1		
2	339°11.5	192°31.7	58.5	273°54.5	53.2	266°12.4	49.3	348°48.4	25.6	2		
3	354°14.0	207°31.1	•• 57.5	288°55.2	•• 53.5	281°14.4	•• 49.3	3°51.0	•• 25.6	3		
4	9°16.4	222°30.6	56.5	303°55.9	53.8	296°16.3	49.4	18°53.5	25.7	4		
5	24°18.9	237°30.1	55.5	318°56.6	54.2	311°18.3	49.4	33°56.1	25.7	5		
6	39°21.4	252°29.5	N14°54.4	333°57.3	N20°54.5	326°20.3	N21°49.5	48°58.7	S06°25.8	6		
7	54°23.8	267°29.0	53.4	348°57.9	54.8	341°22.3	49.5	64°01.2	25.8	7		
8	69°26.3	282°28.5	52.4	3°58.6	55.1	356°24.3	49.6	79°03.8	25.9	8		
9	84°28.7	297°27.9	•• 51.4	18°59.3	•• 55.4	11°26.3	•• 49.6	94°06.4	•• 25.9	9		
10	99°31.2	312°27.4	50.4	34°00.2	55.7	26°28.2	49.7	109°09.0	26.0	10		
11	114°33.7	327°26.9	49.4	49°00.7	56.0	41°30.2	49.7	124°11.5	26.1	11		
12	129°36.1	342°26.3	N14°48.3	64°01.4	N20°56.3	56°32.2	N21°49.8	139°14.1	S06°26.1	12		
13	144°38.6	357°25.8	47.3	79°02.0	56.7	71°34.2	49.8	154°16.7	26.2	13		
14	159°41.1	12°25.3	46.3	94°02.7	57.0	86°36.2	49.9	169°19.3	26.2	14		
15	174°43.5	27°24.7	•• 45.3	109°03.4	•• 57.3	101°38.2	•• 49.9	184°21.8	•• 26.3	15		
16	189°46.0	42°24.2	44.3	124°04.1	57.6	116°40.1	50.0	199°24.4	26.3	16		
17	204°48.5	57°23.7	43.2	139°04.8	57.9	131°42.1	50.0	214°27.0	26.4	17		
18	219°50.9	72°23.1	N14°42.2	154°05.5	N20°58.2	146°44.1	N21°50.1	229°29.6	S06°26.5	18		
19	234°53.4	87°22.6	41.2	169°06.1	58.5	161°46.1	50.1	244°32.1	26.5	19		
20	249°55.8	102°22.1	40.2	184°06.8	58.8	176°48.1	50.2	259°34.7	26.6	20		
21	264°58.3	117°21.6	•• 39.2	199°07.5	•• 59.1	191°50.1	•• 50.2	274°37.3	•• 26.6	21		
22	280°00.8	132°21.0	38.1	214°08.2	59.4	206°52.1	50.3	289°39.9	26.7	22		
23	295°03.2	147°20.5	37.1	229°08.9	59.8	221°54.0	50.3	304°42.4	26.7	23		
Mer.pass. 03:23		ν -0.5' d -1.0' m -3.87		ν 0.7' d 0.3' m 0.87		ν 2.0' d 0.1' m -2.11		ν 2.6' d 0.1' m 0.76				

Jul 29 Mon		SHA	Mer.pass
Venus	215°50.8		13:09
Mars	296°12.1		07:46
Jupiter	287°25.2		08:21
Saturn	9°31.6		02:53

Jul 30 Tue		SHA	Mer.pass
Venus	214°38.4		13:09
Mars	295°29.3		07:45
Jupiter	287°13.5		08:18
Saturn	9°34.1		02:49

Jul 31 Wed		SHA	Mer.pass
Venus	213°26.2		13:10
Mars	294°46.6		07:44
Jupiter	287°01.8		08:14
Saturn	9°36.7		02:45

Horizontal parallax		
Venus:		0.1
Mars:		0.1

Sun				Moon					
Mon	GHA	Dec		GHA	ν	Dec	d	HP	
0	178°22.6	N18°40.7		262°47.9	9.0'	N20°12.0	12.0'	58.4'	
1	193°22.6	40.1		277°15.9	9.0'	20°24.0	11.9'	58.4'	
2	208°22.6	39.5		291°43.8	8.9'	20°35.9	11.8'	58.4'	
3	223°22.7	· · 38.9		306°11.7	8.8'	20°47.7	11.7'	58.4'	
4	238°22.7	38.3		320°39.5	8.8'	20°59.3	11.5'	58.3'	
5	253°22.7	37.7		335°07.3	8.7'	21°10.9	11.4'	58.3'	
6	268°22.7	N18°37.1		349°35.0	8.6'	N21°22.3	11.3'	58.3'	
7	283°22.7	36.5		4°02.6	8.6'	21°33.6	11.1'	58.3'	
8	298°22.8	35.9		18°30.2	8.5'	21°44.7	11.0'	58.3'	
9	313°22.8	· · 35.3		32°57.7	8.4'	21°55.7	10.9'	58.2'	
10	328°22.8	34.7		47°25.1	8.4'	22°06.6	10.8'	58.2'	
11	343°22.8	34.1		61°52.5	8.3'	22°17.3	10.6'	58.2'	
12	358°22.9	N18°33.5		76°19.8	8.2'	N22°28.0	10.5'	58.2'	
13	13°22.9	32.9		90°47.0	8.2'	22°38.4	10.3'	58.1'	
14	28°22.9	32.3		105°14.2	8.1'	22°48.8	10.2'	58.1'	
15	43°22.9	· · 31.7		119°41.3	8.0'	22°59.0	10.1'	58.1'	
16	58°23.0	31.1		134°08.3	8.0'	23°09.1	9.9'	58.1'	
17	73°23.0	30.5		148°35.3	7.9'	23°19.0	9.8'	58.1'	
18	88°23.0	N18°29.9		163°02.2	7.9'	N23°28.8	9.6'	58.0'	
19	103°23.0	29.3		177°29.1	7.8'	23°38.4	9.5'	58.0'	
20	118°23.1	28.7		191°55.9	7.7'	23°48.0	9.4'	58.0'	
21	133°23.1	· · 28.1		206°22.6	7.7'	23°57.3	9.2'	58.0'	
22	148°23.1	27.5		220°49.3	7.6'	24°06.5	9.1'	57.9'	
23	163°23.1	26.9		235°15.9	7.6'	24°15.6	8.9'	57.9'	
SD = 15.7'				$d = -0.6'$				SD = 15.9'	
Tue	GHA	Dec		GHA	ν	Dec	d	HP	
0	178°23.2	N18°26.3		249°42.4	7.5'	N24°24.5	8.8'	57.9'	
1	193°23.2	25.6		264°08.9	7.4'	24°33.3	8.6'	57.9'	
2	208°23.2	25.0		278°35.4	7.4'	24°41.9	8.5'	57.9'	
3	223°23.2	· · 24.4		293°01.7	7.3'	24°50.4	8.3'	57.8'	
4	238°23.3	23.8		307°28.1	7.3'	24°58.8	8.2'	57.8'	
5	253°23.3	23.2		321°54.3	7.2'	25°06.9	8.0'	57.8'	
6	268°23.3	N18°22.6		336°20.5	7.2'	N25°15.0	7.9'	57.8'	
7	283°23.4	22.0		350°46.7	7.1'	25°22.8	7.7'	57.7'	
8	298°23.4	21.4		5°12.8	7.1'	25°30.5	7.6'	57.7'	
9	313°23.4	· · 20.8		19°38.9	7.0'	25°38.1	7.4'	57.7'	
10	328°23.4	20.1		34°04.9	7.0'	25°45.5	7.2'	57.7'	
11	343°23.5	19.5		48°30.8	6.9'	25°52.8	7.1'	57.7'	
12	358°23.5	N18°18.9		62°56.7	6.9'	N25°59.9	6.9'	57.6'	
13	13°23.5	18.3		77°22.6	6.8'	26°06.8	6.8'	57.6'	
14	28°23.6	17.7		91°48.4	6.8'	26°13.6	6.6'	57.6'	
15	43°23.6	· · 17.1		106°14.2	6.7'	26°20.2	6.5'	57.6'	
16	58°23.6	16.5		120°39.9	6.7'	26°26.6	6.3'	57.6'	
17	73°23.6	15.8		135°05.6	6.6'	26°32.9	6.1'	57.5'	
18	88°23.7	N18°15.2		149°31.2	6.6'	N26°39.0	6.0'	57.5'	
19	103°23.7	14.6		163°56.8	6.6'	26°45.0	5.8'	57.5'	
20	118°23.7	14.0		178°22.4	6.5'	26°50.8	5.6'	57.5'	
21	133°23.8	· · 13.4		192°47.9	6.5'	26°56.5	5.5'	57.4'	
22	148°23.8	12.8		207°13.4	6.5'	27°01.9	5.3'	57.4'	
23	163°23.8	12.1		221°38.8	6.4'	27°07.2	5.1'	57.4'	
SD = 15.7'				$d = -0.6'$				SD = 15.8'	
Wed	GHA	Dec		GHA	ν	Dec	d	HP	
0	178°23.9	N18°11.5		236°04.3	6.4'	N27°12.4	5.0'	57.4'	
1	193°23.9	10.9		250°29.7	6.4'	27°17.4	4.8'	57.4'	
2	208°23.9	10.3		264°55.0	6.3'	27°22.2	4.6'	57.3'	
3	223°24.0	· · 09.6		279°20.4	6.3'	27°26.8	4.5'	57.3'	
4	238°24.0	09.0		293°45.7	6.3'	27°31.3	4.3'	57.3'	
5	253°24.0	08.4		308°11.0	6.3'	27°35.6	4.1'	57.3'	
6	268°24.1	N18°07.8		322°36.2	6.2'	N27°39.7	4.0'	57.2'	
7	283°24.1	07.2		337°01.5	6.2'	27°43.7	3.8'	57.2'	
8	298°24.1	06.5		351°26.7	6.2'	27°47.5	3.6'	57.2'	
9	313°24.2	· · 05.9		5°51.9	6.2'	27°51.2	3.5'	57.2'	
10	328°24.2	05.3		20°17.1	6.2'	27°54.6	3.3'	57.2'	
11	343°24.3	04.7		34°42.3	6.2'	27°57.9	3.1'	57.1'	
12	358°24.3	N18°04.0		49°07.5	6.2'	N28°01.0	3.0'	57.1'	
13	13°24.3	03.4		63°32.6	6.2'	28°04.0	2.8'	57.1'	
14	28°24.4	02.8		77°57.8	6.1'	28°06.8	2.6'	57.1'	
15	43°24.4	· · 02.1		92°22.9	6.1'	28°09.4	2.4'	57.0'	
16	58°24.4	01.5		106°48.1	6.1'	28°11.8	2.3'	57.0'	
17	73°24.5	00.9		121°13.2	6.1'	28°14.1	2.1'	57.0'	
18	88°24.5	N18°00.3		135°38.3	6.1'	N28°16.2	1.9'	57.0'	
19	103°24.5	17°59.6		150°03.5	6.1'	28°18.1	1.8'	57.0'	
20	118°24.6	59.0		164°28.6	6.2'	28°19.9	1.6'	56.9'	
21	133°24.6	· · 58.4		178°53.8	6.2'	28°21.5	1.4'	56.9'	
22	148°24.7	57.7		193°19.0	6.2'	28°22.9	1.2'	56.9'	
23	163°24.7	57.1		207°44.1	6.2'	28°24.2	1.1'	56.9'	
SD = 15.7'				$d = -0.6'$				SD = 15.6'	

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°	--:--	--:--	01:11	22:52	--:--	--:--
68°	--:--	--:--	02:07	22:01	--:--	--:--
66°	--:--	--:--	02:39	21:30	--:--	--:--
64°	--:--	01:30	03:03	21:07	22:37	--:--
62°	--:--	02:09	03:22	20:49	22:01	--:--
60°	--:--	02:35	03:37	20:34	21:35	--:--
N 58°	01:22	02:55	03:50	20:22	21:15	22:46
56°	01:57	03:12	04:01	20:11	20:59	22:13
54°	02:22	03:26	04:11	20:01	20:46	21:49
52°	02:41	03:38	04:19	19:52	20:34	21:30
50°	02:56	03:48	04:27	19:45	20:24	21:15
45°	03:27	04:10	04:44	19:29	20:02	20:45
N 40°	03:49	04:26	04:57	19:15	19:46	20:23
35°	04:06	04:40	05:08	19:04	19:32	20:06
30°	04:21	04:52	05:18	18:54	19:20	19:52
20°	04:43	05:11	05:35	18:38	19:01	19:29
N 10°	05:01	05:27	05:49	18:23	18:46	19:12
0°	05:16	05:41	06:03	18:10	18:32	18:57
S 10°	05:29	05:54	06:16	17:57	18:19	18:44
20°	05:41	06:07	06:30	17:43	18:06	18:32
30°	05:52	06:21	06:47	17:27	17:52	18:21
35°	05:58	06:29	06:56	17:17	17:44	18:15
40°	06:05	06:37	07:06	17:07	17:36	18:08
45°	06:12	06:47	07:19	16:55	17:26	18:02
S 50°	06:19	06:58	07:34	16:40	17:15	17:54
52°	06:22	07:03	07:41	16:33	17:10	17:51
54°	06:26	07:09	07:49	16:25	17:04	17:48
56°	06:30	07:15	07:57	16:16	16:58	17:44
58°	06:34	07:22	08:07	16:07	16:52	17:40
S 60°	06:38	07:30	08:18	15:55	16:44	17:36

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°						
N 70°						
68°						
66°	19:54			18:13		
64°	20:55			17:12		
62°	21:30	21:21		16:38	18:45	
60°	21:55	22:05	22:33	16:14	18:01	19:34
N 58°	22:15	22:35	23:12	15:54	17:32	18:56
56°	22:32	22:57	23:39	15:38	17:10	18:29
54°	22:47	23:16	· · ·	15:24	16:52	18:07
52°	22:59	23:32	· · ·	15:12	16:36	17:50
50°	23:10	23:45	· · ·	15:02	16:23	17:35
45°	23:34	· · ·	00:13	14:40	15:55	17:04
N 40°	23:52	· · ·	00:35	14:22	15:34	16:41
35°	· · ·	00:08	00:54	14:07	15:16	16:21
30°	· · ·	00:22	01:10	13:54	15:01	16:04
20°	· · ·	00:45	01:37	13:32	14:35	15:36
N 10°	00:15	01:06	02:00	13:14	14:13	15:12
0°	00:31	01:25	02:22	12:56	13:52	14:49
S 10°	00:46	01:44	02:44	12:39	13:31	14:27
20°	01:03	02:05	03:07	12:20	13:09	14:03
30°	01:22	02:29	03:35	11:59	12:43	13:35
35°	01:34	02:44	03:51	11:46	12:28	13:18
40°	01:47	03:00	04:10	11:32	12:11	12:59
45°	02:02	03:20	04:33	11:15	11:50	12:35
S 50°	02:22	03:45	05:03	10:54	11:24	12:05
52°	02:31	03:58	05:17	10:45	11:11	11:50
54°	02:41	04:12	05:34	10:34	10:57	11:33
56°	02:53	04:28	05:55	10:21	10:40	11:12
58°	03:07	04:48	06:21	10:07	10:20	10:46
S 60°	03:23	05:12	06:57	09:50	09:55	10:10

Day	Sun		Mer.	Moon		Age
	Eqn.of Time		Pass	Mer.Pass.		24-26
	00 ^h	12 ^h	Pass	Upper	Lower	40-20%
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	
29	06:30	06:29	12:06	06:43	19:10	
30	06:27	06:26	12:06	07:38	20:07	
31	06:25	06:23	12:06	08:36	21:05	

August 01, 02, 03 UT (Thu., Fri., Sat.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	310°05.7	162°20.0	N14°36.1	244°09.6	N21°00.1	236°56.0	N21°50.4	319°45.0	S06°26.8	Alpheratz	357°34.9	29°13.5
1	325°08.2	177°19.5	35.0	259°10.2	00.4	251°58.0	50.4	334°47.6	26.8	Ankaa	353°07.2	-42°10.1
2	340°10.6	192°18.9	34.0	274°10.9	00.7	267°00.0	50.5	349°50.2	26.9	Schedar	349°31.2	56°40.1
3	355°13.1	207°18.4	· · 33.0	289°11.6	· · 01.0	282°02.0	· · 50.5	4°52.7	· · 27.0	Diphda	348°47.5	-17°50.9
4	10°15.6	222°17.9	32.0	304°12.3	01.3	297°04.0	50.6	19°55.3	27.0	Achernar	335°20.3	-57°06.4
5	25°18.0	237°17.4	30.9	319°13.0	01.6	312°06.0	50.6	34°57.9	27.1	Hamal	327°51.6	23°34.7
6	40°20.5	252°16.8	N14°29.9	334°13.7	N21°01.9	327°08.0	N21°50.7	50°00.5	S06°27.1	Polaris	314°16.5	89°21.7
7	55°23.0	267°16.3	28.9	349°14.4	02.2	342°09.9	50.8	65°03.0	27.2	Acamar	315°12.1	-40°12.1
8	70°25.4	282°15.8	27.8	4°15.0	02.5	357°11.9	50.8	80°05.6	27.2	Menkar	314°06.6	4°11.2
9	85°27.9	297°15.3	· · 26.8	19°15.7	· · 02.8	12°13.9	· · 50.9	95°08.2	· · 27.3	Mirfak	308°28.9	49°56.7
10	100°30.3	312°14.8	25.8	34°16.4	03.1	27°15.9	50.9	110°10.8	27.4	Aldebaran	290°40.2	16°33.5
11	115°32.8	327°14.2	24.7	49°17.1	03.4	42°17.9	51.0	125°13.3	27.4	Rigel	281°04.4	-8°10.2
12	130°35.3	342°13.7	N14°23.7	64°17.8	N21°03.7	57°19.9	N21°51.0	140°15.9	S06°27.5	Capella	280°22.7	46°01.2
13	145°37.7	357°13.2	22.7	79°18.5	04.0	72°21.9	51.1	155°18.5	27.5	Bellatrix	278°23.5	6°22.4
14	160°40.2	12°12.7	21.6	94°19.2	04.3	87°23.9	51.1	170°21.1	27.6	Elnath	278°02.6	28°37.7
15	175°42.7	27°12.2	· · 20.6	109°19.8	· · 04.6	102°25.8	· · 51.2	185°23.7	· · 27.6	Alnilam	275°38.3	-1°11.1
16	190°45.1	42°11.6	19.6	124°20.5	04.9	117°27.8	51.2	200°26.2	27.7	Betelgeuse	270°52.7	7°24.8
17	205°47.6	57°11.1	18.5	139°21.2	05.2	132°29.8	51.3	215°28.8	27.8	Canopus	263°53.0	-52°42.3
18	220°50.1	72°10.6	N14°17.5	154°21.9	N21°05.6	147°31.8	N21°51.3	230°31.4	S06°27.8	Sirius	258°26.9	-16°44.8
19	235°52.5	87°10.1	16.4	169°22.6	05.9	162°33.8	51.4	245°34.0	27.9	Adhara	255°06.5	-29°00.1
20	250°55.0	102°09.6	15.4	184°23.3	06.2	177°35.8	51.4	260°36.5	27.9	Procyon	244°51.5	5°09.8
21	265°57.5	117°09.1	· · 14.4	199°24.0	· · 06.5	192°37.8	· · 51.5	275°39.1	· · 28.0	Pollux	243°18.1	27°58.1
22	280°59.9	132°08.5	13.3	214°24.6	06.8	207°39.8	51.5	290°41.7	28.0	Avior	234°15.6	-59°35.2
23	296°02.4	147°08.0	12.3	229°25.3	07.1	222°41.8	51.6	305°44.3	28.1	Suhail	222°47.0	-43°31.8
Mer.pass. 03:19		ν -0.5' d -1.0' m -3.87		ν 0.7' d 0.3' m 0.86		ν 2.0' d 0.1' m -2.11		ν 2.6' d 0.1' m 0.75				

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	311°04.8	162°07.5	N14°11.2	244°26.0	N21°07.4	237°43.8	N21°51.6	320°46.9	S06°28.2	Denebola	182°25.5	14°26.2
1	326°07.3	177°07.0	10.2	259°26.7	07.7	252°45.7	51.7	335°49.4	28.2	Gienah	175°44.1	-17°40.7
2	341°09.8	192°06.5	09.1	274°27.4	08.0	267°47.7	51.7	350°52.0	28.3	Acrux	173°00.9	-63°14.3
3	356°12.2	207°06.0	· · 08.1	289°28.1	· · 08.3	282°49.7	· · 51.8	5°54.6	· · 28.3	Gacrux	171°52.4	-57°15.2
4	11°14.7	222°05.5	07.1	304°28.8	08.6	297°51.7	51.8	20°57.2	28.4	Alioth	166°13.5	55°49.9
5	26°17.2	237°04.9	06.0	319°29.4	08.9	312°53.7	51.9	35°59.7	28.4	Spica	158°22.8	-11°17.4
6	41°19.6	252°04.4	N14°05.0	334°30.1	N21°09.2	327°55.7	N21°51.9	51°02.3	S06°28.5	Alkaid	152°52.4	-49°11.7
7	56°22.1	267°03.9	03.9	349°30.8	09.5	342°57.7	52.0	66°04.9	28.6	Hadar	148°36.6	-60°29.7
8	71°24.6	282°03.4	02.9	4°31.5	09.7	357°59.7	52.0	81°07.5	28.6	Menkent	147°58.1	-36°29.6
9	86°27.0	297°02.9	· · 01.8	19°32.2	· · 10.0	13°01.7	· · 52.1	96°10.1	· · 28.7	Arcturus	145°48.3	19°03.4
10	101°29.5	312°02.4	14°00.8	34°32.9	10.3	28°03.7	52.1	111°12.6	28.7	Rigil Kent.	139°40.8	-60°56.4
11	116°32.0	327°01.9	13°59.7	49°33.6	10.6	43°05.7	52.2	126°15.2	28.8	Kochab	137°19.6	74°03.5
12	131°34.4	342°01.4	N13°58.7	64°34.3	N21°10.9	58°07.7	N21°52.2	141°17.8	S06°28.9	Zuben'ubi	136°56.4	-16°08.7
13	146°36.9	357°00.9	57.6	79°34.9	11.2	73°09.6	52.3	156°20.4	28.9	Alphecca	126°03.9	26°38.1
14	161°39.3	12°00.4	56.6	94°35.6	11.5	88°11.6	52.3	171°23.0	29.0	Antares	112°16.1	-26°29.2
15	176°41.8	26°59.9	· · 55.5	109°36.3	· · 11.8	103°13.6	· · 52.4	186°25.5	· · 29.0	Atria	107°10.3	-69°04.5
16	191°44.3	41°59.3	54.5	124°37.0	12.1	118°15.6	52.4	201°28.1	29.1	Sabik	102°03.0	-15°45.3
17	206°46.7	56°58.8	53.4	139°37.7	12.4	133°17.6	52.5	216°30.7	29.1	Shaula	96°10.6	-37°07.4
18	221°49.2	71°58.3	N13°52.4	154°38.4	N21°12.7	148°19.6	N21°52.5	231°33.3	S06°29.2	Rasalhague	95°58.6	12°32.6
19	236°51.7	86°57.8	51.3	169°39.1	13.0	163°21.6	52.6	246°35.9	29.3	Eltanin	90°41.9	51°29.3
20	251°54.1	101°57.3	50.2	184°39.8	13.3	178°23.6	52.6	261°38.5	29.3	Kaus Aust.	83°32.6	-34°22.4
21	266°56.6	116°56.8	· · 49.2	199°40.4	· · 13.6	193°25.6	· · 52.7	276°41.0	· · 29.4	Vega	80°33.1	38°48.5
22	281°59.1	131°56.3	48.1	214°41.1	13.9	208°27.6	52.7	291°43.6	29.4	Nunki	75°47.9	-26°16.0
23	297°01.5	146°55.8	47.1	229°41.8	14.2	223°29.6	52.8	306°46.2	29.5	Altair	62°00.0	8°56.1
Mer.pass. 03:15		ν -0.5' d -1.0' m -3.87		ν 0.7' d 0.3' m 0.86		ν 2.0' d 0.1' m -2.12		ν 2.6' d 0.1' m 0.75				

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	312°04.0	161°55.3	N13°46.0	244°42.5	N21°14.5	238°31.6	N21°52.8	321°48.8	S06°29.6	01 Thu	SHA	Mer.pass
1	327°06.5	176°54.8	45.0	259°43.2	14.8	253°33.6	52.9	336°51.4	29.6	Venus	212°14.3	13:11
2	342°08.9	191°54.3	43.9	274°43.9	15.0	268°35.6	52.9	351°53.9	29.7	Mars	294°03.9	07:43
3	357°11.4	206°53.8	· · 42.8	289°44.6	· · 15.3	283°37.6	· · 53.0	6°56.5	· · 29.7	Jupiter	286°50.3	08:11
4	12°13.8	221°53.3	41.8	304°45.3	15.6	298°39.6	53.0	21°59.1	29.8	Saturn	9°39.3	02:41
5	27°16.3	236°52.8	40.7	319°45.9	15.9	313°41.6	53.0	37°01.7	29.8			
6	42°18.8	251°52.3	N13°39.7	334°46.6	N21°16.2	328°43.6	N21°53.1	52°04.3	S06°29.9			
7	57°21.2	266°51.8	38.6	349°47.3	16.5	343°45.5	53.1	67°06.8	30.0			
8	72°23.7	281°51.3	37.5	4°48.0	16.8	358°47.5	53.2	82°09.4	30.0			
9	87°26.2	296°50.8	· · 36.5	19°48.7	· · 17.1	13°49.5	· · 53.2	97°12.0	· · 30.1			
10	102°28.6	311°50.3	35.4	34°49.4	17.4	28°51.5	53.3	112°14.6	30.1			
11	117°31.1	326°49.8	34.3	49°50.1	17.7	43°53.5	53.3	127°17.2	30.2			
12	132°33.6	341°49.3	N13°33.3	64°50.8	N21°17.9	58°55.5	N21°53.4	142°19.8	S06°30.3			
13	147°36.0	356°48.8	32.2	79°51.5	18.2	73°57.5	53.4	157°22.3	30.3			
14	162°38.5	11°48.3	31.1	94°52.1	18.5	88°59.5	53.5	172°24.9	30.4			
15	177°41.0	26°47.8	· · 30.1	109°52.8	· · 18.8	104°01.5	· · 53.5	187°27.5	· · 30.4			
16	192°43.4	41°47.3	29.0	124°53.5	19.1	119°03.5	53.6	202°30.1	30.5			
17	207°45.9	56°46.8	27.9	139°54.2	19.4	134°05.5	53.6	217°32.7	30.6			
18	222°48.3	71°46.3	N13°26.9	154°54.9	N21°19.7	149°07.5	N21°53.7	232°35.3	S06°30.6			
19	237°50.8	86°45.8	25.8	169°55.6	20.0	164°09.5	53.7	247°37.8	30.7			
20	252°53.3	101°45.3	24.7	184°56.3	20.2	179°11.5	53.8	262°40.4	30.7			
21	267°55.7	116°44.9	· · 23.7	199°57.0	· · 20.5	194°13.5	· · 53.8	277°43.0	· · 30.8			
22	282°58.2	131°44.4	22.6	214°57.7	20.8	209°15.5	53.9	292°45.6	30.9			
23	298°00.7	146°43.9	21.5	229°58.3	21.							

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	178°24.7	N17°56.5	222°09.3	6.2'	N28°25.2	0.9'	56.9'
1	193°24.8	55.8	236°34.5	6.2'	28°26.1	0.7'	56.8'
2	208°24.8	55.2	250°59.7	6.2'	28°26.9	0.6'	56.8'
3	223°24.9	· · 54.6	265°24.9	6.2'	28°27.5	0.4'	56.8'
4	238°24.9	53.9	279°50.2	6.3'	28°27.9	0.2'	56.8'
5	253°24.9	53.3	294°15.5	6.3'	28°28.1	0.1'	56.7'
6	268°25.0	N17°52.7	308°40.8	6.3'	N28°28.1	-0.1'	56.7'
7	283°25.0	52.0	323°06.1	6.3'	28°28.0	-0.3'	56.7'
8	298°25.1	51.4	337°31.4	6.4'	28°27.8	-0.4'	56.7'
9	313°25.1	· · 50.8	351°56.8	6.4'	28°27.3	-0.6'	56.7'
10	328°25.1	50.1	6°22.2	6.4'	28°26.7	-0.8'	56.6'
11	343°25.2	49.5	20°47.6	6.5'	28°25.9	-0.9'	56.6'
12	358°25.2	N17°48.8	35°13.1	6.5'	N28°25.0	-1.1'	56.6'
13	13°25.3	48.2	49°38.6	6.6'	28°23.9	-1.3'	56.6'
14	28°25.3	47.6	64°04.2	6.6'	28°22.6	-1.4'	56.6'
15	43°25.4	· · 46.9	78°29.8	6.6'	28°21.2	-1.6'	56.5'
16	58°25.4	46.3	92°55.4	6.7'	28°19.6	-1.8'	56.5'
17	73°25.4	45.6	107°21.1	6.7'	28°17.8	-1.9'	56.5'
18	88°25.5	N17°45.0	121°46.8	6.8'	N28°15.9	-2.1'	56.5'
19	103°25.5	44.4	136°12.6	6.8'	28°13.8	-2.3'	56.4'
20	118°25.6	43.7	150°38.4	6.9'	28°11.5	-2.4'	56.4'
21	133°25.6	· · 43.1	165°04.3	6.9'	28°09.1	-2.6'	56.4'
22	148°25.7	42.4	179°30.2	7.0'	28°06.5	-2.7'	56.4'
23	163°25.7	41.8	193°56.2	7.0'	28°03.8	-2.9'	56.4'
SD = 15.8'			SD = 15.5'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	178°25.8	N17°41.1	208°22.2	7.1'	N28°00.9	-3.1'	56.3'
1	193°25.8	40.5	222°48.3	7.2'	27°57.8	-3.2'	56.3'
2	208°25.8	39.8	237°14.5	7.2'	27°54.6	-3.4'	56.3'
3	223°25.9	· · 39.2	251°40.7	7.3'	27°51.2	-3.5'	56.3'
4	238°25.9	38.6	266°07.0	7.3'	27°47.7	-3.7'	56.3'
5	253°26.0	37.9	280°33.3	7.4'	27°44.0	-3.8'	56.2'
6	268°26.0	N17°37.3	294°59.7	7.5'	N27°40.2	-4.0'	56.2'
7	283°26.1	36.6	309°26.2	7.5'	27°36.2	-4.1'	56.2'
8	298°26.1	36.0	323°52.7	7.6'	27°32.1	-4.3'	56.2'
9	313°26.2	· · 35.3	338°19.3	7.7'	27°27.8	-4.4'	56.2'
10	328°26.2	34.7	352°46.0	7.8'	27°23.4	-4.6'	56.1'
11	343°26.3	34.0	7°12.8	7.8'	27°18.8	-4.7'	56.1'
12	358°26.3	N17°33.4	21°39.6	7.9'	N27°14.0	-4.9'	56.1'
13	13°26.4	32.7	36°06.5	8.0'	27°09.1	-5.0'	56.1'
14	28°26.4	32.1	50°33.5	8.1'	27°04.1	-5.2'	56.1'
15	43°26.5	· · 31.4	65°00.6	8.1'	26°58.9	-5.3'	56.0'
16	58°26.5	30.8	79°27.7	8.2'	26°53.6	-5.5'	56.0'
17	73°26.6	30.1	93°54.9	8.3'	26°48.2	-5.6'	56.0'
18	88°26.6	N17°29.4	108°22.2	8.4'	N26°42.6	-5.7'	56.0'
19	103°26.7	28.8	122°49.6	8.5'	26°36.8	-5.9'	56.0'
20	118°26.7	28.1	137°17.1	8.5'	26°30.9	-6.0'	55.9'
21	133°26.8	· · 27.5	151°44.6	8.6'	26°24.9	-6.2'	55.9'
22	148°26.8	26.8	166°12.3	8.7'	26°18.8	-6.3'	55.9'
23	163°26.9	26.2	180°40.0	8.8'	26°12.5	-6.4'	55.9'
SD = 15.8'			SD = 15.4'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	178°26.9	N17°25.5	195°07.8	8.9'	N26°06.0	-6.6'	55.9'
1	193°27.0	24.9	209°35.7	9.0'	25°59.5	-6.7'	55.8'
2	208°27.0	24.2	224°03.7	9.1'	25°52.8	-6.8'	55.8'
3	223°27.1	· · 23.5	238°31.7	9.2'	25°45.9	-7.0'	55.8'
4	238°27.1	22.9	252°59.9	9.3'	25°39.0	-7.1'	55.8'
5	253°27.2	22.2	267°28.2	9.3'	25°31.9	-7.2'	55.8'
6	268°27.2	N17°21.6	281°56.5	9.4'	N25°24.7	-7.3'	55.7'
7	283°27.3	20.9	296°24.9	9.5'	25°17.3	-7.5'	55.7'
8	298°27.3	20.2	310°53.5	9.6'	25°09.9	-7.6'	55.7'
9	313°27.4	· · 19.6	325°22.1	9.7'	25°02.3	-7.7'	55.7'
10	328°27.4	18.9	339°50.8	9.8'	24°54.6	-7.8'	55.7'
11	343°27.5	18.3	354°19.6	9.9'	24°46.7	-8.0'	55.6'
12	358°27.6	N17°17.6	8°48.5	10.0'	N24°38.8	-8.1'	55.6'
13	13°27.6	16.9	23°17.5	10.1'	24°30.7	-8.2'	55.6'
14	28°27.7	16.3	37°46.5	10.2'	24°22.5	-8.3'	55.6'
15	43°27.7	· · 15.6	52°15.7	10.3'	24°14.2	-8.4'	55.6'
16	58°27.8	14.9	66°45.0	10.4'	24°05.8	-8.5'	55.5'
17	73°27.8	14.3	81°14.4	10.5'	23°57.3	-8.6'	55.5'
18	88°27.9	N17°13.6	95°43.8	10.6'	N23°48.6	-8.8'	55.5'
19	103°27.9	12.9	110°13.4	10.7'	23°39.9	-8.9'	55.5'
20	118°28.0	12.3	124°43.1	10.8'	23°31.0	-9.0'	55.5'
21	133°28.1	· · 11.6	139°12.8	10.8'	23°22.0	-9.1'	55.4'
22	148°28.1	10.9	153°42.6	10.9'	23°12.9	-9.2'	55.4'
23	163°28.2	10.3	168°12.6	11.0'	23°03.7	-9.3'	55.4'
SD = 15.8'			SD = 15.2'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	01:39	22:27	☐	☐
68°	☐	☐	02:22	21:46	☐	☐
66°	☐	☐	00:45	21:18	23:14	☐
64°	☐	☐	01:48	20:57	22:20	☐
62°	☐	☐	02:21	20:41	21:48	☐
60°	00:41	02:45	03:44	20:27	21:25	23:20
N 58°	01:38	03:03	03:56	20:15	21:07	22:30
56°	02:08	03:19	04:07	20:05	20:52	22:02
54°	02:30	03:32	04:16	19:56	20:39	21:40
52°	02:48	03:43	04:24	19:47	20:28	21:23
50°	03:03	03:53	04:31	19:40	20:18	21:08
45°	03:31	04:13	04:47	19:25	19:58	20:40
N 40°	03:52	04:29	05:00	19:12	19:42	20:19
35°	04:09	04:43	05:10	19:02	19:29	20:03
30°	04:23	04:54	05:20	18:52	19:18	19:49
20°	04:45	05:13	05:36	18:36	19:00	19:27
N 10°	05:02	05:28	05:50	18:23	18:45	19:11
0°	05:16	05:41	06:03	18:10	18:31	18:57
S 10°	05:28	05:54	06:16	17:57	18:19	18:44
20°	05:40	06:06	06:29	17:44	18:07	18:33
30°	05:51	06:19	06:44	17:28	17:54	18:22
35°	05:56	06:27	06:53	17:20	17:46	18:17
40°	06:02	06:35	07:03	17:10	17:38	18:11
45°	06:08	06:44	07:15	16:58	17:29	18:05
S 50°	06:15	06:54	07:29	16:44	17:19	17:58
52°	06:18	06:59	07:36	16:37	17:14	17:55
54°	06:21	07:04	07:43	16:30	17:09	17:52
56°	06:25	07:10	07:51	16:22	17:03	17:48
58°	06:28	07:16	08:01	16:12	16:57	17:45
S 60°	06:32	07:23	08:11	16:02	16:50	17:41

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	☐	☐	☐	☐	☐	22:37
64°	☐	☐	☐	☐	☐	21:49
62°	☐	☐	00:36	☐	21:26	21:19
60°	23:39	· ·	01:14	20:28	20:48	20:56
N 58°	· ·	00:16	01:40	19:52	20:22	20:37
56°	· ·	00:42	02:01	19:25	20:00	20:21
54°	00:00	01:02	02:18	19:05	19:43	20:07
52°	00:18	01:20	02:33	18:47	19:28	19:56
50°	00:33	01:34	02:46	18:32	19:15	19:45
45°	01:04	02:04	03:12	18:02	18:48	19:23
N 40°	01:27	02:27	03:32	17:39	18:26	19:05
35°	01:47	02:47	03:50	17:19	18:09	18:50
30°	02:04	03:03	04:04	17:02	17:53	18:37
20°	02:32	03:31	04:29	16:34	17:27	18:14
N 10°	02:57	03:55	04:51	16:10	17:04	17:54
0°	03:20	04:17	05:11	15:47	16:43	17:36
S 10°	03:42	04:39	05:31	15:25	16:22	17:17
20°	04:07	05:03	05:52	15:00	15:59	16:57
30°	04:36	05:30	06:17	14:32	15:32	16:34
35°	04:53	05:46	06:31	14:15	15:16	16:20
40°	05:13	06:05	06:48	13:55	14:58	16:04
45°	05:37	06:28	07:08	13:31	14:36	15:45
S 50°	06:08	06:57	07:33	13:00	14:07	15:21
52°	06:23	07:12	07:45	12:44	13:53	15:10
54°	06:41	07:28	07:59	12:26	13:37	14:57
56°	07:03	07:48	08:14	12:04	13:17	14:41
58°	07:32	08:13	08:33	11:36	12:53	14:23
S 60°	08:13	08:46	08:56	10:55	12:20	14:01

Day	Sun		Mer. Pass hh:mm	Moon		Age 27-29 13-2%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
01	06:21	06:19	12:06	09:34	22:02	
02	06:17	06:15	12:06	10:30	22:57	
03	06:12	06:10	12:06	11:24	23:49	

August 04, 05, 06 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	313°03.1	161°43.4	N13°20.4	244°59.0	N21°21.4	239°19.5	N21°54.0	322°50.8	S06°31.0	Alpheratz	357°34.9	29°13.5
1	328°05.6	176°42.9		259°59.7		254°21.5	54.0	337°53.4	31.0	Ankaa	353°07.2	-42°10.1
2	343°08.1	191°42.4		275°00.4	21.9	269°23.5	54.1	352°55.9	31.1	Schedar	349°31.2	56°40.1
3	358°10.5	206°41.9	•• 17.2	290°01.1	•• 22.2	284°25.5	•• 54.1	7°58.5	•• 31.2	Diphda	348°47.5	-17°50.9
4	13°13.0	221°41.4		305°01.8	22.5	299°27.5	54.2	23°01.1	31.2	Achernar	335°20.2	-57°06.4
5	28°15.4	236°40.9		320°02.5	22.8	314°29.5	54.2	38°03.7	31.3	Hamal	327°51.5	23°34.7
6	43°17.9	251°40.4	N13°14.0	335°03.2	N21°23.1	329°31.5	N21°54.3	53°06.3	S06°31.3	Polaris	314°15.0	89°21.7
7	58°20.4	266°39.9		350°03.9	23.4	344°33.5	54.3	68°08.9	31.4	Acamar	315°12.0	-40°12.1
8	73°22.8	281°39.5		5°04.6	23.6	359°35.5	54.4	83°11.4	31.5	Menkar	314°06.5	4°11.3
9	88°25.3	296°39.0	•• 10.7	20°05.2	•• 23.9	14°37.5	•• 54.4	98°14.0	•• 31.5	Mirfak	308°28.9	49°56.7
10	103°27.8	311°38.5		35°05.9	24.2	29°39.5	54.4	113°16.6	31.6	Aldebaran	290°40.2	16°33.5
11	118°30.2	326°38.0		50°06.6	24.5	44°41.5	54.5	128°19.2	31.6	Rigel	281°04.4	-8°10.2
12	133°32.7	341°37.5	N13°07.5	65°07.3	N21°24.8	59°43.5	N21°54.5	143°21.8	S06°31.7	Capella	280°22.7	46°01.2
13	148°35.2	356°37.0		80°08.0	25.0	74°45.5	54.6	158°24.4	31.8	Bellatrix	278°23.5	6°22.4
14	163°37.6	11°36.5		95°08.7	25.3	89°47.5	54.6	173°27.0	31.8	Elnath	278°02.5	28°37.7
15	178°40.1	26°36.1	•• 04.3	110°09.4	•• 25.6	104°49.5	•• 54.7	188°29.5	•• 31.9	Alnilam	275°38.3	-1°11.0
16	193°42.6	41°35.6		125°10.1	25.9	119°51.5	54.7	203°32.1	31.9	Betelgeuse	270°52.7	7°24.8
17	208°45.0	56°35.1		140°10.8	26.2	134°53.5	54.8	218°34.7	32.0	Canopus	263°53.0	-52°42.2
18	223°47.5	71°34.6	N13°01.0	155°11.5	N21°26.4	149°55.5	N21°54.8	233°37.3	S06°32.1	Sirius	258°26.8	-16°44.8
19	238°49.9	86°34.1	12°59.9	170°12.2	26.7	164°57.5	54.9	248°39.9	32.1	Adhara	255°06.5	-29°00.1
20	253°52.4	101°33.6		185°12.8	27.0	179°59.5	54.9	263°42.5	32.2	Procyon	244°51.5	5°09.8
21	268°54.9	116°33.2	•• 57.8	200°13.5	•• 27.3	195°01.5	•• 55.0	278°45.1	•• 32.2	Pollux	243°18.1	27°58.1
22	283°57.3	131°32.7		215°14.2	27.5	210°03.5	55.0	293°47.7	32.3	Avior	234°15.5	-59°35.2
23	298°59.8	146°32.2		230°14.9	27.8	225°05.5	55.1	308°50.2	32.4	Suhail	222°47.0	-43°31.8
Mer.pass. 03:07		ν -0.5' d -1.1' m -3.86		ν 0.7' d 0.3' m 0.85		ν 2.0' d 0.0' m -2.13		ν 2.6' d 0.1' m 0.74		Miaplacidus	221°39.3	-69°49.0
										Alphard	217°48.4	-8°45.8
										Regulus	207°35.1	11°51.0
										Dubhe	193°41.9	61°37.3
										Denebola	182°25.5	14°26.2
										Gienah	175°44.2	-17°40.7
										Acrux	173°00.9	-63°14.3
										Gacrux	171°52.4	-57°15.2
										Alioth	166°13.5	55°49.9
										Spica	158°22.8	-11°17.4
										Alkaid	152°52.4	49°11.7
										Hadar	148°36.6	-60°29.7
										Menkent	147°58.1	-36°29.6
										Arcturus	145°48.3	19°03.4
										Rigil Kent.	139°40.8	-60°56.5
										Kochab	137°19.7	74°03.5
										Zuben'ubi	136°56.4	-16°08.7
										Alphecca	126°03.9	26°38.1
										Antares	112°16.1	-26°29.2
										Atria	107°10.3	-69°04.5
										Sabik	102°03.0	-15°45.3
										Shaula	96°10.6	-37°07.4
										Rasalhague	95°58.6	12°32.6
										Eltanin	90°41.9	51°29.3
										Kaus Aust.	83°32.6	-34°22.4
										Vega	80°33.1	38°48.5
										Nunki	75°47.9	-26°16.0
										Altair	62°00.0	8°56.1
										Peacock	53°05.6	-56°39.3
										Deneb	49°25.6	45°22.1
										Enif	33°38.8	9°59.3
										Al Na'ir	27°32.9	-46°50.4
										Fomalhaut	15°14.6	-29°29.4
										Scheat	13°45.2	28°12.9
										Markab	13°30.0	15°20.3
										Aug 04 Sun SHA Mer.pass		
										Venus	208°40.3	13:14
										Mars	291°55.9	07:40
										Jupiter	286°16.4	08:02
										Saturn	9°47.6	02:28
										Aug 05 Mon SHA Mer.pass		
										Venus	207°29.4	13:14
										Mars	291°13.3	07:39
										Jupiter	286°05.3	07:58
										Saturn	9°50.6	02:24
										Aug 06 Tue SHA Mer.pass		
										Venus	206°18.9	13:15
										Mars	290°30.8	07:37
										Jupiter	285°54.3	07:55
										Saturn	9°53.6	02:20
										Horizontal parallax		
										Venus:	0.1	
										Mars:	0.1	
Mer.pass. 02:59		ν -0.5' d -1.1' m -3.86		ν 0.7' d 0.3' m 0.84		ν 2.0' d 0.0' m -2.13		ν 2.6' d 0.1' m 0.73				

Sun				Moon				
Sun	GHA	Dec		GHA	ν	Dec	d	HP
0	178°28.2	N17°09.6		182°42.6	11.1'	N22°54.4	-9.4'	55.4'
1	193°28.3	08.9		197°12.8	11.2'	22°45.0	-9.5'	55.4'
2	208°28.3	08.3		211°43.0	11.3'	22°35.5	-9.6'	55.4'
3	223°28.4	07.6		226°13.3	11.4'	22°25.9	-9.7'	55.3'
4	238°28.5	06.9		240°43.7	11.5'	22°16.2	-9.8'	55.3'
5	253°28.5	06.3		255°14.3	11.6'	22°06.4	-9.9'	55.3'
6	268°28.6	N17°05.6		269°44.9	11.7'	N21°56.5	-10.0'	55.3'
7	283°28.6	04.9		284°15.6	11.8'	21°46.5	-10.1'	55.3'
8	298°28.7	04.2		298°46.4	11.9'	21°36.4	-10.2'	55.2'
9	313°28.8	03.6		313°17.3	12.0'	21°26.3	-10.3'	55.2'
10	328°28.8	02.9		327°48.3	12.1'	21°16.0	-10.4'	55.2'
11	343°28.9	02.2		342°19.3	12.2'	21°05.6	-10.5'	55.2'
12	358°28.9	N17°01.6		356°50.5	12.3'	N20°55.2	-10.5'	55.2'
13	13°29.0	00.9		11°21.8	12.4'	20°44.6	-10.6'	55.2'
14	28°29.1	17°00.2		25°53.2	12.5'	20°34.0	-10.7'	55.1'
15	43°29.1	16°59.5		40°24.6	12.5'	20°23.3	-10.8'	55.1'
16	58°29.2	58.9		54°56.2	12.6'	20°12.5	-10.9'	55.1'
17	73°29.2	58.2		69°27.8	12.7'	20°01.6	-11.0'	55.1'
18	88°29.3	N16°57.5		83°59.5	12.8'	N19°50.6	-11.0'	55.1'
19	103°29.4	56.8		98°31.3	12.9'	19°39.6	-11.1'	55.1'
20	118°29.4	56.1		113°03.2	13.0'	19°28.4	-11.2'	55.0'
21	133°29.5	55.5		127°35.2	13.1'	19°17.2	-11.3'	55.0'
22	148°29.6	54.8		142°07.3	13.2'	19°06.0	-11.4'	55.0'
23	163°29.6	54.1		156°39.5	13.3'	18°54.6	-11.4'	55.0'
SD = 15.8' d = -0.7'				SD = 15.1'				
Mon	GHA	Dec		GHA	ν	Dec	d	HP
0	178°29.7	N16°53.4		171°11.8	13.4'	N18°43.2	-11.5'	55.0'
1	193°29.8	52.7		185°44.1	13.4'	18°31.7	-11.6'	54.9'
2	208°29.8	52.1		200°16.6	13.5'	18°20.1	-11.7'	54.9'
3	223°29.9	51.4		214°49.1	13.6'	18°08.4	-11.7'	54.9'
4	238°29.9	50.7		229°21.7	13.7'	17°56.7	-11.8'	54.9'
5	253°30.0	50.0		243°54.4	13.8'	17°44.9	-11.9'	54.9'
6	268°30.1	N16°49.3		258°27.2	13.9'	N17°33.1	-11.9'	54.9'
7	283°30.1	48.7		273°00.0	13.9'	17°21.1	-12.0'	54.9'
8	298°30.2	48.0		287°33.0	14.0'	17°09.1	-12.1'	54.8'
9	313°30.3	47.3		302°06.0	14.1'	16°57.1	-12.1'	54.8'
10	328°30.3	46.6		316°39.1	14.2'	16°45.0	-12.2'	54.8'
11	343°30.4	45.9		331°12.3	14.3'	16°32.8	-12.2'	54.8'
12	358°30.5	N16°45.2		345°45.5	14.3'	N16°20.6	-12.3'	54.8'
13	13°30.5	44.5		0°18.9	14.4'	16°08.3	-12.4'	54.8'
14	28°30.6	43.9		14°52.3	14.5'	15°55.9	-12.4'	54.7'
15	43°30.7	43.2		29°25.8	14.6'	15°43.5	-12.5'	54.7'
16	58°30.7	42.5		43°59.4	14.6'	15°31.0	-12.5'	54.7'
17	73°30.8	41.8		58°33.0	14.7'	15°18.5	-12.6'	54.7'
18	88°30.9	N16°41.1		73°06.7	14.8'	N15°05.9	-12.6'	54.7'
19	103°31.0	40.4		87°40.5	14.9'	14°53.3	-12.7'	54.7'
20	118°31.0	39.7		102°14.4	14.9'	14°40.6	-12.7'	54.7'
21	133°31.1	39.0		116°48.3	15.0'	14°27.8	-12.8'	54.6'
22	148°31.2	38.4		131°22.4	15.1'	14°15.1	-12.8'	54.6'
23	163°31.2	37.7		145°56.4	15.2'	14°02.2	-12.9'	54.6'
SD = 15.8' d = -0.7'				SD = 15.0'				
Tue	GHA	Dec		GHA	ν	Dec	d	HP
0	178°31.3	N16°37.0		160°30.6	15.2'	N13°49.3	-12.9'	54.6'
1	193°31.4	36.3		175°04.8	15.3'	13°36.4	-13.0'	54.6'
2	208°31.4	35.6		189°39.1	15.4'	13°23.4	-13.0'	54.6'
3	223°31.5	34.9		204°13.4	15.4'	13°10.4	-13.1'	54.6'
4	238°31.6	34.2		218°47.9	15.5'	12°57.3	-13.1'	54.5'
5	253°31.7	33.5		233°22.3	15.5'	12°44.2	-13.2'	54.5'
6	268°31.7	N16°32.8		247°56.9	15.6'	N12°31.1	-13.2'	54.5'
7	283°31.8	32.1		262°31.5	15.7'	12°17.9	-13.2'	54.5'
8	298°31.9	31.4		277°06.2	15.7'	12°04.6	-13.3'	54.5'
9	313°31.9	30.7		291°40.9	15.8'	11°51.3	-13.3'	54.5'
10	328°32.0	30.0		306°15.7	15.8'	11°38.0	-13.4'	54.5'
11	343°32.1	29.3		320°50.5	15.9'	11°24.7	-13.4'	54.5'
12	358°32.2	N16°28.6		335°25.4	16.0'	N11°11.3	-13.4'	54.4'
13	13°32.2	27.9		350°00.4	16.0'	10°57.9	-13.5'	54.4'
14	28°32.3	27.3		4°35.4	16.1'	10°44.4	-13.5'	54.4'
15	43°32.4	26.6		19°10.5	16.1'	10°30.9	-13.5'	54.4'
16	58°32.5	25.9		33°45.6	16.2'	10°17.4	-13.6'	54.4'
17	73°32.5	25.2		48°20.8	16.2'	10°03.8	-13.6'	54.4'
18	88°32.6	N16°24.5		62°56.0	16.3'	N09°50.3	-13.6'	54.4'
19	103°32.7	23.8		77°31.3	16.3'	09°36.6	-13.6'	54.4'
20	118°32.8	23.1		92°06.6	16.4'	09°23.0	-13.7'	54.3'
21	133°32.8	22.4		106°42.0	16.4'	09°09.3	-13.7'	54.3'
22	148°32.9	21.7		121°17.4	16.5'	08°55.6	-13.7'	54.3'
23	163°33.0	21.0		135°52.9	16.5'	08°41.9	-13.8'	54.3'
SD = 15.8' d = -0.7'				SD = 14.9'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—	—	00:48	23:08	—	—
N 70°	—	—	02:01	22:06	—	—
68°	—	—	02:37	21:31	—	—
66°	—	01:19	03:03	21:07	22:45	—
64°	—	02:04	03:22	20:47	22:04	—
62°	—	02:32	03:38	20:32	21:36	—
60°	01:11	02:54	03:51	20:19	21:16	22:54
N 58°	01:52	03:11	04:03	20:08	20:59	22:16
56°	02:18	03:26	04:12	19:58	20:45	21:51
54°	02:39	03:38	04:21	19:50	20:33	21:31
52°	02:55	03:48	04:29	19:42	20:22	21:15
50°	03:09	03:58	04:36	19:35	20:13	21:02
45°	03:36	04:17	04:50	19:21	19:54	20:35
N 40°	03:56	04:33	05:02	19:09	19:39	20:15
35°	04:12	04:45	05:13	18:59	19:26	19:59
30°	04:25	04:56	05:22	18:50	19:15	19:46
20°	04:46	05:14	05:37	18:35	18:58	19:25
N 10°	05:02	05:28	05:50	18:22	18:44	19:09
0°	05:16	05:41	06:03	18:09	18:31	18:56
S 10°	05:27	05:53	06:15	17:57	18:19	18:45
20°	05:38	06:05	06:28	17:45	18:08	18:34
30°	05:49	06:17	06:42	17:30	17:55	18:24
35°	05:54	06:24	06:51	17:22	17:48	18:19
40°	05:59	06:32	07:00	17:12	17:41	18:13
45°	06:05	06:40	07:11	17:01	17:32	18:08
S 50°	06:11	06:50	07:25	16:48	17:23	18:01
52°	06:14	06:54	07:31	16:42	17:18	17:59
54°	06:17	06:59	07:38	16:35	17:13	17:56
56°	06:20	07:05	07:45	16:27	17:08	17:53
58°	06:23	07:10	07:54	16:19	17:02	17:50
S 60°	06:26	07:17	08:04	16:09	16:56	17:46

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	☐	☐	04:29	☐	22:37	21:49
N 70°	☐	02:22	04:54	23:10	22:10	21:36
68°	☐	03:06	05:13	22:24	21:49	21:26
66°	01:15	03:35	05:27	21:54	21:33	21:17
64°	02:01	03:57	05:40	21:31	21:19	21:09
62°	02:31	04:14	05:50	21:13	21:08	21:03
60°	02:53	04:29	05:58	20:58	20:58	20:57
N 58°	03:12	04:41	06:06	20:45	20:49	20:52
56°	03:27	04:51	06:13	20:34	20:42	20:48
54°	03:40	05:01	06:19	20:24	20:35	20:44
52°	03:51	05:09	06:24	20:15	20:29	20:40
50°	04:01	05:16	06:29	20:07	20:23	20:37
45°	04:22	05:32	06:39	19:50	20:11	20:29
N 40°	04:39	05:45	06:48	19:36	20:01	20:23
35°	04:53	05:55	06:55	19:24	19:53	20:18
30°	05:06	06:05	07:01	19:13	19:45	20:13
20°	05:27	06:21	07:13	18:55	19:32	20:05
N 10°	05:45	06:35	07:22	18:39	19:20	19:58
0°	06:02	06:48	07:31	18:24	19:09	19:51
S 10°	06:18	07:01	07:40	18:09	18:58	19:44
20°	06:36	07:15	07:50	17:53	18:46	19:37
30°	06:57	07:31	08:00	17:34	18:32	19:28
35°	07:09	07:40	08:07	17:23	18:24	19:23
40°	07:22	07:50	08:14	17:11	18:15	19:17
45°	07:39	08:02	08:22	16:56	18:05	19:11
S 50°	07:58	08:17	08:32	16:37	17:51	19:03
52°	08:08	08:24	08:36	16:29	17:45	18:59
54°	08:18	08:31	08:41	16:19	17:39	18:55
56°	08:30	08:40	08:46	16:07	17:31	18:51
58°	08:44	08:49	08:53	15:55	17:22	18:46
S 60°	08:59	09:00	08:59	15:39	17:12	18:40

Day	Sun		Mer. Pass hh:mm	Moon		Age 0-2 0-2%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
04	06:07	06:04	12:06	12:13	—	
05	06:01	05:58	12:06	12:59	00:36	
06	05:55	05:51	12:06	13:41	01:20	

August 07, 08, 09 UT (Wed., Thu., Fri.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	316°00.5	161°09.2	N12°01.6	245°48.9	N21°41.0	241°43.9	N21°57.3	325°57.2	S06°35.4	Alpheratz	357°34.9	29°13.5					
1	331°03.0	176°08.7	12°00.5	260°49.6	41.2	256°45.9	57.4	340°59.7	35.5	Ankaa	353°07.2	-42°10.1					
2	346°05.5	191°08.2	11°59.4	275°50.3	41.5	271°48.0	57.4	356°02.3	35.5	Schedar	349°31.1	56°40.2					
3	1°07.9	206°07.8	•• 58.3	290°51.0	•• 41.8	286°50.0	•• 57.5	11°04.9	•• 35.6	Diphda	348°47.4	-17°50.9					
4	16°10.4	221°07.3	57.2	305°51.7	42.0	301°52.0	57.5	26°07.5	35.7	Achernar	335°20.2	-57°06.4					
5	31°12.9	236°06.9	56.0	320°52.4	42.3	316°54.0	57.5	41°10.1	35.7	Hamal	327°51.5	23°34.7					
6	46°15.3	251°06.4	N11°54.9	335°53.1	N21°42.5	331°56.0	N21°57.6	56°12.7	S06°35.8	Polaris	314°13.6	89°21.7					
7	61°17.8	266°05.9	53.8	350°53.8	42.8	346°58.0	57.6	71°15.3	35.8	Acamar	315°12.0	-40°12.0					
8	76°20.3	281°05.5	52.7	5°54.5	43.0	2°00.0	57.7	86°17.9	35.9	Menkar	314°06.5	4°11.3					
9	91°22.7	296°05.0	•• 51.6	20°55.2	•• 43.3	17°02.1	•• 57.7	101°20.5	•• 36.0	Mirfak	308°28.8	49°56.7					
10	106°25.2	311°04.6	50.4	35°55.9	43.6	32°04.1	57.8	116°23.1	36.0	Aldebaran	290°40.2	16°33.6					
11	121°27.6	326°04.1	49.3	50°56.6	43.8	47°06.1	57.8	131°25.7	36.1	Rigel	281°04.4	-8°10.2					
12	136°30.1	341°03.7	N11°48.2	65°57.3	N21°44.1	62°08.1	N21°57.9	146°28.3	S06°36.2	Capella	280°22.7	46°01.2					
13	151°32.6	356°03.2	47.1	80°58.0	44.3	77°10.1	57.9	161°30.9	36.2	Bellatrix	278°23.4	6°22.4					
14	166°35.0	11°02.8	46.0	95°58.7	44.6	92°12.1	57.9	176°33.5	36.3	Elnath	278°02.5	28°37.7					
15	181°37.5	26°02.3	•• 44.8	110°59.4	•• 44.8	107°14.1	•• 58.0	191°36.1	•• 36.4	Alnilam	275°38.3	-1°11.0					
16	196°40.0	41°01.9	43.7	126°00.1	45.1	122°16.2	58.0	206°38.7	36.4	Betelgeuse	270°52.7	7°24.8					
17	211°42.4	56°01.4	42.6	141°00.8	45.4	137°18.2	58.1	221°41.2	36.5	Canopus	263°53.0	-52°42.2					
18	226°44.9	71°01.0	N11°41.5	156°01.5	N21°45.6	152°20.2	N21°58.1	236°43.8	S06°36.5	Sirius	258°26.8	-16°44.8					
19	241°47.4	86°00.5	40.3	171°02.2	45.9	167°22.2	58.2	251°46.4	36.6	Adhara	255°06.5	-29°00.1					
20	256°49.8	101°00.1	39.2	186°02.9	46.1	182°24.2	58.2	266°49.0	36.7	Procyon	244°51.5	5°09.8					
21	271°52.3	115°59.6	•• 38.1	201°03.6	•• 46.4	197°26.2	•• 58.3	281°51.6	•• 36.7	Pollux	243°18.1	27°58.1					
22	286°54.8	130°59.1	37.0	216°04.3	46.6	212°28.3	58.3	296°54.2	36.8	Avior	234°15.5	-59°35.1					
23	301°57.2	145°58.7	35.8	231°05.0	46.9	227°30.3	58.3	311°56.8	36.9	Suhail	222°47.0	-43°31.8					
Mer.pass. 02:55			ν -0.5' d -1.1' m -3.86			ν 0.7' d 0.3' m 0.84			ν 2.0' d 0.0' m -2.14			ν 2.6' d 0.1' m 0.72					

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	316°59.7	160°58.3	N11°34.7	246°05.7	N21°47.1	242°32.3	N21°58.4	326°59.4	S06°36.9					
1	332°02.1	175°57.8	33.6	261°06.4	47.4	257°34.3	58.4	342°02.0	37.0					
2	347°04.6	190°57.4	32.4	276°07.1	47.6	272°36.3	58.5	357°04.6	37.1					
3	2°07.1	205°56.9	•• 31.3	291°07.7	•• 47.9	287°38.4	•• 58.5	12°07.2	•• 37.1					
4	17°09.5	220°56.5	30.2	306°08.4	48.1	302°40.4	58.6	27°09.8	37.2					
5	32°12.0	235°56.0	29.0	321°09.1	48.4	317°42.4	58.6	42°12.4	37.2					
6	47°14.5	250°55.6	N11°27.9	336°09.8	N21°48.6	332°44.4	N21°58.6	57°15.0	S06°37.3					
7	62°16.9	265°55.1	26.8	351°10.5	48.9	347°46.4	58.7	72°17.6	37.4					
8	77°19.4	280°54.7	25.7	6°11.2	49.1	2°48.4	58.7	87°20.2	37.4					
9	92°21.9	295°54.2	•• 24.5	21°11.9	•• 49.4	17°50.5	•• 58.8	102°22.8	•• 37.5					
10	107°24.3	310°53.8	23.4	36°12.6	49.6	32°52.5	58.8	117°25.4	37.6					
11	122°26.8	325°53.3	22.3	51°13.3	49.9	47°54.5	58.9	132°28.0	37.6					
12	137°29.2	340°52.9	N11°21.1	66°14.0	N21°50.1	62°56.5	N21°58.9	147°30.6	S06°37.7					
13	152°31.7	355°52.4	20.0	81°14.7	50.4	77°58.5	59.0	162°33.2	37.8					
14	167°34.2	10°52.0	18.8	96°15.4	50.6	93°00.6	59.0	177°35.8	37.8					
15	182°36.6	25°51.6	•• 17.7	111°16.1	•• 50.9	108°02.6	•• 59.0	192°38.4	•• 37.9					
16	197°39.1	40°51.1	16.6	126°16.8	51.1	123°04.6	59.1	207°41.0	38.0					
17	212°41.6	55°50.7	15.4	141°17.5	51.4	138°06.6	59.1	222°43.6	38.0					
18	227°44.0	70°50.2	N11°14.3	156°18.2	N21°51.6	153°08.6	N21°59.2	237°46.2	S06°38.1					
19	242°46.5	85°49.8	13.2	171°18.9	51.9	168°10.7	59.2	252°48.7	38.2					
20	257°49.0	100°49.3	12.0	186°19.6	52.1	183°12.7	59.3	267°51.3	38.2					
21	272°51.4	115°48.9	•• 10.9	201°20.3	•• 52.4	198°14.7	•• 59.3	282°53.9	•• 38.3					
22	287°53.9	130°48.5	09.7	216°21.0	52.6	213°16.7	59.3	297°56.5	38.3					
23	302°56.4	145°48.0	08.6	231°21.7	52.8	228°18.8	59.4	312°59.1	38.4					
Mer.pass. 02:52			ν -0.4' d -1.1' m -3.86			ν 0.7' d 0.3' m 0.83			ν 2.0' d 0.0' m -2.14			ν 2.6' d 0.1' m 0.72		

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	317°58.8	160°47.6	N11°07.5	246°22.4	N21°53.1	243°20.8	N21°59.4	328°01.7	S06°38.5					
1	333°01.3	175°47.1	06.3	261°23.1	53.3	258°22.8	59.5	343°04.3	38.5					
2	348°03.7	190°46.7	05.2	276°23.8	53.6	273°24.8	59.5	358°06.9	38.6					
3	3°06.2	205°46.3	•• 04.0	291°24.5	•• 53.8	288°26.8	•• 59.6	13°09.5	•• 38.7					
4	18°08.7	220°45.8	02.9	306°25.3	54.1	303°28.9	59.6	28°12.1	38.7					
5	33°11.1	235°45.4	01.8	321°26.0	54.3	318°30.9	59.6	43°14.7	38.8					
6	48°13.6	250°45.0	N11°00.6	336°26.7	N21°54.5	333°32.9	N21°59.7	58°17.3	S06°38.9					
7	63°16.1	265°44.5	10°59.5	351°27.4	54.8	348°34.9	59.7	73°19.9	38.9					
8	78°18.5	280°44.1	58.3	6°28.1	55.0	3°37.0	59.8	88°22.5	39.0					
9	93°21.0	295°43.7	•• 57.2	21°28.8	•• 55.3	18°39.0	•• 59.8	103°25.1	•• 39.1					
10	108°23.5	310°43.2	56.0	36°29.5	55.5	33°41.0	59.9	118°27.7	39.1					
11	123°25.9	325°42.8	54.9	51°30.2	55.8	48°43.0	59.9	133°30.3	39.2					
12	138°28.4	340°42.3	N10°53.7	66°30.9	N21°56.0	63°45.1	N21°59.9	148°32.9	S06°39.3					
13	153°30.8	355°41.9	52.6	81°31.6	56.2	78°47.1	22°00.0	163°35.5	39.3					
14	168°33.3	10°41.5	51.5	96°32.3	56.5	93°49.1	00.0	178°38.1	39.4					
15	183°35.8	25°41.0	•• 50.3	111°33.0	•• 56.7	108°51.1	•• 00.1	193°40.7	•• 39.5					
16	198°38.2	40°40.6	49.2	126°33.7	57.0	123°53.2	00.1	208°43.3	39.5					
17	213°40.7	55°40.2	48.0	141°34.4	57.2	138°55.2	00.1	223°45.9	39.6					
18	228°43.2	70°39.7	N10°46.9	156°35.1	N21°57.4	153°57.2	N22°00.2	238°48.5	S06°39.7					
19	243°45.6	85°39.3	45.7	171°35.8	57.7	168°59.2	00.2	253°51.1	39.7					
20	258°48.1	100°38.9	44.6	186°36.5	57.9	184°01.3	00.3	268°53.7	39.8					
21	273°50.6	115°38.5	•• 43.4	201°37.2	•• 58.1	199°03.3	•• 00.3	283°56.3	•• 39.8					
22	288°53.0	130°38.0	42.3	216°37.9	58.4	214°05.3	00.4	298°58.9	39.9					
23	303°55.5	145°37.6	41.1	231°38.6	58.6	229°07.3	00.4	314°01.5	40.0					
Mer.pass. 02:48			ν -0.4' d -1.1' m -3.86			ν 0.7' d 0.2' m 0.83			ν 2.0' d 0.0' m -2.15			ν 2.6' d 0.1' m 0.71		

Aug 07 Wed			SHA	Mer.pass
Venus	205°08.6		13:16	
Mars	289°48.4		07:36	
Jupiter	285°43.4		07:52	
Saturn	9°56.6		02:16	

Aug 08 Thu			SHA	Mer.pass
Venus	203°58.6		13:17	
Mars	289°06.0		07:35	
Jupiter	285°32.6		07:49	
Saturn	9°59.7		02:12	

Aug 09 Fri			SHA	Mer.pass
Venus	202°48.8		13:17	
Mars	288°23.6		07:34	
Jupiter	285°22.0		07:46	
Saturn	10°02.9		02:08	

Horizontal parallax			
Venus:		0.1	
Mars:		0.1	

Sun				Moon				
Wed	GHA	Dec		GHA	ν	Dec	d	HP
0	178°33.1	N16°20.2		150°28.4	16.6'	N08°28.1	-13.8'	54.3'
1	193°33.1	19.5		165°04.0	16.6'	08°14.4	-13.8'	54.3'
2	208°33.2	18.8		179°39.6	16.6'	08°00.6	-13.8'	54.3'
3	223°33.3	•• 18.1		194°15.3	16.7'	07°46.7	-13.8'	54.3'
4	238°33.4	17.4		208°50.9	16.7'	07°32.9	-13.9'	54.3'
5	253°33.5	16.7		223°26.7	16.8'	07°19.0	-13.9'	54.3'
6	268°33.5	N16°16.0		238°02.4	16.8'	N07°05.1	-13.9'	54.2'
7	283°33.6	15.3		252°38.3	16.8'	06°51.2	-13.9'	54.2'
8	298°33.7	14.6		267°14.1	16.9'	06°37.3	-13.9'	54.2'
9	313°33.8	•• 13.9		281°50.0	16.9'	06°23.4	-14.0'	54.2'
10	328°33.8	13.2		296°25.9	16.9'	06°09.4	-14.0'	54.2'
11	343°33.9	12.5		311°01.8	17.0'	05°55.4	-14.0'	54.2'
12	358°34.0	N16°11.8		325°37.8	17.0'	N05°41.4	-14.0'	54.2'
13	13°34.1	11.1		340°13.8	17.0'	05°27.4	-14.0'	54.2'
14	28°34.2	10.4		354°49.9	17.1'	05°13.4	-14.0'	54.2'
15	43°34.2	•• 09.7		9°26.0	17.1'	04°59.3	-14.1'	54.2'
16	58°34.3	09.0		24°02.1	17.1'	04°45.3	-14.1'	54.2'
17	73°34.4	08.2		38°38.2	17.2'	04°31.2	-14.1'	54.2'
18	88°34.5	N16°07.5		53°14.3	17.2'	N04°17.2	-14.1'	54.1'
19	103°34.6	06.8		67°50.5	17.2'	04°03.1	-14.1'	54.1'
20	118°34.7	06.1		82°26.7	17.2'	03°49.0	-14.1'	54.1'
21	133°34.7	•• 05.4		97°03.0	17.3'	03°34.9	-14.1'	54.1'
22	148°34.8	04.7		111°39.2	17.3'	03°20.8	-14.1'	54.1'
23	163°34.9	04.0		126°15.5	17.3'	03°06.6	-14.1'	54.1'
SD = 15.8' $d = -0.7'$				SD = 14.8'				
Thu	GHA	Dec		GHA	ν	Dec	d	HP
0	178°35.0	N16°03.3		140°51.8	17.3'	N02°52.5	-14.1'	54.1'
1	193°35.1	02.6		155°28.1	17.3'	02°38.4	-14.1'	54.1'
2	208°35.1	01.8		170°04.4	17.3'	02°24.3	-14.1'	54.1'
3	223°35.2	•• 01.1		184°40.7	17.4'	02°10.1	-14.1'	54.1'
4	238°35.3	16°00.4		199°17.1	17.4'	01°56.0	-14.1'	54.1'
5	253°35.4	15°59.7		213°53.5	17.4'	01°41.8	-14.2'	54.1'
6	268°35.5	N15°59.0		228°29.9	17.4'	N01°27.7	-14.2'	54.1'
7	283°35.6	58.3		243°06.3	17.4'	01°13.5	-14.2'	54.1'
8	298°35.7	57.5		257°42.7	17.4'	00°59.4	-14.2'	54.1'
9	313°35.7	•• 56.8		272°19.1	17.4'	00°45.2	-14.2'	54.1'
10	328°35.8	56.1		286°55.5	17.4'	00°31.0	-14.2'	54.1'
11	343°35.9	55.4		301°32.0	17.4'	00°16.9	-14.2'	54.1'
12	358°36.0	N15°54.7		316°08.4	17.4'	N00°02.7	-14.1'	54.1'
13	13°36.1	54.0		330°44.9	17.5'	S00°11.4	14.1'	54.0'
14	28°36.2	53.2		345°21.3	17.5'	00°25.5	14.1'	54.0'
15	43°36.3	•• 52.5		359°57.8	17.5'	00°39.7	14.1'	54.0'
16	58°36.3	51.8		14°34.2	17.5'	00°53.8	14.1'	54.0'
17	73°36.4	51.1		29°10.7	17.5'	01°08.0	14.1'	54.0'
18	88°36.5	N15°50.4		43°47.1	17.5'	S01°22.1	14.1'	54.0'
19	103°36.6	49.6		58°23.6	17.5'	01°36.2	14.1'	54.0'
20	118°36.7	48.9		73°00.1	17.5'	01°50.3	14.1'	54.0'
21	133°36.8	•• 48.2		87°36.5	17.4'	02°04.4	14.1'	54.0'
22	148°36.9	47.5		102°13.0	17.4'	02°18.5	14.1'	54.0'
23	163°37.0	46.7		116°49.4	17.4'	02°32.6	14.1'	54.0'
SD = 15.8' $d = -0.7'$				SD = 14.8'				
Fri	GHA	Dec		GHA	ν	Dec	d	HP
0	178°37.0	N15°46.0		131°25.8	17.4'	S02°46.7	14.1'	54.0'
1	193°37.1	45.3		146°02.3	17.4'	03°00.7	14.1'	54.0'
2	208°37.2	44.6		160°38.7	17.4'	03°14.8	14.0'	54.0'
3	223°37.3	•• 43.8		175°15.1	17.4'	03°28.8	14.0'	54.0'
4	238°37.4	43.1		189°51.5	17.4'	03°42.9	14.0'	54.0'
5	253°37.5	42.4		204°27.9	17.4'	03°56.9	14.0'	54.0'
6	268°37.6	N15°41.7		219°04.3	17.4'	S04°10.9	14.0'	54.0'
7	283°37.7	40.9		233°40.6	17.3'	04°24.9	14.0'	54.0'
8	298°37.8	40.2		248°17.0	17.3'	04°38.8	14.0'	54.0'
9	313°37.9	•• 39.5		262°53.3	17.3'	04°52.8	13.9'	54.0'
10	328°38.0	38.8		277°29.6	17.3'	05°06.7	13.9'	54.0'
11	343°38.0	38.0		292°05.9	17.3'	05°20.6	13.9'	54.0'
12	358°38.1	N15°37.3		306°42.2	17.3'	S05°34.5	13.9'	54.0'
13	13°38.2	36.6		321°18.4	17.2'	05°48.4	13.9'	54.0'
14	28°38.3	35.8		335°54.7	17.2'	06°02.3	13.8'	54.0'
15	43°38.4	•• 35.1		350°30.9	17.2'	06°16.1	13.8'	54.1'
16	58°38.5	34.4		5°07.0	17.2'	06°30.0	13.8'	54.1'
17	73°38.6	33.7		19°43.2	17.1'	06°43.8	13.8'	54.1'
18	88°38.7	N15°32.9		34°19.3	17.1'	S06°57.5	13.8'	54.1'
19	103°38.8	32.2		48°55.4	17.1'	07°11.3	13.7'	54.1'
20	118°38.9	31.5		63°31.5	17.0'	07°25.0	13.7'	54.1'
21	133°39.0	•• 30.7		78°07.5	17.0'	07°38.7	13.7'	54.1'
22	148°39.1	30.0		92°43.6	17.0'	07°52.4	13.7'	54.1'
23	163°39.2	29.3		107°19.5	16.9'	08°06.1	13.6'	54.1'
SD = 15.8' $d = -0.7'$				SD = 14.7'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	—:—	01:30	22:33	—:—	—:—
N 70°	—:—	—:—	02:20	21:46	—:—	—:—
68°	—:—	—:—	02:51	21:17	23:31	—:—
66°	—:—	01:42	03:14	20:55	22:23	—:—
64°	—:—	02:18	03:32	20:37	21:49	—:—
62°	—:—	02:44	03:46	20:23	21:25	23:35
60°	01:32	03:03	03:58	20:11	21:06	22:34
N 58°	02:05	03:19	04:09	20:01	20:50	22:03
56°	02:28	03:32	04:18	19:52	20:37	21:40
54°	02:47	03:44	04:26	19:44	20:26	21:22
52°	03:02	03:54	04:33	19:37	20:16	21:08
50°	03:15	04:03	04:40	19:30	20:07	20:55
45°	03:40	04:21	04:54	19:17	19:49	20:30
N 40°	04:00	04:36	05:05	19:05	19:35	20:11
35°	04:15	04:48	05:15	18:56	19:23	19:56
30°	04:27	04:58	05:23	18:47	19:13	19:43
20°	04:47	05:15	05:38	18:33	18:56	19:23
N 10°	05:03	05:29	05:51	18:21	18:42	19:08
0°	05:16	05:41	06:02	18:09	18:31	18:55
S 10°	05:27	05:52	06:14	17:58	18:19	18:45
20°	05:37	06:03	06:26	17:46	18:08	18:35
30°	05:46	06:15	06:40	17:32	17:57	18:25
35°	05:51	06:21	06:48	17:24	17:50	18:21
40°	05:56	06:28	06:57	17:15	17:43	18:16
45°	06:01	06:36	07:07	17:05	17:36	18:11
S 50°	06:07	06:45	07:20	16:52	17:27	18:05
52°	06:09	06:49	07:26	16:46	17:22	18:03
54°	06:12	06:54	07:32	16:40	17:18	18:00
56°	06:14	06:59	07:39	16:33	17:13	17:58
58°	06:17	07:04	07:47	16:25	17:08	17:55
S 60°	06:20	07:10	07:57	16:16	17:02	17:52

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	06:42	08:41	10:38	21:15	20:43	20:10
N 70°	06:53	08:42	10:30	21:10	20:46	20:21
68°	07:01	08:43	10:24	21:06	20:48	20:29
66°	07:08	08:44	10:18	21:03	20:50	20:37
64°	07:14	08:44	10:14	21:00	20:52	20:43
62°	07:19	08:45	10:10	20:58	20:53	20:48
60°	07:23	08:45	10:07	20:56	20:54	20:53
N 58°	07:27	08:46	10:04	20:54	20:56	20:57
56°	07:31	08:46	10:01	20:52	20:57	21:01
54°	07:34	08:47	09:59	20:51	20:58	21:04
52°	07:36	08:47	09:57	20:50	20:58	21:07
50°	07:39	08:47	09:55	20:48	20:59	21:10
45°	07:44	08:48	09:51	20:45	21:01	21:16
N 40°	07:49	08:48	09:47	20:43	21:02	21:21
35°	07:53	08:49	09:44	20:41	21:03	21:26
30°	07:56	08:49	09:42	20:39	21:04	21:30
20°	08:02	08:50	09:37	20:36	21:06	21:37
N 10°	08:07	08:50	09:33	20:		

August 10, 11, 12 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	318°58.0	160°37.2	N10°40.0	246°39.3	N21°58.9	244°09.4	N22°00.4	329°04.1	S06°40.0	Alpheratz	357°34.9	29°13.5
1	334°00.4	175°36.7	38.8	261°40.0	59.1	259°11.4	00.5	344°06.7	40.1	Ankaa	353°07.2	-42°10.1
2	349°02.9	190°36.3	37.6	276°40.7	59.3	274°13.4	00.5	359°09.3	40.2	Schedar	349°31.1	56°40.2
3	4°05.3	205°35.9	•• 36.5	291°41.4	•• 59.6	289°15.5	•• 00.6	14°11.9	•• 40.2	Diphda	348°47.4	-17°50.9
4	19°07.8	220°35.4	35.3	306°42.1	21°59.8	304°17.5	00.6	29°14.5	40.3	Achernar	335°20.2	-57°06.4
5	34°10.3	235°35.0	34.2	321°42.8	22°00.0	319°19.5	00.7	44°17.1	40.4	Hamal	327°51.5	23°34.7
6	49°12.7	250°34.6	N10°33.0	336°43.5	N22°00.3	334°21.5	N22°00.7	59°19.7	S06°40.4	Polaris	314°12.0	89°21.7
7	64°15.2	265°34.2	31.9	351°44.2	00.5	349°23.6	00.7	74°22.3	40.5	Acamar	315°12.0	-40°12.0
8	79°17.7	280°33.7	30.7	6°44.9	00.7	4°25.6	00.8	89°24.9	40.6	Menkar	314°06.5	4°11.3
9	94°20.1	295°33.3	•• 29.6	21°45.6	•• 01.0	19°27.6	•• 00.8	104°27.5	•• 40.6	Mirfak	308°28.8	49°56.7
10	109°22.6	310°32.9	28.4	36°46.3	01.2	34°29.7	00.9	119°30.1	40.7	Aldebaran	290°40.1	16°33.6
11	124°25.1	325°32.5	27.2	51°47.0	01.4	49°31.7	00.9	134°32.7	40.8	Rigel	281°04.4	-8°10.2
12	139°27.5	340°32.0	N10°26.1	66°47.8	N22°01.7	64°33.7	N22°00.9	149°35.3	S06°40.8	Capella	280°22.6	46°01.2
13	154°30.0	355°31.6	24.9	81°48.5	01.9	79°35.7	01.0	164°37.9	40.9	Bellatrix	278°23.4	6°22.4
14	169°32.5	10°31.2	23.8	96°49.2	02.1	94°37.8	01.0	179°40.5	41.0	Elnath	278°02.5	28°37.7
15	184°34.9	25°30.8	•• 22.6	111°49.9	•• 02.4	109°39.8	•• 01.1	194°43.1	•• 41.0	Alnilam	275°38.3	-1°11.0
16	199°37.4	40°30.3	21.5	126°50.6	02.6	124°41.8	01.1	209°45.7	41.1	Betelgeuse	270°52.7	7°24.8
17	214°39.8	55°29.9	20.3	141°51.3	02.8	139°43.9	01.1	224°48.4	41.2	Canopus	263°53.0	-52°42.2
18	229°42.3	70°29.5	N10°19.1	156°52.0	N22°03.1	154°45.9	N22°01.2	239°51.0	S06°41.2	Sirius	258°26.8	-16°44.8
19	244°44.8	85°29.1	18.0	171°52.7	03.3	169°47.9	01.2	254°53.6	41.3	Adhara	255°06.5	-29°00.1
20	259°47.2	100°28.6	16.8	186°53.4	03.5	184°50.0	01.3	269°56.2	41.4	Procyon	244°51.5	5°09.8
21	274°49.7	115°28.2	•• 15.6	201°54.1	•• 03.7	199°52.0	•• 01.3	284°58.8	•• 41.4	Pollux	243°18.1	27°58.1
22	289°52.2	130°27.8	14.5	216°54.8	04.0	214°54.0	01.4	300°01.4	41.5	Avior	234°15.5	-59°35.1
23	304°54.6	145°27.4	13.3	231°55.5	04.2	229°56.1	01.4	315°04.0	41.6	Suhail	222°47.0	-43°31.8
Mer.pass. 02:44		ν -0.4' d -1.2' m -3.86		ν 0.7' d 0.2' m 0.82		ν 2.0' d 0.0' m -2.15		ν 2.6' d 0.1' m 0.71				

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	319°57.1	160°27.0	N10°12.2	246°56.2	N22°04.4	244°58.1	N22°01.4	330°06.6	S06°41.6
1	334°59.6	175°26.5	11.0	261°56.9	04.7	260°00.1	01.5	345°09.2	41.7
2	350°02.0	190°26.1	09.8	276°57.6	04.9	275°02.1	01.5	0°11.8	41.8
3	5°04.5	205°25.7	•• 08.7	291°58.3	•• 05.1	290°04.2	•• 01.6	15°14.4	•• 41.8
4	20°06.9	220°25.3	07.5	306°59.0	05.3	305°06.2	01.6	30°17.0	41.9
5	35°09.4	235°24.9	06.3	321°59.8	05.6	320°08.2	01.6	45°19.6	42.0
6	50°11.9	250°24.4	N10°05.2	337°00.5	N22°05.8	335°10.3	N22°01.7	60°22.2	S06°42.0
7	65°14.3	265°24.0	04.0	352°01.2	06.0	350°12.3	01.7	75°24.8	42.1
8	80°16.8	280°23.6	02.8	7°01.9	06.3	5°14.3	01.8	90°27.4	42.2
9	95°19.3	295°23.2	•• 01.7	22°02.6	•• 06.5	20°16.4	•• 01.8	105°30.0	•• 42.2
10	110°21.7	310°22.8	10°00.5	37°03.3	06.7	35°18.4	01.8	120°32.6	42.3
11	125°24.2	325°22.4	09°59.3	52°04.0	06.9	50°20.4	01.9	135°35.2	42.4
12	140°26.7	340°21.9	N09°58.2	67°04.7	N22°07.2	65°22.5	N22°01.9	150°37.8	S06°42.4
13	155°29.1	355°21.5	57.0	82°05.4	07.4	80°24.5	02.0	165°40.4	42.5
14	170°31.6	10°21.1	55.8	97°06.1	07.6	95°26.6	02.0	180°43.0	42.6
15	185°34.1	25°20.7	•• 54.6	112°06.8	•• 07.8	110°28.6	•• 02.0	195°45.6	•• 42.6
16	200°36.5	40°20.3	53.5	127°07.5	08.1	125°30.6	02.1	210°48.2	42.7
17	215°39.0	55°19.9	52.3	142°08.2	08.3	140°32.7	02.1	225°50.8	42.8
18	230°41.4	70°19.5	N09°51.1	157°08.9	N22°08.5	155°34.7	N22°02.2	240°53.4	S06°42.8
19	245°43.9	85°19.0	50.0	172°09.7	08.7	170°36.7	02.2	255°56.1	42.9
20	260°46.4	100°18.6	48.8	187°10.4	08.9	185°38.8	02.2	270°58.7	43.0
21	275°48.8	115°18.2	•• 47.6	202°11.1	•• 09.2	200°40.8	•• 02.3	286°01.3	•• 43.0
22	290°51.3	130°17.8	46.4	217°11.8	09.4	215°42.8	02.3	301°03.9	43.1
23	305°53.8	145°17.4	45.3	232°12.5	09.6	230°44.9	02.4	316°06.5	43.2
Mer.pass. 02:40		ν -0.4' d -1.2' m -3.86		ν 0.7' d 0.2' m 0.82		ν 2.0' d 0.0' m -2.15		ν 2.6' d 0.1' m 0.70	

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	320°56.2	160°17.0	N09°44.1	247°13.2	N22°09.8	245°46.9	N22°02.4	331°09.1	S06°43.2
1	335°58.7	175°16.6	42.9	262°13.9	10.0	260°49.0	02.4	346°11.7	43.3
2	351°01.2	190°16.2	41.7	277°14.6	10.3	275°51.0	02.5	1°14.3	43.4
3	6°03.6	205°15.8	•• 40.6	292°15.3	•• 10.5	290°53.0	•• 02.5	16°16.9	•• 43.4
4	21°06.1	220°15.3	39.4	307°16.0	10.7	305°55.1	02.6	31°19.5	43.5
5	36°08.6	235°14.9	38.2	322°16.7	10.9	320°57.1	02.6	46°22.1	43.6
6	51°11.0	250°14.5	N09°37.0	337°17.5	N22°11.1	335°59.1	N22°02.6	61°24.7	S06°43.7
7	66°13.5	265°14.1	35.9	352°18.2	11.4	351°01.2	02.7	76°27.3	43.7
8	81°15.9	280°13.7	34.7	7°18.9	11.6	6°03.2	02.7	91°29.9	43.8
9	96°18.4	295°13.3	•• 33.5	22°19.6	•• 11.8	21°05.3	•• 02.8	106°32.5	•• 43.9
10	111°20.9	310°12.9	32.3	37°20.3	12.0	36°07.3	02.8	121°35.1	43.9
11	126°23.3	325°12.5	31.1	52°21.0	12.2	51°09.3	02.8	136°37.7	44.0
12	141°25.8	340°12.1	N09°30.0	67°21.7	N22°12.5	66°11.4	N22°02.9	151°40.4	S06°44.1
13	156°28.3	355°11.7	28.8	82°22.4	12.7	81°13.4	02.9	166°43.0	44.1
14	171°30.7	10°11.3	27.6	97°23.1	12.9	96°15.5	03.0	181°45.6	44.2
15	186°33.2	25°10.9	•• 26.4	112°23.9	•• 13.1	111°17.5	•• 03.0	196°48.2	•• 44.3
16	201°35.7	40°10.5	25.2	127°24.6	13.3	126°19.5	03.0	211°50.8	44.3
17	216°38.1	55°10.0	24.1	142°25.3	13.5	141°21.6	03.1	226°53.4	44.4
18	231°40.6	70°09.6	N09°22.9	157°26.0	N22°13.7	156°23.6	N22°03.1	241°56.0	S06°44.5
19	246°43.0	85°09.2	21.7	172°26.7	14.0	171°25.7	03.2	256°58.6	44.5
20	261°45.5	100°08.8	20.5	187°27.4	14.2	186°27.7	03.2	272°01.2	44.6
21	276°48.0	115°08.4	•• 19.3	202°28.1	•• 14.4	201°29.7	•• 03.2	287°03.8	•• 44.7
22	291°50.4	130°08.0	18.1	217°28.8	14.6	216°31.8	03.3	302°06.4	44.7
23	306°52.9	145°07.6	17.0	232°29.5	14.8	231°33.8	03.3	317°09.0	44.8
Mer.pass. 02:36		ν -0.4' d -1.2' m -3.86		ν 0.7' d 0.2' m 0.81		ν 2.0' d 0.0' m -2.16		ν 2.6' d 0.1' m 0.70	

Aug 10 Sat	SHA	Mer.pass
Venus	201°39.2	13:18
Mars	287°41.3	07:33
Jupiter	285°11.4	07:42
Saturn	10°06.2	02:03

Aug 11 Sun	SHA	Mer.pass
Venus	200°29.9	13:19
Mars	286°59.1	07:32
Jupiter	285°01.0	07:39
Saturn	10°09.5	01:59

Aug 12 Mon	SHA	Mer.pass
Venus	199°20.8	13:19
Mars	286°17.0	07:31
Jupiter	284°50.7	07:36
Saturn	10°12.8	01:55

Horizontal parallax	
Venus:	0.1
Mars:	0.1

Sun				Moon				
Sat	GHA	Dec		GHA	ν	Dec	d	HP
0	178°39.3	N15°28.5		121°55.5	16.9'	S08°19.7	13.6'	54.1'
1	193°39.4	27.8		136°31.4	16.9'	08°33.3	13.6'	54.1'
2	208°39.4	27.1		151°07.3	16.8'	08°46.9	13.5'	54.1'
3	223°39.5	· · 26.3		165°43.1	16.8'	09°00.4	13.5'	54.1'
4	238°39.6	25.6		180°18.9	16.8'	09°13.9	13.5'	54.1'
5	253°39.7	24.9		194°54.7	16.7'	09°27.4	13.5'	54.1'
6	268°39.8	N15°24.1		209°30.4	16.7'	S09°40.9	13.4'	54.1'
7	283°39.9	23.4		224°06.1	16.6'	09°54.3	13.4'	54.1'
8	298°40.0	22.6		238°41.7	16.6'	10°07.7	13.4'	54.1'
9	313°40.1	· · 21.9		253°17.3	16.6'	10°21.0	13.3'	54.2'
10	328°40.2	21.2		267°52.9	16.5'	10°34.4	13.3'	54.2'
11	343°40.3	20.4		282°28.4	16.5'	10°47.7	13.3'	54.2'
12	358°40.4	N15°19.7		297°03.8	16.4'	S11°00.9	13.2'	54.2'
13	13°40.5	18.9		311°39.2	16.4'	11°14.1	13.2'	54.2'
14	28°40.6	18.2		326°14.6	16.3'	11°27.3	13.1'	54.2'
15	43°40.7	· · 17.5		340°49.9	16.3'	11°40.5	13.1'	54.2'
16	58°40.8	16.7		355°25.2	16.2'	11°53.6	13.1'	54.2'
17	73°40.9	16.0		10°00.4	16.2'	12°06.6	13.0'	54.2'
18	88°41.0	N15°15.2		24°35.5	16.1'	S12°19.7	13.0'	54.2'
19	103°41.1	14.5		39°10.6	16.0'	12°32.6	12.9'	54.3'
20	118°41.2	13.8		53°45.7	16.0'	12°45.6	12.9'	54.3'
21	133°41.3	· · 13.0		68°20.7	15.9'	12°58.5	12.9'	54.3'
22	148°41.4	12.3		82°55.6	15.9'	13°11.3	12.8'	54.3'
23	163°41.5	11.5		97°30.5	15.8'	13°24.2	12.8'	54.3'
SD = 15.8'			$d = -0.7'$	SD = 14.8'				
Sun	GHA	Dec		GHA	ν	Dec	d	HP
0	178°41.6	N15°10.8		112°05.3	15.8'	S13°36.9	12.7'	54.3'
1	193°41.7	10.0		126°40.0	15.7'	13°49.6	12.7'	54.3'
2	208°41.8	09.3		141°14.7	15.6'	14°02.3	12.6'	54.3'
3	223°41.9	· · 08.6		155°49.3	15.6'	14°15.0	12.6'	54.4'
4	238°42.0	07.8		170°23.9	15.5'	14°27.5	12.5'	54.4'
5	253°42.1	07.1		184°58.4	15.4'	14°40.1	12.5'	54.4'
6	268°42.2	N15°06.3		199°32.8	15.4'	S14°52.5	12.4'	54.4'
7	283°42.3	05.6		214°07.2	15.3'	15°05.0	12.4'	54.4'
8	298°42.4	04.8		228°41.5	15.2'	15°17.4	12.3'	54.4'
9	313°42.5	· · 04.1		243°15.7	15.2'	15°29.7	12.3'	54.4'
10	328°42.6	03.3		257°49.9	15.1'	15°42.0	12.2'	54.5'
11	343°42.7	02.6		272°24.0	15.0'	15°54.2	12.2'	54.5'
12	358°42.9	N15°01.8		286°58.0	14.9'	S16°06.3	12.1'	54.5'
13	13°43.0	01.1		301°31.9	14.9'	16°18.4	12.1'	54.5'
14	28°43.1	15°00.3		316°05.8	14.8'	16°30.5	12.0'	54.5'
15	43°43.2	14°59.6		330°39.5	14.7'	16°42.5	11.9'	54.5'
16	58°43.3	58.8		345°13.3	14.6'	16°54.4	11.9'	54.5'
17	73°43.4	58.1		359°46.9	14.6'	17°06.3	11.8'	54.6'
18	88°43.5	N14°57.3		14°20.4	14.5'	S17°18.1	11.8'	54.6'
19	103°43.6	56.6		28°53.9	14.4'	17°29.9	11.7'	54.6'
20	118°43.7	55.8		43°27.3	14.3'	17°41.5	11.6'	54.6'
21	133°43.8	· · 55.1		58°00.6	14.2'	17°53.2	11.6'	54.6'
22	148°43.9	54.3		72°33.8	14.1'	18°04.7	11.5'	54.7'
23	163°44.0	53.6		87°07.0	14.1'	18°16.2	11.4'	54.7'
SD = 15.8'			$d = -0.7'$	SD = 14.8'				
Mon	GHA	Dec		GHA	ν	Dec	d	HP
0	178°44.1	N14°52.8		101°40.1	14.0'	S18°27.6	11.4'	54.7'
1	193°44.2	52.0		116°13.0	13.9'	18°39.0	11.3'	54.7'
2	208°44.3	51.3		130°45.9	13.8'	18°50.3	11.2'	54.7'
3	223°44.4	· · 50.5		145°18.7	13.7'	19°01.5	11.1'	54.8'
4	238°44.6	49.8		159°51.4	13.6'	19°12.7	11.1'	54.8'
5	253°44.7	49.0		174°24.1	13.5'	19°23.7	11.0'	54.8'
6	268°44.8	N14°48.3		188°56.6	13.4'	S19°34.7	10.9'	54.8'
7	283°44.9	47.5		203°29.0	13.4'	19°45.7	10.9'	54.8'
8	298°45.0	46.8		218°01.4	13.3'	19°56.5	10.8'	54.9'
9	313°45.1	· · 46.0		232°33.6	13.2'	20°07.3	10.7'	54.9'
10	328°45.2	45.2		247°05.8	13.1'	20°18.0	10.6'	54.9'
11	343°45.3	44.5		261°37.9	13.0'	20°28.6	10.5'	54.9'
12	358°45.4	N14°43.7		276°09.9	12.9'	S20°39.2	10.5'	54.9'
13	13°45.5	43.0		290°41.8	12.8'	20°49.6	10.4'	55.0'
14	28°45.6	42.2		305°13.5	12.7'	21°00.0	10.3'	55.0'
15	43°45.8	· · 41.4		319°45.2	12.6'	21°10.3	10.2'	55.0'
16	58°45.9	40.7		334°16.8	12.5'	21°20.5	10.1'	55.0'
17	73°46.0	39.9		348°48.3	12.4'	21°30.6	10.0'	55.1'
18	88°46.1	N14°39.2		3°19.7	12.3'	S21°40.7	10.0'	55.1'
19	103°46.2	38.4		17°51.0	12.2'	21°50.6	9.9'	55.1'
20	118°46.3	37.6		32°22.2	12.1'	22°00.5	9.8'	55.1'
21	133°46.4	· · 36.9		46°53.3	12.0'	22°10.3	9.7'	55.2'
22	148°46.5	36.1		61°24.3	11.9'	22°19.9	9.6'	55.2'
23	163°46.7	35.3		75°55.2	11.8'	22°29.5	9.5'	55.2'
SD = 15.8'			$d = -0.8'$	SD = 14.9'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	—:—	01:58	22:06	—:—	—:—
N 70°	—:—	—:—	02:37	21:28	—:—	—:—
68°	—:—	01:10	03:04	21:03	22:50	—:—
66°	—:—	02:01	03:25	20:43	22:04	—:—
64°	—:—	02:32	03:41	20:27	21:35	—:—
62°	01:03	02:54	03:54	20:14	21:13	22:59
60°	01:49	03:12	04:06	20:03	20:56	22:17
N 58°	02:17	03:27	04:15	19:53	20:41	21:50
56°	02:38	03:39	04:24	19:45	20:29	21:30
54°	02:55	03:50	04:31	19:38	20:19	21:13
52°	03:09	03:59	04:38	19:31	20:10	21:00
50°	03:21	04:08	04:44	19:25	20:01	20:48
45°	03:45	04:25	04:57	19:12	19:44	20:24
N 40°	04:03	04:39	05:08	19:01	19:31	20:06
35°	04:18	04:50	05:17	18:52	19:19	19:52
30°	04:30	05:00	05:25	18:45	19:10	19:40
20°	04:49	05:16	05:39	18:31	18:54	19:21
N 10°	05:03	05:29	05:51	18:19	18:41	19:07
0°	05:15	05:40	06:02	18:09	18:30	18:55
S 10°	05:26	05:51	06:13	17:58	18:19	18:45
20°	05:35	06:01	06:24	17:46	18:09	18:35
30°	05:44	06:12	06:37	17:34	17:58	18:27
35°	05:48	06:18	06:44	17:26	17:52	18:22
40°	05:53	06:25	06:53	17:18	17:46	18:18
45°	05:57	06:32	07:03	17:08	17:39	18:14
S 50°	06:02	06:40	07:15	16:56	17:31	18:09
52°	06:04	06:44	07:20	16:51	17:27	18:07
54°	06:06	06:48	07:26	16:45	17:23	18:05
56°	06:09	06:53	07:33	16:38	17:18	18:03
58°	06:11	06:58	07:40	16:31	17:13	18:00
S 60°	06:14	07:03	07:49	16:22	17:08	17:58

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	12:44	15:36	■	19:28	18:04	■
N 70°	12:24	14:40	■	19:50	19:02	■
68°	12:09	14:06	16:59	20:07	19:37	18:17
66°	11:56	13:43	15:50	20:21	20:02	19:28
64°	11:46	13:24	15:14	20:33	20:21	20:05
62°	11:37	13:09	14:49	20:43	20:37	20:31
60°	11:30	12:56	14:29	20:52	20:51	20:52
N 58°	11:23	12:46	14:12	20:59	21:03	21:09
56°	11:17	12:36	13:59	21:06	21:13	21:23
54°	11:12	12:28	13:47	21:12	21:22	21:35
52°	11:08	12:21	13:36	21:18	21:30	21:46
50°	11:03	12:14	13:27	21:22	21:37	21:56
45°	10:54	12:00	13:08	21:33	21:53	22:17
N 40°	10:47	11:48	12:52	21:42	22:06	22:34
35°	10:41	11:38	12:39	21:50	22:17	22:48
30°	10:35	11:30	12:2			

August 13, 14, 15 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	321°55.4	160°07.2	N09°15.8	247°30.3	N22°15.0	246°35.9	N22°03.4	332°11.6	S06°44.9	Alpheratz	357°34.9	29°13.6
1	336°57.8	175°06.8	14.6	262°31.0	15.2	261°37.9	03.4	347°14.2	44.9	Ankaa	353°07.2	-42°10.1
2	352°00.3	190°06.4	13.4	277°31.7	15.5	276°40.0	03.4	2°16.9	45.0	Schedar	349°31.1	56°40.2
3	7°02.8	205°06.0	•• 12.2	292°32.4	•• 15.7	291°42.0	•• 03.5	17°19.5	•• 45.1	Diphda	348°47.4	-17°50.9
4	22°05.2	220°05.6	11.0	307°33.1	15.9	306°44.0	03.5	32°22.1	45.2	Achernar	335°20.1	-57°06.4
5	37°07.7	235°05.2	09.8	322°33.8	16.1	321°46.1	03.5	47°24.7	45.2	Hamal	327°51.5	23°34.7
6	52°10.2	250°04.8	N09°08.7	337°34.5	N22°16.3	336°48.1	N22°03.6	62°27.3	S06°45.3	Polaris	314°10.3	89°21.8
7	67°12.6	265°04.4	07.5	352°35.2	16.5	351°50.2	03.6	77°29.9	45.4	Acamar	315°12.0	-40°12.0
8	82°15.1	280°04.0	06.3	7°36.0	16.7	6°52.2	03.7	92°32.5	45.4	Menkar	314°06.5	4°11.3
9	97°17.5	295°03.6	•• 05.1	22°36.7	•• 16.9	21°54.3	•• 03.7	107°35.1	•• 45.5	Mirfak	308°28.8	49°56.7
10	112°20.0	310°03.2	03.9	37°37.4	17.1	36°56.3	03.7	122°37.7	45.6	Aldebaran	290°40.1	16°33.6
11	127°22.5	325°02.8	02.7	52°38.1	17.4	51°58.3	03.8	137°40.3	45.6	Rigel	281°04.3	-8°10.2
12	142°24.9	340°02.4	N09°01.5	67°38.8	N22°17.6	67°00.4	N22°03.8	152°42.9	S06°45.7	Capella	280°22.6	46°01.2
13	157°27.4	355°02.0	09°00.3	82°39.5	17.8	82°02.4	03.9	167°45.6	45.8	Bellatrix	278°23.4	6°22.4
14	172°29.9	10°01.6	08°59.1	97°40.2	18.0	97°04.5	03.9	182°48.2	45.8	Elnath	278°02.5	28°37.7
15	187°32.3	25°01.2	•• 57.9	112°41.0	•• 18.2	112°06.5	•• 03.9	197°50.8	•• 45.9	Alnilam	275°38.2	-1°11.0
16	202°34.8	40°00.8	56.8	127°41.7	18.4	127°08.6	04.0	212°53.4	46.0	Betelgeuse	270°52.7	7°24.8
17	217°37.3	55°00.4	55.6	142°42.4	18.6	142°10.6	04.0	227°56.0	46.0	Canopus	263°52.9	-52°42.2
18	232°39.7	70°00.0	N08°54.4	157°43.1	N22°18.8	157°12.7	N22°04.0	242°58.6	S06°46.1	Sirius	258°26.8	-16°44.8
19	247°42.2	84°59.6	53.2	172°43.8	19.0	172°14.7	04.1	258°01.2	46.2	Adhara	255°06.4	-29°00.1
20	262°44.7	99°59.2	52.0	187°44.5	19.2	187°16.8	04.1	273°03.8	46.3	Procyon	244°51.5	5°09.8
21	277°47.1	114°58.8	•• 50.8	202°45.2	•• 19.4	202°18.8	•• 04.2	288°06.4	•• 46.3	Pollux	243°18.0	27°58.1
22	292°49.6	129°58.4	49.6	217°46.0	19.6	217°20.8	04.2	303°09.0	46.4	Avior	234°15.5	-59°35.1
23	307°52.0	144°58.1	48.4	232°46.7	19.8	232°22.9	04.2	318°11.6	46.5	Suhail	222°47.0	-43°31.8
Mer.pass. 02:32		ν -0.4' d -1.2' m -3.86		ν 0.7' d 0.2' m 0.81		ν 2.0' d 0.0' m -2.16		ν 2.6' d 0.1' m 0.69		Miaplacidus	221°39.3	-69°49.0
										Alphard	217°48.4	-8°45.8
										Regulus	207°35.1	11°51.0
										Dubhe	193°42.0	61°37.3
										Denebola	182°25.5	14°26.2
										Gienah	175°44.2	-17°40.7
										Acrux	173°01.0	-63°14.3
										Gacrux	171°52.5	-57°15.2
										Alioth	166°13.6	55°49.8
										Spica	158°22.8	-11°17.3
										Alkaid	152°52.4	49°11.7
										Hadar	148°36.7	-60°29.7
										Menkent	147°58.1	-36°29.6
										Arcturus	145°48.3	19°03.4
										Rigel Kent.	139°40.9	-60°56.4
										Kochab	137°19.9	74°03.5
										Zuben'ubi	136°56.4	-16°08.6
										Alphecca	126°04.0	26°38.1
										Antares	112°16.1	-26°29.2
										Atria	107°10.4	-69°04.5
										Sabik	102°03.0	-15°45.3
										Shaula	96°10.6	-37°07.4
										Rasalhague	95°58.7	12°32.6
										Eltanin	90°42.0	51°29.3
										Kaus Aust.	83°32.7	-34°22.4
										Vega	80°33.1	38°48.5
										Nunki	75°47.9	-26°16.0
										Altair	62°00.0	8°56.1
										Peacock	53°05.7	-56°39.4
										Deneb	49°25.6	45°22.1
										Enif	33°38.8	9°59.3
										Al Na'ir	27°32.8	-46°50.4
										Fomalhaut	15°14.5	-29°29.4
										Scheat	13°45.2	28°13.0
										Markab	13°29.9	15°20.3
Aug 13 Tue			SHA		Mer.pass							
Venus			198°11.9		13:20							
Mars			285°34.9		07:30							
Jupiter			284°40.5		07:33							
Saturn			10°16.3		01:51							
Aug 14 Wed			SHA		Mer.pass							
Venus			197°03.2		13:21							
Mars			284°52.9		07:28							
Jupiter			284°30.4		07:29							
Saturn			10°19.8		01:47							
Aug 15 Thu			SHA		Mer.pass							
Venus			195°54.6		13:21							
Mars			284°10.9		07:27							
Jupiter			284°20.5		07:26							
Saturn			10°23.3		01:43							
Horizontal parallax												
Venus: 0.1												
Mars: 0.1												

Sun								Moon							
Tue	GHA	Dec	GHA	ν	Dec	d	HP								
0	178°46.8	N14°34.6	90°26.0	11.7'	S22°39.0	9.4'	55.2'								
1	193°46.9	33.8	104°56.7	11.6'	22°48.4	9.3'	55.3'								
2	208°47.0	33.1	119°27.2	11.5'	22°57.7	9.2'	55.3'								
3	223°47.1	· · 32.3	133°57.7	11.4'	23°06.9	9.1'	55.3'								
4	238°47.2	31.5	148°28.1	11.3'	23°16.0	9.0'	55.3'								
5	253°47.3	30.8	162°58.4	11.2'	23°25.0	8.9'	55.4'								
6	268°47.5	N14°30.0	177°28.5	11.1'	S23°33.9	8.8'	55.4'								
7	283°47.6	29.2	191°58.6	11.0'	23°42.7	8.7'	55.4'								
8	298°47.7	28.5	206°28.5	10.8'	23°51.4	8.6'	55.5'								
9	313°47.8	· · 27.7	220°58.4	10.7'	24°00.0	8.5'	55.5'								
10	328°47.9	26.9	235°28.1	10.6'	24°08.5	8.4'	55.5'								
11	343°48.0	26.2	249°57.8	10.5'	24°16.9	8.3'	55.5'								
12	358°48.1	N14°25.4	264°27.3	10.4'	S24°25.2	8.2'	55.6'								
13	13°48.3	24.6	278°56.7	10.3'	24°33.3	8.0'	55.6'								
14	28°48.4	23.9	293°26.0	10.2'	24°41.4	7.9'	55.6'								
15	43°48.5	· · 23.1	307°55.2	10.1'	24°49.3	7.8'	55.7'								
16	58°48.6	22.3	322°24.3	10.0'	24°57.1	7.7'	55.7'								
17	73°48.7	21.5	336°53.3	9.9'	25°04.8	7.6'	55.7'								
18	88°48.8	N14°20.8	351°22.2	9.8'	S25°12.4	7.5'	55.7'								
19	103°49.0	20.0	5°51.0	9.7'	25°19.9	7.3'	55.8'								
20	118°49.1	19.2	20°19.6	9.6'	25°27.2	7.2'	55.8'								
21	133°49.2	· · 18.5	34°48.2	9.5'	25°34.5	7.1'	55.8'								
22	148°49.3	17.7	49°16.7	9.3'	25°41.6	7.0'	55.9'								
23	163°49.4	16.9	63°45.0	9.2'	25°48.6	6.9'	55.9'								
		SD = 15.8'	d = -0.8'	SD = 15.1'											
Wed	GHA	Dec	GHA	ν	Dec	d	HP								
0	178°49.6	N14°16.1	78°13.3	9.1'	S25°55.4	6.7'	55.9'								
1	193°49.7	15.4	92°41.4	9.0'	26°02.1	6.6'	56.0'								
2	208°49.8	14.6	107°09.4	8.9'	26°08.7	6.5'	56.0'								
3	223°49.9	· · 13.8	121°37.3	8.8'	26°15.2	6.3'	56.0'								
4	238°50.0	13.0	136°05.2	8.7'	26°21.6	6.2'	56.1'								
5	253°50.2	12.3	150°32.9	8.6'	26°27.8	6.1'	56.1'								
6	268°50.3	N14°11.5	165°00.5	8.5'	S26°33.8	5.9'	56.1'								
7	283°50.4	10.7	179°28.0	8.4'	26°39.8	5.8'	56.2'								
8	298°50.5	09.9	193°55.4	8.3'	26°45.6	5.7'	56.2'								
9	313°50.6	· · 09.2	208°22.7	8.2'	26°51.3	5.5'	56.2'								
10	328°50.8	08.4	222°49.9	8.1'	26°56.8	5.4'	56.3'								
11	343°50.9	07.6	237°17.0	8.0'	27°02.2	5.3'	56.3'								
12	358°51.0	N14°06.8	251°44.0	7.9'	S27°07.4	5.1'	56.3'								
13	13°51.1	06.0	266°10.9	7.8'	27°12.6	5.0'	56.4'								
14	28°51.2	05.3	280°37.7	7.7'	27°17.5	4.8'	56.4'								
15	43°51.4	· · 04.5	295°04.4	7.6'	27°22.3	4.7'	56.4'								
16	58°51.5	03.7	309°31.0	7.5'	27°27.0	4.5'	56.5'								
17	73°51.6	02.9	323°57.5	7.4'	27°31.5	4.4'	56.5'								
18	88°51.7	N14°02.2	338°23.9	7.3'	S27°35.9	4.2'	56.5'								
19	103°51.9	01.4	352°50.2	7.2'	27°40.1	4.1'	56.6'								
20	118°52.0	14°00.6	7°16.4	7.1'	27°44.2	3.9'	56.6'								
21	133°52.1	13°59.8	21°42.6	7.0'	27°48.1	3.8'	56.6'								
22	148°52.2	59.0	36°08.6	6.9'	27°51.9	3.6'	56.7'								
23	163°52.4	58.2	50°34.6	6.9'	27°55.5	3.5'	56.7'								
		SD = 15.8'	d = -0.8'	SD = 15.3'											
Thu	GHA	Dec	GHA	ν	Dec	d	HP								
0	178°52.5	N13°57.5	65°00.4	6.8'	S27°59.0	3.3'	56.8'								
1	193°52.6	56.7	79°26.2	6.7'	28°02.3	3.1'	56.8'								
2	208°52.7	55.9	93°51.9	6.6'	28°05.4	3.0'	56.8'								
3	223°52.9	· · 55.1	108°17.5	6.5'	28°08.4	2.8'	56.9'								
4	238°53.0	54.3	122°43.0	6.4'	28°11.2	2.7'	56.9'								
5	253°53.1	53.5	137°08.4	6.4'	28°13.9	2.5'	56.9'								
6	268°53.2	N13°52.8	151°33.8	6.3'	S28°16.4	2.3'	57.0'								
7	283°53.4	52.0	165°59.1	6.2'	28°18.7	2.2'	57.0'								
8	298°53.5	51.2	180°24.3	6.1'	28°20.9	2.0'	57.0'								
9	313°53.6	· · 50.4	194°49.4	6.1'	28°22.9	1.8'	57.1'								
10	328°53.7	49.6	209°14.5	6.0'	28°24.7	1.7'	57.1'								
11	343°53.9	48.8	223°39.4	5.9'	28°26.4	1.5'	57.2'								
12	358°54.0	N13°48.0	238°04.3	5.8'	S28°27.9	1.3'	57.2'								
13	13°54.1	47.3	252°29.2	5.8'	28°29.2	1.2'	57.2'								
14	28°54.3	46.5	266°54.0	5.7'	28°30.3	1.0'	57.3'								
15	43°54.4	· · 45.7	281°18.7	5.6'	28°31.3	0.8'	57.3'								
16	58°54.5	44.9	295°43.3	5.6'	28°32.1	0.6'	57.3'								
17	73°54.6	44.1	310°07.9	5.5'	28°32.7	0.5'	57.4'								
18	88°54.8	N13°43.3	324°32.4	5.5'	S28°33.2	0.3'	57.4'								
19	103°54.9	42.5	338°56.9	5.4'	28°33.5	0.1'	57.5'								
20	118°55.0	41.7	353°21.3	5.4'	28°33.6	-0.1'	57.5'								
21	133°55.2	· · 40.9	7°45.7	5.3'	28°33.5	-0.3'	57.5'								
22	148°55.3	40.2	22°10.0	5.3'	28°33.3	-0.4'	57.6'								
23	163°55.4	39.4	36°34.2	5.2'	28°32.8	-0.6'	57.6'								
		SD = 15.8'	d = -0.8'	SD = 15.5'											

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	—:—	02:21	21:43	—:—	—:—
N 70°	—:—	—:—	02:54	21:12	23:49	—:—
68°	—:—	01:39	03:17	20:49	22:23	—:—
66°	—:—	02:18	03:35	20:31	21:47	—:—
64°	—:—	02:45	03:50	20:17	21:21	23:51
62°	01:28	03:05	04:02	20:05	21:02	22:35
60°	02:04	03:21	04:13	19:55	20:46	22:02
N 58°	02:28	03:35	04:22	19:46	20:33	21:38
56°	02:47	03:46	04:30	19:38	20:21	21:20
54°	03:02	03:56	04:37	19:31	20:12	21:05
52°	03:16	04:05	04:43	19:25	20:03	20:52
50°	03:27	04:13	04:49	19:19	19:55	20:41
45°	03:49	04:29	05:01	19:07	19:39	20:19
N 40°	04:07	04:42	05:11	18:58	19:27	20:02
35°	04:20	04:53	05:20	18:49	19:16	19:48
30°	04:32	05:02	05:27	18:42	19:07	19:37
20°	04:50	05:17	05:40	18:29	18:52	19:19
N 10°	05:04	05:29	05:51	18:18	18:40	19:05
0°	05:15	05:40	06:01	18:08	18:29	18:54
S 10°	05:25	05:50	06:11	17:58	18:20	18:45
20°	05:33	05:59	06:22	17:47	18:10	18:36
30°	05:41	06:10	06:34	17:35	18:00	18:28
35°	05:45	06:15	06:41	17:28	17:55	18:24
40°	05:49	06:21	06:49	17:21	17:49	18:21
45°	05:53	06:28	06:58	17:12	17:42	18:17
S 50°	05:57	06:35	07:09	17:01	17:35	18:13
52°	05:59	06:39	07:14	16:56	17:31	18:11
54°	06:01	06:43	07:20	16:50	17:27	18:0

August 16, 17, 18 UT (Fri., Sat., Sun.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	324°52.8	159°39.1	N07°49.4	248°21.9	N22°29.5	249°03.5	N22°06.1	335°19.7	S06°49.9	Alpheratz	357°34.8	29°13.6
1	339°55.3	174°38.7	48.2	263°22.6	29.7	264°05.5	06.1	350°22.3	49.9	Ankaa	353°07.1	-42°10.1
2	354°57.7	189°38.4	47.0	278°23.3	29.9	279°07.6	06.1	5°24.9	50.0	Schedar	349°31.0	56°40.2
3	10°00.2	204°38.0	· · 45.8	293°24.0	· · 30.1	294°09.6	· · 06.2	20°27.5	· · 50.1	Diphda	348°47.4	-17°50.9
4	25°02.6	219°37.6	44.6	308°24.7	30.3	309°11.7	06.2	35°30.1	50.2	Achernar	335°20.1	-57°06.4
5	40°05.1	234°37.2	43.3	323°25.5	30.5	324°13.8	06.2	50°32.7	50.2	Hamal	327°51.4	23°34.7
6	55°07.6	249°36.9	N07°42.1	338°26.2	N22°30.7	339°15.8	N22°06.3	65°35.3	S06°50.3	Polaris	314°08.5	89°21.8
7	70°10.0	264°36.5	40.9	353°26.9	30.8	354°17.9	06.3	80°38.0	50.4	Acamar	315°11.9	-40°12.0
8	85°12.5	279°36.1	39.7	8°27.6	31.0	9°19.9	06.3	95°40.6	50.4	Menkar	314°06.4	4°11.3
9	100°15.0	294°35.7	· · 38.5	23°28.4	· · 31.2	24°22.0	· · 06.4	110°43.2	· · 50.5	Mirfak	308°28.7	49°56.7
10	115°17.4	309°35.4	37.3	38°29.1	31.4	39°24.1	06.4	125°45.8	50.6	Aldebaran	290°40.1	16°33.6
11	130°19.9	324°35.0	36.0	53°29.8	31.6	54°26.1	06.4	140°48.4	50.7	Rigel	281°04.3	-8°10.2
12	145°22.4	339°34.6	N07°34.8	68°30.5	N22°31.8	69°28.2	N22°06.5	155°51.0	S06°50.7	Capella	280°22.6	46°01.2
13	160°24.8	354°34.2	33.6	83°31.3	32.0	84°30.2	06.5	170°53.7	50.8	Bellatrix	278°23.4	6°22.4
14	175°27.3	9°33.9	32.4	98°32.0	32.1	99°32.3	06.6	185°56.3	50.9	Elnath	278°02.5	28°37.7
15	190°29.8	24°33.5	· · 31.2	113°32.7	· · 32.3	114°34.4	· · 06.6	200°58.9	· · 50.9	Alnilam	275°38.2	-1°11.0
16	205°32.2	39°33.1	30.0	128°33.4	32.5	129°36.4	06.6	216°01.5	51.0	Betelgeuse	270°52.6	7°24.8
17	220°34.7	54°32.7	28.7	143°34.1	32.7	144°38.5	06.7	231°04.1	51.1	Canopus	263°52.9	-52°42.2
18	235°37.1	69°32.4	N07°27.5	158°34.9	N22°32.9	159°40.5	N22°06.7	246°06.7	S06°51.2	Sirius	258°26.8	-16°44.8
19	250°39.6	84°32.0	26.3	173°35.6	33.1	174°42.6	06.7	261°09.3	51.2	Adhara	255°06.4	-29°00.1
20	265°42.1	99°31.6	25.1	188°36.3	33.3	189°44.7	06.8	276°12.0	51.3	Procyon	244°51.5	5°09.8
21	280°44.5	114°31.2	· · 23.9	203°37.0	· · 33.4	204°46.7	· · 06.8	291°14.6	· · 51.4	Pollux	243°18.0	27°58.1
22	295°47.0	129°30.9	22.6	218°37.8	33.6	219°48.8	06.8	306°17.2	51.4	Avior	234°15.5	-59°35.1
23	310°49.5	144°30.5	21.4	233°38.5	33.8	234°50.9	06.9	321°19.8	51.5	Suhail	222°47.0	-43°31.8
Mer.pass. 02:20		ν -0.4' d -1.2' m -3.86		ν 0.7' d 0.2' m 0.79		ν 2.1' d 0.0' m -2.18		ν 2.6' d 0.1' m 0.68				

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	325°51.9	159°30.1	N07°20.2	248°39.2	N22°34.0	249°52.9	N22°06.9	336°22.4	S06°51.6
1	340°54.4	174°29.8	19.0	263°39.9	34.2	264°55.0	06.9	351°25.0	51.7
2	355°56.9	189°29.4	17.7	278°40.7	34.3	279°57.0	07.0	6°27.7	51.7
3	10°59.3	204°29.0	· · 16.5	293°41.4	· · 34.5	294°59.1	· · 07.0	21°30.3	· · 51.8
4	26°01.8	219°28.6	15.3	308°42.1	34.7	310°01.2	07.1	36°32.9	51.9
5	41°04.3	234°28.3	14.1	323°42.8	34.9	325°03.2	07.1	51°35.5	51.9
6	56°06.7	249°27.9	N07°12.9	338°43.6	N22°35.1	340°05.3	N22°07.1	66°38.1	S06°52.0
7	71°09.2	264°27.5	11.6	353°44.3	35.3	355°07.4	07.2	81°40.7	52.1
8	86°11.6	279°27.2	10.4	8°45.0	35.4	10°09.4	07.2	96°43.4	52.1
9	101°14.1	294°26.8	· · 09.2	23°45.7	· · 35.6	25°11.5	· · 07.2	111°46.0	· · 52.2
10	116°16.6	309°26.4	08.0	38°46.5	35.8	40°13.6	07.3	126°48.6	52.3
11	131°19.0	324°26.1	06.7	53°47.2	36.0	55°15.6	07.3	141°51.2	52.4
12	146°21.5	339°25.7	N07°05.5	68°47.9	N22°36.1	70°17.7	N22°07.3	156°53.8	S06°52.4
13	161°24.0	354°25.3	04.3	83°48.7	36.3	85°19.8	07.4	171°56.5	52.5
14	176°26.4	9°25.0	03.1	98°49.4	36.5	100°21.8	07.4	186°59.1	52.6
15	191°28.9	24°24.6	· · 01.8	113°50.1	· · 36.7	115°23.9	· · 07.4	202°01.7	· · 52.7
16	206°31.4	39°24.2	07°00.6	128°50.8	36.9	130°26.0	07.5	217°04.3	52.7
17	221°33.8	54°23.9	06°59.4	143°51.6	37.0	145°28.0	07.5	232°06.9	52.8
18	236°36.3	69°23.5	N06°58.2	158°52.3	N22°37.2	160°30.1	N22°07.5	247°09.5	S06°52.9
19	251°38.8	84°23.1	56.9	173°53.0	37.4	175°32.2	07.6	262°12.2	52.9
20	266°41.2	99°22.8	55.7	188°53.7	37.6	190°34.2	07.6	277°14.8	53.0
21	281°43.7	114°22.4	· · 54.5	203°54.5	· · 37.7	205°36.3	· · 07.6	292°17.4	· · 53.1
22	296°46.1	129°22.0	53.2	218°55.2	37.9	220°38.4	07.7	307°20.0	53.2
23	311°48.6	144°21.7	52.0	233°55.9	38.1	235°40.4	07.7	322°22.6	53.2
Mer.pass. 02:16		ν -0.4' d -1.2' m -3.86		ν 0.7' d 0.2' m 0.79		ν 2.1' d 0.0' m -2.18		ν 2.6' d 0.1' m 0.67	

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	326°51.1	159°21.3	N06°50.8	248°56.7	N22°38.3	250°42.5	N22°07.7	337°25.2	S06°53.3
1	341°53.5	174°20.9	49.6	263°57.4	38.4	265°44.6	07.8	352°27.9	53.4
2	356°56.0	189°20.6	48.3	278°58.1	38.6	280°46.6	07.8	7°30.5	53.4
3	11°58.5	204°20.2	· · 47.1	293°58.8	· · 38.8	295°48.7	· · 07.8	22°33.1	· · 53.5
4	27°00.9	219°19.8	45.9	308°59.6	39.0	310°50.8	07.9	37°35.7	53.6
5	42°03.4	234°19.5	44.6	324°00.3	39.1	325°52.9	07.9	52°38.3	53.7
6	57°05.9	249°19.1	N06°43.4	339°01.0	N22°39.3	340°54.9	N22°08.0	67°41.0	S06°53.7
7	72°08.3	264°18.8	42.2	354°01.8	39.5	355°57.0	08.0	82°43.6	53.8
8	87°10.8	279°18.4	40.9	9°02.5	39.6	10°59.1	08.0	97°46.2	53.9
9	102°13.2	294°18.0	· · 39.7	24°03.2	· · 39.8	26°01.1	· · 08.1	112°48.8	· · 53.9
10	117°15.7	309°17.7	38.5	39°04.0	40.0	41°03.2	08.1	127°51.4	54.0
11	132°18.2	324°17.3	37.2	54°04.7	40.2	56°05.3	08.1	142°54.1	54.1
12	147°20.6	339°16.9	N06°36.0	69°05.4	N22°40.3	71°07.3	N22°08.2	157°56.7	S06°54.2
13	162°23.1	354°16.6	34.8	84°06.1	40.5	86°09.4	08.2	172°59.3	54.2
14	177°25.6	9°16.2	33.5	99°06.9	40.7	101°11.5	08.2	188°01.9	54.3
15	192°28.0	24°15.9	· · 32.3	114°07.6	· · 40.8	116°13.6	· · 08.3	203°04.5	· · 54.4
16	207°30.5	39°15.5	31.1	129°08.3	41.0	131°15.6	08.3	218°07.2	54.4
17	222°33.0	54°15.1	29.8	144°09.1	41.2	146°17.7	08.3	233°09.8	54.5
18	237°35.4	69°14.8	N06°28.6	159°09.8	N22°41.4	161°19.8	N22°08.4	248°12.4	S06°54.6
19	252°37.9	84°14.4	27.4	174°10.5	41.5	176°21.9	08.4	263°15.0	54.7
20	267°40.4	99°14.1	26.1	189°11.3	41.7	191°23.9	08.4	278°17.6	54.7
21	282°42.8	114°13.7	· · 24.9	204°12.0	· · 41.9	206°26.0	· · 08.5	293°20.3	· · 54.8
22	297°45.3	129°13.3	23.7	219°12.7	42.0	221°28.1	08.5	308°22.9	54.9
23	312°47.7	144°13.0	22.4	234°13.5	42.2	236°30.1	08.5	323°25.5	55.0
Mer.pass. 02:12		ν -0.4' d -1.2' m -3.86		ν 0.7' d 0.2' m 0.78		ν 2.1' d 0.0' m -2.19		ν 2.6' d 0.1' m 0.67	

Aug 16 Fri	SHA	Mer.pass
Venus	194°46.3	13:22
Mars	283°29.1	07:26
Jupiter	284°10.7	07:23
Saturn	10°26.9	01:38

Aug 17 Sat	SHA	Mer.pass
Venus	193°38.2	13:22
Mars	282°47.3	07:25
Jupiter	284°01.0	07:19
Saturn	10°30.5	01:34

Aug 18 Sun	SHA	Mer.pass
Venus	192°30.2	13:23
Mars	282°05.6	07:24
Jupiter	283°51.4	07:16
Saturn	10°34.2	01:30

Horizontal parallax	
Venus:	0.1
Mars:	0.1

h	Sun				Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP	
0	178°55.5	N13°38.6	50°58.5	5.2'	S28°32.2	-0.8'	57.7'	
1	193°55.7	37.8	65°22.6	5.1'	28°31.4	-1.0'	57.7'	
2	208°55.8	37.0	79°46.7	5.1'	28°30.5	-1.2'	57.7'	
3	223°55.9	· · 36.2	94°10.8	5.0'	28°29.3	-1.3'	57.8'	
4	238°56.1	35.4	108°34.9	5.0'	28°28.0	-1.5'	57.8'	
5	253°56.2	34.6	122°58.9	5.0'	28°26.5	-1.7'	57.8'	
6	268°56.3	N13°33.8	137°22.9	4.9'	S28°24.8	-1.9'	57.9'	
7	283°56.5	33.0	151°46.8	4.9'	28°22.9	-2.1'	57.9'	
8	298°56.6	32.2	166°10.7	4.9'	28°20.8	-2.2'	58.0'	
9	313°56.7	· · 31.4	180°34.6	4.9'	28°18.6	-2.4'	58.0'	
10	328°56.9	30.6	194°58.5	4.8'	28°16.2	-2.6'	58.0'	
11	343°57.0	29.8	209°22.3	4.8'	28°13.5	-2.8'	58.1'	
12	358°57.1	N13°29.0	223°46.1	4.8'	S28°10.7	-3.0'	58.1'	
13	13°57.3	28.2	238°09.9	4.8'	28°07.7	-3.2'	58.1'	
14	28°57.4	27.5	252°33.7	4.8'	28°04.6	-3.4'	58.2'	
15	43°57.5	· · 26.7	266°57.5	4.8'	28°01.2	-3.5'	58.2'	
16	58°57.7	25.9	281°21.2	4.7'	27°57.7	-3.7'	58.3'	
17	73°57.8	25.1	295°45.0	4.7'	27°53.9	-3.9'	58.3'	
18	88°57.9	N13°24.3	310°08.7	4.7'	S27°50.0	-4.1'	58.3'	
19	103°58.1	23.5	324°32.4	4.7'	27°45.9	-4.3'	58.4'	
20	118°58.2	22.7	338°56.1	4.7'	27°41.6	-4.5'	58.4'	
21	133°58.3	· · 21.9	353°19.9	4.7'	27°37.2	-4.7'	58.4'	
22	148°58.5	21.1	7°43.6	4.7'	27°32.5	-4.8'	58.5'	
23	163°58.6	20.3	22°07.3	4.7'	27°27.7	-5.0'	58.5'	
SD = 15.8'		$d = -0.8'$	SD = 15.7'					
Sat	GHA	Dec	GHA	ν	Dec	d	HP	
0	178°58.7	N13°19.5	36°31.1	4.7'	S27°22.6	-5.2'	58.6'	
1	193°58.9	18.7	50°54.8	4.8'	27°17.4	-5.4'	58.6'	
2	208°59.0	17.9	65°18.6	4.8'	27°12.0	-5.6'	58.6'	
3	223°59.2	· · 17.1	79°42.3	4.8'	27°06.4	-5.8'	58.7'	
4	238°59.3	16.3	94°06.1	4.8'	27°00.7	-6.0'	58.7'	
5	253°59.4	15.5	108°29.9	4.8'	26°54.7	-6.1'	58.7'	
6	268°59.6	N13°14.7	122°53.7	4.8'	S26°48.6	-6.3'	58.8'	
7	283°59.7	13.9	137°17.6	4.9'	26°42.3	-6.5'	58.8'	
8	298°59.8	13.0	151°41.4	4.9'	26°35.7	-6.7'	58.9'	
9	314°00.0	· · 12.2	166°05.3	4.9'	26°29.1	-6.9'	58.9'	
10	329°00.1	11.4	180°29.2	4.9'	26°22.2	-7.0'	58.9'	
11	344°00.3	10.6	194°53.2	5.0'	26°15.1	-7.2'	59.0'	
12	359°00.4	N13°09.8	209°17.1	5.0'	S26°07.9	-7.4'	59.0'	
13	14°00.5	09.0	223°41.1	5.0'	26°00.5	-7.6'	59.0'	
14	29°00.7	08.2	238°05.2	5.1'	25°52.9	-7.8'	59.1'	
15	44°00.8	· · 07.4	252°29.2	5.1'	25°45.2	-7.9'	59.1'	
16	59°00.9	06.6	266°53.3	5.1'	25°37.2	-8.1'	59.1'	
17	74°01.1	05.8	281°17.5	5.2'	25°29.1	-8.3'	59.2'	
18	89°01.2	N13°05.0	295°41.7	5.2'	S25°20.8	-8.5'	59.2'	
19	104°01.4	04.2	310°05.9	5.3'	25°12.3	-8.6'	59.2'	
20	119°01.5	03.4	324°30.2	5.3'	25°03.7	-8.8'	59.3'	
21	134°01.6	· · 02.6	338°54.5	5.4'	24°54.9	-9.0'	59.3'	
22	149°01.8	01.8	353°18.9	5.4'	24°45.9	-9.2'	59.3'	
23	164°01.9	01.0	7°43.3	5.5'	24°36.7	-9.3'	59.4'	
SD = 15.8'		$d = -0.8'$	SD = 16.0'					
Sun	GHA	Dec	GHA	ν	Dec	d	HP	
0	179°02.1	N13°00.1	22°07.7	5.5'	S24°27.4	-9.5'	59.4'	
1	194°02.2	12°59.3	36°32.2	5.6'	24°17.9	-9.7'	59.4'	
2	209°02.4	58.5	50°56.8	5.6'	24°08.2	-9.8'	59.5'	
3	224°02.5	· · 57.7	65°21.4	5.7'	23°58.4	-10.0'	59.5'	
4	239°02.6	56.9	79°46.1	5.7'	23°48.4	-10.2'	59.5'	
5	254°02.8	56.1	94°10.8	5.8'	23°38.2	-10.3'	59.6'	
6	269°02.9	N12°55.3	108°35.6	5.8'	S23°27.9	-10.5'	59.6'	
7	284°03.1	54.5	123°00.4	5.9'	23°17.4	-10.7'	59.6'	
8	299°03.2	53.7	137°25.3	6.0'	23°06.7	-10.8'	59.7'	
9	314°03.4	· · 52.9	151°50.3	6.0'	22°55.9	-11.0'	59.7'	
10	329°03.5	52.0	166°15.3	6.1'	22°44.9	-11.1'	59.7'	
11	344°03.6	51.2	180°40.4	6.1'	22°33.8	-11.3'	59.8'	
12	359°03.8	N12°50.4	195°05.5	6.2'	S22°22.5	-11.4'	59.8'	
13	14°03.9	49.6	209°30.7	6.3'	22°11.1	-11.6'	59.8'	
14	29°04.1	48.8	223°56.0	6.3'	21°59.5	-11.7'	59.8'	
15	44°04.2	· · 48.0	238°21.3	6.4'	21°47.7	-11.9'	59.9'	
16	59°04.4	47.2	252°46.8	6.5'	21°35.8	-12.0'	59.9'	
17	74°04.5	46.3	267°12.2	6.5'	21°23.8	-12.2'	59.9'	
18	89°04.6	N12°45.5	281°37.8	6.6'	S21°11.6	-12.3'	60.0'	
19	104°04.8	44.7	296°03.4	6.7'	20°59.3	-12.5'	60.0'	
20	119°04.9	43.9	310°29.0	6.7'	20°46.8	-12.6'	60.0'	
21	134°05.1	· · 43.1	324°54.8	6.8'	20°34.2	-12.8'	60.0'	
22	149°05.2	42.3	339°20.6	6.9'	20°21.4	-12.9'	60.1'	
23	164°05.4	41.4	353°46.5	6.9'	20°08.5	-13.0'	60.1'	
SD = 15.8'		$d = -0.8'$	SD = 16.2'					

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	—:—	02:41	21:22	—:—	—:—
N 70°	—:—	01:06	03:09	20:55	22:50	—:—
68°	—:—	02:01	03:30	20:35	22:01	—:—
66°	—:—	02:34	03:46	20:19	21:30	—:—
64°	00:58	02:57	03:59	20:06	21:08	22:59
62°	01:48	03:15	04:11	19:55	20:50	22:15
60°	02:17	03:30	04:20	19:46	20:36	21:47
N 58°	02:39	03:42	04:28	19:38	20:24	21:26
56°	02:56	03:53	04:36	19:31	20:13	21:10
54°	03:10	04:02	04:42	19:25	20:04	20:56
52°	03:22	04:10	04:48	19:19	19:56	20:44
50°	03:33	04:18	04:53	19:14	19:49	20:34
45°	03:54	04:33	05:05	19:03	19:34	20:13
N 40°	04:10	04:45	05:14	18:53	19:22	19:57
35°	04:23	04:55	05:22	18:46	19:12	19:44
30°	04:34	05:04	05:29	18:39	19:04	19:33
20°	04:51	05:18	05:41	18:27	18:50	19:17
N 10°	05:04	05:29	05:51	18:17	18:38	19:04
0°	05:15	05:39	06:01	18:07	18:29	18:53
S 10°	05:24	05:49	06:10	17:58	18:20	18:44
20°	05:31	05:57	06:20	17:48	18:11	18:37
30°	05:39	06:07	06:31	17:37	18:02	18:30
35°	05:42	06:12	06:38	17:31	17:57	18:26
40°	05:45	06:17	06:45	17:23	17:51	18:23
45°	05:49	06:23	06:53	17:15	17:45	18:20
S 50°	05:52	06:30	07:04	17:05	17:39	18:17
52°	05:54	06:33	07:08	17:00	17:36	18:15
54°	05:55	06:36	07:13	16:55	17:32	18:14
56°	05:56	06:40	07:19	16:50	17:29	18:12
58°	05:58	06:44	07:25	16:43	17:25	18:11
S 60°	06:00	06:48	07:33	16:36	17:21	18:09

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	21:54	■	■	■
66°	■	■	21:13	■	■	■
64°	■	21:18	20:44	■	23:51	· · ·
62°	20:54	20:31	20:22	22:07	· · ·	00:37
60°	19:49	20:01	20:04	23:13	· · ·	01:06
N 58°	19:14	19:37	19:49	23:47	· · ·	01:29
56°	18:49	19:19	19:36	· · ·	00:12	01:47
54°	18:28	19:03	19:25	· · ·	00:32	02:03
52°	18:11	18:49	19:15	· · ·	00:49	02:16
50°	17:57	18:37	19:06	· · ·	01:04	02:27
45°	17:27	18:12	18:47	00:24	01:33	02:51
N 40°	17:04	17:52	18:31	00:49	01:55	03:11
35°	16:44	17:35	18:18	01:09	02:14	03:26
30°	16:28	17:20	18:06	01:26	02:30	03:40
20°	16:00	16:55	17:46	01:54	02:57	04:03
N 10°	15:36	16:34	17:29	02:19	03:21	04:23
0°	15:13	16:14	17:12	02:42	03:42	04:42
S 10°	14:51	15:53	16:56	03:05	04:04	05:00
20°	14:27	15:32	16:38	03:29	04:27	05:20
30°	13:59	15:06	16:18	03:58	04:53	05:42
35°	13:42	14:52	16:06	04:15	05:09	05:55
40°	13:23	14:34	15:52	04:34	05:27	06:10
45°	12:59	14:13	15:35	04:58	05:48	06:28
S 50°	12:29	13:47	15:15	05:29	06:16	06:50
52°						

August 19, 20, 21 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	327°50.2	159°12.6	N06°21.2	249°14.2	N22°42.4	251°32.2	N22°08.6	338°28.1	S06°55.0	Alpheratz	357°34.8	29°13.6
1	342°52.7	174°12.3	20.0	264°14.9	42.5	266°34.3	08.6	353°30.7	55.1	Ankaa	353°07.1	-42°10.1
2	357°55.1	189°11.9	18.7	279°15.6	42.7	281°36.4	08.6	8°33.4	55.2	Schedar	349°31.0	56°40.2
3	12°57.6	204°11.6	•• 17.5	294°16.4	•• 42.9	296°38.4	•• 08.7	23°36.0	•• 55.2	Diphda	348°47.4	-17°50.9
4	28°00.1	219°11.2	16.2	309°17.1	43.0	311°40.5	08.7	38°38.6	55.3	Achernar	335°20.1	-57°06.4
5	43°02.5	234°10.8	15.0	324°17.8	43.2	326°42.6	08.7	53°41.2	55.4	Hamal	327°51.4	23°34.7
6	58°05.0	249°10.5	N06°13.8	339°18.6	N22°43.4	341°44.7	N22°08.8	68°43.8	S06°55.5	Polaris	314°07.0	89°21.8
7	73°07.5	264°10.1	12.5	354°19.3	43.5	356°46.7	08.8	83°46.5	55.5	Acamar	315°11.9	-40°12.0
8	88°09.9	279°09.8	11.3	9°20.0	43.7	11°48.8	08.8	98°49.1	55.6	Menkar	314°06.4	4°11.3
9	103°12.4	294°09.4	•• 10.0	24°20.8	•• 43.8	26°50.9	•• 08.9	113°51.7	•• 55.7	Mirfak	308°28.7	49°56.8
10	118°14.9	309°09.1	08.8	39°21.5	44.0	41°53.0	08.9	128°54.3	55.8	Aldebaran	290°40.1	16°33.6
11	133°17.3	324°08.7	07.6	54°22.2	44.2	56°55.1	08.9	143°56.9	55.8	Rigel	281°04.3	-8°10.2
12	148°19.8	339°08.4	N06°06.3	69°23.0	N22°44.3	71°57.1	N22°09.0	158°59.6	S06°55.9	Capella	280°22.5	46°01.2
13	163°22.2	354°08.0	05.1	84°23.7	44.5	86°59.2	09.0	174°02.2	56.0	Bellatrix	278°23.4	6°22.4
14	178°24.7	9°07.6	03.8	99°24.4	44.7	102°01.3	09.0	189°04.8	56.0	Elnath	278°02.4	28°37.7
15	193°27.2	24°07.3	•• 02.6	114°25.2	•• 44.8	117°03.4	•• 09.1	204°07.4	•• 56.1	Alnilam	275°38.2	-1°11.0
16	208°29.6	39°06.9	01.4	129°25.9	45.0	132°05.4	09.1	219°10.1	56.2	Betelgeuse	270°52.6	7°24.8
17	223°32.1	54°06.6	06°00.1	144°26.7	45.1	147°07.5	09.1	234°12.7	56.3	Canopus	263°52.9	-52°42.2
18	238°34.6	69°06.2	N05°58.9	159°27.4	N22°45.3	162°09.6	N22°09.2	249°15.3	S06°56.3	Sirius	258°26.8	-16°44.8
19	253°37.0	84°05.9	57.6	174°28.1	45.5	177°11.7	09.2	264°17.9	56.4	Adhara	255°06.4	-29°00.1
20	268°39.5	99°05.5	56.4	189°28.9	45.6	192°13.8	09.2	279°20.5	56.5	Procyon	244°51.4	5°09.8
21	283°42.0	114°05.2	•• 55.2	204°29.6	•• 45.8	207°15.8	•• 09.3	294°23.2	•• 56.6	Pollux	243°18.0	27°58.0
22	298°44.4	129°04.8	53.9	219°30.3	46.0	222°17.9	09.3	309°25.8	56.6	Avior	234°15.5	-59°35.1
23	313°46.9	144°04.5	52.7	234°31.1	46.1	237°20.0	09.3	324°28.4	56.7	Suhail	222°47.0	-43°31.8
Mer.pass. 02:08		ν -0.4' d -1.2' m -3.86		ν 0.7' d 0.2' m 0.78		ν 2.1' d 0.0' m -2.19		ν 2.6' d 0.1' m 0.66				

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	328°49.3	159°04.1	N05°51.4	249°31.8	N22°46.3	252°22.1	N22°09.4	339°31.0	S06°56.8
1	343°51.8	174°03.8	50.2	264°32.5	46.4	267°24.1	09.4	354°33.7	56.8
2	358°54.3	189°03.4	48.9	279°33.3	46.6	282°26.2	09.4	9°36.3	56.9
3	13°56.7	204°03.1	•• 47.7	294°34.0	•• 46.7	297°28.3	•• 09.4	24°38.9	•• 57.0
4	28°59.2	219°02.7	46.5	309°34.7	46.9	312°30.4	09.5	39°41.5	57.1
5	44°01.7	234°02.4	45.2	324°35.5	47.1	327°32.5	09.5	54°44.1	57.1
6	59°04.1	249°02.0	N05°44.0	339°36.2	N22°47.2	342°34.6	N22°09.5	69°46.8	S06°57.2
7	74°06.6	264°01.7	42.7	354°37.0	47.4	357°36.6	09.6	84°49.4	57.3
8	89°09.1	279°01.3	41.5	9°37.7	47.5	12°38.7	09.6	99°52.0	57.4
9	104°11.5	294°01.0	•• 40.2	24°38.4	•• 47.7	27°40.8	•• 09.6	114°54.6	•• 57.4
10	119°14.0	309°00.6	39.0	39°39.2	47.8	42°42.9	09.7	129°57.3	57.5
11	134°16.5	324°00.3	37.7	54°39.9	48.0	57°45.0	09.7	144°59.9	57.6
12	149°18.9	338°59.9	N05°36.5	69°40.6	N22°48.2	72°47.0	N22°09.7	160°02.5	S06°57.7
13	164°21.4	353°59.6	35.2	84°41.4	48.3	87°49.1	09.8	175°05.1	57.7
14	179°23.8	8°59.2	34.0	99°42.1	48.5	102°51.2	09.8	190°07.7	57.8
15	194°26.3	23°58.9	•• 32.7	114°42.9	•• 48.6	117°53.3	•• 09.8	205°10.4	•• 57.9
16	209°28.8	38°58.5	31.5	129°43.6	48.8	132°55.4	09.9	220°13.0	57.9
17	224°31.2	53°58.2	30.2	144°44.3	48.9	147°57.5	09.9	235°15.6	58.0
18	239°33.7	68°57.8	N05°29.0	159°45.1	N22°49.1	162°59.5	N22°09.9	250°18.2	S06°58.1
19	254°36.2	83°57.5	27.8	174°45.8	49.2	178°01.6	10.0	265°20.9	58.2
20	269°38.6	98°57.1	26.5	189°46.5	49.4	193°03.7	10.0	280°23.5	58.2
21	284°41.1	113°56.8	•• 25.3	204°47.3	•• 49.5	208°05.8	•• 10.0	295°26.1	•• 58.3
22	299°43.6	128°56.4	24.0	219°48.0	49.7	223°07.9	10.1	310°28.7	58.4
23	314°46.0	143°56.1	22.8	234°48.8	49.9	238°10.0	10.1	325°31.4	58.5
Mer.pass. 02:04		ν -0.4' d -1.2' m -3.86		ν 0.7' d 0.2' m 0.77		ν 2.1' d 0.0' m -2.20		ν 2.6' d 0.1' m 0.66	

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	329°48.5	158°55.7	N05°21.5	249°49.5	N22°50.0	253°12.1	N22°10.1	340°34.0	S06°58.5
1	344°51.0	173°55.4	20.3	264°50.2	50.2	268°14.1	10.2	355°36.6	58.6
2	359°53.4	188°55.0	19.0	279°51.0	50.3	283°16.2	10.2	10°39.2	58.7
3	14°55.9	203°54.7	•• 17.8	294°51.7	•• 50.5	298°18.3	•• 10.2	25°41.9	•• 58.8
4	29°58.3	218°54.4	16.5	309°52.5	50.6	313°20.4	10.2	40°44.5	58.8
5	45°00.8	233°54.0	15.3	324°53.2	50.8	328°22.5	10.3	55°47.1	58.9
6	60°03.3	248°53.7	N05°14.0	339°53.9	N22°50.9	343°24.6	N22°10.3	70°49.7	S06°59.0
7	75°05.7	263°53.3	12.8	354°54.7	51.1	358°26.7	10.3	85°52.4	59.1
8	90°08.2	278°53.0	11.5	9°55.4	51.2	13°28.7	10.4	100°55.0	59.1
9	105°10.7	293°52.6	•• 10.2	24°56.2	•• 51.4	28°30.8	•• 10.4	115°57.6	•• 59.2
10	120°13.1	308°52.3	09.0	39°56.9	51.5	43°32.9	10.4	131°00.2	59.3
11	135°15.6	323°51.9	07.7	54°57.6	51.7	58°35.0	10.5	146°02.9	59.3
12	150°18.1	338°51.6	N05°06.5	69°58.4	N22°51.8	73°37.1	N22°10.5	161°05.5	S06°59.4
13	165°20.5	353°51.3	05.2	84°59.1	51.9	88°39.2	10.5	176°08.1	59.5
14	180°23.0	8°50.9	04.0	99°59.9	52.1	103°41.3	10.6	191°10.7	59.6
15	195°25.4	23°50.6	•• 02.7	115°00.6	•• 52.2	118°43.4	•• 10.6	206°13.4	•• 59.6
16	210°27.9	38°50.2	01.5	130°01.4	52.4	133°45.5	10.6	221°16.0	59.7
17	225°30.4	53°49.9	05°00.2	145°02.1	52.5	148°47.5	10.7	236°18.6	59.8
18	240°32.8	68°49.5	N04°59.0	160°02.8	N22°52.7	163°49.6	N22°10.7	251°21.2	S06°59.9
19	255°35.3	83°49.2	57.7	175°03.6	52.8	178°51.7	10.7	266°23.9	06°59.9
20	270°37.8	98°48.9	56.5	190°04.3	53.0	193°53.8	10.7	281°26.5	07°00.0
21	285°40.2	113°48.5	•• 55.2	205°05.1	•• 53.1	208°55.9	•• 10.8	296°29.1	•• 00.1
22	300°42.7	128°48.2	53.9	220°05.8	53.3	223°58.0	10.8	311°31.7	00.2
23	315°45.2	143°47.8	52.7	235°06.6	53.4	239°00.1	10.8	326°34.4	00.2
Mer.pass. 02:00		ν -0.3' d -1.3' m -3.86		ν 0.7' d 0.2' m 0.77		ν 2.1' d 0.0' m -2.20		ν 2.6' d 0.1' m 0.65	

Aug 19 Mon	SHA	Mer.pass
Venus	191°22.4	13:23
Mars	281°24.0	07:23
Jupiter	283°42.0	07:13
Saturn	10°37.9	01:26

Aug 20 Tue	SHA	Mer.pass
Venus	190°14.8	13:24
Mars	280°42.4	07:22
Jupiter	283°32.7	07:10
Saturn	10°41.7	01:22

Aug 21 Wed	SHA	Mer.pass
Venus	189°07.3	13:25
Mars	280°01.0	07:20
Jupiter	283°23.6	07:06
Saturn	10°45.5	01:18

Horizontal parallax	
Venus:	0.1
Mars:	0.1

Sun				Moon				
Mon	GHA	Dec		GHA	ν	Dec	d	HP
0	179°05.5	N12°40.6		8°12.4	7.0'	S19°55.5	-13.2'	60.1'
1	194°05.7	39.8		22°38.4	7.1'	19°42.3	-13.3'	60.1'
2	209°05.8	39.0		37°04.5	7.2'	19°29.0	-13.4'	60.2'
3	224°06.0	· · 38.2		51°30.7	7.2'	19°15.6	-13.6'	60.2'
4	239°06.1	37.4		65°56.9	7.3'	19°02.0	-13.7'	60.2'
5	254°06.3	36.5		80°23.2	7.4'	18°48.3	-13.8'	60.2'
6	269°06.4	N12°35.7		94°49.6	7.4'	S18°34.5	-13.9'	60.3'
7	284°06.6	34.9		109°16.0	7.5'	18°20.5	-14.1'	60.3'
8	299°06.7	34.1		123°42.5	7.6'	18°06.4	-14.2'	60.3'
9	314°06.9	· · 33.3		138°09.1	7.6'	17°52.3	-14.3'	60.3'
10	329°07.0	32.4		152°35.7	7.7'	17°37.9	-14.4'	60.3'
11	344°07.1	31.6		167°02.5	7.8'	17°23.5	-14.5'	60.4'
12	359°07.3	N12°30.8		181°29.3	7.9'	S17°09.0	-14.7'	60.4'
13	14°07.4	30.0		195°56.1	7.9'	16°54.3	-14.8'	60.4'
14	29°07.6	29.2		210°23.0	8.0'	16°39.5	-14.9'	60.4'
15	44°07.7	· · 28.3		224°50.0	8.1'	16°24.6	-15.0'	60.4'
16	59°07.9	27.5		239°17.1	8.1'	16°09.6	-15.1'	60.5'
17	74°08.0	26.7		253°44.2	8.2'	15°54.5	-15.2'	60.5'
18	89°08.2	N12°25.9		268°11.4	8.3'	S15°39.3	-15.3'	60.5'
19	104°08.3	25.0		282°38.7	8.3'	15°24.0	-15.4'	60.5'
20	119°08.5	24.2		297°06.1	8.4'	15°08.6	-15.5'	60.5'
21	134°08.6	· · 23.4		311°33.5	8.5'	14°53.1	-15.6'	60.5'
22	149°08.8	22.6		326°00.9	8.5'	14°37.5	-15.7'	60.6'
23	164°08.9	21.7		340°28.5	8.6'	14°21.8	-15.8'	60.6'
SD = 15.8' d = -0.8'				SD = 16.4'				
Tue	GHA	Dec		GHA	ν	Dec	d	HP
0	179°09.1	N12°20.9		354°56.1	8.7'	S14°06.0	-15.9'	60.6'
1	194°09.3	20.1		9°23.7	8.7'	13°50.1	-16.0'	60.6'
2	209°09.4	19.3		23°51.5	8.8'	13°34.2	-16.1'	60.6'
3	224°09.6	· · 18.4		38°19.3	8.9'	13°18.1	-16.1'	60.6'
4	239°09.7	17.6		52°47.1	8.9'	13°02.0	-16.2'	60.6'
5	254°09.9	16.8		67°15.0	9.0'	12°45.7	-16.3'	60.7'
6	269°10.0	N12°15.9		81°43.0	9.0'	S12°29.4	-16.4'	60.7'
7	284°10.2	15.1		96°11.0	9.1'	12°13.0	-16.5'	60.7'
8	299°10.3	14.3		110°39.1	9.2'	11°56.6	-16.5'	60.7'
9	314°10.5	· · 13.5		125°07.3	9.2'	11°40.0	-16.6'	60.7'
10	329°10.6	12.6		139°35.5	9.3'	11°23.4	-16.7'	60.7'
11	344°10.8	11.8		154°03.8	9.3'	11°06.7	-16.8'	60.7'
12	359°10.9	N12°11.0		168°32.1	9.4'	S10°50.0	-16.8'	60.7'
13	14°11.1	10.1		183°00.5	9.4'	10°33.2	-16.9'	60.7'
14	29°11.2	09.3		197°28.9	9.5'	10°16.3	-16.9'	60.7'
15	44°11.4	· · 08.5		211°57.4	9.5'	09°59.3	-17.0'	60.8'
16	59°11.5	07.7		226°26.0	9.6'	09°42.3	-17.1'	60.8'
17	74°11.7	06.8		240°54.6	9.6'	09°25.3	-17.1'	60.8'
18	89°11.9	N12°06.0		255°23.2	9.7'	S09°08.1	-17.2'	60.8'
19	104°12.0	05.2		269°51.9	9.7'	08°51.0	-17.2'	60.8'
20	119°12.2	04.3		284°20.6	9.8'	08°33.7	-17.3'	60.8'
21	134°12.3	· · 03.5		298°49.4	9.8'	08°16.4	-17.3'	60.8'
22	149°12.5	02.7		313°18.3	9.9'	07°59.1	-17.4'	60.8'
23	164°12.6	01.8		327°47.1	9.9'	07°41.7	-17.4'	60.8'
SD = 15.8' d = -0.8'				SD = 16.5'				
Wed	GHA	Dec		GHA	ν	Dec	d	HP
0	179°12.8	N12°01.0		342°16.1	10.0'	S07°24.3	-17.5'	60.8'
1	194°12.9	12°00.2		356°45.0	10.0'	07°06.8	-17.5'	60.8'
2	209°13.1	11°59.3		11°14.0	10.0'	06°49.3	-17.5'	60.8'
3	224°13.3	· · 58.5		25°43.1	10.1'	06°31.8	-17.6'	60.8'
4	239°13.4	57.7		40°12.2	10.1'	06°14.2	-17.6'	60.8'
5	254°13.6	56.8		54°41.3	10.2'	05°56.6	-17.7'	60.8'
6	269°13.7	N11°56.0		69°10.5	10.2'	S05°38.9	-17.7'	60.8'
7	284°13.9	55.2		83°39.7	10.2'	05°21.2	-17.7'	60.8'
8	299°14.0	54.3		98°08.9	10.3'	05°03.5	-17.7'	60.8'
9	314°14.2	· · 53.5		112°38.2	10.3'	04°45.8	-17.8'	60.8'
10	329°14.4	52.6		127°07.5	10.3'	04°28.0	-17.8'	60.8'
11	344°14.5	51.8		141°36.8	10.4'	04°10.2	-17.8'	60.8'
12	359°14.7	N11°51.0		156°06.1	10.4'	S03°52.4	-17.8'	60.8'
13	14°14.8	50.1		170°35.5	10.4'	03°34.6	-17.8'	60.8'
14	29°15.0	49.3		185°05.0	10.4'	03°16.8	-17.9'	60.8'
15	44°15.2	· · 48.5		199°34.4	10.5'	02°58.9	-17.9'	60.8'
16	59°15.3	47.6		214°03.9	10.5'	02°41.1	-17.9'	60.8'
17	74°15.5	46.8		228°33.4	10.5'	02°23.2	-17.9'	60.8'
18	89°15.6	N11°45.9		243°02.9	10.5'	S02°05.3	-17.9'	60.8'
19	104°15.8	45.1		257°32.4	10.6'	01°47.4	-17.9'	60.8'
20	119°16.0	44.3		272°02.0	10.6'	01°29.5	-17.9'	60.7'
21	134°16.1	· · 43.4		286°31.5	10.6'	01°11.6	-17.9'	60.7'
22	149°16.3	42.6		301°01.1	10.6'	00°53.7	-17.9'	60.7'
23	164°16.4	41.7		315°30.7	10.6'	00°35.8	-17.9'	60.7'
SD = 15.8' d = -0.8'				SD = 16.6'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	---	---	02:59	21:03	---	---
N 70°	---	01:40	03:23	20:40	22:19	---
68°	---	02:21	03:42	20:22	21:41	---
66°	---	02:48	03:56	20:08	21:15	---
64°	01:27	03:08	04:08	19:56	20:55	22:32
62°	02:04	03:25	04:19	19:46	20:39	21:58
60°	02:29	03:38	04:27	19:38	20:26	21:34
N 58°	02:49	03:50	04:35	19:30	20:15	21:15
56°	03:04	04:00	04:42	19:24	20:05	21:00
54°	03:18	04:08	04:48	19:18	19:57	20:47
52°	03:29	04:16	04:53	19:13	19:49	20:36
50°	03:39	04:23	04:58	19:08	19:43	20:27
45°	03:58	04:37	05:08	18:58	19:29	20:07
N 40°	04:14	04:48	05:17	18:49	19:18	19:52
35°	04:26	04:58	05:24	18:42	19:08	19:40
30°	04:36	05:06	05:31	18:36	19:00	19:30
20°	04:52	05:19	05:42	18:25	18:47	19:14
N 10°	05:04	05:30	05:51	18:15	18:37	19:02
0°	05:14	05:39	06:00	18:07	18:28	18:52
S 10°	05:22	05:47	06:09	17:58	18:19	18:44
20°	05:29	05:55	06:18	17:49	18:11	18:37
30°	05:36	06:04	06:28	17:39	18:03	18:31
35°	05:39	06:08	06:34	17:33	17:59	18:28
40°	05:41	06:13	06:41	17:26	17:54	18:26
45°	05:44	06:18	06:49	17:19	17:49	18:23
S 50°	05:47	06:25	06:58	17:09	17:43	18:20
52°	05:48	06:27	07:02	17:05	17:40	18:19
54°	05:49	06:30	07:07	17:01	17:37	18:19
56°	05:50	06:33	07:12	16:55	17:34	18:18
58°	05:51	06:37	07:18	16:50	17:31	18:17
S 60°	05:52	06:41	07:24	16:43	17:27	18:16

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	21:54	20:54	20:12	■	03:19	06:11
N 70°	21:23	20:42	20:10	■	03:47	06:20
68°	21:01	20:32	20:08	01:18	04:08	06:27
66°	20:43	20:23	20:07	01:58	04:24	06:33
64°	20:28	20:16	20:06	02:26	04:37	06:38
62°	20:16	20:10	20:05	02:47	04:48	06:42
60°	20:05	20:05	20:04	03:04	04:57	06:46
N 58°	19:56	20:00	20:03	03:18	05:05	06:49
56°	19:47	19:55	20:02	03:30	05:12	06:52
54°	19:40	19:52	20:01	03:40	05:18	06:54
52°	19:34	19:48	20:01	03:50	05:24	06:56
50°	19:28	19:45	20:00	03:58	05:29	06:59
45°	19:15	19:38	19:59	04:15	05:40	07:03
N 40°	19:04	19:32	19:58	04:30	05:49	07:07
35°	18:55	19:27	19:57	04:41	05:56	07:10
30°	18:46	19:22	19:56	04:52	06:03	07:13
20°	18:32	19:15	19:55	05:10	06:15	07:18
N 10°	18:20	19:08	19:53	05:25	06:24	07:22
0°	18:08	19:01	19:52	05:39	06:34	07:26
S 10°	17:56	18:55	19:51	05:53	06:43	07:29
20°	17:44	18:48	19:50	06:08	06:52	07:33
30°	17:29	18:40	19:49	06:25	07:03	07:38
35°	17:21	18:35	19:48	06:35	07:10	07:40
40°	17:11	18:30	19:47	06:46	07:17	07:43
45°	17:00	18:24	19:46	06:59	07:25	07:47
S 50°	16:46	18:16	19:45	07:15	07:34	07:50
52°	16:39	18:13	19:44	07:22	07:39	07:52
54°	16:32	18:09	19:44	07:31	07:44	07:54
56°	16:24	18:05	19:43	07:40	07:49	07:56
58°	16:15	18:00	19:42	07:50	07:55	07:59
S 60°	16:04	17:55	19:42	08:02	08:02	08:01

Day	Sun		Mer. Pass hh:mm	Moon		Age 15-17 99-98%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
19	03:38	03:31	12:04	---	11:54	
20	03:24	03:16	12:03	00:21	12:48	
21	03:09	03:01	12:03	01:13	13:39	

August 22, 23, 24 UT (Thu., Fri., Sat.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	330°47.6	158°47.5	N04°51.4	250°07.3	N22°53.6	254°02.2	N22°10.9	341°37.0	S07°00.3	Alpheratz	357°34.8	29°13.6											
1	345°50.1	173°47.2	50.2	265°08.0	53.7	269°04.3	10.9	356°39.6	00.4	Ankaa	353°07.1	-42°10.1											
2	0°52.6	188°46.8	48.9	280°08.8	53.8	284°06.4	10.9	11°42.2	00.5	Schedar	349°31.0	56°40.2											
3	15°55.0	203°46.5	· · 47.7	295°09.5	· · 54.0	299°08.5	· · 11.0	26°44.9	· · 00.5	Diphda	348°47.4	-17°50.9											
4	30°57.5	218°46.1	46.4	310°10.3	54.1	314°10.5	11.0	41°47.5	00.6	Achernar	335°20.0	-57°06.4											
5	45°59.9	233°45.8	45.2	325°11.0	54.3	329°12.6	11.0	56°50.1	00.7	Hamal	327°51.4	23°34.7											
6	61°02.4	248°45.5	N04°43.9	340°11.8	N22°54.4	344°14.7	N22°11.1	71°52.7	S07°00.8	Polaris	314°05.6	89°21.8											
7	76°04.9	263°45.1	42.6	355°12.5	54.6	359°16.8	11.1	86°55.4	00.8	Acamar	315°11.9	-40°12.0											
8	91°07.3	278°44.8	41.4	10°13.3	54.7	14°18.9	11.1	101°58.0	00.9	Menkar	314°06.4	4°11.3											
9	106°09.8	293°44.4	· · 40.1	25°14.0	· · 54.8	29°21.0	· · 11.1	117°00.6	· · 01.0	Mirfak	308°28.7	49°56.8											
10	121°12.3	308°44.1	38.9	40°14.7	55.0	44°23.1	11.2	132°03.2	01.0	Aldebaran	290°40.0	16°33.6											
11	136°14.7	323°43.8	37.6	55°15.5	55.1	59°25.2	11.2	147°05.9	01.1	Rigel	281°04.3	-8°10.2											
12	151°17.2	338°43.4	N04°36.4	70°16.2	N22°55.3	74°27.3	N22°11.2	162°08.5	S07°01.2	Capella	280°22.5	46°01.2											
13	166°19.7	353°43.1	35.1	85°17.0	55.4	89°29.4	11.3	177°11.1	01.3	Bellatrix	278°23.3	6°22.4											
14	181°22.1	8°42.7	33.8	100°17.7	55.5	104°31.5	11.3	192°13.7	01.3	Elnath	278°02.4	28°37.7											
15	196°24.6	23°42.4	· · 32.6	115°18.5	· · 55.7	119°33.6	· · 11.3	207°16.4	· · 01.4	Anlilam	275°38.2	-1°11.0											
16	211°27.0	38°42.1	31.3	130°19.2	55.8	134°35.7	11.4	222°19.0	01.5	Betelgeuse	270°52.6	7°24.8											
17	226°29.5	53°41.7	30.1	145°20.0	56.0	149°37.8	11.4	237°21.6	01.6	Canopus	263°52.9	-52°42.2											
18	241°32.0	68°41.4	N04°28.8	160°20.7	N22°56.1	164°39.9	N22°11.4	252°24.3	S07°01.6	Sirius	258°26.7	-16°44.8											
19	256°34.4	83°41.1	27.5	175°21.5	56.2	179°42.0	11.4	267°26.9	01.7	Adhara	255°06.4	-29°00.1											
20	271°36.9	98°40.7	26.3	190°22.2	56.4	194°44.1	11.5	282°29.5	01.8	Procyon	244°51.4	5°09.8											
21	286°39.4	113°40.4	· · 25.0	205°23.0	· · 56.5	209°46.1	· · 11.5	297°32.1	· · 01.9	Pollux	243°18.0	27°58.0											
22	301°41.8	128°40.0	23.8	220°23.7	56.7	224°48.2	11.5	312°34.8	01.9	Avior	234°15.5	-59°35.1											
23	316°44.3	143°39.7	22.5	235°24.4	56.8	239°50.3	11.6	327°37.4	02.0	Suhail	222°47.0	-43°31.7											
Mer.pass. 01:57				ν -0.3' d -1.3' m -3.86				ν 0.7' d 0.1' m 0.76				ν 2.1' d 0.0' m -2.21				ν 2.6' d 0.1' m 0.65							

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	331°46.8	158°39.4	N04°21.2	250°25.2	N22°56.9	254°52.4	N22°11.6	342°40.0	S07°02.1										
1	346°49.2	173°39.0	20.0	265°25.9	57.1	269°54.5	11.6	357°42.6	02.2										
2	1°51.7	188°38.7	18.7	280°26.7	57.2	284°56.6	11.7	12°45.3	02.2										
3	16°54.2	203°38.4	· · 17.4	295°27.4	· · 57.3	299°58.7	· · 11.7	27°47.9	· · 02.3										
4	31°56.6	218°38.0	16.2	310°28.2	57.5	315°00.8	11.7	42°50.5	02.4										
5	46°59.1	233°37.7	14.9	325°28.9	57.6	330°02.9	11.7	57°53.2	02.5										
6	62°01.5	248°37.4	N04°13.7	340°29.7	N22°57.7	345°05.0	N22°11.8	72°55.8	S07°02.5										
7	77°04.0	263°37.0	12.4	355°30.4	57.9	0°07.1	11.8	87°58.4	02.6										
8	92°06.5	278°36.7	11.1	10°31.2	58.0	15°09.2	11.8	103°01.0	02.7										
9	107°08.9	293°36.4	· · 09.9	25°31.9	· · 58.1	30°11.3	· · 11.9	118°03.7	· · 02.8										
10	122°11.4	308°36.0	08.6	40°32.7	58.3	45°13.4	11.9	133°06.3	02.8										
11	137°13.9	323°35.7	07.3	55°33.4	58.4	60°15.5	11.9	148°08.9	02.9										
12	152°16.3	338°35.3	N04°06.1	70°34.2	N22°58.5	75°17.6	N22°12.0	163°11.6	S07°03.0										
13	167°18.8	353°35.0	04.8	85°34.9	58.7	90°19.7	12.0	178°14.2	03.1										
14	182°21.3	8°34.7	03.6	100°35.7	58.8	105°21.8	12.0	193°16.8	03.1										
15	197°23.7	23°34.3	· · 02.3	115°36.4	· · 58.9	120°23.9	· · 12.0	208°19.4	· · 03.2										
16	212°26.2	38°34.0	04°01.0	130°37.2	59.1	135°26.0	12.1	223°22.1	03.3										
17	227°28.6	53°33.7	03°59.8	145°37.9	59.2	150°28.1	12.1	238°24.7	03.4										
18	242°31.1	68°33.3	N03°58.5	160°38.7	N22°59.3	165°30.2	N22°12.1	253°27.3	S07°03.4										
19	257°33.6	83°33.0	57.2	175°39.4	59.5	180°32.3	12.2	268°29.9	03.5										
20	272°36.0	98°32.7	56.0	190°40.2	59.6	195°34.4	12.2	283°32.6	03.6										
21	287°38.5	113°32.3	· · 54.7	205°40.9	· · 59.7	210°36.5	· · 12.2	298°35.2	· · 03.7										
22	302°41.0	128°32.0	53.4	220°41.7	22°59.9	225°38.6	12.2	313°37.8	03.7										
23	317°43.4	143°31.7	52.2	235°42.4	23°00.0	240°40.7	12.3	328°40.5	03.8										
Mer.pass. 01:53				ν -0.3' d -1.3' m -3.86				ν 0.7' d 0.1' m 0.76				ν 2.1' d 0.0' m -2.21				ν 2.6' d 0.1' m 0.64			

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	332°45.9	158°31.4	N03°50.9	250°43.2	N23°00.1	255°42.8	N22°12.3	343°43.1	S07°03.9										
1	347°48.4	173°31.0	49.6	265°43.9	00.2	270°44.9	12.3	358°45.7	04.0										
2	2°50.8	188°30.7	48.4	280°44.7	00.4	285°47.0	12.4	13°48.4	04.0										
3	17°53.3	203°30.4	· · 47.1	295°45.4	· · 00.5	300°49.2	· · 12.4	28°51.0	· · 04.1										
4	32°55.8	218°30.0	45.8	310°46.2	00.6	315°51.3	12.4	43°53.6	04.2										
5	47°58.2	233°29.7	44.6	325°47.0	00.8	330°53.4	12.4	58°56.2	04.3										
6	63°00.7	248°29.4	N03°43.3	340°47.7	N23°00.9	345°55.5	N22°12.5	73°58.9	S07°04.3										
7	78°03.1	263°29.0	42.0	355°48.5	01.0	0°57.6	12.5	89°01.5	04.4										
8	93°05.6	278°28.7	40.8	10°49.2	01.1	15°59.7	12.5	104°04.1	04.5										
9	108°08.1	293°28.4	· · 39.5	25°50.0	· · 01.3	31°01.8	· · 12.6	119°06.8	· · 04.6										
10	123°10.5	308°28.0	38.2	40°50.7	01.4	46°03.9	12.6	134°09.4	04.6										
11	138°13.0	323°27.7	37.0	55°51.5	01.5	61°06.0	12.6	149°12.0	04.7										
12	153°15.5	338°27.4	N03°35.7	70°52.2	N23°01.7	76°08.1	N22°12.6	164°14.6	S07°04.8										
13	168°17.9	353°27.1	34.4	85°53.0	01.8	91°10.2	12.7	179°17.3	04.9										
14	183°20.4	8°26.7	33.2	100°53.7	01.9	106°12.3	12.7	194°19.9	04.9										
15	198°22.9	23°26.4	· · 31.9	115°54.5	· · 02.0	121°14.4	· · 12.7	209°22.5	· · 05.0										
16	213°25.3	38°26.1	30.6	130°55.2	02.2	136°16.5	12.8	224°25.2	05.1										
17	228°27.8	53°25.7	29.3	145°56.0	02.3	151°18.6	12.8	239°27.8	05.2										
18	243°30.3	68°25.4	N03°28.1	160°56.8	N23°02.4	166°20.7	N22°12.8	254°30.4	S07°05.2										
19	258°32.7	83°25.1	26.8	175°57.5	02.5	181°22.8	12.8	269°33.1	05.3										
20	273°35.2	98°24.7	25.5	190°58.3	02.6	196°25.0	12.9	284°35.7	05.4										
21	288°37.6	113°24.4	· · 24.3	205°59.0	· · 02.8	211°27.1	· · 12.9	299°38.3	· · 05.5										
22	303°40.1	128°24.1	23.0	220°59.8	02.9	226°29.2	12.9	314°40.9	05.5										
23	318°42.6	143°23.8	21.7	236°00.5	03.0	241°31.3	13.0	329°43.6	05.6										
Mer.pass. 01:49				ν -0.3' d -1.3' m -3.86				ν 0.8' d 0.1' m 0.75				ν 2.1' d 0.0' m -2.22				ν 2.6' d 0.1' m 0.64			

Aug 22 Thu			SHA	Mer.pass
Venus	187°59.9		13:25	
Mars	279°19.7		07:19	
Jupiter	283°14.6		07:03	
Saturn	10°49.4		01:13	

Aug 23 Fri			SHA	Mer.pass
Venus	186°52.6		13:26	
Mars	278°38.4		07:18	
Jupiter	283°05.7		07:00	
Saturn	10°53.3		01:09	

Aug 24 Sat			SHA	Mer.pass
Venus	185°45.5		13:26	
Mars	277°57.3		07:17	
Jupiter	282°56.9		06:56	
Saturn	10°57.2		01:05	

Horizontal parallax			
Venus:		0.1	
Mars:		0.1	

h	Sun				Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP	
0	179°16.6	N11°40.9	330°00.4	10.6'	S00°17.9	-17.9'	60.7'	
1	194°16.8	40.1	344°30.0	10.6'	N00°00.0	17.9'	60.7'	
2	209°16.9	39.2	358°59.6	10.7'	00°17.9	17.9'	60.7'	
3	224°17.1	· · 38.4	13°29.3	10.7'	00°35.8	17.9'	60.7'	
4	239°17.2	37.5	27°59.0	10.7'	00°53.7	17.9'	60.7'	
5	254°17.4	36.7	42°28.7	10.7'	01°11.5	17.8'	60.7'	
6	269°17.6	N11°35.8	56°58.3	10.7'	N01°29.4	17.8'	60.7'	
7	284°17.7	35.0	71°28.0	10.7'	01°47.2	17.8'	60.6'	
8	299°17.9	34.2	85°57.7	10.7'	02°05.0	17.8'	60.6'	
9	314°18.1	· · 33.3	100°27.4	10.7'	02°22.8	17.8'	60.6'	
10	329°18.2	32.5	114°57.1	10.7'	02°40.6	17.7'	60.6'	
11	344°18.4	31.6	129°26.8	10.7'	02°58.3	17.7'	60.6'	
12	359°18.5	N11°30.8	143°56.5	10.7'	N03°16.1	17.7'	60.6'	
13	14°18.7	29.9	158°26.2	10.7'	03°33.7	17.7'	60.6'	
14	29°18.9	29.1	172°55.9	10.7'	03°51.4	17.6'	60.5'	
15	44°19.0	· · 28.2	187°25.6	10.7'	04°09.0	17.6'	60.5'	
16	59°19.2	27.4	201°55.3	10.7'	04°26.7	17.6'	60.5'	
17	74°19.4	26.5	216°25.0	10.7'	04°44.2	17.5'	60.5'	
18	89°19.5	N11°25.7	230°54.7	10.7'	N05°01.8	17.5'	60.5'	
19	104°19.7	24.8	245°24.4	10.7'	05°19.2	17.5'	60.5'	
20	119°19.9	24.0	259°54.0	10.6'	05°36.7	17.4'	60.4'	
21	134°20.0	· · 23.1	274°23.7	10.6'	05°54.1	17.4'	60.4'	
22	149°20.2	22.3	288°53.3	10.6'	06°11.5	17.3'	60.4'	
23	164°20.3	21.5	303°22.9	10.6'	06°28.8	17.3'	60.4'	
SD = 15.8'			$d = -0.8'$		SD = 16.6'			
Fri	GHA	Dec	GHA	ν	Dec	d	HP	
0	179°20.5	N11°20.6	317°52.5	10.6'	N06°46.1	17.2'	60.4'	
1	194°20.7	19.8	332°22.1	10.6'	07°03.3	17.2'	60.4'	
2	209°20.8	18.9	346°51.7	10.5'	07°20.5	17.1'	60.3'	
3	224°21.0	· · 18.1	1°21.2	10.5'	07°37.6	17.1'	60.3'	
4	239°21.2	17.2	15°50.7	10.5'	07°54.7	17.0'	60.3'	
5	254°21.3	16.4	30°20.2	10.5'	08°11.7	17.0'	60.3'	
6	269°21.5	N11°15.5	44°49.7	10.5'	N08°28.7	16.9'	60.3'	
7	284°21.7	14.7	59°19.2	10.4'	08°45.6	16.8'	60.2'	
8	299°21.8	13.8	73°48.6	10.4'	09°02.4	16.8'	60.2'	
9	314°22.0	· · 12.9	88°18.0	10.4'	09°19.2	16.7'	60.2'	
10	329°22.2	12.1	102°47.4	10.4'	09°35.9	16.6'	60.2'	
11	344°22.3	11.2	117°16.8	10.3'	09°52.6	16.6'	60.2'	
12	359°22.5	N11°10.4	131°46.1	10.3'	N10°09.1	16.5'	60.1'	
13	14°22.7	09.5	146°15.4	10.3'	10°25.7	16.4'	60.1'	
14	29°22.8	08.7	160°44.7	10.2'	10°42.1	16.4'	60.1'	
15	44°23.0	· · 07.8	175°13.9	10.2'	10°58.5	16.3'	60.1'	
16	59°23.2	07.0	189°43.1	10.2'	11°14.8	16.2'	60.0'	
17	74°23.4	06.1	204°12.3	10.1'	11°31.0	16.1'	60.0'	
18	89°23.5	N11°05.3	218°41.4	10.1'	N11°47.1	16.1'	60.0'	
19	104°23.7	04.4	233°10.5	10.1'	12°03.2	16.0'	60.0'	
20	119°23.9	03.6	247°39.6	10.0'	12°19.2	15.9'	59.9'	
21	134°24.0	· · 02.7	262°08.6	10.0'	12°35.1	15.8'	59.9'	
22	149°24.2	01.8	276°37.6	9.9'	12°50.9	15.7'	59.9'	
23	164°24.4	01.0	291°06.6	9.9'	13°06.7	15.6'	59.9'	
SD = 15.8'			$d = -0.8'$		SD = 16.5'			
Sat	GHA	Dec	GHA	ν	Dec	d	HP	
0	179°24.5	N11°00.1	305°35.5	9.9'	N13°22.3	15.6'	59.8'	
1	194°24.7	10°59.3	320°04.3	9.8'	13°37.9	15.5'	59.8'	
2	209°24.9	58.4	334°33.2	9.8'	13°53.3	15.4'	59.8'	
3	224°25.0	· · 57.6	349°01.9	9.7'	14°08.7	15.3'	59.8'	
4	239°25.2	56.7	3°30.7	9.7'	14°24.0	15.2'	59.7'	
5	254°25.4	55.8	17°59.4	9.6'	14°39.2	15.1'	59.7'	
6	269°25.6	N10°55.0	32°28.0	9.6'	N14°54.3	15.0'	59.7'	
7	284°25.7	54.1	46°56.6	9.6'	15°09.3	14.9'	59.7'	
8	299°25.9	53.3	61°25.2	9.5'	15°24.2	14.8'	59.6'	
9	314°26.1	· · 52.4	75°53.7	9.5'	15°39.0	14.7'	59.6'	
10	329°26.2	51.6	90°22.1	9.4'	15°53.7	14.6'	59.6'	
11	344°26.4	50.7	104°50.5	9.4'	16°08.3	14.5'	59.5'	
12	359°26.6	N10°49.8	119°18.9	9.3'	N16°22.8	14.4'	59.5'	
13	14°26.8	49.0	133°47.2	9.3'	16°37.2	14.3'	59.5'	
14	29°26.9	48.1	148°15.5	9.2'	16°51.4	14.2'	59.5'	
15	44°27.1	· · 47.2	162°43.7	9.2'	17°05.6	14.1'	59.4'	
16	59°27.3	46.4	177°11.8	9.1'	17°19.7	13.9'	59.4'	
17	74°27.4	45.5	191°39.9	9.0'	17°33.6	13.8'	59.4'	
18	89°27.6	N10°44.7	206°08.0	9.0'	N17°47.5	13.7'	59.3'	
19	104°27.8	43.8	220°36.0	8.9'	18°01.2	13.6'	59.3'	
20	119°28.0	42.9	235°03.9	8.9'	18°14.8	13.5'	59.3'	
21	134°28.1	· · 42.1	249°31.8	8.8'	18°28.3	13.4'	59.3'	
22	149°28.3	41.2	263°59.6	8.8'	18°41.6	13.2'	59.2'	
23	164°28.5	40.4	278°27.4	8.7'	18°54.9	13.1'	59.2'	
SD = 15.8'			$d = -0.9'$		SD = 16.3'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	01:07	03:17	20:44	22:45	—:—
N 70°	—:—	02:05	03:37	20:24	21:54	—:—
68°	—:—	02:38	03:54	20:09	21:23	—:—
66°	00:58	03:01	04:07	19:56	21:00	22:56
64°	01:49	03:20	04:17	19:45	20:43	22:10
62°	02:19	03:34	04:27	19:37	20:28	21:42
60°	02:41	03:47	04:34	19:29	20:16	21:21
N 58°	02:58	03:57	04:41	19:22	20:06	21:04
56°	03:13	04:06	04:47	19:16	19:57	20:50
54°	03:25	04:14	04:53	19:11	19:49	20:38
52°	03:35	04:21	04:58	19:06	19:43	20:28
50°	03:44	04:27	05:02	19:02	19:36	20:19
45°	04:03	04:41	05:12	18:52	19:23	20:01
N 40°	04:17	04:51	05:20	18:45	19:13	19:47
35°	04:29	05:00	05:26	18:38	19:04	19:36
30°	04:38	05:07	05:32	18:32	18:57	19:26
20°	04:53	05:20	05:42	18:22	18:45	19:11
N 10°	05:05	05:30	05:51	18:14	18:35	19:00
0°	05:14	05:38	05:59	18:06	18:27	18:51
S 10°	05:21	05:46	06:07	17:58	18:19	18:44
20°	05:27	05:53	06:16	17:50	18:12	18:38
30°	05:33	06:01	06:25	17:40	18:05	18:33
35°	05:35	06:05	06:30	17:35	18:01	18:30
40°	05:37	06:09	06:37	17:29	17:57	18:28
45°	05:39	06:14	06:44	17:22	17:52	18:26
S 50°	05:41	06:19	06:52	17:14	17:47	18:24
52°	05:42	06:21	06:56	17:10	17:45	18:24
54°	05:43	06:24	07:00	17:06	17:42	18:23
56°	05:43	06:27	07:05	17:01	17:39	18:23
58°	05:44	06:30	07:10	16:56	17:37	18:22
S 60°	05:44	06:33	07:16	16:50	17:33	18:22

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	19:33	18:46	17:12	08:44	11:20	14:45
N 70°	19:40	19:06	18:11	08:41	11:03	13:48
68°	19:46	19:21	18:46	08:38	10:50	13:14
66°	19:51	19:34	19:12	08:36	10:40	12:50
64°	19:55	19:45	19:32	08:34	10:31	12:32
62°	19:59	19:54	19:48	08:33	10:23	12:17
60°	20:03	20:02	20:02	08:31	10:17	12:04
N 58°	20:05	20:09	20:14	08:30	10:11	11:53
56°	20:08	20:15	20:24	08:29	10:06	11:44
54°	20:11	20:21	20:33	08:28	10:02	11:36
52°	20:13	20:26	20:42	08:27	09:58	11:28
50°	20:15	20:30	20:49	08:27	09:54	11:21
45°	20:19	20:40	21:05	08:25	09:46	11:07
N 40°	20:23	20:49	21:18	08:23	09:40	10:56
35°	20:26	20:56	21:29	08:22	09:34	10:46
30°	20:29	21:03	21:39	08:21	09:29	10:37
20°	20:34	21:14	21:56	08:19	09:21	10:23
N 10°	20:38	21:24	22:11	08:18	09:13	10:10
0°	20:43	21:33	22:26	08:16	09:06	09:58
S 10°	20:47	21:43	22:40	08:14	08:59	09:46
20°	20:51	21:53	22:55	08:13	08:52	09:33
30°	20:57	22:05	23:13	08:11	08:44	09:19
35°	21:00	22:12	23:24	08:10	08:39	09:11
40°	21:03	22:19	23:36	08:08	08:34	09:01
45°	21:07	22:29	23:50	08:07	08:28	08:50
S 50°	21:12	22:40	· · ·	08:05	08:20	08:37
52°	21:15	22:45	· · ·	08:04	08:17	08:31
54°	21:17	22:51	· · ·	08:04	08:13	08:24
56°	21:20	22:57	· · ·	08:03	08:09	08:17
58°	21:23	23:05	· · ·	08:02	08:04	08:08
S 60°	21:27	23:13	· · ·	08:00	07:59	07:59

Day	Sun		Mer. Pass	Moon		Age
	Eqn.of Time 00 ^h	12 ^h	Mer. Pass hh:mm	Upper hh:mm	Lower hh:mm	18-20 93-76%
22	02:54	02:46	12:03	02:04	14:29	
23	02:38	02:30	12:02	02:54	15:20	
24	02:22	02:14	12:02	03:45	16:12	

August 25, 26, 27 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	333°45.0	158°23.4	N03°20.5	251°01.3	N23°03.1	256°33.4	N22°13.0	344°46.2	S07°05.7	Alpheratz	357°34.8	29°13.6
1	348°47.5	173°23.1	19.2	266°02.0	03.3	271°35.5	13.0	359°48.8	05.8	Ankaa	353°07.1	-42°10.1
2	3°50.0	188°22.8	17.9	281°02.8	03.4	286°37.6	13.0	14°51.5	05.8	Schedar	349°31.0	56°40.2
3	18°52.4	203°22.4	•• 16.6	296°03.6	•• 03.5	301°39.7	•• 13.1	29°54.1	•• 05.9	Diphda	348°47.3	-17°50.9
4	33°54.9	218°22.1	15.4	311°04.3	03.6	316°41.8	13.1	44°56.7	06.0	Achernar	335°20.0	-57°06.4
5	48°57.4	233°21.8	14.1	326°05.1	03.7	331°43.9	13.1	59°59.4	06.1	Hamal	327°51.4	23°34.7
6	63°59.8	248°21.5	N03°12.8	341°05.8	N23°03.9	346°46.0	N22°13.2	75°02.0	S07°06.1	Polaris	314°03.9	89°21.8
7	79°02.3	263°21.1	11.6	356°06.6	04.0	1°48.2	13.2	90°04.6	06.2	Acamar	315°11.9	-40°12.0
8	94°04.7	278°20.8	10.3	11°07.3	04.1	16°50.3	13.2	105°07.2	06.3	Menkar	314°06.4	4°11.3
9	109°07.2	293°20.5	•• 09.0	26°08.1	•• 04.2	31°52.4	•• 13.2	120°09.9	•• 06.4	Mirfak	308°28.6	49°56.8
10	124°09.7	308°20.2	07.7	41°08.9	04.3	46°54.5	13.3	135°12.5	06.4	Aldebaran	290°40.0	16°33.6
11	139°12.1	323°19.8	06.5	56°09.6	04.5	61°56.6	13.3	150°15.1	06.5	Rigel	281°04.3	-8°10.2
12	154°14.6	338°19.5	N03°05.2	71°10.4	N23°04.6	76°58.7	N22°13.3	165°17.8	S07°06.6	Capella	280°22.5	46°01.2
13	169°17.1	353°19.2	03.9	86°11.1	04.7	92°00.8	13.3	180°20.4	06.7	Bellatrix	278°23.3	6°22.4
14	184°19.5	8°18.9	02.7	101°11.9	04.8	107°02.9	13.4	195°23.0	06.7	Elnath	278°02.4	28°37.7
15	199°22.0	23°18.5	•• 01.4	116°12.7	•• 04.9	122°05.1	•• 13.4	210°25.7	•• 06.8	Alnilam	275°38.2	-1°11.0
16	214°24.5	38°18.2	03°00.1	131°13.4	05.1	137°07.2	13.4	225°28.3	06.9	Betelgeuse	270°52.6	7°24.8
17	229°26.9	53°17.9	02°58.8	146°14.2	05.2	152°09.3	13.5	240°30.9	07.0	Canopus	263°52.8	-52°42.2
18	244°29.4	68°17.6	N02°57.6	161°14.9	N23°05.3	167°11.4	N22°13.5	255°33.6	S07°07.0	Sirius	258°26.7	-16°44.8
19	259°31.9	83°17.2	56.3	176°15.7	05.4	182°13.5	13.5	270°36.2	07.1	Adhara	255°06.4	-29°00.1
20	274°34.3	98°16.9	55.0	191°16.5	05.5	197°15.6	13.5	285°38.8	07.2	Procyon	244°51.4	5°09.8
21	289°36.8	113°16.6	•• 53.7	206°17.2	•• 05.6	212°17.7	•• 13.6	300°41.5	•• 07.3	Pollux	243°18.0	27°58.0
22	304°39.2	128°16.3	52.5	221°18.0	05.7	227°19.8	13.6	315°44.1	07.4	Avior	234°15.5	-59°35.0
23	319°41.7	143°15.9	51.2	236°18.7	05.9	242°22.0	13.6	330°46.7	07.4	Suhail	222°47.0	-43°31.7
Mer.pass. 01:45		ν -0.3' d -1.3' m -3.86		ν 0.8' d 0.1' m 0.74		ν 2.1' d 0.0' m -2.22		ν 2.6' d 0.1' m 0.64				
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	334°44.2	158°15.6	N02°49.9	251°19.5	N23°06.0	257°24.1	N22°13.7	345°49.3	S07°07.5	Alphard	217°48.4	-8°45.8
1	349°46.6	173°15.3	48.6	266°20.3	06.1	272°26.2	13.7	0°52.0	07.6	Regulus	207°35.1	11°50.9
2	4°49.1	188°15.0	47.4	281°21.0	06.2	287°28.3	13.7	15°54.6	07.7	Dubhe	193°42.0	61°37.2
3	19°51.6	203°14.6	•• 46.1	296°21.8	•• 06.3	302°30.4	•• 13.7	30°57.2	•• 07.7	Denebola	182°25.6	14°26.2
4	34°54.0	218°14.3	44.8	311°22.5	06.4	317°32.5	13.8	45°59.9	07.8	Gienah	175°44.2	-17°40.7
5	49°56.5	233°14.0	43.5	326°23.3	06.5	332°34.7	13.8	61°02.5	07.9	Acrux	173°01.1	-63°14.2
6	64°59.0	248°13.7	N02°42.3	341°24.1	N23°06.7	347°36.8	N22°13.8	76°05.1	S07°08.0	Gacrux	171°52.5	-57°15.2
7	80°01.4	263°13.3	41.0	356°24.8	06.8	2°38.9	13.8	91°07.8	08.0	Alioth	166°13.6	55°49.8
8	95°03.9	278°13.0	39.7	11°25.6	06.9	17°41.0	13.9	106°10.4	08.1	Spica	158°22.8	-11°17.3
9	110°06.4	293°12.7	•• 38.4	26°26.4	•• 07.0	32°43.1	•• 13.9	121°13.0	•• 08.2	Alkaid	152°52.5	49°11.7
10	125°08.8	308°12.4	37.2	41°27.1	07.1	47°45.2	13.9	136°15.7	08.3	Hadar	148°36.8	-60°29.7
11	140°11.3	323°12.0	35.9	56°27.9	07.2	62°47.4	13.9	151°18.3	08.3	Menkent	147°58.2	-36°29.5
12	155°13.7	338°11.7	N02°34.6	71°28.6	N23°07.3	77°49.5	N22°14.0	166°20.9	S07°08.4	Arcturus	145°48.3	19°03.4
13	170°16.2	353°11.4	33.3	86°29.4	07.4	92°51.6	14.0	181°23.6	08.5	Rigel Kent.	139°41.0	-60°56.4
14	185°18.7	8°11.1	32.1	101°30.2	07.6	107°53.7	14.0	196°26.2	08.6	Kochab	137°20.1	74°03.5
15	200°21.1	23°10.7	•• 30.8	116°30.9	•• 07.7	122°55.8	•• 14.1	211°28.8	•• 08.6	Zuben'ubi	136°56.5	-16°08.6
16	215°23.6	38°10.4	29.5	131°31.7	07.8	137°58.0	14.1	226°31.5	08.7	Alphecca	126°04.0	26°38.1
17	230°26.1	53°10.1	28.2	146°32.5	07.9	153°00.1	14.1	241°34.1	08.8	Antares	112°16.2	-26°29.2
18	245°28.5	68°09.8	N02°26.9	161°33.2	N23°08.0	168°02.2	N22°14.1	256°36.7	S07°08.9	Atria	107°10.6	-69°04.5
19	260°31.0	83°09.5	25.7	176°34.0	08.1	183°04.3	14.2	271°39.4	08.9	Sabik	102°03.0	-15°45.3
20	275°33.5	98°09.1	24.4	191°34.8	08.2	198°06.4	14.2	286°42.0	09.0	Shaula	96°10.6	-37°07.4
21	290°35.9	113°08.8	•• 23.1	206°35.5	•• 08.3	213°08.6	•• 14.2	301°44.6	•• 09.1	Rasalhague	95°58.7	12°32.7
22	305°38.4	128°08.5	21.8	221°36.3	08.4	228°10.7	14.2	316°47.3	09.2	Eltanin	90°42.1	51°29.4
23	320°40.9	143°08.2	20.6	236°37.1	08.5	243°12.8	14.3	331°49.9	09.2	Kaus Aust.	83°32.7	-34°22.4
Mer.pass. 01:41		ν -0.3' d -1.3' m -3.86		ν 0.8' d 0.1' m 0.74		ν 2.1' d 0.0' m -2.23		ν 2.6' d 0.1' m 0.63				
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	335°43.3	158°07.8	N02°19.3	251°37.8	N23°08.6	258°14.9	N22°14.3	346°52.5	S07°09.3	Vega	80°33.2	38°48.6
1	350°45.8	173°07.5	18.0	266°38.6	08.8	273°17.0	14.3	1°55.2	09.4	Nunki	75°47.9	-26°16.0
2	5°48.2	188°07.2	16.7	281°39.4	08.9	288°19.2	14.3	16°57.8	09.5	Altair	62°00.0	8°56.1
3	20°50.7	203°06.9	•• 15.4	296°40.1	•• 09.0	303°21.3	•• 14.4	32°00.4	•• 09.6	Peacock	53°05.7	-56°39.4
4	35°53.2	218°06.6	14.2	311°40.9	09.1	318°23.4	14.4	47°03.1	09.6	Deneb	49°25.6	45°22.2
5	50°55.6	233°06.2	12.9	326°41.7	09.2	333°25.5	14.4	62°05.7	09.7	Enif	33°38.8	9°59.4
6	65°58.1	248°05.9	N02°11.6	341°42.4	N23°09.3	348°27.6	N22°14.5	77°08.3	S07°09.8	Al Na'ir	27°32.8	-46°50.4
7	81°00.6	263°05.6	10.3	356°43.2	09.4	3°29.8	14.5	92°11.0	09.9	Fomalhaut	15°14.5	-29°29.4
8	96°03.0	278°05.3	09.0	11°44.0	09.5	18°31.9	14.5	107°13.6	09.9	Scheat	13°45.2	28°13.0
9	111°05.5	293°05.0	•• 07.8	26°44.7	•• 09.6	33°34.0	•• 14.5	122°16.2	•• 10.0	Markab	13°29.9	15°20.3
10	126°08.0	308°04.6	06.5	41°45.5	09.7	48°36.1	14.6	137°18.9	10.1	Aug 25 Sun	SHA	Mer.pass
11	141°10.4	323°04.3	05.2	56°46.3	09.8	63°38.3	14.6	152°21.5	10.2	Venus	184°38.4	13:27
12	156°12.9	338°04.0	N02°03.9	71°47.0	N23°09.9	78°40.4	N22°14.6	167°24.1	S07°10.2	Mars	277°16.3	07:16
13	171°15.4	353°03.7	02.6	86°47.8	10.0	93°42.5	14.6	182°26.8	10.3	Jupiter	282°48.3	06:53
14	186°17.8	8°03.4	01.4	101°48.6	10.1	108°44.6	14.7	197°29.4	10.4	Saturn	11°01.2	01:01
15	201°20.3	23°03.0	02°00.1	116°49.3	•• 10.2	123°46.8	•• 14.7	212°32.0	•• 10.5			
16	216°22.7	38°02.7	01°58.8	131°50.1	10.3	138°48.9	14.7	227°34.7	10.5	Aug 26 Mon	SHA	Mer.pass
17	231°25.2	53°02.4	57.5	146°50.9	10.4	153°51.0	14.7	242°37.3	10.6	Venus	183°31.4	13:27
18	246°27.7	68°02.1	N01°56.2	161°51.6	N23°10.5	168°53.1	N22°14.8	257°39.9	S07°10.7	Mars	276°35.3	07:14
19	261°30.1	83°01.8	55.0	176°52.4	10.6	183°55.3	14.8	272°42.6	10.8	Jupiter	282°39.9	06:49
20	276°32.6	98°01.4	53.7	191°53.2	10.7	198°57.4	14.8	287°45.2	10.8	Saturn	11°05.2	00:57
21	291°35.1	113°01.1	•• 52.4	206°54.0	•• 10.8	213°59.5</						

Sun				Moon				
Sun	GHA	Dec		GHA	ν	Dec	d	HP
0	179°28.7	N10°39.5		292°55.1	8.7'	N19°08.0	13.0'	59.2'
1	194°28.8	38.6		307°22.8	8.6'	19°21.0	12.9'	59.1'
2	209°29.0	37.8		321°50.4	8.6'	19°33.9	12.8'	59.1'
3	224°29.2	· · 36.9		336°18.0	8.5'	19°46.7	12.6'	59.1'
4	239°29.4	36.0		350°45.5	8.4'	19°59.3	12.5'	59.0'
5	254°29.5	35.2		5°12.9	8.4'	20°11.8	12.4'	59.0'
6	269°29.7	N10°34.3		19°40.3	8.3'	N20°24.2	12.2'	59.0'
7	284°29.9	33.4		34°07.7	8.3'	20°36.4	12.1'	59.0'
8	299°30.0	32.6		48°34.9	8.2'	20°48.5	12.0'	58.9'
9	314°30.2	· · 31.7		63°02.2	8.2'	21°00.5	11.8'	58.9'
10	329°30.4	30.8		77°29.3	8.1'	21°12.4	11.7'	58.9'
11	344°30.6	30.0		91°56.4	8.1'	21°24.1	11.6'	58.8'
12	359°30.8	N10°29.1		106°23.5	8.0'	N21°35.6	11.4'	58.8'
13	14°30.9	28.2		120°50.5	7.9'	21°47.1	11.3'	58.8'
14	29°31.1	27.4		135°17.4	7.9'	21°58.4	11.2'	58.7'
15	44°31.3	· · 26.5		149°44.3	7.8'	22°09.5	11.0'	58.7'
16	59°31.5	25.6		164°11.1	7.8'	22°20.6	10.9'	58.7'
17	74°31.6	24.8		178°37.9	7.7'	22°31.4	10.7'	58.6'
18	89°31.8	N10°23.9		193°04.6	7.7'	N22°42.2	10.6'	58.6'
19	104°32.0	23.0		207°31.3	7.6'	22°52.8	10.4'	58.6'
20	119°32.2	22.2		221°57.9	7.6'	23°03.2	10.3'	58.6'
21	134°32.3	· · 21.3		236°24.5	7.5'	23°13.5	10.2'	58.5'
22	149°32.5	20.4		250°51.0	7.5'	23°23.6	10.0'	58.5'
23	164°32.7	19.5		265°17.4	7.4'	23°33.7	9.9'	58.5'
SD = 15.8' d = -0.9'				SD = 16.1'				
Mon	GHA	Dec		GHA	ν	Dec	d	HP
0	179°32.9	N10°18.7		279°43.8	7.4'	N23°43.5	9.7'	58.4'
1	194°33.0	17.8		294°10.2	7.3'	23°53.2	9.6'	58.4'
2	209°33.2	16.9		308°36.5	7.2'	24°02.8	9.4'	58.4'
3	224°33.4	· · 16.1		323°02.7	7.2'	24°12.2	9.2'	58.3'
4	239°33.6	15.2		337°28.9	7.2'	24°21.4	9.1'	58.3'
5	254°33.8	14.3		351°55.1	7.1'	24°30.5	8.9'	58.3'
6	269°33.9	N10°13.4		6°21.2	7.1'	N24°39.5	8.8'	58.2'
7	284°34.1	12.6		20°47.2	7.0'	24°48.2	8.6'	58.2'
8	299°34.3	11.7		35°13.3	7.0'	24°56.9	8.5'	58.2'
9	314°34.5	· · 10.8		49°39.2	6.9'	25°05.3	8.3'	58.2'
10	329°34.7	09.9		64°05.1	6.9'	25°13.7	8.2'	58.1'
11	344°34.8	09.1		78°31.0	6.8'	25°21.8	8.0'	58.1'
12	359°35.0	N10°08.2		92°56.8	6.8'	N25°29.8	7.8'	58.1'
13	14°35.2	07.3		107°22.6	6.8'	25°37.6	7.7'	58.0'
14	29°35.4	06.4		121°48.4	6.7'	25°45.3	7.5'	58.0'
15	44°35.6	· · 05.6		136°14.1	6.7'	25°52.8	7.4'	58.0'
16	59°35.7	04.7		150°39.8	6.6'	26°00.2	7.2'	57.9'
17	74°35.9	03.8		165°05.4	6.6'	26°07.4	7.0'	57.9'
18	89°36.1	N10°02.9		179°31.0	6.6'	N26°14.4	6.9'	57.9'
19	104°36.3	02.1		193°56.6	6.5'	26°21.3	6.7'	57.8'
20	119°36.5	01.2		208°22.1	6.5'	26°27.9	6.5'	57.8'
21	134°36.6	10°00.3		222°47.6	6.5'	26°34.5	6.4'	57.8'
22	149°36.8	09°59.4		237°13.1	6.4'	26°40.8	6.2'	57.8'
23	164°37.0	58.6		251°38.5	6.4'	26°47.0	6.0'	57.7'
SD = 15.8' d = -0.9'				SD = 15.9'				
Tue	GHA	Dec		GHA	ν	Dec	d	HP
0	179°37.2	N09°57.7		266°03.9	6.4'	N26°53.1	5.9'	57.7'
1	194°37.4	56.8		280°29.3	6.4'	26°58.9	5.7'	57.7'
2	209°37.5	55.9		294°54.7	6.3'	27°04.6	5.5'	57.6'
3	224°37.7	· · 55.1		309°20.0	6.3'	27°10.2	5.4'	57.6'
4	239°37.9	54.2		323°45.3	6.3'	27°15.5	5.2'	57.6'
5	254°38.1	53.3		338°10.6	6.3'	27°20.7	5.0'	57.5'
6	269°38.3	N09°52.4		352°35.9	6.3'	N27°25.7	4.9'	57.5'
7	284°38.5	51.5		7°01.1	6.2'	27°30.6	4.7'	57.5'
8	299°38.6	50.7		21°26.3	6.2'	27°35.3	4.5'	57.5'
9	314°38.8	· · 49.8		35°51.6	6.2'	27°39.8	4.3'	57.4'
10	329°39.0	48.9		50°16.8	6.2'	27°44.1	4.2'	57.4'
11	344°39.2	48.0		64°42.0	6.2'	27°48.3	4.0'	57.4'
12	359°39.4	N09°47.1		79°07.2	6.2'	N27°52.3	3.8'	57.3'
13	14°39.6	46.3		93°32.3	6.2'	27°56.1	3.7'	57.3'
14	29°39.7	45.4		107°57.5	6.2'	27°59.8	3.5'	57.3'
15	44°39.9	· · 44.5		122°22.7	6.2'	28°03.2	3.3'	57.2'
16	59°40.1	43.6		136°47.8	6.2'	28°06.6	3.1'	57.2'
17	74°40.3	42.7		151°13.0	6.2'	28°09.7	3.0'	57.2'
18	89°40.5	N09°41.8		165°38.2	6.2'	N28°12.7	2.8'	57.2'
19	104°40.7	41.0		180°03.3	6.2'	28°15.5	2.6'	57.1'
20	119°40.8	40.1		194°28.5	6.2'	28°18.1	2.5'	57.1'
21	134°41.0	· · 39.2		208°53.7	6.2'	28°20.6	2.3'	57.1'
22	149°41.2	38.3		223°18.9	6.2'	28°22.8	2.1'	57.0'
23	164°41.4	37.4		237°44.0	6.2'	28°25.0	1.9'	57.0'
SD = 15.8' d = -0.9'				SD = 15.7'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	01:44	03:33	20:26	22:11	—:—
N 70°	—:—	02:26	03:51	20:09	21:32	—:—
68°	—:—	02:54	04:05	19:55	21:06	23:48
66°	01:30	03:14	04:17	19:44	20:46	22:26
64°	02:07	03:30	04:26	19:35	20:30	21:51
62°	02:33	03:44	04:35	19:27	20:17	21:27
60°	02:52	03:55	04:42	19:20	20:06	21:08
N 58°	03:08	04:04	04:48	19:14	19:57	20:53
56°	03:21	04:13	04:53	19:09	19:49	20:40
54°	03:32	04:20	04:58	19:04	19:42	20:30
52°	03:41	04:26	05:03	19:00	19:36	20:20
50°	03:50	04:32	05:07	18:56	19:30	20:12
45°	04:07	04:44	05:15	18:47	19:18	19:55
N 40°	04:21	04:54	05:22	18:40	19:08	19:42
35°	04:31	05:02	05:29	18:34	19:00	19:31
30°	04:40	05:09	05:34	18:29	18:54	19:23
20°	04:54	05:21	05:43	18:20	18:42	19:09
N 10°	05:05	05:30	05:51	18:12	18:34	18:58
0°	05:13	05:37	05:58	18:05	18:26	18:50
S 10°	05:20	05:44	06:06	17:58	18:19	18:44
20°	05:25	05:51	06:13	17:50	18:13	18:38
30°	05:30	05:58	06:22	17:42	18:06	18:34
35°	05:32	06:01	06:27	17:37	18:03	18:32
40°	05:33	06:05	06:32	17:32	17:59	18:31
45°	05:35	06:09	06:38	17:26	17:55	18:29
S 50°	05:36	06:13	06:46	17:18	17:51	18:29
52°	05:36	06:15	06:49	17:15	17:49	18:28
54°	05:36	06:17	06:53	17:11	17:47	18:28
56°	05:36	06:19	06:57	17:07	17:45	18:28
58°	05:37	06:22	07:02	17:02	17:42	18:28
S 60°	05:37	06:25	07:07	16:57	17:40	18:28

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°						
N 70°						
68°						
66°	18:30			15:25		
64°	19:13			14:43		
62°	19:42	19:33		14:15	16:22	
60°	20:04	20:11	20:30	13:54	15:45	17:26
N 58°	20:22	20:38	21:08	13:37	15:18	16:48
56°	20:37	20:59	21:35	13:22	14:57	16:22
54°	20:51	21:17	21:56	13:09	14:40	16:01
52°	21:02	21:32	22:13	12:58	14:25	15:43
50°	21:13	21:45	22:29	12:49	14:13	15:28
45°	21:34	22:12	22:59	12:28	13:47	14:58
N 40°	21:52	22:33	23:23	12:12	13:26	14:35
35°	22:07	22:51	23:42	11:58	13:09	14:15
30°	22:20	23:07	23:59	11:46	12:54	13:59
20°	22:42	23:33	· · ·	11:25	12:29	13:31
N 10°	23:02	23:56	· · ·	11:08	12:07	13:07
0°	23:20	· · ·	00:17	10:51	11:47	12:45
S 10°	23:39	· · ·	00:38	10:35	11:27	12:22
20°	23:58	· · ·	01:01	10:17	11:06	11:58
30°	· · ·	00:22	01:28	09:57	10:41	11:31
35°	· · ·	00:35	01:44	09:46	10:27	11:14
40°	· · ·	00:51	02:03	09:33	10:10	10:55
45°	· · ·	01:10	02:26	09:17	09:50	10:32
S 50°	00:08	01:34	02:54	08:58	09:25	10:03
52°	00:16	01:45	03:09	08:49	09:13	09:48
54°	00:25	01:58	03:25	08:39	09:00	09:31
56°	00:36	02:14	03:45	08:27	08:44	09:11
58°	00:48	02:32	04:10	08:14	08:25	08:46
S 60°	01:02	02:54	04:43	07:59	08:02	08:12

Day	Sun		Mer. Pass hh:mm	Moon		Age 66-43%
	Eqn.of Time	Mer.		Mer.Pass.	Age	
	00 ^h mm:ss	12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
25	02:05	01:57	12:02	04:38	17:06	
26	01:49	01:40	12:02	05:34	18:02	
2						

August 28, 29, 30 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	336°42.5	158°00.2	N01°48.6	251°56.3	N23°11.1	259°05.9	N22°14.9	347°55.7	S07°11.2	Alpheratz	357°34.8	29°13.6
1	351°44.9	172°59.8	47.3	266°57.0	11.2	274°08.0	14.9	2°58.4	11.2	Ankaa	353°07.0	-42°10.1
2	6°47.4	187°59.5	46.0	281°57.8	11.3	289°10.2	15.0	18°01.0	11.3	Schedar	349°30.9	56°40.3
3	21°49.8	202°59.2	•• 44.7	296°58.6	•• 11.4	304°12.3	•• 15.0	33°03.6	•• 11.4	Diphda	348°47.3	-17°50.9
4	36°52.3	217°58.9	43.4	311°59.4	11.5	319°14.4	15.0	48°06.3	11.5	Achernar	335°20.0	-57°06.4
5	51°54.8	232°58.6	42.2	327°00.1	11.6	334°16.5	15.0	63°08.9	11.5	Hamal	327°51.4	23°34.8
6	66°57.2	247°58.2	N01°40.9	342°00.9	N23°11.7	349°18.7	N22°15.1	78°11.5	S07°11.6	Polaris	314°02.2	89°21.8
7	81°59.7	262°57.9	39.6	357°01.7	11.8	4°20.8	15.1	93°14.2	11.7	Acamar	315°11.8	-40°12.0
8	97°02.2	277°57.6	38.3	12°02.4	11.9	19°22.9	15.1	108°16.8	11.8	Menkar	314°06.4	4°11.3
9	112°04.6	292°57.3	•• 37.0	27°03.2	•• 12.0	34°25.1	•• 15.2	123°19.4	•• 11.8	Mirfak	308°28.6	49°56.8
10	127°07.1	307°57.0	35.7	42°04.0	12.1	49°27.2	15.2	138°22.1	11.9	Aldebaran	290°40.0	16°33.6
11	142°09.6	322°56.7	34.5	57°04.8	12.2	64°29.3	15.2	153°24.7	12.0	Rigel	281°04.2	-8°10.2
12	157°12.0	337°56.3	N01°33.2	72°05.5	N23°12.3	79°31.5	N22°15.2	168°27.3	S07°12.1	Capella	280°22.4	46°01.2
13	172°14.5	352°56.0	31.9	87°06.3	12.4	94°33.6	15.3	183°30.0	12.1	Bellatrix	278°23.3	6°22.4
14	187°17.0	7°55.7	30.6	102°07.1	12.5	109°35.7	15.3	198°32.6	12.2	Elnath	278°02.3	28°37.7
15	202°19.4	22°55.4	•• 29.3	117°07.9	•• 12.6	124°37.9	•• 15.3	213°35.2	•• 12.3	Alnilam	275°38.1	-1°11.0
16	217°21.9	37°55.1	28.0	132°08.6	12.7	139°40.0	15.3	228°37.9	12.4	Betelgeuse	270°52.5	7°24.8
17	232°24.3	52°54.7	26.8	147°09.4	12.8	154°42.1	15.4	243°40.5	12.5	Canopus	263°52.8	-52°42.2
18	247°26.8	67°54.4	N01°25.5	162°10.2	N23°12.9	169°44.2	N22°15.4	258°43.2	S07°12.5	Sirius	258°26.7	-16°44.8
19	262°29.3	82°54.1	24.2	177°11.0	13.0	184°46.4	15.4	273°45.8	12.6	Adhara	255°06.3	-29°00.0
20	277°31.7	97°53.8	22.9	192°11.7	13.1	199°48.5	15.4	288°48.4	12.7	Procyon	244°51.4	5°09.8
21	292°34.2	112°53.5	•• 21.6	207°12.5	•• 13.2	214°50.6	•• 15.5	303°51.1	•• 12.8	Pollux	243°17.9	27°58.0
22	307°36.7	127°53.2	20.3	222°13.3	13.3	229°52.8	15.5	318°53.7	12.8	Avior	234°15.4	-59°35.0
23	322°39.1	142°52.8	19.1	237°14.1	13.4	244°54.9	15.5	333°56.3	12.9	Suhail	222°47.0	-43°31.7
Mer.pass. 01:33		ν -0.3' d -1.3' m -3.86		ν 0.8' d 0.1' m 0.73		ν 2.1' d 0.0' m -2.24		ν 2.6' d 0.1' m 0.62		Miaplacidus	221°39.3	-69°48.9
										Alphard	217°48.4	-8°45.8
										Regulus	207°35.1	11°50.9
										Dubhe	193°42.0	61°37.2
										Denebola	182°25.5	14°26.2
										Gienah	175°44.2	-17°40.7
										Acrux	173°01.1	-63°14.2
										Gacrux	171°52.5	-57°15.1
										Alioth	166°13.6	55°49.8
										Spica	158°22.8	-11°17.3
										Alkaid	152°52.5	49°11.7
										Hadar	148°36.8	-60°29.7
										Menkent	147°58.2	-36°29.5
										Arcturus	145°48.3	19°03.4
										Rigel Kent.	139°41.0	-60°56.4
										Kochab	137°20.2	74°03.5
										Zuben'ubi	136°56.5	-16°08.6
										Alphecca	126°04.1	26°38.1
										Antares	112°16.2	-26°29.2
										Atria	107°10.6	-69°04.5
										Sabik	102°03.0	-15°45.3
										Shaula	96°10.6	-37°07.4
										Rasalhague	95°58.7	12°32.7
										Eltanin	90°42.1	51°29.4
										Kaus Aust.	83°32.7	-34°22.4
										Vega	80°33.2	38°48.6
										Nunki	75°47.9	-26°16.0
										Altair	62°00.0	8°56.1
										Peacock	53°05.7	-56°39.4
										Deneb	49°25.6	45°22.2
										Enif	33°38.8	9°59.4
										Al Na'ir	27°32.8	-46°50.4
										Fomalhaut	15°14.5	-29°29.4
										Scheat	13°45.2	28°13.1
										Markab	13°29.9	15°20.4
										Aug 28 Wed	SHA	Mer.pass
										Venus	181°17.7	13:28
										Mars	275°13.8	07:12
										Jupiter	282°23.4	06:43
										Saturn	11°13.3	00:48
										Aug 29 Thu	SHA	Mer.pass
										Venus	180°10.9	13:29
										Mars	274°33.2	07:11
										Jupiter	282°15.5	06:39
										Saturn	11°17.4	00:44
										Aug 30 Fri	SHA	Mer.pass
										Venus	179°04.2	13:29
										Mars	273°52.8	07:09
										Jupiter	282°07.6	06:36
										Saturn	11°21.5	00:40

Sun				Moon				
Wed	GHA	Dec		GHA	ν	Dec	d	HP
0	179°41.6	N09°36.5		252°09.3	6.2'	N28°26.9	1.8'	57.0'
1	194°41.8	35.7		266°34.5	6.2'	28°28.7	1.6'	57.0'
2	209°42.0	34.8		280°59.7	6.2'	28°30.3	1.4'	56.9'
3	224°42.1	· · 33.9		295°24.9	6.3'	28°31.7	1.3'	56.9'
4	239°42.3	33.0		309°50.2	6.3'	28°33.0	1.1'	56.9'
5	254°42.5	32.1		324°15.5	6.3'	28°34.1	0.9'	56.9'
6	269°42.7	N09°31.2		338°40.8	6.3'	N28°35.0	0.8'	56.8'
7	284°42.9	30.3		353°06.1	6.4'	28°35.7	0.6'	56.8'
8	299°43.1	29.5		7°31.5	6.4'	28°36.3	0.4'	56.8'
9	314°43.3	· · 28.6		21°56.9	6.4'	28°36.7	0.2'	56.7'
10	329°43.4	27.7		36°22.3	6.4'	28°37.0	0.1'	56.7'
11	344°43.6	26.8		50°47.8	6.5'	28°37.1	-0.1'	56.7'
12	359°43.8	N09°25.9		65°13.2	6.5'	N28°37.0	-0.3'	56.7'
13	14°44.0	25.0		79°38.8	6.6'	28°36.7	-0.4'	56.6'
14	29°44.2	24.1		94°04.3	6.6'	28°36.3	-0.6'	56.6'
15	44°44.4	· · 23.2		108°29.9	6.6'	28°35.7	-0.8'	56.6'
16	59°44.6	22.4		122°55.5	6.7'	28°34.9	-0.9'	56.6'
17	74°44.7	21.5		137°21.2	6.7'	28°34.0	-1.1'	56.5'
18	89°44.9	N09°20.6		151°46.9	6.8'	N28°32.9	-1.2'	56.5'
19	104°45.1	19.7		166°12.7	6.8'	28°31.7	-1.4'	56.5'
20	119°45.3	18.8		180°38.5	6.9'	28°30.3	-1.6'	56.5'
21	134°45.5	· · 17.9		195°04.3	6.9'	28°28.7	-1.7'	56.4'
22	149°45.7	17.0		209°30.2	7.0'	28°27.0	-1.9'	56.4'
23	164°45.9	16.1		223°56.2	7.0'	28°25.1	-2.1'	56.4'
SD = 15.8' d = -0.9'				SD = 15.5'				
Thu	GHA	Dec		GHA	ν	Dec	d	HP
0	179°46.1	N09°15.2		238°22.2	7.1'	N28°23.0	-2.2'	56.4'
1	194°46.3	14.4		252°48.3	7.1'	28°20.8	-2.4'	56.3'
2	209°46.4	13.5		267°14.4	7.2'	28°18.4	-2.5'	56.3'
3	224°46.6	· · 12.6		281°40.5	7.2'	28°15.9	-2.7'	56.3'
4	239°46.8	11.7		296°06.8	7.3'	28°13.2	-2.8'	56.3'
5	254°47.0	10.8		310°33.1	7.4'	28°10.3	-3.0'	56.2'
6	269°47.2	N09°09.9		324°59.4	7.4'	N28°07.3	-3.2'	56.2'
7	284°47.4	09.0		339°25.9	7.5'	28°04.2	-3.3'	56.2'
8	299°47.6	08.1		353°52.3	7.6'	28°00.9	-3.5'	56.2'
9	314°47.8	· · 07.2		8°18.9	7.6'	27°57.4	-3.6'	56.1'
10	329°48.0	06.3		22°45.5	7.7'	27°53.8	-3.8'	56.1'
11	344°48.1	05.4		37°12.2	7.8'	27°50.0	-3.9'	56.1'
12	359°48.3	N09°04.5		51°39.0	7.8'	N27°46.1	-4.1'	56.1'
13	14°48.5	03.6		66°05.8	7.9'	27°42.0	-4.2'	56.0'
14	29°48.7	02.8		80°32.7	8.0'	27°37.8	-4.4'	56.0'
15	44°48.9	· · 01.9		94°59.7	8.0'	27°33.4	-4.5'	56.0'
16	59°49.1	01.0		109°26.7	8.1'	27°28.9	-4.7'	56.0'
17	74°49.3	09°00.1		123°53.8	8.2'	27°24.2	-4.8'	55.9'
18	89°49.5	N08°59.2		138°21.0	8.3'	N27°19.4	-4.9'	55.9'
19	104°49.7	58.3		152°48.3	8.4'	27°14.5	-5.1'	55.9'
20	119°49.9	57.4		167°15.7	8.4'	27°09.4	-5.2'	55.9'
21	134°50.1	· · 56.5		181°43.1	8.5'	27°04.1	-5.4'	55.8'
22	149°50.2	55.6		196°10.6	8.6'	26°58.8	-5.5'	55.8'
23	164°50.4	54.7		210°38.2	8.7'	26°53.3	-5.7'	55.8'
SD = 15.8' d = -0.9'				SD = 15.4'				
Fri	GHA	Dec		GHA	ν	Dec	d	HP
0	179°50.6	N08°53.8		225°05.9	8.8'	N26°47.6	-5.8'	55.8'
1	194°50.8	52.9		239°33.7	8.8'	26°41.8	-5.9'	55.8'
2	209°51.0	52.0		254°01.5	8.9'	26°35.9	-6.1'	55.7'
3	224°51.2	· · 51.1		268°29.5	9.0'	26°29.8	-6.2'	55.7'
4	239°51.4	50.2		282°57.5	9.1'	26°23.6	-6.3'	55.7'
5	254°51.6	49.3		297°25.6	9.2'	26°17.3	-6.5'	55.7'
6	269°51.8	N08°48.4		311°53.8	9.3'	N26°10.8	-6.6'	55.6'
7	284°52.0	47.5		326°22.1	9.4'	26°04.3	-6.7'	55.6'
8	299°52.2	46.6		340°50.4	9.5'	25°57.5	-6.8'	55.6'
9	314°52.4	· · 45.7		355°18.9	9.5'	25°50.7	-7.0'	55.6'
10	329°52.6	44.8		9°47.4	9.6'	25°43.7	-7.1'	55.6'
11	344°52.8	43.9		24°16.1	9.7'	25°36.6	-7.2'	55.5'
12	359°52.9	N08°43.0		38°44.8	9.8'	N25°29.4	-7.3'	55.5'
13	14°53.1	42.1		53°13.6	9.9'	25°22.0	-7.5'	55.5'
14	29°53.3	41.2		67°42.5	10.0'	25°14.6	-7.6'	55.5'
15	44°53.5	· · 40.3		82°11.5	10.1'	25°07.0	-7.7'	55.5'
16	59°53.7	39.4		96°40.6	10.2'	24°59.3	-7.8'	55.4'
17	74°53.9	38.5		111°09.8	10.3'	24°51.4	-8.0'	55.4'
18	89°54.1	N08°37.6		125°39.1	10.4'	N24°43.5	-8.1'	55.4'
19	104°54.3	36.7		140°08.4	10.5'	24°35.4	-8.2'	55.4'
20	119°54.5	35.8		154°37.9	10.5'	24°27.2	-8.3'	55.4'
21	134°54.7	· · 34.9		169°07.4	10.6'	24°18.9	-8.4'	55.3'
22	149°54.9	34.0		183°37.1	10.7'	24°10.5	-8.5'	55.3'
23	164°55.1	33.1		198°06.8	10.8'	24°02.0	-8.6'	55.3'
SD = 15.8' d = -0.9'				SD = 15.2'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	02:11	03:49	20:09	21:43	—:—
N 70°	—:—	02:45	04:04	19:54	21:12	—:—
68°	01:02	03:08	04:17	19:42	20:50	22:48
66°	01:53	03:26	04:27	19:33	20:32	22:03
64°	02:23	03:41	04:35	19:24	20:18	21:34
62°	02:45	03:53	04:42	19:17	20:06	21:13
60°	03:03	04:03	04:49	19:11	19:57	20:56
N 58°	03:17	04:12	04:54	19:06	19:48	20:43
56°	03:29	04:19	04:59	19:01	19:41	20:31
54°	03:39	04:26	05:04	18:57	19:34	20:21
52°	03:48	04:32	05:08	18:53	19:28	20:12
50°	03:55	04:37	05:11	18:49	19:23	20:05
45°	04:11	04:48	05:19	18:42	19:12	19:49
N 40°	04:24	04:57	05:25	18:36	19:03	19:37
35°	04:34	05:05	05:31	18:30	18:56	19:27
30°	04:42	05:11	05:36	18:26	18:50	19:19
20°	04:55	05:21	05:44	18:17	18:40	19:06
N 10°	05:05	05:30	05:51	18:11	18:32	18:57
0°	05:12	05:37	05:57	18:04	18:25	18:49
S 10°	05:18	05:43	06:04	17:58	18:19	18:44
20°	05:23	05:48	06:11	17:51	18:13	18:39
30°	05:26	05:54	06:18	17:44	18:08	18:36
35°	05:28	05:57	06:23	17:39	18:05	18:34
40°	05:29	06:00	06:28	17:35	18:02	18:33
45°	05:30	06:04	06:33	17:29	17:59	18:33
S 50°	05:30	06:07	06:40	17:23	17:55	18:33
52°	05:30	06:09	06:43	17:20	17:54	18:33
54°	05:30	06:10	06:46	17:16	17:52	18:33
56°	05:29	06:12	06:50	17:13	17:50	18:33
58°	05:29	06:14	06:54	17:09	17:48	18:34
S 60°	05:28	06:16	06:59	17:04	17:46	18:35

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°						
N 70°						
68°						
66°						
64°			23:28			20:14
62°		22:06	· · ·		19:46	19:35
60°	21:24	22:53	· · ·	18:33	18:59	19:08
N 58°	22:04	23:23	· · ·	17:53	18:29	18:46
56°	22:31	23:46	· · ·	17:25	18:05	18:29
54°	22:53	· · ·	00:05	17:03	17:46	18:14
52°	23:11	· · ·	00:21	16:46	17:30	18:01
50°	23:26	· · ·	00:34	16:30	17:17	17:49
45°	23:56	· · ·	01:02	15:59	16:48	17:26
N 40°	· · ·	00:20	01:24	15		

August 31, 01, 02 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	339°39.9	157°37.4	N00°16.1	252°52.3	N23°17.6	261°39.8	N22°16.7	351°05.5	S07°16.7	Alpheratz	357°34.8	29°13.6
1	354°42.3	172°37.1	14.8	267°53.1	17.7	276°41.9	16.7	6°08.1	16.7	Ankaa	353°07.0	-42°10.1
2	9°44.8	187°36.8	13.5	282°53.9	17.8	291°44.1	16.7	21°10.8	16.8	Schedar	349°30.9	56°40.3
3	24°47.3	202°36.4	•• 12.2	297°54.7	•• 17.8	306°46.2	•• 16.8	36°13.4	•• 16.9	Diphda	348°47.3	-17°50.9
4	39°49.7	217°36.1	10.9	312°55.5	17.9	321°48.4	16.8	51°16.1	17.0	Achernar	335°20.0	-57°06.4
5	54°52.2	232°35.8	09.6	327°56.3	18.0	336°50.5	16.8	66°18.7	17.1	Hamal	327°51.3	23°34.8
6	69°54.7	247°35.5	N00°08.3	342°57.1	N23°18.1	351°52.7	N22°16.8	81°21.3	S07°17.1	Polaris	314°00.7	89°21.8
7	84°57.1	262°35.2	07.1	357°57.9	18.2	6°54.8	16.8	96°24.0	17.2	Acamar	315°11.8	-40°12.0
8	99°59.6	277°34.9	05.8	12°58.7	18.2	21°57.0	16.9	111°26.6	17.3	Menkar	314°06.3	4°11.3
9	115°02.1	292°34.6	•• 04.5	27°59.4	•• 18.3	36°59.1	•• 16.9	126°29.2	•• 17.4	Mirfak	308°28.5	49°56.8
10	130°04.5	307°34.2	03.2	43°00.2	18.4	52°01.3	16.9	141°31.9	17.4	Aldebaran	290°40.0	16°33.6
11	145°07.0	322°33.9	01.9	58°01.0	18.5	67°03.4	16.9	156°34.5	17.5	Rigel	281°04.2	-8°10.2
12	160°09.4	337°33.6	N00°00.6	73°01.8	N23°18.5	82°05.6	N22°17.0	171°37.2	S07°17.6	Capella	280°22.4	46°01.2
13	175°11.9	352°33.3	S00°00.7	88°02.6	18.6	97°07.7	17.0	186°39.8	17.7	Bellatrix	278°23.3	6°22.4
14	190°14.4	7°33.0	02.0	103°03.4	18.7	112°09.9	17.0	201°42.4	17.8	Elnath	278°02.3	28°37.7
15	205°16.8	22°32.7	•• 03.2	118°04.2	•• 18.8	127°12.0	•• 17.0	216°45.1	•• 17.8	Alnilam	275°38.1	-1°11.0
16	220°19.3	37°32.4	04.5	133°05.0	18.8	142°14.2	17.1	231°47.7	17.9	Betelgeuse	270°52.5	7°24.8
17	235°21.8	52°32.1	05.8	148°05.8	18.9	157°16.3	17.1	246°50.3	18.0	Canopus	263°52.8	-52°42.1
18	250°24.2	67°31.7	S00°07.1	163°06.6	N23°19.0	172°18.5	N22°17.1	261°53.0	S07°18.1	Sirius	258°26.7	-16°44.7
19	265°26.7	82°31.4	08.4	178°07.3	19.1	187°20.6	17.1	276°55.6	18.1	Adhara	255°06.3	-29°00.0
20	280°29.2	97°31.1	09.7	193°08.1	19.1	202°22.8	17.1	291°58.3	18.2	Procyon	244°51.4	5°09.8
21	295°31.6	112°30.8	•• 11.0	208°08.9	•• 19.2	217°25.0	•• 17.2	307°00.9	•• 18.3	Pollux	243°17.9	27°58.0
22	310°34.1	127°30.5	12.3	223°09.7	19.3	232°27.1	17.2	322°03.5	18.4	Avior	234°15.4	-59°35.0
23	325°36.6	142°30.2	13.6	238°10.5	19.4	247°29.3	17.2	337°06.2	18.4	Suhail	222°47.0	-43°31.7
Mer.pass. 01:21		ν -0.3' d -1.3' m -3.86		ν 0.8' d 0.1' m 0.71		ν 2.1' d 0.0' m -2.26		ν 2.6' d 0.1' m 0.61		Miaplacidus	221°39.3	-69°48.9
										Alphard	217°48.4	-8°45.8
										Regulus	207°35.1	11°50.9
										Dubhe	193°42.0	61°37.2
										Denebola	182°25.5	14°26.2
										Gienah	175°44.2	-17°40.6
										Acrux	173°01.1	-63°14.2
										Gacrux	171°52.5	-57°15.1
										Alioth	166°13.6	55°49.8
										Spica	158°22.8	-11°17.3
										Alkaid	152°52.5	49°11.6
										Hadar	148°36.8	-60°29.7
										Menkent	147°58.2	-36°29.5
										Arcturus	145°48.4	19°03.4
										Rigil Kent.	139°41.1	-60°56.4
										Kochab	137°20.2	74°03.5
										Zuben'ubi	136°56.5	-16°08.6
										Alphecca	126°04.1	26°38.1
										Antares	112°16.2	-26°29.2
										Atria	107°10.6	-69°04.5
										Sabik	102°03.1	-15°45.3
										Shaula	96°10.6	-37°07.4
										Rasalhague	95°58.7	12°32.7
										Eltanin	90°42.1	51°29.4
										Kaus Aust.	83°32.7	-34°22.4
										Vega	80°33.2	38°48.6
										Nunki	75°47.9	-26°16.0
										Altair	62°00.0	8°56.1
										Peacock	53°05.7	-56°39.4
										Deneb	49°25.6	45°22.2
										Enif	33°38.8	9°59.4
										Al Na'ir	27°32.8	-46°50.4
										Fomalhaut	15°14.5	-29°29.4
										Scheat	13°45.1	28°13.1
										Markab	13°29.9	15°20.4
										Aug 31 Sat	SHA	Mer.pass
										Venus	177°57.5	13:30
										Mars	273°12.5	07:08
										Jupiter	281°59.9	06:32
										Saturn	11°25.6	00:36
										Sep 01 Sun	SHA	Mer.pass
										Venus	176°50.8	13:30
										Mars	272°32.3	07:07
										Jupiter	281°52.4	06:29
										Saturn	11°29.8	00:31
										Sep 02 Mon	SHA	Mer.pass
										Venus	175°44.2	13:31
										Mars	271°52.3	07:06
										Jupiter	281°45.0	06:26
										Saturn	11°34.0	00:27

Sun				Moon				
Sat	GHA	Dec		GHA	ν	Dec	d	HP
0	179°55.3	N08°32.2		212°36.6	10.9'	N23°53.4	-8.7'	55.3'
1	194°55.5	31.3		227°06.5	11.0'	23°44.6	-8.8'	55.3'
2	209°55.7	30.4		241°36.6	11.1'	23°35.8	-9.0'	55.2'
3	224°55.9	· · 29.5		256°06.7	11.2'	23°26.8	-9.1'	55.2'
4	239°56.1	28.6		270°36.9	11.3'	23°17.8	-9.2'	55.2'
5	254°56.3	27.7		285°07.2	11.4'	23°08.6	-9.3'	55.2'
6	269°56.5	N08°26.8		299°37.5	11.5'	N22°59.3	-9.4'	55.2'
7	284°56.6	25.9		314°08.0	11.6'	22°50.0	-9.5'	55.2'
8	299°56.8	25.0		328°38.6	11.7'	22°40.5	-9.6'	55.1'
9	314°57.0	· · 24.1		343°09.3	11.8'	22°30.9	-9.7'	55.1'
10	329°57.2	23.2		357°40.0	11.8'	22°21.2	-9.8'	55.1'
11	344°57.4	22.3		12°10.9	11.9'	22°11.5	-9.9'	55.1'
12	359°57.6	N08°21.4		26°41.8	12.0'	N22°01.6	-10.0'	55.1'
13	14°57.8	20.5		41°12.8	12.1'	21°51.7	-10.1'	55.0'
14	29°58.0	19.5		55°43.9	12.2'	21°41.6	-10.1'	55.0'
15	44°58.2	· · 18.6		70°15.1	12.3'	21°31.5	-10.2'	55.0'
16	59°58.4	17.7		84°46.5	12.4'	21°21.2	-10.3'	55.0'
17	74°58.6	16.8		99°17.8	12.5'	21°10.9	-10.4'	55.0'
18	89°58.8	N08°15.9		113°49.3	12.6'	N21°00.5	-10.5'	55.0'
19	104°59.0	15.0		128°20.9	12.7'	20°50.0	-10.6'	54.9'
20	119°59.2	14.1		142°52.5	12.7'	20°39.4	-10.7'	54.9'
21	134°59.4	· · 13.2		157°24.3	12.8'	20°28.7	-10.8'	54.9'
22	149°59.6	12.3		171°56.1	12.9'	20°18.0	-10.8'	54.9'
23	164°59.8	11.4		186°28.0	13.0'	20°07.1	-10.9'	54.9'
SD = 15.8'			d = -0.9'			SD = 15.1'		
Sun	GHA	Dec		GHA	ν	Dec	d	HP
0	180°00.0	N08°10.5		201°00.1	13.1'	N19°56.2	-11.0'	54.9'
1	195°00.2	09.6		215°32.2	13.2'	19°45.2	-11.1'	54.8'
2	210°00.4	08.7		230°04.3	13.3'	19°34.1	-11.2'	54.8'
3	225°00.6	· · 07.8		244°36.6	13.3'	19°23.0	-11.2'	54.8'
4	240°00.8	06.8		259°08.9	13.4'	19°11.7	-11.3'	54.8'
5	255°01.0	05.9		273°41.4	13.5'	19°00.4	-11.4'	54.8'
6	270°01.2	N08°05.0		288°13.9	13.6'	N18°49.0	-11.5'	54.8'
7	285°01.4	04.1		302°46.5	13.7'	18°37.5	-11.5'	54.8'
8	300°01.6	03.2		317°19.2	13.8'	18°26.0	-11.6'	54.7'
9	315°01.8	· · 02.3		331°51.9	13.8'	18°14.4	-11.7'	54.7'
10	330°02.0	01.4		346°24.8	13.9'	18°02.7	-11.7'	54.7'
11	345°02.2	08°00.5		0°57.7	14.0'	17°51.0	-11.8'	54.7'
12	0°02.4	N07°59.6		15°30.7	14.1'	N17°39.2	-11.9'	54.7'
13	15°02.6	58.7		30°03.8	14.2'	17°27.3	-11.9'	54.7'
14	30°02.8	57.7		44°37.0	14.2'	17°15.3	-12.0'	54.7'
15	45°03.0	· · 56.8		59°10.2	14.3'	17°03.3	-12.1'	54.6'
16	60°03.2	55.9		73°43.5	14.4'	16°51.2	-12.1'	54.6'
17	75°03.4	55.0		88°16.9	14.5'	16°39.1	-12.2'	54.6'
18	90°03.6	N07°54.1		102°50.4	14.5'	N16°26.9	-12.3'	54.6'
19	105°03.8	53.2		117°23.9	14.6'	16°14.6	-12.3'	54.6'
20	120°04.0	52.3		131°57.5	14.7'	16°02.3	-12.4'	54.6'
21	135°04.2	· · 51.4		146°31.2	14.8'	15°49.9	-12.4'	54.6'
22	150°04.4	50.4		161°05.0	14.8'	15°37.5	-12.5'	54.5'
23	165°04.6	49.5		175°38.8	14.9'	15°25.0	-12.6'	54.5'
SD = 15.8'			d = -0.9'			SD = 15.0'		
Mon	GHA	Dec		GHA	ν	Dec	d	HP
0	180°04.8	N07°48.6		190°12.7	15.0'	N15°12.4	-12.6'	54.5'
1	195°05.0	47.7		204°46.7	15.0'	14°59.8	-12.7'	54.5'
2	210°05.2	46.8		219°20.7	15.1'	14°47.2	-12.7'	54.5'
3	225°05.4	· · 45.9		233°54.8	15.2'	14°34.4	-12.8'	54.5'
4	240°05.6	45.0		248°29.0	15.2'	14°21.7	-12.8'	54.5'
5	255°05.8	44.0		263°03.2	15.3'	14°08.9	-12.9'	54.5'
6	270°06.0	N07°43.1		277°37.5	15.4'	N13°56.0	-12.9'	54.4'
7	285°06.2	42.2		292°11.9	15.4'	13°43.1	-13.0'	54.4'
8	300°06.4	41.3		306°46.3	15.5'	13°30.1	-13.0'	54.4'
9	315°06.6	· · 40.4		321°20.8	15.6'	13°17.1	-13.1'	54.4'
10	330°06.8	39.5		335°55.4	15.6'	13°04.1	-13.1'	54.4'
11	345°07.0	38.6		350°30.0	15.7'	12°51.0	-13.1'	54.4'
12	0°07.2	N07°37.6		5°04.7	15.7'	N12°37.8	-13.2'	54.4'
13	15°07.4	36.7		19°39.4	15.8'	12°24.6	-13.2'	54.4'
14	30°07.6	35.8		34°14.2	15.8'	12°11.4	-13.3'	54.3'
15	45°07.8	· · 34.9		48°49.0	15.9'	11°58.1	-13.3'	54.3'
16	60°08.0	34.0		63°23.9	16.0'	11°44.8	-13.3'	54.3'
17	75°08.2	33.1		77°58.9	16.0'	11°31.5	-13.4'	54.3'
18	90°08.4	N07°32.1		92°33.9	16.1'	N11°18.1	-13.4'	54.3'
19	105°08.6	31.2		107°09.0	16.1'	11°04.7	-13.5'	54.3'
20	120°08.8	30.3		121°44.1	16.2'	10°51.2	-13.5'	54.3'
21	135°09.0	· · 29.4		136°19.3	16.2'	10°37.7	-13.5'	54.3'
22	150°09.2	28.5		150°54.5	16.3'	10°24.2	-13.6'	54.3'
23	165°09.5	27.5		165°29.8	16.3'	10°10.6	-13.6'	54.3'
SD = 15.8'			d = -0.9'			SD = 14.9'		

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	02:34	04:04	19:53	21:20	—:—
N 70°	—:—	03:02	04:17	19:40	20:53	23:28
68°	01:35	03:22	04:28	19:29	20:34	22:17
66°	02:12	03:38	04:37	19:21	20:19	21:42
64°	02:38	03:51	04:44	19:14	20:06	21:18
62°	02:57	04:02	04:50	19:08	19:56	21:00
60°	03:13	04:11	04:56	19:02	19:47	20:45
N 58°	03:25	04:19	05:01	18:57	19:39	20:32
56°	03:36	04:26	05:05	18:53	19:32	20:22
54°	03:46	04:32	05:09	18:50	19:27	20:12
52°	03:54	04:37	05:12	18:46	19:21	20:04
50°	04:01	04:42	05:16	18:43	19:17	19:57
45°	04:16	04:52	05:22	18:36	19:07	19:43
N 40°	04:27	05:00	05:28	18:31	18:59	19:32
35°	04:37	05:07	05:33	18:26	18:52	19:22
30°	04:44	05:13	05:37	18:22	18:46	19:15
20°	04:56	05:22	05:44	18:15	18:37	19:03
N 10°	05:05	05:30	05:51	18:09	18:30	18:55
0°	05:11	05:36	05:57	18:03	18:24	18:48
S 10°	05:17	05:41	06:02	17:58	18:19	18:43
20°	05:20	05:46	06:08	17:52	18:14	18:40
30°	05:23	05:51	06:15	17:45	18:09	18:37
35°	05:24	05:53	06:19	17:42	18:07	18:36
40°	05:24	05:56	06:23	17:37	18:05	18:36
45°	05:24	05:58	06:28	17:33	18:02	18:36
S 50°	05:24	06:01	06:34	17:27	17:59	18:37
52°	05:23	06:02	06:36	17:24	17:58	18:37
54°	05:23	06:04	06:39	17:21	17:57	18:38
56°	05:22	06:05	06:42	17:18	17:56	18:39
58°	05:21	06:06	06:46	17:15	17:54	18:40
S 60°	05:20	06:08	06:50	17:11	17:53	18:41

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°			01:46		21:14	20:16
N 70°			02:20		20:38	19:59
68°			02:44	20:58	20:12	19:46
66°			01:05	20:18	19:52	19:34
64°	· · ·	01:32	03:18	19:50	19:36	19:25
62°	00:06	01:53	03:30	19:28	19:22	19:17
60°	00:33	02:09	03:40	19:10	19:11	19:10
N 58°	00:53	02:23	03:50	18:56	19:01	19:04
56°	01:10	02:35	03:57	18:43	18:52	18:58
54°	01:25	02:46	04:04	18:32	18:44	18:53
52°	01:37	02:55	04:11	18:22	18:37	18:49
50°	01:48	03:03	04:16	18:13	18:31	18:45
45°	02:11	03:21	04:29	17:54	18:17	18:36
N 40°	02:29	03:35	04:39	17:39	18:06	18:28
35°	02:45	03:47	04:47	17:26	17:56	18:22
30°	02:58	03:57	04:55	17:14	17:47	18:16
20°	03:20	04:15	05:07	16:55	17:32	18:06
N 10°	03:40	04:31	05:19	16:37	17:19	17:57
0°	03:58	04:45	05:29	16:21	17:07	17:49
S 10°	04:16	05:00	05:39	16:05	16:54	17:41
20°	04:35	05:15	05:50	15:47	16:41	17:32
30°	04:57	05:32	06:03	15:27	16:25	17:21
35°	05:10	05:42	06:10	15:15	16:16	17:15
40°	05:25	05:54	06:18	15:01	16:06	17:09
45°	05:42	06:07	06:28	14:45	15:54	17:00
S 50°	06:04	06:24	06:39	14:24	15:39	16:51
52°	06:14	06:32	06:45	14:15	15:32	16:46
54°	06:26	06:40	06:50	14:04	15:24	16:41
56°	06:39	06:50	06:57	13:51	15:15	16:36
58°	06:54	07:00	07:04	13:36	15:05	16:30
S 60°	07:12	07:13	07:12	13:19	14:54	16:22

Day	Sun		Mer. Pass hh:mm	Moon		Age 27-29 9-1%
	Eqn.of Time	Mer. Pass hh:mm		Mer.Pass. Upper hh:mm	Lower hh:mm	
	00 ^h mm:ss					
31	00:19	00:09	12:00	10:10	22:33	
01	00:00	00:10	12:00	10:56	23:18	
02	00:19	00:29	12:00	11:39	—:—	

September 03, 04, 05 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	342° 37.3	157° 14.8	S01° 16.7	253° 49.7	N23° 22.6	264° 15.1	N22° 18.3	354° 15.5	S07° 22.2	Alpheratz	357° 34.7 29° 13.6
1	357° 39.8	172° 14.5	18.0	268° 50.5	22.6	279° 17.3	18.3	9° 18.1	22.3	Ankaa	353° 07.0 -42° 10.1
2	12° 42.2	187° 14.2	19.3	283° 51.3	22.7	294° 19.5	18.3	24° 20.7	22.4	Schedar	349° 30.9 56° 40.3
3	27° 44.7	202° 13.9	•• 20.6	298° 52.1	•• 22.8	309° 21.6	•• 18.3	39° 23.4	•• 22.5	Diphda	348° 47.3 -17° 50.9
4	42° 47.1	217° 13.6	21.9	313° 52.9	22.8	324° 23.8	18.4	54° 26.0	22.5	Achernar	335° 19.9 -57° 06.4
5	57° 49.6	232° 13.3	23.1	328° 53.7	22.9	339° 26.0	18.4	69° 28.7	22.6	Hamal	327° 51.3 23° 34.8
6	72° 52.1	247° 13.0	S01° 24.4	343° 54.5	N23° 22.9	354° 28.1	N22° 18.4	84° 31.3	S07° 22.7	Polaris	313° 59.4 89° 21.8
7	87° 54.5	262° 12.7	25.7	358° 55.3	23.0	9° 30.3	18.4	99° 33.9	22.8	Acamar	315° 11.8 -40° 12.0
8	102° 57.0	277° 12.3	27.0	13° 56.1	23.0	24° 32.5	18.4	114° 36.6	22.8	Menkar	314° 06.3 4° 11.3
9	117° 59.5	292° 12.0	•• 28.3	28° 56.9	•• 23.1	39° 34.6	•• 18.5	129° 39.2	•• 22.9	Mirfak	308° 28.5 49° 56.8
10	133° 01.9	307° 11.7	29.6	43° 57.7	23.2	54° 36.8	18.5	144° 41.9	23.0	Aldebaran	290° 39.9 16° 33.6
11	148° 04.4	322° 11.4	30.9	58° 58.5	23.2	69° 39.0	18.5	159° 44.5	23.1	Rigel	281° 04.2 -8° 10.2
12	163° 06.9	337° 11.1	S01° 32.2	73° 59.3	N23° 23.3	84° 41.2	N22° 18.5	174° 47.1	S07° 23.2	Capella	280° 22.4 46° 01.2
13	178° 09.3	352° 10.8	33.5	89° 00.1	23.3	99° 43.3	18.5	189° 49.8	23.2	Bellatrix	278° 23.2 6° 22.5
14	193° 11.8	7° 10.5	34.7	104° 01.0	23.4	114° 45.5	18.6	204° 52.4	23.3	Elnath	278° 02.3 28° 37.7
15	208° 14.3	22° 10.1	•• 36.0	119° 01.8	•• 23.4	129° 47.7	•• 18.6	219° 55.1	•• 23.4	Alnilam	275° 38.1 -1° 11.0
16	223° 16.7	37° 09.8	37.3	134° 02.6	23.5	144° 49.8	18.6	234° 57.7	23.5	Betelgeuse	270° 52.5 7° 24.8
17	238° 19.2	52° 09.5	38.6	149° 03.4	23.5	159° 52.0	18.6	250° 00.3	23.5	Canopus	263° 52.7 -52° 42.1
18	253° 21.6	67° 09.2	S01° 39.9	164° 04.2	N23° 23.6	174° 54.2	N22° 18.6	265° 03.0	S07° 23.6	Sirius	258° 26.7 -16° 44.7
19	268° 24.1	82° 08.9	41.2	179° 05.0	23.7	189° 56.4	18.7	280° 05.6	23.7	Adhara	255° 06.3 -29° 00.0
20	283° 26.6	97° 08.6	42.5	194° 05.8	23.7	204° 58.5	18.7	295° 08.3	23.8	Procyon	244° 51.4 5° 09.9
21	298° 29.0	112° 08.3	•• 43.8	209° 06.6	•• 23.8	220° 00.7	•• 18.7	310° 10.9	•• 23.8	Pollux	243° 17.9 27° 58.0
22	313° 31.5	127° 08.0	45.1	224° 07.4	23.8	235° 02.9	18.7	325° 13.5	23.9	Avior	234° 15.4 -59° 35.0
23	328° 34.0	142° 07.6	46.3	239° 08.2	23.9	250° 05.1	18.7	340° 16.2	24.0	Suhail	222° 46.9 -43° 31.7
Mer.pass. 01:09		ν -0.3' d 1.3' m -3.86		ν 0.8' d 0.1' m 0.69		ν 2.2' d 0.0' m -2.27		ν 2.6' d 0.1' m 0.60			

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	343° 36.4	157° 07.3	S01° 47.6	254° 09.1	N23° 23.9	265° 07.2	N22° 18.8	355° 18.8	S07° 24.1
1	358° 38.9	172° 07.0	48.9	269° 09.9	24.0	280° 09.4	18.8	10° 21.5	24.2
2	13° 41.4	187° 06.7	50.2	284° 10.7	24.0	295° 11.6	18.8	25° 24.1	24.2
3	28° 43.8	202° 06.4	•• 51.5	299° 11.5	•• 24.1	310° 13.8	•• 18.8	40° 26.7	•• 24.3
4	43° 46.3	217° 06.1	52.8	314° 12.3	24.1	325° 15.9	18.9	55° 29.4	24.4
5	58° 48.7	232° 05.8	54.1	329° 13.1	24.2	340° 18.1	18.9	70° 32.0	24.5
6	73° 51.2	247° 05.4	S01° 55.4	344° 13.9	N23° 24.2	355° 20.3	N22° 18.9	85° 34.7	S07° 24.5
7	88° 53.7	262° 05.1	56.6	359° 14.7	24.3	10° 22.5	18.9	100° 37.3	24.6
8	103° 56.1	277° 04.8	57.9	14° 15.5	24.3	25° 24.6	18.9	115° 39.9	24.7
9	118° 58.6	292° 04.5	01° 59.2	29° 16.4	•• 24.4	40° 26.8	•• 19.0	130° 42.6	•• 24.8
10	134° 01.1	307° 04.2	02° 00.5	44° 17.2	24.4	55° 29.0	19.0	145° 45.2	24.9
11	149° 03.5	322° 03.9	01.8	59° 18.0	24.5	70° 31.2	19.0	160° 47.9	24.9
12	164° 06.0	337° 03.6	S02° 03.1	74° 18.8	N23° 24.5	85° 33.4	N22° 19.0	175° 50.5	S07° 25.0
13	179° 08.5	352° 03.3	04.4	89° 19.6	24.6	100° 35.5	19.0	190° 53.1	25.1
14	194° 10.9	7° 02.9	05.7	104° 20.4	24.6	115° 37.7	19.0	205° 55.8	25.2
15	209° 13.4	22° 02.6	•• 07.0	119° 21.2	•• 24.7	130° 39.9	•• 19.1	220° 58.4	•• 25.2
16	224° 15.9	37° 02.3	08.2	134° 22.1	24.7	145° 42.1	19.1	236° 01.1	25.3
17	239° 18.3	52° 02.0	09.5	149° 22.9	24.8	160° 44.2	19.1	251° 03.7	25.4
18	254° 20.8	67° 01.7	S02° 10.8	164° 23.7	N23° 24.8	175° 46.4	N22° 19.1	266° 06.3	S07° 25.5
19	269° 23.2	82° 01.4	12.1	179° 24.5	24.9	190° 48.6	19.1	281° 09.0	25.5
20	284° 25.7	97° 01.1	13.4	194° 25.3	24.9	205° 50.8	19.2	296° 11.6	25.6
21	299° 28.2	112° 00.7	•• 14.7	209° 26.1	•• 25.0	220° 53.0	•• 19.2	311° 14.3	•• 25.7
22	314° 30.6	127° 00.4	16.0	224° 27.0	25.0	235° 55.1	19.2	326° 16.9	25.8
23	329° 33.1	142° 00.1	17.3	239° 27.8	25.0	250° 57.3	19.2	341° 19.5	25.9
Mer.pass. 01:05		ν -0.3' d 1.3' m -3.86		ν 0.8' d 0.1' m 0.68		ν 2.2' d 0.0' m -2.28		ν 2.6' d 0.1' m 0.59	

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	344° 35.6	156° 59.8	S02° 18.5	254° 28.6	N23° 25.1	265° 59.5	N22° 19.2	356° 22.2	S07° 25.9
1	359° 38.0	171° 59.5	19.8	269° 29.4	25.1	281° 01.7	19.3	11° 24.8	26.0
2	14° 40.5	186° 59.2	21.1	284° 30.2	25.2	296° 03.9	19.3	26° 27.5	26.1
3	29° 43.0	201° 58.9	•• 22.4	299° 31.1	•• 25.2	311° 06.1	•• 19.3	41° 30.1	•• 26.2
4	44° 45.4	216° 58.5	23.7	314° 31.9	25.3	326° 08.2	19.3	56° 32.7	26.2
5	59° 47.9	231° 58.2	25.0	329° 32.7	25.3	341° 10.4	19.3	71° 35.4	26.3
6	74° 50.3	246° 57.9	S02° 26.3	344° 33.5	N23° 25.4	356° 12.6	N22° 19.4	86° 38.0	S07° 26.4
7	89° 52.8	261° 57.6	27.6	359° 34.3	25.4	11° 14.8	19.4	101° 40.7	26.5
8	104° 55.3	276° 57.3	28.8	14° 35.1	25.5	26° 17.0	19.4	116° 43.3	26.5
9	119° 57.7	291° 57.0	•• 30.1	29° 36.0	•• 25.5	41° 19.2	•• 19.4	131° 45.9	•• 26.6
10	135° 00.2	306° 56.7	31.4	44° 36.8	25.5	56° 21.3	19.4	146° 48.6	26.7
11	150° 02.7	321° 56.3	32.7	59° 37.6	25.6	71° 23.5	19.5	161° 51.2	26.8
12	165° 05.1	336° 56.0	S02° 34.0	74° 38.4	N23° 25.6	86° 25.7	N22° 19.5	176° 53.9	S07° 26.9
13	180° 07.6	351° 55.7	35.3	89° 39.3	25.7	101° 27.9	19.5	191° 56.5	26.9
14	195° 10.1	6° 55.4	36.6	104° 40.1	25.7	116° 30.1	19.5	206° 59.1	27.0
15	210° 12.5	21° 55.1	•• 37.8	119° 40.9	•• 25.7	131° 32.3	•• 19.5	222° 01.8	•• 27.1
16	225° 15.0	36° 54.8	39.1	134° 41.7	25.8	146° 34.5	19.6	237° 04.4	27.2
17	240° 17.5	51° 54.5	40.4	149° 42.5	25.8	161° 36.6	19.6	252° 07.1	27.2
18	255° 19.9	66° 54.1	S02° 41.7	164° 43.4	N23° 25.9	176° 38.8	N22° 19.6	267° 09.7	S07° 27.3
19	270° 22.4	81° 53.8	43.0	179° 44.2	25.9	191° 41.0	19.6	282° 12.4	27.4
20	285° 24.8	96° 53.5	44.3	194° 45.0	26.0	206° 43.2	19.6	297° 15.0	27.5
21	300° 27.3	111° 53.2	•• 45.6	209° 45.8	•• 26.0	221° 45.4	•• 19.6	312° 17.6	•• 27.6
22	315° 29.8	126° 52.9	46.9	224° 46.7	26.0	236° 47.6	19.7	327° 20.3	27.6
23	330° 32.2	141° 52.6	48.1	239° 47.5	26.1	251° 49.8	19.7	342° 22.9	27.7
Mer.pass. 01:01		ν -0.3' d 1.3' m -3.86		ν 0.8' d 0.0' m 0.67		ν 2.2' d 0.0' m -2.29		ν 2.6' d 0.1' m 0.59	

Sep 03 Tue	SHA	Mer.pass
Venus	174° 37.6	13:31
Mars	271° 12.4	07:04
Jupiter	281° 37.8	06:22
Saturn	11° 38.2	00:23

Sep 04 Wed	SHA	Mer.pass
Venus	173° 30.9	13:32
Mars	270° 32.6	07:03
Jupiter	281° 30.8	06:19
Saturn	11° 42.4	00:19

Sep 05 Thu	SHA	Mer.pass
Venus	172° 24.2	13:32
Mars	269° 53.0	07:02
Jupiter	281° 23.9	06:15
Saturn	11° 46.6	00:14

Horizontal parallax	
Venus:	0.1
Mars:	0.1

h Sun Moon						
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	180°09.7	N07°26.6	180°05.1	16.4'	N09°57.0	-13.6' 54.2'
1	195°09.9	25.7	194°40.5	16.4'	09°43.4	-13.7' 54.2'
2	210°10.1	24.8	209°15.9	16.5'	09°29.8	-13.7' 54.2'
3	225°10.3	· · 23.9	223°51.3	16.5'	09°16.1	-13.7' 54.2'
4	240°10.5	23.0	238°26.8	16.5'	09°02.4	-13.7' 54.2'
5	255°10.7	22.0	253°02.4	16.6'	08°48.6	-13.8' 54.2'
6	270°10.9	N07°21.1	267°38.0	16.6'	N08°34.9	-13.8' 54.2'
7	285°11.1	20.2	282°13.6	16.7'	08°21.1	-13.8' 54.2'
8	300°11.3	19.3	296°49.3	16.7'	08°07.2	-13.8' 54.2'
9	315°11.5	· · 18.4	311°25.0	16.8'	07°53.4	-13.9' 54.2'
10	330°11.7	17.4	326°00.8	16.8'	07°39.5	-13.9' 54.1'
11	345°11.9	16.5	340°36.5	16.8'	07°25.6	-13.9' 54.1'
12	0°12.1	N07°15.6	355°12.4	16.9'	N07°11.7	-13.9' 54.1'
13	15°12.3	14.7	9°48.2	16.9'	06°57.8	-14.0' 54.1'
14	30°12.5	13.7	24°24.1	16.9'	06°43.8	-14.0' 54.1'
15	45°12.7	· · 12.8	39°00.1	17.0'	06°29.8	-14.0' 54.1'
16	60°12.9	11.9	53°36.0	17.0'	06°15.8	-14.0' 54.1'
17	75°13.1	11.0	68°12.0	17.0'	06°01.8	-14.0' 54.1'
18	90°13.3	N07°10.1	82°48.0	17.1'	N05°47.8	-14.0' 54.1'
19	105°13.6	09.1	97°24.1	17.1'	05°33.7	-14.1' 54.1'
20	120°13.8	08.2	112°00.2	17.1'	05°19.7	-14.1' 54.1'
21	135°14.0	· · 07.3	126°36.3	17.1'	05°05.6	-14.1' 54.1'
22	150°14.2	06.4	141°12.4	17.2'	04°51.5	-14.1' 54.1'
23	165°14.4	05.4	155°48.6	17.2'	04°37.4	-14.1' 54.0'
SD = 15.8' d = -0.9'			SD = 14.8'			
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	180°14.6	N07°04.5	170°24.8	17.2'	N04°23.3	-14.1' 54.0'
1	195°14.8	03.6	185°01.0	17.2'	04°09.1	-14.1' 54.0'
2	210°15.0	02.7	199°37.2	17.3'	03°55.0	-14.2' 54.0'
3	225°15.2	· · 01.7	214°13.5	17.3'	03°40.9	-14.2' 54.0'
4	240°15.4	07°00.8	228°49.8	17.3'	03°26.7	-14.2' 54.0'
5	255°15.6	06°59.9	243°26.1	17.3'	03°12.5	-14.2' 54.0'
6	270°15.8	N06°59.0	258°02.4	17.3'	N02°58.3	-14.2' 54.0'
7	285°16.0	58.1	272°38.7	17.4'	02°44.2	-14.2' 54.0'
8	300°16.2	57.1	287°15.1	17.4'	02°30.0	-14.2' 54.0'
9	315°16.4	· · 56.2	301°51.4	17.4'	02°15.8	-14.2' 54.0'
10	330°16.7	55.3	316°27.8	17.4'	02°01.6	-14.2' 54.0'
11	345°16.9	54.3	331°04.2	17.4'	01°47.4	-14.2' 54.0'
12	0°17.1	N06°53.4	345°40.6	17.4'	N01°33.2	-14.2' 54.0'
13	15°17.3	52.5	0°17.0	17.4'	01°19.0	-14.2' 54.0'
14	30°17.5	51.6	14°53.5	17.4'	01°04.7	-14.2' 54.0'
15	45°17.7	· · 50.6	29°29.9	17.4'	00°50.5	-14.2' 54.0'
16	60°17.9	49.7	44°06.4	17.5'	00°36.3	-14.2' 54.0'
17	75°18.1	48.8	58°42.8	17.5'	00°22.1	-14.2' 54.0'
18	90°18.3	N06°47.9	73°19.3	17.5'	N00°07.9	-14.2' 54.0'
19	105°18.5	46.9	87°55.7	17.5'	S00°06.3	14.2' 53.9'
20	120°18.7	46.0	102°32.2	17.5'	00°20.5	14.2' 53.9'
21	135°18.9	· · 45.1	117°08.7	17.5'	00°34.7	14.2' 53.9'
22	150°19.2	44.2	131°45.2	17.5'	00°48.9	14.2' 53.9'
23	165°19.4	43.2	146°21.6	17.5'	01°03.1	14.2' 53.9'
SD = 15.9' d = -0.9'			SD = 14.7'			
Thu	GHA	Dec	GHA	ν	Dec	d HP
0	180°19.6	N06°42.3	160°58.1	17.5'	S01°17.3	14.2' 53.9'
1	195°19.8	41.4	175°34.6	17.5'	01°31.5	14.2' 53.9'
2	210°20.0	40.4	190°11.1	17.5'	01°45.7	14.2' 53.9'
3	225°20.2	· · 39.5	204°47.6	17.5'	01°59.8	14.2' 53.9'
4	240°20.4	38.6	219°24.0	17.5'	02°14.0	14.1' 53.9'
5	255°20.6	37.7	234°00.5	17.5'	02°28.1	14.1' 53.9'
6	270°20.8	N06°36.7	248°36.9	17.5'	S02°42.3	14.1' 53.9'
7	285°21.0	35.8	263°13.4	17.4'	02°56.4	14.1' 53.9'
8	300°21.2	34.9	277°49.8	17.4'	03°10.5	14.1' 53.9'
9	315°21.5	· · 33.9	292°26.3	17.4'	03°24.6	14.1' 53.9'
10	330°21.7	33.0	307°02.7	17.4'	03°38.7	14.1' 53.9'
11	345°21.9	32.1	321°39.1	17.4'	03°52.8	14.1' 53.9'
12	0°22.1	N06°31.1	336°15.5	17.4'	S04°06.8	14.0' 53.9'
13	15°22.3	30.2	350°51.9	17.4'	04°20.9	14.0' 53.9'
14	30°22.5	29.3	5°28.3	17.4'	04°34.9	14.0' 53.9'
15	45°22.7	· · 28.4	20°04.7	17.3'	04°48.9	14.0' 53.9'
16	60°22.9	27.4	34°41.0	17.3'	05°02.9	14.0' 53.9'
17	75°23.1	26.5	49°17.3	17.3'	05°16.9	14.0' 53.9'
18	90°23.4	N06°25.6	63°53.6	17.3'	S05°30.9	13.9' 53.9'
19	105°23.6	24.6	78°29.9	17.3'	05°44.8	13.9' 53.9'
20	120°23.8	23.7	93°06.2	17.3'	05°58.7	13.9' 53.9'
21	135°24.0	· · 22.8	107°42.5	17.2'	06°12.6	13.9' 53.9'
22	150°24.2	21.8	122°18.7	17.2'	06°26.5	13.9' 53.9'
23	165°24.4	20.9	136°54.9	17.2'	06°40.3	13.8' 53.9'
SD = 15.9' d = -0.9'			SD = 14.7'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	02:55	04:18	19:36	20:58	—:—
N 70°	01:10	03:18	04:30	19:25	20:36	22:36
68°	01:59	03:36	04:39	19:17	20:19	21:52
66°	02:29	03:50	04:46	19:09	20:05	21:24
64°	02:51	04:01	04:53	19:03	19:54	21:03
62°	03:08	04:11	04:58	18:58	19:45	20:47
60°	03:22	04:19	05:03	18:53	19:37	20:33
N 58°	03:34	04:26	05:07	18:49	19:30	20:22
56°	03:44	04:32	05:11	18:46	19:24	20:12
54°	03:52	04:37	05:14	18:42	19:19	20:04
52°	04:00	04:42	05:17	18:39	19:14	19:57
50°	04:06	04:47	05:20	18:37	19:10	19:50
45°	04:20	04:56	05:26	18:31	19:01	19:37
N 40°	04:30	05:03	05:31	18:26	18:54	19:26
35°	04:39	05:09	05:35	18:22	18:48	19:18
30°	04:46	05:15	05:39	18:18	18:43	19:11
20°	04:57	05:23	05:45	18:12	18:35	19:00
N 10°	05:05	05:29	05:51	18:07	18:28	18:53
0°	05:11	05:35	05:56	18:02	18:23	18:47
S 10°	05:15	05:39	06:00	17:57	18:18	18:43
20°	05:18	05:43	06:06	17:52	18:15	18:40
30°	05:20	05:47	06:11	17:47	18:11	18:39
35°	05:20	05:49	06:15	17:44	18:09	18:38
40°	05:20	05:51	06:18	17:40	18:07	18:39
45°	05:19	05:53	06:22	17:36	18:06	18:40
S 50°	05:18	05:55	06:27	17:31	18:04	18:41
52°	05:17	05:56	06:29	17:29	18:03	18:42
54°	05:16	05:57	06:32	17:27	18:02	18:43
56°	05:15	05:57	06:35	17:24	18:01	18:44
58°	05:13	05:58	06:38	17:21	18:01	18:46
S 60°	05:11	05:59	06:41	17:18	18:00	18:48

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	04:11	06:13	08:10	19:38	19:06	18:33
N 70°	04:26	06:17	08:05	19:31	19:06	18:41
68°	04:37	06:21	08:01	19:25	19:06	18:48
66°	04:46	06:23	07:58	19:20	19:06	18:53
64°	04:54	06:26	07:55	19:16	19:07	18:58
62°	05:01	06:28	07:53	19:12	19:07	19:02
60°	05:07	06:29	07:51	19:09	19:07	19:05
N 58°	05:12	06:31	07:49	19:06	19:07	19:08
56°	05:16	06:32	07:48	19:03	19:07	19:11
54°	05:20	06:34	07:46	19:01	19:07	19:14
52°	05:24	06:35	07:45	18:58	19:07	19:16
50°	05:27	06:36	07:44	18:56	19:07	19:18
45°	05:34	06:38	07:41	18:52	19:07	19:23
N 40°	05:40	06:40	07:39	18:49	19:08	19:27
35°	05:45	06:41	07:37	18:45	19:08	19:30
30°	05:50	06:43	07:36	18:43	19:08	19:33
20°	05:57	06:45	07:33	18:38	19:08	19:38
N 10°	06:04	06:47	07:30	18:33	19:08	19:43
0°	06:10	06:49	07:28	18:29	19:08	19:47
S 10°	06:16	06:52	07:26	18:25	19:08	19:51
20°	06:23	06:54	07:24	18:21	19:08	19:56
30°	06:30	06:56	07:21	18:15	19:08	20:01
35°	06:35	06:57	07:20	18:13	19:08	20:04
40°	06:40	06:59	07:18	18:09	19:09	20:08
45°	06:45	07:01	07:16	18:05	19:09	20:12
S 50°	06:52	07:03	07:14	18:00	19:09	20:16
52°	06:55	07:04	07:13	17:58	19:09	20:19
54°	06:58	07:05	07:12	17:56	19:09	20:21
56°	07:02	07:06	07:10	17:53	19:09	20:24
58°	07:06	07:08	07:09	17:50	19:09	20:27
S 60°	07:11	07:09	07:07	17:47	19:09	20:30

Day	Sun		Mer. Pass hh:mm	Moon		Age 0-2 0-3%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
03	00:39	00:48	11:59	12:20	00:00	
04	00:58	01:08	11:59	12:59	00:39	
05	01:18	01:28	11:59	13:38	01:18	

September 06, 07, 08 UT (Fri., Sat., Sun.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	345°34.7	156°52.2	S02°49.4	254°48.3	N23°26.1	266°52.0	N22°19.7	357°25.6	S07°27.8	Alpheratz	357°34.7	29°13.7
1	0°37.2	171°51.9	50.7	269°49.1	26.2	281°54.1	19.7	12°28.2	27.9	Ankaa	353°07.0	-42°10.1
2	15°39.6	186°51.6	52.0	284°50.0	26.2	296°56.3	19.7	27°30.8	27.9	Schedar	349°30.9	56°40.3
3	30°42.1	201°51.3	•• 53.3	299°50.8	•• 26.2	311°58.5	•• 19.8	42°33.5	•• 28.0	Diphda	348°47.3	-17°50.9
4	45°44.6	216°51.0	54.6	314°51.6	26.3	327°00.7	19.8	57°36.1	28.1	Achernar	335°19.9	-57°06.4
5	60°47.0	231°50.7	55.9	329°52.4	26.3	342°02.9	19.8	72°38.8	28.2	Hamal	327°51.3	23°34.8
6	75°49.5	246°50.4	S02°57.1	344°53.3	N23°26.3	357°05.1	N22°19.8	87°41.4	S07°28.2	Polaris	313°58.0	89°21.8
7	90°51.9	261°50.0	58.4	359°54.1	26.4	12°07.3	19.8	102°44.0	28.3	Acamar	315°11.8	-40°12.0
8	105°54.4	276°49.7	02°59.7	14°54.9	26.4	27°09.5	19.9	117°46.7	28.4	Menkar	314°06.3	4°11.3
9	120°56.9	291°49.4	03°01.0	29°55.7	•• 26.5	42°11.7	•• 19.9	132°49.3	•• 28.5	Mirfak	308°28.5	49°56.8
10	135°59.3	306°49.1	02.3	44°56.6	26.5	57°13.9	19.9	147°52.0	28.6	Aldebaran	290°39.9	16°33.6
11	151°01.8	321°48.8	03.6	59°57.4	26.5	72°16.0	19.9	162°54.6	28.6	Rigel	281°04.2	-8°10.2
12	166°04.3	336°48.5	S03°04.8	74°58.2	N23°26.6	87°18.2	N22°19.9	177°57.2	S07°28.7	Capella	280°22.4	46°01.2
13	181°06.7	351°48.1	06.1	89°59.0	26.6	102°20.4	19.9	192°59.9	28.8	Bellatrix	278°23.2	6°22.5
14	196°09.2	6°47.8	07.4	104°59.9	26.6	117°22.6	20.0	208°02.5	28.9	Elnath	278°02.3	28°37.7
15	211°11.7	21°47.5	•• 08.7	120°00.7	•• 26.7	132°24.8	•• 20.0	223°05.2	•• 28.9	Alnilam	275°38.1	-1°11.0
16	226°14.1	36°47.2	10.0	135°01.5	26.7	147°27.0	20.0	238°07.8	29.0	Betelgeuse	270°52.5	7°24.8
17	241°16.6	51°46.9	11.3	150°02.4	26.7	162°29.2	20.0	253°10.4	29.1	Canopus	263°52.7	-52°42.1
18	256°19.1	66°46.6	S03°12.6	165°03.2	N23°26.8	177°31.4	N22°20.0	268°13.1	S07°29.2	Sirius	258°26.6	-16°44.7
19	271°21.5	81°46.2	13.8	180°04.0	26.8	192°33.6	20.1	283°15.7	29.2	Adhara	255°06.3	-29°00.0
20	286°24.0	96°45.9	15.1	195°04.8	26.8	207°35.8	20.1	298°18.4	29.3	Procyon	244°51.3	5°09.9
21	301°26.4	111°45.6	•• 16.4	210°05.7	•• 26.9	222°38.0	•• 20.1	313°21.0	•• 29.4	Pollux	243°17.9	27°58.0
22	316°28.9	126°45.3	17.7	225°06.5	26.9	237°40.2	20.1	328°23.7	29.5	Avior	234°15.4	-59°35.0
23	331°31.4	141°45.0	19.0	240°07.3	26.9	252°42.4	20.1	343°26.3	29.6	Suhail	222°46.9	-43°31.7
Mer.pass. 00:58		ν -0.3' d 1.3' m -3.86		ν 0.8' d 0.0' m 0.67		ν 2.2' d 0.0' m -2.29		ν 2.6' d 0.1' m 0.58				

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	346°33.8	156°44.7	S03°20.3	255°08.2	N23°27.0	267°44.6	N22°20.1	358°28.9	S07°29.6	Regulus	207°35.1	11°50.9
1	1°36.3	171°44.3	21.5	270°09.0	27.0	282°46.8	20.2	13°31.6	29.7	Dubhe	193°42.0	61°37.2
2	16°38.8	186°44.0	22.8	285°09.8	27.0	297°49.0	20.2	28°34.2	29.8	Denebola	182°25.5	14°26.2
3	31°41.2	201°43.7	•• 24.1	300°10.7	•• 27.1	312°51.2	•• 20.2	43°36.9	•• 29.9	Gienah	175°44.2	-17°40.6
4	46°43.7	216°43.4	25.4	315°11.5	27.1	327°53.4	20.2	58°39.5	29.9	Acrux	173°01.1	-63°14.2
5	61°46.2	231°43.1	26.7	330°12.3	27.1	342°55.5	20.2	73°42.1	30.0	Gacrux	171°52.6	-57°15.1
6	76°48.6	246°42.8	S03°28.0	345°13.2	N23°27.2	357°57.7	N22°20.3	88°44.8	S07°30.1	Alioth	166°13.7	55°49.7
7	91°51.1	261°42.4	29.2	0°14.0	27.2	12°59.9	20.3	103°47.4	30.2	Spica	158°22.9	-11°17.3
8	106°53.5	276°42.1	30.5	15°14.8	27.2	28°02.1	20.3	118°50.1	30.2	Alkaid	152°52.5	49°11.6
9	121°56.0	291°41.8	•• 31.8	30°15.7	•• 27.3	43°04.3	•• 20.3	133°52.7	•• 30.3	Hadar	148°36.9	-60°29.7
10	136°58.5	306°41.5	33.1	45°16.5	27.3	58°06.5	20.3	148°55.3	30.4	Menkent	147°58.2	-36°29.5
11	152°00.9	321°41.2	34.4	60°17.3	27.3	73°08.7	20.3	163°58.0	30.5	Arcturus	145°48.4	19°03.4
12	167°03.4	336°40.8	S03°35.7	75°18.2	N23°27.4	88°10.9	N22°20.4	179°00.6	S07°30.6	Rigil Kent.	139°41.1	-60°56.4
13	182°05.9	351°40.5	36.9	90°19.0	27.4	103°13.1	20.4	194°03.3	30.6	Kochab	137°20.3	74°03.5
14	197°08.3	6°40.2	38.2	105°19.8	27.4	118°15.3	20.4	209°05.9	30.7	Zuben'ubi	136°56.5	-16°08.6
15	212°10.8	21°39.9	•• 39.5	120°20.7	•• 27.4	133°17.5	•• 20.4	224°08.6	•• 30.8	Alphecca	126°04.1	26°38.1
16	227°13.3	36°39.6	40.8	135°21.5	27.5	148°19.7	20.4	239°11.2	30.9	Antares	112°16.2	-26°29.2
17	242°15.7	51°39.3	42.1	150°22.3	27.5	163°21.9	20.4	254°13.8	30.9	Atria	107°10.7	-69°04.6
18	257°18.2	66°38.9	S03°43.4	165°23.2	N23°27.5	178°24.1	N22°20.5	269°16.5	S07°31.0	Sabik	102°03.1	-15°45.3
19	272°20.7	81°38.6	44.6	180°24.0	27.6	193°26.3	20.5	284°19.1	31.1	Shaula	96°10.7	-37°07.4
20	287°23.1	96°38.3	45.9	195°24.8	27.6	208°28.5	20.5	299°21.8	31.2	Rasalhague	95°58.8	12°32.7
21	302°25.6	111°38.0	•• 47.2	210°25.7	•• 27.6	223°30.7	•• 20.5	314°24.4	•• 31.2	Eltanin	90°42.2	51°29.4
22	317°28.0	126°37.7	48.5	225°26.5	27.6	238°32.9	20.5	329°27.0	31.3	Kaus Aust.	83°32.7	-34°22.4
23	332°30.5	141°37.3	49.8	240°27.4	27.7	253°35.1	20.5	344°29.7	31.4	Vega	80°33.2	38°48.6
Mer.pass. 00:54		ν -0.3' d 1.3' m -3.86		ν 0.8' d 0.0' m 0.66		ν 2.2' d 0.0' m -2.30		ν 2.6' d 0.1' m 0.58		Nunki	75°47.9	-26°16.0

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	347°33.0	156°37.0	S03°51.0	255°28.2	N23°27.7	268°37.4	N22°20.6	359°32.3	S07°31.5	Scheat	13°45.1	28°13.1
1	2°35.4	171°36.7	52.3	270°29.0	27.7	283°39.6	20.6	14°35.0	31.6	Markab	13°29.9	15°20.4
2	17°37.9	186°36.4	53.6	285°29.9	27.7	298°41.8	20.6	29°37.6	31.6	Sep 06 FriSHA Mer.pass Venus171°17.513:33 Mars269°13.607:00 Jupiter281°17.306:12 Saturn11°50.900:10		
3	32°40.4	201°36.1	•• 54.9	300°30.7	•• 27.8	313°44.0	•• 20.6	44°40.2	•• 31.7			
4	47°42.8	216°35.7	56.2	315°31.6	27.8	328°46.2	20.6	59°42.9	31.8			
5	62°45.3	231°35.4	57.5	330°32.4	27.8	343°48.4	20.7	74°45.5	31.9			
6	77°47.8	246°35.1	S03°58.7	345°33.2	N23°27.8	358°50.6	N22°20.7	89°48.2	S07°31.9	Sep 07 SatSHA Mer.pass Venus170°10.813:33 Mars268°34.306:59 Jupiter281°10.706:08 Saturn11°55.100:06		
7	92°50.2	261°34.8	04°00.0	0°34.1	27.9	13°52.8	20.7	104°50.8	32.0			
8	107°52.7	276°34.5	01.3	15°34.9	27.9	28°55.0	20.7	119°53.5	32.1			
9	122°55.2	291°34.1	•• 02.6	30°35.8	•• 27.9	43°57.2	•• 20.7	134°56.1	•• 32.2			
10	137°57.6	306°33.8	03.9	45°36.6	27.9	58°59.4	20.7	149°58.7	32.2	Sep 08 SunSHA Mer.pass Venus169°04.013:34 Mars267°55.206:58 Jupiter281°04.406:05 Saturn11°59.300:02		
11	153°00.1	321°33.5	05.1	60°37.4	28.0	74°01.6	20.8	165°01.4	32.3			
12	168°02.5	336°33.2	S04°06.4	75°38.3	N23°28.0	89°03.8	N22°20.8	180°04.0	S07°32.4			
13	183°05.0	351°32.9	07.7	90°39.1	28.0	104°06.0	20.8	195°06.7	32.5			
14	198°07.5	6°32.5	09.0	105°40.0	28.0	119°08.2	20.8	210°09.3	32.5	Horizontal parallax Venus:0.1 Mars:0.1		
15	213°09.9	21°32.2	•• 10.3	120°40.8	•• 28.1	134°10.4	•• 20.8	225°11.9	•• 32.6			
16	228°12.4	36°31.9	11.5	135°41.6	28.1	149°12.6	20.8	240°14.6	32.7			
17	243°14.9	51°31.6	12.8	150°42.5	28.1	164°14.8	20.9	255°17.2	32.8			
18	258°17.3	66°31.3	S04°14.1	165°43.3	N23°28.1	179°17.1	N22°20.9	270°19.9	S07°32.9			
19	273°19.8	81°30.9	15.4	180°44.2	28.2	194°19.3	20.9	285°22.5	32.9			
20	288°22.3	96°30.6	16.7	195°45.0	28.2	209°21.5	20.9	300°25.1	33.0			
21	303°24.7	111°30.3	•• 17.9	210°45.9	•• 28.2	224°23.7	•• 20.9	315°27.8	•• 33.1			
22	318°27.2	126°30.0	19.2	225°46.7	28.2	239°25.9	20.9	330°30.4	33.2			
23	333°29.6	141°29.6	20.5	240°47.5	28.2	254°28.1	21.0	345°33.1	33.2			

h	Sun				Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP	
0	180°24.6	N06°20.0	151°31.1	17.2'	S06°54.2	13.8'	53.9'	
1	195°24.8	19.0	166°07.2	17.1'	07°08.0	13.8'	53.9'	
2	210°25.0	18.1	180°43.4	17.1'	07°21.8	13.8'	53.9'	
3	225°25.3	· · 17.2	195°19.5	17.1'	07°35.5	13.7'	53.9'	
4	240°25.5	16.2	209°55.5	17.0'	07°49.2	13.7'	53.9'	
5	255°25.7	15.3	224°31.6	17.0'	08°02.9	13.7'	53.9'	
6	270°25.9	N06°14.4	239°07.6	17.0'	S08°16.6	13.6'	53.9'	
7	285°26.1	13.4	253°43.6	17.0'	08°30.3	13.6'	53.9'	
8	300°26.3	12.5	268°19.5	16.9'	08°43.9	13.6'	53.9'	
9	315°26.5	· · 11.6	282°55.5	16.9'	08°57.5	13.6'	53.9'	
10	330°26.7	10.6	297°31.3	16.8'	09°11.0	13.5'	53.9'	
11	345°26.9	09.7	312°07.2	16.8'	09°24.6	13.5'	53.9'	
12	0°27.2	N06°08.8	326°43.0	16.8'	S09°38.0	13.5'	54.0'	
13	15°27.4	07.8	341°18.8	16.7'	09°51.5	13.4'	54.0'	
14	30°27.6	06.9	355°54.5	16.7'	10°04.9	13.4'	54.0'	
15	45°27.8	· · 06.0	10°30.2	16.7'	10°18.3	13.4'	54.0'	
16	60°28.0	05.0	25°05.9	16.6'	10°31.7	13.3'	54.0'	
17	75°28.2	04.1	39°41.5	16.6'	10°45.0	13.3'	54.0'	
18	90°28.4	N06°03.1	54°17.1	16.5'	S10°58.3	13.2'	54.0'	
19	105°28.6	02.2	68°52.6	16.5'	11°11.5	13.2'	54.0'	
20	120°28.9	01.3	83°28.1	16.4'	11°24.7	13.2'	54.0'	
21	135°29.1	06°00.3	98°03.5	16.4'	11°37.9	13.1'	54.0'	
22	150°29.3	05°59.4	112°38.9	16.3'	11°51.0	13.1'	54.0'	
23	165°29.5	58.5	127°14.3	16.3'	12°04.1	13.0'	54.0'	
SD = 15.9'			$d = -0.9'$		SD = 14.7'			
Sat	GHA	Dec	GHA	ν	Dec	d	HP	
0	180°29.7	N05°57.5	141°49.6	16.2'	S12°17.2	13.0'	54.0'	
1	195°29.9	56.6	156°24.8	16.2'	12°30.2	13.0'	54.0'	
2	210°30.1	55.7	171°00.0	16.1'	12°43.1	12.9'	54.0'	
3	225°30.4	· · 54.7	185°35.2	16.1'	12°56.0	12.9'	54.0'	
4	240°30.6	53.8	200°10.2	16.0'	13°08.9	12.8'	54.0'	
5	255°30.8	52.8	214°45.3	16.0'	13°21.7	12.8'	54.1'	
6	270°31.0	N05°51.9	229°20.3	15.9'	S13°34.5	12.7'	54.1'	
7	285°31.2	51.0	243°55.2	15.9'	13°47.2	12.7'	54.1'	
8	300°31.4	50.0	258°30.1	15.8'	13°59.9	12.6'	54.1'	
9	315°31.6	· · 49.1	273°04.9	15.8'	14°12.5	12.6'	54.1'	
10	330°31.9	48.2	287°39.7	15.7'	14°25.1	12.5'	54.1'	
11	345°32.1	47.2	302°14.4	15.6'	14°37.7	12.5'	54.1'	
12	0°32.3	N05°46.3	316°49.0	15.6'	S14°50.1	12.4'	54.1'	
13	15°32.5	45.3	331°23.6	15.5'	15°02.6	12.4'	54.1'	
14	30°32.7	44.4	345°58.1	15.5'	15°14.9	12.3'	54.1'	
15	45°32.9	· · 43.5	0°32.6	15.4'	15°27.3	12.3'	54.1'	
16	60°33.1	42.5	15°07.0	15.3'	15°39.5	12.2'	54.2'	
17	75°33.4	41.6	29°41.3	15.3'	15°51.7	12.2'	54.2'	
18	90°33.6	N05°40.6	44°15.6	15.2'	S16°03.9	12.1'	54.2'	
19	105°33.8	39.7	58°49.8	15.1'	16°16.0	12.0'	54.2'	
20	120°34.0	38.8	73°23.9	15.1'	16°28.0	12.0'	54.2'	
21	135°34.2	· · 37.8	87°58.0	15.0'	16°40.0	11.9'	54.2'	
22	150°34.4	36.9	102°32.0	14.9'	16°51.9	11.9'	54.2'	
23	165°34.6	35.9	117°05.9	14.9'	17°03.8	11.8'	54.2'	
SD = 15.9'			$d = -0.9'$		SD = 14.7'			
Sun	GHA	Dec	GHA	ν	Dec	d	HP	
0	180°34.9	N05°35.0	131°39.7	14.8'	S17°15.6	11.7'	54.2'	
1	195°35.1	34.1	146°13.5	14.7'	17°27.3	11.7'	54.3'	
2	210°35.3	33.1	160°47.2	14.6'	17°39.0	11.6'	54.3'	
3	225°35.5	· · 32.2	175°20.9	14.6'	17°50.6	11.5'	54.3'	
4	240°35.7	31.2	189°54.4	14.5'	18°02.1	11.5'	54.3'	
5	255°35.9	30.3	204°27.9	14.4'	18°13.6	11.4'	54.3'	
6	270°36.2	N05°29.3	219°01.3	14.3'	S18°25.0	11.3'	54.3'	
7	285°36.4	28.4	233°34.7	14.3'	18°36.3	11.3'	54.3'	
8	300°36.6	27.5	248°07.9	14.2'	18°47.6	11.2'	54.4'	
9	315°36.8	· · 26.5	262°41.1	14.1'	18°58.8	11.1'	54.4'	
10	330°37.0	25.6	277°14.2	14.0'	19°09.9	11.0'	54.4'	
11	345°37.2	24.6	291°47.2	13.9'	19°20.9	11.0'	54.4'	
12	0°37.5	N05°23.7	306°20.2	13.9'	S19°31.9	10.9'	54.4'	
13	15°37.7	22.8	320°53.0	13.8'	19°42.8	10.8'	54.4'	
14	30°37.9	21.8	335°25.8	13.7'	19°53.6	10.7'	54.4'	
15	45°38.1	· · 20.9	349°58.5	13.6'	20°04.4	10.7'	54.5'	
16	60°38.3	19.9	4°31.1	13.5'	20°15.0	10.6'	54.5'	
17	75°38.5	19.0	19°03.6	13.4'	20°25.6	10.5'	54.5'	
18	90°38.8	N05°18.0	33°36.1	13.4'	S20°36.1	10.4'	54.5'	
19	105°39.0	17.1	48°08.4	13.3'	20°46.6	10.3'	54.5'	
20	120°39.2	16.1	62°40.7	13.2'	20°56.9	10.3'	54.5'	
21	135°39.4	· · 15.2	77°12.9	13.1'	21°07.2	10.2'	54.6'	
22	150°39.6	14.3	91°45.0	13.0'	21°17.4	10.1'	54.6'	
23	165°39.8	13.3	106°17.0	12.9'	21°27.4	10.0'	54.6'	
SD = 15.9'			$d = -0.9'$		SD = 14.8'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	00:15	03:13	04:32	19:20	20:38	23:09
N 70°	01:43	03:33	04:42	19:11	20:19	22:05
68°	02:20	03:48	04:50	19:04	20:04	21:31
66°	02:45	04:01	04:56	18:58	19:53	21:07
64°	03:04	04:11	05:01	18:52	19:43	20:49
62°	03:19	04:19	05:06	18:48	19:35	20:34
60°	03:31	04:26	05:10	18:44	19:27	20:22
N 58°	03:42	04:33	05:14	18:41	19:21	20:12
56°	03:51	04:38	05:17	18:38	19:16	20:03
54°	03:59	04:43	05:20	18:35	19:11	19:55
52°	04:05	04:48	05:22	18:32	19:07	19:49
50°	04:11	04:51	05:25	18:30	19:03	19:43
45°	04:24	05:00	05:30	18:25	18:55	19:31
N 40°	04:34	05:06	05:34	18:21	18:49	19:21
35°	04:42	05:12	05:37	18:18	18:43	19:13
30°	04:48	05:16	05:40	18:15	18:39	19:07
20°	04:58	05:24	05:46	18:10	18:32	18:58
N 10°	05:05	05:29	05:50	18:05	18:26	18:51
0°	05:10	05:34	05:55	18:01	18:22	18:46
S 10°	05:13	05:38	05:59	17:57	18:18	18:43
20°	05:15	05:41	06:03	17:53	18:15	18:41
30°	05:16	05:44	06:08	17:48	18:12	18:40
35°	05:16	05:45	06:10	17:46	18:11	18:41
40°	05:15	05:46	06:13	17:43	18:10	18:41
45°	05:14	05:47	06:17	17:40	18:09	18:43
S 50°	05:11	05:49	06:21	17:36	18:08	18:45
52°	05:10	05:49	06:23	17:34	18:08	18:47
54°	05:09	05:49	06:25	17:32	18:07	18:48
56°	05:07	05:50	06:27	17:30	18:07	18:50
58°	05:05	05:50	06:29	17:27	18:07	18:52
S 60°	05:03	05:51	06:32	17:25	18:06	18:55

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	10:12	12:37	■	17:55	16:55	■
N 70°	09:56	12:01	■	18:13	17:33	■
68°	09:44	11:36	13:54	18:27	18:00	17:12
66°	09:34	11:17	13:14	18:38	18:20	17:54
64°	09:26	11:02	12:46	18:48	18:37	18:22
62°	09:19	10:49	12:25	18:56	18:51	18:44
60°	09:13	10:38	12:08	19:04	19:02	19:02
N 58°	09:08	10:29	11:54	19:10	19:13	19:17
56°	09:03	10:21	11:41	19:16	19:22	19:30
54°	08:59	10:14	11:31	19:21	19:29	19:41
52°	08:55	10:07	11:21	19:25	19:37	19:51
50°	08:52	10:01	11:13	19:30	19:43	20:00
45°	08:44	09:49	10:55	19:39	19:57	20:19
N 40°	08:38	09:39	10:41	19:46	20:09	20:34
35°	08:33	09:30	10:29	19:53	20:18	20:48
30°	08:28	09:23	10:19	19:59	20:27	20:59
20°	08:21	09:10	10:01	20:09	20:42	21:19
N 10°	08:14	08:58	09:45	20:18	20:56	21:36
0°	08:07	08:48	09:31	20:26	21:08	21:52
S 10°	08:01	08:37	09:16	20:35	21:20	22:09
20°	07:54	08:26	09:01	20:44	21:34	22:26
30°	07:47	08:14	08:44	20:54	21:49	22:46
35°						

September 09, 10, 11 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	348°32.1	156°29.3	S04°21.8	255°48.4	N23°28.3	269°30.3	N22°21.0	0°35.7	S07°33.3	Alpheratz	357°34.7 29°13.7
1	3°34.6	171°29.0	23.0	270°49.2	28.3	284°32.5	21.0	15°38.4	33.4	Ankaa	353°07.0 -42°10.1
2	18°37.0	186°28.7	24.3	285°50.1	28.3	299°34.7	21.0	30°41.0	33.5	Schedar	349°30.9 56°40.3
3	33°39.5	201°28.4	•• 25.6	300°50.9	•• 28.3	314°36.9	•• 21.0	45°43.6	•• 33.5	Diphda	348°47.3 -17°50.9
4	48°42.0	216°28.0	26.9	315°51.8	28.3	329°39.2	21.0	60°46.3	33.6	Achernar	335°19.9 -57°06.4
5	63°44.4	231°27.7	28.2	330°52.6	28.4	344°41.4	21.1	75°48.9	33.7	Hamal	327°51.3 23°34.8
6	78°46.9	246°27.4	S04°29.4	345°53.5	N23°28.4	359°43.6	N22°21.1	90°51.6	S07°33.8	Polaris	313°56.4 89°21.8
7	93°49.4	261°27.1	30.7	0°54.3	28.4	14°45.8	21.1	105°54.2	33.8	Acamar	315°11.7 -40°12.0
8	108°51.8	276°26.7	32.0	15°55.2	28.4	29°48.0	21.1	120°56.8	33.9	Menkar	314°06.3 4°11.3
9	123°54.3	291°26.4	•• 33.3	30°56.0	•• 28.4	44°50.2	•• 21.1	135°59.5	•• 34.0	Mirfak	308°28.5 49°56.8
10	138°56.8	306°26.1	34.5	45°56.9	28.4	59°52.4	21.1	151°02.1	34.1	Aldebaran	290°39.9 16°33.6
11	153°59.2	321°25.8	35.8	60°57.7	28.5	74°54.6	21.2	166°04.8	34.2	Rigel	281°04.1 -8°10.2
12	169°01.7	336°25.5	S04°37.1	75°58.6	N23°28.5	89°56.9	N22°21.2	181°07.4	S07°34.2	Capella	280°22.3 46°01.2
13	184°04.1	351°25.1	38.4	90°59.4	28.5	104°59.1	21.2	196°10.0	34.3	Bellatrix	278°23.2 6°22.5
14	199°06.6	6°24.8	39.7	106°00.3	28.5	120°01.3	21.2	211°12.7	34.4	Elnath	278°02.3 28°37.7
15	214°09.1	21°24.5	•• 40.9	121°01.1	•• 28.5	135°03.5	•• 21.2	226°15.3	•• 34.5	Alnilam	275°38.1 -1°11.0
16	229°11.5	36°24.2	42.2	136°02.0	28.6	150°05.7	21.2	241°18.0	34.5	Betelgeuse	270°52.5 7°24.8
17	244°14.0	51°23.8	43.5	151°02.8	28.6	165°07.9	21.3	256°20.6	34.6	Canopus	263°52.7 -52°42.1
18	259°16.5	66°23.5	S04°44.8	166°03.7	N23°28.6	180°10.2	N22°21.3	271°23.3	S07°34.7	Sirius	258°26.6 -16°44.7
19	274°18.9	81°23.2	46.0	181°04.5	28.6	195°12.4	21.3	286°25.9	34.8	Adhara	255°06.3 -29°00.0
20	289°21.4	96°22.9	47.3	196°05.4	28.6	210°14.6	21.3	301°28.5	34.8	Procyon	244°51.3 5°09.8
21	304°23.9	111°22.5	•• 48.6	211°06.2	•• 28.6	225°16.8	•• 21.3	316°31.2	•• 34.9	Pollux	243°17.9 27°58.0
22	319°26.3	126°22.2	49.9	226°07.1	28.6	240°19.0	21.3	331°33.8	35.0	Avior	234°15.4 -59°35.0
23	334°28.8	141°21.9	51.1	241°07.9	28.7	255°21.2	21.3	346°36.5	35.1	Suhail	222°46.9 -43°31.7
Mer.pass. 00:46		ν -0.3' d 1.3' m -3.87		ν 0.8' d 0.0' m 0.65		ν 2.2' d 0.0' m -2.31		ν 2.6' d 0.1' m 0.57		Miaplacidus	221°39.2 -69°48.8
Alphard 217°48.3 -8°45.7 Regulus 207°35.0 11°50.9 Dubhe 193°42.0 61°37.1 Denebola 182°25.5 14°26.2 Gienah 175°44.2 -17°40.6 Acrux 173°01.1 -63°14.2 Gacrux 171°52.6 -57°15.1 Alioth 166°13.7 55°49.7 Spica 158°22.9 -11°17.3 Alkaid 152°52.6 49°11.6 Hadar 148°36.9 -60°29.6 Menkent 147°58.2 -36°29.5 Arcturus 145°48.4 19°03.4 Rigel Kent. 139°41.1 -60°56.4 Kochab 137°20.4 74°03.5 Zuben'ubi 136°56.5 -16°08.6 Alphecca 126°04.1 26°38.1 Antares 112°16.2 -26°29.2 Atria 107°10.7 -69°04.5 Sabik 102°03.1 -15°45.3 Shaula 96°10.7 -37°07.4 Rasalhague 95°58.8 12°32.7 Eltanin 90°42.2 51°29.4 Kaus Aust. 83°32.8 -34°22.4 Vega 80°33.3 38°48.6 Nunki 75°47.9 -26°16.0 Altair 62°00.0 8°56.1 Peacock 53°05.7 -56°39.4 Deneb 49°25.7 45°22.3 Enif 33°38.8 9°59.4 Al Na'ir 27°32.8 -46°50.5 Fomalhaut 15°14.4 -29°29.4 Scheat 13°45.1 28°13.1 Markab 13°29.9 15°20.4											
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	349°31.3	156°21.6	S04°52.4	256°08.8	N23°28.7	270°23.5	N22°21.4	1°39.1	S07°35.1		
1	4°33.7	171°21.2	53.7	271°09.6	28.7	285°25.7	21.4	16°41.7	35.2		
2	19°36.2	186°20.9	55.0	286°10.5	28.7	300°27.9	21.4	31°44.4	35.3		
3	34°38.6	201°20.6	•• 56.2	301°11.3	•• 28.7	315°30.1	•• 21.4	46°47.0	•• 35.4		
4	49°41.1	216°20.3	57.5	316°12.2	28.7	330°32.3	21.4	61°49.7	35.4		
5	64°43.6	231°19.9	04°58.8	331°13.0	28.7	345°34.5	21.4	76°52.3	35.5		
6	79°46.0	246°19.6	S05°00.1	346°13.9	N23°28.8	0°36.8	N22°21.5	91°54.9	S07°35.6		
7	94°48.5	261°19.3	01.3	1°14.7	28.8	15°39.0	21.5	106°57.6	35.7		
8	109°51.0	276°19.0	02.6	16°15.6	28.8	30°41.2	21.5	122°00.2	35.8		
9	124°53.4	291°18.6	•• 03.9	31°16.4	•• 28.8	45°43.4	•• 21.5	137°02.9	•• 35.8		
10	139°55.9	306°18.3	05.2	46°17.3	28.8	60°45.6	21.5	152°05.5	35.9		
11	154°58.4	321°18.0	06.4	61°18.2	28.8	75°47.9	21.5	167°08.2	36.0		
12	170°00.8	336°17.7	S05°07.7	76°19.0	N23°28.8	90°50.1	N22°21.6	182°10.8	S07°36.1		
13	185°03.3	351°17.3	09.0	91°19.9	28.8	105°52.3	21.6	197°13.4	36.1		
14	200°05.8	6°17.0	10.2	106°20.7	28.9	120°54.5	21.6	212°16.1	36.2		
15	215°08.2	21°16.7	•• 11.5	121°21.6	•• 28.9	135°56.8	•• 21.6	227°18.7	•• 36.3		
16	230°10.7	36°16.4	12.8	136°22.4	28.9	150°59.0	21.6	242°21.4	36.4		
17	245°13.1	51°16.0	14.1	151°23.3	28.9	166°01.2	21.6	257°24.0	36.4		
18	260°15.6	66°15.7	S05°15.3	166°24.2	N23°28.9	181°03.4	N22°21.6	272°26.6	S07°36.5		
19	275°18.1	81°15.4	16.6	181°25.0	28.9	196°05.7	21.7	287°29.3	36.6		
20	290°20.5	96°15.0	17.9	196°25.9	28.9	211°07.9	21.7	302°31.9	36.7		
21	305°23.0	111°14.7	•• 19.2	211°26.7	•• 28.9	226°10.1	•• 21.7	317°34.6	•• 36.7		
22	320°25.5	126°14.4	20.4	226°27.6	28.9	241°12.3	21.7	332°37.2	36.8		
23	335°27.9	141°14.1	21.7	241°28.4	28.9	256°14.5	21.7	347°39.8	36.9		
Mer.pass. 00:42		ν -0.3' d 1.3' m -3.87		ν 0.9' d 0.0' m 0.64		ν 2.2' d 0.0' m -2.32		ν 2.6' d 0.1' m 0.58			
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	350°30.4	156°13.7	S05°23.0	256°29.3	N23°28.9	271°16.8	N22°21.7	2°42.5	S07°37.0		
1	5°32.9	171°13.4	24.2	271°30.2	29.0	286°19.0	21.8	17°45.1	37.0		
2	20°35.3	186°13.1	25.5	286°31.0	29.0	301°21.2	21.8	32°47.8	37.1		
3	35°37.8	201°12.7	•• 26.8	301°31.9	•• 29.0	316°23.5	•• 21.8	47°50.4	•• 37.2		
4	50°40.2	216°12.4	28.0	316°32.7	29.0	331°25.7	21.8	62°53.0	37.3		
5	65°42.7	231°12.1	29.3	331°33.6	29.0	346°27.9	21.8	77°55.7	37.4		
6	80°45.2	246°11.8	S05°30.6	346°34.5	N23°29.0	1°30.1	N22°21.8	92°58.3	S07°37.4		
7	95°47.6	261°11.4	31.9	1°35.3	29.0	16°32.4	21.8	108°01.0	37.5		
8	110°50.1	276°11.1	33.1	16°36.2	29.0	31°34.6	21.9	123°03.6	37.6		
9	125°52.6	291°10.8	•• 34.4	31°37.1	•• 29.0	46°36.8	•• 21.9	138°06.3	•• 37.7		
10	140°55.0	306°10.4	35.7	46°37.9	29.0	61°39.0	21.9	153°08.9	37.7		
11	155°57.5	321°10.1	36.9	61°38.8	29.0	76°41.3	21.9	168°11.5	37.8		
12	171°00.0	336°09.8	S05°38.2	76°39.6	N23°29.0	91°43.5	N22°21.9	183°14.2	S07°37.9		
13	186°02.4	351°09.5	39.5	91°40.5	29.0	106°45.7	21.9	198°16.8	38.0		
14	201°04.9	6°09.1	40.7	106°41.4	29.0	121°48.0	21.9	213°19.5	38.0		
15	216°07.4	21°08.8	•• 42.0	121°42.2	•• 29.0	136°50.2	•• 22.0	228°22.1	•• 38.1		
16	231°09.8	36°08.5	43.3	136°43.1	29.1	151°52.4	22.0	243°24.7	38.2		
17	246°12.3	51°08.1	44.5	151°44.0	29.1	166°54.7	22.0	258°27.4	38.3		
18	261°14.7	66°07.8	S05°45.8	166°44.8	N23°29.1	181°56.9	N22°22.0	273°30.0	S07°38.3		
19	276°17.2	81°07.5	47.1	181°45.7	29.1	196°59.1	22.0	288°32.7	38.4		
20	291°19.7	96°07.1	48.4	196°46.6	29.1	212°01.3	22.0	303°35.3	38.5		
21	306°22.1	111°06.8	•• 49.6	211°47.4	•• 29.1	227°03.6	•• 22.1	318°37.9	•• 38.6		
22	321°24.6	126°06.5	50.9	226°48.3	29.1	242°05.8	22.1	333°40.6	38.6		
23	336°27.1	141°06.1	52.2	241°49.2	29.1	257°08.0	22.1	348°43.2	38.7		
Mer.pass. 00:38		ν -0.3' d 1.3' m -3.87		ν 0.9' d 0.0' m 0.63		ν 2.2' d 0.0' m -2.32		ν 2.6' d 0.1' m 0.58			

Sep 09 Mon			SHA	Mer.pass
Venus			167°57.2	13:34
Mars			267°16.3	06:56
Jupiter			280°58.2	06:01
Saturn			12°03.6	23:53
Sep 10 Tue			SHA	Mer.pass
Venus			166°50.3	13:35
Mars			266°37.5	06:55
Jupiter			280°52.2	05:58
Saturn			12°07.8	23:49
Sep 11 Wed			SHA	Mer.pass
Venus			165°43.3	13:35
Mars			265°58.9	06:54
Jupiter			280°46.4	05:54
Saturn			12°12.1	23:45
Horizontal parallax				
Venus:			0.1	
Mars:			0.1	

Sun				Moon				
Mon	GHA	Dec		GHA	ν	Dec	d	HP
0	180°40.1	N05°12.4		120°48.9	12.8'	S21°37.5	9.9'	54.6'
1	195°40.3	11.4		135°20.7	12.7'	21°47.4	9.8'	54.6'
2	210°40.5	10.5		149°52.5	12.7'	21°57.2	9.7'	54.7'
3	225°40.7	· · 09.5		164°24.1	12.6'	22°07.0	9.7'	54.7'
4	240°40.9	08.6		178°55.7	12.5'	22°16.6	9.6'	54.7'
5	255°41.1	07.6		193°27.2	12.4'	22°26.2	9.5'	54.7'
6	270°41.4	N05°06.7		207°58.5	12.3'	S22°35.7	9.4'	54.7'
7	285°41.6	05.8		222°29.8	12.2'	22°45.0	9.3'	54.8'
8	300°41.8	04.8		237°01.0	12.1'	22°54.3	9.2'	54.8'
9	315°42.0	· · 03.9		251°32.1	12.0'	23°03.5	9.1'	54.8'
10	330°42.2	02.9		266°03.1	11.9'	23°12.6	9.0'	54.8'
11	345°42.5	02.0		280°34.0	11.8'	23°21.6	8.9'	54.8'
12	0°42.7	N05°01.0		295°04.9	11.7'	S23°30.5	8.8'	54.9'
13	15°42.9	05°00.1		309°35.6	11.6'	23°39.3	8.7'	54.9'
14	30°43.1	04°59.1		324°06.2	11.5'	23°48.0	8.6'	54.9'
15	45°43.3	· · 58.2		338°36.8	11.4'	23°56.5	8.5'	54.9'
16	60°43.5	57.2		353°07.2	11.3'	24°05.0	8.4'	54.9'
17	75°43.8	56.3		7°37.5	11.3'	24°13.4	8.3'	55.0'
18	90°44.0	N04°55.3		22°07.8	11.2'	S24°21.7	8.2'	55.0'
19	105°44.2	54.4		36°38.0	11.1'	24°29.9	8.1'	55.0'
20	120°44.4	53.5		51°08.0	11.0'	24°37.9	7.9'	55.0'
21	135°44.6	· · 52.5		65°38.0	10.9'	24°45.9	7.8'	55.1'
22	150°44.9	51.6		80°07.8	10.8'	24°53.7	7.7'	55.1'
23	165°45.1	50.6		94°37.6	10.7'	25°01.4	7.6'	55.1'
SD = 15.9'			$d = -0.9'$	SD = 14.9'				

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	180°45.3	N04°49.7	109°07.3	10.6'	S25°09.0	7.5'	55.1'
1	195°45.5	48.7	123°36.9	10.5'	25°16.5	7.4'	55.2'
2	210°45.7	47.8	138°06.4	10.4'	25°23.9	7.3'	55.2'
3	225°45.9	· · 46.8	152°35.8	10.3'	25°31.2	7.2'	55.2'
4	240°46.2	45.9	167°05.0	10.2'	25°38.3	7.0'	55.2'
5	255°46.4	44.9	181°34.2	10.1'	25°45.4	6.9'	55.3'
6	270°46.6	N04°44.0	196°03.3	10.0'	S25°52.3	6.8'	55.3'
7	285°46.8	43.0	210°32.3	9.9'	25°59.1	6.7'	55.3'
8	300°47.0	42.1	225°01.3	9.8'	26°05.7	6.5'	55.3'
9	315°47.3	· · 41.1	239°30.1	9.7'	26°12.3	6.4'	55.4'
10	330°47.5	40.2	253°58.8	9.6'	26°18.7	6.3'	55.4'
11	345°47.7	39.2	268°27.4	9.5'	26°25.0	6.2'	55.4'
12	0°47.9	N04°38.3	282°56.0	9.4'	S26°31.2	6.0'	55.5'
13	15°48.1	37.3	297°24.4	9.3'	26°37.2	5.9'	55.5'
14	30°48.4	36.4	311°52.7	9.3'	26°43.1	5.8'	55.5'
15	45°48.6	· · 35.4	326°21.0	9.2'	26°48.9	5.6'	55.5'
16	60°48.8	34.5	340°49.2	9.1'	26°54.5	5.5'	55.6'
17	75°49.0	33.5	355°17.2	9.0'	27°00.1	5.4'	55.6'
18	90°49.2	N04°32.6	9°45.2	8.9'	S27°05.4	5.2'	55.6'
19	105°49.5	31.6	24°13.1	8.8'	27°10.7	5.1'	55.7'
20	120°49.7	30.7	38°40.9	8.7'	27°15.8	5.0'	55.7'
21	135°49.9	· · 29.7	53°08.6	8.6'	27°20.8	4.8'	55.7'
22	150°50.1	28.8	67°36.2	8.5'	27°25.6	4.7'	55.7'
23	165°50.3	27.8	82°03.8	8.4'	27°30.3	4.6'	55.8'
SD = 15.9'			$d = -0.9'$	SD = 15.0'			

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	180°50.6	N04°26.9	96°31.2	8.4'	S27°34.9	4.4'	55.8'
1	195°50.8	25.9	110°58.6	8.3'	27°39.3	4.3'	55.8'
2	210°51.0	25.0	125°25.8	8.2'	27°43.5	4.1'	55.9'
3	225°51.2	· · 24.0	139°53.0	8.1'	27°47.7	4.0'	55.9'
4	240°51.4	23.1	154°20.1	8.0'	27°51.6	3.8'	55.9'
5	255°51.7	22.1	168°47.2	7.9'	27°55.5	3.7'	56.0'
6	270°51.9	N04°21.2	183°14.1	7.9'	S27°59.2	3.5'	56.0'
7	285°52.1	20.2	197°41.0	7.8'	28°02.7	3.4'	56.0'
8	300°52.3	19.3	212°07.7	7.7'	28°06.1	3.2'	56.1'
9	315°52.5	· · 18.3	226°34.4	7.6'	28°09.3	3.1'	56.1'
10	330°52.8	17.3	241°01.1	7.5'	28°12.4	2.9'	56.1'
11	345°53.0	16.4	255°27.6	7.5'	28°15.3	2.8'	56.2'
12	0°53.2	N04°15.4	269°54.1	7.4'	S28°18.1	2.6'	56.2'
13	15°53.4	14.5	284°20.5	7.3'	28°20.8	2.5'	56.2'
14	30°53.7	13.5	298°46.8	7.3'	28°23.2	2.3'	56.3'
15	45°53.9	· · 12.6	313°13.1	7.2'	28°25.5	2.2'	56.3'
16	60°54.1	11.6	327°39.3	7.1'	28°27.7	2.0'	56.3'
17	75°54.3	10.7	342°05.4	7.1'	28°29.7	1.8'	56.4'
18	90°54.5	N04°09.7	356°31.4	7.0'	S28°31.5	1.7'	56.4'
19	105°54.8	08.8	10°57.4	6.9'	28°33.2	1.5'	56.4'
20	120°55.0	07.8	25°23.3	6.9'	28°34.7	1.4'	56.5'
21	135°55.2	· · 06.9	39°49.2	6.8'	28°36.1	1.2'	56.5'
22	150°55.4	05.9	54°15.0	6.7'	28°37.3	1.0'	56.5'
23	165°55.6	05.0	68°40.7	6.7'	28°38.3	0.9'	56.6'
SD = 15.9'			$d = -1.0'$	SD = 15.2'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	01:21	03:31	04:46	19:04	20:19	22:22
N 70°	02:08	03:47	04:54	18:57	20:03	21:39
68°	02:38	04:01	05:00	18:51	19:50	21:11
66°	02:59	04:11	05:06	18:46	19:40	20:51
64°	03:16	04:20	05:10	18:42	19:31	20:35
62°	03:29	04:28	05:14	18:38	19:24	20:22
60°	03:40	04:34	05:17	18:35	19:18	20:11
N 58°	03:50	04:40	05:20	18:32	19:12	20:02
56°	03:58	04:44	05:23	18:30	19:08	19:54
54°	04:05	04:49	05:25	18:27	19:03	19:47
52°	04:11	04:53	05:27	18:25	19:00	19:41
50°	04:17	04:56	05:29	18:24	18:56	19:36
45°	04:28	05:03	05:33	18:20	18:49	19:25
N 40°	04:37	05:09	05:37	18:16	18:44	19:16
35°	04:44	05:14	05:40	18:14	18:39	19:09
30°	04:50	05:18	05:42	18:11	18:35	19:03
20°	04:58	05:24	05:46	18:07	18:29	18:55
N 10°	05:05	05:29	05:50	18:03	18:24	18:49
0°	05:09	05:33	05:54	18:00	18:21	18:45
S 10°	05:11	05:36	05:57	17:57	18:18	18:42
20°	05:13	05:38	06:00	17:54	18:16	18:41
30°	05:12	05:40	06:04	17:50	18:14	18:42
35°	05:12	05:41	06:06	17:48	18:13	18:43
40°	05:10	05:41	06:08	17:46	18:13	18:44
45°	05:08	05:42	06:11	17:43	18:13	18:47
S 50°	05:05	05:42	06:14	17:40	18:12	18:50
52°	05:03	05:42	06:16	17:39	18:12	18:52
54°	05:01	05:42	06:17	17:37	18:13	18:54
56°	04:59	05:42	06:19	17:36	18:13	18:56
58°	04:56	05:42	06:21	17:34	18:13	18:59
S 60°	04:53	05:42	06:23	17:32	18:13	19:02

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	■	■	■	■
66°	■	■	■	■	■	■
64°	14:49	■	■	17:56	■	■
62°	14:10	16:12	■	18:36	18:20	■
60°	13:43	15:22	16:55	19:04	19:10	19:32
N 58°	13:22	14:52	16:13	19:25	19:41	20:14
56°	13:05	14:28	15:44	19:43	20:05	20:43
54°	12:50	14:09	15:22	19:58	20:24	21:05
52°	12:38	13:54	15:04	20:11	20:40	21:23
50°	12:27	13:40	14:48	20:23	20:54	21:39
45°	12:04	13:12	14:17	20:47	21:23	22:10
N 40°	11:45	12:50	13:53	21:06	21:45	22:34
35°	11:30	12:32	13:33	21:22	22:04	22:54
30°	11:17	12:16	13:17	21:36	22:20	23:12
20°	10:54	11:50	12:48	22:00	22:47	23:40
N 10°	10:35	11:28	12:24	22:21	23:11	· · ·
0°	10:17	11:07	12:01	22:40	23:32	· · ·
S 10°	09:59	10:46	11:38	23:00	23:54	· · ·
20°	09:40	10:24	11:14	23:21	· · ·	00:18
30°	09:18	09:58	10:46	23:45	· · ·	00:45
35°	09:05	09:43	10:29	· · ·	00:00	01:02
40°	08:51	09:26	10:10	· · ·	00:16	01:21
45°	08:33	09:05	09:47	· · ·	00:36	01:44
S 50°						

September 12, 13, 14 UT (Thu., Fri., Sat.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	351°29.5	156°05.8	S05°53.4	256°50.0	N23°29.1	272°10.3	N22°22.1	3°45.9	S07°38.8					Alpheratz	357°34.7 29°13.7		
1	6°32.0	171°05.5	54.7	271°50.9	29.1	287°12.5	22.1	18°48.5	38.9					Ankaa	353°07.0 -42°10.1		
2	21°34.5	186°05.1	56.0	286°51.8	29.1	302°14.7	22.1	33°51.1	38.9					Schedar	349°30.9 56°40.3		
3	36°36.9	201°04.8	57.2	301°52.6	29.1	317°17.0	22.1	48°53.8	39.0					Diphda	348°47.3 -17°50.9		
4	51°39.4	216°04.5	58.5	316°53.5	29.1	332°19.2	22.2	63°56.4	39.1					Achernar	335°19.9 -57°06.4		
5	66°41.9	231°04.1	05°59.7	331°54.4	29.1	347°21.4	22.2	78°59.1	39.2					Hamal	327°51.3 23°34.8		
6	81°44.3	246°03.8	S06°01.0	346°55.2	N23°29.1	2°23.7	N22°22.2	94°01.7	S07°39.2					Polaris	313°54.8 89°21.8		
7	96°46.8	261°03.5	02.3	1°56.1	29.1	17°25.9	22.2	109°04.4	39.3					Acamar	315°11.7 -40°12.0		
8	111°49.2	276°03.1	03.5	16°57.0	29.1	32°28.1	22.2	124°07.0	39.4					Menkar	314°06.3 4°11.3		
9	126°51.7	291°02.8	04.8	31°57.8	29.1	47°30.4	22.2	139°09.6	39.5					Mirfak	308°28.4 49°56.8		
10	141°54.2	306°02.5	06.1	46°58.7	29.1	62°32.6	22.2	154°12.3	39.5					Aldebaran	290°39.9 16°33.6		
11	156°56.6	321°02.1	07.3	61°59.6	29.1	77°34.9	22.3	169°14.9	39.6					Rigel	281°04.1 -8°10.2		
12	171°59.1	336°01.8	S06°08.6	77°00.5	N23°29.1	92°37.1	N22°22.3	184°17.6	S07°39.7					Capella	280°22.3 46°01.2		
13	187°01.6	351°01.5	09.9	92°01.3	29.1	107°39.3	22.3	199°20.2	39.8					Bellatrix	278°23.2 6°22.5		
14	202°04.0	6°01.1	11.1	107°02.2	29.1	122°41.6	22.3	214°22.8	39.8					Elnath	278°02.2 28°37.7		
15	217°06.5	21°00.8	12.4	122°03.1	29.1	137°43.8	22.3	229°25.5	39.9					Alnilam	275°38.0 -1°11.0		
16	232°09.0	36°00.5	13.7	137°03.9	29.1	152°46.0	22.3	244°28.1	40.0					Betelgeuse	270°52.4 7°24.8		
17	247°11.4	51°00.1	14.9	152°04.8	29.1	167°48.3	22.3	259°30.8	40.1					Canopus	263°52.7 -52°42.1		
18	262°13.9	65°59.8	S06°16.2	167°05.7	N23°29.1	182°50.5	N22°22.4	274°33.4	S07°40.1					Sirius	258°26.6 -16°44.7		
19	277°16.4	80°59.5	17.4	182°06.6	29.1	197°52.8	22.4	289°36.0	40.2					Adhara	255°06.2 -29°00.0		
20	292°18.8	95°59.1	18.7	197°07.4	29.1	212°55.0	22.4	304°38.7	40.3					Procyon	244°51.3 5°09.8		
21	307°21.3	110°58.8	20.0	212°08.3	29.1	227°57.2	22.4	319°41.3	40.4					Pollux	243°17.8 27°58.0		
22	322°23.7	125°58.5	21.2	227°09.2	29.1	242°59.5	22.4	334°44.0	40.4					Avior	234°15.3 -59°35.0		
23	337°26.2	140°58.1	22.5	242°10.0	29.1	258°01.7	22.4	349°46.6	40.5					Suhail	222°46.9 -43°31.7		
Mer.pass. 00:34			ν -0.3' d 1.3' m -3.87			ν 0.9' d 0.0' m 0.62			ν 2.2' d 0.0' m -2.33			ν 2.6' d 0.1' m 0.58					

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	352°28.7	155°57.8	S06°23.8	257°10.9	N23°29.1	273°04.0	N22°22.4	4°49.2	S07°40.6					Denebola	182°25.5 14°26.2	
1	7°31.1	170°57.5	25.0	272°11.8	29.1	288°06.2	22.5	19°51.9	40.7					Gienah	175°44.2 -17°40.6	
2	22°33.6	185°57.1	26.3	287°12.7	29.1	303°08.4	22.5	34°54.5	40.7					Acrux	173°01.1 -63°14.2	
3	37°36.1	200°56.8	27.5	302°13.5	29.1	318°10.7	22.5	49°57.2	40.8					Gacrux	171°52.6 -57°15.1	
4	52°38.5	215°56.4	28.8	317°14.4	29.1	333°12.9	22.5	64°59.8	40.9					Alioth	166°13.7 55°49.7	
5	67°41.0	230°56.1	30.1	332°15.3	29.1	348°15.2	22.5	80°02.4	41.0					Spica	158°22.9 -11°17.3	
6	82°43.5	245°55.8	S06°31.3	347°16.2	N23°29.1	3°17.4	N22°22.5	95°05.1	S07°41.0					Alkaid	152°52.6 49°11.6	
7	97°45.9	260°55.4	32.6	2°17.1	29.0	18°19.6	22.5	110°07.7	41.1					Hadar	148°36.9 -60°29.6	
8	112°48.4	275°55.1	33.8	17°17.9	29.0	33°21.9	22.5	125°10.4	41.2					Menkent	147°58.3 -36°29.5	
9	127°50.8	290°54.8	35.1	32°18.8	29.0	48°24.1	22.6	140°13.0	41.3					Arcturus	145°48.4 19°03.4	
10	142°53.3	305°54.4	36.4	47°19.7	29.0	63°26.4	22.6	155°15.6	41.3					Rigil Kent.	139°41.2 -60°56.4	
11	157°55.8	320°54.1	37.6	62°20.6	29.0	78°28.6	22.6	170°18.3	41.4					Kochab	137°20.4 74°03.4	
12	172°58.2	335°53.7	S06°38.9	77°21.4	N23°29.0	93°30.9	N22°22.6	185°20.9	S07°41.5					Zuben'ubi	136°56.5 -16°08.6	
13	188°00.7	350°53.4	40.1	92°22.3	29.0	108°33.1	22.6	200°23.6	41.6					Alphecca	126°04.1 26°38.1	
14	203°03.2	5°53.1	41.4	107°23.2	29.0	123°35.4	22.6	215°26.2	41.6					Antares	112°16.3 -26°29.2	
15	218°05.6	20°52.7	42.7	122°24.1	29.0	138°37.6	22.6	230°28.8	41.7					Atria	107°10.8 -69°04.5	
16	233°08.1	35°52.4	43.9	137°25.0	29.0	153°39.8	22.7	245°31.5	41.8					Sabik	102°03.1 -15°45.3	
17	248°10.6	50°52.1	45.2	152°25.8	29.0	168°42.1	22.7	260°34.1	41.9					Shaula	96°10.7 -37°07.4	
18	263°13.0	65°51.7	S06°46.4	167°26.7	N23°29.0	183°44.3	N22°22.7	275°36.8	S07°41.9					Rasalhague	95°58.8 12°32.7	
19	278°15.5	80°51.4	47.7	182°27.6	29.0	198°46.6	22.7	290°39.4	42.0					Eltanin	90°42.2 51°29.4	
20	293°18.0	95°51.0	48.9	197°28.5	29.0	213°48.8	22.7	305°42.0	42.1					Kaus Aust.	83°32.8 -34°22.4	
21	308°20.4	110°50.7	50.2	212°29.4	29.0	228°51.1	22.7	320°44.7	42.2					Vega	80°33.3 38°48.6	
22	323°22.9	125°50.4	51.5	227°30.2	29.0	243°53.3	22.7	335°47.3	42.2					Nunki	75°47.9 -26°16.0	
23	338°25.3	140°50.0	52.7	242°31.1	28.9	258°55.6	22.8	350°50.0	42.3					Altair	62°00.0 8°56.1	
Mer.pass. 00:30			ν -0.3' d 1.3' m -3.87			ν 0.9' d -0.0' m 0.62			ν 2.2' d 0.0' m -2.34			ν 2.6' d 0.1' m 0.59				

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	353°27.8	155°49.7	S06°54.0	257°32.0	N23°28.9	273°57.8	N22°22.8	5°52.6	S07°42.4					Sep 12 Thu	SHA Mer.pass
1	8°30.3	170°49.3	55.2	272°32.9	28.9	289°00.1	22.8	20°55.2	42.5					Venus	164°36.3 13:36
2	23°32.7	185°49.0	56.5	287°33.8	28.9	304°02.3	22.8	35°57.9	42.5					Mars	265°20.5 06:52
3	38°35.2	200°48.6	57.7	302°34.7	28.9	319°04.6	22.8	51°00.5	42.6					Jupiter	280°40.7 05:50
4	53°37.7	215°48.3	06°59.0	317°35.5	28.9	334°06.8	22.8	66°03.2	42.7					Saturn	12°16.3 23:41
5	68°40.1	230°48.0	07°00.2	332°36.4	28.9	349°09.1	22.8	81°05.8	42.8						
6	83°42.6	245°47.6	S07°01.5	347°37.3	N23°28.9	4°11.3	N22°22.8	96°08.4	S07°42.8						
7	98°45.1	260°47.3	02.8	2°38.2	28.9	19°13.6	22.9	111°11.1	42.9						
8	113°47.5	275°46.9	04.0	17°39.1	28.9	34°15.8	22.9	126°13.7	43.0					Sep 13 Fri	SHA Mer.pass
9	128°50.0	290°46.6	05.3	32°40.0	28.8	49°18.1	22.9	141°16.4	43.1					Venus	163°29.1 13:36
10	143°52.5	305°46.3	06.5	47°40.8	28.8	64°20.3	22.9	156°19.0	43.1					Mars	264°42.3 06:51
11	158°54.9	320°45.9	07.8	62°41.7	28.8	79°22.6	22.9	171°21.6	43.2					Jupiter	280°35.3 05:47
12	173°57.4	335°45.6	S07°09.0	77°42.6	N23°28.8	94°24.8	N22°22.9	186°24.3							

Sun				Moon				
Thu	GHA	Dec		GHA	ν	Dec	d	HP
0	180°55.9	N04°04.0		83°06.4	6.6'	S28°39.2	0.7'	56.6'
1	195°56.1	03.0		97°32.0	6.6'	28°39.9	0.5'	56.6'
2	210°56.3	02.1		111°57.6	6.5'	28°40.4	0.4'	56.7'
3	225°56.5	· · 01.1		126°23.1	6.5'	28°40.8	0.2'	56.7'
4	240°56.8	04°00.2		140°48.6	6.4'	28°41.0	0.0'	56.8'
5	255°57.0	03°59.2		155°14.0	6.4'	28°41.0	-0.1'	56.8'
6	270°57.2	N03°58.3		169°39.3	6.3'	S28°40.9	-0.3'	56.8'
7	285°57.4	57.3		184°04.6	6.3'	28°40.6	-0.5'	56.9'
8	300°57.6	56.4		198°29.9	6.2'	28°40.1	-0.6'	56.9'
9	315°57.9	· · 55.4		212°55.1	6.2'	28°39.5	-0.8'	56.9'
10	330°58.1	54.5		227°20.3	6.1'	28°38.7	-1.0'	57.0'
11	345°58.3	53.5		241°45.5	6.1'	28°37.7	-1.2'	57.0'
12	0°58.5	N03°52.5		256°10.6	6.1'	S28°36.6	-1.3'	57.1'
13	15°58.7	51.6		270°35.6	6.0'	28°35.2	-1.5'	57.1'
14	30°59.0	50.6		285°00.7	6.0'	28°33.7	-1.7'	57.1'
15	45°59.2	· · 49.7		299°25.7	6.0'	28°32.0	-1.8'	57.2'
16	60°59.4	48.7		313°50.6	5.9'	28°30.2	-2.0'	57.2'
17	75°59.6	47.8		328°15.6	5.9'	28°28.2	-2.2'	57.2'
18	90°59.9	N03°46.8		342°40.5	5.9'	S28°26.0	-2.4'	57.3'
19	106°00.1	45.8		357°05.4	5.9'	28°23.6	-2.5'	57.3'
20	121°00.3	44.9		11°30.2	5.8'	28°21.1	-2.7'	57.4'
21	136°00.5	· · 43.9		25°55.1	5.8'	28°18.3	-2.9'	57.4'
22	151°00.7	43.0		40°19.9	5.8'	28°15.4	-3.1'	57.4'
23	166°01.0	42.0		54°44.7	5.8'	28°12.4	-3.3'	57.5'
SD = 15.9' d = -1.0'				SD = 15.4'				
Fri	GHA	Dec		GHA	ν	Dec	d	HP
0	181°01.2	N03°41.1		69°09.5	5.8'	S28°09.1	-3.4'	57.5'
1	196°01.4	40.1		83°34.3	5.8'	28°05.7	-3.6'	57.6'
2	211°01.6	39.1		97°59.0	5.7'	28°02.1	-3.8'	57.6'
3	226°01.9	· · 38.2		112°23.8	5.7'	27°58.3	-4.0'	57.6'
4	241°02.1	37.2		126°48.5	5.7'	27°54.3	-4.1'	57.7'
5	256°02.3	36.3		141°13.2	5.7'	27°50.2	-4.3'	57.7'
6	271°02.5	N03°35.3		155°38.0	5.7'	S27°45.9	-4.5'	57.8'
7	286°02.8	34.4		170°02.7	5.7'	27°41.4	-4.7'	57.8'
8	301°03.0	33.4		184°27.4	5.7'	27°36.7	-4.8'	57.8'
9	316°03.2	· · 32.4		198°52.1	5.7'	27°31.9	-5.0'	57.9'
10	331°03.4	31.5		213°16.9	5.7'	27°26.8	-5.2'	57.9'
11	346°03.6	30.5		227°41.6	5.7'	27°21.6	-5.4'	58.0'
12	1°03.9	N03°29.6		242°06.3	5.7'	S27°16.3	-5.6'	58.0'
13	16°04.1	28.6		256°31.1	5.8'	27°10.7	-5.7'	58.0'
14	31°04.3	27.7		270°55.8	5.8'	27°05.0	-5.9'	58.1'
15	46°04.5	· · 26.7		285°20.6	5.8'	26°59.1	-6.1'	58.1'
16	61°04.8	25.7		299°45.4	5.8'	26°53.0	-6.3'	58.2'
17	76°05.0	24.8		314°10.2	5.8'	26°46.7	-6.4'	58.2'
18	91°05.2	N03°23.8		328°35.0	5.8'	S26°40.3	-6.6'	58.2'
19	106°05.4	22.9		342°59.8	5.8'	26°33.7	-6.8'	58.3'
20	121°05.7	21.9		357°24.6	5.9'	26°26.9	-7.0'	58.3'
21	136°05.9	· · 20.9		11°49.5	5.9'	26°20.0	-7.1'	58.4'
22	151°06.1	20.0		26°14.4	5.9'	26°12.8	-7.3'	58.4'
23	166°06.3	19.0		40°39.3	5.9'	26°05.5	-7.5'	58.4'
SD = 15.9' d = -1.0'				SD = 15.7'				
Sat	GHA	Dec		GHA	ν	Dec	d	HP
0	181°06.5	N03°18.1		55°04.2	6.0'	S25°58.0	-7.6'	58.5'
1	196°06.8	17.1		69°29.1	6.0'	25°50.4	-7.8'	58.5'
2	211°07.0	16.1		83°54.1	6.0'	25°42.6	-8.0'	58.6'
3	226°07.2	· · 15.2		98°19.1	6.0'	25°34.6	-8.2'	58.6'
4	241°07.4	14.2		112°44.2	6.1'	25°26.4	-8.3'	58.6'
5	256°07.7	13.3		127°09.3	6.1'	25°18.1	-8.5'	58.7'
6	271°07.9	N03°12.3		141°34.4	6.1'	S25°09.6	-8.7'	58.7'
7	286°08.1	11.3		155°59.5	6.2'	25°00.9	-8.8'	58.8'
8	301°08.3	10.4		170°24.7	6.2'	24°52.1	-9.0'	58.8'
9	316°08.6	· · 09.4		184°49.9	6.2'	24°43.1	-9.2'	58.8'
10	331°08.8	08.5		199°15.1	6.3'	24°33.9	-9.3'	58.9'
11	346°09.0	07.5		213°40.4	6.3'	24°24.6	-9.5'	58.9'
12	1°09.2	N03°06.5		228°05.7	6.4'	S24°15.1	-9.7'	59.0'
13	16°09.5	05.6		242°31.1	6.4'	24°05.4	-9.8'	59.0'
14	31°09.7	04.6		256°56.5	6.5'	23°55.6	-10.0'	59.0'
15	46°09.9	· · 03.7		271°22.0	6.5'	23°45.6	-10.1'	59.1'
16	61°10.1	02.7		285°47.5	6.5'	23°35.4	-10.3'	59.1'
17	76°10.3	01.7		300°13.0	6.6'	23°25.1	-10.5'	59.2'
18	91°10.6	N03°00.8		314°38.6	6.6'	S23°14.7	-10.6'	59.2'
19	106°10.8	02°59.8		329°04.2	6.7'	23°04.0	-10.8'	59.2'
20	121°11.0	58.9		343°29.9	6.7'	22°53.3	-10.9'	59.3'
21	136°11.2	· · 57.9		357°55.7	6.8'	22°42.3	-11.1'	59.3'
22	151°11.5	56.9		12°21.4	6.8'	22°31.2	-11.2'	59.4'
23	166°11.7	56.0		26°47.3	6.9'	22°20.0	-11.4'	59.4'
SD = 15.9' d = -1.0'				SD = 15.9'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	01:54	03:47	05:00	18:49	20:01	21:50
N 70°	02:29	04:01	05:06	18:43	19:47	21:17
68°	02:54	04:12	05:11	18:38	19:36	20:54
66°	03:12	04:22	05:15	18:34	19:27	20:36
64°	03:27	04:29	05:19	18:31	19:20	20:22
62°	03:39	04:36	05:22	18:28	19:14	20:10
60°	03:49	04:42	05:24	18:26	19:08	20:00
N 58°	03:57	04:46	05:26	18:24	19:04	19:52
56°	04:05	04:51	05:28	18:22	18:59	19:45
54°	04:11	04:54	05:30	18:20	18:56	19:39
52°	04:17	04:58	05:32	18:18	18:53	19:33
50°	04:22	05:01	05:33	18:17	18:50	19:28
45°	04:32	05:07	05:37	18:14	18:43	19:18
N 40°	04:40	05:12	05:39	18:11	18:39	19:11
35°	04:46	05:16	05:42	18:09	18:35	19:04
30°	04:52	05:20	05:44	18:07	18:31	18:59
20°	04:59	05:25	05:47	18:04	18:26	18:52
N 10°	05:04	05:29	05:50	18:01	18:22	18:47
0°	05:08	05:32	05:53	17:59	18:20	18:44
S 10°	05:10	05:34	05:55	17:57	18:18	18:42
20°	05:10	05:36	05:58	17:54	18:16	18:42
30°	05:09	05:36	06:00	17:52	18:16	18:43
35°	05:07	05:37	06:02	17:50	18:15	18:45
40°	05:05	05:37	06:04	17:49	18:16	18:47
45°	05:02	05:36	06:05	17:47	18:16	18:50
S 50°	04:58	05:36	06:08	17:45	18:17	18:54
52°	04:56	05:35	06:09	17:44	18:17	18:57
54°	04:54	05:35	06:10	17:43	18:18	18:59
56°	04:51	05:34	06:11	17:41	18:18	19:02
58°	04:48	05:34	06:13	17:40	18:19	19:05
S 60°	04:44	05:33	06:14	17:39	18:20	19:09

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	■	■	■	■
66°	■	■	19:57	■	■	22:36
64°	■	■	19:10	■	■	23:23
62°	■	18:52	18:39	■	21:38	23:53
60°	17:51	18:10	18:16	20:36	22:20	· ·
N 58°	17:10	17:41	17:57	21:17	22:48	· ·
56°	16:42	17:19	17:41	21:45	23:10	· ·
54°	16:20	17:01	17:27	22:07	23:28	· ·
52°	16:02	16:45	17:15	22:25	23:43	· ·
50°	15:47	16:32	17:05	22:40	23:56	· ·
45°	15:16	16:04	16:42	23:11	· ·	00:23
N 40°	14:51	15:42	16:24	23:35	· ·	00:45
35°	14:32	15:24	16:09	23:54	· ·	01:02
30°	14:14	15:08	15:56	· ·	00:11	01:17
20°	13:46	14:41	15:33	· ·	00:39	01:43
N 10°	13:21	14:18	15:14	00:05	01:04	02:04
0°	12:58	13:57	14:55	00:28	01:26	02:25
S 10°	12:35	13:35	14:36	00:51	01:49	02:45
20°	12:11	13:12	14:16	01:16	02:13	03:06
30°	11:42	12:45	13:53	01:45	02:40	03:31
35°	11:25	12:29	13:40	02:02		

September 15, 16, 17 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	354°27.0	155°41.4	S07°24.0	257°53.3	N23°28.7	274°51.9	N22°23.1	6°56.0	S07°44.2	Alpheratz	357°34.7 29°13.7
1	9°29.4	170°41.1	25.3	272°54.2	28.6	289°54.1	23.1	21°58.6	44.3	Ankaa	353°07.0 -42°10.2
2	24°31.9	185°40.7	26.5	287°55.1	28.6	304°56.4	23.1	37°01.2	44.3	Schedar	349°30.8 56°40.4
3	39°34.3	200°40.4	•• 27.8	302°56.0	•• 28.6	319°58.6	•• 23.1	52°03.9	•• 44.4	Diphda	348°47.2 -17°50.9
4	54°36.8	215°40.0	29.0	317°56.8	28.6	335°00.9	23.1	67°06.5	44.5	Achernar	335°19.8 -57°06.5
5	69°39.3	230°39.7	30.3	332°57.7	28.6	350°03.2	23.1	82°09.2	44.6	Hamal	327°51.2 23°34.8
6	84°41.7	245°39.4	S07°31.5	347°58.6	N23°28.6	5°05.4	N22°23.2	97°11.8	S07°44.6	Polaris	313°53.5 89°21.9
7	99°44.2	260°39.0	32.8	2°59.5	28.6	20°07.7	23.2	112°14.4	44.7	Acamar	315°11.7 -40°12.0
8	114°46.7	275°38.7	34.0	18°00.4	28.5	35°09.9	23.2	127°17.1	44.8	Menkar	314°06.2 4°11.3
9	129°49.1	290°38.3	•• 35.3	33°01.3	•• 28.5	50°12.2	•• 23.2	142°19.7	•• 44.9	Mirfak	308°28.4 49°56.8
10	144°51.6	305°38.0	36.5	48°02.2	28.5	65°14.4	23.2	157°22.4	44.9	Aldebaran	290°39.8 16°33.6
11	159°54.1	320°37.6	37.8	63°03.1	28.5	80°16.7	23.2	172°25.0	45.0	Rigel	281°04.1 -8°10.2
12	174°56.5	335°37.3	S07°39.0	78°04.0	N23°28.5	95°19.0	N22°23.2	187°27.6	S07°45.1	Capella	280°22.3 46°01.2
13	189°59.0	350°36.9	40.3	93°04.9	28.5	110°21.2	23.2	202°30.3	45.2	Bellatrix	278°23.2 6°22.5
14	205°01.4	5°36.6	41.5	108°05.8	28.4	125°23.5	23.3	217°32.9	45.2	Elnath	278°02.2 28°37.7
15	220°03.9	20°36.2	•• 42.8	123°06.7	•• 28.4	140°25.7	•• 23.3	232°35.6	•• 45.3	Alnilam	275°38.0 -1°11.0
16	235°06.4	35°35.9	44.0	138°07.6	28.4	155°28.0	23.3	247°38.2	45.4	Betelgeuse	270°52.4 7°24.8
17	250°08.8	50°35.5	45.2	153°08.5	28.4	170°30.3	23.3	262°40.8	45.4	Canopus	263°52.6 -52°42.1
18	265°11.3	65°35.2	S07°46.5	168°09.4	N23°28.4	185°32.5	N22°23.3	277°43.5	S07°45.5	Sirius	258°26.6 -16°44.7
19	280°13.8	80°34.8	47.7	183°10.3	28.3	200°34.8	23.3	292°46.1	45.6	Adhara	255°06.2 -29°00.0
20	295°16.2	95°34.5	49.0	198°11.2	28.3	215°37.1	23.3	307°48.7	45.7	Procyon	244°51.3 5°09.8
21	310°18.7	110°34.1	•• 50.2	213°12.1	•• 28.3	230°39.3	•• 23.3	322°51.4	•• 45.7	Pollux	243°17.8 27°58.0
22	325°21.2	125°33.8	51.5	228°12.9	28.3	245°41.6	23.4	337°54.0	45.8	Avior	234°15.3 -59°35.0
23	340°23.6	140°33.4	52.7	243°13.8	28.3	260°43.8	23.4	352°56.7	45.9	Suhail	222°46.9 -43°31.7
Mer.pass. 00:22		ν -0.3' d 1.2' m -3.87		ν 0.9' d -0.0' m 0.60		ν 2.3' d 0.0' m -2.35		ν 2.6' d 0.1' m 0.60			

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	355°26.1	155°33.1	S07°54.0	258°14.7	N23°28.2	275°46.1	N22°23.4	7°59.3	S07°46.0
1	10°28.6	170°32.7	55.2	273°15.6	28.2	290°48.4	23.4	23°01.9	46.0
2	25°31.0	185°32.4	56.4	288°16.5	28.2	305°50.6	23.4	38°04.6	46.1
3	40°33.5	200°32.0	•• 57.7	303°17.4	•• 28.2	320°52.9	•• 23.4	53°07.2	•• 46.2
4	55°35.9	215°31.7	07°58.9	318°18.3	28.2	335°55.2	23.4	68°09.9	46.3
5	70°38.4	230°31.3	08°00.2	333°19.2	28.1	350°57.4	23.4	83°12.5	46.3
6	85°40.9	245°31.0	S08°01.4	348°20.1	N23°28.1	5°59.7	N22°23.4	98°15.1	S07°46.4
7	100°43.3	260°30.6	02.7	3°21.0	28.1	21°02.0	23.5	113°17.8	46.5
8	115°45.8	275°30.3	03.9	18°21.9	28.1	36°04.2	23.5	128°20.4	46.6
9	130°48.3	290°29.9	•• 05.1	33°22.8	•• 28.1	51°06.5	•• 23.5	143°23.1	•• 46.6
10	145°50.7	305°29.5	06.4	48°23.7	28.0	66°08.8	23.5	158°25.7	46.7
11	160°53.2	320°29.2	07.6	63°24.6	28.0	81°11.0	23.5	173°28.3	46.8
12	175°55.7	335°28.8	S08°08.9	78°25.5	N23°28.0	96°13.3	N22°23.5	188°31.0	S07°46.9
13	190°58.1	350°28.5	10.1	93°26.5	28.0	111°15.6	23.5	203°33.6	46.9
14	206°00.6	5°28.1	11.3	108°27.4	28.0	126°17.8	23.5	218°36.2	47.0
15	221°03.1	20°27.8	•• 12.6	123°28.3	•• 27.9	141°20.1	•• 23.6	233°38.9	•• 47.1
16	236°05.5	35°27.4	13.8	138°29.2	27.9	156°22.4	23.6	248°41.5	47.1
17	251°08.0	50°27.1	15.1	153°30.1	27.9	171°24.6	23.6	263°44.2	47.2
18	266°10.4	65°26.7	S08°16.3	168°31.0	N23°27.9	186°26.9	N22°23.6	278°46.8	S07°47.3
19	281°12.9	80°26.4	17.5	183°31.9	27.8	201°29.2	23.6	293°49.4	47.4
20	296°15.4	95°26.0	18.8	198°32.8	27.8	216°31.4	23.6	308°52.1	47.4
21	311°17.8	110°25.6	•• 20.0	213°33.7	•• 27.8	231°33.7	•• 23.6	323°54.7	•• 47.5
22	326°20.3	125°25.3	21.2	228°34.6	27.8	246°36.0	23.6	338°57.4	47.6
23	341°22.8	140°24.9	22.5	243°35.5	27.7	261°38.3	23.6	354°00.0	47.7
Mer.pass. 00:18		ν -0.4' d 1.2' m -3.87		ν 0.9' d -0.0' m 0.59		ν 2.3' d 0.0' m -2.36		ν 2.6' d 0.1' m 0.60	

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	356°25.2	155°24.6	S08°23.7	258°36.4	N23°27.7	276°40.5	N22°23.7	9°02.6	S07°47.7
1	11°27.7	170°24.2	25.0	273°37.3	27.7	291°42.8	23.7	24°05.3	47.8
2	26°30.2	185°23.9	26.2	288°38.2	27.6	306°45.1	23.7	39°07.9	47.9
3	41°32.6	200°23.5	•• 27.4	303°39.1	•• 27.7	321°47.3	•• 23.7	54°10.5	•• 48.0
4	56°35.1	215°23.1	28.7	318°40.0	27.6	336°49.6	23.7	69°13.2	48.0
5	71°37.5	230°22.8	29.9	333°40.9	27.6	351°51.9	23.7	84°15.8	48.1
6	86°40.0	245°22.4	S08°31.1	348°41.8	N23°27.6	6°54.2	N22°23.7	99°18.5	S07°48.2
7	101°42.5	260°22.1	32.4	3°42.8	27.5	21°56.4	23.7	114°21.1	48.2
8	116°44.9	275°21.7	33.6	18°43.7	27.5	36°58.7	23.7	129°23.7	48.3
9	131°47.4	290°21.3	•• 34.8	33°44.6	•• 27.5	52°01.0	•• 23.8	144°26.4	•• 48.4
10	146°49.9	305°21.0	36.1	48°45.5	27.5	67°03.3	23.8	159°29.0	48.5
11	161°52.3	320°20.6	37.3	63°46.4	27.4	82°05.5	23.8	174°31.6	48.5
12	176°54.8	335°20.3	S08°38.5	78°47.3	N23°27.4	97°07.8	N22°23.8	189°34.3	S07°48.6
13	191°57.3	350°19.9	39.8	93°48.2	27.4	112°10.1	23.8	204°36.9	48.7
14	206°59.7	5°19.5	41.0	108°49.1	27.3	127°12.4	23.8	219°39.6	48.8
15	222°02.2	20°19.2	•• 42.2	123°50.0	•• 27.3	142°14.6	•• 23.8	234°42.2	•• 48.8
16	237°04.7	35°18.8	43.5	138°50.9	27.3	157°16.9	23.8	249°44.8	48.9
17	252°07.1	50°18.5	44.7	153°51.9	27.3	172°19.2	23.8	264°47.5	49.0
18	267°09.6	65°18.1	S08°45.9	168°52.8	N23°27.2	187°21.5	N22°23.9	279°50.1	S07°49.1
19	282°12.0	80°17.7	47.1	183°53.7	27.2	202°23.8	23.9	294°52.7	49.1
20	297°14.5	95°17.4	48.4	198°54.6	27.2	217°26.0	23.9	309°55.4	49.2
21	312°17.0	110°17.0	•• 49.6	213°55.5	•• 27.1	232°28.3	•• 23.9	324°58.0	•• 49.3
22	327°19.4	125°16.7	50.8	228°56.4	27.1	247°30.6	23.9	340°00.7	49.3
23	342°21.9	140°16.3	52.1	243°57.3	27.1	262°32.9	23.9	355°03.3	49.4
Mer.pass. 00:14		ν -0.4' d 1.2' m -3.88		ν 0.9' d -0.0' m 0.58		ν 2.3' d 0.0' m -2.36		ν 2.6' d 0.1' m 0.60	

Sep 15 Sun	SHA	Mer.pass
Venus	161°14.5	13:38
Mars	263°26.3	06:48
Jupiter	280°24.9	05:40
Saturn	12°29.0	23:28

Sep 16 Mon	SHA	Mer.pass
Venus	160°07.0	13:38
Mars	262°48.7	06:47
Jupiter	280°20.0	05:36
Saturn	12°33.2	23:24

Sep 17 Tue	SHA	Mer.pass
Venus	158°59.3	13:39
Mars	262°11.2	06:45
Jupiter	280°15.3	05:32
Saturn	12°37.4	23:20

Horizontal parallax	
Venus:	0.1
Mars:	0.1

2024 September 15 to Sep. 17 UT

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	02:19	04:02	05:14	18:33	19:43	21:24
N 70°	02:48	04:14	05:18	18:29	19:32	20:57
68°	03:09	04:24	05:22	18:26	19:23	20:37
66°	03:25	04:32	05:25	18:23	19:15	20:21
64°	03:38	04:39	05:27	18:21	19:09	20:09
62°	03:48	04:44	05:29	18:19	19:03	19:59
60°	03:57	04:49	05:31	18:17	18:59	19:50
N 58°	04:05	04:53	05:33	18:15	18:55	19:43
56°	04:11	04:57	05:34	18:14	18:51	19:36
54°	04:17	05:00	05:36	18:13	18:48	19:31
52°	04:22	05:03	05:37	18:11	18:45	19:26
50°	04:27	05:05	05:38	18:10	18:43	19:21
45°	04:36	05:11	05:40	18:08	18:38	19:12
N 40°	04:43	05:15	05:42	18:06	18:34	19:05
35°	04:49	05:19	05:44	18:05	18:30	19:00
30°	04:53	05:21	05:45	18:04	18:28	18:55
20°	05:00	05:26	05:48	18:01	18:23	18:49
N 10°	05:04	05:29	05:50	18:00	18:21	18:45
0°	05:07	05:31	05:51	17:58	18:19	18:43
S 10°	05:08	05:32	05:53	17:56	18:17	18:42
20°	05:07	05:33	05:55	17:55	18:17	18:42
30°	05:05	05:33	05:57	17:53	18:17	18:45
35°	05:03	05:32	05:58	17:52	18:18	18:47
40°	05:00	05:32	05:59	17:51	18:18	18:50
45°	04:57	05:31	06:00	17:50	18:20	18:54
S 50°	04:51	05:29	06:01	17:49	18:21	18:59
52°	04:49	05:28	06:02	17:48	18:22	19:02
54°	04:46	05:27	06:03	17:48	18:23	19:04
56°	04:43	05:26	06:03	17:47	18:24	19:08
58°	04:39	05:25	06:04	17:46	18:26	19:12
S 60°	04:35	05:24	06:05	17:46	18:27	19:16

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	21:19	19:33	18:44	23:17		02:58
N 70°	20:12	19:13	18:37	██	00:22	03:14
68°	19:35	18:58	18:32	██	00:57	03:27
66°	19:09	18:45	18:27		01:22	03:38
64°	18:48	18:34	18:23		01:41	03:47
62°	18:31	18:25	18:19		01:57	03:54
60°	18:17	18:17	18:16	00:15	02:10	04:01
N 58°	18:05	18:10	18:13	00:33	02:21	04:07
56°	17:54	18:03	18:10	00:48	02:30	04:12
54°	17:45	17:58	18:08	01:01	02:39	04:16
52°	17:37	17:53	18:06	01:13	02:46	04:20
50°	17:29	17:48	18:04	01:23	02:53	04:24
45°	17:13	17:38	18:00	01:44	03:07	04:31
N 40°	17:00	17:30	17:56	02:01	03:19	04:38
35°	16:48	17:22	17:53	02:15	03:29	04:43
30°	16:38	17:16	17:51	02:27	03:38	04:48
20°	16:21	17:05	17:46	02:48	03:53	04:57
N 10°	16:06	16:55	17:42	03:05	04:05	05:04
0°	15:51	16:46	17:38	03:22	04:17	05:10
S 10°	15:37	16:36	17:34	03:39	04:29	05:17
20°	15:22	16:26	17:30	03:56	04:42	05:24
30°	15:04	16:15	17:25	04:16	04:56	05:32
35°	14:54	16:08	17:22	04:28	05:04	05:36
40°	14:42	16:00	17:19	04:41	05:13	05:41
45°	14:27	15:52	17:15	04:56	05:24	05:47
S 50°	14:10	15:41	17:11	05:15	05:37	05:54
52°	14:02	15:36	17:09	05:24	05:43	05:57
54°	13:53	15:30	17:07	05:34	05:49	06:01
56°	13:42	15:24	17:04	05:45	05:57	06:05
58°	13:30	15:17	17:02	05:58	06:05	06:09
S 60°	13:16	15:09	16:59	06:13	06:14	06:13

Day	Sun			Moon		
	Eqn. of Time		Mer.	Mer. Pass.		Age
	00 ^h mm:ss	12 ^h mm:ss	Pass hh:mm	Upper hh:mm	Lower hh:mm	12-14 86-98%
15	04:48	04:58	11:55	22:04	09:37	
16	05:09	05:20	11:55	22:58	10:32	
17	05:31	05:41	11:54	23:50	11:24	

September 18, 19, 20 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	357°24.4	155°15.9	S08°53.3	258°58.3	N23°27.0	277°35.1	N22°23.9	10°05.9	S07°49.5	Alpheratz	357°34.7 29°13.7
1	12°26.8	170°15.6	54.5	273°59.2	27.0	292°37.4	23.9	25°08.6	49.6	Ankaa	353°07.0 -42°10.2
2	27°29.3	185°15.2	55.8	289°00.1	27.0	307°39.7	23.9	40°11.2	49.6	Schedar	349°30.8 56°40.4
3	42°31.8	200°14.8	· · 57.0	304°01.0	· · 27.0	322°42.0	· · 24.0	55°13.8	· · 49.7	Diphda	348°47.2 -17°50.9
4	57°34.2	215°14.5	58.2	319°01.9	26.9	337°44.3	24.0	70°16.5	49.8	Achernar	335°19.8 -57°06.5
5	72°36.7	230°14.1	08°59.4	334°02.8	26.9	352°46.6	24.0	85°19.1	49.9	Hamal	327°51.2 23°34.8
6	87°39.1	245°13.7	S09°00.7	349°03.7	N23°26.9	7°48.8	N22°24.0	100°21.8	S07°49.9	Polaris	313°52.4 89°21.9
7	102°41.6	260°13.4	01.9	4°04.7	26.8	22°51.1	24.0	115°24.4	50.0	Acamar	315°11.7 -40°12.0
8	117°44.1	275°13.0	03.1	19°05.6	26.8	37°53.4	24.0	130°27.0	50.1	Menkar	314°06.2 4°11.3
9	132°46.5	290°12.6	· · 04.3	34°06.5	· · 26.8	52°55.7	· · 24.0	145°29.7	· · 50.1	Mirfak	308°28.4 49°56.8
10	147°49.0	305°12.3	05.6	49°07.4	26.7	67°58.0	24.0	160°32.3	50.2	Aldebaran	290°39.8 16°33.6
11	162°51.5	320°11.9	06.8	64°08.3	26.7	83°00.2	24.0	175°34.9	50.3	Rigel	281°04.1 -8°10.2
12	177°53.9	335°11.5	S09°08.0	79°09.3	N23°26.7	98°02.5	N22°24.1	190°37.6	S07°50.4	Capella	280°22.2 46°01.3
13	192°56.4	350°11.2	09.2	94°10.2	26.6	113°04.8	24.1	205°40.2	50.4	Bellatrix	278°23.1 6°22.5
14	207°58.9	5°10.8	10.5	109°11.1	26.6	128°07.1	24.1	220°42.9	50.5	Elnath	278°02.2 28°37.7
15	223°01.3	20°10.4	· · 11.7	124°12.0	· · 26.6	143°09.4	· · 24.1	235°45.5	· · 50.6	Alnilam	275°38.0 -1°11.0
16	238°03.8	35°10.1	12.9	139°12.9	26.5	158°11.7	24.1	250°48.1	50.7	Betelgeuse	270°52.4 7°24.8
17	253°06.3	50°09.7	14.1	154°13.9	26.5	173°14.0	24.1	265°50.8	50.7	Canopus	263°52.6 -52°42.1
18	268°08.7	65°09.3	S09°15.4	169°14.8	N23°26.5	188°16.2	N22°24.1	280°53.4	S07°50.8	Sirius	258°26.6 -16°44.7
19	283°11.2	80°09.0	16.6	184°15.7	26.4	203°18.5	24.1	295°56.0	50.9	Adhara	255°06.2 -29°00.0
20	298°13.6	95°08.6	17.8	199°16.6	26.4	218°20.8	24.1	310°58.7	50.9	Procyon	244°51.3 5°09.8
21	313°16.1	110°08.2	· · 19.0	214°17.5	· · 26.4	233°23.1	· · 24.1	326°01.3	· · 51.0	Pollux	243°17.8 27°58.0
22	328°18.6	125°07.9	20.3	229°18.5	26.3	248°25.4	24.2	341°04.0	51.1	Avior	234°15.3 -59°35.0
23	343°21.0	140°07.5	21.5	244°19.4	26.3	263°27.7	24.2	356°06.6	51.2	Suhail	222°46.9 -43°31.6
Mer.pass. 00:10		ν -0.4' d 1.2' m -3.88		ν 0.9' d -0.0' m 0.58		ν 2.3' d 0.0' m -2.37		ν 2.6' d 0.1' m 0.61		Miaplacidus	221°39.1 -69°48.8
										Alphard	217°48.3 -8°45.7
										Regulus	207°35.0 11°50.9
										Dubhe	193°41.9 61°37.1
										Denebola	182°25.5 14°26.2
										Gienah	175°44.2 -17°40.6
										Acrux	173°01.2 -63°14.1
										Gacrux	171°52.6 -57°15.1
										Alioth	166°13.7 55°49.7
										Spica	158°22.9 -11°17.3
										Alkaid	152°52.6 49°11.6
										Hadar	148°37.0 -60°29.6
										Menkent	147°58.3 -36°29.5
										Arcturus	145°48.4 19°03.4
										Rigel Kent.	139°41.2 -60°56.4
										Kochab	137°20.5 74°03.4
										Zuben'ubi	136°56.5 -16°08.6
										Alphecca	126°04.1 26°38.1
										Antares	112°16.3 -26°29.2
										Atria	107°10.9 -69°04.5
										Sabik	102°03.1 -15°45.3
										Shaula	96°10.7 -37°07.4
										Rasalhague	95°58.8 12°32.7
										Eltanin	90°42.3 51°29.4
										Kaus Aust.	83°32.8 -34°22.5
										Vega	80°33.3 38°48.6
										Nunki	75°48.0 -26°16.0
										Altair	62°00.1 8°56.1
										Peacock	53°05.7 -56°39.5
										Deneb	49°25.7 45°22.3
										Enif	33°38.8 9°59.4
										Al Na'ir	27°32.8 -46°50.5
										Fomalhaut	15°14.4 -29°29.4
										Scheat	13°45.1 28°13.1
										Markab	13°29.8 15°20.4
										Sep 18 Wed	SHA Mer.pass
										Venus	157°51.6 13:39
										Mars	261°33.9 06:44
										Jupiter	280°10.8 05:29
										Saturn	12°41.6 23:16
										Sep 19 Thu	SHA Mer.pass
										Venus	156°43.6 13:40
										Mars	260°56.8 06:42
										Jupiter	280°06.5 05:25
										Saturn	12°45.7 23:11
										Sep 20 Fri	SHA Mer.pass
										Venus	155°35.5 13:40
										Mars	260°19.9 06:41
										Jupiter	280°02.3 05:22
										Saturn	12°49.9 23:07
										Horizontal parallax	
										Venus:	0.1
										Mars:	0.1

Sun				Moon				
Wed	GHA	Dec		GHA	ν	Dec	d	HP
0	181°28.1	N01°45.5		2°24.6	9.9'	S03°25.3	-18.2'	61.2'
1	196°28.3	44.6		16°53.6	9.9'	03°07.1	-18.2'	61.2'
2	211°28.5	43.6		31°22.5	10.0'	02°48.9	-18.2'	61.3'
3	226°28.7	· · 42.6		45°51.5	10.0'	02°30.7	-18.3'	61.3'
4	241°29.0	41.7		60°20.4	10.0'	02°12.4	-18.3'	61.3'
5	256°29.2	40.7		74°49.4	10.0'	01°54.1	-18.3'	61.3'
6	271°29.4	N01°39.7		89°18.4	10.0'	S01°35.9	-18.3'	61.3'
7	286°29.6	38.8		103°47.4	10.0'	01°17.6	-18.3'	61.3'
8	301°29.9	37.8		118°16.3	10.0'	00°59.3	-18.3'	61.3'
9	316°30.1	· · 36.8		132°45.3	10.0'	00°40.9	-18.3'	61.3'
10	331°30.3	35.9		147°14.3	10.0'	00°22.6	-18.3'	61.3'
11	346°30.5	34.9		161°43.3	10.0'	S00°04.3	-18.3'	61.3'
12	1°30.7	N01°33.9		176°12.3	10.0'	N00°14.0	18.3'	61.3'
13	16°31.0	32.9		190°41.3	10.0'	00°32.3	18.3'	61.3'
14	31°31.2	32.0		205°10.3	10.0'	00°50.7	18.3'	61.3'
15	46°31.4	· · 31.0		219°39.3	10.0'	01°09.0	18.3'	61.3'
16	61°31.6	30.0		234°08.3	10.0'	01°27.3	18.3'	61.3'
17	76°31.9	29.1		248°37.2	10.0'	01°45.6	18.3'	61.3'
18	91°32.1	N01°28.1		263°06.2	10.0'	N02°03.9	18.3'	61.3'
19	106°32.3	27.1		277°35.1	9.9'	02°22.1	18.3'	61.3'
20	121°32.5	26.2		292°04.1	9.9'	02°40.4	18.2'	61.3'
21	136°32.8	· · 25.2		306°33.0	9.9'	02°58.6	18.2'	61.3'
22	151°33.0	24.2		321°01.9	9.9'	03°16.8	18.2'	61.3'
23	166°33.2	23.3		335°30.8	9.9'	03°35.0	18.2'	61.3'
SD = 15.9' $d = -1.0'$				SD = 16.7'				
Thu	GHA	Dec		GHA	ν	Dec	d	HP
0	181°33.4	N01°22.3		349°59.7	9.9'	N03°53.2	18.1'	61.3'
1	196°33.7	21.3		4°28.6	9.9'	04°11.4	18.1'	61.3'
2	211°33.9	20.4		18°57.5	9.8'	04°29.5	18.1'	61.2'
3	226°34.1	· · 19.4		33°26.3	9.8'	04°47.6	18.1'	61.2'
4	241°34.3	18.4		47°55.1	9.8'	05°05.6	18.0'	61.2'
5	256°34.6	17.4		62°23.9	9.8'	05°23.6	18.0'	61.2'
6	271°34.8	N01°16.5		76°52.7	9.8'	N05°41.6	17.9'	61.2'
7	286°35.0	15.5		91°21.4	9.7'	05°59.6	17.9'	61.2'
8	301°35.2	14.5		105°50.2	9.7'	06°17.5	17.9'	61.2'
9	316°35.5	· · 13.6		120°18.9	9.7'	06°35.3	17.8'	61.2'
10	331°35.7	12.6		134°47.5	9.6'	06°53.1	17.8'	61.2'
11	346°35.9	11.6		149°16.2	9.6'	07°10.9	17.7'	61.2'
12	1°36.1	N01°10.7		163°44.8	9.6'	N07°28.6	17.7'	61.1'
13	16°36.3	09.7		178°13.4	9.6'	07°46.3	17.6'	61.1'
14	31°36.6	08.7		192°41.9	9.5'	08°03.9	17.6'	61.1'
15	46°36.8	· · 07.7		207°10.5	9.5'	08°21.5	17.5'	61.1'
16	61°37.0	06.8		221°39.0	9.5'	08°39.0	17.4'	61.1'
17	76°37.2	05.8		236°07.4	9.4'	08°56.4	17.4'	61.1'
18	91°37.5	N01°04.8		250°35.8	9.4'	N09°13.8	17.3'	61.1'
19	106°37.7	03.9		265°04.2	9.3'	09°31.1	17.3'	61.0'
20	121°37.9	02.9		279°32.6	9.3'	09°48.4	17.2'	61.0'
21	136°38.1	· · 01.9		294°00.9	9.3'	10°05.5	17.1'	61.0'
22	151°38.4	01.0		308°29.2	9.2'	10°22.7	17.0'	61.0'
23	166°38.6	00.0		322°57.4	9.2'	10°39.7	17.0'	61.0'
SD = 15.9' $d = -1.0'$				SD = 16.7'				
Fri	GHA	Dec		GHA	ν	Dec	d	HP
0	181°38.8	N00°59.0		337°25.6	9.1'	N10°56.7	16.9'	61.0'
1	196°39.0	58.0		351°53.7	9.1'	11°13.6	16.8'	60.9'
2	211°39.2	57.1		6°21.8	9.1'	11°30.4	16.7'	60.9'
3	226°39.5	· · 56.1		20°49.9	9.0'	11°47.1	16.7'	60.9'
4	241°39.7	55.1		35°17.9	9.0'	12°03.8	16.6'	60.9'
5	256°39.9	54.2		49°45.9	8.9'	12°20.4	16.5'	60.9'
6	271°40.1	N00°53.2		64°13.8	8.9'	N12°36.9	16.4'	60.8'
7	286°40.4	52.2		78°41.7	8.8'	12°53.3	16.3'	60.8'
8	301°40.6	51.3		93°09.5	8.8'	13°09.6	16.2'	60.8'
9	316°40.8	· · 50.3		107°37.3	8.7'	13°25.8	16.1'	60.8'
10	331°41.0	49.3		122°05.0	8.7'	13°41.9	16.0'	60.7'
11	346°41.3	48.3		136°32.7	8.6'	13°58.0	15.9'	60.7'
12	1°41.5	N00°47.4		151°00.3	8.6'	N14°13.9	15.8'	60.7'
13	16°41.7	46.4		165°27.9	8.5'	14°29.8	15.7'	60.7'
14	31°41.9	45.4		179°55.4	8.5'	14°45.5	15.6'	60.6'
15	46°42.1	· · 44.5		194°22.9	8.4'	15°01.1	15.5'	60.6'
16	61°42.4	43.5		208°50.3	8.4'	15°16.7	15.4'	60.6'
17	76°42.6	42.5		223°17.7	8.3'	15°32.1	15.3'	60.6'
18	91°42.8	N00°41.5		237°45.0	8.3'	N15°47.4	15.2'	60.5'
19	106°43.0	40.6		252°12.3	8.2'	16°02.7	15.1'	60.5'
20	121°43.3	39.6		266°39.5	8.1'	16°17.8	15.0'	60.5'
21	136°43.5	· · 38.6		281°06.6	8.1'	16°32.8	14.9'	60.5'
22	151°43.7	37.7		295°33.7	8.0'	16°47.6	14.8'	60.4'
23	166°43.9	36.7		310°00.7	8.0'	17°02.4	14.6'	60.4'
SD = 15.9' $d = -1.0'$				SD = 16.6'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	02:41	04:17	05:27	18:18	19:27	21:00
N 70°	03:05	04:27	05:30	18:15	19:17	20:38
68°	03:23	04:35	05:32	18:13	19:09	20:21
66°	03:37	04:42	05:34	18:11	19:03	20:08
64°	03:48	04:48	05:36	18:10	18:58	19:57
62°	03:57	04:52	05:37	18:09	18:53	19:48
60°	04:05	04:56	05:38	18:08	18:49	19:40
N 58°	04:12	05:00	05:39	18:07	18:46	19:33
56°	04:18	05:03	05:40	18:06	18:43	19:28
54°	04:23	05:05	05:41	18:05	18:40	19:23
52°	04:28	05:08	05:42	18:04	18:38	19:18
50°	04:32	05:10	05:42	18:04	18:36	19:14
45°	04:40	05:14	05:44	18:03	18:32	19:06
N 40°	04:46	05:18	05:45	18:01	18:28	19:00
35°	04:51	05:21	05:46	18:01	18:26	18:55
30°	04:55	05:23	05:47	18:00	18:24	18:52
20°	05:01	05:26	05:48	17:59	18:21	18:46
N 10°	05:04	05:28	05:49	17:58	18:19	18:43
0°	05:06	05:30	05:50	17:57	18:17	18:41
S 10°	05:06	05:30	05:51	17:56	18:17	18:41
20°	05:04	05:30	05:52	17:55	18:17	18:43
30°	05:01	05:29	05:53	17:55	18:19	18:47
35°	04:59	05:28	05:53	17:55	18:20	18:49
40°	04:55	05:27	05:54	17:54	18:21	18:53
45°	04:51	05:25	05:54	17:54	18:23	18:57
S 50°	04:45	05:22	05:55	17:54	18:26	19:04
52°	04:42	05:21	05:55	17:53	18:27	19:07
54°	04:38	05:20	05:55	17:53	18:28	19:10
56°	04:35	05:18	05:55	17:53	18:30	19:14
58°	04:30	05:17	05:56	17:53	18:32	19:19
S 60°	04:25	05:15	05:56	17:53	18:34	19:24

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	18:04	17:21	16:19	05:39	08:15	11:12
N 70°	18:07	17:34	16:51	05:41	08:06	10:42
68°	18:09	17:45	17:15	05:44	07:58	10:20
66°	18:11	17:54	17:34	05:46	07:52	10:03
64°	18:12	18:01	17:49	05:47	07:46	09:49
62°	18:13	18:08	18:02	05:48	07:42	09:37
60°	18:15	18:14	18:13	05:50	07:38	09:28
N 58°	18:16	18:19	18:23	05:51	07:34	09:19
56°	18:17	18:23	18:31	05:51	07:31	09:12
54°	18:18	18:27	18:39	05:52	07:28	09:05
52°	18:18	18:31	18:46	05:53	07:26	08:59
50°	18:19	18:34	18:52	05:54	07:24	08:54
45°	18:21	18:42	19:05	05:55	07:19	08:43
N 40°	18:22	18:48	19:17	05:56	07:15	08:33
35°	18:23	18:53	19:26	05:57	07:11	08:25
30°	18:24	18:58	19:35	05:58	07:08	08:18
20°	18:26	19:07	19:49	06:00	07:03	08:06
N 10°	18:28	19:14	20:02	06:01	06:58	07:56
0°	18:29	19:21	20:15	06:02	06:54	07:46
S 10°	18:31	19:28	20:27	06:03	06:49	07:36
20°	18:33	19:36	20:40	06:04	06:45	07:26
30°	18:35	19:45	20:56	06:06	06:39	07:14
35°	18:36	19:50	21:05	06:07	06:36	07:08
40°	18:37	19:56	21:15	06:07	06:33	07:00
45°	18:39	20:03	21:27	06:08	06:29	06:51
S 50°	18:41	20:11	21:42	06:09	06:24	06:41
52°	18:42	20:15	21:49	06:10	06:22	06:36
54°	18:43	20:19	21:57	06:10	06:20	06:30
56°	18:44	20:24	22:06	06:11	06:17	06:24
58°	18:45	20:29	22:16	06:12	06:14	06:18
S 60°	18:47	20:35	22:27	06:12	06:11	06:10

Day	Sun		Mer. Pass hh:mm	Moon		Age 15-17 100-95%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
18	05:52	06:03	11:54	--	12:16	
19	06:14	06:24	11:54	00:41	13:07	
20	06:35	06:46	11:53	01:34	14:00	

September 21, 22, 23 UT (Sat., Sun., Mon.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	0°21.8	154°49.0	S10°20.9	260°05.0	N23°24.3	280°20.2	N22°24.6	13°15.7	S07°54.7					Alpheratz	357°34.7	29°13.7	
1	15°24.2	169°48.6	22.1	275°06.0	24.3	295°22.5	24.6	28°18.4	54.7					Ankaa	353°07.0	-42°10.2	
2	30°26.7	184°48.2	23.3	290°06.9	24.2	310°24.8	24.7	43°21.0	54.8					Schedar	349°30.8	56°40.4	
3	45°29.2	199°47.9	•• 24.5	305°07.9	•• 24.2	325°27.1	•• 24.7	58°23.6	•• 54.9					Diphda	348°47.2	-17°50.9	
4	60°31.6	214°47.5	25.7	320°08.8	24.1	340°29.4	24.7	73°26.3	54.9					Achernar	335°19.8	-57°06.5	
5	75°34.1	229°47.1	26.9	335°09.7	24.1	355°31.7	24.7	88°28.9	55.0					Hamal	327°51.2	23°34.8	
6	90°36.6	244°46.7	S10°28.1	350°10.7	N23°24.0	10°34.0	N22°24.7	103°31.5	S07°55.1					Polaris	313°51.0	89°21.9	
7	105°39.0	259°46.3	29.3	5°11.6	24.0	25°36.3	24.7	118°34.2	55.2					Acamar	315°11.7	-40°12.1	
8	120°41.5	274°45.9	30.5	20°12.6	24.0	40°38.6	24.7	133°36.8	55.2					Menkar	314°06.2	4°11.3	
9	135°44.0	289°45.5	•• 31.7	35°13.5	•• 23.9	55°40.9	•• 24.7	148°39.4	•• 55.3					Mirfak	308°28.3	49°56.8	
10	150°46.4	304°45.2	32.9	50°14.5	23.9	70°43.2	24.7	163°42.1	55.4					Aldebaran	290°39.8	16°33.6	
11	165°48.9	319°44.8	34.1	65°15.4	23.8	85°45.5	24.7	178°44.7	55.4					Rigel	281°04.1	-8°10.2	
12	180°51.3	334°44.4	S10°35.3	80°16.3	N23°23.8	100°47.8	N22°24.7	193°47.3	S07°55.5					Capella	280°22.2	46°01.2	
13	195°53.8	349°44.0	36.5	95°17.3	23.7	115°50.1	24.8	208°50.0	55.6					Bellatrix	278°23.1	6°22.5	
14	210°56.3	4°43.6	37.7	110°18.2	23.7	130°52.4	24.8	223°52.6	55.7					Elnath	278°02.1	28°37.7	
15	225°58.7	19°43.2	•• 38.9	125°19.2	•• 23.6	145°54.8	•• 24.8	238°55.3	•• 55.7					Alnilam	275°38.0	-1°11.0	
16	241°01.2	34°42.8	40.1	140°20.1	23.6	160°57.1	24.8	253°57.9	55.8					Betelgeuse	270°52.4	7°24.8	
17	256°03.7	49°42.4	41.3	155°21.1	23.5	175°59.4	24.8	269°00.5	55.9					Canopus	263°52.6	-52°42.1	
18	271°06.1	64°42.0	S10°42.5	170°22.0	N23°23.5	191°01.7	N22°24.8	284°03.2	S07°55.9					Sirius	258°26.5	-16°44.7	
19	286°08.6	79°41.7	43.7	185°23.0	23.4	206°04.0	24.8	299°05.8	56.0					Adhara	255°06.2	-29°00.0	
20	301°11.1	94°41.3	44.9	200°23.9	23.4	221°06.3	24.8	314°08.4	56.1					Procyon	244°51.2	5°09.8	
21	316°13.5	109°40.9	•• 46.1	215°24.9	•• 23.3	236°08.6	•• 24.8	329°11.1	•• 56.1					Pollux	243°17.8	27°58.0	
22	331°16.0	124°40.5	47.3	230°25.8	23.3	251°10.9	24.8	344°13.7	56.2					Avior	234°15.2	-59°35.0	
23	346°18.5	139°40.1	48.4	245°26.8	23.2	266°13.2	24.8	359°16.3	56.3					Suhail	222°46.8	-43°31.6	
Mer.pass. 23:55			ν -0.4' d 1.2' m -3.88			ν 0.9' d -0.0' m 0.55			ν 2.3' d 0.0' m -2.39			ν 2.6' d 0.1' m 0.62					

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	1°20.9	154°39.7	S10°49.6	260°27.7	N23°23.2	281°15.5	N22°24.8	14°19.0	S07°56.4					
1	16°23.4	169°39.3	50.8	275°28.7	23.1	296°17.9	24.9	29°21.6	56.4					
2	31°25.8	184°38.9	52.0	290°29.6	23.1	311°20.2	24.9	44°24.2	56.5					
3	46°28.3	199°38.5	•• 53.2	305°30.6	•• 23.0	326°22.5	•• 24.9	59°26.9	•• 56.6					
4	61°30.8	214°38.1	54.4	320°31.5	23.0	341°24.8	24.9	74°29.5	56.6					
5	76°33.2	229°37.7	55.6	335°32.5	22.9	356°27.1	24.9	89°32.1	56.7					
6	91°35.7	244°37.3	S10°56.8	350°33.4	N23°22.9	11°29.4	N22°24.9	104°34.8	S07°56.8					
7	106°38.2	259°36.9	58.0	5°34.4	22.8	26°31.7	24.9	119°37.4	56.8					
8	121°40.6	274°36.5	10°59.2	20°35.3	22.8	41°34.1	24.9	134°40.0	56.9					
9	136°43.1	289°36.1	11°00.4	35°36.3	•• 22.7	56°36.4	•• 24.9	149°42.7	•• 57.0					
10	151°45.6	304°35.8	01.6	50°37.2	22.7	71°38.7	24.9	164°45.3	57.1					
11	166°48.0	319°35.4	02.7	65°38.2	22.6	86°41.0	24.9	179°47.9	57.1					
12	181°50.5	334°35.0	S11°03.9	80°39.1	N23°22.6	101°43.3	N22°24.9	194°50.6	S07°57.2					
13	196°52.9	349°34.6	05.1	95°40.1	22.5	116°45.6	25.0	209°53.2	57.3					
14	211°55.4	4°34.2	06.3	110°41.0	22.5	131°47.9	25.0	224°55.8	57.3					
15	226°57.9	19°33.8	•• 07.5	125°42.0	•• 22.4	146°50.3	•• 25.0	239°58.5	•• 57.4					
16	242°00.3	34°33.4	08.7	140°42.9	22.4	161°52.6	25.0	255°01.1	57.5					
17	257°02.8	49°33.0	09.9	155°43.9	22.3	176°54.9	25.0	270°03.7	57.5					
18	272°05.3	64°32.6	S11°11.1	170°44.9	N23°22.3	191°57.2	N22°25.0	285°06.3	S07°57.6					
19	287°07.7	79°32.2	12.2	185°45.8	22.2	206°59.5	25.0	300°09.0	57.7					
20	302°10.2	94°31.8	13.4	200°46.8	22.1	222°01.9	25.0	315°11.6	57.8					
21	317°12.7	109°31.4	•• 14.6	215°47.7	•• 22.1	237°04.2	•• 25.0	330°14.2	•• 57.8					
22	332°15.1	124°31.0	15.8	230°48.7	22.0	252°06.5	25.0	345°16.9	57.9					
23	347°17.6	139°30.6	17.0	245°49.6	22.0	267°08.8	25.0	0°19.5	58.0					
Mer.pass. 23:51			ν -0.4' d 1.2' m -3.89			ν 0.9' d -0.1' m 0.54			ν 2.3' d 0.0' m -2.40			ν 2.6' d 0.1' m 0.62		

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	2°20.1	154°30.2	S11°18.2	260°50.6	N23°21.9	282°11.1	N22°25.0	15°22.1	S07°58.0					
1	17°22.5	169°29.8	19.4	275°51.6	21.9	297°13.5	25.1	30°24.8	58.1					
2	32°25.0	184°29.4	20.5	290°52.5	21.8	312°15.8	25.1	45°27.4	58.2					
3	47°27.4	199°29.0	•• 21.7	305°53.5	•• 21.8	327°18.1	•• 25.1	60°30.0	•• 58.2					
4	62°29.9	214°28.6	22.9	320°54.4	21.7	342°20.4	25.1	75°32.7	58.3					
5	77°32.4	229°28.2	24.1	335°55.4	21.7	357°22.7	25.1	90°35.3	58.4					
6	92°34.8	244°27.8	S11°25.3	350°56.4	N23°21.6	12°25.1	N22°25.1	105°37.9	S07°58.4					
7	107°37.3	259°27.4	26.4	5°57.3	21.5	27°27.4	25.1	120°40.6	58.5					
8	122°39.8	274°27.0	27.6	20°58.3	21.5	42°29.7	25.1	135°43.2	58.6					
9	137°42.2	289°26.6	•• 28.8	35°59.2	•• 21.4	57°32.0	•• 25.1	150°45.8	•• 58.7					
10	152°44.7	304°26.1	30.0	51°00.2	21.4	72°34.4	25.1	165°48.5	58.7					
11	167°47.2	319°25.7	31.2	66°01.2	21.3	87°36.7	25.1	180°51.1	58.8					
12	182°49.6	334°25.3	S11°32.3	81°02.1	N23°21.3	102°39.0	N22°25.1	195°53.7	S07°58.9					
13	197°52.1	349°24.9	33.5	96°03.1	21.2	117°41.3	25.1	210°56.4	58.9					
14	212°54.6	4°24.5	34.7	111°04.1	21.1	132°43.7	25.2	225°59.0	59.0					
15	227°57.0	19°24.1	•• 35.9	126°05.0	•• 21.1	147°46.0	•• 25.2	241°01.6	•• 59.1					
16	242°59.5	34°23.7	37.1	141°06.0	21.0	162°48.3	25.2	256°04.3	59.1					
17	258°01.9	49°23.3	38.2	156°07.0	21.0	177°50.6	25.2	271°06.9	59.2					
18	273°04.4	64°22.9	S11°39.4	171°07.9	N23°20.9	192°53.0	N22°25.2	286°09.5	S07°59.3					
19	288°06.9	79°22.5	40.6	186°08.9	20.9	207°55.3	25.2	301°12.1	59.3					
20	303°09.3	94°22.1	41.8	201°09.8	20.8	222°57.6	25.2	316°14.8	59.4					
21	318°11.8	109°21.7	•• 42.9	216°10.8	•• 20.7	237°59.9	•• 25.2	331°17.4	•• 59.5					
22	333°14.3	124°21.3	44.1	231°11.8	20.7	253°02.3	25.2	346°20.0	59.6					
23	348°16.7	139°20.9	45.3	246°12.8	20.6	268°04.6	25.2	1°22.7	59.6					
Mer.pass. 23:47			ν -0.4' d 1.2' m -3.89			ν 1.0' d -0.1' m 0.53			ν 2.3' d 0.0' m -2.41			ν 2.6' d 0.1' m 0.63		

Sep 21 Sat	SHA	Mer.pass
Venus	154°27.2	13:41
Mars	259°43.3	06:39
Jupiter	279°58.4	05:18
Saturn	12°54.0	23:03

Sep 22 Sun	SHA	Mer.pass
Venus	153°18.8	13:42
Mars	259°06.8	06:38
Jupiter	279°54.6	05:14
Saturn	12°58.0	22:59

Sep 23 Mon	SHA	Mer.pass
Venus	152°10.1	13:42
Mars	258°30.5	06:36
Jupiter	279°51.1	05:10
Saturn	13°02.1	22:55

Horizontal parallax	
Venus:	0.1
Mars:	0.1

h Sun Moon						
Sat	GHA	Dec	GHA	ν	Dec	d HP
0	181°44.1	N00°35.7	324°27.7	7.9'	N17°17.1	14.5' 60.4'
1	196°44.4	34.7	338°54.6	7.9'	17°31.6	14.4' 60.3'
2	211°44.6	33.8	353°21.5	7.8'	17°46.0	14.3' 60.3'
3	226°44.8	· · 32.8	7°48.3	7.7'	18°00.3	14.2' 60.3'
4	241°45.0	31.8	22°15.0	7.7'	18°14.4	14.0' 60.3'
5	256°45.3	30.9	36°41.7	7.6'	18°28.5	13.9' 60.2'
6	271°45.5	N00°29.9	51°08.4	7.6'	N18°42.4	13.8' 60.2'
7	286°45.7	28.9	65°34.9	7.5'	18°56.2	13.7' 60.2'
8	301°45.9	27.9	80°01.4	7.5'	19°09.8	13.5' 60.1'
9	316°46.1	· · 27.0	94°27.9	7.4'	19°23.3	13.4' 60.1'
10	331°46.4	26.0	108°54.3	7.3'	19°36.7	13.2' 60.1'
11	346°46.6	25.0	123°20.6	7.3'	19°50.0	13.1' 60.0'
12	1°46.8	N00°24.1	137°46.9	7.2'	N20°03.1	13.0' 60.0'
13	16°47.0	23.1	152°13.1	7.2'	20°16.1	12.8' 60.0'
14	31°47.3	22.1	166°39.3	7.1'	20°28.9	12.7' 59.9'
15	46°47.5	· · 21.1	181°05.4	7.0'	20°41.6	12.6' 59.9'
16	61°47.7	20.2	195°31.4	7.0'	20°54.1	12.4' 59.9'
17	76°47.9	19.2	209°57.4	6.9'	21°06.6	12.3' 59.8'
18	91°48.1	N00°18.2	224°23.4	6.9'	N21°18.8	12.1' 59.8'
19	106°48.4	17.2	238°49.2	6.8'	21°30.9	12.0' 59.8'
20	121°48.6	16.3	253°15.0	6.8'	21°42.9	11.8' 59.7'
21	136°48.8	· · 15.3	267°40.8	6.7'	21°54.7	11.7' 59.7'
22	151°49.0	14.3	282°06.5	6.7'	22°06.4	11.5' 59.7'
23	166°49.2	13.4	296°32.2	6.6'	22°17.9	11.4' 59.6'
SD = 15.9' d = -1.0' SD = 16.5'						

Sun	GHA	Dec	GHA	ν	Dec	d HP
0	181°49.5	N00°12.4	310°57.8	6.5'	N22°29.3	11.2' 59.6'
1	196°49.7	11.4	325°23.3	6.5'	22°40.5	11.1' 59.6'
2	211°49.9	10.4	339°48.8	6.4'	22°51.6	10.9' 59.5'
3	226°50.1	· · 09.5	354°14.2	6.4'	23°02.5	10.7' 59.5'
4	241°50.3	08.5	8°39.6	6.3'	23°13.2	10.6' 59.5'
5	256°50.6	07.5	23°04.9	6.3'	23°23.8	10.4' 59.4'
6	271°50.8	N00°06.6	37°30.2	6.2'	N23°34.2	10.3' 59.4'
7	286°51.0	05.6	51°55.4	6.2'	23°44.5	10.1' 59.4'
8	301°51.2	04.6	66°20.6	6.1'	23°54.6	9.9' 59.3'
9	316°51.5	· · 03.6	80°45.8	6.1'	24°04.5	9.8' 59.3'
10	331°51.7	02.7	95°10.8	6.0'	24°14.3	9.6' 59.2'
11	346°51.9	01.7	109°35.9	6.0'	24°23.9	9.4' 59.2'
12	1°52.1	N00°00.7	124°00.9	6.0'	N24°33.3	9.3' 59.2'
13	16°52.3	S00°00.3	138°25.8	5.9'	24°42.6	9.1' 59.1'
14	31°52.6	01.2	152°50.7	5.9'	24°51.7	8.9' 59.1'
15	46°52.8	· · 02.2	167°15.6	5.8'	25°00.7	8.8' 59.1'
16	61°53.0	03.2	181°40.4	5.8'	25°09.4	8.6' 59.0'
17	76°53.2	04.1	196°05.2	5.7'	25°18.0	8.4' 59.0'
18	91°53.4	S00°05.1	210°30.0	5.7'	N25°26.5	8.3' 58.9'
19	106°53.7	06.1	224°54.7	5.7'	25°34.7	8.1' 58.9'
20	121°53.9	07.1	239°19.4	5.6'	25°42.8	7.9' 58.9'
21	136°54.1	· · 08.0	253°44.0	5.6'	25°50.7	7.7' 58.8'
22	151°54.3	09.0	268°08.6	5.6'	25°58.5	7.6' 58.8'
23	166°54.5	10.0	282°33.2	5.5'	26°06.0	7.4' 58.8'
SD = 15.9' d = -1.0' SD = 16.3'						

Mon	GHA	Dec	GHA	ν	Dec	d HP
0	181°54.8	S00°11.0	296°57.7	5.5'	N26°13.4	7.2' 58.7'
1	196°55.0	11.9	311°22.2	5.5'	26°20.6	7.0' 58.7'
2	211°55.2	12.9	325°46.7	5.5'	26°27.7	6.9' 58.6'
3	226°55.4	· · 13.9	340°11.2	5.4'	26°34.5	6.7' 58.6'
4	241°55.6	14.8	354°35.6	5.4'	26°41.2	6.5' 58.6'
5	256°55.8	15.8	9°00.1	5.4'	26°47.7	6.3' 58.5'
6	271°56.1	S00°16.8	23°24.5	5.4'	N26°54.1	6.1' 58.5'
7	286°56.3	17.8	37°48.8	5.4'	27°00.2	6.0' 58.5'
8	301°56.5	18.7	52°13.2	5.3'	27°06.2	5.8' 58.4'
9	316°56.7	· · 19.7	66°37.5	5.3'	27°12.0	5.6' 58.4'
10	331°56.9	20.7	81°01.9	5.3'	27°17.6	5.4' 58.3'
11	346°57.2	21.7	95°26.2	5.3'	27°23.0	5.3' 58.3'
12	1°57.4	S00°22.6	109°50.5	5.3'	N27°28.3	5.1' 58.3'
13	16°57.6	23.6	124°14.8	5.3'	27°33.3	4.9' 58.2'
14	31°57.8	24.6	138°39.1	5.3'	27°38.2	4.7' 58.2'
15	46°58.0	· · 25.6	153°03.4	5.3'	27°42.9	4.5' 58.2'
16	61°58.3	26.5	167°27.7	5.3'	27°47.5	4.3' 58.1'
17	76°58.5	27.5	181°52.0	5.3'	27°51.8	4.2' 58.1'
18	91°58.7	S00°28.5	196°16.3	5.3'	N27°56.0	4.0' 58.0'
19	106°58.9	29.4	210°40.5	5.3'	28°00.0	3.8' 58.0'
20	121°59.1	30.4	225°04.8	5.3'	28°03.8	3.6' 58.0'
21	136°59.3	· · 31.4	239°29.1	5.3'	28°07.4	3.4' 57.9'
22	151°59.6	32.4	253°53.5	5.3'	28°10.9	3.3' 57.9'
23	166°59.8	33.3	268°17.8	5.3'	28°14.1	3.1' 57.9'
SD = 15.9' d = 1.0' SD = 16.0'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	03:01	04:32	05:40	18:02	19:10	20:39
N 70°	03:21	04:40	05:42	18:01	19:02	20:20
68°	03:36	04:47	05:43	18:00	18:56	20:06
66°	03:48	04:52	05:43	18:00	18:51	19:54
64°	03:58	04:56	05:44	17:59	18:47	19:45
62°	04:06	05:00	05:45	17:59	18:43	19:37
60°	04:13	05:03	05:45	17:58	18:40	19:30
N 58°	04:19	05:06	05:46	17:58	18:37	19:24
56°	04:24	05:09	05:46	17:58	18:35	19:19
54°	04:29	05:11	05:46	17:58	18:33	19:15
52°	04:33	05:13	05:47	17:57	18:31	19:11
50°	04:36	05:15	05:47	17:57	18:29	19:07
45°	04:44	05:18	05:47	17:57	18:26	19:00
N 40°	04:49	05:21	05:48	17:56	18:23	18:55
35°	04:53	05:23	05:48	17:56	18:21	18:51
30°	04:57	05:25	05:49	17:56	18:20	18:48
20°	05:01	05:27	05:49	17:56	18:18	18:43
N 10°	05:04	05:28	05:49	17:56	18:17	18:41
0°	05:05	05:29	05:49	17:56	18:16	18:40
S 10°	05:04	05:28	05:49	17:56	18:17	18:41
20°	05:02	05:27	05:49	17:56	18:18	18:44
30°	04:57	05:25	05:49	17:56	18:20	18:48
35°	04:54	05:24	05:49	17:57	18:22	18:52
40°	04:50	05:22	05:49	17:57	18:24	18:56
45°	04:45	05:19	05:48	17:57	18:27	19:01
S 50°	04:38	05:16	05:48	17:58	18:30	19:08
52°	04:34	05:14	05:48	17:58	18:32	19:12
54°	04:31	05:12	05:48	17:59	18:34	19:16
56°	04:26	05:10	05:47	17:59	18:36	19:20
58°	04:21	05:08	05:47	17:59	18:39	19:26
S 60°	04:15	05:05	05:47	18:00	18:41	19:32

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	16:18	☐	☐	13:13	☐	☐
66°	17:03	☐	☐	12:29	☐	☐
64°	17:33	16:58	☐	12:00	14:35	☐
62°	17:56	17:47	17:26	11:39	13:47	16:13
60°	18:14	18:18	18:32	11:21	13:17	15:07
N 58°	18:30	18:42	19:06	11:07	12:54	14:33
56°	18:43	19:01	19:32	10:54	12:35	14:08
54°	18:55	19:17	19:52	10:43	12:20	13:48
52°	19:05	19:31	20:09	10:34	12:06	13:31
50°	19:14	19:43	20:24	10:25	11:54	13:16
45°	19:34	20:09	20:54	10:07	11:30	12:47
N 40°	19:50	20:29	21:17	09:53	11:11	12:24
35°	20:03	20:46	21:36	09:40	10:55	12:05
30°	20:15	21:01	21:53	09:30	10:41	11:49
20°	20:36	21:26	22:21	09:11	10:17	11:22
N 10°	20:53	21:48	22:45	08:55	09:57	10:59
0°	21:10	22:08	23:08	08:41	09:38	10:37
S 10°	21:27	22:29	23:30	08:26	09:19	10:15
20°	21:46	22:51	23:55	08:10	08:59	09:52
30°	22:07	23:17	· · ·	07:53	08:36	09:25
35°	22:19	23:32	· · ·	07:42	08:22	09:09
40°	22:34	23:50	· · ·	07:30	08:06	08:50
45°	22:51	· · ·	00:12	07:17	07:48	08:28
S 50°	23:13	· · ·	00:39	07:00	07:25	08:00
52°	23:23	· · ·	00:53	06:52	07:14	07:46
54°	23:35	· · ·	01:08	06:43	07:02	07:30
56°	23:48	· · ·	01:27	06:34	06:48	07:11
58°	· · ·	00:04	01:49	06:23	06:31	06:48
S 60°	· · ·	00:23	02:19	06:10	06:12	06:18

Day	Sun		Mer. Pass	Moon		Age 18-20 88-69%
	Eqn. of Time 00 ^h mm:ss	Time 12 ^h mm:ss		Upper Mer. Pass. hh:mm	Lower Mer. Pass. hh:mm	
21	06:57	07:07	11:53	02:28	14:55	
22	07:18	07:28	11:53	03:24	15:53	
23	07:39	07:50	11:52	04:23	16:52	

September 24, 25, 26 UT (Tue., Wed., Thu.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	3°19.2	154°20.5	S11°46.5	261°13.7	N23°20.6	283°06.9	N22°25.2	16°25.3	S07°59.7					Alpheratz	357°34.7	29°13.7	
1	18°21.7	169°20.0	47.6	276°14.7	20.5	298°09.3	25.2	31°27.9	59.8					Ankaa	353°06.9	-42°10.2	
2	33°24.1	184°19.6	48.8	291°15.7	20.4	313°11.6	25.2	46°30.6	59.8					Schedar	349°30.8	56°40.4	
3	48°26.6	199°19.2	· · 50.0	306°16.6	· · 20.4	328°13.9	· · 25.2	61°33.2	07°59.9					Diphda	348°47.2	-17°50.9	
4	63°29.1	214°18.8	51.1	321°17.6	20.3	343°16.2	25.3	76°35.8	08°00.0					Achernar	335°19.8	-57°06.5	
5	78°31.5	229°18.4	52.3	336°18.6	20.3	358°18.6	25.3	91°38.5	00.0					Hamal	327°51.2	23°34.8	
6	93°34.0	244°18.0	S11°53.5	351°19.5	N23°20.2	13°20.9	N22°25.3	106°41.1	S08°00.1					Polaris	313°49.5	89°21.9	
7	108°36.4	259°17.6	54.7	6°20.5	20.1	28°23.2	25.3	121°43.7	00.2					Acamar	315°11.6	-40°12.1	
8	123°38.9	274°17.2	55.8	21°21.5	20.1	43°25.6	25.3	136°46.3	00.2					Menkar	314°06.2	4°11.3	
9	138°41.4	289°16.8	· · 57.0	36°22.4	· · 20.0	58°27.9	· · 25.3	151°49.0	· · 00.3					Mirfak	308°28.3	49°56.8	
10	153°43.8	304°16.3	58.2	51°23.4	20.0	73°30.2	25.3	166°51.6	00.4					Aldebaran	290°39.8	16°33.6	
11	168°46.3	319°15.9	11°59.3	66°24.4	19.9	88°32.6	25.3	181°54.2	00.4					Rigel	281°04.0	-8°10.2	
12	183°48.8	334°15.5	S12°00.5	81°25.4	N23°19.8	103°34.9	N22°25.3	196°56.9	S08°00.5					Capella	280°22.2	46°01.3	
13	198°51.2	349°15.1	01.7	96°26.3	19.8	118°37.2	25.3	211°59.5	00.6					Bellatrix	278°23.1	6°22.5	
14	213°53.7	4°14.7	02.8	111°27.3	19.7	133°39.6	25.3	227°02.1	00.6					Elnath	278°02.1	28°37.7	
15	228°56.2	19°14.3	· · 04.0	126°28.3	· · 19.7	148°41.9	· · 25.3	242°04.8	· · 00.7					Alnilam	275°37.9	-1°11.0	
16	243°58.6	34°13.9	05.2	141°29.3	19.6	163°44.2	25.3	257°07.4	00.8					Betelgeuse	270°52.3	7°24.8	
17	259°01.1	49°13.4	06.3	156°30.2	19.5	178°46.6	25.3	272°10.0	00.9					Canopus	263°52.5	-52°42.1	
18	274°03.5	64°13.0	S12°07.5	171°31.2	N23°19.5	193°48.9	N22°25.4	287°12.7	S08°00.9					Sirius	258°26.5	-16°44.7	
19	289°06.0	79°12.6	08.7	186°32.2	19.4	208°51.2	25.4	302°15.3	01.0					Adhara	255°06.1	-29°00.0	
20	304°08.5	94°12.2	09.8	201°33.2	19.3	223°53.6	25.4	317°17.9	01.1					Procyon	244°51.2	5°09.8	
21	319°10.9	109°11.8	· · 11.0	216°34.1	· · 19.3	238°55.9	· · 25.4	332°20.5	· · 01.1					Pollux	243°17.7	27°58.0	
22	334°13.4	124°11.4	12.2	231°35.1	19.2	253°58.3	25.4	347°23.2	01.2					Avior	234°15.2	-59°34.9	
23	349°15.9	139°10.9	13.3	246°36.1	19.2	269°00.6	25.4	2°25.8	01.3					Suhail	222°46.8	-43°31.6	
Mer.pass. 23:43			ν -0.4' d 1.2' m -3.89	ν 1.0' d -0.1' m 0.52	ν 2.3' d 0.0' m -2.41	ν 2.6' d 0.1' m 0.63											

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	4°18.3	154°10.5	S12°14.5	261°37.1	N23°19.1	284°02.9	N22°25.4	17°28.4	S08°01.3		
1	19°20.8	169°10.1	15.6	276°38.0	19.0	299°05.3	25.4	32°31.1	01.4		
2	34°23.3	184°09.7	16.8	291°39.0	19.0	314°07.6	25.4	47°33.7	01.5		
3	49°25.7	199°09.3	· · 18.0	306°40.0	· · 18.9	329°09.9	· · 25.4	62°36.3	· · 01.5		
4	64°28.2	214°08.8	19.1	321°41.0	18.8	344°12.3	25.4	77°38.9	01.6		
5	79°30.7	229°08.4	20.3	336°42.0	18.8	359°14.6	25.4	92°41.6	01.7		
6	94°33.1	244°08.0	S12°21.4	351°42.9	N23°18.7	14°17.0	N22°25.4	107°44.2	S08°01.7		
7	109°35.6	259°07.6	22.6	6°43.9	18.6	29°19.3	25.4	122°46.8	01.8		
8	124°38.0	274°07.2	23.8	21°44.9	18.6	44°21.6	25.4	137°49.5	01.9		
9	139°40.5	289°06.7	· · 24.9	36°45.9	· · 18.5	59°24.0	· · 25.5	152°52.1	· · 01.9		
10	154°43.0	304°06.3	26.1	51°46.9	18.4	74°26.3	25.5	167°54.7	02.0		
11	169°45.4	319°05.9	27.2	66°47.8	18.4	89°28.7	25.5	182°57.4	02.1		
12	184°47.9	334°05.5	S12°28.4	81°48.8	N23°18.3	104°31.0	N22°25.5	198°00.0	S08°02.1		
13	199°50.4	349°05.0	29.5	96°49.8	18.3	119°33.3	25.5	213°02.6	02.2		
14	214°52.8	4°04.6	30.7	111°50.8	18.2	134°35.7	25.5	228°05.2	02.3		
15	229°55.3	19°04.2	· · 31.9	126°51.8	· · 18.1	149°38.0	· · 25.5	243°07.9	· · 02.3		
16	244°57.8	34°03.8	33.0	141°52.8	18.1	164°40.4	25.5	258°10.5	02.4		
17	260°00.2	49°03.3	34.2	156°53.7	18.0	179°42.7	25.5	273°13.1	02.5		
18	275°02.7	64°02.9	S12°35.3	171°54.7	N23°17.9	194°45.1	N22°25.5	288°15.8	S08°02.5		
19	290°05.2	79°02.5	36.5	186°55.7	17.9	209°47.4	25.5	303°18.4	02.6		
20	305°07.6	94°02.1	37.6	201°56.7	17.8	224°49.7	25.5	318°21.0	02.7		
21	320°10.1	109°01.6	· · 38.8	216°57.7	· · 17.7	239°52.1	· · 25.5	333°23.6	· · 02.7		
22	335°12.5	124°01.2	39.9	231°58.7	17.7	254°54.4	25.5	348°26.3	02.8		
23	350°15.0	139°00.8	41.1	246°59.7	17.6	269°56.8	25.5	3°28.9	02.9		
Mer.pass. 23:39			ν -0.4' d 1.2' m -3.89	ν 1.0' d -0.1' m 0.51	ν 2.3' d 0.0' m -2.42	ν 2.6' d 0.1' m 0.63					

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	5°17.5	154°00.4	S12°42.2	262°00.6	N23°17.5	284°59.1	N22°25.6	18°31.5	S08°02.9		
1	20°19.9	168°59.9	43.4	277°01.6	17.5	300°01.5	25.6	33°34.2	03.0		
2	35°22.4	183°59.5	44.5	292°02.6	17.4	315°03.8	25.6	48°36.8	03.1		
3	50°24.9	198°59.1	· · 45.7	307°03.6	· · 17.3	330°06.2	· · 25.6	63°39.4	· · 03.1		
4	65°27.3	213°58.6	46.8	322°04.6	17.2	345°08.5	25.6	78°42.0	03.2		
5	80°29.8	228°58.2	48.0	337°05.6	17.2	0°10.9	25.6	93°44.7	03.3		
6	95°32.3	243°57.8	S12°49.1	352°06.6	N23°17.1	15°13.2	N22°25.6	108°47.3	S08°03.3		
7	110°34.7	258°57.4	50.3	7°07.6	17.0	30°15.6	25.6	123°49.9	03.4		
8	125°37.2	273°56.9	51.4	22°08.6	17.0	45°17.9	25.6	138°52.5	03.5		
9	140°39.7	288°56.5	· · 52.6	37°09.5	· · 16.9	60°20.3	· · 25.6	153°55.2	· · 03.5		
10	155°42.1	303°56.1	53.7	52°10.5	16.8	75°22.6	25.6	168°57.8	03.6		
11	170°44.6	318°55.6	54.9	67°11.5	16.8	90°25.0	25.6	184°00.4	03.7		
12	185°47.0	333°55.2	S12°56.0	82°12.5	N23°16.7	105°27.3	N22°25.6	199°03.1	S08°03.7		
13	200°49.5	348°54.8	57.1	97°13.5	16.6	120°29.7	25.6	214°05.7	03.8		
14	215°52.0	3°54.3	58.3	112°14.5	16.6	135°32.0	25.6	229°08.3	03.9		
15	230°54.4	18°53.9	12°59.4	127°15.5	· · 16.5	150°34.4	· · 25.6	244°10.9	· · 03.9		
16	245°56.9	33°53.5	13°00.6	142°16.5	16.4	165°36.7	25.6	259°13.6	04.0		
17	260°59.4	48°53.0	01.7	157°17.5	16.3	180°39.1	25.7	274°16.2	04.1		
18	276°01.8	63°52.6	S13°02.9	172°18.5	N23°16.3	195°41.4	N22°25.7	289°18.8	S08°04.1		
19	291°04.3	78°52.2	04.0	187°19.5	16.2	210°43.8	25.7	304°21.4	04.2		
20	306°06.8	93°51.7	05.1	202°20.5	16.1	225°46.1	25.7	319°24.1	04.3		
21	321°09.2	108°51.3	· · 06.3	217°21.5	· · 16.1	240°48.5	· · 25.7	334°26.7	· · 04.3		
22	336°11.7	123°50.9	07.4	232°22.5	16.0	255°50.8	25.7	349°29.3	04.4		
23	351°14.1	138°50.4	08.6	247°23.5	15.9	270°53.2	25.7	4°32.0	04.5		
Mer.pass. 23:35			ν -0.4' d 1.2' m -3.89	ν 1.0' d -0.1' m 0.50	ν 2.3' d 0.0' m -2.43	ν 2.6' d 0.1' m 0.64					

Stars		
Sep 24 Tue	SHA	Mer.pass
Venus	151°01.3	13:43
Mars	257°54.5	06:35
Jupiter	279°47.7	05:07
Saturn	13°06.1	22:50
Sep 25 Wed	SHA	Mer.pass
Venus	149°52.2	13:44
Mars	257°18.7	06:33
Jupiter	279°44.6	05:03
Saturn	13°10.1	22:46
Sep 26 Thu	SHA	Mer.pass
Venus	148°42.9	13:44
Mars	256°43.2	06:32
Jupiter	279°41.6	04:59
Saturn	13°14.0	22:42
Horizontal parallax		
Venus:	0.1	
Mars:	0.1	

Sun								Moon									
Tue	GHA	Dec	GHA	ν	Dec	d	HP	GHA	ν	Dec	d	HP	GHA	ν	Dec	d	HP
0	182°00.0	S00°34.3	282°42.1	5.3'	N28°17.2	2.9'	57.8'	282°42.1	5.3'	N28°17.2	2.9'	57.8'	282°42.1	5.3'	N28°17.2	2.9'	57.8'
1	197°00.2	35.3	297°06.5	5.4'	28°20.1	2.7'	57.8'	297°06.5	5.4'	28°20.1	2.7'	57.8'	297°06.5	5.4'	28°20.1	2.7'	57.8'
2	212°00.4	36.3	311°30.8	5.4'	28°22.8	2.5'	57.7'	311°30.8	5.4'	28°22.8	2.5'	57.7'	311°30.8	5.4'	28°22.8	2.5'	57.7'
3	227°00.7	37.2	325°55.2	5.4'	28°25.4	2.4'	57.7'	325°55.2	5.4'	28°25.4	2.4'	57.7'	325°55.2	5.4'	28°25.4	2.4'	57.7'
4	242°00.9	38.2	340°19.6	5.4'	28°27.7	2.2'	57.7'	340°19.6	5.4'	28°27.7	2.2'	57.7'	340°19.6	5.4'	28°27.7	2.2'	57.7'
5	257°01.1	39.2	354°44.0	5.4'	28°29.9	2.0'	57.6'	354°44.0	5.4'	28°29.9	2.0'	57.6'	354°44.0	5.4'	28°29.9	2.0'	57.6'
6	272°01.3	S00°40.2	9°08.5	5.5'	N28°31.9	1.8'	57.6'	9°08.5	5.5'	N28°31.9	1.8'	57.6'	9°08.5	5.5'	N28°31.9	1.8'	57.6'
7	287°01.5	41.1	23°32.9	5.5'	28°33.8	1.6'	57.6'	23°32.9	5.5'	28°33.8	1.6'	57.6'	23°32.9	5.5'	28°33.8	1.6'	57.6'
8	302°01.7	42.1	37°57.4	5.5'	28°35.4	1.5'	57.5'	37°57.4	5.5'	28°35.4	1.5'	57.5'	37°57.4	5.5'	28°35.4	1.5'	57.5'
9	317°02.0	43.1	52°22.0	5.6'	28°36.9	1.3'	57.5'	52°22.0	5.6'	28°36.9	1.3'	57.5'	52°22.0	5.6'	28°36.9	1.3'	57.5'
10	332°02.2	44.1	66°46.5	5.6'	28°38.2	1.1'	57.4'	66°46.5	5.6'	28°38.2	1.1'	57.4'	66°46.5	5.6'	28°38.2	1.1'	57.4'
11	347°02.4	45.0	81°11.1	5.6'	28°39.3	0.9'	57.4'	81°11.1	5.6'	28°39.3	0.9'	57.4'	81°11.1	5.6'	28°39.3	0.9'	57.4'
12	2°02.6	S00°46.0	95°35.8	5.7'	N28°40.2	0.8'	57.4'	95°35.8	5.7'	N28°40.2	0.8'	57.4'	95°35.8	5.7'	N28°40.2	0.8'	57.4'
13	17°02.8	47.0	110°00.4	5.7'	28°41.0	0.6'	57.3'	110°00.4	5.7'	28°41.0	0.6'	57.3'	110°00.4	5.7'	28°41.0	0.6'	57.3'
14	32°03.0	47.9	124°25.2	5.8'	28°41.6	0.4'	57.3'	124°25.2	5.8'	28°41.6	0.4'	57.3'	124°25.2	5.8'	28°41.6	0.4'	57.3'
15	47°03.3	48.9	138°49.9	5.8'	28°42.0	0.2'	57.3'	138°49.9	5.8'	28°42.0	0.2'	57.3'	138°49.9	5.8'	28°42.0	0.2'	57.3'
16	62°03.5	49.9	153°14.7	5.8'	28°42.2	0.1'	57.2'	153°14.7	5.8'	28°42.2	0.1'	57.2'	153°14.7	5.8'	28°42.2	0.1'	57.2'
17	77°03.7	50.9	167°39.5	5.9'	28°42.3	-0.1'	57.2'	167°39.5	5.9'	28°42.3	-0.1'	57.2'	167°39.5	5.9'	28°42.3	-0.1'	57.2'
18	92°03.9	S00°51.8	182°04.4	5.9'	N28°42.2	-0.3'	57.2'	182°04.4	5.9'	N28°42.2	-0.3'	57.2'	182°04.4	5.9'	N28°42.2	-0.3'	57.2'
19	107°04.1	52.8	196°29.4	6.0'	28°41.9	-0.5'	57.1'	196°29.4	6.0'	28°41.9	-0.5'	57.1'	196°29.4	6.0'	28°41.9	-0.5'	57.1'
20	122°04.3	53.8	210°54.3	6.0'	28°41.4	-0.6'	57.1'	210°54.3	6.0'	28°41.4	-0.6'	57.1'	210°54.3	6.0'	28°41.4	-0.6'	57.1'
21	137°04.6	54.8	225°19.4	6.1'	28°40.8	-0.8'	57.1'	225°19.4	6.1'	28°40.8	-0.8'	57.1'	225°19.4	6.1'	28°40.8	-0.8'	57.1'
22	152°04.8	55.7	239°44.5	6.1'	28°40.0	-1.0'	57.0'	239°44.5	6.1'	28°40.0	-1.0'	57.0'	239°44.5	6.1'	28°40.0	-1.0'	57.0'
23	167°05.0	56.7	254°09.6	6.2'	28°39.1	-1.1'	57.0'	254°09.6	6.2'	28°39.1	-1.1'	57.0'	254°09.6	6.2'	28°39.1	-1.1'	57.0'
SD = 15.9' d = 1.0'								SD = 15.8'									
Wed	GHA	Dec	GHA	ν	Dec	d	HP	GHA	ν	Dec	d	HP	GHA	ν	Dec	d	HP
0	182°05.2	S00°57.7	268°34.8	6.3'	N28°37.9	-1.3'	57.0'	268°34.8	6.3'	N28°37.9	-1.3'	57.0'	268°34.8	6.3'	N28°37.9	-1.3'	57.0'
1	197°05.4	58.7	283°00.1	6.3'	28°36.6	-1.5'	56.9'	283°00.1	6.3'	28°36.6	-1.5'	56.9'	283°00.1	6.3'	28°36.6	-1.5'	56.9'
2	212°05.6	00°59.6	297°25.4	6.4'	28°35.2	-1.6'	56.9'	297°25.4	6.4'	28°35.2	-1.6'	56.9'	297°25.4	6.4'	28°35.2	-1.6'	56.9'
3	227°05.9	01°00.6	311°50.8	6.5'	28°33.5	-1.8'	56.8'	311°50.8	6.5'	28°33.5	-1.8'	56.8'	311°50.8	6.5'	28°33.5	-1.8'	56.8'
4	242°06.1	01.6	326°16.3	6.5'	28°31.7	-2.0'	56.8'	326°16.3	6.5'	28°31.7	-2.0'	56.8'	326°16.3	6.5'	28°31.7	-2.0'	56.8'
5	257°06.3	02.5	340°41.8	6.6'	28°29.8	-2.1'	56.8'	340°41.8	6.6'	28°29.8	-2.1'	56.8'	340°41.8	6.6'	28°29.8	-2.1'	56.8'
6	272°06.5	S01°03.5	355°07.4	6.7'	N28°27.6	-2.3'	56.7'	355°07.4	6.7'	N28°27.6	-2.3'	56.7'	355°07.4	6.7'	N28°27.6	-2.3'	56.7'
7	287°06.7	04.5	9°33.0	6.7'	28°25.3	-2.5'	56.7'	9°33.0	6.7'	28°25.3	-2.5'	56.7'	9°33.0	6.7'	28°25.3	-2.5'	56.7'
8	302°06.9	05.5	23°58.7	6.8'	28°22.9	-2.6'	56.7'	23°58.7	6.8'	28°22.9	-2.6'	56.7'	23°58.7	6.8'	28°22.9	-2.6'	56.7'
9	317°07.2	06.4	38°24.5	6.9'	28°20.3	-2.8'	56.6'	38°24.5	6.9'	28°20.3	-2.8'	56.6'	38°24.5	6.9'	28°20.3	-2.8'	56.6'
10	332°07.4	07.4	52°50.4	6.9'	28°17.5	-2.9'	56.6'	52°50.4	6.9'	28°17.5	-2.9'	56.6'	52°50.4	6.9'	28°17.5	-2.9'	56.6'
11	347°07.6	08.4	67°16.3	7.0'	28°14.5	-3.1'	56.6'	67°16.3	7.0'	28°14.5	-3.1'	56.6'	67°16.3	7.0'	28°14.5	-3.1'	56.6'
12	2°07.8	S01°09.4	81°42.3	7.1'	N28°11.5	-3.3'	56.6'	81°42.3	7.1'	N28°11.5	-3.3'	56.6'	81°42.3	7.1'	N28°11.5	-3.3'	56.6'
13	17°08.0	10.3	96°08.4	7.2'	28°08.2	-3.4'	56.5'	96°08.4	7.2'	28°08.2	-3.4'	56.5'	96°08.4	7.2'	28°08.2	-3.4'	56.5'
14	32°08.2	11.3	110°34.6	7.2'	28°04.8	-3.6'	56.5'	110°34.6	7.2'	28°04.8	-3.6'	56.5'	110°34.6	7.2'	28°04.8	-3.6'	56.5'
15	47°08.4	12.3	125°00.9	7.3'	28°01.2	-3.7'	56.5'	125°00.9	7.3'	28°01.2	-3.7'	56.5'	125°00.9	7.3'	28°01.2	-3.7'	56.5'
16	62°08.7	13.3	139°27.2	7.4'	27°57.5	-3.9'	56.4'	139°27.2	7.4'	27°57.5	-3.9'	56.4'	139°27.2	7.4'	27°57.5	-3.9'	56.4'
17	77°08.9	14.2	153°53.6	7.5'	27°53.7	-4.0'	56.4'	153°53.6	7.5'	27°53.7	-4.0'	56.4'	153°53.6	7.5'	27°53.7	-4.0'	56.4'
18	92°09.1	S01°15.2	168°20.1	7.6'	N27°49.6	-4.2'	56.4'	168°20.1	7.6'	N27°49.6	-4.2'	56.4'	168°20.1	7.6'	N27°49.6	-4.2'	56.4'
19	107°09.3	16.2	182°46.7	7.7'	27°45.5	-4.3'	56.3'	182°46.7	7.7'	27°45.5	-4.3'	56.3'	182°46.7	7.7'	27°45.5	-4.3'	56.3'
20	122°09.5	17.2	197°13.3	7.7'	27°41.1	-4.5'	56.3'	197°13.3	7.7'	27°41.1	-4.5'	56.3'	197°13.3	7.7'	27°41.1	-4.5'	56.3'
21	137°09.7	18.1	211°40.1	7.8'	27°36.7	-4.6'	56.3'	211°40.1	7.8'	27°36.7	-4.6'	56.3'	211°40.1	7.8'	27°36.7	-4.6'	56.3'
22	152°09.9	19.1	226°06.9	7.9'	27°32.1	-4.8'	56.2'	226°06.9	7.9'	27°32.1	-4.8'	56.2'	226°06.9	7.9'	27°32.1	-4.8'	56.2'
23	167°10.2	20.1	240°33.8	8.0'	27°27.3	-4.9'	56.2'	240°33.8	8.0'	27°27.3	-4.9'	56.2'	240°33.8	8.0'	27°27.3	-4.9'	56.2'
SD = 15.9' d = 1.0'								SD = 15.5'									
Thu	GHA	Dec	GHA	ν	Dec	d	HP	GHA	ν	Dec	d	HP	GHA	ν	Dec	d	HP
0	182°10.4	S01°21.0	255°00.8	8.1'	N27°22.4	-5.0'	56.2'	255°00.8	8.1'	N27°22.4	-5.0'	56.2'	255°00.8	8.1'	N27°22.4	-5.0'	56.2'
1	197°10.6	22.0	269°28.0	8.2'	27°17.3	-5.2'	56.1'	269°28.0	8.2'	27°17.3	-5.2'	56.1'	269°28.0	8.2'	27°17.3	-5.2'	56.1'
2	212°10.8	23.0	283°55.1	8.3'	27°12.2	-5.3'	56.1'	283°55.1	8.3'	27°12.2	-5.3'	56.1'	283°55.1	8.3'	27°12.2	-5.3'	56.1'
3	227°11.0	24.0	298°22.4	8.4'	27°06.8	-5.5'	56.1'	298°22.4	8.4'	27°06.8	-5.5'	56.1'	298°22.4	8.4'	27°06.8	-5.5'	56.1'
4	242°11.2	24.9	312°49.8	8.5'	27°01.3	-5.6'	56.1'	312°49.8	8.5'	2							

September 27, 28, 29 UT (Fri., Sat., Sun.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	6°16.6	153°50.0	S13°09.7	262°24.5	N23°15.8	285°55.5	N22°25.7	19°34.6	S08°04.5	Alpheratz	357°34.7	29°13.7					
1	21°19.1	168°49.5	10.8	277°25.5	15.8	300°57.9	25.7	34°37.2	04.6	Ankaa	353°06.9	-42°10.2					
2	36°21.5	183°49.1	12.0	292°26.5	15.7	316°00.2	25.7	49°39.8	04.7	Schedar	349°30.8	56°40.4					
3	51°24.0	198°48.7	•• 13.1	307°27.5	•• 15.6	331°02.6	•• 25.7	64°42.5	•• 04.7	Diphda	348°47.2	-17°50.9					
4	66°26.5	213°48.2	14.3	322°28.5	15.6	346°05.0	25.7	79°45.1	04.8	Achernar	335°19.8	-57°06.5					
5	81°28.9	228°47.8	15.4	337°29.5	15.5	1°07.3	25.7	94°47.7	04.9	Hamal	327°51.2	23°34.8					
6	96°31.4	243°47.3	S13°16.5	352°30.5	N23°15.4	16°09.7	N22°25.7	109°50.3	S08°04.9	Polaris	313°48.3	89°21.9					
7	111°33.9	258°46.9	17.7	7°31.5	15.3	31°12.0	25.7	124°53.0	05.0	Acamar	315°11.6	-40°12.1					
8	126°36.3	273°46.5	18.8	22°32.5	15.3	46°14.4	25.7	139°55.6	05.1	Menkar	314°06.2	4°11.3					
9	141°38.8	288°46.0	•• 19.9	37°33.5	•• 15.2	61°16.7	•• 25.7	154°58.2	•• 05.1	Mirfak	308°28.3	49°56.9					
10	156°41.3	303°45.6	21.1	52°34.5	15.1	76°19.1	25.8	170°00.8	05.2	Aldebaran	290°39.8	16°33.6					
11	171°43.7	318°45.1	22.2	67°35.5	15.1	91°21.5	25.8	185°03.5	05.3	Rigel	281°04.0	-8°10.2					
12	186°46.2	333°44.7	S13°23.3	82°36.5	N23°15.0	106°23.8	N22°25.8	200°06.1	S08°05.3	Capella	280°22.1	46°01.3					
13	201°48.6	348°44.3	24.5	97°37.5	14.9	121°26.2	25.8	215°08.7	05.4	Bellatrix	278°23.1	6°22.5					
14	216°51.1	3°43.8	25.6	112°38.5	14.8	136°28.5	25.8	230°11.3	05.5	Elnath	278°02.1	28°37.7					
15	231°53.6	18°43.4	•• 26.7	127°39.5	•• 14.8	151°30.9	•• 25.8	245°14.0	•• 05.5	Alnilam	275°37.9	-1°11.0					
16	246°56.0	33°42.9	27.9	142°40.5	14.7	166°33.2	25.8	260°16.6	05.6	Betelgeuse	270°52.3	7°24.8					
17	261°58.5	48°42.5	29.0	157°41.5	14.6	181°35.6	25.8	275°19.2	05.7	Canopus	263°52.5	-52°42.1					
18	277°01.0	63°42.0	S13°30.1	172°42.5	N23°14.5	196°38.0	N22°25.8	290°21.8	S08°05.7	Sirius	258°26.5	-16°44.7					
19	292°03.4	78°41.6	31.2	187°43.5	14.5	211°40.3	25.8	305°24.5	05.8	Adhara	255°06.1	-29°00.0					
20	307°05.9	93°41.1	32.4	202°44.5	14.4	226°42.7	25.8	320°27.1	05.9	Procyon	244°51.2	5°09.8					
21	322°08.4	108°40.7	•• 33.5	217°45.5	•• 14.3	241°45.1	•• 25.8	335°29.7	•• 05.9	Pollux	243°17.7	27°58.0					
22	337°10.8	123°40.3	34.6	232°46.5	14.2	256°47.4	25.8	350°32.3	06.0	Avior	234°15.2	-59°34.9					
23	352°13.3	138°39.8	35.7	247°47.5	14.2	271°49.8	25.8	5°35.0	06.1	Suhail	222°46.8	-43°31.6					
Mer.pass. 23:31			ν -0.4' d 1.1' m -3.90			ν 1.0' d -0.1' m 0.50			ν 2.4' d 0.0' m -2.44			ν 2.6' d 0.1' m 0.64					

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	7°15.8	153°39.4	S13°36.9	262°48.5	N23°14.1	286°52.1	N22°25.8	20°37.6	S08°06.1					
1	22°18.2	168°38.9	38.0	277°49.5	14.0	301°54.5	25.8	35°40.2	06.2					
2	37°20.7	183°38.5	39.1	292°50.5	13.9	316°56.9	25.8	50°42.8	06.3					
3	52°23.1	198°38.0	•• 40.3	307°51.6	•• 13.9	331°59.2	•• 25.8	65°45.5	•• 06.3					
4	67°25.6	213°37.6	41.4	322°52.6	13.8	347°01.6	25.8	80°48.1	06.4					
5	82°28.1	228°37.1	42.5	337°53.6	13.7	2°04.0	25.9	95°50.7	06.4					
6	97°30.5	243°36.7	S13°43.6	352°54.6	N23°13.6	17°06.3	N22°25.9	110°53.3	S08°06.5					
7	112°33.0	258°36.2	44.7	7°55.6	13.6	32°08.7	25.9	125°56.0	06.6					
8	127°35.5	273°35.8	45.9	22°56.6	13.5	47°11.1	25.9	140°58.6	06.6					
9	142°37.9	288°35.3	•• 47.0	37°57.6	•• 13.4	62°13.4	•• 25.9	156°01.2	•• 06.7					
10	157°40.4	303°34.9	48.1	52°58.6	13.3	77°15.8	25.9	171°03.8	06.8					
11	172°42.9	318°34.4	49.2	67°59.6	13.2	92°18.2	25.9	186°06.5	06.8					
12	187°45.3	333°34.0	S13°50.3	83°00.6	N23°13.2	107°20.5	N22°25.9	201°09.1	S08°06.9					
13	202°47.8	348°33.5	51.5	98°01.7	13.1	122°22.9	25.9	216°11.7	07.0					
14	217°50.3	3°33.1	52.6	113°02.7	13.0	137°25.3	25.9	231°14.3	07.0					
15	232°52.7	18°32.6	•• 53.7	128°03.7	•• 12.9	152°27.6	•• 25.9	246°17.0	•• 07.1					
16	247°55.2	33°32.1	54.8	143°04.7	12.9	167°30.0	25.9	261°19.6	07.2					
17	262°57.6	48°31.7	55.9	158°05.7	12.8	182°32.4	25.9	276°22.2	07.2					
18	278°00.1	63°31.2	S13°57.1	173°06.7	N23°12.7	197°34.7	N22°25.9	291°24.8	S08°07.3					
19	293°02.6	78°30.8	58.2	188°07.8	12.6	212°37.1	25.9	306°27.5	07.4					
20	308°05.0	93°30.3	13°59.3	203°08.8	12.5	227°39.5	25.9	321°30.1	07.4					
21	323°07.5	108°29.9	14°00.4	218°09.8	•• 12.5	242°41.9	•• 25.9	336°32.7	•• 07.5					
22	338°10.0	123°29.4	01.5	233°10.8	12.4	257°44.2	25.9	351°35.3	07.5					
23	353°12.4	138°29.0	02.6	248°11.8	12.3	272°46.6	25.9	6°37.9	07.6					
Mer.pass. 23:27			ν -0.4' d 1.1' m -3.90			ν 1.0' d -0.1' m 0.49			ν 2.4' d 0.0' m -2.44			ν 2.6' d 0.1' m 0.64		

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	8°14.9	153°28.5	S14°03.7	263°12.8	N23°12.2	287°49.0	N22°25.9	21°40.6	S08°07.7					
1	23°17.4	168°28.0	04.9	278°13.9	12.1	302°51.3	25.9	36°43.2	07.7					
2	38°19.8	183°27.6	06.0	293°14.9	12.1	317°53.7	26.0	51°45.8	07.8					
3	53°22.3	198°27.1	•• 07.1	308°15.9	•• 12.0	332°56.1	•• 26.0	66°48.4	•• 07.9					
4	68°24.7	213°26.7	08.2	323°16.9	11.9	347°58.5	26.0	81°51.1	07.9					
5	83°27.2	228°26.2	09.3	338°17.9	11.8	3°00.8	26.0	96°53.7	08.0					
6	98°29.7	243°25.7	S14°10.4	353°19.0	N23°11.7	18°03.2	N22°26.0	111°56.3	S08°08.1					
7	113°32.1	258°25.3	11.5	8°20.0	11.7	33°05.6	26.0	126°58.9	08.1					
8	128°34.6	273°24.8	12.6	23°21.0	11.6	48°08.0	26.0	142°01.6	08.2					
9	143°37.1	288°24.4	•• 13.7	38°22.0	•• 11.5	63°10.3	•• 26.0	157°04.2	•• 08.3					
10	158°39.5	303°23.9	14.8	53°23.0	11.4	78°12.7	26.0	172°06.8	08.3					
11	173°42.0	318°23.4	15.9	68°24.1	11.3	93°15.1	26.0	187°09.4	08.4					
12	188°44.5	333°23.0	S14°17.1	83°25.1	N23°11.3	108°17.5	N22°26.0	202°12.0	S08°08.4					
13	203°46.9	348°22.5	18.2	98°26.1	11.2	123°19.8	26.0	217°14.7	08.5					
14	218°49.4	3°22.1	19.3	113°27.1	11.1	138°22.2	26.0	232°17.3	08.6					
15	233°51.9	18°21.6	•• 20.4	128°28.2	•• 11.0	153°24.6	•• 26.0	247°19.9	•• 08.6					
16	248°54.3	33°21.1	21.5	143°29.2	10.9	168°27.0	26.0	262°22.5	08.7					
17	263°56.8	48°20.7	22.6	158°30.2	10.9	183°29.3	26.0	277°25.2	08.8					
18	278°59.2	63°20.2	S14°23.7	173°31.2	N23°10.8	198°31.7	N22°26.0	292°27.8	S08°08.8					
19	294°01.7	78°19.7	24.8	188°32.3	10.7	213°34.1	26.0	307°30.4	08.9					
20	309°04.2	93°19.3	25.9	203°33.3	10.6	228°36.5	26.0	322°33.0	09.0					
21	324°06.6	108°18.8	•• 27.0	218°34.3	•• 10.5	243°38.9	•• 26.0	337°35.6	•• 09.0					
22	339°09.1	123°18.3	28.1	233°35.3	10.4	258°41.2	26.0	352°38.3	09.1					
23	354°11.6	138°17.9	29.2	248°36.4	10.4	273°43.6	26.0	7°40.9	09.1					
Mer.pass. 23:23			ν -0.5' d 1.1' m -3.90			ν 1.0' d -0.1' m 0.48			ν 2.4' d 0.0' m -2.45			ν 2.6' d 0.1' m 0.65		

Sep 27 Fri	SHA	Mer.pass
Venus	147°33.4	13:45
Mars	256°07.8	06:30
Jupiter	279°38.9	04:56
Saturn	13°18.0	22:38

Sep 28 Sat	SHA	Mer.pass
Venus	146°23.6	13:46
Mars	255°32.8	06:28
Jupiter	279°36.4	04:52
Saturn	13°21.8	22:34

Sep 29 Sun	SHA	Mer.pass
Venus	145°13.6	13:47
Mars	254°57.9	06:27
Jupiter	279°34.1	04:48
Saturn	13°25.7	22:29

Horizontal parallax		
Venus:	0.1	
Mars:	0.1	

Sun				Moon				
Fri	GHA	Dec		GHA	ν	Dec	d	HP
0	182°15.5	S01°44.4		242°17.5	10.4'	N24°44.3	-8.1'	55.5'
1	197°15.7	45.4		256°46.9	10.5'	24°36.2	-8.2'	55.5'
2	212°15.9	46.4		271°16.4	10.6'	24°27.9	-8.4'	55.5'
3	227°16.1	· · 47.3		285°46.0	10.7'	24°19.5	-8.5'	55.4'
4	242°16.3	48.3		300°15.7	10.8'	24°11.1	-8.6'	55.4'
5	257°16.5	49.3		314°45.5	10.9'	24°02.5	-8.7'	55.4'
6	272°16.8	S01°50.2		329°15.5	11.0'	N23°53.8	-8.8'	55.4'
7	287°17.0	51.2		343°45.5	11.1'	23°45.0	-8.9'	55.3'
8	302°17.2	52.2		358°15.6	11.2'	23°36.1	-9.0'	55.3'
9	317°17.4	· · 53.2		12°45.8	11.3'	23°27.1	-9.1'	55.3'
10	332°17.6	54.1		27°16.1	11.4'	23°18.0	-9.2'	55.3'
11	347°17.8	55.1		41°46.5	11.5'	23°08.8	-9.3'	55.2'
12	2°18.0	S01°56.1		56°17.0	11.6'	N22°59.5	-9.4'	55.2'
13	17°18.2	57.1		70°47.6	11.7'	22°50.1	-9.5'	55.2'
14	32°18.4	58.0		85°18.3	11.8'	22°40.6	-9.6'	55.2'
15	47°18.7	01°59.0		99°49.1	11.9'	22°30.9	-9.7'	55.2'
16	62°18.9	02°00.0		114°20.0	12.0'	22°21.2	-9.8'	55.1'
17	77°19.1	00.9		128°51.0	12.1'	22°11.4	-9.9'	55.1'
18	92°19.3	S02°01.9		143°22.1	12.2'	N22°01.6	-10.0'	55.1'
19	107°19.5	02.9		157°53.3	12.3'	21°51.6	-10.1'	55.1'
20	122°19.7	03.9		172°24.5	12.4'	21°41.5	-10.2'	55.1'
21	137°19.9	· · 04.8		186°55.9	12.5'	21°31.3	-10.3'	55.0'
22	152°20.1	05.8		201°27.4	12.6'	21°21.1	-10.3'	55.0'
23	167°20.3	06.8		215°58.9	12.6'	21°10.7	-10.4'	55.0'
SD = 15.9'			d = 1.0'	SD = 15.1'				
Sat	GHA	Dec		GHA	ν	Dec	d	HP
0	182°20.5	S02°07.8		230°30.6	12.7'	N21°00.3	-10.5'	55.0'
1	197°20.8	08.7		245°02.3	12.8'	20°49.8	-10.6'	54.9'
2	212°21.0	09.7		259°34.1	12.9'	20°39.2	-10.7'	54.9'
3	227°21.2	· · 10.7		274°06.0	13.0'	20°28.5	-10.8'	54.9'
4	242°21.4	11.6		288°38.1	13.1'	20°17.8	-10.8'	54.9'
5	257°21.6	12.6		303°10.1	13.2'	20°06.9	-10.9'	54.9'
6	272°21.8	S02°13.6		317°42.3	13.3'	N19°56.0	-11.0'	54.9'
7	287°22.0	14.6		332°14.6	13.4'	19°45.0	-11.1'	54.8'
8	302°22.2	15.5		346°47.0	13.5'	19°33.9	-11.2'	54.8'
9	317°22.4	· · 16.5		1°19.4	13.5'	19°22.7	-11.2'	54.8'
10	332°22.6	17.5		15°52.0	13.6'	19°11.5	-11.3'	54.8'
11	347°22.8	18.5		30°24.6	13.7'	19°00.2	-11.4'	54.8'
12	2°23.0	S02°19.4		44°57.3	13.8'	N18°48.8	-11.5'	54.7'
13	17°23.3	20.4		59°30.1	13.9'	18°37.4	-11.5'	54.7'
14	32°23.5	21.4		74°03.0	14.0'	18°25.9	-11.6'	54.7'
15	47°23.7	· · 22.3		88°35.9	14.0'	18°14.3	-11.7'	54.7'
16	62°23.9	23.3		103°09.0	14.1'	18°02.6	-11.7'	54.7'
17	77°24.1	24.3		117°42.1	14.2'	17°50.9	-11.8'	54.7'
18	92°24.3	S02°25.3		132°15.3	14.3'	N17°39.1	-11.9'	54.6'
19	107°24.5	26.2		146°48.6	14.4'	17°27.2	-11.9'	54.6'
20	122°24.7	27.2		161°21.9	14.4'	17°15.3	-12.0'	54.6'
21	137°24.9	· · 28.2		175°55.3	14.5'	17°03.3	-12.1'	54.6'
22	152°25.1	29.1		190°28.8	14.6'	16°51.3	-12.1'	54.6'
23	167°25.3	30.1		205°02.4	14.7'	16°39.2	-12.2'	54.6'
SD = 16.0'			d = 1.0'	SD = 15.0'				
Sun	GHA	Dec		GHA	ν	Dec	d	HP
0	182°25.5	S02°31.1		219°36.1	14.7'	N16°27.0	-12.2'	54.5'
1	197°25.7	32.1		234°09.8	14.8'	16°14.7	-12.3'	54.5'
2	212°25.9	33.0		248°43.6	14.9'	16°02.5	-12.3'	54.5'
3	227°26.2	· · 34.0		263°17.5	15.0'	15°50.1	-12.4'	54.5'
4	242°26.4	35.0		277°51.5	15.0'	15°37.7	-12.5'	54.5'
5	257°26.6	36.0		292°25.5	15.1'	15°25.2	-12.5'	54.5'
6	272°26.8	S02°36.9		306°59.6	15.2'	N15°12.7	-12.6'	54.5'
7	287°27.0	37.9		321°33.7	15.2'	15°00.2	-12.6'	54.4'
8	302°27.2	38.9		336°08.0	15.3'	14°47.5	-12.7'	54.4'
9	317°27.4	· · 39.8		350°42.3	15.4'	14°34.9	-12.7'	54.4'
10	332°27.6	40.8		5°16.6	15.4'	14°22.1	-12.8'	54.4'
11	347°27.8	41.8		19°51.0	15.5'	14°09.4	-12.8'	54.4'
12	2°28.0	S02°42.8		34°25.5	15.6'	N13°56.5	-12.9'	54.4'
13	17°28.2	43.7		49°00.1	15.6'	13°43.7	-12.9'	54.4'
14	32°28.4	44.7		63°34.7	15.7'	13°30.7	-13.0'	54.3'
15	47°28.6	· · 45.7		78°09.4	15.7'	13°17.8	-13.0'	54.3'
16	62°28.8	46.6		92°44.1	15.8'	13°04.8	-13.1'	54.3'
17	77°29.0	47.6		107°18.9	15.9'	12°51.7	-13.1'	54.3'
18	92°29.2	S02°48.6		121°53.7	15.9'	N12°38.6	-13.1'	54.3'
19	107°29.4	49.5		136°28.6	16.0'	12°25.5	-13.2'	54.3'
20	122°29.6	50.5		151°03.6	16.0'	12°12.3	-13.2'	54.3'
21	137°29.8	· · 51.5		165°38.6	16.1'	11°59.1	-13.3'	54.3'
22	152°30.1	52.5		180°13.7	16.1'	11°45.8	-13.3'	54.3'
23	167°30.3	53.4		194°48.8	16.2'	11°32.5	-13.3'	54.2'
SD = 16.0'			d = 1.0'	SD = 14.9'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	03:36	05:00	06:07	17:31	18:38	20:01
N 70°	03:50	05:04	06:05	17:34	18:34	19:48
68°	04:01	05:08	06:04	17:35	18:30	19:37
66°	04:10	05:11	06:02	17:37	18:28	19:29
64°	04:17	05:14	06:01	17:38	18:25	19:21
62°	04:23	05:16	06:00	17:39	18:23	19:15
60°	04:29	05:18	05:59	17:40	18:22	19:10
N 58°	04:33	05:19	05:59	17:41	18:20	19:06
56°	04:37	05:21	05:58	17:42	18:19	19:02
54°	04:41	05:22	05:57	17:43	18:18	18:59
52°	04:43	05:23	05:57	17:43	18:17	18:56
50°	04:46	05:24	05:56	17:44	18:16	18:54
45°	04:51	05:25	05:55	17:45	18:15	18:49
N 40°	04:55	05:27	05:54	17:47	18:14	18:45
35°	04:58	05:27	05:53	17:48	18:13	18:42
30°	05:00	05:28	05:52	17:49	18:12	18:40
20°	05:03	05:28	05:50	17:50	18:12	18:38
N 10°	05:03	05:28	05:49	17:52	18:13	18:37
0°	05:03	05:27	05:47	17:54	18:14	18:38
S 10°	05:00	05:25	05:46	17:55	18:16	18:41
20°	04:56	05:22	05:44	17:57	18:19	18:45
30°	04:50	05:18	05:42	18:00	18:24	18:52
35°	04:45	05:15	05:40	18:01	18:27	18:56
40°	04:40	05:12	05:39	18:03	18:30	19:02
45°	04:33	05:08	05:37	18:05	18:34	19:09
S 50°	04:24	05:02	05:35	18:07	18:40	19:18
52°	04:19	05:00	05:34	18:08	18:42	19:23
54°	04:15	04:57	05:33	18:09	18:45	19:28
56°	04:09	04:54	05:31	18:11	18:48	19:34
58°	04:03	04:51	05:30	18:12	18:52	19:40
S 60°	03:55	04:47	05:28	18:14	18:56	19:48

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	☐	22:54	· · ·	☐	20:02	18:44
N 70°	☐	23:45	· · ·	☐	19:09	18:23
68°	21:33	· · ·	00:17	19:46	18:36	18:06
66°	22:34	· · ·	00:40	18:44	18:12	17:52
64°	23:08	· · ·	00:58	18:09	17:53	17:41
62°	23:32	· · ·	01:12	17:43	17:37	17:31
60°	23:52	· · ·	01:24	17:23	17:24	17:23
N 58°	· · ·	00:08	01:35	17:07	17:12	17:16
56°	· · ·	00:21	01:44	16:52	17:02	17:09
54°	· · ·	00:33	01:52	16:40	16:54	17:03
52°	· · ·	00:43	01:59	16:29	16:46	16:58
50°	· · ·	00:52	02:06	16:20	16:39	16:53
45°	00:02	01:12	02:20	15:59	16:23	16:43
N 40°	00:21	01:27	02:31	15:42	16:10	16:34
35°	00:38	01:40	02:41	15:28	16:00	16:27
30°	00:52	01:52	02:49	15:16	15:50	16:20
20°	01:16	02:11	03:04	14:55	15:34	16:08
N 10°	01:36	02:28	03:16	14:36	15:19	15:58
0°	01:55	02:44	03:28	14:1		

September 30, 01, 02 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	9°14.0	153°17.4	S14°30.3	263°37.4	N23°10.3	288°46.0	N22°26.0	22°43.5	S08°09.2	Alpheratz	357°34.7 29°13.7
1	24°16.5	168°16.9	31.4	278°38.4	10.2	303°48.4	26.0	37°46.1	09.3	Ankaa	353°06.9 -42°10.2
2	39°19.0	183°16.5	32.5	293°39.5	10.1	318°50.8	26.1	52°48.7	09.3	Schedar	349°30.8 56°40.4
3	54°21.4	198°16.0	· · 33.6	308°40.5	· · 10.0	333°53.1	· · 26.1	67°51.4	· · 09.4	Diphda	348°47.2 -17°50.9
4	69°23.9	213°15.5	34.7	323°41.5	09.9	348°55.5	26.1	82°54.0	09.5	Achernar	335°19.8 -57°06.5
5	84°26.3	228°15.0	35.8	338°42.6	09.9	3°57.9	26.1	97°56.6	09.5	Hamal	327°51.2 23°34.8
6	99°28.8	243°14.6	S14°36.9	353°43.6	N23°09.8	19°00.3	N22°26.1	112°59.2	S08°09.6	Polaris	313°47.3 89°21.9
7	114°31.3	258°14.1	38.0	8°44.6	09.7	34°02.7	26.1	128°01.9	09.7	Acamar	315°11.6 -40°12.1
8	129°33.7	273°13.6	39.1	23°45.7	09.6	49°05.1	26.1	143°04.5	09.7	Menkar	314°06.1 4°11.3
9	144°36.2	288°13.2	· · 40.1	38°46.7	· · 09.5	64°07.4	· · 26.1	158°07.1	· · 09.8	Mirfak	308°28.3 49°56.9
10	159°38.7	303°12.7	41.2	53°47.7	09.4	79°09.8	26.1	173°09.7	09.8	Aldebaran	290°39.7 16°33.6
11	174°41.1	318°12.2	42.3	68°48.7	09.4	94°12.2	26.1	188°12.3	09.9	Rigel	281°04.0 -8°10.2
12	189°43.6	333°11.7	S14°43.4	83°49.8	N23°09.3	109°14.6	N22°26.1	203°15.0	S08°10.0	Capella	280°22.1 46°01.3
13	204°46.1	348°11.3	44.5	98°50.8	09.2	124°17.0	26.1	218°17.6	10.0	Bellatrix	278°23.0 6°22.5
14	219°48.5	3°10.8	45.6	113°51.9	09.1	139°19.4	26.1	233°20.2	10.1	Elnath	278°02.1 28°37.7
15	234°51.0	18°10.3	· · 46.7	128°52.9	· · 09.0	154°21.7	· · 26.1	248°22.8	· · 10.2	Alnilam	275°37.9 -1°11.0
16	249°53.5	33°09.8	47.8	143°53.9	08.9	169°24.1	26.1	263°25.4	10.2	Betelgeuse	270°52.3 7°24.8
17	264°55.9	48°09.4	48.9	158°55.0	08.8	184°26.5	26.1	278°28.1	10.3	Canopus	263°52.5 -52°42.1
18	279°58.4	63°08.9	S14°50.0	173°56.0	N23°08.8	199°28.9	N22°26.1	293°30.7	S08°10.3	Sirius	258°26.5 -16°44.7
19	295°00.8	78°08.4	51.1	188°57.0	08.7	214°31.3	26.1	308°33.3	10.4	Adhara	255°06.1 -29°00.0
20	310°03.3	93°07.9	52.1	203°58.1	08.6	229°33.7	26.1	323°35.9	10.5	Procyon	244°51.2 5°09.8
21	325°05.8	108°07.5	· · 53.2	218°59.1	· · 08.5	244°36.1	· · 26.1	338°38.5	· · 10.5	Pollux	243°17.7 27°58.0
22	340°08.2	123°07.0	54.3	234°00.2	08.4	259°38.5	26.1	353°41.2	10.6	Avior	234°15.1 -59°34.9
23	355°10.7	138°06.5	55.4	249°01.2	08.3	274°40.9	26.1	8°43.8	10.7	Suhail	222°46.8 -43°31.6
Mer.pass. 23:19		ν -0.5' d 1.1' m -3.90		ν 1.0' d -0.1' m 0.47		ν 2.4' d 0.0' m -2.46		ν 2.6' d 0.1' m 0.65		Miaplacidus	221°39.0 -69°48.8
										Alphard	217°48.2 -8°45.7
										Regulus	207°34.9 11°50.9
										Dubhe	193°41.9 61°37.0
										Denebola	182°25.5 14°26.2
										Gienah	175°44.2 -17°40.6
										Acrux	173°01.1 -63°14.1
										Gacrux	171°52.6 -57°15.0
										Alioth	166°13.7 55°49.6
										Spica	158°22.9 -11°17.3
										Alkaid	152°52.6 49°11.5
										Hadar	148°37.0 -60°29.6
										Menkent	147°58.3 -36°29.5
										Arcturus	145°48.4 19°03.4
										Rigel Kent.	139°41.3 -60°56.3
										Kochab	137°20.7 74°03.4
										Zuben'ubi	136°56.6 -16°08.6
										Alphecca	126°04.2 26°38.1
										Antares	112°16.3 -26°29.2
										Atria	107°11.0 -69°04.5
										Sabik	102°03.2 -15°45.3
										Shaula	96°10.8 -37°07.4
										Rasalhague	95°58.9 12°32.7
										Eltanin	90°42.4 51°29.4
										Kaus Aust.	83°32.8 -34°22.5
										Vega	80°33.4 38°48.6
										Nunki	75°48.0 -26°16.0
										Altair	62°00.1 8°56.1
										Peacock	53°05.8 -56°39.5
										Deneb	49°25.8 45°22.3
										Enif	33°38.8 9°59.4
										Al Na'ir	27°32.8 -46°50.5
										Fomalhaut	15°14.4 -29°29.4
										Scheat	13°45.1 28°13.2
										Markab	13°29.8 15°20.4
Sep 30 Mon		SHA		Mer.pass							
		Venus		144°03.4		13:47					
		Mars		254°23.4		06:25					
		Jupiter		279°32.0		04:44					
		Saturn		13°29.5		22:25					
Oct 01 Tue		SHA		Mer.pass							
		Venus		142°52.9		13:48					
		Mars		253°49.1		06:23					
		Jupiter		279°30.1		04:40					
		Saturn		13°33.2		22:21					
Oct 02 Wed		SHA		Mer.pass							
		Venus		141°42.1		13:49					
		Mars		253°15.0		06:22					
		Jupiter		279°28.4		04:37					
		Saturn		13°36.9		22:17					
		Horizontal parallax									
		Venus:		0.1							
		Mars:		0.1							

Sun				Moon				
Mon	GHA	Dec		GHA	ν	Dec	d	HP
0	182°30.5	S02°54.4		209°24.0	16.2'	N11°19.2	-13.4'	54.2'
1	197°30.7	55.4		223°59.2	16.3'	11°05.8	-13.4'	54.2'
2	212°30.9	56.3		238°34.5	16.3'	10°52.4	-13.5'	54.2'
3	227°31.1	· · 57.3		253°09.8	16.4'	10°38.9	-13.5'	54.2'
4	242°31.3	58.3		267°45.2	16.4'	10°25.4	-13.5'	54.2'
5	257°31.5	02°59.3		282°20.6	16.5'	10°11.9	-13.6'	54.2'
6	272°31.7	S03°00.2		296°56.1	16.5'	N09°58.4	-13.6'	54.2'
7	287°31.9	01.2		311°31.6	16.6'	09°44.8	-13.6'	54.2'
8	302°32.1	02.2		326°07.2	16.6'	09°31.2	-13.6'	54.1'
9	317°32.3	· · 03.1		340°42.8	16.6'	09°17.5	-13.7'	54.1'
10	332°32.5	04.1		355°18.4	16.7'	09°03.8	-13.7'	54.1'
11	347°32.7	05.1		9°54.1	16.7'	08°50.1	-13.7'	54.1'
12	2°32.9	S03°06.0		24°29.8	16.8'	N08°36.4	-13.8'	54.1'
13	17°33.1	07.0		39°05.6	16.8'	08°22.6	-13.8'	54.1'
14	32°33.3	08.0		53°41.4	16.8'	08°08.9	-13.8'	54.1'
15	47°33.5	· · 09.0		68°17.2	16.9'	07°55.0	-13.8'	54.1'
16	62°33.7	09.9		82°53.1	16.9'	07°41.2	-13.9'	54.1'
17	77°33.9	10.9		97°29.0	16.9'	07°27.3	-13.9'	54.1'
18	92°34.1	S03°11.9		112°05.0	17.0'	N07°13.5	-13.9'	54.1'
19	107°34.3	12.8		126°41.0	17.0'	06°59.6	-13.9'	54.1'
20	122°34.5	13.8		141°17.0	17.0'	06°45.6	-13.9'	54.0'
21	137°34.7	· · 14.8		155°53.0	17.1'	06°31.7	-14.0'	54.0'
22	152°34.9	15.7		170°29.1	17.1'	06°17.7	-14.0'	54.0'
23	167°35.1	16.7		185°05.2	17.1'	06°03.7	-14.0'	54.0'
SD = 16.0' d = 1.0'				SD = 14.8'				
Tue	GHA	Dec		GHA	ν	Dec	d	HP
0	182°35.3	S03°17.7		199°41.3	17.2'	N05°49.7	-14.0'	54.0'
1	197°35.5	18.7		214°17.5	17.2'	05°35.7	-14.0'	54.0'
2	212°35.7	19.6		228°53.7	17.2'	05°21.7	-14.1'	54.0'
3	227°35.9	· · 20.6		243°29.9	17.2'	05°07.6	-14.1'	54.0'
4	242°36.1	21.6		258°06.1	17.3'	04°53.5	-14.1'	54.0'
5	257°36.3	22.5		272°42.3	17.3'	04°39.4	-14.1'	54.0'
6	272°36.5	S03°23.5		287°18.6	17.3'	N04°25.3	-14.1'	54.0'
7	287°36.7	24.5		301°54.9	17.3'	04°11.2	-14.1'	54.0'
8	302°36.9	25.4		316°31.2	17.3'	03°57.1	-14.1'	54.0'
9	317°37.1	· · 26.4		331°07.6	17.4'	03°43.0	-14.1'	54.0'
10	332°37.3	27.4		345°43.9	17.4'	03°28.8	-14.2'	54.0'
11	347°37.5	28.3		0°20.3	17.4'	03°14.7	-14.2'	54.0'
12	2°37.7	S03°29.3		14°56.7	17.4'	N03°00.5	-14.2'	53.9'
13	17°37.9	30.3		29°33.1	17.4'	02°46.3	-14.2'	53.9'
14	32°38.1	31.3		44°09.5	17.4'	02°32.1	-14.2'	53.9'
15	47°38.3	· · 32.2		58°45.9	17.4'	02°18.0	-14.2'	53.9'
16	62°38.5	33.2		73°22.4	17.5'	02°03.8	-14.2'	53.9'
17	77°38.7	34.2		87°58.8	17.5'	01°49.6	-14.2'	53.9'
18	92°38.9	S03°35.1		102°35.3	17.5'	N01°35.3	-14.2'	53.9'
19	107°39.1	36.1		117°11.7	17.5'	01°21.1	-14.2'	53.9'
20	122°39.3	37.1		131°48.2	17.5'	01°06.9	-14.2'	53.9'
21	137°39.5	· · 38.0		146°24.7	17.5'	00°52.7	-14.2'	53.9'
22	152°39.7	39.0		161°01.2	17.5'	00°38.5	-14.2'	53.9'
23	167°39.9	40.0		175°37.7	17.5'	00°24.3	-14.2'	53.9'
SD = 16.0' d = 1.0'				SD = 14.7'				
Wed	GHA	Dec		GHA	ν	Dec	d	HP
0	182°40.1	S03°40.9		190°14.2	17.5'	N00°10.1	-14.2'	53.9'
1	197°40.3	41.9		204°50.7	17.5'	S00°04.2	14.2'	53.9'
2	212°40.5	42.9		219°27.2	17.5'	00°18.4	14.2'	53.9'
3	227°40.7	· · 43.8		234°03.7	17.5'	00°32.6	14.2'	53.9'
4	242°40.9	44.8		248°40.2	17.5'	00°46.8	14.2'	53.9'
5	257°41.1	45.8		263°16.7	17.5'	01°01.0	14.2'	53.9'
6	272°41.3	S03°46.7		277°53.2	17.5'	S01°15.2	14.2'	53.9'
7	287°41.5	47.7		292°29.7	17.5'	01°29.4	14.2'	53.9'
8	302°41.7	48.7		307°06.2	17.5'	01°43.6	14.2'	53.9'
9	317°41.9	· · 49.6		321°42.7	17.5'	01°57.8	14.2'	53.9'
10	332°42.1	50.6		336°19.2	17.5'	02°12.0	14.2'	53.9'
11	347°42.3	51.6		350°55.6	17.5'	02°26.1	14.2'	53.9'
12	2°42.5	S03°52.5		5°32.1	17.5'	S02°40.3	14.2'	53.9'
13	17°42.7	53.5		20°08.6	17.4'	02°54.4	14.1'	53.9'
14	32°42.9	54.5		34°45.0	17.4'	03°08.6	14.1'	53.9'
15	47°43.1	· · 55.4		49°21.4	17.4'	03°22.7	14.1'	53.9'
16	62°43.2	56.4		63°57.9	17.4'	03°36.8	14.1'	53.9'
17	77°43.4	57.4		78°34.3	17.4'	03°50.9	14.1'	53.9'
18	92°43.6	S03°58.3		93°10.7	17.4'	S04°05.0	14.1'	53.9'
19	107°43.8	03°59.3		107°47.0	17.4'	04°19.1	14.1'	53.9'
20	122°44.0	04°00.3		122°23.4	17.3'	04°33.2	14.0'	53.9'
21	137°44.2	· · 01.2		136°59.8	17.3'	04°47.2	14.0'	53.9'
22	152°44.4	02.2		151°36.1	17.3'	05°01.2	14.0'	53.9'
23	167°44.6	03.2		166°12.4	17.3'	05°15.2	14.0'	53.9'
SD = 16.0' d = 1.0'				SD = 14.7'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	03:51	05:13	06:21	17:16	18:23	19:44
N 70°	04:03	05:17	06:17	17:20	18:20	19:33
68°	04:13	05:19	06:14	17:23	18:18	19:24
66°	04:20	05:21	06:12	17:25	18:16	19:16
64°	04:27	05:23	06:10	17:28	18:15	19:10
62°	04:32	05:24	06:08	17:30	18:14	19:05
60°	04:36	05:25	06:06	17:31	18:13	19:01
N 58°	04:40	05:26	06:05	17:33	18:12	18:57
56°	04:43	05:27	06:04	17:34	18:11	18:54
54°	04:46	05:27	06:03	17:35	18:10	18:52
52°	04:49	05:28	06:02	17:36	18:10	18:49
50°	04:51	05:28	06:01	17:37	18:10	18:47
45°	04:55	05:29	05:58	17:40	18:09	18:43
N 40°	04:58	05:30	05:57	17:42	18:09	18:40
35°	05:00	05:30	05:55	17:43	18:09	18:38
30°	05:02	05:30	05:54	17:45	18:09	18:37
20°	05:03	05:29	05:51	17:48	18:10	18:35
N 10°	05:03	05:28	05:49	17:50	18:11	18:36
0°	05:02	05:26	05:46	17:53	18:13	18:37
S 10°	04:58	05:23	05:44	17:55	18:16	18:41
20°	04:53	05:19	05:41	17:58	18:20	18:46
30°	04:46	05:14	05:38	18:02	18:26	18:54
35°	04:41	05:11	05:36	18:04	18:29	18:59
40°	04:35	05:07	05:34	18:06	18:33	19:05
45°	04:27	05:02	05:31	18:09	18:38	19:13
S 50°	04:17	04:56	05:28	18:12	18:44	19:23
52°	04:12	04:53	05:27	18:13	18:47	19:28
54°	04:06	04:49	05:25	18:15	18:51	19:34
56°	04:00	04:46	05:23	18:17	18:55	19:41
58°	03:53	04:42	05:22	18:19	18:59	19:48
S 60°	03:45	04:37	05:19	18:21	19:04	19:56

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	01:41	03:47	05:44	18:02	17:29	16:57
N 70°	01:59	03:54	05:42	17:52	17:27	17:02
68°	02:14	04:00	05:41	17:44	17:25	17:06
66°	02:26	04:04	05:39	17:37	17:23	17:10
64°	02:36	04:08	05:38	17:31	17:22	17:13
62°	02:44	04:12	05:37	17:26	17:21	17:16
60°	02:52	04:15	05:37	17:22	17:20	17:18
N 58°	02:58	04:18	05:36	17:18	17:19	17:20
56°	03:03	04:20	05:35	17:14	17:18	17:22
54°	03:08	04:22	05:35	17:11	17:18	17:24
52°	03:13	04:24	05:34	17:08	17:17	17:26
50°	03:17	04:26	05:34	17:05	17:16	17:27
45°	03:26	04:30	05:33	17:00	17:15	17:30
N 40°	03:33	04:33	05:32	16:55	17:14	17:33
35°	03:39	04:36	05:31	16:51	17:13	17:35
30°	03:44	04:38	05:31	16:47	17:12	17:37
20°	03:54	04:42	05:30	16:40	17:11	17:41
N 10°	04:02	04:46	05:29	16:34	17:09	17:44
0°	04:10	04:49	05:28	16:29	17:08	17:47
S 10°	04:17	04:53	05:27	16:23	17:07	17:50
20°	04:25	04:56	05:26	16:17	17:05	17:53
30°	04:34	05:00	05:25	16:11	17:04	17:56
35°	04:40	05:03	05:25	16:07	17:03	17:58
40°	04:46	05:05	05:24	16:02	17:02	18:01
45°	04:53	05:08	05:24	15:57	17:00	18:03
S 50°	05:01	05:12	05:23	15:50	16:59	18:07
52°	05:05	05:14	05:22	15:47	16:58	18:08
54°	05:09	05:16	05:22	15:44	16:57	18:10
56°	05:13	05:18	05:22	15:41	16:56	18:12
58°	05:18	05:20	05:21	15:37	16:56	18:14
S 60°	05:24	05:22	05:20	15:32	16:54	18:16

Day	Sun		Mer. Pass hh:mm	Moon		Age 27-0 7-1%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
30	10:02	10:12	11:50	10:19	22:39	
01	10:21	10:31	11:49	10:59	23:18	
02	10:40	10:50	11:49	11:37	23:57	

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	13° 10.6	152° 30.4	S16° 13.0	265° 18.3	N23° 01.7	292° 36.3	N22° 26.3	26° 54.8	S08° 15.1
1	28° 13.0	167° 29.8	14.0	280° 19.4	01.6	307° 38.7	26.3	41° 57.4	15.2
2	43° 15.5	182° 29.3	15.0	295° 20.5	01.5	322° 41.1	26.3	57° 00.0	15.2
3	58° 18.0	197° 28.8	• • 16.1	310° 21.5	• • 01.4	337° 43.5	• • 26.3	72° 02.6	• • 15.3
4	73° 20.4	212° 28.3	17.1	325° 22.6	01.3	352° 45.9	26.3	87° 05.2	15.3
5	88° 22.9	227° 27.8	18.1	340° 23.7	01.2	7° 48.3	26.3	102° 07.8	15.4
6	103° 25.4	242° 27.3	S16° 19.2	355° 24.8	N23° 01.1	22° 50.8	N22° 26.3	117° 10.4	S08° 15.4
7	118° 27.8	257° 26.8	20.2	10° 25.8	01.0	37° 53.2	26.3	132° 13.1	15.5
8	133° 30.3	272° 26.2	21.2	25° 26.9	00.9	52° 55.6	26.3	147° 15.7	15.6
9	148° 32.8	287° 25.7	• • 22.3	40° 28.0	• • 00.8	67° 58.0	• • 26.3	162° 18.3	• • 15.6
10	163° 35.2	302° 25.2	23.3	55° 29.1	00.7	83° 00.4	26.3	177° 20.9	15.7
11	178° 37.7	317° 24.7	24.3	70° 30.2	00.6	98° 02.9	26.3	192° 23.5	15.7
12	193° 40.1	332° 24.2	S16° 25.4	85° 31.2	N23° 00.5	113° 05.3	N22° 26.4	207° 26.1	S08° 15.8
13	208° 42.6	347° 23.7	26.4	100° 32.3	00.4	128° 07.7	26.4	222° 28.7	15.9
14	223° 45.1	2° 23.1	27.4	115° 33.4	00.3	143° 10.1	26.4	237° 31.3	15.9
15	238° 47.5	17° 22.6	• • 28.5	130° 34.5	• • 00.2	158° 12.5	• • 26.4	252° 33.9	• • 16.0
16	253° 50.0	32° 22.1	29.5	145° 35.6	00.1	173° 15.0	26.4	267° 36.6	16.0
17	268° 52.5	47° 21.6	30.5	160° 36.6	23° 00.0	188° 17.4	26.4	282° 39.2	16.1
18	283° 54.9	62° 21.1	S16° 31.5	175° 37.7	N22° 59.9	203° 19.8	N22° 26.4	297° 41.8	S08° 16.1
19	298° 57.4	77° 20.5	32.6	190° 38.8	59.8	218° 22.2	26.4	312° 44.4	16.2
20	313° 59.9	92° 20.0	33.6	205° 39.9	59.7	233° 24.7	26.4	327° 47.0	16.3
21	329° 02.3	107° 19.5	• • 34.6	220° 41.0	• • 59.6	248° 27.1	• • 26.4	342° 49.6	• • 16.3
22	344° 04.8	122° 19.0	35.6	235° 42.1	59.5	263° 29.5	26.4	357° 52.2	16.4
23	359° 07.2	137° 18.5	36.7	250° 43.1	59.4	278° 31.9	26.4	12° 54.8	16.4
Mer.pass. 23:04		ν -0.5' d 1.0' m -3.91		ν 1.1' d -0.1' m 0.42		ν 2.4' d 0.0' m -2.49		ν 2.6' d 0.1' m 0.67	

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	14°09.7	152°17.9	S16°37.7	265°44.2	N22°59.3	293°34.4	N22°26.4	27°57.4	S08°16.5
1	29°12.2	167°17.4	38.7	280°45.3	59.2	308°36.8	26.4	43°00.1	16.6
2	44°14.6	182°16.9	39.7	295°46.4	59.1	323°39.2	26.4	58°02.7	16.6
3	59°17.1	197°16.4	· · 40.8	310°47.5	· · 59.0	338°41.6	· · 26.4	73°05.3	· · 16.7
4	74°19.6	212°15.8	41.8	325°48.6	58.9	353°44.1	26.4	88°07.9	16.7
5	89°22.0	227°15.3	42.8	340°49.7	58.8	8°46.5	26.4	103°10.5	16.8
6	104°24.5	242°14.8	S16°43.8	355°50.8	N22°58.7	23°48.9	N22°26.4	118°13.1	S08°16.8
7	119°27.0	257°14.2	44.8	10°51.8	58.6	38°51.3	26.4	133°15.7	16.9
8	134°29.4	272°13.7	45.9	25°52.9	58.5	53°53.8	26.4	148°18.3	17.0
9	149°31.9	287°13.2	· · 46.9	40°54.0	· · 58.4	68°56.2	· · 26.4	163°20.9	· · 17.0
10	164°34.4	302°12.7	47.9	55°55.1	58.3	83°58.6	26.4	178°23.5	17.1
11	179°36.8	317°12.1	48.9	70°56.2	58.2	99°01.1	26.4	193°26.2	17.1
12	194°39.3	332°11.6	S16°49.9	85°57.3	N22°58.1	114°03.5	N22°26.4	208°28.8	S08°17.2
13	209°41.7	347°11.1	50.9	100°58.4	58.0	129°05.9	26.4	223°31.4	17.2
14	224°44.2	2°10.5	51.9	115°59.5	57.9	144°08.3	26.4	238°34.0	17.3
15	239°46.7	17°10.0	· · 53.0	131°00.6	· · 57.8	159°10.8	· · 26.4	253°36.6	· · 17.4
16	254°49.1	32°09.5	54.0	146°01.7	57.7	174°13.2	26.4	268°39.2	17.4
17	269°51.6	47°09.0	55.0	161°02.8	57.6	189°15.6	26.4	283°41.8	17.5
18	284°54.1	62°08.4	S16°56.0	176°03.8	N22°57.5	204°18.1	N22°26.4	298°44.4	S08°17.5
19	299°56.5	77°07.9	57.0	191°04.9	57.4	219°20.5	26.4	313°47.0	17.6
20	314°59.0	92°07.4	58.0	206°06.0	57.3	234°22.9	26.4	328°49.6	17.6
21	330°01.5	107°06.8	16°59.0	221°07.1	· · 57.2	249°25.4	· · 26.4	343°52.3	· · 17.7
22	345°03.9	122°06.3	17°00.0	236°08.2	57.1	264°27.8	26.4	358°54.9	17.8
23	0°06.4	137°05.8	01.0	251°09.3	57.0	279°30.2	26.4	13°57.5	17.8
Mer.pass. 23:00		ν -0.5' d 1.0' m -3.92		ν 1.1' d -0.1' m 0.41		ν 2.4' d 0.0' m -2.50		ν 2.6' d 0.1' m 0.67	

	Stars	
	SHA	Dec
Alpheratz	357°34.7	29°13.8
Ankaa	353°06.9	-42°10.2
Schedar	349°30.8	56°40.5
Diphda	348°47.2	-17°50.9
Achernar	335°19.7	-57°06.5
Hamal	327°51.2	23°34.9
Polaris	313°46.3	89°21.9
Acamar	315°11.6	-40°12.1
Menkar	314°06.1	4°11.3
Mirfak	308°28.2	49°56.9
Aldebaran	290°39.7	16°33.6
Rigel	281°04.0	-8°10.2
Capella	280°22.1	46°01.3
Bellatrix	278°23.0	6°22.5
Elnath	278°02.0	28°37.7
Alnilam	275°37.9	-1°11.0
Betelgeuse	270°52.3	7°24.8
Canopus	263°52.4	-52°42.1
Sirius	258°26.4	-16°44.7
Adhara	255°06.1	-29°00.0
Procyon	244°51.2	5°09.8
Pollux	243°17.7	27°58.0
Avior	234°15.1	-59°34.9
Suhail	222°46.8	-43°31.6
Miaplacidus	221°38.9	-69°48.8
Alphard	217°48.2	-8°45.7
Regulus	207°34.9	11°50.9
Dubhe	193°41.8	61°37.0
Denebola	182°25.5	14°26.1
Gienah	175°44.2	-17°40.6
Acrux	173°01.1	-63°14.1
Gacrux	171°52.6	-57°15.0
Alioth	166°13.7	55°49.6
Spica	158°22.9	-11°17.3
Alkaid	152°52.6	49°11.5
Hadar	148°37.0	-60°29.6
Menkent	147°58.3	-36°29.5
Arcturus	145°48.4	19°03.4
Rigel Kent.	139°41.3	-60°56.3
Kochab	137°20.7	74°03.3
Zuben'ubi	136°56.6	-16°08.6
Alphecca	126°04.2	26°38.1
Antares	112°16.3	-26°29.2
Atria	107°11.0	-69°04.5
Sabik	102°03.2	-15°45.3
Shaula	96°10.8	-37°07.4
Rasalhague	95°58.9	12°32.7
Eltanin	90°42.4	51°29.4
Kaus Aust.	83°32.9	-34°22.5
Vega	80°33.4	38°48.6
Nunki	75°48.0	-26°16.0
Altair	62°00.1	8°56.1
Peacock	53°05.8	-56°39.5
Deneb	49°25.8	45°22.3
Enif	33°38.8	9°59.4
Al Na'ir	27°32.8	-46°50.6
Fomalhaut	15°14.4	-29°29.4
Scheat	13°45.1	28°13.2
Markab	13°29.9	15°20.4

Oct 03 Thu	SHA	Mer.pass
Venus	140° 31.1	13:50
Mars	252° 41.2	06:20
Jupiter	279° 26.9	04:33
Saturn	13° 40.6	22:13

Oct 04 Fri	SHA	Mer.pass
Venus	139°19.8	13:50
Mars	252°07.7	06:18
Jupiter	279°25.7	04:29
Saturn	13°44.2	22:08

Oct 05 Sat	SHA	Mer.pass
Venus	138°08.2	13:51
Mars	251°34.5	06:17
Jupiter	279°24.6	04:25
Saturn	13°47.7	22:04

Horizontal parallax	
Venus:	0.1
Mars:	0.1

h	Sun				Moon			
Thu	GHA	Dec		GHA	ν	Dec	d	HP
0	182°44.8	S04°04.1		180°48.7	17.3'	S05°29.2	14.0'	53.9'
1	197°45.0	05.1		195°25.0	17.2'	05°43.2	14.0'	53.9'
2	212°45.2	06.1		210°01.2	17.2'	05°57.2	13.9'	53.9'
3	227°45.4	•• 07.0		224°37.4	17.2'	06°11.1	13.9'	53.9'
4	242°45.6	08.0		239°13.6	17.2'	06°25.0	13.9'	53.9'
5	257°45.8	09.0		253°49.8	17.2'	06°38.9	13.9'	53.9'
6	272°46.0	S04°09.9		268°26.0	17.1'	S06°52.8	13.8'	53.9'
7	287°46.2	10.9		283°02.1	17.1'	07°06.6	13.8'	53.9'
8	302°46.4	11.9		297°38.2	17.1'	07°20.5	13.8'	53.9'
9	317°46.5	•• 12.8		312°14.3	17.0'	07°34.2	13.8'	53.9'
10	332°46.7	13.8		326°50.3	17.0'	07°48.0	13.7'	53.9'
11	347°46.9	14.8		341°26.3	17.0'	08°01.8	13.7'	53.9'
12	2°47.1	S04°15.7		356°02.3	16.9'	S08°15.5	13.7'	53.9'
13	17°47.3	16.7		10°38.2	16.9'	08°29.2	13.7'	53.9'
14	32°47.5	17.7		25°14.1	16.9'	08°42.8	13.6'	53.9'
15	47°47.7	•• 18.6		39°50.0	16.8'	08°56.5	13.6'	53.9'
16	62°47.9	19.6		54°25.8	16.8'	09°10.1	13.6'	53.9'
17	77°48.1	20.5		69°01.6	16.8'	09°23.6	13.5'	53.9'
18	92°48.3	S04°21.5		83°37.4	16.7'	S09°37.2	13.5'	53.9'
19	107°48.5	22.5		98°13.1	16.7'	09°50.7	13.5'	53.9'
20	122°48.7	23.4		112°48.8	16.7'	10°04.1	13.4'	53.9'
21	137°48.9	•• 24.4		127°24.5	16.6'	10°17.6	13.4'	53.9'
22	152°49.0	25.4		142°00.1	16.6'	10°31.0	13.4'	53.9'
23	167°49.2	26.3		156°35.7	16.5'	10°44.3	13.3'	53.9'
SD = 16.0'			$d = 1.0'$	SD = 14.7'				
Fri	GHA	Dec		GHA	ν	Dec	d	HP
0	182°49.4	S04°27.3		171°11.2	16.5'	S10°57.7	13.3'	53.9'
1	197°49.6	28.3		185°46.7	16.4'	11°11.0	13.3'	53.9'
2	212°49.8	29.2		200°22.1	16.4'	11°24.2	13.2'	53.9'
3	227°50.0	•• 30.2		214°57.5	16.3'	11°37.4	13.2'	53.9'
4	242°50.2	31.1		229°32.8	16.3'	11°50.6	13.1'	54.0'
5	257°50.4	32.1		244°08.1	16.2'	12°03.7	13.1'	54.0'
6	272°50.6	S04°33.1		258°43.4	16.2'	S12°16.8	13.0'	54.0'
7	287°50.8	34.0		273°18.6	16.1'	12°29.9	13.0'	54.0'
8	302°50.9	35.0		287°53.7	16.1'	12°42.9	13.0'	54.0'
9	317°51.1	•• 36.0		302°28.8	16.0'	12°55.8	12.9'	54.0'
10	332°51.3	36.9		317°03.9	16.0'	13°08.7	12.9'	54.0'
11	347°51.5	37.9		331°38.9	15.9'	13°21.6	12.8'	54.0'
12	2°51.7	S04°38.9		346°13.8	15.9'	S13°34.4	12.8'	54.0'
13	17°51.9	39.8		0°48.7	15.8'	13°47.2	12.7'	54.0'
14	32°52.1	40.8		15°23.5	15.8'	13°59.9	12.7'	54.0'
15	47°52.3	•• 41.7		29°58.3	15.7'	14°12.6	12.6'	54.0'
16	62°52.5	42.7		44°33.0	15.7'	14°25.2	12.6'	54.0'
17	77°52.6	43.7		59°07.7	15.6'	14°37.7	12.5'	54.0'
18	92°52.8	S04°44.6		73°42.3	15.5'	S14°50.3	12.5'	54.0'
19	107°53.0	45.6		88°16.8	15.5'	15°02.7	12.4'	54.0'
20	122°53.2	46.5		102°51.3	15.4'	15°15.1	12.4'	54.0'
21	137°53.4	•• 47.5		117°25.7	15.4'	15°27.5	12.3'	54.1'
22	152°53.6	48.5		132°00.1	15.3'	15°39.8	12.2'	54.1'
23	167°53.8	49.4		146°34.4	15.2'	15°52.0	12.2'	54.1'
SD = 16.0'			$d = 1.0'$	SD = 14.7'				
Sat	GHA	Dec		GHA	ν	Dec	d	HP
0	182°54.0	S04°50.4		161°08.6	15.2'	S16°04.2	12.1'	54.1'
1	197°54.1	51.4		175°42.8	15.1'	16°16.3	12.1'	54.1'
2	212°54.3	52.3		190°16.9	15.0'	16°28.4	12.0'	54.1'
3	227°54.5	•• 53.3		204°50.9	15.0'	16°40.4	11.9'	54.1'
4	242°54.7	54.2		219°24.9	14.9'	16°52.3	11.9'	54.1'
5	257°54.9	55.2		233°58.8	14.8'	17°04.2	11.8'	54.1'
6	272°55.1	S04°56.2		248°32.6	14.8'	S17°16.0	11.7'	54.1'
7	287°55.3	57.1		263°06.4	14.7'	17°27.7	11.7'	54.1'
8	302°55.4	58.1		277°40.1	14.6'	17°39.4	11.6'	54.2'
9	317°55.6	04°59.0		292°13.7	14.6'	17°51.0	11.5'	54.2'
10	332°55.8	05°00.0		306°47.3	14.5'	18°02.6	11.5'	54.2'
11	347°56.0	01.0		321°20.8	14.4'	18°14.1	11.4'	54.2'
12	2°56.2	S05°01.9		335°54.2	14.3'	S18°25.5	11.3'	54.2'
13	17°56.4	02.9		350°27.5	14.3'	18°36.8	11.3'	54.2'
14	32°56.6	03.8		5°00.8	14.2'	18°48.1	11.2'	54.2'
15	47°56.7	•• 04.8		19°34.0	14.1'	18°59.3	11.1'	54.2'
16	62°56.9	05.8		34°07.1	14.0'	19°10.4	11.0'	54.2'
17	77°57.1	06.7		48°40.1	14.0'	19°21.4	11.0'	54.2'
18	92°57.3	S05°07.7		63°13.1	13.9'	S19°32.4	10.9'	54.3'
19	107°57.5	08.6		77°46.0	13.8'	19°43.3	10.8'	54.3'
20	122°57.7	09.6		92°18.8	13.7'	19°54.1	10.7'	54.3'
21	137°57.8	•• 10.6		106°51.5	13.7'	20°04.9	10.7'	54.3'
22	152°58.0	11.5		121°24.2	13.6'	20°15.5	10.6'	54.3'
23	167°58.2	12.5		135°56.8	13.5'	20°26.1	10.5'	54.3'
SD = 16.0'			$d = 1.0'$	SD = 14.7'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	04:06	05:27	06:34	17:01	18:08	19:27
N 70°	04:16	05:28	06:29	17:06	18:06	19:18
68°	04:24	05:30	06:25	17:10	18:06	19:11
66°	04:30	05:30	06:22	17:14	18:05	19:05
64°	04:36	05:31	06:19	17:17	18:04	19:00
62°	04:40	05:32	06:16	17:20	18:04	18:55
60°	04:44	05:32	06:14	17:22	18:04	18:52
N 58°	04:47	05:32	06:12	17:24	18:03	18:49
56°	04:49	05:33	06:10	17:26	18:03	18:46
54°	04:52	05:33	06:08	17:28	18:03	18:44
52°	04:54	05:33	06:07	17:30	18:03	18:42
50°	04:55	05:33	06:05	17:31	18:03	18:41
45°	04:59	05:33	06:02	17:34	18:03	18:37
N 40°	05:01	05:33	06:00	17:37	18:04	18:35
35°	05:03	05:32	05:57	17:39	18:05	18:34
30°	05:04	05:31	05:55	17:41	18:05	18:33
20°	05:04	05:30	05:52	17:45	18:07	18:33
N 10°	05:03	05:27	05:48	17:49	18:10	18:34
0°	05:00	05:25	05:45	17:52	18:13	18:37
S 10°	04:56	05:21	05:42	17:55	18:16	18:41
20°	04:51	05:16	05:38	17:59	18:21	18:47
30°	04:42	05:10	05:34	18:03	18:27	18:56
35°	04:36	05:06	05:32	18:06	18:31	19:01
40°	04:29	05:02	05:29	18:09	18:36	19:09
45°	04:21	04:56	05:26	18:12	18:42	19:17
S 50°	04:10	04:49	05:22	18:16	18:49	19:29
52°	04:04	04:46	05:20	18:18	18:53	19:34
54°	03:58	04:42	05:18	18:20	18:57	19:41
56°	03:51	04:38	05:16	18:23	19:01	19:48
58°	03:43	04:33	05:13	18:25	19:06	19:56
S 60°	03:34	04:27	05:10	18:28	19:11	20:05

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	07:43	09:57	■	16:21	15:32	■
N 70°	07:31	09:30	12:05	16:35	16:01	14:55
68°	07:22	09:11	11:16	16:46	16:22	15:45
66°	07:15	08:55	10:46	16:56	16:39	16:16
64°	07:08	08:42	10:23	17:04	16:53	16:40
62°	07:03	08:32	10:05	17:11	17:05	16:59
60°	06:58	08:22	09:51	17:17	17:15	17:14
N 58°	06:54	08:15	09:38	17:22	17:24	17:28
56°	06:51	08:08	09:27	17:27	17:32	17:39
54°	06:47	08:01	09:18	17:31	17:39	17:49
52°	06:44	07:56	09:10	17:35	17:45	17:58
50°	06:42	07:51	09:02	17:38	17:51	18:06
45°	06:36	07:40	08:46	17:46	18:03	18:24
N 40°	06:31	07:31	08:33	17:52	18:14	18:38
35°	06:27	07:24	08:22	17:58	18:22	18:50
30°	06:23	07:17	08:13	18:03	18:30	19:01
20°	06:17	07:06	07:56	18:11	18:44	19:19
N 10°	06:12	06:56	07:42	18:19	18:56	19:35
0°	06:07	06:47	07:29	18:26	19:07	19:50
S 10°	06:02	06:38	07:16	18:33	19:18	20:05
20°	05:57	06:28	07:02	18:41	19:30	20:21
30°	05:51	06:17	06:46	18:49	19:44	20:40
35°	05:47	06:11	06:37	18:54	19:52	20:51
40°	05:43	06:04	06:26	19:00	20:01	21:04
45°	05:39	05:55	06:14	19:07	20:12	21:19
S 50°	05:34	05:45	05:59	19:15	20:25	21:37
52°	05:31	05:41	05:53	19:19	20:31	21:46
54°	05:28	05:36	05:45	19:23	20:38	21:56
56°	05:26	05:30	05:37	19:28	20:46	22:07
58°	05:22	05:24	05:27	19:33	20:54	22:19
S 60°	05:19	05:17	05:16	19:38	21:04	22:34

Day	Sun			Moon		
	Eqn.of Time		Mer. Pass hh:mm	Mer.Pass.		Age 1-3 0-4%
	00 ^h mm:ss	12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
03	10:59	11:09	11:49	12:16	—:—	
04	11:18	11:27	11:49	12:57	00:36	
05	11:36	11:45	11:48	13:39	01:18	

October 06, 07, 08 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	15°08.9	152°05.2	S17°02.0	266°10.4	N22°56.9	294°32.7	N22°26.4	29°00.1	S08°17.9	Alpheratz	357°34.7	29°13.8
1	30°11.3	167°04.7	03.0	281°11.5	56.8	309°35.1	26.4	44°02.7	17.9	Ankaa	353°06.9	-42°10.2
2	45°13.8	182°04.1	04.0	296°12.6	56.7	324°37.5	26.4	59°05.3	18.0	Schedar	349°30.8	56°40.5
3	60°16.2	197°03.6	•• 05.0	311°13.7	•• 56.6	339°40.0	•• 26.4	74°07.9	•• 18.0	Diphda	348°47.2	-17°50.9
4	75°18.7	212°03.1	06.0	326°14.8	56.5	354°42.4	26.4	89°10.5	18.1	Achernar	335°19.7	-57°06.5
5	90°21.2	227°02.5	07.0	341°15.9	56.4	9°44.8	26.4	104°13.1	18.2	Hamal	327°51.1	23°34.9
6	105°23.6	242°02.0	S17°08.1	356°17.0	N22°56.3	24°47.3	N22°26.4	119°15.7	S08°18.2	Polaris	313°45.1	89°21.9
7	120°26.1	257°01.5	09.1	11°18.1	56.2	39°49.7	26.4	134°18.3	18.3	Acamar	315°11.6	-40°12.1
8	135°28.6	272°00.9	10.1	26°19.2	56.1	54°52.2	26.4	149°20.9	18.3	Menkar	314°06.1	4°11.3
9	150°31.0	287°00.4	•• 11.1	41°20.3	•• 56.0	69°54.6	•• 26.4	164°23.5	•• 18.4	Mirfak	308°28.2	49°56.9
10	165°33.5	301°59.8	12.0	56°21.4	55.9	84°57.0	26.4	179°26.2	18.4	Aldebaran	290°39.7	16°33.6
11	180°36.0	316°59.3	13.0	71°22.5	55.8	99°59.5	26.4	194°28.8	18.5	Rigel	281°03.9	-8°10.2
12	195°38.4	331°58.8	S17°14.0	86°23.6	N22°55.7	115°01.9	N22°26.4	209°31.4	S08°18.6	Capella	280°22.0	46°01.3
13	210°40.9	346°58.2	15.0	101°24.7	55.6	130°04.3	26.4	224°34.0	18.6	Bellatrix	278°23.0	6°22.5
14	225°43.3	1°57.7	16.0	116°25.8	55.5	145°06.8	26.4	239°36.6	18.7	Elnath	278°02.0	28°37.7
15	240°45.8	16°57.1	•• 17.0	131°26.9	•• 55.4	160°09.2	•• 26.4	254°39.2	•• 18.7	Alnilam	275°37.9	-1°11.0
16	255°48.3	31°56.6	18.0	146°28.0	55.3	175°11.7	26.4	269°41.8	18.8	Betelgeuse	270°52.2	7°24.8
17	270°50.7	46°56.0	19.0	161°29.2	55.2	190°14.1	26.4	284°44.4	18.8	Canopus	263°52.4	-52°42.1
18	285°53.2	61°55.5	S17°20.0	176°30.3	N22°55.1	205°16.5	N22°26.4	299°47.0	S08°18.9	Sirius	258°26.4	-16°44.7
19	300°55.7	76°55.0	21.0	191°31.4	55.0	220°19.0	26.4	314°49.6	18.9	Adhara	255°06.1	-29°00.0
20	315°58.1	91°54.4	22.0	206°32.5	54.9	235°21.4	26.4	329°52.2	19.0	Procyon	244°51.1	5°09.8
21	331°00.6	106°53.9	•• 23.0	221°33.6	•• 54.8	250°23.9	•• 26.4	344°54.8	•• 19.1	Pollux	243°17.7	27°58.0
22	346°03.1	121°53.3	24.0	236°34.7	54.7	265°26.3	26.4	359°57.4	19.1	Avior	234°15.1	-59°34.9
23	1°05.5	136°52.8	25.0	251°35.8	54.6	280°28.8	26.4	15°00.0	19.2	Suhail	222°46.7	-43°31.6
Mer.pass. 22:56		ν -0.5' d 1.0' m -3.92		ν 1.1' d -0.1' m 0.40		ν 2.4' d 0.0' m -2.50		ν 2.6' d 0.1' m 0.68				

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	16°08.0	151°52.2	S17°26.0	266°36.9	N22°54.5	295°31.2	N22°26.4	30°02.6	S08°19.2
1	31°10.5	166°51.7	26.9	281°38.0	54.4	310°33.6	26.4	45°05.3	19.3
2	46°12.9	181°51.1	27.9	296°39.1	54.3	325°36.1	26.4	60°07.9	19.3
3	61°15.4	196°50.6	•• 28.9	311°40.2	•• 54.2	340°38.5	•• 26.4	75°10.5	•• 19.4
4	76°17.8	211°50.0	29.9	326°41.4	54.1	355°41.0	26.4	90°13.1	19.4
5	91°20.3	226°49.5	30.9	341°42.5	54.0	10°43.4	26.4	105°15.7	19.5
6	106°22.8	241°48.9	S17°31.9	356°43.6	N22°53.8	25°45.9	N22°26.4	120°18.3	S08°19.6
7	121°25.2	256°48.4	32.9	11°44.7	53.7	40°48.3	26.4	135°20.9	19.6
8	136°27.7	271°47.8	33.8	26°45.8	53.6	55°50.8	26.4	150°23.5	19.7
9	151°30.2	286°47.3	•• 34.8	41°46.9	•• 53.5	70°53.2	•• 26.4	165°26.1	•• 19.7
10	166°32.6	301°46.7	35.8	56°48.0	53.4	85°55.7	26.4	180°28.7	19.8
11	181°35.1	316°46.2	36.8	71°49.1	53.3	100°58.1	26.4	195°31.3	19.8
12	196°37.6	331°45.6	S17°37.8	86°50.3	N22°53.2	116°00.6	N22°26.4	210°33.9	S08°19.9
13	211°40.0	346°45.1	38.7	101°51.4	53.1	131°03.0	26.4	225°36.5	19.9
14	226°42.5	1°44.5	39.7	116°52.5	53.0	146°05.4	26.4	240°39.1	20.0
15	241°45.0	16°44.0	•• 40.7	131°53.6	•• 52.9	161°07.9	•• 26.4	255°41.7	•• 20.1
16	256°47.4	31°43.4	41.7	146°54.7	52.8	176°10.3	26.4	270°44.3	20.0
17	271°49.9	46°42.9	42.6	161°55.8	52.7	191°12.8	26.4	285°46.9	20.2
18	286°52.3	61°42.3	S17°43.6	176°57.0	N22°52.6	206°15.2	N22°26.4	300°49.5	S08°20.2
19	301°54.8	76°41.7	44.6	191°58.1	52.5	221°17.7	26.4	315°52.1	20.3
20	316°57.3	91°41.2	45.6	206°59.2	52.4	236°20.1	26.4	330°54.7	20.3
21	331°59.7	106°40.6	•• 46.5	222°00.3	•• 52.3	251°22.6	•• 26.4	345°57.4	•• 20.4
22	347°02.2	121°40.1	47.5	237°01.4	52.2	266°25.0	26.4	1°00.0	20.4
23	2°04.7	136°39.5	48.5	252°02.6	52.1	281°27.5	26.4	16°02.6	20.5
Mer.pass. 22:52		ν -0.5' d 1.0' m -3.92		ν 1.1' d -0.1' m 0.39		ν 2.4' d 0.0' m -2.51		ν 2.6' d 0.1' m 0.68	

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	17°07.1	151°39.0	S17°49.5	267°03.7	N22°52.0	296°30.0	N22°26.4	31°05.2	S08°20.5
1	32°09.6	166°38.4	50.4	282°04.8	51.8	311°32.4	26.4	46°07.8	20.6
2	47°12.1	181°37.8	51.4	297°05.9	51.7	326°34.9	26.4	61°10.4	20.6
3	62°14.5	196°37.3	•• 52.4	312°07.1	•• 51.6	341°37.3	•• 26.4	76°13.0	•• 20.7
4	77°17.0	211°36.7	53.3	327°08.2	51.5	356°39.8	26.4	91°15.6	20.8
5	92°19.4	226°36.1	54.3	342°09.3	51.4	11°42.2	26.4	106°18.2	20.8
6	107°21.9	241°35.6	S17°55.3	357°10.4	N22°51.3	26°44.7	N22°26.4	121°20.8	S08°20.9
7	122°24.4	256°35.0	56.2	12°11.6	51.2	41°47.1	26.4	136°23.4	20.9
8	137°26.8	271°34.5	57.2	27°12.7	51.1	56°49.6	26.4	151°26.0	21.0
9	152°29.3	286°33.9	•• 58.2	42°13.8	•• 51.0	71°52.0	•• 26.4	166°28.6	•• 21.0
10	167°31.8	301°33.3	17°59.1	57°14.9	50.9	86°54.5	26.4	181°31.2	21.1
11	182°34.2	316°32.8	18°00.1	72°16.1	50.8	101°57.0	26.4	196°33.8	21.1
12	197°36.7	331°32.2	S18°01.0	87°17.2	N22°50.7	116°59.4	N22°26.4	211°36.4	S08°21.2
13	212°39.2	346°31.6	02.0	102°18.3	50.6	132°01.9	26.4	226°39.0	21.2
14	227°41.6	1°31.1	03.0	117°19.4	50.5	147°04.3	26.4	241°41.6	21.3
15	242°44.1	16°30.5	•• 03.9	132°20.6	•• 50.3	162°06.8	•• 26.4	256°44.2	•• 21.3
16	257°46.6	31°29.9	04.9	147°21.7	50.2	177°09.2	26.4	271°46.8	21.4
17	272°49.0	46°29.4	05.8	162°22.8	50.1	192°11.7	26.4	286°49.4	21.5
18	287°51.5	61°28.8	S18°06.8	177°24.0	N22°50.0	207°14.2	N22°26.4	301°52.0	S08°21.5
19	302°53.9	76°28.2	07.7	192°25.1	49.9	222°16.6	26.4	316°54.6	21.6
20	317°56.4	91°27.7	08.7	207°26.2	49.8	237°19.1	26.4	331°57.2	21.6
21	332°58.9	106°27.1	•• 09.7	222°27.4	•• 49.7	252°21.5	•• 26.4	346°59.8	•• 21.7
22	348°01.3	121°26.5	10.6	237°28.5	49.6	267°24.0	26.4	2°02.4	21.7
23	3°03.8	136°26.0	11.6	252°29.6	49.5	282°26.5	26.4	17°05.0	21.8
Mer.pass. 22:48		ν -0.6' d 1.0' m -3.93		ν 1.1' d -0.1' m 0.38		ν 2.5' d -0.0' m -2.52		ν 2.6' d 0.1' m 0.69	

Oct 06 Sun		SHA	Mer.pass
Venus		136°56.4	13:52
Mars		251°01.6	06:15
Jupiter		279°23.8	04:21
Saturn		13°51.2	22:00
Oct 07 Mon		SHA	Mer.pass
Venus		135°44.2	13:53
Mars		250°28.9	06:13
Jupiter		279°23.2	04:17
Saturn		13°54.7	21:56
Oct 08 Tue		SHA	Mer.pass
Venus		134°31.8	13:54
Mars		249°56.6	06:11
Jupiter		279°22.8	04:13
Saturn		13°58.0	21:52
Horizontal parallax			
Venus:			0.1
Mars:			0.1

Sun				Moon				
Sun	GHA	Dec		GHA	ν	Dec	d	HP
0	182°58.4	S05°13.4		150°29.3	13.4'	S20°36.6	10.4'	54.3'
1	197°58.6	14.4		165°01.7	13.3'	20°47.0	10.3'	54.3'
2	212°58.8	15.4		179°34.0	13.3'	20°57.4	10.2'	54.4'
3	227°58.9	· · 16.3		194°06.3	13.2'	21°07.6	10.2'	54.4'
4	242°59.1	17.3		208°38.4	13.1'	21°17.8	10.1'	54.4'
5	257°59.3	18.2		223°10.5	13.0'	21°27.9	10.0'	54.4'
6	272°59.5	S05°19.2		237°42.5	12.9'	S21°37.8	9.9'	54.4'
7	287°59.7	20.1		252°14.5	12.8'	21°47.7	9.8'	54.4'
8	302°59.9	21.1		266°46.3	12.8'	21°57.6	9.7'	54.4'
9	318°00.0	· · 22.1		281°18.1	12.7'	22°07.3	9.6'	54.4'
10	333°00.2	23.0		295°49.7	12.6'	22°16.9	9.5'	54.5'
11	348°00.4	24.0		310°21.3	12.5'	22°26.4	9.4'	54.5'
12	3°00.6	S05°24.9		324°52.9	12.4'	S22°35.9	9.3'	54.5'
13	18°00.8	25.9		339°24.3	12.3'	22°45.2	9.3'	54.5'
14	33°00.9	26.8		353°55.6	12.3'	22°54.5	9.2'	54.5'
15	48°01.1	· · 27.8		8°26.9	12.2'	23°03.6	9.1'	54.5'
16	63°01.3	28.8		22°58.0	12.1'	23°12.7	9.0'	54.6'
17	78°01.5	29.7		37°29.1	12.0'	23°21.6	8.9'	54.6'
18	93°01.7	S05°30.7		52°00.1	11.9'	S23°30.5	8.8'	54.6'
19	108°01.8	31.6		66°31.0	11.8'	23°39.3	8.7'	54.6'
20	123°02.0	32.6		81°01.9	11.7'	23°47.9	8.5'	54.6'
21	138°02.2	· · 33.5		95°32.6	11.7'	23°56.5	8.4'	54.6'
22	153°02.4	34.5		110°03.2	11.6'	24°04.9	8.3'	54.6'
23	168°02.6	35.5		124°33.8	11.5'	24°13.2	8.2'	54.7'
SD = 16.0' $d = 1.0'$				SD = 14.8'				
Mon	GHA	Dec		GHA	ν	Dec	d	HP
0	183°02.7	S05°36.4		139°04.3	11.4'	S24°21.5	8.1'	54.7'
1	198°02.9	37.4		153°34.7	11.3'	24°29.6	8.0'	54.7'
2	213°03.1	38.3		168°05.0	11.2'	24°37.6	7.9'	54.7'
3	228°03.3	· · 39.3		182°35.2	11.1'	24°45.5	7.8'	54.7'
4	243°03.5	40.2		197°05.4	11.0'	24°53.3	7.7'	54.8'
5	258°03.6	41.2		211°35.4	11.0'	25°01.0	7.6'	54.8'
6	273°03.8	S05°42.1		226°05.4	10.9'	S25°08.6	7.5'	54.8'
7	288°04.0	43.1		240°35.2	10.8'	25°16.0	7.3'	54.8'
8	303°04.2	44.1		255°05.0	10.7'	25°23.4	7.2'	54.8'
9	318°04.3	· · 45.0		269°34.7	10.6'	25°30.6	7.1'	54.8'
10	333°04.5	46.0		284°04.4	10.5'	25°37.7	7.0'	54.9'
11	348°04.7	46.9		298°33.9	10.5'	25°44.7	6.9'	54.9'
12	3°04.9	S05°47.9		313°03.4	10.4'	S25°51.6	6.7'	54.9'
13	18°05.1	48.8		327°32.7	10.3'	25°58.3	6.6'	54.9'
14	33°05.2	49.8		342°02.0	10.2'	26°04.9	6.5'	54.9'
15	48°05.4	· · 50.7		356°31.2	10.1'	26°11.4	6.4'	55.0'
16	63°05.6	51.7		11°00.3	10.0'	26°17.8	6.3'	55.0'
17	78°05.8	52.6		25°29.4	10.0'	26°24.1	6.1'	55.0'
18	93°05.9	S05°53.6		39°58.3	9.9'	S26°30.2	6.0'	55.0'
19	108°06.1	54.6		54°27.2	9.8'	26°36.2	5.9'	55.0'
20	123°06.3	55.5		68°56.0	9.7'	26°42.1	5.7'	55.1'
21	138°06.5	· · 56.5		83°24.7	9.6'	26°47.8	5.6'	55.1'
22	153°06.6	57.4		97°53.3	9.5'	26°53.5	5.5'	55.1'
23	168°06.8	58.4		112°21.9	9.5'	26°58.9	5.4'	55.1'
SD = 16.0' $d = 1.0'$				SD = 14.9'				
Tue	GHA	Dec		GHA	ν	Dec	d	HP
0	183°07.0	S05°59.3		126°50.3	9.4'	S27°04.3	5.2'	55.2'
1	198°07.2	06°00.3		141°18.7	9.3'	27°09.5	5.1'	55.2'
2	213°07.3	01.2		155°47.0	9.2'	27°14.6	5.0'	55.2'
3	228°07.5	· · 02.2		170°15.3	9.2'	27°19.6	4.8'	55.2'
4	243°07.7	03.1		184°43.4	9.1'	27°24.4	4.7'	55.2'
5	258°07.9	04.1		199°11.5	9.0'	27°29.1	4.5'	55.3'
6	273°08.0	S06°05.0		213°39.5	8.9'	S27°33.6	4.4'	55.3'
7	288°08.2	06.0		228°07.5	8.9'	27°38.0	4.3'	55.3'
8	303°08.4	06.9		242°35.3	8.8'	27°42.3	4.1'	55.3'
9	318°08.5	· · 07.9		257°03.1	8.7'	27°46.4	4.0'	55.4'
10	333°08.7	08.8		271°30.8	8.7'	27°50.4	3.8'	55.4'
11	348°08.9	09.8		285°58.5	8.6'	27°54.2	3.7'	55.4'
12	3°09.1	S06°10.7		300°26.1	8.5'	S27°57.9	3.6'	55.4'
13	18°09.2	11.7		314°53.6	8.4'	28°01.5	3.4'	55.5'
14	33°09.4	12.6		329°21.0	8.4'	28°04.9	3.3'	55.5'
15	48°09.6	· · 13.6		343°48.4	8.3'	28°08.2	3.1'	55.5'
16	63°09.8	14.5		358°15.8	8.3'	28°11.3	3.0'	55.5'
17	78°09.9	15.5		12°43.0	8.2'	28°14.3	2.8'	55.6'
18	93°10.1	S06°16.4		27°10.2	8.1'	S28°17.1	2.7'	55.6'
19	108°10.3	17.4		41°37.3	8.1'	28°19.7	2.5'	55.6'
20	123°10.4	18.3		56°04.4	8.0'	28°22.3	2.4'	55.6'
21	138°10.6	· · 19.3		70°31.4	8.0'	28°24.6	2.2'	55.7'
22	153°10.8	20.2		84°58.4	7.9'	28°26.9	2.1'	55.7'
23	168°11.0	21.2		99°25.3	7.8'	28°28.9	1.9'	55.7'
SD = 16.0' $d = 1.0'$				SD = 15.0'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	04:21	05:40	06:48	16:45	17:53	19:11
N 70°	04:29	05:40	06:41	16:52	17:53	19:04
68°	04:35	05:40	06:36	16:58	17:53	18:58
66°	04:40	05:40	06:31	17:03	17:54	18:53
64°	04:45	05:40	06:27	17:07	17:54	18:49
62°	04:48	05:39	06:24	17:10	17:54	18:46
60°	04:51	05:39	06:21	17:13	17:55	18:43
N 58°	04:53	05:39	06:18	17:16	17:55	18:41
56°	04:56	05:39	06:16	17:19	17:56	18:39
54°	04:57	05:38	06:14	17:21	17:56	18:37
52°	04:59	05:38	06:12	17:23	17:56	18:35
50°	05:00	05:38	06:10	17:25	17:57	18:34
45°	05:03	05:37	06:06	17:29	17:58	18:32
N 40°	05:04	05:36	06:03	17:32	17:59	18:30
35°	05:05	05:34	06:00	17:35	18:00	18:30
30°	05:05	05:33	05:57	17:38	18:02	18:29
20°	05:05	05:30	05:53	17:43	18:05	18:30
N 10°	05:03	05:27	05:48	17:47	18:08	18:32
0°	05:00	05:24	05:44	17:51	18:12	18:36
S 10°	04:55	05:19	05:40	17:55	18:16	18:41
20°	04:48	05:14	05:36	18:00	18:22	18:48
30°	04:38	05:07	05:31	18:05	18:29	18:58
35°	04:32	05:02	05:28	18:08	18:34	19:04
40°	04:24	04:57	05:24	18:12	18:39	19:12
45°	04:15	04:50	05:20	18:16	18:46	19:22
S 50°	04:03	04:42	05:15	18:21	18:54	19:34
52°	03:57	04:38	05:13	18:23	18:58	19:40
54°	03:50	04:34	05:10	18:26	19:03	19:47
56°	03:42	04:29	05:08	18:29	19:07	19:55
58°	03:34	04:24	05:05	18:32	19:13	20:04
S 60°	03:23	04:18	05:01	18:36	19:19	20:14

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	■	■	■	■
66°	13:08	■	■	15:29	■	■
64°	12:18	■	■	16:20	■	■
62°	11:47	13:41	■	16:51	16:40	■
60°	11:24	13:02	14:37	17:15	17:19	17:33
N 58°	11:05	12:35	13:59	17:34	17:47	18:12
56°	10:50	12:13	13:32	17:50	18:08	18:39
54°	10:37	11:56	13:11	18:04	18:26	19:01
52°	10:25	11:41	12:53	18:16	18:41	19:18
50°	10:15	11:28	12:38	18:27	18:55	19:34
45°	09:54	11:02	12:08	18:49	19:22	20:04
N 40°	09:37	10:41	11:44	19:07	19:43	20:28
35°	09:22	10:24	11:25	19:22	20:01	20:48
30°	09:10	10:09	11:08	19:36	20:17	21:05
20°	08:49	09:44	10:40	19:58	20:43	21:33
N 10°	08:31	09:22	10:16	20:18	21:06	21:58
0°	08:14	09:02	09:54	20:37	21:27	22:20
S 10°	07:57	08:42	09:32	20:55	21:48	22:43
20°	07:39	08:21	09:08	21:15	22:11	23:08
30°	07:19	07:56	08:41	21:38	22:37	23:36
35°	07:07	07:42	08:25	21:52	22:53	23:53
40°	06:53	07:26	08:06	22:08	23:12	· ·
45°	06:37	07:06	07:43	22:27	23:34	· ·
S 50°	06:17	06:41	07:14	22:50	· ·	00:02
52°	06:08	06:29	07:00	23:02	· ·	00:16
54°	05:58	06:16	06:44	23:15	· ·	00:32
56°	05:46	06:00	06:24	23:30	· ·	00:51
58°	05:32	05:42	06:00	23:48	· ·	01:15
S 60°	05:17	05:19	05:28	· ·	00:10	01:46

Day	Sun		Mer.	Moon		Age
	Eqn.of Time		Pass	Mer.Pass.		
	00 ^h	12 ^h	hh:mm	Upper	Lower	4-6
	mm:ss	mm:ss		hh:mm	hh:mm	9-23%
06	11:54	12:02	11:48	14:25	02:02	
07	12:11	12:19	11:48	15:14	02:49	
08	12:28	12:36	11:47	16:07	03:40	

October 09, 10, 11 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	18°06.3	151°25.4	S18°12.5	267°30.8	N22°49.4	297°28.9	N22°26.4	32°07.6	S08°21.8	Alpheratz	357°34.7	29°13.8
1	33°08.7	166°24.8	13.5	282°31.9	49.3	312°31.4	26.4	47°10.2	21.9	Ankaa	353°06.9	-42°10.2
2	48°11.2	181°24.2	14.4	297°33.0	49.2	327°33.8	26.4	62°12.8	21.9	Schedar	349°30.8	56°40.5
3	63°13.7	196°23.7	· · 15.4	312°34.2	· · 49.0	342°36.3	· · 26.4	77°15.4	· · 22.0	Diphda	348°47.2	-17°50.9
4	78°16.1	211°23.1	16.3	327°35.3	48.9	357°38.8	26.4	92°18.0	22.0	Achernar	335°19.7	-57°06.6
5	93°18.6	226°22.5	17.3	342°36.4	48.8	12°41.2	26.4	107°20.6	22.1	Hamal	327°51.1	23°34.9
6	108°21.1	241°22.0	S18°18.2	357°37.6	N22°48.7	27°43.7	N22°26.4	122°23.2	S08°22.1	Polaris	313°43.9	89°22.0
7	123°23.5	256°21.4	19.2	12°38.7	48.6	42°46.2	26.4	137°25.8	22.2	Acamar	315°11.6	-40°12.1
8	138°26.0	271°20.8	20.1	27°39.9	48.5	57°48.6	26.4	152°28.4	22.2	Menkar	314°06.1	4°11.3
9	153°28.4	286°20.2	· · 21.0	42°41.0	· · 48.4	72°51.1	· · 26.4	167°31.0	· · 22.3	Mirfak	308°28.2	49°56.9
10	168°30.9	301°19.7	22.0	57°42.1	48.3	87°53.6	26.4	182°33.6	22.4	Aldebaran	290°39.7	16°33.6
11	183°33.4	316°19.1	22.9	72°43.3	48.2	102°56.0	26.4	197°36.2	22.4	Rigel	281°03.9	-8°10.2
12	198°35.8	331°18.5	S18°23.9	87°44.4	N22°48.1	117°58.5	N22°26.4	212°38.8	S08°22.5	Capella	280°22.0	46°01.3
13	213°38.3	346°17.9	24.8	102°45.6	48.0	133°00.9	26.4	227°41.4	22.5	Bellatrix	278°23.0	6°22.5
14	228°40.8	1°17.3	25.8	117°46.7	47.8	148°03.4	26.4	242°44.0	22.6	Elnath	278°02.0	28°37.7
15	243°43.2	16°16.8	· · 26.7	132°47.8	· · 47.7	163°05.9	· · 26.4	257°46.6	· · 22.6	Alnilam	275°37.8	-1°11.0
16	258°45.7	31°16.2	27.6	147°49.0	47.6	178°08.3	26.4	272°49.2	22.7	Betelgeuse	270°52.2	7°24.8
17	273°48.2	46°15.6	28.6	162°50.1	47.5	193°10.8	26.4	287°51.8	22.7	Canopus	263°52.4	-52°42.1
18	288°50.6	61°15.0	S18°29.5	177°51.3	N22°47.4	208°13.3	N22°26.4	302°54.4	S08°22.8	Sirius	258°26.4	-16°44.7
19	303°53.1	76°14.5	30.4	192°52.4	47.3	223°15.8	26.4	317°57.0	22.8	Adhara	255°06.0	-29°00.0
20	318°55.6	91°13.9	31.4	207°53.6	47.2	238°18.2	26.4	332°59.6	22.9	Procyon	244°51.1	5°09.8
21	333°58.0	106°13.3	· · 32.3	222°54.7	· · 47.1	253°20.7	· · 26.4	348°02.2	· · 22.9	Pollux	243°17.6	27°58.0
22	349°00.5	121°12.7	33.3	237°55.8	47.0	268°23.2	26.4	3°04.8	23.0	Avior	234°15.0	-59°34.9
23	4°02.9	136°12.1	34.2	252°57.0	46.9	283°25.6	26.4	18°07.4	23.0	Suhail	222°46.7	-43°31.6
Mer.pass. 22:44		ν -0.6' d 1.0' m -3.93		ν 1.1' d -0.1' m 0.37		ν 2.5' d -0.0' m -2.53		ν 2.6' d 0.1' m 0.69				

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	19°05.4	151°11.5	S18°35.1	267°58.1	N22°46.7	298°28.1	N22°26.4	33°10.0	S08°23.1
1	34°07.9	166°11.0	36.1	282°59.3	46.6	313°30.6	26.4	48°12.6	23.1
2	49°10.3	181°10.4	37.0	298°00.4	46.5	328°33.0	26.4	63°15.2	23.2
3	64°12.8	196°09.8	· · 37.9	313°01.6	· · 46.4	343°35.5	· · 26.4	78°17.8	· · 23.2
4	79°15.3	211°09.2	38.8	328°02.7	46.3	358°38.0	26.4	93°20.4	23.3
5	94°17.7	226°08.6	39.8	343°03.9	46.2	13°40.5	26.4	108°23.0	23.3
6	109°20.2	241°08.0	S18°40.7	358°05.0	N22°46.1	28°42.9	N22°26.4	123°25.6	S08°23.4
7	124°22.7	256°07.5	41.6	13°06.2	46.0	43°45.4	26.4	138°28.2	23.4
8	139°25.1	271°06.9	42.6	28°07.3	45.9	58°47.9	26.4	153°30.8	23.5
9	154°27.6	286°06.3	· · 43.5	43°08.5	· · 45.8	73°50.4	· · 26.4	168°33.4	· · 23.5
10	169°30.1	301°05.7	44.4	58°09.6	45.6	88°52.8	26.4	183°36.0	23.6
11	184°32.5	316°05.1	45.3	73°10.8	45.5	103°55.3	26.4	198°38.6	23.6
12	199°35.0	331°04.5	S18°46.3	88°11.9	N22°45.4	118°57.8	N22°26.4	213°41.2	S08°23.7
13	214°37.4	346°03.9	47.2	103°13.1	45.3	134°00.3	26.4	228°43.8	23.7
14	229°39.9	1°03.3	48.1	118°14.3	45.2	149°02.7	26.4	243°46.4	23.8
15	244°42.4	16°02.7	· · 49.0	133°15.4	· · 45.1	164°05.2	· · 26.4	258°49.0	· · 23.9
16	259°44.8	31°02.2	49.9	148°16.6	45.0	179°07.7	26.4	273°51.6	23.9
17	274°47.3	46°01.6	50.9	163°17.7	44.9	194°10.2	26.4	288°54.2	24.0
18	289°49.8	61°01.0	S18°51.8	178°18.9	N22°44.8	209°12.6	N22°26.4	303°56.7	S08°24.0
19	304°52.2	76°00.4	52.7	193°20.0	44.6	224°15.1	26.4	318°59.3	24.1
20	319°54.7	90°59.8	53.6	208°21.2	44.5	239°17.6	26.4	334°01.9	24.1
21	334°57.2	105°59.2	· · 54.5	223°22.3	· · 44.4	254°20.1	· · 26.4	349°04.5	· · 24.2
22	349°59.6	120°58.6	55.4	238°23.5	44.3	269°22.5	26.4	4°07.1	24.2
23	5°02.1	135°58.0	56.4	253°24.7	44.2	284°25.0	26.4	19°09.7	24.3
Mer.pass. 22:40		ν -0.6' d 0.9' m -3.93		ν 1.1' d -0.1' m 0.36		ν 2.5' d -0.0' m -2.53		ν 2.6' d 0.1' m 0.69	

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	20°04.5	150°57.4	S18°57.3	268°25.8	N22°44.1	299°27.5	N22°26.4	34°12.3	S08°24.3
1	35°07.0	165°56.8	58.2	283°27.0	44.0	314°30.0	26.4	49°14.9	24.4
2	50°09.5	180°56.2	18°59.1	298°28.1	43.9	329°32.5	26.4	64°17.5	24.4
3	65°11.9	195°55.6	19°00.0	313°29.3	· · 43.7	344°34.9	· · 26.4	79°20.1	· · 24.5
4	80°14.4	210°55.0	00.9	328°30.5	43.6	359°37.4	26.3	94°22.7	24.5
5	95°16.9	225°54.4	01.8	343°31.6	43.5	14°39.9	26.3	109°25.3	24.6
6	110°19.3	240°53.8	S19°02.7	358°32.8	N22°43.4	29°42.4	N22°26.3	124°27.9	S08°24.6
7	125°21.8	255°53.2	03.6	13°34.0	43.3	44°44.9	26.3	139°30.5	24.7
8	140°24.3	270°52.6	04.5	28°35.1	43.2	59°47.4	26.3	154°33.1	24.7
9	155°26.7	285°52.0	· · 05.4	43°36.3	· · 43.1	74°49.8	· · 26.3	169°35.7	· · 24.8
10	170°29.2	300°51.4	06.4	58°37.4	43.0	89°52.3	26.3	184°38.3	24.8
11	185°31.7	315°50.8	07.3	73°38.6	42.8	104°54.8	26.3	199°40.9	24.9
12	200°34.1	330°50.2	S19°08.2	88°39.8	N22°42.7	119°57.3	N22°26.3	214°43.5	S08°24.9
13	215°36.6	345°49.6	09.1	103°40.9	42.6	134°59.8	26.3	229°46.1	25.0
14	230°39.0	0°49.0	10.0	118°42.1	42.5	150°02.3	26.3	244°48.6	25.0
15	245°41.5	15°48.4	· · 10.9	133°43.3	· · 42.4	165°04.7	· · 26.3	259°51.2	· · 25.1
16	260°44.0	30°47.8	11.8	148°44.4	42.3	180°07.2	26.3	274°53.8	25.1
17	275°46.4	45°47.2	12.7	163°45.6	42.2	195°09.7	26.3	289°56.4	25.2
18	290°48.9	60°46.6	S19°13.6	178°46.8	N22°42.0	210°12.2	N22°26.3	304°59.0	S08°25.2
19	305°51.4	75°46.0	14.5	193°48.0	41.9	225°14.7	26.3	320°01.6	25.3
20	320°53.8	90°45.4	15.4	208°49.1	41.8	240°17.2	26.3	335°04.2	25.3
21	335°56.3	105°44.8	· · 16.3	223°50.3	· · 41.7	255°19.7	· · 26.3	350°06.8	· · 25.4
22	350°58.8	120°44.2	17.1	238°51.5	41.6	270°22.1	26.3	5°09.4	25.4
23	6°01.2	135°43.6	18.0	253°52.6	41.5	285°24.6	26.3	20°12.0	25.5
Mer.pass. 22:36		ν -0.6' d 0.9' m -3.94		ν 1.2' d -0.1' m 0.35		ν 2.5' d -0.0' m -2.54		ν 2.6' d 0.1' m 0.70	

Oct 09 Wed	SHA	Mer.pass
Venus	133°19.1	13:55
Mars	249°24.5	06:09
Jupiter	279°22.6	04:09
Saturn	14°01.3	21:48

Oct 10 Thu	SHA	Mer.pass
Venus	132°06.1	13:56
Mars	248°52.7	06:08
Jupiter	279°22.7	04:05
Saturn	14°04.6	21:44

Oct 11 Fri	SHA	Mer.pass
Venus	130°52.9	13:57
Mars	248°21.3	06:06
Jupiter	279°23.0	04:02
Saturn	14°07.8	21:39

Horizontal parallax		
Venus:	0.1	
Mars:	0.1	

h	Sun				Moon			
Wed	GHA	Dec	GHA	ν	Dec	d	HP	
0	183°11.1	S06°22.1	113°52.1	7.8'	S28°30.9	1.8'	55.7'	
1	198°11.3	23.1	128°18.9	7.7'	28°32.6	1.6'	55.8'	
2	213°11.5	24.0	142°45.7	7.7'	28°34.2	1.5'	55.8'	
3	228°11.6	· · 25.0	157°12.3	7.6'	28°35.7	1.3'	55.8'	
4	243°11.8	25.9	171°39.0	7.6'	28°37.0	1.1'	55.9'	
5	258°12.0	26.9	186°05.6	7.5'	28°38.1	1.0'	55.9'	
6	273°12.1	S06°27.8	200°32.1	7.5'	S28°39.1	0.8'	55.9'	
7	288°12.3	28.8	214°58.6	7.5'	28°40.0	0.7'	55.9'	
8	303°12.5	29.7	229°25.1	7.4'	28°40.6	0.5'	56.0'	
9	318°12.6	· · 30.7	243°51.5	7.4'	28°41.1	0.4'	56.0'	
10	333°12.8	31.6	258°17.9	7.3'	28°41.5	0.2'	56.0'	
11	348°13.0	32.6	272°44.2	7.3'	28°41.7	0.0'	56.1'	
12	3°13.2	S06°33.5	287°10.5	7.3'	S28°41.7	-0.1'	56.1'	
13	18°13.3	34.5	301°36.7	7.2'	28°41.6	-0.3'	56.1'	
14	33°13.5	35.4	316°03.0	7.2'	28°41.3	-0.4'	56.1'	
15	48°13.7	· · 36.4	330°29.2	7.2'	28°40.9	-0.6'	56.2'	
16	63°13.8	37.3	344°55.3	7.1'	28°40.3	-0.8'	56.2'	
17	78°14.0	38.3	359°21.4	7.1'	28°39.5	-0.9'	56.2'	
18	93°14.2	S06°39.2	13°47.5	7.1'	S28°38.6	-1.1'	56.3'	
19	108°14.3	40.1	28°13.6	7.0'	28°37.5	-1.3'	56.3'	
20	123°14.5	41.1	42°39.7	7.0'	28°36.2	-1.4'	56.3'	
21	138°14.7	· · 42.0	57°05.7	7.0'	28°34.8	-1.6'	56.4'	
22	153°14.8	43.0	71°31.7	7.0'	28°33.2	-1.8'	56.4'	
23	168°15.0	43.9	85°57.7	7.0'	28°31.4	-1.9'	56.4'	
SD = 16.0'		$d = 0.9'$		SD = 15.2'				
Thu	GHA	Dec	GHA	ν	Dec	d	HP	
0	183°15.2	S06°44.9	100°23.6	6.9'	S28°29.5	-2.1'	56.5'	
1	198°15.3	45.8	114°49.6	6.9'	28°27.4	-2.2'	56.5'	
2	213°15.5	46.8	129°15.5	6.9'	28°25.2	-2.4'	56.5'	
3	228°15.6	· · 47.7	143°41.4	6.9'	28°22.8	-2.6'	56.6'	
4	243°15.8	48.7	158°07.3	6.9'	28°20.2	-2.7'	56.6'	
5	258°16.0	49.6	172°33.2	6.9'	28°17.5	-2.9'	56.6'	
6	273°16.1	S06°50.5	186°59.1	6.9'	S28°14.6	-3.1'	56.7'	
7	288°16.3	51.5	201°25.0	6.9'	28°11.5	-3.2'	56.7'	
8	303°16.5	52.4	215°50.8	6.9'	28°08.2	-3.4'	56.7'	
9	318°16.6	· · 53.4	230°16.7	6.9'	28°04.8	-3.6'	56.8'	
10	333°16.8	54.3	244°42.6	6.9'	28°01.3	-3.7'	56.8'	
11	348°17.0	55.3	259°08.4	6.9'	27°57.5	-3.9'	56.8'	
12	3°17.1	S06°56.2	273°34.3	6.9'	S27°53.6	-4.1'	56.9'	
13	18°17.3	57.2	288°00.1	6.9'	27°49.6	-4.2'	56.9'	
14	33°17.4	58.1	302°26.0	6.9'	27°45.3	-4.4'	56.9'	
15	48°17.6	06°59.0	316°51.9	6.9'	27°40.9	-4.6'	57.0'	
16	63°17.8	07°00.0	331°17.7	6.9'	27°36.4	-4.7'	57.0'	
17	78°17.9	00.9	345°43.6	6.9'	27°31.6	-4.9'	57.0'	
18	93°18.1	S07°01.9	0°09.5	6.9'	S27°26.7	-5.1'	57.1'	
19	108°18.3	02.8	14°35.4	6.9'	27°21.7	-5.2'	57.1'	
20	123°18.4	03.8	29°01.3	6.9'	27°16.4	-5.4'	57.1'	
21	138°18.6	· · 04.7	43°27.2	6.9'	27°11.0	-5.6'	57.2'	
22	153°18.7	05.6	57°53.2	7.0'	27°05.5	-5.7'	57.2'	
23	168°18.9	06.6	72°19.1	7.0'	26°59.8	-5.9'	57.2'	
SD = 16.0'		$d = 0.9'$		SD = 15.4'				
Fri	GHA	Dec	GHA	ν	Dec	d	HP	
0	183°19.1	S07°07.5	86°45.1	7.0'	S26°53.9	-6.0'	57.3'	
1	198°19.2	08.5	101°11.1	7.0'	26°47.8	-6.2'	57.3'	
2	213°19.4	09.4	115°37.1	7.0'	26°41.6	-6.4'	57.3'	
3	228°19.5	· · 10.3	130°03.1	7.1'	26°35.2	-6.5'	57.4'	
4	243°19.7	11.3	144°29.2	7.1'	26°28.7	-6.7'	57.4'	
5	258°19.9	12.2	158°55.2	7.1'	26°22.0	-6.9'	57.5'	
6	273°20.0	S07°13.2	173°21.3	7.1'	S26°15.1	-7.0'	57.5'	
7	288°20.2	14.1	187°47.5	7.2'	26°08.1	-7.2'	57.5'	
8	303°20.3	15.0	202°13.6	7.2'	26°00.9	-7.3'	57.6'	
9	318°20.5	· · 16.0	216°39.8	7.2'	25°53.6	-7.5'	57.6'	
10	333°20.7	16.9	231°06.0	7.2'	25°46.1	-7.7'	57.6'	
11	348°20.8	17.9	245°32.2	7.3'	25°38.4	-7.8'	57.7'	
12	3°21.0	S07°18.8	259°58.5	7.3'	S25°30.6	-8.0'	57.7'	
13	18°21.1	19.7	274°24.8	7.3'	25°22.6	-8.1'	57.8'	
14	33°21.3	20.7	288°51.1	7.4'	25°14.5	-8.3'	57.8'	
15	48°21.5	· · 21.6	303°17.5	7.4'	25°06.2	-8.5'	57.8'	
16	63°21.6	22.6	317°43.9	7.4'	24°57.7	-8.6'	57.9'	
17	78°21.8	23.5	332°10.3	7.5'	24°49.1	-8.8'	57.9'	
18	93°21.9	S07°24.4	346°36.8	7.5'	S24°40.3	-8.9'	57.9'	
19	108°22.1	25.4	1°03.3	7.5'	24°31.4	-9.1'	58.0'	
20	123°22.2	26.3	15°29.9	7.6'	24°22.3	-9.2'	58.0'	
21	138°22.4	· · 27.3	29°56.5	7.6'	24°13.1	-9.4'	58.1'	
22	153°22.6	28.2	44°23.1	7.7'	24°03.7	-9.5'	58.1'	
23	168°22.7	29.1	58°49.7	7.7'	23°54.2	-9.7'	58.1'	
SD = 16.0'		$d = 0.9'$		SD = 15.6'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	04:35	05:53	07:02	16:29	17:38	18:56
N 70°	04:41	05:52	06:54	16:38	17:40	18:50
68°	04:46	05:51	06:47	16:45	17:41	18:45
66°	04:50	05:49	06:41	16:51	17:43	18:42
64°	04:53	05:48	06:36	16:56	17:44	18:39
62°	04:56	05:47	06:32	17:01	17:45	18:36
60°	04:58	05:46	06:28	17:04	17:46	18:34
N 58°	05:00	05:45	06:25	17:08	17:47	18:32
56°	05:02	05:45	06:22	17:11	17:48	18:31
54°	05:03	05:44	06:19	17:14	17:49	18:30
52°	05:04	05:43	06:17	17:16	17:50	18:29
50°	05:05	05:42	06:15	17:18	17:51	18:28
45°	05:06	05:40	06:10	17:23	17:53	18:27
N 40°	05:07	05:38	06:06	17:27	17:55	18:26
35°	05:07	05:37	06:02	17:31	17:56	18:26
30°	05:07	05:35	05:59	17:34	17:58	18:26
20°	05:06	05:31	05:53	17:40	18:02	18:28
N 10°	05:03	05:27	05:48	17:45	18:06	18:31
0°	04:59	05:23	05:44	17:50	18:11	18:35
S 10°	04:53	05:18	05:39	17:55	18:16	18:41
20°	04:45	05:11	05:33	18:01	18:23	18:49
30°	04:35	05:03	05:27	18:07	18:31	19:00
35°	04:28	04:58	05:24	18:11	18:36	19:07
40°	04:19	04:52	05:19	18:15	18:43	19:15
45°	04:09	04:45	05:15	18:20	18:50	19:26
S 50°	03:55	04:36	05:09	18:26	18:59	19:40
52°	03:49	04:31	05:06	18:29	19:04	19:46
54°	03:42	04:27	05:03	18:32	19:09	19:54
56°	03:33	04:21	05:00	18:35	19:14	20:03
58°	03:23	04:15	04:56	18:39	19:20	20:12
S 60°	03:12	04:08	04:52	18:43	19:27	20:24

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	■	■	■	■
66°	■	■	■	■	■	■
64°	■	■	17:40	■	■	20:24
62°	■	17:17	16:56	■	18:48	21:07
60°	15:48	16:17	16:26	18:18	19:48	21:36
N 58°	15:05	15:44	16:03	19:02	20:21	21:58
56°	14:36	15:19	15:45	19:31	20:45	22:16
54°	14:13	14:59	15:29	19:53	21:05	22:31
52°	13:55	14:42	15:16	20:12	21:22	22:44
50°	13:39	14:28	15:04	20:27	21:36	22:56
45°	13:07	13:58	14:39	20:59	22:05	23:19
N 40°	12:43	13:35	14:19	21:23	22:27	23:38
35°	12:23	13:16	14:03	21:43	22:46	23:54
30°	12:06	13:00	13:48	22:00	23:02	· · ·
20°	11:37	12:32	13:24	22:29	23:29	· · ·
N 10°	11:12	12:08	13:02	22:53	23:52	· · ·
0°	10:49	11:46	12:42	23:16	· · ·	00:13
S 10°	10:26	11:23	12:22	23:39	· · ·	00:34
20°	10:01	10:59	12:01	· · ·	00:04	00:57
30°	09:33	10:32	11:36	· · ·	00:32	01:23
35°	09:16	10:15	11:21	· · ·	00:49	01:39
40°	08:56	09:56	11:04	00:13	01:08	01:56
45°	08:32	09:32	10:43	00:37	01:32	02:18
S 50°	08:01	09:02	10:17	01:08	02:03	02:45</

October 12, 13, 14 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	21°03.7	150°43.0	S19°18.9	268°53.8	N22°41.4	300°27.1	N22°26.3	35°14.6	S08°25.5	Alpheratz	357°34.7	29°13.8
1	36°06.2	165°42.4	19.8	283°55.0	41.3	315°29.6	26.3	50°17.2	25.6	Ankaa	353°06.9	-42°10.2
2	51°08.6	180°41.8	20.7	298°56.2	41.1	330°32.1	26.3	65°19.8	25.6	Schedar	349°30.8	56°40.5
3	66°11.1	195°41.2	· · 21.6	313°57.3	· · 41.0	345°34.6	· · 26.3	80°22.4	· · 25.6	Diphda	348°47.2	-17°50.9
4	81°13.5	210°40.6	22.5	328°58.5	40.9	0°37.1	26.3	95°25.0	25.7	Achernar	335°19.7	-57°06.6
5	96°16.0	225°39.9	23.4	343°59.7	40.8	15°39.6	26.3	110°27.5	25.7	Hamal	327°51.1	23°34.9
6	111°18.5	240°39.3	S19°24.3	359°00.9	N22°40.7	30°42.1	N22°26.3	125°30.1	S08°25.8	Polaris	313°43.0	89°22.0
7	126°20.9	255°38.7	25.2	14°02.0	40.6	45°44.6	26.3	140°32.7	25.8	Acamar	315°11.5	-40°12.1
8	141°23.4	270°38.1	26.1	29°03.2	40.5	60°47.0	26.3	155°35.3	25.9	Menkar	314°06.1	4°11.3
9	156°25.9	285°37.5	· · 26.9	44°04.4	· · 40.3	75°49.5	· · 26.3	170°37.9	· · 25.9	Mirfak	308°28.2	49°56.9
10	171°28.3	300°36.9	27.8	59°05.6	40.2	90°52.0	26.3	185°40.5	26.0	Aldebaran	290°39.6	16°33.6
11	186°30.8	315°36.3	28.7	74°06.8	40.1	105°54.5	26.3	200°43.1	26.0	Rigel	281°03.9	-8°10.2
12	201°33.3	330°35.7	S19°29.6	89°07.9	N22°40.0	120°57.0	N22°26.3	215°45.7	S08°26.1	Capella	280°22.0	46°01.3
13	216°35.7	345°35.1	30.5	104°09.1	39.9	135°59.5	26.3	230°48.3	26.1	Bellatrix	278°23.0	6°22.5
14	231°38.2	0°34.4	31.4	119°10.3	39.8	151°02.0	26.3	245°50.9	26.2	Elnath	278°02.0	28°37.7
15	246°40.7	15°33.8	· · 32.2	134°11.5	· · 39.7	166°04.5	· · 26.3	260°53.5	· · 26.2	Alnilam	275°37.8	-1°11.0
16	261°43.1	30°33.2	33.1	149°12.7	39.5	181°07.0	26.3	275°56.0	26.3	Betelgeuse	270°52.2	7°24.8
17	276°45.6	45°32.6	34.0	164°13.8	39.4	196°09.5	26.3	290°58.6	26.3	Canopus	263°52.4	-52°42.1
18	291°48.0	60°32.0	S19°34.9	179°15.0	N22°39.3	211°12.0	N22°26.3	306°01.2	S08°26.4	Sirius	258°26.4	-16°44.7
19	306°50.5	75°31.4	35.7	194°16.2	39.2	226°14.5	26.3	321°03.8	26.4	Adhara	255°06.0	-29°00.0
20	321°53.0	90°30.8	36.6	209°17.4	39.1	241°17.0	26.3	336°06.4	26.5	Procyon	244°51.1	5°09.8
21	336°55.4	105°30.1	· · 37.5	224°18.6	· · 39.0	256°19.5	· · 26.3	351°09.0	· · 26.5	Pollux	243°17.6	27°58.0
22	351°57.9	120°29.5	38.4	239°19.8	38.8	271°22.0	26.3	6°11.6	26.6	Avior	234°15.0	-59°34.9
23	7°00.4	135°28.9	39.2	254°20.9	38.7	286°24.5	26.3	21°14.2	26.6	Suhail	222°46.7	-43°31.6
Mer.pass. 22:32		ν -0.6' d 0.9' m -3.94		ν 1.2' d -0.1' m 0.33		ν 2.5' d -0.0' m -2.55		ν 2.6' d 0.0' m 0.70		Miaplacidus	221°38.8	-69°48.7
										Alphard	217°48.1	-8°45.8
										Regulus	207°34.9	11°50.9
										Dubhe	193°41.8	61°37.0
										Denebola	182°25.5	14°26.1
										Gienah	175°44.1	-17°40.6
										Acrux	173°01.1	-63°14.0
										Gacrux	171°52.5	-57°15.0
										Alioth	166°13.7	55°49.5
										Spica	158°22.9	-11°17.3
										Alkaid	152°52.6	49°11.5
										Hadar	148°37.0	-60°29.5
										Menkent	147°58.3	-36°29.4
										Arcturus	145°48.4	19°03.3
										Rigel Kent.	139°41.3	-60°56.3
										Kochab	137°20.8	74°03.3
										Zuben'ubi	136°56.6	-16°08.6
										Alphecca	126°04.2	26°38.0
										Antares	112°16.4	-26°29.2
										Atria	107°11.1	-69°04.5
										Sabik	102°03.2	-15°45.3
										Shaula	96°10.9	-37°07.4
										Rasalhague	95°58.9	12°32.7
										Eltanin	90°42.5	51°29.4
										Kaus Aust.	83°32.9	-34°22.5
										Vega	80°33.5	38°48.6
										Nunki	75°48.1	-26°16.0
										Altair	62°00.1	8°56.2
										Peacock	53°05.9	-56°39.5
										Deneb	49°25.8	45°22.4
										Enif	33°38.8	9°59.4
										Al Na'ir	27°32.9	-46°50.6
										Fomalhaut	15°14.5	-29°29.5
										Scheat	13°45.1	28°13.2
										Markab	13°29.9	15°20.5
										Oct 12 Sat	SHA	Mer.pass
										Venus	129°39.3	13:58
										Mars	247°50.1	06:04
										Jupiter	279°23.4	03:58
										Saturn	14°10.9	21:35
										Oct 13 Sun	SHA	Mer.pass
										Venus	128°25.5	13:59
										Mars	247°19.3	06:02
										Jupiter	279°24.1	03:54
										Saturn	14°13.9	21:31
										Oct 14 Mon	SHA	Mer.pass
										Venus	127°11.3	14:00
										Mars	246°48.8	06:00
										Jupiter	279°25.0	03:50
										Saturn	14°16.9	21:27
										Horizontal parallax		
										Venus:	0.1	
										Mars:	0.1	

Sun				Moon				
Sat	GHA	Dec		GHA	ν	Dec	d	HP
0	183°22.9	S07°30.1		73°16.5	7.7'	S23°44.5	-9.8'	58.2'
1	198°23.0	31.0		87°43.2	7.8'	23°34.7	-10.0'	58.2'
2	213°23.2	31.9		102°10.0	7.8'	23°24.7	-10.1'	58.2'
3	228°23.3	· · 32.9		116°36.8	7.9'	23°14.6	-10.3'	58.3'
4	243°23.5	33.8		131°03.7	7.9'	23°04.3	-10.4'	58.3'
5	258°23.6	34.7		145°30.6	8.0'	22°53.9	-10.6'	58.4'
6	273°23.8	S07°35.7		159°57.6	8.0'	S22°43.3	-10.7'	58.4'
7	288°23.9	36.6		174°24.6	8.0'	22°32.6	-10.9'	58.4'
8	303°24.1	37.6		188°51.6	8.1'	22°21.7	-11.0'	58.5'
9	318°24.3	· · 38.5		203°18.7	8.1'	22°10.7	-11.1'	58.5'
10	333°24.4	39.4		217°45.9	8.2'	21°59.6	-11.3'	58.5'
11	348°24.6	40.4		232°13.0	8.2'	21°48.3	-11.4'	58.6'
12	3°24.7	S07°41.3		246°40.3	8.3'	S21°36.9	-11.6'	58.6'
13	18°24.9	42.2		261°07.5	8.3'	21°25.3	-11.7'	58.7'
14	33°25.0	43.2		275°34.9	8.4'	21°13.6	-11.8'	58.7'
15	48°25.2	· · 44.1		290°02.2	8.4'	21°01.8	-12.0'	58.7'
16	63°25.3	45.0		304°29.6	8.5'	20°49.8	-12.1'	58.8'
17	78°25.5	46.0		318°57.1	8.5'	20°37.7	-12.2'	58.8'
18	93°25.6	S07°46.9		333°24.6	8.6'	S20°25.5	-12.4'	58.8'
19	108°25.8	47.8		347°52.2	8.6'	20°13.1	-12.5'	58.9'
20	123°25.9	48.8		2°19.8	8.6'	20°00.6	-12.6'	58.9'
21	138°26.1	· · 49.7		16°47.4	8.7'	19°47.9	-12.8'	59.0'
22	153°26.2	50.6		31°15.1	8.7'	19°35.2	-12.9'	59.0'
23	168°26.4	51.6		45°42.8	8.8'	19°22.3	-13.0'	59.0'
SD = 16.0' $d = 0.9'$				SD = 15.9'				
Sun	GHA	Dec		GHA	ν	Dec	d	HP
0	183°26.5	S07°52.5		60°10.6	8.8'	S19°09.3	-13.1'	59.1'
1	198°26.7	53.4		74°38.4	8.9'	18°56.1	-13.3'	59.1'
2	213°26.8	54.4		89°06.3	8.9'	18°42.8	-13.4'	59.2'
3	228°27.0	· · 55.3		103°34.2	9.0'	18°29.5	-13.5'	59.2'
4	243°27.1	56.2		118°02.2	9.0'	18°15.9	-13.6'	59.2'
5	258°27.3	57.2		132°30.2	9.1'	18°02.3	-13.8'	59.3'
6	273°27.4	S07°58.1		146°58.2	9.1'	S17°48.5	-13.9'	59.3'
7	288°27.6	07°59.0		161°26.3	9.1'	17°34.7	-14.0'	59.3'
8	303°27.7	08°00.0		175°54.5	9.2'	17°20.7	-14.1'	59.4'
9	318°27.9	· · 00.9		190°22.6	9.2'	17°06.6	-14.2'	59.4'
10	333°28.0	01.8		204°50.9	9.3'	16°52.4	-14.3'	59.4'
11	348°28.2	02.8		219°19.1	9.3'	16°38.0	-14.4'	59.5'
12	3°28.3	S08°03.7		233°47.4	9.3'	S16°23.6	-14.6'	59.5'
13	18°28.5	04.6		248°15.8	9.4'	16°09.0	-14.7'	59.6'
14	33°28.6	05.5		262°44.2	9.4'	15°54.4	-14.8'	59.6'
15	48°28.8	· · 06.5		277°12.6	9.5'	15°39.6	-14.9'	59.6'
16	63°28.9	07.4		291°41.1	9.5'	15°24.7	-15.0'	59.7'
17	78°29.1	08.3		306°09.6	9.5'	15°09.7	-15.1'	59.7'
18	93°29.2	S08°09.3		320°38.1	9.6'	S14°54.6	-15.2'	59.7'
19	108°29.4	10.2		335°06.7	9.6'	14°39.4	-15.3'	59.8'
20	123°29.5	11.1		349°35.3	9.7'	14°24.2	-15.4'	59.8'
21	138°29.6	· · 12.0		4°04.0	9.7'	14°08.8	-15.5'	59.8'
22	153°29.8	13.0		18°32.7	9.7'	13°53.3	-15.6'	59.9'
23	168°29.9	13.9		33°01.4	9.8'	13°37.7	-15.7'	59.9'
SD = 16.0' $d = 0.9'$				SD = 16.1'				
Mon	GHA	Dec		GHA	ν	Dec	d	HP
0	183°30.1	S08°14.8		47°30.2	9.8'	S13°22.0	-15.8'	59.9'
1	198°30.2	15.8		61°59.0	9.8'	13°06.3	-15.9'	60.0'
2	213°30.4	16.7		76°27.8	9.9'	12°50.4	-15.9'	60.0'
3	228°30.5	· · 17.6		90°56.7	9.9'	12°34.5	-16.0'	60.0'
4	243°30.7	18.5		105°25.5	9.9'	12°18.4	-16.1'	60.1'
5	258°30.8	19.5		119°54.5	9.9'	12°02.3	-16.2'	60.1'
6	273°31.0	S08°20.4		134°23.4	10.0'	S11°46.1	-16.3'	60.1'
7	288°31.1	21.3		148°52.4	10.0'	11°29.8	-16.4'	60.2'
8	303°31.2	22.3		163°21.4	10.0'	11°13.4	-16.5'	60.2'
9	318°31.4	· · 23.2		177°50.4	10.1'	10°57.0	-16.5'	60.2'
10	333°31.5	24.1		192°19.5	10.1'	10°40.4	-16.6'	60.3'
11	348°31.7	25.0		206°48.5	10.1'	10°23.8	-16.7'	60.3'
12	3°31.8	S08°26.0		221°17.6	10.1'	S10°07.2	-16.8'	60.3'
13	18°32.0	26.9		235°46.8	10.1'	09°50.4	-16.8'	60.3'
14	33°32.1	27.8		250°15.9	10.2'	09°33.6	-16.9'	60.4'
15	48°32.2	· · 28.7		264°45.1	10.2'	09°16.7	-17.0'	60.4'
16	63°32.4	29.7		279°14.3	10.2'	08°59.7	-17.0'	60.4'
17	78°32.5	30.6		293°43.5	10.2'	08°42.7	-17.1'	60.5'
18	93°32.7	S08°31.5		308°12.7	10.2'	S08°25.6	-17.2'	60.5'
19	108°32.8	32.4		322°41.9	10.3'	08°08.4	-17.2'	60.5'
20	123°32.9	33.4		337°11.2	10.3'	07°51.2	-17.3'	60.5'
21	138°33.1	· · 34.3		351°40.4	10.3'	07°33.9	-17.3'	60.6'
22	153°33.2	35.2		6°09.7	10.3'	07°16.5	-17.4'	60.6'
23	168°33.4	36.1		20°39.0	10.3'	06°59.1	-17.5'	60.6'
SD = 16.0' $d = 0.9'$				SD = 16.3'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	04:48	06:07	07:17	16:14	17:23	18:41
N 70°	04:53	06:04	07:06	16:24	17:27	18:37
68°	04:57	06:01	06:58	16:33	17:29	18:33
66°	05:00	05:59	06:51	16:40	17:32	18:31
64°	05:02	05:57	06:45	16:46	17:34	18:29
62°	05:04	05:55	06:40	16:51	17:36	18:27
60°	05:05	05:53	06:35	16:56	17:38	18:26
N 58°	05:06	05:52	06:32	17:00	17:39	18:24
56°	05:07	05:50	06:28	17:03	17:41	18:24
54°	05:08	05:49	06:25	17:06	17:42	18:23
52°	05:09	05:48	06:22	17:09	17:43	18:22
50°	05:09	05:47	06:19	17:12	17:45	18:22
45°	05:10	05:44	06:14	17:18	17:47	18:21
N 40°	05:10	05:42	06:09	17:23	17:50	18:21
35°	05:10	05:39	06:05	17:27	17:53	18:22
30°	05:09	05:37	06:01	17:31	17:55	18:23
20°	05:06	05:32	05:54	17:38	18:00	18:25
N 10°	05:03	05:27	05:48	17:44	18:05	18:29
0°	04:58	05:22	05:43	17:49	18:10	18:35
S 10°	04:51	05:16	05:37	17:55	18:16	18:41
20°	04:43	05:09	05:31	18:02	18:24	18:50
30°	04:31	04:59	05:24	18:09	18:33	19:02
35°	04:23	04:54	05:20	18:13	18:39	19:10
40°	04:14	04:47	05:15	18:18	18:46	19:19
45°	04:03	04:39	05:09	18:24	18:54	19:31
S 50°	03:48	04:29	05:03	18:31	19:04	19:46
52°	03:41	04:24	04:59	18:34	19:09	19:53
54°	03:33	04:19	04:56	18:37	19:15	20:01
56°	03:24	04:13	04:52	18:41	19:21	20:10
58°	03:13	04:06	04:48	18:46	19:28	20:21
S 60°	03:01	03:59	04:43	18:51	19:36	20:34

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	■	18:20	17:19	■	23:36	· · ·
N 70°	■	17:49	17:07	■	■	00:05
68°	18:18	17:26	16:56	21:42	· · ·	00:26
66°	17:37	17:07	16:48	22:22	· · ·	00:43
64°	17:08	16:52	16:40	22:50	· · ·	00:56
62°	16:46	16:40	16:34	23:11	· · ·	01:07
60°	16:29	16:29	16:28	23:27	· · ·	01:17
N 58°	16:14	16:19	16:23	23:42	· · ·	01:25
56°	16:01	16:11	16:19	23:54	· · ·	01:32
54°	15:49	16:04	16:15	· · ·	00:04	01:39
52°	15:39	15:57	16:11	· · ·	00:13	01:44
50°	15:31	15:51	16:08	· · ·	00:22	01:49
45°	15:11	15:38	16:00	· · ·	00:39	02:01
N 40°	14:56	15:27	15:54	· · ·	00:53	02:10
35°	14:43	15:18	15:49	· · ·	01:05	02:17
30°	14:31	15:09	15:44	00:08	01:16	02:24
20°	14:11	14:55	15:36	00:31	01:33	02:36
N 10°	13:54	14:43	15:29	00:50	01:49	02:46
0°	13:37	14:31	15:22	01:09	02:03	02:55
S 10°	13:21	14:19	15:16	01:27	02:17	03:05
20°	13:03	14:06	15:08	01:47	02:32	03:15
30°	12:43	13:52	15:00	02:09	02:49	03:26
35°	12:31	13:43	14:55	02:22	02:59	03:32
40°	12:17	13:33	14:50	02:37	03:10	03:39
45°	12:01	13:22	14:43	02:54	03:23	03:48
S 50°	11:41	13:08	14:36	03:16	03:39	03:57
52°	11:31	13:01	14:32	03:26	03:47	04:02
54°	11:20	12:54	14:28	03:38	03:55	04:07
56°	11:07	12:46	14:24	03:51	04:04	04:13
58°	10:53	12:36	14:19	04:06	04:14	04:19
S 60°	10:36	12:26	14:13	04:24	04:26	04:26

Day	Sun		Mer. Pass	Moon		Age
	Eqn.of Time	00 ^h 12 ^h		Mer.Pass.	10-12	
	mm:ss	mm:ss	hh:mm	Upper hh:mm	Lower hh:mm	63-83%
12	13:31	13:39	11:46	19:50	07:23	
13	13:46	13:53	11:46	20:43	08:17	
14	14:00	14:07	11:46	21:34	09:09	

October 15, 16, 17 UT (Tue., Wed., Thu.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	24°01.1	149°58.0	S20°21.0	270°19.7	N22°33.0	303°27.3	N22°26.1	38°20.9	S08°28.9	Alpheratz	357°34.7	29°13.8											
1	39°03.6	164°57.4	21.8	285°20.9	32.9	318°29.8	26.1	53°23.5	28.9	Ankaa	353°06.9	-42°10.3											
2	54°06.0	179°56.7	22.6	300°22.1	32.8	333°32.3	26.1	68°26.1	29.0	Schedar	349°30.8	56°40.5											
3	69°08.5	194°56.1	•• 23.4	315°23.3	•• 32.7	348°34.8	•• 26.1	83°28.7	•• 29.0	Diphda	348°47.2	-17°51.0											
4	84°11.0	209°55.4	24.3	330°24.6	32.5	3°37.3	26.1	98°31.3	29.1	Achernar	335°19.7	-57°06.6											
5	99°13.4	224°54.8	25.1	345°25.8	32.4	18°39.9	26.1	113°33.9	29.1	Hamal	327°51.1	23°34.9											
6	114°15.9	239°54.1	S20°25.9	0°27.0	N22°32.3	33°42.4	N22°26.1	128°36.4	S08°29.2	Polaris	313°42.4	89°22.0											
7	129°18.4	254°53.5	26.7	15°28.2	32.2	48°44.9	26.1	143°39.0	29.2	Acamar	315°11.5	-40°12.1											
8	144°20.8	269°52.9	27.6	30°29.4	32.1	63°47.4	26.1	158°41.6	29.2	Menkar	314°06.1	4°11.3											
9	159°23.3	284°52.2	•• 28.4	45°30.6	•• 32.0	78°49.9	•• 26.1	173°44.2	•• 29.3	Mirfak	308°28.1	49°56.9											
10	174°25.7	299°51.6	29.2	60°31.9	31.8	93°52.5	26.1	188°46.8	29.3	Aldebaran	290°39.6	16°33.6											
11	189°28.2	314°50.9	30.0	75°33.1	31.7	108°55.0	26.1	203°49.4	29.4	Rigel	281°03.9	-8°10.2											
12	204°30.7	329°50.3	S20°30.9	90°34.3	N22°31.6	123°57.5	N22°26.1	218°51.9	S08°29.4	Capella	280°21.9	46°01.3											
13	219°33.1	344°49.6	31.7	105°35.5	31.5	139°00.0	26.1	233°54.5	29.5	Bellatrix	278°22.9	6°22.5											
14	234°35.6	359°49.0	32.5	120°36.7	31.4	154°02.5	26.1	248°57.1	29.5	Elnath	278°01.9	28°37.7											
15	249°38.1	14°48.3	•• 33.3	135°38.0	•• 31.2	169°05.1	•• 26.1	263°59.7	•• 29.6	Alnilam	275°37.8	-1°11.0											
16	264°40.5	29°47.7	34.1	150°39.2	31.1	184°07.6	26.1	279°02.3	29.6	Betelgeuse	270°52.2	7°24.8											
17	279°43.0	44°47.0	34.9	165°40.4	31.0	199°10.1	26.1	294°04.8	29.6	Canopus	263°52.3	-52°42.1											
18	294°45.5	59°46.4	S20°35.7	180°41.6	N22°30.9	214°12.6	N22°26.0	309°07.4	S08°29.7	Sirius	258°26.3	-16°44.7											
19	309°47.9	74°45.7	36.6	195°42.8	30.8	229°15.1	26.0	324°10.0	29.7	Adhara	255°06.0	-29°00.0											
20	324°50.4	89°45.1	37.4	210°44.1	30.6	244°17.7	26.0	339°12.6	29.8	Procyon	244°51.1	5°09.8											
21	339°52.8	104°44.4	•• 38.2	225°45.3	•• 30.5	259°20.2	•• 26.0	354°15.2	•• 29.8	Pollux	243°17.6	27°58.0											
22	354°55.3	119°43.8	39.0	240°46.5	30.4	274°22.7	26.0	9°17.8	29.9	Avior	234°15.0	-59°34.9											
23	9°57.8	134°43.1	39.8	255°47.7	30.3	289°25.2	26.0	24°20.3	29.9	Suhail	222°46.7	-43°31.6											
Mer.pass. 22:20				ν -0.6' d 0.8' m -3.95				ν 1.2' d -0.1' m 0.30				ν 2.5' d -0.0' m -2.57				ν 2.6' d 0.0' m 0.72							

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	25°00.2	149°42.4	S20°40.6	270°49.0	N22°30.2	304°27.8	N22°26.0	39°22.9	S08°30.0										
1	40°02.7	164°41.8	41.4	285°50.2	30.1	319°30.3	26.0	54°25.5	30.0										
2	55°05.2	179°41.1	42.2	300°51.4	29.9	334°32.8	26.0	69°28.1	30.0										
3	70°07.6	194°40.5	•• 43.0	315°52.7	•• 29.8	349°35.3	•• 26.0	84°30.7	•• 30.1										
4	85°10.1	209°39.8	43.8	330°53.9	29.7	4°37.9	26.0	99°33.2	30.1										
5	100°12.6	224°39.2	44.6	345°55.1	29.6	19°40.4	26.0	114°35.8	30.2										
6	115°15.0	239°38.5	S20°45.4	0°56.3	N22°29.5	34°42.9	N22°26.0	129°38.4	S08°30.2										
7	130°17.5	254°37.9	46.2	15°57.6	29.3	49°45.4	26.0	144°41.0	30.3										
8	145°20.0	269°37.2	47.0	30°58.8	29.2	64°48.0	26.0	159°43.6	30.3										
9	160°22.4	284°36.5	•• 47.8	46°00.0	•• 29.1	79°50.5	•• 26.0	174°46.1	•• 30.3										
10	175°24.9	299°35.9	48.6	61°01.3	29.0	94°53.0	26.0	189°48.7	30.4										
11	190°27.3	314°35.2	49.4	76°02.5	28.9	109°55.6	26.0	204°51.3	30.4										
12	205°29.8	329°34.6	S20°50.2	91°03.7	N22°28.7	124°58.1	N22°26.0	219°53.9	S08°30.5										
13	220°32.3	344°33.9	51.0	106°05.0	28.6	140°00.6	26.0	234°56.5	30.5										
14	235°34.7	359°33.2	51.8	121°06.2	28.5	155°03.1	26.0	249°59.0	30.6										
15	250°37.2	14°32.6	•• 52.6	136°07.4	•• 28.4	170°05.7	•• 26.0	265°01.6	•• 30.6										
16	265°39.7	29°31.9	53.4	151°08.7	28.3	185°08.2	26.0	280°04.2	30.6										
17	280°42.1	44°31.2	54.2	166°09.9	28.1	200°10.7	26.0	295°06.8	30.7										
18	295°44.6	59°30.6	S20°55.0	181°11.2	N22°28.0	215°13.3	N22°25.9	310°09.4	S08°30.7										
19	310°47.1	74°29.9	55.8	196°12.4	27.9	230°15.8	25.9	325°11.9	30.8										
20	325°49.5	89°29.3	56.6	211°13.6	27.8	245°18.3	25.9	340°14.5	30.8										
21	340°52.0	104°28.6	•• 57.4	226°14.9	•• 27.7	260°20.9	•• 25.9	355°17.1	•• 30.9										
22	355°54.4	119°27.9	58.1	241°16.1	27.5	275°23.4	25.9	10°19.7	30.9										
23	10°56.9	134°27.3	58.9	256°17.3	27.4	290°25.9	25.9	25°22.2	30.9										
Mer.pass. 22:16				ν -0.7' d 0.8' m -3.95				ν 1.2' d -0.1' m 0.29				ν 2.5' d -0.0' m -2.58				ν 2.6' d 0.0' m 0.72			

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	25°59.4	149°26.6	S20°59.7	271°18.6	N22°27.3	305°28.5	N22°25.9	40°24.8	S08°31.0										
1	41°01.8	164°25.9	21°00.5	286°19.8	27.2	320°31.0	25.9	55°27.4	31.0										
2	56°04.3	179°25.3	01.3	301°21.1	27.1	335°33.5	25.9	70°30.0	31.1										
3	71°06.8	194°24.6	•• 02.1	316°22.3	•• 26.9	350°36.1	•• 25.9	85°32.6	•• 31.1										
4	86°09.2	209°23.9	02.8	331°23.6	26.8	5°38.6	25.9	100°35.1	31.1										
5	101°11.7	224°23.3	03.6	346°24.8	26.7	20°41.1	25.9	115°37.7	31.2										
6	116°14.2	239°22.6	S21°04.4	1°26.0	N22°26.6	35°43.7	N22°25.9	130°40.3	S08°31.2										
7	131°16.6	254°21.9	05.2	16°27.3	26.5	50°46.2	25.9	145°42.9	31.3										
8	146°19.1	269°21.3	06.0	31°28.5	26.3	65°48.7	25.9	160°45.4	31.3										
9	161°21.6	284°20.6	•• 06.7	46°29.8	•• 26.2	80°51.3	•• 25.9	175°48.0	•• 31.4										
10	176°24.0	299°19.9	07.5	61°31.0	26.1	95°53.8	25.9	190°50.6	31.4										
11	191°26.5	314°19.2	08.3	76°32.3	26.0	110°56.4	25.9	205°53.2	31.4										
12	206°28.9	329°18.6	S21°09.1	91°33.5	N22°25.9	125°58.9	N22°25.9	220°55.7	S08°31.5										
13	221°31.4	344°17.9	09.8	106°34.8	25.7	141°01.4	25.9	235°58.3	31.5										
14	236°33.9	359°17.2	10.6	121°36.0	25.6	156°04.0	25.9	251°00.9	31.6										
15	251°36.3	14°16.6	•• 11.4	136°37.3	•• 25.5	171°06.5	•• 25.9	266°03.5	•• 31.6										
16	266°38.8	29°15.9	12.1	151°38.5	25.4	186°09.1	25.8	281°06.0	31.6										
17	281°41.3	44°15.2	12.9	166°39.8	25.3	201°11.6	25.8	296°08.6	31.7										
18	296°43.7	59°14.5	S21°13.7	181°41.0	N22°25.1	216°14.1	N22°25.8	311°11.2	S08°31.7										
19	311°46.2	74°13.9	14.4	196°42.3	25.0	231°16.7	25.8	326°13.8	31.8										
20	326°48.7	89°13.2	15.2	211°43.5	24.9	246°19.2	25.8	341°16.4	31.8										
21	341°51.1	104°12.5	•• 16.0	226°44.8	•• 24.8	261°21.8	•• 25.8	356°18.9	•• 31.9										
22	356°53.6	119°11.8	16.7	241°46.0	24.7	276°24.3	25.8	11°21.5	31.9										
23	11°56.1	134°11.2	17.5	256°47.3	24.5	291°26.8	25.8	26°24.1	31.9										
Mer.pass. 22:12				ν -0.7' d 0.8' m -3.96				ν 1.2' d -0.1' m 0.27				ν 2.5' d -0.0' m -2.59				ν 2.6' d 0.0' m 0.73			

Oct 15 Tue	SHA	Mer.pass
Venus	125°56.9	14:01
Mars	246°18.6	05:58
Jupiter	279°26.2	03:46
Saturn	14°19.8	21:23

Oct 16 Wed	SHA	Mer.pass
Venus	124°42.2	14:02
Mars	245°48.7	05:56
Jupiter	279°27.5	03:42
Saturn	14°22.7	21:19

Oct 17 Thu	SHA	Mer.pass
Venus	123°27.2	14:03
Mars	245°19.2	05:54
Jupiter	279°29.1	03:37
Saturn	14°25.4	21:15

Horizontal parallax		
Venus:	0.1	
Mars:	0.1	

Sun				Moon				
Tue	GHA	Dec		GHA	ν	Dec	d	HP
0	183°33.5	S08°37.0		35°08.3	10.3'	S06°41.7	-17.5'	60.6'
1	198°33.6	38.0		49°37.6	10.3'	06°24.2	-17.6'	60.7'
2	213°33.8	38.9		64°06.9	10.3'	06°06.6	-17.6'	60.7'
3	228°33.9	· · 39.8		78°36.3	10.3'	05°49.0	-17.7'	60.7'
4	243°34.1	40.7		93°05.6	10.3'	05°31.3	-17.7'	60.7'
5	258°34.2	41.7		107°34.9	10.3'	05°13.6	-17.7'	60.8'
6	273°34.3	S08°42.6		122°04.3	10.3'	S04°55.9	-17.8'	60.8'
7	288°34.5	43.5		136°33.6	10.3'	04°38.1	-17.8'	60.8'
8	303°34.6	44.4		151°03.0	10.3'	04°20.3	-17.9'	60.8'
9	318°34.7	· · 45.3		165°32.3	10.3'	04°02.4	-17.9'	60.9'
10	333°34.9	46.3		180°01.6	10.3'	03°44.5	-17.9'	60.9'
11	348°35.0	47.2		194°31.0	10.3'	03°26.6	-18.0'	60.9'
12	3°35.2	S08°48.1		209°00.3	10.3'	S03°08.6	-18.0'	60.9'
13	18°35.3	49.0		223°29.7	10.3'	02°50.6	-18.0'	60.9'
14	33°35.4	49.9		237°59.0	10.3'	02°32.6	-18.1'	61.0'
15	48°35.6	· · 50.9		252°28.3	10.3'	02°14.5	-18.1'	61.0'
16	63°35.7	51.8		266°57.6	10.3'	01°56.5	-18.1'	61.0'
17	78°35.8	52.7		281°26.9	10.3'	01°38.4	-18.1'	61.0'
18	93°36.0	S08°53.6		295°56.2	10.3'	S01°20.2	-18.1'	61.0'
19	108°36.1	54.5		310°25.5	10.3'	01°02.1	-18.2'	61.1'
20	123°36.2	55.5		324°54.7	10.2'	00°43.9	-18.2'	61.1'
21	138°36.4	· · 56.4		339°24.0	10.2'	00°25.8	-18.2'	61.1'
22	153°36.5	57.3		353°53.2	10.2'	S00°07.6	-18.2'	61.1'
23	168°36.6	58.2		8°22.4	10.2'	N00°10.6	18.2'	61.1'
SD = 16.0'			$d = 0.9'$	SD = 16.5'				
Wed	GHA	Dec		GHA	ν	Dec	d	HP
0	183°36.8	S08°59.1		22°51.6	10.2'	N00°28.8	18.2'	61.1'
1	198°36.9	09°00.0		37°20.8	10.2'	00°47.0	18.2'	61.1'
2	213°37.0	01.0		51°49.9	10.1'	01°05.2	18.2'	61.2'
3	228°37.2	· · 01.9		66°19.1	10.1'	01°23.4	18.2'	61.2'
4	243°37.3	02.8		80°48.2	10.1'	01°41.7	18.2'	61.2'
5	258°37.4	03.7		95°17.3	10.1'	01°59.9	18.2'	61.2'
6	273°37.6	S09°04.6		109°46.3	10.0'	N02°18.1	18.2'	61.2'
7	288°37.7	05.6		124°15.3	10.0'	02°36.3	18.2'	61.2'
8	303°37.8	06.5		138°44.3	10.0'	02°54.5	18.2'	61.2'
9	318°38.0	· · 07.4		153°13.3	9.9'	03°12.7	18.2'	61.2'
10	333°38.1	08.3		167°42.2	9.9'	03°30.8	18.2'	61.2'
11	348°38.2	09.2		182°11.1	9.9'	03°49.0	18.1'	61.3'
12	3°38.4	S09°10.1		196°40.0	9.8'	N04°07.1	18.1'	61.3'
13	18°38.5	11.0		211°08.9	9.8'	04°25.3	18.1'	61.3'
14	33°38.6	12.0		225°37.7	9.8'	04°43.4	18.1'	61.3'
15	48°38.8	· · 12.9		240°06.4	9.7'	05°01.4	18.1'	61.3'
16	63°38.9	13.8		254°35.1	9.7'	05°19.5	18.0'	61.3'
17	78°39.0	14.7		269°03.8	9.6'	05°37.5	18.0'	61.3'
18	93°39.1	S09°15.6		283°32.5	9.6'	N05°55.5	18.0'	61.3'
19	108°39.3	16.5		298°01.1	9.6'	06°13.5	17.9'	61.3'
20	123°39.4	17.4		312°29.6	9.5'	06°31.4	17.9'	61.3'
21	138°39.5	· · 18.4		326°58.2	9.5'	06°49.3	17.9'	61.3'
22	153°39.7	19.3		341°26.6	9.4'	07°07.2	17.8'	61.3'
23	168°39.8	20.2		355°55.0	9.4'	07°25.0	17.8'	61.3'
SD = 16.0'			$d = 0.9'$	SD = 16.7'				
Thu	GHA	Dec		GHA	ν	Dec	d	HP
0	183°39.9	S09°21.1		10°23.4	9.3'	N07°42.8	17.7'	61.3'
1	198°40.0	22.0		24°51.7	9.3'	08°00.5	17.7'	61.3'
2	213°40.2	22.9		39°20.0	9.2'	08°18.2	17.6'	61.3'
3	228°40.3	· · 23.8		53°48.2	9.2'	08°35.8	17.6'	61.3'
4	243°40.4	24.7		68°16.4	9.1'	08°53.4	17.5'	61.3'
5	258°40.5	25.6		82°44.5	9.1'	09°10.9	17.5'	61.3'
6	273°40.7	S09°26.6		97°12.6	9.0'	N09°28.4	17.4'	61.3'
7	288°40.8	27.5		111°40.6	9.0'	09°45.8	17.4'	61.3'
8	303°40.9	28.4		126°08.6	8.9'	10°03.2	17.3'	61.3'
9	318°41.0	· · 29.3		140°36.5	8.8'	10°20.5	17.2'	61.3'
10	333°41.2	30.2		155°04.3	8.8'	10°37.7	17.2'	61.3'
11	348°41.3	31.1		169°32.1	8.7'	10°54.9	17.1'	61.3'
12	3°41.4	S09°32.0		183°59.8	8.7'	N11°12.0	17.0'	61.3'
13	18°41.5	32.9		198°27.5	8.6'	11°29.0	17.0'	61.3'
14	33°41.7	33.8		212°55.0	8.5'	11°46.0	16.9'	61.3'
15	48°41.8	· · 34.7		227°22.6	8.5'	12°02.8	16.8'	61.3'
16	63°41.9	35.7		241°50.1	8.4'	12°19.6	16.7'	61.2'
17	78°42.0	36.6		256°17.5	8.3'	12°36.4	16.6'	61.2'
18	93°42.2	S09°37.5		270°44.8	8.3'	N12°53.0	16.6'	61.2'
19	108°42.3	38.4		285°12.1	8.2'	13°09.6	16.5'	61.2'
20	123°42.4	39.3		299°39.3	8.1'	13°26.0	16.4'	61.2'
21	138°42.5	· · 40.2		314°06.4	8.1'	13°42.4	16.3'	61.2'
22	153°42.7	41.1		328°33.5	8.0'	13°58.7	16.2'	61.2'
23	168°42.8	42.0		343°00.5	7.9'	14°14.9	16.1'	61.2'
SD = 16.0'			$d = 0.9'$	SD = 16.7'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	05:02	06:20	07:31	15:58	17:09	18:27
N 70°	05:05	06:15	07:19	16:10	17:14	18:24
68°	05:07	06:11	07:09	16:20	17:18	18:22
66°	05:09	06:08	07:01	16:29	17:21	18:20
64°	05:10	06:05	06:54	16:36	17:24	18:19
62°	05:11	06:03	06:48	16:42	17:27	18:18
60°	05:12	06:00	06:43	16:47	17:29	18:17
N 58°	05:13	05:58	06:38	16:52	17:31	18:17
56°	05:13	05:56	06:34	16:56	17:33	18:16
54°	05:14	05:55	06:30	16:59	17:35	18:16
52°	05:14	05:53	06:27	17:03	17:37	18:16
50°	05:14	05:51	06:24	17:06	17:39	18:16
45°	05:14	05:48	06:17	17:13	17:42	18:16
N 40°	05:13	05:45	06:12	17:18	17:46	18:17
35°	05:12	05:42	06:07	17:23	17:49	18:18
30°	05:11	05:39	06:03	17:28	17:52	18:20
20°	05:07	05:33	05:55	17:35	17:58	18:23
N 10°	05:03	05:27	05:49	17:42	18:03	18:28
0°	04:57	05:21	05:42	17:49	18:10	18:34
S 10°	04:50	05:14	05:36	17:55	18:17	18:42
20°	04:40	05:06	05:29	18:03	18:25	18:51
30°	04:27	04:56	05:20	18:11	18:35	19:04
35°	04:19	04:50	05:16	18:16	18:42	19:13
40°	04:09	04:42	05:10	18:21	18:49	19:23
45°	03:57	04:34	05:04	18:28	18:58	19:35
S 50°	03:41	04:22	04:56	18:36	19:10	19:51
52°	03:33	04:17	04:53	18:39	19:15	19:59
54°	03:25	04:11	04:49	18:43	19:21	20:08
56°	03:15	04:05	04:45	18:48	19:28	20:18
58°	03:03	03:57	04:40	18:53	19:35	20:30
S 60°	02:49	03:49	04:34	18:58	19:44	20:45

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	16:37	15:57	15:09	02:27	05:01	07:43
N 70°	16:35	16:04	15:29	02:36	04:59	07:26
68°	16:33	16:10	15:44	02:44	04:57	07:13
66°	16:31	16:15	15:57	02:50	04:55	07:03
64°	16:29	16:19	16:08	02:55	04:53	06:54
62°	16:28	16:23	16:17	03:00	04:52	06:46
60°	16:27	16:26	16:25	03:04	04:51	06:40
N 58°	16:26	16:29	16:32	03:07	04:50	06:34
56°	16:25	16:31	16:39	03:10	04:49	06:29
54°	16:24	16:34	16:44	03:13	04:48	06:25
52°	16:23	16:36	16:49	03:15	04:47	06:21
50°	16:23	16:38	16:54	03:18	04:47	06:17
45°	16:21	16:42	17:04	03:22	04:45	06:09
N 40°	16:20	16:45	17:13	03:26	04:44	06:03
35°	16:19	16:49	17:20	03:30	04:43	05:57

Aries Venus M

<https://pypi.org/project/sfalmanac/>

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	183°42.9	S09°42.9	357°27.4	7.9'	N14°31.0	16.0'	61.2'
1	198°43.0	43.8	11°54.3	7.8'	14°47.0	15.9'	61.1'
2	213°43.1	44.7	26°21.1	7.7'	15°02.9	15.8'	61.1'
3	228°43.3	· · 45.6	40°47.8	7.7'	15°18.7	15.7'	61.1'
4	243°43.4	46.5	55°14.5	7.6'	15°34.4	15.6'	61.1'
5	258°43.5	47.4	69°41.1	7.5'	15°50.0	15.5'	61.1'
6	273°43.6	S09°48.4	84°07.6	7.4'	N16°05.5	15.4'	61.1'
7	288°43.7	49.3	98°34.0	7.4'	16°20.9	15.3'	61.0'
8	303°43.9	50.2	113°00.4	7.3'	16°36.2	15.2'	61.0'
9	318°44.0	· · 51.1	127°26.7	7.2'	16°51.3	15.0'	61.0'
10	333°44.1	52.0	141°52.9	7.1'	17°06.4	14.9'	61.0'
11	348°44.2	52.9	156°19.0	7.1'	17°21.3	14.8'	61.0'
12	3°44.3	S09°53.8	170°45.1	7.0'	N17°36.1	14.7'	61.0'
13	18°44.4	54.7	185°11.1	6.9'	17°50.8	14.6'	60.9'
14	33°44.6	55.6	199°37.0	6.8'	18°05.3	14.4'	60.9'
15	48°44.7	· · 56.5	214°02.8	6.8'	18°19.8	14.3'	60.9'
16	63°44.8	57.4	228°28.6	6.7'	18°34.1	14.2'	60.9'
17	78°44.9	58.3	242°54.3	6.6'	18°48.3	14.0'	60.9'
18	93°45.0	S09°59.2	257°19.9	6.5'	N19°02.3	13.9'	60.8'
19	108°45.1	10°00.1	271°45.5	6.5'	19°16.2	13.8'	60.8'
20	123°45.3	01.0	286°11.0	6.4'	19°30.0	13.6'	60.8'
21	138°45.4	· · 01.9	300°36.4	6.3'	19°43.6	13.5'	60.8'
22	153°45.5	02.8	315°01.7	6.3'	19°57.1	13.4'	60.7'
23	168°45.6	03.7	329°26.9	6.2'	20°10.5	13.2'	60.7'
SD = 16.0' $d = 0.9'$ SD = 16.7'							

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	183°45.7	S10°04.6	343°52.1	6.1'	N20°23.7	13.1'	60.7'
1	198°45.8	05.5	358°17.2	6.0'	20°36.8	12.9'	60.7'
2	213°46.0	06.4	12°42.3	6.0'	20°49.7	12.8'	60.6'
3	228°46.1	· · 07.3	27°07.2	5.9'	21°02.5	12.6'	60.6'
4	243°46.2	08.2	41°32.1	5.8'	21°15.1	12.5'	60.6'
5	258°46.3	09.1	55°56.9	5.8'	21°27.6	12.3'	60.6'
6	273°46.4	S10°10.0	70°21.7	5.7'	N21°39.9	12.2'	60.5'
7	288°46.5	10.9	84°46.4	5.6'	21°52.1	12.0'	60.5'
8	303°46.6	11.8	99°11.0	5.5'	22°04.1	11.9'	60.5'
9	318°46.7	· · 12.7	113°35.5	5.5'	22°16.0	11.7'	60.4'
10	333°46.9	13.6	128°00.0	5.4'	22°27.7	11.5'	60.4'
11	348°47.0	14.5	142°24.4	5.3'	22°39.2	11.4'	60.4'
12	3°47.1	S10°15.4	156°48.8	5.3'	N22°50.6	11.2'	60.3'
13	18°47.2	16.3	171°13.0	5.2'	23°01.8	11.0'	60.3'
14	33°47.3	17.2	185°37.3	5.2'	23°12.9	10.9'	60.3'
15	48°47.4	· · 18.1	200°01.4	5.1'	23°23.7	10.7'	60.3'
16	63°47.5	19.0	214°25.5	5.0'	23°34.4	10.5'	60.2'
17	78°47.6	19.9	228°49.5	5.0'	23°45.0	10.4'	60.2'
18	93°47.7	S10°20.8	243°13.5	4.9'	N23°55.4	10.2'	60.2'
19	108°47.9	21.7	257°37.4	4.9'	24°05.6	10.0'	60.1'
20	123°48.0	22.6	272°01.3	4.8'	24°15.6	9.9'	60.1'
21	138°48.1	· · 23.5	286°25.1	4.7'	24°25.4	9.7'	60.1'
22	153°48.2	24.4	300°48.8	4.7'	24°35.1	9.5'	60.0'
23	168°48.3	25.2	315°12.5	4.6'	24°44.6	9.3'	60.0'
SD = 16.1' $d = 0.9'$ SD = 16.6'							

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	183°48.4	S10°26.1	329°36.2	4.6'	N24°53.9	9.1'	60.0'
1	198°48.5	27.0	343°59.8	4.5'	25°03.1	9.0'	59.9'
2	213°48.6	27.9	358°23.3	4.5'	25°12.1	8.8'	59.9'
3	228°48.7	· · 28.8	12°46.8	4.5'	25°20.8	8.6'	59.9'
4	243°48.8	29.7	27°10.3	4.4'	25°29.4	8.4'	59.8'
5	258°48.9	30.6	41°33.7	4.4'	25°37.9	8.2'	59.8'
6	273°49.0	S10°31.5	55°57.1	4.3'	N25°46.1	8.0'	59.7'
7	288°49.1	32.4	70°20.4	4.3'	25°54.1	7.9'	59.7'
8	303°49.2	33.3	84°43.7	4.3'	26°02.0	7.7'	59.7'
9	318°49.4	· · 34.2	99°07.0	4.2'	26°09.7	7.5'	59.6'
10	333°49.5	35.1	113°30.2	4.2'	26°17.2	7.3'	59.6'
11	348°49.6	36.0	127°53.4	4.2'	26°24.5	7.1'	59.6'
12	3°49.7	S10°36.9	142°16.6	4.1'	N26°31.6	6.9'	59.5'
13	18°49.8	37.7	156°39.7	4.1'	26°38.5	6.7'	59.5'
14	33°49.9	38.6	171°02.8	4.1'	26°45.2	6.5'	59.4'
15	48°50.0	· · 39.5	185°25.9	4.1'	26°51.8	6.4'	59.4'
16	63°50.1	40.4	199°49.0	4.0'	26°58.1	6.2'	59.4'
17	78°50.2	41.3	214°12.0	4.0'	27°04.3	6.0'	59.3'
18	93°50.3	S10°42.2	228°35.0	4.0'	N27°10.3	5.8'	59.3'
19	108°50.4	43.1	242°58.1	4.0'	27°16.0	5.6'	59.3'
20	123°50.5	44.0	257°21.1	4.0'	27°21.6	5.4'	59.2'
21	138°50.6	· · 44.9	271°44.1	4.0'	27°27.0	5.2'	59.2'
22	153°50.7	45.8	286°07.0	4.0'	27°32.2	5.0'	59.1'
23	168°50.8	46.6	300°30.0	4.0'	27°37.2	4.8'	59.1'
SD = 16.1' $d = 0.9'$ SD = 16.4'							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	05:15	06:33	07:46	15:42	16:55	18:13
N 70°	05:16	06:27	07:32	15:56	17:01	18:11
68°	05:17	06:22	07:21	16:08	17:06	18:11
66°	05:18	06:18	07:11	16:17	17:11	18:10
64°	05:19	06:14	07:03	16:25	17:15	18:10
62°	05:19	06:10	06:56	16:32	17:18	18:09
60°	05:19	06:07	06:50	16:38	17:21	18:09
N 58°	05:19	06:05	06:45	16:44	17:24	18:09
56°	05:19	06:02	06:40	16:48	17:26	18:09
54°	05:19	06:00	06:36	16:53	17:29	18:10
52°	05:19	05:58	06:32	16:56	17:31	18:10
50°	05:18	05:56	06:29	17:00	17:33	18:10
45°	05:17	05:52	06:21	17:08	17:37	18:12
N 40°	05:16	05:48	06:15	17:14	17:42	18:13
35°	05:14	05:44	06:10	17:20	17:45	18:15
30°	05:13	05:41	06:05	17:25	17:49	18:17
20°	05:08	05:34	05:56	17:33	17:55	18:21
N 10°	05:03	05:27	05:49	17:41	18:02	18:27
0°	04:56	05:21	05:42	17:48	18:09	18:34
S 10°	04:48	05:13	05:34	17:56	18:17	18:42
20°	04:38	05:04	05:26	18:04	18:26	18:53
30°	04:24	04:53	05:17	18:13	18:38	19:07
35°	04:15	04:46	05:12	18:18	18:45	19:16
40°	04:04	04:38	05:06	18:25	18:53	19:27
45°	03:51	04:28	04:59	18:32	19:03	19:40
S 50°	03:34	04:16	04:50	18:41	19:15	19:57
52°	03:25	04:10	04:46	18:45	19:21	20:06
54°	03:16	04:04	04:42	18:49	19:27	20:16
56°	03:05	03:57	04:37	18:54	19:35	20:27
58°	02:52	03:49	04:32	19:00	19:43	20:40
S 60°	02:37	03:39	04:26	19:06	19:53	20:56

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	13:19	□	□	11:30	□	□
N 70°	14:29	□	□	10:22	□	□
68°	15:07	□	□	09:47	□	□
66°	15:33	14:43	□	09:21	12:14	□
64°	15:54	15:33	□	09:02	11:25	□
62°	16:11	16:04	15:53	08:46	10:54	13:13
60°	16:26	16:28	16:37	08:33	10:31	12:30
N 58°	16:38	16:47	17:06	08:22	10:13	12:01
56°	16:48	17:04	17:29	08:12	09:57	11:39
54°	16:58	17:17	17:47	08:04	09:44	11:21
52°	17:06	17:30	18:03	07:56	09:33	11:05
50°	17:14	17:40	18:17	07:49	09:23	10:52
45°	17:30	18:03	18:45	07:35	09:01	10:25
N 40°	17:44	18:21	19:07	07:23	08:44	10:03
35°	17:56	18:37	19:25	07:13	08:30	09:45
30°	18:06	18:50	19:41	07:04	08:17	09:30
20°	18:23	19:13	20:08	06:49	07:56	09:04
N 10°	18:39	19:33	20:32	06:36	07:38	08:42
0°	18:53	19:52	20:53	06:24	07:21	08:21
S 10°	19:08	20:11	21:15	06:11	07:04	08:01
20°	19:24	20:32	21:39	05:59	06:46	07:39
30°	19:42	20:56	22:07	05:44	06:25	07:13
35°	19:53	21:10	22:23	05:35	06:13	06:58
40°	20:06	21:26	22:42	05:26	06:00	06:41
45°	20:20	21:46	23:06	05:15	05:43	06:20
S 50°	20:39	22:11	23:35	05:01	05:24	05:54
52°	20:47	22:23	23:50	04:55	05:14	05:42
54°	20:57	22:37	· · ·	04:48	05:04	05:27
56°	21:08	22:53	· · ·	04:40	04:52	05:11
58°	21:21	23:12	· · ·	04:32	04:38	04:51
S 60°	21:36	23:36	· · ·	04:22	04:23	04:26

Day	Sun		Mer. Pass hh:mm	Moon		Age 16-18 100-91%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
18	14:52	14:57	11:45	00:11	12:38	
19	15:03	15:08	11:45	01:07	13:37	
20	15:14	15:19	11:45	02:07	14:37	

October 21, 22, 23 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	29°55.9	148°20.5	S22°10.5	273°20.5	N22°15.7	309°33.4	N22°25.4	44°31.7	S08°34.7	Alpheratz	357°34.7 29°13.8
1	44°58.4	163°19.7	11.2	288°21.8	15.6	324°36.0	25.4	59°34.2	34.8	Ankaa	353°06.9 -42°10.3
2	60°00.9	178°19.0	11.9	303°23.1	15.5	339°38.6	25.4	74°36.8	34.8	Schedar	349°30.8 56°40.6
3	75°03.3	193°18.3	•• 12.6	318°24.4	•• 15.3	354°41.1	•• 25.4	89°39.4	•• 34.9	Diphda	348°47.2 -17°51.0
4	90°05.8	208°17.6	13.3	333°25.7	15.2	9°43.7	25.4	104°41.9	34.9	Achernar	335°19.7 -57°06.6
5	105°08.3	223°16.9	14.0	348°27.0	15.1	24°46.3	25.4	119°44.5	34.9	Hamal	327°51.1 23°34.9
6	120°10.7	238°16.2	S22°14.7	3°28.3	N22°15.0	39°48.8	N22°25.3	134°47.1	S08°35.0	Polaris	313°40.4 89°22.0
7	135°13.2	253°15.5	15.3	18°29.6	14.9	54°51.4	25.3	149°49.6	35.0	Acamar	315°11.5 -40°12.1
8	150°15.6	268°14.8	16.0	33°30.9	14.7	69°54.0	25.3	164°52.2	35.0	Menkar	314°06.0 4°11.3
9	165°18.1	283°14.0	•• 16.7	48°32.2	•• 14.6	84°56.6	•• 25.3	179°54.8	•• 35.1	Mirfak	308°28.1 49°56.9
10	180°20.6	298°13.3	17.4	63°33.5	14.5	99°59.1	25.3	194°57.3	35.1	Aldebaran	290°39.6 16°33.6
11	195°23.0	313°12.6	18.1	78°34.8	14.4	115°01.7	25.3	209°59.9	35.1	Rigel	281°03.8 -8°10.2
12	210°25.5	328°11.9	S22°18.7	93°36.1	N22°14.3	130°04.3	N22°25.3	225°02.4	S08°35.2	Capella	280°21.9 46°01.3
13	225°28.0	343°11.2	19.4	108°37.4	14.1	145°06.9	25.3	240°05.0	35.2	Bellatrix	278°22.9 6°22.4
14	240°30.4	358°10.5	20.1	123°38.7	14.0	160°09.4	25.3	255°07.6	35.2	Elnath	278°01.9 28°37.7
15	255°32.9	13°09.7	•• 20.8	138°40.0	•• 13.9	175°12.0	•• 25.3	270°10.1	•• 35.3	Alnilam	275°37.7 -1°11.0
16	270°35.4	28°09.0	21.4	153°41.3	13.8	190°14.6	25.3	285°12.7	35.3	Betelgeuse	270°52.1 7°24.8
17	285°37.8	43°08.3	22.1	168°42.6	13.7	205°17.2	25.3	300°15.3	35.3	Canopus	263°52.3 -52°42.1
18	300°40.3	58°07.6	S22°22.8	183°44.0	N22°13.5	220°19.7	N22°25.3	315°17.8	S08°35.4	Sirius	258°26.3 -16°44.8
19	315°42.8	73°06.9	23.5	198°45.3	13.4	235°22.3	25.3	330°20.4	35.4	Adhara	255°05.9 -29°00.0
20	330°45.2	88°06.1	24.1	213°46.6	13.3	250°24.9	25.3	345°22.9	35.5	Procyon	244°51.0 5°09.8
21	345°47.7	103°05.4	•• 24.8	228°47.9	•• 13.2	265°27.5	•• 25.2	0°25.5	•• 35.5	Pollux	243°17.5 27°58.0
22	0°50.1	118°04.7	25.5	243°49.2	13.0	280°30.0	25.2	15°28.1	35.5	Avior	234°14.9 -59°34.9
23	15°52.6	133°04.0	26.1	258°50.5	12.9	295°32.6	25.2	30°30.6	35.6	Suhail	222°46.6 -43°31.6
Mer.pass. 21:57		ν -0.7' d 0.7' m -3.97		ν 1.3' d -0.1' m 0.22		ν 2.6' d -0.0' m -2.61		ν 2.6' d 0.0' m 0.75		Miaplacidus	221°38.7 -69°48.7
										Alphard	217°48.1 -8°45.8
										Regulus	207°34.8 11°50.8
										Dubhe	193°41.7 61°36.9
										Denebola	182°25.4 14°26.1
										Gienah	175°44.1 -17°40.6
										Acrux	173°01.0 -63°14.0
										Gacrux	171°52.5 -57°14.9
										Alioth	166°13.7 55°49.5
										Spica	158°22.8 -11°17.3
										Alkaid	152°52.6 49°11.4
										Hadar	148°37.0 -60°29.5
										Menkent	147°58.3 -36°29.4
										Arcturus	145°48.4 19°03.3
										Rigel Kent.	139°41.3 -60°56.2
										Kochab	137°20.9 74°03.2
										Zuben'ubi	136°56.6 -16°08.6
										Alphecca	126°04.2 26°38.0
										Antares	112°16.4 -26°29.2
										Atria	107°11.2 -69°04.5
										Sabik	102°03.2 -15°45.3
										Shaula	96°10.9 -37°07.4
										Rasalhague	95°58.9 12°32.6
										Eltanin	90°42.6 51°29.3
										Kaus Aust.	83°32.9 -34°22.4
										Vega	80°33.5 38°48.6
										Nunki	75°48.1 -26°16.0
										Altair	62°00.2 8°56.2
										Peacock	53°06.0 -56°39.5
										Deneb	49°25.9 45°22.4
										Enif	33°38.9 9°59.5
										Al Na'ir	27°32.9 -46°50.6
										Fomalhaut	15°14.5 -29°29.5
										Scheat	13°45.2 28°13.2
										Markab	13°29.9 15°20.5
										Oct 21 Mon	SHA Mer.pass
										Venus	118°24.5 14:07
										Mars	243°24.5 05:46
										Jupiter	279°37.5 03:21
										Saturn	14°35.7 20:58
										Oct 22 Tue	SHA Mer.pass
										Venus	117°08.2 14:08
										Mars	242°56.8 05:44
										Jupiter	279°40.1 03:17
										Saturn	14°38.1 20:54
										Oct 23 Wed	SHA Mer.pass
										Venus	115°51.6 14:10
										Mars	242°29.3 05:42
										Jupiter	279°43.0 03:13
										Saturn	14°40.4 20:50
										Horizontal parallax	
										Venus:	0.1
										Mars:	0.1
0	30°55.1	148°03.3	S22°26.8	273°51.8	N22°12.8	310°35.2	N22°25.2	45°33.2	S08°35.6		
1	45°57.5	163°02.5	27.5	288°53.1	12.7	325°37.8	25.2	60°35.8	35.6		
2	61°00.0	178°01.8	28.1	303°54.5	12.6	340°40.3	25.2	75°38.3	35.7		
3	76°02.5	193°01.1	•• 28.8	318°55.8	•• 12.4	355°42.9	•• 25.2	90°40.9	•• 35.7		
4	91°04.9	208°00.4	29.5	333°57.1	12.3	10°45.5	25.2	105°43.4	35.7		
5	106°07.4	222°59.6	30.1	348°58.4	12.2	25°48.1	25.2	120°46.0	35.8		
6	121°09.9	237°58.9	S22°30.8	3°59.7	N22°12.1	40°50.7	N22°25.2	135°48.6	S08°35.8		
7	136°12.3	252°58.2	31.4	19°01.0	12.0	55°53.2	25.2	150°51.1	35.8		
8	151°14.8	267°57.5	32.1	34°02.4	11.8	70°55.8	25.2	165°53.7	35.9		
9	166°17.3	282°56.7	•• 32.7	49°03.7	•• 11.7	85°58.4	•• 25.2	180°56.2	•• 35.9		
10	181°19.7	297°56.0	33.4	64°05.0	11.6	101°01.0	25.2	195°58.8	35.9		
11	196°22.2	312°55.3	34.1	79°06.3	11.5	116°03.6	25.1	211°01.4	36.0		
12	211°24.6	327°54.6	S22°34.7	94°07.6	N22°11.4	131°06.2	N22°25.1	226°03.9	S08°36.0		
13	226°27.1	342°53.8	35.4	109°09.0	11.2	146°08.7	25.1	241°06.5	36.0		
14	241°29.6	357°53.1	36.0	124°10.3	11.1	161°11.3	25.1	256°09.0	36.1		
15	256°32.0	12°52.4	•• 36.7	139°11.6	•• 11.0	176°13.9	•• 25.1	271°11.6	•• 36.1		
16	271°34.5	27°51.6	37.3	154°12.9	10.9	191°16.5	25.1	286°14.2	36.1		
17	286°37.0	42°50.9	38.0	169°14.3	10.7	206°19.1	25.1	301°16.7	36.2		
18	301°39.4	57°50.2	S22°38.6	184°15.6	N22°10.6	221°21.7	N22°25.1	316°19.3	S08°36.2		
19	316°41.9	72°49.4	39.2	199°16.9	10.5	236°24.2	25.1	331°21.8	36.2		
20	331°44.4	87°48.7	39.9	214°18.2	10.4	251°26.8	25.1	346°24.4	36.3		
21	346°46.8	102°48.0	•• 40.5	229°19.6	•• 10.3	266°29.4	•• 25.1	1°27.0	•• 36.3		
22	1°49.3	117°47.3	41.2	244°20.9	10.1	281°32.0	25.1	16°29.5	36.3		
23	16°51.8	132°46.5	41.8	259°22.2	10.0	296°34.6	25.1	31°32.1	36.4		
Mer.pass. 21:53		ν -0.7' d 0.7' m -3.98		ν 1.3' d -0.1' m 0.21		ν 2.6' d -0.0' m -2.62		ν 2.6' d 0.0' m 0.75			

2024 October 21 to Oct. 23 UT

IERS Earth Orientation data as of 01-Aug-2024
 Page 199
 <https://pypi.org/project/sfalmanac/>

October 24, 25, 26 UT (Thu., Fri., Sat.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	32°53.4	147°28.1	S22°57.5	274°55.7	N22°07.0	312°39.4	N22°24.9	47°36.0	S08°37.2	Alpheratz	357°34.7	29°13.8											
1	47°55.8	162°27.3	58.1	289°57.0	06.9	327°42.0	24.9	62°38.5	37.2	Ankaa	353°06.9	-42°10.3											
2	62°58.3	177°26.6	58.7	304°58.4	06.8	342°44.6	24.8	77°41.1	37.2	Schedar	349°30.8	56°40.6											
3	78°00.7	192°25.8	· · 59.3	319°59.7	· · 06.6	357°47.2	· · 24.8	92°43.6	· · 37.3	Diphda	348°47.2	-17°51.0											
4	93°03.2	207°25.1	22°59.9	335°01.1	06.5	12°49.8	24.8	107°46.2	37.3	Achernar	335°19.7	-57°06.6											
5	108°05.7	222°24.4	23°00.6	350°02.4	06.4	27°52.4	24.8	122°48.8	37.3	Hamal	327°51.1	23°34.9											
6	123°08.1	237°23.6	S23°01.2	5°03.8	N22°06.3	42°55.0	N22°24.8	137°51.3	S08°37.4	Polaris	313°39.7	89°22.1											
7	138°10.6	252°22.9	01.8	20°05.1	06.2	57°57.6	24.8	152°53.9	37.4	Acamar	315°11.5	-40°12.2											
8	153°13.1	267°22.1	02.4	35°06.5	06.0	73°00.2	24.8	167°56.4	37.4	Menkar	314°06.0	4°11.3											
9	168°15.5	282°21.4	· · 03.0	50°07.8	· · 05.9	88°02.8	· · 24.8	182°59.0	· · 37.5	Mirfak	308°28.1	49°57.0											
10	183°18.0	297°20.6	03.6	65°09.2	05.8	103°05.4	24.8	198°01.5	37.5	Aldebaran	290°39.6	16°33.6											
11	198°20.5	312°19.9	04.2	80°10.5	05.7	118°08.0	24.8	213°04.1	37.5	Rigel	281°03.8	-8°10.2											
12	213°22.9	327°19.1	S23°04.8	95°11.9	N22°05.6	133°10.6	N22°24.8	228°06.6	S08°37.6	Capella	280°21.8	46°01.3											
13	228°25.4	342°18.4	05.4	110°13.2	05.4	148°13.2	24.8	243°09.2	37.6	Bellatrix	278°22.9	6°22.4											
14	243°27.9	357°17.6	06.0	125°14.6	05.3	163°15.8	24.8	258°11.7	37.6	Elnath	278°01.9	28°37.7											
15	258°30.3	12°16.9	· · 06.6	140°15.9	· · 05.2	178°18.4	· · 24.7	273°14.3	· · 37.6	Alnilam	275°37.7	-1°11.0											
16	273°32.8	27°16.1	07.2	155°17.3	05.1	193°21.0	24.7	288°16.8	37.7	Betelgeuse	270°52.1	7°24.8											
17	288°35.2	42°15.4	07.8	170°18.7	05.0	208°23.6	24.7	303°19.4	37.7	Canopus	263°52.2	-52°42.2											
18	303°37.7	57°14.6	S23°08.4	185°20.0	N22°04.8	223°26.2	N22°24.7	318°21.9	S08°37.7	Sirius	258°26.3	-16°44.8											
19	318°40.2	72°13.9	09.0	200°21.4	04.7	238°28.8	24.7	333°24.5	37.8	Adhara	255°05.9	-29°00.0											
20	333°42.6	87°13.1	09.6	215°22.7	04.6	253°31.4	24.7	348°27.0	37.8	Procyon	244°51.0	5°09.8											
21	348°45.1	102°12.4	· · 10.2	230°24.1	· · 04.5	268°34.0	· · 24.7	3°29.6	· · 37.8	Pollux	243°17.5	27°57.9											
22	3°47.6	117°11.6	10.7	245°25.5	04.4	283°36.6	24.7	18°32.2	37.9	Avior	234°14.9	-59°34.9											
23	18°50.0	132°10.9	11.3	260°26.8	04.2	298°39.2	24.7	33°34.7	37.9	Suhail	222°46.6	-43°31.6											
Mer.pass. 21:45				ν -0.7' d 0.6' m -3.98				ν 1.3' d -0.1' m 0.18				ν 2.6' d -0.0' m -2.64				ν 2.6' d 0.0' m 0.76							

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec						
0	33°52.5	147°10.1	S23°11.9	275°28.2	N22°04.1	313°41.8	N22°24.7	48°37.3	S08°37.9	Denebola	182°25.4	14°26.1							
1	48°55.0	162°09.4	12.5	290°29.5	04.0	328°44.4	24.7	63°39.8	38.0	Gienah	175°44.1	-17°40.6							
2	63°57.4	177°08.6	13.1	305°30.9	03.9	343°47.0	24.6	78°42.4	38.0	Acrux	173°01.0	-63°14.0							
3	78°59.9	192°07.9	· · 13.7	320°32.3	· · 03.8	358°49.6	· · 24.6	93°44.9	· · 38.0	Gacrux	171°52.4	-57°14.9							
4	94°02.4	207°07.1	14.3	335°33.6	03.7	13°52.2	24.6	108°47.5	38.0	Alioth	166°13.6	55°49.5							
5	109°04.8	222°06.3	14.9	350°35.0	03.5	28°54.8	24.6	123°50.0	38.1	Spica	158°22.8	-11°17.3							
6	124°07.3	237°05.6	S23°15.4	5°36.4	N22°03.4	43°57.4	N22°24.6	138°52.6	S08°38.1	Alkaid	152°52.6	49°11.4							
7	139°09.7	252°04.8	16.0	20°37.7	03.3	59°00.0	24.6	153°55.1	38.1	Hadar	148°37.0	-60°29.5							
8	154°12.2	267°04.1	16.6	35°39.1	03.2	74°02.6	24.6	168°57.7	38.2	Menkent	147°58.3	-36°29.4							
9	169°14.7	282°03.3	· · 17.2	50°40.5	· · 03.1	89°05.2	· · 24.6	184°00.2	· · 38.2	Arcturus	145°48.4	19°03.3							
10	184°17.1	297°02.6	17.8	65°41.8	02.9	104°07.9	24.6	199°02.8	38.2	Rigil Kent.	139°41.3	-60°56.2							
11	199°19.6	312°01.8	18.3	80°43.2	02.8	119°10.5	24.6	214°05.3	38.2	Kochab	137°20.9	74°03.2							
12	214°22.1	327°01.0	S23°18.9	95°44.6	N22°02.7	134°13.1	N22°24.6	229°07.9	S08°38.3	Zuben'ubi	136°56.6	-16°08.6							
13	229°24.5	342°00.3	19.5	110°46.0	02.6	149°15.7	24.6	244°10.4	38.3	Alphecca	126°04.2	26°38.0							
14	244°27.0	356°59.5	20.0	125°47.3	02.5	164°18.3	24.5	259°13.0	38.3	Antares	112°16.4	-26°29.2							
15	259°29.5	11°58.8	· · 20.6	140°48.7	· · 02.3	179°20.9	· · 24.5	274°15.5	· · 38.4	Atria	107°11.2	-69°04.5							
16	274°31.9	26°58.0	21.2	155°50.1	02.2	194°23.5	24.5	289°18.1	38.4	Sabik	102°03.3	-15°45.3							
17	289°34.4	41°57.3	21.8	170°51.4	02.1	209°26.1	24.5	304°20.6	38.4	Shaula	96°10.9	-37°07.4							
18	304°36.8	56°56.5	S23°22.3	185°52.8	N22°02.0	224°28.7	N22°24.5	319°23.1	S08°38.5	Rasalhague	95°58.9	12°32.6							
19	319°39.3	71°55.7	22.9	200°54.2	01.9	239°31.3	24.5	334°25.7	38.5	Eltanin	90°42.6	51°29.3							
20	334°41.8	86°55.0	23.5	215°55.6	01.7	254°34.0	24.5	349°28.2	38.5	Kaus Aust.	83°33.0	-34°22.4							
21	349°44.2	101°54.2	· · 24.0	230°56.9	· · 01.6	269°36.6	· · 24.5	4°30.8	· · 38.5	Vega	80°33.5	38°48.6							
22	4°46.7	116°53.4	24.6	245°58.3	01.5	284°39.2	24.5	19°33.3	38.6	Nunki	75°48.1	-26°16.0							
23	19°49.2	131°52.7	25.2	260°59.7	01.4	299°41.8	24.5	34°35.9	38.6	Altair	62°00.2	8°56.1							
Mer.pass. 21:41				ν -0.8' d 0.6' m -3.99				ν 1.4' d -0.1' m 0.17				ν 2.6' d -0.0' m -2.64				ν 2.6' d 0.0' m 0.76			

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec						
0	34°51.6	146°51.9	S23°25.7	276°01.1	N22°01.3	314°44.4	N22°24.5	49°38.4	S08°38.6	Oct 24 Thu	SHA	Mer.pass							
1	49°54.1	161°51.2	26.3	291°02.5	01.2	329°47.0	24.4	64°41.0	38.7	Venus	114°34.7	14:11							
2	64°56.6	176°50.4	26.8	306°03.8	01.0	344°49.6	24.4	79°43.5	38.7	Mars	242°02.3	05:40							
3	79°59.0	191°49.6	· · 27.4	321°05.2	· · 00.9	359°52.2	· · 24.4	94°46.1	· · 38.7	Jupiter	279°46.0	03:09							
4	95°01.5	206°48.9	28.0	336°06.6	00.8	14°54.9	24.4	109°48.6	38.7	Saturn	14°42.6	20:46							
5	110°04.0	221°48.1	28.5	351°08.0	00.7	29°57.5	24.4	124°51.2	38.8										
6	125°06.4	236°47.3	S23°29.1	6°09.4	N22°00.6	45°00.1	N22°24.4	139°53.7	S08°38.8										
7	140°08.9	251°46.6	29.6	21°10.8	00.4	60°02.7	24.4	154°56.3	38.8										
8	155°11.3	266°45.8	30.2	36°12.1	00.3	75°05.3	24.4	169°58.8	38.8										
9	170°13.8	281°45.0	· · 30.7	51°13.5	· · 00.2	90°07.9	· · 24.4	185°01.4	· · 38.9										
10	185°16.3	296°44.3	31.3	66°14.9	00.1	105°10.6	24.4	200°03.9	38.9										
11	200°18.7	311°43.5	31.8	81°16.3	22°00.0	120°13.2	24.4	215°06.5	38.9										
12	215°21.2	326°42.7	S23°32.4	96°17.7	N21°59.9	135°15.8	N22°24.3	230°09.0	S08°39.0										
13	230°23.7	341°42.0	32.9	111°19.1	59.7	150°18.4	24.3	245°11.5	39.0										
14	245°26.1	356°41.2	33.5	126°20.5	59.6	165°21.0	24.3	260°14.1	39.0										
15	260°28.6	11°40.4	· · 34.0	141°21.9	· · 59.5	180°23.6	· · 24.3	275°16.6	· · 39.0										
16	275°31.1	26°39.7	34.6	156°23.2	59.4	195°26.3	24.3	290°19.2	39.1										
17	290°33.5	41°38.9	35.1	171°24.6	59.3	210°28.9	24.3	305°21.7	39.1										
18	305°36.0	56°38.1	S23°35.6	186°26.0	N21°59.1	225°31.5	N22°24.3	320°24.3	S08°39.1										
19	320°38.5	71°37.4	36.2	201°27.4	59.0	240°34.1	24.3	335°26.8	39.2										
20	335°40.9	86°36.6	36.7	216°28.8	58.9	255°36.7	24.3	350°29.4	39.2										
21	350°43.4	101°35.8	· · 37.3	231°30.2	· · 58.8	270°39.4	· · 24.3	5°31.9	· · 39.2										
22	5°45.8	116°35.0	37.8	246°31.6	58.7	285°42.0	24.3	20°34.4	39.2										
23	20°48.3	131°34.3	38.3	261°33.0	58.6	300°44.6	24.2	35°37.0	39.3										
Mer.pass. 21:37				ν -0.8' d 0.6' m -3.99				ν 1.4' d -0.1' m 0.15				ν 2.6' d -0.0' m -2.65				ν 2.5' d 0.0' m 0.77			

Oct 24 Thu			SHA	Mer.pass
Venus	114°34.7		14:11	
Mars	242°02.3		05:40	
Jupiter	279°46.0		03:09	
Saturn	14°42.6		20:46	

Oct 25 Fri			SHA	Mer.pass
Venus	113°17.6		14:12	
Mars				

Sun								Moon							
Thu	GHA	Dec	GHA	ν	Dec	d	HP	Lat.	Twilight		Sunrise	Sunset	Twilight		
									Naut.	Civil			Civil	Naut.	
0	183°57.4	S11°50.7	272°39.1	9.1'	N25°33.0	-7.5'	56.3'	N 72°	05:40	07:00	08:18	15:08	16:26	17:46	
1	198°57.5	51.6	287°07.3	9.2'	25°25.6	-7.6'	56.2'	N 70°	05:38	06:50	07:59	15:27	16:36	17:48	
2	213°57.5	52.4	301°35.5	9.4'	25°18.0	-7.7'	56.2'	68°	05:37	06:43	07:44	15:43	16:44	17:49	
3	228°57.6	· · 53.3	316°03.9	9.5'	25°10.2	-7.9'	56.2'	66°	05:36	06:36	07:32	15:55	16:50	17:50	
4	243°57.7	54.2	330°32.3	9.6'	25°02.4	-8.0'	56.1'	64°	05:35	06:31	07:22	16:05	16:56	17:52	
5	258°57.8	55.0	345°00.9	9.7'	24°54.4	-8.1'	56.1'	62°	05:34	06:26	07:13	16:14	17:01	17:53	
6	273°57.9	S11°55.9	359°29.6	9.8'	N24°46.3	-8.2'	56.1'	60°	05:33	06:22	07:05	16:22	17:05	17:54	
7	288°57.9	56.8	13°58.4	9.9'	24°38.1	-8.3'	56.0'	N 58°	05:32	06:18	06:59	16:28	17:09	17:55	
8	303°58.0	57.6	28°27.4	10.0'	24°29.8	-8.4'	56.0'	56°	05:31	06:14	06:53	16:34	17:13	17:56	
9	318°58.1	· · 58.5	42°56.4	10.1'	24°21.3	-8.5'	56.0'	54°	05:30	06:11	06:48	16:39	17:16	17:57	
10	333°58.2	11°59.4	57°25.5	10.3'	24°12.8	-8.7'	55.9'	52°	05:29	06:08	06:43	16:44	17:19	17:58	
11	348°58.2	12°00.2	71°54.8	10.4'	24°04.1	-8.8'	55.9'	50°	05:28	06:05	06:39	16:49	17:22	18:00	
12	3°58.3	S12°01.1	86°24.2	10.5'	N23°55.4	-8.9'	55.9'	45°	05:25	05:59	06:29	16:58	17:28	18:02	
13	18°58.4	02.0	100°53.7	10.6'	23°46.5	-9.0'	55.8'	N 40°	05:22	05:54	06:22	17:06	17:34	18:05	
14	33°58.5	02.8	115°23.3	10.7'	23°37.5	-9.1'	55.8'	35°	05:19	05:49	06:15	17:13	17:39	18:08	
15	48°58.5	· · 03.7	129°53.0	10.8'	23°28.4	-9.2'	55.8'	30°	05:16	05:45	06:09	17:19	17:43	18:11	
16	63°58.6	04.5	144°22.8	10.9'	23°19.3	-9.3'	55.8'	20°	05:10	05:36	05:59	17:29	17:52	18:18	
17	78°58.7	05.4	158°52.7	11.0'	23°10.0	-9.4'	55.7'	N 10°	05:03	05:28	05:49	17:39	18:00	18:25	
18	93°58.8	S12°06.3	173°22.7	11.1'	N23°00.6	-9.5'	55.7'	0°	04:55	05:19	05:41	17:47	18:09	18:33	
19	108°58.8	07.1	187°52.9	11.3'	22°51.1	-9.6'	55.7'	S 10°	04:45	05:10	05:32	17:56	18:18	18:43	
20	123°58.9	08.0	202°23.2	11.4'	22°41.5	-9.7'	55.6'	20°	04:33	05:00	05:22	18:06	18:29	18:55	
21	138°59.0	· · 08.9	216°53.5	11.5'	22°31.8	-9.8'	55.6'	30°	04:17	04:46	05:11	18:17	18:42	19:12	
22	153°59.0	09.7	231°24.0	11.6'	22°22.0	-9.9'	55.6'	35°	04:07	04:38	05:05	18:24	18:50	19:22	
23	168°59.1	10.6	245°54.6	11.7'	22°12.1	-10.0'	55.5'	40°	03:55	04:29	04:57	18:31	19:00	19:34	
SD = 16.1' d = 0.9'								SD = 15.3'							
Fri	GHA	Dec	GHA	ν	Dec	d	HP	Lat.	Moonrise			Moonset			
									Thu	Fri	Sat	Thu	Fri	Sat	
0	183°59.2	S12°11.4	260°25.3	11.8'	N22°02.2	-10.1'	55.5'	N 72°			23:07			17:14	
1	198°59.3	12.3	274°56.1	11.9'	21°52.1	-10.2'	55.5'	N 70°		21:02	23:32		17:47	16:47	
2	213°59.3	13.2	289°27.0	12.0'	21°42.0	-10.2'	55.5'	68°		21:45	23:51		17:02	16:27	
3	228°59.4	· · 14.0	303°58.0	12.1'	21°31.7	-10.3'	55.4'	66°	19:53	22:14	· ·	17:15	16:32	16:10	
4	243°59.5	14.9	318°29.1	12.2'	21°21.4	-10.4'	55.4'	64°	20:40	22:36	· ·	16:28	16:09	15:57	
5	258°59.5	15.7	333°00.3	12.3'	21°11.0	-10.5'	55.4'	62°	21:10	22:53	· ·	15:57	15:51	15:45	
6	273°59.6	S12°16.6	347°31.6	12.4'	N21°00.5	-10.6'	55.4'	60°	21:32	23:07	· ·	15:34	15:36	15:36	
7	288°59.7	17.4	2°03.1	12.5'	20°49.9	-10.7'	55.3'	N 58°	21:50	23:19	· ·	15:15	15:23	15:27	
8	303°59.7	18.3	16°34.6	12.6'	20°39.2	-10.7'	55.3'	56°	22:05	23:30	· ·	15:00	15:12	15:19	
9	318°59.8	· · 19.2	31°06.2	12.7'	20°28.5	-10.8'	55.3'	54°	22:18	23:39	· ·	14:46	15:02	15:13	
10	333°59.9	20.0	45°37.9	12.8'	20°17.6	-10.9'	55.2'	52°	22:30	23:47	· ·	14:34	14:53	15:07	
11	349°00.0	20.9	60°09.8	12.9'	20°06.7	-11.0'	55.2'	50°	22:40	23:55	· ·	14:24	14:45	15:01	
12	4°00.0	S12°21.7	74°41.7	13.0'	N19°55.7	-11.1'	55.2'	45°	23:01	· ·	00:10	14:01	14:28	14:49	
13	19°00.1	22.6	89°13.7	13.1'	19°44.7	-11.1'	55.2'	N 40°	23:18	· ·	00:23	13:44	14:14	14:39	
14	34°00.2	23.4	103°45.8	13.2'	19°33.5	-11.2'	55.1'	35°	23:32	· ·	00:34	13:28	14:02	14:30	
15	49°00.2	· · 24.3	118°18.0	13.3'	19°22.3	-11.3'	55.1'	30°	23:44	· ·	00:43	13:15	13:52	14:23	
16	64°00.3	25.1	132°50.3	13.4'	19°11.0	-11.4'	55.1'	20°	· ·	00:05	00:59	12:53	13:34	14:10	
17	79°00.4	26.0	147°22.8	13.5'	18°59.7	-11.4'	55.1'	N 10°	· ·	00:23	01:13	12:33	13:18	13:58	
18	94°00.4	S12°26.8	161°55.2	13.6'	N18°48.2	-11.5'	55.0'	0°	· ·	00:40	01:26	12:14	13:03	13:47	
19	109°00.5	27.7	176°27.8	13.7'	18°36.7	-11.6'	55.0'	S 10°	00:10	00:57	01:39	11:56	12:48	13:36	
20	124°00.6	28.6	191°00.5	13.8'	18°25.2	-11.6'	55.0'	20°	00:31	01:15	01:53	11:36	12:32	13:24	
21	139°00.6	· · 29.4	205°33.3	13.9'	18°13.5	-11.7'	55.0'	30°	00:56	01:35	02:09	11:12	12:13	13:10	
22	154°00.7	30.3	220°06.2	13.9'	18°01.8	-11.8'	55.0'	35°	01:10	01:47	02:18	10:59	12:02	13:02	
23	169°00.7	31.1	234°39.1	14.0'	17°50.1	-11.8'	54.9'	40°	01:27	02:01	02:28	10:43	11:49	12:53	
SD = 16.1' d = 0.9'								SD = 15.1'							
Sat	GHA	Dec	GHA	ν	Dec	d	HP	Day	Sun		Mer.	Moon		Age	
									Eqn.of Time	Pass	Upper	Lower	22-24		
									00 ^h	12 ^h	hh:mm	hh:mm	54-34%		
									mm:ss	mm:ss					
0	184°00.8	S12°32.0	249°12.1	14.1'	N17°38.2	-11.9'	54.9'	24	15:50	15:53	11:44	06:02	18:27		
1	199°00.9	32.8	263°45.3	14.2'	17°26.3	-12.0'	54.9'	25	15:57	16:00	11:44	06:52	19:15		
2	214°00.9	33.7	278°18.5	14.3'	17°14.4	-12.0'	54.9'	26	16:03	16:06	11:44	07:37	19:58		
3	229°01.0	· · 34.5	292°51.8	14.4'	17°02.3	-12.1'	54.8'								
4	244°01.1	35.4	307°25.1	14.5'	16°50.3	-12.1'	54.8'								
5	259°01.1	36.2	321°58.6	14.5'	16°38.1	-12.2'	54.8'								
6	274°01.2	S12°37.1	336°32.1	14.6'	N16°25.9	-12.3'	54.8'								
7	289°01.2	37.9	351°05.8	14.7'	16°13.7	-12.3'	54.8'								
8	304°01.3	38.8	5°39.5	14.8'	16°01.3	-12.4'	54.7'								
9	319°01.4	· · 39.6	20°13.2	14.9'	15°49.0	-12.4'	54.7'								
10	334°01.4	40.5	34°47.1	14.9'	15°36.5	-12.5'	54.7'								
11	349°01.5	41.3	49°21.0	15.0'	15°24.1	-12.5'	54.7'								
12	4°01.5	S12°42.2	63°55.0	15.1'	N15°11.5	-12.6'	54.7'								
13	19°01.6	43.0	78°29.1	15.2'	14°59.0	-12.6'	54.6'								
14	34°01.7	43.9	93°03.3	15.2'	14°46.3	-12.7'	54.6'								
15	49°01.7	· · 44.7	107°37.5	15.3'	14°33.6	-12.7'	54.6'								
16	64°01.8	45.5	122°11.8	15.4'	14°20.9	-12.8'	54.6'								
17	79°01.8	46.4	136°46.2	15.4'	14°08.1	-12.8'	54.6'								
18	94°01.9	S12°47.2	151°20.6	15.5'	N13°55.3	-12.9'	54.6'								
19	109°02.0	48.1	165°55.1	15.6'	13°42.4	-12.9'	54.5'								
20	124°02.0	48.9	180°29.7	15.6'	13°29.5										

October 27, 28, 29 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	35°50.8	146°33.5	S23°38.9	276°34.4	N21°58.4	315°47.2	N22°24.2	50°39.5	S08°39.3	Alpheratz	357°34.7 29°13.8
1	50°53.2	161°32.7	39.4	291°35.8	58.3	330°49.9	24.2	65°42.1	39.3	Ankaa	353°06.9 -42°10.3
2	65°55.7	176°32.0	39.9	306°37.2	58.2	345°52.5	24.2	80°44.6	39.3	Schedar	349°30.8 56°40.6
3	80°58.2	191°31.2	· · 40.5	321°38.6	· · 58.1	0°55.1	· · 24.2	95°47.2	· · 39.4	Diphda	348°47.2 -17°51.0
4	96°00.6	206°30.4	41.0	336°40.0	58.0	15°57.7	24.2	110°49.7	39.4	Achernar	335°19.7 -57°06.6
5	111°03.1	221°29.6	41.5	351°41.4	57.9	31°00.3	24.2	125°52.3	39.4	Hamal	327°51.1 23°34.9
6	126°05.6	236°28.9	S23°42.1	6°42.8	N21°57.7	46°03.0	N22°24.2	140°54.8	S08°39.4	Polaris	313°39.2 89°22.1
7	141°08.0	251°28.1	42.6	21°44.2	57.6	61°05.6	24.2	155°57.3	39.5	Acamar	315°11.5 -40°12.2
8	156°10.5	266°27.3	43.1	36°45.6	57.5	76°08.2	24.2	170°59.9	39.5	Menkar	314°06.0 4°11.3
9	171°12.9	281°26.5	· · 43.6	51°47.0	· · 57.4	91°10.8	· · 24.1	186°02.4	· · 39.5	Mirfak	308°28.1 49°57.0
10	186°15.4	296°25.8	44.2	66°48.4	57.3	106°13.5	24.1	201°05.0	39.6	Aldebaran	290°39.6 16°33.6
11	201°17.9	311°25.0	44.7	81°49.8	57.2	121°16.1	24.1	216°07.5	39.6	Rigel	281°03.8 -8°10.2
12	216°20.3	326°24.2	S23°45.2	96°51.2	N21°57.0	136°18.7	N22°24.1	231°10.1	S08°39.6	Capella	280°21.8 46°01.3
13	231°22.8	341°23.4	45.7	111°52.6	56.9	151°21.3	24.1	246°12.6	39.6	Bellatrix	278°22.8 6°22.4
14	246°25.3	356°22.7	46.2	126°54.0	56.8	166°24.0	24.1	261°15.1	39.7	Elnath	278°01.8 28°37.7
15	261°27.7	11°21.9	· · 46.8	141°55.4	· · 56.7	181°26.6	· · 24.1	276°17.7	· · 39.7	Alnilam	275°37.7 -1°11.0
16	276°30.2	26°21.1	47.3	156°56.8	56.6	196°29.2	24.1	291°20.2	39.7	Betelgeuse	270°52.1 7°24.8
17	291°32.7	41°20.3	47.8	171°58.2	56.5	211°31.9	24.1	306°22.8	39.7	Canopus	263°52.2 -52°42.2
18	306°35.1	56°19.5	S23°48.3	186°59.6	N21°56.3	226°34.5	N22°24.1	321°25.3	S08°39.8	Sirius	258°26.3 -16°44.8
19	321°37.6	71°18.8	48.8	202°01.0	56.2	241°37.1	24.0	336°27.8	39.8	Adhara	255°05.9 -29°00.0
20	336°40.1	86°18.0	49.3	217°02.5	56.1	256°39.7	24.0	351°30.4	39.8	Procyon	244°51.0 5°09.8
21	351°42.5	101°17.2	· · 49.8	232°03.9	· · 56.0	271°42.4	· · 24.0	6°32.9	· · 39.8	Pollux	243°17.5 27°57.9
22	6°45.0	116°16.4	50.4	247°05.3	55.9	286°45.0	24.0	21°35.5	39.9	Avior	234°14.8 -59°34.9
23	21°47.4	131°15.7	50.9	262°06.7	55.8	301°47.6	24.0	36°38.0	39.9	Suhail	222°46.5 -43°31.6
Mer.pass. 21:33		ν -0.8' d 0.5' m -4.00		ν 1.4' d -0.1' m 0.14		ν 2.6' d -0.0' m -2.66		ν 2.5' d 0.0' m 0.77		Miaplacidus	221°38.6 -69°48.7
										Alphard	217°48.0 -8°45.8
										Regulus	207°34.8 11°50.8
										Dubhe	193°41.6 61°36.9
										Denebola	182°25.4 14°26.1
										Gienah	175°44.1 -17°40.6
										Acrux	173°01.0 -63°14.0
										Gacrux	171°52.4 -57°14.9
										Alioth	166°13.6 55°49.5
										Spica	158°22.8 -11°17.3
										Alkaid	152°52.6 49°11.4
										Hadar	148°37.0 -60°29.5
										Menkent	147°58.3 -36°29.4
										Arcturus	145°48.4 19°03.3
										Rigel Kent.	139°41.3 -60°56.2
										Kochab	137°20.9 74°03.2
										Zuben'ubi	136°56.6 -16°08.6
										Alphecca	126°04.2 26°38.0
										Antares	112°16.4 -26°29.2
										Atria	107°11.2 -69°04.4
										Sabik	102°03.3 -15°45.3
										Shaula	96°10.9 -37°07.4
										Rasalhague	95°59.0 12°32.6
										Eltanin	90°42.6 51°29.3
										Kaus Aust.	83°33.0 -34°22.4
										Vega	80°33.6 38°48.6
										Nunki	75°48.1 -26°16.0
										Altair	62°00.2 8°56.1
										Peacock	53°06.0 -56°39.5
										Deneb	49°25.9 45°22.4
										Enif	33°38.9 9°59.5
										Al Na'ir	27°32.9 -46°50.6
										Fomalhaut	15°14.5 -29°29.5
										Scheat	13°45.2 28°13.2
										Markab	13°29.9 15°20.5
										Oct 27 Sun	SHA Mer.pass
										Venus	110°42.7 14:15
										Mars	240°43.6 05:33
										Jupiter	279°56.5 02:56
										Saturn	14°48.8 20:34
										Oct 28 Mon	SHA Mer.pass
										Venus	109°25.0 14:16
										Mars	240°18.2 05:31
										Jupiter	280°00.4 02:52
										Saturn	14°50.6 20:30
										Oct 29 Tue	SHA Mer.pass
										Venus	108°07.0 14:17
										Mars	239°53.2 05:29
										Jupiter	280°04.5 02:48
										Saturn	14°52.4 20:26
										Horizontal parallax	
										Venus:	0.1
										Mars:	0.1
Mer.pass. 21:25		ν -0.8' d 0.5' m -4.01		ν 1.4' d -0.1' m 0.11		ν 2.6' d -0.0' m -2.67		ν 2.5' d 0.0' m 0.78			

Sun				Moon				
Sun	GHA	Dec		GHA	ν	Dec	d	HP
0	184°02.2	S12°52.3		238°48.6	15.9'	N12°37.4	-13.1'	54.5'
1	199°02.3	53.1		253°23.5	15.9'	12°24.3	-13.2'	54.4'
2	214°02.4	54.0		267°58.4	16.0'	12°11.1	-13.2'	54.4'
3	229°02.4	54.8		282°33.5	16.1'	11°57.9	-13.2'	54.4'
4	244°02.5	55.7		297°08.5	16.1'	11°44.7	-13.3'	54.4'
5	259°02.5	56.5		311°43.6	16.2'	11°31.4	-13.3'	54.4'
6	274°02.6	S12°57.4		326°18.8	16.2'	N11°18.1	-13.4'	54.4'
7	289°02.6	58.2		340°54.0	16.3'	11°04.7	-13.4'	54.3'
8	304°02.7	59.0		355°29.3	16.3'	10°51.3	-13.4'	54.3'
9	319°02.7	12°59.9		10°04.7	16.4'	10°37.9	-13.5'	54.3'
10	334°02.8	13°00.7		24°40.0	16.4'	10°24.5	-13.5'	54.3'
11	349°02.8	01.6		39°15.5	16.5'	10°11.0	-13.5'	54.3'
12	4°02.9	S13°02.4		53°51.0	16.5'	N09°57.5	-13.5'	54.3'
13	19°02.9	03.2		68°26.5	16.6'	09°43.9	-13.6'	54.3'
14	34°03.0	04.1		83°02.1	16.6'	09°30.3	-13.6'	54.3'
15	49°03.0	04.9		97°37.7	16.7'	09°16.7	-13.6'	54.2'
16	64°03.1	05.8		112°13.4	16.7'	09°03.1	-13.7'	54.2'
17	79°03.1	06.6		126°49.1	16.8'	08°49.4	-13.7'	54.2'
18	94°03.2	S13°07.4		141°24.8	16.8'	N08°35.7	-13.7'	54.2'
19	109°03.2	08.3		156°00.6	16.8'	08°22.0	-13.7'	54.2'
20	124°03.3	09.1		170°36.5	16.9'	08°08.3	-13.8'	54.2'
21	139°03.3	09.9		185°12.3	16.9'	07°54.5	-13.8'	54.2'
22	154°03.4	10.8		199°48.3	16.9'	07°40.7	-13.8'	54.2'
23	169°03.4	11.6		214°24.2	17.0'	07°26.9	-13.8'	54.2'
SD = 16.1' d = 0.8'				SD = 14.9'				
Mon	GHA	Dec		GHA	ν	Dec	d	HP
0	184°03.5	S13°12.4		229°00.2	17.0'	N07°13.1	-13.9'	54.1'
1	199°03.5	13.3		243°36.2	17.1'	06°59.2	-13.9'	54.1'
2	214°03.6	14.1		258°12.3	17.1'	06°45.3	-13.9'	54.1'
3	229°03.6	14.9		272°48.3	17.1'	06°31.4	-13.9'	54.1'
4	244°03.7	15.8		287°24.4	17.1'	06°17.5	-13.9'	54.1'
5	259°03.7	16.6		302°00.6	17.2'	06°03.6	-14.0'	54.1'
6	274°03.8	S13°17.4		316°36.8	17.2'	N05°49.6	-14.0'	54.1'
7	289°03.8	18.3		331°13.0	17.2'	05°35.7	-14.0'	54.1'
8	304°03.9	19.1		345°49.2	17.3'	05°21.7	-14.0'	54.1'
9	319°03.9	19.9		0°25.4	17.3'	05°07.7	-14.0'	54.1'
10	334°03.9	20.8		15°01.7	17.3'	04°53.7	-14.0'	54.1'
11	349°04.0	21.6		29°38.0	17.3'	04°39.6	-14.0'	54.1'
12	4°04.0	S13°22.4		44°14.3	17.3'	N04°25.6	-14.1'	54.0'
13	19°04.1	23.3		58°50.7	17.4'	04°11.5	-14.1'	54.0'
14	34°04.1	24.1		73°27.0	17.4'	03°57.5	-14.1'	54.0'
15	49°04.2	24.9		88°03.4	17.4'	03°43.4	-14.1'	54.0'
16	64°04.2	25.8		102°39.8	17.4'	03°29.3	-14.1'	54.0'
17	79°04.2	26.6		117°16.2	17.4'	03°15.2	-14.1'	54.0'
18	94°04.3	S13°27.4		131°52.7	17.4'	N03°01.1	-14.1'	54.0'
19	109°04.3	28.2		146°29.1	17.5'	02°47.0	-14.1'	54.0'
20	124°04.4	29.1		161°05.6	17.5'	02°32.9	-14.1'	54.0'
21	139°04.4	29.9		175°42.0	17.5'	02°18.7	-14.1'	54.0'
22	154°04.5	30.7		190°18.5	17.5'	02°04.6	-14.1'	54.0'
23	169°04.5	31.5		204°55.0	17.5'	01°50.4	-14.1'	54.0'
SD = 16.1' d = 0.8'				SD = 14.8'				
Tue	GHA	Dec		GHA	ν	Dec	d	HP
0	184°04.5	S13°32.4		219°31.5	17.5'	N01°36.3	-14.2'	54.0'
1	199°04.6	33.2		234°08.0	17.5'	01°22.1	-14.2'	54.0'
2	214°04.6	34.0		248°44.5	17.5'	01°08.0	-14.2'	54.0'
3	229°04.6	34.9		263°21.0	17.5'	00°53.8	-14.2'	54.0'
4	244°04.7	35.7		277°57.6	17.5'	00°39.7	-14.2'	54.0'
5	259°04.7	36.5		292°34.1	17.5'	00°25.5	-14.2'	54.0'
6	274°04.8	S13°37.3		307°10.6	17.5'	N00°11.3	-14.2'	53.9'
7	289°04.8	38.2		321°47.2	17.5'	S00°02.8	14.2'	53.9'
8	304°04.8	39.0		336°23.7	17.5'	00°17.0	14.2'	53.9'
9	319°04.9	39.8		351°00.2	17.5'	00°31.2	14.2'	53.9'
10	334°04.9	40.6		5°36.8	17.5'	00°45.3	14.2'	53.9'
11	349°04.9	41.4		20°13.3	17.5'	00°59.5	14.2'	53.9'
12	4°05.0	S13°42.3		34°49.8	17.5'	S01°13.6	14.1'	53.9'
13	19°05.0	43.1		49°26.3	17.5'	01°27.8	14.1'	53.9'
14	34°05.1	43.9		64°02.8	17.5'	01°41.9	14.1'	53.9'
15	49°05.1	44.7		78°39.3	17.5'	01°56.1	14.1'	53.9'
16	64°05.1	45.5		93°15.8	17.5'	02°10.2	14.1'	53.9'
17	79°05.2	46.4		107°52.3	17.5'	02°24.3	14.1'	53.9'
18	94°05.2	S13°47.2		122°28.8	17.5'	S02°38.4	14.1'	53.9'
19	109°05.2	48.0		137°05.3	17.5'	02°52.5	14.1'	53.9'
20	124°05.3	48.8		151°41.7	17.4'	03°06.6	14.1'	53.9'
21	139°05.3	49.6		166°18.2	17.4'	03°20.7	14.1'	53.9'
22	154°05.3	50.5		180°54.6	17.4'	03°34.8	14.1'	53.9'
23	169°05.4	51.3		195°31.0	17.4'	03°48.9	14.1'	53.9'
SD = 16.1' d = 0.8'				SD = 14.7'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	05:52	07:13	08:35	14:51	16:13	17:34
N 70°	05:49	07:02	08:13	15:13	16:24	17:36
68°	05:47	06:53	07:56	15:30	16:33	17:39
66°	05:45	06:46	07:42	15:44	16:40	17:41
64°	05:43	06:39	07:31	15:55	16:47	17:43
62°	05:41	06:34	07:21	16:05	16:53	17:45
60°	05:40	06:29	07:13	16:13	16:58	17:47
N 58°	05:38	06:24	07:06	16:21	17:02	17:48
56°	05:36	06:20	06:59	16:27	17:06	17:50
54°	05:35	06:17	06:54	16:33	17:10	17:52
52°	05:33	06:13	06:48	16:38	17:13	17:53
50°	05:32	06:10	06:44	16:43	17:17	17:55
45°	05:29	06:03	06:33	16:53	17:24	17:58
N 40°	05:25	05:57	06:25	17:02	17:30	18:02
35°	05:22	05:52	06:18	17:09	17:35	18:05
30°	05:18	05:47	06:11	17:16	17:40	18:09
20°	05:11	05:37	06:00	17:27	17:50	18:16
N 10°	05:03	05:28	05:50	17:38	17:59	18:24
0°	04:54	05:19	05:40	17:47	18:08	18:33
S 10°	04:44	05:09	05:31	17:57	18:19	18:44
20°	04:31	04:58	05:20	18:07	18:30	18:57
30°	04:14	04:43	05:08	18:19	18:45	19:14
35°	04:03	04:35	05:01	18:27	18:53	19:25
40°	03:50	04:25	04:54	18:35	19:04	19:38
45°	03:34	04:13	04:44	18:44	19:16	19:55
S 50°	03:13	03:57	04:33	18:56	19:31	20:17
52°	03:02	03:50	04:28	19:01	19:39	20:27
54°	02:50	03:42	04:22	19:07	19:47	20:39
56°	02:36	03:33	04:15	19:14	19:57	20:54
58°	02:19	03:22	04:08	19:21	20:07	21:12
S 60°	01:58	03:10	04:00	19:29	20:20	21:34

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°		01:20	03:18	16:26	15:52	15:20
N 70°		01:30	03:19	16:13	15:47	15:23
68°		01:39	03:20	16:03	15:43	15:25
66°	00:05	01:45	03:21	15:54	15:40	15:27
64°	00:17	01:51	03:21	15:47	15:37	15:29
62°	00:28	01:56	03:22	15:40	15:35	15:30
60°	00:36	02:01	03:22	15:34	15:33	15:31
N 58°	00:44	02:04	03:23	15:29	15:31	15:32
56°	00:51	02:08	03:23	15:25	15:29	15:33
54°	00:56	02:11	03:23	15:21	15:28	15:34
52°	01:02	02:14	03:24	15:17	15:26	15:35
50°	01:07	02:16	03:24	15:14	15:25	15:36
45°	01:17	02:21	03:25	15:07	15:22	15:38
N 40°	01:25	02:26				

October 30, 31, 01 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	38°48.2	145°37.0	S24°14.4	278°16.8	N21°50.1	318°56.9	N22°23.5	53°42.3	S08°41.0	Alpheratz	357°34.7 29°13.8
1	53°50.6	160°36.2	14.8	293°18.3	50.0	333°59.6	23.5	68°44.8	41.1	Ankaa	353°06.9 -42°10.3
2	68°53.1	175°35.4	15.3	308°19.7	49.9	349°02.2	23.5	83°47.4	41.1	Schedar	349°30.8 56°40.6
3	83°55.6	190°34.6	•• 15.7	323°21.2	•• 49.8	4°04.9	•• 23.4	98°49.9	•• 41.1	Diphda	348°47.2 -17°51.0
4	98°58.0	205°33.8	16.2	338°22.6	49.7	19°07.5	23.4	113°52.4	41.1	Achernar	335°19.7 -57°06.7
5	114°00.5	220°33.0	16.6	353°24.1	49.6	34°10.2	23.4	128°55.0	41.2	Hamal	327°51.1 23°34.9
6	129°03.0	235°32.2	S24°17.1	8°25.5	N21°49.5	49°12.8	N22°23.4	143°57.5	S08°41.2	Polaris	313°38.8 89°22.1
7	144°05.4	250°31.4	17.5	23°27.0	49.4	64°15.5	23.4	159°00.0	41.2	Acamar	315°11.5 -40°12.2
8	159°07.9	265°30.6	18.0	38°28.4	49.2	79°18.1	23.4	174°02.6	41.2	Menkar	314°06.0 4°11.3
9	174°10.4	280°29.8	•• 18.4	53°29.9	•• 49.1	94°20.8	•• 23.4	189°05.1	•• 41.2	Mirfak	308°28.0 49°57.0
10	189°12.8	295°29.0	18.8	68°31.3	49.0	109°23.4	23.4	204°07.6	41.3	Aldebaran	290°39.5 16°33.6
11	204°15.3	310°28.2	19.3	83°32.8	48.9	124°26.1	23.4	219°10.2	41.3	Rigel	281°03.8 -8°10.2
12	219°17.8	325°27.4	S24°19.7	98°34.2	N21°48.8	139°28.7	N22°23.3	234°12.7	S08°41.3	Capella	280°21.8 46°01.3
13	234°20.2	340°26.6	20.2	113°35.7	48.7	154°31.4	23.3	249°15.2	41.3	Bellatrix	278°22.8 6°22.4
14	249°22.7	355°25.8	20.6	128°37.2	48.6	169°34.0	23.3	264°17.8	41.3	Elnath	278°01.8 28°37.7
15	264°25.1	10°25.0	•• 21.0	143°38.6	•• 48.5	184°36.7	•• 23.3	279°20.3	•• 41.4	Alnilam	275°37.7 -1°11.0
16	279°27.6	25°24.2	21.5	158°40.1	48.4	199°39.3	23.3	294°22.8	41.4	Betelgeuse	270°52.1 7°24.8
17	294°30.1	40°23.4	21.9	173°41.5	48.2	214°42.0	23.3	309°25.3	41.4	Canopus	263°52.2 -52°42.2
18	309°32.5	55°22.6	S24°22.3	188°43.0	N21°48.1	229°44.7	N22°23.3	324°27.9	S08°41.4	Sirius	258°26.2 -16°44.8
19	324°35.0	70°21.8	22.8	203°44.5	48.0	244°47.3	23.3	339°30.4	41.5	Adhara	255°05.9 -29°00.0
20	339°37.5	85°21.0	23.2	218°45.9	47.9	259°50.0	23.3	354°32.9	41.5	Procyon	244°50.9 5°09.8
21	354°39.9	100°20.2	•• 23.6	233°47.4	•• 47.8	274°52.6	•• 23.2	9°35.5	•• 41.5	Pollux	243°17.4 27°57.9
22	9°42.4	115°19.4	24.0	248°48.9	47.7	289°55.3	23.2	24°38.0	41.5	Avior	234°14.8 -59°34.9
23	24°44.9	130°18.6	24.5	263°50.3	47.6	304°57.9	23.2	39°40.5	41.5	Suhail	222°46.5 -43°31.6
Mer.pass. 21:21		ν -0.8' d 0.5' m -4.01		ν 1.4' d -0.1' m 0.09		ν 2.6' d -0.0' m -2.68		ν 2.5' d 0.0' m 0.79		Miaplacidus	221°38.5 -69°48.7
Alphard 217°48.0 -8°45.8 Regulus 207°34.8 11°50.8 Dubhe 193°41.6 61°36.9 Denebola 182°25.4 14°26.1 Gienah 175°44.1 -17°40.6 Acrux 173°01.0 -63°14.0 Gacrux 171°52.4 -57°14.9 Alioth 166°13.6 55°49.4 Spica 158°22.8 -11°17.3 Alkaid 152°52.6 49°11.4 Hadar 148°37.0 -60°29.5 Menkent 147°58.3 -36°29.4 Arcturus 145°48.4 19°03.3 Rigel Kent. 139°41.3 -60°56.2 Kochab 137°20.9 74°03.2 Zuben'ubi 136°56.6 -16°08.6 Alphecca 126°04.2 26°38.0 Antares 112°16.4 -26°29.2 Atria 107°11.3 -69°04.4 Sabik 102°03.3 -15°45.3 Shaula 96°10.9 -37°07.4 Rasalhague 95°59.0 12°32.6 Eltanin 90°42.6 51°29.3 Kaus Aust. 83°33.0 -34°22.4 Vega 80°33.6 38°48.6 Nunki 75°48.2 -26°16.0 Altair 62°00.2 8°56.1 Peacock 53°06.0 -56°39.5 Deneb 49°26.0 45°22.4 Enif 33°38.9 9°59.5 Al Na'ir 27°32.9 -46°50.6 Fomalhaut 15°14.5 -29°29.5 Scheat 13°45.2 28°13.2 Markab 13°29.9 15°20.5											
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	39°47.3	145°17.8	S24°24.9	278°51.8	N21°47.5	320°00.6	N22°23.2	54°43.1	S08°41.6		
1	54°49.8	160°17.0	25.3	293°53.3	47.4	335°03.2	23.2	69°45.6	41.6		
2	69°52.2	175°16.2	25.7	308°54.7	47.2	350°05.9	23.2	84°48.1	41.6		
3	84°54.7	190°15.4	•• 26.2	323°56.2	•• 47.1	5°08.5	•• 23.2	99°50.6	•• 41.6		
4	99°57.2	205°14.6	26.6	338°57.7	47.0	20°11.2	23.2	114°53.2	41.6		
5	114°59.6	220°13.8	27.0	353°59.2	46.9	35°13.9	23.1	129°55.7	41.7		
6	130°02.1	235°13.0	S24°27.4	9°00.6	N21°46.8	50°16.5	N22°23.1	144°58.2	S08°41.7		
7	145°04.6	250°12.1	27.8	24°02.1	46.7	65°19.2	23.1	160°00.8	41.7		
8	160°07.0	265°11.3	28.2	39°03.6	46.6	80°21.8	23.1	175°03.3	41.7		
9	175°09.5	280°10.5	•• 28.6	54°05.0	•• 46.5	95°24.5	•• 23.1	190°05.8	•• 41.7		
10	190°12.0	295°09.7	29.1	69°06.5	46.4	110°27.2	23.1	205°08.3	41.8		
11	205°14.4	310°08.9	29.5	84°08.0	46.3	125°29.8	23.1	220°10.9	41.8		
12	220°16.9	325°08.1	S24°29.9	99°09.5	N21°46.1	140°32.5	N22°23.1	235°13.4	S08°41.8		
13	235°19.4	340°07.3	30.3	114°10.9	46.0	155°35.1	23.1	250°15.9	41.8		
14	250°21.8	355°06.5	30.7	129°12.4	45.9	170°37.8	23.0	265°18.4	41.8		
15	265°24.3	10°05.7	•• 31.1	144°13.9	•• 45.8	185°40.5	•• 23.0	280°21.0	•• 41.9		
16	280°26.7	25°04.9	31.5	159°15.4	45.7	200°43.1	23.0	295°23.5	41.9		
17	295°29.2	40°04.1	31.9	174°16.9	45.6	215°45.8	23.0	310°26.0	41.9		
18	310°31.7	55°03.3	S24°32.3	189°18.3	N21°45.5	230°48.4	N22°23.0	325°28.5	S08°41.9		
19	325°34.1	70°02.5	32.7	204°19.8	45.4	245°51.1	23.0	340°31.1	41.9		
20	340°36.6	85°01.6	33.1	219°21.3	45.3	260°53.8	23.0	355°33.6	41.9		
21	355°39.1	100°00.8	•• 33.5	234°22.8	•• 45.2	275°56.4	•• 23.0	10°36.1	•• 42.0		
22	10°41.5	115°00.0	33.9	249°24.3	45.1	290°59.1	22.9	25°38.7	42.0		
23	25°44.0	129°59.2	34.3	264°25.8	44.9	306°01.8	22.9	40°41.2	42.0		
Mer.pass. 21:17		ν -0.8' d 0.4' m -4.02		ν 1.5' d -0.1' m 0.08		ν 2.7' d -0.0' m -2.68		ν 2.5' d 0.0' m 0.79			
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	40°46.5	144°58.4	S24°34.7	279°27.2	N21°44.8	321°04.4	N22°22.9	55°43.7	S08°42.0		
1	55°48.9	159°57.6	35.1	294°28.7	44.7	336°07.1	22.9	70°46.2	42.0		
2	70°51.4	174°56.8	35.5	309°30.2	44.6	351°09.7	22.9	85°48.8	42.1		
3	85°53.8	189°56.0	•• 35.9	324°31.7	•• 44.5	6°12.4	•• 22.9	100°51.3	•• 42.1		
4	100°56.3	204°55.2	36.3	339°33.2	44.4	21°15.1	22.9	115°53.8	42.1		
5	115°58.8	219°54.3	36.7	354°34.7	44.3	36°17.7	22.9	130°56.3	42.1		
6	131°01.2	234°53.5	S24°37.0	9°36.2	N21°44.2	51°20.4	N22°22.8	145°58.9	S08°42.1		
7	146°03.7	249°52.7	37.4	24°37.7	44.1	66°23.1	22.8	161°01.4	42.2		
8	161°06.2	264°51.9	37.8	39°39.2	44.0	81°25.7	22.8	176°03.9	42.2		
9	176°08.6	279°51.1	•• 38.2	54°40.7	•• 43.9	96°28.4	•• 22.8	191°06.4	•• 42.2		
10	191°11.1	294°50.3	38.6	69°42.1	43.8	111°31.1	22.8	206°08.9	42.2		
11	206°13.6	309°49.5	39.0	84°43.6	43.7	126°33.7	22.8	221°11.5	42.2		
12	221°16.0	324°48.7	S24°39.4	99°45.1	N21°43.5	141°36.4	N22°22.8	236°14.0	S08°42.2		
13	236°18.5	339°47.8	39.7	114°46.6	43.4	156°39.1	22.8	251°16.5	42.3		
14	251°21.0	354°47.0	40.1	129°48.1	43.3	171°41.7	22.7	266°19.0	42.3		
15	266°23.4	9°46.2	•• 40.5	144°49.6	•• 43.2	186°44.4	•• 22.7	281°21.6	•• 42.3		
16	281°25.9	24°45.4	40.9	159°51.1	43.1	201°47.1	22.7	296°24.1	42.3		
17	296°28.3	39°44.6	41.2	174°52.6	43.0	216°49.8	22.7	311°26.6	42.3		
18	311°30.8	54°43.8	S24°41.6	189°54.1	N21°42.9	231°52.4	N22°22.7	326°29.1	S08°42.3		
19	326°33.3	69°42.9	42.0	204°55.6	42.8	246°55.1	22.7	341°31.7	42.4		
20	341°35.7	84°42.1	42.4	219°57.1	42.7	261°57.8	22.7	356°34.2	42.4		
21	356°38.2	99°41.3	•• 42.7	234°58.6	•• 42.6	277°00.4	•• 22.7	11°36.7	•• 42.4		
22	11°40.7	114°40.5	43.1	250°00.1	42.5	292°03.1	22.6	26°39.2	42.4		
23	26°43.1	129°39.7	43.5	265°01.6	42.4	307°05.8	22.6	41°41.7	42.4		
Mer.pass. 21:13		ν -0.8' d 0.4' m -4.02		ν 1.5' d -0.1' m 0.06		ν 2.7' d -0.0' m -2.69		ν 2.5' d 0.0' m 0.80			
Horizontal parallax Venus: 0.1 Mars: 0.1											

Sun				Moon				
Wed	GHA	Dec		GHA	ν	Dec	d	HP
0	184°05.4	S13°52.1		210°07.4	17.4'	S04°02.9	14.0'	53.9'
1	199°05.4	52.9		224°43.8	17.4'	04°17.0	14.0'	53.9'
2	214°05.4	53.7		239°20.2	17.3'	04°31.0	14.0'	53.9'
3	229°05.5	· · 54.5		253°56.5	17.3'	04°45.0	14.0'	53.9'
4	244°05.5	55.4		268°32.8	17.3'	04°59.0	14.0'	53.9'
5	259°05.5	56.2		283°09.1	17.3'	05°13.0	14.0'	53.9'
6	274°05.6	S13°57.0		297°45.4	17.3'	S05°27.0	13.9'	53.9'
7	289°05.6	57.8		312°21.7	17.2'	05°40.9	13.9'	53.9'
8	304°05.6	58.6		326°57.9	17.2'	05°54.8	13.9'	53.9'
9	319°05.7	13°59.4		341°34.1	17.2'	06°08.7	13.9'	53.9'
10	334°05.7	14°00.2		356°10.3	17.2'	06°22.6	13.9'	53.9'
11	349°05.7	01.1		10°46.4	17.1'	06°36.5	13.8'	53.9'
12	4°05.7	S14°01.9		25°22.5	17.1'	S06°50.4	13.8'	53.9'
13	19°05.8	02.7		39°58.6	17.1'	07°04.2	13.8'	53.9'
14	34°05.8	03.5		54°34.7	17.0'	07°18.0	13.8'	53.9'
15	49°05.8	· · 04.3		69°10.7	17.0'	07°31.8	13.8'	53.9'
16	64°05.8	05.1		83°46.7	17.0'	07°45.5	13.7'	53.9'
17	79°05.9	05.9		98°22.7	16.9'	07°59.3	13.7'	54.0'
18	94°05.9	S14°06.7		112°58.6	16.9'	S08°13.0	13.7'	54.0'
19	109°05.9	07.6		127°34.5	16.9'	08°26.6	13.7'	54.0'
20	124°05.9	08.4		142°10.4	16.8'	08°40.3	13.6'	54.0'
21	139°06.0	· · 09.2		156°46.2	16.8'	08°53.9	13.6'	54.0'
22	154°06.0	10.0		171°22.0	16.7'	09°07.5	13.6'	54.0'
23	169°06.0	10.8		185°57.7	16.7'	09°21.1	13.5'	54.0'
SD = 16.1'			$d = 0.8'$	SD = 14.7'				
Thu	GHA	Dec		GHA	ν	Dec	d	HP
0	184°06.0	S14°11.6		200°33.4	16.7'	S09°34.6	13.5'	54.0'
1	199°06.1	12.4		215°09.1	16.6'	09°48.1	13.5'	54.0'
2	214°06.1	13.2		229°44.7	16.6'	10°01.6	13.4'	54.0'
3	229°06.1	· · 14.0		244°20.3	16.5'	10°15.0	13.4'	54.0'
4	244°06.1	14.8		258°55.8	16.5'	10°28.4	13.4'	54.0'
5	259°06.2	15.6		273°31.3	16.4'	10°41.8	13.3'	54.0'
6	274°06.2	S14°16.4		288°06.7	16.4'	S10°55.1	13.3'	54.0'
7	289°06.2	17.2		302°42.1	16.3'	11°08.4	13.3'	54.0'
8	304°06.2	18.0		317°17.5	16.3'	11°21.7	13.2'	54.0'
9	319°06.2	· · 18.9		331°52.8	16.2'	11°34.9	13.2'	54.0'
10	334°06.3	19.7		346°28.0	16.2'	11°48.1	13.1'	54.0'
11	349°06.3	20.5		1°03.2	16.1'	12°01.2	13.1'	54.0'
12	4°06.3	S14°21.3		15°38.4	16.1'	S12°14.3	13.1'	54.0'
13	19°06.3	22.1		30°13.5	16.0'	12°27.3	13.0'	54.0'
14	34°06.3	22.9		44°48.5	16.0'	12°40.4	13.0'	54.1'
15	49°06.3	· · 23.7		59°23.5	15.9'	12°53.3	12.9'	54.1'
16	64°06.4	24.5		73°58.4	15.9'	13°06.3	12.9'	54.1'
17	79°06.4	25.3		88°33.3	15.8'	13°19.1	12.8'	54.1'
18	94°06.4	S14°26.1		103°08.1	15.8'	S13°32.0	12.8'	54.1'
19	109°06.4	26.9		117°42.9	15.7'	13°44.8	12.7'	54.1'
20	124°06.4	27.7		132°17.6	15.6'	13°57.5	12.7'	54.1'
21	139°06.5	· · 28.5		146°52.2	15.6'	14°10.2	12.6'	54.1'
22	154°06.5	29.3		161°26.8	15.5'	14°22.8	12.6'	54.1'
23	169°06.5	30.1		176°01.3	15.5'	14°35.4	12.5'	54.1'
SD = 16.1'			$d = 0.8'$	SD = 14.7'				
Fri	GHA	Dec		GHA	ν	Dec	d	HP
0	184°06.5	S14°30.9		190°35.8	15.4'	S14°47.9	12.5'	54.1'
1	199°06.5	31.7		205°10.2	15.3'	15°00.4	12.4'	54.1'
2	214°06.5	32.5		219°44.5	15.3'	15°12.8	12.4'	54.1'
3	229°06.5	· · 33.3		234°18.8	15.2'	15°25.2	12.3'	54.1'
4	244°06.6	34.1		248°53.0	15.1'	15°37.5	12.3'	54.2'
5	259°06.6	34.9		263°27.1	15.1'	15°49.8	12.2'	54.2'
6	274°06.6	S14°35.7		278°01.2	15.0'	S16°02.0	12.1'	54.2'
7	289°06.6	36.5		292°35.2	14.9'	16°14.1	12.1'	54.2'
8	304°06.6	37.2		307°09.1	14.9'	16°26.2	12.0'	54.2'
9	319°06.6	· · 38.0		321°43.0	14.8'	16°38.3	12.0'	54.2'
10	334°06.6	38.8		336°16.8	14.7'	16°50.2	11.9'	54.2'
11	349°06.6	39.6		350°50.5	14.7'	17°02.1	11.8'	54.2'
12	4°06.6	S14°40.4		5°24.2	14.6'	S17°14.0	11.8'	54.2'
13	19°06.7	41.2		19°57.8	14.5'	17°25.7	11.7'	54.2'
14	34°06.7	42.0		34°31.3	14.4'	17°37.4	11.6'	54.2'
15	49°06.7	· · 42.8		49°04.7	14.4'	17°49.1	11.6'	54.3'
16	64°06.7	43.6		63°38.1	14.3'	18°00.6	11.5'	54.3'
17	79°06.7	44.4		78°11.4	14.2'	18°12.2	11.4'	54.3'
18	94°06.7	S14°45.2		92°44.6	14.1'	S18°23.6	11.4'	54.3'
19	109°06.7	46.0		107°17.8	14.1'	18°34.9	11.3'	54.3'
20	124°06.7	46.8		121°50.8	14.0'	18°46.2	11.2'	54.3'
21	139°06.7	· · 47.5		136°23.8	13.9'	18°57.5	11.1'	54.3'
22	154°06.7	48.3		150°56.7	13.8'	19°08.6	11.1'	54.3'
23	169°06.7	49.1		165°29.6	13.8'	19°19.7	11.0'	54.3'
SD = 16.1'			$d = 0.8'$	SD = 14.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	06:04	07:27	08:53	14:32	15:59	17:21
N 70°	06:00	07:14	08:28	14:58	16:12	17:25
68°	05:57	07:04	08:09	15:17	16:22	17:29
66°	05:54	06:55	07:53	15:33	16:31	17:32
64°	05:51	06:48	07:41	15:46	16:38	17:35
62°	05:49	06:41	07:30	15:56	16:45	17:37
60°	05:46	06:36	07:21	16:05	16:50	17:40
N 58°	05:44	06:31	07:13	16:14	16:56	17:42
56°	05:42	06:26	07:06	16:21	17:00	17:44
54°	05:40	06:22	06:59	16:27	17:04	17:46
52°	05:38	06:18	06:54	16:33	17:08	17:48
50°	05:37	06:15	06:49	16:38	17:12	17:50
45°	05:32	06:07	06:38	16:49	17:20	17:54
N 40°	05:28	06:00	06:28	16:58	17:26	17:58
35°	05:24	05:54	06:20	17:06	17:32	18:02
30°	05:20	05:49	06:13	17:13	17:38	18:06
20°	05:12	05:39	06:01	17:26	17:48	18:15
N 10°	05:04	05:29	05:50	17:37	17:58	18:23
0°	04:54	05:19	05:40	17:47	18:08	18:33
S 10°	04:43	05:08	05:30	17:57	18:19	18:45
20°	04:29	04:56	05:19	18:09	18:32	18:59
30°	04:11	04:41	05:06	18:22	18:47	19:17
35°	03:59	04:31	04:58	18:29	18:56	19:29
40°	03:45	04:21	04:50	18:38	19:07	19:43
45°	03:28	04:08	04:40	18:48	19:20	20:00
S 50°	03:06	03:51	04:27	19:01	19:37	20:23
52°	02:54	03:44	04:22	19:07	19:45	20:35
54°	02:41	03:35	04:15	19:13	19:54	20:48
56°	02:26	03:25	04:08	19:20	20:04	21:04
58°	02:07	03:13	04:01	19:28	20:16	21:23
S 60°	01:43	03:00	03:52	19:37	20:30	21:49

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	05:15	07:22	10:15	14:46	14:04	12:38
N 70°	05:07	07:02	09:18	14:57	14:26	13:38
68°	05:01	06:46	08:44	15:06	14:44	14:13
66°	04:55	06:33	08:20	15:13	14:58	14:38
64°	04:51	06:23	08:01	15:19	15:10	14:58
62°	04:47	06:14	07:46	15:25	15:20	15:14
60°	04:44	06:07	07:34	15:30	15:28	15:27
N 58°	04:41	06:00	07:23	15:34	15:36	15:39
56°	04:38	05:54	07:13	15:38	15:43	15:49
54°	04:36	05:49	07:05	15:41	15:49	15:58
52°	04:34	05:45	06:58	15:44	15:54	16:06
50°	04:32	05:40	06:51	15:47	15:59	16:14
45°	04:27	05:31	06:37	15:53	16:10	16:29
N 40°	04:24	05:24	06:25	15:58	16:19	16:42
35°	04:21	05:17	06:15	16:03	16:27	16:53
30°	04:18	05:12	06:07	16:07	16:33	17:03
20°	04:14	05:02	05:52	16:13	16:45	17:20
N 10°	04:10	04:54	05:39	16:19	16:56	17:34
0°	04:06	04:46	05:27	16:25	17:06	17:48
S 10°	04:03	04:38	05:16	16:31	17:15	18:02
20°	03:59	04:30	05:03	16:37	17:26	18:17
30°	03:55	04:21	04:49	16:44	17:38	18:34
35°	03:52	04:15	04:40	16:48	17:45	18:44
40°	03:49	04:09	04:31	16:53	17:53	18:56
45°	03:46	04:02	04:20	16:58	18:03	19:09
S 50°	03:42	03:54	04:07	17:05	18:14	19:26
52°	03:41	03:50	04:01	17:08	18:20	19:34

November 02, 03, 04 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	41°45.6	144°38.9	S24°43.8	280°03.1	N21°42.3	322°08.4	N22°22.6	56°44.3	S08°42.4	Alpheratz	357°34.7	29°13.8
1	56°48.1	159°38.0	44.2	295°04.7	42.2	337°11.1	22.6	71°46.8	42.5	Ankaa	353°07.0	-42°10.3
2	71°50.5	174°37.2	44.6	310°06.2	42.1	352°13.8	22.6	86°49.3	42.5	Schedar	349°30.8	56°40.6
3	86°53.0	189°36.4	· · 44.9	325°07.7	· · 42.0	7°16.5	· · 22.6	101°51.8	· · 42.5	Diphda	348°47.2	-17°51.0
4	101°55.5	204°35.6	45.3	340°09.2	41.8	22°19.1	22.6	116°54.3	42.5	Achernar	335°19.7	-57°06.7
5	116°57.9	219°34.8	45.6	355°10.7	41.7	37°21.8	22.6	131°56.9	42.5	Hamal	327°51.1	23°34.9
6	132°00.4	234°34.0	S24°46.0	10°12.2	N21°41.6	52°24.5	N22°22.5	146°59.4	S08°42.5	Polaris	313°38.2	89°22.1
7	147°02.8	249°33.1	46.4	25°13.7	41.5	67°27.2	22.5	162°01.9	42.6	Acamar	315°11.5	-40°12.2
8	162°05.3	264°32.3	46.7	40°15.2	41.4	82°29.8	22.5	177°04.4	42.6	Menkar	314°06.0	4°11.3
9	177°07.8	279°31.5	· · 47.1	55°16.7	· · 41.3	97°32.5	· · 22.5	192°06.9	· · 42.6	Mirfak	308°28.0	49°57.0
10	192°10.2	294°30.7	47.4	70°18.2	41.2	112°35.2	22.5	207°09.5	42.6	Aldebaran	290°39.5	16°33.6
11	207°12.7	309°29.9	47.8	85°19.8	41.1	127°37.9	22.5	222°12.0	42.6	Rigel	281°03.8	-8°10.2
12	222°15.2	324°29.0	S24°48.1	100°21.3	N21°41.0	142°40.5	N22°22.5	237°14.5	S08°42.6	Capella	280°21.8	46°01.3
13	237°17.6	339°28.2	48.5	115°22.8	40.9	157°43.2	22.4	252°17.0	42.7	Bellatrix	278°22.8	6°22.4
14	252°20.1	354°27.4	48.8	130°24.3	40.8	172°45.9	22.4	267°19.5	42.7	Elnath	278°01.8	28°37.7
15	267°22.6	9°26.6	· · 49.2	145°25.8	· · 40.7	187°48.6	· · 22.4	282°22.1	· · 42.7	Alnilam	275°37.7	-1°11.0
16	282°25.0	24°25.8	49.5	160°27.3	40.6	202°51.2	22.4	297°24.6	42.7	Betelgeuse	270°52.1	7°24.8
17	297°27.5	39°24.9	49.9	175°28.8	40.5	217°53.9	22.4	312°27.1	42.7	Canopus	263°52.1	-52°42.2
18	312°29.9	54°24.1	S24°50.2	190°30.4	N21°40.4	232°56.6	N22°22.4	327°29.6	S08°42.7	Sirius	258°26.2	-16°44.8
19	327°32.4	69°23.3	50.6	205°31.9	40.3	247°59.3	22.4	342°32.1	42.8	Adhara	255°05.8	-29°00.1
20	342°34.9	84°22.5	50.9	220°33.4	40.2	263°02.0	22.4	357°34.7	42.8	Procyon	244°50.9	5°09.8
21	357°37.3	99°21.7	· · 51.2	235°34.9	· · 40.1	278°04.6	· · 22.3	12°37.2	· · 42.8	Pollux	243°17.4	27°57.9
22	12°39.8	114°20.8	51.6	250°36.5	40.0	293°07.3	22.3	27°39.7	42.8	Avior	234°14.8	-59°34.9
23	27°42.3	129°20.0	51.9	265°38.0	39.9	308°10.0	22.3	42°42.2	42.8	Suhail	222°46.5	-43°31.6
Mer.pass. 21:09		ν -0.8' d 0.4' m -4.03		ν 1.5' d -0.1' m 0.04		ν 2.7' d -0.0' m -2.70		ν 2.5' d 0.0' m 0.80				

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	42°44.7	144°19.2	S24°52.3	280°39.5	N21°39.8	323°12.7	N22°22.3	57°44.7	S08°42.8
1	57°47.2	159°18.4	52.6	295°41.0	39.7	338°15.3	22.3	72°47.2	42.9
2	72°49.7	174°17.5	52.9	310°42.6	39.6	353°18.0	22.3	87°49.8	42.9
3	87°52.1	189°16.7	· · 53.3	325°44.1	· · 39.4	8°20.7	· · 22.3	102°52.3	· · 42.9
4	102°54.6	204°15.9	53.6	340°45.6	39.3	23°23.4	22.2	117°54.8	42.9
5	117°57.1	219°15.1	53.9	355°47.1	39.2	38°26.1	22.2	132°57.3	42.9
6	132°59.5	234°14.3	S24°54.3	10°48.7	N21°39.1	53°28.8	N22°22.2	147°59.8	S08°42.9
7	148°02.0	249°13.4	54.6	25°50.2	39.0	68°31.4	22.2	163°02.4	42.9
8	163°04.4	264°12.6	54.9	40°51.7	38.9	83°34.1	22.2	178°04.9	43.0
9	178°06.9	279°11.8	· · 55.2	55°53.3	· · 38.8	98°36.8	· · 22.2	193°07.4	· · 43.0
10	193°09.4	294°11.0	55.6	70°54.8	38.7	113°39.5	22.2	208°09.9	43.0
11	208°11.8	309°10.1	55.9	85°56.3	38.6	128°42.2	22.2	223°12.4	43.0
12	223°14.3	324°09.3	S24°56.2	100°57.9	N21°38.5	143°44.8	N22°22.1	238°14.9	S08°43.0
13	238°16.8	339°08.5	56.5	115°59.4	38.4	158°47.5	22.1	253°17.4	43.0
14	253°19.2	354°07.7	56.9	131°00.9	38.3	173°50.2	22.1	268°20.0	43.0
15	268°21.7	9°06.8	· · 57.2	146°02.5	· · 38.2	188°52.9	· · 22.1	283°22.5	· · 43.1
16	283°24.2	24°06.0	57.5	161°04.0	38.1	203°55.6	22.1	298°25.0	43.1
17	298°26.6	39°05.2	57.8	176°05.5	38.0	218°58.3	22.1	313°27.5	43.1
18	313°29.1	54°04.4	S24°58.1	191°07.1	N21°37.9	234°01.0	N22°22.1	328°30.0	S08°43.1
19	328°31.6	69°03.5	58.4	206°08.6	37.8	249°03.6	22.0	343°32.5	43.1
20	343°34.0	84°02.7	58.7	221°10.2	37.7	264°06.3	22.0	358°35.0	43.1
21	358°36.5	99°01.9	· · 59.1	236°11.7	· · 37.6	279°09.0	· · 22.0	13°37.6	· · 43.1
22	13°38.9	114°01.0	59.4	251°13.2	37.5	294°11.7	22.0	28°40.1	43.2
23	28°41.4	129°00.2	59.7	266°14.8	37.4	309°14.4	22.0	43°42.6	43.2
Mer.pass. 21:06		ν -0.8' d 0.3' m -4.03		ν 1.5' d -0.1' m 0.03		ν 2.7' d -0.0' m -2.70		ν 2.5' d 0.0' m 0.81	

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	43°43.9	143°59.4	S25°00.0	281°16.3	N21°37.3	324°17.1	N22°22.0	58°45.1	S08°43.2
1	58°46.3	158°58.6	00.3	296°17.9	37.2	339°19.8	22.0	73°47.6	43.2
2	73°48.8	173°57.7	00.6	311°19.4	37.1	354°22.5	21.9	88°50.1	43.2
3	88°51.3	188°56.9	· · 00.9	326°21.0	· · 37.0	9°25.1	· · 21.9	103°52.6	· · 43.2
4	103°53.7	203°56.1	01.2	341°22.5	36.9	24°27.8	21.9	118°55.2	43.2
5	118°56.2	218°55.3	01.5	356°24.1	36.8	39°30.5	21.9	133°57.7	43.2
6	133°58.7	233°54.4	S25°01.8	11°25.6	N21°36.7	54°33.2	N22°21.9	149°00.2	S08°43.3
7	149°01.8	248°53.6	02.1	26°27.2	36.6	69°35.9	21.9	164°02.7	43.3
8	164°03.6	263°52.8	02.4	41°28.7	36.5	84°38.6	21.9	179°05.2	43.3
9	179°06.1	278°51.9	· · 02.7	56°30.3	· · 36.4	99°41.3	· · 21.8	194°07.7	· · 43.3
10	194°08.5	293°51.1	03.0	71°31.8	36.3	114°44.0	21.8	209°10.2	43.3
11	209°11.0	308°50.3	03.3	86°33.4	36.2	129°46.7	21.8	224°12.7	43.3
12	224°13.4	323°49.5	S25°03.6	101°34.9	N21°36.1	144°49.4	N22°21.8	239°15.3	S08°43.3
13	239°15.9	338°48.6	03.9	116°36.5	36.0	159°52.0	21.8	254°17.8	43.4
14	254°18.4	353°47.8	04.2	131°38.0	35.9	174°54.7	21.8	269°20.3	43.4
15	269°20.8	8°47.0	· · 04.5	146°39.6	· · 35.8	189°57.4	· · 21.8	284°22.8	· · 43.4
16	284°23.3	23°46.1	04.7	161°41.1	35.7	205°00.1	21.7	299°25.3	43.4
17	299°25.8	38°45.3	05.0	176°42.7	35.6	220°02.8	21.7	314°27.8	43.4
18	314°28.2	53°44.5	S25°05.3	191°44.3	N21°35.5	235°05.5	N22°21.7	329°30.3	S08°43.4
19	329°30.7	68°43.6	05.6	206°45.8	35.4	250°08.2	21.7	344°32.8	43.4
20	344°33.2	83°42.8	05.9	221°47.4	35.3	265°10.9	21.7	359°35.3	43.4
21	359°35.6	98°42.0	· · 06.2	236°48.9	· · 35.2	280°13.6	· · 21.7	14°37.9	· · 43.4
22	14°38.1	113°41.2	06.4	251°50.5	35.1	295°16.3	21.7	29°40.4	43.5
23	29°40.6	128°40.3	06.7	266°52.1	35.0	310°19.0	21.6	44°42.9	43.5
Mer.pass. 21:02		ν -0.8' d 0.3' m -4.04		ν 1.5' d -0.1' m 0.01		ν 2.7' d -0.0' m -2.71		ν 2.5' d 0.0' m 0.81	

Nov 02 Sat	SHA	Mer.pass
Venus	102°53.3	14:22
Mars	238°17.5	05:19
Jupiter	280°22.9	02:31
Saturn	14°58.7	20:10

Nov 03 Sun	SHA	Mer.pass
Venus	101°34.5	14:24
Mars	237°54.8	05:17
Jupiter	280°27.9	02:27
Saturn	15°00.0	20:06

Nov 04 Mon	SHA	Mer.pass
Venus	100°15.5	14:25
Mars	237°32.5	05:14
Jupiter	280°33.2	02:22
Saturn	15°01.2	20:02

Horizontal parallax		
Venus:	0.1	
Mars:	0.1	

Sun				Moon														
Sat	GHA	Dec		GHA	ν	Dec	d	HP	Lat.	Twilight		Sunrise	Sunset	Twilight				
										Naut.	Civil			Civil	Naut.			
0	184°06.8	S14°49.9		180°02.3	13.7'	S19°30.7	10.9'	54.3'	N 72°	06:16	07:40	09:13	14:13	15:45	17:09			
1	199°06.8	50.7		194°35.0	13.6'	19°41.6	10.8'	54.4'		N 70°	06:11	07:26	08:43	14:43	16:00	17:15		
2	214°06.8	51.5		209°07.6	13.5'	19°52.4	10.8'	54.4'			68°	06:06	07:14	08:21	15:05	16:12	17:19	
3	229°06.8	52.3		223°40.1	13.4'	20°03.2	10.7'	54.4'			66°	06:02	07:04	08:04	15:22	16:21	17:24	
4	244°06.8	53.1		238°12.6	13.4'	20°13.9	10.6'	54.4'			64°	05:59	06:56	07:50	15:36	16:30	17:27	
5	259°06.8	53.9		252°44.9	13.3'	20°24.5	10.5'	54.4'			62°	05:56	06:49	07:38	15:48	16:37	17:30	
6	274°06.8	S14°54.6		267°17.2	13.2'	S20°35.0	10.4'	54.4'			60°	05:53	06:43	07:28	15:58	16:43	17:33	
7	289°06.8	55.4		281°49.4	13.1'	20°45.4	10.3'	54.4'			N 58°	05:50	06:37	07:20	16:06	16:49	17:36	
8	304°06.8	56.2		296°21.5	13.0'	20°55.7	10.3'	54.4'				56°	05:48	06:32	07:12	16:14	16:54	17:38
9	319°06.8	57.0		310°53.6	13.0'	21°06.0	10.2'	54.4'				54°	05:45	06:27	07:05	16:21	16:59	17:41
10	334°06.8	57.8		325°25.5	12.9'	21°16.2	10.1'	54.5'	52°			05:43	06:23	06:59	16:27	17:03	17:43	
11	349°06.8	58.6		339°57.4	12.8'	21°26.3	10.0'	54.5'	50°	05:41		06:19	06:54	16:33	17:07	17:45		
12	4°06.8	S14°59.3		354°29.2	12.7'	S21°36.3	9.9'	54.5'	45°	05:36		06:11	06:42	16:45	17:16	17:50		
13	19°06.8	15°00.1		9°00.9	12.6'	21°46.2	9.8'	54.5'	N 40°	05:31		06:04	06:32	16:55	17:23	17:55		
14	34°06.8	00.9		23°32.5	12.5'	21°56.0	9.7'	54.5'		35°		05:27	05:57	06:23	17:03	17:30	18:00	
15	49°06.8	01.7		38°04.0	12.5'	22°05.7	9.6'	54.5'		30°		05:22	05:51	06:16	17:11	17:36	18:04	
16	64°06.8	02.5		52°35.5	12.4'	22°15.3	9.5'	54.5'		20°		05:14	05:40	06:03	17:24	17:47	18:13	
17	79°06.8	03.3		67°06.9	12.3'	22°24.9	9.4'	54.5'		N 10°	05:04	05:29	05:51	17:36	17:58	18:23		
18	94°06.8	S15°04.0		81°38.1	12.2'	S22°34.3	9.3'	54.6'			0°	04:54	05:19	05:40	17:47	18:08	18:33	
19	109°06.8	04.8		96°09.3	12.1'	22°43.7	9.2'	54.6'			S 10°	04:42	05:07	05:29	17:58	18:20	18:46	
20	124°06.8	05.6		110°40.4	12.0'	22°52.9	9.1'	54.6'				20°	04:27	04:54	05:17	18:10	18:33	19:01
21	139°06.8	06.4		125°11.5	11.9'	23°02.1	9.0'	54.6'				30°	04:08	04:38	05:03	18:24	18:50	19:20
22	154°06.8	07.2		139°42.4	11.9'	23°11.1	8.9'	54.6'				35°	03:56	04:28	04:55	18:32	18:59	19:32
23	169°06.8	07.9		154°13.3	11.8'	23°20.1	8.8'	54.6'	40°			03:41	04:17	04:46	18:42	19:11	19:47	
SD = 16.1'			$d = 0.8'$	SD = 14.8'														
Sun	GHA	Dec		GHA	ν	Dec	d	HP										
0	184°06.8	S15°08.7		168°44.0	11.7'	S23°28.9	8.7'	54.6'	S 50°			02:59	03:46	04:22	19:06	19:43	20:30	
1	199°06.8	09.5		183°14.7	11.6'	23°37.6	8.6'	54.7'		52°		02:47	03:37	04:16	19:12	19:51	20:42	
2	214°06.8	10.3		197°45.3	11.5'	23°46.3	8.5'	54.7'		54°		02:33	03:28	04:09	19:19	20:01	20:57	
3	229°06.8	11.0		212°15.8	11.4'	23°54.8	8.4'	54.7'		56°	02:16	03:17	04:02	19:27	20:12	21:14		
4	244°06.8	11.8		226°46.3	11.3'	24°03.2	8.3'	54.7'		58°	01:55	03:05	03:53	19:35	20:24	21:36		
5	259°06.8	12.6		241°16.6	11.3'	24°11.6	8.2'	54.7'		S 60°	01:27	02:50	03:43	19:45	20:39	22:05		
6	274°06.8	S15°13.4		255°46.9	11.2'	S24°19.8	8.1'	54.7'										
7	289°06.8	14.2		270°17.1	11.1'	24°27.9	8.0'	54.7'										
8	304°06.8	14.9		284°47.2	11.0'	24°35.8	7.9'	54.8'										
9	319°06.8	15.7		299°17.2	10.9'	24°43.7	7.8'	54.8'										
10	334°06.8	16.5		313°47.1	10.8'	24°51.5	7.6'	54.8'										
11	349°06.8	17.2		328°16.9	10.8'	24°59.1	7.5'	54.8'										
12	4°06.8	S15°18.0		342°46.7	10.7'	S25°06.7	7.4'	54.8'										
13	19°06.7	18.8		357°16.4	10.6'	25°14.1	7.3'	54.8'										
14	34°06.7	19.6		11°45.9	10.5'	25°21.4	7.2'	54.8'										
15	49°06.7	20.3		26°15.5	10.4'	25°28.6	7.1'	54.9'										
16	64°06.7	21.1		40°44.9	10.3'	25°35.6	6.9'	54.9'										
17	79°06.7	21.9		55°14.2	10.3'	25°42.6	6.8'	54.9'										
18	94°06.7	S15°22.7		69°43.5	10.2'	S25°49.4	6.7'	54.9'										
19	109°06.7	23.4		84°12.7	10.1'	25°56.1	6.6'	54.9'										
20	124°06.7	24.2		98°41.8	10.0'	26°02.6	6.4'	54.9'										
21	139°06.7	25.0		113°10.8	9.9'	26°09.1	6.3'	55.0'										
22	154°06.7	25.7		127°39.8	9.9'	26°15.4	6.2'	55.0'										
23	169°06.7	26.5		142°08.6	9.8'	26°21.6	6.1'	55.0'										
SD = 16.1'			$d = 0.8'$	SD = 14.9'														
Mon	GHA	Dec		GHA	ν	Dec	d	HP										
0	184°06.7	S15°27.3		156°37.4	9.7'	S26°27.7	5.9'	55.0'	S 10°	05:56	06:40	07:29	18:52	19:44	20:39			
1	199°06.6	28.0		171°06.1	9.6'	26°33.6	5.8'	55.0'		20°	05:39	06:20	07:06	19:10	20:06	21:03		
2	214°06.6	28.8		185°34.8	9.6'	26°39.4	5.7'	55.0'		30°	05:20	05:57	06:39	19:32	20:31	21:31		
3	229°06.6	29.6		200°03.3	9.5'	26°45.1	5.5'	55.1'		35°	05:09	05:43	06:24	19:45	20:47	21:47		
4	244°06.6	30.3		214°31.8	9.4'	26°50.6	5.4'	55.1'		40°	04:57	05:28	06:06	20:00	21:04	22:07		
5	259°06.6	31.1		229°00.3	9.3'	26°56.0	5.3'	55.1'		45°	04:42	05:09	05:44	20:17	21:25	22:30		
6	274°06.6	S15°31.9		243°28.6	9.3'	S27°01.3	5.1'	55.1'		S 50°	04:24	04:46	05:16	20:39	21:52	23:00		
7	289°06.6	32.6		257°56.9	9.2'	27°06.5	5.0'	55.1'			52°	04:15	04:35	05:03	20:50	22:05	23:15	
8	304°06.6	33.4		272°25.1	9.1'	27°11.5	4.9'	55.1'			54°	04:06	04:22	04:47	21:02	22:20	23:33	
9	319°06.5	34.2		286°53.2	9.1'	27°16.3	4.7'	55.2'			56°	03:55	04:08	04:29	21:15	22:38	23:54	
10	334°06.5	34.9		301°21.3	9.0'	27°21.1	4.6'	55.2'	58°		03:43	03:52	04:07	21:32	23:00	...		
11	349°06.5	35.7		315°49.3	8.9'	27°25.7	4.5'	55.2'	S 60°		03:29	03:32	03:39	21:51	23:28	...		
12	4°06.5	S15°36.4		330°17.2	8.9'	S27°30.1	4.3'	55.2'										
13	19°06.5	37.2		344°45.0	8.8'	27°34.4	4.2'	55.2'										
14	34°06.5	38.0		359°12.8	8.7'	27°38.6	4.0'	55.3'										
15	49°06.5	38.7		13°40.6	8.7'	27°42.6	3.9'	55.3'										
16	64°06.4	39.5		28°08.2	8.6'	27°46.5	3.7'	55.3'										
17	79°06.4	40.3		42°35.8	8.5'	27°50.3	3.6'	55.3'										
18	94°06.4	S15°41.0		57°03.4	8.5'	S27°53.9	3.5'	55.3'										
19	109°06.4	41.8		71°30.9	8.4'	27°57.3	3.3'	55.3'										
20	124°06.4	42.5		85°58.3	8.4'	28°00.6	3.2'	55.4'										
21	139°06.4	43.3		100°25.7	8.3'	28°03.8	3.0'	55.4'										
22	154°06.3	44.0		114°53.0	8.3'	28°06.8	2.9'	55.4'										
23	169°06.3	44.8		129°20.3	8.2'	28°09.7	2.7'	55.4'										
SD = 16.1'			$d = 0.8'$	SD = 15.0'														
Day		Sun			Mer.		Moon			Age								
		Eqn.of Time	12h	Mer.	Mer.Pass.		Lower		1-3									
		00h	mm:ss	Pass	Upper	Lower	hh:mm	hh:mm	hh:mm	0-6%								
02		16:27	16:27	11:44	12:23	00:00												
03		16:27	16:27	11:44	13:11	00:47												
04		16:27	16:26	11:44	14:03	01:37												

November 05, 06, 07 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	44°43.0	143°39.5	S25°07.0	281°53.6	N21°34.9	325°21.7	N22°21.6	59°45.4	S08°43.5	Alpheratz	357°34.7	29°13.8
1	59°45.5	158°38.7	07.3	296°55.2	34.8	340°24.4	21.6	74°47.9	43.5	Ankaa	353°07.0	-42°10.3
2	74°47.9	173°37.8	07.6	311°56.8	34.8	355°27.1	21.6	89°50.4	43.5	Schedar	349°30.8	56°40.6
3	89°50.4	188°37.0	· · 07.8	326°58.3	· · 34.7	10°29.8	· · 21.6	104°52.9	· · 43.5	Diphda	348°47.2	-17°51.0
4	104°52.9	203°36.2	08.1	341°59.9	34.6	25°32.5	21.6	119°55.4	43.5	Achernar	335°19.7	-57°06.7
5	119°55.3	218°35.3	08.4	357°01.5	34.5	40°35.2	21.6	134°57.9	43.5	Hamal	327°51.1	23°34.9
6	134°57.8	233°34.5	S25°08.6	12°03.0	N21°34.4	55°37.8	N22°21.5	150°00.4	S08°43.6	Polaris	313°37.6	89°22.1
7	150°00.3	248°33.7	08.9	27°04.6	34.3	70°40.5	21.5	165°02.9	43.6	Acamar	315°11.5	-40°12.2
8	165°02.7	263°32.8	09.2	42°06.2	34.2	85°43.2	21.5	180°05.5	43.6	Menkar	314°06.0	4°11.3
9	180°05.2	278°32.0	· · 09.5	57°07.7	· · 34.1	100°45.9	· · 21.5	195°08.0	· · 43.6	Mirfak	308°28.0	49°57.0
10	195°07.7	293°31.2	09.7	72°09.3	34.0	115°48.6	21.5	210°10.5	43.6	Aldebaran	290°39.5	16°33.6
11	210°10.1	308°30.3	10.0	87°10.9	33.9	130°51.3	21.5	225°13.0	43.6	Rigel	281°03.8	-8°10.2
12	225°12.6	323°29.5	S25°10.2	102°12.5	N21°33.8	145°54.0	N22°21.5	240°15.5	S08°43.6	Capella	280°21.7	46°01.3
13	240°15.0	338°28.7	10.5	117°14.0	33.7	160°56.7	21.4	255°18.0	43.6	Bellatrix	278°22.8	6°22.4
14	255°17.5	353°27.8	10.8	132°15.6	33.6	175°59.4	21.4	270°20.5	43.6	Elnath	278°01.8	28°37.7
15	270°20.0	8°27.0	· · 11.0	147°17.2	· · 33.5	191°02.1	· · 21.4	285°23.0	· · 43.7	Alnilam	275°37.6	-1°11.0
16	285°22.4	23°26.2	11.3	162°18.8	33.4	206°04.8	21.4	300°25.5	43.7	Betelgeuse	270°52.0	7°24.8
17	300°24.9	38°25.3	11.5	177°20.3	33.3	221°07.5	21.4	315°28.0	43.7	Canopus	263°52.1	-52°42.2
18	315°27.4	53°24.5	S25°11.8	192°21.9	N21°33.2	236°10.2	N22°21.4	330°30.5	S08°43.7	Sirius	258°26.2	-16°44.8
19	330°29.8	68°23.7	12.1	207°23.5	33.1	251°12.9	21.4	345°33.0	43.7	Adhara	255°05.8	-29°00.1
20	345°32.3	83°22.8	12.3	222°25.1	33.0	266°15.6	21.3	0°35.5	43.7	Procyon	244°50.9	5°09.8
21	0°34.8	98°22.0	· · 12.6	237°26.7	· · 32.9	281°18.3	· · 21.3	15°38.1	· · 43.7	Pollux	243°17.4	27°57.9
22	15°37.2	113°21.2	12.8	252°28.3	32.8	296°21.0	21.3	30°40.6	43.7	Avior	234°14.7	-59°34.9
23	30°39.7	128°20.3	13.1	267°29.8	32.7	311°23.7	21.3	45°43.1	43.7	Suhail	222°46.5	-43°31.6
Mer.pass. 20:58		ν -0.8' d 0.3' m -4.04		ν 1.6' d -0.1' m -0.00		ν 2.7' d -0.0' m -2.71		ν 2.5' d 0.0' m 0.82				

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	45°42.2	143°19.5	S25°13.3	282°31.4	N21°32.7	326°26.4	N22°21.3	60°45.6	S08°43.7
1	60°44.6	158°18.7	13.6	297°33.0	32.6	341°29.1	21.3	75°48.1	43.8
2	75°47.1	173°17.8	13.8	312°34.6	32.5	356°31.9	21.3	90°50.6	43.8
3	90°49.5	188°17.0	· · 14.0	327°36.2	· · 32.4	11°34.6	· · 21.2	105°53.1	· · 43.8
4	105°52.0	203°16.2	14.3	342°37.8	32.3	26°37.3	21.2	120°55.6	43.8
5	120°54.5	218°15.3	14.5	357°39.4	32.2	41°40.0	21.2	135°58.1	43.8
6	135°56.9	233°14.5	S25°14.8	12°40.9	N21°32.1	56°42.7	N22°21.2	151°00.6	S08°43.8
7	150°59.4	248°13.6	15.0	27°42.5	32.0	71°45.4	21.2	166°03.1	43.8
8	166°01.9	263°12.8	15.3	42°44.1	31.9	86°48.1	21.2	181°05.6	43.8
9	181°04.3	278°12.0	· · 15.5	57°45.7	· · 31.8	101°50.8	· · 21.1	196°08.1	· · 43.8
10	196°06.8	293°11.1	15.7	72°47.3	31.7	116°53.5	21.1	211°10.6	43.8
11	211°09.3	308°10.3	16.0	87°48.9	31.6	131°56.2	21.1	226°13.1	43.9
12	226°11.7	323°09.5	S25°16.2	102°50.5	N21°31.5	146°58.9	N22°21.1	241°15.6	S08°43.9
13	241°14.2	338°08.6	16.4	117°52.1	31.4	162°01.6	21.1	256°18.1	43.9
14	256°16.7	353°07.8	16.7	132°53.7	31.4	177°04.3	21.1	271°20.6	43.9
15	271°19.1	8°07.0	· · 16.9	147°55.3	· · 31.3	192°07.0	· · 21.1	286°23.1	· · 43.9
16	286°21.6	23°06.1	17.1	162°56.9	31.2	207°09.7	21.0	301°25.6	43.9
17	301°24.0	38°05.3	17.3	177°58.5	31.1	222°12.4	21.0	316°28.1	43.9
18	316°26.5	53°04.4	S25°17.6	193°00.1	N21°31.0	237°15.1	N22°21.0	331°30.6	S08°43.9
19	331°29.0	68°03.6	17.8	208°01.7	30.9	252°17.9	21.0	346°33.1	43.9
20	346°31.4	83°02.8	18.0	223°03.3	30.8	267°20.6	21.0	1°35.7	43.9
21	1°33.9	98°01.9	· · 18.2	238°04.9	· · 30.7	282°23.3	· · 21.0	16°38.2	· · 43.9
22	16°36.4	113°01.1	18.5	253°06.5	30.6	297°26.0	20.9	31°40.7	44.0
23	31°38.8	128°00.3	18.7	268°08.1	30.5	312°28.7	20.9	46°43.2	44.0
Mer.pass. 20:54		ν -0.8' d 0.2' m -4.05		ν 1.6' d -0.1' m -0.02		ν 2.7' d -0.0' m -2.72		ν 2.5' d 0.0' m 0.82	

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	46°41.3	142°59.4	S25°18.9	283°09.7	N21°30.4	327°31.4	N22°20.9	61°45.7	S08°44.0
1	61°43.8	157°58.6	19.1	298°11.3	30.4	342°34.1	20.9	76°48.2	44.0
2	76°46.2	172°57.7	19.3	313°12.9	30.3	357°36.8	20.9	91°50.7	44.0
3	91°48.7	187°56.9	· · 19.5	328°14.5	· · 30.2	12°39.5	· · 20.9	106°53.2	· · 44.0
4	106°51.2	202°56.1	19.8	343°16.1	30.1	27°42.2	20.9	121°55.7	44.0
5	121°53.6	217°55.2	20.0	358°17.7	30.0	42°45.0	20.8	136°58.2	44.0
6	136°56.1	232°54.4	S25°20.2	13°19.3	N21°29.9	57°47.7	N22°20.8	152°00.7	S08°44.0
7	151°58.5	247°53.6	20.4	28°21.0	29.8	72°50.4	20.8	167°03.2	44.0
8	167°01.0	262°52.7	20.6	43°22.6	29.7	87°53.1	20.8	182°05.7	44.0
9	182°03.5	277°51.9	· · 20.8	58°24.2	· · 29.6	102°55.8	· · 20.8	197°08.2	· · 44.0
10	197°05.9	292°51.0	21.0	73°25.8	29.6	117°58.5	20.8	212°10.7	44.1
11	212°08.4	307°50.2	21.2	88°27.4	29.5	133°01.2	20.7	227°13.2	44.1
12	227°10.9	322°49.4	S25°21.4	103°29.0	N21°29.4	148°03.9	N22°20.7	242°15.7	S08°44.1
13	242°13.3	337°48.5	21.6	118°30.6	29.3	163°06.7	20.7	257°18.2	44.1
14	257°15.8	352°47.7	21.8	133°32.3	29.2	178°09.4	20.7	272°20.7	44.1
15	272°18.3	7°46.8	· · 22.0	148°33.9	· · 29.1	193°12.1	· · 20.7	287°23.2	· · 44.1
16	287°20.7	22°46.0	22.2	163°35.5	29.0	208°14.8	20.7	302°25.7	44.1
17	302°23.2	37°45.2	22.4	178°37.1	28.9	223°17.5	20.7	317°28.2	44.1
18	317°25.7	52°44.3	S25°22.6	193°38.7	N21°28.8	238°20.2	N22°20.6	332°30.7	S08°44.1
19	332°28.1	67°43.5	22.8	208°40.4	28.8	253°22.9	20.6	347°33.2	44.1
20	347°30.6	82°42.7	23.0	223°42.0	28.7	268°25.7	20.6	2°35.7	44.1
21	2°33.0	97°41.8	· · 23.2	238°43.6	· · 28.6	283°28.4	· · 20.6	17°38.2	· · 44.1
22	17°35.5	112°41.0	23.4	253°45.2	28.5	298°31.1	20.6	32°40.7	44.1
23	32°38.0	127°40.1	23.6	268°46.9	28.4	313°33.8	20.6	47°43.2	44.1
Mer.pass. 20:50		ν -0.8' d 0.2' m -4.05		ν 1.6' d -0.1' m -0.04		ν 2.7' d -0.0' m -2.73		ν 2.5' d 0.0' m 0.83	

Nov 05 Tue		SHA	Mer.pass
Venus		98°56.5	14:26
Mars		237°10.6	05:12
Jupiter		280°38.7	02:18
Saturn		15°02.4	19:58

Nov 06 Wed		SHA	Mer.pass
Venus		97°37.3	14:28
Mars		236°49.3	05:09
Jupiter		280°44.3	02:14
Saturn		15°03.4	19:54

Nov 07 Thu		SHA	Mer.pass
Venus		96°18.1	14:29
Mars		236°28.4	05:07
Jupiter		280°50.1	02:10
Saturn		15°04.4	19:50

Horizontal parallax		
Venus:		0.1
Mars:		0.2

2024 November 05 to Nov. 07 UT

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	06:28	07:54	09:34	13:52	15:32	16:58
N 70°	06:21	07:38	08:59	14:27	15:48	17:05
68°	06:15	07:25	08:34	14:52	16:01	17:10
66°	06:11	07:14	08:15	15:11	16:12	17:15
64°	06:06	07:04	08:00	15:27	16:22	17:20
62°	06:03	06:57	07:47	15:39	16:30	17:24
60°	05:59	06:50	07:36	15:50	16:37	17:27
N 58°	05:56	06:43	07:27	16:00	16:43	17:30
56°	05:53	06:38	07:19	16:08	16:49	17:33
54°	05:50	06:33	07:11	16:15	16:54	17:36
52°	05:48	06:28	07:05	16:22	16:58	17:39
50°	05:45	06:24	06:59	16:28	17:03	17:41
45°	05:40	06:15	06:46	16:41	17:12	17:47
N 40°	05:34	06:07	06:35	16:52	17:20	17:52
35°	05:29	06:00	06:26	17:01	17:27	17:57
30°	05:25	05:53	06:18	17:09	17:34	18:02
20°	05:15	05:41	06:04	17:23	17:46	18:12
N 10°	05:05	05:30	05:52	17:35	17:57	18:22
0°	04:54	05:19	05:40	17:47	18:09	18:34
S 10°	04:41	05:06	05:28	17:59	18:21	18:47
20°	04:25	04:53	05:16	18:12	18:35	19:02
30°	04:05	04:36	05:01	18:27	18:52	19:23
35°	03:52	04:25	04:53	18:35	19:03	19:36
40°	03:37	04:13	04:43	18:45	19:15	19:51
45°	03:18	03:59	04:31	18:57	19:30	20:11
S 50°	02:52	03:40	04:17	19:11	19:48	20:37
52°	02:39	03:31	04:11	19:18	19:57	20:50
54°	02:24	03:21	04:03	19:25	20:08	21:06
56°	02:06	03:10	03:55	19:33	20:19	21:25
58°	01:42	02:56	03:46	19:43	20:33	21:49
S 60°	01:09	02:40	03:36	19:54	20:49	22:24

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	██	██	██	██	██	██
N 70°	██	██	██	██	██	██
68°	██	██	██	██	██	██
66°	██	██	██	██	██	██
64°	██	██	16:15	██	██	17:33
62°	██	██	15:09	██	██	18:39
60°	13:40	14:21	14:34	16:15	17:31	19:13
N 58°	12:58	13:44	14:09	16:58	18:07	19:38
56°	12:29	13:18	13:49	17:26	18:33	19:58
54°	12:07	12:57	13:32	17:49	18:54	20:15
52°	11:49	12:40	13:17	18:07	19:11	20:29
50°	11:33	12:25	13:04	18:22	19:26	20:41
45°	11:02	11:55	12:38	18:54	19:56	21:06
N 40°	10:38	11:31	12:17	19:18	20:19	21:26
35°	10:18	11:12	12:00	19:38	20:38	21:43
30°	10:01	10:55	11:45	19:55	20:54	21:58
20°	09:32	10:27	11:19	20:24	21:22	22:22
N 10°	09:07	10:03	10:57	20:48	21:45	22:43
0°	08:45	09:40	10:36	21:11	22:07	23:02
S 10°	08:22	09:18	10:15	21:34	22:29	23:22
20°	07:57	08:53	09:53	21:59	22:52	23:42
30°	07:29	08:25	09:27	22:27	23:19	
35°	07:12	08:08	09:11	22:44	23:35	
40°	06:52	07:49	08:54	23:04	23:54	
45°	06:29	07:25	08:32	23:28		00:16
S 50°	05:58	06:55	08:05	23:59		00:44
52°	05:43	06:39	07:51		00:14	00:58
54°	05:25	06:22	07:36		00:32	01:14
56°	05:04	06:00	07:17		00:53	01:32
58°	04:37	05:33	06:55	00:21	01:21	01:55
S 60°	03:59	04:54	06:25	00:59	02:00	02:25

Day	Sun			Moon		
	Eqn. of Time		Mer.	Mer. Pass.		Age
	00 ^h mm:ss	12 ^h mm:ss	Pass hh:mm	Upper hh:mm	Lower hh:mm	4-6 11-27%
05	16:25	16:24	11:44	14:58	02:30	
06	16:23	16:22	11:44	15:54	03:26	
07	16:20	16:18	11:44	16:49	04:22	

November 08, 09, 10 UT (Fri., Sat., Sun.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	47°40.4	142°39.3	S25°23.8	283°48.5	N21°28.3	328°36.5	N22°20.5	62°45.7	S08°44.2	Alpheratz	357°34.7	29°13.8											
1	62°42.9	157°38.5	24.0	298°50.1	28.2	343°39.2	20.5	77°48.1	44.2	Ankaa	353°07.0	-42°10.3											
2	77°45.4	172°37.6	24.1	313°51.7	28.2	358°42.0	20.5	92°50.6	44.2	Schedar	349°30.8	56°40.6											
3	92°47.8	187°36.8	•• 24.3	328°53.4	•• 28.1	13°44.7	•• 20.5	107°53.1	•• 44.2	Diphda	348°47.2	-17°51.0											
4	107°50.3	202°35.9	24.5	343°55.0	28.0	28°47.4	20.5	122°55.6	44.2	Achernar	335°19.7	-57°06.7											
5	122°52.8	217°35.1	24.7	358°56.6	27.9	43°50.1	20.5	137°58.1	44.2	Hamal	327°51.0	23°34.9											
6	137°55.2	232°34.3	S25°24.9	13°58.3	N21°27.8	58°52.8	N22°20.4	153°00.6	S08°44.2	Polaris	313°37.2	89°22.1											
7	152°57.7	247°33.4	25.0	28°59.9	27.7	73°55.6	20.4	168°03.1	44.2	Acamar	315°11.5	-40°12.2											
8	168°00.1	262°32.6	25.2	44°01.5	27.6	88°58.3	20.4	183°05.6	44.2	Menkar	314°06.0	4°11.3											
9	183°02.6	277°31.7	•• 25.4	59°03.2	•• 27.6	104°01.0	•• 20.4	198°08.1	•• 44.2	Mirfak	308°28.0	49°57.0											
10	198°05.1	292°30.9	25.6	74°04.8	27.5	119°03.7	20.4	213°10.6	44.2	Aldebaran	290°39.5	16°33.6											
11	213°07.5	307°30.1	25.8	89°06.4	27.4	134°06.4	20.4	228°13.1	44.2	Rigel	281°03.7	-8°10.2											
12	228°10.0	322°29.2	S25°25.9	104°08.1	N21°27.3	149°09.2	N22°20.3	243°15.6	S08°44.2	Capella	280°21.7	46°01.3											
13	243°12.5	337°28.4	26.1	119°09.7	27.2	164°11.9	20.3	258°18.1	44.2	Bellatrix	278°22.8	6°22.4											
14	258°14.9	352°27.5	26.3	134°11.3	27.1	179°14.6	20.3	273°20.6	44.2	Elnath	278°01.8	28°37.7											
15	273°17.4	7°26.7	•• 26.4	149°13.0	•• 27.0	194°17.3	•• 20.3	288°23.1	•• 44.2	Alnilam	275°37.6	-1°11.0											
16	288°19.9	22°25.9	26.6	164°14.6	27.0	209°20.0	20.3	303°25.6	44.2	Betelgeuse	270°52.0	7°24.8											
17	303°22.3	37°25.0	26.8	179°16.3	26.9	224°22.8	20.3	318°28.1	44.3	Canopus	263°52.1	-52°42.2											
18	318°24.8	52°24.2	S25°26.9	194°17.9	N21°26.8	239°25.5	N22°20.2	333°30.6	S08°44.3	Sirius	258°26.2	-16°44.8											
19	333°27.3	67°23.3	27.1	209°19.5	26.7	254°28.2	20.2	348°33.1	44.3	Adhara	255°05.8	-29°00.1											
20	348°29.7	82°22.5	27.3	224°21.2	26.6	269°30.9	20.2	3°35.6	44.3	Procyon	244°50.9	5°09.8											
21	3°32.2	97°21.6	•• 27.4	239°22.8	•• 26.5	284°33.6	•• 20.2	18°38.1	•• 44.3	Pollux	243°17.4	27°57.9											
22	18°34.6	112°20.8	27.6	254°24.5	26.5	299°36.4	20.2	33°40.6	44.3	Avior	234°14.7	-59°34.9											
23	33°37.1	127°20.0	27.7	269°26.1	26.4	314°39.1	20.2	48°43.1	44.3	Suhail	222°46.4	-43°31.6											
Mer.pass. 20:46				ν -0.8' d 0.2' m -4.06				ν 1.6' d -0.1' m -0.05				ν 2.7' d -0.0' m -2.73				ν 2.5' d 0.0' m 0.83							

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	48°39.6	142°19.1	S25°27.9	284°27.8	N21°26.3	329°41.8	N22°20.2	63°45.5	S08°44.3										
1	63°42.0	157°18.3	28.1	299°29.4	26.2	344°44.5	20.1	78°48.0	44.3										
2	78°44.5	172°17.4	28.2	314°31.1	26.1	359°47.3	20.1	93°50.5	44.3										
3	93°47.0	187°16.6	•• 28.4	329°32.7	•• 26.1	14°50.0	•• 20.1	108°53.0	•• 44.3										
4	108°49.4	202°15.8	28.5	344°34.4	26.0	29°52.7	20.1	123°55.5	44.3										
5	123°51.9	217°14.9	28.7	359°36.0	25.9	44°55.4	20.1	138°58.0	44.3										
6	138°54.4	232°14.1	S25°28.8	14°37.7	N21°25.8	59°58.2	N22°20.1	154°00.5	S08°44.3										
7	153°56.8	247°13.2	29.0	29°39.3	25.7	75°00.9	20.0	169°03.0	44.3										
8	168°59.3	262°12.4	29.1	44°41.0	25.6	90°03.6	20.0	184°05.5	44.3										
9	184°01.8	277°11.6	•• 29.3	59°42.6	•• 25.6	105°06.3	•• 20.0	199°08.0	•• 44.3										
10	199°04.2	292°10.7	29.4	74°44.3	25.5	120°09.1	20.0	214°10.5	44.3										
11	214°06.7	307°09.9	29.6	89°46.0	25.4	135°11.8	20.0	229°13.0	44.3										
12	229°09.1	322°09.0	S25°29.7	104°47.6	N21°25.3	150°14.5	N22°20.0	244°15.5	S08°44.3										
13	244°11.6	337°08.2	29.8	119°49.3	25.2	165°17.3	19.9	259°18.0	44.4										
14	259°14.1	352°07.4	30.0	134°50.9	25.2	180°20.0	19.9	274°20.4	44.4										
15	274°16.5	7°06.5	•• 30.1	149°52.6	•• 25.1	195°22.7	•• 19.9	289°22.9	•• 44.4										
16	289°19.0	22°05.7	30.3	164°54.3	25.0	210°25.4	19.9	304°25.4	44.4										
17	304°21.5	37°04.8	30.4	179°55.9	24.9	225°28.2	19.9	319°27.9	44.4										
18	319°23.9	52°04.0	S25°30.5	194°57.6	N21°24.8	240°30.9	N22°19.9	334°30.4	S08°44.4										
19	334°26.4	67°03.1	30.7	209°59.2	24.8	255°33.6	19.8	349°32.9	44.4										
20	349°28.9	82°02.3	30.8	225°00.9	24.7	270°36.4	19.8	4°35.4	44.4										
21	4°31.3	97°01.5	•• 30.9	240°02.6	•• 24.6	285°39.1	•• 19.8	19°37.9	•• 44.4										
22	19°33.8	112°00.6	31.1	255°04.2	24.5	300°41.8	19.8	34°40.4	44.4										
23	34°36.3	126°59.8	31.2	270°05.9	24.5	315°44.5	19.8	49°42.9	44.4										
Mer.pass. 20:42				ν -0.8' d 0.2' m -4.06				ν 1.6' d -0.1' m -0.07				ν 2.7' d -0.0' m -2.74				ν 2.5' d 0.0' m 0.84			

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	49°38.7	141°58.9	S25°31.3	285°07.6	N21°24.4	330°47.3	N22°19.7	64°45.3	S08°44.4										
1	64°41.2	156°58.1	31.4	300°09.3	24.3	345°50.0	19.7	79°47.8	44.4										
2	79°43.6	171°57.3	31.6	315°10.9	24.2	0°52.7	19.7	94°50.3	44.4										
3	94°46.1	186°56.4	•• 31.7	330°12.6	•• 24.1	15°55.5	•• 19.7	109°52.8	•• 44.4										
4	109°48.6	201°55.6	31.8	345°14.3	24.1	30°58.2	19.7	124°55.3	44.4										
5	124°51.0	216°54.7	31.9	0°15.9	24.0	46°00.9	19.7	139°57.8	44.4										
6	139°53.5	231°53.9	S25°32.1	15°17.6	N21°23.9	61°03.7	N22°19.6	155°00.3	S08°44.4										
7	154°56.0	246°53.1	32.2	30°19.3	23.8	76°06.4	19.6	170°02.8	44.4										
8	169°58.4	261°52.2	32.3	45°21.0	23.8	91°09.1	19.6	185°05.3	44.4										
9	185°00.9	276°51.4	•• 32.4	60°22.6	•• 23.7	106°11.9	•• 19.6	200°07.7	•• 44.4										
10	200°03.4	291°50.5	32.5	75°24.3	23.6	121°14.6	19.6	215°10.2	44.4										
11	215°05.8	306°49.7	32.6	90°26.0	23.5	136°17.3	19.6	230°12.7	44.4										
12	230°08.3	321°48.8	S25°32.8	105°27.7	N21°23.4	151°20.1	N22°19.5	245°15.2	S08°44.4										
13	245°10.7	336°48.0	32.9	120°29.4	23.4	166°22.8	19.5	260°17.7	44.4										
14	260°13.2	351°47.2	33.0	135°31.0	23.3	181°25.5	19.5	275°20.2	44.4										
15	275°15.7	6°46.3	•• 33.1	150°32.7	•• 23.2	196°28.3	•• 19.5	290°22.7	•• 44.4										
16	290°18.1	21°45.5	33.2	165°34.4	23.1	211°31.0	19.5	305°25.2	44.4										
17	305°20.6	36°44.6	33.3	180°36.1	23.1	226°33.7	19.5	320°27.6	44.4										
18	320°23.1	51°43.8	S25°33.4	195°37.8	N21°23.0	241°36.5	N22°19.4	335°30.1	S08°44.4										
19	335°25.5	66°43.0	33.5	210°39.5	22.9	256°39.2	19.4	350°32.6	44.4										
20	350°28.0	81°42.1	33.6	225°41.2	22.8	271°42.0	19.4	5°35.1	44.4										
21	5°30.5	96°41.3	•• 33.7	240°42.8	•• 22.8	286°44.7	•• 19.4	20°37.6	•• 44.4										
22	20°32.9	111°40.4	33.8	255°44.5	22.7	301°47.4	19.4	35°40.1	44.4										
23	35°35.4	126°39.6	33.9	270°46.2	22.6	316°50.2	19.4	50°42.6	44.4										
Mer.pass. 20:38				ν -0.8' d 0.1' m -4.07				ν 1.7' d -0.1' m -0.09				ν 2.7' d -0.0' m -2.74				ν 2.5' d 0.0' m 0.84			

Nov 08 Fri			SHA	Mer.pass
			Venus	94°58.9 14:30
			Mars	236°08.0 05:04
			Jupiter	280°56.1 02:05
			Saturn	15°05.2 19:46
			</	

2024 November 08 to Nov. 10 UT

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	06:39	08:08	09:58	13:29	15:18	16:47
N 70°	06:31	07:50	09:16	14:11	15:37	16:55
68°	06:25	07:35	08:47	14:39	15:52	17:02
66°	06:19	07:23	08:26	15:00	16:04	17:08
64°	06:14	07:13	08:09	15:17	16:14	17:13
62°	06:10	07:04	07:56	15:31	16:23	17:17
60°	06:06	06:56	07:44	15:43	16:30	17:21
N 58°	06:02	06:50	07:34	15:53	16:37	17:25
56°	05:59	06:44	07:25	16:02	16:43	17:28
54°	05:56	06:38	07:17	16:10	16:49	17:31
52°	05:53	06:33	07:10	16:17	16:54	17:34
50°	05:50	06:29	07:04	16:23	16:58	17:37
45°	05:43	06:19	06:50	16:37	17:09	17:44
N 40°	05:38	06:10	06:39	16:49	17:17	17:50
35°	05:32	06:02	06:29	16:58	17:25	17:55
30°	05:27	05:55	06:20	17:07	17:32	18:01
20°	05:16	05:43	06:06	17:22	17:45	18:11
N 10°	05:05	05:31	05:53	17:35	17:57	18:22
0°	04:54	05:19	05:40	17:47	18:09	18:34
S 10°	04:40	05:06	05:28	18:00	18:22	18:48
20°	04:24	04:51	05:15	18:13	18:37	19:04
30°	04:03	04:33	04:59	18:29	18:55	19:26
35°	03:49	04:23	04:50	18:38	19:06	19:39
40°	03:33	04:10	04:40	18:49	19:19	19:56
45°	03:13	03:54	04:27	19:01	19:34	20:16
S 50°	02:46	03:35	04:13	19:16	19:54	20:44
52°	02:32	03:25	04:06	19:23	20:04	20:58
54°	02:15	03:15	03:58	19:31	20:15	21:15
56°	01:55	03:02	03:49	19:40	20:27	21:36
58°	01:28	02:48	03:39	19:50	20:42	22:04
S 60°	00:47	02:31	03:28	20:02	21:00	22:49

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	■	17:13	15:51	■	20:22	23:30
N 70°	■	16:23	15:33	■	21:10	23:45
68°	17:22	15:51	15:18	18:21	21:40	23:58
66°	16:04	15:27	15:06	19:38	22:02	
64°	15:26	15:08	14:56	20:14	22:20	
62°	15:00	14:53	14:47	20:40	22:34	
60°	14:39	14:40	14:39	21:00	22:46	
N 58°	14:22	14:28	14:33	21:17	22:56	
56°	14:07	14:19	14:27	21:31	23:05	
54°	13:54	14:10	14:21	21:43	23:13	
52°	13:43	14:02	14:17	21:53	23:20	
50°	13:33	13:55	14:12	22:03	23:27	
45°	13:12	13:40	14:03	22:22	23:40	
N 40°	12:55	13:27	13:55	22:38	23:51	
35°	12:41	13:16	13:48	22:51		00:01
30°	12:28	13:07	13:41	23:03		00:09
20°	12:07	12:50	13:31	23:23		00:23
N 10°	11:48	12:36	13:21	23:40		00:35
0°	11:30	12:22	13:13	23:55		00:46
S 10°	11:13	12:09	13:04		00:11	00:58
20°	10:54	11:54	12:54		00:28	01:10
30°	10:32	11:37	12:43	00:06	00:47	01:23
35°	10:19	11:28	12:37	00:20	00:58	01:31
40°	10:04	11:16	12:30	00:35	01:10	01:40
45°	09:46	11:03	12:21	00:54	01:25	01:50
S 50°	09:23	10:47	12:11	01:18	01:43	02:02
52°	09:13	10:39	12:06	01:29	01:51	02:08
54°	09:01	10:30	12:01	01:42	02:00	02:14
56°	08:47	10:21	11:55	01:56	02:11	02:21
58°	08:30	10:10	11:48	02:13	02:23	02:28
S 60°	08:10	09:57	11:41	02:34	02:37	02:37

Day	Sun			Moon		Age 7-9 37-58%
	Egn. of Time 00 ^h mm:ss	12 ^h mm:ss	Mer. Pass hh:mm	Mer. Pass. Upper hh:mm	Lower hh:mm	
08	16:16	16:14	11:44	17:43	05:16	
09	16:11	16:09	11:44	18:34	06:09	
10	16:06	16:03	11:44	19:24	07:00	

November 11, 12, 13 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	50°37.9	141°38.8	S25°34.0	285°47.9	N21°22.5	331°52.9	N22°19.3	65°45.0	S08°44.4	Alpheratz	357°34.7	29°13.8
1	65°40.3	156°37.9	34.1	300°49.6	22.5	346°55.6	19.3	80°47.5	44.5	Ankaa	353°07.0	-42°10.4
2	80°42.8	171°37.1	34.2	315°51.3	22.4	1°58.4	19.3	95°50.0	44.5	Schedar	349°30.8	56°40.6
3	95°45.2	186°36.2	· · 34.3	330°53.0	· · 22.3	17°01.1	· · 19.3	110°52.5	· · 44.5	Diphda	348°47.2	-17°51.0
4	110°47.7	201°35.4	34.4	345°54.7	22.3	32°03.9	19.3	125°55.0	44.5	Achernar	335°19.7	-57°06.7
5	125°50.2	216°34.6	34.5	0°56.4	22.2	47°06.6	19.2	140°57.5	44.5	Hamal	327°51.0	23°34.9
6	140°52.6	231°33.7	S25°34.6	15°58.1	N21°22.1	62°09.3	N22°19.2	156°00.0	S08°44.5	Polaris	313°37.2	89°22.2
7	155°55.1	246°32.9	34.6	30°59.8	22.0	77°12.1	19.2	171°02.4	44.5	Acamar	315°11.4	-40°12.2
8	170°57.6	261°32.0	34.7	46°01.5	22.0	92°14.8	19.2	186°04.9	44.5	Menkar	314°06.0	4°11.3
9	186°00.0	276°31.2	· · 34.8	61°03.2	· · 21.9	107°17.5	· · 19.2	201°07.4	· · 44.5	Mirfak	308°28.0	49°57.0
10	201°02.5	291°30.3	34.9	76°04.9	21.8	122°20.3	19.2	216°09.9	44.5	Aldebaran	290°39.5	16°33.6
11	216°05.0	306°29.5	35.0	91°06.6	21.7	137°23.0	19.1	231°12.4	44.5	Rigel	281°03.7	-8°10.2
12	231°07.4	321°28.7	S25°35.1	106°08.3	N21°21.7	152°25.8	N22°19.1	246°14.9	S08°44.5	Capella	280°21.7	46°01.3
13	246°09.9	336°27.8	35.2	121°10.0	21.6	167°28.5	19.1	261°17.3	44.5	Bellatrix	278°22.8	6°22.4
14	261°12.4	351°27.0	35.2	136°11.7	21.5	182°31.3	19.1	276°19.8	44.5	Elnath	278°01.7	28°37.7
15	276°14.8	6°26.1	· · 35.3	151°13.4	· · 21.5	197°34.0	· · 19.1	291°22.3	· · 44.5	Alnilam	275°37.6	-1°11.0
16	291°17.3	21°25.3	35.4	166°15.1	21.4	212°36.7	19.0	306°24.8	44.5	Betelgeuse	270°52.0	7°24.8
17	306°19.7	36°24.5	35.5	181°16.8	21.3	227°39.5	19.0	321°27.3	44.5	Canopus	263°52.1	-52°42.2
18	321°22.2	51°23.6	S25°35.5	196°18.5	N21°21.3	242°42.2	N22°19.0	336°29.8	S08°44.5	Sirius	258°26.1	-16°44.8
19	336°24.7	66°22.8	35.6	211°20.2	21.2	257°45.0	19.0	351°32.2	44.5	Adhara	255°05.8	-29°00.1
20	351°27.1	81°21.9	35.7	226°21.9	21.1	272°47.7	19.0	6°34.7	44.5	Procyon	244°50.8	5°09.8
21	6°29.6	96°21.1	· · 35.8	241°23.7	· · 21.0	287°50.4	· · 19.0	21°37.2	· · 44.5	Pollux	243°17.3	27°57.9
22	21°32.1	111°20.3	35.8	256°25.4	21.0	302°53.2	18.9	36°39.7	44.5	Avior	234°14.6	-59°35.0
23	36°34.5	126°19.4	35.9	271°27.1	20.9	317°55.9	18.9	51°42.2	44.5	Suhail	222°46.4	-43°31.6
Mer.pass. 20:34		ν -0.8' d 0.1' m -4.07		ν 1.7' d -0.1' m -0.11		ν 2.7' d -0.0' m -2.75		ν 2.5' d 0.0' m 0.85		Miaplacidus	221°38.3	-69°48.7
										Alphard	217°47.9	-8°45.8
										Regulus	207°34.7	11°50.8
										Dubhe	193°41.4	61°36.8
										Denebola	182°25.3	14°26.0
										Gienah	175°44.0	-17°40.6
										Acrux	173°00.8	-63°13.9
										Gacrux	171°52.3	-57°14.9
										Alioth	166°13.5	55°49.4
										Spica	158°22.8	-11°17.3
										Alkaid	152°52.6	49°11.3
										Hadar	148°36.9	-60°29.4
										Menkent	147°58.2	-36°29.4
										Arcturus	145°48.4	19°03.2
										Rigil Kent.	139°41.2	-60°56.2
										Kochab	137°20.9	74°03.1
										Zuben'ubi	136°56.5	-16°08.6
										Alphecca	126°04.2	26°37.9
										Antares	112°16.4	-26°29.2
										Atria	107°11.3	-69°04.4
										Sabik	102°03.3	-15°45.3
										Shaula	96°10.9	-37°07.3
										Rasalhague	95°59.0	12°32.6
										Eltanin	90°42.7	51°29.3
										Kaus Aust.	83°33.0	-34°22.4
										Vega	80°33.6	38°48.6
										Nunki	75°48.2	-26°16.0
										Altair	62°00.3	8°56.1
										Peacock	53°06.1	-56°39.5
										Deneb	49°26.0	45°22.4
										Enif	33°38.9	9°59.4
										Al Na'ir	27°33.0	-46°50.6
										Fomalhaut	15°14.5	-29°29.5
										Scheat	13°45.2	28°13.3
										Markab	13°29.9	15°20.5
										Nov 11 Mon SHA Mer.pass		
										Venus	91°00.9	14:34
										Mars	235°10.1	04:56
										Jupiter	281°15.0	01:52
										Saturn	15°07.2	19:34
										Nov 12 Tue SHA Mer.pass		
										Venus	89°41.6	14:36
										Mars	234°51.8	04:54
										Jupiter	281°21.7	01:48
										Saturn	15°07.7	19:30
										Nov 13 Wed SHA Mer.pass		
										Venus	88°22.3	14:37
										Mars	234°34.1	04:51
										Jupiter	281°28.5	01:43
										Saturn	15°08.0	19:26
										Horizontal parallax Venus: 0.1		
										Mars:		0.2

Sun							Moon								
Mon	GHA	Dec	GHA	ν	Dec	d	HP	Mon	GHA	ν	Dec	d	HP		
0	183°59.9	S17°29.6	66°40.1	11.4'	S09°07.2	-16.3'	59.4'	0	183°59.9	S17°29.6	66°40.1	11.4'	S09°07.2	-16.3'	59.4'
1	198°59.8	30.3	81°10.5	11.4'	08°50.8	-16.4'	59.4'	1	198°59.8	30.3	81°10.5	11.4'	08°50.8	-16.4'	59.4'
2	213°59.7	31.0	95°40.9	11.4'	08°34.5	-16.4'	59.4'	2	213°59.7	31.0	95°40.9	11.4'	08°34.5	-16.4'	59.4'
3	228°59.6	· · 31.7	110°11.3	11.4'	08°18.0	-16.5'	59.5'	3	228°59.6	· · 31.7	110°11.3	11.4'	08°18.0	-16.5'	59.5'
4	243°59.6	32.3	124°41.8	11.4'	08°01.5	-16.6'	59.5'	4	243°59.6	32.3	124°41.8	11.4'	08°01.5	-16.6'	59.5'
5	258°59.5	33.0	139°12.2	11.4'	07°45.0	-16.6'	59.5'	5	258°59.5	33.0	139°12.2	11.4'	07°45.0	-16.6'	59.5'
6	273°59.4	S17°33.7	153°42.6	11.5'	S07°28.3	-16.7'	59.5'	6	273°59.4	S17°33.7	153°42.6	11.5'	S07°28.3	-16.7'	59.5'
7	288°59.3	34.4	168°13.1	11.5'	07°11.7	-16.7'	59.6'	7	288°59.3	34.4	168°13.1	11.5'	07°11.7	-16.7'	59.6'
8	303°59.3	35.1	182°43.6	11.5'	06°54.9	-16.8'	59.6'	8	303°59.3	35.1	182°43.6	11.5'	06°54.9	-16.8'	59.6'
9	318°59.2	· · 35.8	197°14.0	11.5'	06°38.2	-16.8'	59.6'	9	318°59.2	· · 35.8	197°14.0	11.5'	06°38.2	-16.8'	59.6'
10	333°59.1	36.4	211°44.5	11.5'	06°21.3	-16.9'	59.7'	10	333°59.1	36.4	211°44.5	11.5'	06°21.3	-16.9'	59.7'
11	348°59.1	37.1	226°15.0	11.5'	06°04.4	-16.9'	59.7'	11	348°59.1	37.1	226°15.0	11.5'	06°04.4	-16.9'	59.7'
12	3°59.0	S17°37.8	240°45.5	11.5'	S05°47.5	-17.0'	59.7'	12	3°59.0	S17°37.8	240°45.5	11.5'	S05°47.5	-17.0'	59.7'
13	18°58.9	38.5	255°15.9	11.5'	05°30.5	-17.0'	59.7'	13	18°58.9	38.5	255°15.9	11.5'	05°30.5	-17.0'	59.7'
14	33°58.8	39.2	269°46.4	11.5'	05°13.5	-17.1'	59.8'	14	33°58.8	39.2	269°46.4	11.5'	05°13.5	-17.1'	59.8'
15	48°58.8	· · 39.8	284°16.9	11.5'	04°56.5	-17.1'	59.8'	15	48°58.8	· · 39.8	284°16.9	11.5'	04°56.5	-17.1'	59.8'
16	63°58.7	40.5	298°47.4	11.5'	04°39.4	-17.1'	59.8'	16	63°58.7	40.5	298°47.4	11.5'	04°39.4	-17.1'	59.8'
17	78°58.6	41.2	313°17.8	11.5'	04°22.2	-17.2'	59.8'	17	78°58.6	41.2	313°17.8	11.5'	04°22.2	-17.2'	59.8'
18	93°58.5	S17°41.9	327°48.3	11.5'	S04°05.0	-17.2'	59.9'	18	93°58.5	S17°41.9	327°48.3	11.5'	S04°05.0	-17.2'	59.9'
19	108°58.4	42.5	342°18.8	11.4'	03°47.8	-17.3'	59.9'	19	108°58.4	42.5	342°18.8	11.4'	03°47.8	-17.3'	59.9'
20	123°58.4	43.2	356°49.2	11.4'	03°30.6	-17.3'	59.9'	20	123°58.4	43.2	356°49.2	11.4'	03°30.6	-17.3'	59.9'
21	138°58.3	· · 43.9	11°19.6	11.4'	03°13.3	-17.3'	59.9'	21	138°58.3	· · 43.9	11°19.6	11.4'	03°13.3	-17.3'	59.9'
22	153°58.2	44.6	25°50.1	11.4'	02°56.0	-17.3'	60.0'	22	153°58.2	44.6	25°50.1	11.4'	02°56.0	-17.3'	60.0'
23	168°58.1	45.2	40°20.5	11.4'	02°38.6	-17.4'	60.0'	23	168°58.1	45.2	40°20.5	11.4'	02°38.6	-17.4'	60.0'
SD = 16.1' d = 0.7'							SD = 16.2'								
Tue	GHA	Dec	GHA	ν	Dec	d	HP	Tue	GHA	ν	Dec	d	HP		
0	183°58.1	S17°45.9	54°50.9	11.4'	S02°21.2	-17.4'	60.0'	0	183°58.1	S17°45.9	54°50.9	11.4'	S02°21.2	-17.4'	60.0'
1	198°58.0	46.6	69°21.3	11.4'	02°03.8	-17.4'	60.0'	1	198°58.0	46.6	69°21.3	11.4'	02°03.8	-17.4'	60.0'
2	213°57.9	47.3	83°51.7	11.4'	01°46.4	-17.5'	60.1'	2	213°57.9	47.3	83°51.7	11.4'	01°46.4	-17.5'	60.1'
3	228°57.8	· · 47.9	98°22.0	11.3'	01°28.9	-17.5'	60.1'	3	228°57.8	· · 47.9	98°22.0	11.3'	01°28.9	-17.5'	60.1'
4	243°57.7	48.6	112°52.3	11.3'	01°11.5	-17.5'	60.1'	4	243°57.7	48.6	112°52.3	11.3'	01°11.5	-17.5'	60.1'
5	258°57.7	49.3	127°22.7	11.3'	00°54.0	-17.5'	60.1'	5	258°57.7	49.3	127°22.7	11.3'	00°54.0	-17.5'	60.1'
6	273°57.6	S17°50.0	141°53.0	11.3'	S00°36.4	-17.5'	60.2'	6	273°57.6	S17°50.0	141°53.0	11.3'	S00°36.4	-17.5'	60.2'
7	288°57.5	50.6	156°23.2	11.2'	00°18.9	-17.6'	60.2'	7	288°57.5	50.6	156°23.2	11.2'	00°18.9	-17.6'	60.2'
8	303°57.4	51.3	170°53.5	11.2'	S00°01.4	-17.6'	60.2'	8	303°57.4	51.3	170°53.5	11.2'	S00°01.4	-17.6'	60.2'
9	318°57.3	· · 52.0	185°23.7	11.2'	N00°16.2	17.6'	60.2'	9	318°57.3	· · 52.0	185°23.7	11.2'	N00°16.2	17.6'	60.2'
10	333°57.2	52.6	199°53.9	11.2'	00°33.8	17.6'	60.3'	10	333°57.2	52.6	199°53.9	11.2'	00°33.8	17.6'	60.3'
11	348°57.2	53.3	214°24.1	11.1'	00°51.4	17.6'	60.3'	11	348°57.2	53.3	214°24.1	11.1'	00°51.4	17.6'	60.3'
12	3°57.1	S17°54.0	228°54.2	11.1'	N01°09.0	17.6'	60.3'	12	3°57.1	S17°54.0	228°54.2	11.1'	N01°09.0	17.6'	60.3'
13	18°57.0	54.6	243°24.3	11.1'	01°26.6	17.6'	60.3'	13	18°57.0	54.6	243°24.3	11.1'	01°26.6	17.6'	60.3'
14	33°56.9	55.3	257°54.4	11.0'	01°44.2	17.6'	60.3'	14	33°56.9	55.3	257°54.4	11.0'	01°44.2	17.6'	60.3'
15	48°56.8	· · 56.0	272°24.4	11.0'	02°01.8	17.6'	60.4'	15	48°56.8	· · 56.0	272°24.4	11.0'	02°01.8	17.6'	60.4'
16	63°56.7	56.6	286°54.4	11.0'	02°19.4	17.6'	60.4'	16	63°56.7	56.6	286°54.4	11.0'	02°19.4	17.6'	60.4'
17	78°56.7	57.3	301°24.4	10.9'	02°37.0	17.6'	60.4'	17	78°56.7	57.3	301°24.4	10.9'	02°37.0	17.6'	60.4'
18	93°56.6	S17°58.0	315°54.3	10.9'	N02°54.6	17.6'	60.4'	18	93°56.6	S17°58.0	315°54.3	10.9'	N02°54.6	17.6'	60.4'
19	108°56.5	58.6	330°24.2	10.8'	03°12.2	17.6'	60.4'	19	108°56.5	58.6	330°24.2	10.8'	03°12.2	17.6'	60.4'
20	123°56.4	59.3	344°54.0	10.8'	03°29.8	17.6'	60.4'	20	123°56.4	59.3	344°54.0	10.8'	03°29.8	17.6'	60.4'
21	138°56.3	17°59.9	359°23.9	10.8'	03°47.4	17.6'	60.5'	21	138°56.3	17°59.9	359°23.9	10.8'	03°47.4	17.6'	60.5'
22	153°56.2	18°00.6	13°53.6	10.7'	04°05.0	17.6'	60.5'	22	153°56.2	18°00.6	13°53.6	10.7'	04°05.0	17.6'	60.5'
23	168°56.1	01.3	28°23.3	10.7'	04°22.6	17.6'	60.5'	23	168°56.1	01.3	28°23.3	10.7'	04°22.6	17.6'	60.5'
SD = 16.2' d = 0.7'							SD = 16.4'								
Wed	GHA	Dec	GHA	ν	Dec	d	HP	Wed	GHA	ν	Dec	d	HP		
0	183°56.0	S18°01.9	42°53.0	10.6'	N04°40.2	17.5'	60.5'	0	183°56.0	S18°01.9	42°53.0	10.6'	N04°40.2	17.5'	60.5'
1	198°56.0	02.6	57°22.6	10.6'	04°57.7	17.5'	60.5'	1	198°56.0	02.6	57°22.6	10.6'	04°57.7	17.5'	60.5'
2	213°55.9	03.3	71°52.2	10.5'	05°15.2	17.5'	60.5'	2	213°55.9	03.3	71°52.2	10.5'	05°15.2	17.5'	60.5'
3	228°55.8	· · 03.9	86°21.7	10.5'	05°32.7	17.5'	60.6'	3	228°55.8	· · 03.9	86°21.7	10.5'	05°32.7	17.5'	60.6'
4	243°55.7	04.6	100°51.2	10.4'	05°50.2	17.5'	60.6'	4	243°55.7	04.6	100°51.2	10.4'	05°50.2	17.5'	60.6'
5	258°55.6	05.2	115°20.6	10.4'	06°07.7	17.4'	60.6'	5	258°55.6	05.2	115°20.6	10.4'	06°07.7	17.4'	60.6'
6	273°55.5	S18°05.9	129°50.0	10.3'	N06°25.1	17.4'	60.6'	6	273°55.5	S18°05.9	129°50.0	10.3'	N06°25.1	17.4'	60.6'
7	288°55.4	06.5	144°19.3	10.3'	06°42.5	17.4'	60.6'	7	288°55.4	06.5	144°19.3	10.3'	06°42.5	17.4'	60.6'
8	303°55.3	07.2	158°48.6	10.2'	06°59.9	17.4'	60.6'	8	303°55.3	07.2	158°48.6	10.2'	06°59.9	17.4'	60.6'
9	318°55.2	· · 07.9	173°17.8	10.1'	07°17.3	17.3'	60.6'	9	318°55.2	· · 07.9	173°17.8	10.1'	07°17.3	17.3'	60.6'
10	333°55.1	08.5	187°46.9	10.1'	07°34.6	17.3'	<								

November 14, 15, 16 UT (Thu., Fri., Sat.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	53°35.3	140°38.4	S25°37.7	287°52.2	N21°17.7	335°10.7	N22°18.0	68°43.6	S08°44.4	Alpheratz	357°34.7	29°13.8											
1	68°37.7	155°37.5	37.7	302°53.9	17.7	350°13.4	18.0	83°46.0	44.4	Ankaa	353°07.0	-42°10.4											
2	83°40.2	170°36.7	37.7	317°55.7	17.6	5°16.2	18.0	98°48.5	44.4	Schedar	349°30.8	56°40.7											
3	98°42.7	185°35.9	•• 37.7	332°57.5	•• 17.6	20°19.0	•• 17.9	113°51.0	•• 44.4	Diphda	348°47.2	-17°51.0											
4	113°45.1	200°35.0	37.7	347°59.2	17.5	35°21.7	17.9	128°53.5	44.4	Achernar	335°19.7	-57°06.7											
5	128°47.6	215°34.2	37.7	3°01.0	17.4	50°24.5	17.9	143°55.9	44.4	Hamal	327°51.0	23°34.9											
6	143°50.1	230°33.4	S25°37.7	18°02.7	N21°17.4	65°27.2	N22°17.9	158°58.4	S08°44.4	Polaris	313°37.0	89°22.2											
7	158°52.5	245°32.5	37.7	33°04.5	17.3	80°30.0	17.9	174°00.9	44.3	Acamar	315°11.4	-40°12.3											
8	173°55.0	260°31.7	37.7	48°06.3	17.3	95°32.7	17.9	189°03.4	44.3	Menkar	314°06.0	4°11.3											
9	188°57.4	275°30.9	•• 37.7	63°08.1	•• 17.2	110°35.5	•• 17.8	204°05.8	•• 44.3	Mirfak	308°28.0	49°57.0											
10	203°59.9	290°30.0	37.6	78°09.8	17.2	125°38.3	17.8	219°08.3	44.3	Aldebaran	290°39.5	16°33.6											
11	219°02.4	305°29.2	37.6	93°11.6	17.1	140°41.0	17.8	234°10.8	44.3	Rigel	281°03.7	-8°10.2											
12	234°04.8	320°28.4	S25°37.6	108°13.4	N21°17.0	155°43.8	N22°17.8	249°13.2	S08°44.3	Capella	280°21.7	46°01.3											
13	249°07.3	335°27.5	37.6	123°15.1	17.0	170°46.5	17.8	264°15.7	44.3	Bellatrix	278°22.7	6°22.4											
14	264°09.8	350°26.7	37.6	138°16.9	16.9	185°49.3	17.7	279°18.2	44.3	Elnath	278°01.7	28°37.7											
15	279°12.2	5°25.9	•• 37.6	153°18.7	•• 16.9	200°52.1	•• 17.7	294°20.7	•• 44.3	Alnilam	275°37.6	-1°11.0											
16	294°14.7	20°25.0	37.6	168°20.5	16.8	215°54.8	17.7	309°23.1	44.3	Betelgeuse	270°52.0	7°24.8											
17	309°17.2	35°24.2	37.6	183°22.2	16.8	230°57.6	17.7	324°25.6	44.3	Canopus	263°52.0	-52°42.2											
18	324°19.6	50°23.4	S25°37.5	198°24.0	N21°16.7	246°00.3	N22°17.7	339°28.1	S08°44.3	Sirius	258°26.1	-16°44.8											
19	339°22.1	65°22.5	37.5	213°25.8	16.7	261°03.1	17.6	354°30.5	44.3	Adhara	255°05.7	-29°00.1											
20	354°24.5	80°21.7	37.5	228°27.6	16.6	276°05.9	17.6	9°33.0	44.3	Procyon	244°50.8	5°09.8											
21	9°27.0	95°20.9	•• 37.5	243°29.4	•• 16.5	291°08.6	•• 17.6	24°35.5	•• 44.3	Pollux	243°17.3	27°57.9											
22	24°29.5	110°20.0	37.5	258°31.1	16.5	306°11.4	17.6	39°37.9	44.3	Avior	234°14.6	-59°35.0											
23	39°31.9	125°19.2	37.4	273°32.9	16.4	321°14.1	17.6	54°40.4	44.3	Suhail	222°46.4	-43°31.6											
Mer.pass. 20:22				ν -0.8' d 0.0' m -4.09				ν 1.8' d -0.1' m -0.16				ν 2.8' d -0.0' m -2.76				ν 2.5' d -0.0' m 0.86							

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	54°34.4	140°18.4	S25°37.4	288°34.7	N21°16.4	336°16.9	N22°17.5	69°42.9	S08°44.3	Denebola	182°25.3	14°26.0							
1	69°36.9	155°17.5	37.4	303°36.5	16.3	351°19.7	17.5	84°45.4	44.3	Gienah	175°44.0	-17°40.6							
2	84°39.3	170°16.7	37.4	318°38.3	16.3	6°22.4	17.5	99°47.8	44.3	Acrux	173°00.8	-63°13.9							
3	99°41.8	185°15.9	•• 37.3	333°40.1	•• 16.2	21°25.2	•• 17.5	114°50.3	•• 44.3	Gacrux	171°52.3	-57°14.9							
4	114°44.3	200°15.0	37.3	348°41.8	16.2	36°28.0	17.5	129°52.8	44.2	Alioth	166°13.5	55°49.4							
5	129°46.7	215°14.2	37.3	3°43.6	16.1	51°30.7	17.4	144°55.2	44.2	Spica	158°22.8	-11°17.3							
6	144°49.2	230°13.4	S25°37.2	18°45.4	N21°16.1	66°33.5	N22°17.4	159°57.7	S08°44.2	Alkaid	152°52.6	49°11.3							
7	159°51.7	245°12.5	37.2	33°47.2	16.0	81°36.2	17.4	175°00.2	44.2	Hadar	148°36.9	-60°29.4							
8	174°54.1	260°11.7	37.2	48°49.0	15.9	96°39.0	17.4	190°02.6	44.2	Menkent	147°58.2	-36°29.4							
9	189°56.6	275°10.9	•• 37.1	63°50.8	•• 15.9	111°41.8	•• 17.4	205°05.1	•• 44.2	Arcturus	145°48.4	19°03.2							
10	204°59.0	290°10.1	37.1	78°52.6	15.8	126°44.5	17.3	220°07.6	44.2	Rigil Kent.	139°41.2	-60°56.1							
11	220°01.5	305°09.2	37.1	93°54.4	15.8	141°47.3	17.3	235°10.0	44.2	Kochab	137°20.9	74°03.1							
12	235°04.0	320°08.4	S25°37.0	108°56.2	N21°15.7	156°50.1	N22°17.3	250°12.5	S08°44.2	Zuben'ubi	136°56.5	-16°08.6							
13	250°06.4	335°07.6	37.0	123°58.0	15.7	171°52.8	17.3	265°15.0	44.2	Alphecca	126°04.2	26°37.9							
14	265°08.9	350°06.7	36.9	138°59.8	15.6	186°55.6	17.3	280°17.4	44.2	Antares	112°16.4	-26°29.2							
15	280°11.4	5°05.9	•• 36.9	154°01.6	•• 15.6	201°58.4	•• 17.2	295°19.9	•• 44.2	Atria	107°11.3	-69°04.4							
16	295°13.8	20°05.1	36.8	169°03.4	15.5	217°01.1	17.2	310°22.4	44.2	Sabik	102°03.3	-15°45.3							
17	310°16.3	35°04.2	36.8	184°05.2	15.5	232°03.9	17.2	325°24.8	44.2	Shaula	96°10.9	-37°07.3							
18	325°18.8	50°03.4	S25°36.7	199°07.0	N21°15.4	247°06.7	N22°17.2	340°27.3	S08°44.2	Rasalhague	95°59.0	12°32.6							
19	340°21.2	65°02.6	36.7	214°08.8	15.4	262°09.4	17.2	355°29.8	44.2	Eltanin	90°42.7	51°29.3							
20	355°23.7	80°01.8	36.6	229°10.6	15.3	277°12.2	17.1	10°32.2	44.1	Kaus Aust.	83°33.0	-34°22.4							
21	10°26.2	95°00.9	•• 36.6	244°12.4	•• 15.3	292°15.0	•• 17.1	25°34.7	•• 44.1	Vega	80°33.6	38°48.5							
22	25°28.6	110°00.1	36.5	259°14.2	15.2	307°17.7	17.1	40°37.2	44.1	Nunki	75°48.2	-26°16.0							
23	40°31.1	124°59.3	36.5	274°16.0	15.2	322°20.5	17.1	55°39.6	44.1	Altair	62°00.3	8°56.1							
Mer.pass. 20:18				ν -0.8' d -0.0' m -4.09				ν 1.8' d -0.1' m -0.18				ν 2.8' d -0.0' m -2.77				ν 2.5' d -0.0' m 0.87			

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	55°33.5	139°58.4	S25°36.4	289°17.8	N21°15.1	337°23.3	N22°17.1	70°42.1	S08°44.1	Fomalhaut	15°14.6	-29°29.5					
1	70°36.0	154°57.6	36.4	304°19.6	15.1	352°26.0	17.0	85°44.6	44.1	Scheat	13°45.2	28°13.3					
2	85°38.5	169°56.8	36.3	319°21.4	15.0	7°28.8	17.0	100°47.0	44.1	Markab	13°29.9	15°20.5					
3	100°40.9	184°56.0	•• 36.3	334°23.2	•• 15.0	22°31.6	•• 17.0	115°49.5	•• 44.1								
4	115°43.4	199°55.1	36.2	349°25.0	14.9	37°34.3	17.0	130°52.0	44.1								
5	130°45.9	214°54.3	36.1	4°26.9	14.9	52°37.1	17.0	145°54.4	44.1								
6	145°48.3	229°53.5	S25°36.1	19°28.7	N21°14.9	67°39.9	N22°16.9	160°56.9	S08°44.1								
7	160°50.8	244°52.7	36.0	34°30.5	14.8	82°42.6	16.9	175°59.3	44.1								
8	175°53.3	259°51.8	35.9	49°32.3	14.8	97°45.4	16.9	191°01.8	44.1								
9	190°55.7	274°51.0	•• 35.9	64°34.1	•• 14.7	112°48.2	•• 16.9	206°04.3	•• 44.1								
10	205°58.2	289°50.2	35.8	79°35.9	14.7	127°50.9	16.9	221°06.7	44.0								
11	221°00.7	304°49.4	35.7	94°37.7	14.6	142°53.7	16.8	236°09.2	44.0								
12	236°03.1	319°48.5	S25°35.7	109°39.6	N21°14.6	157°56.5	N22°16.8	251°11.7	S08°44.0								
13	251°05.6	334°47.7	35.6	124°41.4	14.5	172°59.3	16.8	266°14.1	44.0								
14	266°08.0	349°46.9	35.5	139°43.2	14.5	188°02.0	16.8	281°16.6	44.0								
15	281°10.5	4°46.1	•• 35.4	154°45.0	•• 14.4	203°04.8	•• 16.8	296°19.1	•• 44.0								
16	296°13.0	19°45.2	35.4	169°46.9	14.4	218°07.6	16.7	311°21.5	44.0								
17	311°15.4	34°44.4	35.3	184°48.7	14.3	233°10.3	16.7	326°24.0	44.0								
18	326°17.9	49°43.6	S25°35.2	199°50.5	N21°14.3												

h	Sun				Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP	
0	183°53.8	S18°17.6	30°28.0	9.1'	N11°32.6	16.6'	60.8'	
1	198°53.7	18.3	44°56.1	9.0'	11°49.2	16.5'	60.8'	
2	213°53.6	18.9	59°24.1	9.0'	12°05.7	16.5'	60.8'	
3	228°53.5	· · 19.6	73°52.1	8.9'	12°22.2	16.4'	60.8'	
4	243°53.4	20.2	88°20.0	8.8'	12°38.5	16.3'	60.8'	
5	258°53.3	20.9	102°47.8	8.7'	12°54.9	16.2'	60.8'	
6	273°53.2	S18°21.5	117°15.5	8.6'	N13°11.1	16.2'	60.8'	
7	288°53.1	22.1	131°43.1	8.6'	13°27.3	16.1'	60.8'	
8	303°53.0	22.8	146°10.7	8.5'	13°43.4	16.0'	60.8'	
9	318°52.9	· · 23.4	160°38.2	8.4'	13°59.4	15.9'	60.8'	
10	333°52.8	24.1	175°05.5	8.3'	14°15.3	15.8'	60.8'	
11	348°52.7	24.7	189°32.8	8.2'	14°31.1	15.8'	60.8'	
12	3°52.6	S18°25.4	204°00.1	8.1'	N14°46.9	15.7'	60.8'	
13	18°52.5	26.0	218°27.2	8.0'	15°02.5	15.6'	60.8'	
14	33°52.4	26.6	232°54.2	8.0'	15°18.1	15.5'	60.8'	
15	48°52.3	· · 27.3	247°21.2	7.9'	15°33.6	15.4'	60.8'	
16	63°52.2	27.9	261°48.0	7.8'	15°49.0	15.3'	60.8'	
17	78°52.1	28.6	276°14.8	7.7'	16°04.3	15.2'	60.8'	
18	93°52.0	S18°29.2	290°41.5	7.6'	N16°19.4	15.1'	60.8'	
19	108°51.9	29.8	305°08.1	7.5'	16°34.5	15.0'	60.8'	
20	123°51.8	30.5	319°34.6	7.4'	16°49.5	14.9'	60.8'	
21	138°51.7	· · 31.1	334°01.0	7.3'	17°04.4	14.8'	60.8'	
22	153°51.6	31.7	348°27.4	7.2'	17°19.1	14.7'	60.8'	
23	168°51.5	32.4	2°53.6	7.1'	17°33.8	14.5'	60.8'	
SD = 16.2'		d = 0.6'	SD = 16.6'					
Fri	GHA	Dec	GHA	ν	Dec	d	HP	
0	183°51.4	S18°33.0	17°19.7	7.1'	N17°48.3	14.4'	60.8'	
1	198°51.3	33.6	31°45.8	7.0'	18°02.8	14.3'	60.8'	
2	213°51.2	34.3	46°11.7	6.9'	18°17.1	14.2'	60.8'	
3	228°51.1	· · 34.9	60°37.6	6.8'	18°31.2	14.1'	60.7'	
4	243°51.0	35.5	75°03.4	6.7'	18°45.3	13.9'	60.7'	
5	258°50.9	36.2	89°29.1	6.6'	18°59.3	13.8'	60.7'	
6	273°50.8	S18°36.8	103°54.7	6.5'	N19°13.1	13.7'	60.7'	
7	288°50.7	37.4	118°20.1	6.4'	19°26.8	13.6'	60.7'	
8	303°50.6	38.1	132°45.6	6.3'	19°40.3	13.4'	60.7'	
9	318°50.4	· · 38.7	147°10.9	6.2'	19°53.8	13.3'	60.7'	
10	333°50.3	39.3	161°36.1	6.1'	20°07.0	13.2'	60.7'	
11	348°50.2	39.9	176°01.2	6.0'	20°20.2	13.0'	60.7'	
12	3°50.1	S18°40.6	190°26.3	5.9'	N20°33.2	12.9'	60.7'	
13	18°50.0	41.2	204°51.2	5.9'	20°46.1	12.7'	60.6'	
14	33°49.9	41.8	219°16.1	5.8'	20°58.9	12.6'	60.6'	
15	48°49.8	· · 42.4	233°40.8	5.7'	21°11.4	12.5'	60.6'	
16	63°49.7	43.1	248°05.5	5.6'	21°23.9	12.3'	60.6'	
17	78°49.6	43.7	262°30.1	5.5'	21°36.2	12.2'	60.6'	
18	93°49.5	S18°44.3	276°54.6	5.4'	N21°48.4	12.0'	60.6'	
19	108°49.3	44.9	291°19.0	5.3'	22°00.4	11.8'	60.6'	
20	123°49.2	45.6	305°43.3	5.2'	22°12.2	11.7'	60.5'	
21	138°49.1	· · 46.2	320°07.6	5.2'	22°23.9	11.5'	60.5'	
22	153°49.0	46.8	334°31.7	5.1'	22°35.4	11.4'	60.5'	
23	168°48.9	47.4	348°55.8	5.0'	22°46.8	11.2'	60.5'	
SD = 16.2'		d = 0.6'	SD = 16.6'					
Sat	GHA	Dec	GHA	ν	Dec	d	HP	
0	183°48.8	S18°48.0	3°19.8	4.9'	N22°58.0	11.1'	60.5'	
1	198°48.7	48.7	17°43.7	4.8'	23°09.1	10.9'	60.5'	
2	213°48.5	49.3	32°07.5	4.7'	23°20.0	10.7'	60.4'	
3	228°48.4	· · 49.9	46°31.2	4.7'	23°30.7	10.6'	60.4'	
4	243°48.3	50.5	60°54.9	4.6'	23°41.3	10.4'	60.4'	
5	258°48.2	51.1	75°18.5	4.5'	23°51.6	10.2'	60.4'	
6	273°48.1	S18°51.8	89°42.0	4.4'	N24°01.9	10.0'	60.4'	
7	288°48.0	52.4	104°05.4	4.4'	24°11.9	9.9'	60.3'	
8	303°47.8	53.0	118°28.8	4.3'	24°21.8	9.7'	60.3'	
9	318°47.7	· · 53.6	132°52.1	4.2'	24°31.5	9.5'	60.3'	
10	333°47.6	54.2	147°15.3	4.1'	24°41.0	9.3'	60.3'	
11	348°47.5	54.8	161°38.4	4.1'	24°50.4	9.2'	60.2'	
12	3°47.4	S18°55.4	176°01.5	4.0'	N24°59.5	9.0'	60.2'	
13	18°47.3	56.1	190°24.5	4.0'	25°08.5	8.8'	60.2'	
14	33°47.1	56.7	204°47.5	3.9'	25°17.3	8.6'	60.2'	
15	48°47.0	· · 57.3	219°10.4	3.8'	25°25.9	8.4'	60.1'	
16	63°46.9	57.9	233°33.2	3.8'	25°34.4	8.2'	60.1'	
17	78°46.8	58.5	247°56.0	3.7'	25°42.6	8.1'	60.1'	
18	93°46.7	S18°59.1	262°18.7	3.7'	N25°50.7	7.9'	60.1'	
19	108°46.5	18°59.7	276°41.4	3.6'	25°58.5	7.7'	60.0'	
20	123°46.4	19°00.3	291°04.0	3.6'	26°06.2	7.5'	60.0'	
21	138°46.3	· · 00.9	305°26.5	3.5'	26°13.7	7.3'	60.0'	
22	153°46.2	01.5	319°49.1	3.5'	26°21.0	7.1'	60.0'	
23	168°46.1	02.2	334°11.5	3.4'	26°28.1	6.9'	59.9'	
SD = 16.2'		d = 0.6'	SD = 16.5'					

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:01	08:36	11:13	12:16	14:52	16:27
N 70°	06:51	08:13	09:53	13:36	15:15	16:37
68°	06:42	07:55	09:15	14:13	15:33	16:46
66°	06:35	07:41	08:49	14:40	15:47	16:53
64°	06:29	07:29	08:29	15:00	15:59	17:00
62°	06:23	07:19	08:13	15:16	16:10	17:06
60°	06:18	07:10	07:59	15:30	16:19	17:11
N 58°	06:13	07:02	07:47	15:41	16:27	17:15
56°	06:09	06:55	07:37	15:51	16:34	17:19
54°	06:05	06:49	07:29	16:00	16:40	17:23
52°	06:02	06:43	07:21	16:08	16:46	17:27
50°	05:58	06:38	07:13	16:15	16:51	17:30
45°	05:51	06:26	06:58	16:31	17:03	17:38
N 40°	05:44	06:17	06:46	16:43	17:12	17:45
35°	05:37	06:08	06:35	16:54	17:21	17:52
30°	05:31	06:00	06:25	17:04	17:29	17:58
20°	05:19	05:46	06:09	17:20	17:43	18:10
N 10°	05:07	05:33	05:55	17:35	17:57	18:22
0°	04:54	05:19	05:41	17:48	18:10	18:35
S 10°	04:39	05:05	05:27	18:02	18:24	18:50
20°	04:21	04:49	05:13	18:17	18:40	19:08
30°	03:58	04:30	04:56	18:34	19:00	19:32
35°	03:44	04:18	04:46	18:44	19:12	19:46
40°	03:26	04:04	04:34	18:56	19:26	20:04
45°	03:04	03:47	04:21	19:09	19:44	20:27
S 50°	02:33	03:25	04:04	19:26	20:06	20:58
52°	02:17	03:14	03:56	19:34	20:16	21:14
54°	01:58	03:02	03:48	19:43	20:29	21:34
56°	01:33	02:48	03:38	19:53	20:43	22:00
58°	00:57	02:32	03:26	20:04	21:00	22:39
S 60°	—:—	02:11	03:13	20:18	21:21	—:—

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	12:48	☐	☐	07:22	☐	☐
N 70°	13:18	☐	☐	06:54	☐	☐
68°	13:41	12:48	☐	06:34	09:25	☐
66°	13:59	13:29	☐	06:17	08:44	☐
64°	14:14	13:58	13:26	06:04	08:17	10:54
62°	14:26	14:20	14:13	05:53	07:56	10:08
60°	14:37	14:38	14:44	05:44	07:39	09:38
N 58°	14:46	14:54	15:07	05:36	07:24	09:15
56°	14:54	15:07	15:26	05:29	07:12	08:57
54°	15:02	15:18	15:42	05:23	07:02	08:41
52°	15:08	15:28	15:56	05:17	06:52	08:28
50°	15:14	15:37	16:08	05:12	06:44	08:16
45°	15:27	15:56	16:34	05:01	06:26	07:52
N 40°	15:38	16:12	16:54	04:52	06:12	07:33
35°	15:48	16:26	17:11	04:44	06:00	07:17
30°	15:56	16:37	17:25	04:37	05:49	07:03
20°	16:10	16:58	17:51	04:26	05:31	06:40
N 10°	16:23	17:15	18:12	04:16	05:16	06:19
0°	16:35	17:32	18:33	04:06	05:01	06:00
S 10°	16:47	17:49	18:53	03:57	04:47	05:42
20°	17:00	18:07	19:16	03:47	04:32	05:22
30°	17:15	18:28	19:41	03:35	04:14	04:59
35°	17:24	18:40	19:57	03:29	04:04	04:45
40°	17:34	18:55	20:15	03:22	03:52	04:30
45°	17:46	19:12	20:36	03:13	03:39	04:11
S 50°	18:00	19:33	21:04	03:03	03:22	03:49
52°	18:07	19:43	21:17	02:58	03:15	03:38
54°	18:15	19:55	21:33	02:53	03:06	03:26
56°	18:23	20:08	21:51	02:47	02:57	03:12
58°	18:33	20:24	22:14	02:41	02:46	02:55
S 60°	18:44	20:42	22:43	02:34	02:34	02:36

Day	Sun			Moon		
	Eqn.of Time	Mer.		Mer.Pass.	Age	
	00 ^h	12 ^h	Pass	Upper	Lower	
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	
14	15:35	15:31	11:44	22:48	10:20	
15	15:26	15:20	11:45	23:46	11:17	
16	15:15	15:10	11:45	—:—	12:17	

November 17, 18, 19 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	56°32.7	139°38.6	S25°34.7	290°01.5	N21°14.0	338°29.7	N22°16.6	71°41.2	S08°43.9	Alpheratz	357°34.7	29°13.9
1	71°35.1	154°37.8	34.6	305°03.3	14.0	353°32.5	16.5	86°43.7	43.9	Ankaa	353°07.0	-42°10.4
2	86°37.6	169°37.0	34.5	320°05.2	14.0	8°35.3	16.5	101°46.1	43.9	Schedar	349°30.8	56°40.7
3	101°40.1	184°36.2	•• 34.5	335°07.0	•• 13.9	23°38.1	•• 16.5	116°48.6	•• 43.9	Diphda	348°47.2	-17°51.0
4	116°42.5	199°35.3	34.4	350°08.8	13.9	38°40.8	16.5	131°51.1	43.9	Achernar	335°19.8	-57°06.7
5	131°45.0	214°34.5	34.3	5°10.7	13.8	53°43.6	16.5	146°53.5	43.9	Hamal	327°51.0	23°34.9
6	146°47.5	229°33.7	S25°34.2	20°12.5	N21°13.8	68°46.4	N22°16.4	161°56.0	S08°43.9	Polaris	313°36.6	89°22.2
7	161°49.9	244°32.9	34.1	35°14.3	13.7	83°49.2	16.4	176°58.4	43.9	Acamar	315°11.4	-40°12.3
8	176°52.4	259°32.1	34.0	50°16.2	13.7	98°51.9	16.4	192°00.9	43.9	Menkar	314°06.0	4°11.3
9	191°54.9	274°31.2	•• 33.9	65°18.0	•• 13.7	113°54.7	•• 16.4	207°03.4	•• 43.8	Mirfak	308°28.0	49°57.0
10	206°57.3	289°30.4	33.8	80°19.9	13.6	128°57.5	16.4	222°05.8	43.8	Aldebaran	290°39.4	16°33.6
11	221°59.8	304°29.6	33.7	95°21.7	13.6	144°00.3	16.3	237°08.3	43.8	Rigel	281°03.7	-8°10.3
12	237°02.3	319°28.8	S25°33.6	110°23.6	N21°13.5	159°03.0	N22°16.3	252°10.7	S08°43.8	Capella	280°21.7	46°01.3
13	252°04.7	334°28.0	33.5	125°25.4	13.5	174°05.8	16.3	267°13.2	43.8	Bellatrix	278°22.7	6°22.4
14	267°07.2	349°27.1	33.4	140°27.2	13.5	189°08.6	16.3	282°15.7	43.8	Elnath	278°01.7	28°37.7
15	282°09.6	4°26.3	•• 33.3	155°29.1	•• 13.4	204°11.4	•• 16.2	297°18.1	•• 43.8	Alnilam	275°37.6	-1°11.1
16	297°12.1	19°25.5	33.2	170°30.9	13.4	219°14.1	16.2	312°20.6	43.8	Betelgeuse	270°52.0	7°24.8
17	312°14.6	34°24.7	33.1	185°32.8	13.3	234°16.9	16.2	327°23.0	43.8	Canopus	263°52.0	-52°42.3
18	327°17.0	49°23.9	S25°33.0	200°34.6	N21°13.3	249°19.7	N22°16.2	342°25.5	S08°43.8	Sirius	258°26.1	-16°44.8
19	342°19.5	64°23.0	32.8	215°36.5	13.3	264°22.5	16.2	357°27.9	43.7	Adhara	255°05.7	-29°00.1
20	357°22.0	79°22.2	32.7	230°38.3	13.2	279°25.3	16.1	12°30.4	43.7	Procyon	244°50.8	5°09.7
21	12°24.4	94°21.4	•• 32.6	245°40.2	•• 13.2	294°28.0	•• 16.1	27°32.9	•• 43.7	Pollux	243°17.3	27°57.9
22	27°26.9	109°20.6	32.5	260°42.1	13.1	309°30.8	16.1	42°35.3	43.7	Avior	234°14.6	-59°35.0
23	42°29.4	124°19.8	32.4	275°43.9	13.1	324°33.6	16.1	57°37.8	43.7	Suhail	222°46.3	-43°31.6
Mer.pass. 20:11		ν -0.8' d -0.1' m -4.11		ν 1.8' d -0.0' m -0.22		ν 2.8' d -0.0' m -2.77		ν 2.5' d -0.0' m 0.88		Miaplacidus	221°38.2	-69°48.7
										Alphard	217°47.9	-8°45.8
										Regulus	207°34.6	11°50.8
										Dubhe	193°41.4	61°36.8
										Denebola	182°25.3	14°26.0
										Gienah	175°43.9	-17°40.6
										Acrux	173°00.8	-63°13.9
										Gacrux	171°52.2	-57°14.9
										Alioth	166°13.5	55°49.3
										Spica	158°22.7	-11°17.4
										Alkaid	152°52.5	49°11.2
										Hadar	148°36.8	-60°29.4
										Menkent	147°58.2	-36°29.4
										Arcturus	145°48.4	19°03.2
										Rigil Kent.	139°41.2	-60°56.1
										Kochab	137°20.9	74°03.1
										Zuben'ubi	136°56.5	-16°08.6
										Alphecca	126°04.2	26°37.9
										Antares	112°16.4	-26°29.2
										Atria	107°11.3	-69°04.4
										Sabik	102°03.3	-15°45.3
										Shaula	96°10.9	-37°07.3
										Rasalhague	95°59.0	12°32.6
										Eltanin	90°42.7	51°29.2
										Kaus Aust.	83°33.0	-34°22.4
										Vega	80°33.6	38°48.5
										Nunki	75°48.2	-26°16.0
										Altair	62°00.3	8°56.1
										Peacock	53°06.2	-56°39.5
										Deneb	49°26.1	45°22.4
										Enif	33°39.0	9°59.4
										Al Na'ir	27°33.0	-46°50.6
										Fomalhaut	15°14.6	-29°29.5
										Scheat	13°45.2	28°13.3
										Markab	13°29.9	15°20.5
										Nov 17 Sun	SHA	Mer.pass
										Venus	83°06.0	14:42
										Mars	233°28.8	04:39
										Jupiter	281°57.1	01:26
										Saturn	15°08.5	19:10
										Nov 18 Mon	SHA	Mer.pass
										Venus	81°47.1	14:44
										Mars	233°13.9	04:36
										Jupiter	282°04.5	01:21
										Saturn	15°08.4	19:06
										Nov 19 Tue	SHA	Mer.pass
										Venus	80°28.4	14:45
										Mars	232°59.7	04:33
										Jupiter	282°12.1	01:17
										Saturn	15°08.2	19:02
										</		

Sun							Moon							
Sun	GHA	Dec	GHA	ν	Dec	d	HP	Lat.	Twilight		Sunrise	Sunset	Twilight	
									Naut.	Civil			Civil	Naut.
0	183°45.9	S19°02.8	348°34.0	3.4'	N26°35.0	6.7'	59.9'	N 72°	07:12	08:51	■	■	14:39	16:18
1	198°45.8	03.4	2°56.3	3.4'	26°41.8	6.5'	59.9'	N 70°	07:00	08:25	10:14	13:16	15:04	16:29
2	213°45.7	04.0	17°18.7	3.3'	26°48.3	6.3'	59.9'	68°	06:51	08:05	09:29	14:00	15:24	16:39
3	228°45.6	· · 04.6	31°41.0	3.3'	26°54.6	6.1'	59.8'	66°	06:42	07:50	09:00	14:29	15:40	16:47
4	243°45.4	05.2	46°03.3	3.3'	27°00.8	5.9'	59.8'	64°	06:35	07:37	08:38	14:51	15:53	16:54
5	258°45.3	05.8	60°25.5	3.2'	27°06.7	5.7'	59.8'	62°	06:29	07:26	08:21	15:09	16:04	17:00
6	273°45.2	S19°06.4	74°47.8	3.2'	N27°12.5	5.5'	59.7'	60°	06:24	07:16	08:06	15:23	16:14	17:06
7	288°45.1	07.0	89°10.0	3.2'	27°18.0	5.3'	59.7'	N 58°	06:19	07:08	07:54	15:36	16:22	17:11
8	303°44.9	07.6	103°32.1	3.2'	27°23.3	5.1'	59.7'	56°	06:14	07:01	07:43	15:46	16:29	17:16
9	318°44.8	· · 08.2	117°54.3	3.1'	27°28.5	4.9'	59.6'	54°	06:10	06:54	07:34	15:56	16:36	17:20
10	333°44.7	08.8	132°16.4	3.1'	27°33.4	4.7'	59.6'	52°	06:06	06:48	07:26	16:04	16:42	17:24
11	348°44.6	09.4	146°38.6	3.1'	27°38.2	4.5'	59.6'	50°	06:03	06:42	07:18	16:12	16:48	17:27
12	3°44.4	S19°10.0	161°00.7	3.1'	N27°42.7	4.3'	59.5'	45°	05:54	06:30	07:02	16:28	17:00	17:36
13	18°44.3	10.6	175°22.8	3.1'	27°47.1	4.1'	59.5'	N 40°	05:47	06:20	06:49	16:41	17:10	17:43
14	33°44.2	11.2	189°44.9	3.1'	27°51.2	3.9'	59.5'	35°	05:40	06:11	06:38	16:52	17:20	17:50
15	48°44.0	· · 11.8	204°07.0	3.1'	27°55.2	3.7'	59.4'	30°	05:33	06:02	06:28	17:02	17:28	17:57
16	63°43.9	12.4	218°29.1	3.1'	27°58.9	3.5'	59.4'	20°	05:21	05:47	06:11	17:20	17:43	18:10
17	78°43.8	13.0	232°51.2	3.1'	28°02.5	3.3'	59.4'	N 10°	05:08	05:34	05:56	17:35	17:57	18:22
18	93°43.7	S19°13.6	247°13.3	3.1'	N28°05.8	3.1'	59.3'	0°	04:54	05:20	05:42	17:49	18:11	18:36
19	108°43.5	14.2	261°35.4	3.1'	28°09.0	2.9'	59.3'	S 10°	04:39	05:05	05:27	18:03	18:26	18:52
20	123°43.4	14.8	275°57.5	3.1'	28°11.9	2.7'	59.3'	20°	04:20	04:48	05:12	18:19	18:42	19:11
21	138°43.3	· · 15.4	290°19.6	3.1'	28°14.6	2.5'	59.2'	30°	03:56	04:28	04:54	18:37	19:03	19:35
22	153°43.1	16.0	304°41.8	3.2'	28°17.2	2.3'	59.2'	35°	03:41	04:16	04:44	18:47	19:15	19:50
23	168°43.0	16.5	319°03.9	3.2'	28°19.5	2.1'	59.2'	40°	03:23	04:01	04:32	18:59	19:30	20:08
	SD = 16.2'	d = 0.6'			SD = 16.3'			45°	02:59	03:43	04:18	19:13	19:48	20:32
Mon	GHA	Dec	GHA	ν	Dec	d	HP	S 50°	02:27	03:21	04:00	19:31	20:11	21:05
0	183°42.9	S19°17.1	333°26.1	3.2'	N28°21.7	1.9'	59.1'	52°	02:10	03:09	03:52	19:39	20:22	21:22
1	198°42.7	17.7	347°48.3	3.2'	28°23.6	1.7'	59.1'	54°	01:49	02:57	03:43	19:49	20:35	21:44
2	213°42.6	18.3	2°10.6	3.3'	28°25.3	1.5'	59.1'	56°	01:21	02:42	03:32	19:59	20:51	22:13
3	228°42.5	· · 18.9	16°32.8	3.3'	28°26.9	1.3'	59.0'	58°	00:36	02:24	03:21	20:11	21:09	23:03
4	243°42.3	19.5	30°55.1	3.3'	28°28.2	1.1'	59.0'	S 60°	—:—	02:02	03:07	20:26	21:32	—:—
5	258°42.2	20.1	45°17.5	3.4'	28°29.4	0.9'	59.0'							
6	273°42.1	S19°20.7	59°39.9	3.4'	N28°30.3	0.7'	58.9'							
7	288°41.9	21.3	74°02.3	3.5'	28°31.0	0.6'	58.9'							
8	303°41.8	21.9	88°24.7	3.5'	28°31.6	0.4'	58.9'							
9	318°41.7	· · 22.4	102°47.3	3.6'	28°32.0	0.2'	58.8'							
10	333°41.5	23.0	117°09.8	3.6'	28°32.1	-0.0'	58.8'							
11	348°41.4	23.6	131°32.4	3.7'	28°32.1	-0.2'	58.7'							
12	3°41.3	S19°24.2	145°55.1	3.7'	N28°31.9	-0.4'	58.7'							
13	18°41.1	24.8	160°17.8	3.8'	28°31.4	-0.6'	58.7'							
14	33°41.0	25.4	174°40.6	3.8'	28°30.8	-0.8'	58.6'							
15	48°40.9	· · 26.0	189°03.5	3.9'	28°30.0	-1.0'	58.6'							
16	63°40.7	26.5	203°26.4	4.0'	28°29.0	-1.2'	58.6'							
17	78°40.6	27.1	217°49.3	4.0'	28°27.8	-1.4'	58.5'							
18	93°40.4	S19°27.7	232°12.4	4.1'	N28°26.5	-1.6'	58.5'							
19	108°40.3	28.3	246°35.5	4.2'	28°24.9	-1.7'	58.4'							
20	123°40.2	28.9	260°58.7	4.3'	28°23.2	-1.9'	58.4'							
21	138°40.0	· · 29.4	275°22.0	4.4'	28°21.2	-2.1'	58.4'							
22	153°39.9	30.0	289°45.3	4.4'	28°19.1	-2.3'	58.3'							
23	168°39.8	30.6	304°08.8	4.5'	28°16.8	-2.5'	58.3'							
	SD = 16.2'	d = 0.6'			SD = 16.1'									
Tue	GHA	Dec	GHA	ν	Dec	d	HP	S 10°	19:59	21:01	21:59	06:41	07:43	08:45
0	183°39.6	S19°31.2	318°32.3	4.6'	N28°14.3	-2.7'	58.3'	20°	20:23	21:26	22:21	06:18	07:18	08:21
1	198°39.5	31.7	332°55.9	4.7'	28°11.7	-2.8'	58.2'	30°	20:51	21:54	22:47	05:51	06:49	07:53
2	213°39.3	32.3	347°19.6	4.8'	28°08.8	-3.0'	58.2'	35°	21:08	22:11	23:03	05:35	06:32	07:37
3	228°39.2	· · 32.9	1°43.4	4.9'	28°05.8	-3.2'	58.1'	40°	21:28	22:31	23:20	05:16	06:12	07:17
4	243°39.0	33.5	16°07.2	5.0'	28°02.6	-3.4'	58.1'	45°	21:52	22:54	23:42	04:54	05:48	06:54
5	258°38.9	34.1	30°31.2	5.1'	27°59.3	-3.5'	58.1'	S 50°	22:23	23:25	· · ·	04:25	05:17	06:23
6	273°38.8	S19°34.6	44°55.3	5.2'	N27°55.7	-3.7'	58.0'	52°	22:39	23:40	· · ·	04:12	05:01	06:08
7	288°38.6	35.2	59°19.4	5.3'	27°52.0	-3.9'	58.0'	54°	22:57	23:58	· · ·	03:56	04:43	05:51
8	303°38.5	35.8	73°43.7	5.4'	27°48.1	-4.1'	57.9'	56°	23:19	· · ·	00:19	03:37	04:21	05:30
9	318°38.3	· · 36.3	88°08.0	5.5'	27°44.1	-4.2'	57.9'	58°	23:47	· · ·	00:47	03:14	03:53	05:03
10	333°38.2	36.9	102°32.5	5.6'	27°39.9	-4.4'	57.9'	S 60°	· · ·	00:29	01:25	02:44	03:11	04:24
11	348°38.0	37.5	116°57.1	5.7'	27°35.5	-4.6'	57.8'							
12	3°37.9	S19°38.1	131°21.7	5.8'	N27°30.9	-4.7'	57.8'							
13	18°37.8	38.6	145°46.5	5.9'	27°26.2	-4.9'	57.8'							
14	33°37.6	39.2	160°11.4	6.0'	27°21.3	-5.0'	57.7'							
15	48°37.5	· · 39.8	174°36.4	6.1'	27°16.3	-5.2'	57.7'							
16	63°37.3	40.3	189°01.5	6.2'	27°11.1	-5.4'	57.6'							
17	78°37.2	40.9	203°26.7	6.3'	27°05.7	-5.5'	57.6'							
18	93°37.0	S19°41.5	217°52.0	6.4'	N27°00.2	-5.7'	57.6'							
19	108°36.9	42.0	232°17.5	6.6'	26°54.6	-5.8'	57.5'							
20	123°36.7	42.6	246°43.0	6.7'	26°48.7	-6.0'	57.5'							
21	138°36.6	· · 43.2	261°08.7	6.8'	26°42.8	-6.1'	57.4'							

November 20, 21, 22 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	59°30.1	138°40.0	S25°25.2	292°16.2	N21°11.6	341°50.0	N22°15.0	74°38.0	S08°43.1	Alpheratz	357°34.7	29°13.9
1	74°32.6	153°39.2	25.0	307°18.1	11.6	356°52.8	15.0	89°40.4	43.1	Ankaa	353°07.0	-42°10.4
2	89°35.0	168°38.4	24.9	322°20.0	11.5	11°55.5	15.0	104°42.9	43.1	Schedar	349°30.8	56°40.7
3	104°37.5	183°37.6	•• 24.7	337°21.9	•• 11.5	26°58.3	•• 14.9	119°45.3	•• 43.1	Diphda	348°47.2	-17°51.0
4	119°40.0	198°36.8	24.5	352°23.8	11.5	42°01.1	14.9	134°47.8	43.1	Achernar	335°19.8	-57°06.7
5	134°42.4	213°36.0	24.3	7°25.7	11.5	57°03.9	14.9	149°50.2	43.0	Hamal	327°51.0	23°34.9
6	149°44.9	228°35.2	S25°24.1	22°27.7	N21°11.4	72°06.7	N22°14.9	164°52.7	S08°43.0	Polaris	313°36.5	89°22.2
7	164°47.4	243°34.4	24.0	37°29.6	11.4	87°09.5	14.9	179°55.1	43.0	Acamar	315°11.4	-40°12.3
8	179°49.8	258°33.6	23.8	52°31.5	11.4	102°12.3	14.8	194°57.6	43.0	Menkar	314°06.0	4°11.3
9	194°52.3	273°32.8	•• 23.6	67°33.4	•• 11.4	117°15.1	•• 14.8	210°00.0	•• 43.0	Mirfak	308°27.9	49°57.1
10	209°54.8	288°32.0	23.4	82°35.3	11.4	132°17.9	14.8	225°02.5	43.0	Aldebaran	290°39.4	16°33.6
11	224°57.2	303°31.2	23.2	97°37.2	11.3	147°20.7	14.8	240°04.9	43.0	Rigel	281°03.7	-8°10.3
12	239°59.7	318°30.4	S25°23.0	112°39.2	N21°11.3	162°23.4	N22°14.7	255°07.3	S08°42.9	Capella	280°21.6	46°01.3
13	255°02.1	333°29.6	22.8	127°41.1	11.3	177°26.2	14.7	270°09.8	42.9	Bellatrix	278°22.7	6°22.4
14	270°04.6	348°28.8	22.6	142°43.0	11.3	192°29.0	14.7	285°12.2	42.9	Elnath	278°01.7	28°37.7
15	285°07.1	3°28.0	•• 22.4	157°44.9	•• 11.2	207°31.8	•• 14.7	300°14.7	•• 42.9	Alnilam	275°37.6	-1°11.1
16	300°09.5	18°27.2	22.2	172°46.9	11.2	222°34.6	14.7	315°17.1	42.9	Betelgeuse	270°51.9	7°24.8
17	315°12.0	33°26.4	22.0	187°48.8	11.2	237°37.4	14.6	330°19.6	42.9	Canopus	263°52.0	-52°42.3
18	330°14.5	48°25.6	S25°21.8	202°50.7	N21°11.2	252°40.2	N22°14.6	345°22.0	S08°42.9	Sirius	258°26.1	-16°44.8
19	345°16.9	63°24.8	21.6	217°52.7	11.2	267°43.0	14.6	0°24.5	42.8	Adhara	255°05.7	-29°00.1
20	0°19.4	78°24.0	21.4	232°54.6	11.1	282°45.8	14.6	15°26.9	42.8	Procyon	244°50.8	5°09.7
21	15°21.9	93°23.2	•• 21.2	247°56.5	•• 11.1	297°48.6	•• 14.5	30°29.4	•• 42.8	Pollux	243°17.2	27°57.9
22	30°24.3	108°22.4	21.0	262°58.5	11.1	312°51.4	14.5	45°31.8	42.8	Avior	234°14.5	-59°35.0
23	45°26.8	123°21.6	20.8	278°00.4	11.1	327°54.1	14.5	60°34.2	42.8	Suhail	222°46.3	-43°31.7
Mer.pass. 19:59		ν -0.8' d -0.2' m -4.12		ν 1.9' d -0.0' m -0.27		ν 2.8' d -0.0' m -2.78		ν 2.4' d -0.0' m 0.89		Miaplacidus	221°38.1	-69°48.8
										Alphard	217°47.8	-8°45.8
										Regulus	207°34.6	11°50.8
										Dubhe	193°41.3	61°36.8
										Denebola	182°25.2	14°26.0
										Gienah	175°43.9	-17°40.6
										Acrux	173°00.7	-63°13.9
										Gacrux	171°52.2	-57°14.9
										Alioth	166°13.5	55°49.3
										Spica	158°22.7	-11°17.4
										Alkaid	152°52.5	49°11.2
										Hadar	148°36.8	-60°29.4
										Menkent	147°58.2	-36°29.4
										Arcturus	145°48.3	19°03.2
										Rigel Kent.	139°41.1	-60°56.1
										Kochab	137°20.9	74°03.0
										Zuben'ubi	136°56.5	-16°08.6
										Alphecca	126°04.2	26°37.9
										Antares	112°16.4	-26°29.2
										Atria	107°11.3	-69°04.3
										Sabik	102°03.3	-15°45.3
										Shaula	96°10.9	-37°07.3
										Rasalhague	95°59.0	12°32.6
										Eltanin	90°42.7	51°29.2
										Kaus Aust.	83°33.0	-34°22.4
										Vega	80°33.7	38°48.5
										Nunki	75°48.2	-26°16.0
										Altair	62°00.3	8°56.1
										Peacock	53°06.2	-56°39.5
										Deneb	49°26.1	45°22.4
										Enif	33°39.0	9°59.4
										Al Na'ir	27°33.1	-46°50.6
										Fomalhaut	15°14.6	-29°29.5
										Scheat	13°45.2	28°13.3
										Markab	13°29.9	15°20.5
Nov 20 Wed		SHA		Mer.pass								
Venus		79°09.9		14:46								
Mars		232°46.1		04:30								
Jupiter		282°19.9		01:12								
Saturn		15°07.9		18:58								
Nov 21 Thu		SHA		Mer.pass								
Venus		77°51.6		14:47								
Mars		232°33.1		04:28								
Jupiter		282°27.7		01:07								
Saturn		15°07.4		18:54								
Nov 22 Fri		SHA		Mer.pass								
Venus		76°33.4		14:49								
Mars		232°20.7		01:24								
Jupiter		282°35.6		01:04								
Saturn		15°06.9		18:51								
Horizontal parallax												
Venus: 0.1												
Mars: 0.2												

Sun				Moon				
Wed	GHA	Dec		GHA	ν	Dec	d	HP
0	183° 36.1	S19° 44.9		304° 26.4	7.1'	N26° 24.0	-6.6'	57.3'
1	198° 36.0	45.4		318° 52.6	7.3'	26° 17.4	-6.7'	57.3'
2	213° 35.8	46.0		333° 18.8	7.4'	26° 10.7	-6.8'	57.3'
3	228° 35.7	· · 46.5		347° 45.2	7.5'	26° 03.9	-7.0'	57.2'
4	243° 35.5	47.1		2° 11.7	7.6'	25° 56.9	-7.1'	57.2'
5	258° 35.4	47.7		16° 38.4	7.8'	25° 49.8	-7.3'	57.1'
6	273° 35.2	S19° 48.2		31° 05.1	7.9'	N25° 42.5	-7.4'	57.1'
7	288° 35.1	48.8		45° 32.0	8.0'	25° 35.2	-7.5'	57.1'
8	303° 34.9	49.3		59° 59.0	8.1'	25° 27.6	-7.7'	57.0'
9	318° 34.8	· · 49.9		74° 26.1	8.2'	25° 20.0	-7.8'	57.0'
10	333° 34.6	50.4		88° 53.4	8.4'	25° 12.2	-7.9'	57.0'
11	348° 34.5	51.0		103° 20.7	8.5'	25° 04.3	-8.0'	56.9'
12	3° 34.3	S19° 51.6		117° 48.2	8.6'	N24° 56.3	-8.2'	56.9'
13	18° 34.2	52.1		132° 15.8	8.7'	24° 48.1	-8.3'	56.8'
14	33° 34.0	52.7		146° 43.6	8.9'	24° 39.8	-8.4'	56.8'
15	48° 33.9	· · 53.2		161° 11.4	9.0'	24° 31.4	-8.5'	56.8'
16	63° 33.7	53.8		175° 39.4	9.1'	24° 22.9	-8.6'	56.7'
17	78° 33.6	54.3		190° 07.6	9.2'	24° 14.3	-8.8'	56.7'
18	93° 33.4	S19° 54.9		204° 35.8	9.4'	N24° 05.5	-8.9'	56.7'
19	108° 33.2	55.4		219° 04.2	9.5'	23° 56.6	-9.0'	56.6'
20	123° 33.1	56.0		233° 32.7	9.6'	23° 47.7	-9.1'	56.6'
21	138° 32.9	· · 56.5		248° 01.3	9.7'	23° 38.6	-9.2'	56.6'
22	153° 32.8	57.1		262° 30.0	9.9'	23° 29.3	-9.3'	56.5'
23	168° 32.6	57.6		276° 58.9	10.0'	23° 20.0	-9.4'	56.5'
SD = 16.2'		d = 0.6'	SD = 15.6'					
Thu	GHA	Dec		GHA	ν	Dec	d	HP
0	183° 32.5	S19° 58.2		291° 27.9	10.1'	N23° 10.6	-9.5'	56.4'
1	198° 32.3	58.7		305° 57.0	10.2'	23° 01.1	-9.6'	56.4'
2	213° 32.1	59.3		320° 26.2	10.4'	22° 51.5	-9.7'	56.4'
3	228° 32.0	19° 59.8		334° 55.5	10.5'	22° 41.7	-9.8'	56.3'
4	243° 31.8	20° 00.4		349° 25.0	10.6'	22° 31.9	-9.9'	56.3'
5	258° 31.7	00.9		3° 54.6	10.7'	22° 22.0	-10.0'	56.3'
6	273° 31.5	S20° 01.5		18° 24.3	10.8'	N22° 11.9	-10.1'	56.2'
7	288° 31.4	02.0		32° 54.1	11.0'	22° 01.8	-10.2'	56.2'
8	303° 31.2	02.5		47° 24.1	11.1'	21° 51.6	-10.3'	56.2'
9	318° 31.0	· · 03.1		61° 54.2	11.2'	21° 41.3	-10.4'	56.1'
10	333° 30.9	03.6		76° 24.4	11.3'	21° 30.9	-10.5'	56.1'
11	348° 30.7	04.2		90° 54.7	11.4'	21° 20.4	-10.6'	56.1'
12	3° 30.5	S20° 04.7		105° 25.1	11.5'	N21° 09.8	-10.7'	56.0'
13	18° 30.4	05.2		119° 55.6	11.7'	20° 59.1	-10.8'	56.0'
14	33° 30.2	05.8		134° 26.3	11.8'	20° 48.3	-10.8'	56.0'
15	48° 30.1	· · 06.3		148° 57.0	11.9'	20° 37.5	-10.9'	55.9'
16	63° 29.9	06.9		163° 27.9	12.0'	20° 26.6	-11.0'	55.9'
17	78° 29.7	07.4		177° 58.9	12.1'	20° 15.6	-11.1'	55.9'
18	93° 29.6	S20° 07.9		192° 30.0	12.2'	N20° 04.5	-11.2'	55.8'
19	108° 29.4	08.5		207° 01.2	12.3'	19° 53.3	-11.2'	55.8'
20	123° 29.2	09.0		221° 32.6	12.4'	19° 42.1	-11.3'	55.8'
21	138° 29.1	· · 09.5		236° 04.0	12.5'	19° 30.7	-11.4'	55.8'
22	153° 28.9	10.1		250° 35.5	12.6'	19° 19.3	-11.5'	55.7'
23	168° 28.7	10.6		265° 07.2	12.8'	19° 07.9	-11.5'	55.7'
SD = 16.2'		d = 0.5'	SD = 15.4'					
Fri	GHA	Dec		GHA	ν	Dec	d	HP
0	183° 28.6	S20° 11.1		279° 38.9	12.9'	N18° 56.3	-11.6'	55.7'
1	198° 28.4	11.7		294° 10.8	13.0'	18° 44.7	-11.7'	55.6'
2	213° 28.3	12.2		308° 42.8	13.1'	18° 33.0	-11.8'	55.6'
3	228° 28.1	· · 12.7		323° 14.8	13.2'	18° 21.3	-11.8'	55.6'
4	243° 27.9	13.3		337° 47.0	13.3'	18° 09.5	-11.9'	55.5'
5	258° 27.7	13.8		352° 19.3	13.4'	17° 57.6	-11.9'	55.5'
6	273° 27.6	S20° 14.3		6° 51.6	13.5'	N17° 45.6	-12.0'	55.5'
7	288° 27.4	14.9		21° 24.1	13.6'	17° 33.6	-12.1'	55.5'
8	303° 27.2	15.4		35° 56.7	13.7'	17° 21.5	-12.1'	55.4'
9	318° 27.1	· · 15.9		50° 29.3	13.8'	17° 09.4	-12.2'	55.4'
10	333° 26.9	16.4		65° 02.1	13.9'	16° 57.2	-12.3'	55.4'
11	348° 26.7	17.0		79° 34.9	13.9'	16° 45.0	-12.3'	55.3'
12	3° 26.6	S20° 17.5		94° 07.9	14.0'	N16° 32.6	-12.4'	55.3'
13	18° 26.4	18.0		108° 40.9	14.1'	16° 20.3	-12.4'	55.3'
14	33° 26.2	18.5		123° 14.1	14.2'	16° 07.8	-12.5'	55.3'
15	48° 26.1	· · 19.1		137° 47.3	14.3'	15° 55.4	-12.5'	55.2'
16	63° 25.9	19.6		152° 20.6	14.4'	15° 42.8	-12.6'	55.2'
17	78° 25.7	20.1		166° 54.0	14.5'	15° 30.2	-12.6'	55.2'
18	93° 25.5	S20° 20.6		181° 27.5	14.6'	N15° 17.6	-12.7'	55.2'
19	108° 25.4	21.1		196° 01.0	14.7'	15° 04.9	-12.7'	55.1'
20	123° 25.2	21.7		210° 34.7	14.7'	14° 52.2	-12.8'	55.1'
21	138° 25.0	· · 22.2		225° 08.4	14.8'	14° 39.4	-12.8'	55.1'
22	153° 24.8	22.7		239° 42.2	14.9'	14° 26.6	-12.9'	55.1'
23	168° 24.7	23.2		254° 16.1	15.0'	14° 13.7	-12.9'	55.0'
SD = 16.2'		d = 0.5'	SD = 15.2'					

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:22	09:05	■	■	14:26	16:09
N 70°	07:09	08:36	■	■	14:55	16:22
68°	06:59	08:15	■	■	15:16	16:32
66°	06:50	07:58	■	■	15:33	16:41
64°	06:42	07:44	■	■	15:47	16:49
62°	06:35	07:33	■	■	15:02	16:56
60°	06:29	07:23	■	■	15:18	17:02
N 58°	06:24	07:14	■	■	16:18	17:07
56°	06:19	07:06	■	■	15:42	17:12
54°	06:15	06:59	■	■	15:52	16:33
52°	06:11	06:52	■	■	16:01	16:39
50°	06:07	06:47	■	■	16:09	16:45
45°	05:58	06:34	■	■	16:25	16:58
N 40°	05:50	06:23	■	■	16:39	17:09
35°	05:42	06:13	■	■	16:51	17:18
30°	05:36	06:05	■	■	17:01	17:27
20°	05:22	05:49	■	■	17:19	17:43
N 10°	05:09	05:35	■	■	17:35	17:57
0°	04:55	05:20	■	■	17:50	18:12
S 10°	04:39	05:05	■	■	18:04	18:27
20°	04:20	04:48	■	■	18:21	18:44
30°	03:55	04:27	■	■	18:39	19:06
35°	03:39	04:14	■	■	18:50	19:19
40°	03:20	03:59	■	■	19:03	19:34
45°	02:56	03:40	■	■	19:18	19:53
S 50°	02:22	03:16	■	■	19:36	20:17
52°	02:04	03:05	■	■	19:45	20:28
54°	01:41	02:51	■	■	19:54	20:42
56°	01:09	02:36	■	■	20:06	20:58
58°	—	02:16	■	■	20:18	21:18
S 60°	—	01:52	■	■	20:33	21:43

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	□	□	20:23	□	□	15:49
N 70°	□	□	20:57	□	□	15:13
68°	□	19:00	21:21	□	15:34	14:47
66°	□	19:40	21:40	□	14:53	14:27
64°	17:59	20:07	21:54	14:49	14:25	14:11
62°	18:37	20:28	22:07	14:10	14:03	13:58
60°	19:05	20:44	22:17	13:42	13:46	13:46
N 58°	19:26	20:58	22:26	13:20	13:31	13:36
56°	19:43	21:11	22:34	13:03	13:18	13:28
54°	19:57	21:21	22:41	12:48	13:07	13:20
52°	20:10	21:30	22:47	12:34	12:57	13:13
50°	20:21	21:39	22:53	12:23	12:48	13:06
45°	20:44	21:56	23:05	11:59	12:29	12:53
N 40°	21:03	22:10	23:15	11:39	12:14	12:41
35°	21:18	22:22	23:23	11:23	12:01	12:32
30°	21:31	22:33	23:31	11:09	11:49	12:23
20°	21:54	22:51	23:44	10:45	11:29	12:08
N 10°	22:13	23:06	23:55	10:24	11:12	11:55
0°	22:32	23:21	· ·	10:04	10:56	11:43
S 10°	22:50	23:35	· ·	09:44	10:39	11:30
20°	23:09	23:50	· ·	09:23	10:22	11:17
30°	23:31	· ·	00:08	08:58	10:01	11:01
35°	23:44	· ·	00:18	08:43	09:49	10:52
40°	23:59	· ·	00:30	08:26	09:35	10:42
45°	· ·	00:17	00:43	08:06	09:19	10:30
S 50°	00:09	00:38	00:59	07:40	08:58	10:15
52°	00:22	00:49	01:07	07:27	08:48	10:08
54°	00:37	01:00	01:16	07:12	08:37	10:00
56°	00:54	01:14	01:25	06:55	08:25	09:51
58°	01:15	01:29	01:36	06:35	08:10	09:41
S 60°	01:42	01:47	01:48	06:08	07:52	09:30

Day	Sun		Mer. Pass	Moon		Age
	Eqn.of Time	Mer. Pass		Mer. Pass	Age	
	00 ^h	12 ^h	Pass	Upper	Lower	19-21
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	79-61%
20	14:25	14:17</				

November 23, 24, 25 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn	
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	62°27.5	137°43.0	S25°09.3	294°36.6	N21°10.5	345°11.2	N22°13.3	77°33.8	S08°41.9
1	77°30.0	152°42.2	09.0	309°38.6	10.5	0°14.0	13.3	92°36.3	41.9
2	92°32.5	167°41.4	08.7	324°40.6	10.5	15°16.8	13.3	107°38.7	41.9
3	107°34.9	182°40.7	•• 08.5	339°42.6	•• 10.5	30°19.6	•• 13.3	122°41.2	•• 41.9
4	122°37.4	197°39.9	08.2	354°44.5	10.5	45°22.4	13.3	137°43.6	41.9
5	137°39.8	212°39.1	07.9	9°46.5	10.5	60°25.2	13.2	152°46.0	41.8
6	152°42.3	227°38.3	S25°07.6	24°48.5	N21°10.5	75°28.0	N22°13.2	167°48.5	S08°41.8
7	167°44.8	242°37.6	07.4	39°50.5	10.5	90°30.8	13.2	182°50.9	41.8
8	182°47.2	257°36.8	07.1	54°52.5	10.5	105°33.6	13.2	197°53.3	41.8
9	197°49.7	272°36.0	•• 06.8	69°54.5	•• 10.5	120°36.4	•• 13.1	212°55.8	•• 41.8
10	212°52.2	287°35.2	06.5	84°56.5	10.5	135°39.2	13.1	227°58.2	41.7
11	227°54.6	302°34.5	06.3	99°58.5	10.5	150°42.0	13.1	243°00.6	41.7
12	242°57.1	317°33.7	S25°06.0	115°00.6	N21°10.5	165°44.8	N22°13.1	258°03.1	S08°41.7
13	257°59.6	332°32.9	05.7	130°02.6	10.5	180°47.6	13.0	273°05.5	41.7
14	273°02.0	347°32.1	05.4	145°04.6	10.5	195°50.4	13.0	288°07.9	41.7
15	288°04.5	2°31.4	•• 05.1	160°06.6	•• 10.5	210°53.2	•• 13.0	303°10.4	•• 41.6
16	303°07.0	17°30.6	04.8	175°08.6	10.5	225°56.1	13.0	318°12.8	41.6
17	318°09.4	32°29.8	04.6	190°10.6	10.5	240°58.9	12.9	333°15.2	41.6
18	333°11.9	47°29.1	S25°04.3	205°12.6	N21°10.5	256°01.7	N22°12.9	348°17.7	S08°41.6
19	348°14.3	62°28.3	04.0	220°14.6	10.5	271°04.5	12.9	3°20.1	41.6
20	3°16.8	77°27.5	03.7	235°16.6	10.5	286°07.3	12.9	18°22.5	41.5
21	18°19.3	92°26.7	•• 03.4	250°18.6	•• 10.5	301°10.1	•• 12.8	33°25.0	•• 41.5
22	33°21.7	107°26.0	03.1	265°20.7	10.5	316°12.9	12.8	48°27.4	41.5
23	48°24.2	122°25.2	02.8	280°22.7	10.5	331°15.7	12.8	63°29.8	41.5
Mer.pass. 19:47		ν -0.8' d -0.3' m -4.14		ν 2.0' d -0.0' m -0.33		ν 2.8' d -0.0' m -2.79		ν 2.4' d -0.0' m 0.91	

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	63°26.7	137°24.4	S25°02.5	295°24.7	N21°10.5	346°18.5	N22°12.8	78°32.3	S08°41.5
1	78°29.1	152°23.7	02.2	310°26.7	10.5	1°21.3	12.7	93°34.7	41.4
2	93°31.6	167°22.9	01.9	325°28.7	10.5	16°24.1	12.7	108°37.1	41.4
3	108°34.1	182°22.1	•• 01.6	340°30.8	•• 10.5	31°26.9	•• 12.7	123°39.6	•• 41.4
4	123°36.5	197°21.4	01.3	355°32.8	10.5	46°29.7	12.7	138°42.0	41.4
5	138°39.0	212°20.6	01.0	10°34.8	10.5	61°32.5	12.6	153°44.4	41.3
6	153°41.5	227°19.8	S25°00.7	25°36.8	N21°10.5	76°35.3	N22°12.6	168°46.9	S08°41.3
7	168°43.9	242°19.1	00.4	40°38.9	10.5	91°38.1	12.6	183°49.3	41.3
8	183°46.4	257°18.3	25°00.1	55°40.9	10.5	106°40.9	12.6	198°51.7	41.3
9	198°48.8	272°17.5	24°59.8	70°42.9	•• 10.6	121°43.7	•• 12.6	213°54.2	•• 41.3
10	213°51.3	287°16.8	59.5	85°45.0	10.6	136°46.6	12.5	228°56.6	41.2
11	228°53.8	302°16.0	59.2	100°47.0	10.6	151°49.4	12.5	243°59.0	41.2
12	243°56.2	317°15.2	S24°58.9	115°49.0	N21°10.6	166°52.2	N22°12.5	259°01.4	S08°41.2
13	258°58.7	332°14.5	58.6	130°51.1	10.6	181°55.0	12.5	274°03.9	41.2
14	274°01.2	347°13.7	58.3	145°53.1	10.6	196°57.8	12.4	289°06.3	41.2
15	289°03.6	2°12.9	•• 57.9	160°55.1	•• 10.6	212°00.6	•• 12.4	304°08.7	•• 41.1
16	304°06.1	17°12.2	57.6	175°57.2	10.6	227°03.4	12.4	319°11.2	41.1
17	319°08.6	32°11.4	57.3	190°59.2	10.6	242°06.2	12.4	334°13.6	41.1
18	334°11.0	47°10.7	S24°57.0	206°01.2	N21°10.6	257°09.0	N22°12.3	349°16.0	S08°41.1
19	349°13.5	62°09.9	56.7	221°03.3	10.6	272°11.8	12.3	4°18.5	41.0
20	4°15.9	77°09.1	56.4	236°05.3	10.6	287°14.6	12.3	19°20.9	41.0
21	19°18.4	92°08.4	•• 56.0	251°07.4	•• 10.6	302°17.4	•• 12.3	34°23.3	•• 41.0
22	34°20.9	107°07.6	55.7	266°09.4	10.7	317°20.2	12.2	49°25.7	41.0
23	49°23.3	122°06.9	55.4	281°11.5	10.7	332°23.1	12.2	64°28.2	41.0
Mer.pass. 19:43		ν -0.8' d -0.3' m -4.15		ν 2.0' d 0.0' m -0.35		ν 2.8' d -0.0' m -2.80		ν 2.4' d -0.0' m 0.91	

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	64°25.8	137°06.1	S24°55.1	296°13.5	N21°10.7	347°25.9	N22°12.2	79°30.6	S08°40.9
1	79°28.3	152°05.3	54.7	311°15.6	10.7	2°28.7	12.2	94°33.0	40.9
2	94°30.7	167°04.6	54.4	326°17.6	10.7	17°31.5	12.1	109°35.5	40.9
3	109°33.2	182°03.8	•• 54.1	341°19.7	•• 10.7	32°34.3	•• 12.1	124°37.9	•• 40.9
4	124°35.7	197°03.1	53.8	356°21.7	10.7	47°37.1	12.1	139°40.3	40.8
5	139°38.1	212°02.3	53.4	11°23.8	10.7	62°39.9	12.1	154°42.7	40.8
6	154°40.6	227°01.6	S24°53.1	26°25.8	N21°10.7	77°42.7	N22°12.0	169°45.2	S08°40.8
7	169°43.1	242°00.8	52.8	41°27.9	10.8	92°45.5	12.0	184°47.6	40.8
8	184°45.5	257°00.0	52.4	56°29.9	10.8	107°48.3	12.0	199°50.0	40.8
9	199°48.0	271°59.3	•• 52.1	71°32.0	•• 10.8	122°51.1	•• 12.0	214°52.4	•• 40.7
10	214°50.4	286°58.5	51.8	86°34.1	10.8	137°54.0	11.9	229°54.9	40.7
11	229°52.9	301°57.8	51.4	101°36.1	10.8	152°56.8	11.9	244°57.3	40.7
12	244°55.4	316°57.0	S24°51.1	116°38.2	N21°10.8	167°59.6	N22°11.9	259°59.7	S08°40.7
13	259°57.8	331°56.3	50.8	131°40.3	10.8	183°02.4	11.9	275°02.1	40.6
14	275°00.3	346°55.5	50.4	146°42.3	10.8	198°05.2	11.8	290°04.6	40.6
15	290°02.8	1°54.8	•• 50.1	161°44.4	•• 10.9	213°08.0	•• 11.8	305°07.0	•• 40.6
16	305°05.2	16°54.0	49.7	176°46.5	10.9	228°10.8	11.8	320°09.4	40.6
17	320°07.7	31°53.3	49.4	191°48.5	10.9	243°13.6	11.8	335°11.9	40.5
18	335°10.2	46°52.5	S24°49.0	206°50.6	N21°10.9	258°16.4	N22°11.7	350°14.3	S08°40.5
19	350°12.6	61°51.8	48.7	221°52.7	10.9	273°19.3	11.7	5°16.7	40.5
20	5°15.1	76°51.0	48.3	236°54.7	10.9	288°22.1	11.7	20°19.1	40.5
21	20°17.5	91°50.3	•• 48.0	251°56.8	•• 11.0	303°24.9	•• 11.7	35°21.6	•• 40.5
22	35°20.0	106°49.5	47.6	266°58.9	11.0	318°27.7	11.6	50°24.0	40.4
23	50°22.5	121°48.8	47.3	282°01.0	11.0	333°30.5	11.6	65°26.4	40.4
Mer.pass. 19:39		ν -0.8' d -0.3' m -4.16		ν 2.0' d 0.0' m -0.37		ν 2.8' d -0.0' m -2.80		ν 2.4' d -0.0' m 0.92	

Stars		
	SHA	Dec
Alpheratz	357°34.7	29°13.9
Ankaa	353°07.0	-42°10.4
Schedar	349°30.8	56°40.7
Diphda	348°47.2	-17°51.0
Achernar	335°19.8	-57°06.8
Hamal	327°51.0	23°34.9
Polaris	313°36.7	89°22.2
Acamar	315°11.4	-40°12.3
Menkar	314°06.0	4°11.3
Mirfak	308°27.9	49°57.1
Aldebaran	290°39.4	16°33.6
Rigel	281°03.7	-8°10.3
Capella	280°21.6	46°01.4
Bellatrix	278°22.7	6°22.4
Elnath	278°01.7	28°37.7
Alnilam	275°37.5	-1°11.1
Betelgeuse	270°51.9	7°24.8
Canopus	263°52.0	-52°42.3
Sirius	258°26.1	-16°44.9
Adhara	255°05.7	-29°00.1
Procyon	244°50.7	5°09.7
Pollux	243°17.2	27°57.9
Avior	234°14.5	-59°35.0
Suhail	222°46.3	-43°31.7
Miaplacidus	221°38.1	-69°48.8
Alphard	217°47.8	-8°45.8
Regulus	207°34.6	11°50.7
Dubhe	193°41.3	61°36.8
Denebola	182°25.2	14°26.0
Gienah	175°43.9	-17°40.6
Acrux	173°00.7	-63°13.9
Gacrux	171°52.2	-57°14.9
Alioth	166°13.4	55°49.3
Spica	158°22.7	-11°17.4
Alkaid	152°52.5	49°11.2
Hadar	148°36.8	-60°29.4
Menkent	147°58.1	-36°29.4
Arcturus	145°48.3	19°03.2
Rigil Kent.	139°41.1	-60°56.1
Kochab	137°20.9	74°03.0
Zuben'ubi	136°56.5	-16°08.6
Alphecca	126°04.2	26°37.8
Antares	112°16.4	-26°29.2
Atria	107°11.3	-69°04.3
Sabik	102°03.3	-15°45.3
Shaula	96°10.9	-37°07.3
Rasalhague	95°59.0	12°32.6
Eltanin	90°42.7	51°29.2
Kaus Aust.	83°33.0	-34°22.4
Vega	80°33.7	38°48.5
Nunki	75°48.2	-26°16.0
Altair	62°00.3	8°56.1
Peacock	53°06.2	-56°39.5
Deneb	49°26.1	45°22.4
Enif	33°39.0	9°59.4
Al Na'ir	27°33.1	-46°50.6
Fomalhaut	15°14.6	-29°29.5
Scheat	13°45.3	28°13.3
Markab	13°30.0	15°20.5

Nov 23 Sat	SHA	Mer.pass
Venus	75°15.5	14:50
Mars	232°09.0	04:21
Jupiter	282°43.7	00:59
Saturn	15°06.3	18:47

Nov 24 Sun	SHA	Mer.pass
Venus	73°57.8	14:51
Mars	231°58.0	04:18
Jupiter	282°51.8	00:55
Saturn	15°05.6	18:43

Nov 25 Mon	SHA	Mer.pass
Venus	72°40.3	14:52
Mars	231°47.7	04:15
Jupiter	283°00.1	00:50
Saturn	15°04.8	18:39

Horizontal parallax		
Venus:	0.1	
Mars:	0.2	

Sun							Moon							
Sat	GHA	Dec	GHA	ν	Dec	d	HP	Lat.	Twilight		Sunrise	Sunset	Twilight	
									Naut.	Civil			Civil	Naut.
0	183°24.5	S20°23.7	268°50.1	15.1'	N14°00.8	-13.0'	55.0'	N 72°	07:32	09:19	■	■	14:13	16:01
1	198°24.3	24.2	283°24.2	15.1'	13°47.8	-13.0'	55.0'	N 70°	07:18	08:48	11:19	12:14	14:45	16:15
2	213°24.1	24.8	297°58.3	15.2'	13°34.8	-13.1'	55.0'	68°	07:06	08:25	10:00	13:33	15:08	16:26
3	228°24.0	· · 25.3	312°32.5	15.3'	13°21.7	-13.1'	54.9'	66°	06:57	08:06	09:23	14:10	15:26	16:36
4	243°23.8	25.8	327°06.8	15.4'	13°08.6	-13.1'	54.9'	64°	06:48	07:52	08:57	14:36	15:41	16:44
5	258°23.6	26.3	341°41.1	15.4'	12°55.5	-13.2'	54.9'	62°	06:41	07:39	08:37	14:56	15:54	16:52
6	273°23.4	S20°26.8	356°15.6	15.5'	N12°42.3	-13.2'	54.9'	60°	06:35	07:29	08:21	15:12	16:04	16:58
7	288°23.3	27.3	10°50.1	15.6'	12°29.1	-13.2'	54.9'	N 58°	06:29	07:19	08:07	15:26	16:14	17:04
8	303°23.1	27.8	25°24.6	15.6'	12°15.9	-13.3'	54.8'	56°	06:24	07:11	07:55	15:38	16:22	17:09
9	318°22.9	· · 28.4	39°59.3	15.7'	12°02.6	-13.3'	54.8'	54°	06:19	07:04	07:45	15:48	16:30	17:14
10	333°22.7	28.9	54°34.0	15.8'	11°49.3	-13.4'	54.8'	52°	06:15	06:57	07:36	15:57	16:36	17:18
11	348°22.6	29.4	69°08.7	15.8'	11°35.9	-13.4'	54.8'	50°	06:11	06:51	07:27	16:06	16:42	17:23
12	3°22.4	S20°29.9	83°43.6	15.9'	N11°22.5	-13.4'	54.7'	45°	06:01	06:37	07:10	16:23	16:56	17:32
13	18°22.2	30.4	98°18.5	16.0'	11°09.1	-13.5'	54.7'	N 40°	05:53	06:26	06:56	16:38	17:07	17:41
14	33°22.0	30.9	112°53.4	16.0'	10°55.6	-13.5'	54.7'	35°	05:45	06:16	06:43	16:50	17:17	17:48
15	48°21.8	· · 31.4	127°28.4	16.1'	10°42.1	-13.5'	54.7'	30°	05:38	06:07	06:33	17:01	17:26	17:56
16	63°21.7	31.9	142°03.5	16.1'	10°28.6	-13.5'	54.7'	20°	05:24	05:51	06:14	17:19	17:43	18:10
17	78°21.5	32.4	156°38.7	16.2'	10°15.1	-13.6'	54.7'	N 10°	05:10	05:36	05:58	17:35	17:58	18:23
18	93°21.3	S20°32.9	171°13.9	16.2'	N10°01.5	-13.6'	54.6'	0°	04:55	05:21	05:43	17:50	18:13	18:38
19	108°21.1	33.4	185°49.1	16.3'	09°47.9	-13.6'	54.6'	S 10°	04:39	05:05	05:28	18:06	18:28	18:55
20	123°20.9	33.9	200°24.4	16.4'	09°34.3	-13.7'	54.6'	20°	04:19	04:47	05:11	18:22	18:46	19:15
21	138°20.8	· · 34.4	214°59.8	16.4'	09°20.6	-13.7'	54.6'	30°	03:54	04:26	04:52	18:42	19:08	19:40
22	153°20.6	34.9	229°35.2	16.5'	09°06.9	-13.7'	54.6'	35°	03:37	04:12	04:41	18:53	19:22	19:57
23	168°20.4	35.4	244°10.6	16.5'	08°53.2	-13.7'	54.5'	40°	03:18	03:57	04:28	19:06	19:37	20:17
	SD = 16.2'	d = 0.5'			SD = 15.0'			45°	02:52	03:37	04:13	19:21	19:57	20:43
S 50°	02:16	03:13	03:54	19:41	20:22	21:19								
52°	01:57	03:01	03:45	19:50	20:34	21:38								
54°	01:32	02:46	03:35	20:00	20:49	22:04								
56°	00:56	02:30	03:23	20:11	21:06	22:42								
58°	—:—	02:09	03:10	20:25	21:26	—:—								
S 60°	—:—	01:43	02:54	20:41	21:54	—:—								
Lat.	Moonrise			Moonset										
	Sat	Sun	Mon	Sat	Sun	Mon								
N 72°	22:49	· · ·	00:50	14:51	14:14	13:42								
N 70°	23:03	· · ·	00:54	14:34	14:07	13:42								
68°	23:14	· · ·	00:57	14:21	14:01	13:42								
66°	23:24	· · ·	01:00	14:10	13:56	13:43								
64°	23:31	· · ·	01:03	14:01	13:52	13:43								
62°	23:38	· · ·	01:04	13:53	13:48	13:43								
60°	23:44	· · ·	01:06	13:46	13:45	13:43								
N 58°	23:49	· · ·	01:08	13:40	13:42	13:43								
56°	23:53	· · ·	01:09	13:34	13:39	13:43								
54°	23:57	· · ·	01:10	13:29	13:37	13:43								
52°	· · ·	00:01	01:11	13:25	13:35	13:44								
50°	· · ·	00:04	01:12	13:21	13:33	13:44								
45°	· · ·	00:11	01:15	13:12	13:28	13:44								
N 40°	· · ·	00:17	01:16	13:04	13:25	13:44								
35°	· · ·	00:22	01:18	12:58	13:22	13:44								
30°	· · ·	00:26	01:19	12:52	13:19	13:44								
20°	· · ·	00:34	01:22	12:42	13:14	13:44								
N 10°	· · ·	00:40	01:24	12:34	13:10	13:44								
0°	00:05	00:47	01:26	12:25	13:06	13:45								
S 10°	00:16	00:53	01:28	12:17	13:02	13:45								
20°	00:27	00:59	01:30	12:08	12:57	13:45								
30°	00:39	01:07	01:32	11:58	12:52	13:45								
35°	00:46	01:11	01:34	11:52	12:49	13:45								
40°	00:54	01:16	01:35	11:45	12:46	13:45								
45°	01:04	01:21	01:37	11:37	12:42	13:45								
S 50°	01:15	01:28	01:39	11:27	12:37	13:45								
52°	01:21	01:31	01:40	11:23	12:35	13:45								
54°	01:26	01:34	01:41	11:18	12:33	13:45								
56°	01:33	01:38	01:43	11:12	12:30	13:45								
58°	01:40	01:42	01:44	11:06	12:27	13:45								
S 60°	01:48	01:47	01:45	10:59	12:24	13:45								
Day	Sun		Mer.	Moon		Age								
	Eqn.of Time	Mer.	Pass	Upper	Lower	22-24								
	00 ^h	12 ^h	Pass	hh:mm	hh:mm	51-32%								
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm									
23	13:38	13:30	11:47	06:15	18:36									
24	13:21	13:12	11:47	06:56	19:16									
25	13:03	12:54	11:47	07:35	19:55									

November 26, 27, 28 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	65°24.9	136°48.0	S24°46.9	297°03.0	N21°11.0	348°33.3	N22°11.6	80°28.8	S08°40.4	Alpheratz	357°34.7	29°13.9
1	80°27.4	151°47.3	46.6	312°05.1	11.0	3°36.1	11.6	95°31.2	40.4	Ankaa	353°07.0	-42°10.4
2	95°29.9	166°46.5	46.2	327°07.2	11.0	18°38.9	11.5	110°33.7	40.3	Schedar	349°30.9	56°40.7
3	110°32.3	181°45.8	· · 45.9	342°09.3	· · 11.1	33°41.8	· · 11.5	125°36.1	· · 40.3	Diphda	348°47.2	-17°51.0
4	125°34.8	196°45.0	45.5	357°11.4	11.1	48°44.6	11.5	140°38.5	40.3	Achernar	335°19.8	-57°06.8
5	140°37.3	211°44.3	45.2	12°13.4	11.1	63°47.4	11.5	155°40.9	40.3	Hamal	327°51.0	23°34.9
6	155°39.7	226°43.5	S24°44.8	27°15.5	N21°11.1	78°50.2	N22°11.4	170°43.4	S08°40.2	Polaris	313°36.9	89°22.3
7	170°42.2	241°42.8	44.4	42°17.6	11.1	93°53.0	11.4	185°45.8	40.2	Acamar	315°11.4	-40°12.3
8	185°44.7	256°42.1	44.1	57°19.7	11.2	108°55.8	11.4	200°48.2	40.2	Menkar	314°06.0	4°11.3
9	200°47.1	271°41.3	· · 43.7	72°21.8	· · 11.2	123°58.6	· · 11.4	215°50.6	· · 40.2	Mirfak	308°27.9	49°57.1
10	215°49.6	286°40.6	43.3	87°23.9	11.2	139°01.4	11.3	230°53.1	40.1	Aldebaran	290°39.4	16°33.6
11	230°52.0	301°39.8	43.0	102°26.0	11.2	154°04.3	11.3	245°55.5	40.1	Rigel	281°03.6	-8°10.3
12	245°54.5	316°39.1	S24°42.6	117°28.1	N21°11.3	169°07.1	N22°11.3	260°57.9	S08°40.1	Capella	280°21.6	46°01.4
13	260°57.0	331°38.3	42.2	132°30.2	11.3	184°09.9	11.3	276°00.3	40.1	Bellatrix	278°22.7	6°22.4
14	275°59.4	346°37.6	41.9	147°32.2	11.3	199°12.7	11.2	291°02.7	40.0	Elnath	278°01.6	28°37.7
15	291°01.9	1°36.9	· · 41.5	162°34.3	· · 11.3	214°15.5	· · 11.2	306°05.2	· · 40.0	Alnilam	275°37.5	-1°11.1
16	306°04.4	16°36.1	41.1	177°36.4	11.3	229°18.3	11.2	321°07.6	40.0	Betelgeuse	270°51.9	7°24.8
17	321°06.8	31°35.4	40.8	192°38.5	11.4	244°21.1	11.1	336°10.0	40.0	Canopus	263°52.0	-52°42.3
18	336°09.3	46°34.6	S24°40.4	207°40.6	N21°11.4	259°24.0	N22°11.1	351°12.4	S08°39.9	Sirius	258°26.1	-16°44.9
19	351°11.8	61°33.9	40.0	222°42.7	11.4	274°26.8	11.1	6°14.9	39.9	Adhara	255°05.7	-29°00.2
20	6°14.2	76°33.2	39.6	237°44.8	11.4	289°29.6	11.1	21°17.3	39.9	Procyon	244°50.7	5°09.7
21	21°16.7	91°32.4	· · 39.2	252°46.9	· · 11.5	304°32.4	· · 11.0	36°19.7	· · 39.9	Pollux	243°17.2	27°57.9
22	36°19.2	106°31.7	38.9	267°49.1	11.5	319°35.2	11.0	51°22.1	39.8	Avior	234°14.5	-59°35.0
23	51°21.6	121°31.0	38.5	282°51.2	11.5	334°38.0	11.0	66°24.5	39.8	Suhail	222°46.3	-43°31.7
Mer.pass. 19:35		ν -0.7' d -0.4' m -4.16		ν 2.1' d 0.0' m -0.39		ν 2.8' d -0.0' m -2.80		ν 2.4' d -0.0' m 0.92		Miaplacidus	221°38.0	-69°48.8
										Alphard	217°47.8	-8°45.9
										Regulus	207°34.5	11°50.7
										Dubhe	193°41.2	61°36.8
										Denebola	182°25.2	24°26.0
										Gienah	175°43.9	-17°40.7
										Acrux	173°00.7	-63°13.9
										Gacrux	171°52.1	-57°14.9
										Alioth	166°13.4	55°49.3
										Spica	158°22.7	-11°17.4
										Alkaid	152°52.5	49°11.2
										Hadar	148°36.8	-60°29.4
										Menkent	147°58.1	-36°29.4
										Arcturus	145°48.3	19°03.1
										Rigel Kent.	139°41.1	-60°56.1
										Kochab	137°20.9	74°03.0
										Zuben'ubi	136°56.5	-16°08.6
										Alphecca	126°04.2	26°37.8
										Antares	112°16.4	-26°29.2
										Atria	107°11.3	-69°04.3
										Sabik	102°03.3	-15°45.3
										Shaula	96°10.9	-37°07.3
										Rasalhague	95°59.0	12°32.6
										Eltanin	90°42.7	51°29.2
										Kaus Aust.	83°33.0	-34°22.4
										Vega	80°33.7	38°48.5
										Nunki	75°48.2	-26°16.0
										Altair	62°00.3	8°56.1
										Peacock	53°06.2	-56°39.5
										Deneb	49°26.1	45°22.4
										Enif	33°39.0	9°59.4
										Al Na'ir	27°33.1	-46°50.6
										Fomalhaut	15°14.6	-29°29.6
										Scheat	13°45.3	28°13.3
										Markab	13°30.0	15°20.5
										Nov 26 Tue	SHA	Mer.pass
										Venus	71°23.1	14:54
										Mars	231°38.1	04:11
										Jupiter	283°08.4	00:46
										Saturn	15°03.9	18:35
										Nov 27 Wed	SHA	Mer.pass
										Venus	70°06.1	14:55
										Mars	231°29.2	04:08
										Jupiter	283°16.8	00:41
										Saturn	15°02.9	18:31
										Nov 28 Thu	SHA	Mer.pass
										Venus	68°49.5	14:56
										Mars	231°21.0	04:04
										Jupiter	283°25.2	00:37
										Saturn	15°01.8	18:27

Sun				Moon				
Tue	GHA	Dec		GHA	ν	Dec	d	HP
0	183°11.1	S20°59.2		239°45.2	17.5'	S02°34.0	14.1'	54.1'
1	198°10.9	20°59.7		254°21.6	17.5'	02°48.1	14.0'	54.1'
2	213°10.7	21°00.2		268°58.1	17.4'	03°02.1	14.0'	54.0'
3	228°10.5	· · 00.6		283°34.5	17.4'	03°16.2	14.0'	54.0'
4	243°10.3	01.1		298°10.9	17.4'	03°30.2	14.0'	54.0'
5	258°10.1	01.6		312°47.4	17.4'	03°44.2	14.0'	54.0'
6	273°09.9	S21°02.0		327°23.8	17.4'	S03°58.2	14.0'	54.0'
7	288°09.7	02.5		342°00.1	17.4'	04°12.2	14.0'	54.0'
8	303°09.5	03.0		356°36.5	17.3'	04°26.1	14.0'	54.0'
9	318°09.3	· · 03.4		11°12.9	17.3'	04°40.1	13.9'	54.0'
10	333°09.1	03.9		25°49.2	17.3'	04°54.0	13.9'	54.0'
11	348°08.9	04.4		40°25.5	17.3'	05°08.0	13.9'	54.0'
12	3°08.7	S21°04.8		55°01.8	17.3'	S05°21.9	13.9'	54.0'
13	18°08.5	05.3		69°38.0	17.2'	05°35.8	13.9'	54.0'
14	33°08.3	05.7		84°14.3	17.2'	05°49.6	13.9'	54.0'
15	48°08.0	· · 06.2		98°50.5	17.2'	06°03.5	13.8'	54.0'
16	63°07.8	06.6		113°26.7	17.2'	06°17.3	13.8'	54.0'
17	78°07.6	07.1		128°02.9	17.1'	06°31.1	13.8'	54.0'
18	93°07.4	S21°07.6		142°39.0	17.1'	S06°44.9	13.8'	54.0'
19	108°07.2	08.0		157°15.1	17.1'	06°58.7	13.7'	54.0'
20	123°07.0	08.5		171°51.2	17.0'	07°12.4	13.7'	54.0'
21	138°06.8	· · 08.9		186°27.2	17.0'	07°26.1	13.7'	54.0'
22	153°06.6	09.4		201°03.3	17.0'	07°39.8	13.7'	54.0'
23	168°06.4	09.8		215°39.2	16.9'	07°53.5	13.6'	54.1'
SD = 16.2' d = 0.5'				SD = 14.7'				
Wed	GHA	Dec		GHA	ν	Dec	d	HP
0	183°06.2	S21°10.3		230°15.2	16.9'	S08°07.2	13.6'	54.1'
1	198°06.0	10.7		244°51.1	16.9'	08°20.8	13.6'	54.1'
2	213°05.8	11.2		259°27.0	16.8'	08°34.4	13.6'	54.1'
3	228°05.6	· · 11.6		274°02.8	16.8'	08°47.9	13.5'	54.1'
4	243°05.4	12.1		288°38.6	16.8'	09°01.5	13.5'	54.1'
5	258°05.2	12.5		303°14.4	16.7'	09°15.0	13.5'	54.1'
6	273°05.0	S21°13.0		317°50.1	16.7'	S09°28.5	13.4'	54.1'
7	288°04.8	13.4		332°25.8	16.6'	09°41.9	13.4'	54.1'
8	303°04.5	13.9		347°01.4	16.6'	09°55.3	13.4'	54.1'
9	318°04.3	· · 14.3		1°37.0	16.5'	10°08.7	13.4'	54.1'
10	333°04.1	14.8		16°12.5	16.5'	10°22.1	13.3'	54.1'
11	348°03.9	15.2		30°48.0	16.5'	10°35.4	13.3'	54.1'
12	3°03.7	S21°15.7		45°23.5	16.4'	S10°48.7	13.2'	54.1'
13	18°03.5	16.1		59°58.9	16.4'	11°01.9	13.2'	54.1'
14	33°03.3	16.6		74°34.3	16.3'	11°15.1	13.2'	54.1'
15	48°03.1	· · 17.0		89°09.6	16.3'	11°28.3	13.1'	54.1'
16	63°02.9	17.4		103°44.8	16.2'	11°41.5	13.1'	54.1'
17	78°02.7	17.9		118°20.0	16.1'	11°54.6	13.1'	54.1'
18	93°02.4	S21°18.3		132°55.2	16.1'	S12°07.6	13.0'	54.1'
19	108°02.2	18.8		147°30.3	16.0'	12°20.6	13.0'	54.1'
20	123°02.0	19.2		162°05.3	16.0'	12°33.6	12.9'	54.2'
21	138°01.8	· · 19.6		176°40.3	15.9'	12°46.5	12.9'	54.2'
22	153°01.6	20.1		191°15.2	15.9'	12°59.4	12.8'	54.2'
23	168°01.4	20.5		205°50.1	15.8'	13°12.3	12.8'	54.2'
SD = 16.2' d = 0.5'				SD = 14.7'				
Thu	GHA	Dec		GHA	ν	Dec	d	HP
0	183°01.2	S21°20.9		220°24.9	15.7'	S13°25.1	12.8'	54.2'
1	198°00.9	21.4		234°59.6	15.7'	13°37.8	12.7'	54.2'
2	213°00.7	21.8		249°34.3	15.6'	13°50.6	12.7'	54.2'
3	228°00.5	· · 22.3		264°08.9	15.6'	14°03.2	12.6'	54.2'
4	243°00.3	22.7		278°43.5	15.5'	14°15.8	12.6'	54.2'
5	258°00.1	23.1		293°18.0	15.4'	14°28.4	12.5'	54.2'
6	272°59.9	S21°23.6		307°52.4	15.4'	S14°40.9	12.5'	54.2'
7	287°59.7	24.0		322°26.8	15.3'	14°53.4	12.4'	54.2'
8	302°59.4	24.4		337°01.1	15.2'	15°05.8	12.4'	54.3'
9	317°59.2	· · 24.8		351°35.4	15.2'	15°18.1	12.3'	54.3'
10	332°59.0	25.3		6°09.5	15.1'	15°30.4	12.2'	54.3'
11	347°58.8	25.7		20°43.6	15.0'	15°42.7	12.2'	54.3'
12	2°58.6	S21°26.1		35°17.7	15.0'	S15°54.9	12.1'	54.3'
13	17°58.4	26.6		49°51.6	14.9'	16°07.0	12.1'	54.3'
14	32°58.1	27.0		64°25.5	14.8'	16°19.1	12.0'	54.3'
15	47°57.9	· · 27.4		78°59.3	14.7'	16°31.1	12.0'	54.3'
16	62°57.7	27.8		93°33.1	14.7'	16°43.1	11.9'	54.3'
17	77°57.5	28.3		108°06.7	14.6'	16°55.0	11.8'	54.3'
18	92°57.3	S21°28.7		122°40.3	14.5'	S17°06.8	11.8'	54.4'
19	107°57.0	29.1		137°13.9	14.4'	17°18.6	11.7'	54.4'
20	122°56.8	29.5		151°47.3	14.4'	17°30.3	11.6'	54.4'
21	137°56.6	· · 29.9		166°20.7	14.3'	17°41.9	11.6'	54.4'
22	152°56.4	30.4		180°53.9	14.2'	17°53.5	11.5'	54.4'
23	167°56.2	30.8		195°27.2	14.1'	18°05.0	11.4'	54.4'
SD = 16.2' d = 0.4'				SD = 14.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:41	09:34	■	■	14:01	15:54
N 70°	07:26	08:59	■	■	14:36	16:09
68°	07:14	08:34	10:16	13:19	15:01	16:21
66°	07:03	08:14	09:34	14:01	15:21	16:31
64°	06:55	07:59	09:06	14:29	15:36	16:40
62°	06:47	07:46	08:45	14:50	15:49	16:48
60°	06:40	07:34	08:27	15:08	16:01	16:55
N 58°	06:34	07:25	08:13	15:22	16:10	17:01
56°	06:28	07:16	08:01	15:34	16:19	17:07
54°	06:23	07:08	07:50	15:45	16:27	17:12
52°	06:19	07:01	07:40	15:55	16:34	17:16
50°	06:14	06:55	07:32	16:03	16:40	17:21
45°	06:04	06:41	07:14	16:22	16:54	17:31
N 40°	05:56	06:29	06:59	16:36	17:06	17:40
35°	05:48	06:19	06:46	16:49	17:17	17:48
30°	05:40	06:09	06:35	17:00	17:26	17:55
20°	05:26	05:53	06:16	17:19	17:43	18:10
N 10°	05:11	05:37	06:00	17:36	17:58	18:24
0°	04:56	05:22	05:44	17:51	18:14	18:39
S 10°	04:39	05:06	05:28	18:07	18:30	18:57
20°	04:19	04:47	05:11	18:24	18:48	19:17
30°	03:53	04:25	04:52	18:44	19:11	19:43
35°	03:36	04:11	04:40	18:56	19:25	20:00
40°	03:15	03:55	04:27	19:09	19:41	20:21
45°	02:49	03:35	04:11	19:25	20:01	20:48
S 50°	02:12	03:10	03:51	19:45	20:27	21:25
52°	01:51	02:57	03:42	19:54	20:40	21:46
54°	01:24	02:42	03:31	20:05	20:55	22:14
56°	00:41	02:24	03:19	20:17	21:13	23:00
58°	—:—	02:02	03:05	20:31	21:35	—:—
S 60°	—:—	01:33	02:49	20:48	22:05	—:—

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	02:46	04:48	07:13	13:10	12:32	11:33
N 70°	02:42	04:32	06:38	13:18	12:50	12:10
68°	02:38	04:20	06:12	13:24	13:04	12:37
66°	02:34	04:11	05:53	13:29	13:15	12:58
64°	02:32	04:02	05:38	13:34	13:25	13:14
62°	02:29	03:55	05:25	13:38	13:33	13:28
60°	02:27	03:49	05:15	13:42	13:40	13:40
N 58°	02:26	03:44	05:05	13:45	13:47	13:50
56°	02:24	03:40	04:57	13:48	13:52	13:59
54°	02:23	03:35	04:50	13:50	13:57	14:07
52°	02:21	03:32	04:44	13:52	14:02	14:14
50°	02:20	03:28	04:38	13:55	14:06	14:20
45°	02:18	03:21	04:26	13:59	14:15	14:34
N 40°	02:15	03:15	04:16	14:03	14:23	14:46
35°	02:14	03:09	04:07	14:06	14:30	14:56
30°	02:12	03:05	03:59	14:09	14:36	15:04
20°	02:09	02:57	03:46	14:14	14:46	15:19
N 10°	02:07	02:50	03:35	14:19	14:55	15:33
0°	02:05	02:44	03:24	14:23	15:03	15:45
S 10°	02:02	02:37	03:14	14:28	15:12	15:57
20°	02:00	02:31	03:03	14:32	15:21	16:11
30°	01:57	02:23	02:50	14:37	15:31	16:26
35°	01:56	02:19	02:43	14:41	15:37	16:35
40°	01:54	02:14	02:35	14:44	15:44	16:46
45°	01:52	02:08	02:26	14:48	15:52	16:58
S 50°	01:50	02:01	02:14	14:53	16:02	17:13
52°	01:49	01:58	02:09			

November 29, 30, 01 UT (Fri., Sat., Sun.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	68°22.4	135°55.5	S24°18.4	299°35.9	N21°13.2	351°56.1	N22°09.7	83°22.9	S08°38.5	Alpheratz	357°34.8	29°13.9
1	83°24.8	150°54.8	18.0	314°38.1	13.2	6°58.9	09.7	98°25.3	38.4	Ankaa	353°07.0	-42°10.4
2	98°27.3	165°54.1	17.5	329°40.3	13.2	22°01.8	09.7	113°27.7	38.4	Schedar	349°30.9	56°40.7
3	113°29.7	180°53.4	•• 17.1	344°42.4	•• 13.3	37°04.6	•• 09.6	128°30.2	•• 38.4	Diphda	348°47.2	-17°51.0
4	128°32.2	195°52.7	16.7	359°44.6	13.3	52°07.4	09.6	143°32.6	38.3	Achernar	335°19.8	-57°06.8
5	143°34.7	210°52.0	16.2	14°46.8	13.4	67°10.2	09.6	158°35.0	38.3	Hamal	327°51.0	23°34.9
6	158°37.1	225°51.3	S24°15.8	29°49.0	N21°13.4	82°13.0	N22°09.6	173°37.4	S08°38.3	Polaris	313°37.0	89°22.3
7	173°39.6	240°50.5	15.3	44°51.1	13.5	97°15.9	09.5	188°39.8	38.3	Acamar	315°11.4	-40°12.3
8	188°42.1	255°49.8	14.9	59°53.3	13.5	112°18.7	09.5	203°42.2	38.2	Menkar	314°06.0	4°11.3
9	203°44.5	270°49.1	•• 14.4	74°55.5	•• 13.5	127°21.5	•• 09.5	218°44.6	•• 38.2	Mirfak	308°27.9	49°57.1
10	218°47.0	285°48.4	14.0	89°57.7	13.6	142°24.3	09.4	233°47.0	38.2	Aldebaran	290°39.4	16°33.6
11	233°49.5	300°47.7	13.5	104°59.9	13.6	157°27.2	09.4	248°49.4	38.1	Rigel	281°03.6	-8°10.3
12	248°51.9	315°47.0	S24°13.1	120°02.0	N21°13.7	172°30.0	N22°09.4	263°51.9	S08°38.1	Capella	280°21.6	46°01.4
13	263°54.4	330°46.3	12.6	135°04.2	13.7	187°32.8	09.4	278°54.3	38.1	Bellatrix	278°22.7	6°22.4
14	278°56.9	345°45.6	12.2	150°06.4	13.8	202°35.6	09.3	293°56.7	38.0	Elnath	278°01.6	28°37.7
15	293°59.3	0°44.9	•• 11.7	165°08.6	•• 13.8	217°38.4	•• 09.3	308°59.1	•• 38.0	Alnilam	275°37.5	-1°11.1
16	309°01.8	15°44.2	11.3	180°10.8	13.9	232°41.3	09.3	324°01.5	38.0	Betelgeuse	270°51.9	7°24.8
17	324°04.2	30°43.5	10.8	195°13.0	13.9	247°44.1	09.3	339°03.9	38.0	Canopus	263°51.9	-52°42.3
18	339°06.7	45°42.8	S24°10.3	210°15.2	N21°14.0	262°46.9	N22°09.2	354°06.3	S08°37.9	Sirius	258°26.0	-16°44.9
19	354°09.2	60°42.1	09.9	225°17.4	14.0	277°49.7	09.2	9°08.7	37.9	Adhara	255°05.6	-29°00.2
20	9°11.6	75°41.4	09.4	240°19.6	14.1	292°52.6	09.2	24°11.1	37.9	Procyon	244°50.7	5°09.7
21	24°14.1	90°40.7	•• 09.0	255°21.8	•• 14.1	307°55.4	•• 09.2	39°13.5	•• 37.8	Pollux	243°17.2	27°57.9
22	39°16.6	105°40.0	08.5	270°24.0	14.2	322°58.2	09.1	54°15.9	37.8	Avior	234°14.4	-59°35.0
23	54°19.0	120°39.3	08.0	285°26.2	14.2	338°01.0	09.1	69°18.3	37.8	Suhail	222°46.2	-43°31.7
Mer.pass. 19:23		ν -0.7' d -0.4' m -4.18		ν 2.2' d 0.0' m -0.46		ν 2.8' d -0.0' m -2.81		ν 2.4' d -0.0' m 0.93		Miaplacidus	221°38.0	-69°48.8
Alphard 217°47.8 -8°45.9 Regulus 207°34.5 11°50.7 Dubhe 193°41.2 61°36.7 Denebola 182°25.2 14°26.0 Gienah 175°43.9 -17°40.7 Acrux 173°00.6 -63°13.9 Gacrux 171°52.1 -57°14.9 Alioth 166°13.4 55°49.3 Spica 158°22.7 -11°17.4 Alkaid 152°52.5 49°11.2 Hadar 148°36.7 -60°29.4 Menkent 147°58.1 -36°29.4 Arcturus 145°48.3 19°03.1 Rigel Kent. 139°41.1 -60°56.1 Kochab 137°20.9 74°03.0 Zuben'ubi 136°56.5 -16°08.6 Alphecca 126°04.2 26°37.8 Antares 112°16.4 -26°29.2 Atria 107°11.3 -69°04.3 Sabik 102°03.3 -15°45.3 Shaula 96°10.9 -37°07.3 Rasalhague 95°59.0 12°32.5 Eltanin 90°42.8 51°29.2 Kaus Aust. 83°33.0 -34°22.4 Vega 80°33.7 38°48.5 Nunki 75°48.2 -26°16.0 Altair 62°00.3 8°56.1 Peacock 53°06.2 -56°39.5 Deneb 49°26.1 45°22.3 Enif 33°39.0 9°59.4 Al Na'ir 27°33.1 -46°50.6 Fomalhaut 15°14.6 -29°29.6 Scheat 13°45.3 28°13.3 Markab 13°30.0 15°20.5												
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	69°21.5	135°38.6	S24°07.6	300°28.4	N21°14.3	353°03.8	N22°09.1	84°20.8	S08°37.7			
1	84°24.0	150°37.9	07.1	315°30.6	14.3	8°06.7	09.0	99°23.2	37.7			
2	99°26.4	165°37.2	06.6	330°32.8	14.4	23°09.5	09.0	114°25.6	37.7			
3	114°28.9	180°36.5	•• 06.2	345°35.0	•• 14.4	38°12.3	•• 09.0	129°28.0	•• 37.6			
4	129°31.4	195°35.8	05.7	0°37.2	14.5	53°15.1	09.0	144°30.4	37.6			
5	144°33.8	210°35.1	05.2	15°39.4	14.5	68°18.0	08.9	159°32.8	37.6			
6	159°36.3	225°34.4	S24°04.8	30°41.6	N21°14.6	83°20.8	N22°08.9	174°35.2	S08°37.6			
7	174°38.7	240°33.8	04.3	45°43.8	14.6	98°23.6	08.9	189°37.6	37.5			
8	189°41.2	255°33.1	03.8	60°46.0	14.7	113°26.4	08.9	204°40.0	37.5			
9	204°43.7	270°32.4	•• 03.3	75°48.2	•• 14.7	128°29.3	•• 08.8	219°42.4	•• 37.5			
10	219°46.1	285°31.7	02.9	90°50.4	14.8	143°32.1	08.8	234°44.8	37.4			
11	234°48.6	300°31.0	02.4	105°52.6	14.8	158°34.9	08.8	249°47.2	37.4			
12	249°51.1	315°30.3	S24°01.9	120°54.9	N21°14.9	173°37.7	N22°08.8	264°49.6	S08°37.4			
13	264°53.5	330°29.6	01.4	135°57.1	14.9	188°40.6	08.7	279°52.0	37.3			
14	279°56.0	345°28.9	00.9	150°59.3	15.0	203°43.4	08.7	294°54.5	37.3			
15	294°58.5	0°28.2	•• 00.5	166°01.5	•• 15.1	218°46.2	•• 08.7	309°56.9	•• 37.3			
16	310°00.9	15°27.5	24°00.0	181°03.7	15.1	233°49.0	08.6	324°59.3	37.2			
17	325°03.4	30°26.9	23°59.5	196°06.0	15.2	248°51.9	08.6	340°01.7	37.2			
18	340°05.9	45°26.2	S23°59.0	211°08.2	N21°15.2	263°54.7	N22°08.6	355°04.1	S08°37.2			
19	355°08.3	60°25.5	58.5	226°10.4	15.3	278°57.5	08.6	10°06.5	37.1			
20	10°10.8	75°24.8	58.0	241°12.6	15.3	294°00.3	08.5	25°08.9	37.1			
21	25°13.2	90°24.1	•• 57.5	256°14.9	•• 15.4	309°03.2	•• 08.5	40°11.3	•• 37.1			
22	40°15.7	105°23.4	57.0	271°17.1	15.4	324°06.0	08.5	55°13.7	37.0			
23	55°18.2	120°22.7	56.6	286°19.3	15.5	339°08.8	08.5	70°16.1	37.0			
Mer.pass. 19:19		ν -0.7' d -0.5' m -4.19		ν 2.2' d 0.1' m -0.48		ν 2.8' d -0.0' m -2.81		ν 2.4' d -0.0' m 0.94				
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	70°20.6	135°22.1	S23°56.1	301°21.6	N21°15.6	354°11.6	N22°08.4	85°18.5	S08°37.0			
1	85°23.1	150°21.4	55.6	316°23.8	15.6	9°14.5	08.4	100°20.9	36.9			
2	100°25.6	165°20.7	55.1	331°26.0	15.7	24°17.3	08.4	115°23.3	36.9			
3	115°28.0	180°20.0	•• 54.6	346°28.3	•• 15.7	39°20.1	•• 08.3	130°25.7	•• 36.9			
4	130°30.5	195°19.3	54.1	1°30.5	15.8	54°22.9	08.3	145°28.1	36.9			
5	145°33.0	210°18.7	53.6	16°32.7	15.9	69°25.8	08.3	160°30.5	36.8			
6	160°35.4	225°18.0	S23°53.1	31°35.0	N21°15.9	84°28.6	N22°08.3	175°32.9	S08°36.8			
7	175°37.9	240°17.3	52.6	46°37.2	16.0	99°31.4	08.2	190°35.3	36.8			
8	190°40.4	255°16.6	52.1	61°39.5	16.0	114°34.2	08.2	205°37.7	36.7			
9	205°42.8	270°15.9	•• 51.6	76°41.7	•• 16.1	129°37.1	•• 08.2	220°40.1	•• 36.7			
10	220°45.3	285°15.3	51.1	91°44.0	16.2	144°39.9	08.2	235°42.5	36.7			
11	235°47.7	300°14.6	50.6	106°46.2	16.2	159°42.7	08.1	250°44.9	36.6			
12	250°50.2	315°13.9	S23°50.1	121°48.4	N21°16.3	174°45.5	N22°08.1	265°47.3	S08°36.6			
13	265°52.7	330°13.2	49.6	136°50.7	16.4	189°48.4	08.1	280°49.7	36.6			
14	280°55.1	345°12.6	49.1	151°52.9	16.4	204°51.2	08.0	295°52.1	36.5			
15	295°57.6	0°11.9	•• 48.5	166°55.2	•• 16.5	219°54.0	•• 08.0	310°54.5	•• 36.5			
16	311°00.1	15°11.2	48.0	181°57.5	16.5	234°56.8	08.0	325°56.9	36.5			
17	326°02.5	30°10.6	47.5	196°59.7	16.6	249°59.7	08.0	340°59.3	36.4			
18	341°05.0	45°09.9	S23°47.0	212°02.0	N21°16.7	265°02.5	N22°07.9	356°01.7	S08°36.4			
19	356°07.5	60°09.2	46.5	227°04.2	16.7	280°05.3	07.9	11°04.1	36.4			
20	11°09.9	75°08.5	46.0	242°06.5	16.8	295°08.1	07.9	26°06.5	36.3			
21	26°12.4	90°07.9	•• 45.5	257°08.7	•• 16.9	310°11.0	•• 07.8	41°08.9	•• 36.3			
22	41°14.8	105°07.2	44.9	272°11.0	16.9	325°13.8	07.8	56°11.3	36.2			
23	56°17.3	120°06.5	44.4	287°13.3	17.0	340°16.6	07.8	71°13.7	3			

Sun				Moon				
Fri	GHA	Dec		GHA	ν	Dec	d	HP
0	182°55.9	S21°31.2		210°00.3	14.1'	S18°16.5	11.4'	54.4'
1	197°55.7	31.6		224°33.3	14.0'	18°27.9	11.3'	54.4'
2	212°55.5	32.0		239°06.3	13.9'	18°39.2	11.2'	54.5'
3	227°55.3	· · 32.5		253°39.2	13.8'	18°50.4	11.2'	54.5'
4	242°55.1	32.9		268°12.0	13.7'	19°01.6	11.1'	54.5'
5	257°54.8	33.3		282°44.7	13.6'	19°12.6	11.0'	54.5'
6	272°54.6	S21°33.7		297°17.4	13.6'	S19°23.7	10.9'	54.5'
7	287°54.4	34.1		311°49.9	13.5'	19°34.6	10.9'	54.5'
8	302°54.2	34.5		326°22.4	13.4'	19°45.5	10.8'	54.5'
9	317°53.9	· · 34.9		340°54.8	13.3'	19°56.2	10.7'	54.5'
10	332°53.7	35.4		355°27.1	13.2'	20°06.9	10.6'	54.6'
11	347°53.5	35.8		9°59.3	13.1'	20°17.6	10.5'	54.6'
12	2°53.3	S21°36.2		24°31.5	13.0'	S20°28.1	10.5'	54.6'
13	17°53.0	36.6		39°03.5	13.0'	20°38.6	10.4'	54.6'
14	32°52.8	37.0		53°35.5	12.9'	20°48.9	10.3'	54.6'
15	47°52.6	· · 37.4		68°07.3	12.8'	20°59.2	10.2'	54.6'
16	62°52.4	37.8		82°39.1	12.7'	21°09.4	10.1'	54.6'
17	77°52.1	38.2		97°10.8	12.6'	21°19.6	10.0'	54.7'
18	92°51.9	S21°38.6		111°42.4	12.5'	S21°29.6	9.9'	54.7'
19	107°51.7	39.0		126°14.0	12.4'	21°39.5	9.9'	54.7'
20	122°51.5	39.4		140°45.4	12.3'	21°49.4	9.8'	54.7'
21	137°51.2	· · 39.8		155°16.7	12.3'	21°59.1	9.7'	54.7'
22	152°51.0	40.2		169°48.0	12.2'	22°08.8	9.6'	54.7'
23	167°50.8	40.6		184°19.1	12.1'	22°18.4	9.5'	54.7'
SD = 16.2'		$d = 0.4'$		SD = 14.8'				
Sat	GHA	Dec		GHA	ν	Dec	d	HP
0	182°50.5	S21°41.1		198°50.2	12.0'	S22°27.8	9.4'	54.8'
1	197°50.3	41.5		213°21.2	11.9'	22°37.2	9.3'	54.8'
2	212°50.1	41.9		227°52.1	11.8'	22°46.5	9.2'	54.8'
3	227°49.9	· · 42.3		242°22.9	11.7'	22°55.7	9.1'	54.8'
4	242°49.6	42.7		256°53.6	11.6'	23°04.8	9.0'	54.8'
5	257°49.4	43.1		271°24.2	11.5'	23°13.8	8.9'	54.8'
6	272°49.2	S21°43.4		285°54.7	11.4'	S23°22.6	8.8'	54.9'
7	287°48.9	43.8		300°25.2	11.3'	23°31.4	8.7'	54.9'
8	302°48.7	44.2		314°55.5	11.3'	23°40.1	8.6'	54.9'
9	317°48.5	· · 44.6		329°25.8	11.2'	23°48.7	8.5'	54.9'
10	332°48.2	45.0		343°55.9	11.1'	23°57.1	8.4'	54.9'
11	347°48.0	45.4		358°26.0	11.0'	24°05.5	8.2'	54.9'
12	2°47.8	S21°45.8		12°56.0	10.9'	S24°13.7	8.1'	54.9'
13	17°47.6	46.2		27°25.9	10.8'	24°21.9	8.0'	55.0'
14	32°47.3	46.6		41°55.7	10.7'	24°29.9	7.9'	55.0'
15	47°47.1	· · 47.0		56°25.4	10.6'	24°37.8	7.8'	55.0'
16	62°46.9	47.4		70°55.0	10.5'	24°45.6	7.7'	55.0'
17	77°46.6	47.8		85°24.6	10.4'	24°53.3	7.6'	55.0'
18	92°46.4	S21°48.2		99°54.0	10.4'	S25°00.8	7.4'	55.0'
19	107°46.2	48.6		114°23.4	10.3'	25°08.3	7.3'	55.1'
20	122°45.9	48.9		128°52.7	10.2'	25°15.6	7.2'	55.1'
21	137°45.7	· · 49.3		143°21.8	10.1'	25°22.8	7.1'	55.1'
22	152°45.5	49.7		157°50.9	10.0'	25°29.9	7.0'	55.1'
23	167°45.2	50.1		172°19.9	9.9'	25°36.9	6.8'	55.1'
SD = 16.2'		$d = 0.4'$		SD = 14.9'				
Sun	GHA	Dec		GHA	ν	Dec	d	HP
0	182°45.0	S21°50.5		186°48.9	9.8'	S25°43.7	6.7'	55.2'
1	197°44.7	50.9		201°17.7	9.8'	25°50.4	6.6'	55.2'
2	212°44.5	51.3		215°46.5	9.7'	25°57.0	6.5'	55.2'
3	227°44.3	· · 51.6		230°15.1	9.6'	26°03.5	6.3'	55.2'
4	242°44.0	52.0		244°43.7	9.5'	26°09.8	6.2'	55.2'
5	257°43.8	52.4		259°12.2	9.4'	26°16.0	6.1'	55.2'
6	272°43.6	S21°52.8		273°40.6	9.3'	S26°22.1	5.9'	55.3'
7	287°43.3	53.2		288°09.0	9.3'	26°28.0	5.8'	55.3'
8	302°43.1	53.5		302°37.2	9.2'	26°33.9	5.7'	55.3'
9	317°42.9	· · 53.9		317°05.4	9.1'	26°39.5	5.5'	55.3'
10	332°42.6	54.3		331°33.5	9.0'	26°45.1	5.4'	55.3'
11	347°42.4	54.7		346°01.5	8.9'	26°50.5	5.3'	55.3'
12	2°42.1	S21°55.0		0°29.5	8.9'	S26°55.8	5.1'	55.4'
13	17°41.9	55.4		14°57.4	8.8'	27°00.9	5.0'	55.4'
14	32°41.7	55.8		29°25.2	8.7'	27°05.9	4.9'	55.4'
15	47°41.4	· · 56.2		43°52.9	8.7'	27°10.8	4.7'	55.4'
16	62°41.2	56.5		58°20.5	8.6'	27°15.5	4.6'	55.4'
17	77°40.9	56.9		72°48.1	8.5'	27°20.1	4.4'	55.5'
18	92°40.7	S21°57.3		87°15.6	8.4'	S27°24.5	4.3'	55.5'
19	107°40.5	57.7		101°43.1	8.4'	27°28.8	4.2'	55.5'
20	122°40.2	58.0		116°10.4	8.3'	27°33.0	4.0'	55.5'
21	137°40.0	· · 58.4		130°37.7	8.2'	27°37.0	3.9'	55.5'
22	152°39.7	58.8		145°05.0	8.2'	27°40.8	3.7'	55.5'
23	167°39.5	59.1		159°32.2	8.1'	27°44.5	3.6'	55.6'
SD = 16.2'		$d = 0.4'$		SD = 15.0'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:50	09:48	■	■	13:49	15:47
N 70°	07:33	09:09	■	■	14:28	16:03
68°	07:20	08:42	10:33	13:04	14:55	16:17
66°	07:09	08:22	09:45	13:52	15:15	16:28
64°	07:00	08:05	09:15	14:22	15:32	16:37
62°	06:52	07:51	08:52	14:45	15:46	16:45
60°	06:45	07:40	08:34	15:03	15:57	16:52
N 58°	06:39	07:30	08:19	15:18	16:08	16:59
56°	06:33	07:21	08:06	15:31	16:17	17:04
54°	06:27	07:12	07:55	15:43	16:25	17:10
52°	06:22	07:05	07:45	15:52	16:32	17:15
50°	06:18	06:59	07:36	16:01	16:39	17:19
45°	06:08	06:44	07:17	16:20	16:53	17:30
N 40°	05:58	06:32	07:02	16:35	17:05	17:39
35°	05:50	06:21	06:49	16:48	17:16	17:47
30°	05:42	06:12	06:38	17:00	17:26	17:55
20°	05:27	05:55	06:18	17:19	17:43	18:10
N 10°	05:13	05:39	06:01	17:36	17:59	18:25
0°	04:57	05:23	05:45	17:52	18:15	18:41
S 10°	04:39	05:06	05:29	18:09	18:32	18:58
20°	04:19	04:47	05:12	18:26	18:50	19:19
30°	03:52	04:24	04:51	18:47	19:14	19:46
35°	03:35	04:10	04:39	18:59	19:28	20:03
40°	03:14	03:54	04:26	19:12	19:44	20:25
45°	02:46	03:33	04:09	19:29	20:05	20:52
S 50°	02:07	03:07	03:49	19:49	20:32	21:32
52°	01:46	02:53	03:39	19:59	20:45	21:54
54°	01:16	02:38	03:28	20:10	21:01	22:24
56°	00:22	02:20	03:16	20:23	21:20	23:25
58°	—:—	01:56	03:02	20:37	21:43	—:—
S 60°	—:—	01:24	02:44	20:55	22:16	—:—

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	08:31	■	■	11:50	■	■
66°	07:50	■	■	12:31	■	■
64°	07:23	09:26	■	13:00	12:35	■
62°	07:02	08:47	10:47	13:22	13:15	13:02
60°	06:45	08:20	09:59	13:40	13:42	13:51
N 58°	06:31	08:00	09:29	13:55	14:04	14:21
56°	06:19	07:42	09:06	14:08	14:21	14:45
54°	06:08	07:28	08:47	14:19	14:36	15:04
52°	05:59	07:15	08:31	14:29	14:50	15:20
50°	05:50	07:04	08:18	14:38	15:01	15:33
45°	05:33	06:41	07:50	14:57	15:25	16:02
N 40°	05:18	06:23	07:28	15:12	15:44	16:24
35°	05:06	06:08	07:10	15:25	16:00	16:43
30°	04:56	05:54	06:55	15:37	16:14	16:59
20°	04:38	05:32	06:28	15:56	16:38	17:26
N 10°	04:22	05:13	06:06	16:14	16:59	17:49
0°	04:08	04:55	05:45	16:30	17:19	18:11
S 10°	03:53	04:37	05:25	16:46	17:38	18:33
20°	03:38	04:18	05:02	17:04	17:59	18:56
30°	03:21	03:56	04:37	17:24	18:23	19:23
35°	03:11	03:43	04:22	17:36	18:38	19:40
40°	02:59	03:29	04:04	17:49	18:54	19:58
45°	02:46	03:11	03:44	18:06	19:14	20:21
S 50°	02:30	02:50	03:18	18:26	19:40	20:50
52°	02:22	02:40	03:05	18:35	19:52	21:05
54°	02:14	02:29	02:51	18:46	20:06	21:22
56°	02:04	02:16	02:34	18:58	20:22	21:42
58°	01:54	02:01	02:14	19:13	20:42	22:07
S 60°	01:41	01:43	01:49	19:30	21:06	22:41

Day	Sun		Mer. Pass	Moon		Age 28-0 5-0%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
29	11:44	11:33	11:48	10:19	22:42	
30	11:22	11:11	11:49	11:06	23:32	
01	11:00	10:49	11:49	11:58	—:—	

December 02, 03, 04 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars			
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	71°19.8	135°05.9	S23°43.9	302°15.5	N21°17.1	355°19.4	N22°07.8	86°16.1	S08°36.2	Alpheratz	357°34.8	29°13.9	
1	86°22.2	150°05.2	43.4	317°17.8	17.1	10°22.3	07.7	101°18.5	36.1	Ankaa	353°07.0	-42°10.4	
2	101°24.7	165°04.5	42.9	332°20.1	17.2	25°25.1	07.7	116°20.9	36.1	Schedar	349°30.9	56°40.7	
3	116°27.2	180°03.9	•• 42.3	347°22.3	•• 17.3	40°27.9	•• 07.7	131°23.3	•• 36.1	Diphda	348°47.2	-17°51.0	
4	131°29.6	195°03.2	41.8	2°24.6	17.3	55°30.8	07.7	146°25.7	36.0	Achernar	335°19.8	-57°06.8	
5	146°32.1	210°02.5	41.3	17°26.9	17.4	70°33.6	07.6	161°28.1	36.0	Hamal	327°51.0	23°34.9	
6	161°34.6	225°01.9	S23°40.8	32°29.2	N21°17.5	85°36.4	N22°07.6	176°30.5	S08°36.0	Polaris	313°37.1	89°22.3	
7	176°37.0	240°01.2	40.2	47°31.4	17.6	100°39.2	07.6	191°32.9	35.9	Acamar	315°11.4	-40°12.3	
8	191°39.5	255°00.5	39.7	62°33.7	17.6	115°42.1	07.5	206°35.3	35.9	Menkar	314°06.0	4°11.3	
9	206°42.0	269°59.9	•• 39.2	77°36.0	•• 17.7	130°44.9	•• 07.5	221°37.7	•• 35.9	Mirfak	308°27.9	49°57.1	
10	221°44.4	284°59.2	38.7	92°38.3	17.8	145°47.7	07.5	236°40.1	35.8	Aldebaran	290°39.4	16°33.6	
11	236°46.9	299°58.6	38.1	107°40.5	17.8	160°50.5	07.5	251°42.5	35.8	Rigel	281°03.6	-8°10.3	
12	251°49.3	314°57.9	S23°37.6	122°42.8	N21°17.9	175°53.4	N22°07.4	266°44.9	S08°35.8	Capella	280°21.6	46°01.4	
13	266°51.8	329°57.2	37.1	137°45.1	18.0	190°56.2	07.4	281°47.3	35.7	Bellatrix	278°22.7	6°22.4	
14	281°54.3	344°56.6	36.5	152°47.4	18.0	205°59.0	07.4	296°49.7	35.7	Elnath	278°01.6	28°37.7	
15	296°56.7	359°55.9	•• 36.0	167°49.7	•• 18.1	221°01.9	•• 07.4	311°52.1	•• 35.7	Alnilam	275°37.5	-1°11.1	
16	311°59.2	14°55.3	35.4	182°52.0	18.2	236°04.7	07.3	326°54.5	35.6	Betelgeuse	270°51.9	7°24.8	
17	327°01.7	29°54.6	34.9	197°54.2	18.3	251°07.5	07.3	341°56.9	35.6	Canopus	263°51.9	-52°42.3	
18	342°04.1	44°54.0	S23°34.4	212°56.5	N21°18.3	266°10.3	N22°07.3	356°59.3	S08°35.6	Sirius	258°26.0	-16°44.9	
19	357°06.6	59°53.3	33.8	227°58.8	18.4	281°13.2	07.2	12°01.7	35.5	Adhara	255°05.6	-29°00.2	
20	12°09.1	74°52.6	33.3	243°01.1	18.5	296°16.0	07.2	27°04.1	35.5	Procyon	244°50.7	5°09.7	
21	27°11.5	89°52.0	•• 32.7	258°03.4	•• 18.6	311°18.8	•• 07.2	42°06.5	•• 35.4	Pollux	243°17.2	27°57.9	
22	42°14.0	104°51.3	32.2	273°05.7	18.6	326°21.7	07.2	57°08.9	35.4	Avior	234°14.4	-59°35.0	
23	57°16.5	119°50.7	31.7	288°08.0	18.7	341°24.5	07.1	72°11.3	35.4	Suhail	222°46.2	-43°31.7	
Mer.pass. 19:12		ν -0.7' d -0.5' m -4.20		ν 2.3' d 0.1' m -0.52		ν 2.8' d -0.0' m -2.81		ν 2.4' d -0.0' m 0.95		Miaplacidus		221°37.9	-69°48.8
										Alphard		217°47.7	-8°45.9
										Regulus		207°34.5	11°50.7
										Dubhe		193°41.2	61°36.7
										Denebola		182°25.1	14°25.9
										Gienah		175°43.8	-17°40.7
										Acrux		173°00.6	-63°13.9
										Gacrux		171°52.1	-57°14.9
										Alioth		166°13.4	55°49.3
										Spica		158°22.6	-11°17.4
										Alkaid		152°52.4	49°11.2
										Hadar		148°36.7	-60°29.4
										Menkent		147°58.1	-36°29.4
										Arcturus		145°48.3	19°03.1
										Rigel Kent.		139°41.0	-60°56.1
										Kochab		137°20.8	74°03.0
										Zuben'ubi		136°56.4	-16°08.6
										Alphecca		126°04.2	26°37.8
										Antares		112°16.3	-26°29.2
										Atria		107°11.2	-69°04.3
										Sabik		102°03.3	-15°45.3
										Shaula		96°10.9	-37°07.3
										Rasalhague		95°59.0	12°32.5
										Eltanin		90°42.8	51°29.2
										Kaus Aust.		83°33.0	-34°22.4
										Vega		80°33.7	38°48.5
										Nunki		75°48.2	-26°16.0
										Altair		62°00.3	8°56.1
										Peacock		53°06.2	-56°39.5
										Deneb		49°26.1	45°22.3
										Enif		33°39.0	9°59.4
										Al Na'ir		27°33.1	-46°50.6
										Fomalhaut		15°14.6	-29°29.6
										Scheat		13°45.3	28°13.3
										Markab		13°30.0	15°20.5
										Dec 02 Mon		SHA	Mer.pass
										Venus		63°46.1	15:00
										Mars		230°55.8	03:50
										Jupiter		283°59.7	00:19
										Saturn		14°56.4	18:12
										Dec 03 Tue		SHA	Mer.pass
										Venus		62°31.1	15:01
										Mars		230°51.4	03:47
										Jupiter		284°08.4	00:14
										Saturn		14°54.8	18:08
										Dec 04 Wed		SHA	Mer.pass
										Venus		61°16.5	15:02
										Mars		230°47.8	03:43
										Jupiter		284°17.1	00:10
										Saturn		14°53.1	18:04

Sun				Moon				
Mon	GHA	Dec		GHA	ν	Dec	d	HP
0	182°39.3	S21°59.5		173°59.3	8.1'	S27°48.1	3.4'	55.6'
1	197°39.0	21°59.9		188°26.3	8.0'	27°51.5	3.3'	55.6'
2	212°38.8	22°00.2		202°53.3	7.9'	27°54.8	3.1'	55.6'
3	227°38.5	· · 00.6		217°20.3	7.9'	27°57.9	3.0'	55.6'
4	242°38.3	01.0		231°47.1	7.8'	28°00.9	2.8'	55.7'
5	257°38.0	01.3		246°14.0	7.8'	28°03.7	2.7'	55.7'
6	272°37.8	S22°01.7		260°40.7	7.7'	S28°06.4	2.5'	55.7'
7	287°37.6	02.0		275°07.4	7.7'	28°08.9	2.4'	55.7'
8	302°37.3	02.4		289°34.1	7.6'	28°11.2	2.2'	55.7'
9	317°37.1	· · 02.8		304°00.7	7.6'	28°13.4	2.0'	55.8'
10	332°36.8	03.1		318°27.3	7.5'	28°15.5	1.9'	55.8'
11	347°36.6	03.5		332°53.8	7.5'	28°17.4	1.7'	55.8'
12	2°36.3	S22°03.8		347°20.3	7.4'	S28°19.1	1.6'	55.8'
13	17°36.1	04.2		1°46.8	7.4'	28°20.7	1.4'	55.8'
14	32°35.9	04.6		16°13.2	7.4'	28°22.1	1.3'	55.9'
15	47°35.6	· · 04.9		30°39.5	7.3'	28°23.4	1.1'	55.9'
16	62°35.4	05.3		45°05.8	7.3'	28°24.5	0.9'	55.9'
17	77°35.1	05.6		59°32.1	7.3'	28°25.4	0.8'	55.9'
18	92°34.9	S22°06.0		73°58.4	7.2'	S28°26.2	0.6'	55.9'
19	107°34.6	06.3		88°24.6	7.2'	28°26.8	0.5'	56.0'
20	122°34.4	06.7		102°50.8	7.2'	28°27.3	0.3'	56.0'
21	137°34.1	· · 07.0		117°17.0	7.1'	28°27.6	0.1'	56.0'
22	152°33.9	07.4		131°43.1	7.1'	28°27.7	-0.0'	56.0'
23	167°33.6	07.7		146°09.2	7.1'	28°27.7	-0.2'	56.0'
SD = 16.2'			d = 0.4'	SD = 15.2'				
Tue	GHA	Dec		GHA	ν	Dec	d	HP
0	182°33.4	S22°08.1		160°35.3	7.1'	S28°27.5	-0.3'	56.0'
1	197°33.1	08.4		175°01.4	7.1'	28°27.2	-0.5'	56.1'
2	212°32.9	08.8		189°27.5	7.0'	28°26.7	-0.7'	56.1'
3	227°32.6	· · 09.1		203°53.5	7.0'	28°26.0	-0.8'	56.1'
4	242°32.4	09.5		218°19.5	7.0'	28°25.2	-1.0'	56.1'
5	257°32.1	09.8		232°45.5	7.0'	28°24.2	-1.2'	56.1'
6	272°31.9	S22°10.2		247°11.5	7.0'	S28°23.1	-1.3'	56.2'
7	287°31.6	10.5		261°37.5	7.0'	28°21.8	-1.5'	56.2'
8	302°31.4	10.9		276°03.5	7.0'	28°20.3	-1.6'	56.2'
9	317°31.1	· · 11.2		290°29.5	7.0'	28°18.6	-1.8'	56.2'
10	332°30.9	11.5		304°55.5	7.0'	28°16.8	-2.0'	56.2'
11	347°30.6	11.9		319°21.4	7.0'	28°14.9	-2.1'	56.3'
12	2°30.4	S22°12.2		333°47.4	7.0'	S28°12.7	-2.3'	56.3'
13	17°30.1	12.6		348°13.4	7.0'	28°10.4	-2.5'	56.3'
14	32°29.9	12.9		2°39.4	7.0'	28°08.0	-2.6'	56.3'
15	47°29.6	· · 13.2		17°05.3	7.0'	28°05.4	-2.8'	56.3'
16	62°29.4	13.6		31°31.3	7.0'	28°02.6	-2.9'	56.4'
17	77°29.1	13.9		45°57.3	7.0'	27°59.6	-3.1'	56.4'
18	92°28.9	S22°14.2		60°23.3	7.0'	S27°56.5	-3.3'	56.4'
19	107°28.6	14.6		74°49.4	7.0'	27°53.3	-3.4'	56.4'
20	122°28.4	14.9		89°15.4	7.1'	27°49.8	-3.6'	56.4'
21	137°28.1	· · 15.3		103°41.5	7.1'	27°46.2	-3.8'	56.5'
22	152°27.9	15.6		118°07.5	7.1'	27°42.5	-3.9'	56.5'
23	167°27.6	15.9		132°33.6	7.1'	27°38.5	-4.1'	56.5'
SD = 16.2'			d = 0.3'	SD = 15.3'				
Wed	GHA	Dec		GHA	ν	Dec	d	HP
0	182°27.4	S22°16.2		146°59.7	7.1'	S27°34.5	-4.2'	56.5'
1	197°27.1	16.6		161°25.9	7.2'	27°30.2	-4.4'	56.5'
2	212°26.9	16.9		175°52.0	7.2'	27°25.8	-4.6'	56.6'
3	227°26.6	· · 17.2		190°18.2	7.2'	27°21.3	-4.7'	56.6'
4	242°26.3	17.6		204°44.4	7.2'	27°16.5	-4.9'	56.6'
5	257°26.1	17.9		219°10.7	7.3'	27°11.7	-5.0'	56.6'
6	272°25.8	S22°18.2		233°36.9	7.3'	S27°06.6	-5.2'	56.6'
7	287°25.6	18.5		248°03.2	7.3'	27°01.4	-5.4'	56.7'
8	302°25.3	18.9		262°29.6	7.4'	26°56.1	-5.5'	56.7'
9	317°25.1	· · 19.2		276°56.0	7.4'	26°50.6	-5.7'	56.7'
10	332°24.8	19.5		291°22.4	7.4'	26°44.9	-5.8'	56.7'
11	347°24.6	19.8		305°48.8	7.5'	26°39.1	-6.0'	56.7'
12	2°24.3	S22°20.2		320°15.3	7.5'	S26°33.1	-6.1'	56.8'
13	17°24.0	20.5		334°41.8	7.6'	26°27.0	-6.3'	56.8'
14	32°23.8	20.8		349°08.4	7.6'	26°20.7	-6.4'	56.8'
15	47°23.5	· · 21.1		3°35.0	7.7'	26°14.3	-6.6'	56.8'
16	62°23.3	21.5		18°01.7	7.7'	26°07.7	-6.7'	56.9'
17	77°23.0	21.8		32°28.4	7.7'	26°00.9	-6.9'	56.9'
18	92°22.7	S22°22.1		46°55.1	7.8'	S25°54.0	-7.0'	56.9'
19	107°22.5	22.4		61°21.9	7.8'	25°47.0	-7.2'	56.9'
20	122°22.2	22.7		75°48.8	7.9'	25°39.8	-7.3'	56.9'
21	137°22.0	· · 23.0		90°15.7	7.9'	25°32.4	-7.5'	57.0'
22	152°21.7	23.4		104°42.6	8.0'	25°25.0	-7.6'	57.0'
23	167°21.5	23.7		119°09.6	8.0'	25°17.3	-7.8'	57.0'
SD = 16.2'			d = 0.3'	SD = 15.4'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:57	10:02	■	■	13:38	15:42
N 70°	07:40	09:19	■	■	14:20	15:59
68°	07:27	08:50	10:52	12:48	14:49	16:13
66°	07:15	08:29	09:55	13:44	15:11	16:24
64°	07:05	08:11	09:23	14:17	15:28	16:34
62°	06:57	07:57	08:59	14:41	15:43	16:43
60°	06:49	07:45	08:40	15:00	15:55	16:50
N 58°	06:43	07:34	08:24	15:16	16:05	16:57
56°	06:37	07:25	08:11	15:29	16:15	17:03
54°	06:31	07:17	07:59	15:40	16:23	17:08
52°	06:26	07:09	07:49	15:51	16:31	17:14
50°	06:21	07:02	07:40	16:00	16:37	17:18
45°	06:11	06:47	07:21	16:19	16:52	17:29
N 40°	06:01	06:35	07:05	16:35	17:05	17:39
35°	05:52	06:24	06:52	16:48	17:16	17:47
30°	05:44	06:14	06:40	17:00	17:26	17:55
20°	05:29	05:56	06:20	17:20	17:43	18:11
N 10°	05:14	05:40	06:03	17:37	18:00	18:26
0°	04:58	05:24	05:46	17:54	18:16	18:42
S 10°	04:40	05:07	05:30	18:10	18:33	19:00
20°	04:19	04:48	05:12	18:28	18:53	19:21
30°	03:51	04:24	04:51	18:49	19:16	19:49
35°	03:34	04:10	04:39	19:01	19:31	20:07
40°	03:12	03:53	04:25	19:15	19:48	20:28
45°	02:44	03:32	04:08	19:32	20:09	20:57
S 50°	02:04	03:04	03:47	19:53	20:36	21:38
52°	01:41	02:51	03:37	20:03	20:50	22:01
54°	01:08	02:35	03:26	20:15	21:06	22:34
56°	—:—	02:15	03:13	20:28	21:26	—:—
58°	—:—	01:51	02:58	20:43	21:51	—:—
S 60°	—:—	01:16	02:40	21:01	22:27	—:—

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	■	■	■	■
66°	■	■	■	■	■	■
64°	■	■	■	■	■	■
62°	■	■	13:22	■	■	16:17
60°	11:28	12:21	12:41	14:16	15:20	16:57
N 58°	10:48	11:43	12:13	14:56	15:58	17:25
56°	10:20	11:16	11:51	15:24	16:25	17:46
54°	09:58	10:54	11:34	15:45	16:46	18:04
52°	09:40	10:37	11:18	16:03	17:04	18:18
50°	09:25	10:22	11:05	16:19	17:19	18:31
45°	08:54	09:51	10:38	16:50	17:49	18:58
N 40°	08:31	09:27	10:16	17:14	18:12	19:19
35°	08:11	09:08	09:58	17:33	18:32	19:36
30°	07:54	08:51	09:42	17:50	18:48	19:51
20°	07:26	08:22	09:16	18:19	19:16	20:17
N 10°	07:02	07:58	08:53	18:43	19:40	20:38
0°	06:39	07:35	08:32	19:06	20:03	20:58
S 10°	06:17	07:13	08:10	19:29	20:25	21:18
20°	05:53	06:48	07:47	19:53	20:48	21:40
30°	05:25	06:20	07:21	20:22	21:16	22:04
35°	05:08	06:03	07:05	20:39	21:32	22:19
40°						

December 05, 06, 07 UT (Thu., Fri., Sat.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	74°17.2	134°19.5	S23°03.7	305°02.2	N21°22.8	358°43.1	N22°05.7	89°08.5	S08°33.5	Alpheratz	357°34.8	29°13.9					
1	89°19.7	149°18.9	03.1	320°04.6	22.9	13°45.9	05.7	104°10.9	33.5	Ankaa	353°07.1	-42°10.4					
2	104°22.1	164°18.3	02.5	335°06.9	23.0	28°48.8	05.7	119°13.3	33.5	Schedar	349°30.9	56°40.7					
3	119°24.6	179°17.6	•• 01.9	350°09.3	•• 23.1	43°51.6	•• 05.7	134°15.6	•• 33.4	Diphda	348°47.2	-17°51.1					
4	134°27.1	194°17.0	01.3	5°11.7	23.2	58°54.4	05.6	149°18.0	33.4	Achernar	335°19.8	-57°06.8					
5	149°29.5	209°16.4	00.6	20°14.1	23.3	73°57.3	05.6	164°20.4	33.3	Hamal	327°51.0	23°34.9					
6	164°32.0	224°15.8	S23°00.0	35°16.4	N21°23.4	89°00.1	N22°05.6	179°22.8	S08°33.3	Polaris	313°37.5	89°22.3					
7	179°34.5	239°15.2	22°59.4	50°18.8	23.5	104°02.9	05.6	194°25.2	33.3	Acamar	315°11.4	-40°12.3					
8	194°36.9	254°14.6	58.8	65°21.2	23.6	119°05.7	05.5	209°27.6	33.2	Menkar	314°05.9	4°11.3					
9	209°39.4	269°13.9	•• 58.2	80°23.6	•• 23.7	134°08.6	•• 05.5	224°30.0	•• 33.2	Mirfak	308°27.9	49°57.1					
10	224°41.8	284°13.3	57.6	95°25.9	23.8	149°11.4	05.5	239°32.3	33.1	Aldebaran	290°39.4	16°33.6					
11	239°44.3	299°12.7	57.0	110°28.3	23.9	164°14.2	05.4	254°34.7	33.1	Rigel	281°03.6	-8°10.3					
12	254°46.8	314°12.1	S22°56.4	125°30.7	N21°24.0	179°17.1	N22°05.4	269°37.1	S08°33.1	Capella	280°21.5	46°01.4					
13	269°49.2	329°11.5	55.8	140°33.1	24.1	194°19.9	05.4	284°39.5	33.0	Bellatrix	278°22.6	6°22.4					
14	284°51.7	344°10.9	55.2	155°35.5	24.2	209°22.7	05.4	299°41.9	33.0	Elnath	278°01.6	28°37.7					
15	299°54.2	359°10.3	•• 54.6	170°37.8	•• 24.3	224°25.6	•• 05.3	314°44.3	•• 32.9	Alnilam	275°37.5	-1°11.1					
16	314°56.6	14°09.7	54.0	185°40.2	24.4	239°28.4	05.3	329°46.7	32.9	Betelgeuse	270°51.8	7°24.7					
17	329°59.1	29°09.1	53.3	200°42.6	24.5	254°31.2	05.3	344°49.0	32.9	Canopus	263°51.9	-52°42.3					
18	345°01.6	44°08.4	S22°52.7	215°45.0	N21°24.6	269°34.1	N22°05.2	359°51.4	S08°32.8	Sirius	258°26.0	-16°44.9					
19	0°04.0	59°07.8	52.1	230°47.4	24.7	284°36.9	05.2	14°53.8	32.8	Adhara	255°05.6	-29°00.2					
20	15°06.5	74°07.2	51.5	245°49.8	24.8	299°39.7	05.2	29°56.2	32.7	Procyon	244°50.7	5°09.7					
21	30°08.9	89°06.6	•• 50.9	260°52.2	•• 24.9	314°42.5	•• 05.2	44°58.6	•• 32.7	Pollux	243°17.1	27°57.9					
22	45°11.4	104°06.0	50.2	275°54.6	25.0	329°45.4	05.1	60°01.0	32.7	Avior	234°14.4	-59°35.1					
23	60°13.9	119°05.4	49.6	290°57.0	25.1	344°48.2	05.1	75°03.3	32.6	Suhail	222°46.2	-43°31.7					
Mer.pass. 19:00			ν -0.6' d -0.6' m -4.22			ν 2.4' d 0.1' m -0.59			ν 2.8' d -0.0' m -2.81			ν 2.4' d -0.0' m 0.96					

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	75°16.3	134°04.8	S22°49.0	305°59.4	N21°25.2	359°51.0	N22°05.1	90°05.7	S08°32.6					
1	90°18.8	149°04.2	48.4	321°01.8	25.3	14°53.9	05.0	105°08.1	32.5					
2	105°21.3	164°03.6	47.8	336°04.2	25.4	29°56.7	05.0	120°10.5	32.5					
3	120°23.7	179°03.0	•• 47.1	351°06.6	•• 25.5	44°59.5	•• 05.0	135°12.9	•• 32.5					
4	135°26.2	194°02.4	46.5	6°09.0	25.6	60°02.4	04.9	150°15.3	32.4					
5	150°28.7	209°01.8	45.9	21°11.4	25.7	75°05.2	04.9	165°17.6	32.4					
6	165°31.1	224°01.2	S22°45.2	36°13.8	N21°25.8	90°08.0	N22°04.9	180°20.0	S08°32.3					
7	180°33.6	239°00.6	44.6	51°16.2	25.9	105°10.8	04.9	195°22.4	32.3					
8	195°36.1	254°00.0	44.0	66°18.6	26.0	120°13.7	04.8	210°24.8	32.3					
9	210°38.5	268°59.4	•• 43.4	81°21.0	•• 26.1	135°16.5	•• 04.8	225°27.2	•• 32.2					
10	225°41.0	283°58.8	42.7	96°23.4	26.2	150°19.3	04.8	240°29.6	32.2					
11	240°43.4	298°58.2	42.1	111°25.9	26.3	165°22.2	04.7	255°31.9	32.1					
12	255°45.9	313°57.6	S22°41.5	126°28.3	N21°26.4	180°25.0	N22°04.7	270°34.3	S08°32.1					
13	270°48.4	328°57.0	40.8	141°30.7	26.5	195°27.8	04.7	285°36.7	32.0					
14	285°50.8	343°56.4	40.2	156°33.1	26.7	210°30.7	04.7	300°39.1	32.0					
15	300°53.3	358°55.9	•• 39.5	171°35.5	•• 26.8	225°33.5	•• 04.6	315°41.5	•• 32.0					
16	315°55.8	13°55.3	38.9	186°38.0	26.9	240°36.3	04.6	330°43.8	31.9					
17	330°58.2	28°54.7	38.3	201°40.4	27.0	255°39.2	04.6	345°46.2	31.9					
18	346°00.7	43°54.1	S22°37.6	216°42.8	N21°27.1	270°42.0	N22°04.5	0°48.6	S08°31.8					
19	1°03.2	58°53.5	37.0	231°45.2	27.2	285°44.8	04.5	15°51.0	31.8					
20	16°05.6	73°52.9	36.3	246°47.7	27.3	300°47.6	04.5	30°53.4	31.8					
21	31°08.1	88°52.3	•• 35.7	261°50.1	•• 27.4	315°50.5	•• 04.5	45°55.8	•• 31.7					
22	46°10.6	103°51.7	35.1	276°52.5	27.5	330°53.3	04.4	60°58.1	31.7					
23	61°13.0	118°51.1	34.4	291°54.9	27.6	345°56.1	04.4	76°00.5	31.6					
Mer.pass. 18:56			ν -0.6' d -0.6' m -4.23			ν 2.4' d 0.1' m -0.61			ν 2.8' d -0.0' m -2.81			ν 2.4' d -0.0' m 0.96		

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	76°15.5	133°50.6	S22°33.8	306°57.4	N21°27.7	0°59.0	N22°04.4	91°02.9	S08°31.6					
1	91°17.9	148°50.0	33.1	321°59.8	27.9	16°01.8	04.3	106°05.3	31.5					
2	106°20.4	163°49.4	32.5	337°02.3	28.0	31°04.6	04.3	121°07.7	31.5					
3	121°22.9	178°48.8	•• 31.8	352°04.7	•• 28.1	46°07.5	•• 04.3	136°10.0	•• 31.5					
4	136°25.3	193°48.2	31.2	7°07.1	28.2	61°10.3	04.3	151°12.4	31.4					
5	151°27.8	208°47.6	30.5	22°09.6	28.3	76°13.1	04.2	166°14.8	31.4					
6	166°30.3	223°47.1	S22°29.9	37°12.0	N21°28.4	91°16.0	N22°04.2	181°17.2	S08°31.3					
7	181°32.7	238°46.5	29.2	52°14.5	28.5	106°18.8	04.2	196°19.5	31.3					
8	196°35.2	253°45.9	28.5	67°16.9	28.6	121°21.6	04.1	211°21.9	31.2					
9	211°37.7	268°45.3	•• 27.9	82°19.3	•• 28.8	136°24.4	•• 04.1	226°24.3	•• 31.2					
10	226°40.1	283°44.7	27.2	97°21.8	28.9	151°27.3	04.1	241°26.7	31.2					
11	241°42.6	298°44.2	26.6	112°24.2	29.0	166°30.1	04.1	256°29.1	31.1					
12	256°45.1	313°43.6	S22°25.9	127°26.7	N21°29.1	181°32.9	N22°04.0	271°31.4	S08°31.1					
13	271°47.5	328°43.0	25.3	142°29.1	29.2	196°35.8	04.0	286°33.8	31.0					
14	286°50.0	343°42.4	24.6	157°31.6	29.3	211°38.6	04.0	301°36.2	31.0					
15	301°52.4	358°41.9	•• 23.9	172°34.0	•• 29.5	226°41.4	•• 03.9	316°38.6	•• 30.9					
16	316°54.9	13°41.3	23.3	187°36.5	29.6	241°44.3	03.9	331°40.9	30.9					
17	331°57.4	28°40.7	22.6	202°39.0	29.7	256°47.1	03.9	346°43.3	30.9					
18	346°59.8	43°40.1	S22°21.9	217°41.4	N21°29.8	271°49.9	N22°03.9	1°45.7	S08°30.8					
19	2°02.3	58°39.6	21.3	232°43.9	29.9	286°52.8	03.8	16°48.1	30.8					
20	17°04.8	73°39.0	20.6	247°46.3	30.0	301°55.6	03.8	31°50.5	30.7					
21	32°07.2	88°38.4	•• 19.9	262°48.8	•• 30.2	316°58.4	•• 03.8	46°52.8	•• 30.7					
22	47°09.7	103°37.9	19.3	277°51.3	30.3	332°01.2	03.7	61°55.2	30.6					
23	62°12.2	118°37.3	18.6	292°53.7	30.4	347°04.1	03.7	76°57.6	30.6					
Mer.pass. 18:52			ν -0.6' d -0.6' m -4.24			ν 2.4' d 0.1' m -0.64			ν 2.8' d -0.0' m -2.81			ν 2.4' d -0.0' m 0.97		

Dec 05 Thu	SHA	Mer.pass
Venus	60°02.3	15:03
Mars	230°45.0	03:39
Jupiter	284°25.9	00:05
Saturn	14°51.3	18:01

Dec 06 Fri	SHA	Mer.pass
Venus	58°48.5	15:04
Mars	230°43.0	03:35
Jupiter	284°34.7	00:01
Saturn	14°49.4	17:57

Dec 07 Sat	SHA	Mer.pass
Venus	57°35.1	15:05
Mars	230°41.9	03:32
Jupiter	284°43.5	23:52
Saturn	14°47.4	17:53

Horizontal parallax		
Venus:	0.2	
Mars:	0.2	

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	182°21.2	S22°24.0	133°36.6	8.1'	S25°09.5	-7.9'	57.0'
1	197°20.9	24.3	148°03.8	8.2'	25°01.6	-8.1'	57.0'
2	212°20.7	24.6	162°30.9	8.2'	24°53.5	-8.2'	57.1'
3	227°20.4	· · 24.9	176°58.1	8.3'	24°45.3	-8.4'	57.1'
4	242°20.2	25.2	191°25.4	8.3'	24°36.9	-8.5'	57.1'
5	257°19.9	25.5	205°52.7	8.4'	24°28.4	-8.6'	57.1'
6	272°19.6	S22°25.8	220°20.1	8.4'	S24°19.8	-8.8'	57.1'
7	287°19.4	26.2	234°47.5	8.5'	24°11.0	-8.9'	57.2'
8	302°19.1	26.5	249°15.0	8.6'	24°02.1	-9.1'	57.2'
9	317°18.8	· · 26.8	263°42.6	8.6'	23°53.0	-9.2'	57.2'
10	332°18.6	27.1	278°10.2	8.7'	23°43.8	-9.3'	57.2'
11	347°18.3	27.4	292°37.9	8.7'	23°34.5	-9.5'	57.2'
12	2°18.1	S22°27.7	307°05.6	8.8'	S23°25.0	-9.6'	57.3'
13	17°17.8	28.0	321°33.4	8.9'	23°15.4	-9.7'	57.3'
14	32°17.5	28.3	336°01.3	8.9'	23°05.7	-9.9'	57.3'
15	47°17.3	· · 28.6	350°29.2	9.0'	22°55.8	-10.0'	57.3'
16	62°17.0	28.9	4°57.2	9.0'	22°45.8	-10.1'	57.3'
17	77°16.7	29.2	19°25.3	9.1'	22°35.7	-10.3'	57.4'
18	92°16.5	S22°29.5	33°53.4	9.2'	S22°25.4	-10.4'	57.4'
19	107°16.2	29.8	48°21.5	9.2'	22°15.0	-10.5'	57.4'
20	122°16.0	30.1	62°49.8	9.3'	22°04.5	-10.6'	57.4'
21	137°15.7	· · 30.4	77°18.1	9.4'	21°53.9	-10.8'	57.4'
22	152°15.4	30.7	91°46.4	9.4'	21°43.1	-10.9'	57.5'
23	167°15.2	31.0	106°14.9	9.5'	21°32.2	-11.0'	57.5'
SD = 16.2'			SD = 15.5'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	182°14.9	S22°31.3	120°43.4	9.6'	S21°21.2	-11.1'	57.5'
1	197°14.6	31.6	135°11.9	9.6'	21°10.1	-11.2'	57.5'
2	212°14.4	31.9	149°40.5	9.7'	20°58.8	-11.4'	57.6'
3	227°14.1	· · 32.2	164°09.2	9.7'	20°47.5	-11.5'	57.6'
4	242°13.8	32.4	178°38.0	9.8'	20°36.0	-11.6'	57.6'
5	257°13.6	32.7	193°06.8	9.9'	20°24.4	-11.7'	57.6'
6	272°13.3	S22°33.0	207°35.7	9.9'	S20°12.7	-11.8'	57.6'
7	287°13.0	33.3	222°04.6	10.0'	20°00.8	-11.9'	57.7'
8	302°12.8	33.6	236°33.6	10.1'	19°48.9	-12.1'	57.7'
9	317°12.5	· · 33.9	251°02.6	10.1'	19°36.8	-12.2'	57.7'
10	332°12.2	34.2	265°31.8	10.2'	19°24.7	-12.3'	57.7'
11	347°12.0	34.5	280°01.0	10.2'	19°12.4	-12.4'	57.7'
12	2°11.7	S22°34.8	294°30.2	10.3'	S19°00.0	-12.5'	57.8'
13	17°11.4	35.0	308°59.5	10.4'	18°47.5	-12.6'	57.8'
14	32°11.2	35.3	323°28.9	10.4'	18°34.9	-12.7'	57.8'
15	47°10.9	· · 35.6	337°58.3	10.5'	18°22.2	-12.8'	57.8'
16	62°10.6	35.9	352°27.8	10.5'	18°09.4	-12.9'	57.8'
17	77°10.4	36.2	6°57.3	10.6'	17°56.5	-13.0'	57.9'
18	92°10.1	S22°36.5	21°26.9	10.7'	S17°43.5	-13.1'	57.9'
19	107°09.8	36.7	35°56.6	10.7'	17°30.4	-13.2'	57.9'
20	122°09.6	37.0	50°26.3	10.8'	17°17.2	-13.3'	57.9'
21	137°09.3	· · 37.3	64°56.1	10.8'	17°03.9	-13.4'	57.9'
22	152°09.0	37.6	79°25.9	10.9'	16°50.5	-13.5'	58.0'
23	167°08.7	37.9	93°55.8	10.9'	16°37.0	-13.6'	58.0'
SD = 16.2'			SD = 15.7'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	182°08.5	S22°38.1	108°25.8	11.0'	S16°23.4	-13.7'	58.0'
1	197°08.2	38.4	122°55.8	11.1'	16°09.7	-13.8'	58.0'
2	212°07.9	38.7	137°25.8	11.1'	15°55.9	-13.9'	58.1'
3	227°07.7	· · 39.0	151°55.9	11.2'	15°42.1	-14.0'	58.1'
4	242°07.4	39.2	166°26.1	11.2'	15°28.1	-14.0'	58.1'
5	257°07.1	39.5	180°56.3	11.3'	15°14.1	-14.1'	58.1'
6	272°06.8	S22°39.8	195°26.5	11.3'	S15°00.0	-14.2'	58.1'
7	287°06.6	40.0	209°56.8	11.4'	14°45.8	-14.3'	58.2'
8	302°06.3	40.3	224°27.2	11.4'	14°31.5	-14.4'	58.2'
9	317°06.0	· · 40.6	238°57.6	11.5'	14°17.1	-14.5'	58.2'
10	332°05.8	40.9	253°28.1	11.5'	14°02.6	-14.5'	58.2'
11	347°05.5	41.1	267°58.6	11.5'	13°48.1	-14.6'	58.2'
12	2°05.2	S22°41.4	282°29.1	11.6'	S13°33.5	-14.7'	58.3'
13	17°04.9	41.7	296°59.7	11.6'	13°18.8	-14.8'	58.3'
14	32°04.7	41.9	311°30.3	11.7'	13°04.0	-14.8'	58.3'
15	47°04.4	· · 42.2	326°01.0	11.7'	12°49.2	-14.9'	58.3'
16	62°04.1	42.4	340°31.7	11.8'	12°34.3	-15.0'	58.3'
17	77°03.9	42.7	355°02.5	11.8'	12°19.3	-15.1'	58.4'
18	92°03.6	S22°43.0	9°33.3	11.8'	S12°04.2	-15.1'	58.4'
19	107°03.3	43.2	24°04.1	11.9'	11°49.1	-15.2'	58.4'
20	122°03.0	43.5	38°35.0	11.9'	11°33.9	-15.3'	58.4'
21	137°02.8	· · 43.8	53°05.9	11.9'	11°18.6	-15.3'	58.4'
22	152°02.5	44.0	67°36.8	12.0'	11°03.3	-15.4'	58.5'
23	167°02.2	44.3	82°07.8	12.0'	10°47.9	-15.5'	58.5'
SD = 16.2'			SD = 15.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:05	10:15	■	■	13:27	15:37
N 70°	07:47	09:28	■	■	14:14	15:55
68°	07:32	08:58	11:14	12:28	14:44	16:10
66°	07:20	08:35	10:05	13:37	15:07	16:22
64°	07:10	08:17	09:30	14:12	15:25	16:32
62°	07:01	08:02	09:05	14:37	15:40	16:41
60°	06:54	07:49	08:45	14:57	15:53	16:48
N 58°	06:47	07:38	08:29	15:13	16:04	16:55
56°	06:40	07:29	08:15	15:27	16:13	17:02
54°	06:35	07:20	08:03	15:39	16:22	17:08
52°	06:29	07:13	07:53	15:49	16:30	17:13
50°	06:24	07:06	07:43	15:59	16:37	17:18
45°	06:13	06:50	07:24	16:18	16:52	17:29
N 40°	06:04	06:37	07:08	16:35	17:05	17:39
35°	05:55	06:26	06:54	16:48	17:16	17:47
30°	05:46	06:16	06:42	17:00	17:26	17:56
20°	05:31	05:58	06:22	17:20	17:44	18:12
N 10°	05:15	05:42	06:04	17:38	18:01	18:27
0°	04:59	05:25	05:48	17:55	18:17	18:43
S 10°	04:41	05:08	05:31	18:12	18:35	19:02
20°	04:19	04:48	05:12	18:30	18:54	19:23
30°	03:51	04:24	04:51	18:51	19:19	19:52
35°	03:33	04:10	04:39	19:04	19:33	20:09
40°	03:11	03:52	04:25	19:18	19:51	20:32
45°	02:42	03:31	04:07	19:35	20:12	21:01
S 50°	02:01	03:03	03:46	19:57	20:40	21:43
52°	01:36	02:49	03:36	20:07	20:55	22:08
54°	01:01	02:32	03:24	20:19	21:11	22:44
56°	—:—	02:12	03:11	20:32	21:32	—:—
58°	—:—	01:46	02:55	20:48	21:58	—:—
S 60°	—:—	01:08	02:37	21:07	22:37	—:—

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	■	■	14:18	■	■	20:53
N 70°	■	14:56	13:55	■	18:29	21:13
68°	■	14:14	13:37	■	19:09	21:30
66°	14:32	13:45	13:22	17:02	19:37	21:43
64°	13:43	13:23	13:10	17:50	19:58	21:53
62°	13:11	13:05	12:59	18:20	20:14	22:02
60°	12:48	12:50	12:50	18:43	20:28	22:10
N 58°	12:29	12:37	12:42	19:02	20:40	22:17
56°	12:13	12:26	12:35	19:17	20:50	22:22
54°	11:59	12:16	12:29	19:30	20:59	22:28
52°	11:47	12:08	12:23	19:42	21:07	22:32
50°	11:36	12:00	12:18	19:52	21:14	22:37
45°	11:14	11:43	12:07	20:13	21:29	22:46
N 40°	10:56	11:29	11:57	20:30	21:42	22:54
35°	10:41	11:18	11:49	20:44	21:52	23:00
30°	10:28	11:07	11:42	20:56	22:01	23:06
20°	10:05	10:49	11:30	21:17	22:17	23:16
N 10°	09:45	10:34	11:19	21:35	22:31	23:24
0°	09:27	10:19	11:09	21:52	22:43	23:32
S 10°	09:08	10:04	10:59	22:09	22:56	23:40
20°	08:48	09:49	10:48	22:26	23:09	23:48
30°	08:25	09:30	10:35	22:47	23:24	23:57
35°	08:11	09:20	10:28	22:58	23:33	· · ·
40°	07:56	09:07	10:19	23:12	23:42	· · ·
45°	07:37	08:53	10:09	23:28	23:54	· · ·
S 50°	07:13	08:35	09:58	23:47	· · ·	00:08
52°	07:02	08:26	09:52	23:56	· · ·	00:14
54°	06:49	08:17	09:46	· · ·	00:06	00:21
56°	06:34	08:06	09:39	00:01	00:18	00:29
58°	06:16	07:54	09:31	00:19	00:31	00:38
S 60°	05:54	07:39	09:22	00:42	00:46	00:48

Day	Sun		Mer.	Moon		Age
	Eqn. of Time	12 ^h	Pass	Upper	Lower	4-6
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	14-32%
05	09:25	09:12	11:51	15:39	03:13	
06	09:00	08:47	11:51	16:31	04:06	
07	08:34	08:21	11:52	17:21	04:56	

December 08, 09, 10 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars			
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	77°14.6	133°36.7	S22°17.9	307°56.2	N21°30.5	2°06.9	N22°03.7	92°00.0	S08°30.6	Alpheratz	357°34.8	29°13.9	
1	92°17.1	148°36.2	17.3	322°58.7	30.6	17°09.7	03.6	107°02.3	30.5	Ankaa	353°07.1	-42°10.4	
2	107°19.5	163°35.6	16.6	338°01.1	30.8	32°12.6	03.6	122°04.7	30.5	Schedar	349°30.9	56°40.7	
3	122°22.0	178°35.0	•• 15.9	353°03.6	•• 30.9	47°15.4	•• 03.6	137°07.1	•• 30.4	Diphda	348°47.2	-17°51.1	
4	137°24.5	193°34.5	15.2	8°06.1	31.0	62°18.2	03.6	152°09.5	30.4	Achernar	335°19.9	-57°06.8	
5	152°26.9	208°33.9	14.6	23°08.6	31.1	77°21.1	03.5	167°11.8	30.3	Hamal	327°51.0	23°34.9	
6	167°29.4	223°33.3	S22°13.9	38°11.0	N21°31.2	92°23.9	N22°03.5	182°14.2	S08°30.3	Polaris	313°38.1	89°22.3	
7	182°31.9	238°32.8	13.2	53°13.5	31.4	107°26.7	03.5	197°16.6	30.2	Acamar	315°11.5	-40°12.4	
8	197°34.3	253°32.2	12.5	68°16.0	31.5	122°29.6	03.4	212°19.0	30.2	Menkar	314°05.9	4°11.3	
9	212°36.8	268°31.6	•• 11.8	83°18.5	•• 31.6	137°32.4	•• 03.4	227°21.3	•• 30.2	Mirfak	308°27.9	49°57.1	
10	227°39.3	283°31.1	11.2	98°21.0	31.7	152°35.2	03.4	242°23.7	30.1	Aldebaran	290°39.4	16°33.6	
11	242°41.7	298°30.5	10.5	113°23.4	31.9	167°38.0	03.4	257°26.1	30.1	Rigel	281°03.6	-8°10.3	
12	257°44.2	313°30.0	S22°09.8	128°25.9	N21°32.0	182°40.9	N22°03.3	272°28.5	S08°30.0	Capella	280°21.5	46°01.4	
13	272°46.7	328°29.4	09.1	143°28.4	32.1	197°43.7	03.3	287°30.8	30.0	Bellatrix	278°22.6	6°22.4	
14	287°49.1	343°28.9	08.4	158°30.9	32.2	212°46.5	03.3	302°33.2	29.9	Elnath	278°01.6	28°37.7	
15	302°51.6	358°28.3	•• 07.7	173°33.4	•• 32.4	227°49.4	•• 03.2	317°35.6	•• 29.9	Alnilam	275°37.5	-1°11.1	
16	317°54.0	13°27.7	07.1	188°35.9	32.5	242°52.2	03.2	332°37.9	29.8	Betelgeuse	270°51.8	7°24.7	
17	332°56.5	28°27.2	06.4	203°38.4	32.6	257°55.0	03.2	347°40.3	29.8	Canopus	263°51.9	-52°42.4	
18	347°59.0	43°26.6	S22°05.7	218°40.9	N21°32.7	272°57.9	N22°03.2	2°42.7	S08°29.8	Sirius	258°26.0	-16°44.9	
19	3°01.4	58°26.1	05.0	233°43.4	32.9	288°00.7	03.1	17°45.1	29.7	Adhara	255°05.6	-29°00.2	
20	18°03.9	73°25.5	04.3	248°45.9	33.0	303°03.5	03.1	32°47.4	29.7	Procyon	244°50.6	5°09.7	
21	33°06.4	88°25.0	•• 03.6	263°48.4	•• 33.1	318°06.3	•• 03.1	47°49.8	•• 29.6	Pollux	243°17.1	27°57.9	
22	48°08.8	103°24.4	02.9	278°50.9	33.2	333°09.2	03.0	62°52.2	29.6	Avior	234°14.3	-59°35.1	
23	63°11.3	118°23.9	02.2	293°53.4	33.4	348°12.0	03.0	77°54.6	29.5	Suhail	222°46.1	-43°31.7	
Mer.pass. 18:48		ν -0.6' d -0.7' m -4.25		ν 2.5' d 0.1' m -0.66		ν 2.8' d -0.0' m -2.81		ν 2.4' d -0.0' m 0.97		Miaplacidus		221°37.8	-69°48.8
										Alphard		217°47.7	-8°45.9
										Regulus		207°34.4	11°50.7
										Dubhe		193°41.1	61°36.7
										Denebola		182°25.1	14°25.9
										Gienah		175°43.8	-17°40.7
										Acrux		173°00.5	-63°13.9
										Gacrux		171°52.0	-57°14.9
										Alioth		166°13.3	55°49.2
										Spica		158°22.6	-11°17.4
										Alkaid		152°52.4	49°11.1
										Hadar		148°36.6	-60°29.4
										Menkent		147°58.0	-36°29.4
										Arcturus		145°48.2	19°03.1
										Rigil Kent.		139°41.0	-60°56.1
										Kochab		137°20.8	74°02.9
										Zuben'ubi		136°56.4	-16°08.6
										Alphecca		126°04.2	26°37.8
										Antares		112°16.3	-26°29.2
										Atria		107°11.2	-69°04.3
										Sabik		102°03.2	-15°45.3
										Shaula		96°10.9	-37°07.3
										Rasalhague		95°59.0	12°32.5
										Eltanin		90°42.8	51°29.1
										Kaus Aust.		83°33.0	-34°22.4
										Vega		80°33.7	38°48.4
										Nunki		75°48.2	-26°16.0
										Altair		62°00.3	8°56.1
										Peacock		53°06.3	-56°39.5
										Deneb		49°26.2	45°22.3
										Enif		33°39.0	9°59.4
										Al Na'ir		27°33.1	-46°50.6
										Fomalhaut		15°14.6	-29°29.6
										Scheat		13°45.3	28°13.3
										Markab		13°30.0	15°20.5
										Dec 08 Sun		SHA	Mer.pass
										Venus		56°22.1	15:06
										Mars		230°41.6	03:28
										Jupiter		284°52.3	23:47
										Saturn		14°45.3	17:49
										Dec 09 Mon		SHA	Mer.pass
										Venus		55°09.6	15:07
										Mars		230°42.1	03:24
										Jupiter		285°01.1	23:43
										Saturn		14°43.2	17:45
										Dec 10 Tue		SHA	Mer.pass
										Venus		53°57.5	15:08
										Mars		230°43.5	03:20
										Jupiter		285°09.9	23:38
										Saturn		14°40.9	17:42

Sun				Moon										
Sun	GHA	Dec		GHA	ν	Dec	d	HP						
0	182°01.9	S22°44.5		96°38.8	12.0'	S10°32.4	-15.5'	58.5'						
1	197°01.7	44.8		111°09.8	12.1'	10°16.9	-15.6'	58.5'						
2	212°01.4	45.0		125°40.9	12.1'	10°01.3	-15.6'	58.5'						
3	227°01.1	· · 45.3		140°12.0	12.1'	09°45.7	-15.7'	58.6'						
4	242°00.8	45.6		154°43.1	12.2'	09°30.0	-15.8'	58.6'						
5	257°00.6	45.8		169°14.3	12.2'	09°14.2	-15.8'	58.6'						
6	272°00.3	S22°46.1		183°45.5	12.2'	S08°58.4	-15.9'	58.6'						
7	287°00.0	46.3		198°16.7	12.2'	08°42.6	-15.9'	58.6'						
8	301°59.7	46.6		212°47.9	12.3'	08°26.6	-16.0'	58.7'						
9	316°59.5	· · 46.8		227°19.2	12.3'	08°10.7	-16.0'	58.7'						
10	331°59.2	47.1		241°50.4	12.3'	07°54.6	-16.1'	58.7'						
11	346°58.9	47.3		256°21.7	12.3'	07°38.6	-16.1'	58.7'						
12	1°58.6	S22°47.6		270°53.0	12.3'	S07°22.5	-16.2'	58.7'						
13	16°58.3	47.8		285°24.4	12.3'	07°06.3	-16.2'	58.8'						
14	31°58.1	48.1		299°55.7	12.4'	06°50.1	-16.3'	58.8'						
15	46°57.8	· · 48.3		314°27.1	12.4'	06°33.8	-16.3'	58.8'						
16	61°57.5	48.6		328°58.4	12.4'	06°17.5	-16.3'	58.8'						
17	76°57.2	48.8		343°29.8	12.4'	06°01.2	-16.4'	58.8'						
18	91°57.0	S22°49.0		358°01.2	12.4'	S05°44.8	-16.4'	58.9'						
19	106°56.7	49.3		12°32.6	12.4'	05°28.4	-16.5'	58.9'						
20	121°56.4	49.5		27°04.0	12.4'	05°11.9	-16.5'	58.9'						
21	136°56.1	· · 49.8		41°35.4	12.4'	04°55.4	-16.5'	58.9'						
22	151°55.8	50.0		56°06.9	12.4'	04°38.9	-16.6'	58.9'						
23	166°55.6	50.3		70°38.3	12.4'	04°22.3	-16.6'	59.0'						
SD = 16.2'		$d = 0.3'$		SD = 16.0'										
Mon	GHA	Dec		GHA	ν	Dec	d	HP						
0	181°55.3	S22°50.5		85°09.7	12.4'	S04°05.7	-16.6'	59.0'						
1	196°55.0	50.7		99°41.1	12.4'	03°49.1	-16.7'	59.0'						
2	211°54.7	51.0		114°12.6	12.4'	03°32.5	-16.7'	59.0'						
3	226°54.4	· · 51.2		128°44.0	12.4'	03°15.8	-16.7'	59.0'						
4	241°54.2	51.4		143°15.4	12.4'	02°59.1	-16.7'	59.0'						
5	256°53.9	51.7		157°46.9	12.4'	02°42.4	-16.8'	59.1'						
6	271°53.6	S22°51.9		172°18.3	12.4'	S02°25.6	-16.8'	59.1'						
7	286°53.3	52.2		186°49.7	12.4'	02°08.8	-16.8'	59.1'						
8	301°53.0	52.4		201°21.1	12.4'	01°52.0	-16.8'	59.1'						
9	316°52.8	· · 52.6		215°52.5	12.4'	01°35.2	-16.8'	59.1'						
10	331°52.5	52.8		230°23.9	12.4'	01°18.4	-16.9'	59.2'						
11	346°52.2	53.1		244°55.2	12.4'	01°01.5	-16.9'	59.2'						
12	1°51.9	S22°53.3		259°26.6	12.3'	S00°44.6	-16.9'	59.2'						
13	16°51.6	53.5		273°57.9	12.3'	00°27.7	-16.9'	59.2'						
14	31°51.4	53.8		288°29.2	12.3'	S00°10.9	-16.9'	59.2'						
15	46°51.1	· · 54.0		303°00.5	12.3'	N00°06.1	16.9'	59.2'						
16	61°50.8	54.2		317°31.8	12.3'	00°23.0	16.9'	59.3'						
17	76°50.5	54.4		332°03.1	12.2'	00°39.9	16.9'	59.3'						
18	91°50.2	S22°54.7		346°34.3	12.2'	N00°56.8	16.9'	59.3'						
19	106°49.9	54.9		1°05.5	12.2'	01°13.8	16.9'	59.3'						
20	121°49.7	55.1		15°36.7	12.2'	01°30.7	16.9'	59.3'						
21	136°49.4	· · 55.3		30°07.9	12.1'	01°47.7	16.9'	59.3'						
22	151°49.1	55.6		44°39.0	12.1'	02°04.6	16.9'	59.4'						
23	166°48.8	55.8		59°10.1	12.1'	02°21.6	16.9'	59.4'						
SD = 16.2'		$d = 0.2'$		SD = 16.1'										
Tue	GHA	Dec		GHA	ν	Dec	d	HP						
0	181°48.5	S22°56.0		73°41.2	12.0'	N02°38.5	16.9'	59.4'						
1	196°48.2	56.2		88°12.3	12.0'	02°55.4	16.9'	59.4'						
2	211°48.0	56.4		102°43.3	12.0'	03°12.4	16.9'	59.4'						
3	226°47.7	· · 56.7		117°14.3	11.9'	03°29.3	16.9'	59.4'						
4	241°47.4	56.9		131°45.2	11.9'	03°46.2	16.9'	59.5'						
5	256°47.1	57.1		146°16.1	11.9'	04°03.1	16.9'	59.5'						
6	271°46.8	S22°57.3		160°46.9	11.8'	N04°20.0	16.9'	59.5'						
7	286°46.5	57.5		175°17.8	11.8'	04°36.9	16.9'	59.5'						
8	301°46.3	57.7		189°48.5	11.7'	04°53.8	16.9'	59.5'						
9	316°46.0	· · 58.0		204°19.3	11.7'	05°10.6	16.8'	59.5'						
10	331°45.7	58.2		218°50.0	11.6'	05°27.5	16.8'	59.5'						
11	346°45.4	58.4		233°20.6	11.6'	05°44.3	16.8'	59.6'						
12	1°45.1	S22°58.6		247°51.2	11.5'	N06°01.1	16.8'	59.6'						
13	16°44.8	58.8		262°21.7	11.5'	06°17.9	16.8'	59.6'						
14	31°44.5	59.0		276°52.2	11.4'	06°34.6	16.7'	59.6'						
15	46°44.3	· · 59.2		291°22.7	11.4'	06°51.3	16.7'	59.6'						
16	61°44.0	59.4		305°53.1	11.3'	07°08.0	16.7'	59.6'						
17	76°43.7	59.6		320°23.4	11.3'	07°24.7	16.6'	59.6'						
18	91°43.4	S22°59.8		334°53.7	11.2'	N07°41.4	16.6'	59.6'						
19	106°43.1	23°00.0		349°23.9	11.2'	07°58.0	16.6'	59.7'						
20	121°42.8	00.3		3°54.1	11.1'	08°14.5	16.5'	59.7'						
21	136°42.5	· · 00.5		18°24.2	11.0'	08°31.1	16.5'	59.7'						
22	151°42.2	00.7		32°54.2	11.0'	08°47.6	16.5'	59.7'						
23	166°42.0	00.9		47°24.2	10.9'	09°04.0	16.4'	59.7'						
SD = 16.2'		$d = 0.2'$		SD = 16.2'										

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:11	10:27	■	■	13:17	15:34
N 70°	07:52	09:36	■	■	14:09	15:52
68°	07:37	09:04	■	■	14:41	16:07
66°	07:25	08:40	10:14	13:31	15:04	16:20
64°	07:14	08:22	09:37	14:08	15:23	16:30
62°	07:05	08:06	09:10	14:34	15:38	16:39
60°	06:57	07:53	08:50	14:55	15:51	16:47
N 58°	06:50	07:42	08:33	15:12	16:03	16:55
56°	06:44	07:32	08:19	15:26	16:12	17:01
54°	06:38	07:24	08:07	15:38	16:21	17:07
52°	06:32	07:16	07:56	15:49	16:29	17:12
50°	06:27	07:09	07:47	15:58	16:36	17:18
45°	06:16	06:53	07:27	16:18	16:52	17:29
N 40°	06:06	06:40	07:10	16:35	17:05	17:39
35°	05:57	06:29	06:57	16:48	17:16	17:48
30°	05:49	06:18	06:45	17:00	17:27	17:56
20°	05:33	06:00	06:24	17:21	17:45	18:12
N 10°	05:17	05:43	06:06	17:39	18:02	18:28
0°	05:00	05:26	05:49	17:56	18:19	18:45
S 10°	04:42	05:09	05:32	18:13	18:36	19:03
20°	04:20	04:49	05:13	18:32	18:56	19:26
30°	03:51	04:24	04:52	18:53	19:21	19:54
35°	03:33	04:10	04:39	19:06	19:36	20:12
40°	03:11	03:52	04:24	19:21	19:53	20:35
45°	02:41	03:30	04:07	19:38	20:15	21:04
S 50°	01:58	03:01	03:45	20:00	20:44	21:48
52°	01:33	02:47	03:35	20:11	20:59	22:14
54°	00:54	02:30	03:23	20:23	21:16	22:53
56°	—:—	02:09	03:09	20:36	21:37	—:—
58°	—:—	01:42	02:53	20:52	22:05	—:—
S 60°	—:—	01:00	02:34	21:12	22:47	—:—

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	13:30	12:52	12:15	23:23	· · ·	01:46
N 70°	13:20	12:51	12:23	23:30	· · ·	01:41
68°	13:12	12:50	12:29	23:35	· · ·	01:38
66°	13:05	12:50	12:35	23:39	· · ·	01:35
64°	12:59	12:49	12:40	23:43	· · ·	01:32
62°	12:54	12:49	12:44	23:46	· · ·	01:30
60°	12:49	12:49	12:48	23:49	· · ·	01:28
N 58°	12:45	12:48	12:51	23:51	· · ·	01:26
56°	12:42	12:48	12:54	23:54	· · ·	01:25
54°	12:39	12:48	12:57	23:55	· · ·	01:24
52°	12:36	12:47	12:59	23:57	· · ·	01:22
50°	12:33	12:47	13:01	23:59	· · ·	01:21
45°	12:27	12:47	13:06	· · ·	00:02	01:19
N 40°	12:23	12:46	13:10	· · ·	00:05	01:17
35°	12:18	12:46	13:14	· · ·	00:07	01:15
30°	12:15	12:46	13:17	· · ·	00:10	01:14
20°	12:08	12:45	13:23	· · ·		

December 11, 12, 13 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	80°12.0	132°57.9	S21°27.0	310°57.7	N21°40.1	5°30.7	N22°01.6	94°50.6	S08°27.2	Alpheratz	357°34.8	29°13.9
1	95°14.5	147°57.4	26.3	326°00.3	40.3	20°33.5	01.5	109°53.0	27.2	Ankaa	353°07.1	-42°10.4
2	110°17.0	162°56.9	25.6	341°02.9	40.4	35°36.3	01.5	124°55.3	27.1	Schedar	349°30.9	56°40.7
3	125°19.4	177°56.3	•• 24.8	356°05.5	•• 40.5	50°39.1	•• 01.5	139°57.7	•• 27.1	Diphda	348°47.2	-17°51.1
4	140°21.9	192°55.8	24.1	11°08.0	40.7	65°42.0	01.4	155°00.0	27.0	Achernar	335°19.9	-57°06.8
5	155°24.4	207°55.3	23.3	26°10.6	40.8	80°44.8	01.4	170°02.4	27.0	Hamal	327°51.0	23°34.9
6	170°26.8	222°54.8	S21°22.6	41°13.2	N21°41.0	95°47.6	N22°01.4	185°04.8	S08°26.9	Polaris	313°38.7	89°22.3
7	185°29.3	237°54.3	21.8	56°15.8	41.1	110°50.5	01.4	200°07.1	26.9	Acamar	315°11.5	-40°12.4
8	200°31.7	252°53.8	21.1	71°18.4	41.3	125°53.3	01.3	215°09.5	26.8	Menkar	314°06.0	4°11.3
9	215°34.2	267°53.3	•• 20.3	86°21.0	•• 41.4	140°56.1	•• 01.3	230°11.9	•• 26.8	Mirfak	308°27.9	49°57.1
10	230°36.7	282°52.8	19.6	101°23.6	41.6	155°58.9	01.3	245°14.2	26.7	Aldebaran	290°39.4	16°33.6
11	245°39.1	297°52.3	18.8	116°26.1	41.7	171°01.8	01.2	260°16.6	26.7	Rigel	281°03.6	-8°10.3
12	260°41.6	312°51.8	S21°18.1	131°28.7	N21°41.9	186°04.6	N22°01.2	275°19.0	S08°26.6	Capella	280°21.5	46°01.4
13	275°44.1	327°51.3	17.3	146°31.3	42.0	201°07.4	01.2	290°21.3	26.6	Bellatrix	278°22.6	6°22.4
14	290°46.5	342°50.8	16.6	161°33.9	42.2	216°10.3	01.1	305°23.7	26.5	Elnath	278°01.6	28°37.7
15	305°49.0	357°50.3	•• 15.8	176°36.5	•• 42.3	231°13.1	•• 01.1	320°26.0	•• 26.5	Alnilam	275°37.5	-1°11.1
16	320°51.5	12°49.8	15.1	191°39.1	42.5	246°15.9	01.1	335°28.4	26.4	Betelgeuse	270°51.8	7°24.7
17	335°53.9	27°49.3	14.3	206°41.7	42.6	261°18.7	01.1	350°30.8	26.4	Canopus	263°51.9	-52°42.4
18	350°56.4	42°48.8	S21°13.5	221°44.3	N21°42.8	276°21.6	N22°01.0	5°33.1	S08°26.3	Sirius	258°26.0	-16°44.9
19	5°58.9	57°48.3	12.8	236°46.9	43.0	291°24.4	01.0	20°35.5	26.3	Adhara	255°05.6	-29°00.2
20	21°01.3	72°47.8	12.0	251°49.5	43.1	306°27.2	01.0	35°37.8	26.2	Procyon	244°50.6	5°09.7
21	36°03.8	87°47.3	•• 11.3	266°52.1	•• 43.3	321°30.1	•• 00.9	50°40.2	•• 26.2	Pollux	243°17.1	27°57.9
22	51°06.2	102°46.8	10.5	281°54.7	43.4	336°32.9	00.9	65°42.6	26.1	Avior	234°14.3	-59°35.1
23	66°08.7	117°46.3	09.7	296°57.3	43.6	351°35.7	00.9	80°44.9	26.1	Suhail	222°46.1	-43°31.7
Mer.pass. 18:36		ν -0.5' d -0.7' m -4.27			ν 2.6' d 0.1' m -0.73		ν 2.8' d -0.0' m -2.81		ν 2.4' d -0.0' m 0.98			

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	81°11.2	132°45.8	S21°09.0	312°00.0	N21°43.7	6°38.5	N22°00.9	95°47.3	S08°26.0	Denebola	182°25.1	14°25.9
1	96°13.6	147°45.3	08.2	327°02.6	43.9	21°41.4	00.8	110°49.6	26.0	Gienah	175°43.8	-17°40.7
2	111°16.1	162°44.9	07.5	342°05.2	44.0	36°44.2	00.8	125°52.0	25.9	Acrux	173°00.4	-63°13.9
3	126°18.6	177°44.4	•• 06.7	357°07.8	•• 44.2	51°47.0	•• 00.8	140°54.4	•• 25.9	Gacrux	171°51.9	-57°14.9
4	141°21.0	192°43.9	05.9	12°10.4	44.3	66°49.8	00.7	155°56.7	25.8	Alioth	166°13.3	55°49.2
5	156°23.5	207°43.4	05.2	27°13.0	44.5	81°52.7	00.7	170°59.1	25.8	Spica	158°22.6	-11°17.4
6	171°26.0	222°42.9	S21°04.4	42°15.6	N21°44.7	96°55.5	N22°00.7	186°01.4	S08°25.7	Alkaid	152°52.4	49°11.1
7	186°28.4	237°42.4	03.6	57°18.3	44.8	111°58.3	00.6	201°03.8	25.7	Hadar	148°36.6	-60°29.3
8	201°30.9	252°41.9	02.8	72°20.9	45.0	127°01.2	00.6	216°06.2	25.6	Menkent	147°58.0	-36°29.4
9	216°33.4	267°41.4	•• 02.1	87°23.5	•• 45.1	142°04.0	•• 00.6	231°08.5	•• 25.6	Arcturus	145°48.2	19°03.1
10	231°35.8	282°41.0	01.3	102°26.1	45.3	157°06.8	00.6	246°10.9	25.5	Rigil Kent.	139°41.0	-60°56.1
11	246°38.3	297°40.5	21°00.5	117°28.8	45.4	172°09.6	00.5	261°13.2	25.5	Kochab	137°20.7	74°02.9
12	261°40.7	312°40.0	S20°59.8	132°31.4	N21°45.6	187°12.5	N22°00.5	276°15.6	S08°25.4	Zuben'ubi	136°56.4	-16°08.6
13	276°43.2	327°39.5	59.0	147°34.0	45.8	202°15.3	00.5	291°18.0	25.4	Alphecca	126°04.1	26°37.8
14	291°45.7	342°39.0	58.2	162°36.6	45.9	217°18.1	00.4	306°20.3	25.3	Antares	112°16.3	-26°29.2
15	306°48.1	357°38.6	•• 57.4	177°39.3	•• 46.1	232°20.9	•• 00.4	321°22.7	•• 25.3	Atria	107°11.2	-69°04.3
16	321°50.6	12°38.1	56.7	192°41.9	46.2	247°23.8	00.4	336°25.0	25.2	Sabik	102°03.2	-15°45.3
17	336°53.1	27°37.6	55.9	207°44.5	46.4	262°26.6	00.3	351°27.4	25.2	Shaula	96°10.9	-37°07.3
18	351°55.5	42°37.1	S20°55.1	222°47.2	N21°46.6	277°29.4	N22°00.3	6°29.7	S08°25.1	Rasalhague	95°59.0	12°32.5
19	6°58.0	57°36.6	54.3	237°49.8	46.7	292°32.2	00.3	21°32.1	25.1	Eltanin	90°42.8	51°29.1
20	22°00.5	72°36.2	53.5	252°52.5	46.9	307°35.1	00.3	36°34.5	25.0	Kaus Aust.	83°33.0	-34°22.4
21	37°02.9	87°35.7	•• 52.8	267°55.1	•• 47.1	322°37.9	•• 00.2	51°36.8	•• 25.0	Vega	80°33.7	38°48.4
22	52°05.4	102°35.2	52.0	282°57.7	47.2	337°40.7	00.2	66°39.2	24.9	Nunki	75°48.2	-26°16.0
23	67°07.8	117°34.8	51.2	298°00.4	47.4	352°43.6	00.2	81°41.5	24.9	Altair	62°00.3	8°56.1
Mer.pass. 18:32		ν -0.5' d -0.8' m -4.28			ν 2.6' d 0.2' m -0.75		ν 2.8' d -0.0' m -2.81		ν 2.4' d -0.1' m 0.99			

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	82°10.3	132°34.3	S20°50.4	313°03.0	N21°47.5	7°46.4	N22°00.1	96°43.9	S08°24.8	Dec 11 Wed	SHA	Mer.pass
1	97°12.8	147°33.8	49.6	328°05.7	47.7	22°49.2	00.1	111°46.2	24.8	Venus	52°45.8	15:09
2	112°15.2	162°33.3	48.8	343°08.3	47.9	37°52.0	00.1	126°48.6	24.7	Mars	230°45.7	03:16
3	127°17.7	177°32.9	•• 48.0	358°11.0	•• 48.0	52°54.9	•• 00.0	141°51.0	•• 24.7	Jupiter	285°18.6	23:34
4	142°20.2	192°32.4	47.3	13°13.6	48.2	67°57.7	00.0	156°53.3	24.6	Saturn	14°38.6	17:38
5	157°22.6	207°31.9	46.5	28°16.3	48.4	83°00.5	00.0	171°55.7	24.6			
6	172°25.1	222°31.5	S20°45.7	43°18.9	N21°48.5	98°03.3	N22°00.0	186°58.0	S08°24.5			
7	187°27.6	237°31.0	44.9	58°21.6	48.7	113°06.2	21°59.9	202°00.4	24.5			
8	202°30.0	252°30.5	44.1	73°24.3	48.9	128°09.0	59.9	217°02.7	24.4			
9	217°32.5	267°30.1	•• 43.3	88°26.9	•• 49.0	143°11.8	•• 59.9	232°05.1	•• 24.3			
10	232°35.0	282°29.6	42.5	103°29.6	49.2	158°14.6	59.8	247°07.4	24.3			
11	247°37.4	297°29.1	41.7	118°32.2	49.4	173°17.5	59.8	262°09.8	24.2			
12	262°39.9	312°28.7	S20°40.9	133°34.9	N21°49.5	188°20.3	N21°59.8	277°12.2	S08°24.2			
13	277°42.3	327°28.2	40.1	148°37.6	49.7	203°23.1	59.8	292°14.5	24.1			
14	292°44.8	342°27.8	39.3	163°40.2	49.9	218°25.9	59.7	307°16.9	24.1			
15	307°47.3	357°27.3	•• 38.5	178°42.9	•• 50.0	233°28.8	•• 59.7	322°19.2	•• 24.0			
16	322°49.7	12°26.8	37.7	193°45.6	50.2	248°31.6	59.7	337°21.6	24.0			
17	337°52.2	27°26.4	36.9	208°48.2	50.4	263°34.4	59.6	352°23.9	23.9			
18	352°54.7	42°25.9	S20°36.1	223°50.9	N21°50.5	278°37.2	N21°59.6	7°26.3	S08°23.9			
19	7°57.1	57°25.5	35.3	238°53.6	50.7	293°40.1	59.6	22°28.6	23.8			
20	22°59.6	72°25.0	34.5	253°56.3	50.9	308°42.9	59.5	37°31.0	23.8			
21	38°02.1	87°24.6	•• 33.7	268°58.9	•• 51.1	323°45.7	•• 59.5	52°33.3	•• 23.7			
22	53°04.5	102°24.1	32.9	284°01.6	51.2	338°48.5	59.5	67°35.7	23.7			
23	68°07.0	117°23.6	32.1	299°04.3	51.4	353°51.4	59.5					

Sun				Moon				
Wed	GHA	Dec		GHA	ν	Dec	d	HP
0	181°41.7	S23°01.1		61°54.1	10.8'	N09°20.5	16.4'	59.7'
1	196°41.4	01.3		76°23.9	10.8'	09°36.8	16.3'	59.7'
2	211°41.1	01.5		90°53.7	10.7'	09°53.2	16.3'	59.7'
3	226°40.8	· · 01.7		105°23.4	10.6'	10°09.5	16.2'	59.7'
4	241°40.5	01.9		119°53.1	10.6'	10°25.7	16.2'	59.8'
5	256°40.2	02.1		134°22.6	10.5'	10°41.9	16.1'	59.8'
6	271°39.9	S23°02.3		148°52.1	10.4'	N10°58.0	16.1'	59.8'
7	286°39.7	02.5		163°21.6	10.4'	11°14.1	16.0'	59.8'
8	301°39.4	02.6		177°50.9	10.3'	11°30.1	16.0'	59.8'
9	316°39.1	· · 02.8		192°20.2	10.2'	11°46.1	15.9'	59.8'
10	331°38.8	03.0		206°49.4	10.1'	12°02.0	15.9'	59.8'
11	346°38.5	03.2		221°18.5	10.0'	12°17.9	15.8'	59.8'
12	1°38.2	S23°03.4		235°47.5	10.0'	N12°33.7	15.7'	59.8'
13	16°37.9	03.6		250°16.5	9.9'	12°49.4	15.7'	59.8'
14	31°37.6	03.8		264°45.4	9.8'	13°05.1	15.6'	59.8'
15	46°37.3	· · 04.0		279°14.2	9.7'	13°20.7	15.5'	59.9'
16	61°37.1	04.2		293°42.9	9.6'	13°36.2	15.5'	59.9'
17	76°36.8	04.4		308°11.6	9.6'	13°51.6	15.4'	59.9'
18	91°36.5	S23°04.6		322°40.1	9.5'	N14°07.0	15.3'	59.9'
19	106°36.2	04.7		337°08.6	9.4'	14°22.3	15.2'	59.9'
20	121°35.9	04.9		351°36.9	9.3'	14°37.6	15.1'	59.9'
21	136°35.6	· · 05.1		6°05.2	9.2'	14°52.7	15.1'	59.9'
22	151°35.3	05.3		20°33.4	9.1'	15°07.8	15.0'	59.9'
23	166°35.0	05.5		35°01.5	9.0'	15°22.8	14.9'	59.9'
SD = 16.2' d = 0.2'				SD = 16.3'				
Thu	GHA	Dec		GHA	ν	Dec	d	HP
0	181°34.7	S23°05.7		49°29.6	8.9'	N15°37.7	14.8'	59.9'
1	196°34.4	05.8		63°57.5	8.8'	15°52.5	14.7'	59.9'
2	211°34.1	06.0		78°25.3	8.7'	16°07.2	14.6'	59.9'
3	226°33.9	· · 06.2		92°53.1	8.7'	16°21.8	14.5'	59.9'
4	241°33.6	06.4		107°20.7	8.6'	16°36.4	14.4'	59.9'
5	256°33.3	06.6		121°48.3	8.5'	16°50.8	14.4'	59.9'
6	271°33.0	S23°06.7		136°15.8	8.4'	N17°05.2	14.3'	59.9'
7	286°32.7	06.9		150°43.1	8.3'	17°19.4	14.2'	59.9'
8	301°32.4	07.1		165°10.4	8.2'	17°33.6	14.0'	59.9'
9	316°32.1	· · 07.3		179°37.6	8.1'	17°47.6	13.9'	59.9'
10	331°31.8	07.4		194°04.6	8.0'	18°01.6	13.8'	59.9'
11	346°31.5	07.6		208°31.6	7.9'	18°15.4	13.7'	59.9'
12	1°31.2	S23°07.8		222°58.5	7.8'	N18°29.1	13.6'	59.9'
13	16°30.9	08.0		237°25.3	7.7'	18°42.7	13.5'	59.9'
14	31°30.6	08.1		251°52.0	7.6'	18°56.3	13.4'	59.9'
15	46°30.3	· · 08.3		266°18.6	7.5'	19°09.6	13.3'	59.9'
16	61°30.1	08.5		280°45.1	7.4'	19°22.9	13.2'	59.9'
17	76°29.8	08.6		295°11.5	7.3'	19°36.1	13.0'	59.9'
18	91°29.5	S23°08.8		309°37.7	7.2'	N19°49.1	12.9'	59.9'
19	106°29.2	09.0		324°03.9	7.1'	20°02.0	12.8'	59.9'
20	121°28.9	09.2		338°30.0	7.0'	20°14.8	12.7'	59.9'
21	136°28.6	· · 09.3		352°56.0	6.9'	20°27.5	12.5'	59.9'
22	151°28.3	09.5		7°21.9	6.8'	20°40.0	12.4'	59.9'
23	166°28.0	09.6		21°47.7	6.7'	20°52.5	12.3'	59.9'
SD = 16.2' d = 0.2'				SD = 16.3'				
Fri	GHA	Dec		GHA	ν	Dec	d	HP
0	181°27.7	S23°09.8		36°13.4	6.6'	N21°04.7	12.1'	59.9'
1	196°27.4	10.0		50°39.0	6.5'	21°16.9	12.0'	59.9'
2	211°27.1	10.1		65°04.5	6.4'	21°28.9	11.9'	59.9'
3	226°26.8	· · 10.3		79°29.9	6.3'	21°40.8	11.7'	59.9'
4	241°26.5	10.5		93°55.2	6.2'	21°52.5	11.6'	59.9'
5	256°26.2	10.6		108°20.4	6.1'	22°04.1	11.5'	59.9'
6	271°25.9	S23°10.8		122°45.5	6.0'	N22°15.5	11.3'	59.9'
7	286°25.6	10.9		137°10.5	5.9'	22°26.9	11.2'	59.9'
8	301°25.3	11.1		151°35.4	5.8'	22°38.0	11.0'	59.9'
9	316°25.1	· · 11.2		166°00.3	5.7'	22°49.0	10.9'	59.9'
10	331°24.8	11.4		180°25.0	5.6'	22°59.9	10.7'	59.9'
11	346°24.5	11.6		194°49.6	5.5'	23°10.6	10.6'	59.9'
12	1°24.2	S23°11.7		209°14.2	5.4'	N23°21.2	10.4'	59.8'
13	16°23.9	11.9		223°38.6	5.4'	23°31.6	10.2'	59.8'
14	31°23.6	12.0		238°03.0	5.3'	23°41.8	10.1'	59.8'
15	46°23.3	· · 12.2		252°27.2	5.2'	23°51.9	9.9'	59.8'
16	61°23.0	12.3		266°51.4	5.1'	24°01.8	9.8'	59.8'
17	76°22.7	12.5		281°15.5	5.0'	24°11.6	9.6'	59.8'
18	91°22.4	S23°12.6		295°39.5	4.9'	N24°21.2	9.4'	59.8'
19	106°22.1	12.8		310°03.4	4.8'	24°30.6	9.3'	59.8'
20	121°21.8	12.9		324°27.3	4.8'	24°39.9	9.1'	59.8'
21	136°21.5	· · 13.1		338°51.0	4.7'	24°49.0	8.9'	59.8'
22	151°21.2	13.2		353°14.7	4.6'	24°57.9	8.8'	59.7'
23	166°20.9	13.3		7°38.3	4.5'	25°06.7	8.6'	59.7'
SD = 16.2' d = 0.2'				SD = 16.3'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:16	10:39	■	■	13:09	15:31
N 70°	07:57	09:43	■	■	14:05	15:50
68°	07:42	09:09	■	■	14:38	16:06
66°	07:29	08:45	10:22	13:26	15:02	16:19
64°	07:18	08:26	09:42	14:05	15:22	16:29
62°	07:09	08:10	09:15	14:33	15:37	16:39
60°	07:01	07:57	08:54	14:53	15:51	16:47
N 58°	06:53	07:46	08:37	15:11	16:02	16:54
56°	06:47	07:36	08:23	15:25	16:12	17:01
54°	06:41	07:27	08:10	15:37	16:21	17:07
52°	06:35	07:19	07:59	15:48	16:29	17:13
50°	06:30	07:11	07:50	15:58	16:36	17:18
45°	06:18	06:56	07:29	16:18	16:52	17:29
N 40°	06:08	06:42	07:13	16:35	17:05	17:39
35°	05:59	06:31	06:59	16:49	17:17	17:49
30°	05:50	06:20	06:47	17:01	17:27	17:57
20°	05:34	06:02	06:26	17:22	17:46	18:14
N 10°	05:18	05:45	06:07	17:40	18:03	18:30
0°	05:02	05:28	05:50	17:58	18:20	18:46
S 10°	04:43	05:10	05:33	18:15	18:38	19:05
20°	04:21	04:50	05:14	18:34	18:58	19:27
30°	03:52	04:25	04:52	18:56	19:23	19:56
35°	03:33	04:10	04:40	19:08	19:38	20:15
40°	03:11	03:52	04:25	19:23	19:56	20:37
45°	02:41	03:30	04:07	19:41	20:18	21:08
S 50°	01:56	03:01	03:45	20:03	20:47	21:52
52°	01:30	02:46	03:34	20:14	21:02	22:19
54°	00:49	02:29	03:22	20:26	21:20	23:01
56°	—:—	02:07	03:08	20:40	21:42	—:—
58°	—:—	01:39	02:52	20:56	22:10	—:—
S 60°	—:—	00:54	02:32	21:16	22:56	—:—

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	11:28	09:39	□	04:18	07:58	□
N 70°	11:49	10:51	□	04:00	06:48	□
68°	12:05	11:29	□	03:46	06:12	□
66°	12:18	11:56	11:10	03:34	05:46	08:31
64°	12:30	12:17	11:57	03:25	05:27	07:44
62°	12:39	12:34	12:28	03:17	05:11	07:14
60°	12:47	12:48	12:52	03:10	04:58	06:51
N 58°	12:55	13:01	13:11	03:04	04:46	06:33
56°	13:01	13:11	13:27	02:59	04:37	06:18
54°	13:07	13:21	13:40	02:54	04:28	06:05
52°	13:12	13:29	13:52	02:50	04:20	05:54
50°	13:17	13:37	14:03	02:46	04:14	05:44
45°	13:28	13:53	14:25	02:38	03:59	05:22
N 40°	13:36	14:07	14:44	02:31	03:47	05:06
35°	13:44	14:18	14:59	02:25	03:37	04:51
30°	13:51	14:28	15:12	02:20	03:28	04:39
20°	14:02	14:46	15:35	02:11	03:13	04:18
N 10°	14:13	15:01	15:55	02:03	03:00	04:00
0°	14:22	15:16	16:14	01:56	02:47	03:43
S 10°	14:32	15:31	16:33	01:48	02:35	03:

December 14, 15, 16 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars			
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	83°09.5	132°23.2	S20°31.3	314°07.0	N21°51.6	8°54.2	N21°59.4	97°40.4	S08°23.6	Alpheratz	357°34.8	29°13.9	
1	98°11.9	147°22.7	30.5	329°09.7	51.7	23°57.0	59.4	112°42.8	23.5	Ankaa	353°07.1	-42°10.4	
2	113°14.4	162°22.3	29.7	344°12.4	51.9	38°59.8	59.4	127°45.1	23.4	Schedar	349°31.0	56°40.7	
3	128°16.8	177°21.8	•• 28.9	359°15.0	•• 52.1	54°02.7	•• 59.3	142°47.5	•• 23.4	Diphda	348°47.3	-17°51.1	
4	143°19.3	192°21.4	28.1	14°17.7	52.3	69°05.5	59.3	157°49.8	23.3	Achernar	335°19.9	-57°06.8	
5	158°21.8	207°20.9	27.3	29°20.4	52.4	84°08.3	59.3	172°52.2	23.3	Hamal	327°51.0	23°34.9	
6	173°24.2	222°20.5	S20°26.5	44°23.1	N21°52.6	99°11.1	N21°59.2	187°54.5	S08°23.2	Polaris	313°39.0	89°22.4	
7	188°26.7	237°20.1	25.7	59°25.8	52.8	114°14.0	59.2	202°56.9	23.2	Acamar	315°11.5	-40°12.4	
8	203°29.2	252°19.6	24.8	74°28.5	53.0	129°16.8	59.2	217°59.2	23.1	Menkar	314°05.9	4°11.3	
9	218°31.6	267°19.2	•• 24.0	89°31.2	•• 53.1	144°19.6	•• 59.2	233°01.6	•• 23.1	Mirfak	308°27.9	49°57.1	
10	233°34.1	282°18.7	23.2	104°33.9	53.3	159°22.4	59.1	248°03.9	23.0	Aldebaran	290°39.4	16°33.6	
11	248°36.6	297°18.3	22.4	119°36.6	53.5	174°25.2	59.1	263°06.3	23.0	Rigel	281°03.6	-8°10.3	
12	263°39.0	312°17.8	S20°21.6	134°39.3	N21°53.7	189°28.1	N21°59.1	278°08.6	S08°22.9	Capella	280°21.5	46°01.4	
13	278°41.5	327°17.4	20.8	149°42.0	53.8	204°30.9	59.0	293°11.0	22.9	Bellatrix	278°22.6	6°22.4	
14	293°44.0	342°17.0	19.9	164°44.7	54.0	219°33.7	59.0	308°13.3	22.8	Elnath	278°01.6	28°37.7	
15	308°46.4	357°16.5	•• 19.1	179°47.4	•• 54.2	234°36.5	•• 59.0	323°15.7	•• 22.7	Alnilam	275°37.5	-1°11.1	
16	323°48.9	12°16.1	18.3	194°50.1	54.4	249°39.4	58.9	338°18.0	22.7	Betelgeuse	270°51.8	7°24.7	
17	338°51.3	27°15.6	17.5	209°52.8	54.5	264°42.2	58.9	353°20.4	22.6	Canopus	263°51.9	-52°42.4	
18	353°53.8	42°15.2	S20°16.7	224°55.5	N21°54.7	279°45.0	N21°58.9	8°22.7	S08°22.6	Sirius	258°26.0	-16°44.9	
19	8°56.3	57°14.8	15.8	239°58.2	54.9	294°47.8	58.9	23°25.1	22.5	Adhara	255°05.6	-29°00.2	
20	23°58.7	72°14.3	15.0	255°00.9	55.1	309°50.7	58.8	38°27.4	22.5	Procyon	244°50.6	5°09.7	
21	39°01.2	87°13.9	•• 14.2	270°03.7	•• 55.3	324°53.5	•• 58.8	53°29.8	•• 22.4	Pollux	243°17.1	27°57.9	
22	54°03.7	102°13.5	13.4	285°06.4	55.4	339°56.3	58.8	68°32.1	22.4	Avior	234°14.3	-59°35.1	
23	69°06.1	117°13.0	12.6	300°09.1	55.6	354°59.1	58.7	83°34.5	22.3	Suhail	222°46.1	-43°31.8	
Mer.pass. 18:24		ν -0.5' d -0.8' m -4.29		ν 2.7' d 0.2' m -0.80		ν 2.8' d -0.0' m -2.81		ν 2.4' d -0.1' m 0.99		Miaplacidus		221°37.7	-69°48.8
										Alphard		217°47.6	-8°45.9
										Regulus		207°34.4	11°50.7
										Dubhe		193°41.0	61°36.7
										Denebola		182°25.0	14°25.9
										Gienah		175°43.7	-17°40.7
										Acrux		173°00.4	-63°13.9
										Gacrux		171°51.9	-57°14.9
										Alioth		166°13.2	55°49.2
										Spica		158°22.5	-11°17.4
										Alkaid		152°52.3	49°11.1
										Hadar		148°36.5	-60°29.3
										Menkent		147°58.0	-36°29.4
										Arcturus		145°48.2	19°03.1
										Rigil Kent.		139°40.9	-60°56.1
										Kochab		137°20.7	74°02.9
										Zuben'ubi		136°56.4	-16°08.7
										Alphecca		126°04.1	26°37.7
										Antares		112°16.3	-26°29.2
										Atria		107°11.2	-69°04.2
										Sabik		102°03.2	-15°45.3
										Shaula		96°10.9	-37°07.3
										Rasalhague		95°59.0	12°32.5
										Eltanin		90°42.8	51°29.1
										Kaus Aust.		83°33.0	-34°22.4
										Vega		80°33.7	38°48.4
										Nunki		75°48.2	-26°16.0
										Altair		62°00.3	8°56.1
										Peacock		53°06.3	-56°39.5
										Deneb		49°26.2	45°22.3
										Enif		33°39.0	9°59.4
										Al Na'ir		27°33.2	-46°50.6
										Fomalhaut		15°14.7	-29°29.6
										Scheat		13°45.3	28°13.2
										Markab		13°30.0	15°20.5
										Dec 14 Sat		SHA	Mer.pass
										Venus		49°13.7	15:11
										Mars		230°57.5	03:03
										Jupiter		285°44.7	23:20
										Saturn		14°30.9	17:27
										Dec 15 Sun		SHA	Mer.pass
										Venus		48°04.0	15:12
										Mars		231°03.2	02:59
										Jupiter		285°53.3	23:15
										Saturn		14°28.2	17:23
										Dec 16 Mon		SHA	Mer.pass
										Venus		46°54.8	15:12
										Mars		231°09.8	02:54
										Jupiter		286°01.9	23:11
										Saturn		14°25.4	17:19

Sun							Moon								
Sat	GHA	Dec	GHA	ν	Dec	d	HP								
0	181°20.6	S23°13.5	22°01.8	4.4'	N25°15.2	8.4'	59.7'								
1	196°20.3	13.6	36°25.2	4.4'	25°23.7	8.2'	59.7'								
2	211°20.0	13.8	50°48.6	4.3'	25°31.9	8.1'	59.7'								
3	226°19.7	· · 13.9	65°11.9	4.2'	25°40.0	7.9'	59.7'								
4	241°19.4	14.1	79°35.1	4.1'	25°47.8	7.7'	59.7'								
5	256°19.1	14.2	93°58.2	4.1'	25°55.5	7.5'	59.7'								
6	271°18.8	S23°14.3	108°21.3	4.0'	N26°03.1	7.3'	59.6'								
7	286°18.5	14.5	122°44.3	3.9'	26°10.4	7.2'	59.6'								
8	301°18.2	14.6	137°07.3	3.9'	26°17.5	7.0'	59.6'								
9	316°17.9	· · 14.8	151°30.1	3.8'	26°24.5	6.8'	59.6'								
10	331°17.6	14.9	165°53.0	3.8'	26°31.3	6.6'	59.6'								
11	346°17.3	15.0	180°15.7	3.7'	26°37.9	6.4'	59.6'								
12	1°17.0	S23°15.2	194°38.5	3.7'	N26°44.3	6.2'	59.5'								
13	16°16.7	15.3	209°01.1	3.6'	26°50.5	6.0'	59.5'								
14	31°16.4	15.4	223°23.7	3.6'	26°56.6	5.8'	59.5'								
15	46°16.1	· · 15.6	237°46.3	3.5'	27°02.4	5.6'	59.5'								
16	61°15.8	15.7	252°08.8	3.5'	27°08.0	5.5'	59.5'								
17	76°15.5	15.8	266°31.3	3.4'	27°13.5	5.3'	59.5'								
18	91°15.2	S23°16.0	280°53.8	3.4'	N27°18.8	5.1'	59.4'								
19	106°14.9	16.1	295°16.2	3.4'	27°23.8	4.9'	59.4'								
20	121°14.6	16.2	309°38.5	3.3'	27°28.7	4.7'	59.4'								
21	136°14.3	· · 16.3	324°00.9	3.3'	27°33.4	4.5'	59.4'								
22	151°14.0	16.5	338°23.2	3.3'	27°37.9	4.3'	59.4'								
23	166°13.7	16.6	352°45.4	3.3'	27°42.2	4.1'	59.3'								
SD = 16.2'			$d = 0.1'$			SD = 16.3'									
Sun	GHA	Dec	GHA	ν	Dec	d	HP								
0	181°13.4	S23°16.7	7°07.7	3.2'	N27°46.3	3.9'	59.3'								
1	196°13.1	16.8	21°29.9	3.2'	27°50.1	3.7'	59.3'								
2	211°12.8	17.0	35°52.1	3.2'	27°53.8	3.5'	59.3'								
3	226°12.5	· · 17.1	50°14.3	3.2'	27°57.3	3.3'	59.3'								
4	241°12.2	17.2	64°36.5	3.2'	28°00.6	3.1'	59.2'								
5	256°11.9	17.3	78°58.7	3.2'	28°03.7	2.9'	59.2'								
6	271°11.6	S23°17.4	93°20.8	3.2'	N28°06.6	2.7'	59.2'								
7	286°11.3	17.6	107°43.0	3.2'	28°09.4	2.5'	59.2'								
8	301°11.0	17.7	122°05.2	3.2'	28°11.9	2.3'	59.1'								
9	316°10.7	· · 17.8	136°27.3	3.2'	28°14.2	2.1'	59.1'								
10	331°10.4	17.9	150°49.5	3.2'	28°16.3	1.9'	59.1'								
11	346°10.1	18.0	165°11.6	3.2'	28°18.2	1.7'	59.1'								
12	1°09.8	S23°18.2	179°33.8	3.2'	N28°19.9	1.5'	59.1'								
13	16°09.5	18.3	193°56.0	3.2'	28°21.4	1.3'	59.0'								
14	31°09.2	18.4	208°18.2	3.2'	28°22.7	1.1'	59.0'								
15	46°08.9	· · 18.5	222°40.5	3.3'	28°23.8	0.9'	59.0'								
16	61°08.6	18.6	237°02.7	3.3'	28°24.7	0.7'	59.0'								
17	76°08.3	18.7	251°25.0	3.3'	28°25.4	0.5'	58.9'								
18	91°08.0	S23°18.8	265°47.3	3.3'	N28°26.0	0.3'	58.9'								
19	106°07.7	18.9	280°09.6	3.4'	28°26.3	0.1'	58.9'								
20	121°07.4	19.0	294°32.0	3.4'	28°26.4	-0.1'	58.8'								
21	136°07.1	· · 19.2	308°54.4	3.4'	28°26.3	-0.3'	58.8'								
22	151°06.8	19.3	323°16.8	3.5'	28°26.1	-0.5'	58.8'								
23	166°06.5	19.4	337°39.3	3.5'	28°25.6	-0.7'	58.8'								
SD = 16.2'			$d = 0.1'$			SD = 16.2'									
Mon	GHA	Dec	GHA	ν	Dec	d	HP								
0	181°06.2	S23°19.5	352°01.9	3.6'	N28°25.0	-0.9'	58.7'								
1	196°05.9	19.6	6°24.4	3.6'	28°24.1	-1.0'	58.7'								
2	211°05.6	19.7	20°47.1	3.7'	28°23.1	-1.2'	58.7'								
3	226°05.3	· · 19.8	35°09.8	3.7'	28°21.8	-1.4'	58.7'								
4	241°05.0	19.9	49°32.5	3.8'	28°20.4	-1.6'	58.6'								
5	256°04.7	20.0	63°55.3	3.9'	28°18.8	-1.8'	58.6'								
6	271°04.4	S23°20.1	78°18.1	3.9'	N28°17.0	-2.0'	58.6'								
7	286°04.1	20.2	92°41.1	4.0'	28°15.0	-2.2'	58.5'								
8	301°03.8	20.3	107°04.1	4.1'	28°12.8	-2.4'	58.5'								
9	316°03.5	· · 20.4	121°27.1	4.1'	28°10.4	-2.6'	58.5'								
10	331°03.2	20.5	135°50.2	4.2'	28°07.9	-2.7'	58.5'								
11	346°02.9	20.6	150°13.4	4.3'	28°05.1	-2.9'	58.4'								
12	1°02.6	S23°20.7	164°36.7	4.4'	N28°02.2	-3.1'	58.4'								
13	16°02.3	20.8	179°00.1	4.4'	27°59.1	-3.3'	58.4'								
14	31°02.0	20.9	193°23.5	4.5'	27°55.8	-3.5'	58.3'								
15	46°01.7	· · 21.0	207°47.0	4.6'	27°52.4	-3.6'	58.3'								
16	61°01.4	21.1	222°10.6	4.7'	27°48.7	-3.8'	58.3'								
17	76°01.0	21.1	236°34.3	4.8'	27°44.9	-4.0'	58.2'								
18	91°00.7	S23°21.2	250°58.1	4.9'	N27°40.9	-4.2'	58.2'								
19	106°00.4	21.3	265°22.0	5.0'	27°36.7	-4.3'	58.2'								
20	121°00.1	21.4	279°45.9	5.1'	27°32.4	-4.5'	58.2'								
21	135°59.8	· · 21.5	294°10.0	5.2'	27°27.9	-4.7'	58.1'								
22	150°59.5	21.6	308°34.2	5.3'	27°23.2	-4.9'	58.1'								
23	165°59.2	21.7	322°58.4	5.4'	27°18.3	-5.0'	58.1'								
SD = 16.2'			$d = 0.1'$			SD = 16.0'									

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:21	10:48	■	■	13:02	15:30
N 70°	08:01	09:48	■	■	14:02	15:50
68°	07:45	09:14	■	■	14:37	16:05
66°	07:32	08:49	10:28	13:22	15:02	16:18
64°	07:21	08:30	09:47	14:04	15:21	16:29
62°	07:12	08:14	09:19	14:32	15:37	16:39
60°	07:03	08:00	08:58	14:53	15:50	16:47
N 58°	06:56	07:49	08:40	15:10	16:02	16:55
56°	06:49	07:38	08:26	15:25	16:12	17:01
54°	06:43	07:29	08:13	15:37	16:21	17:07
52°	06:38	07:21	08:02	15:49	16:29	17:13
50°	06:32	07:14	07:52	15:58	16:37	17:18
45°	06:21	06:58	07:32	16:19	16:53	17:30
N 40°	06:10	06:44	07:15	16:36	17:06	17:40
35°	06:01	06:33	07:01	16:50	17:18	17:50
30°	05:52	06:22	06:49	17:02	17:28	17:58
20°	05:36	06:03	06:27	17:23	17:47	18:15
N 10°	05:20	05:46	06:09	17:42	18:05	18:31
0°	05:03	05:29	05:52	17:59	18:22	18:48
S 10°	04:44	05:11	05:34	18:17	18:40	19:07
20°	04:22	04:51	05:15	18:35	19:00	19:29
30°	03:53	04:26	04:53	18:57	19:25	19:58
35°	03:34	04:11	04:40	19:10	19:40	20:17
40°	03:11	03:53	04:25	19:25	19:58	20:40
45°	02:41					

December 17, 18, 19 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	86°06.9	131°52.9	S19°31.1	317°24.1	N22°04.8	12°17.3	N21°57.3	100°29.4	S08°19.6	Alpheratz	357°34.8	29°13.9
1	101°09.3	146°52.5	30.2	332°26.9	05.0	27°20.1	57.3	115°31.7	19.5	Ankaa	353°07.1	-42°10.4
2	116°11.8	161°52.1	29.4	347°29.7	05.2	42°22.9	57.2	130°34.1	19.4	Schedar	349°31.0	56°40.7
3	131°14.3	176°51.7	•• 28.5	2°32.5	•• 05.4	57°25.8	•• 57.2	145°36.4	•• 19.4	Diphda	348°47.3	-17°51.1
4	146°16.7	191°51.3	27.6	17°35.3	05.6	72°28.6	57.2	160°38.8	19.3	Achernar	335°19.9	-57°06.8
5	161°19.2	206°50.9	26.8	32°38.1	05.8	87°31.4	57.1	175°41.1	19.3	Hamal	327°51.0	23°35.0
6	176°21.7	221°50.6	S19°25.9	47°40.9	N22°06.0	102°34.2	N21°57.1	190°43.5	S08°19.2	Polaris	313°39.6	89°22.4
7	191°24.1	236°50.2	25.0	62°43.7	06.2	117°37.0	57.1	205°45.8	19.2	Acamar	315°11.5	-40°12.4
8	206°26.6	251°49.8	24.2	77°46.5	06.4	132°39.8	57.0	220°48.1	19.1	Menkar	314°05.9	4°11.3
9	221°29.1	266°49.4	•• 23.3	92°49.3	•• 06.6	147°42.7	•• 57.0	235°50.5	•• 19.0	Mirfak	308°27.9	49°57.1
10	236°31.5	281°49.0	22.4	107°52.1	06.8	162°45.5	57.0	250°52.8	19.0	Aldebaran	290°39.3	16°33.6
11	251°34.0	296°48.6	21.5	122°54.9	07.0	177°48.3	57.0	265°55.2	18.9	Rigel	281°03.6	-8°10.3
12	266°36.5	311°48.3	S19°20.7	137°57.7	N22°07.2	192°51.1	N21°56.9	280°57.5	S08°18.9	Capella	280°21.5	46°01.4
13	281°38.9	326°47.9	19.8	153°00.5	07.4	207°53.9	56.9	295°59.8	18.8	Bellatrix	278°22.6	6°22.4
14	296°41.4	341°47.5	18.9	168°03.4	07.6	222°56.7	56.9	311°02.2	18.7	Elnath	278°01.5	28°37.7
15	311°43.8	356°47.1	•• 18.0	183°06.2	•• 07.8	237°59.5	•• 56.8	326°04.5	•• 18.7	Alnilam	275°37.4	-1°11.1
16	326°46.3	11°46.7	17.2	198°09.0	08.0	253°02.4	56.8	341°06.9	18.6	Betelgeuse	270°51.8	7°24.7
17	341°48.8	26°46.4	16.3	213°11.8	08.2	268°05.2	56.8	356°09.2	18.6	Canopus	263°51.9	-52°42.4
18	356°51.2	41°46.0	S19°15.4	228°14.6	N22°08.4	283°08.0	N21°56.7	11°11.5	S08°18.5	Sirius	258°25.9	-16°45.0
19	11°53.7	56°45.6	14.5	243°17.4	08.6	298°10.8	56.7	26°13.9	18.5	Adhara	255°05.5	-29°00.3
20	26°56.2	71°45.2	13.6	258°20.3	08.8	313°13.6	56.7	41°16.2	18.4	Procyon	244°50.6	5°09.7
21	41°58.6	86°44.9	•• 12.8	273°23.1	•• 09.1	328°16.4	•• 56.7	56°18.6	•• 18.3	Pollux	243°17.0	27°57.9
22	57°01.1	101°44.5	11.9	288°25.9	09.3	343°19.3	56.6	71°20.9	18.3	Avior	234°14.2	-59°35.1
23	72°03.6	116°44.1	11.0	303°28.7	09.5	358°22.1	56.6	86°23.2	18.2	Suhail	222°46.1	-43°31.8
Mer.pass. 18:13		ν -0.4' d -0.9' m -4.32		ν 2.8' d 0.2' m -0.87		ν 2.8' d -0.0' m -2.80		ν 2.3' d -0.1' m 1.01				

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	87°06.0	131°43.8	S19°10.1	318°31.6	N22°09.7	13°24.9	N21°56.6	101°25.6	S08°18.2	Denebola	182°25.0	14°25.9
1	102°08.5	146°43.4	09.2	333°34.4	09.9	28°27.7	56.5	116°27.9	18.1	Gienah	175°43.7	-17°40.7
2	117°10.9	161°43.0	08.3	348°37.2	10.1	43°30.5	56.5	131°30.2	18.0	Acrux	173°00.3	-63°13.9
3	132°13.4	176°42.7	•• 07.4	3°40.1	•• 10.3	58°33.3	•• 56.5	146°32.6	•• 18.0	Gacrux	171°51.9	-57°14.9
4	147°15.9	191°42.3	06.6	18°42.9	10.5	73°36.1	56.5	161°34.9	17.9	Alioth	166°13.2	55°49.2
5	162°18.3	206°41.9	05.7	33°45.7	10.7	88°39.0	56.4	176°37.3	17.9	Spica	158°22.5	-11°17.4
6	177°20.8	221°41.6	S19°04.8	48°48.6	N22°10.9	103°41.8	N21°56.4	191°39.6	S08°17.8	Alkaid	152°52.3	49°11.1
7	192°23.3	236°41.2	03.9	63°51.4	11.1	118°44.6	56.4	206°41.9	17.7	Hadar	148°36.5	-60°29.3
8	207°25.7	251°40.8	03.0	78°54.3	11.3	133°47.4	56.3	221°44.3	17.7	Menkent	147°57.9	-36°29.4
9	222°28.2	266°40.5	•• 02.1	93°57.1	•• 11.5	148°50.2	•• 56.3	236°46.6	•• 17.6	Arcturus	145°48.2	19°03.0
10	237°30.7	281°40.1	01.2	108°59.9	11.7	163°53.0	56.3	251°48.9	17.6	Rigil Kent.	139°40.9	-60°56.1
11	252°33.1	296°39.8	19°00.3	124°02.8	11.9	178°55.8	56.2	266°51.3	17.5	Kochab	137°20.7	74°02.9
12	267°35.6	311°39.4	S18°59.4	139°05.6	N22°12.1	193°58.7	N21°56.2	281°53.6	S08°17.4	Zuben'ubi	136°56.3	-16°08.7
13	282°38.1	326°39.0	58.5	154°08.5	12.3	209°01.5	56.2	296°55.9	17.4	Alphecca	126°04.1	26°37.7
14	297°40.5	341°38.7	57.6	169°11.3	12.6	224°04.3	56.2	311°58.3	17.3	Antares	112°16.3	-26°29.2
15	312°43.0	356°38.3	•• 56.7	184°14.2	•• 12.8	239°07.1	•• 56.1	327°00.6	•• 17.3	Atria	107°11.1	-69°04.2
16	327°45.4	11°38.0	55.8	199°17.0	13.0	254°09.9	56.1	342°03.0	17.2	Sabik	102°03.2	-15°45.3
17	342°47.9	26°37.6	55.0	214°19.9	13.2	269°12.7	56.1	357°05.3	17.1	Shaula	96°10.9	-37°07.3
18	357°50.4	41°37.3	S18°54.1	229°22.8	N22°13.4	284°15.5	N21°56.0	12°07.6	S08°17.1	Rasalhague	95°58.9	12°32.5
19	12°52.8	56°36.9	53.2	244°25.6	13.6	299°18.3	56.0	27°10.0	17.0	Eltanin	90°42.8	51°29.1
20	27°55.3	71°36.6	52.3	259°28.5	13.8	314°21.2	56.0	42°12.3	17.0	Kaus Aust.	83°33.0	-34°22.4
21	42°57.8	86°36.2	•• 51.4	274°31.3	•• 14.0	329°24.0	•• 56.0	57°14.6	•• 16.9	Vega	80°33.7	38°48.4
22	58°00.2	101°35.8	50.5	289°34.2	14.2	344°26.8	55.9	72°17.0	16.8	Nunki	75°48.2	-26°16.0
23	73°02.7	116°35.5	49.6	304°37.1	14.5	359°29.6	55.9	87°19.3	16.8	Altair	62°00.3	8°56.1
Mer.pass. 18:09		ν -0.4' d -0.9' m -4.33		ν 2.8' d 0.2' m -0.89		ν 2.8' d -0.0' m -2.80		ν 2.3' d -0.1' m 1.01				

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	88°05.2	131°35.2	S18°48.6	319°39.9	N22°14.7	14°32.4	N21°55.9	102°21.6	S08°16.7	Fomalhaut	15°14.7	-29°29.6
1	103°07.6	146°34.8	47.7	334°42.8	14.9	29°35.2	55.8	117°24.0	16.7	Scheat	13°45.3	28°13.2
2	118°10.1	161°34.5	46.8	349°45.7	15.1	44°38.0	55.8	132°26.3	16.6	Markab	13°30.0	15°20.5
3	133°12.6	176°34.1	•• 45.9	4°48.5	•• 15.3	59°40.8	•• 55.8	147°28.6	•• 16.5			
4	148°15.0	191°33.8	45.0	19°51.4	15.5	74°43.7	55.7	162°31.0	16.5	Dec 17 Tue	SHA	Mer.pass
5	163°17.5	206°33.4	44.1	34°54.3	15.7	89°46.5	55.7	177°33.3	16.4	Venus	45°46.0	15:13
6	178°19.9	221°33.1	S18°43.2	49°57.2	N22°15.9	104°49.3	N21°55.7	192°35.6	S08°16.4	Mars	231°17.2	02:50
7	193°22.4	236°32.7	42.3	65°00.0	16.2	119°52.1	55.7	207°38.0	16.3	Jupiter	286°10.4	23:07
8	208°24.9	251°32.4	41.4	80°02.9	16.4	134°54.9	55.6	222°40.3	16.2	Saturn	14°22.5	17:15
9	223°27.3	266°32.1	•• 40.5	95°05.8	•• 16.6	149°57.7	•• 55.6	237°42.6	•• 16.2			
10	238°29.8	281°31.7	39.6	110°08.7	16.8	165°00.5	55.6	252°45.0	16.1	Dec 18 Wed	SHA	Mer.pass
11	253°32.3	296°31.4	38.7	125°11.6	17.0	180°03.3	55.5	267°47.3	16.1	Venus	44°37.7	15:13
12	268°34.7	311°31.0	S18°37.8	140°14.4	N22°17.2	195°06.1	N21°55.5	282°49.6	S08°16.0	Mars	231°25.6	02:45
13	283°37.2	326°30.7	36.8	155°17.3	17.4	210°08.9	55.5	297°52.0	15.9	Jupiter	286°18.9	23:02
14	298°39.7	341°30.4	35.9	170°20.2	17.7	225°11.8	55.5	312°54.3	15.9	Saturn	14°19.5	17:12
15	313°42.1	356°30.0	•• 35.0	185°23.1	•• 17.9	240°14.6	•• 55.4	327°56.6	•• 15.8			
16	328°44.6	11°29.7	34.1	200°26.0	18.1	255°17.4	55.4	342°59.0	15.7	Dec 19 Thu	SHA	Mer.pass
17	343°47.1	26°29.4	33.2	215°28.9	18.3	270°20.2	55.4	358°01.3	15.7	Venus	43°30.0	15:14
18	358°49.5	41°29.0	S18°32.3	230°31.8	N22°18.5	285°23.0	N21°55.3	13°03.6	S08°15.6	Mars	231°34.8	02:41
19	13°52.0	56°28.7	31.4	245°34.7	18.7	300°25.8	55.3	28°06.0	15.6	Jupiter	286°27.2	22:58
20	28°54.4	71°28.4	30.4	260°37.6	19.0	315°28.6	55.3	43°08.3	15.5	Saturn	14°16.5	17:08
21	43°56.9	86°28.0	•• 29.5	275°40.5	•• 19.2	330°31.4	•• 55.2	58°10.6	•• 15.4			
22	58°59.4	101°27.7	28.6	290°43.4	19.4	345°34.2	55.2	73°13.0	15.4			
23	74°01.8	116°27.4	27.7	305°46.3	19.6	0°37.0	55.2	88°15.3	15.3			
Mer.pass. 18:05		ν -0.3' d -0.9' m -4.33		ν 2.9' d 0.2' m -0.92		ν 2.8' d -0.0' m -2.79		ν 2.3' d -0.1' m 1.01				

Horizontal parallax		
Venus:		0.2
Mars:		0.2

Sun							Moon						
Tue	GHA	Dec	GHA	ν	Dec	d	HP						
0	180°58.9	S23°21.8	337°22.8	5.5'	N27°13.3	-5.2'	58.0'						
1	195°58.6	21.9	351°47.2	5.6'	27°08.1	-5.4'	58.0'						
2	210°58.3	21.9	6°11.8	5.7'	27°02.7	-5.5'	58.0'						
3	225°58.0	· · 22.0	20°36.4	5.8'	26°57.2	-5.7'	57.9'						
4	240°57.7	22.1	35°01.2	5.9'	26°51.5	-5.8'	57.9'						
5	255°57.4	22.2	49°26.1	6.0'	26°45.7	-6.0'	57.9'						
6	270°57.1	S23°22.3	63°51.1	6.1'	N26°39.7	-6.2'	57.8'						
7	285°56.8	22.3	78°16.2	6.2'	26°33.6	-6.3'	57.8'						
8	300°56.5	22.4	92°41.4	6.3'	26°27.2	-6.5'	57.8'						
9	315°56.2	· · 22.5	107°06.7	6.4'	26°20.8	-6.6'	57.7'						
10	330°55.9	22.6	121°32.2	6.6'	26°14.2	-6.8'	57.7'						
11	345°55.6	22.7	135°57.7	6.7'	26°07.4	-6.9'	57.7'						
12	0°55.3	S23°22.7	150°23.4	6.8'	N26°00.5	-7.1'	57.6'						
13	15°54.9	22.8	164°49.2	6.9'	25°53.4	-7.2'	57.6'						
14	30°54.6	22.9	179°15.1	7.0'	25°46.2	-7.3'	57.6'						
15	45°54.3	· · 23.0	193°41.1	7.1'	25°38.9	-7.5'	57.5'						
16	60°54.0	23.0	208°07.3	7.3'	25°31.4	-7.6'	57.5'						
17	75°53.7	23.1	222°33.5	7.4'	25°23.8	-7.8'	57.5'						
18	90°53.4	S23°23.2	236°59.9	7.5'	N25°16.0	-7.9'	57.4'						
19	105°53.1	23.3	251°26.4	7.6'	25°08.1	-8.0'	57.4'						
20	120°52.8	23.3	265°53.1	7.8'	25°00.1	-8.2'	57.4'						
21	135°52.5	· · 23.4	280°19.8	7.9'	24°51.9	-8.3'	57.3'						
22	150°52.2	23.5	294°46.7	8.0'	24°43.6	-8.4'	57.3'						
23	165°51.9	23.5	309°13.7	8.1'	24°35.2	-8.6'	57.3'						
SD = 16.2'		d = 0.1'		SD = 15.8'									
Wed	GHA	Dec	GHA	ν	Dec	d	HP						
0	180°51.6	S23°23.6	323°40.8	8.2'	N24°26.6	-8.7'	57.2'						
1	195°51.3	23.7	338°08.1	8.4'	24°17.9	-8.8'	57.2'						
2	210°51.0	23.7	352°35.4	8.5'	24°09.1	-8.9'	57.2'						
3	225°50.7	· · 23.8	7°02.9	8.6'	24°00.2	-9.1'	57.1'						
4	240°50.4	23.9	21°30.6	8.7'	23°51.1	-9.2'	57.1'						
5	255°50.0	23.9	35°58.3	8.9'	23°42.0	-9.3'	57.1'						
6	270°49.7	S23°24.0	50°26.2	9.0'	N23°32.7	-9.4'	57.0'						
7	285°49.4	24.0	64°54.2	9.1'	23°23.3	-9.5'	57.0'						
8	300°49.1	24.1	79°22.3	9.2'	23°13.7	-9.6'	57.0'						
9	315°48.8	· · 24.2	93°50.6	9.4'	23°04.1	-9.7'	56.9'						
10	330°48.5	24.2	108°18.9	9.5'	22°54.4	-9.9'	56.9'						
11	345°48.2	24.3	122°47.4	9.6'	22°44.5	-10.0'	56.9'						
12	0°47.9	S23°24.3	137°16.0	9.7'	N22°34.6	-10.1'	56.8'						
13	15°47.6	24.4	151°44.8	9.9'	22°24.5	-10.2'	56.8'						
14	30°47.3	24.4	166°13.7	10.0'	22°14.3	-10.3'	56.8'						
15	45°47.0	· · 24.5	180°42.7	10.1'	22°04.1	-10.4'	56.7'						
16	60°46.7	24.6	195°11.8	10.2'	21°53.7	-10.5'	56.7'						
17	75°46.4	24.6	209°41.0	10.4'	21°43.2	-10.6'	56.7'						
18	90°46.1	S23°24.7	224°10.4	10.5'	N21°32.7	-10.7'	56.6'						
19	105°45.7	24.7	238°39.9	10.6'	21°22.0	-10.8'	56.6'						
20	120°45.4	24.8	253°09.5	10.7'	21°11.2	-10.8'	56.6'						
21	135°45.1	· · 24.8	267°39.2	10.8'	21°00.4	-10.9'	56.5'						
22	150°44.8	24.9	282°09.0	11.0'	20°49.5	-11.0'	56.5'						
23	165°44.5	24.9	296°39.0	11.1'	20°38.4	-11.1'	56.5'						
SD = 16.2'		d = 0.1'		SD = 15.6'									
Thu	GHA	Dec	GHA	ν	Dec	d	HP						
0	180°44.2	S23°25.0	311°09.1	11.2'	N20°27.3	-11.2'	56.5'						
1	195°43.9	25.0	325°39.3	11.3'	20°16.1	-11.3'	56.4'						
2	210°43.6	25.0	340°09.6	11.4'	20°04.8	-11.4'	56.4'						
3	225°43.3	· · 25.1	354°40.0	11.5'	19°53.4	-11.5'	56.4'						
4	240°43.0	25.1	9°10.6	11.7'	19°42.0	-11.5'	56.3'						
5	255°42.7	25.2	23°41.2	11.8'	19°30.5	-11.6'	56.3'						
6	270°42.4	S23°25.2	38°12.0	11.9'	N19°18.8	-11.7'	56.3'						
7	285°42.0	25.3	52°42.9	12.0'	19°07.2	-11.8'	56.2'						
8	300°41.7	25.3	67°13.9	12.1'	18°55.4	-11.8'	56.2'						
9	315°41.4	· · 25.3	81°45.0	12.2'	18°43.6	-11.9'	56.2'						
10	330°41.1	25.4	96°16.2	12.3'	18°31.6	-12.0'	56.1'						
11	345°40.8	25.4	110°47.6	12.4'	18°19.7	-12.1'	56.1'						
12	0°40.5	S23°25.5	125°19.0	12.6'	N18°07.6	-12.1'	56.1'						
13	15°40.2	25.5	139°50.6	12.7'	17°55.5	-12.2'	56.0'						
14	30°39.9	25.5	154°22.2	12.8'	17°43.3	-12.3'	56.0'						
15	45°39.6	· · 25.6	168°54.0	12.9'	17°31.1	-12.3'	56.0'						
16	60°39.3	25.6	183°25.9	13.0'	17°18.7	-12.4'	56.0'						
17	75°39.0	25.6	197°57.8	13.1'	17°06.4	-12.4'	55.9'						
18	90°38.6	S23°25.7	212°29.9	13.2'	N16°53.9	-12.5'	55.9'						
19	105°38.3	25.7	227°02.1	13.3'	16°41.4	-12.6'	55.9'						
20	120°38.0	25.7	241°34.4	13.4'	16°28.8	-12.6'	55.8'						
21	135°37.7	· · 25.8	256°06.7	13.5'	16°16.2	-12.7'	55.8'						
22	150°37.4	25.8	270°39.2	13.6'	16°03.5	-12.7'	55.8'						
23	165°37.1	25.8	285°11.8	13.7'	15°50.8	-12.8'	55.7'						
SD = 16.2'		d = 0.0'		SD = 15.4'									

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:24	10:55	■	■	12:59	15:30
N 70°	08:04	09:52	■	■	14:01	15:50
68°	07:48	09:17	■	■	14:36	16:06
66°	07:35	08:52	10:33	13:21	15:02	16:19
64°	07:24	08:32	09:50	14:03	15:21	16:30
62°	07:14	08:16	09:22	14:32	15:37	16:39
60°	07:06	08:03	09:00	14:53	15:51	16:48
N 58°	06:58	07:51	08:43	15:11	16:03	16:55
56°	06:51	07:41	08:28	15:25	16:13	17:02
54°	06:45	07:32	08:15	15:38	16:22	17:08
52°	06:40	07:23	08:04	15:49	16:30	17:14
50°	06:34	07:16	07:54	15:59	16:38	17:19
45°	06:23	07:00	07:34	16:20	16:54	17:31
N 40°	06:12	06:46	07:17	16:37	17:07	17:41
35°	06:03	06:34	07:03	16:51	17:19	17:51
30°	05:54	06:24	06:50	17:03	17:30	18:00
20°	05:38	06:05	06:29	17:25	17:49	18:16
N 10°	05:21	05:48	06:11	17:43	18:06	18:32
0°	05:04	05:31	05:53	18:01	18:23	18:49
S 10°	04:45	05:12	05:35	18:18	18:41	19:08
20°	04:23	04:52	05:17	18:37	19:02	19:31
30°	03:54	04:27	04:54	18:59	19:27	20:00
35°	03:35	04:12	04:41	19:12	19:42	20:19
40°	03:12	03:54	04:26	19:27	20:00	20:42
45°	02:41	03:31	04:08	19:45	20:23	21:13
S 50°	01:56	03:01	03:46	20:08	20:53	21:58
52°	01:28	02:46	03:35	20:19	21:08	22:26
54°	00:42	02:28	03:23	20:31	21:26	23:13
56°	--	02:06	03:08	20:45	21:48	--
58°	--	01:36	02:52	21:02	22:18	--
S 60°	--	00:46	02:31	21:23	23:09	--

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	□	□	17:09	□	□	14:46
N 70°	□	□	18:07	□	□	13:46
68°	□	15:23	18:40	□	14:48	13:11
66°	□	16:49	19:04	□	13:20	12:46
64°	14:51	17:26	19:23	13:25	12:42	12:26
62°	15:54	17:53	19:38	12:22	12:15	12:09
60°	16:28	18:13	19:51	11:47	11:54	11:56
N 58°	16:53	18:30	20:02	11:21	11:36	11:44
56°	17:13	18:44	20:11	11:01	11:22	11:34
54°	17:29	18:56	20:19	10:44	11:09	11:25
52°	17:44	19:06	20:27	10:30	10:58	11:17
50°	17:56	19:16	20:34	10:17	10:48	11:09
45°	18:21	19:36	20:48	09:51	10:26	10:53
N 40°	18:41	19:52	20:59	09:30	10:09	10:40
35°	18:58	20:05	21:09	09:12	09:55	10:29
30°	19:13	20:17	21:18	08:57	09:42	10:19
20°	19:37	20:37	21:33	08:31	09:20	10:02
N 10°	19:58	20:54	21:46	08:09	09:01	09:48
0°	20:18	21:10	21:58	07:48	08:44	09:34
S 10°	20:37	21:26	22:10	07:27	08:26	09:19
20°	20:58	21:43	22:22	07:05	08:06	09:04
30°	21:22	22:03	22:37	06:38	07:44	08:47
35°	21:36	22:14	22:45	06:22	07:31	08:36
40°	21:52	22:27	22:55	06:04	07:15	08:25
45°	22:12	22:42	23:06	05:42	06:57	08:11
S 50°	22:36	23:01	23:19	05:14	06:34	07:53
52°	22:47	23:10	23:25	05:00	06:23	07:45
54°	23:00	23:20	23:32	04:44	06:10	07:36
56°	23:15	23:31	23:40	04:25	05:56	07:26
58°	23:33	23:43	23:48	04:01	05:39	07:14
S 60°	23:55	23:58				

December 20, 21, 22 UT (Fri., Sat., Sun.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	89°04.3	131°27.0	S18°26.8	320°49.2	N22°19.8	15°39.8	N21°55.2	103°17.6	S08°15.3	Alpheratz	357°34.8	29°13.9											
1	104°06.8	146°26.7	25.8	335°52.1	20.1	30°42.7	55.1	118°20.0	15.2	Ankaa	353°07.1	-42°10.4											
2	119°09.2	161°26.4	24.9	350°55.0	20.3	45°45.5	55.1	133°22.3	15.1	Schedar	349°31.0	56°40.7											
3	134°11.7	176°26.1	•• 24.0	5°57.9	•• 20.5	60°48.3	•• 55.1	148°24.6	•• 15.1	Diphda	348°47.3	-17°51.1											
4	149°14.2	191°25.7	23.1	21°00.8	20.7	75°51.1	55.0	163°26.9	15.0	Achernar	335°19.9	-57°06.8											
5	164°16.6	206°25.4	22.2	206°25.4	20.9	90°53.9	55.0	178°29.3	14.9	Hamal	327°51.1	23°35.0											
6	179°19.1	221°25.1	S18°21.2	51°06.6	N22°21.2	105°56.7	N21°55.0	193°31.6	S08°14.9	Polaris	313°40.5	89°22.4											
7	194°21.5	236°24.8	20.3	66°09.5	21.4	120°59.5	55.0	208°33.9	14.8	Acamar	315°11.5	-40°12.4											
8	209°24.0	251°24.5	19.4	81°12.5	21.6	136°02.3	54.9	223°36.3	14.8	Menkar	314°05.9	4°11.3											
9	224°26.5	266°24.1	•• 18.4	96°15.4	•• 21.8	151°05.1	•• 54.9	238°38.6	•• 14.7	Mirfak	308°27.9	49°57.1											
10	239°28.9	281°23.8	17.5	111°18.3	22.1	166°07.9	54.9	253°40.9	14.6	Aldebaran	290°39.3	16°33.6											
11	254°31.4	296°23.5	16.6	126°21.2	22.3	181°10.7	54.8	268°43.3	14.6	Rigel	281°03.6	-8°10.3											
12	269°33.9	311°23.2	S18°15.7	141°24.1	N22°22.5	196°13.5	N21°54.8	283°45.6	S08°14.5	Capella	280°21.5	46°01.4											
13	284°36.3	326°22.9	14.7	156°27.1	22.7	211°16.3	54.8	298°47.9	14.4	Bellatrix	278°22.6	6°22.4											
14	299°38.8	341°22.6	13.8	171°30.0	22.9	226°19.2	54.7	313°50.2	14.4	Elnath	278°01.5	28°37.7											
15	314°41.3	356°22.2	•• 12.9	186°32.9	•• 23.2	241°22.0	•• 54.7	328°52.6	•• 14.3	Alnilam	275°37.4	-1°11.1											
16	329°43.7	11°21.9	11.9	201°35.8	23.4	256°24.8	54.7	343°54.9	14.3	Betelgeuse	270°51.8	7°24.7											
17	344°46.2	26°21.6	11.0	216°38.8	23.6	271°27.6	54.7	358°57.2	14.2	Canopus	263°51.9	-52°42.4											
18	359°48.7	41°21.3	S18°10.1	231°41.7	N22°23.8	286°30.4	N21°54.6	13°59.6	S08°14.1	Sirius	258°25.9	-16°45.0											
19	14°51.1	56°21.0	09.1	246°44.6	24.1	301°33.2	54.6	29°01.9	14.1	Adhara	255°05.5	-29°00.3											
20	29°53.6	71°20.7	08.2	261°47.6	24.3	316°36.0	54.6	44°04.2	14.0	Procyon	244°50.6	5°09.7											
21	44°56.0	86°20.4	•• 07.3	276°50.5	•• 24.5	331°38.8	•• 54.5	59°06.5	•• 13.9	Pollux	243°17.0	27°57.9											
22	59°58.5	101°20.1	06.3	291°53.4	24.7	346°41.6	54.5	74°08.9	13.9	Avior	234°14.2	-59°35.1											
23	75°01.0	116°19.8	05.4	306°56.4	25.0	1°44.4	54.5	89°11.2	13.8	Suhail	222°46.0	-43°31.8											
Mer.pass. 18:01				ν -0.3' d -0.9' m -4.34				ν 2.9' d 0.2' m -0.94				ν 2.8' d -0.0' m -2.79				ν 2.3' d -0.1' m 1.02							

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec						
0	90°03.4	131°19.4	S18°04.5	321°59.3	N22°25.2	16°47.2	N21°54.5	104°13.5	S08°13.7						
1	105°05.9	146°19.1	03.5	337°02.2	25.4	31°50.0	54.4	119°15.9	13.7						
2	120°08.4	161°18.8	02.6	352°05.2	25.6	46°52.8	54.4	134°18.2	13.6						
3	135°10.8	176°18.5	•• 01.6	7°08.1	•• 25.9	61°55.6	•• 54.4	149°20.5	•• 13.6						
4	150°13.3	191°18.2	18°00.7	22°11.1	26.1	76°58.4	54.3	164°22.8	13.5						
5	165°15.8	206°17.9	17°59.8	37°14.0	26.3	92°01.2	54.3	179°25.2	13.4						
6	180°18.2	221°17.6	S17°58.8	52°17.0	N22°26.6	107°04.0	N21°54.3	194°27.5	S08°13.4						
7	195°20.7	236°17.3	57.9	67°19.9	26.8	122°06.8	54.3	209°29.8	13.3						
8	210°23.2	251°17.0	56.9	82°22.9	27.0	137°09.6	54.2	224°32.1	13.2						
9	225°25.6	266°16.7	•• 56.0	97°25.8	•• 27.2	152°12.4	•• 54.2	239°34.5	•• 13.2						
10	240°28.1	281°16.4	55.0	112°28.8	27.5	167°15.2	54.2	254°36.8	13.1						
11	255°30.5	296°16.1	54.1	127°31.7	27.7	182°18.1	54.1	269°39.1	13.0						
12	270°33.0	311°15.8	S17°53.2	142°34.7	N22°27.9	197°20.9	N21°54.1	284°41.4	S08°13.0						
13	285°35.5	326°15.5	52.2	157°37.7	28.2	212°23.7	54.1	299°43.8	12.9						
14	300°37.9	341°15.3	51.3	172°40.6	28.4	227°26.5	54.0	314°46.1	12.9						
15	315°40.4	356°15.0	•• 50.3	187°43.6	•• 28.6	242°29.3	•• 54.0	329°48.4	•• 12.8						
16	330°42.9	11°14.7	49.4	202°46.5	28.9	257°32.1	54.0	344°50.7	12.7						
17	345°45.3	26°14.4	48.4	217°49.5	29.1	272°34.9	54.0	359°53.1	12.7						
18	0°47.8	41°14.1	S17°47.5	232°52.5	N22°29.3	287°37.7	N21°53.9	14°55.4	S08°12.6						
19	15°50.3	56°13.8	46.5	247°55.4	29.5	302°40.5	53.9	29°57.7	12.5						
20	30°52.7	71°13.5	45.6	262°58.4	29.8	317°43.3	53.9	45°00.0	12.5						
21	45°55.2	86°13.2	•• 44.6	278°01.4	•• 30.0	332°46.1	•• 53.8	60°02.4	•• 12.4						
22	60°57.6	101°12.9	43.7	293°04.4	30.2	347°48.9	53.8	75°04.7	12.3						
23	76°00.1	116°12.7	42.7	308°07.3	30.5	2°51.7	53.8	90°07.0	12.3						
Mer.pass. 17:57				ν -0.3' d -0.9' m -4.35				ν 2.8' d -0.0' m -2.79				ν 2.3' d -0.1' m 1.02			

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	91°02.6	131°12.4	S17°41.7	323°10.3	N22°30.7	17°54.5	N21°53.8	105°09.3	S08°12.2										
1	106°05.0	146°12.1	40.8	338°13.3	30.9	32°57.3	53.7	120°11.7	12.1										
2	121°07.5	161°11.8	39.8	353°16.3	31.2	48°00.1	53.7	135°14.0	12.1										
3	136°10.0	176°11.5	•• 38.9	8°19.3	•• 31.4	63°02.9	•• 53.7	150°16.3	•• 12.0										
4	151°12.4	191°11.2	37.9	23°22.2	31.6	78°05.7	53.6	165°18.6	11.9										
5	166°14.9	206°11.0	37.0	38°25.2	31.9	93°08.5	53.6	180°21.0	11.9										
6	181°17.4	221°10.7	S17°36.0	53°28.2	N22°32.1	108°11.3	N21°53.6	195°23.3	S08°11.8										
7	196°19.8	236°10.4	35.0	68°31.2	32.3	123°14.1	53.6	210°25.6	11.8										
8	211°22.3	251°10.1	34.1	83°34.2	32.6	138°16.9	53.5	225°27.9	11.7										
9	226°24.8	266°09.8	•• 33.1	98°37.2	•• 32.8	153°19.7	•• 53.5	240°30.2	•• 11.6										
10	241°27.2	281°09.6	32.2	113°40.2	33.1	168°22.5	53.5	255°32.6	11.6										
11	256°29.7	296°09.3	31.2	128°43.2	33.3	183°25.3	53.4	270°34.9	11.5										
12	271°32.1	311°09.0	S17°30.2	143°46.2	N22°33.5	198°28.1	N21°53.4	285°37.2	S08°11.4										
13	286°34.6	326°08.7	29.3	158°49.1	33.8	213°30.9	53.4	300°39.5	11.4										
14	301°37.1	341°08.5	28.3	173°52.1	34.0	228°33.7	53.4	315°41.9	11.3										
15	316°39.5	356°08.2	•• 27.3	188°55.1	•• 34.2	243°36.5	•• 53.3	330°44.2	•• 11.2										
16	331°42.0	11°07.9	26.4	203°58.1	34.5	258°39.3	53.3	345°46.5	11.2										
17	346°44.5	26°07.7	25.4	219°01.1	34.7	273°42.1	53.3	0°48.8	11.1										
18	1°46.9	41°07.4	S17°24.4	234°04.1	N22°35.0	288°44.9	N21°53.2	15°51.1	S08°11.0										
19	16°49.4	56°07.1	23.5	249°07.2	35.2	303°47.7	53.2	30°53.5	11.0										
20	31°51.9	71°06.9	22.5	264°10.2	35.4	318°50.5	53.2	45°55.8	10.9										
21	46°54.3	86°06.6	•• 21.5	279°13.2	•• 35.7	333°53.3	•• 53.2	60°58.1	•• 10.8										
22	61°56.8	101°06.3	20.6	294°16.2	35.9	348°56.1	53.1	76°00.4	10.8										
23	76°59.3	116°06.1	19.6	309°19.2	36.1	3°58.9	53.1	91°02.7	10.7										
Mer.pass. 17:53				ν -0.3' d -1.0' m -4.36				ν 3.0' d 0.2' m -0.99				ν 2.8' d -0.0' m -2.79				ν 2.3' d -0.1' m 1.02			

Dec 20 Fri	SHA	Mer.pass
Venus	42°22.7	15:15
Mars	231°44.9	02:36
Jupiter	286°35.5	22:53
Saturn	14°13.3	17:04

Dec 21 Sat	SHA	Mer.pass
Venus	41°16.0	15:15
Mars	231°55.9	02:32
Jupiter	286°43.8	22:49
Saturn	14°10.1	17:00

Dec 22 Sun	SHA	Mer.pass
Venus	40°09.8	15:15
Mars	232°07.7	02:27
Jupiter	286°51.9	22:44
Saturn	14°06.8	16:57

Horizontal parallax		
Venus:		0.2
Mars:		0.2

h	Sun				Moon			
Fri	GHA	Dec		GHA	ν	Dec	d	HP
0	180°36.8	S23°25.8		299°44.5	13.8'	N15°38.0	-12.8'	55.7'
1	195°36.5	25.9		314°17.2	13.9'	15°25.2	-12.9'	55.7'
2	210°36.2	25.9		328°50.1	14.0'	15°12.3	-12.9'	55.7'
3	225°35.9	•• 25.9		343°23.0	14.0'	14°59.4	-13.0'	55.6'
4	240°35.6	25.9		357°56.1	14.1'	14°46.4	-13.0'	55.6'
5	255°35.2	26.0		12°29.2	14.2'	14°33.3	-13.1'	55.6'
6	270°34.9	S23°26.0		27°02.4	14.3'	N14°20.2	-13.1'	55.6'
7	285°34.6	26.0		41°35.7	14.4'	14°07.1	-13.2'	55.5'
8	300°34.3	26.0		56°09.1	14.5'	13°53.9	-13.2'	55.5'
9	315°34.0	•• 26.1		70°42.6	14.6'	13°40.7	-13.3'	55.5'
10	330°33.7	26.1		85°16.2	14.7'	13°27.5	-13.3'	55.4'
11	345°33.4	26.1		99°49.8	14.7'	13°14.2	-13.3'	55.4'
12	0°33.1	S23°26.1		114°23.6	14.8'	N13°00.8	-13.4'	55.4'
13	15°32.8	26.1		128°57.4	14.9'	12°47.4	-13.4'	55.4'
14	30°32.5	26.1		143°31.3	15.0'	12°34.0	-13.5'	55.3'
15	45°32.2	•• 26.2		158°05.2	15.0'	12°20.5	-13.5'	55.3'
16	60°31.8	26.2		172°39.3	15.1'	12°07.1	-13.5'	55.3'
17	75°31.5	26.2		187°13.4	15.2'	11°53.5	-13.6'	55.3'
18	90°31.2	S23°26.2		201°47.6	15.3'	N11°40.0	-13.6'	55.2'
19	105°30.9	26.2		216°21.9	15.3'	11°26.4	-13.6'	55.2'
20	120°30.6	26.2		230°56.2	15.4'	11°12.7	-13.7'	55.2'
21	135°30.3	•• 26.2		245°30.6	15.5'	10°59.1	-13.7'	55.2'
22	150°30.0	26.3		260°05.1	15.6'	10°45.4	-13.7'	55.1'
23	165°29.7	26.3		274°39.7	15.6'	10°31.7	-13.7'	55.1'
SD = 16.2'			d = 0.0'	SD = 15.2'				
Sat	GHA	Dec		GHA	ν	Dec	d	HP
0	180°29.4	S23°26.3		289°14.3	15.7'	N10°17.9	-13.8'	55.1'
1	195°29.1	26.3		303°49.0	15.7'	10°04.1	-13.8'	55.1'
2	210°28.7	26.3		318°23.7	15.8'	09°50.3	-13.8'	55.1'
3	225°28.4	•• 26.3		332°58.6	15.9'	09°36.5	-13.9'	55.0'
4	240°28.1	26.3		347°33.4	15.9'	09°22.7	-13.9'	55.0'
5	255°27.8	26.3		2°08.4	16.0'	09°08.8	-13.9'	55.0'
6	270°27.5	S23°26.3		16°43.4	16.1'	N08°54.9	-13.9'	55.0'
7	285°27.2	26.3		31°18.4	16.1'	08°41.0	-13.9'	54.9'
8	300°26.9	26.3		45°53.5	16.2'	08°27.0	-14.0'	54.9'
9	315°26.6	•• 26.3		60°28.7	16.2'	08°13.1	-14.0'	54.9'
10	330°26.3	26.3		75°03.9	16.3'	07°59.1	-14.0'	54.9'
11	345°25.9	26.3		89°39.2	16.3'	07°45.1	-14.0'	54.9'
12	0°25.6	S23°26.3		104°14.5	16.4'	N07°31.1	-14.0'	54.8'
13	15°25.3	26.3		118°49.9	16.4'	07°17.1	-14.0'	54.8'
14	30°25.0	26.3		133°25.3	16.5'	07°03.0	-14.1'	54.8'
15	45°24.7	•• 26.3		148°00.8	16.5'	06°48.9	-14.1'	54.8'
16	60°24.4	26.3		162°36.3	16.6'	06°34.9	-14.1'	54.8'
17	75°24.1	26.3		177°11.9	16.6'	06°20.8	-14.1'	54.7'
18	90°23.8	S23°26.3		191°47.5	16.7'	N06°06.7	-14.1'	54.7'
19	105°23.5	26.3		206°23.1	16.7'	05°52.6	-14.1'	54.7'
20	120°23.2	26.3		220°58.8	16.7'	05°38.4	-14.1'	54.7'
21	135°22.8	•• 26.2		235°34.5	16.8'	05°24.3	-14.1'	54.7'
22	150°22.5	26.2		250°10.3	16.8'	05°10.1	-14.2'	54.6'
23	165°22.2	26.2		264°46.1	16.8'	04°56.0	-14.2'	54.6'
SD = 16.3'			d = 0.0'	SD = 15.0'				
Sun	GHA	Dec		GHA	ν	Dec	d	HP
0	180°21.9	S23°26.2		279°22.0	16.9'	N04°41.8	-14.2'	54.6'
1	195°21.6	26.2		293°57.8	16.9'	04°27.6	-14.2'	54.6'
2	210°21.3	26.2		308°33.8	16.9'	04°13.5	-14.2'	54.6'
3	225°21.0	•• 26.2		323°09.7	17.0'	03°59.3	-14.2'	54.6'
4	240°20.7	26.2		337°45.7	17.0'	03°45.1	-14.2'	54.5'
5	255°20.4	26.1		352°21.7	17.0'	03°30.9	-14.2'	54.5'
6	270°20.0	S23°26.1		6°57.7	17.1'	N03°16.7	-14.2'	54.5'
7	285°19.7	26.1		21°33.8	17.1'	03°02.5	-14.2'	54.5'
8	300°19.4	26.1		36°09.9	17.1'	02°48.3	-14.2'	54.5'
9	315°19.1	•• 26.1		50°46.0	17.1'	02°34.1	-14.2'	54.5'
10	330°18.8	26.1		65°22.2	17.2'	02°19.8	-14.2'	54.5'
11	345°18.5	26.0		79°58.3	17.2'	02°05.6	-14.2'	54.4'
12	0°18.2	S23°26.0		94°34.5	17.2'	N01°51.4	-14.2'	54.4'
13	15°17.9	26.0		109°10.7	17.2'	01°37.2	-14.2'	54.4'
14	30°17.6	26.0		123°46.9	17.2'	01°23.0	-14.2'	54.4'
15	45°17.2	•• 25.9		138°23.2	17.3'	01°08.8	-14.2'	54.4'
16	60°16.9	25.9		152°59.4	17.3'	00°54.6	-14.2'	54.4'
17	75°16.6	25.9		167°35.7	17.3'	00°40.4	-14.2'	54.4'
18	90°16.3	S23°25.9		182°12.0	17.3'	N00°26.2	-14.2'	54.4'
19	105°16.0	25.8		196°48.3	17.3'	N00°12.0	-14.2'	54.3'
20	120°15.7	25.8		211°24.6	17.3'	S00°02.2	14.2'	54.3'
21	135°15.4	•• 25.8		226°00.9	17.3'	00°16.4	14.2'	54.3'
22	150°15.1	25.8		240°37.3	17.3'	00°30.6	14.2'	54.3'
23	165°14.8	25.7		255°13.6	17.3'	00°44.7	14.2'	54.3'
SD = 16.3'			d = -0.0'	SD = 14.9'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:26	10:58	■	■	12:58	15:31
N 70°	08:06	09:55	■	■	14:02	15:51
68°	07:50	09:19	■	■	14:37	16:07
66°	07:37	08:54	10:35	13:21	15:03	16:20
64°	07:26	08:34	09:52	14:04	15:22	16:31
62°	07:16	08:18	09:24	14:33	15:39	16:41
60°	07:07	08:04	09:02	14:54	15:52	16:49
N 58°	07:00	07:53	08:45	15:12	16:04	16:57
56°	06:53	07:42	08:30	15:27	16:14	17:03
54°	06:47	07:33	08:17	15:39	16:23	17:10
52°	06:41	07:25	08:06	15:51	16:32	17:15
50°	06:36	07:18	07:56	16:00	16:39	17:21
45°	06:24	07:01	07:35	16:21	16:55	17:32
N 40°	06:14	06:48	07:18	16:38	17:09	17:43
35°	06:04	06:36	07:04	16:52	17:21	17:52
30°	05:56	06:25	06:52	17:05	17:31	18:01
20°	05:39	06:07	06:31	17:26	17:50	18:18
N 10°	05:23	05:49	06:12	17:45	18:07	18:34
0°	05:06	05:32	05:55	18:02	18:25	18:51
S 10°	04:47	05:14	05:37	18:20	18:43	19:10
20°	04:24	04:53	05:18	18:39	19:03	19:32
30°	03:55	04:28	04:56	19:01	19:28	20:02
35°	03:36	04:13	04:43	19:14	19:44	20:20
40°	03:13	03:55	04:28	19:29	20:02	20:44
45°	02:42	03:32	04:10	19:47	20:24	21:14
S 50°	01:57	03:02	03:47	20:10	20:54	22:00
52°	01:29	02:47	03:36	20:20	21:09	22:28
54°	00:41	02:29	03:24	20:33	21:27	23:15
56°	—:—	02:07	03:10	20:47	21:50	—:—
58°	—:—	01:37	02:53	21:04	22:20	—:—
S 60°	—:—	00:46	02:32	21:24	23:11	—:—

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	20:08	22:17	•• ••	13:19	12:37	12:03
N 70°	20:28	22:25	•• ••	12:57	12:26	12:01
68°	20:43	22:31	•• ••	12:40	12:18	11:59
66°	20:55	22:36	•• ••	12:26	12:11	11:58
64°	21:06	22:40	•• ••	12:14	12:05	11:56
62°	21:14	22:43	•• ••	12:04	12:00	11:55
60°	21:21	22:47	•• ••	11:56	11:55	11:54
N 58°	21:28	22:49	•• ••	11:48	11:51	11:53
56°	21:34	22:52	•• ••	11:42	11:47	11:52
54°	21:39	22:54	•• ••	11:36	11:44	11:51
52°	21:43	22:56	•• ••	11:30	11:41	11:51
50°	21:47	22:58	•• ••	11:25	11:38	11:50
45°	21:56	23:02	•• ••	11:15	11:32	11:48
N 40°	22:04	23:05	•• ••	11:06	11:27	11:47
35°	22:10	23:08	•• ••	10:58	11:23	11:46
30°	22:16	23:10	•• ••	10:51	11:19	11:45
20°	22:25	23:15	•• ••	10:39	11:13	11:44
N 10°	22:33	23:18	•• ••	10:29	11:07	11:42
0°	22:41	23:22	•• ••	10:19	11:01	11:41
S 10°	22:49	23:25	•• ••	10:09	10:55	11:39
20°	22:57	23:29	•• ••	09:58	10:49	11:38
30°	23:07	23:33	23:59	09:46	10:42	11:36
35°	23:12	23:36	23:58	09:39	10:38	11:35
40°	23:18	23:39	23:58	09:31	10:34	11:34
45°	23:25	23:42	23:57	09:21	10:28	11:33
S 50°	23:33	23:45	23:57	09:09	10:22	11:31
52°	23:37	23:47	23:56	09:04	10:19	11:30
54°	23:42	23:49	23:56	08:58	10:15	11:29
56°	23:46	23:51	23:55	08:51	10:12	11:28
58°	23:51	23:53	23:55	08:44	10:07	11:27
S 60°	23:57	23:56	23:55	08:35	1	

Aries Venus Mars

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	94°00.0	130°54.2	S16°31.3	326°48.6	N22°48.2	21°15.7	N21°51.7	107°56.3	S08°07.4
1	109°02.5	145°54.0	30.3	341°51.7	48.4	36°18.5	51.7	122°58.6	07.3
2	124°04.9	160°53.8	29.3	356°54.7	48.7	51°21.3	51.6	138°00.9	07.3
3	139°07.4	175°53.6	•• 28.3	11°57.8	•• 48.9	66°24.1	•• 51.6	153°03.2	•• 07.2
4	154°09.8	190°53.3	27.3	27°00.9	49.2	81°26.9	51.6	168°05.5	07.1
5	169°12.3	205°53.1	26.3	42°04.0	49.4	96°29.6	51.6	183°07.8	07.0
6	184°14.8	220°52.9	S16°25.3	57°07.1	N22°49.7	111°32.4	N21°51.5	198°10.2	S08°07.0
7	199°17.2	235°52.7	24.3	72°10.2	49.9	126°35.2	51.5	213°12.5	06.9
8	214°19.7	250°52.5	23.3	87°13.3	50.2	141°38.0	51.5	228°14.8	06.8
9	229°22.2	265°52.3	•• 22.3	102°16.4	•• 50.4	156°40.8	•• 51.5	243°17.1	•• 06.8
10	244°24.6	280°52.1	21.2	117°19.5	50.7	171°43.6	51.4	258°19.4	06.7
11	259°27.1	295°51.9	20.2	132°22.6	50.9	186°46.4	51.4	273°21.7	06.6
12	274°29.6	310°51.6	S16°19.2	147°25.7	N22°51.2	201°49.1	N21°51.4	288°24.0	S08°06.6
13	289°32.0	325°51.4	18.2	162°28.8	51.4	216°51.9	51.3	303°26.3	06.5
14	304°34.5	340°51.2	17.2	177°31.9	51.7	231°54.7	51.3	318°28.7	06.4
15	319°37.0	355°51.0	•• 16.2	192°35.0	•• 52.0	246°57.5	•• 51.3	333°31.0	•• 06.3
16	334°39.4	10°50.8	15.2	207°38.1	52.2	262°00.3	51.3	348°33.3	06.3
17	349°41.9	25°50.6	14.2	222°41.2	52.5	277°03.1	51.2	3°35.6	06.2
18	4°44.3	40°50.4	S16°13.2	237°44.3	N22°52.7	292°05.9	N21°51.2	18°37.9	S08°06.1
19	19°46.8	55°50.2	12.2	252°47.4	53.0	307°08.6	51.2	33°40.2	06.1
20	34°49.3	70°50.0	11.1	267°50.6	53.2	322°11.4	51.1	48°42.5	06.0
21	49°51.7	85°49.8	•• 10.1	282°53.7	•• 53.5	337°14.2	•• 51.1	63°44.8	•• 05.9
22	64°54.2	100°49.6	09.1	297°56.8	53.7	352°17.0	51.1	78°47.1	05.9
23	79°56.7	115°49.4	08.1	312°59.9	54.0	7°19.8	51.1	93°49.5	05.8
Mer.pass. 17:41		ν -0.2' d -1.0' m -4.38		ν 3.1' d 0.3' m -1.06		ν 2.8' d -0.0' m -2.77		ν 2.3' d -0.1' m 1.03	

Dec 23 Mon	SHA	Mer.pass
Venus	39° 04.1	15:16
Mars	232° 20.5	02:22
Jupiter	286° 59.9	22:40
Saturn	14° 03.4	16:53

Dec 24 Tue	SHA	Mer.pass
Venus	37° 58.9	15:16
Mars	232° 34.1	02:17
Jupiter	287° 07.9	22:35
Saturn	13° 59.9	16:49

Dec 25 Wed	SHA	Mer.pass
Venus	36° 54.2	15:17
Mars	232° 48.6	02:12
Jupiter	287° 15.7	22:31
Saturn	13° 56.3	16:46

Horizontal parallax	
Venus:	0.2
Mars:	0.2

Sun							Moon							
Mon	GHA	Dec	GHA	ν	Dec	d	HP	Lat.	Twilight		Sunrise	Sunset	Twilight	
								Naut.	Civil				Civil	Naut.
0	180°14.4	S23°25.7	269°49.9	17.4'	S00°58.9	14.2'	54.3'	N 72°	08:27	10:58	■	■	13:02	15:33
1	195°14.1	25.7	284°26.3	17.4'	01°13.1	14.1'	54.3'	N 70°	08:07	09:55	■	■	14:04	15:53
2	210°13.8	25.6	299°02.7	17.4'	01°27.2	14.1'	54.3'	68°	07:51	09:20	■	■	14:39	16:09
3	225°13.5	· · 25.6	313°39.0	17.4'	01°41.3	14.1'	54.3'	66°	07:38	08:55	10:36	13:24	15:05	16:22
4	240°13.2	25.6	328°15.4	17.4'	01°55.5	14.1'	54.3'	64°	07:27	08:35	09:53	14:06	15:24	16:33
5	255°12.9	25.5	342°51.7	17.4'	02°09.6	14.1'	54.3'	62°	07:17	08:19	09:25	14:35	15:41	16:42
6	270°12.6	S23°25.5	357°28.1	17.4'	S02°23.7	14.1'	54.2'	60°	07:09	08:06	09:03	14:56	15:54	16:51
7	285°12.3	25.5	12°04.5	17.4'	02°37.8	14.1'	54.2'	N 58°	07:01	07:54	08:46	15:14	16:06	16:58
8	300°12.0	25.4	26°40.8	17.4'	02°51.9	14.1'	54.2'	56°	06:54	07:44	08:31	15:29	16:16	17:05
9	315°11.7	· · 25.4	41°17.2	17.3'	03°05.9	14.1'	54.2'	54°	06:48	07:35	08:18	15:41	16:25	17:11
10	330°11.3	25.3	55°53.5	17.3'	03°20.0	14.0'	54.2'	52°	06:43	07:26	08:07	15:52	16:33	17:17
11	345°11.0	25.3	70°29.9	17.3'	03°34.0	14.0'	54.2'	50°	06:37	07:19	07:57	16:02	16:41	17:22
12	0°10.7	S23°25.3	85°06.2	17.3'	S03°48.0	14.0'	54.2'	45°	06:26	07:03	07:37	16:23	16:57	17:34
13	15°10.4	25.2	99°42.5	17.3'	04°02.0	14.0'	54.2'	N 40°	06:15	06:49	07:20	16:40	17:10	17:44
14	30°10.1	25.2	114°18.8	17.3'	04°16.0	14.0'	54.2'	35°	06:06	06:37	07:06	16:54	17:22	17:54
15	45°09.8	· · 25.1	128°55.1	17.3'	04°30.0	14.0'	54.2'	30°	05:57	06:27	06:53	17:06	17:33	18:03
16	60°09.5	25.1	143°31.4	17.3'	04°44.0	13.9'	54.2'	20°	05:40	06:08	06:32	17:28	17:52	18:19
17	75°09.2	25.0	158°07.7	17.3'	04°57.9	13.9'	54.2'	N 10°	05:24	05:51	06:13	17:46	18:09	18:35
18	90°08.9	S23°25.0	172°44.0	17.2'	S05°11.8	13.9'	54.2'	0°	05:07	05:33	05:56	18:04	18:26	18:52
19	105°08.5	25.0	187°20.2	17.2'	05°25.7	13.9'	54.2'	S 10°	04:48	05:15	05:38	18:21	18:44	19:11
20	120°08.2	24.9	201°56.4	17.2'	05°39.6	13.9'	54.2'	20°	04:26	04:55	05:19	18:40	19:05	19:34
21	135°07.9	· · 24.9	216°32.6	17.2'	05°53.4	13.8'	54.2'	30°	03:57	04:30	04:57	19:02	19:30	20:03
22	150°07.6	24.8	231°08.8	17.2'	06°07.3	13.8'	54.2'	35°	03:38	04:15	04:44	19:15	19:45	20:22
23	165°07.3	24.8	245°45.0	17.1'	06°21.1	13.8'	54.2'	40°	03:15	03:56	04:29	19:30	20:03	20:45
	SD = 16.3'	d = -0.0'	SD = 14.8'											
Tue	GHA	Dec	GHA	ν	Dec	d	HP	Lat.	Moonrise		Moonset			
									Mon	Tue	Wed	Mon	Tue	Wed
0	180°07.0	S23°24.7	260°21.1	17.1'	S06°34.9	13.8'	54.2'	N 72°	00:15	02:13	04:26	11:32	10:57	10:10
1	195°06.7	24.7	274°57.3	17.1'	06°48.6	13.7'	54.2'	N 70°	00:13	02:02	04:00	11:37	11:11	10:37
2	210°06.4	24.6	289°33.4	17.1'	07°02.4	13.7'	54.2'	68°	00:12	01:54	03:41	11:41	11:22	10:58
3	225°06.1	· · 24.5	304°09.4	17.0'	07°16.1	13.7'	54.2'	66°	00:11	01:46	03:26	11:45	11:31	11:15
4	240°05.7	24.5	318°45.5	17.0'	07°29.8	13.7'	54.2'	64°	00:10	01:40	03:13	11:48	11:39	11:29
5	255°05.4	24.4	333°21.5	17.0'	07°43.4	13.6'	54.1'	62°	00:09	01:35	03:03	11:50	11:45	11:40
6	270°05.1	S23°24.4	347°57.5	17.0'	S07°57.1	13.6'	54.1'	60°	00:09	01:30	02:54	11:52	11:51	11:50
7	285°04.8	24.3	2°33.4	16.9'	08°10.7	13.6'	54.1'	N 58°	00:08	01:26	02:46	11:54	11:56	11:59
8	300°04.5	24.3	17°09.4	16.9'	08°24.2	13.6'	54.1'	56°	00:08	01:23	02:39	11:56	12:01	12:07
9	315°04.2	· · 24.2	31°45.3	16.9'	08°37.8	13.5'	54.1'	54°	00:07	01:20	02:33	11:58	12:05	12:13
10	330°03.9	24.2	46°21.1	16.8'	08°51.3	13.5'	54.1'	52°	00:07	01:17	02:28	11:59	12:09	12:20
11	345°03.6	24.1	60°57.0	16.8'	09°04.8	13.5'	54.2'	50°	00:06	01:14	02:23	12:01	12:12	12:25
12	0°03.3	S23°24.0	75°32.7	16.8'	S09°18.3	13.4'	54.2'	45°	00:05	01:08	02:13	12:04	12:20	12:37
13	15°03.0	24.0	90°08.5	16.7'	09°31.7	13.4'	54.2'	N 40°	00:05	01:04	02:04	12:06	12:26	12:48
14	30°02.6	23.9	104°44.2	16.7'	09°45.1	13.4'	54.2'	35°	00:04	01:00	01:56	12:09	12:32	12:56
15	45°02.3	· · 23.8	119°19.9	16.6'	09°58.4	13.3'	54.2'	30°	00:03	00:56	01:50	12:11	12:36	13:04
16	60°02.0	23.8	133°55.5	16.6'	10°11.8	13.3'	54.2'	20°	00:03	00:50	01:39	12:14	12:45	13:17
17	75°01.7	23.7	148°31.1	16.6'	10°25.1	13.3'	54.2'	N 10°	00:02	00:45	01:29	12:17	12:52	13:29
18	90°01.4	S23°23.7	163°06.7	16.5'	S10°38.3	13.2'	54.2'	0°	00:01	00:40	01:20	12:20	12:59	13:40
19	105°01.1	23.6	177°42.2	16.5'	10°51.5	13.2'	54.2'	S 10°	00:00	00:35	01:11	12:23	13:06	13:51
20	120°00.8	23.5	192°17.6	16.4'	11°04.7	13.1'	54.2'	20°	00:00	00:30	01:01	12:26	13:13	14:03
21	135°00.5	· · 23.5	206°53.1	16.4'	11°17.9	13.1'	54.2'	30°	· · ·	00:24	00:51	12:29	13:22	14:16
22	150°00.2	23.4	221°28.4	16.3'	11°31.0	13.1'	54.2'	35°	· · ·	00:21	00:45	12:31	13:27	14:24
23	164°59.8	23.3	236°03.8	16.3'	11°44.1	13.0'	54.2'	40°	· · ·	00:17	00:38	12:33	13:33	14:33
	SD = 16.3'	d = -0.1'	SD = 14.8'											
Wed	GHA	Dec	GHA	ν	Dec	d	HP	Day	Sun		Mer.	Moon		Age
									Eqn.of Time	12h	Pass	Upper	Lower	22-24
									00h		hh:mm	hh:mm	hh:mm	49-31%
									mm:ss	mm:ss				
0	179°59.5	S23°23.2	250°39.0	16.2'	S11°57.1	13.0'	54.2'	23	00:58	00:43	11:59	06:10	18:30	
1	194°59.2	23.2	265°14.3	16.2'	12°10.1	12.9'	54.2'	24	00:28	00:13	12:00	06:49	19:09	
2	209°58.9	23.1	279°49.4	16.1'	12°23.0	12.9'	54.2'	25	00:02	00:17	12:00	07:30	19:51	
3	224°58.6	· · 23.0	294°24.5	16.1'	12°35.9	12.9'	54.2'							
4	239°58.3	23.0	308°59.6	16.0'	12°48.8	12.8'	54.2'							
5	254°58.0	22.9	323°34.6	16.0'	13°01.6	12.8'	54.2'							
6	269°57.7	S23°22.8	338°09.6	15.9'	S13°14.4	12.7'	54.2'							
7	284°57.4	22.7	352°44.5	15.8'	13°27.1	12.7'	54.2'							
8	299°57.1	22.6	7°19.3	15.8'	13°39.8	12.6'	54.2'							
9	314°56.7	· · 22.6	21°54.1	15.7'	13°52.4	12.6'	54.2'							
10	329°56.4	22.5	36°28.8	15.7'	14°05.0	12.5'	54.3'							
11	344°56.1	22.4	51°03.5	15.6'	14°17.6	12.5'	54.3'							
12	359°55.8	S23°22.3	65°38.1	15.5'	S14°30.0	12.4'	54.3'							
13	14°55.5	22.3	80°12.6	15.5'	14°42.5	12.4'	54.3'							
14	29°55.2	22.2	94°47.1	15.4'	14°54.9	12.3'	54.3'							
15	44°54.9	· · 22.1	109°21.5	15.3'	15°07.2	12.3'	54.3'							
16	59°54.6	22.0	123°55.8	15.3'	15°19.5	12.2'	54.3'							
17	74°54.3	21.9	138°30.1	15.2'	15°31.7	12.2'	54.3'							
18	89°54.0	S23°21.8	153°04.3	15.1'	S15°43.9	12.1'	54.3'							
19	104°53.7	21.8	167°38.5	15.1'	15°56.0	12.1'	54.3'							
20	119°53.3	21.7	182°12.5	15.0'	16°08.1	12.0'	54.3'							
21	134°53.0	· · 21.6	196°46.5	14.9'	16°20.1	11.9'	54.3'							
22	149°52.7	21.5	211°20.5	14.9'	16°32.0	11.9'	54.4'							
23	164°52.4	21.4	225°54.3	14.8'	16°43.9	11.8'	54.4'							
	SD = 16.3'	d = -0.1'	SD = 14.8'											

December 26, 27, 28 UT (Thu., Fri., Sat.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	94°59.1	130°49.2	S16°07.1	328°03.0	N22°54.2	22°22.6	N21°51.0	108°51.8	S08°05.7	Alpheratz	357°34.8	29°13.9
1	110°01.6	145°49.0	06.1	343°06.1	54.5	37°25.3	51.0	123°54.1	05.6	Ankaa	353°07.2	-42°10.4
2	125°04.1	160°48.8	05.1	358°09.3	54.8	52°28.1	51.0	138°56.4	05.6	Schedar	349°31.0	56°40.7
3	140°06.5	175°48.6	•• 04.0	13°12.4	•• 55.0	67°30.9	•• 51.0	153°58.7	•• 05.5	Diphda	348°47.3	-17°51.1
4	155°09.0	190°48.4	03.0	28°15.5	55.3	82°33.7	50.9	169°01.0	05.4	Achernar	335°20.0	-57°06.9
5	170°11.5	205°48.2	02.0	43°18.6	55.5	97°36.5	50.9	184°03.3	05.4	Hamal	327°51.1	23°34.9
6	185°13.9	220°48.0	S16°01.0	58°21.8	N22°55.8	112°39.3	N21°50.9	199°05.6	S08°05.3	Polaris	313°42.3	89°22.4
7	200°16.4	235°47.8	16°00.0	73°24.9	56.0	127°42.0	50.8	214°07.9	05.2	Acamar	315°11.5	-40°12.4
8	215°18.8	250°47.7	15°58.9	88°28.0	56.3	142°44.8	50.8	229°10.2	05.2	Menkar	314°06.0	4°11.3
9	230°21.3	265°47.5	•• 57.9	103°31.2	•• 56.6	157°47.6	•• 50.8	244°12.6	•• 05.1	Mirfak	308°27.9	49°57.2
10	245°23.8	280°47.3	56.9	118°34.3	56.8	172°50.4	50.8	259°14.9	05.0	Aldebaran	290°39.3	16°33.6
11	260°26.2	295°47.1	55.9	133°37.4	57.1	187°53.2	50.7	274°17.2	04.9	Rigel	281°03.6	-8°10.4
12	275°28.7	310°46.9	S15°54.9	148°40.6	N22°57.3	202°55.9	N21°50.7	289°19.5	S08°04.9	Capella	280°21.5	46°01.4
13	290°31.2	325°46.7	53.8	163°43.7	57.6	217°58.7	50.7	304°21.8	04.8	Bellatrix	278°22.6	6°22.4
14	305°33.6	340°46.5	52.8	178°46.8	57.8	233°01.5	50.6	319°24.1	04.7	Elnath	278°01.5	28°37.7
15	320°36.1	355°46.3	•• 51.8	193°50.0	•• 58.1	248°04.3	•• 50.6	334°26.4	•• 04.7	Alnilam	275°37.4	-1°11.1
16	335°38.6	10°46.2	50.8	208°53.1	58.4	263°07.1	50.6	349°28.7	04.6	Betelgeuse	270°51.8	7°24.7
17	350°41.0	25°46.0	49.7	223°56.3	58.6	278°09.8	50.6	4°31.0	04.5	Canopus	263°51.8	-52°42.5
18	5°43.5	40°45.8	S15°48.7	238°59.4	N22°58.9	293°12.6	N21°50.5	19°33.3	S08°04.4	Sirius	258°25.9	-16°45.0
19	20°45.9	55°45.6	47.7	254°02.6	59.1	308°15.4	50.5	34°35.6	04.4	Adhara	255°05.5	-29°00.3
20	35°48.4	70°45.4	46.7	269°05.7	59.4	323°18.2	50.5	49°37.9	04.3	Procyon	244°50.5	5°09.6
21	50°50.9	85°45.3	•• 45.6	284°08.9	•• 59.7	338°21.0	•• 50.5	64°40.2	•• 04.2	Pollux	243°17.0	27°57.9
22	65°53.3	100°45.1	44.6	299°12.0	22°59.9	353°23.7	50.4	79°42.6	04.2	Avior	234°14.2	-59°35.2
23	80°55.8	115°44.9	43.6	314°15.2	23°00.2	8°26.5	50.4	94°44.9	04.1	Suhail	222°46.0	-43°31.8
Mer.pass. 17:37		ν -0.2' d -1.0' m -4.39		ν 3.1' d 0.3' m -1.08		ν 2.8' d -0.0' m -2.77		ν 2.3' d -0.1' m 1.04				

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	95°58.3	130°44.7	S15°42.5	329°18.3	N23°00.4	23°29.3	N21°50.4	109°47.2	S08°04.0	Alphard	217°47.6
1	111°00.7	145°44.6	41.5	344°21.5	00.7	38°32.1	50.3	124°49.5	03.9	Regulus	207°34.3
2	126°03.2	160°44.4	40.5	359°24.6	01.0	53°34.9	50.3	139°51.8	03.9	Dubhe	193°40.8
3	141°05.7	175°44.2	•• 39.4	14°27.8	•• 01.2	68°37.6	•• 50.3	154°54.1	•• 03.8	Denebola	175°43.6
4	156°08.1	190°44.0	38.4	29°30.9	01.5	83°40.4	50.3	169°56.4	03.7	Gienah	173°00.2
5	171°10.6	205°43.9	37.4	44°34.1	01.8	98°43.2	50.2	184°58.7	03.7	Acrux	173°00.2
6	186°13.1	220°43.7	S15°36.3	59°37.3	N23°02.0	113°46.0	N21°50.2	200°01.0	S08°03.6	Gacrux	171°51.7
7	201°15.5	235°43.5	35.3	74°40.4	02.3	128°48.7	50.2	215°03.3	03.5	Alioth	166°13.1
8	216°18.0	250°43.3	34.3	89°43.6	02.5	143°51.5	50.2	230°05.6	03.4	Spica	158°22.4
9	231°20.4	265°43.2	•• 33.2	104°46.7	•• 02.8	158°54.3	•• 50.1	245°07.9	•• 03.4	Alkaid	152°52.2
10	246°22.9	280°43.0	32.2	119°49.9	03.1	173°57.1	50.1	260°10.2	03.3	49°11.0	
11	261°25.4	295°42.8	31.2	134°53.1	03.3	188°59.9	50.1	275°12.5	03.2	Hadar	148°36.4
12	276°27.8	310°42.7	S15°30.1	149°56.3	N23°03.6	204°02.6	N21°50.0	290°14.8	S08°03.1	Menkent	147°57.9
13	291°30.3	325°42.5	29.1	164°59.4	03.8	219°05.4	50.0	305°17.1	03.1	Arcturus	145°48.1
14	306°32.8	340°42.4	28.1	180°02.6	04.1	234°08.2	50.0	320°19.4	03.0	Rigel Kent.	139°40.8
15	321°35.2	355°42.2	•• 27.0	195°05.8	•• 04.4	249°11.0	•• 50.0	335°21.8	•• 02.9	Kochab	137°20.5
16	336°37.7	10°42.0	26.0	210°08.9	04.6	264°13.7	49.9	350°24.1	02.9	Zuben'ubi	136°56.3
17	351°40.2	25°41.9	24.9	225°12.1	04.9	279°16.5	49.9	5°26.4	02.8	Alphecca	126°04.1
18	6°42.6	40°41.7	S15°23.9	240°15.3	N23°05.2	294°19.3	N21°49.9	20°28.7	S08°02.7	Antares	112°16.2
19	21°45.1	55°41.6	22.9	255°18.5	05.4	309°22.1	49.9	35°31.0	02.6	Atria	107°11.0
20	36°47.6	70°41.4	21.8	270°21.7	05.7	324°24.8	49.8	50°33.3	02.6	Sabik	102°03.2
21	51°50.0	85°41.2	•• 20.8	285°24.8	•• 06.0	339°27.6	•• 49.8	65°35.6	•• 02.5	Shaula	96°10.8
22	66°52.5	100°41.1	19.7	300°28.0	06.2	354°30.4	49.8	80°37.9	02.4	Rasalhague	95°58.9
23	81°54.9	115°40.9	18.7	315°31.2	06.5	9°33.1	49.8	95°40.2	02.3	Eltanin	90°42.7
Mer.pass. 17:33		ν -0.2' d -1.0' m -4.40		ν 3.2' d 0.3' m -1.10		ν 2.8' d -0.0' m -2.77		ν 2.3' d -0.1' m 1.04		Kaus Aust.	83°33.0

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	96°57.4	130°40.8	S15°17.7	330°34.4	N23°06.7	24°35.9	N21°49.7	110°42.5	S08°02.3	Venus	35°50.1
1	111°59.9	145°40.6	16.6	345°37.6	07.0	39°38.7	49.7	125°44.8	02.2	Mars	233°03.9
2	127°02.3	160°40.5	15.6	0°40.8	07.3	54°41.5	49.7	140°47.1	02.1	Jupiter	287°23.4
3	142°04.8	175°40.3	•• 14.5	15°44.0	•• 07.5	69°44.2	•• 49.6	155°49.4	•• 02.1	Saturn	13°52.6
4	157°07.3	190°40.2	13.5	30°47.2	07.8	84°47.0	49.6	170°51.7	02.0		
5	172°09.7	205°40.0	12.4	45°50.4	08.1	99°49.8	49.6	185°54.0	01.9		
6	187°12.2	220°39.9	S15°11.4	60°53.5	N23°08.3	114°52.6	N21°49.6	200°56.3	S08°01.8		
7	202°14.7	235°39.7	10.3	75°56.7	08.6	129°55.3	49.5	215°58.6	01.8		
8	217°17.1	250°39.6	09.3	90°59.9	08.9	144°58.1	49.5	231°00.9	01.7		
9	232°19.6	265°39.4	•• 08.3	106°03.1	•• 09.1	160°00.9	•• 49.5	246°03.2	•• 01.6		
10	247°22.1	280°39.3	07.2	121°06.3	09.4	175°03.6	49.5	261°05.5	01.5		
11	262°24.5	295°39.1	06.2	136°09.5	09.7	190°06.4	49.4	276°07.8	01.5		
12	277°27.0	310°39.0	S15°05.1	151°12.7	N23°09.9	205°09.2	N21°49.4	291°10.1	S08°01.4		
13	292°29.4	325°38.9	04.1	166°15.9	10.2	220°12.0	49.4	306°12.4	01.3		
14	307°31.9	340°38.7	03.0	181°19.2	10.5	235°14.7	49.3	321°14.7	01.2		
15	322°34.4	355°38.6	•• 02.0	196°22.4	•• 10.7	250°17.5	•• 49.3	336°17.0	•• 01.2		
16	337°36.8	10°38.4	15°00.9	211°25.6	11.0	265°20.3	49.3	351°19.3	01.1		
17	352°39.3	25°38.3	14°59.9	226°28.8	11.3	280°23.0	49.3	6°21.6	01.0		
18	7°41.8	40°38.2	S14°58.8	241°32.0	N23°11.5	295°25.8	N21°49.2	21°23.9	S08°00.9		
19	22°44.2	55°38.0	57.7	256°35.2	11.8	310°28.6	49.2	36°26.2	00.9		
20	37°46.7	70°37.9	56.7	271°38.4	12.1	325°31.3	49.2	51°28.5	00.8		
21	52°49.2	85°37.7	•• 55.6	286°41.6	•• 12.3	340°34.1	•• 49.2	66°30.8	•• 00.7		
22	67°51.6	100°37.6	54.6	301°44.8	12.6	355°36.9	49.1	81°33.1	00.7		
23	82°54.1	115°37.5	53.5	316°48.1	12.9	10°39.6	49.1	96°35.4	00.6		
Mer.pass. 17:29		ν -0.2' d -1.0' m -4.41		ν 3.2' d 0.3' m -1.13		ν 2.8' d -0.0' m -2.76		ν 2.3' d -0.1' m 1.04			

Dec 26 Thu			SHA	Mer.pass
Venus			35°50.1	15:17
Mars			233°03.9	02:07
Jupiter			287°23.4	22:26
Saturn			13°52.6	16:42

Dec 27 Fri			SHA	Mer.pass
Venus			34°46.5	15:17
Mars			233°20.0	02:02
Jupiter			287°31.0	22:22
Saturn			13°48.9	16:38

Dec 28 Sat			SHA	Mer.pass

Sun							Moon								
Thu	GHA	Dec	GHA	ν	Dec	d	HP								
0	179°52.1	S23°21.3	240°28.1	14.7'	S16°55.7	11.8'	54.4'								
1	194°51.8	21.2	255°01.8	14.6'	17°07.5	11.7'	54.4'								
2	209°51.5	21.1	269°35.4	14.6'	17°19.2	11.6'	54.4'								
3	224°51.2	· · 21.0	284°09.0	14.5'	17°30.8	11.6'	54.4'								
4	239°50.9	20.9	298°42.5	14.4'	17°42.4	11.5'	54.4'								
5	254°50.6	20.8	313°15.9	14.3'	17°53.9	11.4'	54.4'								
6	269°50.3	S23°20.8	327°49.2	14.2'	S18°05.4	11.4'	54.5'								
7	284°49.9	20.7	342°22.4	14.2'	18°16.7	11.3'	54.5'								
8	299°49.6	20.6	356°55.6	14.1'	18°28.0	11.2'	54.5'								
9	314°49.3	· · 20.5	11°28.7	14.0'	18°39.3	11.2'	54.5'								
10	329°49.0	20.4	26°01.7	13.9'	18°50.5	11.1'	54.5'								
11	344°48.7	20.3	40°34.6	13.8'	19°01.6	11.0'	54.5'								
12	359°48.4	S23°20.2	55°07.4	13.7'	S19°12.6	11.0'	54.5'								
13	14°48.1	20.1	69°40.1	13.7'	19°23.5	10.9'	54.5'								
14	29°47.8	20.0	84°12.8	13.6'	19°34.4	10.8'	54.6'								
15	44°47.5	· · 19.9	98°45.4	13.5'	19°45.2	10.7'	54.6'								
16	59°47.2	19.8	113°17.9	13.4'	19°55.9	10.6'	54.6'								
17	74°46.9	19.7	127°50.3	13.3'	20°06.6	10.6'	54.6'								
18	89°46.5	S23°19.6	142°22.6	13.2'	S20°17.2	10.5'	54.6'								
19	104°46.2	19.4	156°54.8	13.1'	20°27.7	10.4'	54.6'								
20	119°45.9	19.3	171°27.0	13.0'	20°38.1	10.3'	54.7'								
21	134°45.6	· · 19.2	185°59.0	13.0'	20°48.4	10.2'	54.7'								
22	149°45.3	19.1	200°31.0	12.9'	20°58.6	10.2'	54.7'								
23	164°45.0	19.0	215°02.8	12.8'	21°08.8	10.1'	54.7'								
SD = 16.3'			$d = -0.1'$		SD = 14.8'										
Fri	GHA	Dec	GHA	ν	Dec	d	HP								
0	179°44.7	S23°18.9	229°34.6	12.7'	S21°18.9	10.0'	54.7'								
1	194°44.4	18.8	244°06.3	12.6'	21°28.9	9.9'	54.7'								
2	209°44.1	18.7	258°37.9	12.5'	21°38.8	9.8'	54.8'								
3	224°43.8	· · 18.6	273°09.4	12.4'	21°48.6	9.7'	54.8'								
4	239°43.5	18.5	287°40.8	12.3'	21°58.3	9.6'	54.8'								
5	254°43.2	18.3	302°12.1	12.2'	22°07.9	9.5'	54.8'								
6	269°42.9	S23°18.2	316°43.3	12.1'	S22°17.5	9.4'	54.8'								
7	284°42.5	18.1	331°14.5	12.0'	22°26.9	9.4'	54.8'								
8	299°42.2	18.0	345°45.5	11.9'	22°36.3	9.3'	54.9'								
9	314°41.9	· · 17.9	0°16.5	11.8'	22°45.5	9.2'	54.9'								
10	329°41.6	17.8	14°47.3	11.7'	22°54.7	9.1'	54.9'								
11	344°41.3	17.6	29°18.0	11.7'	23°03.8	9.0'	54.9'								
12	359°41.0	S23°17.5	43°48.7	11.6'	S23°12.7	8.9'	54.9'								
13	14°40.7	17.4	58°19.3	11.5'	23°21.6	8.8'	54.9'								
14	29°40.4	17.3	72°49.7	11.4'	23°30.4	8.7'	55.0'								
15	44°40.1	· · 17.2	87°20.1	11.3'	23°39.0	8.6'	55.0'								
16	59°39.8	17.0	101°50.4	11.2'	23°47.6	8.4'	55.0'								
17	74°39.5	16.9	116°20.5	11.1'	23°56.0	8.3'	55.0'								
18	89°39.2	S23°16.8	130°50.6	11.0'	S24°04.4	8.2'	55.0'								
19	104°38.9	16.7	145°20.6	10.9'	24°12.6	8.1'	55.1'								
20	119°38.6	16.5	159°50.5	10.8'	24°20.7	8.0'	55.1'								
21	134°38.2	· · 16.4	174°20.3	10.7'	24°28.7	7.9'	55.1'								
22	149°37.9	16.3	188°50.0	10.6'	24°36.6	7.8'	55.1'								
23	164°37.6	16.2	203°19.6	10.5'	24°44.4	7.7'	55.1'								
SD = 16.3'			$d = -0.1'$		SD = 14.9'										
Sat	GHA	Dec	GHA	ν	Dec	d	HP								
0	179°37.3	S23°16.0	217°49.1	10.4'	S24°52.1	7.6'	55.2'								
1	194°37.0	15.9	232°18.5	10.3'	24°59.7	7.4'	55.2'								
2	209°36.7	15.8	246°47.8	10.2'	25°07.1	7.3'	55.2'								
3	224°36.4	· · 15.6	261°17.0	10.1'	25°14.5	7.2'	55.2'								
4	239°36.1	15.5	275°46.1	10.0'	25°21.7	7.1'	55.2'								
5	254°35.8	15.4	290°15.2	9.9'	25°28.8	7.0'	55.3'								
6	269°35.5	S23°15.2	304°44.1	9.8'	S25°35.7	6.8'	55.3'								
7	284°35.2	15.1	319°12.9	9.7'	25°42.6	6.7'	55.3'								
8	299°34.9	15.0	333°41.7	9.7'	25°49.3	6.6'	55.3'								
9	314°34.6	· · 14.8	348°10.3	9.6'	25°55.9	6.5'	55.4'								
10	329°34.3	14.7	2°38.9	9.5'	26°02.4	6.3'	55.4'								
11	344°34.0	14.6	17°07.4	9.4'	26°08.7	6.2'	55.4'								
12	359°33.7	S23°14.4	31°35.8	9.3'	S26°14.9	6.1'	55.4'								
13	14°33.3	14.3	46°04.0	9.2'	26°21.0	6.0'	55.4'								
14	29°33.0	14.1	60°32.2	9.1'	26°27.0	5.8'	55.5'								
15	44°32.7	· · 14.0	75°00.4	9.0'	26°32.8	5.7'	55.5'								
16	59°32.4	13.9	89°28.4	8.9'	26°38.5	5.6'	55.5'								
17	74°32.1	13.7	103°56.3	8.9'	26°44.1	5.4'	55.5'								
18	89°31.8	S23°13.6	118°24.2	8.8'	S26°49.5	5.3'	55.6'								
19	104°31.5	13.4	132°51.9	8.7'	26°54.8	5.2'	55.6'								
20	119°31.2	13.3	147°19.6	8.6'	26°59.9	5.0'	55.6'								
21	134°30.9	· · 13.1	161°47.2	8.5'	27°05.0	4.9'	55.6'								
22	149°30.6	13.0	176°14.7	8.4'	27°09.8	4.7'	55.6'								
23	164°30.3	12.8	190°42.2	8.4'	27°14.6	4.6'	55.7'								
SD = 16.3'			$d = -0.1'$		SD = 15.0'										

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:27	10:54	■	■	13:09	15:36
N 70°	08:07	09:54	■	■	14:08	15:56
68°	07:51	09:20	■	■	14:43	16:12
66°	07:38	08:55	10:34	13:29	15:08	16:25
64°	07:27	08:35	09:53	14:10	15:27	16:35
62°	07:18	08:20	09:25	14:38	15:43	16:45
60°	07:09	08:06	09:04	14:59	15:57	16:53
N 58°	07:02	07:54	08:46	15:16	16:08	17:01
56°	06:55	07:44	08:32	15:31	16:18	17:07
54°	06:49	07:35	08:19	15:44	16:27	17:14
52°	06:43	07:27	08:08	15:55	16:35	17:19
50°	06:38	07:20	07:58	16:04	16:43	17:24
45°	06:27	07:04	07:38	16:25	16:59	17:36
N 40°	06:16	06:50	07:21	16:42	17:12	17:46
35°	06:07	06:39	07:07	16:56	17:24	17:56
30°	05:58	06:28	06:54	17:08	17:34	18:04
20°	05:42	06:09	06:33	17:29	17:53	18:21
N 10°	05:26	05:52	06:15	17:48	18:10	18:37
0°	05:09	05:35	05:58	18:05	18:28	18:54
S 10°	04:50	05:17	05:40	18:22	18:46	19:13
20°	04:27	04:57	05:21	18:41	19:06	19:35
30°	03:58	04:32	04:59	19:03	19:31	20:04
35°	03:40	04:17	04:46	19:16	19	

December 29, 30, 31 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	97°56.6	130°37.3	S14°52.5	331°51.3	N23°13.1	25°42.4	N21°49.1	111°37.7	S08°00.5	Alpheratz	357°34.8	29°13.9
1	112°59.0	145°37.2	51.4	346°54.5	13.4	40°45.2	49.1	126°40.0	00.4	Ankaa	353°07.2	-42°10.4
2	128°01.5	160°37.1	50.4	1°57.7	13.7	55°47.9	49.0	141°42.3	00.4	Schedar	349°31.1	56°40.7
3	143°03.9	175°37.0	•• 49.3	17°00.9	•• 13.9	70°50.7	•• 49.0	156°44.6	•• 00.3	Diphda	348°47.3	-17°51.1
4	158°06.4	190°36.8	48.2	32°04.2	14.2	85°53.5	49.0	171°46.9	00.2	Achernar	335°20.0	-57°06.9
5	173°08.9	205°36.7	47.2	47°07.4	14.5	100°56.2	49.0	186°49.2	00.1	Hamal	327°51.1	23°34.9
6	188°11.3	220°36.6	S14°46.1	62°10.6	N23°14.7	115°59.0	N21°48.9	201°51.5	S08°00.1	Polaris	313°43.0	89°22.4
7	203°13.8	235°36.4	45.1	77°13.9	15.0	131°01.8	48.9	216°53.8	08°00.0	Acamar	315°11.5	-40°12.4
8	218°16.3	250°36.3	44.0	92°17.1	15.3	146°04.5	48.9	231°56.1	07°59.9	Menkar	314°06.0	4°11.3
9	233°18.7	265°36.2	•• 43.0	107°20.3	•• 15.6	161°07.3	•• 48.8	246°58.4	•• 59.8	Mirfak	308°27.9	49°57.2
10	248°21.2	280°36.1	41.9	122°23.5	15.8	176°10.1	48.8	262°00.7	59.8	Aldebaran	290°39.3	16°33.6
11	263°23.7	295°36.0	40.8	137°26.8	16.1	191°12.8	48.8	277°03.0	59.7	Rigel	281°03.6	-8°10.4
12	278°26.1	310°35.8	S14°39.8	152°30.0	N23°16.4	206°15.6	N21°48.8	292°05.3	S07°59.6	Capella	280°21.5	46°01.4
13	293°28.6	325°35.7	38.7	167°33.3	16.6	221°18.4	48.7	307°07.6	59.5	Bellatrix	278°22.6	6°22.3
14	308°31.1	340°35.6	37.6	182°36.5	16.9	236°21.1	48.7	322°09.9	59.5	Elnath	278°01.5	28°37.7
15	323°33.5	355°35.5	•• 36.6	197°39.7	•• 17.2	251°23.9	•• 48.7	337°12.2	•• 59.4	Alnilam	275°37.4	-1°11.2
16	338°36.0	10°35.4	35.5	212°43.0	17.4	266°26.7	48.7	352°14.5	59.3	Betelgeuse	270°51.8	7°24.7
17	353°38.4	25°35.2	34.5	227°46.2	17.7	281°29.4	48.6	7°16.8	59.2	Canopus	263°51.8	-52°42.5
18	8°40.9	40°35.1	S14°33.4	242°49.5	N23°18.0	296°32.2	N21°48.6	22°19.1	S07°59.2	Sirius	258°25.9	-16°45.0
19	23°43.4	55°35.0	32.3	257°52.7	18.3	311°34.9	48.6	37°21.4	59.1	Adhara	255°05.5	-29°00.3
20	38°45.8	70°34.9	31.3	272°55.9	18.5	326°37.7	48.6	52°23.7	59.0	Procyon	244°50.5	5°09.6
21	53°48.3	85°34.8	•• 30.2	287°59.2	•• 18.8	341°40.5	•• 48.5	67°26.0	•• 58.9	Pollux	243°17.0	27°57.9
22	68°50.8	100°34.7	29.1	303°02.4	19.1	356°43.2	48.5	82°28.3	58.9	Avior	234°14.2	-59°35.2
23	83°53.2	115°34.6	28.1	318°05.7	19.3	11°46.0	48.5	97°30.6	58.8	Suhail	222°46.0	-43°31.8
Mer.pass. 17:25		ν -0.1' d -1.1' m -4.42		ν 3.2' d 0.3' m -1.15		ν 2.8' d -0.0' m -2.76		ν 2.3' d -0.1' m 1.05				

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	98°55.7	130°34.5	S14°27.0	333°08.9	N23°19.6	26°48.8	N21°48.5	112°32.9	S07°58.7	Alphard	217°47.5	-8°46.0
1	113°58.2	145°34.3	25.9	348°12.2	19.9	41°51.5	48.4	127°35.2	58.6	Regulus	207°34.3	11°50.6
2	129°00.6	160°34.2	24.9	3°15.4	20.1	56°54.3	48.4	142°37.5	58.5	Dubhe	193°40.7	61°36.7
3	144°03.1	175°34.1	•• 23.8	18°18.7	•• 20.4	71°57.0	•• 48.4	157°39.8	•• 58.5	Denebola	182°24.9	14°25.8
4	159°05.6	190°34.0	22.7	33°21.9	20.7	86°59.8	48.4	172°42.1	58.4	Gienah	175°43.6	-17°40.8
5	174°08.0	205°33.9	21.6	48°25.2	21.0	102°02.6	48.3	187°44.4	58.3	Acrux	173°00.1	-63°14.0
6	189°10.5	220°33.8	S14°20.6	63°28.5	N23°21.2	117°05.3	N21°48.3	202°46.7	S07°58.2	Gacrux	171°51.7	-57°14.9
7	204°12.9	235°33.7	19.5	78°31.7	21.5	132°08.1	48.3	217°49.0	58.2	Alioth	166°13.0	55°49.1
8	219°15.4	250°33.6	18.4	93°35.0	21.8	147°10.8	48.3	232°51.3	58.1	Spica	158°22.4	-11°17.5
9	234°17.9	265°33.5	•• 17.4	108°38.2	•• 22.0	162°13.6	•• 48.2	247°53.6	•• 58.0	Alkaid	152°52.2	49°11.0
10	249°20.3	280°33.4	16.3	123°41.5	22.3	177°16.4	48.2	262°55.9	57.9	Hadar	148°36.3	-60°29.3
11	264°22.8	295°33.3	15.2	138°44.8	22.6	192°19.1	48.2	277°58.2	57.9	Menkent	147°57.8	-36°29.4
12	279°25.3	310°33.2	S14°14.1	153°48.0	N23°22.9	207°21.9	N21°48.1	293°00.4	S07°57.8	Arcturus	145°48.1	19°03.0
13	294°27.7	325°33.1	13.1	168°51.3	23.1	222°24.6	48.1	308°02.7	57.7	Rigil Kent.	139°40.7	-60°56.1
14	309°30.2	340°33.0	12.0	183°54.6	23.4	237°27.4	48.1	323°05.0	57.6	Kochab	137°20.5	74°02.8
15	324°32.7	355°32.9	•• 10.9	198°57.8	•• 23.7	252°30.2	•• 48.1	338°07.3	•• 57.6	Zuben'ubi	136°56.3	-16°08.7
16	339°35.1	10°32.8	09.8	214°01.1	24.0	267°32.9	48.0	353°09.6	57.5	Alphecca	126°04.0	26°37.7
17	354°37.6	25°32.7	08.8	229°04.4	24.2	282°35.7	48.0	8°11.9	57.4	Antares	112°16.2	-26°29.2
18	9°40.0	40°32.6	S14°07.7	244°07.7	N23°24.5	297°38.4	N21°48.0	23°14.2	S07°57.3	Atria	107°11.0	-69°04.2
19	24°42.5	55°32.5	06.6	259°10.9	24.8	312°41.2	48.0	38°16.5	57.3	Sabik	102°03.1	-15°45.3
20	39°45.0	70°32.5	05.5	274°14.2	25.0	327°43.9	47.9	53°18.8	57.2	Shaula	96°10.8	-37°07.3
21	54°47.4	85°32.4	•• 04.5	289°17.5	•• 25.3	342°46.7	•• 47.9	68°21.1	•• 57.1	Rasalhague	95°58.9	12°32.4
22	69°49.9	100°32.3	03.4	304°20.8	25.6	357°49.5	47.9	83°23.4	57.0	Eltanin	90°42.7	51°29.0
23	84°52.4	115°32.2	02.3	319°24.0	25.9	12°52.2	47.9	98°25.7	56.9	Kaus Aust.	83°33.0	-34°22.4
Mer.pass. 17:21		ν -0.1' d -1.1' m -4.43		ν 3.3' d 0.3' m -1.17		ν 2.8' d -0.0' m -2.75		ν 2.3' d -0.1' m 1.05		Vega	80°33.7	38°48.3

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	99°54.8	130°32.1	S14°01.2	334°27.3	N23°26.1	27°55.0	N21°47.8	113°28.0	S07°56.9	Nunki	75°48.2	-26°16.0
1	114°57.3	145°32.0	14°00.1	349°30.6	26.4	42°57.7	47.8	128°30.3	56.8	Altair	62°00.3	8°56.0
2	129°59.8	160°31.9	13°59.1	4°33.9	26.7	58°00.5	47.8	143°32.6	56.7	Peacock	53°06.3	-56°39.4
3	145°02.2	175°31.8	•• 58.0	19°37.2	•• 27.0	73°03.2	•• 47.8	158°34.9	•• 56.6	Deneb	49°26.2	45°22.2
4	160°04.7	190°31.8	56.9	34°40.5	27.2	88°06.0	47.7	173°37.2	56.6	Enif	33°39.1	9°59.4
5	175°07.2	205°31.7	55.8	49°43.7	27.5	103°08.7	47.7	188°39.4	56.5	Al Na'ir	27°33.2	-46°50.6
6	190°09.6	220°31.6	S13°54.7	64°47.0	N23°27.8	118°11.5	N21°47.7	203°41.7	S07°56.4	Fomalhaut	15°14.7	-29°29.6
7	205°12.1	235°31.5	53.7	79°50.3	28.1	133°14.3	47.7	218°44.0	56.3	Scheat	13°45.4	28°13.2
8	220°14.5	250°31.4	52.6	94°53.6	28.3	148°17.0	47.6	233°46.3	56.3	Markab	13°30.1	15°20.4
9	235°17.0	265°31.4	•• 51.5	109°56.9	•• 28.6	163°19.8	•• 47.6	248°48.6	•• 56.2			
10	250°19.5	280°31.3	50.4	125°00.2	28.9	178°22.5	47.6	263°50.9	56.1	Dec 29 Sun	SHA	Mer.pass
11	265°21.9	295°31.2	49.3	140°03.5	29.1	193°25.3	47.6	278°53.2	56.0	Venus	32°40.8	15:18
12	280°24.4	310°31.1	S13°48.2	155°06.8	N23°29.4	208°28.0	N21°47.5	293°55.5	S07°55.9	Mars	233°54.7	01:52
13	295°26.9	325°31.1	47.2	170°10.1	29.7	223°30.8	47.5	308°57.8	55.9	Jupiter	287°45.9	22:13
14	310°29.3	340°31.0	46.1	185°13.4	30.0	238°33.5	47.5	324°00.1	55.8	Saturn	13°41.2	16:31
15	325°31.8	355°30.9	•• 45.0	200°16.7	•• 30.2	253°36.3	•• 47.5	339°02.4	•• 55.7			
16	340°34.3	10°30.8	43.9	215°20.0	30.5	268°39.0	47.4	354°04.7	55.6	Dec 30 Mon	SHA	Mer.pass
17	355°36.7	25°30.8	42.8	230°23.3	30.8	283°41.8	47.4	9°07.0	55.6	Venus	31°38.8	15:18
18	10°39.2	40°30.7	S13°41.7	245°26.6	N23°31.1	298°44.5	N21°47.4	24°09.2	S07°55.5	Mars	234°32.5	01:42
19	25°41.7	55°30.6	40.6	260°29.9	31.3	313°47.3	47.4	39°11.5	55.4	Jupiter	288°00.1	22:04
20	40°44.1	70°30.6	39.5	275°33.2	31.6	328°50.0	47.3	54°13.8	55.3	Saturn	13°33.1	16:24
21	55°46.6	85°30.5	•• 38.5	290°36.5								

Sun							Moon								
Sun	GHA	Dec	GHA	ν	Dec	d	HP	Lat.	Twilight		Sunrise	Sunset	Twilight		
									Naut.	Civil			Civil	Naut.	
0	179°30.0	S23°12.7	205°09.5	8.3'	S27°19.2	4.4'	55.7'	N 72°	08:25	10:47	■	■	13:19	15:41	
1	194°29.7	12.5	219°36.8	8.2'	27°23.6	4.3'	55.7'	N 70°	08:06	09:52	■	■	14:14	16:00	
2	209°29.4	12.4	234°04.0	8.1'	27°27.9	4.2'	55.7'	68°	07:51	09:18	■	■	14:47	16:15	
3	224°29.1	· · 12.2	248°31.1	8.0'	27°32.1	4.0'	55.8'	66°	07:38	08:54	10:30	13:35	15:12	16:28	
4	239°28.8	12.1	262°58.2	8.0'	27°36.1	3.9'	55.8'	64°	07:27	08:35	09:51	14:15	15:31	16:39	
5	254°28.5	11.9	277°25.1	7.9'	27°39.9	3.7'	55.8'	62°	07:18	08:19	09:24	14:42	15:46	16:48	
6	269°28.2	S23°11.8	291°52.0	7.8'	S27°43.7	3.6'	55.8'	60°	07:10	08:06	09:03	15:03	16:00	16:56	
7	284°27.9	11.6	306°18.9	7.8'	27°47.2	3.4'	55.9'	N 58°	07:02	07:55	08:46	15:20	16:11	17:03	
8	299°27.6	11.5	320°45.6	7.7'	27°50.6	3.3'	55.9'	56°	06:56	07:45	08:32	15:34	16:21	17:10	
9	314°27.3	· · 11.3	335°12.3	7.6'	27°53.9	3.1'	55.9'	54°	06:50	07:36	08:19	15:46	16:30	17:16	
10	329°27.0	11.2	349°38.9	7.6'	27°57.0	3.0'	55.9'	52°	06:44	07:28	08:08	15:57	16:38	17:22	
11	344°26.7	11.0	4°05.5	7.5'	28°00.0	2.8'	56.0'	50°	06:39	07:20	07:59	16:07	16:45	17:27	
12	359°26.4	S23°10.9	18°32.0	7.4'	S28°02.8	2.7'	56.0'	45°	06:27	07:04	07:38	16:27	17:01	17:38	
13	14°26.0	10.7	32°58.4	7.4'	28°05.4	2.5'	56.0'	N 40°	06:17	06:51	07:22	16:44	17:14	17:48	
14	29°25.7	10.5	47°24.8	7.3'	28°07.9	2.3'	56.0'	35°	06:08	06:40	07:08	16:58	17:26	17:58	
15	44°25.4	· · 10.4	61°51.1	7.2'	28°10.3	2.2'	56.0'	30°	05:59	06:29	06:55	17:10	17:36	18:06	
16	59°25.1	10.2	76°17.3	7.2'	28°12.5	2.0'	56.1'	20°	05:43	06:11	06:35	17:31	17:55	18:22	
17	74°24.8	10.0	90°43.5	7.1'	28°14.5	1.9'	56.1'	N 10°	05:27	05:53	06:16	17:49	18:12	18:38	
18	89°24.5	S23°09.9	105°09.7	7.1'	S28°16.4	1.7'	56.1'	0°	05:10	05:37	05:59	18:06	18:29	18:55	
19	104°24.2	09.7	119°35.7	7.0'	28°18.1	1.6'	56.1'	S 10°	04:52	05:19	05:42	18:24	18:47	19:14	
20	119°23.9	09.6	134°01.8	7.0'	28°19.6	1.4'	56.2'	20°	04:29	04:58	05:23	18:42	19:07	19:36	
21	134°23.6	· · 09.4	148°27.8	6.9'	28°21.0	1.2'	56.2'	30°	04:01	04:34	05:01	19:04	19:32	20:05	
22	149°23.3	09.2	162°53.7	6.9'	28°22.3	1.1'	56.2'	35°	03:42	04:19	04:48	19:17	19:47	20:23	
23	164°23.0	09.1	177°19.6	6.8'	28°23.3	0.9'	56.2'	40°	03:19	04:01	04:33	19:32	20:04	20:46	
SD = 16.3' d = -0.1'									45°	02:49	03:38	04:16	19:50	20:27	21:16
									S 50°	02:05	03:09	03:54	20:12	20:56	22:00
									52°	01:38	02:54	03:43	20:22	21:10	22:26
									54°	00:56	02:37	03:31	20:34	21:28	23:08
									56°	—:—	02:15	03:17	20:48	21:49	—:—
									58°	—:—	01:47	03:00	21:04	22:18	—:—
									S 60°	—:—	01:01	02:40	21:24	23:02	—:—
Mon	GHA	Dec	GHA	ν	Dec	d	HP	Lat.	Moonrise			Moonset			
									Sun	Mon	Tue	Sun	Mon	Tue	
0	179°22.7	S23°08.9	191°45.4	6.8'	S28°24.2	0.7'	56.3'	N 72°	■	■	■	■	■	■	
1	194°22.4	08.7	206°11.2	6.8'	28°25.0	0.6'	56.3'	N 70°	■	■	■	■	■	■	
2	209°22.1	08.6	220°37.0	6.7'	28°25.5	0.4'	56.3'	68°	■	■	■	■	■	■	
3	224°21.8	· · 08.4	235°02.7	6.7'	28°26.0	0.2'	56.3'	66°	■	■	■	■	■	■	
4	239°21.5	08.2	249°28.4	6.7'	28°26.2	0.1'	56.4'	64°	■	■	■	■	■	■	
5	254°21.2	08.0	263°54.1	6.6'	28°26.3	-0.1'	56.4'	62°	■	■	11:37	■	■	13:44	
6	269°20.9	S23°07.9	278°19.7	6.6'	S28°26.2	-0.2'	56.4'	60°	09:08	10:18	10:47	12:14	13:03	14:34	
7	284°20.6	07.7	292°45.3	6.6'	28°26.0	-0.4'	56.4'	N 58°	08:31	09:37	10:16	12:51	13:44	15:05	
8	299°20.3	07.5	307°10.8	6.5'	28°25.5	-0.6'	56.5'	56°	08:05	09:09	09:52	13:18	14:12	15:28	
9	314°20.0	· · 07.3	321°36.4	6.5'	28°25.0	-0.7'	56.5'	54°	07:44	08:47	09:33	13:39	14:34	15:47	
10	329°19.7	07.2	336°01.9	6.5'	28°24.2	-0.9'	56.5'	52°	07:27	08:29	09:16	13:56	14:52	16:03	
11	344°19.4	07.0	350°27.4	6.5'	28°23.3	-1.1'	56.5'	50°	07:12	08:14	09:02	14:11	15:07	16:17	
12	359°19.1	S23°06.8	4°52.9	6.5'	S28°22.2	-1.3'	56.5'	45°	06:42	07:43	08:34	14:42	15:38	16:45	
13	14°18.8	06.6	19°18.3	6.4'	28°21.0	-1.4'	56.6'	N 40°	06:19	07:19	08:11	15:05	16:02	17:07	
14	29°18.5	06.5	33°43.8	6.4'	28°19.5	-1.6'	56.6'	35°	06:00	06:59	07:52	15:25	16:21	17:25	
15	44°18.2	· · 06.3	48°09.2	6.4'	28°17.9	-1.8'	56.6'	30°	05:43	06:42	07:36	15:41	16:38	17:41	
16	59°17.9	06.1	62°34.6	6.4'	28°16.2	-1.9'	56.6'	20°	05:15	06:13	07:09	16:10	17:07	18:07	
17	74°17.6	05.9	77°00.0	6.4'	28°14.3	-2.1'	56.7'	N 10°	04:52	05:49	06:45	16:34	17:31	18:30	
18	89°17.3	S23°05.7	91°25.4	6.4'	S28°12.2	-2.3'	56.7'	0°	04:30	05:26	06:23	16:56	17:54	18:51	
19	104°17.0	05.5	105°50.8	6.4'	28°09.9	-2.4'	56.7'	S 10°	04:08	05:03	06:01	17:19	18:16	19:12	
20	119°16.7	05.4	120°16.1	6.4'	28°07.5	-2.6'	56.7'	20°	03:44	04:39	05:38	17:43	18:40	19:34	
21	134°16.4	· · 05.2	134°41.5	6.4'	28°04.9	-2.8'	56.8'	30°	03:17	04:10	05:10	18:12	19:08	20:00	
22	149°16.1	05.0	149°06.9	6.4'	28°02.1	-2.9'	56.8'	35°	03:01	03:53	04:54	18:28	19:25	20:15	
23	164°15.8	04.8	163°32.3	6.4'	27°59.2	-3.1'	56.8'	40°	02:42	03:34	04:35	18:48	19:44	20:32	
SD = 16.3' d = -0.2'									45°	02:20	03:10	04:12	19:12	20:07	20:53
									S 50°	01:51	02:39	03:43	19:42	20:37	21:19
									52°	01:37	02:24	03:28	19:57	20:52	21:32
									54°	01:21	02:06	03:11	20:15	21:09	21:47
									56°	01:02	01:45	02:51	20:36	21:30	22:03
									58°	00:39	01:18	02:26	21:03	21:55	22:24
									S 60°	00:08	00:39	01:51	21:42	22:30	22:49
Tue	GHA	Dec	GHA	ν	Dec	d	HP	Day	Sun		Mer. Pass	Moon		Age	
									Eqn. of Time	12 ^h	Mer. Pass	Upper Mer. Pass.	Lower	28-1	
									00 ^h	mm:ss	hh:mm	hh:mm	hh:mm	4-0%	
0	179°15.5	S23°04.6	177°57.7	6.4'	S27°56.1	-3.3'	56.8'	29	02:00	02:15	12:02	10:43	23:11		
1	194°15.2	04.4	192°23.1	6.4'	27°52.8	-3.4'	56.9'	30	02:29	02:44	12:03	11:40	—:—		
2	209°14.9	04.2	206°48.5	6.4'	27°49.3	-3.6'	56.9'	31	02:58	03:12	12:03	12:37	00:08		
3	224°14.6	· · 04.1	221°13.9	6.4'	27°45.7	-3.8'	56.9'								
4	239°14.3	03.9	235°39.3	6.4'	27°41.9	-3.9'	56.9'								
5	254°14.0	03.7	250°04.8	6.5'	27°38.0	-4.1'	56.9'								
6	269°13.7	S23°03.5	264°30.2	6.5'	S27°33.9	-4.3'	57.0'								
7	284°13.4	03.3	278°55.7	6.5'	27°29.6	-4.5'	57.0'								
8	299°13.1	03.1	293°21.2	6.5'	27°25.1	-4.6'	57.0'								
9	314°12.8	· · 02.9	307°46.7	6.5'	27°20.5	-4.8'	57.0'								
10	329°12.5	02.7	322°12.2	6.6'	27°15.7	-4.9'	57.1'								
11	344°12.2	02.5	336°37.8	6.6'	27°10.8	-5.1'	57.1'								
12	359°11.9	S23°02.3	351°03.4	6.6'	S27°05.7	-5.3'	57.1'								
13	14°11.6	02.1	5°29.0	6.6'	27°00.4	-5.4'	57.1'								
14	29°11.3	01.9	19°54.6	6.7'	26°55.0	-5.6'	57.1'								
15	44°11.0	· · 01.7	34°20.3	6.7'	26°49.3	-5.8'	57.2'								
16	59°10.7	01.5	48°46.0	6.7'	26°43.6	-5.9'	57.2'								